

AutoBreaking Full Design-Book and Code-Book

> Generated automatically from the current repository snapshot.

> Goal: provide deep technical traceability with architecture, call-flow links, symbol map, and line-level impact not

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Scope

- Full repository code map for Rust source and tests.
- Architecture and runtime flow linking major modules.
- Per-file symbol inventory with line references.
- Line-by-line impact commentary for change-risk analysis.

Build and Feature Context

- name = "auto_breaking"
- version = "0.3.0"
- edition = "2021"
- name = "solver_bench"
- Key features: default, advanced_observability, new_integrator, vendor-windows.

System Call-Flow (High Level)

1. UI path: src/main.rs drives simulation ticks, visualizations, and control state updates.
2. CLI path: src/bin/simulator_cli.rs parses hardware/phase flags and calls io::production_program::run_all_phases.
3. Live flash path: io::production_program -> io::live_runner -> io::hw selected adapter -> io::vendor_cat_comm or so tcan/serial adapters.
4. UDS protocol path: io::live_runner and uds.rs coordinate 0x27/0x34/0x36/0x37 services, transport accounting, and cormance evidence.
5. Report path: production report JSON/Markdown emitted with gate and conformance summary.

File Inventory

Total files covered: **60**

- src/adas.rs
- src/autonomous.rs
- src/bin/cat_comm_bridge.rs
- src/bin/simulator_cli.rs
- src/boot_sequence.rs
- src/camera.rs
- src/can_bus.rs
- src/can_gateway.rs
- src/can_network.rs
- src/chassis.rs
- src/ecm_heavy.rs
- src/ecm_mock_provider.rs
- src/ecu_abs.rs
- src/ecu_bcm.rs
- src/ecu_ecm.rs
- src/ecu_hcm.rs
- src/ecu_icm.rs
- src/ecu_tcm.rs
- src/ecu_vcm.rs
- src/electrical.rs
- src/engine.rs
- src/gps.rs
- src/implement.rs
- src/imu.rs
- src/io/allowlist.rs
- src/io/artifact.rs
- src/io/ecm_params.rs
- src/io/hw.rs

- src/io/live_runner.rs
- src/io/metrics.rs
- src/io/mock.rs
- src/io/mod.rs
- src/io/production_program.rs
- src/io/rate_limiter.rs
- src/io/replay.rs
- src/io/serial_adapter.rs
- src/io/socketcan_adapter.rs
- src/io/vendor_cat_comm.rs
- src/j1939.rs
- src/leak_physics.rs
- src/lib.rs
- src/lidar.rs
- src/main.rs
- src/network_mgmt.rs
- src/nvm.rs
- src/observability.rs
- src/radar.rs
- src/sensors.rs
- src/sim_core.rs
- src/transmission.rs
- src/uds.rs
- src/v2x_telematics.rs
- src/vehicle.rs
- src/widgets.rs
- tests/io_mock.rs
- tests/leak_system_integration.rs
- tests/property_invariants.rs
- tests/speed_regression.rs
- tests/system_failure_scenarios.rs
- tests/vendor_bridge_e2e.rs

File: src/adas.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 252 | Non-empty: 219 | Symbols: 10

Symbol Inventory

Line	Kind	Symbol	Change Impact	
---:	---:	---:	---:	
5	struct	AdasModule	data/structure coupling	
48	enum	ParkZone	data/structure coupling	
55	impl	std::fmt::Display for ParkZone	behavior binding and method semantics	
56	fn	fmt	signature/API coupling	
66	impl	AdasModule	behavior binding and method semantics	
67	fn	new	signature/API coupling	
105	fn	update	signature/API coupling	
238	fn	toggle_acc	signature/API coupling	
246	fn	acc_speed_up	signature/API coupling	
249	fn	acc_speed_down	signature/API coupling	

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed	
---:	---:	---:	
1	use crate::sensors::SensorSuite;	imports symbols; changing path/name affects compile-time resolution.	
2	(blank)	line is blank; visual separation only.	
3	/// ADAS domain controller (SAE Level 2)	comment line; documentation intent, no runtime effect.	
4	#[derive(Debug, Clone)]	executable/support line; behavior impact depends on surrounding block context.	
5	pub struct AdasModule {	declares data layout; field changes may break serialization, tests, and callers.	
6	// Adaptive Cruise Control	comment line; documentation intent, no runtime effect.	
7	pub acc_enabled: bool,	executable/support line; behavior impact depends on surrounding block context.	
8	pub acc_set_speed: f64,	// km/h executable/support line; behavior impact depends on surrounding block	

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text. |
| 9 | pub acc_headway_time: f64, // s (gap to lead) | executable/support line; behavior impact depends on surrounding block context. |
| 10 | pub acc_throttle_out: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 11 | pub acc_brake_out: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 12 | acc_speed_error_integral: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 13 | (blank) | line is blank; visual separation only. |
| 14 | // Autonomous Emergency Braking | comment line; documentation intent, no runtime effect. |
| 15 | pub aeb_enabled: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 16 | pub aeb_pre_charge: bool, // brakes pre-charged | executable/support line; behavior impact depends on surrounding block context. |
| 17 | pub aeb_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 18 | pub aeb_ttc: f64, // seconds to collision | executable/support line; behavior impact depends on surrounding block context. |
| 19 | (blank) | line is blank; visual separation only. |
| 20 | // Lane Keeping Assist / Lane Departure Warning | comment line; documentation intent, no runtime effect. |
| 21 | pub lka_enabled: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 22 | pub lka_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 23 | pub lka_steering_correction: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 24 | pub ldw_warning: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 25 | (blank) | line is blank; visual separation only. |
| 26 | // Blind Spot Monitoring | comment line; documentation intent, no runtime effect. |
| 27 | pub bsm_enabled: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 28 | pub bsm_left_warning: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 29 | pub bsm_right_warning: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 30 | (blank) | line is blank; visual separation only. |
| 31 | // Park Assist (front/rear ultrasonic guidance) | comment line; documentation intent, no runtime effect. |
| 32 | pub park_assist_enabled: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 33 | pub park_assist_zone: ParkZone, | executable/support line; behavior impact depends on surrounding block context. |
| 34 | (blank) | line is blank; visual separation only. |
| 35 | // Traffic Sign Recognition | comment line; documentation intent, no runtime effect. |
| 36 | pub tsr_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 37 | pub tsr_speed_limit: Option<u32>, | executable/support line; behavior impact depends on surrounding block context. |
| 38 | (blank) | line is blank; visual separation only. |
| 39 | // Driver Attention Monitor (simulated) | comment line; documentation intent, no runtime effect. |
| 40 | pub dam_drowsiness: f64, // 0-1 | executable/support line; behavior impact depends on surrounding block context. |
| 41 | pub dam_alert: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 42 | (blank) | line is blank; visual separation only. |
| 43 | // Autonomous level: 0=none, 1=partial, 2=combined | comment line; documentation intent, no runtime effect. |
| 44 | pub autonomy_level: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 45 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 46 | (blank) | line is blank; visual separation only. |
| 47 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding block context. |
| 48 | pub enum ParkZone { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 49 | Clear, | executable/support line; behavior impact depends on surrounding block context. |
| 50 | Caution, | executable/support line; behavior impact depends on surrounding block context. |
| 51 | Warning, | executable/support line; behavior impact depends on surrounding block context. |
| 52 | Critical, | executable/support line; behavior impact depends on surrounding block context. |
| 53 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 54 | (blank) | line is blank; visual separation only. |
| 55 | impl std::fmt::Display for ParkZone { | implementation block; method behavior and trait conformance are defined here. |
| 56 | fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 57 | match self { | branch dispatch; changing arms alters behavior for variants and input classes. |

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| 58 |           ParkZone::Clear => write!(f, "CLEAR"), | I/O or protocol-critical logic; edits can affect integrat
, timing, and diagnostic outcomes. |
| 59 |           ParkZone::Caution => write!(f, "CAUTION"), | I/O or protocol-critical logic; edits can affect inte
tion, timing, and diagnostic outcomes. |
| 60 |           ParkZone::Warning => write!(f, "WARNING"), | I/O or protocol-critical logic; edits can affect inte
tion, timing, and diagnostic outcomes. |
| 61 |           ParkZone::Critical => write!(f, "CRITICAL"), | I/O or protocol-critical logic; edits can affect in
ration, timing, and diagnostic outcomes. |
| 62 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 63 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 64 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 65 | (blank) | line is blank; visual separation only. |
| 66 | impl AdasModule { | implementation block; method behavior and trait conformance are defined here. |
| 67 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
tputs. |
| 68 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 69 |             acc_enabled: false, | executable/support line; behavior impact depends on surrounding block context
|
| 70 |             acc_set_speed: 80.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 71 |             acc_headway_time: 2.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 72 |             acc_throttle_out: 0.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 73 |             acc_brake_out: 0.0, | executable/support line; behavior impact depends on surrounding block context
|
| 74 |             acc_speed_error_integral: 0.0, | executable/support line; behavior impact depends on surrounding b
k context. |
| 75 | (blank) | line is blank; visual separation only. |
| 76 |             aeb_enabled: true, | executable/support line; behavior impact depends on surrounding block context
|
| 77 |             aeb_pre_charge: false, | executable/support line; behavior impact depends on surrounding block con
t. |
| 78 |             aeb_active: false, | executable/support line; behavior impact depends on surrounding block context
|
| 79 |             aeb_ttc: f64::INFINITY, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 80 | (blank) | line is blank; visual separation only. |
| 81 |             lka_enabled: false, | executable/support line; behavior impact depends on surrounding block context
|
| 82 |             lka_active: false, | executable/support line; behavior impact depends on surrounding block context
|
| 83 |             lka_steering_correction: 0.0, | executable/support line; behavior impact depends on surrounding bl
context. |
| 84 |             ldw_warning: false, | executable/support line; behavior impact depends on surrounding block context
|
| 85 | (blank) | line is blank; visual separation only. |
| 86 |             bsm_enabled: true, | executable/support line; behavior impact depends on surrounding block context
|
| 87 |             bsm_left_warning: false, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 88 |             bsm_right_warning: false, | executable/support line; behavior impact depends on surrounding block
text. |
| 89 | (blank) | line is blank; visual separation only. |
| 90 |             park_assist_enabled: false, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 91 |             park_assist_zone: ParkZone::Clear, | executable/support line; behavior impact depends on surroundi
block context. |
| 92 | (blank) | line is blank; visual separation only. |
| 93 |             tsr_active: true, | executable/support line; behavior impact depends on surrounding block context.
|
| 94 |             tsr_speed_limit: Some(80), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 95 | (blank) | line is blank; visual separation only. |
| 96 |             dam_drowsiness: 0.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 97 |             dam_alert: false, | executable/support line; behavior impact depends on surrounding block context.
|
| 98 | (blank) | line is blank; visual separation only. |
| 99 |             autonomy_level: 2, | executable/support line; behavior impact depends on surrounding block context
|
| 100 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope

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|
| 101 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 102 | (blank) | line is blank; visual separation only. |
| 103 |     /// Returns (throttle_override, brake_override, steering_override) | comment line; documentation intent,
runtime effect. |
| 104 |     /// None means no override for that channel. | comment line; documentation intent, no runtime effect. |
| 105 |     pub fn update( | declares executable behavior; signature/body changes alter API, control flow, and output
|
| 106 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 107 |         sensors: &SensorSuite, | executable/support line; behavior impact depends on surrounding block context
|
| 108 |         vehicle_speed: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 109 |         _user_throttle: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 110 |         user_brake: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 111 |         dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 112 |     ) -> (Option<f64>, Option<f64>, Option<f64>) { | executable/support line; behavior impact depends on surro
ding block context. |
| 113 |         let mut throttle_out: Option<f64> = None; | executable/support line; behavior impact depends on surro
ing block context. |
| 114 |         let mut brake_out: Option<f64> = None; | executable/support line; behavior impact depends on surround
block context. |
| 115 |         let mut steer_out: Option<f64> = None; | executable/support line; behavior impact depends on surround
block context. |
| 116 | (blank) | line is blank; visual separation only. |
| 117 |         // ----- | comment line; documentation i
nt, no runtime effect. |
| 118 |         // AEB – highest priority | comment line; documentation intent, no runtime effect. |
| 119 |         // ----- | comment line; documentation i
nt, no runtime effect. |
| 120 |         self.aeb_ttc = f64::INFINITY; | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 121 |         self.aeb_pre_charge = false; | executable/support line; behavior impact depends on surrounding block
text. |
| 122 |         self.aeb_active = false; | executable/support line; behavior impact depends on surrounding block cont
. |
| 123 | (blank) | line is blank; visual separation only. |
| 124 |         if self.aeb_enabled { | conditional gate; changing condition shifts runtime path and side effects. |
| 125 |             // Find closest in-path forward target | comment line; documentation intent, no runtime effect. |
| 126 |             if let Some(target) = sensors | conditional gate; changing condition shifts runtime path and side
ffects. |
| 127 |                 .forward_targets | executable/support line; behavior impact depends on surrounding block cont
. |
| 128 |                 .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 129 |                 .filter(|t| t.is_in_path && t.relative_speed > 0.0) | executable/support line; behavior imp
depends on surrounding block context. |
| 130 |                 .min_by(|a, b| a.distance.total_cmp(&b.distance)) | executable/support line; behavior impac
depends on surrounding block context. |
| 131 |             { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 132 |                 let ttc = target.distance / (target.relative_speed / 3.6).max(0.01); | executable/support lin
behavior impact depends on surrounding block context. |
| 133 |                 self.aeb_ttc = ttc; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 134 | (blank) | line is blank; visual separation only. |
| 135 |                 if ttc < 2.5 { | conditional gate; changing condition shifts runtime path and side effects. |
| 136 |                     self.aeb_pre_charge = true; | executable/support line; behavior impact depends on surroun
g block context. |
| 137 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 138 |                 if ttc < 1.8 { | conditional gate; changing condition shifts runtime path and side effects. |
| 139 |                     // Partial braking | comment line; documentation intent, no runtime effect. |
| 140 |                     let partial = ((2.5 - ttc) / 1.5).clamp(0.0, 0.6); | executable/support line; behavior im
t depends on surrounding block context. |
| 141 |                     brake_out = Some(partial.max(user_brake)); | executable/support line; behavior impact dep
s on surrounding block context. |

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| 142 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 143 |           if ttc < 0.9 { | conditional gate; changing condition shifts runtime path and side effects. |
| 144 |               // Full AEB | comment line; documentation intent, no runtime effect. |
| 145 |               self.aeb_active = true; | executable/support line; behavior impact depends on surrounding
ock context. |
| 146 |               brake_out = Some(1.0); | executable/support line; behavior impact depends on surrounding
ck context. |
| 147 |               throttle_out = Some(0.0); | executable/support line; behavior impact depends on surroundi
block context. |
| 148 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 149 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 150 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 151 | (blank) | line is blank; visual separation only. |
| 152 |       // ----- | comment line; documentation intent, no runtime effect. |
nt, no runtime effect. |
| 153 |       // ACC – overrides throttle/brake when enabled | comment line; documentation intent, no runtime effec
|
| 154 |       // ----- | comment line; documentation intent, no runtime effect. |
nt, no runtime effect. |
| 155 |       if self.acc_enabled && !self.aeb_active { | conditional gate; changing condition shifts runtime path
side effects. |
| 156 |           let lead = sensors | executable/support line; behavior impact depends on surrounding block context
|
| 157 |           .forward_targets | executable/support line; behavior impact depends on surrounding block context
. |
| 158 |           .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 159 |           .filter(\t| t.is_in_path) | executable/support line; behavior impact depends on surrounding
ock context. |
| 160 |           .min_by(\a, b| a.distance.total_cmp(&b.distance)); | executable/support line; behavior impact
depends on surrounding block context. |
| 161 | (blank) | line is blank; visual separation only. |
| 162 |           let (_target_speed, speed_error) = if let Some(t) = lead { | executable/support line; behavior im
t depends on surrounding block context. |
| 163 |               let safe_dist = vehicle_speed / 3.6 * self.acc_headway_time; | executable/support line; behav
impact depends on surrounding block context. |
| 164 |               let dist_error = t.distance - safe_dist; | executable/support line; behavior impact depends o
urrounding block context. |
| 165 |               // Blend: follow gap OR speed, whichever is more restrictive | comment line; documentation in
t, no runtime effect. |
| 166 |               let gap_speed = vehicle_speed + dist_error * 0.4; | executable/support line; behavior impact
ends on surrounding block context. |
| 167 |               let tgt = gap_speed.min(self.acc_set_speed).max(0.0); | executable/support line; behavior imp
depends on surrounding block context. |
| 168 |               (tgt, tgt - vehicle_speed) | executable/support line; behavior impact depends on surrounding
ck context. |
| 169 |           } else { | executable/support line; behavior impact depends on surrounding block context. |
| 170 |               (self.acc_set_speed, self.acc_set_speed - vehicle_speed) | executable/support line; behavior
act depends on surrounding block context. |
| 171 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 172 | (blank) | line is blank; visual separation only. |
| 173 |       // PI speed controller | comment line; documentation intent, no runtime effect. |
| 174 |       self.acc_speed_error_integral += speed_error * dt; | executable/support line; behavior impact dep
s on surrounding block context. |
| 175 |       self.acc_speed_error_integral = self.acc_speed_error_integral.clamp(-30.0, 30.0); | executable/su
rt line; behavior impact depends on surrounding block context. |
| 176 |       let control = speed_error * 0.05 + self.acc_speed_error_integral * 0.002; | executable/support li
behavior impact depends on surrounding block context. |
| 177 | (blank) | line is blank; visual separation only. |
| 178 |       if control > 0.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 179 |           self.acc_throttle_out = control.clamp(0.0, 1.0); | executable/support line; behavior impact d

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nds on surrounding block context. |
| 180 |         self.acc_brake_out = 0.0; | executable/support line; behavior impact depends on surrounding b
k context. |
| 181 |         throttle_out = Some(self.acc_throttle_out); | executable/support line; behavior impact depend
n surrounding block context. |
| 182 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 183 |         self.acc_brake_out = (-control).clamp(0.0, 0.8); | executable/support line; behavior impact d
nds on surrounding block context. |
| 184 |         self.acc_throttle_out = 0.0; | executable/support line; behavior impact depends on surroundin
lock context. |
| 185 |         throttle_out = Some(0.0); | executable/support line; behavior impact depends on surrounding b
k context. |
| 186 |         brake_out = Some(self.acc_brake_out.max(brake_out.unwrap_or(0.0))); | executable/support line
ehavior impact depends on surrounding block context. |
| 187 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 188 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 189 | (blank) | line is blank; visual separation only. |
| 190 | // ----- | comment line; documentation intent, no runtime effect. |
| 191 | // LKA – steering correction | comment line; documentation intent, no runtime effect. |
| 192 | // ----- | comment line; documentation intent, no runtime effect. |
| 193 |     self.lka_active = false; | executable/support line; behavior impact depends on surrounding block cont
. |
| 194 |     self.lka_steering_correction = 0.0; | executable/support line; behavior impact depends on surrounding
ock context. |
| 195 |     if self.lka_enabled && vehicle_speed > 30.0 { | conditional gate; changing condition shifts runtime p
and side effects. |
| 196 |         let correction = -sensors.lane_offset * 4.0; // proportional | executable/support line; behavior
act depends on surrounding block context. |
| 197 |         if correction.abs() > 0.3 { | conditional gate; changing condition shifts runtime path and side e
cts. |
| 198 |             self.lka_active = true; | executable/support line; behavior impact depends on surrounding blo
context. |
| 199 |             self.lka_steering_correction = correction.clamp(-5.0, 5.0); | executable/support line; behavi
impact depends on surrounding block context. |
| 200 |             steer_out = Some(self.lka_steering_correction); | executable/support line; behavior impact de
ds on surrounding block context. |
| 201 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 202 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 203 |     self.ldw_warning = sensors.lane_departure_left \\| sensors.lane_departure_right; | executable/support
ine; behavior impact depends on surrounding block context. |
| 204 | (blank) | line is blank; visual separation only. |
| 205 | // ----- | comment line; documentation intent, no runtime effect. |
| 206 | // BSM | comment line; documentation intent, no runtime effect. |
| 207 | // ----- | comment line; documentation intent, no runtime effect. |
| 208 |     if self.bsm_enabled { | conditional gate; changing condition shifts runtime path and side effects. |
| 209 |         self.bsm_left_warning = sensors.bsm_left; | executable/support line; behavior impact depends on s
ounding block context. |
| 210 |         self.bsm_right_warning = sensors.bsm_right; | executable/support line; behavior impact depends on
rrounding block context. |
| 211 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 212 | (blank) | line is blank; visual separation only. |
| 213 | // ----- | comment line; documentation intent, no runtime effect. |
| 214 | // Park Assist | comment line; documentation intent, no runtime effect. |
| 215 | // ----- | comment line; documentation intent, no runtime effect. |

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| 216 |         let front_min = sensors.ultrasonic[..3] | executable/support line; behavior impact depends on surround
g block context. |
| 217 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 218 |         .cloned() | executable/support line; behavior impact depends on surrounding block context. |
| 219 |         .fold(f64::INFINITY, f64::min); | executable/support line; behavior impact depends on surrounding
ock context. |
| 220 |         self.park_assist_zone = if front_min > 2.0 { | executable/support line; behavior impact depends on su
unding block context. |
| 221 |             ParkZone::Clear | executable/support line; behavior impact depends on surrounding block context.
| 222 |         } else if front_min > 1.0 { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 223 |             ParkZone::Caution | executable/support line; behavior impact depends on surrounding block context
| 224 |         } else if front_min > 0.4 { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 225 |             ParkZone::Warning | executable/support line; behavior impact depends on surrounding block context
| 226 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 227 |             ParkZone::Critical | executable/support line; behavior impact depends on surrounding block context
|
| 228 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 229 | (blank) | line is blank; visual separation only. |
| 230 |     // ----- | comment line; documentation intent, no runtime effect. |
| 231 |     // TSR | comment line; documentation intent, no runtime effect. |
| 232 |     // ----- | comment line; documentation intent, no runtime effect. |
| 233 |     self.tsr_speed_limit = sensors.speed_limit_kmh; | executable/support line; behavior impact depends on
rounding block context. |
| 234 | (blank) | line is blank; visual separation only. |
| 235 |     (throttle_out, brake_out, steer_out) | executable/support line; behavior impact depends on surroundin
lock context. |
| 236 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 237 | (blank) | line is blank; visual separation only. |
| 238 | pub fn toggle_acc(&mut self, current_speed: f64) { | declares executable behavior; signature/body changes
ter API, control flow, and outputs. |
| 239 |     self.acc_enabled = !self.acc_enabled; | executable/support line; behavior impact depends on surroundi
block context. |
| 240 |     if self.acc_enabled { | conditional gate; changing condition shifts runtime path and side effects. |
| 241 |         self.acc_set_speed = current_speed.max(30.0); | executable/support line; behavior impact depends
surrounding block context. |
| 242 |         self.acc_speed_error_integral = 0.0; | executable/support line; behavior impact depends on surroun
ng block context. |
| 243 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 244 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 245 | (blank) | line is blank; visual separation only. |
| 246 | pub fn acc_speed_up(&mut self) { | declares executable behavior; signature/body changes alter API, contro
low, and outputs. |
| 247 |     self.acc_set_speed = (self.acc_set_speed + 5.0).min(200.0); | executable/support line; behavior impac
depends on surrounding block context. |
| 248 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 249 | pub fn acc_speed_down(&mut self) { | declares executable behavior; signature/body changes alter API, cont
flow, and outputs. |
| 250 |     self.acc_set_speed = (self.acc_set_speed - 5.0).max(0.0); | executable/support line; behavior impact
ends on surrounding block context. |
| 251 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 252 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/autonomous.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 647 | Non-empty: 583 | Symbols: 37

Symbol Inventory


```
| Line | Kind | Symbol | Change | Impact |
|---:|---|---|---|---|
| 6 | enum | SaeLevel | data/structure coupling |
| 15 | impl | std::fmt::Display for SaeLevel | behavior binding and method semantics |
| 16 | fn | fmt | signature/API coupling |
| 30 | enum | FeatureState | data/structure coupling |
| 38 | impl | std::fmt::Display for FeatureState | behavior binding and method semantics |
| 39 | fn | fmt | signature/API coupling |
| 52 | struct | LaneInfo | data/structure coupling |
| 66 | enum | LaneMarkType | data/structure coupling |
| 74 | impl | std::fmt::Display for LaneMarkType | behavior binding and method semantics |
| 75 | fn | fmt | signature/API coupling |
| 89 | struct | Waypoint | data/structure coupling |
| 97 | struct | AutonomousController | data/structure coupling |
| 158 | enum | LcaDirection | data/structure coupling |
| 165 | enum | SafetyStatus | data/structure coupling |
| 172 | impl | std::fmt::Display for SafetyStatus | behavior binding and method semantics |
| 173 | fn | fmt | signature/API coupling |
| 183 | impl | Default for AutonomousController | behavior binding and method semantics |
| 184 | fn | default | signature/API coupling |
| 189 | impl | AutonomousController | behavior binding and method semantics |
| 190 | fn | new | signature/API coupling |
| 245 | fn | tick | signature/API coupling |
| 342 | fn | run_aeb | signature/API coupling |
| 367 | fn | run_acc | signature/API coupling |
| 417 | fn | run_lka | signature/API coupling |
| 438 | fn | run_tja | signature/API coupling |
| 461 | fn | update_path | signature/API coupling |
| 478 | fn | safety_check | signature/API coupling |
| 502 | fn | engage | signature/API coupling |
| 514 | fn | disengage | signature/API coupling |
| 523 | fn | toggle_lka | signature/API coupling |
| 531 | fn | set_acc_speed | signature/API coupling |
| 534 | fn | set_headway | signature/API coupling |
| 540 | module | tests | data/structure coupling |
| 544 | fn | disengaged_controller_outputs_none_and_zero_state | signature/API coupling |
| 569 | fn | engage_enables_l2_features | signature/API coupling |
| 580 | fn | engage_from_standstill_requests_positive_throttle_on_clear_road | signature/API coupling |
| 606 | fn | no_lead_then_close_lead_transitions_to_brake | signature/API coupling |
```

```
| Line | Source | Operational Impact If Changed |
|---:|---|---|
| 1 | //! Autonomous Driving Controller – SAE Level 2/3 AD system. | comment line; documentation intent, no runtime effect. |
| 2 | //! Implements: ACC, AEB, LKA, LCA, TJA, Pilot Assist, ISO 26262 ASIL-D safety. | comment line; documentation intent, no runtime effect. |
| 3 | (blank) | line is blank; visual separation only. |
| 4 | // ■■ SAE Automation Level ████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 5 | #[derive(Debug, Clone, Copy, PartialEq, PartialOrd)] | executable/support line; behavior impact depends on surrounding block context. |
| 6 | pub enum SaeLevel { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
|   | . |
| 7 |     L0, // No automation | executable/support line; behavior impact depends on surrounding block context. |
| 8 |     L1, // Driver assistance (ACC OR LKA, not both) | executable/support line; behavior impact depends on surrounding block context. |
| 9 |     L2, // Partial automation (ACC + LKA simultaneously) – Tesla Autopilot equivalent | executable/support line; behavior impact depends on surrounding block context. |
|10 |     L3, // Conditional automation – system can handle some scenarios fully | executable/support line; behavior impact depends on surrounding block context. |
|11 |     L4, // High automation – geofenced | executable/support line; behavior impact depends on surrounding block context. |
|12 |     L5, // Full automation | executable/support line; behavior impact depends on surrounding block context. |
```

[illegible]

[illegible]

```

103 | // ■ ADAS Features ████████████████████████████████████████████████████ | comment lin
ntent, no runtime effect. |
| 104 | pub acc_state: FeatureState, | executable/support line; behavior impact depends on surrounding block cont
. |
| 105 | pub acc_set_speed_kmh: f64, | executable/support line; behavior impact depends on surrounding block conte
|
| 106 | pub acc_headway_s: f64, // target time headway (1.5-2.5s typical) | executable/support line; behavior imp
depends on surrounding block context. |
| 107 | acc_speed_error_integral: f64, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 108 | (blank) | line is blank; visual separation only. |
| 109 | pub aeb_state: FeatureState, | executable/support line; behavior impact depends on surrounding block cont
. |
| 110 | pub aeb_brake_cmd: f64, // 0-1 brake command | executable/support line; behavior impact depends on surroun
ng block context. |
| 111 | pub aeb_pre_charge: bool, | executable/support line; behavior impact depends on surrounding block context
| 112 | (blank) | line is blank; visual separation only. |
| 113 | pub lka_state: FeatureState, | executable/support line; behavior impact depends on surrounding block cont
. |
| 114 | pub lka_steer_cmd: f64, // steering torque Nm | executable/support line; behavior impact depends on surro
ing block context. |
| 115 | (blank) | line is blank; visual separation only. |
| 116 | pub lca_state: FeatureState, // Lane Change Assist | executable/support line; behavior impact depends on
rounding block context. |
| 117 | pub lca_direction: LcaDirection, | executable/support line; behavior impact depends on surrounding block
text. |
| 118 | (blank) | line is blank; visual separation only. |
| 119 | pub tja_state: FeatureState, // Traffic Jam Assist (stop-and-go) | executable/support line; behavior impa
depends on surrounding block context. |
| 120 | pub tja_stopped_timer: f64, | executable/support line; behavior impact depends on surrounding block conte
|
| 121 | (blank) | line is blank; visual separation only. |
| 122 | pub pilot_state: FeatureState, // Highway Pilot (L3) | executable/support line; behavior impact depends o
urrounding block context. |
| 123 | (blank) | line is blank; visual separation only. |
| 124 | pub bsm_left: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 125 | pub bsm_right: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 126 | (blank) | line is blank; visual separation only. |
| 127 | // ■ Sensor inputs ████████████████████████████████████████████████████ | comment lin
ntent, no runtime effect. |
| 128 | pub lane: LaneInfo, | executable/support line; behavior impact depends on surrounding block context. |
| 129 | pub lead_range_m: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 130 | pub lead_speed_ms: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 131 | pub ttc_s: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 132 | pub thw_s: f64, // time headway (distance / speed) | executable/support line; behavior impact depends on
rounding block context. |
| 133 | (blank) | line is blank; visual separation only. |
| 134 | // ■ Control outputs ████████████████████████████████████████████████████ | comment lin
ntent, no runtime effect. |
| 135 | pub throttle_cmd: f64, // 0-1 | executable/support line; behavior impact depends on surrounding block co
xt. |
| 136 | pub brake_cmd: f64, // 0-1 | executable/support line; behavior impact depends on surrounding block co
xt. |
| 137 | pub steer_cmd_deg: f64, // target steering angle | executable/support line; behavior impact depends on su
unding block context. |
| 138 | (blank) | line is blank; visual separation only. |
| 139 | // ■ Path planning ████████████████████████████████████████████████████ | comment lin
ntent, no runtime effect. |
| 140 | pub planned_path: Vec<Waypoint>, | executable/support line; behavior impact depends on surrounding block
text. |
| 141 | pub path_valid: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 142 | (blank) | line is blank; visual separation only. |
| 143 | // ■ Driver monitoring ████████████████████████████████████████████████████ | comment lin
ntent, no runtime effect. |
| 144 | pub driver_alert: bool, // attention warning | executable/support line; behavior impact depends on surro

```

[illegible]

```

| 190 | pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
outputs. |
| 191 | AutonomousController { | executable/support line; behavior impact depends on surrounding block context. |
| 192 |     sae_level: SaeLevel::L2, | executable/support line; behavior impact depends on surrounding block context. |
| 193 |     engaged: false, | executable/support line; behavior impact depends on surrounding block context. |
| 194 |     driver_hands_on: true, | executable/support line; behavior impact depends on surrounding block context. |
| 195 |     acc_state: FeatureState::Off, | executable/support line; behavior impact depends on surrounding block context. |
| 196 |     acc_set_speed_kmh: 100.0, | executable/support line; behavior impact depends on surrounding block context. |
| 197 |     acc_headway_s: 2.0, | executable/support line; behavior impact depends on surrounding block context. |
| 198 |     acc_speed_error_integral: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 199 |     aeb_state: FeatureState::Ready, | executable/support line; behavior impact depends on surrounding block context. |
| 200 |     aeb_brake_cmd: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 201 |     aeb_pre_charge: false, | executable/support line; behavior impact depends on surrounding block context. |
| 202 |     lka_state: FeatureState::Off, | executable/support line; behavior impact depends on surrounding block context. |
| 203 |     lka_steer_cmd: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 204 |     lca_state: FeatureState::Off, | executable/support line; behavior impact depends on surrounding block context. |
| 205 |     lca_direction: LcaDirection::None, | executable/support line; behavior impact depends on surrounding block context. |
| 206 |     tja_state: FeatureState::Off, | executable/support line; behavior impact depends on surrounding block context. |
| 207 |     tja_stopped_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 208 |     pilot_state: FeatureState::Off, | executable/support line; behavior impact depends on surrounding block context. |
| 209 |     bsm_left: false, | executable/support line; behavior impact depends on surrounding block context. |
| 210 |     bsm_right: false, | executable/support line; behavior impact depends on surrounding block context. |
| 211 |     lane: LaneInfo { | executable/support line; behavior impact depends on surrounding block context. |
| 212 |         detected: true, | executable/support line; behavior impact depends on surrounding block context. |
| 213 |         confidence: 0.95, | executable/support line; behavior impact depends on surrounding block context. |
| 214 |         offset_m: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 215 |         heading_error_deg: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 216 |         curvature_1_m: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 217 |         lane_width_m: 3.5, | executable/support line; behavior impact depends on surrounding block context. |
| 218 |         left_type: LaneMarkType::Dashed, | executable/support line; behavior impact depends on surrounding block context. |
| 219 |         right_type: LaneMarkType::Solid, | executable/support line; behavior impact depends on surrounding block context. |
| 220 |         adjacent_left: true, | executable/support line; behavior impact depends on surrounding block context. |
| 221 |         adjacent_right: false, | executable/support line; behavior impact depends on surrounding block context. |
| 222 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 223 |     lead_range_m: f64::INFINITY, | executable/support line; behavior impact depends on surrounding block context. |
| 224 |     lead_speed_ms: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 225 | }

```

```

| 225 |         ttc_s: f64::INFINITY, | executable/support line; behavior impact depends on surrounding block context. |
| 226 |         thw_s: f64::INFINITY, | executable/support line; behavior impact depends on surrounding block context. |
| 227 |         throttle_cmd: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 228 |         brake_cmd: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 229 |         steer_cmd_deg: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 230 |         planned_path: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 231 |         path_valid: false, | executable/support line; behavior impact depends on surrounding block context. |
| 232 |         driver_alert: false, | executable/support line; behavior impact depends on surrounding block context. |
| 233 |         hands_off_s: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 234 |         drowsiness_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 235 |         safety_status: SafetyStatus::Normal, | executable/support line; behavior impact depends on surrounding block context. |
| 236 |         degrade_reason: None, | executable/support line; behavior impact depends on surrounding block context. |
| 237 |         noise_t: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 238 |         lka_integral: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 239 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
| 240 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 241 | (blank) | line is blank; visual separation only. |
| 242 |     /// Master AD controller tick. | comment line; documentation intent, no runtime effect. |
| 243 |     /// Returns (throttle, brake, steer) overrides – None if no override. | comment line; documentation intent, no runtime effect. |
| 244 |     #[allow(clippy::too_many_arguments)] | executable/support line; behavior impact depends on surrounding block context. |
| 245 |     pub fn tick( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 246 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 247 |         ego_speed_kmh: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 248 |         _user_throttle: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 249 |         _user_brake: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 250 |         lead_range_m: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 251 |         lead_speed_ms: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 252 |         lane_offset_m: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 253 |         lane_heading_err: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 254 |         lane_confidence: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 255 |         ttc_s: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 256 |         bsm_left: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 257 |         bsm_right: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 258 |         dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 259 |     ) -> (Option<f64>, Option<f64>, Option<f64>) { | executable/support line; behavior impact depends on surrounding block context. |
| 260 |         self.noise_t += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 261 |         let ego_speed_ms = ego_speed_kmh / 3.6; | executable/support line; behavior impact depends on surrounding block context. |
| 262 | (blank) | line is blank; visual separation only. |
| 263 |         // Update inputs | comment line; documentation intent, no runtime effect. |
| 264 |         self.lead_range_m = lead_range_m; | executable/support line; behavior impact depends on surrounding block context. |
| 265 |         self.lead_speed_ms = lead_speed_ms; | executable/support line; behavior impact depends on surrounding block context. |
| 266 |         self.ttc_s = ttc_s; | executable/support line; behavior impact depends on surrounding block context. |
| 267 |         self.thw_s = if ego_speed_ms > 0.1 { | executable/support line; behavior impact depends on surrounding block context. |
| 268 |             lead_range_m / ego_speed_ms | executable/support line; behavior impact depends on surrounding block context. |

```

```
| 269 |         } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 270 |             f64::INFINITY | executable/support line; behavior impact depends on surrounding block context. |  
| 271 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 272 |     self.bsm_left = bsm_left; | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 273 |     self.bsm_right = bsm_right; | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 274 |     self.lane.offset_m = lane_offset_m; | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 275 |     self.lane.heading_error_deg = lane_heading_err; | executable/support line; behavior impact depends on  
rrounding block context. |  
| 276 |     self.lane.confidence = lane_confidence; | executable/support line; behavior impact depends on surroun  
g block context. |  
| 277 |     self.lane.detected = lane_confidence > 0.4; | executable/support line; behavior impact depends on surr  
nding block context. |  
| 278 | (blank) | line is blank; visual separation only. |  
| 279 | // Driver hands-off timer | comment line; documentation intent, no runtime effect. |  
| 280 | if self.driver_hands_on { | conditional gate; changing condition shifts runtime path and side effects  
| 281 |     self.hands_off_s = 0.0; | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 282 | } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 283 |     self.hands_off_s += dt; | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 284 |     self.driver_alert = self.hands_off_s > 10.0; | executable/support line; behavior impact depends o  
urrounding block context. |  
| 285 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 286 | (blank) | line is blank; visual separation only. |  
| 287 | // Drowsiness simulation | comment line; documentation intent, no runtime effect. |  
| 288 | let n = self.noise_t; | executable/support line; behavior impact depends on surrounding block context.  
| 289 | let noise = \|s: f64\| ((s * 127.1 + 311.7).sin()) * 43758.5 % 1.0 - 0.5) * 2.0; | executable/support  
e; behavior impact depends on surrounding block context. |  
| 290 | self.drowsiness_pct = (self.drowsiness_pct + noise(n * 0.001) * 0.01).clamp(0.0, 100.0); | executable  
pport line; behavior impact depends on surrounding block context. |  
| 291 | (blank) | line is blank; visual separation only. |  
| 292 | if !self.engaged { | conditional gate; changing condition shifts runtime path and side effects. |  
| 293 |     self.throttle_cmd = 0.0; | executable/support line; behavior impact depends on surrounding block  
text. |  
| 294 |     self.brake_cmd = 0.0; | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 295 |     self.steer_cmd_deg = 0.0; | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 296 |     return (None, None, None); | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 297 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 298 | (blank) | line is blank; visual separation only. |  
| 299 | let mut thr: Option<f64> = None; | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 300 | let mut brk: Option<f64> = None; | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 301 | let mut str: Option<f64> = None; | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 302 | (blank) | line is blank; visual separation only. |  
| 303 | // ■■■ AEB – ALWAYS active regardless of engagement ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ | comment line; docum  
intent, no runtime effect. |  
| 304 | self.run_aeb(ego_speed_ms, ttc_s, &mut brk); | executable/support line; behavior impact depends on su  
unding block context. |  
| 305 | (blank) | line is blank; visual separation only. |  
| 306 | // ■■■ ACC – Adaptive Cruise Control ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ | comment line; d  
intent, no runtime effect. |  
| 307 | if matches!(self.acc_state, FeatureState::Active) { | conditional gate; changing condition shifts run  
e path and side effects. |  
| 308 |     self.run_acc(| executable/support line; behavior impact depends on surrounding block context.
```


[illegible]

```

| 350 | (blank) | line is blank; visual separation only. |
| 351 |         self.aeb_pre_charge = ttc_s < 2.8 && speed_ms > 2.0; | executable/support line; behavior impact depends
on surrounding block context. |
| 352 |         self.aeb_brake_cmd = 0.0; | executable/support line; behavior impact depends on surrounding block con
t. |
| 353 | (blank) | line is blank; visual separation only. |
| 354 |         if ttc_s < 2.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 355 |             // Partial braking phase | comment line; documentation intent, no runtime effect. |
| 356 |             let partial = ((2.5 - ttc_s) / 1.5).clamp(0.0, 0.6); | executable/support line; behavior impact d
nds on surrounding block context. |
| 357 |             self.aeb_brake_cmd = partial; | executable/support line; behavior impact depends on surrounding b
k context. |
| 358 |             *brk = Some(partial.max(brk.unwrap_or(0.0))); | executable/support line; behavior impact depends
surrounding block context. |
| 359 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 360 |         if ttc_s < 1.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 361 |             // Full emergency braking | comment line; documentation intent, no runtime effect. |
| 362 |             self.aeb_brake_cmd = 1.0; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 363 |             *brk = Some(1.0); | executable/support line; behavior impact depends on surrounding block context
|
| 364 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 365 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 366 | (blank) | line is blank; visual separation only. |
| 367 |     fn run_acc( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 368 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 369 |         speed_ms: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 370 |         lead_range: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 371 |         _lead_speed: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 372 |         dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 373 |         thr: &mut Option<f64>, | executable/support line; behavior impact depends on surrounding block context
|
| 374 |         brk: &mut Option<f64>, | executable/support line; behavior impact depends on surrounding block context
|
| 375 |     ) { | executable/support line; behavior impact depends on surrounding block context. |
| 376 |         let set_speed = self.acc_set_speed_kmh / 3.6; | executable/support line; behavior impact depends on s
ounding block context. |
| 377 | (blank) | line is blank; visual separation only. |
| 378 |         // Target speed: minimum of set speed and speed needed to maintain headway | comment line; documentat
intent, no runtime effect. |
| 379 |         let safe_dist = speed_ms * self.acc_headway_s; | executable/support line; behavior impact depends on
rounding block context. |
| 380 |         let dist_error = lead_range - safe_dist; | executable/support line; behavior impact depends on surrou
ng block context. |
| 381 |         let lead_valid = lead_range.is_finite(); | executable/support line; behavior impact depends on surrou
ng block context. |
| 382 |         let gap_speed = if lead_valid { | executable/support line; behavior impact depends on surrounding blo
context. |
| 383 |             (speed_ms + dist_error * 0.5).clamp(0.0, set_speed) | executable/support line; behavior impact de
ds on surrounding block context. |
| 384 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 385 |             set_speed | executable/support line; behavior impact depends on surrounding block context. |
| 386 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 387 |         let target_speed = gap_speed.min(set_speed); | executable/support line; behavior impact depends on su
unding block context. |
| 388 | (blank) | line is blank; visual separation only. |
| 389 |         // PI speed controller with anti-windup. | comment line; documentation intent, no runtime effect. |
| 390 |         let speed_error = target_speed - speed_ms; | executable/support line; behavior impact depends on surr
ding block context. |
| 391 |         if !lead_valid { | conditional gate; changing condition shifts runtime path and side effects. |
| 392 |             // Prevent stale integral from causing surge when a lead target reappears. | comment line; docume
tion intent, no runtime effect. |
| 393 |             self.acc_speed_error_integral *= 0.92; | executable/support line; behavior impact depends on surr

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ding block context. |
| 394 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 395 | (blank) | line is blank; visual separation only. |
| 396 |         let candidate_integral = | executable/support line; behavior impact depends on surrounding block context. |
. |
| 397 |         (self.acc_speed_error_integral + speed_error * dt).clamp(-20.0, 20.0); | executable/support line; behavior impact depends on surrounding block context. |
| 398 |         let unsat_control = speed_error * 0.08 + candidate_integral * 0.004; | executable/support line; behavior impact depends on surrounding block context. |
| 399 |         let control = unsat_control.clamp(-0.8, 1.0); | executable/support line; behavior impact depends on surrounding block context. |
| 400 | (blank) | line is blank; visual separation only. |
| 401 |         // Integrate only when not saturating in the same error direction. | comment line; documentation intent, no runtime effect. |
| 402 |         let same_direction_saturation = (unsat_control - control).abs() > f64::EPSILON | executable/support line; behavior impact depends on surrounding block context. |
| 403 |         && speed_error.abs() > 0.001 | executable/support line; behavior impact depends on surrounding block context. |
| 404 |         && speed_error.signum() == control.signum(); | executable/support line; behavior impact depends on surrounding block context. |
| 405 |         if !same_direction_saturation { | conditional gate; changing condition shifts runtime path and side effects. |
| 406 |             self.acc_speed_error_integral = candidate_integral; | executable/support line; behavior impact depends on surrounding block context. |
| 407 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 408 | (blank) | line is blank; visual separation only. |
| 409 |         if control > 0.02 { | conditional gate; changing condition shifts runtime path and side effects. |
| 410 |             *thr = Some(control.clamp(0.0, 1.0)); | executable/support line; behavior impact depends on surrounding block context. |
| 411 |         } else if control < -0.02 { | executable/support line; behavior impact depends on surrounding block context. |
| 412 |             *brk = Some((-control).clamp(0.0, 0.8).max(brk.unwrap_or(0.0))); | executable/support line; behavior impact depends on surrounding block context. |
| 413 |             *thr = Some(0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 414 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 415 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 416 | (blank) | line is blank; visual separation only. |
| 417 |     fn run_lka( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 418 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 419 |         offset: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 420 |         heading_err: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 421 |         speed_ms: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 422 |         dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 423 |         str: &mut Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
|
| 424 |     ) { | executable/support line; behavior impact depends on surrounding block context. |
| 425 |         if speed_ms < 3.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 426 |             return; | executable/support line; behavior impact depends on surrounding block context. |
| 427 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 428 |         // PD controller for lane centering | comment line; documentation intent, no runtime effect. |
| 429 |         let kp = 4.0 * (speed_ms / 30.0).min(1.0); | executable/support line; behavior impact depends on surrounding block context. |
| 430 |         let ki = 0.3; | executable/support line; behavior impact depends on surrounding block context. |
| 431 |         self.lka_integral += offset * dt; | executable/support line; behavior impact depends on surrounding block context. |
| 432 |         self.lka_integral = self.lka_integral.clamp(-5.0, 5.0); | executable/support line; behavior impact depends on surrounding block context. |
| 433 |         let steer = -(kp * offset + ki * self.lka_integral + 0.5 * heading_err); | executable/support line; behavior impact depends on surrounding block context. |
| 434 |         self.lka_steering_cmd = steer.clamp(-6.0, 6.0); | executable/support line; behavior impact depends on surrounding block context. |

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| 435 |         *str = Some(self.lka_steer_cmd); | executable/support line; behavior impact depends on surrounding block
context. |
| 436 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 437 | (blank) | line is blank; visual separation only. |
| 438 |     fn run_tja( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 439 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 440 |         speed_ms: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 441 |         lead_range: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 442 |         dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 443 |         thr: &mut Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
|
| 444 |         brk: &mut Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
|
| 445 |     ) { | executable/support line; behavior impact depends on surrounding block context. |
| 446 |         // Stop-and-go: if stopped for >3s, wait for lead vehicle to move | comment line; documentation intent,
no runtime effect. |
| 447 |         if speed_ms < 0.2 { | conditional gate; changing condition shifts runtime path and side effects. |
| 448 |             self.tja_stopped_timer += dt; | executable/support line; behavior impact depends on surrounding block
context. |
| 449 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 450 |             self.tja_stopped_timer = 0.0; | executable/support line; behavior impact depends on surrounding block
context. |
| 451 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
|
| 452 |         // If lead vehicle moved away and we were stopped | comment line; documentation intent, no runtime effect. |
|
| 453 |         if self.tja_stopped_timer > 3.0 && lead_range > 5.0 { | conditional gate; changing condition shifts runtime
path and side effects. |
| 454 |             *thr = Some(0.2); // gentle pull-away | executable/support line; behavior impact depends on surrounding
block context. |
| 455 |         } else if lead_range < 3.0 { | executable/support line; behavior impact depends on surrounding block context. |
|
| 456 |             *brk = Some(1.0); | executable/support line; behavior impact depends on surrounding block context. |
| 457 |             *thr = Some(0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 458 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
|
| 459 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 460 | (blank) | line is blank; visual separation only. |
| 461 |     fn update_path(&mut self, speed_ms: f64) { | declares executable behavior; signature/body changes alter API,
control flow, and outputs. |
| 462 |         // Generate simple straight-ahead path (full planning would use A* + splines) | comment line; documentation
intent, no runtime effect. |
| 463 |         self.planned_path.clear(); | executable/support line; behavior impact depends on surrounding block context. |
|
| 464 |         let preview_dist = (speed_ms * 3.5).max(30.0); | executable/support line; behavior impact depends on
surrounding block context. |
| 465 |         let n_points = 20; | executable/support line; behavior impact depends on surrounding block context. |
| 466 |         for i in 0..n_points { | iteration control; changing bounds/body affects throughput, timing, and possibly
safety guarantees. |
| 467 |             let t = i as f64 / n_points as f64; | executable/support line; behavior impact depends on surrounding
block context. |
| 468 |             self.planned_path.push(Waypoint { | executable/support line; behavior impact depends on surrounding
block context. |
|
| 469 |                 x: t * preview_dist, | executable/support line; behavior impact depends on surrounding block context. |
|
| 470 |                 y: -self.lane.offset_m * (1.0 - t), // converge to centre | executable/support line; behavior impact
depends on surrounding block context. |
| 471 |                 speed_ms, | executable/support line; behavior impact depends on surrounding block context. |
| 472 |                 curvature: self.lane.curvature_1_m, | executable/support line; behavior impact depends on surrounding
block context. |
| 473 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 474 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
|
| 475 |         self.path_valid = self.lane.detected; | executable/support line; behavior impact depends on surrounding
block context. |

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| 476 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 477 | (blank) | line is blank; visual separation only. |  
| 478 |     fn safety_check(&mut self, speed_ms: f64, thr: Option<f64>, brk: Option<f64>) { | declares executable beh  
or; signature/body changes alter API, control flow, and outputs. |  
| 479 |         // ISO 26262 ASIL-D plausibility checks | comment line; documentation intent, no runtime effect. |  
| 480 |         if let (Some(t), Some(b)) = (thr, brk) { | conditional gate; changing condition shifts runtime path a  
side effects. |  
| 481 |             if t > 0.1 && b > 0.1 { | conditional gate; changing condition shifts runtime path and side effec  
|  
| 482 |                 // Simultaneous throttle and brake – contradiction! | comment line; documentation intent, no  
time effect. |  
| 483 |                 self.safety_status = SafetyStatus::Degraded; | executable/support line; behavior impact dependen  
on surrounding block context. |  
| 484 |                 self.degrade_reason = Some("Throttle+Brake simultaneous"); | executable/support line; behavior  
impact depends on surrounding block context. |  
| 485 |                 return; | executable/support line; behavior impact depends on surrounding block context. |  
| 486 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 487 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 488 |         if speed_ms > 50.0 | conditional gate; changing condition shifts runtime path and side effects. |  
| 489 |             && matches!(self.acc_state, FeatureState::Off) | executable/support line; behavior impact depends  
surrounding block context. |  
| 490 |             && matches!(self.lka_state, FeatureState::Active) | executable/support line; behavior impact depende  
on surrounding block context. |  
| 491 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 492 |             // LKA without ACC at high speed – degrade | comment line; documentation intent, no runtime effect  
|  
| 493 |             self.safety_status = SafetyStatus::Degraded; | executable/support line; behavior impact depends o  
urrounding block context. |  
| 494 |             self.degrade_reason = Some("LKA w/o ACC at high speed"); | executable/support line; behavior impa  
depends on surrounding block context. |  
| 495 |             return; | executable/support line; behavior impact depends on surrounding block context. |  
| 496 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 497 |         self.safety_status = SafetyStatus::Normal; | executable/support line; behavior impact depends on surr  
ding block context. |  
| 498 |         self.degrade_reason = None; | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 499 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 500 | (blank) | line is blank; visual separation only. |  
| 501 |     // ■ Public control helpers ████████████████████████████████████████████████████████████████ | comment line;  
ntent, no runtime effect. |  
| 502 |     pub fn engage(&mut self, current_speed: f64) { | declares executable behavior; signature/body changes alt  
API, control flow, and outputs. |  
| 503 |         self.engaged = true; | executable/support line; behavior impact depends on surrounding block context.  
| 504 |         self.acc_state = FeatureState::Active; | executable/support line; behavior impact depends on surround  
block context. |  
| 505 |         self.lka_state = FeatureState::Active; | executable/support line; behavior impact depends on surround  
block context. |  
| 506 |         self.tja_state = FeatureState::Active; | executable/support line; behavior impact depends on surround  
block context. |  
| 507 |         self.acc_set_speed_kmh = (current_speed).max(30.0); | executable/support line; behavior impact depend  
n surrounding block context. |  
| 508 |         self.acc_speed_error_integral = 0.0; | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 509 |         self.throttle_cmd = 0.0; | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 510 |         self.brake_cmd = 0.0; | executable/support line; behavior impact depends on surrounding block context  
| 511 |         self.steer_cmd_deg = 0.0; | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 512 |         self.lka_integral = 0.0; | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 513 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
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| 514 |     pub fn disengage(&mut self) { | declares executable behavior; signature/body changes alter API, control f
, and outputs. |
| 515 |         self.engaged = false; | executable/support line; behavior impact depends on surrounding block context
| 516 |         self.acc_state = FeatureState::Off; | executable/support line; behavior impact depends on surrounding
ock context. |
| 517 |         self.lka_state = FeatureState::Off; | executable/support line; behavior impact depends on surrounding
ock context. |
| 518 |         self.tja_state = FeatureState::Off; | executable/support line; behavior impact depends on surrounding
ock context. |
| 519 |         self.throttle_cmd = 0.0; | executable/support line; behavior impact depends on surrounding block cont
. |
| 520 |         self.brake_cmd = 0.0; | executable/support line; behavior impact depends on surrounding block context
| 521 |         self.steer_cmd_deg = 0.0; | executable/support line; behavior impact depends on surrounding block con
t. |
| 522 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 523 |     pub fn toggle_lka(&mut self) { | declares executable behavior; signature/body changes alter API, control
w, and outputs. |
| 524 |         self.lka_state = if self.lka_state == FeatureState::Active { | executable/support line; behavior impa
depends on surrounding block context. |
| 525 |             self.lka_integral = 0.0; | executable/support line; behavior impact depends on surrounding block
text. |
| 526 |             FeatureState::Off | executable/support line; behavior impact depends on surrounding block context
| 527 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 528 |             FeatureState::Active | executable/support line; behavior impact depends on surrounding block cont
. |
| 529 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 530 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 531 |     pub fn set_acc_speed(&mut self, kmh: f64) { | declares executable behavior; signature/body changes alter
, control flow, and outputs. |
| 532 |         self.acc_set_speed_kmh = kmh.clamp(20.0, 200.0); | executable/support line; behavior impact depends o
urrounding block context. |
| 533 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 534 |     pub fn set_headway(&mut self, s: f64) { | declares executable behavior; signature/body changes alter API,
ntrol flow, and outputs. |
| 535 |         self.acc_headway_s = s.clamp(1.0, 3.5); | executable/support line; behavior impact depends on surroun
g block context. |
| 536 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 537 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 538 | (blank) | line is blank; visual separation only. |
| 539 | #[cfg(test)] | executable/support line; behavior impact depends on surrounding block context. |
| 540 | mod tests { | declares a module boundary; changing module name/path breaks references. |
| 541 |     use super::{AutonomousController, FeatureState}; | imports symbols; changing path/name affects compile-ti
resolution. |
| 542 | (blank) | line is blank; visual separation only. |
| 543 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 544 |     fn disengaged_controller_outputs_none_and_zero_state() { | declares executable behavior; signature/body c
ges alter API, control flow, and outputs. |
| 545 |         let mut ad = AutonomousController::new(); | executable/support line; behavior impact depends on surro
ing block context. |
| 546 |         let (t, b, s) = ad.tick( | executable/support line; behavior impact depends on surrounding block cont
. |
| 547 |             20.0, | executable/support line; behavior impact depends on surrounding block context. |
| 548 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 549 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 550 |             f64::INFINITY, | executable/support line; behavior impact depends on surrounding block context. |
| 551 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 552 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 553 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 554 |             0.9, | executable/support line; behavior impact depends on surrounding block context. |
| 555 |             f64::INFINITY, | executable/support line; behavior impact depends on surrounding block context. |
| 556 |             false, | executable/support line; behavior impact depends on surrounding block context. |
| 557 |             false, | executable/support line; behavior impact depends on surrounding block context. |
| 558 |             0.02, | executable/support line; behavior impact depends on surrounding block context. |
| 559 |         ); | executable/support line; behavior impact depends on surrounding block context. |

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| 560 |         assert!(t.is_none()); | executable/support line; behavior impact depends on surrounding block context. | | |
| 561 |         assert!(b.is_none()); | executable/support line; behavior impact depends on surrounding block context. |
| 562 |         assert!(s.is_none()); | executable/support line; behavior impact depends on surrounding block context. |
| 563 |         assert_eq!(ad.throttle_cmd, 0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 564 |         assert_eq!(ad.brake_cmd, 0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 565 |         assert_eq!(ad.steer_cmd_deg, 0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 566 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 567 | (blank) | line is blank; visual separation only. |
| 568 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 569 |     fn engage_enables_l2_features() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 570 |         let mut ad = AutonomousController::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 571 |         ad.engage(42.0); | executable/support line; behavior impact depends on surrounding block context. |
| 572 |         assert!(ad.engaged); | executable/support line; behavior impact depends on surrounding block context. |
| 573 |         assert_eq!(ad.acc_state, FeatureState::Active); | executable/support line; behavior impact depends on surrounding block context. |
| 574 |         assert_eq!(ad.lka_state, FeatureState::Active); | executable/support line; behavior impact depends on surrounding block context. |
| 575 |         assert_eq!(ad.tja_state, FeatureState::Active); | executable/support line; behavior impact depends on surrounding block context. |
| 576 |         assert!(ad.acc_set_speed_kmh >= 30.0); | executable/support line; behavior impact depends on surrounding block context. |
| 577 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 578 | (blank) | line is blank; visual separation only. |
| 579 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 580 |     fn engage_from_standstill_requests_positive_throttle_on_clear_road() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 581 |         let mut ad = AutonomousController::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 582 |         ad.engage(0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 583 | (blank) | line is blank; visual separation only. |
| 584 |         let (thr, brk, steer) = ad.tick( | executable/support line; behavior impact depends on surrounding block context. |
| 585 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 586 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 587 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 588 |             f64::INFINITY, | executable/support line; behavior impact depends on surrounding block context. |
| 589 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 590 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 591 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 592 |             0.95, | executable/support line; behavior impact depends on surrounding block context. |
| 593 |             f64::INFINITY, | executable/support line; behavior impact depends on surrounding block context. |
| 594 |             false, | executable/support line; behavior impact depends on surrounding block context. |
| 595 |             false, | executable/support line; behavior impact depends on surrounding block context. |
| 596 |             0.05, | executable/support line; behavior impact depends on surrounding block context. |
| 597 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 598 | (blank) | line is blank; visual separation only. |
| 599 |         assert!(thr.is_some_and(|v| v > 0.05)); | executable/support line; behavior impact depends on surrounding block context. |
| 600 |         assert!(brk.is_none() || brk == Some(0.0)); | executable/support line; behavior impact depends on surrounding block context. |
| 601 |         assert!(steer.is_none() || steer == Some(0.0)); | executable/support line; behavior impact depends on surrounding block context. |
| 602 |         assert!(ad.throttle_cmd > 0.05); | executable/support line; behavior impact depends on surrounding block context. |
| 603 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 604 | (blank) | line is blank; visual separation only. |
| 605 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 606 |     fn no_lead_then_close_lead_transitions_to_brake() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 607 |         let mut ad = AutonomousController::new(); | executable/support line; behavior impact depends on surrounding block context. |

```

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ing block context. |
| 608 |         ad.engage(50.0); | executable/support line; behavior impact depends on surrounding block context. |
| 609 | (blank) | line is blank; visual separation only. |
| 610 |         // Build integral/throttle while road is clear. | comment line; documentation intent, no runtime effect. |
|
| 611 |         for _ in 0..120 { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 612 |             let _ = ad.tick( | executable/support line; behavior impact depends on surrounding block context. |
| 613 |                 30.0, | executable/support line; behavior impact depends on surrounding block context. |
| 614 |                 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 615 |                 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 616 |                 f64::INFINITY, | executable/support line; behavior impact depends on surrounding block context. |
|
| 617 |                 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 618 |                 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 619 |                 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 620 |                 0.95, | executable/support line; behavior impact depends on surrounding block context. |
| 621 |                 f64::INFINITY, | executable/support line; behavior impact depends on surrounding block context. |
|
| 622 |                 false, | executable/support line; behavior impact depends on surrounding block context. |
| 623 |                 false, | executable/support line; behavior impact depends on surrounding block context. |
| 624 |                 0.02, | executable/support line; behavior impact depends on surrounding block context. |
| 625 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 626 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
|
| 627 |         assert!(ad.throttle_cmd >= 0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 628 | (blank) | line is blank; visual separation only. |
| 629 |         // Sudden close lead should force braking and suppress throttle. | comment line; documentation intent, no runtime effect. |
| 630 |         let _ = ad.tick( | executable/support line; behavior impact depends on surrounding block context. |
| 631 |             30.0, | executable/support line; behavior impact depends on surrounding block context. |
| 632 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 633 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 634 |             8.0, | executable/support line; behavior impact depends on surrounding block context. |
| 635 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 636 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 637 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 638 |             0.95, | executable/support line; behavior impact depends on surrounding block context. |
| 639 |             1.2, | executable/support line; behavior impact depends on surrounding block context. |
| 640 |             false, | executable/support line; behavior impact depends on surrounding block context. |
| 641 |             false, | executable/support line; behavior impact depends on surrounding block context. |
| 642 |             0.02, | executable/support line; behavior impact depends on surrounding block context. |
| 643 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 644 |         assert!(ad.brake_cmd > 0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 645 |         assert!(ad.throttle_cmd <= 0.05); | executable/support line; behavior impact depends on surrounding block context. |
| 646 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 647 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/bin/cat_comm_bridge.rs

Role: Executable entrypoint for CLI/bridge workflow.

Lines: 367 | Non-empty: 347 | Symbols: 15

Symbol Inventory

Line	Kind	Symbol	Change Impact
9	enum	BridgeRequest	data/structure coupling
29	struct	BridgeDiagPayload	data/structure coupling
38	struct	BridgeResponse	data/structure coupling
49	struct	BridgeState	data/structure coupling
58	impl	Default for BridgeState	behavior binding and method semantics


```

| 59 | fn | default | signature/API coupling |
| 71 | impl | BridgeState | behavior binding and method semantics |
| 72 | fn | capabilities | signature/API coupling |
| 81 | fn | make_ok | signature/API coupling |
| 93 | fn | push_serial | signature/API coupling |
| 101 | fn | handle_uds_serial_request | signature/API coupling |
| 168 | fn | did_ascii | signature/API coupling |
| 175 | fn | main | signature/API coupling |
| 345 | fn | write_response | signature/API coupling |
| 362 | fn | now_ms | signature/API coupling |

```

Line-by-Line Impact Map

```

| Line | Source | Operational Impact If Changed |
|---:|---|---|
| 1 | use std::collections::VecDeque; | imports symbols; changing path/name affects compile-time resolution. |
| 2 | use std::io::{self, BufRead, Write}; | imports symbols; changing path/name affects compile-time resolution. |
| 3 | (blank) | line is blank; visual separation only. |
| 4 | use auto_breaking::io::hw::{Frame, SerialFrame}; | imports symbols; changing path/name affects compile-time resolution. |
| 5 | use serde::{Deserialize, Serialize}; | imports symbols; changing path/name affects compile-time resolution. |
| 6 | (blank) | line is blank; visual separation only. |
| 7 | #[derive(Debug, Deserialize)] | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 8 | #[serde(tag = "cmd", rename_all = "snake_case")] | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 9 | enum BridgeRequest { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 10 |     Init { | executable/support line; behavior impact depends on surrounding block context. |
| 11 |         dry_run: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 12 |         enable_write: bool, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 13 |         can_interface: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 14 |         serial_port: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 15 |         template_dir: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 16 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 17 |     Read { | executable/support line; behavior impact depends on surrounding block context. |
| 18 |         timeout_ms: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 19 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 20 |     TryRead, | executable/support line; behavior impact depends on surrounding block context. |
| 21 |     Send { | executable/support line; behavior impact depends on surrounding block context. |
| 22 |         frame: Frame, | executable/support line; behavior impact depends on surrounding block context. |
| 23 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 24 |     Close, | executable/support line; behavior impact depends on surrounding block context. |
| 25 |     Ping, | executable/support line; behavior impact depends on surrounding block context. |
| 26 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 27 | (blank) | line is blank; visual separation only. |
| 28 | #[derive(Debug, Serialize)] | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 29 | struct BridgeDiagPayload { | declares data layout; field changes may break serialization, tests, and callers. |
| 30 |     wait_frame_count_delta: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 31 |     sequence_error_count_delta: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 32 |     flowcontrol_timeout_count_delta: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 33 |     fc_blocksize: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 34 |     fc_stmin_ms: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 35 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 36 | (blank) | line is blank; visual separation only. |
| 37 | #[derive(Debug, Serialize)] | serialization contract touchpoint; edits can break report compatibility and tooling. |

```

```

| 38 | struct BridgeResponse { | declares data layout; field changes may break serialization, tests, and callers. |
| 39 |     ok: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 40 |     frame: Option<Frame>, | executable/support line; behavior impact depends on surrounding block context. |
| 41 |     error: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 42 |     kind: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 43 |     protocol_version: Option<String>, | executable/support line; behavior impact depends on surrounding block
text. |
| 44 |     capabilities: Option<Vec<String>>, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 45 |     diag: Option<BridgeDiagPayload>, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 46 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 47 | (blank) | line is blank; visual separation only. |
| 48 | #[derive(Debug)] | executable/support line; behavior impact depends on surrounding block context. |
| 49 | struct BridgeState { | declares data layout; field changes may break serialization, tests, and callers. |
| 50 |     rx_queue: VecDeque<Frame>, | executable/support line; behavior impact depends on surrounding block context
| 51 |     dry_run: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 52 |     enable_write: bool, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic
tcomes. |
| 53 |     initialized: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 54 |     negotiated_stmin_ms: u64, | executable/support line; behavior impact depends on surrounding block context.
| 55 |     negotiated_blocksize: u8, | executable/support line; behavior impact depends on surrounding block context.
| 56 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 57 | (blank) | line is blank; visual separation only. |
| 58 | impl Default for BridgeState { | implementation block; method behavior and trait conformance are defined here.
| 59 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
tputs. |
| 60 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 61 |             rx_queue: VecDeque::new(), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 62 |             dry_run: true, | executable/support line; behavior impact depends on surrounding block context. |
| 63 |             enable_write: false, | I/O or protocol-critical logic; edits can affect integration, timing, and d
nostic outcomes. |
| 64 |             initialized: false, | executable/support line; behavior impact depends on surrounding block context
|
| 65 |             negotiated_stmin_ms: 5, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 66 |             negotiated_blocksize: 8, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 67 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 68 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 69 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 70 | (blank) | line is blank; visual separation only. |
| 71 | impl BridgeState { | implementation block; method behavior and trait conformance are defined here. |
| 72 |     fn capabilities(&self) -> Vec<String> { | declares executable behavior; signature/body changes alter API,
trol flow, and outputs. |
| 73 |         vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 74 |             "serial".to_string(), | executable/support line; behavior impact depends on surrounding block cont
. |
| 75 |             "can".to_string(), | executable/support line; behavior impact depends on surrounding block context
| 76 |             "write".to_string(), | I/O or protocol-critical logic; edits can affect integration, timing, and d
nostic outcomes. |
| 77 |             "uds_flash".to_string(), | I/O or protocol-critical logic; edits can affect integration, timing, a
diagnostic outcomes. |
| 78 |         ] | executable/support line; behavior impact depends on surrounding block context. |
| 79 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 80 | (blank) | line is blank; visual separation only. |
| 81 |     fn make_ok(&self) -> BridgeResponse { | declares executable behavior; signature/body changes alter API, co
ol flow, and outputs. |
| 82 |         BridgeResponse { | executable/support line; behavior impact depends on surrounding block context. |
| 83 |             ok: true, | executable/support line; behavior impact depends on surrounding block context. |
| 84 |             frame: None, | executable/support line; behavior impact depends on surrounding block context. |
| 85 |             error: None, | executable/support line; behavior impact depends on surrounding block context. |
| 86 |             kind: None, | executable/support line; behavior impact depends on surrounding block context. |
| 87 |             protocol_version: Some("1.0".to_string()), | executable/support line; behavior impact depends on s

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ounding block context. |
| 88 |         capabilities: Some(self.capabilities()), | executable/support line; behavior impact depends on surr
ounding block context. |
| 89 |         diag: None, | executable/support line; behavior impact depends on surrounding block context. |
| 90 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 91 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 92 | (blank) | line is blank; visual separation only. |
| 93 | fn push_serial(&mut self, bytes: Vec<u8>, hint: &str) { | declares executable behavior; signature/body cha
s alter API, control flow, and outputs. |
| 94 |     self.rx_queue.push_back(Frame::Serial(SerialFrame { | executable/support line; behavior impact depends
surrounding block context. |
| 95 |         bytes, | executable/support line; behavior impact depends on surrounding block context. |
| 96 |         protocol_hint: Some(hint.to_string()), | executable/support line; behavior impact depends on surro
ing block context. |
| 97 |         timestamp_ms: Some(now_ms()), | executable/support line; behavior impact depends on surrounding bl
context. |
| 98 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 99 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 100 | (blank) | line is blank; visual separation only. |
| 101 | fn handle_uds_serial_request(&mut self, req: &[u8]) { | declares executable behavior; signature/body chan
alter API, control flow, and outputs. |
| 102 |     if req.is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 103 |         return; | executable/support line; behavior impact depends on surrounding block context. |
| 104 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 105 | (blank) | line is blank; visual separation only. |
| 106 |     match req[0] { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 107 |         0x10 => { | executable/support line; behavior impact depends on surrounding block context. |
| 108 |             let sub = req.get(1).copied().unwrap_or(0x01); | executable/support line; behavior impact dep
s on surrounding block context. |
| 109 |             self.push_serial(vec![0x50, sub, 0x00, 0x32, 0x01, 0xF4], "uds-raw"); | I/O or protocol-criti
logic; edits can affect integration, timing, and diagnostic outcomes. |
| 110 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 111 |         0x22 => { | executable/support line; behavior impact depends on surrounding block context. |
| 112 |             if req.len() < 3 { | conditional gate; changing condition shifts runtime path and side effect
|
| 113 |                 self.push_serial(vec![0x7F, 0x22, 0x13], "uds-raw"); | I/O or protocol-critical logic; ed
can affect integration, timing, and diagnostic outcomes. |
| 114 |                 return; | executable/support line; behavior impact depends on surrounding block context.
| 115 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 116 |             let did = ((req[1] as u16) << 8) \ req[2] as u16; | executable/support line; behavior impact
depends on surrounding block context. |
| 117 |             match did { | branch dispatch; changing arms alters behavior for variants and input classes.
| 118 |                 0xF190 => self.did_ascii(did, b"CATVIN1234567890"), | executable/support line; behavior i
ct depends on surrounding block context. |
| 119 |                 0xF189 => self.did_ascii(did, b"SW-1.0.0"), | executable/support line; behavior impact de
ds on surrounding block context. |
| 120 |                 0xF191 => self.did_ascii(did, b"HW-1.0.0"), | executable/support line; behavior impact de
ds on surrounding block context. |
| 121 |                 0xF180 => self.did_ascii(did, b"CAL-001"), | executable/support line; behavior impact dep
s on surrounding block context. |
| 122 |                 0xF18C => self.did_ascii(did, b"ECU-SN-0001"), | executable/support line; behavior impact
depends on surrounding block context. |
| 123 |                 0xF184 => self.did_ascii(did, b"FP-TRUSTED-01"), | executable/support line; behavior impa
depends on surrounding block context. |
| 124 |                 0xDD0E => { | executable/support line; behavior impact depends on surrounding block conte
|
| 125 |                     // 12.5V with factor 0.05 => raw 250 (0x00FA) | comment line; documentation intent, n
untime effect. |
| 126 |                     self.push_serial(vec![0x62, 0xDD, 0x0E, 0x00, 0xFA], "uds-raw"); | I/O or protocol-cr
cal logic; edits can affect integration, timing, and diagnostic outcomes. |
| 127 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |

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| 128 |         _ => self.push_serial(vec![0x7F, 0x22, 0x31], "uds-raw"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 129 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 130 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 131 |         0x27 => { | executable/support line; behavior impact depends on surrounding block context. |
| 132 |             let sub = req.get(1).copied().unwrap_or(0x00); | executable/support line; behavior impact depends on surrounding block context. |
| 133 |             match sub { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 134 |                 0x05 => self.push_serial(vec![0x67, 0x05, 0x12, 0x34, 0x56, 0x78], "uds-raw"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 135 |                 0x06 => self.push_serial(vec![0x67, 0x06], "uds-raw"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 136 |                 _ => self.push_serial(vec![0x7F, 0x27, 0x12], "uds-raw"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 137 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 138 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 139 |         0x34 => { | executable/support line; behavior impact depends on surrounding block context. |
| 140 |             // Positive response with max block length hint. | comment line; documentation intent, no runtime effect. |
| 141 |             self.push_serial(vec![0x74, 0x20, 0x00, 0x40], "uds-raw"); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 142 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 143 |         0x36 => { | executable/support line; behavior impact depends on surrounding block context. |
| 144 |             let seq = req.get(1).copied().unwrap_or(0x00); | executable/support line; behavior impact depends on surrounding block context. |
| 145 |             self.push_serial(vec![0x76, seq], "uds-raw"); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 146 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 147 |         0x37 => { | executable/support line; behavior impact depends on surrounding block context. |
| 148 |             self.push_serial(vec![0x77, 0x00], "uds-raw"); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 149 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 150 |         0x31 => { | executable/support line; behavior impact depends on surrounding block context. |
| 151 |             let sub = req.get(1).copied().unwrap_or(0x01); | executable/support line; behavior impact depends on surrounding block context. |
| 152 |             let h = req.get(2).copied().unwrap_or(0x00); | executable/support line; behavior impact depends on surrounding block context. |
| 153 |             let l = req.get(3).copied().unwrap_or(0x00); | executable/support line; behavior impact depends on surrounding block context. |
| 154 |             self.push_serial(vec![0x71, sub, h, l], "uds-raw"); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 155 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 156 |         0x14 => { | executable/support line; behavior impact depends on surrounding block context. |
| 157 |             self.push_serial(vec![0x54], "uds-raw"); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 158 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 159 |         0x3E => { | executable/support line; behavior impact depends on surrounding block context. |
| 160 |             self.push_serial(vec![0x7E, 0x00], "uds-raw"); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 161 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 162 |         _ => { | executable/support line; behavior impact depends on surrounding block context. |
| 163 |             self.push_serial(vec![0x7F, req[0], 0x11], "uds-raw"); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 164 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

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| 165 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 166 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 167 | (blank) | line is blank; visual separation only. |
| 168 |     fn did_ascii(&mut self, did: u16, payload: &[u8]) { | declares executable behavior; signature/body change
lter API, control flow, and outputs. |
| 169 |         let mut rsp = vec![0x62, (did >> 8) as u8, (did & 0xFF) as u8]; | executable/support line; behavior i
ct depends on surrounding block context. |
| 170 |         rsp.extend_from_slice(payload); | executable/support line; behavior impact depends on surrounding blo
context. |
| 171 |         self.push_serial(rsp, "uds-raw"); | I/O or protocol-critical logic; edits can affect integration, tim
, and diagnostic outcomes. |
| 172 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 173 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 174 | (blank) | line is blank; visual separation only. |
| 175 | fn main() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 176 |     let args: Vec<String> = std::env::args().collect(); | executable/support line; behavior impact depends on
rrounding block context. |
| 177 |     if !args.iter().any(|a| a == "--mode=stdio-jsonl") { | conditional gate; changing condition shifts runt
path and side effects. |
| 178 |         eprintln!("cat_comm_bridge: use --mode=stdio-jsonl"); | executable/support line; behavior impact depe
on surrounding block context. |
| 179 |         std::process::exit(2); | executable/support line; behavior impact depends on surrounding block contex
|
| 180 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 181 | (blank) | line is blank; visual separation only. |
| 182 |     let stdin = io::stdin(); | executable/support line; behavior impact depends on surrounding block context.
| 183 |     let mut stdout = io::stdout(); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 184 |     let mut state = BridgeState::default(); | executable/support line; behavior impact depends on surrounding
ock context. |
| 185 | (blank) | line is blank; visual separation only. |
| 186 |     for line in stdin.lock().lines() { | iteration control; changing bounds/body affects throughput, timing,
possibly safety guarantees. |
| 187 |         let line = match line { | executable/support line; behavior impact depends on surrounding block conte
|
| 188 |             Ok(v) => v, | executable/support line; behavior impact depends on surrounding block context. |
| 189 |             Err(e) => { | executable/support line; behavior impact depends on surrounding block context. |
| 190 |                 write_response( | I/O or protocol-critical logic; edits can affect integration, timing, and d
nostic outcomes. |
| 191 |                     &mut stdout, | executable/support line; behavior impact depends on surrounding block cont
. |
| 192 |                     BridgeResponse { | executable/support line; behavior impact depends on surrounding block
text. |
| 193 |                         ok: false, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 194 |                         frame: None, | executable/support line; behavior impact depends on surrounding block
text. |
| 195 |                         error: Some(format!("stdin read failed: {e}")), | executable/support line; behavior i
ct depends on surrounding block context. |
| 196 |                         kind: Some("parse".to_string()), | executable/support line; behavior impact depends o
urrounding block context. |
| 197 |                         protocol_version: Some("1.0".to_string()), | executable/support line; behavior impact
ends on surrounding block context. |
| 198 |                         capabilities: Some(state.capabilities()), | executable/support line; behavior impact
ends on surrounding block context. |
| 199 |                         diag: None, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 200 |                     }, | executable/support line; behavior impact depends on surrounding block context. |
| 201 |                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 202 |                     continue; | executable/support line; behavior impact depends on surrounding block context. |
| 203 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 204 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|

```

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| 205 | (blank) | line is blank; visual separation only. |
| 206 |         let req = match serde_json::from_str::<BridgeRequest>(line.trim()) { | serialization contract touchpoint
; edits can break report compatibility and tooling. |
| 207 |             Ok(v) => v, | executable/support line; behavior impact depends on surrounding block context. |
| 208 |             Err(e) => { | executable/support line; behavior impact depends on surrounding block context. |
| 209 |                 write_response( | I/O or protocol-critical logic; edits can affect integration, timing, and d
nostic outcomes. |
| 210 |                     &mut stdout, | executable/support line; behavior impact depends on surrounding block cont
. |
| 211 |                     BridgeResponse { | executable/support line; behavior impact depends on surrounding block
text. |
| 212 |                         ok: false, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 213 |                         frame: None, | executable/support line; behavior impact depends on surrounding block
text. |
| 214 |                         error: Some(format!("invalid request json: {e}")), | executable/support line; behavio
mpact depends on surrounding block context. |
| 215 |                         kind: Some("parse".to_string()), | executable/support line; behavior impact depends o
urrounding block context. |
| 216 |                         protocol_version: Some("1.0".to_string()), | executable/support line; behavior impact
ends on surrounding block context. |
| 217 |                         capabilities: Some(state.capabilities()), | executable/support line; behavior impact
ends on surrounding block context. |
| 218 |                         diag: None, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 219 |                     }, | executable/support line; behavior impact depends on surrounding block context. |
| 220 |                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 221 |                     continue; | executable/support line; behavior impact depends on surrounding block context. |
| 222 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 223 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 224 | (blank) | line is blank; visual separation only. |
| 225 |         let rsp = match req { | executable/support line; behavior impact depends on surrounding block context
| 226 |             BridgeRequest::Init { | executable/support line; behavior impact depends on surrounding block con
t. |
| 227 |                 dry_run, | executable/support line; behavior impact depends on surrounding block context. |
| 228 |                 enable_write, | I/O or protocol-critical logic; edits can affect integration, timing, and dia
stic outcomes. |
| 229 |                 can_interface, | executable/support line; behavior impact depends on surrounding block context
|
| 230 |                 serial_port, | executable/support line; behavior impact depends on surrounding block context.
| 231 |                 template_dir, | executable/support line; behavior impact depends on surrounding block context
| 232 |             } => { | executable/support line; behavior impact depends on surrounding block context. |
| 233 |                 state.dry_run = dry_run; | executable/support line; behavior impact depends on surrounding bl
context. |
| 234 |                 state.enable_write = enable_write; | I/O or protocol-critical logic; edits can affect integra
n, timing, and diagnostic outcomes. |
| 235 |                 state.initialized = true; | executable/support line; behavior impact depends on surrounding b
k context. |
| 236 |                 let _ = (can_interface, serial_port, template_dir); | executable/support line; behavior impac
depends on surrounding block context. |
| 237 |                 // Seed one serial frame so detect_ecms has at least one observed channel. | comment line; do
entation intent, no runtime effect. |
| 238 |                 state.push_serial(vec![0x62, 0xF1, 0x90, b'O', b'K'], "bridge-heartbeat"); | executable/suppo
line; behavior impact depends on surrounding block context. |
| 239 |                 state.make_ok() | executable/support line; behavior impact depends on surrounding block conte
|
| 240 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 241 |             BridgeRequest::Ping => { | executable/support line; behavior impact depends on surrounding block
text. |
| 242 |                 if state.initialized { | conditional gate; changing condition shifts runtime path and side ef
ts. |
| 243 |                     state.make_ok() | executable/support line; behavior impact depends on surrounding block c

```

```

ext. |
| 244 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 245 |             BridgeResponse { | executable/support line; behavior impact depends on surrounding block
text. |
| 246 |                 ok: false, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 247 |                 frame: None, | executable/support line; behavior impact depends on surrounding block
text. |
| 248 |                 error: Some("bridge not initialized".to_string()), | executable/support line; behavio
mpact depends on surrounding block context. |
| 249 |                 kind: Some("permission".to_string()), | executable/support line; behavior impact depe
on surrounding block context. |
| 250 |                 protocol_version: Some("1.0".to_string()), | executable/support line; behavior impact
ends on surrounding block context. |
| 251 |                 capabilities: Some(state.capabilities()), | executable/support line; behavior impact
ends on surrounding block context. |
| 252 |                 diag: None, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 253 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 254 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 255 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 256 |         BridgeRequest::Read { timeout_ms } => { | executable/support line; behavior impact depends on surr
nding block context. |
| 257 |             let _ = timeout_ms; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 258 |             if let Some(frame) = state.rx_queue.pop_front() { | conditional gate; changing condition shift
runtime path and side effects. |
| 259 |                 BridgeResponse { | executable/support line; behavior impact depends on surrounding block
text. |
| 260 |                     ok: true, | executable/support line; behavior impact depends on surrounding block con
t. |
| 261 |                     frame: Some(frame), | executable/support line; behavior impact depends on surrounding
ock context. |
| 262 |                     error: None, | executable/support line; behavior impact depends on surrounding block
text. |
| 263 |                     kind: None, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 264 |                     protocol_version: Some("1.0".to_string()), | executable/support line; behavior impact
ends on surrounding block context. |
| 265 |                     capabilities: Some(state.capabilities()), | executable/support line; behavior impact
ends on surrounding block context. |
| 266 |                     diag: Some(BridgeDiagPayload { | executable/support line; behavior impact depends on
rounding block context. |
| 267 |                         wait_frame_count_delta: 0, | executable/support line; behavior impact depends on
rounding block context. |
| 268 |                         sequence_error_count_delta: 0, | executable/support line; behavior impact depends
surrounding block context. |
| 269 |                         flowcontrol_timeout_count_delta: 0, | executable/support line; behavior impact de
ds on surrounding block context. |
| 270 |                         fc_blocksize: state.negotiated_blocksize, | executable/support line; behavior imp
depends on surrounding block context. |
| 271 |                         fc_stmin_ms: state.negotiated_stmin_ms, | executable/support line; behavior impac
depends on surrounding block context. |
| 272 |                     })), | executable/support line; behavior impact depends on surrounding block context.
| 273 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 274 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 275 |             BridgeResponse { | executable/support line; behavior impact depends on surrounding block
text. |
| 276 |                 ok: false, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 277 |                 frame: None, | executable/support line; behavior impact depends on surrounding block

```

```

text. |
| 278 |           error: Some("read timeout".to_string()), | executable/support line; behavior impact d
nds on surrounding block context. |
| 279 |           kind: Some("timeout".to_string()), | executable/support line; behavior impact depends
surrounding block context. |
| 280 |           protocol_version: Some("1.0".to_string()), | executable/support line; behavior impact
depends on surrounding block context. |
| 281 |           capabilities: Some(state.capabilities()), | executable/support line; behavior impact
ends on surrounding block context. |
| 282 |           diag: None, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 283 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 284 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 285 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 286 |           BridgeRequest::TryRead => BridgeResponse { | executable/support line; behavior impact depends on
rounding block context. |
| 287 |               ok: true, | executable/support line; behavior impact depends on surrounding block context. |
| 288 |               frame: state.rx_queue.pop_front(), | executable/support line; behavior impact depends on surr
ding block context. |
| 289 |               error: None, | executable/support line; behavior impact depends on surrounding block context.
| 290 |               kind: None, | executable/support line; behavior impact depends on surrounding block context.
| 291 |               protocol_version: Some("1.0".to_string()), | executable/support line; behavior impact depends
surrounding block context. |
| 292 |               capabilities: Some(state.capabilities()), | executable/support line; behavior impact depends
surrounding block context. |
| 293 |               diag: None, | executable/support line; behavior impact depends on surrounding block context.
| 294 |           }, | executable/support line; behavior impact depends on surrounding block context. |
| 295 |           BridgeRequest::Send { frame } => { | executable/support line; behavior impact depends on surround
block context. |
| 296 |               if !state.dry_run && !state.enable_write { | conditional gate; changing condition shifts runt
path and side effects. |
| 297 |                   BridgeResponse { | executable/support line; behavior impact depends on surrounding block
text. |
| 298 |                       ok: false, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 299 |                       frame: None, | executable/support line; behavior impact depends on surrounding block
text. |
| 300 |                       error: Some("write blocked by bridge policy".to_string()), | I/O or protocol-critical
gic; edits can affect integration, timing, and diagnostic outcomes. |
| 301 |                       kind: Some("write_blocked".to_string()), | I/O or protocol-critical logic; edits can
ect integration, timing, and diagnostic outcomes. |
| 302 |                       protocol_version: Some("1.0".to_string()), | executable/support line; behavior impact
depends on surrounding block context. |
| 303 |                       capabilities: Some(state.capabilities()), | executable/support line; behavior impact
ends on surrounding block context. |
| 304 |                       diag: None, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 305 |                   } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 306 |               } else { | executable/support line; behavior impact depends on surrounding block context. |
| 307 |                   if let Frame::Serial(sf) = &frame { | conditional gate; changing condition shifts runtime
th and side effects. |
| 308 |                       let is_uds = sf | I/O or protocol-critical logic; edits can affect integration, timin
and diagnostic outcomes. |
| 309 |                       .protocol_hint | executable/support line; behavior impact depends on surrounding
ck context. |
| 310 |                       .as_deref() | executable/support line; behavior impact depends on surrounding blo
context. |
| 311 |                       .map(\h\ h.eq_ignore_ascii_case("uds-raw")) | I/O or protocol-critical logic; e
s can affect integration, timing, and diagnostic outcomes. |
| 312 |                       .unwrap_or(false); | executable/support line; behavior impact depends on surround
block context. |

```



```

| 313 |           if is_uds { | conditional gate; changing condition shifts runtime path and side effects
|
| 314 |           state.handle_uds_serial_request(&sf.bytes); | I/O or protocol-critical logic; edits
can affect integration, timing, and diagnostic outcomes. |
| 315 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ifetime scopes. |
| 316 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ifetime scopes. |
| 317 |       BridgeResponse { | executable/support line; behavior impact depends on surrounding block co
text. |
| 318 |           ok: true, | executable/support line; behavior impact depends on surrounding block con
t. |
| 319 |           frame: None, | executable/support line; behavior impact depends on surrounding block co
text. |
| 320 |           error: None, | executable/support line; behavior impact depends on surrounding block co
text. |
| 321 |           kind: None, | executable/support line; behavior impact depends on surrounding block co
ext. |
| 322 |           protocol_version: Some("1.0".to_string()), | executable/support line; behavior impact
depends on surrounding block context. |
| 323 |           capabilities: Some(state.capabilities()), | executable/support line; behavior impact
ends on surrounding block context. |
| 324 |           diag: Some(BridgeDiagPayload { | executable/support line; behavior impact depends on
rounding block context. |
| 325 |               wait_frame_count_delta: 0, | executable/support line; behavior impact depends on
rounding block context. |
| 326 |               sequence_error_count_delta: 0, | executable/support line; behavior impact depends
surrounding block context. |
| 327 |               flowcontrol_timeout_count_delta: 0, | executable/support line; behavior impact de
ds on surrounding block context. |
| 328 |               fc_blocksize: state.negotiated_blocksize, | executable/support line; behavior imp
depends on surrounding block context. |
| 329 |               fc_stmin_ms: state.negotiated_stmin_ms, | executable/support line; behavior impac
depends on surrounding block context. |
| 330 |           } | executable/support line; behavior impact depends on surrounding block context.
| 331 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ifetime scopes. |
| 332 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 333 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 334 |   BridgeRequest::Close => { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 335 |       let rsp = state.make_ok(); | executable/support line; behavior impact depends on surrounding
ck context. |
| 336 |       write_response(&mut stdout, rsp); | I/O or protocol-critical logic; edits can affect integrat
, timing, and diagnostic outcomes. |
| 337 |       break; | executable/support line; behavior impact depends on surrounding block context. |
| 338 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 339 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 340 | (blank) | line is blank; visual separation only. |
| 341 |     write_response(&mut stdout, rsp); | I/O or protocol-critical logic; edits can affect integration, tim
, and diagnostic outcomes. |
| 342 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 343 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 344 | (blank) | line is blank; visual separation only. |
| 345 | fn write_response(stdout: &mut io::Stdout, rsp: BridgeResponse) { | declares executable behavior; signature/b
changes alter API, control flow, and outputs. |
| 346 |     let line = match serde_json::to_string(&rsp) { | serialization contract touchpoint; edits can break repor
ompatibility and tooling. |
| 347 |         Ok(v) => v, | executable/support line; behavior impact depends on surrounding block context. |
| 348 |         Err(e) => { | executable/support line; behavior impact depends on surrounding block context. |
| 349 |             let fallback = format!( | executable/support line; behavior impact depends on surrounding block c

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ext. |
| 350 |         "{{\"ok\":false,\"error\": \"serialize failed: {}\", \"kind\": \"parse\"}}", | executable/support
ine; behavior impact depends on surrounding block context. |
| 351 |         e | executable/support line; behavior impact depends on surrounding block context. |
| 352 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 353 |     let _ = writeln!(stdout, "{fallback}"); | I/O or protocol-critical logic; edits can affect integr
on, timing, and diagnostic outcomes. |
| 354 |     let _ = stdout.flush(); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 355 |     return; | executable/support line; behavior impact depends on surrounding block context. |
| 356 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 357 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 358 | let _ = writeln!(stdout, "{line}"); | I/O or protocol-critical logic; edits can affect integration, timin
and diagnostic outcomes. |
| 359 | let _ = stdout.flush(); | executable/support line; behavior impact depends on surrounding block context.
| 360 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 361 | (blank) | line is blank; visual separation only. |
| 362 | fn now_ms() -> u64 { | declares executable behavior; signature/body changes alter API, control flow, and outp
. |
| 363 |     std::time::SystemTime::now() | executable/support line; behavior impact depends on surrounding block cont
. |
| 364 |     .duration_since(std::time::UNIX_EPOCH) | executable/support line; behavior impact depends on surround
block context. |
| 365 |     .map(|d| d.as_millis() as u64) | executable/support line; behavior impact depends on surrounding bl
context. |
| 366 |     .unwrap_or(0) | executable/support line; behavior impact depends on surrounding block context. |
| 367 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/bin/simulator_cli.rs

Role: Executable entrypoint for CLI/bridge workflow.

Lines: 142 | Non-empty: 127 | Symbols: 4

Symbol Inventory

Line	Kind	Symbol	Change Impact
8	fn	arg_value	signature/API coupling
15	fn	has_flag	signature/API coupling
19	fn	arg_value_prefixed	signature/API coupling
24	fn	main	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	use std::env;	imports symbols; changing path/name affects compile-time resolution.
2	use std::path::Path;	imports symbols; changing path/name affects compile-time resolution.
3	use std::path::PathBuf;	imports symbols; changing path/name affects compile-time resolution.
4	(blank)	line is blank; visual separation only.
5	use auto_breaking::HeavyMachinery;	imports symbols; changing path/name affects compile-time resolution.
6	use auto_breaking::io;	imports symbols; changing path/name affects compile-time resolution.
7	(blank)	line is blank; visual separation only.
8	fn arg_value(args: &[String], name: &str) -> Option<String> {	declares executable behavior; signature/body changes alter API, control flow, and outputs.
9	args.iter()	executable/support line; behavior impact depends on surrounding block context.
10	.position(a a == name)	executable/support line; behavior impact depends on surrounding block context.
11	.and_then(i args.get(i + 1))	executable/support line; behavior impact depends on surrounding block context.
12	.cloned()	executable/support line; behavior impact depends on surrounding block context.
13	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
14	(blank)	line is blank; visual separation only.
15	fn has_flag(args: &[String], name: &str) -> bool {	declares executable behavior; signature/body changes alter API, control flow, and outputs.

```

PI, control flow, and outputs. |
| 16 |     args.iter().any(\|a\| a == name) | executable/support line; behavior impact depends on surrounding block context. |
| 17 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 18 | (blank) | line is blank; visual separation only. |
| 19 | fn arg_value_prefixed(args: &[String], prefix: &str) -> Option<String> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 20 |     args.iter() | executable/support line; behavior impact depends on surrounding block context. |
| 21 |     .find_map(\|a\| a.strip_prefix(prefix).map(\|v\| v.to_string())) | executable/support line; behavior impact depends on surrounding block context. |
| 22 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 23 | (blank) | line is blank; visual separation only. |
| 24 | fn main() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 25 |     let args: Vec<String> = env::args().collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 26 | (blank) | line is blank; visual separation only. |
| 27 |     if has_flag(&args, "--validate-cat-bridge") { | conditional gate; changing condition shifts runtime path and side effects. |
| 28 |         let cfg = match io::hw::HwConfig::from_cli_args(&args) { | executable/support line; behavior impact depends on surrounding block context. |
| 29 |             Ok(v) => v, | executable/support line; behavior impact depends on surrounding block context. |
| 30 |             Err(e) => { | executable/support line; behavior impact depends on surrounding block context. |
| 31 |                 eprintln!("bridge validate config error: {}", e); | executable/support line; behavior impact depends on surrounding block context. |
| 32 |                 std::process::exit(2); | executable/support line; behavior impact depends on surrounding block context. |
| 33 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 34 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 35 | (blank) | line is blank; visual separation only. |
| 36 |         match io::hw::probe_live_adapter(&cfg) { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 37 |             Ok(info) => { | executable/support line; behavior impact depends on surrounding block context. |
| 38 |                 println!( | executable/support line; behavior impact depends on surrounding block context. |
| 39 |                     "bridge_validate=ok mode={:?} capabilities={} adapter_info={}", | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 40 |                     cfg.mode, | executable/support line; behavior impact depends on surrounding block context. |
| 41 |                     cfg.capabilities_summary(), | executable/support line; behavior impact depends on surrounding block context. |
| 42 |                     info | executable/support line; behavior impact depends on surrounding block context. |
| 43 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 44 |                 return; | executable/support line; behavior impact depends on surrounding block context. |
| 45 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 46 |             Err(e) => { | executable/support line; behavior impact depends on surrounding block context. |
| 47 |                 eprintln!("bridge_validate=failed error={}", e); | executable/support line; behavior impact depends on surrounding block context. |
| 48 |                 std::process::exit(1); | executable/support line; behavior impact depends on surrounding block context. |
| 49 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 50 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 51 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 52 | (blank) | line is blank; visual separation only. |
| 53 |     if has_flag(&args, "--run-production-phases") { | conditional gate; changing condition shifts runtime path and side effects. |
| 54 |         let cfg = match io::hw::HwConfig::from_cli_args(&args) { | executable/support line; behavior impact depends on surrounding block context. |
| 55 |             Ok(v) => v, | executable/support line; behavior impact depends on surrounding block context. |
| 56 |             Err(e) => { | executable/support line; behavior impact depends on surrounding block context. |
| 57 |                 eprintln!("production-phase config error: {}", e); | executable/support line; behavior impact depends on surrounding block context. |
| 58 |                 std::process::exit(2); | executable/support line; behavior impact depends on surrounding block context. |

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```

| 59 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
s. |
| 60 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 61 |     let target_sa = arg_value_prefixed(&args, "--target-sa=") | executable/support line; behavior impact d
nds on surrounding block context. |
| 62 |         .and_then(|v| u8::from_str_radix(v.trim_start_matches("0x"), 16).ok()) | executable/support line
ehavior impact depends on surrounding block context. |
| 63 |         .or_else(|_| arg_value_prefixed(&args, "--target-sa-dec=").and_then(|v| v.parse::<u8>().ok()));
executable/support line; behavior impact depends on surrounding block context. |
| 64 |     let firmware = arg_value_prefixed(&args, "--firmware=").map(PathBuf::from); | executable/support line;
havior impact depends on surrounding block context. |
| 65 |     let report_dir = arg_value_prefixed(&args, "--phase-report-dir=") | executable/support line; behavior
act depends on surrounding block context. |
| 66 |         .map(PathBuf::from) | executable/support line; behavior impact depends on surrounding block context
|
| 67 |         .unwrap_or_else(|_| PathBuf::from("reports")); | executable/support line; behavior impact depends
surrounding block context. |
| 68 |     let options = io::production_program::ProgramOptions { | executable/support line; behavior impact depende
on surrounding block context. |
| 69 |         strict_mode: !has_flag(&args, "--allow-blocked-phases"), | executable/support line; behavior impac
depends on surrounding block context. |
| 70 |         execute_flash: has_flag(&args, "--execute-flash"), | I/O or protocol-critical logic; edits can aff
integration, timing, and diagnostic outcomes. |
| 71 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 72 | (blank) | line is blank; visual separation only. |
| 73 |     let result = io::production_program::run_all_phases( | executable/support line; behavior impact depend
n surrounding block context. |
| 74 |         &cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 75 |         target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 76 |         firmware.as_deref().map(Path::new), | executable/support line; behavior impact depends on surround
block context. |
| 77 |         &report_dir, | executable/support line; behavior impact depends on surrounding block context. |
| 78 |         options, | executable/support line; behavior impact depends on surrounding block context. |
| 79 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 80 |     match result { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 81 |         Ok(path) => { | executable/support line; behavior impact depends on surrounding block context. |
| 82 |             println!( | executable/support line; behavior impact depends on surrounding block context. |
| 83 |                 "production_phases=done report_json={?} mode={?}"; | error-propagating operation; changing
impacts failure semantics and caller recovery paths. |
| 84 |             path.display(), | executable/support line; behavior impact depends on surrounding block co
xt. |
| 85 |             cfg.mode | executable/support line; behavior impact depends on surrounding block context.
| 86 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 87 |         return; | executable/support line; behavior impact depends on surrounding block context. |
| 88 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
s. |
| 89 |     Err(e) => { | executable/support line; behavior impact depends on surrounding block context. |
| 90 |         eprintln!("production phases failed: {}", e); | executable/support line; behavior impact depende
on surrounding block context. |
| 91 |         std::process::exit(1); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 92 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
s. |
| 93 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 94 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 95 | (blank) | line is blank; visual separation only. |
| 96 | let seed = arg_value(&args, "--seed") | executable/support line; behavior impact depends on surrounding bl
context. |
| 97 |         .and_then(|s| s.parse::<u64>().ok()) | executable/support line; behavior impact depends on surroundi
block context. |
| 98 |         .unwrap_or(1); | executable/support line; behavior impact depends on surrounding block context. |
| 99 | let model = arg_value(&args, "--model").unwrap_or_else(|_| "reduced".to_string()); | executable/support l
; behavior impact depends on surrounding block context. |

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```

| 100 |     let steps = arg_value(&args, "--steps") | executable/support line; behavior impact depends on surrounding
ock context. |
| 101 |         .and_then(\|s\| s.parse::<usize>().ok()) | executable/support line; behavior impact depends on surrou
ng block context. |
| 102 |         .unwrap_or(10_000); | executable/support line; behavior impact depends on surrounding block context.
| 103 | (blank) | line is blank; visual separation only. |
| 104 |     if let Some(replay) = arg_value(&args, "--replay") { | conditional gate; changing condition shifts runtim
ath and side effects. |
| 105 |         let verify = has_flag(&args, "--verify"); | executable/support line; behavior impact depends on surro
ing block context. |
| 106 |         println!("replay_file={{} verify={} status=ok", replay, verify); | executable/support line; behavior i
ct depends on surrounding block context. |
| 107 |         return; | executable/support line; behavior impact depends on surrounding block context. |
| 108 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 109 | (blank) | line is blank; visual separation only. |
| 110 |     let mut sim = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding bl
context. |
| 111 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 112 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 113 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 114 | (blank) | line is blank; visual separation only. |
| 115 |     let mut lcg = seed; | executable/support line; behavior impact depends on surrounding block context. |
| 116 |     for i in 0..steps { | iteration control; changing bounds/body affects throughput, timing, and possibly sa
y guarantees. |
| 117 |         // Simple deterministic pseudo-randomized demand profile by seed. | comment line; documentation inten
no runtime effect. |
| 118 |         lcg = lcg.wrapping_mul(6364136223846793005).wrapping_add(1); | executable/support line; behavior impa
depends on surrounding block context. |
| 119 |         let n = ((lcg >> 33) as f64) / ((1u64 << 31) as f64); | executable/support line; behavior impact depe
on surrounding block context. |
| 120 |         sim.throttle_pct = (25.0 + n * 55.0).clamp(0.0, 100.0); | executable/support line; behavior impact de
ds on surrounding block context. |
| 121 |         sim.brake_pct = if i % 400 > 360 { 20.0 } else { 0.0 }; | executable/support line; behavior impact de
ds on surrounding block context. |
| 122 |         sim.loader_lift_cmd = if i % 140 < 70 { 0.5 } else { -0.4 }; | executable/support line; behavior impa
depends on surrounding block context. |
| 123 |         sim.loader_tilt_cmd = if i % 100 < 50 { 0.3 } else { -0.2 }; | executable/support line; behavior impa
depends on surrounding block context. |
| 124 |         sim.hitch_joystick = if i % 120 < 60 { 0.4 } else { -0.3 }; | executable/support line; behavior impac
depends on surrounding block context. |
| 125 |         sim.tick(1.0 / 60.0); | executable/support line; behavior impact depends on surrounding block context
| 126 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 127 | (blank) | line is blank; visual separation only. |
| 128 |     let summary = serde_json::json!({ | serialization contract touchpoint; edits can break report compatibility
and tooling. |
| 129 |         "seed": seed, | executable/support line; behavior impact depends on surrounding block context. |
| 130 |         "model": model, | executable/support line; behavior impact depends on surrounding block context. |
| 131 |         "steps": steps, | executable/support line; behavior impact depends on surrounding block context. |
| 132 |         "elapsed_s": sim.elapsed, | executable/support line; behavior impact depends on surrounding block con
t. |
| 133 |         "fuel_pct": sim.ecm.fuel_level_pct, | executable/support line; behavior impact depends on surrounding
ock context. |
| 134 |         "oil_kpa": sim.ecm.oil_pressure_kpa, | executable/support line; behavior impact depends on surroundin
lock context. |
| 135 |         "can_health": sim.can_net.network_health_score_01(), | executable/support line; behavior impact depen
on surrounding block context. |
| 136 |         "can_errors": sim.can_net.total_errors(), | executable/support line; behavior impact depends on surro
ing block context. |
| 137 |         "loop_duration_ms": sim.metrics.loop_duration_ms, | executable/support line; behavior impact depends
surrounding block context. |
| 138 |         "steps_completed": sim.metrics.steps_completed, | executable/support line; behavior impact depends on
rrounding block context. |
| 139 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 140 | (blank) | line is blank; visual separation only. |
| 141 |     println!("{}", summary); | executable/support line; behavior impact depends on surrounding block context.

```

| 142 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

File: src/boot_sequence.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 517 | Non-empty: 468 | Symbols: 29

Symbol Inventory

Line	Kind	Symbol	Change Impact
---:	---	---	---
19	enum	IgnitionState	data/structure coupling
32	impl	std::fmt::Display for IgnitionState	behavior binding and method semantics
33	fn	fmt	signature/API coupling
47	enum	EcuBootStage	data/structure coupling
66	impl	std::fmt::Display for EcuBootStage	behavior binding and method semantics
67	fn	fmt	signature/API coupling
85	struct	EcuBootRecord	data/structure coupling
98	impl	EcuBootRecord	behavior binding and method semantics
99	fn	new	signature/API coupling
111	fn	is_online	signature/API coupling
115	fn	is_faulted	signature/API coupling
123	struct	BootEvent	data/structure coupling
131	enum	BootEventKind	data/structure coupling
146	struct	SafetyInterlock	data/structure coupling
155	struct	BootSequence	data/structure coupling
177	impl	Default for BootSequence	behavior binding and method semantics
178	fn	default	signature/API coupling
183	impl	BootSequence	behavior binding and method semantics
184	fn	new	signature/API coupling
271	fn	key_advance	signature/API coupling
296	fn	key_off	signature/API coupling
313	fn	tick	signature/API coupling
360	fn	boot_ecu	signature/API coupling
453	fn	update_interlocks	signature/API coupling
461	fn	check_crank_inhibit	signature/API coupling
473	fn	check_all_critical_online	signature/API coupling
486	fn	push_event	signature/API coupling
500	fn	ecu_by_sa	signature/API coupling
508	fn	build_address_claim	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
---:	---	---
1	/// Boot Sequence Module – models every step from turning the key to full operation.	comment line; documentation intent, no runtime effect.
2	///	comment line; documentation intent, no runtime effect.
3	/// Implements the real power-up sequence of a heavy machine:	comment line; documentation intent, no runtime effect.
4	/// OFF → ACCESSORY → IGN_ON → SELF_TEST → ADDRESS_CLAIM → PRE_START_CHECKS	comment line; documentation intent, no runtime effect.
5	/// → CRANKING → START_CONFIRM → RUNNING	comment line; documentation intent, no runtime effect.
6	///	comment line; documentation intent, no runtime effect.
7	/// J1939 Address Claiming (AC) procedure per SAE J1939-81:	comment line; documentation intent, no runtime effect.
8	/// 1. Node broadcasts its 64-bit NAME on null address (0xFE)	comment line; documentation intent, no runtime effect.
9	/// 2. Waits 250 ms for conflicts	comment line; documentation intent, no runtime effect.
10	/// 3. If no conflict: claims the desired SA	comment line; documentation intent, no runtime effect.
11	/// 4. If conflict: lower NAME wins; loser tries next available SA or goes null	comment line; documentation intent, no runtime effect.
12	(blank)	line is blank; visual separation only.
13	use crate::can_gateway::CanGateway;	imports symbols; changing path/name affects compile-time resolution.
14	use crate::j1939::{self, addr, J1939Frame};	imports symbols; changing path/name affects compile-time resolution.

[illegible]

```
| 60 |      /// Fully operational | comment line; documentation intent, no runtime effect. |  
| 61 | Running, | executable/support line; behavior impact depends on surrounding block context. |  
| 62 |    /// Fault – cannot enter normal operation | comment line; documentation intent, no runtime effect. |  
| 63 | Fault, | executable/support line; behavior impact depends on surrounding block context. |  
| 64 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 65 | (blank) | line is blank; visual separation only. |  
| 66 | impl std::fmt::Display for EcuBootStage { | implementation block; method behavior and trait conformance are de  
ed here. |  
| 67 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signa  
e/body changes alter API, control flow, and outputs. |  
| 68 |         let s = match self { | executable/support line; behavior impact depends on surrounding block context.  
| 69 |             EcuBootStage::Unpowered => "UNPOWERED          ", | executable/support line; behavior impact depends o  
urrounding block context. |  
| 70 |             EcuBootStage::HardwareInit => "HW INIT           ", | executable/support line; behavior impact depend  
n surrounding block context. |  
| 71 |             EcuBootStage::LoadingCalibration => "LOADING CAL   ", | executable/support line; behavior impact  
ends on surrounding block context. |  
| 72 |             EcuBootStage::AddressClaiming => "ADDR CLAIM       ", | executable/support line; behavior impact dep  
s on surrounding block context. |  
| 73 |             EcuBootStage::Handshaking => "HANDSHAKING        ", | executable/support line; behavior impact depende  
surrounding block context. |  
| 74 |             EcuBootStage::PreOperational => "PRE-OPER CHECK ", | executable/support line; behavior impact depende  
on surrounding block context. |  
| 75 |             EcuBootStage::Running => "RUNNING ✓              ", | executable/support line; behavior impact depends on  
rounding block context. |  
| 76 |             EcuBootStage::Fault => "FAULT ✗                  ", | executable/support line; behavior impact depends on su  
unding block context. |  
| 77 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 78 |         write!(f, "{}", s) | I/O or protocol-critical logic; edits can affect integration, timing, and diagnos  
outcomes. |  
| 79 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 80 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 81 | (blank) | line is blank; visual separation only. |  
| 82 | // ■■ Per-ECU boot record ████████████████████████████████████████████████████████████████ | comment line  
tent, no runtime effect. |  
| 83 | (blank) | line is blank; visual separation only. |  
| 84 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |  
| 85 | pub struct EcuBootRecord { | declares data layout; field changes may break serialization, tests, and callers.  
| 86 |     pub name: &'static str, | executable/support line; behavior impact depends on surrounding block context. |  
| 87 |     pub sa: u8, | executable/support line; behavior impact depends on surrounding block context. |  
| 88 |     /// 64-bit J1939 NAME (unique per device class/instance) | comment line; documentation intent, no runtime  
ect. |  
| 89 |     pub j1939_name: u64, | executable/support line; behavior impact depends on surrounding block context. |  
| 90 |     pub stage: EcuBootStage, | executable/support line; behavior impact depends on surrounding block context.  
| 91 |     /// Accumulated time in the current stage | comment line; documentation intent, no runtime effect. |  
| 92 |     pub stage_timer: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 93 |     /// When the ECU first transitioned to Running | comment line; documentation intent, no runtime effect. |  
| 94 |     pub online_at: Option<f64>, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 95 |     pub boot_error: Option<&'static str>, | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 96 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 97 | (blank) | line is blank; visual separation only. |  
| 98 | impl EcuBootRecord { | implementation block; method behavior and trait conformance are defined here. |  
| 99 |     pub fn new(name: &'static str, sa: u8, j1939_name: u64) -> Self { | declares executable behavior; signatur  
ody changes alter API, control flow, and outputs. |  
| 100 |         EcuBootRecord { | executable/support line; behavior impact depends on surrounding block context. |  
| 101 |             name, | executable/support line; behavior impact depends on surrounding block context. |  
| 102 |             sa, | executable/support line; behavior impact depends on surrounding block context. |  
| 103 |             j1939_name, | executable/support line; behavior impact depends on surrounding block context. |  
| 104 |             stage: EcuBootStage::Unpowered, | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 105 |             stage_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context  
| 106 |             online_at: None, | executable/support line; behavior impact depends on surrounding block context.
```


[illegible]

```

t. |
| 158 | pub event_log: Vec<BootEvent>, | executable/support line; behavior impact depends on surrounding block context. |
| 159 | pub safety_interlocks: Vec<SafetyInterlock>, | executable/support line; behavior impact depends on surrounding block context. |
| 160 | (blank) | line is blank; visual separation only. |
| 161 | // Global sequence timer (resets each state transition) | comment line; documentation intent, no runtime effect. |
| 162 | sequence_timer: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 163 | (blank) | line is blank; visual separation only. |
| 164 | // Are all critical ECUs online? | comment line; documentation intent, no runtime effect. |
| 165 | pub all_critical_online: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 166 | (blank) | line is blank; visual separation only. |
| 167 | // Crank inhibit | comment line; documentation intent, no runtime effect. |
| 168 | pub crank_inhibited: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 169 | pub crank_inhibit_reason: Option<&'static str>, | executable/support line; behavior impact depends on surrounding block context. |
| 170 | (blank) | line is blank; visual separation only. |
| 171 | // Engine actually running (>400 RPM) – set by ECM | comment line; documentation intent, no runtime effect. |
| 172 | pub engine_running: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 173 | (blank) | line is blank; visual separation only. |
| 174 | pub elapsed: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 175 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 176 | (blank) | line is blank; visual separation only. |
| 177 | impl Default for BootSequence { | implementation block; method behavior and trait conformance are defined here. |
| 178 | fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 179 |     Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 180 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 181 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 182 | (blank) | line is blank; visual separation only. |
| 183 | impl BootSequence { | implementation block; method behavior and trait conformance are defined here. |
| 184 | pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 185 | // Register all ECUs with their J1939 NAMES. | comment line; documentation intent, no runtime effect. |
| 186 | // NAME bits: [62:60]=industry(0=global,2=ag), [59:56]=vehicle-system, | comment line; documentation intent, no runtime effect. |
| 187 | // [55:52]=function, [47:40]=manufacturer, [28:21]=ECU-instance | comment line; documentation intent, no runtime effect. |
| 188 | let ecus = vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 189 |     EcuBootRecord::new("ECM #1", addr::ECM_1, 0x0000_6000_0000_0000), | executable/support line; behavior impact depends on surrounding block context. |
| 190 |     EcuBootRecord::new("TCM", addr::TRANSMISSION, 0x0000_6000_0100_0000), | executable/support line; behavior impact depends on surrounding block context. |
| 191 |     EcuBootRecord::new("BCM", addr::CAB, 0x0000_6000_0200_0000), | executable/support line; behavior impact depends on surrounding block context. |
| 192 |     EcuBootRecord::new("ICM/DASH", addr::INSTRUMENT, 0x0000_6000_0300_0000), | executable/support line; behavior impact depends on surrounding block context. |
| 193 |     EcuBootRecord::new("HCM/HITCH", addr::HITCH, 0x0000_6000_0400_0000), | executable/support line; behavior impact depends on surrounding block context. |
| 194 |     EcuBootRecord::new("ABS/STAB", addr::BRAKES, 0x0000_6000_0500_0000), | executable/support line; behavior impact depends on surrounding block context. |
| 195 |     EcuBootRecord::new("ISOBUS-VT", addr::ISOBUS_VT, 0x0000_6000_0600_0000), | executable/support line; behavior impact depends on surrounding block context. |
| 196 | ]; | executable/support line; behavior impact depends on surrounding block context. |
| 197 | (blank) | line is blank; visual separation only. |
| 198 | // Real machine safety interlocks (prevents engine start if not met) | comment line; documentation intent, no runtime effect. |
| 199 | let interlocks = vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 200 |     SafetyInterlock { | executable/support line; behavior impact depends on surrounding block context.

```

```
| 201 |         id: "TRANS_NEUTRAL", | executable/support line; behavior impact depends on surrounding block  
text. |  
| 202 |         description: "Transmission in neutral/park", | executable/support line; behavior impact depen  
on surrounding block context. |  
| 203 |         satisfied: false, | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 204 |         blocks_start: true, | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 205 |     }, | executable/support line; behavior impact depends on surrounding block context. |  
| 206 |     SafetyInterlock { | executable/support line; behavior impact depends on surrounding block context  
| 207 |         id: "PARK_BRAKE", | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 208 |         description: "Parking brake applied", | executable/support line; behavior impact depends on s  
urrounding block context. |  
| 209 |         satisfied: true, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 210 |         blocks_start: false, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 211 |     }, | executable/support line; behavior impact depends on surrounding block context. |  
| 212 |     SafetyInterlock { | executable/support line; behavior impact depends on surrounding block context  
| 213 |         id: "SEAT_OCCUPIED", | executable/support line; behavior impact depends on surrounding block  
text. |  
| 214 |         description: "Operator seat occupied", | executable/support line; behavior impact depends on  
rounding block context. |  
| 215 |         satisfied: true, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 216 |         blocks_start: true, | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 217 |     }, | executable/support line; behavior impact depends on surrounding block context. |  
| 218 |     SafetyInterlock { | executable/support line; behavior impact depends on surrounding block context  
| 219 |         id: "DOOR_CLOSED", | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 220 |         description: "Cab door closed", | executable/support line; behavior impact depends on surroun  
g block context. |  
| 221 |         satisfied: true, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 222 |         blocks_start: false, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 223 |     }, | executable/support line; behavior impact depends on surrounding block context. |  
| 224 |     SafetyInterlock { | executable/support line; behavior impact depends on surrounding block context  
| 225 |         id: "PTO_OFF", | executable/support line; behavior impact depends on surrounding block contex  
|  
| 226 |         description: "PTO disengaged", | executable/support line; behavior impact depends on surround  
block context. |  
| 227 |         satisfied: true, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 228 |         blocks_start: false, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 229 |     }, | executable/support line; behavior impact depends on surrounding block context. |  
| 230 |     SafetyInterlock { | executable/support line; behavior impact depends on surrounding block context  
| 231 |         id: "HITCH_UP", | executable/support line; behavior impact depends on surrounding block conte  
|  
| 232 |         description: "Rear hitch raised for road", | executable/support line; behavior impact depends  
surrounding block context. |  
| 233 |         satisfied: true, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 234 |         blocks_start: false, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 235 |     }, | executable/support line; behavior impact depends on surrounding block context. |  
| 236 |     SafetyInterlock { | executable/support line; behavior impact depends on surrounding block context  
| 237 |         id: "OIL_PRESS_OK", | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 238 |         description: "Engine oil pressure in range", | executable/support line; behavior impact depen  
on surrounding block context. |  
| 239 |         satisfied: true, | executable/support line; behavior impact depends on surrounding block cont
```

```

. |
| 240 |         blocks_start: false, | executable/support line; behavior impact depends on surrounding block
text. |
| 241 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 242 |     SafetyInterlock { | executable/support line; behavior impact depends on surrounding block context.
| 243 |         id: "COOLANT_OK", | executable/support line; behavior impact depends on surrounding block con
t. |
| 244 |         description: "Coolant level OK", | executable/support line; behavior impact depends on surrou
ng block context. |
| 245 |         satisfied: true, | executable/support line; behavior impact depends on surrounding block cont
. |
| 246 |         blocks_start: false, | executable/support line; behavior impact depends on surrounding block
text. |
| 247 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 248 |     SafetyInterlock { | executable/support line; behavior impact depends on surrounding block context
| 249 |         id: "DEF_LEVEL_OK", | executable/support line; behavior impact depends on surrounding block c
ext. |
| 250 |         description: "DEF level ≥ 5%", | executable/support line; behavior impact depends on surround
block context. |
| 251 |         satisfied: true, | executable/support line; behavior impact depends on surrounding block cont
. |
| 252 |         blocks_start: false, | executable/support line; behavior impact depends on surrounding block
text. |
| 253 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 254 | ]; | executable/support line; behavior impact depends on surrounding block context. |
| 255 | (blank) | line is blank; visual separation only. |
| 256 | BootSequence { | executable/support line; behavior impact depends on surrounding block context. |
| 257 |     ignition: IgnitionState::Off, | executable/support line; behavior impact depends on surrounding b
k context. |
| 258 |     ecus, | executable/support line; behavior impact depends on surrounding block context. |
| 259 |     event_log: Vec::new(), | executable/support line; behavior impact depends on surrounding block co
xt. |
| 260 |     safety_interlocks: interlocks, | executable/support line; behavior impact depends on surrounding
ck context. |
| 261 |     sequence_timer: 0.0, | executable/support line; behavior impact depends on surrounding block cont
. |
| 262 |     all_critical_online: false, | executable/support line; behavior impact depends on surrounding blo
context. |
| 263 |     crank_inhibited: false, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 264 |     crank_inhibit_reason: None, | executable/support line; behavior impact depends on surrounding blo
context. |
| 265 |     engine_running: false, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 266 |     elapsed: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 267 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 268 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 269 | (blank) | line is blank; visual separation only. |
| 270 |     /// Advance the ignition key one step | comment line; documentation intent, no runtime effect. |
| 271 |     pub fn key_advance(&mut self) { | declares executable behavior; signature/body changes alter API, control
ow, and outputs. |
| 272 |         self.ignition = match self.ignition { | executable/support line; behavior impact depends on surroundi
block context. |
| 273 |             IgnitionState::Off => IgnitionState::Accessory, | executable/support line; behavior impact depend
n surrounding block context. |
| 274 |             IgnitionState::Accessory => IgnitionState::On, | executable/support line; behavior impact depends
surrounding block context. |
| 275 |             IgnitionState::On => { | executable/support line; behavior impact depends on surrounding block co
xt. |
| 276 |                 if self.crank_inhibited { | conditional gate; changing condition shifts runtime path and side
fects. |
| 277 |                     IgnitionState::On | executable/support line; behavior impact depends on surrounding block
ntext. |
| 278 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti

```

```

scopes. |
| 279 |         // can't start | comment line; documentation intent, no runtime effect. |
| 280 |         else { | executable/support line; behavior impact depends on surrounding block context. |
| 281 |             IgnitionState::Cranking | executable/support line; behavior impact depends on surrounding
ock context. |
| 282 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
scopes. |
| 283 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 284 |         IgnitionState::Cranking => IgnitionState::On, // spring-return | executable/support line; behavior
mpact depends on surrounding block context. |
| 285 |         IgnitionState::Running => IgnitionState::On, // engine still running | executable/support line;
avior impact depends on surrounding block context. |
| 286 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 287 |         self.sequence_timer = 0.0; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 288 |         self.push_event( | executable/support line; behavior impact depends on surrounding block context. |
| 289 |             "IGNITION", | executable/support line; behavior impact depends on surrounding block context. |
| 290 |             BootEventKind::PowerOn, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 291 |             format!("Key advanced to {:?}", self.ignition), | error-propagating operation; changing it impact
ailure semantics and caller recovery paths. |
| 292 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 293 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 294 | (blank) | line is blank; visual separation only. |
| 295 |     /// Turn key off | comment line; documentation intent, no runtime effect. |
| 296 |     pub fn key_off(&mut self) { | declares executable behavior; signature/body changes alter API, control flo
and outputs. |
| 297 |         self.ignition = IgnitionState::Off; | executable/support line; behavior impact depends on surrounding
ock context. |
| 298 |         self.engine_running = false; | executable/support line; behavior impact depends on surrounding block
text. |
| 299 |         // Power down all ECUs | comment line; documentation intent, no runtime effect. |
| 300 |         for ecu in &mut self.ecus { | iteration control; changing bounds/body affects throughput, timing, and
ssibly safety guarantees. |
| 301 |             ecu.stage = EcuBootStage::Unpowered; | executable/support line; behavior impact depends on surrou
ng block context. |
| 302 |             ecu.stage_timer = 0.0; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 303 |             ecu.online_at = None; | executable/support line; behavior impact depends on surrounding block con
t. |
| 304 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 305 |         self.push_event( | executable/support line; behavior impact depends on surrounding block context. |
| 306 |             "IGNITION", | executable/support line; behavior impact depends on surrounding block context. |
| 307 |             BootEventKind::PowerOn, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 308 |             "Key OFF – all ECUs unpowered".into(), | executable/support line; behavior impact depends on surr
ding block context. |
| 309 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 310 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 311 | (blank) | line is blank; visual separation only. |
| 312 |     /// Update the boot sequence state machine. Returns CAN frames to transmit. | comment line; documentation
tent, no runtime effect. |
| 313 |     pub fn tick( | declares executable behavior; signature/body changes alter API, control flow, and outputs.
| 314 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 315 |         dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 316 |         gateway: &mut CanGateway, | executable/support line; behavior impact depends on surrounding block con
t. |
| 317 |         trans_neutral: bool, | executable/support line; behavior impact depends on surrounding block context.
| 318 |         powered_sas: &[u8], | executable/support line; behavior impact depends on surrounding block context.
| 319 |     ) -> Vec<J1939Frame> { | executable/support line; behavior impact depends on surrounding block context. |
| 320 |         self.elapsed += dt; | executable/support line; behavior impact depends on surrounding block context.
| 321 |         self.sequence_timer += dt; | executable/support line; behavior impact depends on surrounding block co

```

```

xt. |
| 322 | (blank) | line is blank; visual separation only. | | |
| 323 |         // Update safety interlocks | comment line; documentation intent, no runtime effect. |
| 324 |         self.update_interlocks(trans_neutral); | executable/support line; behavior impact depends on surrounding block context. |
| 325 |         self.check_crank_inhibit(); | executable/support line; behavior impact depends on surrounding block context. |
| 326 | (blank) | line is blank; visual separation only. |
| 327 |         let mut frames: Vec<J1939Frame> = Vec::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 328 | (blank) | line is blank; visual separation only. |
| 329 |         for i in 0..self.ecus.len() { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 330 |             let sa = self.ecus[i].sa; | executable/support line; behavior impact depends on surrounding block context. |
| 331 |             if powered_sas.contains(&sa) { | conditional gate; changing condition shifts runtime path and side effects. |
| 332 |                 self.boot_ecu(sa, dt, &mut frames); | executable/support line; behavior impact depends on surrounding block context. |
| 333 |             } else if self.ecus[i].stage != EcuBootStage::Unpowered { | executable/support line; behavior impact depends on surrounding block context. |
| 334 |                 self.ecus[i].stage = EcuBootStage::Unpowered; | executable/support line; behavior impact depends on surrounding block context. |
| 335 |                 self.ecus[i].stage_timer = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 336 |                 self.ecus[i].online_at = None; | executable/support line; behavior impact depends on surrounding block context. |
| 337 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 338 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 339 | (blank) | line is blank; visual separation only. |
| 340 |         self.check_all_critical_online(); | executable/support line; behavior impact depends on surrounding block context. |
| 341 | (blank) | line is blank; visual separation only. |
| 342 |         // Crank → Running transition (engine fires) | comment line; documentation intent, no runtime effect. |
| 343 |         if self.ignition == IgnitionState::Cranking && self.engine_running { | conditional gate; changing condition shifts runtime path and side effects. |
| 344 |             self.ignition = IgnitionState::Running; | executable/support line; behavior impact depends on surrounding block context. |
| 345 |             self.push_event( | executable/support line; behavior impact depends on surrounding block context. |
| 346 |                 "IGNITION", | executable/support line; behavior impact depends on surrounding block context. |
| 347 |                 BootEventKind::Running, | executable/support line; behavior impact depends on surrounding block context. |
| 348 |                 "Engine fired – transition to RUNNING".into(), | executable/support line; behavior impact depends on surrounding block context. |
| 349 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 350 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 351 | (blank) | line is blank; visual separation only. |
| 352 |         // Dispatch frames to gateway | comment line; documentation intent, no runtime effect. |
| 353 |         for f in &frames { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 354 |             gateway.transmit(f.clone()); | executable/support line; behavior impact depends on surrounding block context. |
| 355 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 356 |         frames | executable/support line; behavior impact depends on surrounding block context. |
| 357 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 358 | (blank) | line is blank; visual separation only. |
| 359 |     /// Run one ECU through its boot state machine | comment line; documentation intent, no runtime effect. |
| 360 |     fn boot_ecu(&mut self, sa: u8, dt: f64, frames: &mut Vec<J1939Frame>) { | declares executable behavior; structure/body changes alter API, control flow, and outputs. |
| 361 |         let idx = match self.ecus.iter().position(|e| e.sa == sa) { | executable/support line; behavior impact depends on surrounding block context. |

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```

| 362 |         Some(i) => i, | executable/support line; behavior impact depends on surrounding block context. |
| 363 |         None => return, | executable/support line; behavior impact depends on surrounding block context. |
| 364 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 365 |         let ts = self.elapsed; | executable/support line; behavior impact depends on surrounding block contex
|
| 366 | (blank) | line is blank; visual separation only. |
| 367 |         // Copy all data we need BEFORE taking a mutable borrow of self | comment line; documentation intent,
runtime effect. |
| 368 |         let stage = self.ecus[idx].stage; | executable/support line; behavior impact depends on surrounding b
k context. |
| 369 |         let stage_timer = self.ecus[idx].stage_timer; | executable/support line; behavior impact depends on s
ounding block context. |
| 370 |         let ecu_name = self.ecus[idx].name; | executable/support line; behavior impact depends on surrounding
ock context. |
| 371 |         let ecu_sa = self.ecus[idx].sa; | executable/support line; behavior impact depends on surrounding blo
context. |
| 372 |         let j1939_name = self.ecus[idx].j1939_name; | executable/support line; behavior impact depends on sur
nding block context. |
| 373 | (blank) | line is blank; visual separation only. |
| 374 |         self.ecus[idx].stage_timer += dt; | executable/support line; behavior impact depends on surrounding b
k context. |
| 375 | (blank) | line is blank; visual separation only. |
| 376 |         match stage { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 377 |             EcuBootStage::Unpowered => { | executable/support line; behavior impact depends on surrounding bl
context. |
| 378 |                 self.ecus[idx].stage = EcuBootStage::HardwareInit; | executable/support line; behavior impact
ends on surrounding block context. |
| 379 |                 self.ecus[idx].stage_timer = 0.0; | executable/support line; behavior impact depends on surro
ing block context. |
| 380 |                 self.ecus[idx].boot_error = None; | executable/support line; behavior impact depends on surro
ing block context. |
| 381 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 382 |             EcuBootStage::HardwareInit => { | executable/support line; behavior impact depends on surrounding
ock context. |
| 383 |                 if stage_timer >= 0.050 { | conditional gate; changing condition shifts runtime path and side
fects. |
| 384 |                     self.ecus[idx].stage = EcuBootStage::LoadingCalibration; | executable/support line; behav
impact depends on surrounding block context. |
| 385 |                     self.ecus[idx].stage_timer = 0.0; | executable/support line; behavior impact depends on s
ounding block context. |
| 386 |                     self.push_event( | executable/support line; behavior impact depends on surrounding block
text. |
| 387 |                         ecu_name, | executable/support line; behavior impact depends on surrounding block con
t. |
| 388 |                         BootEventKind::StageChange, | executable/support line; behavior impact depends on sur
nding block context. |
| 389 |                         format!("{: HW init OK → loading calibration", ecu_name), | executable/support line
havior impact depends on surrounding block context. |
| 390 |                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 391 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 392 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 393 |             EcuBootStage::LoadingCalibration => { | executable/support line; behavior impact depends on surro
ing block context. |
| 394 |                 if stage_timer >= 0.080 { | conditional gate; changing condition shifts runtime path and side
fects. |
| 395 |                     self.ecus[idx].stage = EcuBootStage::AddressClaiming; | executable/support line; behavior
pact depends on surrounding block context. |
| 396 |                     self.ecus[idx].stage_timer = 0.0; | executable/support line; behavior impact depends on s
ounding block context. |
| 397 |                     self.push_event( | executable/support line; behavior impact depends on surrounding block
text. |

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```

| 398 |          ecu_name, | executable/support line; behavior impact depends on surrounding block con
t. |
| 399 |          BootEventKind::StageChange, | executable/support line; behavior impact depends on sur
nding block context. |
| 400 |          format!( | executable/support line; behavior impact depends on surrounding block cont
. |
| 401 |          "{}: Calibration loaded → claiming SA 0x{:02X}", | executable/support line; behav
impact depends on surrounding block context. |
| 402 |          ecu_name, ecu_sa | executable/support line; behavior impact depends on surroundin
lock context. |
| 403 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 404 |          ); | executable/support line; behavior impact depends on surrounding block context. |
| 405 |          frames.push(build_address_claim(ts, j1939_name, ecu_sa)); | executable/support line; beha
r impact depends on surrounding block context. |
| 406 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 407 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 408 |          EcuBootStage::AddressClaiming => { | executable/support line; behavior impact depends on surround
block context. |
| 409 |              if stage_timer >= 0.250 { | conditional gate; changing condition shifts runtime path and side
fects. |
| 410 |                  self.ecus[idx].stage = EcuBootStage::Handshaking; | executable/support line; behavior imp
depends on surrounding block context. |
| 411 |                  self.ecus[idx].stage_timer = 0.0; | executable/support line; behavior impact depends on s
ounding block context. |
| 412 |                  self.push_event( | executable/support line; behavior impact depends on surrounding block
text. |
| 413 |                      ecu_name, | executable/support line; behavior impact depends on surrounding block con
t. |
| 414 |                      BootEventKind::AddressClaimed, | executable/support line; behavior impact depends on
rounding block context. |
| 415 |                      format!( | executable/support line; behavior impact depends on surrounding block cont
. |
| 416 |                      "{}: Address 0x{:02X} claimed successfully", | executable/support line; behavior
act depends on surrounding block context. |
| 417 |                      ecu_name, ecu_sa | executable/support line; behavior impact depends on surroundin
lock context. |
| 418 |                      ), | executable/support line; behavior impact depends on surrounding block context. |
| 419 |                      ); | executable/support line; behavior impact depends on surrounding block context. |
| 420 |                      frames.push(build_address_claim(ts, j1939_name, ecu_sa)); | executable/support line; beha
r impact depends on surrounding block context. |
| 421 |                  } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 422 |              } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 423 |          EcuBootStage::Handshaking => { | executable/support line; behavior impact depends on surrounding
ck context. |
| 424 |              if stage_timer >= 0.150 { | conditional gate; changing condition shifts runtime path and side
fects. |
| 425 |                  self.ecus[idx].stage = EcuBootStage::PreOperational; | executable/support line; behavior
act depends on surrounding block context. |
| 426 |                  self.ecus[idx].stage_timer = 0.0; | executable/support line; behavior impact depends on s
ounding block context. |
| 427 |                  self.push_event( | executable/support line; behavior impact depends on surrounding block
text. |
| 428 |                      ecu_name, | executable/support line; behavior impact depends on surrounding block con
t. |
| 429 |                      BootEventKind::HandshakeOk, | executable/support line; behavior impact depends on sur
nding block context. |
| 430 |                      format!("{}", Peer handshake OK", ecu_name), | executable/support line; behavior impac
depends on surrounding block context. |
| 431 |                      ); | executable/support line; behavior impact depends on surrounding block context. |
| 432 |                  } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |

```



```

| 433 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 434 |         EcuBootStage::PreOperational => { | executable/support line; behavior impact depends on surroundi
block context. |
| 435 |             if stage_timer >= 0.200 { | conditional gate; changing condition shifts runtime path and side
fects. |
| 436 |                 self.ecus[idx].stage = EcuBootStage::Running; | executable/support line; behavior impact
ends on surrounding block context. |
| 437 |                 self.ecus[idx].stage_timer = 0.0; | executable/support line; behavior impact depends on s
ounding block context. |
| 438 |                 self.ecus[idx].online_at = Some(ts); | executable/support line; behavior impact depends o
urrounding block context. |
| 439 |                 self.push_event( | executable/support line; behavior impact depends on surrounding block
text. |
| 440 |                     ecu_name, | executable/support line; behavior impact depends on surrounding block con
t. |
| 441 |                     BootEventKind::Running, | executable/support line; behavior impact depends on surroun
g block context. |
| 442 |                     format!("{: Pre-op checks passed → RUNNING at {:.3}s", ecu_name, ts), | executable/s
ort line; behavior impact depends on surrounding block context. |
| 443 |                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 444 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 445 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 446 | (blank) | line is blank; visual separation only. |
| 447 |         EcuBootStage::Running \ EcuBootStage::Fault => { | executable/support line; behavior impact depe
on surrounding block context. |
| 448 |             // Nothing more to do in boot state machine | comment line; documentation intent, no runtime
ect. |
| 449 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 450 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 451 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 452 | (blank) | line is blank; visual separation only. |
| 453 | fn update_interlocks(&mut self, trans_neutral: bool) { | declares executable behavior; signature/body cha
s alter API, control flow, and outputs. |
| 454 |     for il in &mut self.safety_interlocks { | iteration control; changing bounds/body affects throughput,
ming, and possibly safety guarantees. |
| 455 |         if il.id == "TRANS_NEUTRAL" { | conditional gate; changing condition shifts runtime path and side
fects. |
| 456 |             il.satisfied = trans_neutral; | executable/support line; behavior impact depends on surroundi
block context. |
| 457 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 458 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 459 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 460 | (blank) | line is blank; visual separation only. |
| 461 | fn check_crank_inhibit(&mut self) { | declares executable behavior; signature/body changes alter API, con
l flow, and outputs. |
| 462 |     self.crank_inhibited = false; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 463 |     self.crank_inhibit_reason = None; | executable/support line; behavior impact depends on surrounding b
k context. |
| 464 |     for il in &self.safety_interlocks { | iteration control; changing bounds/body affects throughput, tim
, and possibly safety guarantees. |
| 465 |         if il.blocks_start && !il.satisfied { | conditional gate; changing condition shifts runtime path
side effects. |
| 466 |             self.crank_inhibited = true; | executable/support line; behavior impact depends on surroundin
lock context. |
| 467 |             self.crank_inhibit_reason = Some(il.id); | executable/support line; behavior impact depends o
urrounding block context. |
| 468 |             break; | executable/support line; behavior impact depends on surrounding block context. |

```

[illegible]

```
| 514 |      // PGN 60928 = 0xEE00, priority 6 | comment line; documentation intent, no runtime effect. |
| 515 |      let raw_id = J1939Frame::build_id(6, j1939::pgn::ECAN, sa, 0xFF); | executable/support line; behavior imp
depends on surrounding block context. |
| 516 |      J1939Frame::from_raw(ts, raw_id, &data) | executable/support line; behavior impact depends on surrounding
ock context. |
| 517 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
```

File: src/camera.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 602 | Non-empty: 556 | Symbols: 38

Symbol Inventory

Line	Kind	Symbol	Change Impact
8	enum	MarkType	data/structure coupling
17	impl	std::fmt::Display for MarkType	behavior binding and method semantics
18	fn	fmt	signature/API coupling
33	enum	SignType	data/structure coupling
45	impl	std::fmt::Display for SignType	behavior binding and method semantics
46	fn	fmt	signature/API coupling
63	struct	DetectedSign	data/structure coupling
74	enum	ObjectClass	data/structure coupling
87	impl	std::fmt::Display for ObjectClass	behavior binding and method semantics
88	fn	fmt	signature/API coupling
105	struct	DetectedObject	data/structure coupling
122	struct	LaneDetection	data/structure coupling
141	impl	Default for LaneDetection	behavior binding and method semantics
142	fn	default	signature/API coupling
147	impl	LaneDetection	behavior binding and method semantics
148	fn	new	signature/API coupling
171	struct	CameraSpec	data/structure coupling
181	impl	CameraSpec	behavior binding and method semantics
182	fn	forward_adas	signature/API coupling
193	fn	surround_wide	signature/API coupling
207	struct	CameraSystem	data/structure coupling
245	impl	Default for CameraSystem	behavior binding and method semantics
246	fn	default	signature/API coupling
251	impl	CameraSystem	behavior binding and method semantics
252	fn	new	signature/API coupling
279	fn	update	signature/API coupling
407	fn	closest_object	signature/API coupling
419	struct	FusedObject	data/structure coupling
435	impl	FusedObject	behavior binding and method semantics
436	fn	new	signature/API coupling
458	fn	predict	signature/API coupling
467	fn	update_with_measurement	signature/API coupling
492	struct	SensorFusion	data/structure coupling
499	impl	Default for SensorFusion	behavior binding and method semantics
500	fn	default	signature/API coupling
505	impl	SensorFusion	behavior binding and method semantics
506	fn	new	signature/API coupling
515	fn	fuse	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	///! Camera-based perception – simulates automotive vision system.	comment line; documentation intent, no runtime effect.
2	#![allow(dead_code)]	executable/support line; behavior impact depends on surrounding block context.
3	///! Models: forward camera (ADAS), surround-view, traffic sign recognition,	comment line; documentation intent, no runtime effect.
4	///! object detection (YOLOv8-style outputs), lane detection with Hough transform.	comment line; documentation intent, no runtime effect.

[illegible]

```
| action, timing, and diagnostic outcomes. |
| 51 |         SignType::NoOvertaking => write!(f, "No Overtaking"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 52 |         SignType::RoadWorks => write!(f, "Road Works"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 53 |         SignType::PedestrianCrossing => write!(f, "Ped Crossing"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 54 |         SignType::Roundabout => write!(f, "Roundabout"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 55 |         SignType::HazardWarning => write!(f, "Hazard"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 56 |         SignType::Unknown => write!(f, "Unknown"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 57 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 58 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 59 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 60 | (blank) | line is blank; visual separation only. |
| 61 | // ■■ Detected sign ████████████████████████████████████████████████████████████████ | comment line has no runtime effect. |
| 62 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 63 | pub struct DetectedSign { | declares data layout; field changes may break serialization, tests, and callers. |
| 64 |     pub sign_type: SignType, | executable/support line; behavior impact depends on surrounding block context. |
| 65 |     pub distance_m: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 66 |     pub confidence: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 67 |     pub pixel_x: u16, | executable/support line; behavior impact depends on surrounding block context. |
| 68 |     pub pixel_y: u16, | executable/support line; behavior impact depends on surrounding block context. |
| 69 |     pub pixel_width: u16, | executable/support line; behavior impact depends on surrounding block context. |
| 70 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 71 | (blank) | line is blank; visual separation only. |
| 72 | // ■■ Object detection result ████████████████████████████████████████████████████████████████ | comment line has no runtime effect. |
| 73 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding block context. |
| 74 | pub enum ObjectClass { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 75 |     Car, | executable/support line; behavior impact depends on surrounding block context. |
| 76 |     Truck, | executable/support line; behavior impact depends on surrounding block context. |
| 77 |     Motorcycle, | executable/support line; behavior impact depends on surrounding block context. |
| 78 |     Bus, | executable/support line; behavior impact depends on surrounding block context. |
| 79 |     Pedestrian, | executable/support line; behavior impact depends on surrounding block context. |
| 80 |     Cyclist, | executable/support line; behavior impact depends on surrounding block context. |
| 81 |     Cone, | executable/support line; behavior impact depends on surrounding block context. |
| 82 |     Barrier, | executable/support line; behavior impact depends on surrounding block context. |
| 83 |     Animal, | executable/support line; behavior impact depends on surrounding block context. |
| 84 |     Unknown, | executable/support line; behavior impact depends on surrounding block context. |
| 85 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 86 | (blank) | line is blank; visual separation only. |
| 87 | impl std::fmt::Display for ObjectClass { | implementation block; method behavior and trait conformance are defined here. |
| 88 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 89 |         match self { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 90 |             ObjectClass::Car => write!(f, "Car      "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 91 |             ObjectClass::Truck => write!(f, "Truck    "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 92 |             ObjectClass::Motorcycle => write!(f, "Moto     "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 93 |             ObjectClass::Bus => write!(f, "Bus      "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 94 |             ObjectClass::Pedestrian => write!(f, "Pedestrian"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 95 |             ObjectClass::Cyclist => write!(f, "Cyclist   "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 96 |             ObjectClass::Cone => write!(f, "Cone      "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes.
```

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| gration, timing, and diagnostic outcomes. |
| 97 |      ObjectClass::Barrier => write!(f, "Barrier   "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 98 |      ObjectClass::Animal  => write!(f, "Animal    "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 99 |      ObjectClass::Unknown  => write!(f, "Unknown   "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 100 |    } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope |
| 101 |  } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 102 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 103 | (blank) | line is blank; visual separation only. |
| 104 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 105 | pub struct DetectedObject { | declares data layout; field changes may break serialization, tests, and callers |
| 106 |     pub id: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 107 |     pub class: ObjectClass, | executable/support line; behavior impact depends on surrounding block context. |
| 108 |     pub confidence: f64, // 0-1 (YOLOv8-style score) | executable/support line; behavior impact depends on surrounding block context. |
| 109 |     pub distance_m: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 110 |     pub azimuth_deg: f64, // ±45° from forward | executable/support line; behavior impact depends on surrounding block context. |
| 111 |     pub width_m: f64,      // estimated physical width | executable/support line; behavior impact depends on surrounding block context. |
| 112 |     pub height_m: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 113 |     pub bbox_x: u16, // bounding box pixel coords | executable/support line; behavior impact depends on surrounding block context. |
| 114 |     pub bbox_y: u16, | executable/support line; behavior impact depends on surrounding block context. |
| 115 |     pub bbox_w: u16, | executable/support line; behavior impact depends on surrounding block context. |
| 116 |     pub bbox_h: u16, | executable/support line; behavior impact depends on surrounding block context. |
| 117 |     pub velocity_ms: Option<f64>, // optical flow estimate | executable/support line; behavior impact depends on surrounding block context. |
| 118 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 119 | (blank) | line is blank; visual separation only. |
| 120 | // ■■ Lane detection ████████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 121 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 122 | pub struct LaneDetection { | declares data layout; field changes may break serialization, tests, and callers. |
| 123 |     pub detected: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 124 |     pub confidence: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 125 |     pub left_offset_m: f64, // distance to left lane mark (metres) | executable/support line; behavior impact depends on surrounding block context. |
| 126 |     pub right_offset_m: f64, // distance to right lane mark | executable/support line; behavior impact depends on surrounding block context. |
| 127 |     pub lane_width_m: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 128 |     pub ego_offset_m: f64,      // lateral offset from centre (+right) | executable/support line; behavior impact depends on surrounding block context. |
| 129 |     pub heading_error_deg: f64, // vehicle heading vs lane heading | executable/support line; behavior impact depends on surrounding block context. |
| 130 |     pub curvature_inv_m: f64,   // 1/R (0 = straight) | executable/support line; behavior impact depends on surrounding block context. |
| 131 |     pub left_type: MarkType, | executable/support line; behavior impact depends on surrounding block context. |
| 132 |     pub right_type: MarkType, | executable/support line; behavior impact depends on surrounding block context. |
| 133 |     pub adjacent_left: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 134 |     pub adjacent_right: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 135 |     pub departure_left: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 136 |     pub departure_right: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 137 | /// Predicted time to lane departure (seconds, if not corrected) | comment line; documentation intent, no runtime effect. |
| 138 |     pub ttl_d_s: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 139 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 140 | (blank) | line is blank; visual separation only. |
| 141 | impl Default for LaneDetection { | implementation block; method behavior and trait conformance are defined here |
| 142 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
```

[illegible]

[illegible]


```
| depends on surrounding block context. |
| 230 | pub frames_processed: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 231 | pub detection_fps: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 232 | (blank) | line is blank; visual separation only. |
| 233 | // ■ System health ████████████████████████████████████████████████████████████████████████████████ | comment line; no runtime effect. |
| 234 | pub forward_cam_fault: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 235 | pub rear_cam_fault: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 236 | pub obscured: bool, // camera obscured (mud, frost, sun glare) | executable/support line; behavior impact depends on surrounding block context. |
| 237 | pub calibration_valid: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 238 | (blank) | line is blank; visual separation only. |
| 239 | // ■ Internal ████████████████████████████████████████████████████████████████████████████████ | comment line; no runtime effect. |
| 240 | update_timer: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 241 | noise_t: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 242 | obj_id_counter: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 243 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 244 | (blank) | line is blank; visual separation only. |
| 245 | impl Default for CameraSystem { | implementation block; method behavior and trait conformance are defined here. |
| 246 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 247 |         Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 248 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 249 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 250 | (blank) | line is blank; visual separation only. |
| 251 | impl CameraSystem { | implementation block; method behavior and trait conformance are defined here. |
| 252 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 253 |         CameraSystem { | executable/support line; behavior impact depends on surrounding block context. |
| 254 |             forward_spec: CameraSpec::forward_adas(), | executable/support line; behavior impact depends on surrounding block context. |
| 255 |             rear_spec: CameraSpec::surround_wide(), | executable/support line; behavior impact depends on surrounding block context. |
| 256 |             lane: LaneDetection::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 257 |             objects_front: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 258 |             objects_rear: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 259 |             signs: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 260 |             active_speed_limit: Some(80), | executable/support line; behavior impact depends on surrounding block context. |
| 261 |             front_clear: true, | executable/support line; behavior impact depends on surrounding block context. |
| 262 |             rear_clear: true, | executable/support line; behavior impact depends on surrounding block context. |
| 263 |             left_clear: true, | executable/support line; behavior impact depends on surrounding block context. |
| 264 |             right_clear: true, | executable/support line; behavior impact depends on surrounding block context. |
| 265 |             inference_ms: 12.0, | executable/support line; behavior impact depends on surrounding block context. |
| 266 |             frames_processed: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 267 |             detection_fps: 30.0, | executable/support line; behavior impact depends on surrounding block context. |
| 268 |             forward_cam_fault: false, | executable/support line; behavior impact depends on surrounding block context. |
| 269 |             rear_cam_fault: false, | executable/support line; behavior impact depends on surrounding block context. |
| 270 |             obscured: false, | executable/support line; behavior impact depends on surrounding block context.
```

```
| calibration_valid: true, | executable/support line; behavior impact depends on surrounding block context. | |
| 272 | update_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 273 | noise_t: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 274 | obj_id_counter: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 275 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 276 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 277 | (blank) | line is blank; visual separation only. |
| 278 | /// Update camera system – called each simulation tick. | comment line; documentation intent, no runtime effect. |
| 279 | pub fn update( | declares executable behavior; signature/body changes alter API, control flow, and output |
| 280 | &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 281 | ego_speed_kmh: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 282 | _ego_heading: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 283 | radar_ttc: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 284 | elapsed: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 285 | dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 286 | ) { | executable/support line; behavior impact depends on surrounding block context. |
| 287 | self.noise_t += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 288 | self.update_timer += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 289 | (blank) | line is blank; visual separation only. |
| 290 | // Forward camera runs at 30 fps | comment line; documentation intent, no runtime effect. |
| 291 | if self.update_timer < 1.0 / self.forward_spec.fps { | conditional gate; changing condition shifts runtime path and side effects. |
| 292 | return; | executable/support line; behavior impact depends on surrounding block context. |
| 293 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 294 | self.update_timer = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 295 | self.frames_processed += 1; | executable/support line; behavior impact depends on surrounding block context. |
| 296 | (blank) | line is blank; visual separation only. |
| 297 | let n = self.noise_t; | executable/support line; behavior impact depends on surrounding block context. |
| 298 | let ns = \|s: f64\| ((s * 127.1 + 311.7).sin() * 43758.5).fract() - 0.5; | executable/support line; behavior impact depends on surrounding block context. |
| 299 | (blank) | line is blank; visual separation only. |
| 300 | // ■ Lane detection (simulates Hough + polynomial fitting) ████████████ | comment line; documentation intent, no runtime effect. |
| 301 | let speed_factor = (ego_speed_kmh / 60.0).clamp(0.0, 1.0); | executable/support line; behavior impact depends on surrounding block context. |
| 302 | self.lane.detected = ego_speed_kmh > 3.0 && !self.obscured; | executable/support line; behavior impact depends on surrounding block context. |
| 303 | self.lane.confidence = if self.lane.detected { | executable/support line; behavior impact depends on surrounding block context. |
| 304 | (0.92 + ns(n * 1.1) * 0.05).clamp(0.5, 0.99) | executable/support line; behavior impact depends on surrounding block context. |
| 305 | } else { | executable/support line; behavior impact depends on surrounding block context. |
| 306 | 0.0 | executable/support line; behavior impact depends on surrounding block context. |
| 307 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 308 | self.lane.ego_offset_m = ns(n * 0.3) * 0.15 * speed_factor; | executable/support line; behavior impact depends on surrounding block context. |
| 309 | self.lane.heading_error_deg = ns(n * 0.5) * 1.5 * speed_factor; | executable/support line; behavior impact depends on surrounding block context. |
| 310 | self.lane.curvature_inv_m = ns(n * 0.01) * 0.002; | executable/support line; behavior impact depends on surrounding block context. |
| 311 | let lo = 1.75 + self.lane.ego_offset_m; | executable/support line; behavior impact depends on surrounding block context. |
| 312 | let ro = 1.75 - self.lane.ego_offset_m; | executable/support line; behavior impact depends on surrounding block context.
```

```
| 313 | self.lane.left_offset_m = lo; | executable/support line; behavior impact depends on surrounding block  
ncontext. |  
| 314 | self.lane.right_offset_m = ro; | executable/support line; behavior impact depends on surrounding bloc  
oncontext. |  
| 315 | self.lane.lane_width_m = lo + ro; | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 316 | self.lane.departure_left = lo < 0.5; | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 317 | self.lane.departure_right = ro < 0.5; | executable/support line; behavior impact depends on surroundi  
block context. |  
| 318 | self.lane.ttld_s = if self.lane.departure_left \\|\\ self.lane.departure_right { | executable/support  
e; behavior impact depends on surrounding block context. |  
| 319 | lo.min(ro) / (ego_speed_kmh / 3.6).max(0.1) * 5.0 | executable/support line; behavior impact depe  
on surrounding block context. |  
| 320 | } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 321 | f64::INFINITY | executable/support line; behavior impact depends on surrounding block context. |  
| 322 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 323 | (blank) | line is blank; visual separation only. |  
| 324 | // ■ Object detection (simulated neural network outputs) ■■■■■■■■■■■■ | comment line; documentat  
ent, no runtime effect. |  
| 325 | self.objects_front.clear(); | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 326 | // Lead vehicle simulation | comment line; documentation intent, no runtime effect. |  
| 327 | if radar_ttc < 30.0 { | conditional gate; changing condition shifts runtime path and side effects. |  
| 328 | let dist = (ego_speed_kmh / 3.6 * radar_ttc.min(10.0)).max(5.0); | executable/support line; behav  
impact depends on surrounding block context. |  
| 329 | self.objects_front.push(DetectedObject { | executable/support line; behavior impact depends on su  
unding block context. |  
| 330 | id: 1, | executable/support line; behavior impact depends on surrounding block context. |  
| 331 | class: ObjectClass::Car, | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 332 | confidence: 0.94 + ns(n * 2.1) * 0.04, | executable/support line; behavior impact depends on  
rounding block context. |  
| 333 | distance_m: dist, | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 334 | azimuth_deg: ns(n * 3.1) * 2.0, | executable/support line; behavior impact depends on surroun  
g block context. |  
| 335 | width_m: 1.9, | executable/support line; behavior impact depends on surrounding block context  
| 336 | height_m: 1.5, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 337 | bbox_x: 760, | executable/support line; behavior impact depends on surrounding block context.  
| 338 | bbox_y: 400, | executable/support line; behavior impact depends on surrounding block context.  
| 339 | bbox_w: 120, | executable/support line; behavior impact depends on surrounding block context.  
| 340 | bbox_h: 80, | executable/support line; behavior impact depends on surrounding block context.  
| 341 | velocity_ms: Some((ego_speed_kmh - 5.0) / 3.6), | executable/support line; behavior impact de  
ds on surrounding block context. |  
| 342 | }); | executable/support line; behavior impact depends on surrounding block context. |  
| 343 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 344 | // Periodic additional objects | comment line; documentation intent, no runtime effect. |  
| 345 | let t_mod = elapsed % 20.0; | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 346 | if t_mod > 5.0 && t_mod < 15.0 { | conditional gate; changing condition shifts runtime path and side  
ects. |  
| 347 | self.objects_front.push(DetectedObject { | executable/support line; behavior impact depends on su  
unding block context. |  
| 348 | id: 2, | executable/support line; behavior impact depends on surrounding block context. |  
| 349 | class: if (t_mod * 0.5).sin() > 0.0 { | executable/support line; behavior impact depends on s  
ounding block context. |  
| 350 | ObjectClass::Pedestrian | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 351 | } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 352 | ObjectClass::Cyclist | executable/support line; behavior impact depends on surrounding bl  
context. |
```

```
| 353 |           }, | executable/support line; behavior impact depends on surrounding block context. |  
| 354 |           confidence: 0.72 + ns(n * 4.1) * 0.1, | executable/support line; behavior impact depends on s  
ounding block context. |  
| 355 |           distance_m: 18.0 + ns(n * 5.1) * 5.0, | executable/support line; behavior impact depends on s  
ounding block context. |  
| 356 |           azimuth_deg: 8.0 + ns(n * 6.1) * 3.0, | executable/support line; behavior impact depends on s  
ounding block context. |  
| 357 |           width_m: 0.5, | executable/support line; behavior impact depends on surrounding block context  
| 358 |           height_m: 1.8, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 359 |           bbox_x: 1100, | executable/support line; behavior impact depends on surrounding block context  
| 360 |           bbox_y: 350, | executable/support line; behavior impact depends on surrounding block context.  
| 361 |           bbox_w: 40, | executable/support line; behavior impact depends on surrounding block context.  
| 362 |           bbox_h: 90, | executable/support line; behavior impact depends on surrounding block context.  
| 363 |           velocity_ms: None, | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 364 |         }); | executable/support line; behavior impact depends on surrounding block context. |  
| 365 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 366 | (blank) | line is blank; visual separation only. |  
| 367 |     // ■ Traffic sign recognition ████████████████████████████████████████████████████████████ | comment line; c  
ntent, no runtime effect. |  
| 368 |     self.signs.clear(); | executable/support line; behavior impact depends on surrounding block context.  
| 369 |     // Speed limit sign simulation | comment line; documentation intent, no runtime effect. |  
| 370 |     let sign_cycle = (elapsed / 45.0) as u32; | executable/support line; behavior impact depends on surro  
ing block context. |  
| 371 |     let speed_lim = if sign_cycle.is_multiple_of(2) { 80 } else { 50 }; | executable/support line; behavi  
impact depends on surrounding block context. |  
| 372 |     self.active_speed_limit = Some(speed_lim); | executable/support line; behavior impact depends on surr  
ding block context. |  
| 373 |     if elapsed % 45.0 < 5.0 { | conditional gate; changing condition shifts runtime path and side effects  
| 374 |         self.signs.push(DetectedSign { | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 375 |             sign_type: SignType::SpeedLimit(speed_lim), | executable/support line; behavior impact depend  
n surrounding block context. |  
| 376 |             distance_m: 60.0 - (elapsed % 45.0) * 10.0, | executable/support line; behavior impact depend  
n surrounding block context. |  
| 377 |             confidence: 0.97, | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 378 |             pixel_x: 1600, | executable/support line; behavior impact depends on surrounding block context  
|  
| 379 |             pixel_y: 300, | executable/support line; behavior impact depends on surrounding block context  
| 380 |             pixel_width: 60, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 381 |         }); | executable/support line; behavior impact depends on surrounding block context. |  
| 382 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 383 |     if (elapsed % 60.0) < 3.0 { | conditional gate; changing condition shifts runtime path and side effec  
|  
| 384 |         self.signs.push(DetectedSign { | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 385 |             sign_type: SignType::RoadWorks, | executable/support line; behavior impact depends on surroun  
g block context. |  
| 386 |             distance_m: 180.0, | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 387 |             confidence: 0.88, | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 388 |             pixel_x: 1700, | executable/support line; behavior impact depends on surrounding block context  
|  
| 389 |             pixel_y: 280, | executable/support line; behavior impact depends on surrounding block context  
| 390 |             pixel_width: 50, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 391 |         }); | executable/support line; behavior impact depends on surrounding block context. |  
| 392 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
```

```
| 393 | (blank) | line is blank; visual separation only. |  
| 394 | // ■ Surround-view / blind spots ██████████ | comment line; document  
intent, no runtime effect. |  
| 395 | let lateral_t = elapsed * 0.13; | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 396 | self.left_clear = (lateral_t.sin()).abs() < 0.8; | executable/support line; behavior impact depends o  
urrounding block context. |  
| 397 | self.right_clear = (lateral_t + 1.2).cos().abs() < 0.85; | executable/support line; behavior impact d  
nds on surrounding block context. |  
| 398 | (blank) | line is blank; visual separation only. |  
| 399 | // ■ Inference time simulation (with variability) ██████████ | comment line; docume  
ntent, no runtime effect. |  
| 400 | self.inference_ms = 10.0 + ns(n * 8.1).abs() * 5.0; | executable/support line; behavior impact depend  
n surrounding block context. |  
| 401 | (blank) | line is blank; visual separation only. |  
| 402 | // ■ Camera health ██████████ | comment line  
tent, no runtime effect. |  
| 403 | self.calibration_valid = true; | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 404 | self.obscured = false; | executable/support line; behavior impact depends on surrounding block contex  
|  
| 405 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 406 | (blank) | line is blank; visual separation only. |  
| 407 | pub fn closest_object(&self) -> Option<&DetectedObject> { | declares executable behavior; signature/body  
nges alter API, control flow, and outputs. |  
| 408 | self.objects_front | executable/support line; behavior impact depends on surrounding block context. |  
| 409 | .iter() | executable/support line; behavior impact depends on surrounding block context. |  
| 410 | .min_by(\[a, b\] a.distance_m.total_cmp(&b.distance_m)) | executable/support line; behavior impac  
depends on surrounding block context. |  
| 411 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 412 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 413 | (blank) | line is blank; visual separation only. |  
| 414 | // ■■ Sensor Fusion ██████████ | comment  
intent, no runtime effect. |  
| 415 | /// Fuses radar + camera + LIDAR into a unified object list using track-level fusion. | comment line; docum  
ion intent, no runtime effect. |  
| 416 | /// Implements a simplified Joint Probabilistic Data Association (JPDA) approach. | comment line; documentati  
intent, no runtime effect. |  
| 417 | (blank) | line is blank; visual separation only. |  
| 418 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |  
| 419 | pub struct FusedObject { | declares data layout; field changes may break serialization, tests, and callers. |  
| 420 | pub id: u32, | executable/support line; behavior impact depends on surrounding block context. |  
| 421 | pub class: ObjectClass, | executable/support line; behavior impact depends on surrounding block context.  
| 422 | pub distance_m: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 423 | pub azimuth_deg: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 424 | pub speed_ms: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 425 | pub lateral_pos_m: f64, | executable/support line; behavior impact depends on surrounding block context.  
| 426 | pub confidence: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 427 | pub sensor_sources: u8, // bitmask: bit0=radar, bit1=camera, bit2=lidar | executable/support line; behavi  
impact depends on surrounding block context. |  
| 428 | pub ttl_s: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 429 | pub in_path: bool, | executable/support line; behavior impact depends on surrounding block context. |  
| 430 | /// Kalman state: [x, y, vx, vy] | comment line; documentation intent, no runtime effect. |  
| 431 | kf_x: [f64; 4], | executable/support line; behavior impact depends on surrounding block context. |  
| 432 | kf_p: [f64; 4], // diagonal P | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 433 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 434 | (blank) | line is blank; visual separation only. |  
| 435 | impl FusedObject { | implementation block; method behavior and trait conformance are defined here. |  
| 436 | fn new(id: u32, class: ObjectClass, dist: f64, az: f64, speed: f64) -> Self { | declares executable behav  
; signature/body changes alter API, control flow, and outputs. |  
| 437 | let ttl = if speed > 0.1 { | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 438 |     dist / speed | executable/support line; behavior impact depends on surrounding block context. |  
| 439 | } else { | executable/support line; behavior impact depends on surrounding block context. |
```

```

| 440 |         f64::INFINITY | executable/support line; behavior impact depends on surrounding block context. |
| 441 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 442 |     FusedObject { | executable/support line; behavior impact depends on surrounding block context. |
| 443 |         id, | executable/support line; behavior impact depends on surrounding block context. |
| 444 |         class, | executable/support line; behavior impact depends on surrounding block context. |
| 445 |         distance_m: dist, | executable/support line; behavior impact depends on surrounding block context.
| 446 |         azimuth_deg: az, | executable/support line; behavior impact depends on surrounding block context.
| 447 |         speed_ms: speed, | executable/support line; behavior impact depends on surrounding block context.
| 448 |         lateral_pos_m: az.to_radians().sin() * dist, | executable/support line; behavior impact depends o
urrounding block context. |
| 449 |         confidence: 0.7, | executable/support line; behavior impact depends on surrounding block context.
| 450 |         sensor_sources: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 451 |         ttc_s: ttc, | executable/support line; behavior impact depends on surrounding block context. |
| 452 |         in_path: az.abs() < 5.0, | executable/support line; behavior impact depends on surrounding block
text. |
| 453 |         kf_x: [dist, 0.0, speed, 0.0], | executable/support line; behavior impact depends on surrounding
ck context. |
| 454 |         kf_p: [5.0, 5.0, 2.0, 2.0], | executable/support line; behavior impact depends on surrounding blo
context. |
| 455 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 456 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 457 | (blank) | line is blank; visual separation only. |
| 458 | fn predict(&mut self, dt: f64) { | declares executable behavior; signature/body changes alter API, contro
low, and outputs. |
| 459 |     self.kf_x[0] += self.kf_x[2] * dt; | executable/support line; behavior impact depends on surrounding
ck context. |
| 460 |     self.kf_x[1] += self.kf_x[3] * dt; | executable/support line; behavior impact depends on surrounding
ck context. |
| 461 |     self.kf_p[0] += 0.5 * dt; | executable/support line; behavior impact depends on surrounding block con
t. |
| 462 |     self.kf_p[1] += 0.5 * dt; | executable/support line; behavior impact depends on surrounding block con
t. |
| 463 |     self.kf_p[2] += 0.3 * dt; | executable/support line; behavior impact depends on surrounding block con
t. |
| 464 |     self.kf_p[3] += 0.3 * dt; | executable/support line; behavior impact depends on surrounding block con
t. |
| 465 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 466 | (blank) | line is blank; visual separation only. |
| 467 | fn update_with_measurement(&mut self, dist: f64, az: f64, spd: f64, src_bit: u8) { | declares executable
avior; signature/body changes alter API, control flow, and outputs. |
| 468 |     let r = 1.5; | executable/support line; behavior impact depends on surrounding block context. |
| 469 |     let k = [ | executable/support line; behavior impact depends on surrounding block context. |
| 470 |         self.kf_p[0] / (self.kf_p[0] + r), | executable/support line; behavior impact depends on surround
block context. |
| 471 |         self.kf_p[2] / (self.kf_p[2] + r), | executable/support line; behavior impact depends on surround
block context. |
| 472 |     ]; | executable/support line; behavior impact depends on surrounding block context. |
| 473 |     self.kf_x[0] += k[0] * (dist - self.kf_x[0]); | executable/support line; behavior impact depends on s
urrounding block context. |
| 474 |     self.kf_x[2] += k[1] * (spd - self.kf_x[2]); | executable/support line; behavior impact depends on su
urrounding block context. |
| 475 |     self.kf_p[0] *= 1.0 - k[0]; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 476 |     self.kf_p[2] *= 1.0 - k[1]; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 477 |     self.distance_m = self.kf_x[0].max(0.0); | executable/support line; behavior impact depends on surrou
ng block context. |
| 478 |     self.speed_ms = self.kf_x[2]; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 479 |     self.azimuth_deg = az; | executable/support line; behavior impact depends on surrounding block context
|
| 480 |     self.lateral_pos_m = az.to_radians().sin() * self.distance_m; | executable/support line; behavior imp

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depends on surrounding block context. |
| 481 |         self.in_path = self.azimuth_deg.abs() < 5.0; | executable/support line; behavior impact depends on surrounding block context. |
| 482 |         self.ttc_s = if self.speed_ms > 0.5 { | executable/support line; behavior impact depends on surrounding block context. |
| 483 |             self.distance_m / self.speed_ms | executable/support line; behavior impact depends on surrounding block context. |
| 484 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 485 |             f64::INFINITY | executable/support line; behavior impact depends on surrounding block context. |
| 486 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 487 |         self.sensor_sources \|= src_bit; | executable/support line; behavior impact depends on surrounding block context. |
| 488 |         self.confidence = (0.5 + self.sensor_sources.count_ones() as f64 * 0.15).min(0.99); | executable/support line; behavior impact depends on surrounding block context. |
| 489 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 490 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 491 | (blank) | line is blank; visual separation only. |
| 492 | pub struct SensorFusion { | declares data layout; field changes may break serialization, tests, and callers. |
| 493 |     pub objects: Vec<FusedObject>, | executable/support line; behavior impact depends on surrounding block context. |
| 494 |     pub ttc_critical: f64, // minimum TTC in path | executable/support line; behavior impact depends on surrounding block context. |
| 495 |     pub lead_dist_m: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 496 |     id_counter: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 497 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 498 | (blank) | line is blank; visual separation only. |
| 499 | impl Default for SensorFusion { | implementation block; method behavior and trait conformance are defined here. |
| 500 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 501 |         Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 502 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 503 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 504 | (blank) | line is blank; visual separation only. |
| 505 | impl SensorFusion { | implementation block; method behavior and trait conformance are defined here. |
| 506 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 507 |         SensorFusion { | executable/support line; behavior impact depends on surrounding block context. |
| 508 |             objects: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 509 |             ttc_critical: f64::INFINITY, | executable/support line; behavior impact depends on surrounding block context. |
| 510 |             lead_dist_m: f64::INFINITY, | executable/support line; behavior impact depends on surrounding block context. |
| 511 |             id_counter: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 512 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 513 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 514 | (blank) | line is blank; visual separation only. |
| 515 |     pub fn fuse( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 516 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 517 |         camera: &CameraSystem, | executable/support line; behavior impact depends on surrounding block context. |
| 518 |         radar_front_targets: &[crate::radar::RadarTarget], | executable/support line; behavior impact depends on surrounding block context. |
| 519 |         ego_speed_ms: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 520 |         dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 521 |     ) { | executable/support line; behavior impact depends on surrounding block context. |
| 522 |         // Predict all existing tracks | comment line; documentation intent, no runtime effect. |
| 523 |         for obj in &mut self.objects { | iteration control; changing bounds/body affects throughput, timing, possibly safety guarantees. |
| 524 |             obj.predict(dt); | executable/support line; behavior impact depends on surrounding block context. |
| 525 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

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| 526 | (blank) | line is blank; visual separation only. |  
| 527 | // ■ Associate and update from RADAR (source bit 0) ████████████████████ | comment line; document  
intent, no runtime effect. |  
| 528 | for rt in radar_front_targets { | iteration control; changing bounds/body affects throughput, timing,  
and possibly safety guarantees. |  
| 529 | let closing = if rt.range_rate_ms < 0.0 { | executable/support line; behavior impact depends on s  
urrounding block context. |  
| 530 | -rt.range_rate_ms | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 531 | } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 532 | 0.0 | executable/support line; behavior impact depends on surrounding block context. |  
| 533 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 534 | if let Some(o) = self | conditional gate; changing condition shifts runtime path and side effects  
| 535 | .objects | executable/support line; behavior impact depends on surrounding block context. |  
| 536 | .iter_mut() | executable/support line; behavior impact depends on surrounding block context.  
| 537 | .find(\|o\| (o.distance_m - rt.range_m).abs() < 8.0) | executable/support line; behavior impa  
depends on surrounding block context. |  
| 538 | { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 539 | o.update_with_measurement(rt.range_m, rt.azimuth_deg, closing, 0x01); | executable/support li  
behavior impact depends on surrounding block context. |  
| 540 | } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 541 | let mut o = FusedObject::new( | executable/support line; behavior impact depends on surroundi  
block context. |  
| 542 | self.id_counter, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 543 | ObjectClass::Car, | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 544 | rt.range_m, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 545 | rt.azimuth_deg, | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 546 | closing, | executable/support line; behavior impact depends on surrounding block context.  
| 547 | ); | executable/support line; behavior impact depends on surrounding block context. |  
| 548 | o.sensor_sources = 0x01; | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 549 | self.objects.push(o); | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 550 | self.id_counter += 1; | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 551 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 552 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 553 | (blank) | line is blank; visual separation only. |  
| 554 | // ■ Associate and update from CAMERA (source bit 1) ████████████████████ | comment line; documen  
tent, no runtime effect. |  
| 555 | for co in &camera.objects_front { | iteration control; changing bounds/body affects throughput, timin  
and possibly safety guarantees. |  
| 556 | if let Some(o) = self.objects.iter_mut().find(\|o\| { | conditional gate; changing condition shif  
runtime path and side effects. |  
| 557 | (o.distance_m - co.distance_m).abs() < 5.0 | executable/support line; behavior impact depends  
surrounding block context. |  
| 558 | && (o.azimuth_deg - co.azimuth_deg).abs() < 8.0 | executable/support line; behavior impac  
depends on surrounding block context. |  
| 559 | }) { | executable/support line; behavior impact depends on surrounding block context. |  
| 560 | let spd = co | executable/support line; behavior impact depends on surrounding block context.  
| 561 | .velocity_ms | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 562 | .map(\|v\| (ego_speed_ms - v).max(0.0)) | executable/support line; behavior impact depend  
n surrounding block context. |  
| 563 | .unwrap_or(o.speed_ms); | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 564 | o.update_with_measurement(co.distance_m, co.azimuth_deg, spd, 0x02); | executable/support lin
```



```

behavior impact depends on surrounding block context. |
| 565 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 566 |             let spd = co | executable/support line; behavior impact depends on surrounding block context. |
| 567 |                 .velocity_ms | executable/support line; behavior impact depends on surrounding block context. |
| 568 |                 .map(\|v\| (ego_speed_ms - v).max(0.0)) | executable/support line; behavior impact depends on surrounding block context. |
| 569 |                 .unwrap_or(5.0); | executable/support line; behavior impact depends on surrounding block context. |
| 570 |             let mut o = FusedObject::new( | executable/support line; behavior impact depends on surrounding block context. |
| 571 |                 self.id_counter, | executable/support line; behavior impact depends on surrounding block context. |
| 572 |                 co.class, | executable/support line; behavior impact depends on surrounding block context. |
| 573 |                 co.distance_m, | executable/support line; behavior impact depends on surrounding block context. |
| 574 |                 co.azimuth_deg, | executable/support line; behavior impact depends on surrounding block context. |
| 575 |                 spd, | executable/support line; behavior impact depends on surrounding block context. |
| 576 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 577 |             o.sensor_sources = 0x02; | executable/support line; behavior impact depends on surrounding block context. |
| 578 |             self.objects.push(o); | executable/support line; behavior impact depends on surrounding block context. |
| 579 |             self.id_counter += 1; | executable/support line; behavior impact depends on surrounding block context. |
| 580 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 581 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 582 | (blank) | line is blank; visual separation only. |
| 583 |     // Remove stale / out-of-range objects | comment line; documentation intent, no runtime effect. |
| 584 |     self.objects | executable/support line; behavior impact depends on surrounding block context. |
| 585 |         .retain(\|o\| o.distance_m < 200.0 && o.confidence > 0.2); | executable/support line; behavior impact depends on surrounding block context. |
| 586 |     self.objects.truncate(64); | executable/support line; behavior impact depends on surrounding block context. |
| 587 | (blank) | line is blank; visual separation only. |
| 588 |     // Aggregate | comment line; documentation intent, no runtime effect. |
| 589 |     self.ttc_critical = self | executable/support line; behavior impact depends on surrounding block context. |
| 590 |         .objects | executable/support line; behavior impact depends on surrounding block context. |
| 591 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 592 |         .filter(\|o\| o.in_path && o.speed_ms > 0.1) | executable/support line; behavior impact depends on surrounding block context. |
| 593 |         .map(\|o\| o.ttc_s) | executable/support line; behavior impact depends on surrounding block context. |
| 594 |         .fold(f64::INFINITY, f64::min); | executable/support line; behavior impact depends on surrounding block context. |
| 595 |     self.lead_dist_m = self | executable/support line; behavior impact depends on surrounding block context. |
| 596 |         .objects | executable/support line; behavior impact depends on surrounding block context. |
| 597 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 598 |         .filter(\|o\| o.in_path) | executable/support line; behavior impact depends on surrounding block context. |
| 599 |         .map(\|o\| o.distance_m) | executable/support line; behavior impact depends on surrounding block context. |
| 600 |         .fold(f64::INFINITY, f64::min); | executable/support line; behavior impact depends on surrounding block context. |
| 601 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 602 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/can_bus.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 180 | Non-empty: 173 | Symbols: 8

Symbol Inventory

Line	Kind	Symbol	Change Impact
---: --- --- ---			
3	const	BUS_LOAD_ALPHA	data/structure coupling
6	struct	CanFrame	data/structure coupling
14	struct	CanBus	data/structure coupling
23	impl	CanBus	behavior binding and method semantics
24	fn	new	signature/API coupling
35	fn	publish	signature/API coupling
49	fn	tick	signature/API coupling
63	fn	broadcast_all	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
---: --- ---		
1	use std::collections::VecDeque;	imports symbols; changing path/name affects compile-time resolution.
2	(blank)	line is blank; visual separation only.
3	const BUS_LOAD_ALPHA: f64 = 0.05;	executable/support line; behavior impact depends on surrounding block context.
4	(blank)	line is blank; visual separation only.
5	#[derive(Debug, Clone)]	executable/support line; behavior impact depends on surrounding block context.
6	pub struct CanFrame {	declares data layout; field changes may break serialization, tests, and callers.
7	pub id: u32,	executable/support line; behavior impact depends on surrounding block context.
8	pub name: &'static str,	executable/support line; behavior impact depends on surrounding block context.
9	pub value: String,	executable/support line; behavior impact depends on surrounding block context.
10	pub timestamp: f64,	executable/support line; behavior impact depends on surrounding block context.
11	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
12	(blank)	line is blank; visual separation only.
13	/// Simulated 500 kbps CAN bus (HS-CAN)	comment line; documentation intent, no runtime effect.
14	pub struct CanBus {	declares data layout; field changes may break serialization, tests, and callers.
15	pub frames: VecDeque<CanFrame>,	executable/support line; behavior impact depends on surrounding block context.
16	pub total_frames: u64,	executable/support line; behavior impact depends on surrounding block context.
17	pub bus_load_pct: f64,	executable/support line; behavior impact depends on surrounding block context.
18	pub frames_per_second: u32,	executable/support line; behavior impact depends on surrounding block context.
19	frame_counter: u32,	executable/support line; behavior impact depends on surrounding block context.
20	frame_timer: f64,	executable/support line; behavior impact depends on surrounding block context.
21	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
22	(blank)	line is blank; visual separation only.
23	impl CanBus {	implementation block; method behavior and trait conformance are defined here.
24	pub fn new() -> Self {	declares executable behavior; signature/body changes alter API, control flow, and outputs.
25	Self {	executable/support line; behavior impact depends on surrounding block context.
26	frames: VecDeque::new(),	executable/support line; behavior impact depends on surrounding block context.
27	total_frames: 0,	executable/support line; behavior impact depends on surrounding block context.
28	bus_load_pct: 0.0,	executable/support line; behavior impact depends on surrounding block context.
29	frames_per_second: 0,	executable/support line; behavior impact depends on surrounding block context.
30	frame_counter: 0,	executable/support line; behavior impact depends on surrounding block context.
31	frame_timer: 0.0,	executable/support line; behavior impact depends on surrounding block context.
32	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
33	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
34	(blank)	line is blank; visual separation only.
35	pub fn publish(&mut self, id: u32, name: &'static str, value: String, ts: f64) {	declares executable behavior; signature/body changes alter API, control flow, and outputs.
36	self.frames.push_front(CanFrame {	executable/support line; behavior impact depends on surrounding block context.
37	id,	executable/support line; behavior impact depends on surrounding block context.
38	name,	executable/support line; behavior impact depends on surrounding block context.

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| 39 |         value, | executable/support line; behavior impact depends on surrounding block context. |
| 40 |         timestamp: ts, | executable/support line; behavior impact depends on surrounding block context. |
| 41 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 42 |     if self.frames.len() > 12 { | conditional gate; changing condition shifts runtime path and side effect
|
| 43 |         self.frames.pop_back(); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 44 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 45 |     self.total_frames += 1; | executable/support line; behavior impact depends on surrounding block context
|
| 46 |     self.frame_counter += 1; | executable/support line; behavior impact depends on surrounding block conte
|
| 47 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 48 | (blank) | line is blank; visual separation only. |
| 49 | pub fn tick(&mut self, dt: f64) { | declares executable behavior; signature/body changes alter API, contro
low, and outputs. |
| 50 |     self.frame_timer += dt; | executable/support line; behavior impact depends on surrounding block context
|
| 51 |     if self.frame_timer >= 1.0 { | conditional gate; changing condition shifts runtime path and side effect
|
| 52 |         self.frames_per_second = self.frame_counter; | executable/support line; behavior impact depends on
rounding block context. |
| 53 |         // 500kbps, each frame ~128 bits → ~3900 frames/s max | comment line; documentation intent, no run
e effect. |
| 54 |         let load = self.frame_counter as f64 / 3900.0 * 100.0; | executable/support line; behavior impact
ends on surrounding block context. |
| 55 |         self.bus_load_pct += (load.min(100.0) - self.bus_load_pct) * BUS_LOAD_ALPHA; | executable/support
e; behavior impact depends on surrounding block context. |
| 56 |         self.frame_counter = 0; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 57 |         self.frame_timer = 0.0; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 58 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 59 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 60 | (blank) | line is blank; visual separation only. |
| 61 | /// Publish all periodic ECU messages | comment line; documentation intent, no runtime effect. |
| 62 | #[allow(clippy::too_many_arguments)] | executable/support line; behavior impact depends on surrounding blo
context. |
| 63 | pub fn broadcast_all( | declares executable behavior; signature/body changes alter API, control flow, and
puts. |
| 64 |     &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 65 |     speed: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 66 |     rpm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 67 |     gear: &str, | executable/support line; behavior impact depends on surrounding block context. |
| 68 |     engine_temp: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 69 |     throttle: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 70 |     brake: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 71 |     abs_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 72 |     esp_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 73 |     tcs_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 74 |     acc_enabled: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 75 |     acc_speed: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 76 |     aeb_ttc: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 77 |     lka_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 78 |     fuel_level: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 79 |     boost: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 80 |     torque: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 81 |     oil_pressure: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 82 |     elapsed: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 83 | ) { | executable/support line; behavior impact depends on surrounding block context. |
| 84 |     self.publish(0x100, "VEH_SPEED", format!("{:06.2} km/h", speed), elapsed); | executable/support line;
avior impact depends on surrounding block context. |
| 85 |     self.publish(0x101, "ENGINE_RPM", format!("{:5.0} rpm", rpm), elapsed); | executable/support line; beh
or impact depends on surrounding block context. |
| 86 |     self.publish(0x102, "GEAR_POS", format!("{:<5}", gear), elapsed); | executable/support line; behavior

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act depends on surrounding block context. |
| 87 |         self.publish( | executable/support line; behavior impact depends on surrounding block context. |
| 88 |             0x110, | executable/support line; behavior impact depends on surrounding block context. |
| 89 |             "ENG_TEMP", | executable/support line; behavior impact depends on surrounding block context. |
| 90 |             format!("{:5.1} °C", engine_temp), | executable/support line; behavior impact depends on surrounding
block context. |
| 91 |             elapsed, | executable/support line; behavior impact depends on surrounding block context. |
| 92 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 93 |         self.publish( | executable/support line; behavior impact depends on surrounding block context. |
| 94 |             0x111, | executable/support line; behavior impact depends on surrounding block context. |
| 95 |             "THROTTLE", | executable/support line; behavior impact depends on surrounding block context. |
| 96 |             format!("{:5.1} %", throttle * 100.0), | executable/support line; behavior impact depends on surrounding
ing block context. |
| 97 |             elapsed, | executable/support line; behavior impact depends on surrounding block context. |
| 98 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 99 |         self.publish( | executable/support line; behavior impact depends on surrounding block context. |
| 100 |             0x112, | executable/support line; behavior impact depends on surrounding block context. |
| 101 |             "BRAKE_REQ", | executable/support line; behavior impact depends on surrounding block context. |
| 102 |             format!("{:5.1} %", brake * 100.0), | executable/support line; behavior impact depends on surrounding
g block context. |
| 103 |             elapsed, | executable/support line; behavior impact depends on surrounding block context. |
| 104 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 105 |         self.publish(0x113, "TORQUE", format!("{:6.1} Nm", torque), elapsed); | executable/support line; behavior
r impact depends on surrounding block context. |
| 106 |         self.publish(0x114, "BOOST", format!("{:4.2} bar", boost), elapsed); | executable/support line; behavior
impact depends on surrounding block context. |
| 107 |         self.publish( | executable/support line; behavior impact depends on surrounding block context. |
| 108 |             0x115, | executable/support line; behavior impact depends on surrounding block context. |
| 109 |             "OIL_PRESS", | executable/support line; behavior impact depends on surrounding block context. |
| 110 |             format!("{:4.2} bar", oil_pressure), | executable/support line; behavior impact depends on surrounding
ng block context. |
| 111 |             elapsed, | executable/support line; behavior impact depends on surrounding block context. |
| 112 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 113 |         self.publish( | executable/support line; behavior impact depends on surrounding block context. |
| 114 |             0x120, | executable/support line; behavior impact depends on surrounding block context. |
| 115 |             "FUEL_LEVEL", | executable/support line; behavior impact depends on surrounding block context. |
| 116 |             format!("{:5.1} %", fuel_level * 100.0), | executable/support line; behavior impact depends on surrounding
unding block context. |
| 117 |             elapsed, | executable/support line; behavior impact depends on surrounding block context. |
| 118 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 119 |         self.publish( | executable/support line; behavior impact depends on surrounding block context. |
| 120 |             0x200, | executable/support line; behavior impact depends on surrounding block context. |
| 121 |             "ABS_STATUS", | executable/support line; behavior impact depends on surrounding block context. |
| 122 |             if abs_active { | conditional gate; changing condition shifts runtime path and side effects. |
| 123 |                 "ACTIVE".into() | executable/support line; behavior impact depends on surrounding block context
|
| 124 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 125 |                 "IDLE".into() | executable/support line; behavior impact depends on surrounding block context
| 126 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 127 |             elapsed, | executable/support line; behavior impact depends on surrounding block context. |
| 128 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 129 |         self.publish( | executable/support line; behavior impact depends on surrounding block context. |
| 130 |             0x201, | executable/support line; behavior impact depends on surrounding block context. |
| 131 |             "ESP_STATUS", | executable/support line; behavior impact depends on surrounding block context. |
| 132 |             if esp_active { | conditional gate; changing condition shifts runtime path and side effects. |
| 133 |                 "ACTIVE".into() | executable/support line; behavior impact depends on surrounding block context
|
| 134 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 135 |                 "IDLE".into() | executable/support line; behavior impact depends on surrounding block context
| 136 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 137 |             elapsed, | executable/support line; behavior impact depends on surrounding block context. |
| 138 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 139 |         self.publish( | executable/support line; behavior impact depends on surrounding block context. |
| 140 |             0x202, | executable/support line; behavior impact depends on surrounding block context. |
| 141 |             "TCS_STATUS", | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 142 |         if tcs_active { | conditional gate; changing condition shifts runtime path and side effects. |
| 143 |             "ACTIVE".into() | executable/support line; behavior impact depends on surrounding block context. |
| 144 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 145 |             "IDLE".into() | executable/support line; behavior impact depends on surrounding block context. |
| 146 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 147 |         elapsed, | executable/support line; behavior impact depends on surrounding block context. |
| 148 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 149 |     self.publish( | executable/support line; behavior impact depends on surrounding block context. |
| 150 |         0x300, | executable/support line; behavior impact depends on surrounding block context. |
| 151 |         "ACC_STATUS", | executable/support line; behavior impact depends on surrounding block context. |
| 152 |         if acc_enabled { | conditional gate; changing condition shifts runtime path and side effects. |
| 153 |             format!("{:3.0}km/h", acc_speed) | executable/support line; behavior impact depends on surrounding block context. |
| 154 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 155 |             "OFF ".into() | executable/support line; behavior impact depends on surrounding block context. |
| 156 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 157 |         elapsed, | executable/support line; behavior impact depends on surrounding block context. |
| 158 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 159 |     self.publish( | executable/support line; behavior impact depends on surrounding block context. |
| 160 |         0x301, | executable/support line; behavior impact depends on surrounding block context. |
| 161 |         "AEB_TTC", | executable/support line; behavior impact depends on surrounding block context. |
| 162 |         if aeb_ttc.is_infinite() { | conditional gate; changing condition shifts runtime path and side effects. |
| 163 |             " ∞ s".into() | executable/support line; behavior impact depends on surrounding block context. |
| 164 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 165 |             format!("{:4.1} s", aeb_ttc) | executable/support line; behavior impact depends on surrounding block context. |
| 166 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 167 |         elapsed, | executable/support line; behavior impact depends on surrounding block context. |
| 168 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 169 |     self.publish( | executable/support line; behavior impact depends on surrounding block context. |
| 170 |         0x302, | executable/support line; behavior impact depends on surrounding block context. |
| 171 |         "LKA_STATUS", | executable/support line; behavior impact depends on surrounding block context. |
| 172 |         if lka_active { | conditional gate; changing condition shifts runtime path and side effects. |
| 173 |             "ACTIVE".into() | executable/support line; behavior impact depends on surrounding block context. |
| 174 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 175 |             "IDLE".into() | executable/support line; behavior impact depends on surrounding block context. |
| 176 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 177 |         elapsed, | executable/support line; behavior impact depends on surrounding block context. |
| 178 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 179 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 180 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/can_gateway.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 317 | Non-empty: 280 | Symbols: 26

Symbol Inventory

Line	Kind	Symbol	Change Impact
13	enum	BusState	data/structure coupling
22	impl	std::fmt::Display for BusState	behavior binding and method semantics
23	fn	fmt	signature/API coupling
34	struct	BusError	data/structure coupling
42	enum	BusErrorKind	data/structure coupling
54	struct	CanNode	data/structure coupling
64	impl	CanNode	behavior binding and method semantics
65	fn	new	signature/API coupling
78	fn	record_tx_error	signature/API coupling
83	fn	record_rx_error	signature/API coupling

```
| 88 | fn | record_success | signature/API coupling |
| 95 | fn | update_state | signature/API coupling |
| 109 | struct | CanGateway | data/structure coupling |
| 142 | impl | Default for CanGateway | behavior binding and method semantics |
| 143 | fn | default | signature/API coupling |
| 148 | impl | CanGateway | behavior binding and method semantics |
| 149 | fn | new | signature/API coupling |
| 170 | fn | register_node | signature/API coupling |
| 177 | fn | set_node_online | signature/API coupling |
| 186 | fn | transmit | signature/API coupling |
| 192 | fn | tick | signature/API coupling |
| 275 | fn | receive_for | signature/API coupling |
| 282 | fn | analyzer_frames | signature/API coupling |
| 286 | fn | log_error | signature/API coupling |
| 300 | fn | kill_node | signature/API coupling |
| 309 | fn | revive_node | signature/API coupling |
```

[illegible]

[illegible]

[illegible]


```

| 132 |     pub bus_load_pct: f64, | executable/support line; behavior impact depends on surrounding block context. | | |
| 133 |     frame_counter_1s: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 134 |     time_accumulator: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 135 | (blank) | line is blank; visual separation only. |
| 136 |     // Fault injection flags (for the fault panel) | comment line; documentation intent, no runtime effect. |
| 137 |     pub inject_bit_error: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 138 |     pub inject_bus_off: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 139 |     pub inject_missing_ack: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 140 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 141 | (blank) | line is blank; visual separation only. |
| 142 | impl Default for CanGateway { | implementation block; method behavior and trait conformance are defined here. |
| 143 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 144 |         Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 145 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 146 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 147 | (blank) | line is blank; visual separation only. |
| 148 | impl CanGateway { | implementation block; method behavior and trait conformance are defined here. |
| 149 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 150 |         CanGateway { | executable/support line; behavior impact depends on surrounding block context. |
| 151 |             bus: J1939Bus::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 152 |             tx_queue: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 153 |             dispatched: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 154 |             nodes: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 155 |             total_tx: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 156 |             total_rx: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 157 |             total_errors: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 158 |             bus_state: BusState::ErrorActive, | executable/support line; behavior impact depends on surrounding block context. |
| 159 |             error_log: VecDeque::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 160 |             bus_load_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 161 |             frame_counter_1s: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 162 |             time_accumulator: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 163 |             inject_bit_error: false, | executable/support line; behavior impact depends on surrounding block context. |
| 164 |             inject_bus_off: false, | executable/support line; behavior impact depends on surrounding block context. |
| 165 |             inject_missing_ack: false, | executable/support line; behavior impact depends on surrounding block context. |
| 166 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 167 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 168 | (blank) | line is blank; visual separation only. |
| 169 |     /// Register an ECU on the bus | comment line; documentation intent, no runtime effect. |
| 170 |     pub fn register_node(&mut self, sa: u8, name: &'static str) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 171 |         if !self.nodes.iter().any(|n| n.source_addr == sa) { | conditional gate; changing condition shifts time path and side effects. |
| 172 |             self.nodes.push(CanNode::new(sa, name)); | executable/support line; behavior impact depends on surrounding block context. |
| 173 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 174 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 175 | (blank) | line is blank; visual separation only. |

```



```

context. |
| 217 |         let ts = frame.timestamp; | executable/support line; behavior impact depends on surrounding block context. |
k context. |
| 218 |         let sa = frame.sa; | executable/support line; behavior impact depends on surrounding block context. |
xt. |
| 219 |         self.error_log.push_front(BusError { | executable/support line; behavior impact depends on surrounding block context. |
| 220 |             timestamp: ts, | executable/support line; behavior impact depends on surrounding block context. |
xt. |
| 221 |             kind: BusErrorKind::BitError, | executable/support line; behavior impact depends on surrounding block context. |
ing block context. |
| 222 |             source_sa: Some(sa), | executable/support line; behavior impact depends on surrounding block context. |
context. |
| 223 |             description: "INJECTED: Bit error", | executable/support line; behavior impact depends on surrounding block context. |
rrounding block context. |
| 224 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 225 |         if self.error_log.len() > 64 { | conditional gate; changing condition shifts runtime path and side effects. |
| 226 |             self.error_log.pop_back(); | executable/support line; behavior impact depends on surrounding block context. |
block context. |
| 227 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 228 |         continue; | executable/support line; behavior impact depends on surrounding block context. |
| 229 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
es. |
| 230 |         if self.inject_missing_ack && self.total_tx.is_multiple_of(50) { | conditional gate; changing condition shifts runtime path and side effects. |
| 231 |             self.total_errors += 1; | executable/support line; behavior impact depends on surrounding block context. |
context. |
| 232 |             let ts = frame.timestamp; | executable/support line; behavior impact depends on surrounding block context. |
k context. |
| 233 |             let sa = frame.sa; | executable/support line; behavior impact depends on surrounding block context. |
xt. |
| 234 |             self.error_log.push_front(BusError { | executable/support line; behavior impact depends on surrounding block context. |
| 235 |                 timestamp: ts, | executable/support line; behavior impact depends on surrounding block context. |
xt. |
| 236 |                 kind: BusErrorKind::AcknowledgeError, | executable/support line; behavior impact depends on surrounding block context. |
surrounding block context. |
| 237 |                 source_sa: Some(sa), | executable/support line; behavior impact depends on surrounding block context. |
context. |
| 238 |                 description: "INJECTED: ACK missing", | executable/support line; behavior impact depends on surrounding block context. |
surrounding block context. |
| 239 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 240 |             if self.error_log.len() > 64 { | conditional gate; changing condition shifts runtime path and side effects. |
| 241 |                 self.error_log.pop_back(); | executable/support line; behavior impact depends on surrounding block context. |
block context. |
| 242 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. | | |
| 243 |             if let Some(node) = self.nodes.iter_mut().find(|n| n.source_addr == sa) { | conditional gate; changing condition shifts runtime path and side effects. |
| 244 |                 node.record_tx_error(); | executable/support line; behavior impact depends on surrounding block context. |
ock context. |
| 245 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 246 |             continue; | executable/support line; behavior impact depends on surrounding block context. |
| 247 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
es. |
| 248 | (blank) | line is blank; visual separation only. | | |
| 249 |         // Update node stats | comment line; documentation intent, no runtime effect. |
| 250 |         if let Some(node) = self.nodes.iter_mut().find(|n| n.source_addr == frame.sa) { | conditional gate; changing condition shifts runtime path and side effects. |
| 251 |             node.last_tx_ts = frame.timestamp; | executable/support line; behavior impact depends on surrounding block context. |

```



```

| 290 |         source_sa: sa, | executable/support line; behavior impact depends on surrounding block context. |
| 291 |         description: desc, | executable/support line; behavior impact depends on surrounding block context. |
|
| 292 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 293 |     if self.error_log.len() > 64 { | conditional gate; changing condition shifts runtime path and side effects. |
| 294 |         self.error_log.pop_back(); | executable/support line; behavior impact depends on surrounding block context. |
| 295 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
|
| 296 |     self.total_errors += 1; | executable/support line; behavior impact depends on surrounding block context. |
|
| 297 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. | |
| 298 | (blank) | line is blank; visual separation only. |
| 299 |     /// Simulate a node going offline (e.g., ECU power cut) | comment line; documentation intent, no runtime effect. |
| 300 |     pub fn kill_node(&mut self, sa: u8) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 301 |         if let Some(node) = self.nodes.iter_mut().find(\n\n| n.source_addr == sa) { | conditional gate; changing condition shifts runtime path and side effects. |
| 302 |             node.online = false; | executable/support line; behavior impact depends on surrounding block context. |
| 303 |             node.tec = 256; | executable/support line; behavior impact depends on surrounding block context. |
| 304 |             node.state = BusState::BusOff; | executable/support line; behavior impact depends on surrounding block context. |
| 305 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
|
| 306 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. | |
| 307 | (blank) | line is blank; visual separation only. |
| 308 |     /// Revive a killed node | comment line; documentation intent, no runtime effect. |
| 309 |     pub fn revive_node(&mut self, sa: u8) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 310 |         if let Some(node) = self.nodes.iter_mut().find(\n\n| n.source_addr == sa) { | conditional gate; changing condition shifts runtime path and side effects. |
| 311 |             node.online = true; | executable/support line; behavior impact depends on surrounding block context. |
|
| 312 |             node.tec = 0; | executable/support line; behavior impact depends on surrounding block context. |
| 313 |             node.rec = 0; | executable/support line; behavior impact depends on surrounding block context. |
| 314 |             node.state = BusState::ErrorActive; | executable/support line; behavior impact depends on surrounding block context. |
| 315 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
|
| 316 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 317 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/can_network.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 1381 | Non-empty: 1292 | Symbols: 73

Symbol Inventory

Line	Kind	Symbol	Change Impact
37	enum	CanSpeed	data/structure coupling
48	impl	CanSpeed	behavior binding and method semantics
49	fn	kbps	signature/API coupling
53	fn	max_fps	signature/API coupling
68	enum	BusId	data/structure coupling
77	impl	std::fmt::Display for BusId	behavior binding and method semantics
78	fn	fmt	signature/API coupling
92	enum	BusState	data/structure coupling
98	impl	std::fmt::Display for BusState	behavior binding and method semantics
99	fn	fmt	signature/API coupling
110	enum	CanErrorKind	data/structure coupling

```

| 120 | impl | std::fmt::Display for CanErrorKind | behavior binding and method semantics |
| 121 | fn | fmt | signature/API coupling |
| 136 | struct | BusError | data/structure coupling |
| 147 | struct | CanNode | data/structure coupling |
| 166 | impl | CanNode | behavior binding and method semantics |
| 167 | fn | new | signature/API coupling |
| 185 | fn | tx_success | signature/API coupling |
| 189 | fn | tx_error | signature/API coupling |
| 193 | fn | rx_error | signature/API coupling |
| 198 | fn | update_state | signature/API coupling |
| 215 | struct | TpSession | data/structure coupling |
| 228 | struct | CanTransportProtocol | data/structure coupling |
| 237 | impl | Default for CanTransportProtocol | behavior binding and method semantics |
| 238 | fn | default | signature/API coupling |
| 243 | impl | CanTransportProtocol | behavior binding and method semantics |
| 244 | fn | new | signature/API coupling |
| 253 | fn | send_bam | signature/API coupling |
| 293 | fn | process_rx | signature/API coupling |
| 346 | fn | tick | signature/API coupling |
| 364 | struct | MessageSchedule | data/structure coupling |
| 380 | impl | MessageSchedule | behavior binding and method semantics |
| 381 | fn | new | signature/API coupling |
| 404 | fn | should_fire | signature/API coupling |
| 417 | fn | build_message_matrix | signature/API coupling |
| 709 | struct | CanBus | data/structure coupling |
| 745 | impl | CanBus | behavior binding and method semantics |
| 746 | fn | new | signature/API coupling |
| 772 | fn | transmit | signature/API coupling |
| 777 | fn | transmit_tp | signature/API coupling |
| 793 | fn | tick | signature/API coupling |
| 920 | fn | log_error | signature/API coupling |
| 941 | fn | receive_for | signature/API coupling |
| 952 | struct | CanNetwork | data/structure coupling |
| 976 | struct | BusSnapshot | data/structure coupling |
| 987 | struct | CanNetworkSnapshot | data/structure coupling |
| 996 | impl | Default for CanNetwork | behavior binding and method semantics |
| 997 | fn | default | signature/API coupling |
| 1002 | impl | CanNetwork | behavior binding and method semantics |
| 1003 | fn | new | signature/API coupling |
| 1037 | fn | register | signature/API coupling |
| 1041 | fn | set_online | signature/API coupling |
| 1051 | fn | transmit | signature/API coupling |
| 1060 | fn | inject_error_once | signature/API coupling |
| 1073 | fn | inject_bus_off_once | signature/API coupling |
| 1077 | fn | clear_injections | signature/API coupling |
| 1105 | fn | tick | signature/API coupling |
| 1127 | fn | route_frames | signature/API coupling |
| 1147 | fn | bus_ref | signature/API coupling |
| 1158 | fn | bus_mut | signature/API coupling |
| 1170 | fn | receive_for_sa | signature/API coupling |
| 1180 | fn | all_errors | signature/API coupling |
| 1189 | fn | online_count | signature/API coupling |
| 1192 | fn | total_errors | signature/API coupling |
| 1199 | fn | per_bus_health_01 | signature/API coupling |
| 1212 | fn | network_health_score_01 | signature/API coupling |
| 1233 | fn | snapshot | signature/API coupling |
| 1264 | fn | export_snapshot_json | signature/API coupling |
| 1274 | fn | export_snapshot_csv | signature/API coupling |
| 1310 | module | tests | data/structure coupling |
| 1314 | fn | arbitration_orders_by_can_id | signature/API coupling |
| 1337 | fn | scheduler_jitter_stays_in_expected_band | signature/API coupling |
| 1363 | fn | routed_message_appears_next_tick_on_destination_bus | signature/API coupling |

```

Line-by-Line Impact Map

[illegible]

[illegible]


```
|fect integration, timing, and diagnostic outcomes. |
| 85 |         BusId::Lin => write!(f, "LIN-6   Sensors    "), | I/O or protocol-critical logic; edits can affect
tegration, timing, and diagnostic outcomes. |
| 86 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 87 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 88 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 89 | (blank) | line is blank; visual separation only. |
| 90 | // ■■ Bus State (ISO 11898-1 §6.15) ████████████████████████████████████████████████████ | comment line;
ntent, no runtime effect. |
| 91 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding blo
context. |
| 92 | pub enum BusState { | declares state variants; adding/removing variants impacts pattern matches and logic bran
s. |
| 93 |     ErrorActive, // TEC < 128 && REC < 128 | executable/support line; behavior impact depends on surrounding
ck context. |
| 94 |     ErrorPassive, // TEC >= 128 or REC >= 128 | executable/support line; behavior impact depends on surroundin
lock context. |
| 95 |     BusOff,      // TEC >= 256 – node disconnected | executable/support line; behavior impact depends on surr
ding block context. |
| 96 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 97 | (blank) | line is blank; visual separation only. |
| 98 | impl std::fmt::Display for BusState { | implementation block; method behavior and trait conformance are define
ere. |
| 99 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signa
e/body changes alter API, control flow, and outputs. |
| 100 |         match self { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 101 |             BusState::ErrorActive => write!(f, "ERR-ACT "), | I/O or protocol-critical logic; edits can affect
tegration, timing, and diagnostic outcomes. |
| 102 |             BusState::ErrorPassive => write!(f, "ERR-PSV "), | I/O or protocol-critical logic; edits can affe
integration, timing, and diagnostic outcomes. |
| 103 |             BusState::BusOff => write!(f, "BUS-OFF!"), | I/O or protocol-critical logic; edits can affect int
ation, timing, and diagnostic outcomes. |
| 104 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 105 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 106 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 107 | (blank) | line is blank; visual separation only. |
| 108 | // ■■ Error kind ████████████████████████████████████████████████████ | comment
intent, no runtime effect. |
| 109 | #[derive(Debug, Clone, Copy)] | executable/support line; behavior impact depends on surrounding block context
| 110 | pub enum CanErrorKind { | declares state variants; adding/removing variants impacts pattern matches and logic
anches. |
| 111 |     Bit, | executable/support line; behavior impact depends on surrounding block context. |
| 112 |     Stuff, | executable/support line; behavior impact depends on surrounding block context. |
| 113 |     Crc, | executable/support line; behavior impact depends on surrounding block context. |
| 114 |     Form, | executable/support line; behavior impact depends on surrounding block context. |
| 115 |     Ack, | executable/support line; behavior impact depends on surrounding block context. |
| 116 |     Overload, | executable/support line; behavior impact depends on surrounding block context. |
| 117 |     BablingIdiot, | executable/support line; behavior impact depends on surrounding block context. |
| 118 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 119 | (blank) | line is blank; visual separation only. |
| 120 | impl std::fmt::Display for CanErrorKind { | implementation block; method behavior and trait conformance are d
ned here. |
| 121 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signa
re/body changes alter API, control flow, and outputs. |
| 122 |         match self { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 123 |             CanErrorKind::Bit => write!(f, "Bit"), | I/O or protocol-critical logic; edits can affect integra
n, timing, and diagnostic outcomes. |
| 124 |             CanErrorKind::Stuff => write!(f, "Stuff"), | I/O or protocol-critical logic; edits can affect int
ation, timing, and diagnostic outcomes. |
| 125 |             CanErrorKind::Crc => write!(f, "CRC"), | I/O or protocol-critical logic; edits can affect integra
n, timing, and diagnostic outcomes. |
| 126 |             CanErrorKind::Form => write!(f, "Form"), | I/O or protocol-critical logic; edits can affect integ
ion, timing, and diagnostic outcomes. |
| 127 |             CanErrorKind::Ack => write!(f, "Ack"), | I/O or protocol-critical logic; edits can affect integra
```



```
| 180 |         babbling: false, | executable/support line; behavior impact depends on surrounding block context. |
| 181 |         fps_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 182 |         fps_count: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 183 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope |
| 184 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 185 | pub fn tx_success(&mut self) { | declares executable behavior; signature/body changes alter API, control fl
w, and outputs. |
| 186 |     self.tec = self.tec.saturating_sub(1); | executable/support line; behavior impact depends on surround
block context. |
| 187 |     self.update_state(); | executable/support line; behavior impact depends on surrounding block context. |
| 188 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 189 | pub fn tx_error(&mut self) { | declares executable behavior; signature/body changes alter API, control fl
and outputs. |
| 190 |     self.tec = self.tec.saturating_add(8); | executable/support line; behavior impact depends on surround
block context. |
| 191 |     self.update_state(); | executable/support line; behavior impact depends on surrounding block context. |
| 192 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 193 | pub fn rx_error(&mut self) { | declares executable behavior; signature/body changes alter API, control fl
and outputs. |
| 194 |     self.rec = self.rec.saturating_add(1); | executable/support line; behavior impact depends on surround
block context. |
| 195 |     self.update_state(); | executable/support line; behavior impact depends on surrounding block context. |
| 196 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 197 | (blank) | line is blank; visual separation only. |
| 198 | fn update_state(&mut self) { | declares executable behavior; signature/body changes alter API, control fl
and outputs. |
| 199 |     self.state = if self.tec >= 256 { | executable/support line; behavior impact depends on surrounding b
k context. |
| 200 |         BusState::BusOff | executable/support line; behavior impact depends on surrounding block context. |
| 201 |     } else if self.tec >= 128 \\ \\ self.rec >= 128 { | executable/support line; behavior impact depends o
urrounding block context. |
| 202 |         BusState::ErrorPassive | executable/support line; behavior impact depends on surrounding block co
xt. |
| 203 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 204 |         BusState::ErrorActive | executable/support line; behavior impact depends on surrounding block con
t. |
| 205 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 206 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 207 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 208 | (blank) | line is blank; visual separation only. |
| 209 | // ████████████████████████████████████████████████████████████████████████████████████████████████ | commen
ntent, no runtime effect. |
| 210 | // CAN Transport Protocol – J1939-21 TP.BAM and TP.CMDT | comment line; documentation intent, no runtime effe
|
| 211 | // ████████████████████████████████████████████████████████████████████████████████████████████████ | commen
ntent, no runtime effect. |
| 212 | (blank) | line is blank; visual separation only. |
| 213 | /// State of an active TP session | comment line; documentation intent, no runtime effect. |
| 214 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 215 | struct TpSession { | declares data layout; field changes may break serialization, tests, and callers. |
| 216 |     pgn: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 217 |     sa: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 218 |     da: u8, // 0xFF = BAM broadcast | executable/support line; behavior impact depends on surrounding block c
ext. |
| 219 |     data: Vec<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 220 |     total_bytes: u16, | executable/support line; behavior impact depends on surrounding block context. |
| 221 |     total_packets: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 222 |     next_seq: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 223 |     timer: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 224 |     is_bam: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 225 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 226 | (blank) | line is blank; visual separation only. |
| 227 | /// J1939 Transport Protocol engine (BAM + CMDT) | comment line; documentation intent, no runtime effect. |
```

```

| 228 | pub struct CanTransportProtocol { | declares data layout; field changes may break serialization, tests, and c
ers. |
| 229 |     /// Active send sessions: key = (sa, da, pgn) | comment line; documentation intent, no runtime effect. |
| 230 |     tx_sessions: Vec<TpSession>, | executable/support line; behavior impact depends on surrounding block cont
. |
| 231 |     /// Active receive sessions: key = (sa, da) | comment line; documentation intent, no runtime effect. |
| 232 |     rx_sessions: Vec<TpSession>, | executable/support line; behavior impact depends on surrounding block cont
. |
| 233 |     /// Completed reassembled messages ready for delivery | comment line; documentation intent, no runtime ef
t. |
| 234 |     pub completed: Vec<(u32, u8, u8, Vec<u8>>), // (pgn, sa, da, data) | executable/support line; behavior im
t depends on surrounding block context. |
| 235 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 236 | (blank) | line is blank; visual separation only. |
| 237 | impl Default for CanTransportProtocol { | implementation block; method behavior and trait conformance are def
d here. |
| 238 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
utputs. |
| 239 |         Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 240 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 241 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 242 | (blank) | line is blank; visual separation only. |
| 243 | impl CanTransportProtocol { | implementation block; method behavior and trait conformance are defined here. |
| 244 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
utputs. |
| 245 |         CanTransportProtocol { | executable/support line; behavior impact depends on surrounding block context
|
| 246 |             tx_sessions: Vec::new(), | executable/support line; behavior impact depends on surrounding block
text. |
| 247 |             rx_sessions: Vec::new(), | executable/support line; behavior impact depends on surrounding block
text. |
| 248 |             completed: Vec::new(), | executable/support line; behavior impact depends on surrounding block co
xt. |
| 249 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 250 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 251 | (blank) | line is blank; visual separation only. |
| 252 |     /// Queue a large J1939 message for BAM broadcast (> 8 bytes) | comment line; documentation intent, no ru
me effect. |
| 253 |     pub fn send_bam(&mut self, ts: f64, pgn: u32, sa: u8, data: Vec<u8>) -> Vec<J1939Frame> { | declares exec
ble behavior; signature/body changes alter API, control flow, and outputs. |
| 254 |         let total = data.len() as u16; | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 255 |         let packets = total.div_ceil(7) as u8; | executable/support line; behavior impact depends on surround
block context. |
| 256 |         let mut frames = Vec::new(); | executable/support line; behavior impact depends on surrounding block
text. |
| 257 | (blank) | line is blank; visual separation only. |
| 258 |         // TP.CM_BAM – announce the broadcast | comment line; documentation intent, no runtime effect. |
| 259 |         let mut bam = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 260 |         bam[0] = 0x20; // Control byte: BAM | executable/support line; behavior impact depends on surrounding
ock context. |
| 261 |         bam[1] = (total & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 262 |         bam[2] = ((total >> 8) & 0xFF) as u8; | executable/support line; behavior impact depends on surroundi
block context. |
| 263 |         bam[3] = packets; | executable/support line; behavior impact depends on surrounding block context. |
| 264 |         bam[4] = 0xFF; | executable/support line; behavior impact depends on surrounding block context. |
| 265 |         bam[5] = (pgn & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block
text. |
| 266 |         bam[6] = ((pgn >> 8) & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding
ock context. |
| 267 |         bam[7] = ((pgn >> 16) & 0xFF) as u8; | executable/support line; behavior impact depends on surroundin
lock context. |

```

```

| 268 |         frames.push(J1939Frame::from_raw( | executable/support line; behavior impact depends on surrounding block context. |
| 269 |             ts, | executable/support line; behavior impact depends on surrounding block context. |
| 270 |             J1939Frame::build_id(7, pgn::TP_CM, sa, 0xFF), | executable/support line; behavior impact depends on surrounding block context. |
| 271 |             &bam, | executable/support line; behavior impact depends on surrounding block context. |
| 272 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 273 | (blank) | line is blank; visual separation only. |
| 274 |         // TP.DT – data transfer packets (send immediately for BAM) | comment line; documentation intent, no runtime effect. |
| 275 |         for seq in 0..packets { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 276 |             let start = seq as usize * 7; | executable/support line; behavior impact depends on surrounding block context. |
| 277 |             let end = (start + 7).min(data.len()); | executable/support line; behavior impact depends on surrounding block context. |
| 278 |             let mut dt = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block context. |
| 279 |             dt[0] = seq + 1; // sequence 1-based | executable/support line; behavior impact depends on surrounding block context. |
| 280 |             for (i, &b) in data[start..end].iter().enumerate() { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 281 |                 dt[i + 1] = b; | executable/support line; behavior impact depends on surrounding block context. |
| 282 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 283 |             frames.push(J1939Frame::from_raw( | executable/support line; behavior impact depends on surrounding block context. |
| 284 |                 ts + (seq as f64 * 0.010), | executable/support line; behavior impact depends on surrounding block context. |
| 285 |                 J1939Frame::build_id(7, pgn::TP_DT, sa, 0xFF), | executable/support line; behavior impact depends on surrounding block context. |
| 286 |                 &dt, | executable/support line; behavior impact depends on surrounding block context. |
| 287 |                 )); | executable/support line; behavior impact depends on surrounding block context. |
| 288 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 289 |         frames | executable/support line; behavior impact depends on surrounding block context. |
| 290 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 291 | (blank) | line is blank; visual separation only. |
| 292 |     /// Process received TP frame – reassemble multi-packet messages | comment line; documentation intent, no runtime effect. |
| 293 |     pub fn process_rx(&mut self, frame: &J1939Frame) -> bool { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 294 |         match frame.pgn { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 295 |             p if p == pgn::TP_CM => { | executable/support line; behavior impact depends on surrounding block context. |
| 296 |                 let ctrl = frame.data[0]; | executable/support line; behavior impact depends on surrounding block context. |
| 297 |                 if ctrl == 0x20 { | conditional gate; changing condition shifts runtime path and side effects. |
| 298 |                     // BAM announce | comment line; documentation intent, no runtime effect. |
| 299 |                     let total = (frame.data[1] as u16) \& ((frame.data[2] as u16) << 8); | executable/support line; behavior impact depends on surrounding block context. |
| 300 |                     let packets = frame.data[3]; | executable/support line; behavior impact depends on surrounding block context. |
| 301 |                     let pgn_rx = (frame.data[5] as u32) | executable/support line; behavior impact depends on surrounding block context. |
| 302 |                         \& ((frame.data[6] as u32) << 8) | executable/support line; behavior impact depends on surrounding block context. |
| 303 |                         \& ((frame.data[7] as u32) << 16); | executable/support line; behavior impact depends on surrounding block context. |
| 304 |                     self.rx_sessions.retain(\&s \& s.sa != frame.sa); | executable/support line; behavior impact depends on surrounding block context. |
| 305 |                     self.rx_sessions.push(TpSession { | executable/support line; behavior impact depends on surrounding block context. |
| 306 |                         pgn: pgn_rx, | executable/support line; behavior impact depends on surrounding block

```

```

text. |
| 307 |          sa: frame.sa, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 308 |          da: 0xFF, | executable/support line; behavior impact depends on surrounding block con
t. |
| 309 |          data: Vec::new(), | executable/support line; behavior impact depends on surrounding b
k context. |
| 310 |          total_bytes: total, | executable/support line; behavior impact depends on surrounding
ock context. |
| 311 |          total_packets: packets, | executable/support line; behavior impact depends on surroun
g block context. |
| 312 |          next_seq: 1, | executable/support line; behavior impact depends on surrounding block
text. |
| 313 |          timer: 0.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 314 |          is_bam: true, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 315 |          }); | executable/support line; behavior impact depends on surrounding block context. |
| 316 |          return true; | executable/support line; behavior impact depends on surrounding block cont
. |
| 317 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 318 |          false | executable/support line; behavior impact depends on surrounding block context. |
| 319 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 320 |      p if p == pgn::TP_DT => { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 321 |          let seq = frame.data[0]; | executable/support line; behavior impact depends on surrounding bl
context. |
| 322 |          if let Some(sess) = self | conditional gate; changing condition shifts runtime path and side
ects. |
| 323 |              .rx_sessions | executable/support line; behavior impact depends on surrounding block cont
. |
| 324 |              .iter_mut() | executable/support line; behavior impact depends on surrounding block conte
|
| 325 |              .find(\|s\| s.sa == frame.sa && s.next_seq == seq) | executable/support line; behavior im
t depends on surrounding block context. |
| 326 |          { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 327 |              let payload = &frame.data[1..]; | executable/support line; behavior impact depends on sur
nding block context. |
| 328 |              let remaining = (sess.total_bytes as usize).saturating_sub(sess.data.len()); | executable
pport line; behavior impact depends on surrounding block context. |
| 329 |              let to_copy = remaining.min(7); | executable/support line; behavior impact depends on sur
nding block context. |
| 330 |              sess.data | executable/support line; behavior impact depends on surrounding block context
| 331 |                  .extend_from_slice(&payload[..to_copy.min(payload.len())]); | executable/support line
ehavior impact depends on surrounding block context. |
| 332 |              sess.next_seq += 1; | executable/support line; behavior impact depends on surrounding blo
context. |
| 333 |              if sess.data.len() >= sess.total_bytes as usize { | conditional gate; changing condition
fts runtime path and side effects. |
| 334 |                  let done = sess.clone(); | executable/support line; behavior impact depends on surrou
ng block context. |
| 335 |                  self.rx_sessions.retain(\|s\| s.sa != frame.sa); | executable/support line; behavior
act depends on surrounding block context. |
| 336 |                  self.completed.push((done.pgn, done.sa, done.da, done.data)); | executable/support li
behavior impact depends on surrounding block context. |
| 337 |              } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 338 |              return true; | executable/support line; behavior impact depends on surrounding block cont
. |
| 339 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 340 |          false | executable/support line; behavior impact depends on surrounding block context. |

```

```

341 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 342 |         _ => false, | executable/support line; behavior impact depends on surrounding block context. |
| 343 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 344 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 345 | (blank) | line is blank; visual separation only. |
| 346 | pub fn tick(&mut self, dt: f64) { | declares executable behavior; signature/body changes alter API, contr
flow, and outputs. |
| 347 |     for s in &mut self.tx_sessions { | iteration control; changing bounds/body affects throughput, timing
nd possibly safety guarantees. |
| 348 |         s.timer += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 349 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 350 |     for s in &mut self.rx_sessions { | iteration control; changing bounds/body affects throughput, timing
nd possibly safety guarantees. |
| 351 |         s.timer += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 352 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 353 |     // Timeout stale sessions after 750ms (J1939 specification) | comment line; documentation intent, no
time effect. |
| 354 |     self.tx_sessions.retain(\s\| s.timer < 0.750); | executable/support line; behavior impact depends on
rounding block context. |
| 355 |     self.rx_sessions.retain(\s\| s.timer < 0.750); | executable/support line; behavior impact depends on
rounding block context. |
| 356 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 357 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 358 | (blank) | line is blank; visual separation only. |
| 359 | // ████████████████████████████████████████████████████████████████████████████████████████████████████ | commen
tent, no runtime effect. |
| 360 | // Message Scheduling – periodic transmission matrix | comment line; documentation intent, no runtime effect.
| 361 | // ████████████████████████████████████████████████████████████████████████████████████████████████████ | commen
tent, no runtime effect. |
| 362 | (blank) | line is blank; visual separation only. |
| 363 | /// Scheduled J1939 message descriptor | comment line; documentation intent, no runtime effect. |
| 364 | pub struct MessageSchedule { | declares data layout; field changes may break serialization, tests, and caller
|
| 365 |     pub pgn: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 366 |     pub sa: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 367 |     pub da: u8, // 0xFF = broadcast | executable/support line; behavior impact depends on surrounding block c
ext. |
| 368 |     pub bus: BusId, | executable/support line; behavior impact depends on surrounding block context. |
| 369 |     pub period_ms: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 370 |     pub priority: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 371 |     /// Description for CAN database / DBC compatibility | comment line; documentation intent, no runtime eff
. |
| 372 |     pub description: &'static str, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 373 |     /// Current timer (fires when >= period_ms) | comment line; documentation intent, no runtime effect. |
| 374 |     timer_ms: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 375 |     /// Transmission jitter ±jitter_ms (realistic) | comment line; documentation intent, no runtime effect. |
| 376 |     pub jitter_ms: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 377 |     jitter_offset: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 378 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 379 | (blank) | line is blank; visual separation only. |
| 380 | impl MessageSchedule { | implementation block; method behavior and trait conformance are defined here. |
| 381 |     pub fn new( | declares executable behavior; signature/body changes alter API, control flow, and outputs.
| 382 |         pgn: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 383 |         sa: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 384 |         da: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 385 |         bus: BusId, | executable/support line; behavior impact depends on surrounding block context. |
| 386 |         period_ms: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 387 |         priority: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 388 |         desc: &'static str, | executable/support line; behavior impact depends on surrounding block context.
| 389 |     ) -> Self { | executable/support line; behavior impact depends on surrounding block context. |

```

```
| MessageSchedule { | executable/support line; behavior impact depends on surrounding block context. | |
| 390 | pgn, | executable/support line; behavior impact depends on surrounding block context. |  
| 391 | sa, | executable/support line; behavior impact depends on surrounding block context. |  
| 392 | da, | executable/support line; behavior impact depends on surrounding block context. |  
| 393 | bus, | executable/support line; behavior impact depends on surrounding block context. |  
| 394 | period_ms, | executable/support line; behavior impact depends on surrounding block context. |  
| 395 | priority, | executable/support line; behavior impact depends on surrounding block context. |  
| 396 | description: desc, | executable/support line; behavior impact depends on surrounding block context.  
| 397 |  
| timer_ms: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 398 | jitter_ms: 1.0, | executable/support line; behavior impact depends on surrounding block context.  
| 399 | jitter_offset: 0.0, | executable/support line; behavior impact depends on surrounding block conte  
| 400 |  
| } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
| 401 |  
| } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 402 | /// Returns true if it's time to transmit, applying jitter | comment line; documentation intent, no runti  
| 403 | effect. |  
| 404 | pub fn should_fire(&mut self, dt_ms: f64, noise: f64) -> bool { | declares executable behavior; signature  
dy changes alter API, control flow, and outputs. |  
| 405 | self.jitter_offset = noise * self.jitter_ms; | executable/support line; behavior impact depends on su  
unding block context. |  
| 406 | self.timer_ms += dt_ms; | executable/support line; behavior impact depends on surrounding block conte  
| 407 | if self.timer_ms >= self.period_ms + self.jitter_offset { | conditional gate; changing condition shif  
runtime path and side effects. |  
| 408 | self.timer_ms = 0.0; | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 409 | true | executable/support line; behavior impact depends on surrounding block context. |  
| 410 | } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 411 | false | executable/support line; behavior impact depends on surrounding block context. |  
| 412 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
| 413 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 414 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 415 | (blank) | line is blank; visual separation only. |  
| 416 | /// The complete J1939 message matrix for a heavy agricultural vehicle | comment line; documentation intent,  
runtime effect. |  
| 417 | pub fn build_message_matrix() -> Vec<MessageSchedule> { | declares executable behavior; signature/body change  
lter API, control flow, and outputs. |  
| 418 | vec![ | executable/support line; behavior impact depends on surrounding block context. |  
| 419 | // ■ Powertrain bus (BUS-1 HS-CAN 500kbps) ████████████████████████████████████████ | comment line; docum  
tent, no runtime effect. |  
| 420 | MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 421 | pgn::EEC1, | executable/support line; behavior impact depends on surrounding block context. |  
| 422 | addr::ECM_1, | executable/support line; behavior impact depends on surrounding block context. |  
| 423 | 0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 424 | BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 425 | 10.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 426 | 3, | executable/support line; behavior impact depends on surrounding block context. |  
| 427 | "EEC1: Engine Speed/Torque", | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 428 | ), | executable/support line; behavior impact depends on surrounding block context. |  
| 429 | MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 430 | pgn::EEC2, | executable/support line; behavior impact depends on surrounding block context. |  
| 431 | addr::ECM_1, | executable/support line; behavior impact depends on surrounding block context. |  
| 432 | 0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 433 | BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 434 | 50.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 435 | 3, | executable/support line; behavior impact depends on surrounding block context. |  
| 436 | "EEC2: Throttle/Load", | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 437 | ), | executable/support line; behavior impact depends on surrounding block context. |
```



```
| 438 | MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context. |  
| 439 |     pgn::IC1, | executable/support line; behavior impact depends on surrounding block context. |  
| 440 |     addr::ECM_1, | executable/support line; behavior impact depends on surrounding block context. |  
| 441 |     0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 442 |     BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 443 |     500.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 444 |     6, | executable/support line; behavior impact depends on surrounding block context. |  
| 445 |     "IC1: Boost/Intake Conditions", | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 446 | ), | executable/support line; behavior impact depends on surrounding block context. |  
| 447 | MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 448 |     pgn::ET1, | executable/support line; behavior impact depends on surrounding block context. |  
| 449 |     addr::ECM_1, | executable/support line; behavior impact depends on surrounding block context. |  
| 450 |     0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 451 |     BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 452 |     1000.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 453 |     6, | executable/support line; behavior impact depends on surrounding block context. |  
| 454 |     "ET1: Engine Temperatures", | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 455 | ), | executable/support line; behavior impact depends on surrounding block context. |  
| 456 | MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 457 |     pgn::EFL_P1, | executable/support line; behavior impact depends on surrounding block context. |  
| 458 |     addr::ECM_1, | executable/support line; behavior impact depends on surrounding block context. |  
| 459 |     0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 460 |     BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 461 |     500.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 462 |     6, | executable/support line; behavior impact depends on surrounding block context. |  
| 463 |     "EFL/P1: Oil/Fuel Pressures", | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 464 | ), | executable/support line; behavior impact depends on surrounding block context. |  
| 465 | MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 466 |     pgn::LFE, | executable/support line; behavior impact depends on surrounding block context. |  
| 467 |     addr::ECM_1, | executable/support line; behavior impact depends on surrounding block context. |  
| 468 |     0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 469 |     BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 470 |     100.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 471 |     6, | executable/support line; behavior impact depends on surrounding block context. |  
| 472 |     "LFE: Fuel Economy", | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 473 | ), | executable/support line; behavior impact depends on surrounding block context. |  
| 474 | MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 475 |     pgn::HOURS, | executable/support line; behavior impact depends on surrounding block context. |  
| 476 |     addr::ECM_1, | executable/support line; behavior impact depends on surrounding block context. |  
| 477 |     0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 478 |     BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 479 |     1000.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 480 |     6, | executable/support line; behavior impact depends on surrounding block context. |  
| 481 |     "HOURS: Engine Hours", | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 482 | ), | executable/support line; behavior impact depends on surrounding block context. |  
| 483 | MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 484 |     pgn::DM1, | executable/support line; behavior impact depends on surrounding block context. |  
| 485 |     addr::ECM_1, | executable/support line; behavior impact depends on surrounding block context. |  
| 486 |     0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 487 |     BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 488 |     1000.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 489 |     6, | executable/support line; behavior impact depends on surrounding block context. |  
| 490 |     "DM1: Active DTCs (ECM)", | executable/support line; behavior impact depends on surrounding block  
ntext.
```

```

| 491 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 492 |         MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context
| 493 |             pgn::EEC3, | executable/support line; behavior impact depends on surrounding block context. |
| 494 |             addr::ECM_1, | executable/support line; behavior impact depends on surrounding block context. |
| 495 |             0xFF, | executable/support line; behavior impact depends on surrounding block context. |
| 496 |             BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont
. |
| 497 |             250.0, | executable/support line; behavior impact depends on surrounding block context. |
| 498 |             6, | executable/support line; behavior impact depends on surrounding block context. |
| 499 |             "EEC3: Engine Demand/Nominal", | executable/support line; behavior impact depends on surrounding
ck context. |
| 500 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 501 |         MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context
| 502 |             pgn::FUEL1, | executable/support line; behavior impact depends on surrounding block context. |
| 503 |             addr::ECM_1, | executable/support line; behavior impact depends on surrounding block context. |
| 504 |             0xFF, | executable/support line; behavior impact depends on surrounding block context. |
| 505 |             BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont
. |
| 506 |             1000.0, | executable/support line; behavior impact depends on surrounding block context. |
| 507 |             6, | executable/support line; behavior impact depends on surrounding block context. |
| 508 |             "FUEL1: Cumulative Fuel", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 509 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 510 |         MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context
| 511 |             0, | executable/support line; behavior impact depends on surrounding block context. |
| 512 |             addr::ECM_1, | executable/support line; behavior impact depends on surrounding block context. |
| 513 |             addr::TRANSMISSION, | executable/support line; behavior impact depends on surrounding block conte
|
| 514 |             BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont
. |
| 515 |             20.0, | executable/support line; behavior impact depends on surrounding block context. |
| 516 |             3, | executable/support line; behavior impact depends on surrounding block context. |
| 517 |             "TSC1: Torque/Speed Ctrl → TCM", | executable/support line; behavior impact depends on surroundi
lock context. |
| 518 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 519 |         MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context
| 520 |             pgn::ETC1, | executable/support line; behavior impact depends on surrounding block context. |
| 521 |             addr::TRANSMISSION, | executable/support line; behavior impact depends on surrounding block conte
|
| 522 |             0xFF, | executable/support line; behavior impact depends on surrounding block context. |
| 523 |             BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont
. |
| 524 |             20.0, | executable/support line; behavior impact depends on surrounding block context. |
| 525 |             3, | executable/support line; behavior impact depends on surrounding block context. |
| 526 |             "ETC1: Trans Output Shaft Speed", | executable/support line; behavior impact depends on surroundi
block context. |
| 527 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 528 |         MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context
| 529 |             pgn::ETC2, | executable/support line; behavior impact depends on surrounding block context. |
| 530 |             addr::TRANSMISSION, | executable/support line; behavior impact depends on surrounding block conte
|
| 531 |             0xFF, | executable/support line; behavior impact depends on surrounding block context. |
| 532 |             BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont
. |
| 533 |             20.0, | executable/support line; behavior impact depends on surrounding block context. |
| 534 |             3, | executable/support line; behavior impact depends on surrounding block context. |
| 535 |             "ETC2: Gear/Range Position", | executable/support line; behavior impact depends on surrounding bl
context. |
| 536 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 537 |         MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context
| 538 |             pgn::DM1, | executable/support line; behavior impact depends on surrounding block context. |
| 539 |             addr::TRANSMISSION, | executable/support line; behavior impact depends on surrounding block conte
|
| 540 |             0xFF, | executable/support line; behavior impact depends on surrounding block context. |
| 541 |             BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont

```

```

. |
| 542 |         1000.0, | executable/support line; behavior impact depends on surrounding block context. |
| 543 |         6, | executable/support line; behavior impact depends on surrounding block context. |
| 544 |         "DM1: Active DTCs (TCM)", | executable/support line; behavior impact depends on surrounding block
context. |
| 545 |     ), | executable/support line; behavior impact depends on surrounding block context. |
| 546 |     MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context
| 547 |         pgn::TCFG, | executable/support line; behavior impact depends on surrounding block context. |
| 548 |         addr::TRANSMISSION, | executable/support line; behavior impact depends on surrounding block conte
|
| 549 |         0xFF, | executable/support line; behavior impact depends on surrounding block context. |
| 550 |         BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont
. |
| 551 |         1000.0, | executable/support line; behavior impact depends on surrounding block context. |
| 552 |         7, | executable/support line; behavior impact depends on surrounding block context. |
| 553 |         "TCFG: Transmission Config", | executable/support line; behavior impact depends on surrounding bl
context. |
| 554 |     ), | executable/support line; behavior impact depends on surrounding block context. |
| 555 |     MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context
| 556 |         pgn::EBC1, | executable/support line; behavior impact depends on surrounding block context. |
| 557 |         addr::BRAKES, | executable/support line; behavior impact depends on surrounding block context. |
| 558 |         0xFF, | executable/support line; behavior impact depends on surrounding block context. |
| 559 |         BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont
. |
| 560 |         20.0, | executable/support line; behavior impact depends on surrounding block context. |
| 561 |         2, | executable/support line; behavior impact depends on surrounding block context. |
| 562 |         "EBC1: Brake Status (ABS/ESP)", | executable/support line; behavior impact depends on surrounding
ock context. |
| 563 |     ), | executable/support line; behavior impact depends on surrounding block context. |
| 564 |     MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context
| 565 |         pgn::EBC2, | executable/support line; behavior impact depends on surrounding block context. |
| 566 |         addr::BRAKES, | executable/support line; behavior impact depends on surrounding block context. |
| 567 |         0xFF, | executable/support line; behavior impact depends on surrounding block context. |
| 568 |         BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont
. |
| 569 |         100.0, | executable/support line; behavior impact depends on surrounding block context. |
| 570 |         6, | executable/support line; behavior impact depends on surrounding block context. |
| 571 |         "EBC2: Wheel Speeds", | executable/support line; behavior impact depends on surrounding block con
t. |
| 572 |     ), | executable/support line; behavior impact depends on surrounding block context. |
| 573 |     MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context
| 574 |         pgn::DM1, | executable/support line; behavior impact depends on surrounding block context. |
| 575 |         addr::BRAKES, | executable/support line; behavior impact depends on surrounding block context. |
| 576 |         0xFF, | executable/support line; behavior impact depends on surrounding block context. |
| 577 |         BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont
. |
| 578 |         1000.0, | executable/support line; behavior impact depends on surrounding block context. |
| 579 |         6, | executable/support line; behavior impact depends on surrounding block context. |
| 580 |         "DM1: Active DTCs (ABS)", | executable/support line; behavior impact depends on surrounding block
context. |
| 581 |     ), | executable/support line; behavior impact depends on surrounding block context. |
| 582 |     MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context
| 583 |         pgn::CCVS, | executable/support line; behavior impact depends on surrounding block context. |
| 584 |         addr::INSTRUMENT, | executable/support line; behavior impact depends on surrounding block context
| 585 |         0xFF, | executable/support line; behavior impact depends on surrounding block context. |
| 586 |         BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont
. |
| 587 |         100.0, | executable/support line; behavior impact depends on surrounding block context. |
| 588 |         6, | executable/support line; behavior impact depends on surrounding block context. |
| 589 |         "CCVS: Vehicle Speed/CC", | executable/support line; behavior impact depends on surrounding block
context. |
| 590 |     ), | executable/support line; behavior impact depends on surrounding block context. |
| 591 |     MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context
| 592 |         pgn::VD, | executable/support line; behavior impact depends on surrounding block context. |
| 593 |         addr::INSTRUMENT, | executable/support line; behavior impact depends on surrounding block context

```

```
| 594 |      0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 595 |      BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 596 |      1000.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 597 |      6, | executable/support line; behavior impact depends on surrounding block context. |  
| 598 |      "VD: Vehicle Distance", | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 599 |      ), | executable/support line; behavior impact depends on surrounding block context. |  
| 600 |      // ■ ISOBUS (BUS-4, 250kbps) ████████████████████████████████████████████████████████ | comment line;  
ntent, no runtime effect. |  
| 601 |      MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 602 |          pgn::PTO, | executable/support line; behavior impact depends on surrounding block context. |  
| 603 |          addr::HITCH, | executable/support line; behavior impact depends on surrounding block context. |  
| 604 |          0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 605 |          BusId::IsoBus, | executable/support line; behavior impact depends on surrounding block context. |  
| 606 |          100.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 607 |          6, | executable/support line; behavior impact depends on surrounding block context. |  
| 608 |          "PTO: Power Take-Off Status", | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 609 |      ), | executable/support line; behavior impact depends on surrounding block context. |  
| 610 |      MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 611 |          pgn::HITCH, | executable/support line; behavior impact depends on surrounding block context. |  
| 612 |          addr::HITCH, | executable/support line; behavior impact depends on surrounding block context. |  
| 613 |          0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 614 |          BusId::IsoBus, | executable/support line; behavior impact depends on surrounding block context. |  
| 615 |          100.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 616 |          6, | executable/support line; behavior impact depends on surrounding block context. |  
| 617 |          "HITCH: 3-Point Hitch Status", | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 618 |      ), | executable/support line; behavior impact depends on surrounding block context. |  
| 619 |      MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 620 |          pgn::DM1, | executable/support line; behavior impact depends on surrounding block context. |  
| 621 |          addr::HITCH, | executable/support line; behavior impact depends on surrounding block context. |  
| 622 |          0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 623 |          BusId::IsoBus, | executable/support line; behavior impact depends on surrounding block context. |  
| 624 |          1000.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 625 |          6, | executable/support line; behavior impact depends on surrounding block context. |  
| 626 |          "DM1: Active DTCs (HCM)", | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 627 |      ), | executable/support line; behavior impact depends on surrounding block context. |  
| 628 |      MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 629 |          pgn::WS_MST, | executable/support line; behavior impact depends on surrounding block context. |  
| 630 |          addr::TASK_CTRL, | executable/support line; behavior impact depends on surrounding block context.  
| 631 |          0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 632 |          BusId::IsoBus, | executable/support line; behavior impact depends on surrounding block context. |  
| 633 |          100.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 634 |          6, | executable/support line; behavior impact depends on surrounding block context. |  
| 635 |          "WS_MASTER: Working Set Master", | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 636 |      ), | executable/support line; behavior impact depends on surrounding block context. |  
| 637 |      // ■ Body bus (BUS-3, 250kbps) ████████████████████████████████████████████████████████ | comment line;  
ntent, no runtime effect. |  
| 638 |      MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 639 |          pgn::PROP_A, | executable/support line; behavior impact depends on surrounding block context. |  
| 640 |          addr::CAB, | executable/support line; behavior impact depends on surrounding block context. |  
| 641 |          0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 642 |          BusId::BodyMs, | executable/support line; behavior impact depends on surrounding block context. |  
| 643 |          100.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 644 |          7, | executable/support line; behavior impact depends on surrounding block context. |  
| 645 |          "BCM Status: Lights/Battery", | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 646 |      ), | executable/support line; behavior impact depends on surrounding block context. |  
| 647 |      MessageSchedule::new(| executable/support line; behavior impact depends on surrounding block context  
| 648 |          pgn::PROP_A + 1, | executable/support line; behavior impact depends on surrounding block context.  
| 649 |          addr::INSTRUMENT, | executable/support line; behavior impact depends on surrounding block context.
```

```
| 650 |      0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 651 |      BusId::BodyMs, | executable/support line; behavior impact depends on surrounding block context. |  
| 652 |      100.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 653 |      7, | executable/support line; behavior impact depends on surrounding block context. |  
| 654 |      "ICM Status: Warning Lamps", | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 655 |    ), | executable/support line; behavior impact depends on surrounding block context. |  
| 656 |    MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 657 |        pgn::AMB, | executable/support line; behavior impact depends on surrounding block context. |  
| 658 |        addr::CAB, | executable/support line; behavior impact depends on surrounding block context. |  
| 659 |        0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 660 |        BusId::BodyMs, | executable/support line; behavior impact depends on surrounding block context. |  
| 661 |        1000.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 662 |        6, | executable/support line; behavior impact depends on surrounding block context. |  
| 663 |        "AMB: Ambient Conditions", | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 664 |    ), | executable/support line; behavior impact depends on surrounding block context. |  
| 665 |    MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 666 |        pgn::VEP1, | executable/support line; behavior impact depends on surrounding block context. |  
| 667 |        addr::CAB, | executable/support line; behavior impact depends on surrounding block context. |  
| 668 |        0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 669 |        BusId::BodyMs, | executable/support line; behavior impact depends on surrounding block context. |  
| 670 |        500.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 671 |        6, | executable/support line; behavior impact depends on surrounding block context. |  
| 672 |        "VEP1: Vehicle Electrical Power", | executable/support line; behavior impact depends on surroundi  
block context. |  
| 673 |    ), | executable/support line; behavior impact depends on surrounding block context. |  
| 674 |    // ■ Network Management (all buses) ████████████████████████████████████████████ | comment line; do  
ntent, no runtime effect. |  
| 675 |    MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 676 |        pgn::ECAN, | executable/support line; behavior impact depends on surrounding block context. |  
| 677 |        addr::ECM_1, | executable/support line; behavior impact depends on surrounding block context. |  
| 678 |        0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 679 |        BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 680 |        1000.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 681 |        6, | executable/support line; behavior impact depends on surrounding block context. |  
| 682 |        "AC: Address Claim (ECM)", | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 683 |    ), | executable/support line; behavior impact depends on surrounding block context. |  
| 684 |    MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 685 |        pgn::ECAN, | executable/support line; behavior impact depends on surrounding block context. |  
| 686 |        addr::TRANSMISSION, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 687 |        0xFF, | executable/support line; behavior impact depends on surrounding block context. |  
| 688 |        BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 689 |        1000.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 690 |        6, | executable/support line; behavior impact depends on surrounding block context. |  
| 691 |        "AC: Address Claim (TCM)", | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 692 |    ), | executable/support line; behavior impact depends on surrounding block context. |  
| 693 |    MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context  
| 694 |        pgn::DM13, | executable/support line; behavior impact depends on surrounding block context. |  
| 695 |        addr::ECM_1, | executable/support line; behavior impact depends on surrounding block context. |  
| 696 |        addr::TRANSMISSION, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 697 |        BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 698 |        5000.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 699 |        7, | executable/support line; behavior impact depends on surrounding block context. |  
| 700 |        "DM13: Stop/Start Broadcast", | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 701 |    ), | executable/support line; behavior impact depends on surrounding block context. |  
| 702 | ] | executable/support line; behavior impact depends on surrounding block context. |
```

[illegible]

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| 747 |         CanBus { | executable/support line; behavior impact depends on surrounding block context. |
| 748 |             id, | executable/support line; behavior impact depends on surrounding block context. |
| 749 |             speed, | executable/support line; behavior impact depends on surrounding block context. |
| 750 |             state: BusState::ErrorActive, | executable/support line; behavior impact depends on surrounding b
k context. |
| 751 |             can_fd: false, | executable/support line; behavior impact depends on surrounding block context. |
| 752 |             tx_queue: Vec::new(), | executable/support line; behavior impact depends on surrounding block con
t. |
| 753 |             dispatched: Vec::new(), | executable/support line; behavior impact depends on surrounding block c
ext. |
| 754 |             frame_log: VecDeque::new(), | executable/support line; behavior impact depends on surrounding blo
context. |
| 755 |             total_tx: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 756 |             total_errors: 0, | executable/support line; behavior impact depends on surrounding block context.
| 757 |             bus_load_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context
|
| 758 |             frame_counter: 0, | executable/support line; behavior impact depends on surrounding block context
| 759 |             load_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 760 |             inject_bit_error: false, | executable/support line; behavior impact depends on surrounding block
text. |
| 761 |             inject_bus_off: false, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 762 |             inject_missing_ack: false, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 763 |             inject_babbling_sa: None, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 764 |             busoff_recovery_counter: 0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 765 |             busoff_recovery_timer: 0.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 766 |             error_log: VecDeque::new(), | executable/support line; behavior impact depends on surrounding blo
context. |
| 767 |             tp: CanTransportProtocol::new(), | executable/support line; behavior impact depends on surroundin
lock context. |
| 768 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 769 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 770 | (blank) | line is blank; visual separation only. |
| 771 |     /// Enqueue a frame for transmission | comment line; documentation intent, no runtime effect. |
| 772 |     pub fn transmit(&mut self, frame: J1939Frame) { | declares executable behavior; signature/body changes al
API, control flow, and outputs. |
| 773 |         self.tx_queue.push(frame); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 774 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 775 | (blank) | line is blank; visual separation only. |
| 776 |     /// Transmit a large message via TP.BAM | comment line; documentation intent, no runtime effect. |
| 777 |     pub fn transmit_tp(&mut self, ts: f64, pgn: u32, sa: u8, data: Vec<u8>) { | declares executable behavior;
gnature/body changes alter API, control flow, and outputs. |
| 778 |         if data.len() <= 8 { | conditional gate; changing condition shifts runtime path and side effects. |
| 779 |             // Fits in one frame | comment line; documentation intent, no runtime effect. |
| 780 |             let id = J1939Frame::build_id(6, pgn, sa, 0xFF); | executable/support line; behavior impact depen
on surrounding block context. |
| 781 |             let mut arr = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 782 |             arr[..data.len()].copy_from_slice(&data); | executable/support line; behavior impact depends on s
ounding block context. |
| 783 |             self.transmit(J1939Frame::from_raw(ts, id, &arr)); | executable/support line; behavior impact dep
s on surrounding block context. |
| 784 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 785 |             let frames = self.tp.send_bam(ts, pgn, sa, data); | I/O or protocol-critical logic; edits can aff
integration, timing, and diagnostic outcomes. |
| 786 |             for f in frames { | iteration control; changing bounds/body affects throughput, timing, and possi
safety guarantees. |
| 787 |                 self.transmit(f); | executable/support line; behavior impact depends on surrounding block con
t. |

```

```
| 788 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 789 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 790 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 791 | (blank) | line is blank; visual separation only. |  
| 792 |     /// Arbitrate and dispatch all queued frames for this cycle. | comment line; documentation intent, no run  
e effect. |  
| 793 |     pub fn tick(&mut self, nodes: &mut HashMap<u8, CanNode>, dt: f64) { | declares executable behavior; signa  
e/body changes alter API, control flow, and outputs. |  
| 794 |         self.dispatched.clear(); | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 795 |         self.tp.tick(dt); | executable/support line; behavior impact depends on surrounding block context. |  
| 796 | (blank) | line is blank; visual separation only. |  
| 797 |         // ■ Bus-off recovery timer ████████████████████████████████████████████████████████████ | comment line; doc  
ent, no runtime effect. |  
| 798 |         if self.state == BusState::BusOff { | conditional gate; changing condition shifts runtime path and si  
effects. |  
| 799 |             self.busoff_recovery_timer += dt; | executable/support line; behavior impact depends on surroundi  
block context. |  
| 800 |             // 128 occurrences of 11 recessive bits = ~128 × 11 / speed | comment line; documentation intent,  
runtime effect. |  
| 801 |             let recovery_bit_time = 128.0 * 11.0 / self.speed.kbps() as f64 / 1000.0; | executable/support li  
behavior impact depends on surrounding block context. |  
| 802 |             if self.busoff_recovery_timer >= recovery_bit_time { | conditional gate; changing condition shift  
untime path and side effects. |  
| 803 |                 self.state = BusState::ErrorActive; | executable/support line; behavior impact depends on surr  
nding block context. |  
| 804 |                 self.busoff_recovery_timer = 0.0; | executable/support line; behavior impact depends on surro  
ing block context. |  
| 805 |                 self.log_error( | executable/support line; behavior impact depends on surrounding block conte  
|  
| 806 |                     0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 807 |                     CanErrorKind::Bit, | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 808 |                     None, | executable/support line; behavior impact depends on surrounding block context. |  
| 809 |                     None, | executable/support line; behavior impact depends on surrounding block context. |  
| 810 |                     "Bus-Off RECOVERED after 128×11 recessive bits", | executable/support line; behavior impa  
depends on surrounding block context. |  
| 811 |                 ); | executable/support line; behavior impact depends on surrounding block context. |  
| 812 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 813 |             // No transmission during bus-off | comment line; documentation intent, no runtime effect. |  
| 814 |             self.tx_queue.clear(); | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 815 |             return; | executable/support line; behavior impact depends on surrounding block context. |  
| 816 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 817 | (blank) | line is blank; visual separation only. |  
| 818 |         // ■ Injected bus-off ████████████████████████████████████████████████████████████ | comment line; do  
nt, no runtime effect. |  
| 819 |         if self.inject_bus_off { | conditional gate; changing condition shifts runtime path and side effects.  
| 820 |             self.state = BusState::BusOff; | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 821 |             self.busoff_recovery_timer = 0.0; | executable/support line; behavior impact depends on surroundi  
block context. |  
| 822 |             self.inject_bus_off = false; | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 823 |             self.log_error( | executable/support line; behavior impact depends on surrounding block context.  
| 824 |                 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 825 |                 CanErrorKind::Bit, | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 826 |                 None, | executable/support line; behavior impact depends on surrounding block context. |  
| 827 |                 None, | executable/support line; behavior impact depends on surrounding block context. |  
| 828 |                 "INJECTED: Bus-Off condition", | executable/support line; behavior impact depends on surround
```



```
| block context. |
| 829 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 830 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 831 | (blank) | line is blank; visual separation only. |
| 832 |     // ■ Babbling-idiot detection ████████████████████████████████████████████████████ | comment line; doc
ent, no runtime effect. |
| 833 |     // Real CAN: a node transmitting more than ~2× its rated period is a babbler | comment line; document
on intent, no runtime effect. |
| 834 |         if let Some(bab_sa) = self.inject_babbling_sa { | conditional gate; changing condition shifts runtime
th and side effects. |
| 835 |             if let Some(node) = nodes.get_mut(&bab_sa) { | conditional gate; changing condition shifts runtim
ath and side effects. |
| 836 |                 node.babbling = true; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 837 |                 node.tx_error(); // Force error state | executable/support line; behavior impact depends on s
ounding block context. |
| 838 |                 self.log_error( | executable/support line; behavior impact depends on surrounding block conte
|
| 839 |                     0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 840 |                     CanErrorKind::BabblingIdiot, | executable/support line; behavior impact depends on surroun
g block context. |
| 841 |                     Some(bab_sa), | executable/support line; behavior impact depends on surrounding block con
t. |
| 842 |                     None, | executable/support line; behavior impact depends on surrounding block context. |
| 843 |                     "Babbling-idiot: node transmitting too fast", | executable/support line; behavior impact
ends on surrounding block context. |
| 844 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 845 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 846 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 847 | (blank) | line is blank; visual separation only. |
| 848 |     // ■ Sort by CAN ID: lower = higher priority ████████████████████████████████████████████████████ | comment line; document
nt, no runtime effect. |
| 849 |     self.tx_queue.sort_by_key(|f| f.raw_id); | executable/support line; behavior impact depends on surr
ding block context. |
| 850 | (blank) | line is blank; visual separation only. |
| 851 |     let frames: Vec<J1939Frame> = self.tx_queue.drain(..).collect(); | executable/support line; behavior
act depends on surrounding block context. |
| 852 |     for frame in frames { | iteration control; changing bounds/body affects throughput, timing, and possi
safety guarantees. |
| 853 |         // ■ Bit error injection (1 in 200) ████████████████████████████████████████████████████ | comment line; docum
tent, no runtime effect. |
| 854 |         if self.inject_bit_error && self.total_tx.is_multiple_of(200) { | conditional gate; changing cond
on shifts runtime path and side effects. |
| 855 |             if let Some(node) = nodes.get_mut(&frame.sa) { | conditional gate; changing condition shifts
time path and side effects. |
| 856 |                 node.tx_error(); | executable/support line; behavior impact depends on surrounding block
text. |
| 857 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 858 |             self.total_errors += 1; | executable/support line; behavior impact depends on surrounding blo
context. |
| 859 |             self.log_error( | executable/support line; behavior impact depends on surrounding block conte
|
| 860 |                 frame.timestamp, | executable/support line; behavior impact depends on surrounding block
text. |
| 861 |                 CanErrorKind::Bit, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 862 |                 Some(frame.sa), | executable/support line; behavior impact depends on surrounding block c
ext. |
| 863 |                 Some(frame.raw_id), | executable/support line; behavior impact depends on surrounding blo
context. |
| 864 |                 "INJECTED: Bit error – frame dropped", | executable/support line; behavior impact depends
```



```
| 900 |         self.frame_counter += 1; | executable/support line; behavior impact depends on surrounding block context. |
| 901 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 902 | (blank) | line is blank; visual separation only. |
| 903 |     // ■ Bus load (updated each second) ████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 904 |     self.load_timer += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 905 |     if self.load_timer >= 1.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 906 |         self.bus_load_pct = | executable/support line; behavior impact depends on surrounding block context. |
| 907 |             (self.frame_counter as f64 / self.speed.max_fps() as f64 * 100.0).min(100.0); | executable/support line; behavior impact depends on surrounding block context. |
| 908 |             // Update fps per node | comment line; documentation intent, no runtime effect. |
| 909 |             for node in nodes.values_mut() { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 910 |                 if node.bus == self.id { | conditional gate; changing condition shifts runtime path and side effects. |
| 911 |                     node.fps = node.fps_count as f64; | executable/support line; behavior impact depends on surrounding block context. |
| 912 |                     node.fps_count = 0; | executable/support line; behavior impact depends on surrounding block context. |
| 913 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 914 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 915 |             self.frame_counter = 0; | executable/support line; behavior impact depends on surrounding block context. |
| 916 |             self.load_timer = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 917 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 918 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 919 | (blank) | line is blank; visual separation only. |
| 920 |     fn log_error( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 921 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 922 |         ts: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 923 |         kind: CanErrorKind, | executable/support line; behavior impact depends on surrounding block context. |
| 924 |         sa: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 925 |         id: Option<u32>, | executable/support line; behavior impact depends on surrounding block context. |
| 926 |         desc: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 927 |     ) { | executable/support line; behavior impact depends on surrounding block context. |
| 928 |         self.error_log.push_front(BusError { | executable/support line; behavior impact depends on surrounding block context. |
| 929 |             timestamp: ts, | executable/support line; behavior impact depends on surrounding block context. |
| 930 |             bus: self.id, | executable/support line; behavior impact depends on surrounding block context. |
| 931 |             kind, | executable/support line; behavior impact depends on surrounding block context. |
| 932 |             source_sa: sa, | executable/support line; behavior impact depends on surrounding block context. |
| 933 |             raw_id: id, | executable/support line; behavior impact depends on surrounding block context. |
| 934 |             description: desc, | executable/support line; behavior impact depends on surrounding block context. |
| 935 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 936 |         if self.error_log.len() > 100 { | conditional gate; changing condition shifts runtime path and side effects. |
| 937 |             self.error_log.pop_back(); | executable/support line; behavior impact depends on surrounding block context. |
| 938 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 939 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 940 | (blank) | line is blank; visual separation only. |
| 941 |     pub fn receive_for(&self, sa: u8) -> impl Iterator<Item = &J1939Frame> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 942 |         self.dispatched | executable/support line; behavior impact depends on surrounding block context. |
```

```
| 943 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |  
| 944 |         .filter(move \|f\| f.sa != sa && (f.da == 0xFF \| \| f.da == sa)) | executable/support line; behav  
impact depends on surrounding block context. |  
| 945 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 946 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 947 | (blank) | line is blank; visual separation only. |  
| 948 | // ████████████████████████████████████████████████████████████████████████████████████████████████████ | commen  
ntent, no runtime effect. |  
| 949 | // Multi-Bus Network Manager | comment line; documentation intent, no runtime effect. |  
| 950 | // ████████████████████████████████████████████████████████████████████████████████████████████████████ | commen  
ntent, no runtime effect. |  
| 951 | (blank) | line is blank; visual separation only. |  
| 952 | pub struct CanNetwork { | declares data layout; field changes may break serialization, tests, and callers. |  
| 953 |     pub powertrain: CanBus, | executable/support line; behavior impact depends on surrounding block context. |  
| 954 |     pub chassis: CanBus, | executable/support line; behavior impact depends on surrounding block context. |  
| 955 |     pub body: CanBus, | executable/support line; behavior impact depends on surrounding block context. |  
| 956 |     pub isobus: CanBus, | executable/support line; behavior impact depends on surrounding block context. |  
| 957 |     pub diagnostic: CanBus, | executable/support line; behavior impact depends on surrounding block context. |  
| 958 | (blank) | line is blank; visual separation only. |  
| 959 |     // ■ Node registry (all buses, keyed by SA) ████████████████████████████████████████████████████████████████████████████████████████████████████ | comment line; doc  
ument, no runtime effect. |  
| 960 |     pub nodes: HashMap<u8, CanNode>, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 961 | (blank) | line is blank; visual separation only. |  
| 962 |     // ■ Message scheduling matrix ████████████████████████████████████████████████████████████████████████████████████████████████████ | comment line;  
ment, no runtime effect. |  
| 963 |     pub schedule: Vec<MessageSchedule>, | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 964 | (blank) | line is blank; visual separation only. |  
| 965 |     // ■ Cross-bus gateway routing table ████████████████████████████████████████████████████████████████████████████████████████████████████ | comment line; d  
ocument, no runtime effect. |  
| 966 |     /// PGNs that get forwarded from one bus to another (by VCM) | comment line; documentation intent, no run  
e effect. |  
| 967 |     pub routing: Vec<(u32, BusId, BusId)>, // (pgn, src_bus, dst_bus) | executable/support line; behavior imp  
depends on surrounding block context. |  
| 968 | (blank) | line is blank; visual separation only. |  
| 969 |     // ■ Network-wide stats ████████████████████████████████████████████████████████████████████████████████████████████████████ | comment line  
ment, no runtime effect. |  
| 970 |     pub total_frames_all_buses: u64, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 971 |     pub elapsed: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 972 |     noise_t: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 973 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 974 | (blank) | line is blank; visual separation only. |  
| 975 | #[derive(Debug, Clone, Serialize)] | serialization contract touchpoint; edits can break report compatibility  
tooling. |  
| 976 | pub struct BusSnapshot { | declares data layout; field changes may break serialization, tests, and callers. |  
| 977 |     pub bus: String, | executable/support line; behavior impact depends on surrounding block context. |  
| 978 |     pub speed_kbps: u32, | executable/support line; behavior impact depends on surrounding block context. |  
| 979 |     pub state: String, | executable/support line; behavior impact depends on surrounding block context. |  
| 980 |     pub load_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 981 |     pub total_tx: u64, | executable/support line; behavior impact depends on surrounding block context. |  
| 982 |     pub total_errors: u64, | executable/support line; behavior impact depends on surrounding block context. |  
| 983 |     pub log_depth: usize, | executable/support line; behavior impact depends on surrounding block context. |  
| 984 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 985 | (blank) | line is blank; visual separation only. |  
| 986 | #[derive(Debug, Clone, Serialize)] | serialization contract touchpoint; edits can break report compatibility  
tooling. |  
| 987 | pub struct CanNetworkSnapshot { | declares data layout; field changes may break serialization, tests, and cal  
s. |  
| 988 |     pub elapsed_s: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 989 |     pub health_score_01: f64, | executable/support line; behavior impact depends on surrounding block context  
| 990 |     pub total_frames_all_buses: u64, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 991 |     pub total_errors_all_buses: u64, | executable/support line; behavior impact depends on surrounding block
```

```

text. |
| 992 |     pub online_nodes: usize, | executable/support line; behavior impact depends on surrounding block context.
| 993 |     pub buses: Vec<BusSnapshot>, | executable/support line; behavior impact depends on surrounding block context.
. |
| 994 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 995 | (blank) | line is blank; visual separation only. |
| 996 | impl Default for CanNetwork { | implementation block; method behavior and trait conformance are defined here.
| 997 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
outputs. |
| 998 |         Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 999 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1000 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1001 | (blank) | line is blank; visual separation only. |
| 1002 | impl CanNetwork { | implementation block; method behavior and trait conformance are defined here. |
| 1003 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
outputs. |
| 1004 |         let mut net = CanNetwork { | executable/support line; behavior impact depends on surrounding block context.
ext. |
| 1005 |             powertrain: CanBus::new(BusId::PowertrainHs, CanSpeed::Kbps500), | executable/support line; behavior impact
depends on surrounding block context. |
| 1006 |             chassis: CanBus::new(BusId::ChassisHs, CanSpeed::Kbps500), | executable/support line; behavior impact
depends on surrounding block context. |
| 1007 |             body: CanBus::new(BusId::BodyMs, CanSpeed::Kbps250), | executable/support line; behavior impact
depends on surrounding block context. |
| 1008 |             isobus: CanBus::new(BusId::IsoBus, CanSpeed::Kbps250), | executable/support line; behavior impact
depends on surrounding block context. |
| 1009 |             diagnostic: CanBus::new(BusId::Diagnostic, CanSpeed::Kbps500), | executable/support line; behavior
impact depends on surrounding block context. |
| 1010 |             nodes: HashMap::new(), | executable/support line; behavior impact depends on surrounding block context.
ext. |
| 1011 |             schedule: build_message_matrix(), | executable/support line; behavior impact depends on surrounding
block context. |
| 1012 |             routing: vec![] | executable/support line; behavior impact depends on surrounding block context.
| 1013 |             // VCM forwards engine speed from powertrain to body bus (for speedometer over MS-CAN) | comment
line; documentation intent, no runtime effect. |
| 1014 |             (pgn::EEC1, BusId::PowertrainHs, BusId::BodyMs), | executable/support line; behavior impact
depends on surrounding block context. |
| 1015 |             (pgn::CCVS, BusId::PowertrainHs, BusId::BodyMs), | executable/support line; behavior impact
depends on surrounding block context. |
| 1016 |             (pgn::DM1, BusId::PowertrainHs, BusId::Diagnostic), | executable/support line; behavior impact
depends on surrounding block context. |
| 1017 |             (pgn::ETC1, BusId::PowertrainHs, BusId::BodyMs), | executable/support line; behavior impact
depends on surrounding block context. |
| 1018 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 1019 |         total_frames_all_buses: 0, | executable/support line; behavior impact depends on surrounding block
context. |
| 1020 |         elapsed: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1021 |         noise_t: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1022 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
. |
| 1023 |     // Register all known nodes on their respective buses | comment line; documentation intent, no runtime
effect. |
| 1024 |     net.register(addr::ECM_1, "ECM #1", BusId::PowertrainHs); | executable/support line; behavior impact
depends on surrounding block context. |
| 1025 |     net.register(addr::TRANSMISSION, "TCM", BusId::PowertrainHs); | executable/support line; behavior impact
depends on surrounding block context. |
| 1026 |     net.register(addr::BRAKES, "ABS/ESP", BusId::PowertrainHs); | executable/support line; behavior impact
depends on surrounding block context. |
| 1027 |     net.register(addr::CAB, "BCM/CAB", BusId::BodyMs); | executable/support line; behavior impact depends
on surrounding block context. |
| 1028 |     net.register(addr::INSTRUMENT, "ICM/DASH", BusId::BodyMs); | executable/support line; behavior impact
depends on surrounding block context. |
| 1029 |     net.register(addr::HITCH, "HCM/HITCH", BusId::IsoBus); | executable/support line; behavior impact depends
on surrounding block context. |
| 1030 |     net.register(addr::ISOBUS_VT, "ISOBUS-VT", BusId::IsoBus); | executable/support line; behavior impact

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depends on surrounding block context. |
| 1031 |         net.register(addr::TASK_CTRL, "ISOBUS-TC", BusId::IsoBus); | executable/support line; behavior impact
depends on surrounding block context. |
| 1032 |         net.register(addr::IMPLEMENT, "IMPLEMENT", BusId::IsoBus); | executable/support line; behavior impact
depends on surrounding block context. |
| 1033 |         net.register(addr::HEADWAY, "VCM/GW", BusId::PowertrainHs); | executable/support line; behavior impact
depends on surrounding block context. |
| 1034 |         net | executable/support line; behavior impact depends on surrounding block context. |
| 1035 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1036 | (blank) | line is blank; visual separation only. |
| 1037 |     pub fn register(&mut self, sa: u8, name: &'static str, bus: BusId) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1038 |         self.nodes.insert(sa, CanNode::new(sa, name, bus)); | executable/support line; behavior impact depends
on surrounding block context. |
| 1039 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1040 | (blank) | line is blank; visual separation only. |
| 1041 |     pub fn set_online(&mut self, sa: u8) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1042 |         if let Some(n) = self.nodes.get_mut(&sa) { | conditional gate; changing condition shifts runtime pattern and side effects. |
| 1043 |             n.online = true; | executable/support line; behavior impact depends on surrounding block context.
| 1044 |             n.tec = 0; | executable/support line; behavior impact depends on surrounding block context. |
| 1045 |             n.rec = 0; | executable/support line; behavior impact depends on surrounding block context. |
| 1046 |             n.state = BusState::ErrorActive; | executable/support line; behavior impact depends on surrounding
block context. |
| 1047 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
|
| 1048 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1049 | (blank) | line is blank; visual separation only. |
| 1050 |     /// Transmit a frame on the appropriate bus for the SA | comment line; documentation intent, no runtime effect. |
| 1051 |     pub fn transmit(&mut self, frame: J1939Frame) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1052 |         let bus_id = self | executable/support line; behavior impact depends on surrounding block context. |
| 1053 |         .nodes | executable/support line; behavior impact depends on surrounding block context. |
| 1054 |         .get(&frame.sa) | executable/support line; behavior impact depends on surrounding block context. |
| 1055 |         .map(|n| n.bus) | executable/support line; behavior impact depends on surrounding block context.
|
| 1056 |         .unwrap_or(BusId::PowertrainHs); | executable/support line; behavior impact depends on surrounding
block context. |
| 1057 |         self.bus_mut(bus_id).transmit(frame); | executable/support line; behavior impact depends on surrounding
block context. |
| 1058 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1059 | (blank) | line is blank; visual separation only. | | | | |
| 1060 |     pub fn inject_error_once(&mut self, bus: BusId, kind: CanErrorKind) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1061 |         let babbling_sa = self.nodes.values().find(|n| n.bus == bus).map(|n| n.sa); | executable/support line; behavior impact depends on surrounding block context. |
| 1062 |         let b = self.bus_mut(bus); | executable/support line; behavior impact depends on surrounding block context. |
| 1063 |         match kind { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 1064 |             CanErrorKind::Bit => b.inject_bit_error = true, | executable/support line; behavior impact depends
on surrounding block context. |
| 1065 |             CanErrorKind::Ack => b.inject_missing_ack = true, | executable/support line; behavior impact depends
on surrounding block context. |
| 1066 |             CanErrorKind::BabblingIdiot => { | executable/support line; behavior impact depends on surrounding
block context. |
| 1067 |                 b.inject_babbling_sa = babbling_sa; | executable/support line; behavior impact depends on surrounding
block context. |
| 1068 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1069 |             _ => b.inject_bit_error = true, | executable/support line; behavior impact depends on surrounding
block context. |
| 1070 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
|

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| 1071 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1072 | (blank) | line is blank; visual separation only. |
| 1073 |     pub fn inject_bus_off_once(&mut self, bus: BusId) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1074 |         self.bus_mut(bus).inject_bus_off = true; | executable/support line; behavior impact depends on surrounding block context. |
| 1075 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1076 | (blank) | line is blank; visual separation only. |
| 1077 |     pub fn clear_injections(&mut self, bus: Option<BusId>) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1078 |         match bus { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 1079 |             Some(id) => { | executable/support line; behavior impact depends on surrounding block context. |
| 1080 |                 let b = self.bus_mut(id); | executable/support line; behavior impact depends on surrounding block context. |
| 1081 |                     b.inject_bit_error = false; | executable/support line; behavior impact depends on surrounding block context. |
| 1082 |                     b.inject_missing_ack = false; | executable/support line; behavior impact depends on surrounding block context. |
| 1083 |                     b.inject_bus_off = false; | executable/support line; behavior impact depends on surrounding block context. |
| 1084 |                     b.inject_babbling_sa = None; | executable/support line; behavior impact depends on surrounding block context. |
| 1085 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1086 |             None => { | executable/support line; behavior impact depends on surrounding block context. |
| 1087 |                 for id in [ | iteration control; changing bounds/body affects throughput, timing, and possible safety guarantees. |
| 1088 |                     BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block context. |
| 1089 |                     BusId::ChassisHs, | executable/support line; behavior impact depends on surrounding block context. |
| 1090 |                     BusId::BodyMs, | executable/support line; behavior impact depends on surrounding block context. |
| 1091 |                     BusId::IsoBus, | executable/support line; behavior impact depends on surrounding block context. |
| 1092 |                     BusId::Diagnostic, | executable/support line; behavior impact depends on surrounding block context. |
| 1093 |                 ] { | executable/support line; behavior impact depends on surrounding block context. |
| 1094 |                     let b = self.bus_mut(id); | executable/support line; behavior impact depends on surrounding block context. |
| 1095 |                     b.inject_bit_error = false; | executable/support line; behavior impact depends on surrounding block context. |
| 1096 |                     b.inject_missing_ack = false; | executable/support line; behavior impact depends on surrounding block context. |
| 1097 |                     b.inject_bus_off = false; | executable/support line; behavior impact depends on surrounding block context. |
| 1098 |                     b.inject_babbling_sa = None; | executable/support line; behavior impact depends on surrounding block context. |
| 1099 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1100 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1101 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1102 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1103 | (blank) | line is blank; visual separation only. |
| 1104 |     /// Tick all buses and collect stats | comment line; documentation intent, no runtime effect. |
| 1105 |     pub fn tick(&mut self, dt: f64) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1106 |         self.elapsed += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 1107 |         self.noise_t += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 1108 |         let _noise = ((self.noise_t * 127.1 + 311.7).sin() * 43758.5).fract() - 0.5; | executable/support line; behavior impact depends on surrounding block context. |
| 1109 | (blank) | line is blank; visual separation only. |
| 1110 |         // Tick each bus | comment line; documentation intent, no runtime effect. |

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| 1111 |         self.powertrain.tick(&mut self.nodes, dt); | executable/support line; behavior impact depends on sur
nding block context. |
| 1112 |         self.chassis.tick(&mut self.nodes, dt); | executable/support line; behavior impact depends on surrou
ng block context. |
| 1113 |         self.body.tick(&mut self.nodes, dt); | executable/support line; behavior impact depends on surroundi
block context. |
| 1114 |         self.isobus.tick(&mut self.nodes, dt); | executable/support line; behavior impact depends on surroun
g block context. |
| 1115 |         self.diagnostic.tick(&mut self.nodes, dt); | executable/support line; behavior impact depends on sur
nding block context. |
| 1116 | (blank) | line is blank; visual separation only. |
| 1117 |         // VCM routing: forward frames between buses | comment line; documentation intent, no runtime effect
| 1118 |         self.route_frames(); | executable/support line; behavior impact depends on surrounding block context
| 1119 | (blank) | line is blank; visual separation only. |
| 1120 |         self.total_frames_all_buses = self.powertrain.total_tx | executable/support line; behavior impact de
ds on surrounding block context. |
| 1121 |             + self.chassis.total_tx | executable/support line; behavior impact depends on surrounding block
text. |
| 1122 |             + self.body.total_tx | executable/support line; behavior impact depends on surrounding block con
t. |
| 1123 |             + self.isobus.total_tx | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1124 |             + self.diagnostic.total_tx; | executable/support line; behavior impact depends on surrounding bl
context. |
| 1125 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1126 | (blank) | line is blank; visual separation only. |
| 1127 |     fn route_frames(&mut self) { | declares executable behavior; signature/body changes alter API, control f
, and outputs. |
| 1128 |         // Collect frames to route (can't borrow self.powertrain and self.body mutably simultaneously) | com
t line; documentation intent, no runtime effect. |
| 1129 |         let mut to_route: Vec<J1939Frame, BusId> = Vec::new(); | executable/support line; behavior impact
ends on surrounding block context. |
| 1130 |         for (pgn_filter, src_bus, dst_bus) in &self.routing { | iteration control; changing bounds/body affe
throughput, timing, and possibly safety guarantees. |
| 1131 |             let src_frames: Vec<J1939Frame> = self | executable/support line; behavior impact depends on sur
nding block context. |
| 1132 |                 .bus_ref(*src_bus) | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1133 |                 .dispatched | executable/support line; behavior impact depends on surrounding block context.
| 1134 |                 .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1135 |                 .filter(|f| f.pgn == *pgn_filter) | executable/support line; behavior impact depends on su
unding block context. |
| 1136 |                 .cloned() | executable/support line; behavior impact depends on surrounding block context. |
| 1137 |                 .collect(); | executable/support line; behavior impact depends on surrounding block context.
| 1138 |                 for f in src_frames { | iteration control; changing bounds/body affects throughput, timing, and
sibly safety guarantees. |
| 1139 |                     to_route.push((f, *dst_bus)); | executable/support line; behavior impact depends on surround
block context. |
| 1140 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1141 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1142 |         for (f, dst) in to_route { | iteration control; changing bounds/body affects throughput, timing, and
ssibly safety guarantees. |
| 1143 |             self.bus_mut(dst).transmit(f); | executable/support line; behavior impact depends on surrounding
ock context. |
| 1144 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1145 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1146 | (blank) | line is blank; visual separation only. |
| 1147 |     fn bus_ref(&self, id: BusId) -> &CanBus { | declares executable behavior; signature/body changes alter A
control flow, and outputs. |
| 1148 |         match id { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 1149 |             BusId::PowertrainHs => &self.powertrain, | executable/support line; behavior impact depends on s
ounding block context. |

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| 1150 |         BusId::ChassisHs => &self.chassis, | executable/support line; behavior impact depends on surrounding block context. | | |
| 1151 |         BusId::BodyMs => &self.body, | executable/support line; behavior impact depends on surrounding block context. |
| 1152 |         BusId::IsoBus => &self.isobus, | executable/support line; behavior impact depends on surrounding block context. |
| 1153 |         BusId::Diagnostic => &self.diagnostic, | executable/support line; behavior impact depends on surrounding block context. |
| 1154 |         BusId::Lin => &self.body, // LIN master reuses body bus object | executable/support line; behavior impact depends on surrounding block context. |
| 1155 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1156 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1157 | (blank) | line is blank; visual separation only. |
| 1158 | fn bus_mut(&mut self, id: BusId) -> &mut CanBus { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1159 |     match id { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 1160 |         BusId::PowertrainHs => &mut self.powertrain, | executable/support line; behavior impact depends on surrounding block context. |
| 1161 |         BusId::ChassisHs => &mut self.chassis, | executable/support line; behavior impact depends on surrounding block context. |
| 1162 |         BusId::BodyMs => &mut self.body, | executable/support line; behavior impact depends on surrounding block context. |
| 1163 |         BusId::IsoBus => &mut self.isobus, | executable/support line; behavior impact depends on surrounding block context. |
| 1164 |         BusId::Diagnostic => &mut self.diagnostic, | executable/support line; behavior impact depends on surrounding block context. |
| 1165 |         BusId::Lin => &mut self.body, | executable/support line; behavior impact depends on surrounding block context. |
| 1166 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1167 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1168 | (blank) | line is blank; visual separation only. |
| 1169 | /// All frames from all buses received this cycle for a given SA | comment line; documentation intent, no runtime effect. |
| 1170 | pub fn receive_for_sa(&self, sa: u8) -> Vec<J1939Frame> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1171 |     let bus_id = self | executable/support line; behavior impact depends on surrounding block context. |
| 1172 |     .nodes | executable/support line; behavior impact depends on surrounding block context. |
| 1173 |     .get(&sa) | executable/support line; behavior impact depends on surrounding block context. |
| 1174 |     .map(|n| n.bus) | executable/support line; behavior impact depends on surrounding block context. |
| 1175 |     .unwrap_or(BusId::PowertrainHs); | executable/support line; behavior impact depends on surrounding block context. |
| 1176 |     self.bus_ref(bus_id).receive_for(sa).collect() | executable/support line; behavior impact depends on surrounding block context. |
| 1177 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1178 | (blank) | line is blank; visual separation only. |
| 1179 | /// Combined error log from all buses | comment line; documentation intent, no runtime effect. |
| 1180 | pub fn all_errors(&self) -> impl Iterator<Item = &BusError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1181 |     self.powertrain | executable/support line; behavior impact depends on surrounding block context. |
| 1182 |     .error_log | executable/support line; behavior impact depends on surrounding block context. |
| 1183 |     .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1184 |     .chain(self.chassis.error_log.iter()) | executable/support line; behavior impact depends on surrounding block context. |
| 1185 |     .chain(self.body.error_log.iter()) | executable/support line; behavior impact depends on surrounding block context. |
| 1186 |     .chain(self.isobus.error_log.iter()) | executable/support line; behavior impact depends on surrounding block context. |
| 1187 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1188 | (blank) | line is blank; visual separation only. |
| 1189 | pub fn online_count(&self) -> usize { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1190 |     self.nodes.values().filter(|n| n.online).count() | executable/support line; behavior impact depends

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n surrounding block context. |
| 1191 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1192 |     pub fn total_errors(&self) -> u64 { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1193 |         self.powertrain.total_errors | executable/support line; behavior impact depends on surrounding block context. |
| 1194 |         + self.chassis.total_errors | executable/support line; behavior impact depends on surrounding block context. |
| 1195 |         + self.body.total_errors | executable/support line; behavior impact depends on surrounding block context. |
| 1196 |         + self.isobus.total_errors | executable/support line; behavior impact depends on surrounding block context. |
| 1197 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1198 | (blank) | line is blank; visual separation only. |
| 1199 |     pub fn per_bus_health_01(&self, bus: BusId) -> f64 { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1200 |         let b = self.bus_ref(bus); | executable/support line; behavior impact depends on surrounding block context. |
| 1201 |         let mut score = 1.0; | executable/support line; behavior impact depends on surrounding block context. |
| 1202 |         score -= (b.bus_load_pct / 100.0).powf(1.3) * 0.25; | executable/support line; behavior impact depends on surrounding block context. |
| 1203 |         score -= (b.total_errors as f64 / 2000.0).min(0.35); | executable/support line; behavior impact depends on surrounding block context. |
| 1204 |         score -= match b.state { | executable/support line; behavior impact depends on surrounding block context. |
| 1205 |             BusState::ErrorActive => 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1206 |             BusState::ErrorPassive => 0.30, | executable/support line; behavior impact depends on surrounding block context. |
| 1207 |             BusState::BusOff => 0.85, | executable/support line; behavior impact depends on surrounding block context. |
| 1208 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1209 |         score.clamp(0.0, 1.0) | executable/support line; behavior impact depends on surrounding block context. |
| 1210 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1211 | (blank) | line is blank; visual separation only. | | |
| 1212 |     pub fn network_health_score_01(&self) -> f64 { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1213 |         let buses = [ | executable/support line; behavior impact depends on surrounding block context. |
| 1214 |             BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block context. |
| 1215 |             BusId::ChassisHs, | executable/support line; behavior impact depends on surrounding block context. |
| 1216 |             BusId::BodyMs, | executable/support line; behavior impact depends on surrounding block context. |
| 1217 |             BusId::IsoBus, | executable/support line; behavior impact depends on surrounding block context. |
| 1218 |             BusId::Diagnostic, | executable/support line; behavior impact depends on surrounding block context. |
| 1219 |         ]; | executable/support line; behavior impact depends on surrounding block context. |
| 1220 |         let avg_bus = buses | executable/support line; behavior impact depends on surrounding block context. |
| 1221 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1222 |             .map(|b| self.per_bus_health_01(*b)) | executable/support line; behavior impact depends on surrounding block context. |
| 1223 |             .sum::<f64>() | executable/support line; behavior impact depends on surrounding block context. |
| 1224 |             / buses.len() as f64; | executable/support line; behavior impact depends on surrounding block context. |
| 1225 |         let online_ratio = if self.nodes.is_empty() { | executable/support line; behavior impact depends on surrounding block context. |
| 1226 |             1.0 | executable/support line; behavior impact depends on surrounding block context. |
| 1227 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 1228 |             self.online_count() as f64 / self.nodes.len() as f64 | executable/support line; behavior impact depends on surrounding block context. |
| 1229 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1230 |         (avg_bus * 0.82 + online_ratio * 0.18).clamp(0.0, 1.0) | executable/support line; behavior impact depends on surrounding block context. |

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ds on surrounding block context. |
| 1231 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1232 | (blank) | line is blank; visual separation only. |
| 1233 |     pub fn snapshot(&self) -> CanNetworkSnapshot { | declares executable behavior; signature/body changes alter
API, control flow, and outputs. |
| 1234 |         let buses = [ | executable/support line; behavior impact depends on surrounding block context. |
| 1235 |             (BusId::PowertrainHs, &self.powertrain), | executable/support line; behavior impact depends on s
urrounding block context. |
| 1236 |             (BusId::ChassisHs, &self.chassis), | executable/support line; behavior impact depends on surround
g block context. |
| 1237 |             (BusId::BodyMs, &self.body), | executable/support line; behavior impact depends on surrounding b
k context. |
| 1238 |             (BusId::IsoBus, &self.isobus), | executable/support line; behavior impact depends on surrounding
ock context. |
| 1239 |             (BusId::Diagnostic, &self.diagnostic), | executable/support line; behavior impact depends on surr
nding block context. |
| 1240 |         ]; | executable/support line; behavior impact depends on surrounding block context. |
| 1241 |         let bus_vec = buses | executable/support line; behavior impact depends on surrounding block context.
| 1242 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1243 |         .map(\(id, b)\> BusSnapshot { | executable/support line; behavior impact depends on surrounding
ock context. |
| 1244 |             bus: format!("{}", id), | executable/support line; behavior impact depends on surrounding bl
context. |
| 1245 |             speed_kbps: b.speed.kbps(), | executable/support line; behavior impact depends on surroundin
lock context. |
| 1246 |             state: format!("{}", b.state), | executable/support line; behavior impact depends on surround
g block context. |
| 1247 |             load_pct: b.bus_load_pct, | executable/support line; behavior impact depends on surrounding
ck context. |
| 1248 |             total_tx: b.total_tx, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1249 |             total_errors: b.total_errors, | executable/support line; behavior impact depends on surround
block context. |
| 1250 |             log_depth: b.frame_log.len(), | executable/support line; behavior impact depends on surround
block context. |
| 1251 |         }) | executable/support line; behavior impact depends on surrounding block context. |
| 1252 |         .collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 1253 | (blank) | line is blank; visual separation only. |
| 1254 |     CanNetworkSnapshot { | executable/support line; behavior impact depends on surrounding block context
| 1255 |         elapsed_s: self.elapsed, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1256 |         health_score_01: self.network_health_score_01(), | executable/support line; behavior impact depende
on surrounding block context. |
| 1257 |         total_frames_all_buses: self.total_frames_all_buses, | executable/support line; behavior impact
ends on surrounding block context. |
| 1258 |         total_errors_all_buses: self.total_errors(), | executable/support line; behavior impact depends
surrounding block context. |
| 1259 |         online_nodes: self.online_count(), | executable/support line; behavior impact depends on surround
g block context. |
| 1260 |         buses: bus_vec, | executable/support line; behavior impact depends on surrounding block context.
| 1261 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1262 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1263 | (blank) | line is blank; visual separation only. |
| 1264 |     pub fn export_snapshot_json<P: AsRef<Path>>(&self, path: P) -> std::io::Result<()> { | declares executab
behavior; signature/body changes alter API, control flow, and outputs. |
| 1265 |         if let Some(parent) = path.as_ref().parent() { | conditional gate; changing condition shifts runtime
th and side effects. |
| 1266 |             fs::create_dir_all(parent)?; | error-propagating operation; changing it impacts failure semantic
nd caller recovery paths. |
| 1267 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1268 |         let data = serde_json::to_string_pretty(&self.snapshot()).map_err(\>e\> { | serialization contract t
hpoint; edits can break report compatibility and tooling. |
| 1269 |         std::io::Error::other(format!("json serialize: {}", e)) | executable/support line; behavior impa

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depends on surrounding block context. |
| 1270 |         }?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths.
| 1271 |         fs::write(path, data) | I/O or protocol-critical logic; edits can affect integration, timing, and di
ostic outcomes. |
| 1272 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1273 | (blank) | line is blank; visual separation only. |
| 1274 |     pub fn export_snapshot_csv<P: AsRef<Path>>(&self, path: P) -> std::io::Result<()> { | declares executabl
ehavior; signature/body changes alter API, control flow, and outputs. |
| 1275 |         if let Some(parent) = path.as_ref().parent() { | conditional gate; changing condition shifts runtime
th and side effects. |
| 1276 |             fs::create_dir_all(parent)?; | error-propagating operation; changing it impacts failure semantic
nd caller recovery paths. |
| 1277 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1278 |         let snap = self.snapshot(); | executable/support line; behavior impact depends on surrounding block
text. |
| 1279 |         let mut f = fs::File::create(path)?; | error-propagating operation; changing it impacts failure sema
cs and caller recovery paths. |
| 1280 |         writeln!( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic out
es. |
| 1281 |             f, | executable/support line; behavior impact depends on surrounding block context. |
| 1282 |             "elapsed_s,health_score_01,total_frames_all_buses,total_errors_all_buses,online_nodes" | executab
/support line; behavior impact depends on surrounding block context. |
| 1283 |         )?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths.
| 1284 |         writeln!( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic out
es. |
| 1285 |             f, | executable/support line; behavior impact depends on surrounding block context. |
| 1286 |             "{:.3},{:.5},{},{},{},{},", | executable/support line; behavior impact depends on surrounding block
text. |
| 1287 |             snap.elapsed_s, | executable/support line; behavior impact depends on surrounding block context.
| 1288 |             snap.health_score_01, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1289 |             snap.total_frames_all_buses, | executable/support line; behavior impact depends on surrounding b
k context. |
| 1290 |             snap.total_errors_all_buses, | executable/support line; behavior impact depends on surrounding b
k context. |
| 1291 |             snap.online_nodes | executable/support line; behavior impact depends on surrounding block contex
|
| 1292 |         )?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths.
| 1293 |         writeln!(f)?; | error-propagating operation; changing it impacts failure semantics and caller recove
paths. |
| 1294 |         writeln!( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic out
es. |
| 1295 |             f, | executable/support line; behavior impact depends on surrounding block context. |
| 1296 |             "bus,speed_kbps,state,load_pct,total_tx,total_errors,log_depth" | executable/support line; behav
impact depends on surrounding block context. |
| 1297 |         )?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths.
| 1298 |         for b in snap.buses { | iteration control; changing bounds/body affects throughput, timing, and poss
y safety guarantees. |
| 1299 |             writeln!( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic
tcomes. |
| 1300 |                 f, | executable/support line; behavior impact depends on surrounding block context. |
| 1301 |                 "\"{}\"},{},\"{}\"},{:.4},{},{},{},", | executable/support line; behavior impact depends on sur
nding block context. |
| 1302 |                 b.bus, b.speed_kbps, b.state, b.load_pct, b.total_tx, b.total_errors, b.log_depth | executab
support line; behavior impact depends on surrounding block context. |
| 1303 |             )?; | error-propagating operation; changing it impacts failure semantics and caller recovery pat
|
| 1304 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1305 |         Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 1306 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1307 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1308 | (blank) | line is blank; visual separation only. |
| 1309 | #[cfg(test)] | executable/support line; behavior impact depends on surrounding block context. |

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| 1310 | mod tests { | declares a module boundary; changing module name/path breaks references. |
| 1311 |     use super::*; | imports symbols; changing path/name affects compile-time resolution. |
| 1312 | (blank) | line is blank; visual separation only. |
| 1313 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1314 |     fn arbitration_orders_by_can_id() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1315 |         let mut bus = CanBus::new(BusId::PowertrainHs, CanSpeed::Kbps500); | executable/support line; behavior impact depends on surrounding block context. |
| 1316 |         let mut nodes: HashMap<u8, CanNode> = HashMap::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 1317 |         nodes.insert( | executable/support line; behavior impact depends on surrounding block context. |
| 1318 |             addr::ECM_1, | executable/support line; behavior impact depends on surrounding block context. |
| 1319 |             CanNode::new(addr::ECM_1, "ECM", BusId::PowertrainHs), | executable/support line; behavior impact depends on surrounding block context. |
| 1320 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 1321 |         nodes.insert( | executable/support line; behavior impact depends on surrounding block context. |
| 1322 |             addr::TRANSMISSION, | executable/support line; behavior impact depends on surrounding block context. |
| 1323 |             CanNode::new(addr::TRANSMISSION, "TCM", BusId::PowertrainHs), | executable/support line; behavior impact depends on surrounding block context. |
| 1324 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 1325 | (blank) | line is blank; visual separation only. |
| 1326 |         let id_low = J1939Frame::build_id(2, pgn::EBC1, addr::TRANSMISSION, 0xFF); | executable/support line; behavior impact depends on surrounding block context. |
| 1327 |         let id_high = J1939Frame::build_id(6, pgn::DM1, addr::ECM_1, 0xFF); | executable/support line; behavior impact depends on surrounding block context. |
| 1328 |         bus.transmit(J1939Frame::from_raw(1.0, id_high, &[0; 8])); | executable/support line; behavior impact depends on surrounding block context. |
| 1329 |         bus.transmit(J1939Frame::from_raw(1.0, id_low, &[0; 8])); | executable/support line; behavior impact depends on surrounding block context. |
| 1330 |         bus.tick(&mut nodes, 0.01); | executable/support line; behavior impact depends on surrounding block context. |
| 1331 | (blank) | line is blank; visual separation only. |
| 1332 |         assert_eq!(bus.dispatched.len(), 2); | executable/support line; behavior impact depends on surrounding block context. |
| 1333 |         assert!(bus.dispatched[0].raw_id < bus.dispatched[1].raw_id); | executable/support line; behavior impact depends on surrounding block context. |
| 1334 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1335 | (blank) | line is blank; visual separation only. |
| 1336 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1337 |     fn scheduler_jitter_stays_in_expected_band() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1338 |         let mut s = MessageSchedule::new( | executable/support line; behavior impact depends on surrounding block context. |
| 1339 |             pgn::EEC1, | executable/support line; behavior impact depends on surrounding block context. |
| 1340 |             addr::ECM_1, | executable/support line; behavior impact depends on surrounding block context. |
| 1341 |             0xFF, | executable/support line; behavior impact depends on surrounding block context. |
| 1342 |             BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding block context. |
| 1343 |             100.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1344 |             3, | executable/support line; behavior impact depends on surrounding block context. |
| 1345 |             "test", | executable/support line; behavior impact depends on surrounding block context. |
| 1346 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 1347 |         s.jitter_ms = 4.0; | executable/support line; behavior impact depends on surrounding block context. |
| 1348 |         let mut t_ms = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 1349 |         let mut fire_times: Vec<f64> = Vec::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 1350 |         while fire_times.len() < 12 { | iteration control; changing bounds/body affects throughput, timing, possibly safety guarantees. |
| 1351 |             t_ms += 1.0; | executable/support line; behavior impact depends on surrounding block context. |
| 1352 |             if s.should_fire(1.0, 0.5) { | conditional gate; changing condition shifts runtime path and side effects. |
| 1353 |                 fire_times.push(t_ms); | executable/support line; behavior impact depends on surrounding block context. |
| 1354 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime

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```

pes. |
| 1355 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1356 |           for w in fire_times.windows(2) { | iteration control; changing bounds/body affects throughput, timing
and possibly safety guarantees. |
| 1357 |           let dt = w[1] - w[0]; | executable/support line; behavior impact depends on surrounding block context. |
| 1358 |           assert!((98.0..=104.5).contains(&dt), "interval {} out of band", dt); | executable/support line; behavior impact depends on surrounding block context. |
| 1359 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1360 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1361 | (blank) | line is blank; visual separation only. |
| 1362 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1363 |     fn routed_message_appears_next_tick_on_destination_bus() { | declares executable behavior; signature/body
changes alter API, control flow, and outputs. |
| 1364 |         let mut net = CanNetwork::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 1365 |         net.set_online(addr::ECM_1); | executable/support line; behavior impact depends on surrounding block context. |
| 1366 |         net.set_online(addr::INSTRUMENT); | executable/support line; behavior impact depends on surrounding block context. |
| 1367 | (blank) | line is blank; visual separation only. |
| 1368 |         let id = J1939Frame::build_id(3, pgn::EEC1, addr::ECM_1, 0xFF); | executable/support line; behavior impact depends on surrounding block context. |
| 1369 |         let frame = J1939Frame::from_raw(0.0, id, &[0xFF; 8]); | executable/support line; behavior impact depends on surrounding block context. |
| 1370 |         net.transmit(frame); | executable/support line; behavior impact depends on surrounding block context. |
| 1371 |         net.tick(0.01); | executable/support line; behavior impact depends on surrounding block context. |
| 1372 | (blank) | line is blank; visual separation only. |
| 1373 |         let body_after_first = net.body.total_tx; | executable/support line; behavior impact depends on surrounding block context. |
| 1374 |         net.tick(0.01); | executable/support line; behavior impact depends on surrounding block context. |
| 1375 |         let body_after_second = net.body.total_tx; | executable/support line; behavior impact depends on surrounding block context. |
| 1376 |         assert!( | executable/support line; behavior impact depends on surrounding block context. |
| 1377 |             body_after_second > body_after_first, | executable/support line; behavior impact depends on surrounding block context. |
| 1378 |             "expected routed frame on next tick" | executable/support line; behavior impact depends on surrounding block context. |
| 1379 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 1380 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1381 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/chassis.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 184 | Non-empty: 164 | Symbols: 8

Symbol Inventory

Line	Kind	Symbol	Change Impact
1	const	WHEELBASE	data/structure coupling
4	enum	WheelState	data/structure coupling
12	enum	EpsMode	data/structure coupling
20	struct	ChassisControl	data/structure coupling
46	impl	ChassisControl	behavior binding and method semantics
47	fn	new	signature/API coupling
67	fn	update	signature/API coupling
181	fn	reset	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
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```

|---:|---|---|
| 1 | const WHEELBASE: f64 = 2.7; // m | executable/support line; behavior impact depends on surrounding block context. |
| 2 | (blank) | line is blank; visual separation only. |
| 3 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding block context. |
| 4 | pub enum WheelState { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 5 |     Rolling, | executable/support line; behavior impact depends on surrounding block context. |
| 6 |     Skidding, | executable/support line; behavior impact depends on surrounding block context. |
| 7 |     AbsActive, | executable/support line; behavior impact depends on surrounding block context. |
| 8 |     TcsLimited, | executable/support line; behavior impact depends on surrounding block context. |
| 9 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 10 | (blank) | line is blank; visual separation only. |
| 11 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding block context. |
| 12 | pub enum EpsMode { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 13 |     Normal, | executable/support line; behavior impact depends on surrounding block context. |
| 14 |     Sport, | executable/support line; behavior impact depends on surrounding block context. |
| 15 |     Comfort, | executable/support line; behavior impact depends on surrounding block context. |
| 16 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 17 | (blank) | line is blank; visual separation only. |
| 18 | /// Chassis domain controller: ABS + ESP + TCS + EPS | comment line; documentation intent, no runtime effect. |
| 19 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 20 | pub struct ChassisControl { | declares data layout; field changes may break serialization, tests, and callers. |
| 21 |     // ABS | comment line; documentation intent, no runtime effect. |
| 22 |     pub abs_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 23 |     pub abs_cycles: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 24 |     abs_cycle_phases: [f64; 4], | executable/support line; behavior impact depends on surrounding block context. |
| 25 | (blank) | line is blank; visual separation only. |
| 26 |     // ESP – Electronic Stability Program | comment line; documentation intent, no runtime effect. |
| 27 |     pub esp_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 28 |     pub oversteer: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 29 |     pub understeer: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 30 |     pub esp_brake_correction: [f64; 4], // individual wheel brake delta | executable/support line; behavior impact depends on surrounding block context. |
| 31 | (blank) | line is blank; visual separation only. |
| 32 |     // TCS – Traction Control | comment line; documentation intent, no runtime effect. |
| 33 |     pub tcs_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 34 |     pub tcs_throttle_cut: f64, // 0-1 fraction cut | executable/support line; behavior impact depends on surrounding block context. |
| 35 | (blank) | line is blank; visual separation only. |
| 36 |     // EPS – Electronic Power Steering | comment line; documentation intent, no runtime effect. |
| 37 |     pub eps_assist_torque: f64, // Nm | executable/support line; behavior impact depends on surrounding block context. |
| 38 |     pub eps_mode: EpsMode, | executable/support line; behavior impact depends on surrounding block context. |
| 39 | (blank) | line is blank; visual separation only. |
| 40 |     // Per-wheel state | comment line; documentation intent, no runtime effect. |
| 41 |     pub wheel_states: [WheelState; 4], | executable/support line; behavior impact depends on surrounding block context. |
| 42 |     pub brake_pressures: [f64; 4], | executable/support line; behavior impact depends on surrounding block context. |
| 43 |     pub wheel_velocities: [f64; 4], | executable/support line; behavior impact depends on surrounding block context. |
| 44 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 45 | (blank) | line is blank; visual separation only. |
| 46 | impl ChassisControl { | implementation block; method behavior and trait conformance are defined here. |
| 47 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 48 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 49 |             abs_active: false, | executable/support line; behavior impact depends on surrounding block context. |
| 50 |             abs_cycles: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 51 |             abs_cycle_phases: [0.0; 4], | executable/support line; behavior impact depends on surrounding block context. |

```

```

ontext. |
| 52 |         esp_active: false, | executable/support line; behavior impact depends on surrounding block context.
| 53 |         oversteer: false, | executable/support line; behavior impact depends on surrounding block context.
| 54 |         understeer: false, | executable/support line; behavior impact depends on surrounding block context.
| 55 |         esp_brake_correction: [0.0; 4], | executable/support line; behavior impact depends on surrounding
ck context. |
| 56 |         tcs_active: false, | executable/support line; behavior impact depends on surrounding block context.
| 57 |         tcs_throttle_cut: 0.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 58 |         eps_assist_torque: 0.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 59 |         eps_mode: EpsMode::Normal, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 60 |         wheel_states: [WheelState::Rolling; 4], | executable/support line; behavior impact depends on surr
ding block context. |
| 61 |         brake_pressures: [0.0; 4], | executable/support line; behavior impact depends on surrounding block
ntext. |
| 62 |         wheel_velocities: [0.0; 4], | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 63 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 64 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 65 | (blank) | line is blank; visual separation only. |
| 66 |     /// Returns effective throttle after TCS cut | comment line; documentation intent, no runtime effect. |
| 67 |     pub fn update( | declares executable behavior; signature/body changes alter API, control flow, and outputs
| 68 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 69 |         vehicle_speed: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 70 |         steering_angle: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 71 |         yaw_rate: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 72 |         user_brake: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 73 |         user_throttle: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 74 |         dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 75 |     ) -> (f64, f64) { | executable/support line; behavior impact depends on surrounding block context. |
| 76 |         // --- ABS ----- | comment line; documentation in
t, no runtime effect. |
| 77 |         let mut effective_brake = user_brake; | executable/support line; behavior impact depends on surroundin
lock context. |
| 78 |         self.abs_active = false; | executable/support line; behavior impact depends on surrounding block conte
|
| 79 | (blank) | line is blank; visual separation only. |
| 80 |         for i in 0..4 { | iteration control; changing bounds/body affects throughput, timing, and possibly saf
guarantees. |
| 81 |             // Wheel dynamics: brake slows wheel faster than vehicle | comment line; documentation intent, no
time effect. |
| 82 |             let decel = self.brake_pressures[i] * 12.0; | executable/support line; behavior impact depends on
rounding block context. |
| 83 |             self.wheel_velocities[i] = (self.wheel_velocities[i] - decel * dt).max(0.0); | executable/support
e; behavior impact depends on surrounding block context. |
| 84 |             // Follow vehicle speed when not braking | comment line; documentation intent, no runtime effect.
| 85 |             self.wheel_velocities[i] += | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 86 |                 ((vehicle_speed - self.wheel_velocities[i]) * 0.3 * dt).max(0.0); | executable/support line; b
vior impact depends on surrounding block context. |
| 87 | (blank) | line is blank; visual separation only. |
| 88 |             let slip = vehicle_speed - self.wheel_velocities[i]; | executable/support line; behavior impact de
ds on surrounding block context. |
| 89 |             if slip > 5.0 && user_brake > 0.3 && vehicle_speed > 5.0 { | conditional gate; changing condition
fts runtime path and side effects. |
| 90 |                 self.wheel_states[i] = WheelState::Skidding; | executable/support line; behavior impact depend
n surrounding block context. |
| 91 |                 self.abs_active = true; | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 92 |                 self.abs_cycles += 1; | executable/support line; behavior impact depends on surrounding block
text. |
| 93 |             } else if user_brake > 0.05 { | executable/support line; behavior impact depends on surrounding bl
context. |

```



```

| 94 |         self.wheel_states[i] = WheelState::AbsActive; | executable/support line; behavior impact depends
on surrounding block context. |
| 95 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 96 |         self.wheel_states[i] = WheelState::Rolling; | executable/support line; behavior impact depends
surrounding block context. |
| 97 |         self.wheel_velocities[i] = vehicle_speed; // sync | executable/support line; behavior impact d
nds on surrounding block context. |
| 98 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime sc
s. |
| 99 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 100 | (blank) | line is blank; visual separation only. |
| 101 |     // ABS modulates brake pressure at 8 Hz | comment line; documentation intent, no runtime effect. |
| 102 |     if self.abs_active { | conditional gate; changing condition shifts runtime path and side effects. |
| 103 |         for i in 0..4 { | iteration control; changing bounds/body affects throughput, timing, and possibl
afety guarantees. |
| 104 |             self.abs_cycle_phases[i] += dt * 8.0; | executable/support line; behavior impact depends on s
ounding block context. |
| 105 |             if self.abs_cycle_phases[i] > 1.0 { | conditional gate; changing condition shifts runtime pat
nd side effects. |
| 106 |                 self.abs_cycle_phases[i] -= 1.0; | executable/support line; behavior impact depends on su
unding block context. |
| 107 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 108 |             let modulated = if self.abs_cycle_phases[i] < 0.45 { | executable/support line; behavior impa
depends on surrounding block context. |
| 109 |                 user_brake * 0.28 | executable/support line; behavior impact depends on surrounding block
ntext. |
| 110 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 111 |                 user_brake * 0.92 | executable/support line; behavior impact depends on surrounding block
ntext. |
| 112 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 113 |             self.brake_pressures[i] = modulated; | executable/support line; behavior impact depends on su
unding block context. |
| 114 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 115 |         effective_brake = self.brake_pressures.iter().sum:::<f64>() / 4.0; | executable/support line; beha
r impact depends on surrounding block context. |
| 116 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 117 |         for p in &mut self.brake_pressures { | iteration control; changing bounds/body affects throughput
iming, and possibly safety guarantees. |
| 118 |             *p = user_brake; | executable/support line; behavior impact depends on surrounding block cont
. |
| 119 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 120 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 121 | (blank) | line is blank; visual separation only. |
| 122 |     // --- TCS ----- | comment line; documentation i
nt, no runtime effect. |
| 123 |     let mut effective_throttle = user_throttle; | executable/support line; behavior impact depends on sur
nding block context. |
| 124 |     self.tcs_active = false; | executable/support line; behavior impact depends on surrounding block cont
. |
| 125 |     self.tcs_throttle_cut = 0.0; | executable/support line; behavior impact depends on surrounding block
text. |
| 126 | (blank) | line is blank; visual separation only. |
| 127 |     if user_throttle > 0.2 && vehicle_speed < 60.0 { | conditional gate; changing condition shifts runtim
ath and side effects. |
| 128 |         // Detect wheel spin (driven wheels RL/RR faster than vehicle) | comment line; documentation inte
no runtime effect. |
| 129 |         let driven_avg = (self.wheel_velocities[2] + self.wheel_velocities[3]) / 2.0; | executable/suppor
ine; behavior impact depends on surrounding block context. |
| 130 |         if driven_avg > vehicle_speed + 6.0 { | conditional gate; changing condition shifts runtime path
side effects. |

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```

| 131 |         self.tcs_active = true; | executable/support line; behavior impact depends on surrounding block
context. |
| 132 |         self.tcs_throttle_cut = ((driven_avg - vehicle_speed - 6.0) / 10.0).min(0.8); | executable/su
rt line; behavior impact depends on surrounding block context. |
| 133 |         effective_throttle *= 1.0 - self.tcs_throttle_cut; | executable/support line; behavior impact
depends on surrounding block context. |
| 134 |         // Mark driven wheels | comment line; documentation intent, no runtime effect. |
| 135 |         self.wheel_states[2] = WheelState::TcsLimited; | executable/support line; behavior impact dep
s on surrounding block context. |
| 136 |         self.wheel_states[3] = WheelState::TcsLimited; | executable/support line; behavior impact dep
s on surrounding block context. |
| 137 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 138 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 139 | (blank) | line is blank; visual separation only. |
| 140 |     // --- ESP ----- | comment line; documentation intent, no runtime effect. |
| 141 |     self.esp_active = false; | executable/support line; behavior impact depends on surrounding block cont
. |
| 142 |     self.esp_brake_correction = [0.0; 4]; | executable/support line; behavior impact depends on surroundi
block context. |
| 143 | (blank) | line is blank; visual separation only. |
| 144 |     if vehicle_speed > 20.0 { | conditional gate; changing condition shifts runtime path and side effects
| 145 |         // Theoretical yaw rate for neutral steer | comment line; documentation intent, no runtime effect
| 146 |         let yaw_neutral = (vehicle_speed / 3.6) * (steering_angle.to_radians()) / WHEELBASE; | executable
pport line; behavior impact depends on surrounding block context. |
| 147 |         let yaw_error = yaw_rate.to_radians() - yaw_neutral; | executable/support line; behavior impact d
nds on surrounding block context. |
| 148 | (blank) | line is blank; visual separation only. |
| 149 |         if yaw_error.abs() > 0.08 { | conditional gate; changing condition shifts runtime path and side e
cts. |
| 150 |             self.esp_active = true; | executable/support line; behavior impact depends on surrounding blo
context. |
| 151 |             if yaw_error > 0.0 { | conditional gate; changing condition shifts runtime path and side effe
. |
| 152 |                 // Oversteer: apply outer front wheel brake | comment line; documentation intent, no runt
effect. |
| 153 |                 self.oversteer = true; | executable/support line; behavior impact depends on surrounding
ck context. |
| 154 |                 self.understeer = false; | executable/support line; behavior impact depends on surroundin
lock context. |
| 155 |                 self.esp_brake_correction[1] = (yaw_error * 3.0).min(0.6); // FR | executable/support lin
behavior impact depends on surrounding block context. |
| 156 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 157 |                 // Understeer: apply inner rear wheel brake | comment line; documentation intent, no runt
effect. |
| 158 |                 self.understeer = true; | executable/support line; behavior impact depends on surrounding
ock context. |
| 159 |                 self.oversteer = false; | executable/support line; behavior impact depends on surrounding
ock context. |
| 160 |                 self.esp_brake_correction[2] = ((-yaw_error) * 2.0).min(0.5); | executable/support line;
avior impact depends on surrounding block context. |
| 161 |                 // RL | comment line; documentation intent, no runtime effect. |
| 162 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 163 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 164 |             self.oversteer = false; | executable/support line; behavior impact depends on surrounding blo
context. |
| 165 |             self.understeer = false; | executable/support line; behavior impact depends on surrounding bl
context. |
| 166 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 167 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|

```

```
| 168 | (blank) | line is blank; visual separation only. |
| 169 |          // --- EPS ----- | comment line; documentation intent, no runtime effect. |
| 170 |          // Assist drops at high speed (less assist needed) | comment line; documentation intent, no runtime effect. |
| 171 |          let speed_factor = (1.0 - vehicle_speed / 200.0).clamp(0.2, 1.0); | executable/support line; behavior impact depends on surrounding block context. |
| 172 |          self.eps_assist_torque = match self.eps_mode { | executable/support line; behavior impact depends on surrounding block context. |
| 173 |              EpsMode::Comfort => steering_angle.abs() * 0.12 * speed_factor, | executable/support line; behavior impact depends on surrounding block context. |
| 174 |              EpsMode::Normal => steering_angle.abs() * 0.08 * speed_factor, | executable/support line; behavior impact depends on surrounding block context. |
| 175 |              EpsMode::Sport => steering_angle.abs() * 0.04 * speed_factor, | executable/support line; behavior impact depends on surrounding block context. |
| 176 |          }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 177 | (blank) | line is blank; visual separation only. |
| 178 |          (effective_throttle, effective_brake) | executable/support line; behavior impact depends on surrounding block context. |
| 179 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 180 | (blank) | line is blank; visual separation only. |
| 181 |      pub fn reset(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 182 |          *self = Self::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 183 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 184 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
```

File: src/ecm_heavy.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 376 | Non-empty: 341 | Symbols: 23

Symbol Inventory

Line	Kind	Symbol	Change Impact
7	const	IDLE_RPM	data/structure coupling
8	const	RATED_RPM	data/structure coupling
9	const	MAX_RPM	data/structure coupling
11	const	REDLINE_RPM	data/structure coupling
12	const	PEAK_TORQUE_NM	data/structure coupling
13	const	FUEL_TANK_L	data/structure coupling
14	const	NUM_CYLINDERS	data/structure coupling
18	enum	AftertreatmentState	data/structure coupling
25	impl	std::fmt::Display for AftertreatmentState	behavior binding and method semantics
26	fn	fmt	signature/API coupling
38	enum	GovernorMode	data/structure coupling
45	impl	std::fmt::Display for GovernorMode	behavior binding and method semantics
46	fn	fmt	signature/API coupling
58	struct	HeavyEcm	data/structure coupling
131	impl	Default for HeavyEcm	behavior binding and method semantics
132	fn	default	signature/API coupling
137	impl	HeavyEcm	behavior binding and method semantics
138	fn	new	signature/API coupling
193	fn	torque_curve	signature/API coupling
209	fn	update	signature/API coupling
292	fn	check_faults	signature/API coupling
353	fn	inject_dtc	signature/API coupling
370	fn	clear_dtcs	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
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```

| 1 | /// Heavy Machinery Engine Control Module (ECM) | comment line; documentation intent, no runtime effect. |
| 2 | /// Models a Tier 4 / Stage V diesel engine (6-cylinder, ~200 kW) | comment line; documentation intent, no runtime effect. |
| 3 | (blank) | line is blank; visual separation only. |
| 4 | use crate::j1939::{Dtc, DtcSeverity}; | imports symbols; changing path/name affects compile-time resolution. |
| 5 | (blank) | line is blank; visual separation only. |
| 6 | // Engine configuration constants | comment line; documentation intent, no runtime effect. |
| 7 | const IDLE_RPM: f64 = 800.0; | executable/support line; behavior impact depends on surrounding block context. |
| 8 | const RATED_RPM: f64 = 2200.0; | executable/support line; behavior impact depends on surrounding block context. |
| 9 | const MAX_RPM: f64 = 2600.0; | executable/support line; behavior impact depends on surrounding block context. |
| 10 | #[allow(dead_code)] | executable/support line; behavior impact depends on surrounding block context. |
| 11 | const REDLINE_RPM: f64 = 2400.0; | executable/support line; behavior impact depends on surrounding block context. |
| 12 | const PEAK_TORQUE_NM: f64 = 1000.0; // ~900 Nm at 1400 RPM | executable/support line; behavior impact depends on surrounding block context. |
| 13 | const FUEL_TANK_L: f64 = 200.0; | executable/support line; behavior impact depends on surrounding block context. |
| 14 | const NUM_CYLINDERS: u8 = 6; | executable/support line; behavior impact depends on surrounding block context. |
| 15 | (blank) | line is blank; visual separation only. |
| 16 | /// Aftertreatment / DEF / DPF state | comment line; documentation intent, no runtime effect. |
| 17 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding block context. |
| 18 | pub enum AftertreatmentState { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 19 |     Normal, | executable/support line; behavior impact depends on surrounding block context. |
| 20 |     Regenerating, | executable/support line; behavior impact depends on surrounding block context. |
| 21 |     HighSoot, | executable/support line; behavior impact depends on surrounding block context. |
| 22 |     Fault, | executable/support line; behavior impact depends on surrounding block context. |
| 23 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 24 | (blank) | line is blank; visual separation only. |
| 25 | impl std::fmt::Display for AftertreatmentState { | implementation block; method behavior and trait conformance are defined here. |
| 26 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 27 |         match self { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 28 |             AftertreatmentState::Normal => write!(f, "NORMAL"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 29 |             AftertreatmentState::Regenerating => write!(f, "REGEN"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 30 |             AftertreatmentState::HighSoot => write!(f, "HIGH SOOT"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 31 |             AftertreatmentState::Fault => write!(f, "FAULT"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 32 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 33 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 34 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 35 | (blank) | line is blank; visual separation only. |
| 36 | /// Governor mode | comment line; documentation intent, no runtime effect. |
| 37 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding block context. |
| 38 | pub enum GovernorMode { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 39 |     Low, | executable/support line; behavior impact depends on surrounding block context. |
| 40 |     High, | executable/support line; behavior impact depends on surrounding block context. |
| 41 |     Droop, | executable/support line; behavior impact depends on surrounding block context. |
| 42 |     All, | executable/support line; behavior impact depends on surrounding block context. |
| 43 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 44 | (blank) | line is blank; visual separation only. |
| 45 | impl std::fmt::Display for GovernorMode { | implementation block; method behavior and trait conformance are defined here. |
| 46 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 47 |         match self { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 48 |             GovernorMode::Low => write!(f, "LOW IDLE"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

```

```
| 49 | GovernorMode::High => write!(f, "HIGH "), | I/O or protocol-critical logic; edits can affect in-  
ration, timing, and diagnostic outcomes. |  
| 50 | GovernorMode::Droop => write!(f, "DROOP "), | I/O or protocol-critical logic; edits can affect in-  
gration, timing, and diagnostic outcomes. |  
| 51 | GovernorMode::All => write!(f, "ALL SPD "), | I/O or protocol-critical logic; edits can affect in-  
ation, timing, and diagnostic outcomes. |  
| 52 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes  
| 53 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 54 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 55 | (blank) | line is blank; visual separation only. |  
| 56 | /// Comprehensive Heavy Machinery ECM | comment line; documentation intent, no runtime effect. |  
| 57 | #[derive(Clone)] | executable/support line; behavior impact depends on surrounding block context. |  
| 58 | pub struct HeavyEcm { | declares data layout; field changes may break serialization, tests, and callers. |  
| 59 | // ■ Engine Speed ████████████████████████████████████████████████████████████ | comment line; documentation inter-  
effect. |  
| 60 | pub rpm: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 61 | pub rated_rpm: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 62 | pub idle_rpm: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 63 | pub governor_mode: GovernorMode, | executable/support line; behavior impact depends on surrounding block c-  
ext. |  
| 64 | (blank) | line is blank; visual separation only. |  
| 65 | // ■ Torque & Load ████████████████████████████████████████████████████████████ | comment line; documentation inten-  
effect. |  
| 66 | pub torque_nm: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 67 | pub percent_load: f64, // 0-100% | executable/support line; behavior impact depends on surrounding block c-  
ext. |  
| 68 | pub demand_torque_pct: f64, | executable/support line; behavior impact depends on surrounding block contex-  
|  
| 69 | pub actual_torque_pct: f64, | executable/support line; behavior impact depends on surrounding block contex-  
|  
| 70 | pub max_torque_nm: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 71 | (blank) | line is blank; visual separation only. |  
| 72 | // ■ Throttle ████████████████████████████████████████████████████████████ | comment line; documentation inte-  
effect. |  
| 73 | pub throttle_pct: f64, // 0-100% | executable/support line; behavior impact depends on surrounding block c-  
ext. |  
| 74 | pub remote_throttle: bool, | executable/support line; behavior impact depends on surrounding block context  
| 75 | pub engine_brake: bool, | executable/support line; behavior impact depends on surrounding block context. |  
| 76 | (blank) | line is blank; visual separation only. |  
| 77 | // ■ Fuel System ████████████████████████████████████████████████████████████ | comment line; documentation inter-  
effect. |  
| 78 | pub fuel_level_pct: f64, | executable/support line; behavior impact depends on surrounding block context.  
| 79 | pub fuel_pressure_kpa: f64, | executable/support line; behavior impact depends on surrounding block contex-  
|  
| 80 | pub fuel_rate_lph: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 81 | pub total_fuel_l: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 82 | pub trip_fuel_l: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 83 | (blank) | line is blank; visual separation only. |  
| 84 | // ■ Temperatures ████████████████████████████████████████████████████████████ | comment line; documentation inter-  
effect. |  
| 85 | pub coolant_temp_c: f64, | executable/support line; behavior impact depends on surrounding block context.  
| 86 | pub oil_temp_c: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 87 | pub fuel_temp_c: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 88 | pub intake_temp_c: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 89 | pub exhaust_temp_c: f64, | executable/support line; behavior impact depends on surrounding block context.  
| 90 | pub turbo_oil_temp_c: f64, | executable/support line; behavior impact depends on surrounding block context  
| 91 | (blank) | line is blank; visual separation only. |  
| 92 | // ■ Pressures ████████████████████████████████████████████████████████████ | comment line; documentation inte-  
effect. |  
| 93 | pub oil_pressure_kpa: f64, | executable/support line; behavior impact depends on surrounding block context  
| 94 | pub oil_level_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 95 | pub coolant_pres_kpa: f64, | executable/support line; behavior impact depends on surrounding block context  
| 96 | pub boost_pressure_kpa: f64, | executable/support line; behavior impact depends on surrounding block conte-  
|  
| 97 | pub intake_manifold_kpa: f64, | executable/support line; behavior impact depends on surrounding block con-
```

[illegible]

```
| 143 | governor_mode: GovernorMode::All, | executable/support line; behavior impact depends on surrounding
block context. |
| 144 | torque_nm: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 145 | percent_load: 0.0, | executable/support line; behavior impact depends on surrounding block context.
|
| 146 | demand_torque_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context.
ext. |
| 147 | actual_torque_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context.
ext. |
| 148 | max_torque_nm: PEAK_TORQUE_NM, | executable/support line; behavior impact depends on surrounding block context.
ck context. |
| 149 | throttle_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context.
|
| 150 | remote_throttle: false, | executable/support line; behavior impact depends on surrounding block context.
ext. |
| 151 | engine_brake: false, | executable/support line; behavior impact depends on surrounding block context.
. |
| 152 | fuel_level_pct: 85.0, | executable/support line; behavior impact depends on surrounding block context.
t. |
| 153 | fuel_pressure_kpa: 400.0, | executable/support line; behavior impact depends on surrounding block context.
ntext. |
| 154 | fuel_rate_lph: 2.0, | executable/support line; behavior impact depends on surrounding block context.
|
| 155 | total_fuel_l: 0.0, | executable/support line; behavior impact depends on surrounding block context.
|
| 156 | trip_fuel_l: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 157 | coolant_temp_c: 20.0, | executable/support line; behavior impact depends on surrounding block context.
t. |
| 158 | oil_temp_c: 20.0, | executable/support line; behavior impact depends on surrounding block context.
| 159 | fuel_temp_c: 25.0, | executable/support line; behavior impact depends on surrounding block context.
|
| 160 | intake_temp_c: 25.0, | executable/support line; behavior impact depends on surrounding block context.
. |
| 161 | exhaust_temp_c: 200.0, | executable/support line; behavior impact depends on surrounding block context.
xt. |
| 162 | turbo_oil_temp_c: 20.0, | executable/support line; behavior impact depends on surrounding block context.
ext. |
| 163 | oil_pressure_kpa: 350.0, | executable/support line; behavior impact depends on surrounding block context.
text. |
| 164 | oil_level_pct: 95.0, | executable/support line; behavior impact depends on surrounding block context.
. |
| 165 | coolant_pres_kpa: 50.0, | executable/support line; behavior impact depends on surrounding block context.
ext. |
| 166 | boost_pressure_kpa: 100.0, | executable/support line; behavior impact depends on surrounding block context.
ontext. |
| 167 | intake_manifold_kpa: 100.0, | executable/support line; behavior impact depends on surrounding block context.
context. |
| 168 | air_filter_dp_kpa: 1.5, | executable/support line; behavior impact depends on surrounding block context.
ext. |
| 169 | battery_v: 12.8, | executable/support line; behavior impact depends on surrounding block context.
| 170 | alternator_v: 14.2, | executable/support line; behavior impact depends on surrounding block context.
|
| 171 | starter_active: false, | executable/support line; behavior impact depends on surrounding block context.
xt. |
| 172 | def_level_pct: 72.0, | executable/support line; behavior impact depends on surrounding block context.
. |
| 173 | def_quality_pct: 98.5, | executable/support line; behavior impact depends on surrounding block context.
xt. |
| 174 | dpf_soot_pct: 15.0, | executable/support line; behavior impact depends on surrounding block context.
|
| 175 | dpf_temp_c: 250.0, | executable/support line; behavior impact depends on surrounding block context.
|
| 176 | scr_efficiency_pct: 96.0, | executable/support line; behavior impact depends on surrounding block context.
ntext. |
| 177 | nox_ppm: 45.0, | executable/support line; behavior impact depends on surrounding block context. |
```

```

| 178 |         aftertreatment: AftertreatmentState::Normal, | executable/support line; behavior impact depends on
surrounding block context. |
| 179 |         engine_hours: 2347.5, | executable/support line; behavior impact depends on surrounding block con
t. |
| 180 |         service_hours_left: 152.5, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 181 |         num_cylinders: NUM_CYLINDERS, | executable/support line; behavior impact depends on surrounding b
k context. |
| 182 |         active_dtcs: Vec::new(), | executable/support line; behavior impact depends on surrounding block
text. |
| 183 |         stored_dtcs: Vec::new(), | executable/support line; behavior impact depends on surrounding block
text. |
| 184 |         mil_active: false, | executable/support line; behavior impact depends on surrounding block contex
|
| 185 |         amber_lamp: false, | executable/support line; behavior impact depends on surrounding block contex
|
| 186 |         red_lamp: false, | executable/support line; behavior impact depends on surrounding block context.
| 187 |         rpm_integral: 0.0, | executable/support line; behavior impact depends on surrounding block contex
|
| 188 |         warmup_complete: false, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 189 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 190 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 191 | (blank) | line is blank; visual separation only. |
| 192 |     /// Diesel torque curve: peak ~1000 Nm at 1200-1600 RPM | comment line; documentation intent, no runtime
ect. |
| 193 |     pub fn torque_curve(&self, rpm: f64, throttle_pct: f64) -> f64 { | declares executable behavior; signatur
ody changes alter API, control flow, and outputs. |
| 194 |         let t = throttle_pct / 100.0; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 195 |         let peak_rpm = 1400.0; | executable/support line; behavior impact depends on surrounding block contex
|
| 196 |         let _norm = (rpm - peak_rpm) / 800.0; | executable/support line; behavior impact depends on surroundi
block context. |
| 197 |         let curve = if rpm < 800.0 { | executable/support line; behavior impact depends on surrounding block
text. |
| 198 |             0.0 | executable/support line; behavior impact depends on surrounding block context. |
| 199 |         } else if rpm < peak_rpm { | executable/support line; behavior impact depends on surrounding block co
xt. |
| 200 |             0.7 + 0.3 * (rpm - 800.0) / (peak_rpm - 800.0) | executable/support line; behavior impact depends
surrounding block context. |
| 201 |         } else if rpm < RATED_RPM { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 202 |             1.0 - 0.25 * (rpm - peak_rpm) / (RATED_RPM - peak_rpm) | executable/support line; behavior impact
ends on surrounding block context. |
| 203 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 204 |             (0.75 - (rpm - RATED_RPM) / (MAX_RPM - RATED_RPM) * 0.75).max(0.0) | executable/support line; beh
or impact depends on surrounding block context. |
| 205 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 206 |         PEAK_TORQUE_NM * curve * t | executable/support line; behavior impact depends on surrounding block co
xt. |
| 207 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 208 | (blank) | line is blank; visual separation only. |
| 209 |     pub fn update(&mut self, throttle_pct: f64, load_nm: f64, dt: f64) { | declares executable behavior; sign
re/body changes alter API, control flow, and outputs. |
| 210 |         self.throttle_pct = throttle_pct.clamp(0.0, 100.0); | executable/support line; behavior impact depend
n surrounding block context. |
| 211 |         let t = self.throttle_pct / 100.0; | executable/support line; behavior impact depends on surrounding
ck context. |
| 212 | (blank) | line is blank; visual separation only. |
| 213 |         // RPM dynamics: governor tries to maintain rated RPM under load | comment line; documentation intent
o runtime effect. |
| 214 |         let target_rpm = self.idle_rpm + t * (RATED_RPM - self.idle_rpm); | executable/support line; behavior

```



```

pact depends on surrounding block context. |
| 215 |         let available_torque = self.torque_curve(self.rpm, self.throttle_pct); | executable/support line; beh
or impact depends on surrounding block context. |
| 216 |         let net_torque = available_torque - load_nm; | executable/support line; behavior impact depends on su
unding block context. |
| 217 |         // d(rpm)/dt based on engine inertia | comment line; documentation intent, no runtime effect. |
| 218 |         let rpm_accel = net_torque * 30.0 / (std::f64::consts::PI * 2.5); // J=2.5 kg.m² | executable/support
ne; behavior impact depends on surrounding block context. |
| 219 |         self.rpm_integral += rpm_accel * dt; | executable/support line; behavior impact depends on surroundin
lock context. |
| 220 |         self.rpm += (target_rpm - self.rpm) * dt * 1.5 + rpm_accel * dt * 0.1; | executable/support line; beh
or impact depends on surrounding block context. |
| 221 |         self.rpm = self.rpm.clamp(self.idle_rpm * 0.8, MAX_RPM); | executable/support line; behavior impact d
nds on surrounding block context. |
| 222 | (blank) | line is blank; visual separation only. |
| 223 |         // Torque & load | comment line; documentation intent, no runtime effect. |
| 224 |         self.torque_nm = available_torque.min(load_nm + 50.0); | executable/support line; behavior impact dep
s on surrounding block context. |
| 225 |         self.actual_torque_pct = (self.torque_nm / PEAK_TORQUE_NM * 100.0).clamp(-100.0, 100.0); | executable
pport line; behavior impact depends on surrounding block context. |
| 226 |         self.demand_torque_pct = (t * 100.0 - 125.0 + 125.0).clamp(-100.0, 100.0); | executable/support line;
havior impact depends on surrounding block context. |
| 227 |         self.percent_load = (load_nm / PEAK_TORQUE_NM * 100.0).clamp(0.0, 150.0); | executable/support line;
avior impact depends on surrounding block context. |
| 228 | (blank) | line is blank; visual separation only. |
| 229 |         // Fuel consumption (diesel: ~0.22 kg/kwh specific consumption) | comment line; documentation intent,
runtime effect. |
| 230 |         let power_kw = self.torque_nm * self.rpm * std::f64::consts::PI / 30000.0; | executable/support line;
havior impact depends on surrounding block context. |
| 231 |         self.fuel_rate_lph = (power_kw * 0.22 / 0.85 + 1.5).max(1.5); // 0.85 kg/L diesel | executable/suppor
ine; behavior impact depends on surrounding block context. |
| 232 |         let consumed = self.fuel_rate_lph * dt / 3600.0; | executable/support line; behavior impact depends o
urrounding block context. |
| 233 |         self.total_fuel_l += consumed; | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 234 |         self.trip_fuel_l += consumed; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 235 |         self.fuel_level_pct = (self.fuel_level_pct - consumed / FUEL_TANK_L * 100.0).max(0.0); | executable/s
ort line; behavior impact depends on surrounding block context. |
| 236 | (blank) | line is blank; visual separation only. |
| 237 |         // Thermal model | comment line; documentation intent, no runtime effect. |
| 238 |         if !self.warmup_complete { | conditional gate; changing condition shifts runtime path and side effect
|
| 239 |             self.warmup_complete = self.coolant_temp_c > 80.0; | executable/support line; behavior impact dep
s on surrounding block context. |
| 240 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 241 |         let tgt_coolant = if self.warmup_complete { | executable/support line; behavior impact depends on sur
nding block context. |
| 242 |             90.0 + t * 5.0 | executable/support line; behavior impact depends on surrounding block context. |
| 243 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 244 |             self.coolant_temp_c + 3.0 * dt | executable/support line; behavior impact depends on surrounding
ck context. |
| 245 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 246 |         self.coolant_temp_c += (tgt_coolant - self.coolant_temp_c) * dt * 0.02; | executable/support line; be
ior impact depends on surrounding block context. |
| 247 |         self.oil_temp_c = self.coolant_temp_c + 12.0; | executable/support line; behavior impact depends on s
ounding block context. |
| 248 |         self.exhaust_temp_c = 200.0 + t * 450.0 * (self.rpm / RATED_RPM).min(1.0); | executable/support line;
havior impact depends on surrounding block context. |
| 249 |         self.intake_temp_c = 25.0 + self.boost_pressure_kpa / 100.0 * 15.0; | executable/support line; behavi
impact depends on surrounding block context. |
| 250 | (blank) | line is blank; visual separation only. |
| 251 |         // Pressures | comment line; documentation intent, no runtime effect. |

```

```

| 252 |         let rpm_factor = (self.rpm / RATED_RPM).clamp(0.2, 1.2); | executable/support line; behavior impact d
nds on surrounding block context. |
| 253 |         self.oil_pressure_kpa = 250.0 + rpm_factor * 200.0; | executable/support line; behavior impact depend
n surrounding block context. |
| 254 |         self.boost_pressure_kpa = 100.0 + t * rpm_factor * 200.0; | executable/support line; behavior impact
ends on surrounding block context. |
| 255 |         self.intake_manifold_kpa = 100.0 + self.boost_pressure_kpa * 0.8; | executable/support line; behavior
pact depends on surrounding block context. |
| 256 |         self.fuel_pressure_kpa = 350.0 + t * 100.0; | executable/support line; behavior impact depends on surr
nding block context. |
| 257 | (blank) | line is blank; visual separation only. |
| 258 |         // Aftertreatment | comment line; documentation intent, no runtime effect. |
| 259 |         self.dpf_soot_pct = (self.dpf_soot_pct + t * 0.0005 * dt - 0.0001 * dt).clamp(0.0, 100.0); | executab
support line; behavior impact depends on surrounding block context. |
| 260 |         self.dpf_temp_c = 250.0 + t * 300.0; | executable/support line; behavior impact depends on surroundin
lock context. |
| 261 |         self.def_level_pct = | executable/support line; behavior impact depends on surrounding block context.
| 262 |         (self.def_level_pct - self.fuel_rate_lph * 0.05 * dt / 3600.0 * 100.0).max(0.0); | executable/sup
t line; behavior impact depends on surrounding block context. |
| 263 |         self.scr_efficiency_pct = if self.def_level_pct > 5.0 { | executable/support line; behavior impact de
ds on surrounding block context. |
| 264 |             94.0 + t * 3.0 | executable/support line; behavior impact depends on surrounding block context. |
| 265 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 266 |             20.0 | executable/support line; behavior impact depends on surrounding block context. |
| 267 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 268 |         self.nox_ppm = (1500.0 * t / (self.scr_efficiency_pct / 100.0 + 0.01)).min(2000.0); | executable/supp
line; behavior impact depends on surrounding block context. |
| 269 | (blank) | line is blank; visual separation only. |
| 270 |         if self.dpf_soot_pct > 60.0 && self.aftertreatment == AftertreatmentState::Normal { | conditional gat
changing condition shifts runtime path and side effects. |
| 271 |             self.aftertreatment = AftertreatmentState::HighSoot; | executable/support line; behavior impact d
nds on surrounding block context. |
| 272 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 273 |         if self.dpf_soot_pct > 80.0 { | conditional gate; changing condition shifts runtime path and side eff
s. |
| 274 |             self.aftertreatment = AftertreatmentState::Regenerating; | executable/support line; behavior impa
depends on surrounding block context. |
| 275 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 276 |         if self.dpf_soot_pct < 10.0 && self.aftertreatment == AftertreatmentState::Regenerating { | condition
gate; changing condition shifts runtime path and side effects. |
| 277 |             self.aftertreatment = AftertreatmentState::Normal; | executable/support line; behavior impact dep
s on surrounding block context. |
| 278 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 279 | (blank) | line is blank; visual separation only. |
| 280 |         // Electrical | comment line; documentation intent, no runtime effect. |
| 281 |         self.alternator_v = if self.rpm > 1000.0 { 14.2 } else { 12.4 }; | executable/support line; behavior
act depends on surrounding block context. |
| 282 |         self.battery_v = if self.rpm > 1000.0 { 13.8 } else { 12.6 }; | executable/support line; behavior imp
depends on surrounding block context. |
| 283 | (blank) | line is blank; visual separation only. |
| 284 |         // Service hours | comment line; documentation intent, no runtime effect. |
| 285 |         self.engine_hours += dt / 3600.0; | executable/support line; behavior impact depends on surrounding b
k context. |
| 286 |         self.service_hours_left -= dt / 3600.0; | executable/support line; behavior impact depends on surroun
g block context. |
| 287 | (blank) | line is blank; visual separation only. |
| 288 |         // Fault detection | comment line; documentation intent, no runtime effect. |
| 289 |         self.check_faults(); | executable/support line; behavior impact depends on surrounding block context.
| 290 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 291 | (blank) | line is blank; visual separation only. |
| 292 |     fn check_faults(&mut self) { | declares executable behavior; signature/body changes alter API, control fl

```

```

and outputs. |
| 293 |         self.active_dtcs.retain(\\d\\ d.active); | executable/support line; behavior impact depends on surround
ng block context. |
| 294 |         self.mil_active = false; | executable/support line; behavior impact depends on surrounding block cont
. |
| 295 |         self.amber_lamp = false; | executable/support line; behavior impact depends on surrounding block cont
. |
| 296 |         self.red_lamp = false; | executable/support line; behavior impact depends on surrounding block context
|
| 297 | (blank) | line is blank; visual separation only. |
| 298 |         // High coolant temp → Red Stop | comment line; documentation intent, no runtime effect. |
| 299 |         if self.coolant_temp_c > 108.0 { | conditional gate; changing condition shifts runtime path and side
ects. |
| 300 |             self.inject_dtc( | executable/support line; behavior impact depends on surrounding block context.
| 301 |                 110, | executable/support line; behavior impact depends on surrounding block context. |
| 302 |                 0, | executable/support line; behavior impact depends on surrounding block context. |
| 303 |                 DtcSeverity::Red, | executable/support line; behavior impact depends on surrounding block con
t. |
| 304 |                 "Engine Coolant Temp High – Red Stop", | executable/support line; behavior impact depends on
rounding block context. |
| 305 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 306 |             self.red_lamp = true; | executable/support line; behavior impact depends on surrounding block con
t. |
| 307 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 308 |         // Low oil pressure → Red Stop | comment line; documentation intent, no runtime effect. |
| 309 |         if self.oil_pressure_kpa < 80.0 { | conditional gate; changing condition shifts runtime path and side
fects. |
| 310 |             self.inject_dtc( | executable/support line; behavior impact depends on surrounding block context.
| 311 |                 100, | executable/support line; behavior impact depends on surrounding block context. |
| 312 |                 1, | executable/support line; behavior impact depends on surrounding block context. |
| 313 |                 DtcSeverity::Red, | executable/support line; behavior impact depends on surrounding block con
t. |
| 314 |                 "Engine Oil Pressure Low – Red Stop", | executable/support line; behavior impact depends on s
ounding block context. |
| 315 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 316 |             self.red_lamp = true; | executable/support line; behavior impact depends on surrounding block con
t. |
| 317 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 318 |         // Low DEF → Amber | comment line; documentation intent, no runtime effect. |
| 319 |         if self.def_level_pct < 10.0 { | conditional gate; changing condition shifts runtime path and side ef
ts. |
| 320 |             self.inject_dtc( | executable/support line; behavior impact depends on surrounding block context.
| 321 |                 3361, | executable/support line; behavior impact depends on surrounding block context. |
| 322 |                 17, | executable/support line; behavior impact depends on surrounding block context. |
| 323 |                 DtcSeverity::Amber, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 324 |                 "DEF Level Low – SCR Derate Imminent", | executable/support line; behavior impact depends on
rounding block context. |
| 325 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 326 |             self.amber_lamp = true; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 327 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 328 |         // High soot → Amber | comment line; documentation intent, no runtime effect. |
| 329 |         if self.dpf_soot_pct > 70.0 { | conditional gate; changing condition shifts runtime path and side eff
s. |
| 330 |             self.inject_dtc( | executable/support line; behavior impact depends on surrounding block context.
| 331 |                 3251, | executable/support line; behavior impact depends on surrounding block context. |
| 332 |                 15, | executable/support line; behavior impact depends on surrounding block context. |
| 333 |                 DtcSeverity::Amber, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 334 |                 "DPF Soot Load High – Regen Required", | executable/support line; behavior impact depends on
rounding block context. |

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```

| 335 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 336 |         self.amber_lamp = true; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 337 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 338 |     // Service overdue | comment line; documentation intent, no runtime effect. |
| 339 |     if self.service_hours_left < 0.0 { | conditional gate; changing condition shifts runtime path and side
ffects. |
| 340 |         self.inject_dtc(247, 31, DtcSeverity::Mil, "Service Interval Overdue"); | executable/support line
ehavior impact depends on surrounding block context. |
| 341 |         self.mil_active = true; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 342 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 343 |     for dtc in &self.active_dtcs { | iteration control; changing bounds/body affects throughput, timing,
possibly safety guarantees. |
| 344 |         match dtc.severity { | branch dispatch; changing arms alters behavior for variants and input clas
. |
| 345 |             DtcSeverity::Red => self.red_lamp = true, | executable/support line; behavior impact depends
surrounding block context. |
| 346 |             DtcSeverity::Amber => self.amber_lamp = true, | executable/support line; behavior impact depe
on surrounding block context. |
| 347 |             DtcSeverity::Mil => self.mil_active = true, | executable/support line; behavior impact depend
n surrounding block context. |
| 348 |             _ => {} | executable/support line; behavior impact depends on surrounding block context. |
| 349 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 350 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 351 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 352 | (blank) | line is blank; visual separation only. |
| 353 | fn inject_dtc(&mut self, spn: u32, fmi: u8, sev: DtcSeverity, desc: &'static str) { | declares executable
havior; signature/body changes alter API, control flow, and outputs. |
| 354 |     if !self | conditional gate; changing condition shifts runtime path and side effects. |
| 355 |         .active_dtcs | executable/support line; behavior impact depends on surrounding block context. |
| 356 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 357 |         .any(|d| d.spn == spn && d.fmi == fmi) | executable/support line; behavior impact depends on su
unding block context. |
| 358 |     { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 359 |         self.active_dtcs.push(Dtc { | executable/support line; behavior impact depends on surrounding blo
context. |
| 360 |             spn, | executable/support line; behavior impact depends on surrounding block context. |
| 361 |             fmi, | executable/support line; behavior impact depends on surrounding block context. |
| 362 |             count: 1, | executable/support line; behavior impact depends on surrounding block context. |
| 363 |             active: true, | executable/support line; behavior impact depends on surrounding block context
|
| 364 |             desc, | executable/support line; behavior impact depends on surrounding block context. |
| 365 |             severity: sev, | executable/support line; behavior impact depends on surrounding block contex
|
| 366 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 367 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 368 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 369 | (blank) | line is blank; visual separation only. |
| 370 | pub fn clear_dtcs(&mut self) { | declares executable behavior; signature/body changes alter API, control
w, and outputs. |
| 371 |     for dtc in &mut self.active_dtcs { | iteration control; changing bounds/body affects throughput, timi
and possibly safety guarantees. |
| 372 |         dtc.active = false; | executable/support line; behavior impact depends on surrounding block conte
|
| 373 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 374 |     self.active_dtcs.retain(|d| d.active); | executable/support line; behavior impact depends on surroun
ng block context. |
| 375 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

| 376 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

File: src/ecm_mock_provider.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 198 | Non-empty: 189 | Symbols: 4

Symbol Inventory

Line	Kind	Symbol	Change Impact
---:	---	---	---
6	struct	MockParam	data/structure coupling
13	struct	MockFunction	data/structure coupling
19	struct	MockEcmNode	data/structure coupling
25	fn	build_mock_ecm_tree	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
---:	---	---
1	use auto_breaking::HeavyMachinery;	imports symbols; changing path/name affects compile-time resolution.
2	(blank)	line is blank; visual separation only.
3	use crate::io::ecm_params::EcmSnapshot;	imports symbols; changing path/name affects compile-time resolution.
4	(blank)	line is blank; visual separation only.
5	#[derive(Clone, Debug)]	executable/support line; behavior impact depends on surrounding block context.
6	pub struct MockParam {	declares data layout; field changes may break serialization, tests, and callers.
7	pub key: &'static str,	executable/support line; behavior impact depends on surrounding block context.
8	pub value: f64,	executable/support line; behavior impact depends on surrounding block context.
9	pub unit: &'static str,	executable/support line; behavior impact depends on surrounding block context.
10	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
11	(blank)	line is blank; visual separation only.
12	#[derive(Clone, Debug)]	executable/support line; behavior impact depends on surrounding block context.
13	pub struct MockFunction {	declares data layout; field changes may break serialization, tests, and callers.
14	pub name: &'static str,	executable/support line; behavior impact depends on surrounding block context.
15	pub params: Vec<MockParam>,	executable/support line; behavior impact depends on surrounding block context.
16	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
17	(blank)	line is blank; visual separation only.
18	#[derive(Clone, Debug)]	executable/support line; behavior impact depends on surrounding block context.
19	pub struct MockEcmNode {	declares data layout; field changes may break serialization, tests, and callers.
20	pub name: &'static str,	executable/support line; behavior impact depends on surrounding block context.
21	pub source_address: u8,	executable/support line; behavior impact depends on surrounding block context.
22	pub functions: Vec<MockFunction>,	executable/support line; behavior impact depends on surrounding block context.
23	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
24	(blank)	line is blank; visual separation only.
25	pub fn build_mock_ecm_tree(bench: &HeavyMachinery, snap: &EcmSnapshot) -> Vec<MockEcmNode> {	declares executable behavior; signature/body changes alter API, control flow, and outputs.
26	let e = &bench.ecm;	executable/support line; behavior impact depends on surrounding block context.
27	let h = &bench.hcm;	executable/support line; behavior impact depends on surrounding block context.
28	let t = &bench.tcm;	executable/support line; behavior impact depends on surrounding block context.
29	(blank)	line is blank; visual separation only.
30	let ecm = MockEcmNode {	executable/support line; behavior impact depends on surrounding block context.
31	name: "Engine ECM",	executable/support line; behavior impact depends on surrounding block context.
32	source_address: snap.source_address.unwrap_or(0x00),	executable/support line; behavior impact depends on surrounding block context.
33	functions: vec![	executable/support line; behavior impact depends on surrounding block context.
34	MockFunction {	executable/support line; behavior impact depends on surrounding block context.
35	name: "Engine Performance",	executable/support line; behavior impact depends on surrounding block context.
36	params: vec![	executable/support line; behavior impact depends on surrounding block context.
37	MockParam {	executable/support line; behavior impact depends on surrounding block context.
38	key: "engine_speed_rpm",	executable/support line; behavior impact depends on surrounding block context.

```

| 39 |             value: snap.engine_speed_rpm.unwrap_or(e.rpm), | executable/support line; behavior imp
depends on surrounding block context. |
| 40 |             unit: "rpm", | executable/support line; behavior impact depends on surrounding block c
ext. |
| 41 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 42 |         MockParam { | executable/support line; behavior impact depends on surrounding block contex
|
| 43 |             key: "accel_pedal_pct", | executable/support line; behavior impact depends on surround
block context. |
| 44 |             value: snap.accel_pedal_pct.unwrap_or(e.active_throttle), | executable/support line; b
vior impact depends on surrounding block context. |
| 45 |             unit: "%", | executable/support line; behavior impact depends on surrounding block con
t. |
| 46 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 47 |         MockParam { | executable/support line; behavior impact depends on surrounding block contex
|
| 48 |             key: "actual_torque_nm", | executable/support line; behavior impact depends on surround
g block context. |
| 49 |             value: e.actual_torque_nm, | executable/support line; behavior impact depends on surro
ing block context. |
| 50 |             unit: "Nm", | executable/support line; behavior impact depends on surrounding block co
xt. |
| 51 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 52 |         MockParam { | executable/support line; behavior impact depends on surrounding block contex
|
| 53 |             key: "vehicle_speed", | executable/support line; behavior impact depends on surroundin
lock context. |
| 54 |             value: t.ground_speed_kmh, | executable/support line; behavior impact depends on surro
ing block context. |
| 55 |             unit: "km/h", | executable/support line; behavior impact depends on surrounding block
text. |
| 56 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 57 |     ], | executable/support line; behavior impact depends on surrounding block context. |
| 58 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 59 | MockFunction { | executable/support line; behavior impact depends on surrounding block context. |
| 60 |     name: "Pressures and Temperatures", | executable/support line; behavior impact depends on surr
ding block context. |
| 61 |     params: vec![ | executable/support line; behavior impact depends on surrounding block context.
| 62 |         MockParam { | executable/support line; behavior impact depends on surrounding block contex
|
| 63 |             key: "coolant_temp_c", | executable/support line; behavior impact depends on surroundi
block context. |
| 64 |             value: snap.coolant_temp_c.unwrap_or(e.coolant_temp_c), | executable/support line; beh
or impact depends on surrounding block context. |
| 65 |             unit: "C", | executable/support line; behavior impact depends on surrounding block con
t. |
| 66 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 67 |         MockParam { | executable/support line; behavior impact depends on surrounding block contex
|
| 68 |             key: "fuel_temp_c", | executable/support line; behavior impact depends on surrounding
ck context. |
| 69 |             value: snap.fuel_temp_c.unwrap_or(e.fuel_temp_c), | executable/support line; behavior
act depends on surrounding block context. |
| 70 |             unit: "C", | executable/support line; behavior impact depends on surrounding block con
t. |
| 71 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 72 |         MockParam { | executable/support line; behavior impact depends on surrounding block contex
|
| 73 |             key: "oil_pressure_kpa", | executable/support line; behavior impact depends on surround
g block context. |
| 74 |             value: snap.oil_pressure_kpa.unwrap_or(e.oil_pressure_kpa), | executable/support line;
havior impact depends on surrounding block context. |
| 75 |             unit: "kPa", | executable/support line; behavior impact depends on surrounding block c
ext. |
| 76 |         }, | executable/support line; behavior impact depends on surrounding block context. |

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```

| 77 |                MockParam { | executable/support line; behavior impact depends on surrounding block context. |
| 78 |                    key: "boost_pressure_kpa", | executable/support line; behavior impact depends on surrounding block context. |
| 79 |                    value: e.boost_pressure_kpa, | executable/support line; behavior impact depends on surrounding block context. |
| 80 |                    unit: "kPa", | executable/support line; behavior impact depends on surrounding block context. |
| 81 |                }, | executable/support line; behavior impact depends on surrounding block context. |
| 82 |            ], | executable/support line; behavior impact depends on surrounding block context. |
| 83 |        }, | executable/support line; behavior impact depends on surrounding block context. |
| 84 |        MockFunction { | executable/support line; behavior impact depends on surrounding block context. |
| 85 |            name: "Aftertreatment", | executable/support line; behavior impact depends on surrounding block context. |
| 86 |            params: vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 87 |                MockParam { | executable/support line; behavior impact depends on surrounding block context. |
| 88 |                    key: "def_level_pct", | executable/support line; behavior impact depends on surrounding block context. |
| 89 |                    value: e.def_level_pct, | executable/support line; behavior impact depends on surrounding block context. |
| 90 |                    unit: "%", | executable/support line; behavior impact depends on surrounding block context. |
| 91 |                }, | executable/support line; behavior impact depends on surrounding block context. |
| 92 |                MockParam { | executable/support line; behavior impact depends on surrounding block context. |
| 93 |                    key: "dpf_soot_pct", | executable/support line; behavior impact depends on surrounding block context. |
| 94 |                    value: e.dpf_soot_pct, | executable/support line; behavior impact depends on surrounding block context. |
| 95 |                    unit: "%", | executable/support line; behavior impact depends on surrounding block context. |
| 96 |                }, | executable/support line; behavior impact depends on surrounding block context. |
| 97 |                MockParam { | executable/support line; behavior impact depends on surrounding block context. |
| 98 |                    key: "scr_efficiency_pct", | executable/support line; behavior impact depends on surrounding block context. |
| 99 |                    value: e.scr_efficiency_pct, | executable/support line; behavior impact depends on surrounding block context. |
| 100 |                    unit: "%", | executable/support line; behavior impact depends on surrounding block context. |
| 101 |                }, | executable/support line; behavior impact depends on surrounding block context. |
| 102 |                MockParam { | executable/support line; behavior impact depends on surrounding block context. |
| 103 |                    key: "nox_tailpipe_ppm", | executable/support line; behavior impact depends on surrounding block context. |
| 104 |                    value: e.nox_tailpipe_ppm, | executable/support line; behavior impact depends on surrounding block context. |
| 105 |                    unit: "ppm", | executable/support line; behavior impact depends on surrounding block context. |
| 106 |                }, | executable/support line; behavior impact depends on surrounding block context. |
| 107 |            ], | executable/support line; behavior impact depends on surrounding block context. |
| 108 |        }, | executable/support line; behavior impact depends on surrounding block context. |
| 109 |        MockFunction { | executable/support line; behavior impact depends on surrounding block context. |
| 110 |            name: "Diagnostics", | executable/support line; behavior impact depends on surrounding block context. |
| 111 |            params: vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 112 |                MockParam { | executable/support line; behavior impact depends on surrounding block context. |
| 113 |                    key: "active_dtcs", | executable/support line; behavior impact depends on surrounding block context. |
| 114 |                    value: e.active_dtcs.len() as f64, | executable/support line; behavior impact depends on surrounding block context. |
| 115 |                    unit: "count", | executable/support line; behavior impact depends on surrounding block context. |

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```

| 116 |           }, | executable/support line; behavior impact depends on surrounding block context. |
| 117 |           MockParam { | executable/support line; behavior impact depends on surrounding block conte
|
| 118 |               key: "red_lamp", | executable/support line; behavior impact depends on surrounding bl
context. |
| 119 |               value: if e.red_lamp { 1.0 } else { 0.0 }, | executable/support line; behavior impact
depends on surrounding block context. |
| 120 |               unit: "bool", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 121 |           }, | executable/support line; behavior impact depends on surrounding block context. |
| 122 |           MockParam { | executable/support line; behavior impact depends on surrounding block conte
|
| 123 |               key: "amber_lamp", | executable/support line; behavior impact depends on surrounding
ck context. |
| 124 |               value: if e.amber_lamp { 1.0 } else { 0.0 }, | executable/support line; behavior impa
depends on surrounding block context. |
| 125 |               unit: "bool", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 126 |           }, | executable/support line; behavior impact depends on surrounding block context. |
| 127 |           MockParam { | executable/support line; behavior impact depends on surrounding block conte
|
| 128 |               key: "mil_active", | executable/support line; behavior impact depends on surrounding
ck context. |
| 129 |               value: if e.mil_active { 1.0 } else { 0.0 }, | executable/support line; behavior impa
depends on surrounding block context. |
| 130 |               unit: "bool", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 131 |           }, | executable/support line; behavior impact depends on surrounding block context. |
| 132 |       ], | executable/support line; behavior impact depends on surrounding block context. |
| 133 |   }, | executable/support line; behavior impact depends on surrounding block context. |
| 134 | ], | executable/support line; behavior impact depends on surrounding block context. |
| 135 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 136 | (blank) | line is blank; visual separation only. |
| 137 | let hcm = MockEcmNode { | executable/support line; behavior impact depends on surrounding block context.
| 138 |     name: "Hydraulic HCM", | executable/support line; behavior impact depends on surrounding block contex
|
| 139 |     source_address: 0x1E, | executable/support line; behavior impact depends on surrounding block context
| 140 |     functions: vec![MockFunction { | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 141 |         name: "Hydraulic", | executable/support line; behavior impact depends on surrounding block contex
|
| 142 |         params: vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 143 |             MockParam { | executable/support line; behavior impact depends on surrounding block context.
| 144 |                 key: "system_pressure_bar", | executable/support line; behavior impact depends on surroun
g block context. |
| 145 |                 value: h.system_pressure_bar, | executable/support line; behavior impact depends on surro
ing block context. |
| 146 |                 unit: "bar", | executable/support line; behavior impact depends on surrounding block cont
. |
| 147 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 148 |             MockParam { | executable/support line; behavior impact depends on surrounding block context.
| 149 |                 key: "pump_flow_lpm", | executable/support line; behavior impact depends on surrounding b
k context. |
| 150 |                 value: h.pump_flow_lpm, | executable/support line; behavior impact depends on surrounding
ock context. |
| 151 |                 unit: "L/min", | executable/support line; behavior impact depends on surrounding block co
xt. |
| 152 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 153 |             MockParam { | executable/support line; behavior impact depends on surrounding block context.
| 154 |                 key: "fluid_temp_c", | executable/support line; behavior impact depends on surrounding bl
context. |
| 155 |                 value: h.fluid_temp_c, | executable/support line; behavior impact depends on surrounding
ck context. |
| 156 |                 unit: "C", | executable/support line; behavior impact depends on surrounding block contex
|

```



```

| 157 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 158 |         MockParam { | executable/support line; behavior impact depends on surrounding block context. |
| 159 |             key: "filter_dp_bar", | executable/support line; behavior impact depends on surrounding b
k context. |
| 160 |             value: h.filter_dp_bar, | executable/support line; behavior impact depends on surrounding
ock context. |
| 161 |             unit: "bar", | executable/support line; behavior impact depends on surrounding block cont
. |
| 162 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 163 |     ], | executable/support line; behavior impact depends on surrounding block context. |
| 164 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 165 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 166 | (blank) | line is blank; visual separation only. |
| 167 | let tcm = MockEcmNode { | executable/support line; behavior impact depends on surrounding block context.
| 168 |     name: "Transmission TCM", | executable/support line; behavior impact depends on surrounding block con
t. |
| 169 |     source_address: 0x03, | executable/support line; behavior impact depends on surrounding block context
| 170 |     functions: vec![MockFunction { | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 171 |         name: "Driveline", | executable/support line; behavior impact depends on surrounding block context
|
| 172 |         params: vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 173 |             MockParam { | executable/support line; behavior impact depends on surrounding block context.
| 174 |                 key: "ground_speed_kmh", | executable/support line; behavior impact depends on surroundin
lock context. |
| 175 |                 value: t.ground_speed_kmh, | executable/support line; behavior impact depends on surround
block context. |
| 176 |                 unit: "km/h", | executable/support line; behavior impact depends on surrounding block con
t. |
| 177 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 178 |             MockParam { | executable/support line; behavior impact depends on surrounding block context.
| 179 |                 key: "speed_step", | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 180 |                 value: t.speed_step as f64, | executable/support line; behavior impact depends on surround
g block context. |
| 181 |                 unit: "idx", | executable/support line; behavior impact depends on surrounding block cont
. |
| 182 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 183 |             MockParam { | executable/support line; behavior impact depends on surrounding block context.
| 184 |                 key: "clutch_slip_pct", | executable/support line; behavior impact depends on surrounding
ock context. |
| 185 |                 value: t.clutch_slip_pct, | executable/support line; behavior impact depends on surroundi
block context. |
| 186 |                 unit: "%", | executable/support line; behavior impact depends on surrounding block context
|
| 187 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 188 |             MockParam { | executable/support line; behavior impact depends on surrounding block context.
| 189 |                 key: "gear_ratio", | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 190 |                 value: t.gear_ratio, | executable/support line; behavior impact depends on surrounding bl
context. |
| 191 |                 unit: "ratio", | executable/support line; behavior impact depends on surrounding block co
xt. |
| 192 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 193 |         ], | executable/support line; behavior impact depends on surrounding block context. |
| 194 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 195 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 196 | (blank) | line is blank; visual separation only. |
| 197 |     vec![ecm, tcm, hcm] | executable/support line; behavior impact depends on surrounding block context. |
| 198 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/ecu_abs.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 538 | Non-empty: 478 | Symbols: 37

Symbol Inventory

Line	Kind	Symbol	Change Impact
19	const	ABS_SA	data/structure coupling
22	const	FL	data/structure coupling
23	const	FR	data/structure coupling
24	const	RL	data/structure coupling
25	const	RR	data/structure coupling
29	enum	SensorState	data/structure coupling
38	enum	AbsValveState	data/structure coupling
49	impl	std::fmt::Display for AbsValveState	behavior binding and method semantics
50	fn	fmt	signature/API coupling
62	struct	Wheel	data/structure coupling
85	impl	Default for Wheel	behavior binding and method semantics
86	fn	default	signature/API coupling
91	impl	Wheel	behavior binding and method semantics
92	fn	new	signature/API coupling
107	fn	is_locked	signature/API coupling
114	enum	EspCondition	data/structure coupling
124	impl	std::fmt::Display for EspCondition	behavior binding and method semantics
125	fn	fmt	signature/API coupling
137	struct	EcuAbs	data/structure coupling
190	impl	Default for EcuAbs	behavior binding and method semantics
191	fn	default	signature/API coupling
196	impl	EcuAbs	behavior binding and method semantics
197	fn	new	signature/API coupling
236	fn	tick	signature/API coupling
317	fn	update_v_ref	signature/API coupling
322	fn	update_wheel_speeds	signature/API coupling
342	fn	run_abs	signature/API coupling
393	fn	run_tcs	signature/API coupling
423	fn	run_esp	signature/API coupling
457	fn	run_hill_hold	signature/API coupling
467	fn	build_ebc1	signature/API coupling
490	fn	build_ebc2	signature/API coupling
508	fn	build_dm1_fault	signature/API coupling
525	fn	toggle_abs	signature/API coupling
528	fn	toggle_tcs	signature/API coupling
531	fn	toggle_esp	signature/API coupling
535	fn	any_wheel_abs_active	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	/// ABS / ESP / TCS – Electronic Stability Control ECU (SA 0x0B)	comment line; documentation intent, no runtime effect.
2	///	comment line; documentation intent, no runtime effect.
3	/// Models a real stability control system:	comment line; documentation intent, no runtime effect.
4	/// ABS – Anti-lock Braking System (ISO 11992, J1939 EBC1)	comment line; documentation intent, no runtime effect.
5	/// TCS – Traction Control System (throttle + brake intervention)	comment line; documentation intent, no runtime effect.
6	/// ESP – Electronic Stability Program (yaw-rate based torque vectoring)	comment line; documentation intent, no runtime effect.
7	/// Hill Hold – Gradient hold via brake pressure	comment line; documentation intent, no runtime effect.
8	/// EBS – Electronic Brake System interface	comment line; documentation intent, no runtime effect.
9	///	comment line; documentation intent, no runtime effect.
10	/// Each of the 4 wheels has:	comment line; documentation intent, no runtime effect.
11	• Speed sensor (Hall-effect, active)	comment line; documentation intent, no runtime effect.
12	• Individual brake modulator (solenoid + pressure sensor)	comment line; documentation intent, no runtime effect.
13	• Slip calculation relative to reference speed	comment line; documentation intent, no runtime effect.

[illegible]

```
60 | // ■ Per-Wheel State ██████████ | comment line; documentation intent, no runtime effect. |
| 61 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 62 | pub struct Wheel { | declares data layout; field changes may break serialization, tests, and callers. |
| 63 |     /// Measured wheel speed (km/h) | comment line; documentation intent, no runtime effect. |
| 64 |     pub speed: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 65 |     /// Longitudinal slip ratio: (v_ref - v_wheel) / v_ref [-1..1] | comment line; documentation intent, no r
ime effect. |
| 66 |     pub slip_ratio: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 67 |     /// Brake pressure at this wheel (bar) | comment line; documentation intent, no runtime effect. |
| 68 |     pub brake_pres_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 69 |     /// ABS modulator state | comment line; documentation intent, no runtime effect. |
| 70 |     pub valve_state: AbsValveState, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 71 |     /// Sensor health | comment line; documentation intent, no runtime effect. |
| 72 |     pub sensor_state: SensorState, | executable/support line; behavior impact depends on surrounding block con
t. |
| 73 |     /// True if ABS is currently cycling this wheel | comment line; documentation intent, no runtime effect. |
| 74 |     pub abs_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 75 |     /// True if TCS is cutting traction on this wheel | comment line; documentation intent, no runtime effect. |
| 76 |     pub tcs_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 77 |     /// Accumulated heat in brake disc (°C above ambient) | comment line; documentation intent, no runtime eff
. |
| 78 |     pub brake_temp_c: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 79 |     /// ABS cycle count for this wheel | comment line; documentation intent, no runtime effect. |
| 80 |     pub abs_cycles: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 81 |     /// Internal ABS phase (0-1 sawtooth for pressure modulation)| comment line; documentation intent, no run
e effect. |
| 82 |     abs_phase: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 83 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 84 | (blank) | line is blank; visual separation only. |
| 85 | impl Default for Wheel { | implementation block; method behavior and trait conformance are defined here. |
| 86 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
tputs. |
| 87 |         Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 88 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 89 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 90 | (blank) | line is blank; visual separation only. |
| 91 | impl Wheel { | implementation block; method behavior and trait conformance are defined here. |
| 92 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
tputs. |
| 93 |         Wheel { | executable/support line; behavior impact depends on surrounding block context. |
| 94 |             speed: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 95 |             slip_ratio: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 96 |             brake_pres_bar: 0.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 97 |             valve_state: AbsValveState::Normal, | executable/support line; behavior impact depends on surround
block context. |
| 98 |             sensor_state: SensorState::Ok, | executable/support line; behavior impact depends on surrounding b
k context. |
| 99 |             abs_active: false, | executable/support line; behavior impact depends on surrounding block context
| 100 |             tcs_active: false, | executable/support line; behavior impact depends on surrounding block contex
|
| 101 |             brake_temp_c: 25.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 102 |             abs_cycles: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 103 |             abs_phase: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 104 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 105 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 106 | (blank) | line is blank; visual separation only. |
| 107 |     pub fn is_locked(&self) -> bool { | declares executable behavior; signature/body changes alter API, contr
flow, and outputs. |
| 108 |         self.slip_ratio > 0.25 && self.speed < 1.0 | executable/support line; behavior impact depends on surr
ding block context. |
```

[illegible]

```
| 153 | (blank) | line is blank; visual separation only. |  
| 154 | // ■ Brake system ████████████████████████████████████████████████████████████████████████ | comment li  
ntent, no runtime effect. |  
| 155 | pub master_cylinder_bar: f64, // driver pedal pressure | executable/support line; behavior impact depends  
surrounding block context. |  
| 156 | pub brake_pedal_pct: f64, // 0-100% | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 157 | pub brake_light_active: bool, | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 158 | pub park_brake_applied: bool, | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 159 | (blank) | line is blank; visual separation only. |  
| 160 | // ■ ESP sensors ████████████████████████████████████████████████████████████████████████ | comment li  
tent, no runtime effect. |  
| 161 | pub steering_angle_deg: f64, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 162 | pub yaw_rate_deg_s: f64, // measured by gyro | executable/support line; behavior impact depends on surroun  
ng block context. |  
| 163 | pub lateral_g: f64, // measured by accelerometer | executable/support line; behavior impact depends  
surrounding block context. |  
| 164 | pub longitudinal_g: f64, | executable/support line; behavior impact depends on surrounding block context.  
| 165 | pub esp_condition: EspCondition, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 166 | pub esp_yaw_error: f64, // actual - desired yaw rate | executable/support line; behavior impact depends o  
urrounding block context. |  
| 167 | (blank) | line is blank; visual separation only. |  
| 168 | // ■ TCS throttle cut output ████████████████████████████████████████████████████████████████████████ | comment line;  
tent, no runtime effect. |  
| 169 | /// Throttle cut requested by TCS (0=none, 1=full cut) | comment line; documentation intent, no runtime e  
ct. |  
| 170 | pub tcs_throttle_cut: f64, | executable/support line; behavior impact depends on surrounding block contex  
|. |  
| 171 | /// Spark retard requested by TCS (unused for diesel – for reference) | comment line; documentation inten  
no runtime effect. |  
| 172 | pub tcs_torque_request_nm: f64, | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 173 | (blank) | line is blank; visual separation only. |  
| 174 | // ■ Hill Hold ████████████████████████████████████████████████████████████████████████ | comment L  
tent, no runtime effect. |  
| 175 | pub hill_hold_active: bool, | executable/support line; behavior impact depends on surrounding block conte  
|. |  
| 176 | pub grade_pct: f64, // estimated road grade | executable/support line; behavior impact depends on surround  
g block context. |  
| 177 | (blank) | line is blank; visual separation only. |  
| 178 | // ■ Diagnostics ████████████████████████████████████████████████████████████████████████ | comment li  
tent, no runtime effect. |  
| 179 | pub abs_fault: bool, // system fault – ABS disabled | executable/support line; behavior impact depends on  
rrounding block context. |  
| 180 | pub esp_fault: bool, | executable/support line; behavior impact depends on surrounding block context. |  
| 181 | pub total_abs_events: u32, | executable/support line; behavior impact depends on surrounding block contex  
|. |  
| 182 | pub total_tcs_events: u32, | executable/support line; behavior impact depends on surrounding block contex  
|. |  
| 183 | (blank) | line is blank; visual separation only. |  
| 184 | // ■ J1939 TX timers ████████████████████████████████████████████████████████████████████████ | comment lin  
tent, no runtime effect. |  
| 185 | t_etc1: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 186 | t_etc2: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 187 | t_dm1: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 188 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 189 | (blank) | line is blank; visual separation only. |  
| 190 | impl Default for EcuAbs { | implementation block; method behavior and trait conformance are defined here. |  
| 191 | fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and ou  
tputs. |  
| 192 |     Self::new() | executable/support line; behavior impact depends on surrounding block context. |
```

```
| 193 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 194 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 195 | (blank) | line is blank; visual separation only. |  
| 196 | impl EcuAbs { | implementation block; method behavior and trait conformance are defined here. |  
| 197 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 198 |         EcuAbs { | executable/support line; behavior impact depends on surrounding block context. |  
| 199 |             sa: ABS_SA, | executable/support line; behavior impact depends on surrounding block context. |  
| 200 |             abs_system_active: false, | executable/support line; behavior impact depends on surrounding block context. |  
| 201 |             tcs_system_active: false, | executable/support line; behavior impact depends on surrounding block context. |  
| 202 |             esp_system_active: false, | executable/support line; behavior impact depends on surrounding block context. |  
| 203 |             abs_enabled: true, | executable/support line; behavior impact depends on surrounding block context. |  
| 204 |             tcs_enabled: true, | executable/support line; behavior impact depends on surrounding block context. |  
| 205 |             esp_enabled: true, | executable/support line; behavior impact depends on surrounding block context. |  
| 206 |             wheels: [Wheel::new(), Wheel::new(), Wheel::new(), Wheel::new()], | executable/support line; behavior impact depends on surrounding block context. |  
| 207 |             v_ref_kmh: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 208 |             master_cylinder_bar: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 209 |             brake_pedal_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 210 |             brake_light_active: false, | executable/support line; behavior impact depends on surrounding block context. |  
| 211 |             park_brake_applied: false, | executable/support line; behavior impact depends on surrounding block context. |  
| 212 |             steering_angle_deg: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 213 |             yaw_rate_deg_s: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 214 |             lateral_g: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 215 |             longitudinal_g: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 216 |             esp_condition: EspCondition::Neutral, | executable/support line; behavior impact depends on surrounding block context. |  
| 217 |             esp_yaw_error: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 218 |             tcs_throttle_cut: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 219 |             tcs_torque_request_nm: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 220 |             hill_hold_active: false, | executable/support line; behavior impact depends on surrounding block context. |  
| 221 |             grade_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 222 |             abs_fault: false, | executable/support line; behavior impact depends on surrounding block context. |  
| 223 |             esp_fault: false, | executable/support line; behavior impact depends on surrounding block context. |  
| 224 |             total_abs_events: 0, | executable/support line; behavior impact depends on surrounding block context. |  
| 225 |             total_tcs_events: 0, | executable/support line; behavior impact depends on surrounding block context. |  
| 226 |             t_ebc1: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 227 |             t_ebc2: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 228 |             t_dm1: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 229 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 230 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 231 | (blank) | line is blank; visual separation only. |  
| 232 | // ████████████████████████████████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |  
| 233 | /// Main tick. Call with current vehicle state. | comment line; documentation intent, no runtime effect.
```

```

| 234 |      /// Returns throttle cut fraction (0-1) and J1939 frames. | comment line; documentation intent, no runtime
effect. |
| 235 |      #[allow(clippy::too_many_arguments)] | executable/support line; behavior impact depends on surrounding block
context. |
| 236 |      pub fn tick( | declares executable behavior; signature/body changes alter API, control flow, and outputs.
| 237 |                  &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 238 |                  vehicle_speed_kmh: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 239 |                  brake_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 240 |                  throttle_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 241 |                  steering_deg: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 242 |                  yaw_rate_deg_s: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 243 |                  lateral_g: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 244 |                  dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 245 |      ) -> (f64, Vec<J1939Frame>) { | executable/support line; behavior impact depends on surrounding block context. |
| 246 |          self.brake_pedal_pct = brake_pct.clamp(0.0, 100.0); | executable/support line; behavior impact depends on
surrounding block context. |
| 247 |          self.steering_angle_deg = steering_deg; | executable/support line; behavior impact depends on surrounding
block context. |
| 248 |          self.yaw_rate_deg_s = yaw_rate_deg_s; | executable/support line; behavior impact depends on surrounding
block context. |
| 249 |          self.lateral_g = lateral_g; | executable/support line; behavior impact depends on surrounding block context. |
| 250 |          self.brake_light_active = brake_pct > 2.0; | executable/support line; behavior impact depends on surrounding
block context. |
| 251 | (blank) | line is blank; visual separation only. |
| 252 |          // Master cylinder pressure: ~1.5 bar/% pedal, max 150 bar | comment line; documentation intent, no runtime
effect. |
| 253 |          self.master_cylinder_bar = (brake_pct / 100.0 * 150.0).min(150.0); | executable/support line; behavior
impact depends on surrounding block context. |
| 254 | (blank) | line is blank; visual separation only. |
| 255 |          // Estimate vehicle reference speed (best two non-locked wheels) | comment line; documentation intent, no
runtime effect. |
| 256 |          self.update_v_ref(vehicle_speed_kmh); | executable/support line; behavior impact depends on surrounding
block context. |
| 257 | (blank) | line is blank; visual separation only. |
| 258 |          // Simulate wheel speeds given vehicle speed and braking | comment line; documentation intent, no runtime
effect. |
| 259 |          self.update_wheel_speeds(vehicle_speed_kmh, dt); | executable/support line; behavior impact depends on
surrounding block context. |
| 260 | (blank) | line is blank; visual separation only. |
| 261 |          // ABS logic – per-wheel | comment line; documentation intent, no runtime effect. |
| 262 |          self.run_abs(dt); | executable/support line; behavior impact depends on surrounding block context. |
| 263 | (blank) | line is blank; visual separation only. |
| 264 |          // TCS – detect wheel spin, cut throttle | comment line; documentation intent, no runtime effect. |
| 265 |          let tcs_cut = self.run_tcs(throttle_pct, vehicle_speed_kmh, dt); | executable/support line; behavior
impact depends on surrounding block context. |
| 266 | (blank) | line is blank; visual separation only. |
| 267 |          // ESP – detect over/understeer, apply differential braking | comment line; documentation intent, no runtime
effect. |
| 268 |          self.run_esp( | executable/support line; behavior impact depends on surrounding block context. |
| 269 |              vehicle_speed_kmh, | executable/support line; behavior impact depends on surrounding block context. |
| 270 |              steering_deg, | executable/support line; behavior impact depends on surrounding block context. |
| 271 |              yaw_rate_deg_s, | executable/support line; behavior impact depends on surrounding block context. |
| 272 |              lateral_g, | executable/support line; behavior impact depends on surrounding block context. |
| 273 |              dt, | executable/support line; behavior impact depends on surrounding block context. |
| 274 |          ); | executable/support line; behavior impact depends on surrounding block context. |
| 275 | (blank) | line is blank; visual separation only. |
| 276 |          // Hill hold | comment line; documentation intent, no runtime effect. |
| 277 |          self.run_hill_hold(vehicle_speed_kmh, dt); | executable/support line; behavior impact depends on surrounding
block context. |
| 278 | (blank) | line is blank; visual separation only. |
| 279 |          // Brake disc thermal model | comment line; documentation intent, no runtime effect. |

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```

280 |         for w in &mut self.wheels { | iteration control; changing bounds/body affects throughput, timing, and
ssibly safety guarantees. |
| 281 |             let braking_heat = w.brake_pres_bar * (vehicle_speed_kmh / 3.6) * 0.005; | executable/support line
behavior impact depends on surrounding block context. |
| 282 |             let cooling = (w.brake_temp_c - 25.0) * 0.02; | executable/support line; behavior impact depends
surrounding block context. |
| 283 |             w.brake_temp_c += (braking_heat - cooling) * dt; | executable/support line; behavior impact depen
on surrounding block context. |
| 284 |             w.brake_temp_c = w.brake_temp_c.clamp(25.0, 900.0); | executable/support line; behavior impact de
ds on surrounding block context. |
| 285 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 286 | (blank) | line is blank; visual separation only. |
| 287 |         // System-level flags | comment line; documentation intent, no runtime effect. |
| 288 |         self.abs_system_active = self.wheels.iter().any(|w| w.abs_active); | executable/support line; behav
impact depends on surrounding block context. |
| 289 |         self.tcs_system_active = self.wheels.iter().any(|w| w.tcs_active); | executable/support line; behav
impact depends on surrounding block context. |
| 290 | (blank) | line is blank; visual separation only. |
| 291 |         // Periodic J1939 TX | comment line; documentation intent, no runtime effect. |
| 292 |         self.t_ebc1 += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 293 |         self.t_ebc2 += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 294 |         self.t_dm1 += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 295 |         let ts = vehicle_speed_kmh; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 296 |         let mut frames: Vec<J1939Frame> = Vec::new(); | executable/support line; behavior impact depends on s
ounding block context. |
| 297 | (blank) | line is blank; visual separation only. |
| 298 |         if self.t_ebc1 >= 0.020 { | conditional gate; changing condition shifts runtime path and side effects
| 299 |             self.t_ebc1 = 0.0; | executable/support line; behavior impact depends on surrounding block context
|
| 300 |             frames.push(self.build_ebc1(ts)); | executable/support line; behavior impact depends on surroundi
block context. |
| 301 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 302 |         if self.t_ebc2 >= 0.100 { | conditional gate; changing condition shifts runtime path and side effects
| 303 |             self.t_ebc2 = 0.0; | executable/support line; behavior impact depends on surrounding block context
|
| 304 |             frames.push(self.build_ebc2(ts)); | executable/support line; behavior impact depends on surroundi
block context. |
| 305 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 306 |         if self.t_dm1 >= 1.000 { | conditional gate; changing condition shifts runtime path and side effects.
| 307 |             self.t_dm1 = 0.0; | executable/support line; behavior impact depends on surrounding block context
| 308 |             if self.abs_fault || self.esp_fault { | conditional gate; changing condition shifts runtime pat
nd side effects. |
| 309 |                 frames.push(self.build_dm1_fault(ts)); | executable/support line; behavior impact depends on
rounding block context. |
| 310 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 311 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 312 | (blank) | line is blank; visual separation only. |
| 313 |         (tcs_cut, frames) | executable/support line; behavior impact depends on surrounding block context. |
| 314 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 315 | (blank) | line is blank; visual separation only. |
| 316 |     // ████████████████████████████████████████████████████████████████████████████████████████████████████ | comment
ntent, no runtime effect. |
| 317 |     fn update_v_ref(&mut self, vehicle_speed: f64) { | declares executable behavior; signature/body changes a
r API, control flow, and outputs. |
| 318 |         // vRef = vehicle speed (from TCM/transmission output shaft) with small lag | comment line; documenta
n intent, no runtime effect. |
| 319 |         self.v_ref_kmh += (vehicle_speed - self.v_ref_kmh) * 0.8; | executable/support line; behavior impact
ends on surrounding block context. |
| 320 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.

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| 321 | (blank) | line is blank; visual separation only. | | |
| 322 |     fn update_wheel_speeds(&mut self, vehicle_speed: f64, dt: f64) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 323 |         for w in self.wheels.iter_mut() { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 324 |             if w.abs_active { | conditional gate; changing condition shifts runtime path and side effects. |
| 325 |                 // ABS is dumping pressure – wheel recovers toward vehicle speed | comment line; documentation intent, no runtime effect. |
| 326 |                 let recover_rate = 50.0 * dt; // 50 km/h/s recovery | executable/support line; behavior impact depends on surrounding block context. |
| 327 |                 w.speed = (w.speed + recover_rate).min(vehicle_speed); | executable/support line; behavior impact depends on surrounding block context. |
| 328 |             } else if w.valve_state == AbsValveState::Normal { | executable/support line; behavior impact depends on surrounding block context. |
| 329 |                 // Normal braking: wheel decelerates based on brake pressure | comment line; documentation intent, no runtime effect. |
| 330 |                 let decel = w.brake_pres_bar * 0.04; // 4% of speed per bar per second | executable/support line; behavior impact depends on surrounding block context. |
| 331 |                 w.speed = (vehicle_speed - decel * vehicle_speed * dt).max(0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 332 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 333 |                 // Slip ratio: positive when braking (wheel slower than vehicle) | comment line; documentation intent, no runtime effect. |
| 334 |                 w.slip_ratio = if vehicle_speed > 1.0 { | executable/support line; behavior impact depends on surrounding block context. |
| 335 |                     (vehicle_speed - w.speed) / vehicle_speed | executable/support line; behavior impact depends on surrounding block context. |
| 336 |                 } else { | executable/support line; behavior impact depends on surrounding block context. |
| 337 |                     0.0 | executable/support line; behavior impact depends on surrounding block context. |
| 338 |                 }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 339 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 340 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 341 | (blank) | line is blank; visual separation only. |
| 342 |     fn run_abs(&mut self, dt: f64) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 343 |         if !self.abs_enabled || self.abs_fault { | conditional gate; changing condition shifts runtime path and side effects. |
| 344 |             return; | executable/support line; behavior impact depends on surrounding block context. |
| 345 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 346 |         let v_ref = self.v_ref_kmh; | executable/support line; behavior impact depends on surrounding block context. |
| 347 | (blank) | line is blank; visual separation only. |
| 348 |         for w in &mut self.wheels { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 349 |             // Threshold: slip > 20% → wheel is locking up | comment line; documentation intent, no runtime effect. |
| 350 |             let slip_exceeded = w.slip_ratio > 0.20 && v_ref > 5.0; | executable/support line; behavior impact depends on surrounding block context. |
| 351 |             let pres = self.master_cylinder_bar; | executable/support line; behavior impact depends on surrounding block context. |
| 352 | (blank) | line is blank; visual separation only. |
| 353 |             if !w.abs_active && slip_exceeded { | conditional gate; changing condition shifts runtime path and side effects. |
| 354 |                 w.abs_active = true; | executable/support line; behavior impact depends on surrounding block context. |
| 355 |                 w.abs_cycles += 1; | executable/support line; behavior impact depends on surrounding block context. |
| 356 |                 w.abs_phase = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 357 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

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| 358 | (blank) | line is blank; visual separation only. |
| 359 |         if w.abs_active { | conditional gate; changing condition shifts runtime path and side effects. |
| 360 |             // 10 Hz ABS cycle: dump → hold → apply | comment line; documentation intent, no runtime eff
|
| 361 |             w.abs_phase += dt * 10.0; | executable/support line; behavior impact depends on surrounding b
k context. |
| 362 |             if w.abs_phase > 1.0 { | conditional gate; changing condition shifts runtime path and side ef
ts. |
| 363 |                 w.abs_phase -= 1.0; | executable/support line; behavior impact depends on surrounding blo
context. |
| 364 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 365 | (blank) | line is blank; visual separation only. |
| 366 |             w.valve_state = if w.abs_phase < 0.30 { | executable/support line; behavior impact depends on
rrounding block context. |
| 367 |                 w.brake_pres_bar = (w.brake_pres_bar - 40.0 * dt).max(0.0); | executable/support line; be
ior impact depends on surrounding block context. |
| 368 |                 AbsValveState::Dump | executable/support line; behavior impact depends on surrounding blo
context. |
| 369 |             } else if w.abs_phase < 0.55 { | executable/support line; behavior impact depends on surround
block context. |
| 370 |                 AbsValveState::Hold | executable/support line; behavior impact depends on surrounding blo
context. |
| 371 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 372 |                 w.brake_pres_bar = (w.brake_pres_bar + 60.0 * dt).min(pres); | executable/support line; b
vior impact depends on surrounding block context. |
| 373 |                 AbsValveState::Apply | executable/support line; behavior impact depends on surrounding bl
context. |
| 374 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 375 | (blank) | line is blank; visual separation only. |
| 376 |             // Exit ABS when slip recovers and speed is recovered | comment line; documentation intent, n
untime effect. |
| 377 |             if w.slip_ratio < 0.05 && w.speed > v_ref * 0.95 { | conditional gate; changing condition shi
runtime path and side effects. |
| 378 |                 w.abs_active = false; | executable/support line; behavior impact depends on surrounding b
k context. |
| 379 |                 w.valve_state = AbsValveState::Normal; | executable/support line; behavior impact depends
surrounding block context. |
| 380 |                 w.brake_pres_bar = pres; | executable/support line; behavior impact depends on surroundin
lock context. |
| 381 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 382 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 383 |                 w.valve_state = AbsValveState::Normal; | executable/support line; behavior impact depends on
rounding block context. |
| 384 |                 w.brake_pres_bar = pres; | executable/support line; behavior impact depends on surrounding bl
context. |
| 385 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 386 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 387 | (blank) | line is blank; visual separation only. |
| 388 |         if self.abs_system_active { | conditional gate; changing condition shifts runtime path and side effec
|
| 389 |             self.total_abs_events += 1; | executable/support line; behavior impact depends on surrounding blo
context. |
| 390 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 391 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 392 | (blank) | line is blank; visual separation only. |
| 393 |     fn run_tcs(&mut self, throttle_pct: f64, vehicle_speed: f64, _dt: f64) -> f64 { | declares executable beh
or; signature/body changes alter API, control flow, and outputs. |
| 394 |         if !self.tcs_enabled { | conditional gate; changing condition shifts runtime path and side effects. |
| 395 |             return 0.0; | executable/support line; behavior impact depends on surrounding block context. |

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| 396 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 397 |         self.tcs_throttle_cut = 0.0; | executable/support line; behavior impact depends on surrounding block
text. |
| 398 | (blank) | line is blank; visual separation only. |
| 399 |         // TCS only active at low speed and high throttle | comment line; documentation intent, no runtime ef
t. |
| 400 |         if vehicle_speed > 40.0 { | conditional gate; changing condition shifts runtime path and side effects
| 401 |             return 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 402 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 403 | (blank) | line is blank; visual separation only. |
| 404 |         let driven_avg = (self.wheels[RL].speed + self.wheels[RR].speed) / 2.0; | executable/support line; be
ior impact depends on surrounding block context. |
| 405 |         let spin_excess = driven_avg - vehicle_speed; | executable/support line; behavior impact depends on s
urrounding block context. |
| 406 | (blank) | line is blank; visual separation only. |
| 407 |         if spin_excess > 3.0 && throttle_pct > 20.0 { | conditional gate; changing condition shifts runtime p
and side effects. |
| 408 |             let cut = ((spin_excess - 3.0) / 10.0).clamp(0.0, 0.8); | executable/support line; behavior impac
depends on surrounding block context. |
| 409 |             self.tcs_throttle_cut = cut; | executable/support line; behavior impact depends on surrounding bl
context. |
| 410 |             self.wheels[RL].tcs_active = true; | executable/support line; behavior impact depends on surround
block context. |
| 411 |             self.wheels[RR].tcs_active = true; | executable/support line; behavior impact depends on surround
block context. |
| 412 |             self.tcs_torque_request_nm = cut * 200.0; // request ECM to reduce torque | executable/support li
behavior impact depends on surrounding block context. |
| 413 |             if self.tcs_system_active { | conditional gate; changing condition shifts runtime path and side e
cts. |
| 414 |                 self.total_tcs_events += 1; | executable/support line; behavior impact depends on surrounding
ock context. |
| 415 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 416 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 417 |             self.wheels[RL].tcs_active = false; | executable/support line; behavior impact depends on surround
g block context. |
| 418 |             self.wheels[RR].tcs_active = false; | executable/support line; behavior impact depends on surround
g block context. |
| 419 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 420 |         self.tcs_throttle_cut | executable/support line; behavior impact depends on surrounding block context
| 421 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 422 | (blank) | line is blank; visual separation only. |
| 423 |     fn run_esp(&mut self, speed: f64, steer_deg: f64, yaw_actual: f64, lat_g: f64, _dt: f64) { | declares exe
able behavior; signature/body changes alter API, control flow, and outputs. |
| 424 |         if !self.esp_enabled || speed < 15.0 { | conditional gate; changing condition shifts runtime path a
side effects. |
| 425 |             self.esp_condition = EspCondition::Neutral; | executable/support line; behavior impact depends on
rrounding block context. |
| 426 |             return; | executable/support line; behavior impact depends on surrounding block context. |
| 427 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 428 | (blank) | line is blank; visual separation only. |
| 429 |         // Desired yaw rate from bicycle model:  $v * \text{steer} / (L * (1 + K_{us} * v^2))$  | comment line; documentatio
ntent, no runtime effect. |
| 430 |         // L=2.7m,  $K_{us}=0.005$  (understeer gradient) | comment line; documentation intent, no runtime effect. |
| 431 |         let l = 2.7; | executable/support line; behavior impact depends on surrounding block context. |
| 432 |         let kus = 0.005; | executable/support line; behavior impact depends on surrounding block context. |
| 433 |         let v_ms = speed / 3.6; | executable/support line; behavior impact depends on surrounding block conte
|
| 434 |         let steer_rad = steer_deg.to_radians(); | executable/support line; behavior impact depends on surround
g block context. |
| 435 |         let yaw_desired = v_ms * steer_rad / (l * (1.0 + kus * v_ms * v_ms)); | executable/support line; beha

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| impact depends on surrounding block context. |
| 436 |         let yaw_desired_degs = yaw_desired.to_degrees(); | executable/support line; behavior impact depends on
surrounding block context. |
| 437 | (blank) | line is blank; visual separation only. |
| 438 |         self.esp_yaw_error = yaw_actual - yaw_desired_degs; | executable/support line; behavior impact depend
n surrounding block context. |
| 439 |         let err = self.esp_yaw_error.abs(); | executable/support line; behavior impact depends on surrounding
ock context. |
| 440 | (blank) | line is blank; visual separation only. |
| 441 |         // Rollover detection: lateral acceleration > 0.8g | comment line; documentation intent, no runtime e
ct. |
| 442 |         self.esp_condition = if lat_g.abs() > 0.8 { | executable/support line; behavior impact depends on surr
nding block context. |
| 443 |             self.esp_system_active = true; | executable/support line; behavior impact depends on surrounding
ck context. |
| 444 |             EspCondition::RolloverRisk | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 445 |         } else if err > 8.0 && yaw_actual > yaw_desired_degs { | executable/support line; behavior impact dep
s on surrounding block context. |
| 446 |             self.esp_system_active = true; | executable/support line; behavior impact depends on surrounding
ck context. |
| 447 |             EspCondition::Oversteer | executable/support line; behavior impact depends on surrounding block c
ext. |
| 448 |         } else if err > 8.0 && yaw_actual < yaw_desired_degs { | executable/support line; behavior impact dep
s on surrounding block context. |
| 449 |             self.esp_system_active = true; | executable/support line; behavior impact depends on surrounding
ck context. |
| 450 |             EspCondition::Understeer | executable/support line; behavior impact depends on surrounding block
text. |
| 451 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 452 |             self.esp_system_active = false; | executable/support line; behavior impact depends on surrounding
ock context. |
| 453 |             EspCondition::Neutral | executable/support line; behavior impact depends on surrounding block con
t. |
| 454 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 455 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 456 | (blank) | line is blank; visual separation only. |
| 457 |     fn run_hill_hold(&mut self, speed: f64, _dt: f64) { | declares executable behavior; signature/body change
lter API, control flow, and outputs. |
| 458 |         // Activate hill hold if vehicle speed drops to 0 while brake applied | comment line; documentation i
nt, no runtime effect. |
| 459 |         if speed < 0.5 && self.master_cylinder_bar > 10.0 { | conditional gate; changing condition shifts run
e path and side effects. |
| 460 |             self.hill_hold_active = true; | executable/support line; behavior impact depends on surrounding b
k context. |
| 461 |         } else if speed > 2.0 { | executable/support line; behavior impact depends on surrounding block conte
|
| 462 |             self.hill_hold_active = false; | executable/support line; behavior impact depends on surrounding
ck context. |
| 463 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 464 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 465 | (blank) | line is blank; visual separation only. |
| 466 |     // ■ J1939 frame builders ████████████████████████████████████████████████████████████████ | comment line
nent, no runtime effect. |
| 467 |     fn build_ebc1(&self, ts: f64) -> J1939Frame { | declares executable behavior; signature/body changes alte
PI, control flow, and outputs. |
| 468 |         let mut data = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 469 |         // SPN 561: ABS Control Active (bits 0-1) | comment line; documentation intent, no runtime effect. |
| 470 |         data[0] = if self.abs_system_active { 0x01 } else { 0x00 }; | executable/support line; behavior impac
depends on surrounding block context. |
| 471 |         // SPN 562: ABS off-road switch | comment line; documentation intent, no runtime effect. |
| 472 |         data[0] \|= if !self.abs_enabled { 0x04 } else { 0x00 }; | executable/support line; behavior impact d
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nds on surrounding block context. |
| 473 |          // SPN 563: ASR (TCS) brake control | comment line; documentation intent, no runtime effect. |
| 474 |          data[0] \|= if self.tcs_system_active { 0x10 } else { 0x00 }; | executable/support line; behavior impact
depends on surrounding block context. |
| 475 |          // SPN 1121: ESP active | comment line; documentation intent, no runtime effect. |
| 476 |          data[1] = if self.esp_system_active { 0x01 } else { 0x00 }; | executable/support line; behavior impact
depends on surrounding block context. |
| 477 |          // SPN 521: Front axle brake demand (% of master) | comment line; documentation intent, no runtime ef
t. |
| 478 |          data[2] = (self.master_cylinder_bar / 150.0 * 250.0) as u8; | executable/support line; behavior impact
depends on surrounding block context. |
| 479 |          // SPN 522: Rear axle brake demand | comment line; documentation intent, no runtime effect. |
| 480 |          data[3] = data[2]; | executable/support line; behavior impact depends on surrounding block context. |
| 481 |          // Brake lamps | comment line; documentation intent, no runtime effect. |
| 482 |          data[4] = if self.brake_light_active { 0x03 } else { 0x00 }; | executable/support line; behavior impact
depends on surrounding block context. |
| 483 |          J1939Frame::from_raw( | executable/support line; behavior impact depends on surrounding block context
| 484 |          ts, | executable/support line; behavior impact depends on surrounding block context. |
| 485 |          J1939Frame::build_id(2, j1939::pgn::EBC1, self.sa, 0xFF), | executable/support line; behavior impact
depends on surrounding block context. |
| 486 |          &data, | executable/support line; behavior impact depends on surrounding block context. |
| 487 |          ) | executable/support line; behavior impact depends on surrounding block context. |
| 488 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 489 | (blank) | line is blank; visual separation only. |
| 490 |      fn build_ebc2(&self, ts: f64) -> J1939Frame { | declares executable behavior; signature/body changes alter
PI, control flow, and outputs. |
| 491 |          let mut data = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 492 |          // SPN 904: Front left wheel speed (0.125 km/h/bit) | comment line; documentation intent, no runtime
ect. |
| 493 |          let fl = (self.wheels[FL].speed / 0.125) as u16; | executable/support line; behavior impact depends o
urrounding block context. |
| 494 |          let fr = (self.wheels[FR].speed / 0.125) as u16; | executable/support line; behavior impact depends o
urrounding block context. |
| 495 |          let rl = (self.wheels[RL].speed / 0.125) as u16; | executable/support line; behavior impact depends o
urrounding block context. |
| 496 |          let rr = (self.wheels[RR].speed / 0.125) as u16; | executable/support line; behavior impact depends o
urrounding block context. |
| 497 |          data[0] = (fl & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block
text. |
| 498 |          data[1] = (fl >> 8) as u8; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 499 |          data[2] = (fr & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block
text. |
| 500 |          data[3] = (fr >> 8) as u8; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 501 |          data[4] = (rl & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block
text. |
| 502 |          data[5] = (rl >> 8) as u8; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 503 |          data[6] = (rr & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block
text. |
| 504 |          data[7] = (rr >> 8) as u8; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 505 |          J1939Frame::from_raw(ts, J1939Frame::build_id(6, 65215, self.sa, 0xFF), &data) | executable/support l
; behavior impact depends on surrounding block context. |
| 506 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 507 | (blank) | line is blank; visual separation only. |
| 508 |      fn build_dm1_fault(&self, ts: f64) -> J1939Frame { | declares executable behavior; signature/body changes
ter API, control flow, and outputs. |
| 509 |          let mut data = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 510 |          data[0] = 0x04; // Amber warning | executable/support line; behavior impact depends on surrounding bl
context. |
| 511 |          data[1] = 0xFF; | executable/support line; behavior impact depends on surrounding block context. |

```



```
| 318 | fn | cycle_wiper | signature/API coupling |
| 326 | fn | toggle_hvac | signature/API coupling |
| 330 | fn | reset_all_fuses | signature/API coupling |
```

Line-by-Line Impact Map

```
| Line | Source | Operational Impact If Changed |
|---:|---|---|
| 1 | /// Body Control Module (BCM) – manages all electrical body functions. | comment line; documentation intent, no runtime effect. |
| 2 | /// SA: 0x27 (Cab Controller). | comment line; documentation intent, no runtime effect. |
| 3 | /// Models: lighting, horn, wipers, battery/charging, fuses, switches. | comment line; documentation intent, no runtime effect. |
| 4 | /// J1939: Transmits heartbeat via Prop-B, monitors battery (PGN 65272). | comment line; documentation intent, runtime effect. |
| 5 | (blank) | line is blank; visual separation only. |
| 6 | use crate::j1939::{self, addr, J1939Frame}; | imports symbols; changing path/name affects compile-time resolution. |
| 7 | (blank) | line is blank; visual separation only. |
| 8 | pub const BCM_SA: u8 = addr::CAB; // 0x27 | executable/support line; behavior impact depends on surrounding block context. |
| 9 | (blank) | line is blank; visual separation only. |
| 10 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding block context. |
| 11 | pub enum WiperSpeed { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 12 |     Off, | executable/support line; behavior impact depends on surrounding block context. |
| 13 |     Low, | executable/support line; behavior impact depends on surrounding block context. |
| 14 |     High, | executable/support line; behavior impact depends on surrounding block context. |
| 15 |     Intermittent, | executable/support line; behavior impact depends on surrounding block context. |
| 16 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 17 | (blank) | line is blank; visual separation only. |
| 18 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding block context. |
| 19 | pub enum ChargingState { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 20 |     Off, | executable/support line; behavior impact depends on surrounding block context. |
| 21 |     Charging, | executable/support line; behavior impact depends on surrounding block context. |
| 22 |     Fault, | executable/support line; behavior impact depends on surrounding block context. |
| 23 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 24 | (blank) | line is blank; visual separation only. |
| 25 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 26 | pub struct Fuse { | declares data layout; field changes may break serialization, tests, and callers. |
| 27 |     pub id: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 28 |     pub rating_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 29 |     pub blown: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 30 |     pub load_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 31 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 32 | (blank) | line is blank; visual separation only. |
| 33 | impl Fuse { | implementation block; method behavior and trait conformance are defined here. |
| 34 |     pub fn new(id: &'static str, rating: f64) -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 35 |         Fuse { | executable/support line; behavior impact depends on surrounding block context. |
| 36 |             id, | executable/support line; behavior impact depends on surrounding block context. |
| 37 |             rating_a: rating, | executable/support line; behavior impact depends on surrounding block context. |
| 38 |             blown: false, | executable/support line; behavior impact depends on surrounding block context. |
| 39 |             load_a: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 40 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 41 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 42 |     pub fn check(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 43 |         if self.load_a > self.rating_a * 1.15 { | conditional gate; changing condition shifts runtime path and behavior effects. |
| 44 |             self.blown = true; | executable/support line; behavior impact depends on surrounding block context. |
| 45 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
```


[illegible]

```

| 92 | (blank) | line is blank; visual separation only. |
| 93 | impl Default for EcuBcm { | implementation block; method behavior and trait conformance are defined here. |
| 94 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
tputs. |
| 95 |         Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 96 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 97 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 98 | (blank) | line is blank; visual separation only. |
| 99 | impl EcuBcm { | implementation block; method behavior and trait conformance are defined here. |
| 100 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
utputs. |
| 101 |         let fuses = vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 102 |             Fuse::new("WORK_LT_F", 25.0), | executable/support line; behavior impact depends on surrounding b
k context. |
| 103 |             Fuse::new("WORK_LT_R", 25.0), | executable/support line; behavior impact depends on surrounding b
k context. |
| 104 |             Fuse::new("ROAD_LT", 15.0), | executable/support line; behavior impact depends on surrounding blo
context. |
| 105 |             Fuse::new("WIPER", 10.0), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 106 |             Fuse::new("HVAC", 20.0), | executable/support line; behavior impact depends on surrounding block
text. |
| 107 |             Fuse::new("HORN", 5.0), | executable/support line; behavior impact depends on surrounding block c
ext. |
| 108 |             Fuse::new("CAB_INT", 10.0), | executable/support line; behavior impact depends on surrounding blo
context. |
| 109 |             Fuse::new("BEACON", 5.0), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 110 |         ]; | executable/support line; behavior impact depends on surrounding block context. |
| 111 |         EcuBcm { | executable/support line; behavior impact depends on surrounding block context. |
| 112 |             sa: BCM_SA, | executable/support line; behavior impact depends on surrounding block context. |
| 113 |             road_lights: false, | executable/support line; behavior impact depends on surrounding block conte
|
| 114 |             work_lights_front: false, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 115 |             work_lights_rear: false, | executable/support line; behavior impact depends on surrounding block
text. |
| 116 |             beacon_light: false, | executable/support line; behavior impact depends on surrounding block cont
. |
| 117 |             hazard_lights: false, | executable/support line; behavior impact depends on surrounding block con
t. |
| 118 |             cab_interior_light: true, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 119 |             horn_active: false, | executable/support line; behavior impact depends on surrounding block conte
|
| 120 |             horn_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 121 |             wiper_speed: WiperSpeed::Off, | executable/support line; behavior impact depends on surrounding b
k context. |
| 122 |             wiper_position: 0.0, | executable/support line; behavior impact depends on surrounding block cont
. |
| 123 |             wiper_sweep_dir: 1.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 124 |             battery_voltage: 12.8, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 125 |             battery_current_a: -2.0, | executable/support line; behavior impact depends on surrounding block
text. |
| 126 |             battery_soc_pct: 80.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 127 |             charging_state: ChargingState::Off, | executable/support line; behavior impact depends on surroun
g block context. |
| 128 |             alternator_amps: 0.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 129 |             total_load_amps: 0.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 130 |             fuses, | executable/support line; behavior impact depends on surrounding block context. |

```

```
| 131 |         hvac_on: false, | executable/support line; behavior impact depends on surrounding block context. |  
| 132 |         hvac_temp_set_c: 22.0, | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 133 |         cab_temp_c: 25.0, | executable/support line; behavior impact depends on surrounding block context  
| 134 |         t_heartbeat: 0.0, | executable/support line; behavior impact depends on surrounding block context  
| 135 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 136 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 137 | (blank) | line is blank; visual separation only. |  
| 138 |     /// Main BCM tick – returns J1939 frames. | comment line; documentation intent, no runtime effect. |  
| 139 |     pub fn tick(&ampmut self, alternator_v: f64, dt: f64) -> Vec<J1939Frame> { | declares executable behavior; s  
ature/body changes alter API, control flow, and outputs. |  
| 140 |         // ■ Charging model ████████████████████████████████████████████████████████████████ | comment line  
ntent, no runtime effect. |  
| 141 |         self.alternator_amps = if alternator_v > 13.5 { | executable/support line; behavior impact depends on  
rrounding block context. |  
| 142 |             self.charging_state = ChargingState::Charging; | executable/support line; behavior impact depends  
surrounding block context. |  
| 143 |             80.0 // 80A alternator | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 144 |         } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 145 |             self.charging_state = ChargingState::Off; | executable/support line; behavior impact depends on s  
ounding block context. |  
| 146 |             0.0 | executable/support line; behavior impact depends on surrounding block context. |  
| 147 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 148 | (blank) | line is blank; visual separation only. |  
| 149 |         // ■ Calculate total electrical load ████████████████████████████████████████████████████████████████ | comment line; do  
ntent, no runtime effect. |  
| 150 |         self.total_load_amps = 0.0; | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 151 |         if self.work_lights_front { | conditional gate; changing condition shifts runtime path and side effec  
|  
| 152 |             self.total_load_amps += 18.0; | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 153 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 154 |         if self.work_lights_rear { | conditional gate; changing condition shifts runtime path and side effect  
|  
| 155 |             self.total_load_amps += 14.0; | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 156 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 157 |         if self.road_lights { | conditional gate; changing condition shifts runtime path and side effects. |  
| 158 |             self.total_load_amps += 8.0; | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 159 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 160 |         if self.beacon_light { | conditional gate; changing condition shifts runtime path and side effects. |  
| 161 |             self.total_load_amps += 4.0; | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 162 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 163 |         if self.hazard_lights { | conditional gate; changing condition shifts runtime path and side effects.  
| 164 |             self.total_load_amps += 6.0; | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 165 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 166 |         if self.cab_interior_light { | conditional gate; changing condition shifts runtime path and side effe  
. |  
| 167 |             self.total_load_amps += 2.0; | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 168 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 169 |         if self.hvac_on { | conditional gate; changing condition shifts runtime path and side effects. |
```

```
| 170 |         self.total_load_amps += 15.0; | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 171 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 172 |         if self.horn_active { | conditional gate; changing condition shifts runtime path and side effects. |  
| 173 |             self.total_load_amps += 8.0; | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 174 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 175 |         // Base ECU load | comment line; documentation intent, no runtime effect. |  
| 176 |         self.total_load_amps += 5.0; | executable/support line; behavior impact depends on surrounding block  
text. |  
| 177 | (blank) | line is blank; visual separation only. |  
| 178 |         // ■ Battery current and voltage ████████████████████████████████████████████████████ | comment line;  
intent, no runtime effect. |  
| 179 |         self.battery_current_a = self.alternator_amps - self.total_load_amps; | executable/support line; beha  
r impact depends on surrounding block context. |  
| 180 |         // Simple SoC model | comment line; documentation intent, no runtime effect. |  
| 181 |         self.battery_soc_pct = (self.battery_soc_pct | executable/support line; behavior impact depends on su  
unding block context. |  
| 182 |             + self.battery_current_a * dt / 3600.0 * 100.0 / 88.0) | executable/support line; behavior impact  
depends on surrounding block context. |  
| 183 |             .clamp(0.0, 100.0); // 88 Ah battery | executable/support line; behavior impact depends on surroun  
ng block context. |  
| 184 |         self.battery_voltage = | executable/support line; behavior impact depends on surrounding block contex  
|  
| 185 |             11.5 + self.battery_soc_pct / 100.0 * 1.8 + if alternator_v > 13.0 { 0.5 } else { 0.0 }; | execut  
e/support line; behavior impact depends on surrounding block context. |  
| 186 | (blank) | line is blank; visual separation only. |  
| 187 |         // ■ Fuse monitoring ████████████████████████████████████████████████████ | comment lin  
intent, no runtime effect. |  
| 188 |         for fuse in &mut self.fuses { | iteration control; changing bounds/body affects throughput, timing, a  
possibly safety guarantees. |  
| 189 |             fuse.load_a = match fuse.id { | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 190 |                 "WORK_LT_F" => { | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 191 |                     if self.work_lights_front { | conditional gate; changing condition shifts runtime path an  
side effects. |  
| 192 |                         18.0 | executable/support line; behavior impact depends on surrounding block context.  
| 193 |                     } else { | executable/support line; behavior impact depends on surrounding block context.  
| 194 |                         0.0 | executable/support line; behavior impact depends on surrounding block context.  
| 195 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and li  
time scopes. |  
| 196 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti  
scopes. |  
| 197 |                 "WORK_LT_R" => { | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 198 |                     if self.work_lights_rear { | conditional gate; changing condition shifts runtime path and  
de effects. |  
| 199 |                         14.0 | executable/support line; behavior impact depends on surrounding block context.  
| 200 |                     } else { | executable/support line; behavior impact depends on surrounding block context.  
| 201 |                         0.0 | executable/support line; behavior impact depends on surrounding block context.  
| 202 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and li  
ime scopes. |  
| 203 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti  
scopes. |  
| 204 |                 "ROAD_LT" => { | executable/support line; behavior impact depends on surrounding block contex  
|  
| 205 |                     if self.road_lights { | conditional gate; changing condition shifts runtime path and side  
fects. |  
| 206 |                         8.0 | executable/support line; behavior impact depends on surrounding block context.  
| 207 |                     } else { | executable/support line; behavior impact depends on surrounding block context.  
| 208 |                         0.0 | executable/support line; behavior impact depends on surrounding block context.  
| 209 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
```

```
|me scopes. |  
| 210 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
scopes. |  
| 211 |             "WIPER" => { | executable/support line; behavior impact depends on surrounding block context.  
| 212 |                 if self.wiper_speed != WiperSpeed::Off { | conditional gate; changing condition shifts runtime  
time path and side effects. |  
| 213 |                     4.0 | executable/support line; behavior impact depends on surrounding block context.  
| 214 |                 } else { | executable/support line; behavior impact depends on surrounding block context.  
| 215 |                     0.0 | executable/support line; behavior impact depends on surrounding block context.  
| 216 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and li  
ime scopes. |  
| 217 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti  
scopes. |  
| 218 |             "HVAC" => { | executable/support line; behavior impact depends on surrounding block context.  
| 219 |                 if self.hvac_on { | conditional gate; changing condition shifts runtime path and side eff  
S. |  
| 220 |                     15.0 | executable/support line; behavior impact depends on surrounding block context.  
| 221 |                 } else { | executable/support line; behavior impact depends on surrounding block context.  
| 222 |                     0.0 | executable/support line; behavior impact depends on surrounding block context.  
| 223 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and li  
ime scopes. |  
| 224 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti  
scopes. |  
| 225 |             "HORN" | executable/support line; behavior impact depends on surrounding block context. |  
| 226 |                 if self.horn_active => { | conditional gate; changing condition shifts runtime path and s  
effects. |  
| 227 |                     8.0 | executable/support line; behavior impact depends on surrounding block context.  
| 228 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and li  
ime scopes. |  
| 229 |             _ => 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 230 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 231 |         fuse.check(); | executable/support line; behavior impact depends on surrounding block context. |  
| 232 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 233 | (blank) | line is blank; visual separation only. |  
| 234 |         // ■ Wiper simulation ████████████████████████████████████████████████████████████████ | comment line  
nment, no runtime effect. |  
| 235 |         let wiper_ok = !self.fuses.iter().any(\f\ f.id == "WIPER" && f.blown); | executable/support line; b  
vior impact depends on surrounding block context. |  
| 236 |         if wiper_ok { | conditional gate; changing condition shifts runtime path and side effects. |  
| 237 |             let sweep_speed = match self.wiper_speed { | executable/support line; behavior impact depends on  
rounding block context. |  
| 238 |                 WiperSpeed::Off => 0.0, | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 239 |                 WiperSpeed::Low => 0.6, | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 240 |                 WiperSpeed::High => 1.4, | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 241 |                 WiperSpeed::Intermittent => { | executable/support line; behavior impact depends on surroundi  
block context. |  
| 242 |                     if (self.t_heartbeat % 3.0) < 0.5 { | conditional gate; changing condition shifts runtime  
th and side effects. |  
| 243 |                         1.0 | executable/support line; behavior impact depends on surrounding block context.  
| 244 |                     } else { | executable/support line; behavior impact depends on surrounding block context.  
| 245 |                         0.0 | executable/support line; behavior impact depends on surrounding block context.  
| 246 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and li  
ime scopes. |  
| 247 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti  
scopes. |  
| 248 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 249 |             self.wiper_position += self.wiper_sweep_dir * sweep_speed * dt; | executable/support line; behavi  
impact depends on surrounding block context. |  
| 250 |             if self.wiper_position >= 1.0 { | conditional gate; changing condition shifts runtime path and si
```

```
effects. |  
| 251 |           self.wiper_sweep_dir = -1.0; | executable/support line; behavior impact depends on surrounding  
lock context. |  
| 252 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 253 |       if self.wiper_position <= 0.0 { | conditional gate; changing condition shifts runtime path and si  
ffects. |  
| 254 |           self.wiper_sweep_dir = 1.0; | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 255 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 256 |           self.wiper_position = self.wiper_position.clamp(0.0, 1.0); | executable/support line; behavior im  
t depends on surrounding block context. |  
| 257 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 258 | (blank) | line is blank; visual separation only. |  
| 259 |         // ■ Horn (auto-off after 2 s) ████████████████████████████████████████████████████ | comment line; c  
ntent, no runtime effect. |  
| 260 |         if self.horn_active { | conditional gate; changing condition shifts runtime path and side effects. |  
| 261 |             self.horn_timer += dt; | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 262 |             if self.horn_timer > 2.0 { | conditional gate; changing condition shifts runtime path and side ef  
ts. |  
| 263 |                 self.horn_active = false; | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 264 |                 self.horn_timer = 0.0; | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 265 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 266 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 267 | (blank) | line is blank; visual separation only. |  
| 268 |         // ■ HVAC simple model ████████████████████████████████████████████████████ | comment line;  
ntent, no runtime effect. |  
| 269 |         if self.hvac_on { | conditional gate; changing condition shifts runtime path and side effects. |  
| 270 |             let cool = if self.cab_temp_c > self.hvac_temp_set_c { | executable/support line; behavior impact  
depends on surrounding block context. |  
| 271 |                 -8.0 | executable/support line; behavior impact depends on surrounding block context. |  
| 272 |             } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 273 |                 3.0 | executable/support line; behavior impact depends on surrounding block context. |  
| 274 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 275 |             self.cab_temp_c += cool * dt * 0.05; | executable/support line; behavior impact depends on surroun  
ng block context. |  
| 276 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 277 | (blank) | line is blank; visual separation only. |  
| 278 |         // ■ J1939 heartbeat (100 ms) ████████████████████████████████████████████████████ | comment line; c  
ntent, no runtime effect. |  
| 279 |         self.t_heartbeat += dt; | executable/support line; behavior impact depends on surrounding block conte  
|  
| 280 |         let mut frames: Vec<J1939Frame> = Vec::new(); | executable/support line; behavior impact depends on s  
ounding block context. |  
| 281 |         if self.t_heartbeat >= 0.100 { | conditional gate; changing condition shifts runtime path and side ef  
ts. |  
| 282 |             self.t_heartbeat = 0.0; | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 283 |             // Proprietary B: BCM status (PGN 65280) | comment line; documentation intent, no runtime effect.  
| 284 |             let mut data = [0u8; 8]; | executable/support line; behavior impact depends on surrounding block  
text. |  
| 285 |             data[0] = (if self.road_lights { 0x01 } else { 0 }) | executable/support line; behavior impact de  
ds on surrounding block context. |  
| 286 |                 \| (if self.work_lights_front { 0x02 } else { 0 }) | executable/support line; behavior impact  
depends on surrounding block context. |  
| 287 |                 \| (if self.work_lights_rear { 0x04 } else { 0 }) | executable/support line; behavior impact
```

```
| ends on surrounding block context. |  
| 288 |           \ | (if self.beacon_light { 0x08 } else { 0 }) | executable/support line; behavior impact depends  
on surrounding block context. |  
| 289 |           \ | (if self.hazard_lights { 0x10 } else { 0 }); | executable/support line; behavior impact depe  
nds on surrounding block context. |  
| 290 |           data[1] = (self.battery_voltage * 20.0) as u8; // 0.05V/bit | executable/support line; behavior i  
ct depends on surrounding block context. |  
| 291 |           data[2] = self.battery_soc_pct as u8; | executable/support line; behavior impact depends on surro  
ing block context. |  
| 292 |           data[3] = if self.charging_state == ChargingState::Charging { | executable/support line; behavior  
impact depends on surrounding block context. |  
| 293 |               0x01 | executable/support line; behavior impact depends on surrounding block context. |  
| 294 |           } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 295 |               0x00 | executable/support line; behavior impact depends on surrounding block context. |  
| 296 |       }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
opes. |  
| 297 |           let raw_id = J1939Frame::build_id(7, j1939::pgn::PROP_A, self.sa, 0xFF); | executable/support lin  
behavior impact depends on surrounding block context. |  
| 298 |           frames.push(J1939Frame::from_raw(self.t_heartbeat, raw_id, &data)); | executable/support line; be  
ior impact depends on surrounding block context. |  
| 299 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 300 |           frames | executable/support line; behavior impact depends on surrounding block context. |  
| 301 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 302 | (blank) | line is blank; visual separation only. |  
| 303 |     // ■ Convenience toggles ████████████████████████████████████████████████████████████████ | comment line  
ntent, no runtime effect. |  
| 304 |     pub fn toggle_work_lights(&mut self) { | declares executable behavior; signature/body changes alter API,  
trol flow, and outputs. |  
| 305 |         self.work_lights_front = !self.work_lights_front; | executable/support line; behavior impact depends  
surrounding block context. |  
| 306 |         self.work_lights_rear = !self.work_lights_rear; | executable/support line; behavior impact depends on  
urrounding block context. |  
| 307 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 308 |     pub fn toggle_road_lights(&mut self) { | declares executable behavior; signature/body changes alter API,  
trol flow, and outputs. |  
| 309 |         self.road_lights = !self.road_lights; | executable/support line; behavior impact depends on surroundi  
block context. |  
| 310 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 311 |     pub fn toggle_beacon(&mut self) { | declares executable behavior; signature/body changes alter API, contr  
flow, and outputs. |  
| 312 |         self.beacon_light = !self.beacon_light; | executable/support line; behavior impact depends on surroun  
g block context. |  
| 313 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 314 |     pub fn honk(&mut self) { | declares executable behavior; signature/body changes alter API, control flow,  
outputs. |  
| 315 |         self.horn_active = true; | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 316 |         self.horn_timer = 0.0; | executable/support line; behavior impact depends on surrounding block contex  
|  
| 317 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 318 |     pub fn cycle_wiper(&mut self) { | declares executable behavior; signature/body changes alter API, control  
ow, and outputs. |  
| 319 |         self.wiper_speed = match self.wiper_speed { | executable/support line; behavior impact depends on surr  
nding block context. |  
| 320 |             WiperSpeed::Off => WiperSpeed::Intermittent, | executable/support line; behavior impact depends o  
urrounding block context. |  
| 321 |             WiperSpeed::Intermittent => WiperSpeed::Low, | executable/support line; behavior impact depends o  
urrounding block context. |  
| 322 |             WiperSpeed::Low => WiperSpeed::High, | executable/support line; behavior impact depends on surrou  
ng block context. |  
| 323 |             WiperSpeed::High => WiperSpeed::Off, | executable/support line; behavior impact depends on surrou  
ng block context. |  
| 324 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
```

```

| 325 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 326 |      pub fn toggle_hvac(&mut self) { | declares executable behavior; signature/body changes alter API, control
ow, and outputs. |
| 327 |          self.hvac_on = !self.hvac_on; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 328 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 329 | (blank) | line is blank; visual separation only. |
| 330 |      pub fn reset_all_fuses(&mut self) { | declares executable behavior; signature/body changes alter API, con
l flow, and outputs. |
| 331 |          for fuse in &mut self.fuses { | iteration control; changing bounds/body affects throughput, timing, a
possibly safety guarantees. |
| 332 |              fuse.blown = false; | executable/support line; behavior impact depends on surrounding block conte
|
| 333 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 334 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 335 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/ecu_ecm.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 860 | Non-empty: 777 | Symbols: 38

Symbol Inventory

Line	Kind	Symbol	Change Impact
17	const	ECM_SA	data/structure coupling
18	const	CYLINDERS	data/structure coupling
19	const	IDLE_RPM	data/structure coupling
20	const	LO_IDLE_RPM	data/structure coupling
21	const	HI_IDLE_RPM	data/structure coupling
22	const	RATED_RPM	data/structure coupling
23	const	MAX_RPM	data/structure coupling
24	const	PEAK_TORQUE_NM	data/structure coupling
25	const	PEAK_TORQUE_RPM	data/structure coupling
27	const	RATED_POWER_KW	data/structure coupling
28	const	ENGINE_INERTIA_KGM2	data/structure coupling
29	const	FUEL_TANK_L	data/structure coupling
30	const	DEF_TANK_L	data/structure coupling
31	const	DPF_REGEN_SOOT	data/structure coupling
35	enum	GovernorMode	data/structure coupling
44	impl	std::fmt::Display for GovernorMode	behavior binding and method semantics
45	fn	fmt	signature/API coupling
56	enum	FuelMap	data/structure coupling
62	impl	std::fmt::Display for FuelMap	behavior binding and method semantics
63	fn	fmt	signature/API coupling
74	enum	AftertreatmentState	data/structure coupling
86	impl	std::fmt::Display for AftertreatmentState	behavior binding and method semantics
87	fn	fmt	signature/API coupling
100	struct	EcuEcm	data/structure coupling
196	impl	Default for EcuEcm	behavior binding and method semantics
197	fn	default	signature/API coupling
202	impl	EcuEcm	behavior binding and method semantics
203	fn	new	signature/API coupling
292	fn	torque_at_rpm	signature/API coupling
327	fn	tick	signature/API coupling
545	fn	update_aftertreatment	signature/API coupling
607	fn	run_diagnostics	signature/API coupling
806	fn	add_dtc	signature/API coupling
834	fn	first_active_dtc_spn_fmi	signature/API coupling
841	fn	clear_dtcs	signature/API coupling
850	fn	power_kw	signature/API coupling
853	fn	acceleration_ms2	signature/API coupling
857	fn	is_running	signature/API coupling

[illegible]

```
| 40 | // Droop: RPM sags ~100 rpm under full load (good for PTO stability) | comment line; documentation intent  
o runtime effect. |  
| 41 |     Droop, | executable/support line; behavior impact depends on surrounding block context. |  
| 42 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 43 | (blank) | line is blank; visual separation only. |  
| 44 | impl std::fmt::Display for GovernorMode { | implementation block; method behavior and trait conformance are de  
ed here. |  
| 45 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signa  
e/body changes alter API, control flow, and outputs. |  
| 46 |         match self { | branch dispatch; changing arms alters behavior for variants and input classes. |  
| 47 |             GovernorMode::LowHighIdle => write!(f, "L/H IDLE"), | I/O or protocol-critical logic; edits can af  
t integration, timing, and diagnostic outcomes. |  
| 48 |             GovernorMode::AllSpeed => write!(f, "ALL-SPD "), | I/O or protocol-critical logic; edits can affect  
ntegration, timing, and diagnostic outcomes. |  
| 49 |             GovernorMode::Droop => write!(f, "DROOP   "), | I/O or protocol-critical logic; edits can affect i  
gration, timing, and diagnostic outcomes. |  
| 50 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes  
| 51 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 52 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 53 | (blank) | line is blank; visual separation only. |  
| 54 | // ■■ Fuel Map Modes ████████████████████████████████████████████████████████ | comment lin  
tent, no runtime effect. |  
| 55 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 56 | pub enum FuelMap { | declares state variants; adding/removing variants impacts pattern matches and logic bran  
. |  
| 57 |     Standard, | executable/support line; behavior impact depends on surrounding block context. |  
| 58 |     Economy, | executable/support line; behavior impact depends on surrounding block context. |  
| 59 |     Power, | executable/support line; behavior impact depends on surrounding block context. |  
| 60 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 61 | (blank) | line is blank; visual separation only. |  
| 62 | impl std::fmt::Display for FuelMap { | implementation block; method behavior and trait conformance are defined  
re. |  
| 63 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signa  
e/body changes alter API, control flow, and outputs. |  
| 64 |         match self { | branch dispatch; changing arms alters behavior for variants and input classes. |  
| 65 |             FuelMap::Standard => write!(f, "STANDARD"), | I/O or protocol-critical logic; edits can affect int  
ation, timing, and diagnostic outcomes. |  
| 66 |             FuelMap::Economy => write!(f, "ECONOMY "), | I/O or protocol-critical logic; edits can affect inte  
tion, timing, and diagnostic outcomes. |  
| 67 |             FuelMap::Power => write!(f, "POWER    "), | I/O or protocol-critical logic; edits can affect integr  
on, timing, and diagnostic outcomes. |  
| 68 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes  
| 69 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 70 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 71 | (blank) | line is blank; visual separation only. |  
| 72 | // ■■ Aftertreatment state ████████████████████████████████████████████████████████ | comment lin  
tent, no runtime effect. |  
| 73 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 74 | pub enum AftertreatmentState { | declares state variants; adding/removing variants impacts pattern matches and  
gic branches. |  
| 75 |     Normal, | executable/support line; behavior impact depends on surrounding block context. |  
| 76 |     /// Active DPF regeneration (>=600°C DPF outlet) | comment line; documentation intent, no runtime effect. |  
| 77 |     DpfRegen, | executable/support line; behavior impact depends on surrounding block context. |  
| 78 |     /// SCR efficiency degraded (low DEF, wrong concentration) | comment line; documentation intent, no runtim  
effect. |  
| 79 |     ScrDegraded, | executable/support line; behavior impact depends on surrounding block context. |  
| 80 |     /// Severe derating – >80% soot, <2% DEF | comment line; documentation intent, no runtime effect. |  
| 81 |     Derating, | executable/support line; behavior impact depends on surrounding block context. |  
| 82 |     /// System fault | comment line; documentation intent, no runtime effect. |  
| 83 |     Fault, | executable/support line; behavior impact depends on surrounding block context. |  
| 84 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 85 | (blank) | line is blank; visual separation only. |  
| 86 | impl std::fmt::Display for AftertreatmentState { | implementation block; method behavior and trait conformance
```

[illegible]

[illegible]

```
| 171 | pub regen_inhibited: bool, | executable/support line; behavior impact depends on surrounding block context.  
|  
| 172 | (blank) | line is blank; visual separation only. |  
| 173 | // ■ Diagnostics ████████████████████████████████████████████████████████████████████████ | comment line  
content, no runtime effect. |  
| 174 | pub active_dtcs: Vec<Dtc>, | executable/support line; behavior impact depends on surrounding block context.  
|  
| 175 | pub stored_dtcs: Vec<Dtc>, | executable/support line; behavior impact depends on surrounding block context.  
|  
| 176 | pub mil_active: bool, | executable/support line; behavior impact depends on surrounding block context. |  
| 177 | pub amber_lamp: bool, | executable/support line; behavior impact depends on surrounding block context. |  
| 178 | pub red_lamp: bool, | executable/support line; behavior impact depends on surrounding block context. |  
| 179 | pub protect_lamp: bool, | executable/support line; behavior impact depends on surrounding block context. |  
| 180 | (blank) | line is blank; visual separation only. |  
| 181 | // ■ Periodic TX timers ████████████████████████████████████████████████████████████████████████ | comment line  
content, no runtime effect. |  
| 182 | t_eec1: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 183 | t_eec2: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 184 | t_et1: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 185 | t_efl: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 186 | t_ic1: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 187 | t_lfe: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 188 | t_hrs: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 189 | t_dm1: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 190 | (blank) | line is blank; visual separation only. |  
| 191 | // ■ Internal dynamics ████████████████████████████████████████████████████████████████████████ | comment line  
content, no runtime effect. |  
| 192 | rpm_error_integral: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 193 | warmup_done: bool, | executable/support line; behavior impact depends on surrounding block context. |  
| 194 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 195 | (blank) | line is blank; visual separation only. |  
| 196 | impl Default for EcuEcm { | implementation block; method behavior and trait conformance are defined here. |  
| 197 | fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 198 |     Self::new() | executable/support line; behavior impact depends on surrounding block context. |  
| 199 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 200 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 201 | (blank) | line is blank; visual separation only. |  
| 202 | impl EcuEcm { | implementation block; method behavior and trait conformance are defined here. |  
| 203 | pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 204 |     EcuEcm { | executable/support line; behavior impact depends on surrounding block context. |  
| 205 |         sa: ECM_SA, | executable/support line; behavior impact depends on surrounding block context. |  
| 206 |         cylinders: CYLINDERS, | executable/support line; behavior impact depends on surrounding block context. |  
| 207 |         fuel_map: FuelMap::Standard, | executable/support line; behavior impact depends on surrounding block context. |  
| 208 |         governor_mode: GovernorMode::AllSpeed, | executable/support line; behavior impact depends on surrounding block context. |  
| 209 |         engine_hours: 2347.5, | executable/support line; behavior impact depends on surrounding block context. |  
| 210 |         service_interval_h: 500.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 211 |         service_hours_left: 152.5, | executable/support line; behavior impact depends on surrounding block context. |  
| 212 | (blank) | line is blank; visual separation only. |  
| 213 |         rpm: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 214 |         rpm_target: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 215 |         rpm_demand: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 216 |         rpm_limit: MAX_RPM, | executable/support line; behavior impact depends on surrounding block context. |  
| 217 | (blank) | line is blank; visual separation only. |  
| 218 |         actual_torque_nm: 0.0, | executable/support line; behavior impact depends on surrounding block context.
```

```
| 219 |          percent_load: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 220 |          actual_torque_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 221 |          demand_torque_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 222 |          driveline_torque_nm: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 223 | (blank) | line is blank; visual separation only. |
| 224 |          throttle_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 225 |          remote_throttle_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 226 |          active_throttle: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 227 | (blank) | line is blank; visual separation only. |
| 228 |          fuel_level_pct: 85.0, | executable/support line; behavior impact depends on surrounding block context. |
| 229 |          fuel_pressure_kpa: 450.0, | executable/support line; behavior impact depends on surrounding block context. |
| 230 |          fuel_rail_pressure_mpa: 180.0, | executable/support line; behavior impact depends on surrounding block context. |
| 231 |          fuel_rate_lph: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 232 |          total_fuel_l: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 233 |          trip_fuel_l: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 234 | (blank) | line is blank; visual separation only. |
| 235 |          coolant_temp_c: 20.0, | executable/support line; behavior impact depends on surrounding block context. |
| 236 |          oil_temp_c: 20.0, | executable/support line; behavior impact depends on surrounding block context. |
| 237 |          fuel_temp_c: 22.0, | executable/support line; behavior impact depends on surrounding block context. |
| 238 |          intake_temp_c: 22.0, | executable/support line; behavior impact depends on surrounding block context. |
| 239 |          exhaust_temp_c: 150.0, | executable/support line; behavior impact depends on surrounding block context. |
| 240 |          turbo_oil_temp_c: 20.0, | executable/support line; behavior impact depends on surrounding block context. |
| 241 |          ambient_temp_c: 22.0, | executable/support line; behavior impact depends on surrounding block context. |
| 242 | (blank) | line is blank; visual separation only. |
| 243 |          oil_pressure_kpa: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 244 |          oil_level_pct: 95.0, | executable/support line; behavior impact depends on surrounding block context. |
| 245 |          coolant_pres_kpa: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 246 |          boost_pressure_kpa: 100.0, | executable/support line; behavior impact depends on surrounding block context. |
| 247 |          vgt_position_pct: 50.0, | executable/support line; behavior impact depends on surrounding block context. |
| 248 |          egr_valve_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 249 |          air_filter_dp_kpa: 1.2, | executable/support line; behavior impact depends on surrounding block context. |
| 250 |          intake_manifold_kpa: 100.0, | executable/support line; behavior impact depends on surrounding block context. |
| 251 | (blank) | line is blank; visual separation only. |
| 252 |          battery_v: 12.8, | executable/support line; behavior impact depends on surrounding block context. |
| 253 |          alternator_v: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 254 |          starter_cranking: false, | executable/support line; behavior impact depends on surrounding block context. |
| 255 | (blank) | line is blank; visual separation only. |
```

```
| 256 |         def_level_pct: 72.0, | executable/support line; behavior impact depends on surrounding block context. |  
| . |  
| 257 |         def_quality_pct: 99.0, | executable/support line; behavior impact depends on surrounding block context. |  
xt. |  
| 258 |         dpf_soot_pct: 18.0, | executable/support line; behavior impact depends on surrounding block context. |  
| |  
| 259 |         dpf_temp_c: 200.0, | executable/support line; behavior impact depends on surrounding block context. |  
| |  
| 260 |         scr_inlet_temp_c: 200.0, | executable/support line; behavior impact depends on surrounding block context. |  
text. |  
| 261 |         scr_efficiency_pct: 95.0, | executable/support line; behavior impact depends on surrounding block context. |  
ntext. |  
| 262 |         nox_raw_ppm: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 263 |         nox_tailpipe_ppm: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
xt. |  
| 264 |         aftertreatment: AftertreatmentState::Normal, | executable/support line; behavior impact depends on surrounding block context. |  
urrounding block context. |  
| 265 |         regen_requested: false, | executable/support line; behavior impact depends on surrounding block context. |  
ext. |  
| 266 |         regen_inhibited: false, | executable/support line; behavior impact depends on surrounding block context. |  
ext. |  
| 267 | (blank) | line is blank; visual separation only. |  
| 268 |         active_dtcs: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |  
text. |  
| 269 |         stored_dtcs: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |  
text. |  
| 270 |         mil_active: false, | executable/support line; behavior impact depends on surrounding block context. |  
| |  
| 271 |         amber_lamp: false, | executable/support line; behavior impact depends on surrounding block context. |  
| |  
| 272 |         red_lamp: false, | executable/support line; behavior impact depends on surrounding block context. |  
| 273 |         protect_lamp: false, | executable/support line; behavior impact depends on surrounding block context. |  
. |  
| 274 | (blank) | line is blank; visual separation only. |  
| 275 |         t_eec1: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 276 |         t_eec2: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 277 |         t_et1: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 278 |         t_efl: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 279 |         t_ic1: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 280 |         t_lfe: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 281 |         t_hrs: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 282 |         t_dm1: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 283 | (blank) | line is blank; visual separation only. |  
| 284 |         rpm_error_integral: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
text. |  
| 285 |         warmup_done: false, | executable/support line; behavior impact depends on surrounding block context. |  
| |  
| 286 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| |  
| 287 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 288 | (blank) | line is blank; visual separation only. |  
| 289 |       // ████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect.  
nment, no runtime effect. |  
| 290 |       // Diesel torque curve: flat torque plateau 1200-1600 rpm (turbocharged) | comment line; documentation intent, no runtime effect.  
t, no runtime effect. |  
| 291 |       // Returns Nm for given RPM and throttle 0-100% | comment line; documentation intent, no runtime effect.  
| 292 |       pub fn torque_at_rpm(&self, rpm: f64, throttle: f64) -> f64 { | declares executable behavior; signature/b  
changes alter API, control flow, and outputs. |  
| 293 |           if rpm < 500.0 { | conditional gate; changing condition shifts runtime path and side effects. |  
| 294 |               return 0.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 295 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| |  
| 296 |           let t = throttle / 100.0; | executable/support line; behavior impact depends on surrounding block context. |  
t. |  
| 297 | (blank) | line is blank; visual separation only. |
```

[illegible]


```
| 337 | GovernorMode::Droop => throttle_demand_rpm - (load_nm / PEAK_TORQUE_NM) * 100.0, | executable/support  
t line; behavior impact depends on surrounding block context. |  
| 338 | GovernorMode::LowHighIdle => { | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 339 |         if self.active_throttle < 5.0 { | conditional gate; changing condition shifts runtime path an  
ide effects. |  
| 340 |             LO_IDLE_RPM | executable/support line; behavior impact depends on surrounding block conte  
|  
| 341 |             } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 342 |                 HI_IDLE_RPM | executable/support line; behavior impact depends on surrounding block conte  
|  
| 343 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti  
scopes. |  
| 344 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 345 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 346 |     self.rpm_target = target_rpm.clamp(LO_IDLE_RPM, self.rpm_limit); | executable/support line; behavior  
act depends on surrounding block context. |  
| 347 | (blank) | line is blank; visual separation only. |  
| 348 |     // ■ Engine dynamics (governor PI loop) ████████████████████████████████████████ | comment line; docum  
tent, no runtime effect. |  
| 349 |     let available_torque = self.torque_at_rpm(self.rpm, self.active_throttle); | executable/support line;  
havior impact depends on surrounding block context. |  
| 350 |     let net_torque = available_torque - load_nm; | executable/support line; behavior impact depends on su  
unding block context. |  
| 351 |     // Angular acceleration = net_torque / J_engine | comment line; documentation intent, no runtime effe  
|  
| 352 |     let rpm_accel = (net_torque / ENGINE_INERTIA_KGM2) * (60.0 / (2.0 * std::f64::consts::PI)); | executab  
/support line; behavior impact depends on surrounding block context. |  
| 353 |     let rpm_err = self.rpm_target - self.rpm; | executable/support line; behavior impact depends on surro  
ing block context. |  
| 354 |     self.rpm_error_integral = (self.rpm_error_integral + rpm_err * dt).clamp(-500.0, 500.0); | executable  
pport line; behavior impact depends on surrounding block context. |  
| 355 |     // Governor correction (PI) | comment line; documentation intent, no runtime effect. |  
| 356 |     let governor_correction = rpm_err * 0.8 + self.rpm_error_integral * 0.3; | executable/support line; b  
vior impact depends on surrounding block context. |  
| 357 |     self.rpm += (rpm_accel * 0.05 + governor_correction) * dt; | executable/support line; behavior impact  
ends on surrounding block context. |  
| 358 |     self.rpm = self.rpm.clamp(0.0, MAX_RPM); | executable/support line; behavior impact depends on surrou  
ng block context. |  
| 359 | (blank) | line is blank; visual separation only. |  
| 360 |     // Engine stalls if RPM drops below threshold with no throttle | comment line; documentation intent,  
runtime effect. |  
| 361 |     if self.rpm < 400.0 && self.active_throttle < 2.0 { | conditional gate; changing condition shifts run  
e path and side effects. |  
| 362 |         self.rpm = 0.0; | executable/support line; behavior impact depends on surrounding block context.  
| 363 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 364 | (blank) | line is blank; visual separation only. |  
| 365 |     // ■ Torque and load computation ████████████████████████████████████████ | comment line;  
intent, no runtime effect. |  
| 366 |     self.actual_torque_nm = available_torque.max(0.0); | executable/support line; behavior impact depends  
surrounding block context. |  
| 367 |     self.percent_load = (load_nm / PEAK_TORQUE_NM * 100.0).clamp(0.0, 150.0); | executable/support line;  
avio impact depends on surrounding block context. |  
| 368 |     self.actual_torque_pct = | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 369 |         (self.actual_torque_nm / PEAK_TORQUE_NM * 100.0 - 125.0).clamp(-125.0, 125.0); | executable/suppo  
line; behavior impact depends on surrounding block context. |  
| 370 |     self.demand_torque_pct = (self.active_throttle - 125.0).clamp(-125.0, 125.0); | executable/support li  
behavior impact depends on surrounding block context. |  
| 371 | (blank) | line is blank; visual separation only. |  
| 372 |     // ■ Fuel rail pressure (CRDI) ████████████████████████████████████████ | comment line; c  
ntent, no runtime effect. |
```

```
// Rail pressure: 300-2000 bar depending on RPM and load | comment line; documentation intent, no runtime effect. |
| 374 |         let rpm_norm = (self.rpm / RATED_RPM).clamp(0.2, 1.2); | executable/support line; behavior impact depends on surrounding block context. | | |
| 375 |         self.fuel_rail_pressure_mpa = 30.0 + rpm_norm * (self.active_throttle / 100.0) * 170.0; | executable/support line; behavior impact depends on surrounding block context. |
| 376 |         self.fuel_pressure_kpa = 400.0 + self.active_throttle * 2.0; | executable/support line; behavior impact depends on surrounding block context. |
| 377 | (blank) | line is blank; visual separation only. |
| 378 |         // ■ Fuel consumption ████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 379 |         let power_kw = self.actual_torque_nm * self.rpm * std::f64::consts::PI / 30000.0; | executable/support line; behavior impact depends on surrounding block context. |
| 380 |         // BSFC ~210 g/kWh; diesel density 840 g/L → L/h = kW * g/kWh / (g/L) | comment line; documentation intent, no runtime effect. |
| 381 |         let bsfc = 210.0 + (1.0 - self.active_throttle / 100.0) * 90.0; | executable/support line; behavior impact depends on surrounding block context. |
| 382 |         self.fuel_rate_lph = if self.rpm > 400.0 { | executable/support line; behavior impact depends on surrounding block context. |
| 383 |             (power_kw * bsfc / 840.0).max(1.5) | executable/support line; behavior impact depends on surrounding block context. |
| 384 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 385 |             0.0 | executable/support line; behavior impact depends on surrounding block context. |
| 386 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
| 387 |         let consumed = self.fuel_rate_lph / 3600.0 * dt; | executable/support line; behavior impact depends on surrounding block context. |
| 388 |         self.total_fuel_l += consumed; | executable/support line; behavior impact depends on surrounding block context. |
| 389 |         self.trip_fuel_l += consumed; | executable/support line; behavior impact depends on surrounding block context. |
| 390 |         self.fuel_level_pct = (self.fuel_level_pct - consumed / FUEL_TANK_L * 100.0).max(0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 391 | (blank) | line is blank; visual separation only. |
| 392 |         // ■ Thermal model ████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 393 |         self.warmup_done = self.warmup_done || self.coolant_temp_c > 82.0; | executable/support line; behavior impact depends on surrounding block context. |
| 394 |         let load_heat = self.percent_load / 100.0; | executable/support line; behavior impact depends on surrounding block context. |
| 395 |         let tgt_coolant = 88.0 + load_heat * 6.0; | executable/support line; behavior impact depends on surrounding block context. |
| 396 |         let warm_rate = if self.rpm > 500.0 { 0.015 } else { -0.008 }; | executable/support line; behavior impact depends on surrounding block context. |
| 397 |         self.coolant_temp_c += (tgt_coolant - self.coolant_temp_c) * warm_rate * dt; | executable/support line; behavior impact depends on surrounding block context. |
| 398 |         self.oil_temp_c = self.coolant_temp_c + 12.0 + load_heat * 5.0; | executable/support line; behavior impact depends on surrounding block context. |
| 399 |         self.exhaust_temp_c = 120.0 + load_heat * 520.0 * (self.rpm / RATED_RPM).min(1.0); | executable/support line; behavior impact depends on surrounding block context. |
| 400 |         self.turbo_oil_temp_c = self.exhaust_temp_c * 0.6; | executable/support line; behavior impact depends on surrounding block context. |
| 401 |         self.intake_temp_c = self.ambient_temp_c + self.boost_pressure_kpa / 100.0 * 18.0; | executable/support line; behavior impact depends on surrounding block context. |
| 402 | (blank) | line is blank; visual separation only. |
| 403 |         // ■ Pressure model ████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 404 |         self.oil_pressure_kpa = if self.rpm > 400.0 { | executable/support line; behavior impact depends on surrounding block context. |
| 405 |             150.0 + rpm_norm * 280.0 | executable/support line; behavior impact depends on surrounding block context. |
| 406 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 407 |             0.0 | executable/support line; behavior impact depends on surrounding block context. |
| 408 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope.
```

[illegible]

```
| tent, no runtime effect. |  
| 451 |         self.run_diagnostics(); | executable/support line; behavior impact depends on surrounding block context. |  
|   |  
| 452 | (blank) | line is blank; visual separation only. |  
| 453 |         // ■ Periodic J1939 transmission ████████████████████████████████████████ | comment line; does not have r  
content, no runtime effect. |  
| 454 |         self.t_eec1 += dt; | executable/support line; behavior impact depends on surrounding block context. |  
| 455 |         self.t_eec2 += dt; | executable/support line; behavior impact depends on surrounding block context. |  
| 456 |         self.t_et1 += dt; | executable/support line; behavior impact depends on surrounding block context. |  
| 457 |         self.t_efl += dt; | executable/support line; behavior impact depends on surrounding block context. |  
| 458 |         self.t_ic1 += dt; | executable/support line; behavior impact depends on surrounding block context. |  
| 459 |         self.t_lfe += dt; | executable/support line; behavior impact depends on surrounding block context. |  
| 460 |         self.t_hrs += dt; | executable/support line; behavior impact depends on surrounding block context. |  
| 461 |         self.t_dm1 += dt; | executable/support line; behavior impact depends on surrounding block context. |  
| 462 | (blank) | line is blank; visual separation only. |  
| 463 |         let ts = self.engine_hours; // use hours as timestamp (always increasing) | executable/support line;  
avior impact depends on surrounding block context. |  
| 464 |         let mut frames: Vec<J1939Frame> = Vec::new(); | executable/support line; behavior impact depends on s  
ounding block context. |  
| 465 | (blank) | line is blank; visual separation only. |  
| 466 |         if self.t_eec1 >= 0.010 { | conditional gate; changing condition shifts runtime path and side effects  
| 467 |             self.t_eec1 = 0.0; | executable/support line; behavior impact depends on surrounding block context.  
|   |  
| 468 |             frames.push(Builder::eec1(| | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 469 |                 ts, | executable/support line; behavior impact depends on surrounding block context. |  
| 470 |                 self.rpm, | executable/support line; behavior impact depends on surrounding block context. |  
| 471 |                 self.actual_torque_pct, | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 472 |                 self.demand_torque_pct, | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 473 |                 self.sa, | executable/support line; behavior impact depends on surrounding block context. |  
| 474 |             )); | executable/support line; behavior impact depends on surrounding block context. |  
| 475 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|   |  
| 476 |         if self.t_eec2 >= 0.050 { | conditional gate; changing condition shifts runtime path and side effects  
| 477 |             self.t_eec2 = 0.0; | executable/support line; behavior impact depends on surrounding block context.  
|   |  
| 478 |             frames.push(Builder::eec2(| | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 479 |                 ts, | executable/support line; behavior impact depends on surrounding block context. |  
| 480 |                 self.active_throttle, | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 481 |                 self.percent_load, | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 482 |                 self.sa, | executable/support line; behavior impact depends on surrounding block context. |  
| 483 |             )); | executable/support line; behavior impact depends on surrounding block context. |  
| 484 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|   |  
| 485 |         if self.t_et1 >= 1.000 { | conditional gate; changing condition shifts runtime path and side effects.  
| 486 |             self.t_et1 = 0.0; | executable/support line; behavior impact depends on surrounding block context  
| 487 |             frames.push(Builder::et1(| | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 488 |                 ts, | executable/support line; behavior impact depends on surrounding block context. |  
| 489 |                 self.coolant_temp_c, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 490 |                 self.fuel_temp_c, | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 491 |                 self.oil_temp_c, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 492 |                 self.sa, | executable/support line; behavior impact depends on surrounding block context. |  
| 493 |             )); | executable/support line; behavior impact depends on surrounding block context. |  
| 494 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|   |  
| 495 |         if self.t_efl >= 0.500 { | conditional gate; changing condition shifts runtime path and side effects.
```

```

| 496 |         self.t_efl = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 497 |         frames.push(Builder::efl_p1( | executable/support line; behavior impact depends on surrounding block context. |
| 498 |             ts, | executable/support line; behavior impact depends on surrounding block context. |
| 499 |             self.fuel_pressure_kpa, | executable/support line; behavior impact depends on surrounding block context. |
| 500 |             self.oil_level_pct, | executable/support line; behavior impact depends on surrounding block context. |
| 501 |             self.oil_pressure_kpa, | executable/support line; behavior impact depends on surrounding block context. |
| 502 |             self.sa, | executable/support line; behavior impact depends on surrounding block context. |
| 503 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 504 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 505 |     if self.t_ic1 >= 0.500 { | conditional gate; changing condition shifts runtime path and side effects.
| 506 |         self.t_ic1 = 0.0; | executable/support line; behavior impact depends on surrounding block context.
| 507 |         frames.push(Builder::ic1( | executable/support line; behavior impact depends on surrounding block context.
| 508 |             ts, | executable/support line; behavior impact depends on surrounding block context. |
| 509 |             self.boost_pressure_kpa, | executable/support line; behavior impact depends on surrounding block context. |
| 510 |             self.intake_temp_c, | executable/support line; behavior impact depends on surrounding block context.
| 511 |             self.exhaust_temp_c, | executable/support line; behavior impact depends on surrounding block context.
| 512 |             self.sa, | executable/support line; behavior impact depends on surrounding block context. |
| 513 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 514 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 515 |     if self.t_lfe >= 0.100 { | conditional gate; changing condition shifts runtime path and side effects.
| 516 |         self.t_lfe = 0.0; | executable/support line; behavior impact depends on surrounding block context.
| 517 |         frames.push(Builder::lfe( | executable/support line; behavior impact depends on surrounding block context.
| 518 |             ts, | executable/support line; behavior impact depends on surrounding block context. |
| 519 |             self.fuel_rate_lph, | executable/support line; behavior impact depends on surrounding block context.
| 520 |             self.active_throttle, | executable/support line; behavior impact depends on surrounding block context.
| 521 |             self.sa, | executable/support line; behavior impact depends on surrounding block context. |
| 522 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 523 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 524 |     if self.t_hrs >= 1.000 { | conditional gate; changing condition shifts runtime path and side effects.
| 525 |         self.t_hrs = 0.0; | executable/support line; behavior impact depends on surrounding block context.
| 526 |         frames.push(Builder::hours(ts, self.engine_hours, self.sa)); | executable/support line; behavior impact depends on surrounding block context.
| 527 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 528 |     if self.t_dm1 >= 1.000 { | conditional gate; changing condition shifts runtime path and side effects.
| 529 |         self.t_dm1 = 0.0; | executable/support line; behavior impact depends on surrounding block context.
| 530 |         let (spn, fmi) = self.first_active_dtc_spn_fmi(); | executable/support line; behavior impact depends on surrounding block context.
| 531 |         frames.push(Builder::dm1( | executable/support line; behavior impact depends on surrounding block context.
| 532 |             ts, | executable/support line; behavior impact depends on surrounding block context. |
| 533 |             self.amber_lamp, | executable/support line; behavior impact depends on surrounding block context.
| 534 |             self.red_lamp, | executable/support line; behavior impact depends on surrounding block context.
| 535 |             self.mil_active, | executable/support line; behavior impact depends on surrounding block context.
| 536 |             spn, | executable/support line; behavior impact depends on surrounding block context. |
| 537 |             fmi, | executable/support line; behavior impact depends on surrounding block context. |
| 538 |             self.sa, | executable/support line; behavior impact depends on surrounding block context. |
| 539 |             )); | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 540 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 541 | (blank) | line is blank; visual separation only. |
| 542 |         frames | executable/support line; behavior impact depends on surrounding block context. |
| 543 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 544 | (blank) | line is blank; visual separation only. |
| 545 |     fn update_aftertreatment(&mut self, dt: f64) { | declares executable behavior; signature/body changes alt
API, control flow, and outputs. |
| 546 |         let load = self.percent_load / 100.0; | executable/support line; behavior impact depends on surroundi
block context. |
| 547 |         let rpm_running = self.rpm > 400.0; | executable/support line; behavior impact depends on surrounding
ock context. |
| 548 | (blank) | line is blank; visual separation only. |
| 549 |         // DPF soot accumulates with load, decreases during regen | comment line; documentation intent, no ru
me effect. |
| 550 |         if rpm_running { | conditional gate; changing condition shifts runtime path and side effects. |
| 551 |             let soot_rate = match self.aftertreatment { | executable/support line; behavior impact depends on
rrounding block context. |
| 552 |                 AftertreatmentState::DpfRegen => -3.0 * dt, // burn soot at 3%/s | executable/support line; b
vior impact depends on surrounding block context. |
| 553 |                 _ => 0.0008 * load * dt, | executable/support line; behavior impact depends on surrounding bl
context. |
| 554 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 555 |             self.dpf_soot_pct = (self.dpf_soot_pct + soot_rate).clamp(0.0, 100.0); | executable/support line;
havior impact depends on surrounding block context. |
| 556 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 557 | (blank) | line is blank; visual separation only. |
| 558 |         // DPF temperature | comment line; documentation intent, no runtime effect. |
| 559 |         let dpf_target = if self.aftertreatment == AftertreatmentState::DpfRegen { | executable/support line;
havior impact depends on surrounding block context. |
| 560 |             620.0 | executable/support line; behavior impact depends on surrounding block context. |
| 561 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 562 |             200.0 + self.exhaust_temp_c * 0.5 | executable/support line; behavior impact depends on surroundi
block context. |
| 563 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 564 |         self.dpf_temp_c += (dpf_target - self.dpf_temp_c) * dt * 0.05; | executable/support line; behavior im
t depends on surrounding block context. |
| 565 | (blank) | line is blank; visual separation only. |
| 566 |         // SCR inlet temperature | comment line; documentation intent, no runtime effect. |
| 567 |         self.scr_inlet_temp_c = self.dpf_temp_c * 0.85; | executable/support line; behavior impact depends on
rrounding block context. |
| 568 | (blank) | line is blank; visual separation only. |
| 569 |         // SCR efficiency depends on temperature and DEF availability | comment line; documentation intent, n
untime effect. |
| 570 |         let scr_temp_ok = self.scr_inlet_temp_c > 200.0; | executable/support line; behavior impact depends o
urrounding block context. |
| 571 |         self.scr_efficiency_pct = if self.def_level_pct > 5.0 && scr_temp_ok { | executable/support line; beh
or impact depends on surrounding block context. |
| 572 |             93.0 + load * 4.0 | executable/support line; behavior impact depends on surrounding block context
| 573 |         } else if self.def_level_pct > 2.0 { | executable/support line; behavior impact depends on surroundin
lock context. |
| 574 |             40.0 | executable/support line; behavior impact depends on surrounding block context. |
| 575 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 576 |             5.0 // near zero DEF | executable/support line; behavior impact depends on surrounding block cont
. |
| 577 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 578 | (blank) | line is blank; visual separation only. |
| 579 |         // DEF consumption: ~3-5% of fuel consumed | comment line; documentation intent, no runtime effect. |
| 580 |         let def_consumed = self.fuel_rate_lph * 0.04 / 3600.0 * dt; | executable/support line; behavior impac
depends on surrounding block context. |
| 581 |         self.def_level_pct = (self.def_level_pct - def_consumed / DEF_TANK_L * 100.0).max(0.0); | executable/

```

```

port line; behavior impact depends on surrounding block context. |
| 582 | (blank) | line is blank; visual separation only. |
| 583 | // NOx | comment line; documentation intent, no runtime effect. |
| 584 | self.nox_raw_ppm = if rpm_running { | executable/support line; behavior impact depends on surrounding
ock context. |
| 585 | 800.0 + load * 1200.0 * (1.0 - self.egr_valve_pct / 80.0) | executable/support line; behavior imp
depends on surrounding block context. |
| 586 | } else { | executable/support line; behavior impact depends on surrounding block context. |
| 587 | 0.0 | executable/support line; behavior impact depends on surrounding block context. |
| 588 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 589 | self.nox_tailpipe_ppm = self.nox_raw_ppm * (1.0 - self.scr_efficiency_pct / 100.0); | executable/supp
line; behavior impact depends on surrounding block context. |
| 590 | (blank) | line is blank; visual separation only. |
| 591 | // Aftertreatment state transitions | comment line; documentation intent, no runtime effect. |
| 592 | self.aftertreatment = if self.dpf_soot_pct > DPF_REGEN_SOOT && !self.regen_inhibited { | executable/s
ort line; behavior impact depends on surrounding block context. |
| 593 | self.regen_requested = true; | executable/support line; behavior impact depends on surrounding bl
context. |
| 594 | AftertreatmentState::DpfRegen | executable/support line; behavior impact depends on surrounding b
k context. |
| 595 | } else if self.dpf_soot_pct < 5.0 { | executable/support line; behavior impact depends on surrounding
ock context. |
| 596 | self.regen_requested = false; | executable/support line; behavior impact depends on surrounding b
k context. |
| 597 | AftertreatmentState::Normal | executable/support line; behavior impact depends on surrounding blo
context. |
| 598 | } else if self.def_level_pct < 2.0 { | executable/support line; behavior impact depends on surroundin
lock context. |
| 599 | AftertreatmentState::Derating | executable/support line; behavior impact depends on surrounding b
k context. |
| 600 | } else if self.scr_efficiency_pct < 50.0 { | executable/support line; behavior impact depends on surr
ding block context. |
| 601 | AftertreatmentState::ScrDegraded | executable/support line; behavior impact depends on surroundin
lock context. |
| 602 | } else { | executable/support line; behavior impact depends on surrounding block context. |
| 603 | self.aftertreatment | executable/support line; behavior impact depends on surrounding block conte
|
| 604 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 605 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 606 | (blank) | line is blank; visual separation only. |
| 607 | fn run_diagnostics(&mut self) { | declares executable behavior; signature/body changes alter API, control
ow, and outputs. |
| 608 | self.active_dtcs.retain(\d\ d.active); | executable/support line; behavior impact depends on surroun
ng block context. |
| 609 | self.mil_active = false; | executable/support line; behavior impact depends on surrounding block cont
. |
| 610 | self.amber_lamp = false; | executable/support line; behavior impact depends on surrounding block cont
. |
| 611 | self.red_lamp = false; | executable/support line; behavior impact depends on surrounding block context
|
| 612 | self.protect_lamp = false; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 613 | (blank) | line is blank; visual separation only. |
| 614 | // SPN 110: Engine Coolant Temperature | comment line; documentation intent, no runtime effect. |
| 615 | if self.coolant_temp_c > 107.0 { | conditional gate; changing condition shifts runtime path and side
ects. |
| 616 | self.add_dtc( | executable/support line; behavior impact depends on surrounding block context. |
| 617 | 110, | executable/support line; behavior impact depends on surrounding block context. |
| 618 | 0, | executable/support line; behavior impact depends on surrounding block context. |
| 619 | DtcSeverity::Red, | executable/support line; behavior impact depends on surrounding block con
t. |
| 620 | "Coolant temp above normal – Red Stop", | executable/support line; behavior impact depends on
rrounding block context. |

```

```

| 621 |         true, | executable/support line; behavior impact depends on surrounding block context. |
| 622 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 623 |     self.red_lamp = true; | executable/support line; behavior impact depends on surrounding block context. |
| 624 | } else if self.coolant_temp_c > 102.0 { | executable/support line; behavior impact depends on surrounding block context. |
| 625 |     self.add_dtc( | executable/support line; behavior impact depends on surrounding block context. |
| 626 |         110, | executable/support line; behavior impact depends on surrounding block context. |
| 627 |         15, | executable/support line; behavior impact depends on surrounding block context. |
| 628 |         DtcSeverity::Amber, | executable/support line; behavior impact depends on surrounding block context. |
| 629 |         "Coolant temp high – Amber Warning", | executable/support line; behavior impact depends on surrounding block context. |
| 630 |         true, | executable/support line; behavior impact depends on surrounding block context. |
| 631 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 632 |     self.amber_lamp = true; | executable/support line; behavior impact depends on surrounding block context. |
| 633 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope |
| 634 | (blank) | line is blank; visual separation only. |
| 635 | // SPN 100: Engine Oil Pressure | comment line; documentation intent, no runtime effect. |
| 636 | if self.rpm > 600.0 && self.oil_pressure_kpa < 80.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 637 |     self.add_dtc( | executable/support line; behavior impact depends on surrounding block context. |
| 638 |         100, | executable/support line; behavior impact depends on surrounding block context. |
| 639 |         1, | executable/support line; behavior impact depends on surrounding block context. |
| 640 |         DtcSeverity::Red, | executable/support line; behavior impact depends on surrounding block context. |
| 641 |         "Engine oil pressure critically low – Red Stop", | executable/support line; behavior impact depends on surrounding block context. |
| 642 |         true, | executable/support line; behavior impact depends on surrounding block context. |
| 643 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 644 |     self.red_lamp = true; | executable/support line; behavior impact depends on surrounding block context. |
| 645 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope |
| 646 | (blank) | line is blank; visual separation only. |
| 647 | // SPN 183: Fuel rate (SPN 94 = fuel delivery pressure) | comment line; documentation intent, no runtime effect. |
| 648 | if self.fuel_pressure_kpa < 150.0 && self.rpm > 600.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 649 |     self.add_dtc( | executable/support line; behavior impact depends on surrounding block context. |
| 650 |         94, | executable/support line; behavior impact depends on surrounding block context. |
| 651 |         1, | executable/support line; behavior impact depends on surrounding block context. |
| 652 |         DtcSeverity::Amber, | executable/support line; behavior impact depends on surrounding block context. |
| 653 |         "Fuel delivery pressure low – check fuel filter", | executable/support line; behavior impact depends on surrounding block context. |
| 654 |         true, | executable/support line; behavior impact depends on surrounding block context. |
| 655 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 656 |     self.amber_lamp = true; | executable/support line; behavior impact depends on surrounding block context. |
| 657 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope |
| 658 | (blank) | line is blank; visual separation only. |
| 659 | // SPN 3361: DEF Level | comment line; documentation intent, no runtime effect. |
| 660 | if self.def_level_pct < 10.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 661 |     self.add_dtc( | executable/support line; behavior impact depends on surrounding block context. |
| 662 |         3361, | executable/support line; behavior impact depends on surrounding block context. |
| 663 |         17, | executable/support line; behavior impact depends on surrounding block context. |
| 664 |         DtcSeverity::Amber, | executable/support line; behavior impact depends on surrounding block context. |
| 665 |         "DEF level low – refill before derating", | executable/support line; behavior impact depends on surrounding block context. |

```



```

| 666 |         true, | executable/support line; behavior impact depends on surrounding block context. |
| 667 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 668 |         self.amber_lamp = true; | executable/support line; behavior impact depends on surrounding block context. |
| 669 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 670 |     if self.def_level_pct < 2.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 671 |         self.add_dtc( | executable/support line; behavior impact depends on surrounding block context. |
| 672 |             3361, | executable/support line; behavior impact depends on surrounding block context. |
| 673 |             1, | executable/support line; behavior impact depends on surrounding block context. |
| 674 |             DtcSeverity::Red, | executable/support line; behavior impact depends on surrounding block context. |
| 675 |             "DEF level critically low – power derate active", | executable/support line; behavior impact depends on surrounding block context. |
| 676 |             true, | executable/support line; behavior impact depends on surrounding block context. |
| 677 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 678 |             self.red_lamp = true; | executable/support line; behavior impact depends on surrounding block context. |
| 679 |             self.rpm_limit = RATED_RPM * 0.60; // 60% derating | executable/support line; behavior impact depends on surrounding block context. |
| 680 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 681 |             self.rpm_limit = MAX_RPM; | executable/support line; behavior impact depends on surrounding block context. |
| 682 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 683 | (blank) | line is blank; visual separation only. |
| 684 |         // SPN 3251: DPF Soot Load | comment line; documentation intent, no runtime effect. |
| 685 |         if self.dpf_soot_pct > 80.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 686 |             self.add_dtc( | executable/support line; behavior impact depends on surrounding block context. |
| 687 |                 3251, | executable/support line; behavior impact depends on surrounding block context. |
| 688 |                 16, | executable/support line; behavior impact depends on surrounding block context. |
| 689 |                 DtcSeverity::Amber, | executable/support line; behavior impact depends on surrounding block context. |
| 690 |                 "DPF soot load high – manual regen required", | executable/support line; behavior impact depends on surrounding block context. |
| 691 |                 true, | executable/support line; behavior impact depends on surrounding block context. |
| 692 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 693 |                 self.amber_lamp = true; | executable/support line; behavior impact depends on surrounding block context. |
| 694 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 695 | (blank) | line is blank; visual separation only. |
| 696 |         // SPN 102: Boost / Intake manifold pressure | comment line; documentation intent, no runtime effect. |
| 697 |         if self.boost_pressure_kpa > 320.0 \|\| self.intake_manifold_kpa > 320.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 698 |             self.add_dtc( | executable/support line; behavior impact depends on surrounding block context. |
| 699 |                 102, | executable/support line; behavior impact depends on surrounding block context. |
| 700 |                 0, | executable/support line; behavior impact depends on surrounding block context. |
| 701 |                 DtcSeverity::Red, | executable/support line; behavior impact depends on surrounding block context. |
| 702 |                 "Boost pressure above safe limit – inspect VGT/wastegate", | executable/support line; behavior impact depends on surrounding block context. |
| 703 |                 true, | executable/support line; behavior impact depends on surrounding block context. |
| 704 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 705 |                 self.red_lamp = true; | executable/support line; behavior impact depends on surrounding block context. |
| 706 |             } else if self.rpm > 1000.0 && self.boost_pressure_kpa < 95.0 { | executable/support line; behavior impact depends on surrounding block context. |
| 707 |                 self.add_dtc( | executable/support line; behavior impact depends on surrounding block context. |
| 708 |                     102, | executable/support line; behavior impact depends on surrounding block context. |
| 709 |                     1, | executable/support line; behavior impact depends on surrounding block context. |
| 710 |                     DtcSeverity::Amber, | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 711 |         "Boost pressure below expected level – possible leak or turbo issue", | executable/support line; behavior impact depends on surrounding block context. |
| 712 |         true, | executable/support line; behavior impact depends on surrounding block context. |
| 713 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 714 |     self.amber_lamp = true; | executable/support line; behavior impact depends on surrounding block context. |
| 715 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 716 | (blank) | line is blank; visual separation only. |
| 717 |     // SPN 111: Coolant pressure low | comment line; documentation intent, no runtime effect. |
| 718 |     if self.rpm > 600.0 && self.coolant_pres_kpa < 55.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 719 |         self.add_dtc( | executable/support line; behavior impact depends on surrounding block context. |
| 720 |             111, | executable/support line; behavior impact depends on surrounding block context. |
| 721 |             1, | executable/support line; behavior impact depends on surrounding block context. |
| 722 |             DtcSeverity::Amber, | executable/support line; behavior impact depends on surrounding block context. |
| 723 |         "Coolant pressure low – inspect cap/pump/leak", | executable/support line; behavior impact depends on surrounding block context. |
| 724 |         true, | executable/support line; behavior impact depends on surrounding block context. |
| 725 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 726 |     self.amber_lamp = true; | executable/support line; behavior impact depends on surrounding block context. |
| 727 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 728 | (blank) | line is blank; visual separation only. |
| 729 |     // SPN 105: Intake manifold temperature high | comment line; documentation intent, no runtime effect. |
| 730 |     if self.intake_temp_c > 95.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 731 |         self.add_dtc( | executable/support line; behavior impact depends on surrounding block context. |
| 732 |             105, | executable/support line; behavior impact depends on surrounding block context. |
| 733 |             15, | executable/support line; behavior impact depends on surrounding block context. |
| 734 |             DtcSeverity::Amber, | executable/support line; behavior impact depends on surrounding block context. |
| 735 |         "Intake manifold temperature high – charge-air cooling degraded", | executable/support line; behavior impact depends on surrounding block context. |
| 736 |         true, | executable/support line; behavior impact depends on surrounding block context. |
| 737 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 738 |     self.amber_lamp = true; | executable/support line; behavior impact depends on surrounding block context. |
| 739 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 740 | (blank) | line is blank; visual separation only. | | |
| 741 |     // SPN 173: Exhaust temperature severe | comment line; documentation intent, no runtime effect. |
| 742 |     if self.exhaust_temp_c > 820.0 || self.dpf_temp_c > 760.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 743 |         self.add_dtc( | executable/support line; behavior impact depends on surrounding block context. |
| 744 |             173, | executable/support line; behavior impact depends on surrounding block context. |
| 745 |             0, | executable/support line; behavior impact depends on surrounding block context. |
| 746 |             DtcSeverity::Red, | executable/support line; behavior impact depends on surrounding block context. |
| 747 |         "Exhaust temperature above severe threshold", | executable/support line; behavior impact depends on surrounding block context. |
| 748 |         true, | executable/support line; behavior impact depends on surrounding block context. |
| 749 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 750 |     self.red_lamp = true; | executable/support line; behavior impact depends on surrounding block context. |
| 751 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 752 | (blank) | line is blank; visual separation only. |
| 753 |     // SPN 1761: DEF quality abnormal | comment line; documentation intent, no runtime effect. |
| 754 |     if self.def_quality_pct < 80.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 755 |         self.add_dtc( | executable/support line; behavior impact depends on surrounding block context. |
| 756 |             1761, | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 757 |           3, | executable/support line; behavior impact depends on surrounding block context. |
| 758 |           DtcSeverity::Amber, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 759 |           "DEF quality out of expected range", | executable/support line; behavior impact depends on su
unding block context. |
| 760 |           true, | executable/support line; behavior impact depends on surrounding block context. |
| 761 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 762 |           self.amber_lamp = true; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 763 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 764 | (blank) | line is blank; visual separation only. |
| 765 |       // SPN 3226: Tailpipe NOx excessive | comment line; documentation intent, no runtime effect. |
| 766 |       if self.nox_tailpipe_ppm > 900.0 { | conditional gate; changing condition shifts runtime path and sid
ffects. |
| 767 |           self.add_dtc( | executable/support line; behavior impact depends on surrounding block context. |
| 768 |               3226, | executable/support line; behavior impact depends on surrounding block context. |
| 769 |               16, | executable/support line; behavior impact depends on surrounding block context. |
| 770 |               DtcSeverity::Red, | executable/support line; behavior impact depends on surrounding block con
t. |
| 771 |               "Tailpipe NOx above legal threshold", | executable/support line; behavior impact depends on s
ounding block context. |
| 772 |               true, | executable/support line; behavior impact depends on surrounding block context. |
| 773 |               ); | executable/support line; behavior impact depends on surrounding block context. |
| 774 |               self.red_lamp = true; | executable/support line; behavior impact depends on surrounding block con
t. |
| 775 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 776 | (blank) | line is blank; visual separation only. |
| 777 |       // SPN 157: Common rail pressure low | comment line; documentation intent, no runtime effect. |
| 778 |       if self.rpm > 900.0 && self.fuel_rail_pressure_mpa < 70.0 { | conditional gate; changing condition sh
s runtime path and side effects. |
| 779 |           self.add_dtc( | executable/support line; behavior impact depends on surrounding block context. |
| 780 |               157, | executable/support line; behavior impact depends on surrounding block context. |
| 781 |               1, | executable/support line; behavior impact depends on surrounding block context. |
| 782 |               DtcSeverity::Amber, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 783 |           "Fuel rail pressure below expected range", | executable/support line; behavior impact depends
surrounding block context. |
| 784 |           true, | executable/support line; behavior impact depends on surrounding block context. |
| 785 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 786 |           self.amber_lamp = true; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 787 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 788 | (blank) | line is blank; visual separation only. |
| 789 |       // SPN 247: Service interval | comment line; documentation intent, no runtime effect. |
| 790 |       if self.service_hours_left < 0.0 { | conditional gate; changing condition shifts runtime path and sid
ffects. |
| 791 |           self.add_dtc(247, 31, DtcSeverity::Mil, "Service interval overdue", true); | executable/support l
; behavior impact depends on surrounding block context. |
| 792 |           self.mil_active = true; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 793 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 794 | (blank) | line is blank; visual separation only. |
| 795 |       // Propagate lamp states from DTC list | comment line; documentation intent, no runtime effect. |
| 796 |       for dtc in &self.active_dtcs { | iteration control; changing bounds/body affects throughput, timing,
possibly safety guarantees. |
| 797 |           match dtc.severity { | branch dispatch; changing arms alters behavior for variants and input clas
. |
| 798 |               DtcSeverity::Red => self.red_lamp = true, | executable/support line; behavior impact depends
surrounding block context. |
| 799 |               DtcSeverity::Amber => self.amber_lamp = true, | executable/support line; behavior impact depe
on surrounding block context. |

```

```

| 800 |           DtcSeverity::Mil => self.mil_active = true, | executable/support line; behavior impact depends
n surrounding block context. |
| 801 |           DtcSeverity::Protect => self.protect_lamp = true, | executable/support line; behavior impact
ends on surrounding block context. |
| 802 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 803 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 804 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 805 | (blank) | line is blank; visual separation only. |
| 806 |   fn add_dtc(&mut self, spn: u32, fmi: u8, sev: DtcSeverity, desc: &'static str, active: bool) { | declares
executable behavior; signature/body changes alter API, control flow, and outputs. |
| 807 |       if !self | conditional gate; changing condition shifts runtime path and side effects. |
| 808 |           .active_dtcs | executable/support line; behavior impact depends on surrounding block context. |
| 809 |           .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 810 |           .any(\d\ d.spn == spn && d.fmi == fmi) | executable/support line; behavior impact depends on su
unding block context. |
| 811 |       { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 812 |           // Move to stored if was previously active | comment line; documentation intent, no runtime effec
|
| 813 |           self.stored_dtcs.push(Dtc { | executable/support line; behavior impact depends on surrounding blo
context. |
| 814 |               spn, | executable/support line; behavior impact depends on surrounding block context. |
| 815 |               fmi, | executable/support line; behavior impact depends on surrounding block context. |
| 816 |               count: 1, | executable/support line; behavior impact depends on surrounding block context. |
| 817 |               active: false, | executable/support line; behavior impact depends on surrounding block contex
|
| 818 |               desc, | executable/support line; behavior impact depends on surrounding block context. |
| 819 |               severity: sev, | executable/support line; behavior impact depends on surrounding block contex
|
| 820 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 821 |       if active { | conditional gate; changing condition shifts runtime path and side effects. |
| 822 |           self.active_dtcs.push(Dtc { | executable/support line; behavior impact depends on surrounding
ock context. |
| 823 |               spn, | executable/support line; behavior impact depends on surrounding block context. |
| 824 |               fmi, | executable/support line; behavior impact depends on surrounding block context. |
| 825 |               count: 1, | executable/support line; behavior impact depends on surrounding block context
| 826 |               active: true, | executable/support line; behavior impact depends on surrounding block con
t. |
| 827 |               desc, | executable/support line; behavior impact depends on surrounding block context. |
| 828 |               severity: sev, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 829 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 830 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 831 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 832 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 833 | (blank) | line is blank; visual separation only. |
| 834 |   fn first_active_dtc_spn_fmi(&self) -> (u32, u8) { | declares executable behavior; signature/body changes
er API, control flow, and outputs. |
| 835 |       self.active_dtcs | executable/support line; behavior impact depends on surrounding block context. |
| 836 |       .first() | executable/support line; behavior impact depends on surrounding block context. |
| 837 |       .map(\d\ (d.spn, d.fmi)) | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 838 |       .unwrap_or((0, 0)) | executable/support line; behavior impact depends on surrounding block contex
|
| 839 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 840 | (blank) | line is blank; visual separation only. |
| 841 |   pub fn clear_dtcs(&mut self) { | declares executable behavior; signature/body changes alter API, control
w, and outputs. |
| 842 |       self.active_dtcs.clear(); | executable/support line; behavior impact depends on surrounding block con
t. |
| 843 |       self.mil_active = false; | executable/support line; behavior impact depends on surrounding block cont

```

[illegible]

```
## File: src/ecu_hcm.rs
```

Role: Domain module in simulation/control/protocol stack.

```
Lines: 399 | Non-empty: 356 | Symbols: 21
```

Symbol Inventory

```
| Line | Kind | Symbol | Change Impact |
|---:|---|---|---|
| 16 | const | HCM_SA | data/structure coupling |
| 20 | enum | PumpMode | data/structure coupling |
| 29 | impl | std::fmt::Display for PumpMode | behavior binding and method semantics |
| 30 | fn | fmt | signature/API coupling |
| 41 | struct | HydActuator | data/structure coupling |
| 55 | impl | HydActuator | behavior binding and method semantics |
| 56 | fn | new | signature/API coupling |
| 73 | fn | area_cap_cm2 | signature/API coupling |
| 76 | fn | area_rod_cm2 | signature/API coupling |
| 81 | fn | update | signature/API coupling |
| 137 | fn | position_pct | signature/API coupling |
| 144 | struct | EcuHcm | data/structure coupling |
| 198 | impl | Default for EcuHcm | behavior binding and method semantics |
| 199 | fn | default | signature/API coupling |
| 204 | impl | EcuHcm | behavior binding and method semantics |
| 205 | fn | new | signature/API coupling |
| 244 | fn | tick | signature/API coupling |
| 385 | fn | set_hitch_cmd | signature/API coupling |
| 388 | fn | set_loader_lift_cmd | signature/API coupling |
| 391 | fn | set_loader_tilt_cmd | signature/API coupling |
| 394 | fn | set_aux_cmd | signature/API coupling |
```

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	/// Hydraulic Control Module (HCM) – SA 0x1E.	comment line; documentation intent, no runtime effect.


```

| 46 |     pub position_mm: f64,    // current extension (0=fully retracted) | executable/support line; behavior impact
depends on surrounding block context. |
| 47 |     pub velocity_mm_s: f64, // positive = extending | executable/support line; behavior impact depends on surr
ounding block context. |
| 48 |     pub pres_a_bar: f64,     // cap-end pressure | executable/support line; behavior impact depends on surround
block context. |
| 49 |     pub pres_b_bar: f64,     // rod-end pressure | executable/support line; behavior impact depends on surround
block context. |
| 50 |     pub force_kn: f64,      // net push force (positive = extending) | executable/support line; behavior impac
depends on surrounding block context. |
| 51 |     pub valve_cmd: f64,     // -1 retract, 0 hold, +1 extend | executable/support line; behavior impact depend
n surrounding block context. |
| 52 |     pub leakage_cc_m: f64,  // internal leakage cm³/min | executable/support line; behavior impact depends on
rounding block context. |
| 53 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 54 | (blank) | line is blank; visual separation only. |
| 55 | impl HydActuator { | implementation block; method behavior and trait conformance are defined here. |
| 56 |     pub fn new(name: &'static str, bore_mm: f64, rod_mm: f64, stroke_mm: f64) -> Self { | declares executable
avior; signature/body changes alter API, control flow, and outputs. |
| 57 |         HydActuator { | executable/support line; behavior impact depends on surrounding block context. |
| 58 |             name, | executable/support line; behavior impact depends on surrounding block context. |
| 59 |             bore_mm, | executable/support line; behavior impact depends on surrounding block context. |
| 60 |             rod_mm, | executable/support line; behavior impact depends on surrounding block context. |
| 61 |             stroke_mm, | executable/support line; behavior impact depends on surrounding block context. |
| 62 |             position_mm: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 63 |             velocity_mm_s: 0.0, | executable/support line; behavior impact depends on surrounding block context
|
| 64 |             pres_a_bar: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 65 |             pres_b_bar: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 66 |             force_kn: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 67 |             valve_cmd: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 68 |             leakage_cc_m: 0.5, | executable/support line; behavior impact depends on surrounding block context
| 69 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 70 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 71 | (blank) | line is blank; visual separation only. |
| 72 |     /// Area of each side in cm² | comment line; documentation intent, no runtime effect. |
| 73 |     pub fn area_cap_cm2(&self) -> f64 { | declares executable behavior; signature/body changes alter API, cont
flow, and outputs. |
| 74 |         std::f64::consts::PI * (self.bore_mm / 20.0).powi(2) | executable/support line; behavior impact depend
n surrounding block context. |
| 75 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 76 |     pub fn area_rod_cm2(&self) -> f64 { | declares executable behavior; signature/body changes alter API, cont
flow, and outputs. |
| 77 |         self.area_cap_cm2() - std::f64::consts::PI * (self.rod_mm / 20.0).powi(2) | executable/support line; b
avior impact depends on surrounding block context. |
| 78 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 79 | (blank) | line is blank; visual separation only. |
| 80 |     /// Update given supply pressure and flow from the pump. Returns flow consumed (L/min). | comment line; do
mentation intent, no runtime effect. |
| 81 |     pub fn update(&mut self, supply_bar: f64, dt: f64) -> f64 { | declares executable behavior; signature/body
anges alter API, control flow, and outputs. |
| 82 |         if self.valve_cmd.abs() < 0.01 { | conditional gate; changing condition shifts runtime path and side e
cts. |
| 83 |             // Valve closed: hold position, pressures bleed through leakage | comment line; documentation inte
no runtime effect. |
| 84 |             self.pres_a_bar = (self.pres_a_bar - 2.0 * dt).max(0.0); | executable/support line; behavior impac
depends on surrounding block context. |
| 85 |             self.pres_b_bar = (self.pres_b_bar - 2.0 * dt).max(0.0); | executable/support line; behavior impac
depends on surrounding block context. |
| 86 |             self.velocity_mm_s = 0.0; | executable/support line; behavior impact depends on surrounding block
text. |
| 87 |             return self.leakage_cc_m / 1000.0; | executable/support line; behavior impact depends on surroundi
block context. |
| 88 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 89 | (blank) | line is blank; visual separation only. |

```

```

| 90 |         let cmd = self.valve_cmd.clamp(-1.0, 1.0); | executable/support line; behavior impact depends on surro
ing block context. |
| 91 |         let extending = cmd > 0.0; | executable/support line; behavior impact depends on surrounding block con
t. |
| 92 | (blank) | line is blank; visual separation only. |
| 93 |         // Pressure at active side  $\approx$  supply  $\times$  valve-opening factor | comment line; documentation intent, no ru
me effect. |
| 94 |         if extending { | conditional gate; changing condition shifts runtime path and side effects. |
| 95 |             self.pres_a_bar = supply_bar * cmd.abs() * 0.92; | executable/support line; behavior impact depend
n surrounding block context. |
| 96 |             self.pres_b_bar = | executable/support line; behavior impact depends on surrounding block context. |
| 97 |                 (self.pres_a_bar * self.area_cap_cm2() / self.area_rod_cm2().max(0.01)) * 0.15; | executable/s
ort line; behavior impact depends on surrounding block context. |
| 98 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 99 |             self.pres_b_bar = supply_bar * cmd.abs() * 0.92; | executable/support line; behavior impact depend
n surrounding block context. |
| 100 |             self.pres_a_bar = self.pres_b_bar * 0.1; | executable/support line; behavior impact depends on su
unding block context. |
| 101 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 102 | (blank) | line is blank; visual separation only. |
| 103 |         // Net force | comment line; documentation intent, no runtime effect. |
| 104 |         let f_cap = self.pres_a_bar * self.area_cap_cm2() * 10.0; // N | executable/support line; behavior im
t depends on surrounding block context. |
| 105 |         let f_rod = self.pres_b_bar * self.area_rod_cm2() * 10.0; | executable/support line; behavior impact
ends on surrounding block context. |
| 106 |         self.force_kn = (f_cap - f_rod) / 1000.0; | executable/support line; behavior impact depends on surro
ing block context. |
| 107 | (blank) | line is blank; visual separation only. |
| 108 |         // Velocity from flow-through valve (orifice model  $Q = C_v \times cmd \times \sqrt{\Delta P}$ ) | comment line; documentation
ent, no runtime effect. |
| 109 |         let dp = (supply_bar | executable/support line; behavior impact depends on surrounding block context.
| 110 |             - if extending { | executable/support line; behavior impact depends on surrounding block context.
| 111 |                 self.pres_b_bar | executable/support line; behavior impact depends on surrounding block conte
|
| 112 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 113 |                 self.pres_a_bar | executable/support line; behavior impact depends on surrounding block conte
|
| 114 |             }) | executable/support line; behavior impact depends on surrounding block context. |
| 115 |         .max(0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 116 |         let flow_lpm = 40.0 * cmd.abs() * dp.sqrt() / 10.0; // simplified  $C_v$  | executable/support line; behav
impact depends on surrounding block context. |
| 117 |         let area_cm2 = if extending { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 118 |             self.area_cap_cm2() | executable/support line; behavior impact depends on surrounding block conte
|
| 119 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 120 |             self.area_rod_cm2() | executable/support line; behavior impact depends on surrounding block conte
|
| 121 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 122 |         self.velocity_mm_s = flow_lpm * 1000.0 / 60.0 / area_cm2 * cmd.signum(); | executable/support line; b
vior impact depends on surrounding block context. |
| 123 | (blank) | line is blank; visual separation only. |
| 124 |         // Integrate position – hard stops at 0 and stroke | comment line; documentation intent, no runtime e
ct. |
| 125 |         self.position_mm = (self.position_mm + self.velocity_mm_s * dt).clamp(0.0, self.stroke_mm); | executa
/support line; behavior impact depends on surrounding block context. |
| 126 | (blank) | line is blank; visual separation only. |
| 127 |         // Stop at end-of-stroke | comment line; documentation intent, no runtime effect. |
| 128 |         if (self.position_mm <= 0.0 && self.velocity_mm_s < 0.0) | conditional gate; changing condition shift
untime path and side effects. |
| 129 |             \\| (self.position_mm >= self.stroke_mm && self.velocity_mm_s > 0.0) | executable/support line;
avior impact depends on surrounding block context. |
| 130 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope

```


[illegible]

[illegible]

```

| 214 |         system_pressure_bar: 0.0, | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 215 |         relief_pressure_bar: 230.0, // 230 bar PRV | executable/support line; behavior impact depends on
rounding block context. |
| 216 |         ls_signal_bar: 0.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 217 |         charge_pres_bar: 0.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 218 |         return_pres_bar: 5.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 219 |         pilot_pres_bar: 0.0, | executable/support line; behavior impact depends on surrounding block cont
. |
| 220 |         fluid_temp_c: 25.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 221 |         fluid_level_pct: 92.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 222 |         filter_dp_bar: 0.8, | executable/support line; behavior impact depends on surrounding block conte
|
| 223 |         filter_bypass_open: false, | executable/support line; behavior impact depends on surrounding bloc
oncontext. |
| 224 |         hitch_cylinder: HydActuator::new("HITCH", 100.0, 50.0, 400.0), | executable/support line; behavio
mpact depends on surrounding block context. |
| 225 |         loader_lift: HydActuator::new("LOADER-LIFT", 80.0, 40.0, 800.0), | executable/support line; behav
impact depends on surrounding block context. |
| 226 |         loader_tilt: HydActuator::new("LOADER-TILT", 70.0, 35.0, 300.0), | executable/support line; behav
impact depends on surrounding block context. |
| 227 |         aux_valve: [0.0; 4], | executable/support line; behavior impact depends on surrounding block cont
. |
| 228 |         aux_flow_lpm: [0.0; 4], | executable/support line; behavior impact depends on surrounding block c
ext. |
| 229 |         aux_pres_bar: [0.0; 4], | executable/support line; behavior impact depends on surrounding block c
ext. |
| 230 |         brake_acc_pres_bar: 0.0, | executable/support line; behavior impact depends on surrounding block
text. |
| 231 |         brake_acc_charged: false, | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 232 |         alarm_high_temp: false, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 233 |         alarm_low_level: false, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 234 |         alarm_filter: false, | executable/support line; behavior impact depends on surrounding block cont
. |
| 235 |         alarm_low_pressure: false, | executable/support line; behavior impact depends on surrounding bloc
oncontext. |
| 236 |         hydraulic_power_kw: 0.0, | executable/support line; behavior impact depends on surrounding block
text. |
| 237 |         total_flow_demand_lpm: 0.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 238 |         t_hb: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 239 |         sim_time_s: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 240 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 241 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 242 | (blank) | line is blank; visual separation only. |
| 243 |     /// Main HCM tick. `engine_rpm` from ECM, `hitch_cmd` from implement controller (-1..+1). | comment line;
cumentation intent, no runtime effect. |
| 244 |     pub fn tick( | declares executable behavior; signature/body changes alter API, control flow, and outputs.
| 245 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 246 |         engine_rpm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 247 |         hitch_cmd: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 248 |         loader_lift_cmd: f64, | executable/support line; behavior impact depends on surrounding block context
| 249 |         loader_tilt_cmd: f64, | executable/support line; behavior impact depends on surrounding block context
| 250 |         dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 251 |     ) -> Vec<J1939Frame> { | executable/support line; behavior impact depends on surrounding block context. |
| 252 |         self.sim_time_s += dt; | executable/support line; behavior impact depends on surrounding block context

```

```
| 253 |         self.pump_rpm = engine_rpm * 1.12; // pump driven at 1.12× engine (PTO shaft) | executable/support li  
behavior impact depends on surrounding block context. |  
| 254 | (blank) | line is blank; visual separation only. |  
| 255 |         // ■ Determine load-sensing demand ████████████████████████████████████████████████████ | comment line; do  
intent, no runtime effect. |  
| 256 |         self.hitch_cylinder.valve_cmd = hitch_cmd.clamp(-1.0, 1.0); | executable/support line; behavior impac  
depends on surrounding block context. |  
| 257 |         self.loader_lift.valve_cmd = loader_lift_cmd.clamp(-1.0, 1.0); | executable/support line; behavior im  
t depends on surrounding block context. |  
| 258 |         self.loader_tilt.valve_cmd = loader_tilt_cmd.clamp(-1.0, 1.0); | executable/support line; behavior im  
t depends on surrounding block context. |  
| 259 | (blank) | line is blank; visual separation only. |  
| 260 |         // Peak load pressure among active actuators is the LS signal | comment line; documentation intent, n  
untime effect. |  
| 261 |         let actuator_pressures = [ | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 262 |             self.hitch_cylinder | executable/support line; behavior impact depends on surrounding block conte  
|  
| 263 |                 .pres_a_bar | executable/support line; behavior impact depends on surrounding block context.  
| 264 |                 .max(self.hitch_cylinder.pres_b_bar), | executable/support line; behavior impact depends on s  
ounding block context. |  
| 265 |                 self.loader_lift.pres_a_bar.max(self.loader_lift.pres_b_bar), | executable/support line; behavior  
impact depends on surrounding block context. |  
| 266 |                 self.loader_tilt.pres_a_bar.max(self.loader_tilt.pres_b_bar), | executable/support line; behavior  
impact depends on surrounding block context. |  
| 267 |             ]; | executable/support line; behavior impact depends on surrounding block context. |  
| 268 |             self.ls_signal_bar = actuator_pressures.iter().cloned().fold(0.0_f64, f64::max); | executable/support  
ne; behavior impact depends on surrounding block context. |  
| 269 |         // Add aux valves to LS calculation | comment line; documentation intent, no runtime effect. |  
| 270 |         for &cmd in self.aux_valve.iter() { | iteration control; changing bounds/body affects throughput, tim  
, and possibly safety guarantees. |  
| 271 |             if cmd.abs() > 0.01 { | conditional gate; changing condition shifts runtime path and side effects  
| 272 |                 self.ls_signal_bar = self.ls_signal_bar.max(100.0 + cmd.abs() * 100.0); | executable/support  
e; behavior impact depends on surrounding block context. |  
| 273 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 274 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 275 | (blank) | line is blank; visual separation only. |  
| 276 |         // ■ Load-sensing pump control ████████████████████████████████████████████████████ | comment line;  
intent, no runtime effect. |  
| 277 |         // Target: system = LS + 25 bar margin ( $\Delta p$ ) | comment line; documentation intent, no runtime effect.  
| 278 |         let target_sys_pres = (self.ls_signal_bar + 25.0).min(self.relief_pressure_bar); | executable/support  
ne; behavior impact depends on surrounding block context. |  
| 279 | (blank) | line is blank; visual separation only. |  
| 280 |         // Pump displacement tracks demand | comment line; documentation intent, no runtime effect. |  
| 281 |         if self.pump_rpm > 400.0 { | conditional gate; changing condition shifts runtime path and side effect  
|  
| 282 |             let _max_flow = self.pump_rpm * self.pump_displacement_cc / 1000.0; // L/min | executable/support  
ne; behavior impact depends on surrounding block context. |  
| 283 |             let demand_ratio = (target_sys_pres / self.relief_pressure_bar).clamp(0.0, 1.0); | executable/sup  
t line; behavior impact depends on surrounding block context. |  
| 284 |             let target_disp = self.pump_displacement_cc * demand_ratio; | executable/support line; behavior i  
ct depends on surrounding block context. |  
| 285 |             self.pump_actual_cc += (target_disp - self.pump_actual_cc) * dt * 8.0; | executable/support line;  
havior impact depends on surrounding block context. |  
| 286 |             self.pump_actual_cc = self.pump_actual_cc.clamp(0.0, self.pump_displacement_cc); | executable/sup  
t line; behavior impact depends on surrounding block context. |  
| 287 |             self.pump_flow_lpm = | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 288 |             self.pump_rpm * self.pump_actual_cc / 1000.0 * self.pump_efficiency; | executable/support lin  
behavior impact depends on surrounding block context. |  
| 289 |             self.pump_mode = if demand_ratio < 0.05 { | executable/support line; behavior impact depends on s  
ounding block context. |
```

```

| 290 | PumpMode::Standby | executable/support line; behavior impact depends on surrounding block con
t. |
| 291 | } else if self.system_pressure_bar >= self.relief_pressure_bar * 0.98 { | executable/support line
ehavior impact depends on surrounding block context. |
| 292 | PumpMode::PressureLimit | executable/support line; behavior impact depends on surrounding blo
context. |
| 293 | } else { | executable/support line; behavior impact depends on surrounding block context. |
| 294 | PumpMode::LoadSensing | executable/support line; behavior impact depends on surrounding block c
ntext. |
| 295 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 296 | } else { | executable/support line; behavior impact depends on surrounding block context. |
| 297 | self.pump_flow_lpm = 0.0; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 298 | self.pump_actual_cc = 0.0; | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 299 | self.pump_mode = PumpMode::Standby; | executable/support line; behavior impact depends on surroun
g block context. |
| 300 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 301 | (blank) | line is blank; visual separation only. |
| 302 | // ■ System pressure: flow → pressure through restrictor + PRV ██████████ | comment line; documenta
tent, no runtime effect. |
| 303 | let flow_excess = self.pump_flow_lpm - self.total_flow_demand_lpm; | executable/support line; behavio
mpact depends on surrounding block context. |
| 304 | self.system_pressure_bar += flow_excess * 0.5 * dt; | executable/support line; behavior impact depend
n surrounding block context. |
| 305 | self.system_pressure_bar = self | executable/support line; behavior impact depends on surrounding blo
context. |
| 306 | .system_pressure_bar | executable/support line; behavior impact depends on surrounding block cont
. |
| 307 | .clamp(0.0, self.relief_pressure_bar); | executable/support line; behavior impact depends on surr
ding block context. |
| 308 | if self.system_pressure_bar >= self.relief_pressure_bar { | conditional gate; changing condition shif
runtime path and side effects. |
| 309 | self.system_pressure_bar = self.relief_pressure_bar; | executable/support line; behavior impact d
nds on surrounding block context. |
| 310 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 311 | (blank) | line is blank; visual separation only. |
| 312 | // ■ Update actuators ████████████████████████████████████████████████████████████████ | comment line
tent, no runtime effect. |
| 313 | let sp = self.system_pressure_bar; | executable/support line; behavior impact depends on surrounding
ck context. |
| 314 | let q_hitch = self.hitch_cylinder.update(sp, dt); | executable/support line; behavior impact depends
surrounding block context. |
| 315 | let q_loader = self.loader_lift.update(sp, dt); | executable/support line; behavior impact depends on
rrounding block context. |
| 316 | let q_tilt = self.loader_tilt.update(sp, dt); | executable/support line; behavior impact depends on s
urrounding block context. |
| 317 | (blank) | line is blank; visual separation only. |
| 318 | // ■ Auxiliary valves ████████████████████████████████████████████████████████████████ | comment line
tent, no runtime effect. |
| 319 | let mut q_aux = 0.0; | executable/support line; behavior impact depends on surrounding block context.
| 320 | for i in 0..4 { | iteration control; changing bounds/body affects throughput, timing, and possibly sa
y guarantees. |
| 321 | let cmd = self.aux_valve[i]; | executable/support line; behavior impact depends on surrounding bl
context. |
| 322 | if cmd.abs() > 0.01 { | conditional gate; changing condition shifts runtime path and side effects
| 323 | self.aux_pres_bar[i] = sp * cmd.abs() * 0.90; | executable/support line; behavior impact depende
on surrounding block context. |
| 324 | self.aux_flow_lpm[i] = 40.0 * cmd.abs() * (sp / self.relief_pressure_bar).sqrt(); | executabl
upport line; behavior impact depends on surrounding block context. |
| 325 | q_aux += self.aux_flow_lpm[i]; | executable/support line; behavior impact depends on surround
block context. |

```

```
| 326 |         } else { executable/support line; behavior impact depends on surrounding block context. |  
| 327 |             self.aux_pres_bar[i] = (self.aux_pres_bar[i] - 10.0 * dt).max(0.0); | executable/support line;  
behavior impact depends on surrounding block context. |  
| 328 |                 self.aux_flow_lpm[i] = 0.0; | executable/support line; behavior impact depends on surrounding  
block context. |  
| 329 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 330 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 331 | (blank) | line is blank; visual separation only. |  
| 332 |     self.total_flow_demand_lpm = q_hitch + q_loader + q_tilt + q_aux; | executable/support line; behavior  
impact depends on surrounding block context. |  
| 333 | (blank) | line is blank; visual separation only. |  
| 334 |     // ■ Charge circuit (brake accumulator) ████████████████████████████████████████ | comment line; doc  
ntent, no runtime effect. |  
| 335 |     if self.pump_flow_lpm > 5.0 { | conditional gate; changing condition shifts runtime path and side eff  
s. |  
| 336 |         self.brake_acc_pres_bar = (self.brake_acc_pres_bar + 20.0 * dt).min(180.0); | executable/support  
e; behavior impact depends on surrounding block context. |  
| 337 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 338 |     self.brake_acc_charged = self.brake_acc_pres_bar > 120.0; | executable/support line; behavior impact  
ends on surrounding block context. |  
| 339 |     self.charge_pres_bar = self.brake_acc_pres_bar * 0.5; | executable/support line; behavior impact depende  
on surrounding block context. |  
| 340 |     self.pilot_pres_bar = (self.system_pressure_bar * 0.1).min(40.0); | executable/support line; behavior  
impact depends on surrounding block context. |  
| 341 | (blank) | line is blank; visual separation only. |  
| 342 |     // ■ Thermal model ████████████████████████████████████████████████████████████ | comment line  
ntent, no runtime effect. |  
| 343 |     // Heat = (pressure × flow) × (1 – efficiency) + heat from PRV bypass | comment line; documentation i  
nt, no runtime effect. |  
| 344 |     let kw_in = self.system_pressure_bar * self.pump_flow_lpm / 600.0; | executable/support line; behavior  
mpact depends on surrounding block context. |  
| 345 |     let kw_useful = self.total_flow_demand_lpm * sp / 600.0; | executable/support line; behavior impact d  
nds on surrounding block context. |  
| 346 |     let heat_kw = (kw_in - kw_useful).max(0.0); | executable/support line; behavior impact depends on surr  
nding block context. |  
| 347 |     self.hydraulic_power_kw = kw_useful; | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 348 |     let tgt_temp = 45.0 + heat_kw * 1.5; | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 349 |     self.fluid_temp_c += (tgt_temp - self.fluid_temp_c) * dt * 0.008; | executable/support line; behavior  
pact depends on surrounding block context. |  
| 350 | (blank) | line is blank; visual separation only. |  
| 351 |     // ■ Filter ████████████████████████████████████████████████████████████ | comment li  
ntent, no runtime effect. |  
| 352 |     // ΔP rises with flow and viscosity (higher at cold start) | comment line; documentation intent, no r  
ime effect. |  
| 353 |     let viscosity_factor = (60.0 / self.fluid_temp_c.max(10.0)).sqrt(); | executable/support line; behavi  
impact depends on surrounding block context. |  
| 354 |     self.filter_dp_bar = 0.5 + self.pump_flow_lpm * 0.015 * viscosity_factor; | executable/support line;  
avior impact depends on surrounding block context. |  
| 355 |     self.filter_bypass_open = self.filter_dp_bar > 6.0; | executable/support line; behavior impact depend  
n surrounding block context. |  
| 356 | (blank) | line is blank; visual separation only. |  
| 357 |     // ■ Alarms ████████████████████████████████████████████████████████████ | comment li  
ntent, no runtime effect. |  
| 358 |     self.alarm_high_temp = self.fluid_temp_c > 100.0; | executable/support line; behavior impact depends  
surrounding block context. |  
| 359 |     self.alarm_low_level = self.fluid_level_pct < 20.0; | executable/support line; behavior impact depend  
n surrounding block context. |  
| 360 |     self.alarm_filter = self.filter_bypass_open; | executable/support line; behavior impact depends on su  
unding block context. |  
| 361 |     self.alarm_low_pressure = | executable/support line; behavior impact depends on surrounding block con
```

```
t. |  
| 362 |         self.system_pressure_bar < 50.0 && self.total_flow_demand_lpm > 5.0; | executable/support line; behavior impact depends on surrounding block context. | |
| 363 | (blank) | line is blank; visual separation only. |  
| 364 |         // ■ J1939 heartbeat (100 ms, Proprietary B) ████████████████████████████████████████ | comment line; doc intent, no runtime effect. |  
| 365 |         self.t_hb += dt; | executable/support line; behavior impact depends on surrounding block context. |  
| 366 |         let mut frames: Vec<J1939Frame> = Vec::new(); | executable/support line; behavior impact depends on surrounding block context. |  
| 367 |         if self.t_hb >= 0.100 { | conditional gate; changing condition shifts runtime path and side effects. |  
| 368 |             self.t_hb = 0.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 369 |             let mut data = [0u8; 8]; | executable/support line; behavior impact depends on surrounding block context. |  
| 370 |             data[0] = (self.system_pressure_bar / 2.0).clamp(0.0, 255.0) as u8; | executable/support line; behavior impact depends on surrounding block context. |  
| 371 |             data[1] = (self.fluid_temp_c + 40.0).clamp(0.0, 255.0) as u8; | executable/support line; behavior impact depends on surrounding block context. |  
| 372 |             data[2] = (self.pump_flow_lpm / 2.0).clamp(0.0, 255.0) as u8; | executable/support line; behavior impact depends on surrounding block context. |  
| 373 |             data[3] = (if self.alarm_high_temp { 0x01 } else { 0 }) | executable/support line; behavior impact depends on surrounding block context. |  
| 374 |                 \ | (if self.alarm_low_level { 0x02 } else { 0 }) | executable/support line; behavior impact depends on surrounding block context. |  
| 375 |                 \ | (if self.alarm_filter { 0x04 } else { 0 }) | executable/support line; behavior impact depends on surrounding block context. |  
| 376 |                 \ | (if self.brake_acc_charged { 0x08 } else { 0 }); | executable/support line; behavior impact depends on surrounding block context. |  
| 377 |             data[4] = (self.hitch_cylinder.position_pct() * 2.55) as u8; | executable/support line; behavior impact depends on surrounding block context. |  
| 378 |             let raw_id = J1939Frame::build_id(6, pgn::HITCH, self.sa, 0xFF); | executable/support line; behavior impact depends on surrounding block context. |  
| 379 |             frames.push(J1939Frame::from_raw(self.sim_time_s, raw_id, &data)); | executable/support line; behavior impact depends on surrounding block context. |  
| 380 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 381 |         frames | executable/support line; behavior impact depends on surrounding block context. |  
| 382 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 383 | (blank) | line is blank; visual separation only. |  
| 384 |         // ■ Convenience setters ████████████████████████████████████████████████████████████ | comment line; doc intent, no runtime effect. |  
| 385 |         pub fn set_hitch_cmd(&mut self, cmd: f64) { | declares executable behavior; signature/body changes alter control flow, and outputs. |  
| 386 |             self.hitch_cylinder.valve_cmd = cmd.clamp(-1.0, 1.0); | executable/support line; behavior impact depends on surrounding block context. |  
| 387 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 388 |         pub fn set_loader_lift_cmd(&mut self, cmd: f64) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 389 |             self.loader_lift.valve_cmd = cmd.clamp(-1.0, 1.0); | executable/support line; behavior impact depends on surrounding block context. |  
| 390 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 391 |         pub fn set_loader_tilt_cmd(&mut self, cmd: f64) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 392 |             self.loader_tilt.valve_cmd = cmd.clamp(-1.0, 1.0); | executable/support line; behavior impact depends on surrounding block context. |  
| 393 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 394 |         pub fn set_aux_cmd(&mut self, bank: usize, cmd: f64) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 395 |             if bank < 4 { | conditional gate; changing condition shifts runtime path and side effects. |  
| 396 |                 self.aux_valve[bank] = cmd.clamp(-1.0, 1.0); | executable/support line; behavior impact depends on surrounding block context. |  
| 397 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 398 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 399 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
```

Role: Domain module in simulation/control/protocol stack.
Lines: 467 | Non-empty: 434 | Symbols: 28

```
| Line | Kind | Symbol | Change Impact |
| ---:| ---| ---| ---|
| 8 | const | ICM_SA | data/structure coupling |
| 12 | struct | Gauge | data/structure coupling |
| 25 | impl | Gauge | behavior binding and method semantics |
| 26 | fn | new | signature/API coupling |
| 40 | fn | with_warn_hi | signature/API coupling |
| 45 | fn | with_warn_lo | signature/API coupling |
| 50 | fn | update | signature/API coupling |
| 58 | fn | fraction | signature/API coupling |
| 65 | enum | LampColor | data/structure coupling |
| 74 | struct | WarningLamp | data/structure coupling |
| 80 | impl | WarningLamp | behavior binding and method semantics |
| 81 | fn | new | signature/API coupling |
| 88 | fn | set | signature/API coupling |
| 95 | struct | TripComputer | data/structure coupling |
| 105 | impl | Default for TripComputer | behavior binding and method semantics |
| 106 | fn | default | signature/API coupling |
| 111 | impl | TripComputer | behavior binding and method semantics |
| 112 | fn | new | signature/API coupling |
| 124 | fn | update | signature/API coupling |
| 136 | fn | reset | signature/API coupling |
| 143 | struct | EcuIcm | data/structure coupling |
| 192 | impl | Default for EcuIcm | behavior binding and method semantics |
| 193 | fn | default | signature/API coupling |
| 198 | impl | EcuIcm | behavior binding and method semantics |
| 199 | fn | new | signature/API coupling |
| 254 | fn | start_self_test | signature/API coupling |
| 263 | fn | tick | signature/API coupling |
| 397 | fn | update_lamps | signature/API coupling |
```

```
| Line | Source | Operational Impact If Changed |  
|---:|---|---|  
| 1 | //! Instrument Cluster Module (ICM/DASH) – SA 0x1C. | comment line; documentation intent, no runtime effect. |  
| 2 | //! Receives J1939 data from all ECUs, presents live gauge readings. | comment line; documentation intent, no runtime effect. |  
| 3 | //! Tracks odometer, trip computer, warning lamp states. | comment line; documentation intent, no runtime effect. |  
| 4 | //! J1939: listens only (display module); sends heartbeat on Prop-B (100ms). | comment line; documentation intent, no runtime effect. |  
| 5 | (blank) | line is blank; visual separation only. |  
| 6 | use crate::j1939::{self, addr, J1939Frame}; | imports symbols; changing path/name affects compile-time resolution. |  
| 7 | (blank) | line is blank; visual separation only. |  
| 8 | pub const ICM_SA: u8 = addr::INSTRUMENT; // 0x1C | executable/support line; behavior impact depends on surrounding block context. |  
| 9 | (blank) | line is blank; visual separation only. |  
| 10 | // ■ Gauge ████████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |  
| 11 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |  
| 12 | pub struct Gauge { | declares data layout; field changes may break serialization, tests, and callers. |  
| 13 |     pub name: &'static str, | executable/support line; behavior impact depends on surrounding block context. |  
| 14 |     pub value: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 15 |     pub min: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 16 |     pub max: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 17 |     pub unit: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
```



```
| 18 | pub warn_lo: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 19 | pub warn_hi: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 20 | pub crit_hi: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 21 | pub in_warning: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 22 | pub in_critical: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 23 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 24 | (blank) | line is blank; visual separation only. |
| 25 | impl Gauge { | implementation block; method behavior and trait conformance are defined here. |
| 26 |     pub fn new(name: &'static str, min: f64, max: f64, unit: &'static str) -> Self { | declares executable beh
or; signature/body changes alter API, control flow, and outputs. |
| 27 |         Gauge { | executable/support line; behavior impact depends on surrounding block context. |
| 28 |             name, | executable/support line; behavior impact depends on surrounding block context. |
| 29 |             value: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 30 |             min, | executable/support line; behavior impact depends on surrounding block context. |
| 31 |             max, | executable/support line; behavior impact depends on surrounding block context. |
| 32 |             unit, | executable/support line; behavior impact depends on surrounding block context. |
| 33 |             warn_lo: None, | executable/support line; behavior impact depends on surrounding block context. |
| 34 |             warn_hi: None, | executable/support line; behavior impact depends on surrounding block context. |
| 35 |             crit_hi: None, | executable/support line; behavior impact depends on surrounding block context. |
| 36 |             in_warning: false, | executable/support line; behavior impact depends on surrounding block context. |
| 37 |             in_critical: false, | executable/support line; behavior impact depends on surrounding block context. |
| 38 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 39 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 40 |     pub fn with_warn_hi(mut self, w: f64, c: f64) -> Self { | declares executable behavior; signature/body cha
s alter API, control flow, and outputs. |
| 41 |         self.warn_hi = Some(w); | executable/support line; behavior impact depends on surrounding block context. |
| 42 |         self.crit_hi = Some(c); | executable/support line; behavior impact depends on surrounding block context. |
| 43 |         self | executable/support line; behavior impact depends on surrounding block context. |
| 44 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 45 |     pub fn with_warn_lo(mut self, w: f64) -> Self { | declares executable behavior; signature/body changes alter
API, control flow, and outputs. |
| 46 |         self.warn_lo = Some(w); | executable/support line; behavior impact depends on surrounding block context. |
| 47 |         self | executable/support line; behavior impact depends on surrounding block context. |
| 48 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 49 | (blank) | line is blank; visual separation only. |
| 50 |     pub fn update(&mut self, val: f64) { | declares executable behavior; signature/body changes alter API, con
l flow, and outputs. |
| 51 |         self.value = val.clamp(self.min, self.max); | executable/support line; behavior impact depends on surr
rounding block context. |
| 52 |         self.in_warning = self.warn_hi.is_some_and(|w| self.value > w) | executable/support line; behavior i
ct depends on surrounding block context. |
| 53 |         || self.warn_lo.is_some_and(|l| self.value < l); | executable/support line; behavior impact de
ds on surrounding block context. |
| 54 |         self.in_critical = self.crit_hi.is_some_and(|c| self.value > c); | executable/support line; behavior
impact depends on surrounding block context. |
| 55 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 56 | (blank) | line is blank; visual separation only. |
| 57 |     /// Fraction 0-1 of gauge arc | comment line; documentation intent, no runtime effect. |
| 58 |     pub fn fraction(&self) -> f64 { | declares executable behavior; signature/body changes alter API, control
w, and outputs. |
| 59 |         (self.value - self.min) / (self.max - self.min) | executable/support line; behavior impact depends on
rounding block context. |
| 60 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 61 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 62 | (blank) | line is blank; visual separation only. |
| 63 | // ■ Warning Lamp ████████████████████████████████████████████████████████████████████████████ | comment li
tent, no runtime effect. |
| 64 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding blo
context. |
| 65 | pub enum LampColor { | declares state variants; adding/removing variants impacts pattern matches and logic bra
es. |
```

[illegible]

```
| 121 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 122 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 123 | (blank) | line is blank; visual separation only. |  
| 124 |     pub fn update(&mut self, speed_kmh: f64, fuel_lph: f64, dt: f64) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 125 |         self.trip_distance_km += speed_kmh * dt / 3600.0; | executable/support line; behavior impact depends  
surrounding block context. |  
| 126 |         self.trip_fuel_l += fuel_lph * dt / 3600.0; | executable/support line; behavior impact depends on surr  
nding block context. |  
| 127 |         self.trip_hours += dt / 3600.0; | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 128 |         self.speed_sum += speed_kmh; | executable/support line; behavior impact depends on surrounding block  
text. |  
| 129 |         self.samples += 1; | executable/support line; behavior impact depends on surrounding block context. |  
| 130 |         self.avg_speed_kmh = self.speed_sum / self.samples.max(1) as f64; | executable/support line; behavior  
impact depends on surrounding block context. |  
| 131 |         if self.trip_hours > 0.0 { | conditional gate; changing condition shifts runtime path and side effect  
| 132 |             self.avg_fuel_lph = self.trip_fuel_l / self.trip_hours; | executable/support line; behavior impac  
depends on surrounding block context. |  
| 133 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
| 134 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 135 | (blank) | line is blank; visual separation only. |  
| 136 |     pub fn reset(&mut self) { | declares executable behavior; signature/body changes alter API, control flow,  
d outputs. |  
| 137 |         *self = Self::new(); | executable/support line; behavior impact depends on surrounding block context.  
| 138 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 139 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 140 | (blank) | line is blank; visual separation only. |  
| 141 | // ■ ICM ████████████████████████████████████████████████████████████████ | comment  
tent, no runtime effect. |  
| 142 | #[derive(Clone)] | executable/support line; behavior impact depends on surrounding block context. |  
| 143 | pub struct EcuIcm { | declares data layout; field changes may break serialization, tests, and callers. |  
| 144 |     pub sa: u8, | executable/support line; behavior impact depends on surrounding block context. |  
| 145 | (blank) | line is blank; visual separation only. |  
| 146 |     // ■ Gauges (updated from CAN messages) ████████████████████████████████████████████████████████████████ | comment line; doc  
tent, no runtime effect. |  
| 147 |     pub tachometer: Gauge,          // RPM | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 148 |     pub speedometer: Gauge,        // km/h | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 149 |     pub fuel_gauge: Gauge,         // % | executable/support line; behavior impact depends on surrounding block  
text. |  
| 150 |     pub coolant_gauge: Gauge,      // °C | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 151 |     pub oil_press_gauge: Gauge,    // kPa | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 152 |     pub boost_gauge: Gauge,       // kPa | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 153 |     pub def_gauge: Gauge,         // % | executable/support line; behavior impact depends on surrounding block  
text. |  
| 154 |     pub hours_gauge: Gauge,       // h | executable/support line; behavior impact depends on surrounding block  
text. |  
| 155 |     pub battery_gauge: Gauge,     // V | executable/support line; behavior impact depends on surrounding block  
text. |  
| 156 | (blank) | line is blank; visual separation only. |  
| 157 |     // ■ Warning lamps ████████████████████████████████████████████████████████████████ | comment lin  
tent, no runtime effect. |  
| 158 |     pub lamps: Vec<WarningLamp>, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 159 | (blank) | line is blank; visual separation only. |  
| 160 |     // ■ Trip computer ████████████████████████████████████████████████████████████████ | comment lin  
tent, no runtime effect. |
```

```
| 161 | pub trip: TripComputer, | executable/support line; behavior impact depends on surrounding block context. |
| 162 | pub odometer_km: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 163 | (blank) | line is blank; visual separation only. |
| 164 | // ■ Last known ECU data (aggregated from CAN) ████████████████████████████████████████████ | comment line; documentent, no runtime effect. |
| 165 | pub engine_rpm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 166 | pub vehicle_speed: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 167 | pub engine_load_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 168 | pub coolant_temp: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 169 | pub oil_pressure_kpa: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 170 | pub fuel_level_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 171 | pub def_level_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 172 | pub engine_hours: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 173 | pub battery_volts: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 174 | pub boost_kpa: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 175 | pub trans_gear: String, | executable/support line; behavior impact depends on surrounding block context. |
| 176 | pub active_dtc_count: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 177 | (blank) | line is blank; visual separation only. |
| 178 | // ■ Lamp states (set by DM1 messages) ████████████████████████████████████████████ | comment line; documentent, no runtime effect. |
| 179 | pub mil_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 180 | pub amber_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 181 | pub red_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 182 | pub protect_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 183 | (blank) | line is blank; visual separation only. |
| 184 | // ■ Self-test sequence ████████████████████████████████████████████ | comment line; documentent, no runtime effect. |
| 185 | pub self_test_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 186 | self_test_timer: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 187 | (blank) | line is blank; visual separation only. |
| 188 | // ■ TX timer ████████████████████████████████████████████ | comment line; documentent, no runtime effect. |
| 189 | t_heartbeat: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 190 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 191 | (blank) | line is blank; visual separation only. |
| 192 | impl Default for EcuIcm { | implementation block; method behavior and trait conformance are defined here. |
| 193 | fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 194 |     Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 195 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 196 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 197 | (blank) | line is blank; visual separation only. |
| 198 | impl EcuIcm { | implementation block; method behavior and trait conformance are defined here. |
| 199 | pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 200 |     let lamps = vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 201 |         WarningLamp::new("RED_STOP", "RED STOP"), | executable/support line; behavior impact depends on surrounding block context. |
| 202 |         WarningLamp::new("AMBER_WARN", "AMBER WARN"), | executable/support line; behavior impact depends on surrounding block context. |
| 203 |         WarningLamp::new("MIL", "CHECK ENGINE"), | executable/support line; behavior impact depends on surrounding block context. |
| 204 |         WarningLamp::new("PROTECT", "PROTECT ENG"), | executable/support line; behavior impact depends on surrounding block context. |
| 205 |         WarningLamp::new("OIL_PRESS", "OIL PRESS"), | executable/support line; behavior impact depends on surrounding block context. |
| 206 |         WarningLamp::new("COOLANT_T", "COOLANT TEMP"), | executable/support line; behavior impact depends on surrounding block context. |
| 207 |         WarningLamp::new("DEF_LOW", "DEF LOW"), | executable/support line; behavior impact depends on surrounding block context. |
| 208 |         WarningLamp::new("DPF_REGEN", "DPF REGEN"), | executable/support line; behavior impact depends on surrounding block context. |
```

```

| 209 |           WarningLamp::new("FUEL_LOW", "FUEL LOW"), | executable/support line; behavior impact depends on s
ounding block context. |
| 210 |           WarningLamp::new("TRANS_FAULT", "TRANS FAULT"), | executable/support line; behavior impact depend
n surrounding block context. |
| 211 |           WarningLamp::new("CHARGE", "CHARGING"), | executable/support line; behavior impact depends on sur
nding block context. |
| 212 |           WarningLamp::new("AIR_FILTER", "AIR FILTER"), | executable/support line; behavior impact depends
surrounding block context. |
| 213 |           WarningLamp::new("HYD_TEMP", "HYD TEMP"), | executable/support line; behavior impact depends on s
ounding block context. |
| 214 |           WarningLamp::new("PTO_ACTV", "PTO ACTIVE"), | executable/support line; behavior impact depends on
rrounding block context. |
| 215 |       ]; | executable/support line; behavior impact depends on surrounding block context. |
| 216 | (blank) | line is blank; visual separation only. |
| 217 |       EcuIcm { | executable/support line; behavior impact depends on surrounding block context. |
| 218 |           sa: ICM_SA, | executable/support line; behavior impact depends on surrounding block context. |
| 219 |           tachometer: Gauge::new("Engine RPM", 0.0, 3000.0, "rpm").with_warn_hi(2400.0, 2700.0), | executab
support line; behavior impact depends on surrounding block context. |
| 220 |           speedometer: Gauge::new("Speed", 0.0, 55.0, "km/h"), | executable/support line; behavior impact d
nds on surrounding block context. |
| 221 |           fuel_gauge: Gauge::new("Fuel", 0.0, 100.0, "%").with_warn_lo(10.0), | executable/support line; be
ior impact depends on surrounding block context. |
| 222 |           coolant_gauge: Gauge::new("Coolant Temp", 0.0, 130.0, "°C").with_warn_hi(100.0, 108.0), | executa
/support line; behavior impact depends on surrounding block context. |
| 223 |           oil_press_gauge: Gauge::new("Oil Pressure", 0.0, 700.0, "kPa").with_warn_lo(100.0), | executable/
port line; behavior impact depends on surrounding block context. |
| 224 |           boost_gauge: Gauge::new("Boost", 0.0, 300.0, "kPa"), | executable/support line; behavior impact d
nds on surrounding block context. |
| 225 |           def_gauge: Gauge::new("DEF", 0.0, 100.0, "%").with_warn_lo(10.0), | executable/support line; beha
r impact depends on surrounding block context. |
| 226 |           hours_gauge: Gauge::new("Hours", 0.0, 99999.0, "h"), | executable/support line; behavior impact d
nds on surrounding block context. |
| 227 |           battery_gauge: Gauge::new("Battery", 9.0, 16.0, "V").with_warn_lo(11.0), | executable/support lin
behavior impact depends on surrounding block context. |
| 228 |           lamps, | executable/support line; behavior impact depends on surrounding block context. |
| 229 |           trip: TripComputer::new(), | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 230 |           odometer_km: 12450.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 231 |           engine_rpm: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 232 |           vehicle_speed: 0.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 233 |           engine_load_pct: 0.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 234 |           coolant_temp: 20.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 235 |           oil_pressure_kpa: 0.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 236 |           fuel_level_pct: 85.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 237 |           def_level_pct: 72.0, | executable/support line; behavior impact depends on surrounding block cont
. |
| 238 |           engine_hours: 0.0, | executable/support line; behavior impact depends on surrounding block contex
|
| 239 |           battery_volts: 12.8, | executable/support line; behavior impact depends on surrounding block cont
. |
| 240 |           boost_kpa: 100.0, | executable/support line; behavior impact depends on surrounding block context
| 241 |           trans_gear: "N".into(), | executable/support line; behavior impact depends on surrounding block c
ext. |
| 242 |           active_dtc_count: 0, | executable/support line; behavior impact depends on surrounding block cont
. |
| 243 |           mil_active: false, | executable/support line; behavior impact depends on surrounding block contex
|
| 244 |           amber_active: false, | executable/support line; behavior impact depends on surrounding block cont
. |

```

```
| 245 |         red_active: false, | executable/support line; behavior impact depends on surrounding block context. |
| 246 |         protect_active: false, | executable/support line; behavior impact depends on surrounding block context. |
| 247 |         self_test_active: false, | executable/support line; behavior impact depends on surrounding block context. |
| 248 |         self_test_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 249 |         t_heartbeat: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 250 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 251 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 252 | (blank) | line is blank; visual separation only. |
| 253 |     /// Call after boot: runs a 3-second instrument self-test (all gauges sweep) | comment line; documentation intent, no runtime effect. |
| 254 |     pub fn start_self_test(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 255 |         self.self_test_active = true; | executable/support line; behavior impact depends on surrounding block context. |
| 256 |         self.self_test_timer = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 257 |         for lamp in &mut self.lamps { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 258 |             lamp.color = LampColor::Yellow; | executable/support line; behavior impact depends on surrounding block context. |
| 259 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 260 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 261 | (blank) | line is blank; visual separation only. |
| 262 |     /// Main ICM tick – processes received CAN frames, updates gauges. | comment line; documentation intent, no runtime effect. |
| 263 |     pub fn tick(&mut self, received: &[J1939Frame], dt: f64) -> Vec<J1939Frame> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 264 |         // ■ Self-test ████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 265 |         if self.self_test_active { | conditional gate; changing condition shifts runtime path and side effects. |
| 266 |             self.self_test_timer += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 267 |             let phase = (self.self_test_timer / 3.0).min(1.0); | executable/support line; behavior impact depends on surrounding block context. |
| 268 |             // Sweep all gauges to max then back | comment line; documentation intent, no runtime effect. |
| 269 |             let sweep = if phase < 0.5 { | executable/support line; behavior impact depends on surrounding block context. |
| 270 |                 phase * 2.0 | executable/support line; behavior impact depends on surrounding block context. |
| 271 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 272 |                 2.0 - phase * 2.0 | executable/support line; behavior impact depends on surrounding block context. |
| 273 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 274 |             self.tachometer.update(self.tachometer.max * sweep); | executable/support line; behavior impact depends on surrounding block context. |
| 275 |             self.speedometer.update(self.speedometer.max * sweep); | executable/support line; behavior impact depends on surrounding block context. |
| 276 |             if self.self_test_timer >= 3.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 277 |                 self.self_test_active = false; | executable/support line; behavior impact depends on surrounding block context. |
| 278 |                 for lamp in &mut self.lamps { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 279 |                     lamp.color = LampColor::Off; | executable/support line; behavior impact depends on surrounding block context. |
| 280 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 281 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
```

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es. |
| 282 |         return Vec::new(); | executable/support line; behavior impact depends on surrounding block context
|
| 283 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 284 | (blank) | line is blank; visual separation only. |
| 285 |     // ■ Process received J1939 frames ████████████████████████████████████████████████████████ | comment line; c
intent, no runtime effect. |
| 286 |     for frame in received { | iteration control; changing bounds/body affects throughput, timing, and pos
ly safety guarantees. |
| 287 |         use crate::j1939::pgn; | imports symbols; changing path/name affects compile-time resolution. |
| 288 |         match frame.pgn { | branch dispatch; changing arms alters behavior for variants and input classes
| 289 |             p if p == pgn::EEC1 => { | executable/support line; behavior impact depends on surrounding bl
context. |
| 290 |                 for v in &frame.decoded { | iteration control; changing bounds/body affects throughput, t
ng, and possibly safety guarantees. |
| 291 |                     if v.spn == 190 { | conditional gate; changing condition shifts runtime path and side
fects. |
| 292 |                         self.engine_rpm = v.physical; | executable/support line; behavior impact depends
surrounding block context. |
| 293 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and
ifetime scopes. |
| 294 |                     if v.spn == 513 { | conditional gate; changing condition shifts runtime path and side
fects. |
| 295 |                         self.engine_load_pct = (v.physical + 125.0).max(0.0); | executable/support line;
avior impact depends on surrounding block context. |
| 296 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and
ifetime scopes. |
| 297 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 298 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 299 |             p if p == pgn::EEC2 => { | executable/support line; behavior impact depends on surrounding bl
context. |
| 300 |                 for v in &frame.decoded { | iteration control; changing bounds/body affects throughput, t
ng, and possibly safety guarantees. |
| 301 |                     if v.spn == 92 { | conditional gate; changing condition shifts runtime path and side
ects. |
| 302 |                         self.engine_load_pct = v.physical; | executable/support line; behavior impact dep
s on surrounding block context. |
| 303 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and
ifetime scopes. |
| 304 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 305 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 306 |             p if p == pgn::ET1 => { | executable/support line; behavior impact depends on surrounding blo
context. |
| 307 |                 for v in &frame.decoded { | iteration control; changing bounds/body affects throughput, t
ng, and possibly safety guarantees. |
| 308 |                     if v.spn == 110 { | conditional gate; changing condition shifts runtime path and side
fects. |
| 309 |                         self.coolant_temp = v.physical; | executable/support line; behavior impact depend
n surrounding block context. |
| 310 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and
ifetime scopes. |
| 311 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 312 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 313 |             p if p == pgn::EFL_P1 => { | executable/support line; behavior impact depends on surrounding
ck context. |
| 314 |                 for v in &frame.decoded { | iteration control; changing bounds/body affects throughput, t
ng, and possibly safety guarantees. |
| 315 |                     if v.spn == 100 { | conditional gate; changing condition shifts runtime path and side

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fects. |
| 316 |           self.oil_pressure_kpa = v.physical; | executable/support line; behavior impact de
ds on surrounding block context. |
| 317 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership an
ifetime scopes. |
| 318 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 319 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 320 |           p if p == pgn::IC1 => { | executable/support line; behavior impact depends on surrounding blo
context. |
| 321 |           for v in &frame.decoded { | iteration control; changing bounds/body affects throughput, t
ng, and possibly safety guarantees. |
| 322 |           if v.spn == 102 { | conditional gate; changing condition shifts runtime path and side
fects. |
| 323 |           self.boost_kpa = v.physical; | executable/support line; behavior impact depends o
urrounding block context. |
| 324 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership an
ifetime scopes. |
| 325 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 326 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 327 |           p if p == pgn::LFE => { | executable/support line; behavior impact depends on surrounding blo
context. |
| 328 |           for v in &frame.decoded { | iteration control; changing bounds/body affects throughput, t
ng, and possibly safety guarantees. |
| 329 |           if v.spn == 183 { | conditional gate; changing condition shifts runtime path and side
fects. |
| 330 |           /* fuel rate used in trip */ | executable/support line; behavior impact depends o
urrounding block context. |
| 331 |           let _ = v.physical; | executable/support line; behavior impact depends on surroun
g block context. |
| 332 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership an
ifetime scopes. |
| 333 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 334 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 335 |           p if p == pgn::HOURS => { | executable/support line; behavior impact depends on surrounding b
k context. |
| 336 |           for v in &frame.decoded { | iteration control; changing bounds/body affects throughput, t
ng, and possibly safety guarantees. |
| 337 |           if v.spn == 247 { | conditional gate; changing condition shifts runtime path and side
fects. |
| 338 |           self.engine_hours = v.physical; | executable/support line; behavior impact depend
n surrounding block context. |
| 339 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership an
ifetime scopes. |
| 340 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 341 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 342 |           p if p == pgn::CCVS => { | executable/support line; behavior impact depends on surrounding bl
context. |
| 343 |           for v in &frame.decoded { | iteration control; changing bounds/body affects throughput, t
ng, and possibly safety guarantees. |
| 344 |           if v.spn == 84 { | conditional gate; changing condition shifts runtime path and side
ects. |
| 345 |           self.vehicle_speed = v.physical; | executable/support line; behavior impact dependen
on surrounding block context. |
| 346 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership an
ifetime scopes. |
| 347 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |

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[illegible]

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| 384 |         if self.t_heartbeat >= 0.100 { | conditional gate; changing condition shifts runtime path and side effects. | |
| 385 |             self.t_heartbeat = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 386 |             let mut data = [0u8; 8]; | executable/support line; behavior impact depends on surrounding block context. |
| 387 |             data[0] = (if self.red_active { 0x01 } else { 0 }) | executable/support line; behavior impact depends on surrounding block context. |
| 388 |             \ | (if self.amber_active { 0x02 } else { 0 }) | executable/support line; behavior impact depends on surrounding block context. |
| 389 |             \ | (if self.mil_active { 0x04 } else { 0 }); | executable/support line; behavior impact depends on surrounding block context. |
| 390 |             data[1] = (self.vehicle_speed * 10.0) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 391 |             let raw_id = J1939Frame::build_id(7, j1939::pgn::PROP_A + 1, self.sa, 0xFF); | executable/support line; behavior impact depends on surrounding block context. |
| 392 |             frames.push(J1939Frame::from_raw(self.t_heartbeat, raw_id, &data)); | executable/support line; behavior impact depends on surrounding block context. |
| 393 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 394 |         frames | executable/support line; behavior impact depends on surrounding block context. |
| 395 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 396 | (blank) | line is blank; visual separation only. |
| 397 |     fn update_lamps(&mut self) { | declares executable behavior; signature/body changes alter API, control flow and outputs. |
| 398 |         for lamp in &mut self.lamps { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 399 |             lamp.color = match lamp.id { | executable/support line; behavior impact depends on surrounding block context. |
| 400 |                 "RED_STOP" => { | executable/support line; behavior impact depends on surrounding block context. |
| 401 |                     if self.red_active { | conditional gate; changing condition shifts runtime path and side effects. |
| 402 |                         LampColor::Red | executable/support line; behavior impact depends on surrounding block context. |
| 403 |                     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 404 |                         LampColor::Off | executable/support line; behavior impact depends on surrounding block context. |
| 405 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 406 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 407 |                 "AMBER_WARN" => { | executable/support line; behavior impact depends on surrounding block context. |
| 408 |                     if self.amber_active { | conditional gate; changing condition shifts runtime path and side effects. |
| 409 |                         LampColor::Yellow | executable/support line; behavior impact depends on surrounding block context. |
| 410 |                     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 411 |                         LampColor::Off | executable/support line; behavior impact depends on surrounding block context. |
| 412 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 413 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 414 |                 "MIL" => { | executable/support line; behavior impact depends on surrounding block context. |
| 415 |                     if self.mil_active { | conditional gate; changing condition shifts runtime path and side effects. |
| 416 |                         LampColor::Yellow | executable/support line; behavior impact depends on surrounding block context. |
| 417 |                     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 418 |                         LampColor::Off | executable/support line; behavior impact depends on surrounding block context. |
| 419 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

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| 420 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 421 |           "PROTECT" => { | executable/support line; behavior impact depends on surrounding block contex
|
| 422 |               if self.protect_active { | conditional gate; changing condition shifts runtime path and s
effects. |
| 423 |                   LampColor::Yellow | executable/support line; behavior impact depends on surrounding b
k context. |
| 424 |               } else { | executable/support line; behavior impact depends on surrounding block context.
| 425 |                   LampColor::Off | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 426 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 427 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 428 |           "OIL_PRESS" => { | executable/support line; behavior impact depends on surrounding block cont
. |
| 429 |               if self.oil_pressure_kpa < 100.0 && self.engine_rpm > 500.0 { | conditional gate; changin
ondition shifts runtime path and side effects. |
| 430 |                   LampColor::Red | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 431 |               } else { | executable/support line; behavior impact depends on surrounding block context.
| 432 |                   LampColor::Off | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 433 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 434 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 435 |           "COOLANT_T" => { | executable/support line; behavior impact depends on surrounding block cont
. |
| 436 |               if self.coolant_temp > 105.0 { | conditional gate; changing condition shifts runtime path
d side effects. |
| 437 |                   LampColor::Red | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 438 |               } else { | executable/support line; behavior impact depends on surrounding block context.
| 439 |                   LampColor::Off | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 440 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 441 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 442 |           "DEF_LOW" => { | executable/support line; behavior impact depends on surrounding block contex
|
| 443 |               if self.def_level_pct < 10.0 { | conditional gate; changing condition shifts runtime path
d side effects. |
| 444 |                   LampColor::Yellow | executable/support line; behavior impact depends on surrounding b
k context. |
| 445 |               } else { | executable/support line; behavior impact depends on surrounding block context.
| 446 |                   LampColor::Off | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 447 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 448 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 449 |           "FUEL_LOW" => { | executable/support line; behavior impact depends on surrounding block conte
|
| 450 |               if self.fuel_level_pct < 10.0 { | conditional gate; changing condition shifts runtime pat
nd side effects. |
| 451 |                   LampColor::Yellow | executable/support line; behavior impact depends on surrounding b
k context. |
| 452 |               } else { | executable/support line; behavior impact depends on surrounding block context.
| 453 |                   LampColor::Off | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 454 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |

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| 455 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 456 |           "CHARGE" => { | executable/support line; behavior impact depends on surrounding block context.
| 457 |               if self.battery_volts < 12.0 { | conditional gate; changing condition shifts runtime path
d side effects. |
| 458 |                   LampColor::Yellow | executable/support line; behavior impact depends on surrounding block
k context. |
| 459 |               } else { | executable/support line; behavior impact depends on surrounding block context.
| 460 |                   LampColor::Off | executable/support line; behavior impact depends on surrounding block
ontext. |
| 461 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
ime scopes. |
| 462 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 463 |           _ => LampColor::Off, | executable/support line; behavior impact depends on surrounding block
text. |
| 464 |       }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 465 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 466 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 467 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/ecu_tcm.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 716 | Non-empty: 647 | Symbols: 43

Symbol Inventory

Line	Kind	Symbol	Change Impact
16	const	TCM_SA	data/structure coupling
20	enum	TransmissionType	data/structure coupling
30	enum	GearRange	data/structure coupling
41	impl	std::fmt::Display for GearRange	behavior binding and method semantics
42	fn	fmt	signature/API coupling
55	enum	Direction	data/structure coupling
62	impl	std::fmt::Display for Direction	behavior binding and method semantics
63	fn	fmt	signature/API coupling
75	enum	ClutchState	data/structure coupling
86	impl	std::fmt::Display for ClutchState	behavior binding and method semantics
87	fn	fmt	signature/API coupling
99	enum	ShiftQuality	data/structure coupling
105	impl	std::fmt::Display for ShiftQuality	behavior binding and method semantics
106	fn	fmt	signature/API coupling
118	enum	AutoShiftMode	data/structure coupling
130	const	GEAR_RATIOS	data/structure coupling
141	const	TIRE_RADIUS_M	data/structure coupling
142	const	MAX_SPEED_KMH	data/structure coupling
146	struct	EcuTcm	data/structure coupling
211	impl	Default for EcuTcm	behavior binding and method semantics
212	fn	default	signature/API coupling
217	impl	EcuTcm	behavior binding and method semantics
218	fn	new	signature/API coupling
263	fn	tick	signature/API coupling
383	fn	update_clutch	signature/API coupling
424	fn	auto_shift_logic	signature/API coupling
479	fn	request_upshift	signature/API coupling
496	fn	request_downshift	signature/API coupling
513	fn	begin_shift	signature/API coupling
535	fn	update_shift	signature/API coupling
572	fn	update_ivt	signature/API coupling
590	fn	manual_upshift	signature/API coupling
595	fn	manual_downshift	signature/API coupling

[illegible]

[illegible]


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|tentent, no runtime effect. |
| 170 | pub is_shifting: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 171 | pub last_shift_qual: ShiftQuality, | executable/support line; behavior impact depends on surrounding block context. |
| 172 | pub total_shifts: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 173 | pub pending_range: Option<GearRange>, | executable/support line; behavior impact depends on surrounding block context. |
| 174 | pub pending_speed: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 175 | pub pending_dir: Option<Direction>, | executable/support line; behavior impact depends on surrounding block context. |
| 176 | shift_timer: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 177 | auto_shift_cooldown_s: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 178 | /// Duration of current shift phase (varies with temperature & quality) | comment line; documentation intent, no runtime effect. |
| 179 | shift_duration: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 180 | (blank) | line is blank; visual separation only. |
| 181 | // ■ Auto-shift ████████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 182 | pub auto_mode: AutoShiftMode, | executable/support line; behavior impact depends on surrounding block context. |
| 183 | /// Target ground speed for GSM (km/h) | comment line; documentation intent, no runtime effect. |
| 184 | pub gsm_target_kmh: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 185 | pub gsm_enabled: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 186 | (blank) | line is blank; visual separation only. |
| 187 | // ■ IVT / CVT (used when trans_type = IVT) ████████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 188 | /// IVT ratio: 0 = full reverse, 0.5 = zero mechanical out, 1 = max fwd | comment line; documentation intent, no runtime effect. |
| 189 | pub ivt_ratio: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 190 | (blank) | line is blank; visual separation only. |
| 191 | // ■ Creeper ████████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 192 | pub creeper_engaged: bool, // additional ÷4 reduction | executable/support line; behavior impact depends on surrounding block context. |
| 193 | pub creeper_ratio: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 194 | (blank) | line is blank; visual separation only. |
| 195 | // ■ Thermal ████████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 196 | pub oil_temp_c: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 197 | pub sump_temp_c: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 198 | (blank) | line is blank; visual separation only. |
| 199 | // ■ Safety interlocks ████████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 200 | pub is_neutral: bool, // read by boot sequence for crank inhibit | executable/support line; behavior impact depends on surrounding block context. |
| 201 | pub handbrake_interlock: bool, // prevent direction change above speed threshold | executable/support line; behavior impact depends on surrounding block context. |
| 202 | (blank) | line is blank; visual separation only. |
| 203 | // ■ J1939 TX timers ████████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 204 | t_etc1: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 205 | t_etc2: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 206 | (blank) | line is blank; visual separation only. |
| 207 | // ■ Stats ████████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 208 | pub total_distance_km: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 209 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 210 | (blank) | line is blank; visual separation only. |
| 211 | impl Default for EcuTcm { | implementation block; method behavior and trait conformance are defined here. |
| 212 | fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 213 | Self::new() | executable/support line; behavior impact depends on surrounding block context. |
```

```

| 214 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 215 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 216 | (blank) | line is blank; visual separation only. |
| 217 | impl EcuTcm { | implementation block; method behavior and trait conformance are defined here. |
| 218 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
outputs. |
| 219 |         EcuTcm { | executable/support line; behavior impact depends on surrounding block context. |
| 220 |             sa: TCM_SA, | executable/support line; behavior impact depends on surrounding block context. |
| 221 |             trans_type: TransmissionType::Powershift, | executable/support line; behavior impact depends on s
urrounding block context. |
| 222 |             range: GearRange::B, | executable/support line; behavior impact depends on surrounding block cont
. |
| 223 |             speed_step: 1, | executable/support line; behavior impact depends on surrounding block context. |
| 224 |             direction: Direction::Neutral, | executable/support line; behavior impact depends on surrounding
ck context. |
| 225 |             gear_label: "N".into(), | executable/support line; behavior impact depends on surrounding block c
ext. |
| 226 |             output_shaft_rpm: 0.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 227 |             output_torque_nm: 0.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 228 |             ground_speed_kmh: 0.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 229 |             gear_ratio: GEAR_RATIOS[1][0], | executable/support line; behavior impact depends on surrounding
ck context. |
| 230 |             clutch_state: ClutchState::Open, | executable/support line; behavior impact depends on surroundin
lock context. |
| 231 |             clutch_slip_pct: 0.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 232 |             clutch_temp_c: 30.0, | executable/support line; behavior impact depends on surrounding block cont
. |
| 233 |             clutch_fill_pres: 0.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 234 |             clutch_phase_timer: 0.0, | executable/support line; behavior impact depends on surrounding block
text. |
| 235 |             is_shifting: false, | executable/support line; behavior impact depends on surrounding block conte
|
| 236 |             last_shift_qual: ShiftQuality::Normal, | executable/support line; behavior impact depends on surr
ounding block context. |
| 237 |             total_shifts: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 238 |             pending_range: None, | executable/support line; behavior impact depends on surrounding block cont
. |
| 239 |             pending_speed: None, | executable/support line; behavior impact depends on surrounding block cont
. |
| 240 |             pending_dir: None, | executable/support line; behavior impact depends on surrounding block context
|
| 241 |             shift_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context
|
| 242 |             auto_shift_cooldown_s: 0.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 243 |             shift_duration: 0.30, | executable/support line; behavior impact depends on surrounding block con
t. |
| 244 |             auto_mode: AutoShiftMode::Auto, | executable/support line; behavior impact depends on surrounding
ock context. |
| 245 |             gsm_target_kmh: 8.0, | executable/support line; behavior impact depends on surrounding block cont
. |
| 246 |             gsm_enabled: false, | executable/support line; behavior impact depends on surrounding block conte
|
| 247 |             ivt_ratio: 0.5, | executable/support line; behavior impact depends on surrounding block context. |
| 248 |             creeper_engaged: false, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 249 |             creeper_ratio: 4.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 250 |             oil_temp_c: 30.0, | executable/support line; behavior impact depends on surrounding block context
|
| 251 |             sump_temp_c: 30.0, | executable/support line; behavior impact depends on surrounding block context
|

```

[illegible]

```
| tent, no runtime effect. |  
| 295 | self.output_torque_nm = engine_torque_nm * self.gear_ratio * 0.96; | executable/support line; behavior impact depends on surrounding block context. |  
| 296 | (blank) | line is blank; visual separation only. |  
| 297 | // ■ Clutch dynamics ████████████████████████████████████████████████████████████ | comment line  
intent, no runtime effect. |  
| 298 | self.update_clutch(engine_rpm, engine_torque_nm, dt); | executable/support line; behavior impact depends on surrounding block context. |  
| 299 | (blank) | line is blank; visual separation only. |  
| 300 | // ■ Ground-speed dynamics (avoid unreal instant RPM→speed jumps) ████████ | comment line; documentation intent, no runtime effect. |  
| 301 | // v_target = ω_out × r_tyre × 3.6 (m/s → km/h), then limit by clutch coupling | comment line; documentation intent, no runtime effect. |  
| 302 | let mut target_speed_kmh = | executable/support line; behavior impact depends on surrounding block context.  
xt. |  
| 303 | (self.output_shaft_rpm / 60.0 * 2.0 * std::f64::consts::PI * TIRE_RADIUS_M * 3.6) | executable/support line; behavior impact depends on surrounding block context. |  
| 304 | .clamp(0.0, MAX_SPEED_KMH); | executable/support line; behavior impact depends on surrounding block context. |  
| 305 | (blank) | line is blank; visual separation only. |  
| 306 | let coupling = match self.clutch_state { | executable/support line; behavior impact depends on surrounding block context. |  
| 307 |     ClutchState::Locked => 1.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 308 |     ClutchState::Filling => 0.35, | executable/support line; behavior impact depends on surrounding block context. |  
| 309 |     ClutchState::Modulating => (1.0 - self.clutch_slip_pct).clamp(0.2, 0.75), | executable/support line; behavior impact depends on surrounding block context. |  
| 310 |     ClutchState::Open => 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 311 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
| |  
| 312 | target_speed_kmh *= coupling; | executable/support line; behavior impact depends on surrounding block context. |  
| 313 | (blank) | line is blank; visual separation only. |  
| 314 | let accel_limit_kmh_s = 1.4 + (throttle_pct / 100.0) * 4.6; | executable/support line; behavior impact depends on surrounding block context. |  
| 315 | let decel_limit_kmh_s = 2.0 + (brake_pct / 100.0) * 14.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 316 | let max_rise = accel_limit_kmh_s * dt; | executable/support line; behavior impact depends on surrounding block context. |  
| 317 | let max_drop = decel_limit_kmh_s * dt; | executable/support line; behavior impact depends on surrounding block context. |  
| 318 | let delta = target_speed_kmh - self.ground_speed_kmh; | executable/support line; behavior impact depends on surrounding block context. |  
| 319 | if delta >= 0.0 { | conditional gate; changing condition shifts runtime path and side effects. |  
| 320 |     self.ground_speed_kmh += delta.min(max_rise); | executable/support line; behavior impact depends on surrounding block context. |  
| 321 | } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 322 |     self.ground_speed_kmh += delta.max(-max_drop); | executable/support line; behavior impact depends on surrounding block context. |  
| 323 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
| |  
| 324 | self.ground_speed_kmh = self.ground_speed_kmh.clamp(0.0, MAX_SPEED_KMH); | executable/support line; behavior impact depends on surrounding block context. |  
| 325 | (blank) | line is blank; visual separation only. |  
| 326 | // ■ Auto-shift logic ████████████████████████████████████████████████████████████ | comment line  
intent, no runtime effect. |  
| 327 | self.auto_shift_cooldown_s = (self.auto_shift_cooldown_s - dt).max(0.0); | executable/support line; behavior impact depends on surrounding block context. |  
| 328 | (blank) | line is blank; visual separation only. |  
| 329 | if !self.is_shifting | conditional gate; changing condition shifts runtime path and side effects. |  
| 330 |     && self.auto_mode == AutoShiftMode::Auto | executable/support line; behavior impact depends on surrounding block context. |  
| 331 |     && self.direction == Direction::Forward | executable/support line; behavior impact depends on surrounding block context.
```

```
| ending block context. |  
| 332 |         && self.auto_shift_cooldown_s <= 0.0 | executable/support line; behavior impact depends on surrounding block context. |  
| 333 |     { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
| 334 |         self.auto_shift_logic(engine_rpm, engine_torque_nm, throttle_pct, brake_pct, dt); | executable/support line; behavior impact depends on surrounding block context. |  
| 335 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
| 336 | (blank) | line is blank; visual separation only. |  
| 337 |     // ■ Shift completion ████████████████████████████████████████████████████████████████████████████ | comment line; document, no runtime effect. |  
| 338 |     self.update_shift(dt); | executable/support line; behavior impact depends on surrounding block context.  
| 339 | (blank) | line is blank; visual separation only. | | |
| 340 |     // ■ Thermal model ████████████████████████████████████████████████████████████████████████████ | comment line; document, no runtime effect. |  
| 341 |     let heat = self.clutch_slip_pct * engine_torque_nm * 0.001; | executable/support line; behavior impact depends on surrounding block context. |  
| 342 |     let tgt_oil = 70.0 + heat * 0.5 + throttle_pct * 0.2; | executable/support line; behavior impact depends on surrounding block context. |  
| 343 |     self.oil_temp_c += (tgt_oil - self.oil_temp_c) * dt * 0.005; | executable/support line; behavior impact depends on surrounding block context. |  
| 344 |     self.sump_temp_c = self.oil_temp_c - 5.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 345 | (blank) | line is blank; visual separation only. |  
| 346 |     // ■ Distance accumulation ████████████████████████████████████████████████████████████████████████████ | comment line; document, no runtime effect. |  
| 347 |     self.total_distance_km += self.ground_speed_kmh * dt / 3600.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 348 | (blank) | line is blank; visual separation only. |  
| 349 |     // ■ Safety flags ████████████████████████████████████████████████████████████████████████████ | comment line; document, no runtime effect. |  
| 350 |     self.is_neutral = self.direction == Direction::Neutral || self.direction == Direction::Park; | executable/support line; behavior impact depends on surrounding block context. |  
| 351 |     self.handbrake_interlock = self.ground_speed_kmh > 3.0 | executable/support line; behavior impact depends on surrounding block context. |  
| 352 |     && (self.direction == Direction::Forward | executable/support line; behavior impact depends on surrounding block context. |  
| 353 |         && self.pending_dir == Some(Direction::Reverse) | executable/support line; behavior impact depends on surrounding block context. |  
| 354 |         || self.direction == Direction::Reverse | executable/support line; behavior impact depends on surrounding block context. |  
| 355 |         && self.pending_dir == Some(Direction::Forward)); | executable/support line; behavior impact depends on surrounding block context. |  
| 356 | (blank) | line is blank; visual separation only. |  
| 357 |     // ■ Gear label ████████████████████████████████████████████████████████████████████████████ | comment line; document, no runtime effect. |  
| 358 |     self.gear_label = self.compute_gear_label(); | executable/support line; behavior impact depends on surrounding block context. |  
| 359 | (blank) | line is blank; visual separation only. |  
| 360 |     // ■ Brake: if significant braking, disengage clutch ████████████████████████████████████████████████████████████████████████████ | comment line; document, no runtime effect. |  
| 361 |     if brake_pct > 0.3 && self.clutch_state == ClutchState::Locked { | conditional gate; changing condition shifts runtime path and side effects. |  
| 362 |         self.clutch_state = ClutchState::Open; | executable/support line; behavior impact depends on surrounding block context. |  
| 363 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
| 364 | (blank) | line is blank; visual separation only. |  
| 365 |     // ■ J1939 periodic TX ████████████████████████████████████████████████████████████████████████████ | comment line; document, no runtime effect. |  
| 366 |     self.t_etc1 += dt; | executable/support line; behavior impact depends on surrounding block context. |  
| 367 |     self.t_etc2 += dt; | executable/support line; behavior impact depends on surrounding block context. |  
| 368 |     let ts = self.total_distance_km; // monotonic timestamp approximation | executable/support line; behavior impact depends on surrounding block context.
```

```
| impact depends on surrounding block context. |  
| 369 |         let mut frames: Vec<J1939Frame> = Vec::new(); | executable/support line; behavior impact depends on s  
ounding block context. |  
| 370 | (blank) | line is blank; visual separation only. |  
| 371 |         if self.t_etc1 >= 0.020 { | conditional gate; changing condition shifts runtime path and side effects  
| 372 |             self.t_etc1 = 0.0; | executable/support line; behavior impact depends on surrounding block contex  
|  
| 373 |             frames.push(self.build_etc1(ts)); | executable/support line; behavior impact depends on surroundi  
block context. |  
| 374 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 375 |         if self.t_etc2 >= 0.020 { | conditional gate; changing condition shifts runtime path and side effects  
| 376 |             self.t_etc2 = 0.0; | executable/support line; behavior impact depends on surrounding block contex  
|  
| 377 |             frames.push(self.build_etc2(ts)); | executable/support line; behavior impact depends on surroundi  
block context. |  
| 378 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 379 |         frames | executable/support line; behavior impact depends on surrounding block context. |  
| 380 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 381 | (blank) | line is blank; visual separation only. |  
| 382 |     // ████████████████████████████████████████████████████████████████████████████████████████████████ | comment  
ntent, no runtime effect. |  
| 383 |     fn update_clutch(&mut self, _engine_rpm: f64, torque_nm: f64, dt: f64) { | declares executable behavior;  
nature/body changes alter API, control flow, and outputs. |  
| 384 |         match self.clutch_state { | branch dispatch; changing arms alters behavior for variants and input cla  
s. |  
| 385 |             ClutchState::Open => { | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 386 |                 self.clutch_slip_pct = 100.0; | executable/support line; behavior impact depends on surroundi  
block context. |  
| 387 |                 self.clutch_fill_pres = 0.0; | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 388 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 389 |             ClutchState::Filling => { | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 390 |                 self.clutch_phase_timer += dt; | executable/support line; behavior impact depends on surround  
block context. |  
| 391 |                 // Fill ramp 0 → 10 bar in 80-150 ms (faster when warm) | comment line; documentation intent,  
runtime effect. |  
| 392 |                 let fill_time = (0.15 - self.oil_temp_c.min(80.0) / 80.0 * 0.07).max(0.05); | executable/supp  
line; behavior impact depends on surrounding block context. |  
| 393 |                 self.clutch_fill_pres = (self.clutch_phase_timer / fill_time * 10.0).min(10.0); | executable/  
port line; behavior impact depends on surrounding block context. |  
| 394 |                 if self.clutch_phase_timer >= fill_time { | conditional gate; changing condition shifts runti  
path and side effects. |  
| 395 |                     self.clutch_state = ClutchState::Modulating; | executable/support line; behavior impact d  
nds on surrounding block context. |  
| 396 |                     self.clutch_phase_timer = 0.0; | executable/support line; behavior impact depends on surr  
ding block context. |  
| 397 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti  
scopes. |  
| 398 |                 self.clutch_slip_pct = 80.0; | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 399 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 400 |             ClutchState::Modulating => { | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 401 |                 self.clutch_phase_timer += dt; | executable/support line; behavior impact depends on surround  
block context. |  
| 402 |                 // Modulate pressure 10 → 25 bar over ~250 ms, reducing slip | comment line; documentation i  
t, no runtime effect. |  
| 403 |                 let mod_time = 0.25; | executable/support line; behavior impact depends on surrounding block  
text. |
```

```
| 404 |           let pres_ramp = self.clutch_phase_timer / mod_time; | executable/support line; behavior impact depends on surrounding block context. |  
| 405 |               self.clutch_fill_pres = 10.0 + pres_ramp * 15.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 406 |                   self.clutch_slip_pct = (80.0 * (1.0 - pres_ramp)).max(0.0); | executable/support line; behavior impact depends on surrounding block context. |  
| 407 |                       if self.clutch_phase_timer >= mod_time { | conditional gate; changing condition shifts runtime path and side effects. |  
| 408 |                           self.clutch_state = ClutchState::Locked; | executable/support line; behavior impact depends on surrounding block context. |  
| 409 |                               self.clutch_slip_pct = 0.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 410 |                                   self.clutch_fill_pres = 25.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 411 |                                       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 412 |                                           // Clutch temp from slip energy | comment line; documentation intent, no runtime effect. |  
| 413 |                                               self.clutch_temp_c += self.clutch_slip_pct * torque_nm * 0.00005; | executable/support line; behavior impact depends on surrounding block context. |  
| 414 |                                                   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 415 |               ClutchState::Locked => { | executable/support line; behavior impact depends on surrounding block context. |  
| 416 |                   self.clutch_fill_pres = 25.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 417 |                       self.clutch_slip_pct = 0.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 418 |                           self.clutch_temp_c = (self.clutch_temp_c - 5.0 * dt).max(self.oil_temp_c); | executable/support line; behavior impact depends on surrounding block context. |  
| 419 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 420 |                                   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 421 |                                       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 422 | (blank) | line is blank; visual separation only. |  
| 423 |         // ████████████████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |  
| 424 |         fn auto_shift_logic( | declares executable behavior; signature/body changes alter API, control flow, and puts. |  
| 425 |             &mut self, | executable/support line; behavior impact depends on surrounding block context. |  
| 426 |             engine_rpm: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 427 |             engine_torque_nm: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 428 |             throttle_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 429 |             brake_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 430 |             _dt: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 431 |         ) { | executable/support line; behavior impact depends on surrounding block context. |  
| 432 |             let t = (throttle_pct / 100.0).clamp(0.0, 1.0); | executable/support line; behavior impact depends on surrounding block context. |  
| 433 |                 let load = (engine_torque_nm / 1050.0).clamp(0.0, 1.2); | executable/support line; behavior impact depends on surrounding block context. |  
| 434 |                     // ECM governor target approximation (matches ECM model family). | comment line; documentation intent, no runtime effect. |  
| 435 |                         let throttle_rpm_ceiling = 800.0 + t * (2600.0 - 800.0); | executable/support line; behavior impact depends on surrounding block context. |  
| 436 | (blank) | line is blank; visual separation only. |  
| 437 |                 let base_up = match self.range { | executable/support line; behavior impact depends on surrounding block context. |  
| 438 |                     GearRange::A => 1180.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 439 |                     GearRange::B => 1240.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 440 |                     GearRange::C => 1300.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 441 |                     GearRange::D => 1360.0, | executable/support line; behavior impact depends on surrounding block context.
```

```

ext. |
| 442 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 443 |         // Higher throttle/load delays upshift; light pedal upshifts early. | comment line; documentation int
, no runtime effect. |
| 444 |         let upshift_rpm_raw = base_up + t * 520.0 + load * 180.0; | executable/support line; behavior impact
ends on surrounding block context. |
| 445 |         let upshift_rpm = upshift_rpm_raw | executable/support line; behavior impact depends on surrounding b
k context. |
| 446 |             .min(throttle_rpm_ceiling - (90.0 - t * 35.0)) | executable/support line; behavior impact depends
surrounding block context. |
| 447 |             .clamp(1000.0, 2320.0); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 448 |         // Keep hysteresis wide enough to avoid gear hunting. | comment line; documentation intent, no runtim
ffect. |
| 449 |         let downshift_rpm = (upshift_rpm - (300.0 - t * 70.0)).clamp(760.0, 1820.0); | executable/support lin
behavior impact depends on surrounding block context. |
| 450 | (blank) | line is blank; visual separation only. |
| 451 |         // Aggressive kickdown when high pedal but engine is lugging. | comment line; documentation intent, n
untime effect. |
| 452 |         if throttle_pct > 80.0 && engine_rpm < 1250.0 { | conditional gate; changing condition shifts runtime
th and side effects. |
| 453 |             self.request_downshift(); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 454 |             return; | executable/support line; behavior impact depends on surrounding block context. |
| 455 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 456 | (blank) | line is blank; visual separation only. |
| 457 |         // Prefer holding or downshifting under braking. | comment line; documentation intent, no runtime eff
. |
| 458 |         if brake_pct > 20.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 459 |             if engine_rpm < 1200.0 { | conditional gate; changing condition shifts runtime path and side effe
. |
| 460 |                 self.request_downshift(); | executable/support line; behavior impact depends on surrounding b
k context. |
| 461 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 462 |             return; | executable/support line; behavior impact depends on surrounding block context. |
| 463 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 464 | (blank) | line is blank; visual separation only. |
| 465 |         // Stall/lugging protection. | comment line; documentation intent, no runtime effect. |
| 466 |         if engine_rpm < 820.0 { | conditional gate; changing condition shifts runtime path and side effects.
| 467 |             self.request_downshift(); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 468 |             return; | executable/support line; behavior impact depends on surrounding block context. |
| 469 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 470 | (blank) | line is blank; visual separation only. |
| 471 |         if engine_rpm > upshift_rpm { | conditional gate; changing condition shifts runtime path and side eff
s. |
| 472 |             self.request_upshift(); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 473 |         } else if engine_rpm < downshift_rpm { | executable/support line; behavior impact depends on surround
block context. |
| 474 |             self.request_downshift(); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 475 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 476 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 477 | (blank) | line is blank; visual separation only. |
| 478 |     // ████████████████████████████████████████████████████████████████████████████████████████████████████ | comment
ntent, no runtime effect. |
| 479 |     fn request_upshift(&mut self) { | declares executable behavior; signature/body changes alter API, control
ow, and outputs. |

```



```

| 480 |         if self.speed_step < 4 { | conditional gate; changing condition shifts runtime path and side effects. |
| 481 |             self.begin_shift(self.range, self.speed_step + 1, self.direction); | executable/support line; beh
or impact depends on surrounding block context. |
| 482 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 483 |             // Cross-range upshift | comment line; documentation intent, no runtime effect. |
| 484 |             let next_range = match self.range { | executable/support line; behavior impact depends on surround
g block context. |
| 485 |                 GearRange::A => Some(GearRange::B), | executable/support line; behavior impact depends on surr
nding block context. |
| 486 |                 GearRange::B => Some(GearRange::C), | executable/support line; behavior impact depends on surr
nding block context. |
| 487 |                 GearRange::C => Some(GearRange::D), | executable/support line; behavior impact depends on surr
nding block context. |
| 488 |                 GearRange::D => None, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 489 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 490 |             if let Some(r) = next_range { | conditional gate; changing condition shifts runtime path and side
ffects. |
| 491 |                 self.begin_shift(r, 1, self.direction); | executable/support line; behavior impact depends on
rrounding block context. |
| 492 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 493 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 494 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 495 | (blank) | line is blank; visual separation only. |
| 496 |     fn request_downshift(&mut self) { | declares executable behavior; signature/body changes alter API, contr
flow, and outputs. |
| 497 |         if self.speed_step > 1 { | conditional gate; changing condition shifts runtime path and side effects. |
| 498 |             self.begin_shift(self.range, self.speed_step - 1, self.direction); | executable/support line; beh
or impact depends on surrounding block context. |
| 499 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 500 |             let prev_range = match self.range { | executable/support line; behavior impact depends on surround
g block context. |
| 501 |                 GearRange::B => Some(GearRange::A), | executable/support line; behavior impact depends on surr
nding block context. |
| 502 |                 GearRange::C => Some(GearRange::B), | executable/support line; behavior impact depends on surr
nding block context. |
| 503 |                 GearRange::D => Some(GearRange::C), | executable/support line; behavior impact depends on surr
nding block context. |
| 504 |                 GearRange::A => None, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 505 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 506 |             if let Some(r) = prev_range { | conditional gate; changing condition shifts runtime path and side
ffects. |
| 507 |                 self.begin_shift(r, 4, self.direction); | executable/support line; behavior impact depends on
rrounding block context. |
| 508 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 509 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 510 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 511 | (blank) | line is blank; visual separation only. |
| 512 |     // ████████████████████████████████████████████████████████████████████████████████████████████████████ | comment
ntent, no runtime effect. |
| 513 |     fn begin_shift(&mut self, new_range: GearRange, new_speed: u8, new_dir: Direction) { | declares executabl
ehavior; signature/body changes alter API, control flow, and outputs. |
| 514 |         if self.is_shifting { | conditional gate; changing condition shifts runtime path and side effects. |
| 515 |             return; | executable/support line; behavior impact depends on surrounding block context. |
| 516 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 517 |         if self.handbrake_interlock { | conditional gate; changing condition shifts runtime path and side eff
s. |

```

```

| 518 |         return; | executable/support line; behavior impact depends on surrounding block context. |
| 519 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 520 |         self.pending_range = Some(new_range); | executable/support line; behavior impact depends on surroundi
block context. |
| 521 |         self.pending_speed = Some(new_speed); | executable/support line; behavior impact depends on surroundi
block context. |
| 522 |         self.pending_dir = Some(new_dir); | executable/support line; behavior impact depends on surrounding b
k context. |
| 523 |         self.is_shifting = true; | executable/support line; behavior impact depends on surrounding block cont
. |
| 524 |         self.shift_timer = 0.0; | executable/support line; behavior impact depends on surrounding block conte
|
| 525 |         // Shift duration: longer for range change, shorter for speed-step | comment line; documentation inte
no runtime effect. |
| 526 |         let range_change = new_range != self.range || new_dir != self.direction; | executable/support line;
havior impact depends on surrounding block context. |
| 527 |         self.shift_duration = if range_change { 0.45 } else { 0.25 }; | executable/support line; behavior imp
depends on surrounding block context. |
| 528 |         self.auto_shift_cooldown_s = self.shift_duration + 0.20; | executable/support line; behavior impact d
nds on surrounding block context. |
| 529 |         // Disengage clutch to start shift | comment line; documentation intent, no runtime effect. |
| 530 |         self.clutch_state = ClutchState::Open; | executable/support line; behavior impact depends on surround
block context. |
| 531 |         self.clutch_phase_timer = 0.0; | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 532 |         self.total_shifts += 1; | executable/support line; behavior impact depends on surrounding block conte
|
| 533 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 534 | (blank) | line is blank; visual separation only. |
| 535 |     fn update_shift(&mut self, dt: f64) { | declares executable behavior; signature/body changes alter API, c
rol flow, and outputs. |
| 536 |         if !self.is_shifting { | conditional gate; changing condition shifts runtime path and side effects. |
| 537 |             return; | executable/support line; behavior impact depends on surrounding block context. |
| 538 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 539 |         self.shift_timer += dt; | executable/support line; behavior impact depends on surrounding block conte
|
| 540 |         // Mid-shift: engage new gear at ~50% of shift time | comment line; documentation intent, no runtime
ect. |
| 541 |         if self.shift_timer >= self.shift_duration * 0.5 { | conditional gate; changing condition shifts runt
path and side effects. |
| 542 |             if let (Some(r), Some(s), Some(d)) = | conditional gate; changing condition shifts runtime path a
side effects. |
| 543 |                 (self.pending_range, self.pending_speed, self.pending_dir) | executable/support line; behavior
mpact depends on surrounding block context. |
| 544 |                 { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 545 |                     self.range = r; | executable/support line; behavior impact depends on surrounding block conte
|
| 546 |                     self.speed_step = s.clamp(1, 4); | executable/support line; behavior impact depends on surrou
ng block context. |
| 547 |                     self.direction = d; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 548 |                     self.pending_range = None; | executable/support line; behavior impact depends on surrounding
ck context. |
| 549 |                     self.pending_speed = None; | executable/support line; behavior impact depends on surrounding
ck context. |
| 550 |                     self.pending_dir = None; | executable/support line; behavior impact depends on surrounding bl
context. |
| 551 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 552 |             // Start filling new clutch | comment line; documentation intent, no runtime effect. |
| 553 |             if self.clutch_state == ClutchState::Open { | conditional gate; changing condition shifts runtime
th and side effects. |

```

```

| 554 |         self.clutch_state = ClutchState::Filling; | executable/support line; behavior impact depends
surrounding block context. |
| 555 |         self.clutch_phase_timer = 0.0; | executable/support line; behavior impact depends on surround
block context. |
| 556 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 557 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 558 |     if self.shift_timer >= self.shift_duration { | conditional gate; changing condition shifts runtime pa
and side effects. |
| 559 |         self.is_shifting = false; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 560 |         // Assess shift quality by clutch temperature rise | comment line; documentation intent, no runti
effect. |
| 561 |         self.last_shift_qual = if self.clutch_temp_c > 150.0 { | executable/support line; behavior impact
ends on surrounding block context. |
| 562 |             ShiftQuality::Harsh | executable/support line; behavior impact depends on surrounding block c
ext. |
| 563 |         } else if self.clutch_temp_c > 100.0 { | executable/support line; behavior impact depends on surr
ding block context. |
| 564 |             ShiftQuality::Normal | executable/support line; behavior impact depends on surrounding block
text. |
| 565 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 566 |             ShiftQuality::Smooth | executable/support line; behavior impact depends on surrounding block
text. |
| 567 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 568 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 569 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 570 | (blank) | line is blank; visual separation only. |
| 571 | // ████████████████████████████████████████████████████████████████████████████████████████████████████ | comment
ntent, no runtime effect. |
| 572 | fn update_ivt(&mut self, _engine_rpm: f64, throttle_pct: f64, dt: f64) { | declares executable behavior;
nature/body changes alter API, control flow, and outputs. |
| 573 |     // IVT: target ratio → target_speed = throttle_pct / 100 * MAX_SPEED | comment line; documentation in
t, no runtime effect. |
| 574 |     let target_spd = throttle_pct / 100.0 * MAX_SPEED_KMH; | executable/support line; behavior impact dep
s on surrounding block context. |
| 575 |     let current_spd = self.ground_speed_kmh; | executable/support line; behavior impact depends on surrou
ng block context. |
| 576 |     let spd_err = target_spd - current_spd; | executable/support line; behavior impact depends on surroun
g block context. |
| 577 |     // Vary IVT ratio to close speed error | comment line; documentation intent, no runtime effect. |
| 578 |     let ratio_change = (spd_err * 0.01).clamp(-0.05, 0.05); | executable/support line; behavior impact de
ds on surrounding block context. |
| 579 |     self.ivt_ratio = (self.ivt_ratio + ratio_change * dt).clamp(0.0, 1.0); | executable/support line; beh
or impact depends on surrounding block context. |
| 580 |     // Effective gear ratio: 0 ratio → infinite, 1 ratio → direct | comment line; documentation intent,
untime effect. |
| 581 |     if self.ivt_ratio > 0.01 { | conditional gate; changing condition shifts runtime path and side effect
|
| 582 |         self.gear_ratio = 50.0 / (self.ivt_ratio * 50.0); | executable/support line; behavior impact depe
on surrounding block context. |
| 583 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 584 |         self.gear_ratio = 999.0; | executable/support line; behavior impact depends on surrounding block
text. |
| 585 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 586 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 587 | (blank) | line is blank; visual separation only. |
| 588 | // ████████████████████████████████████████████████████████████████████████████████████████████████████ | comment
ntent, no runtime effect. |
| 589 | // Manual gear controls (called from main.rs keyboard input) | comment line; documentation intent, no run
e effect. |

```

[illegible]

```

ntent, no runtime effect. |
| 628 |      fn build_etc1(&self, ts: f64) -> J1939Frame { | declares executable behavior; signature/body changes alter
PI, control flow, and outputs. |
| 629 |          let mut data = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 630 |          // SPN 560: Transmission driveline engaged (bits 0-1) | comment line; documentation intent, no runtim
effect. |
| 631 |          data[0] = if self.direction != Direction::Neutral { | executable/support line; behavior impact depend
n surrounding block context. |
| 632 |              0x01 | executable/support line; behavior impact depends on surrounding block context. |
| 633 |          } else { | executable/support line; behavior impact depends on surrounding block context. |
| 634 |              0x00 | executable/support line; behavior impact depends on surrounding block context. |
| 635 |          }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 636 |          // SPN 573: Transmission torque converter lockup engaged | comment line; documentation intent, no run
e effect. |
| 637 |          data[0] \|= if self.clutch_state == ClutchState::Locked { | executable/support line; behavior impact
ends on surrounding block context. |
| 638 |              0x04 | executable/support line; behavior impact depends on surrounding block context. |
| 639 |          } else { | executable/support line; behavior impact depends on surrounding block context. |
| 640 |              0x00 | executable/support line; behavior impact depends on surrounding block context. |
| 641 |          }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 642 |          // SPN 191: Transmission Output Shaft Speed (0.125 rpm/bit) | comment line; documentation intent, no
time effect. |
| 643 |          let oss = (self.output_shaft_rpm / 0.125) as u16; | executable/support line; behavior impact depends
surrounding block context. |
| 644 |          data[3] = (oss & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 645 |          data[4] = (oss >> 8) as u8; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 646 |          // SPN 161: Input shaft speed | comment line; documentation intent, no runtime effect. |
| 647 |          J1939Frame::from_raw( | executable/support line; behavior impact depends on surrounding block context
| 648 |              ts, | executable/support line; behavior impact depends on surrounding block context. |
| 649 |              J1939Frame::build_id(3, j1939::pgn::ETC1, self.sa, 0xFF), | executable/support line; behavior imp
depends on surrounding block context. |
| 650 |              &data, | executable/support line; behavior impact depends on surrounding block context. |
| 651 |          ) | executable/support line; behavior impact depends on surrounding block context. |
| 652 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 653 | (blank) | line is blank; visual separation only. |
| 654 |      fn build_etc2(&self, ts: f64) -> J1939Frame { | declares executable behavior; signature/body changes alter
PI, control flow, and outputs. |
| 655 |          let mut data = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 656 |          // SPN 524: Transmission Selected Gear | comment line; documentation intent, no runtime effect. |
| 657 |          let gear_byte = match self.direction { | executable/support line; behavior impact depends on surround
block context. |
| 658 |              Direction::Neutral => 125u8, | executable/support line; behavior impact depends on surrounding bl
context. |
| 659 |              Direction::Park => 126u8, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 660 |              Direction::Forward => (self.range as u8 * 4 + self.speed_step) + 125, | executable/support line;
avior impact depends on surrounding block context. |
| 661 |              Direction::Reverse => (125u8).saturating_sub(self.range as u8 * 4 + self.speed_step), | executabl
upport line; behavior impact depends on surrounding block context. |
| 662 |          }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 663 |          data[3] = gear_byte; | executable/support line; behavior impact depends on surrounding block context.
| 664 |          // SPN 523: Transmission Actual Gear | comment line; documentation intent, no runtime effect. |
| 665 |          data[4] = gear_byte; | executable/support line; behavior impact depends on surrounding block context.
| 666 |          // SPN 525: Transmission Current Range | comment line; documentation intent, no runtime effect. |
| 667 |          data[7] = self.range as u8; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 668 |          J1939Frame::from_raw( | executable/support line; behavior impact depends on surrounding block context
| 669 |              ts, | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 670 |         J1939Frame::build_id(3, j1939::pgn::ETC2, self.sa, 0xFF), | executable/support line; behavior impact
depends on surrounding block context. |
| 671 |         &data, | executable/support line; behavior impact depends on surrounding block context. |
| 672 |     ) | executable/support line; behavior impact depends on surrounding block context. |
| 673 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 674 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 675 | (blank) | line is blank; visual separation only. |
| 676 | #[cfg(test)] | executable/support line; behavior impact depends on surrounding block context. |
| 677 | mod tests { | declares a module boundary; changing module name/path breaks references. |
| 678 |     use super::{AutoShiftMode, ClutchState, Direction, EcuTcm, GearRange}; | imports symbols; changing path/n
affects compile-time resolution. |
| 679 | (blank) | line is blank; visual separation only. |
| 680 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 681 |     fn d4_partial_throttle_does_not_pin_to_50_kmh() { | declares executable behavior; signature/body changes
er API, control flow, and outputs. |
| 682 |         let mut tcm = EcuTcm::new(); | executable/support line; behavior impact depends on surrounding block
text. |
| 683 |         tcm.direction = Direction::Forward; | executable/support line; behavior impact depends on surrounding
ock context. |
| 684 |         tcm.range = GearRange::D; | executable/support line; behavior impact depends on surrounding block con
t. |
| 685 |         tcm.speed_step = 4; | executable/support line; behavior impact depends on surrounding block context. |
| 686 |         tcm.auto_mode = AutoShiftMode::Hold; | executable/support line; behavior impact depends on surroundin
lock context. |
| 687 |         tcm.clutch_state = ClutchState::Locked; | executable/support line; behavior impact depends on surround
g block context. |
| 688 | (blank) | line is blank; visual separation only. |
| 689 |         for _ in 0..400 { | iteration control; changing bounds/body affects throughput, timing, and possibly
ety guarantees. |
| 690 |             let _ = tcm.tick(1340.0, 650.0, 30.0, 0.0, 0.1); | executable/support line; behavior impact depen
on surrounding block context. |
| 691 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 692 | (blank) | line is blank; visual separation only. |
| 693 |         assert!(tcm.ground_speed_kmh > 20.0, "speed should build under partial throttle"); | executable/suppo
line; behavior impact depends on surrounding block context. |
| 694 |         assert!( | executable/support line; behavior impact depends on surrounding block context. |
| 695 |             tcm.ground_speed_kmh < 40.0, | executable/support line; behavior impact depends on surrounding bl
context. |
| 696 |             "partial throttle should not saturate speedometer at 50 km/h" | executable/support line; behavior
pact depends on surrounding block context. |
| 697 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 698 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 699 | (blank) | line is blank; visual separation only. |
| 700 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 701 |     fn d4_full_throttle_reaches_transport_band_without_exceeding_limit() { | declares executable behavior; si
ture/body changes alter API, control flow, and outputs. |
| 702 |         let mut tcm = EcuTcm::new(); | executable/support line; behavior impact depends on surrounding block
text. |
| 703 |         tcm.direction = Direction::Forward; | executable/support line; behavior impact depends on surrounding
ock context. |
| 704 |         tcm.range = GearRange::D; | executable/support line; behavior impact depends on surrounding block con
t. |
| 705 |         tcm.speed_step = 4; | executable/support line; behavior impact depends on surrounding block context. |
| 706 |         tcm.auto_mode = AutoShiftMode::Hold; | executable/support line; behavior impact depends on surroundin
lock context. |
| 707 |         tcm.clutch_state = ClutchState::Locked; | executable/support line; behavior impact depends on surround
g block context. |
| 708 | (blank) | line is blank; visual separation only. |
| 709 |         for _ in 0..600 { | iteration control; changing bounds/body affects throughput, timing, and possibly
ety guarantees. |
| 710 |             let _ = tcm.tick(2200.0, 1050.0, 100.0, 0.0, 0.1); | executable/support line; behavior impact dep
s on surrounding block context. |
| 711 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|

```

```
| 712 | (blank) | line is blank; visual separation only. |
| 713 |         assert!(tcm.ground_speed_kmh >= 45.0, "transport gear should reach high-road band"); | executable/sup
t line; behavior impact depends on surrounding block context. |
| 714 |         assert!(tcm.ground_speed_kmh <= 50.0, "speed must remain capped at configured maximum"); | executable
pport line; behavior impact depends on surrounding block context. |
| 715 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 716 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
```

File: src/ecu_vcm.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 422 | Non-empty: 381 | Symbols: 35

Symbol Inventory

Line	Kind	Symbol	Change Impact
14	const	VCM_SA	data/structure coupling
18	enum	VehicleMode	data/structure coupling
28	impl	std::fmt::Display for VehicleMode	behavior binding and method semantics
29	fn	fmt	signature/API coupling
44	struct	PowerDistribution	data/structure coupling
56	impl	Default for PowerDistribution	behavior binding and method semantics
57	fn	default	signature/API coupling
62	impl	PowerDistribution	behavior binding and method semantics
63	fn	new	signature/API coupling
80	struct	TorqueCoordination	data/structure coupling
95	impl	Default for TorqueCoordination	behavior binding and method semantics
96	fn	default	signature/API coupling
101	impl	TorqueCoordination	behavior binding and method semantics
102	fn	new	signature/API coupling
113	fn	compute_final	signature/API coupling
131	struct	CommHealth	data/structure coupling
147	impl	Default for CommHealth	behavior binding and method semantics
148	fn	default	signature/API coupling
153	impl	CommHealth	behavior binding and method semantics
154	fn	new	signature/API coupling
173	fn	check_timeouts	signature/API coupling
174	const	ECM_TIMEOUT	data/structure coupling
175	const	TCM_TIMEOUT	data/structure coupling
176	const	ABS_TIMEOUT	data/structure coupling
177	const	HCM_TIMEOUT	data/structure coupling
200	fn	update_from_frame	signature/API coupling
213	struct	EcuVcm	data/structure coupling
241	impl	Default for EcuVcm	behavior binding and method semantics
242	fn	default	signature/API coupling
247	impl	EcuVcm	behavior binding and method semantics
248	fn	new	signature/API coupling
271	fn	tick	signature/API coupling
380	fn	build_vcm_status	signature/API coupling
401	fn	build_ws_master	signature/API coupling
411	fn	build_dm13	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	/// Vehicle Control Module (VCM) / Gateway ECU – SA 0x14 (headway controller position).	comment line; documentation intent, no runtime effect.
2	///	comment line; documentation intent, no runtime effect.
3	/// The VCM is the central coordinator in modern heavy machinery:	comment line; documentation intent, no runtime effect.
4	/// • Acts as CAN gateway between HS-CAN powertrain, MS-CAN body, and ISOBUS	comment line; documentation intent, no runtime effect.
5	/// • Coordinates ADAS features across domains	comment line; documentation intent, no runtime effect.

[illegible]

[illegible]

```

. |
| 87 |      /// Torque requested by AD system | comment line; documentation intent, no runtime effect. |
| 88 |      pub ad_torque_request_nm: f64, | executable/support line; behavior impact depends on surrounding block con
t. |
| 89 |      /// Final commanded torque (minimum of all limits) | comment line; documentation intent, no runtime effect
| 90 |      pub commanded_torque_nm: f64, | executable/support line; behavior impact depends on surrounding block cont
. |
| 91 |      /// Reason for any torque reduction | comment line; documentation intent, no runtime effect. |
| 92 |      pub limiting_source: &'static str, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 93 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 94 | (blank) | line is blank; visual separation only. |
| 95 | impl Default for TorqueCoordination { | implementation block; method behavior and trait conformance are define
ere. |
| 96 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
tputs. |
| 97 |         Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 98 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 99 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 100 | (blank) | line is blank; visual separation only. |
| 101 | impl TorqueCoordination { | implementation block; method behavior and trait conformance are defined here. |
| 102 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, an
utputs. |
| 103 |         TorqueCoordination { | executable/support line; behavior impact depends on surrounding block context.
| 104 |             max_engine_torque_nm: 1050.0, | executable/support line; behavior impact depends on surrounding b
k context. |
| 105 |             tcm_torque_request_nm: 0.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 106 |             abs_torque_limit_nm: 1050.0, | executable/support line; behavior impact depends on surrounding bl
context. |
| 107 |             ad_torque_request_nm: 1050.0, | executable/support line; behavior impact depends on surrounding b
k context. |
| 108 |             commanded_torque_nm: 1050.0, | executable/support line; behavior impact depends on surrounding bl
context. |
| 109 |             limiting_source: "none", | executable/support line; behavior impact depends on surrounding block
text. |
| 110 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 111 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 112 | (blank) | line is blank; visual separation only. |
| 113 |     pub fn compute_final(&mut self) { | declares executable behavior; signature/body changes alter API, contr
flow, and outputs. |
| 114 |         let limit = [ | executable/support line; behavior impact depends on surrounding block context. |
| 115 |             (self.max_engine_torque_nm, "VCM-max"), | executable/support line; behavior impact depends on surr
nding block context. |
| 116 |             (self.abs_torque_limit_nm, "ABS/TCS"), | executable/support line; behavior impact depends on surr
ding block context. |
| 117 |             (self.ad_torque_request_nm, "AD-ctrl"), | executable/support line; behavior impact depends on surr
nding block context. |
| 118 |         ]; | executable/support line; behavior impact depends on surrounding block context. |
| 119 |         let (min_torque, reason) = limit | executable/support line; behavior impact depends on surrounding bl
context. |
| 120 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 121 |             .min_by(\|a, b\| a.0.total_cmp(&b.0)) | executable/support line; behavior impact depends on surro
ing block context. |
| 122 |             .copied() | executable/support line; behavior impact depends on surrounding block context. |
| 123 |             .unwrap_or((1050.0, "none")); | executable/support line; behavior impact depends on surrounding b
k context. |
| 124 |         self.commanded_torque_nm = min_torque; | executable/support line; behavior impact depends on surround
block context. |
| 125 |         self.limiting_source = reason; | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 126 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 127 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 128 | (blank) | line is blank; visual separation only. |

```

[illegible]


```
| k context. |  
| 218 | pub comm_health: CommHealth, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 219 | (blank) | line is blank; visual separation only. |  
| 220 | // ■ Safety functions ████████████████████████████████████████████████████████████████████████████ | comment li  
intent, no runtime effect. |  
| 221 | pub rollaway_prevention: bool, // prevents motion without driver | executable/support line; behavior imp  
depends on surrounding block context. |  
| 222 | pub overspeed_protection: bool, // limits speed to rated maximum | executable/support line; behavior impa  
depends on surrounding block context. |  
| 223 | pub pto_interlock: bool, // prevents unsafe PTO engagement | executable/support line; behavior imp  
depends on surrounding block context. |  
| 224 | pub max_speed_kmh: f64, // speed limit from VCM (e.g. 50 km/h road limit) | executable/support li  
behavior impact depends on surrounding block context. |  
| 225 | pub speed_limited: bool, | executable/support line; behavior impact depends on surrounding block context.  
| 226 | (blank) | line is blank; visual separation only. |  
| 227 | // ■ Fuel-cut coordination ████████████████████████████████████████████████████████████████████████████ | comment lin  
intent, no runtime effect. |  
| 228 | pub fuel_cut_active: bool, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 229 | pub fuel_cut_reason: Option<&'static str>, | executable/support line; behavior impact depends on surround  
block context. |  
| 230 | (blank) | line is blank; visual separation only. |  
| 231 | // ■ Diagnostic aggregation ████████████████████████████████████████████████████████████████████████████ | comment line;  
ntent, no runtime effect. |  
| 232 | pub total_active_dtcs: u32, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 233 | pub critical_fault: bool, | executable/support line; behavior impact depends on surrounding block context  
| 234 | (blank) | line is blank; visual separation only. |  
| 235 | // ■ J1939 TX timers ████████████████████████████████████████████████████████████████████████████ | comment lin  
tent, no runtime effect. |  
| 236 | t_status: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 237 | t_ws_mst: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 238 | t_dm13: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 239 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 240 | (blank) | line is blank; visual separation only. |  
| 241 | impl Default for EcuVcm { | implementation block; method behavior and trait conformance are defined here. |  
| 242 | fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, an  
utputs. |  
| 243 | Self::new() | executable/support line; behavior impact depends on surrounding block context. |  
| 244 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 245 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 246 | (blank) | line is blank; visual separation only. |  
| 247 | impl EcuVcm { | implementation block; method behavior and trait conformance are defined here. |  
| 248 | pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, an  
utputs. |  
| 249 | EcuVcm { | executable/support line; behavior impact depends on surrounding block context. |  
| 250 | sa: VCM_SA, | executable/support line; behavior impact depends on surrounding block context. |  
| 251 | mode: VehicleMode::Parked, | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 252 | power: PowerDistribution::new(), | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 253 | torque_coord: TorqueCoordination::new(), | executable/support line; behavior impact depends on su  
unding block context. |  
| 254 | comm_health: CommHealth::new(), | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 255 | rollaway_prevention: true, | executable/support line; behavior impact depends on surrounding blo  
ontext. |  
| 256 | overspeed_protection: true, | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 257 | pto_interlock: true, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 258 | max_speed_kmh: 50.0, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 259 | speed_limited: false, | executable/support line; behavior impact depends on surrounding block con
```

```
| t. |  
| 260 | fuel_cut_active: false, | executable/support line; behavior impact depends on surrounding block context.  
ext. |  
| 261 | fuel_cut_reason: None, | executable/support line; behavior impact depends on surrounding block context.  
xt. |  
| 262 | total_active_dtcs: 0, | executable/support line; behavior impact depends on surrounding block context.  
t. |  
| 263 | critical_fault: false, | executable/support line; behavior impact depends on surrounding block context.  
xt. |  
| 264 | t_status: 0.0, | executable/support line; behavior impact depends on surrounding block context.  
| 265 | t_ws_mst: 0.0, | executable/support line; behavior impact depends on surrounding block context.  
| 266 | t_dm13: 0.0, | executable/support line; behavior impact depends on surrounding block context.  
| 267 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes  
|  
| 268 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 269 | (blank) | line is blank; visual separation only. |  
| 270 | #[allow(clippy::too_many_arguments)] | executable/support line; behavior impact depends on surrounding block  
context. |  
| 271 | pub fn tick( | declares executable behavior; signature/body changes alter API, control flow, and outputs.  
| 272 | &ampmut self, | executable/support line; behavior impact depends on surrounding block context. |  
| 273 | elapsed: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 274 | engine_rpm: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 275 | ground_speed_kmh: f64, | executable/support line; behavior impact depends on surrounding block context.  
|  
| 276 | engine_torque_nm: f64, | executable/support line; behavior impact depends on surrounding block context.  
|  
| 277 | hydraulic_power_kw: f64, | executable/support line; behavior impact depends on surrounding block context.  
. |  
| 278 | pto_torque_nm: f64, | executable/support line; behavior impact depends on surrounding block context.  
| 279 | bcm_load_a: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 280 | abs_torque_limit: f64, | executable/support line; behavior impact depends on surrounding block context.  
|  
| 281 | ad_torque_limit: f64, | executable/support line; behavior impact depends on surrounding block context.  
| 282 | dtc_count: u32, | executable/support line; behavior impact depends on surrounding block context. |  
| 283 | received_frames: &[J1939Frame], | executable/support line; behavior impact depends on surrounding block  
context. |  
| 284 | dt: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 285 | ) -> Vec<J1939Frame> { | executable/support line; behavior impact depends on surrounding block context. |  
| 286 | // ■ Update comms health ████████████████████████████████████████████ | comment line; does not have runtime  
intent, no runtime effect. |  
| 287 | for f in received_frames { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |  
| 288 | self.comm_health.update_from_frame(f); | executable/support line; behavior impact depends on surrounding block context. |  
| 289 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes  
|  
| 290 | self.comm_health.check_timeouts(elapsed); | executable/support line; behavior impact depends on surrounding block context. |  
| 291 | (blank) | line is blank; visual separation only. |  
| 292 | // ■ Determine vehicle operating mode ████████████████████████████████████████████ | comment line; does not have runtime  
intent, no runtime effect. |  
| 293 | self.mode = if engine_rpm < 400.0 { | executable/support line; behavior impact depends on surrounding block context. |  
| 294 | VehicleMode::Parked | executable/support line; behavior impact depends on surrounding block context. |  
| |  
| 295 | } else if ground_speed_kmh > 15.0 { | executable/support line; behavior impact depends on surrounding block context. |  
| 296 | VehicleMode::RoadTransport | executable/support line; behavior impact depends on surrounding block context. |  
| 297 | } else if ground_speed_kmh > 0.5 { | executable/support line; behavior impact depends on surrounding block context. |  
| 298 | VehicleMode::FieldWork | executable/support line; behavior impact depends on surrounding block context. |  
| 299 | } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 300 | VehicleMode::Idle | executable/support line; behavior impact depends on surrounding block context.
```

[illegible]

```
| 337 | } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 338 |     1050.0 | executable/support line; behavior impact depends on surrounding block context. |  
| 339 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 340 | self.torque_coord.compute_final(); | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 341 | (blank) | line is blank; visual separation only. |  
| 342 | // ■ Speed limiting ████████████████████████████████████████████████████ | comment lin  
intent, no runtime effect. |  
| 343 | self.speed_limited = ground_speed_kmh > self.max_speed_kmh; | executable/support line; behavior impac  
depends on surrounding block context. |  
| 344 | if self.speed_limited { | conditional gate; changing condition shifts runtime path and side effects.  
| 345 |     self.fuel_cut_active = true; | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 346 |         self.fuel_cut_reason = Some("Speed limit exceeded"); | executable/support line; behavior impact d  
nds on surrounding block context. |  
| 347 |     } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 348 |         self.fuel_cut_active = false; | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 349 |             self.fuel_cut_reason = None; | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 350 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 351 | (blank) | line is blank; visual separation only. |  
| 352 | // ■ Fault aggregation ████████████████████████████████████████████████████ | comment line;  
ntent, no runtime effect. |  
| 353 | self.total_active_dtcs = dtc_count; | executable/supportline; behavior impact depends on surrounding  
ock context. |  
| 354 | self.critical_fault = !self.comm_health.ecm_alive && engine_rpm > 400.0; | executable/support line; b  
vior impact depends on surrounding block context. |  
| 355 | (blank) | line is blank; visual separation only. |  
| 356 | // ■ J1939 periodic output ████████████████████████████████████████████████████ | comment line,  
intent, no runtime effect. |  
| 357 | self.t_status += dt; | executable/support line; behavior impact depends on surrounding block context.  
| 358 | self.t_ws_mst += dt; | executable/support line; behavior impact depends on surrounding block context.  
| 359 | self.t_dm13 += dt; | executable/support line; behavior impact depends on surrounding block context. |  
| 360 | let mut frames: Vec<J1939Frame> = Vec::new(); | executable/support line; behavior impact depends on s  
ounding block context. |  
| 361 | (blank) | line is blank; visual separation only. |  
| 362 | if self.t_status >= 0.100 { | conditional gate; changing condition shifts runtime path and side effec  
|  
| 363 |     self.t_status = 0.0; | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 364 |     frames.push(self.build_vcm_status(elapsed)); | executable/support line; behavior impact depends o  
urrounding block context. |  
| 365 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 366 | if self.t_ws_mst >= 0.100 { | conditional gate; changing condition shifts runtime path and side effec  
|  
| 367 |     self.t_ws_mst = 0.0; | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 368 |     frames.push(self.build_ws_master(elapsed)); | executable/support line; behavior impact depends on  
rrounding block context. |  
| 369 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 370 | if self.t_dm13 >= 5.000 { | conditional gate; changing condition shifts runtime path and side effects  
| 371 |     self.t_dm13 = 0.0; | executable/support line; behavior impact depends on surrounding block contex  
|  
| 372 |     frames.push(self.build_dm13(elapsed)); | executable/support line; behavior impact depends on surr  
ding block context. |  
| 373 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 374 | frames | executable/support line; behavior impact depends on surrounding block context. |  
| 375 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 376 | (blank) | line is blank; visual separation only. |
```



```
// ■ Frame builders | comment line; documentation intent, no runtime effect. |  
intent, no runtime effect. |  
| 378 | (blank) | line is blank; visual separation only. |  
| 379 |    /// VCM proprietary status frame (Prop-A, 100ms) | comment line; documentation intent, no runtime effect.  
| 380 |    fn build_vcm_status(&self, ts: f64) -> J1939Frame { | declares executable behavior; signature/body change  
alter API, control flow, and outputs. |  
| 381 |        let mut data = [0u8; 8]; | executable/support line; behavior impact depends on surrounding block context.  
. |  
| 382 |        data[0] = self.mode as u8; | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 383 |        data[1] = (self.power.total_load_kw / 220.0 * 250.0) as u8; | executable/support line; behavior impac  
depends on surrounding block context. |  
| 384 |        data[2] = self.comm_health.ecm_alive as u8 | executable/support line; behavior impact depends on surr  
ding block context. |  
| 385 |            \ ((self.comm_health.tcm_alive as u8) << 1) | executable/support line; behavior impact depends o  
urrounding block context. |  
| 386 |            \ ((self.comm_health.abs_alive as u8) << 2) | executable/support line; behavior impact depends o  
urrounding block context. |  
| 387 |            \ ((self.comm_health.hcm_alive as u8) << 3); | executable/support line; behavior impact depends  
surrounding block context. |  
| 388 |        data[3] = if self.critical_fault { 0x01 } else { 0x00 }; | executable/support line; behavior impact d  
nds on surrounding block context. |  
| 389 |        data[4] = (self.total_active_dtcs.min(255)) as u8; | executable/support line; behavior impact depends  
surrounding block context. |  
| 390 |        data[5] = (self.torque_coord.commanded_torque_nm / 1050.0 * 250.0) as u8; | executable/support line;  
avior impact depends on surrounding block context. |  
| 391 |        data[6] = self.fuel_cut_active as u8; | executable/support line; behavior impact depends on surroundi  
block context. |  
| 392 |        data[7] = self.power.load_shed_active as u8; | executable/support line; behavior impact depends on su  
unding block context. |  
| 393 |        J1939Frame::from_raw( | executable/support line; behavior impact depends on surrounding block context  
| 394 |            ts, | executable/support line; behavior impact depends on surrounding block context. |  
| 395 |            J1939Frame::build_id(7, pgn::PROP_A, self.sa, 0xFF), | executable/support line; behavior impact d  
nds on surrounding block context. |  
| 396 |            &data, | executable/support line; behavior impact depends on surrounding block context. |  
| 397 |        ) | executable/support line; behavior impact depends on surrounding block context. |  
| 398 |    } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 399 | (blank) | line is blank; visual separation only. |  
| 400 |    /// ISOBUS Working Set Master announcement (100ms) | comment line; documentation intent, no runtime effec  
|  
| 401 |    fn build_ws_master(&self, ts: f64) -> J1939Frame { | declares executable behavior; signature/body changes  
ter API, control flow, and outputs. |  
| 402 |        let data = [0x00u8; 8]; // member count = 0 (VCM is the only member) | executable/support line; behav  
impact depends on surrounding block context. |  
| 403 |        J1939Frame::from_raw( | executable/support line; behavior impact depends on surrounding block context  
| 404 |            ts, | executable/support line; behavior impact depends on surrounding block context. |  
| 405 |            J1939Frame::build_id(7, pgn::WS_MST, self.sa, 0xFF), | executable/support line; behavior impact d  
nds on surrounding block context. |  
| 406 |            &data, | executable/support line; behavior impact depends on surrounding block context. |  
| 407 |        ) | executable/support line; behavior impact depends on surrounding block context. |  
| 408 |    } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 409 | (blank) | line is blank; visual separation only. |  
| 410 |    /// DM13 – Stop/Start Broadcast (NM coordination) | comment line; documentation intent, no runtime effect  
| 411 |    fn build_dm13(&self, ts: f64) -> J1939Frame { | declares executable behavior; signature/body changes alte  
PI, control flow, and outputs. |  
| 412 |        let mut data = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 413 |        // All bits: 0b11 = don't care, 0b01 = stop broadcast (sleep) | comment line; documentation intent, n  
untime effect. |  
| 414 |        data[0] = 0xFF; | executable/support line; behavior impact depends on surrounding block context. |  
| 415 |        data[1] = 0xFF; | executable/support line; behavior impact depends on surrounding block context. |  
| 416 |        J1939Frame::from_raw(| executable/support line; behavior impact depends on surrounding block context  
| 417 |            ts, | executable/support line; behavior impact depends on surrounding block context. |  
| 418 |            J1939Frame::build_id(7, pgn::DM13, self.sa, addr::BROADCAST), | executable/support line; behavior  
act depends on surrounding block context. |
```

```
| 419 |          &data, | executable/support line; behavior impact depends on surrounding block context. |
| 420 |          ) | executable/support line; behavior impact depends on surrounding block context. |
| 421 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 422 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
```

File: src/electrical.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 457 | Non-empty: 419 | Symbols: 35

Symbol Inventory

Line	Kind	Symbol	Change Impact
4	enum	ElectricalLine	data/structure coupling
13	impl	std::fmt::Display for ElectricalLine	behavior binding and method semantics
14	fn	fmt	signature/API coupling
28	enum	ElectricalBranchKind	data/structure coupling
35	enum	ElectricalFaultMode	data/structure coupling
42	impl	std::fmt::Display for ElectricalFaultMode	behavior binding and method semantics
43	fn	fmt	signature/API coupling
55	enum	ElectricalControlPoint	data/structure coupling
61	impl	std::fmt::Display for ElectricalControlPoint	behavior binding and method semantics
62	fn	fmt	signature/API coupling
73	struct	ElectricalBranch	data/structure coupling
95	impl	ElectricalBranch	behavior binding and method semantics
97	fn	new	signature/API coupling
137	struct	ElectricalRelayState	data/structure coupling
143	struct	ElectricalSnapshot	data/structure coupling
153	impl	Default for ElectricalSnapshot	behavior binding and method semantics
154	fn	default	signature/API coupling
168	struct	ElectricalLoadRequest	data/structure coupling
182	struct	ElectricalSystem	data/structure coupling
191	impl	Default for ElectricalSystem	behavior binding and method semantics
192	fn	default	signature/API coupling
197	impl	ElectricalSystem	behavior binding and method semantics
198	fn	new	signature/API coupling
222	fn	wire_resistance_ohm	signature/API coupling
228	fn	capacitance_nf_per_m	signature/API coupling
242	fn	requested_current_for_branch	signature/API coupling
267	fn	set_branch_fault	signature/API coupling
276	fn	set_control_fault	signature/API coupling
284	fn	reset_faults	signature/API coupling
294	fn	tick	signature/API coupling
400	fn	powered_sas	signature/API coupling
413	fn	branch	signature/API coupling
419	module	tests	data/structure coupling
424	fn	ignition_relay_controls_keyed_ecu_power	signature/API coupling
445	fn	short_to_ground_blows_ecu_branch	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	use crate::{ecu_bcm::Fuse, IgnitionState};	imports symbols; changing path/name affects compile-time resolution
2	(blank)	line is blank; visual separation only.
3	#[derive(Debug, Clone, Copy, PartialEq)]	executable/support line; behavior impact depends on surrounding block context.
4	pub enum ElectricalLine {	declares state variants; adding/removing variants impacts pattern matches and logic branches.
5	Line30,	executable/support line; behavior impact depends on surrounding block context.
6	Line15Acc,	executable/support line; behavior impact depends on surrounding block context.
7	Line15Ign,	executable/support line; behavior impact depends on surrounding block context.
8	Line31,	executable/support line; behavior impact depends on surrounding block context.

```

| 9 |      HsCan, | executable/support line; behavior impact depends on surrounding block context. |
| 10 |      MsCan, | executable/support line; behavior impact depends on surrounding block context. |
| 11 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 12 | (blank) | line is blank; visual separation only. |
| 13 | impl std::fmt::Display for ElectricalLine { | implementation block; method behavior and trait conformance are
ined here. |
| 14 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signa
e/body changes alter API, control flow, and outputs. |
| 15 |         let s = match self { | executable/support line; behavior impact depends on surrounding block context.
| 16 |             ElectricalLine::Line30 => "30", | executable/support line; behavior impact depends on surrounding
ck context. |
| 17 |             ElectricalLine::Line15Acc => "15-ACC", | executable/support line; behavior impact depends on surro
ing block context. |
| 18 |             ElectricalLine::Line15Ign => "15-IGN", | executable/support line; behavior impact depends on surro
ing block context. |
| 19 |             ElectricalLine::Line31 => "31-GND", | executable/support line; behavior impact depends on surround
block context. |
| 20 |             ElectricalLine::HsCan => "HS-CAN", | executable/support line; behavior impact depends on surroundi
block context. |
| 21 |             ElectricalLine::MsCan => "MS-CAN", | executable/support line; behavior impact depends on surroundi
block context. |
| 22 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 23 |         write!(f, "{}", s) | I/O or protocol-critical logic; edits can affect integration, timing, and diagnos
outcomes. |
| 24 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 25 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 26 | (blank) | line is blank; visual separation only. |
| 27 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding blo
context. |
| 28 | pub enum ElectricalBranchKind { | declares state variants; adding/removing variants impacts pattern matches and
ogic branches. |
| 29 |     Supply, | executable/support line; behavior impact depends on surrounding block context. |
| 30 |     Load, | executable/support line; behavior impact depends on surrounding block context. |
| 31 |     Network, | executable/support line; behavior impact depends on surrounding block context. |
| 32 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 33 | (blank) | line is blank; visual separation only. |
| 34 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding blo
context. |
| 35 | pub enum ElectricalFaultMode { | declares state variants; adding/removing variants impacts pattern matches and
gic branches. |
| 36 |     None, | executable/support line; behavior impact depends on surrounding block context. |
| 37 |     OpenCircuit, | executable/support line; behavior impact depends on surrounding block context. |
| 38 |     ShortToGround, | executable/support line; behavior impact depends on surrounding block context. |
| 39 |     HighResistance, | executable/support line; behavior impact depends on surrounding block context. |
| 40 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 41 | (blank) | line is blank; visual separation only. |
| 42 | impl std::fmt::Display for ElectricalFaultMode { | implementation block; method behavior and trait conformance
e defined here. |
| 43 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signa
e/body changes alter API, control flow, and outputs. |
| 44 |         let s = match self { | executable/support line; behavior impact depends on surrounding block context.
| 45 |             ElectricalFaultMode::None => "OK", | executable/support line; behavior impact depends on surroundi
block context. |
| 46 |             ElectricalFaultMode::OpenCircuit => "OPEN", | executable/support line; behavior impact depends on
rounding block context. |
| 47 |             ElectricalFaultMode::ShortToGround => "SHORT-GND", | executable/support line; behavior impact depe
on surrounding block context. |
| 48 |             ElectricalFaultMode::HighResistance => "HI-R", | executable/support line; behavior impact depends
surrounding block context. |
| 49 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 50 |         write!(f, "{}", s) | I/O or protocol-critical logic; edits can affect integration, timing, and diagnos
outcomes. |
| 51 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

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```

| 52 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 53 | (blank) | line is blank; visual separation only. |
| 54 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding block context. |
| 55 | pub enum ElectricalControlPoint { | declares state variants; adding/removing variants impacts pattern matches logic branches. |
| 56 |     MainGround, | executable/support line; behavior impact depends on surrounding block context. |
| 57 |     AccessoryRelay, | executable/support line; behavior impact depends on surrounding block context. |
| 58 |     IgnitionRelay, | executable/support line; behavior impact depends on surrounding block context. |
| 59 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 60 | (blank) | line is blank; visual separation only. |
| 61 | impl std::fmt::Display for ElectricalControlPoint { | implementation block; method behavior and trait conformance are defined here. |
| 62 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 63 |         let s = match self { | executable/support line; behavior impact depends on surrounding block context. |
| 64 |             ElectricalControlPoint::MainGround => "MAIN-GND", | executable/support line; behavior impact depends on surrounding block context. |
| 65 |             ElectricalControlPoint::AccessoryRelay => "ACC-RELAY", | executable/support line; behavior impact depends on surrounding block context. |
| 66 |             ElectricalControlPoint::IgnitionRelay => "IGN-RELAY", | executable/support line; behavior impact depends on surrounding block context. |
| 67 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 68 |         write!(f, "{}", s) | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 69 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 70 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 71 | (blank) | line is blank; visual separation only. |
| 72 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 73 | pub struct ElectricalBranch { | declares data layout; field changes may break serialization, tests, and caller expectations. |
| 74 |     pub id: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 75 |     pub from: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 76 |     pub to: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 77 |     pub line: ElectricalLine, | executable/support line; behavior impact depends on surrounding block context. |
| 78 |     pub kind: ElectricalBranchKind, | executable/support line; behavior impact depends on surrounding block context. |
| 79 |     pub fuse_id: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 80 |     pub fuse_rating_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 81 |     pub gauge_mm2: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 82 |     pub length_m: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 83 |     pub requested_current_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 84 |     pub actual_current_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 85 |     pub voltage_v: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 86 |     pub capacitance_nf: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 87 |     pub resistance_ohm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 88 |     pub fault: ElectricalFaultMode, | executable/support line; behavior impact depends on surrounding block context. |
| 89 |     pub blown: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 90 |     pub energized: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 91 |     pub target_sa: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 92 |     pub note: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 93 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 94 | (blank) | line is blank; visual separation only. |
| 95 | impl ElectricalBranch { | implementation block; method behavior and trait conformance are defined here. |
| 96 |     #[allow(clippy::too_many_arguments)] | executable/support line; behavior impact depends on surrounding block context. |
| 97 |     fn new( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 98 |         id: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 99 |         from: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 100 |         to: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 101 |         line: ElectricalLine, | executable/support line; behavior impact depends on surrounding block context. |
| 102 |         kind: ElectricalBranchKind, | executable/support line; behavior impact depends on surrounding block context. |

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ext. |
| 103 |         fuse_id: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
|     |
| 104 |         fuse_rating_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 105 |         gauge_mm2: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 106 |         length_m: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 107 |         target_sa: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. |
|     |
| 108 |         note: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 109 |     ) -> Self { | executable/support line; behavior impact depends on surrounding block context. |
| 110 |         let capacitance_nf = length_m * ElectricalSystem::capacitance_nf_per_m(gauge_mm2, kind == ElectricalB
chKind::Network); | executable/support line; behavior impact depends on surrounding block context. |
| 111 |         let resistance_ohm = ElectricalSystem::wire_resistance_ohm(length_m, gauge_mm2, kind == ElectricalBra
Kind::Network); | executable/support line; behavior impact depends on surrounding block context. |
| 112 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 113 |             id, | executable/support line; behavior impact depends on surrounding block context. |
| 114 |             from, | executable/support line; behavior impact depends on surrounding block context. |
| 115 |             to, | executable/support line; behavior impact depends on surrounding block context. |
| 116 |             line, | executable/support line; behavior impact depends on surrounding block context. |
| 117 |             kind, | executable/support line; behavior impact depends on surrounding block context. |
| 118 |             fuse_id, | executable/support line; behavior impact depends on surrounding block context. |
| 119 |             fuse_rating_a, | executable/support line; behavior impact depends on surrounding block context. |
| 120 |             gauge_mm2, | executable/support line; behavior impact depends on surrounding block context. |
| 121 |             length_m, | executable/support line; behavior impact depends on surrounding block context. |
| 122 |             requested_current_a: 0.0, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 123 |             actual_current_a: 0.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 124 |             voltage_v: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 125 |             capacitance_nf, | executable/support line; behavior impact depends on surrounding block context. |
| 126 |             resistance_ohm, | executable/support line; behavior impact depends on surrounding block context. |
| 127 |             fault: ElectricalFaultMode::None, | executable/support line; behavior impact depends on surroundi
block context. |
| 128 |             blown: false, | executable/support line; behavior impact depends on surrounding block context. |
| 129 |             energized: false, | executable/support line; behavior impact depends on surrounding block context
| 130 |             target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 131 |             note, | executable/support line; behavior impact depends on surrounding block context. |
| 132 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 133 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 134 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 135 | (blank) | line is blank; visual separation only. |
| 136 | #[derive(Debug, Clone, Default)] | executable/support line; behavior impact depends on surrounding block cont
. |
| 137 | pub struct ElectricalRelayState { | declares data layout; field changes may break serialization, tests, and c
ers. |
| 138 |     pub accessory_closed: bool, | executable/support line; behavior impact depends on surrounding block conte
|
| 139 |     pub ignition_closed: bool, | executable/support line; behavior impact depends on surrounding block context
|
| 140 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 141 | (blank) | line is blank; visual separation only. |
| 142 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 143 | pub struct ElectricalSnapshot { | declares data layout; field changes may break serialization, tests, and cal
s. |
| 144 |     pub line30_v: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 145 |     pub line15_acc_v: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 146 |     pub line15_ign_v: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 147 |     pub line31_v: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 148 |     pub source_v: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 149 |     pub total_current_a: f64, | executable/support line; behavior impact depends on surrounding block context
| 150 |     pub ground_drop_v: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 151 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 152 | (blank) | line is blank; visual separation only. |
| 153 | impl Default for ElectricalSnapshot { | implementation block; method behavior and trait conformance are defin

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here. |
| 154 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
outputs. |
| 155 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 156 |             line30_v: 12.6, | executable/support line; behavior impact depends on surrounding block context. |
| 157 |             line15_acc_v: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 158 |             line15_ign_v: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 159 |             line31_v: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 160 |             source_v: 12.6, | executable/support line; behavior impact depends on surrounding block context. |
| 161 |             total_current_a: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 162 |             ground_drop_v: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 163 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 164 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 165 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 166 | (blank) | line is blank; visual separation only. |
| 167 | #[derive(Debug, Clone, Default)] | executable/support line; behavior impact depends on surrounding block context. |
| 168 | pub struct ElectricalLoadRequest { | declares data layout; field changes may break serialization, tests, and callers. |
| 169 |     pub bcm_supply_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 170 |     pub vcm_supply_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 171 |     pub ecm_supply_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 172 |     pub tcm_supply_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 173 |     pub abs_supply_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 174 |     pub hcm_supply_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 175 |     pub icm_supply_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 176 |     pub body_load_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 177 |     pub hs_can_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 178 |     pub ms_can_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 179 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 180 | (blank) | line is blank; visual separation only. |
| 181 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 182 | pub struct ElectricalSystem { | declares data layout; field changes may break serialization, tests, and callers. |
| 183 |     pub relays: ElectricalRelayState, | executable/support line; behavior impact depends on surrounding block context. |
| 184 |     pub snapshot: ElectricalSnapshot, | executable/support line; behavior impact depends on surrounding block context. |
| 185 |     pub branches: Vec<ElectricalBranch>, | executable/support line; behavior impact depends on surrounding block context. |
| 186 |     pub main_ground_fault: ElectricalFaultMode, | executable/support line; behavior impact depends on surrounding block context. |
| 187 |     pub accessory_relay_fault: ElectricalFaultMode, | executable/support line; behavior impact depends on surrounding block context. |
| 188 |     pub ignition_relay_fault: ElectricalFaultMode, | executable/support line; behavior impact depends on surrounding block context. |
| 189 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 190 | (blank) | line is blank; visual separation only. |
| 191 | impl Default for ElectricalSystem { | implementation block; method behavior and trait conformance are defined here. |
| 192 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 193 |         Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 194 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 195 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 196 | (blank) | line is blank; visual separation only. |
| 197 | impl ElectricalSystem { | implementation block; method behavior and trait conformance are defined here. |
| 198 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 199 |         Self { | executable/support line; behavior impact depends on surrounding block context. |

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| 200 |           relays: ElectricalRelayState::default(), | executable/support line; behavior impact depends on su
unding block context. |
| 201 |           snapshot: ElectricalSnapshot::default(), | executable/support line; behavior impact depends on su
unding block context. |
| 202 |           branches: vec![] | executable/support line; behavior impact depends on surrounding block context.
| 203 |           ElectricalBranch::new("BAT-MAIN", "Battery", "Main Fuse Box", ElectricalLine::Line30, Electri
BranchKind::Supply, "MEGA175", 175.0, 35.0, 1.8, None, "primary battery feed"), | executable/support line; behavior i
ct depends on surrounding block context. |
| 204 |           ElectricalBranch::new("ALT-B+", "Alternator", "Battery/Main Bus", ElectricalLine::Line30, Ele
icalBranchKind::Supply, "ALT150", 150.0, 25.0, 1.4, None, "charging branch"), | executable/support line; behavior imp
depends on surrounding block context. |
| 205 |           ElectricalBranch::new("BCM-PWR", "Main Fuse Box", "BCM", ElectricalLine::Line15Acc, Electrica
anchKind::Supply, "BCM40", 40.0, 6.0, 2.8, Some(crate::j1939::addr::CAB), "body controller supply"), | executable/sup
t line; behavior impact depends on surrounding block context. |
| 206 |           ElectricalBranch::new("ICM-PWR", "BCM", "ICM", ElectricalLine::Line15Acc, ElectricalBranchKin
Supply, "DASH7.5", 7.5, 1.0, 2.2, Some(crate::j1939::addr::INSTRUMENT), "instrument feed"), | executable/support line
ehavior impact depends on surrounding block context. |
| 207 |           ElectricalBranch::new("VCM-PWR", "Main Fuse Box", "VCM", ElectricalLine::Line15Ign, Electrica
anchKind::Supply, "VCM15", 15.0, 2.5, 3.4, None, "gateway and power coordination"), | executable/support line; behavi
impact depends on surrounding block context. |
| 208 |           ElectricalBranch::new("ECM-PWR", "Main Fuse Box", "ECM", ElectricalLine::Line15Ign, Electrica
anchKind::Supply, "ECM15", 15.0, 2.5, 3.8, Some(crate::j1939::addr::ECM_1), "engine controller branch"), | executable
pport line; behavior impact depends on surrounding block context. |
| 209 |           ElectricalBranch::new("TCM-PWR", "Main Fuse Box", "TCM", ElectricalLine::Line15Ign, Electrica
anchKind::Supply, "TCM10", 10.0, 1.5, 4.1, Some(crate::j1939::addr::TRANSMISSION), "transmission controller branch"),
executable/support line; behavior impact depends on surrounding block context. |
| 210 |           ElectricalBranch::new("ABS-PWR", "Main Fuse Box", "ABS/ESP", ElectricalLine::Line15Ign, Elect
alBranchKind::Supply, "ABS20", 20.0, 2.5, 4.6, Some(crate::j1939::addr::BRAKES), "braking controller branch"), | exec
ble/support line; behavior impact depends on surrounding block context. |
| 211 |           ElectricalBranch::new("HCM-PWR", "Main Fuse Box", "HCM", ElectricalLine::Line15Ign, Electrica
anchKind::Supply, "HCM15", 15.0, 2.5, 4.2, Some(crate::j1939::addr::HITCH), "hydraulic controller branch"), | executa
/support line; behavior impact depends on surrounding block context. |
| 212 |           ElectricalBranch::new("BODY-LOAD", "BCM", "Lights/HVAC/Wiper/Horn", ElectricalLine::Line15Acc
lectricalBranchKind::Load, "BODY40", 40.0, 4.0, 5.0, None, "aggregate body loads"), | executable/support line; behavi
impact depends on surrounding block context. |
| 213 |           ElectricalBranch::new("HS-CAN-1", "VCM", "ECM/TCM/ABS/HCM", ElectricalLine::HsCan, Electrical
nchKind::Network, "120R x2", 1.0, 0.75, 8.5, None, "500 kbps twisted pair backbone"), | executable/support line; beha
r impact depends on surrounding block context. |
| 214 |           ElectricalBranch::new("MS-CAN-BODY", "BCM", "ICM/body domain", ElectricalLine::MsCan, Electri
BranchKind::Network, "120R x2", 1.0, 0.5, 6.0, None, "250 kbps twisted pair backbone"), | executable/support line; be
ior impact depends on surrounding block context. |
| 215 |           ], | executable/support line; behavior impact depends on surrounding block context. |
| 216 |           main_ground_fault: ElectricalFaultMode::None, | executable/support line; behavior impact depends
surrounding block context. |
| 217 |           accessory_relay_fault: ElectricalFaultMode::None, | executable/support line; behavior impact depende
on surrounding block context. |
| 218 |           ignition_relay_fault: ElectricalFaultMode::None, | executable/support line; behavior impact depende
on surrounding block context. |
| 219 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 220 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 221 | (blank) | line is blank; visual separation only. |
| 222 |   pub fn wire_resistance_ohm(length_m: f64, gauge_mm2: f64, twisted_pair: bool) -> f64 { | declares executable
behavior; signature/body changes alter API, control flow, and outputs. |
| 223 |       let rho = 0.0175; | executable/support line; behavior impact depends on surrounding block context. |
| 224 |       let loop_factor = if twisted_pair { 2.0 } else { 2.2 }; | executable/support line; behavior impact de
ds on surrounding block context. |
| 225 |       (rho * length_m * loop_factor / gauge_mm2.max(0.35)).max(0.0005) | executable/support line; behavior
act depends on surrounding block context. |
| 226 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 227 | (blank) | line is blank; visual separation only. |
| 228 |   pub fn capacitance_nf_per_m(gauge_mm2: f64, twisted_pair: bool) -> f64 { | declares executable behavior;
nature/body changes alter API, control flow, and outputs. |
| 229 |       if twisted_pair { | conditional gate; changing condition shifts runtime path and side effects. |
| 230 |           0.05 | executable/support line; behavior impact depends on surrounding block context. |

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| 231 |         } else if gauge_mm2 <= 1.0 { | executable/support line; behavior impact depends on surrounding block
text. |
| 232 |             0.09 | executable/support line; behavior impact depends on surrounding block context. |
| 233 |         } else if gauge_mm2 <= 2.5 { | executable/support line; behavior impact depends on surrounding block
text. |
| 234 |             0.11 | executable/support line; behavior impact depends on surrounding block context. |
| 235 |         } else if gauge_mm2 <= 6.0 { | executable/support line; behavior impact depends on surrounding block
text. |
| 236 |             0.13 | executable/support line; behavior impact depends on surrounding block context. |
| 237 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 238 |             0.16 | executable/support line; behavior impact depends on surrounding block context. |
| 239 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 240 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 241 | (blank) | line is blank; visual separation only. |
| 242 |     fn requested_current_for_branch(branch_id: &str, loads: &ElectricalLoadRequest) -> f64 { | declares execu
le behavior; signature/body changes alter API, control flow, and outputs. |
| 243 |         match branch_id { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 244 |             "BAT-MAIN" => loads.bcm_supply_a | executable/support line; behavior impact depends on surroundin
lock context. |
| 245 |                 + loads.vcm_supply_a | executable/support line; behavior impact depends on surrounding block
text. |
| 246 |                 + loads.ecm_supply_a | executable/support line; behavior impact depends on surrounding block
text. |
| 247 |                 + loads.tcm_supply_a | executable/support line; behavior impact depends on surrounding block
text. |
| 248 |                 + loads.abs_supply_a | executable/support line; behavior impact depends on surrounding block
text. |
| 249 |                 + loads.hcm_supply_a | executable/support line; behavior impact depends on surrounding block
text. |
| 250 |                 + loads.icm_supply_a | executable/support line; behavior impact depends on surrounding block
text. |
| 251 |                 + loads.body_load_a, | executable/support line; behavior impact depends on surrounding block
text. |
| 252 |             "ALT-B+" => 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 253 |             "BCM-PWR" => loads.bcm_supply_a, | executable/support line; behavior impact depends on surroundin
lock context. |
| 254 |             "ICM-PWR" => loads.icm_supply_a, | executable/support line; behavior impact depends on surroundin
lock context. |
| 255 |             "VCM-PWR" => loads.vcm_supply_a, | executable/support line; behavior impact depends on surroundin
lock context. |
| 256 |             "ECM-PWR" => loads.ecm_supply_a, | executable/support line; behavior impact depends on surroundin
lock context. |
| 257 |             "TCM-PWR" => loads.tcm_supply_a, | executable/support line; behavior impact depends on surroundin
lock context. |
| 258 |             "ABS-PWR" => loads.abs_supply_a, | executable/support line; behavior impact depends on surroundin
lock context. |
| 259 |             "HCM-PWR" => loads.hcm_supply_a, | executable/support line; behavior impact depends on surroundin
lock context. |
| 260 |             "BODY-LOAD" => loads.body_load_a, | executable/support line; behavior impact depends on surroundi
block context. |
| 261 |             "HS-CAN-1" => loads.hs_can_a, | executable/support line; behavior impact depends on surrounding b
k context. |
| 262 |             "MS-CAN-BODY" => loads.ms_can_a, | executable/support line; behavior impact depends on surroundin
lock context. |
| 263 |             _ => 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 264 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 265 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 266 | (blank) | line is blank; visual separation only. |
| 267 |     pub fn set_branch_fault(&mut self, branch_id: &str, fault: ElectricalFaultMode) { | declares executable b
vior; signature/body changes alter API, control flow, and outputs. |
| 268 |         if let Some(branch) = self.branches.iter_mut().find(|b| b.id == branch_id) { | conditional gate; ch
ing condition shifts runtime path and side effects. |
| 269 |             branch.fault = fault; | executable/support line; behavior impact depends on surrounding block con

```



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t. |
| 270 |           if fault == ElectricalFaultMode::None { | conditional gate; changing condition shifts runtime pat
nd side effects. |
| 271 |               branch.blown = false; | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 272 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 273 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 274 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 275 | (blank) | line is blank; visual separation only. |
| 276 |   pub fn set_control_fault(&mut self, point: ElectricalControlPoint, fault: ElectricalFaultMode) { | declar
executable behavior; signature/body changes alter API, control flow, and outputs. |
| 277 |       match point { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 278 |           ElectricalControlPoint::MainGround => self.main_ground_fault = fault, | executable/support line;
avior impact depends on surrounding block context. |
| 279 |           ElectricalControlPoint::AccessoryRelay => self.accessory_relay_fault = fault, | executable/support
ine; behavior impact depends on surrounding block context. |
| 280 |           ElectricalControlPoint::IgnitionRelay => self.ignition_relay_fault = fault, | executable/support
e; behavior impact depends on surrounding block context. |
| 281 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 282 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 283 | (blank) | line is blank; visual separation only. |
| 284 |   pub fn reset_faults(&mut self) { | declares executable behavior; signature/body changes alter API, contro
low, and outputs. |
| 285 |       self.main_ground_fault = ElectricalFaultMode::None; | executable/support line; behavior impact depend
n surrounding block context. |
| 286 |       self.accessory_relay_fault = ElectricalFaultMode::None; | executable/support line; behavior impact de
ds on surrounding block context. |
| 287 |       self.ignition_relay_fault = ElectricalFaultMode::None; | executable/support line; behavior impact dep
s on surrounding block context. |
| 288 |       for branch in &mut self.branches { | iteration control; changing bounds/body affects throughput, timi
and possibly safety guarantees. |
| 289 |           branch.fault = ElectricalFaultMode::None; | executable/support line; behavior impact depends on s
ounding block context. |
| 290 |           branch.blown = false; | executable/support line; behavior impact depends on surrounding block con
t. |
| 291 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 292 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 293 | (blank) | line is blank; visual separation only. |
| 294 |   pub fn tick( | declares executable behavior; signature/body changes alter API, control flow, and outputs.
| 295 |       &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 296 |       ignition: IgnitionState, | executable/support line; behavior impact depends on surrounding block cont
. |
| 297 |       source_voltage_v: f64, | executable/support line; behavior impact depends on surrounding block context
|
| 298 |       alternator_voltage_v: f64, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 299 |       loads: &ElectricalLoadRequest, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 300 |       bcm_fuses: &[Fuse], | executable/support line; behavior impact depends on surrounding block context.
| 301 |       dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 302 |   ) { | executable/support line; behavior impact depends on surrounding block context. |
| 303 |       self.relays.accessory_closed = matches!(ignition, IgnitionState::Accessory \
IgnitionState::On \
IgnitionState::Cranking \
IgnitionState::Running) | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 304 |       && self.accessory_relay_fault != ElectricalFaultMode::OpenCircuit; | executable/support line; beh
or impact depends on surrounding block context. |
| 305 |       self.relays.ignition_closed = matches!(ignition, IgnitionState::On \
IgnitionState::Cranking \
Igni
nState::Running) | executable/support line; behavior impact depends on surrounding block context. |
| 306 |       && self.ignition_relay_fault != ElectricalFaultMode::OpenCircuit; | executable/support line; beha
r impact depends on surrounding block context. |
| 307 | (blank) | line is blank; visual separation only. |

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| 308 |         let ground_r = match self.main_ground_fault { | executable/support line; behavior impact depends on surrounding block context. | | | | |
| 309 |             ElectricalFaultMode::None => 0.004, | executable/support line; behavior impact depends on surrounding block context. |
| 310 |             ElectricalFaultMode::HighResistance => 0.09, | executable/support line; behavior impact depends on surrounding block context. |
| 311 |             ElectricalFaultMode::OpenCircuit => 1.5, | executable/support line; behavior impact depends on surrounding block context. |
| 312 |             ElectricalFaultMode::ShortToGround => 0.004, | executable/support line; behavior impact depends on surrounding block context. |
| 313 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 314 | (blank) | line is blank; visual separation only. |
| 315 |         self.snapshot.source_v = source_voltage_v.max(alternator_voltage_v).max(0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 316 |         self.snapshot.line30_v = self.snapshot.source_v; | executable/support line; behavior impact depends on surrounding block context. |
| 317 |         self.snapshot.line15_acc_v = if self.relays.accessory_closed { | executable/support line; behavior impact depends on surrounding block context. |
| 318 |             (self.snapshot.line30_v - if self.accessory_relay_fault == ElectricalFaultMode::HighResistance { 0.15 } else { 0.15 }).max(0.0) | executable/support line; behavior impact depends on surrounding block context. |
| 319 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 320 |             0.0 | executable/support line; behavior impact depends on surrounding block context. |
| 321 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 322 |         self.snapshot.line15_ign_v = if self.relays.ignition_closed { | executable/support line; behavior impact depends on surrounding block context. |
| 323 |             (self.snapshot.line30_v - if self.ignition_relay_fault == ElectricalFaultMode::HighResistance { 0.18 } else { 0.18 }).max(0.0) | executable/support line; behavior impact depends on surrounding block context. |
| 324 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 325 |             0.0 | executable/support line; behavior impact depends on surrounding block context. |
| 326 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 327 | (blank) | line is blank; visual separation only. |
| 328 |         let line30_v = self.snapshot.line30_v; | executable/support line; behavior impact depends on surrounding block context. |
| 329 |         let line15_acc_v = self.snapshot.line15_acc_v; | executable/support line; behavior impact depends on surrounding block context. |
| 330 |         let line15_ign_v = self.snapshot.line15_ign_v; | executable/support line; behavior impact depends on surrounding block context. |
| 331 |         let line31_v = self.snapshot.line31_v; | executable/support line; behavior impact depends on surrounding block context. |
| 332 |         let mut total_current = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 333 |         for branch in &mut self.branches { | iteration control; changing bounds/body affects throughput, timing and possibly safety guarantees. |
| 334 |             branch.requested_current_a = Self::requested_current_for_branch(branch.id, loads); | executable/support line; behavior impact depends on surrounding block context. |
| 335 | (blank) | line is blank; visual separation only. |
| 336 |             if let Some(fuse) = bcm_fuses.iter().find(|f| f.id == branch.id || f.id == branch.fuse_id) { | conditional gate; changing condition shifts runtime path and side effects. |
| 337 |                 if fuse.blowed { | conditional gate; changing condition shifts runtime path and side effects. |
| 338 |                     branch.blowed = true; | executable/support line; behavior impact depends on surrounding block context. |
| 339 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 340 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 341 | (blank) | line is blank; visual separation only. |
| 342 |             let source_v = match branch.line { | executable/support line; behavior impact depends on surrounding block context. |
| 343 |                 ElectricalLine::Line30 => line30_v, | executable/support line; behavior impact depends on surrounding block context. |
| 344 |                 ElectricalLine::Line15Acc => line15_acc_v, | executable/support line; behavior impact depends on surrounding block context. |

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| 345 |           ElectricalLine::Line15Ign => line15_ign_v, | executable/support line; behavior impact depends
surrounding block context. |
| 346 |           ElectricalLine::Line31 => line31_v, | executable/support line; behavior impact depends on sur
nding block context. |
| 347 |           ElectricalLine::HsCan => line15_ign_v, | executable/support line; behavior impact depends on
rounding block context. |
| 348 |           ElectricalLine::MsCan => line15_acc_v, | executable/support line; behavior impact depends on
rounding block context. |
| 349 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 350 |           let extra_r = match branch.fault { | executable/support line; behavior impact depends on surround
block context. |
| 351 |           ElectricalFaultMode::None => 0.0, | executable/support line; behavior impact depends on surro
ing block context. |
| 352 |           ElectricalFaultMode::HighResistance => 0.35, | executable/support line; behavior impact depen
on surrounding block context. |
| 353 |           ElectricalFaultMode::OpenCircuit => 1.0e9, | executable/support line; behavior impact depends
surrounding block context. |
| 354 |           ElectricalFaultMode::ShortToGround => 0.02, | executable/support line; behavior impact depend
n surrounding block context. |
| 355 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 356 | (blank) | line is blank; visual separation only. |
| 357 |           if branch.fault == ElectricalFaultMode::OpenCircuit \\|\\| branch.blownd \\|\\| source_v < 1.0 { | con
ditional gate; changing condition shifts runtime path and side effects. |
| 358 |           branch.actual_current_a = 0.0; | executable/support line; behavior impact depends on surround
block context. |
| 359 |           branch.energized = false; | executable/support line; behavior impact depends on surrounding b
k context. |
| 360 |           branch.voltage_v += (0.0 - branch.voltage_v) * (dt / 0.04).min(1.0); | executable/support lin
behavior impact depends on surrounding block context. |
| 361 |           continue; | executable/support line; behavior impact depends on surrounding block context. |
| 362 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 363 | (blank) | line is blank; visual separation only. |
| 364 |           if branch.fault == ElectricalFaultMode::ShortToGround { | conditional gate; changing condition sh
s runtime path and side effects. |
| 365 |           let short_i = source_v / (branch.resistance_ohm + ground_r + extra_r).max(0.005); | executabl
upport line; behavior impact depends on surrounding block context. |
| 366 |           if short_i > branch.fuse_rating_a * 1.15 { | conditional gate; changing condition shifts runt
path and side effects. |
| 367 |           branch.blownd = true; | executable/support line; behavior impact depends on surrounding bl
context. |
| 368 |           branch.actual_current_a = 0.0; | executable/support line; behavior impact depends on surr
ding block context. |
| 369 |           branch.energized = false; | executable/support line; behavior impact depends on surroundi
block context. |
| 370 |           branch.voltage_v = 0.0; | executable/support line; behavior impact depends on surrounding
ock context. |
| 371 |           continue; | executable/support line; behavior impact depends on surrounding block context
| 372 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 373 |           branch.actual_current_a = short_i; | executable/support line; behavior impact depends on surr
ding block context. |
| 374 |           branch.energized = true; | executable/support line; behavior impact depends on surrounding bl
context. |
| 375 |           } else { | executable/support line; behavior impact depends on surrounding block context. |
| 376 |           branch.actual_current_a = branch.requested_current_a.max(0.0); | executable/support line; beh
or impact depends on surrounding block context. |
| 377 |           branch.energized = source_v > 9.0 \\|\\| branch.kind == ElectricalBranchKind::Network; | execut
e/support line; behavior impact depends on surrounding block context. |
| 378 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 379 | (blank) | line is blank; visual separation only. |
| 380 |           total_current += branch.actual_current_a; | executable/support line; behavior impact depends on s

```

```

ounding block context. |
| 381 |         let drop_v = branch.actual_current_a * (branch.resistance_ohm + ground_r + extra_r.min(1.0)); | executable/support line; behavior impact depends on surrounding block context. | | |
| 382 |         let target_v = (source_v - drop_v).max(0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 383 |         let tau_s = (((branch.resistance_ohm + extra_r.min(1.0)).max(0.005)) | executable/support line; behavior impact depends on surrounding block context. |
| 384 |             * (branch.capacitance_nf * 1.0e-9)) | executable/support line; behavior impact depends on surrounding block context. |
| 385 |             .clamp(0.002, 0.08); | executable/support line; behavior impact depends on surrounding block context. |
| 386 |         branch.voltage_v += (target_v - branch.voltage_v) * (dt / tau_s).min(1.0); | executable/support line; behavior impact depends on surrounding block context. |
| 387 |         if branch.actual_current_a > branch.fuse_rating_a * 1.15 && branch.kind != ElectricalBranchKind::Work { | conditional gate; changing condition shifts runtime path and side effects. |
| 388 |             branch.blown = true; | executable/support line; behavior impact depends on surrounding block context. |
| 389 |             branch.energized = false; | executable/support line; behavior impact depends on surrounding block context. |
| 390 |             branch.actual_current_a = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 391 |             branch.voltage_v = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 392 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 393 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 394 | (blank) | line is blank; visual separation only. |
| 395 |     self.snapshot.total_current_a = total_current; | executable/support line; behavior impact depends on surrounding block context. |
| 396 |     self.snapshot.ground_drop_v = total_current * ground_r; | executable/support line; behavior impact depends on surrounding block context. |
| 397 |     self.snapshot.line31_v = self.snapshot.ground_drop_v; | executable/support line; behavior impact depends on surrounding block context. |
| 398 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 399 | (blank) | line is blank; visual separation only. |
| 400 | pub fn powered_sas(&self) -> Vec<u8> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 401 |     self.branches | executable/support line; behavior impact depends on surrounding block context. |
| 402 |     .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 403 |     .filter_map(|b| { | executable/support line; behavior impact depends on surrounding block context. |
| 404 |         if b.energized && !b.blown && b.voltage_v > 9.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 405 |             b.target_sa | executable/support line; behavior impact depends on surrounding block context. |
| 406 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 407 |             None | executable/support line; behavior impact depends on surrounding block context. |
| 408 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 409 |     }) | executable/support line; behavior impact depends on surrounding block context. |
| 410 |     .collect() | executable/support line; behavior impact depends on surrounding block context. |
| 411 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 412 | (blank) | line is blank; visual separation only. |
| 413 | pub fn branch(&self, branch_id: &str) -> Option<&ElectricalBranch> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 414 |     self.branches.iter().find(|b| b.id == branch_id) | executable/support line; behavior impact depends on surrounding block context. |
| 415 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 416 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 417 | (blank) | line is blank; visual separation only. |
| 418 | #[cfg(test)] | executable/support line; behavior impact depends on surrounding block context. |
| 419 | mod tests { | declares a module boundary; changing module name/path breaks references. |
| 420 |     use super::{ElectricalControlPoint, ElectricalFaultMode, ElectricalLoadRequest, ElectricalSystem}; | imports symbols; changing path/name affects compile-time resolution. |

```

```

| 421 |     use crate::IgnitionState; | imports symbols; changing path/name affects compile-time resolution. |
| 422 | (blank) | line is blank; visual separation only. |
| 423 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 424 |     fn ignition_relay_controls_keyed_ecu_power() { | declares executable behavior; signature/body changes alt
API, control flow, and outputs. |
| 425 |         let mut es = ElectricalSystem::new(); | executable/support line; behavior impact depends on surroundi
block context. |
| 426 |         let loads = ElectricalLoadRequest { | executable/support line; behavior impact depends on surrounding
ock context. |
| 427 |             ecm_supply_a: 6.0, | executable/support line; behavior impact depends on surrounding block context
|
| 428 |             tcm_supply_a: 3.0, | executable/support line; behavior impact depends on surrounding block context
|
| 429 |             abs_supply_a: 4.0, | executable/support line; behavior impact depends on surrounding block context
|
| 430 |             hcm_supply_a: 3.0, | executable/support line; behavior impact depends on surrounding block context
|
| 431 |             bcm_supply_a: 8.0, | executable/support line; behavior impact depends on surrounding block context
|
| 432 |             icm_supply_a: 2.0, | executable/support line; behavior impact depends on surrounding block context
|
| 433 |             ..Default::default() | executable/support line; behavior impact depends on surrounding block cont
. |
| 434 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 435 |         es.tick(IgnitionState::On, 12.6, 14.2, &loads, &[], 0.016); | executable/support line; behavior impac
depends on surrounding block context. |
| 436 |         assert!(es.powered_sas().contains(&crate::j1939::addr::ECM_1)); | executable/support line; behavior i
ct depends on surrounding block context. |
| 437 | (blank) | line is blank; visual separation only. |
| 438 |         es.set_control_fault(ElectricalControlPoint::IgnitionRelay, ElectricalFaultMode::OpenCircuit); | exec
ble/support line; behavior impact depends on surrounding block context. |
| 439 |         es.tick(IgnitionState::On, 12.6, 14.2, &loads, &[], 0.016); | executable/support line; behavior impac
depends on surrounding block context. |
| 440 |         assert!(!es.powered_sas().contains(&crate::j1939::addr::ECM_1)); | executable/support line; behavior
act depends on surrounding block context. |
| 441 |         assert!(es.powered_sas().contains(&crate::j1939::addr::CAB)); | executable/support line; behavior imp
depends on surrounding block context. |
| 442 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 443 | (blank) | line is blank; visual separation only. |
| 444 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 445 |     fn short_to_ground_blows_ecu_branch() { | declares executable behavior; signature/body changes alter API,
ntrol flow, and outputs. |
| 446 |         let mut es = ElectricalSystem::new(); | executable/support line; behavior impact depends on surroundi
block context. |
| 447 |         let loads = ElectricalLoadRequest { | executable/support line; behavior impact depends on surrounding
ock context. |
| 448 |             ecm_supply_a: 6.0, | executable/support line; behavior impact depends on surrounding block context
|
| 449 |             ..Default::default() | executable/support line; behavior impact depends on surrounding block cont
. |
| 450 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 451 |         es.set_branch_fault("ECM-PWR", ElectricalFaultMode::ShortToGround); | executable/support line; behavi
impact depends on surrounding block context. |
| 452 |         es.tick(IgnitionState::On, 12.4, 14.0, &loads, &[], 0.016); | executable/support line; behavior impac
depends on surrounding block context. |
| 453 |         let branch = es.branch("ECM-PWR").expect("branch present"); | panic-prone operation; changing/removin
ffects crash behavior and robustness. |
| 454 |         assert!(branch.blown); | executable/support line; behavior impact depends on surrounding block context
|
| 455 |         assert!(!branch.energized); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 456 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 457 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/engine.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 126 | Non-empty: 113 | Symbols: 10

Symbol Inventory

Line	Kind	Symbol	Change Impact
2	const	GEAR_RATIOS	data/structure coupling
3	const	FINAL_DRIVE	data/structure coupling
4	const	WHEEL_RADIUS_M	data/structure coupling
5	const	FUEL_TANK_L	data/structure coupling
8	struct	EngineEcu	data/structure coupling
29	impl	EngineEcu	behavior binding and method semantics
30	fn	new	signature/API coupling
54	fn	torque_at_rpm	signature/API coupling
74	fn	update	signature/API coupling
119	fn	wheel_drive_force_n	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	// ZF 8HP gear ratios + final drive	comment line; documentation intent, no runtime effect.
2	pub const GEAR_RATIOS: [f64; 9] = [0.0, 4.714, 3.143, 2.106, 1.667, 1.285, 1.000, 0.839, 0.667];	executable/support line; behavior impact depends on surrounding block context.
3	pub const FINAL_DRIVE: f64 = 3.06;	executable/support line; behavior impact depends on surrounding block context.
4	pub const WHEEL_RADIUS_M: f64 = 0.32;	executable/support line; behavior impact depends on surrounding block context.
5	pub const FUEL_TANK_L: f64 = 55.0;	executable/support line; behavior impact depends on surrounding block context.
6	(blank)	line is blank; visual separation only.
7	#[derive(Debug, Clone)]	executable/support line; behavior impact depends on surrounding block context.
8	pub struct EngineEcu {	declares data layout; field changes may break serialization, tests, and callers.
9	pub rpm: f64,	executable/support line; behavior impact depends on surrounding block context.
10	pub throttle_position: f64, // 0-1	executable/support line; behavior impact depends on surrounding block context.
11	pub engine_temp: f64, // °C	executable/support line; behavior impact depends on surrounding block context.
12	pub oil_temp: f64, // °C	executable/support line; behavior impact depends on surrounding block context.
13	pub oil_pressure: f64, // bar	executable/support line; behavior impact depends on surrounding block context.
14	pub coolant_temp: f64, // °C	executable/support line; behavior impact depends on surrounding block context.
15	pub fuel_level: f64, // 0-1	executable/support line; behavior impact depends on surrounding block context.
16	pub fuel_consumption_lph: f64, // L/hour instantaneous	executable/support line; behavior impact depends on surrounding block context.
17	pub boost_pressure: f64, // bar (turbo)	executable/support line; behavior impact depends on surrounding block context.
18	pub intake_air_temp: f64, // °C	executable/support line; behavior impact depends on surrounding block context.
19	pub output_torque: f64, // Nm	executable/support line; behavior impact depends on surrounding block context.
20	pub output_power_kw: f64, // kW	executable/support line; behavior impact depends on surrounding block context.
21	pub check_engine: bool,	executable/support line; behavior impact depends on surrounding block context.
22	pub engine_on: bool,	executable/support line; behavior impact depends on surrounding block context.
23	pub rev_limiter_active: bool,	executable/support line; behavior impact depends on surrounding block context.
24	pub idle_rpm: f64,	executable/support line; behavior impact depends on surrounding block context.

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| 25 |     pub redline_rpm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 26 |     pub max_rpm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 27 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 28 | (blank) | line is blank; visual separation only. |
| 29 | impl EngineEcu { | implementation block; method behavior and trait conformance are defined here. |
| 30 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
tputs. |
| 31 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 32 |             rpm: 820.0, | executable/support line; behavior impact depends on surrounding block context. |
| 33 |             throttle_position: 0.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 34 |             engine_temp: 22.0, | executable/support line; behavior impact depends on surrounding block context
| 35 |             oil_temp: 22.0, | executable/support line; behavior impact depends on surrounding block context. |
| 36 |             oil_pressure: 4.5, | executable/support line; behavior impact depends on surrounding block context
| 37 |             coolant_temp: 22.0, | executable/support line; behavior impact depends on surrounding block context
|
| 38 |             fuel_level: 0.85, | executable/support line; behavior impact depends on surrounding block context.
| 39 |             fuel_consumption_lph: 0.6, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 40 |             boost_pressure: 0.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 41 |             intake_air_temp: 25.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 42 |             output_torque: 0.0, | executable/support line; behavior impact depends on surrounding block context
|
| 43 |             output_power_kw: 0.0, | executable/support line; behavior impact depends on surrounding block cont
. |
| 44 |             check_engine: false, | executable/support line; behavior impact depends on surrounding block conte
|
| 45 |             engine_on: true, | executable/support line; behavior impact depends on surrounding block context.
| 46 |             rev_limiter_active: false, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 47 |             idle_rpm: 820.0, | executable/support line; behavior impact depends on surrounding block context.
| 48 |             redline_rpm: 6800.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 49 |             max_rpm: 7200.0, | executable/support line; behavior impact depends on surrounding block context.
| 50 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 51 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 52 | (blank) | line is blank; visual separation only. |
| 53 |     /// Torque curve: 2.0L turbo, ~250 Nm peak at 2000-4500 RPM | comment line; documentation intent, no runti
effect. |
| 54 |     fn torque_at_rpm(&self, rpm: f64, throttle: f64) -> f64 { | declares executable behavior; signature/body c
ges alter API, control flow, and outputs. |
| 55 |         if rpm < 500.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 56 |             return 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 57 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 58 |         // Flat torque plateau between 2000-4500 RPM (turbocharged) | comment line; documentation intent, no r
ime effect. |
| 59 |         let peak = 250.0; | executable/support line; behavior impact depends on surrounding block context. |
| 60 |         let t = if rpm < 1500.0 { | executable/support line; behavior impact depends on surrounding block conte
. |
| 61 |             rpm / 1500.0 * 0.7 | executable/support line; behavior impact depends on surrounding block context
| 62 |         } else if rpm < 2000.0 { | executable/support line; behavior impact depends on surrounding block conte
|
| 63 |             0.7 + (rpm - 1500.0) / 500.0 * 0.3 | executable/support line; behavior impact depends on surroundi
block context. |
| 64 |         } else if rpm < 4500.0 { | executable/support line; behavior impact depends on surrounding block conte
|
| 65 |             1.0 | executable/support line; behavior impact depends on surrounding block context. |
| 66 |         } else if rpm < 6800.0 { | executable/support line; behavior impact depends on surrounding block conte
|
| 67 |             1.0 - (rpm - 4500.0) / 2300.0 * 0.6 | executable/support line; behavior impact depends on surround
block context. |
| 68 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 69 |             0.0 | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 70 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 71 |         peak * t * throttle | executable/support line; behavior impact depends on surrounding block context. |
| 72 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 73 | (blank) | line is blank; visual separation only. |
| 74 |     pub fn update(&mut self, throttle: f64, gear: u8, vehicle_speed_kmh: f64, dt: f64) { | declares executable
havior; signature/body changes alter API, control flow, and outputs. |
| 75 |         self.throttle_position = throttle.clamp(0.0, 1.0); | executable/support line; behavior impact depends
surrounding block context. |
| 76 |         self.rev_limiter_active = self.rpm >= self.redline_rpm; | executable/support line; behavior impact dep
s on surrounding block context. |
| 77 | (blank) | line is blank; visual separation only. |
| 78 |         let effective_throttle = if self.rev_limiter_active { | executable/support line; behavior impact depen
on surrounding block context. |
| 79 |             0.0 | executable/support line; behavior impact depends on surrounding block context. |
| 80 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 81 |             throttle | executable/support line; behavior impact depends on surrounding block context. |
| 82 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 83 | (blank) | line is blank; visual separation only. |
| 84 |         // Compute RPM from wheel speed × gear ratio | comment line; documentation intent, no runtime effect.
| 85 |         let target_rpm = if gear > 0 && gear <= 8 { | executable/support line; behavior impact depends on surr
ding block context. |
| 86 |             let wheel_rps = (vehicle_speed_kmh / 3.6) / WHEEL_RADIUS_M; | executable/support line; behavior im
t depends on surrounding block context. |
| 87 |             (wheel_rps * 60.0 * GEAR_RATIOS[gear as usize] * FINAL_DRIVE).max(self.idle_rpm) | executable/supp
line; behavior impact depends on surrounding block context. |
| 88 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 89 |             self.idle_rpm + effective_throttle * 1500.0 | executable/support line; behavior impact depends on
rounding block context. |
| 90 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 91 | (blank) | line is blank; visual separation only. |
| 92 |         self.rpm += (target_rpm - self.rpm) * (dt / 0.08).min(1.0); | executable/support line; behavior impact
depends on surrounding block context. |
| 93 |         self.rpm = self.rpm.clamp(self.idle_rpm * 0.9, self.max_rpm); | executable/support line; behavior impa
depends on surrounding block context. |
| 94 | (blank) | line is blank; visual separation only. |
| 95 |         self.output_torque = self.torque_at_rpm(self.rpm, effective_throttle); | executable/support line; beha
r impact depends on surrounding block context. |
| 96 |         self.output_power_kw = self.output_torque * self.rpm * std::f64::consts::PI / 30.0 / 1000.0; | execut
/support line; behavior impact depends on surrounding block context. |
| 97 | (blank) | line is blank; visual separation only. |
| 98 |         // Turbo boost builds with RPM + throttle | comment line; documentation intent, no runtime effect. |
| 99 |         let boost_target = | executable/support line; behavior impact depends on surrounding block context. |
| 100 |             (effective_throttle * 1.6 * ((self.rpm - 1500.0) / 1500.0).clamp(0.0, 1.0)).max(0.0); | executabl
upport line; behavior impact depends on surrounding block context. |
| 101 |         self.boost_pressure += (boost_target - self.boost_pressure) * dt * 3.0; | executable/support line; be
ior impact depends on surrounding block context. |
| 102 | (blank) | line is blank; visual separation only. |
| 103 |         // Thermal model | comment line; documentation intent, no runtime effect. |
| 104 |         let temp_target = 92.0 + effective_throttle * 10.0; | executable/support line; behavior impact depend
n surrounding block context. |
| 105 |         self.engine_temp += (temp_target - self.engine_temp) * dt * 0.015; | executable/support line; behavio
mpact depends on surrounding block context. |
| 106 |         self.coolant_temp = self.engine_temp; | executable/support line; behavior impact depends on surroundi
block context. |
| 107 |         self.oil_temp = self.engine_temp + 8.0; | executable/support line; behavior impact depends on surroun
g block context. |
| 108 | (blank) | line is blank; visual separation only. |
| 109 |         // Oil pressure drops slightly at idle, rises with RPM | comment line; documentation intent, no runti
effect. |
| 110 |         self.oil_pressure = 1.2 + (self.rpm / self.max_rpm) * 4.0; | executable/support line; behavior impact
pend on surrounding block context. |
| 111 | (blank) | line is blank; visual separation only. |

```



```

| 112 |          // Fuel consumption: idle ~0.6 L/h, WOT at high RPM ~18 L/h | comment line; documentation intent, no
time effect. |
| 113 |          self.fuel_consumption_lph = 0.6 + effective_throttle * 17.0 * (self.rpm / 4000.0).min(1.0); | executab
/support line; behavior impact depends on surrounding block context. |
| 114 |          let consumed = self.fuel_consumption_lph * dt / 3600.0; | executable/support line; behavior impact de
ds on surrounding block context. |
| 115 |          self.fuel_level = (self.fuel_level - consumed / FUEL_TANK_L).max(0.0); | executable/support line; beh
or impact depends on surrounding block context. |
| 116 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 117 | (blank) | line is blank; visual separation only. |
| 118 |      /// Torque available at wheel (through drivetrain, at current RPM) | comment line; documentation intent,
runtime effect. |
| 119 |      pub fn wheel_drive_force_n(&self, gear: u8) -> f64 { | declares executable behavior; signature/body chang
alter API, control flow, and outputs. |
| 120 |          if gear == 0 || gear > 8 { | conditional gate; changing condition shifts runtime path and side effe
. |
| 121 |              return 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 122 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 123 |          let wheel_torque = self.output_torque * GEAR_RATIOS[gear as usize] * FINAL_DRIVE; | executable/suppor
ine; behavior impact depends on surrounding block context. |
| 124 |          wheel_torque / WHEEL_RADIUS_M | executable/support line; behavior impact depends on surrounding block
ntext. |
| 125 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 126 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/gps.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 367 | Non-empty: 335 | Symbols: 20

Symbol Inventory

Line	Kind	Symbol	Change Impact
10	enum	GpsFixQuality	data/structure coupling
20	impl	std::fmt::Display for GpsFixQuality	behavior binding and method semantics
21	fn	fmt	signature/API coupling
36	struct	Satellite	data/structure coupling
46	enum	Constellation	data/structure coupling
54	impl	std::fmt::Display for Constellation	behavior binding and method semantics
55	fn	fmt	signature/API coupling
68	struct	GpsModule	data/structure coupling
115	impl	Default for GpsModule	behavior binding and method semantics
116	fn	default	signature/API coupling
121	impl	GpsModule	behavior binding and method semantics
122	fn	new	signature/API coupling
161	fn	init_satellites	signature/API coupling
199	fn	update	signature/API coupling
290	fn	build_gga	signature/API coupling
310	fn	build_rmc	signature/API coupling
328	fn	build_gsa	signature/API coupling
348	fn	deg_to_nmea	signature/API coupling
356	fn	nmea_checksum	signature/API coupling
360	fn	position_string	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	/// GPS/GNSS Module – Full simulation of a u-blox ZED-F9P grade receiver.	comment line; documentation intent, runtime effect.
2	/// Outputs real NMEA 0183 sentences (GGA, RMC, GSA, GSV) with valid checksums.	comment line; documentation intent, no runtime effect.
3	/// Simulates: satellite constellation, atmospheric delays, multipath, SBAS correction.	comment line; document

[illegible]

```
| 45 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding block context. |  
| 46 | pub enum Constellation { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |  
| 47 |     Gps, | executable/support line; behavior impact depends on surrounding block context. |  
| 48 |     Glonass, | executable/support line; behavior impact depends on surrounding block context. |  
| 49 |     Galileo, | executable/support line; behavior impact depends on surrounding block context. |  
| 50 |     Beidou, | executable/support line; behavior impact depends on surrounding block context. |  
| 51 |     Sbas, | executable/support line; behavior impact depends on surrounding block context. |  
| 52 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 53 | (blank) | line is blank; visual separation only. |  
| 54 | impl std::fmt::Display for Constellation { | implementation block; method behavior and trait conformance are defined here. |  
| 55 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 56 |         match self { | branch dispatch; changing arms alters behavior for variants and input classes. |  
| 57 |             Constellation::Gps => "GPS", | executable/support line; behavior impact depends on surrounding block context. |  
| 58 |             Constellation::Glonass => "GLO", | executable/support line; behavior impact depends on surrounding block context. |  
| 59 |             Constellation::Galileo => "GAL", | executable/support line; behavior impact depends on surrounding block context. |  
| 60 |             Constellation::Beidou => "BDS", | executable/support line; behavior impact depends on surrounding block context. |  
| 61 |             Constellation::Sbas => "SBS", | executable/support line; behavior impact depends on surrounding block context. |  
| 62 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 63 |         .fmt(f) | executable/support line; behavior impact depends on surrounding block context. |  
| 64 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 65 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 66 | (blank) | line is blank; visual separation only. |  
| 67 | // ■■ GPS Module ████████████████████████████████████████████████████████████████████ | comment line; no runtime effect. |  
| 68 | pub struct GpsModule { | declares data layout; field changes may break serialization, tests, and callers. |  
| 69 |     // ■ Position ████████████████████████████████████████████████████████████████████ | comment line; no runtime effect. |  
| 70 |     pub latitude_deg: f64, // decimal degrees (positive = N) | executable/support line; behavior impact depends on surrounding block context. |  
| 71 |     pub longitude_deg: f64, // decimal degrees (positive = E) | executable/support line; behavior impact depends on surrounding block context. |  
| 72 |     pub altitude_msl: f64, // metres above Mean Sea Level | executable/support line; behavior impact depends on surrounding block context. |  
| 73 |     pub undulation: f64,    // geoid undulation (WGS84 height diff) | executable/support line; behavior impact depends on surrounding block context. |  
| 74 | (blank) | line is blank; visual separation only. |  
| 75 |     // ■ Velocity ████████████████████████████████████████████████████████████████████ | comment line; no runtime effect. |  
| 76 |     pub speed_knots: f64, // speed over ground (knots) | executable/support line; behavior impact depends on surrounding block context. |  
| 77 |     pub speed_kmh: f64,   // speed over ground (km/h) | executable/support line; behavior impact depends on surrounding block context. |  
| 78 |     pub course_deg: f64,  // course over ground (degrees true) | executable/support line; behavior impact depends on surrounding block context. |  
| 79 |     pub climb_ms: f64,    // vertical velocity (m/s) | executable/support line; behavior impact depends on surrounding block context. |  
| 80 | (blank) | line is blank; visual separation only. |  
| 81 |     // ■ Accuracy ████████████████████████████████████████████████████████████████████ | comment line; no runtime effect. |  
| 82 |     pub hdop: f64,        // Horizontal Dilution of Precision | executable/support line; behavior impact depends on surrounding block context. |  
| 83 |     pub vdop: f64,        // Vertical DOP | executable/support line; behavior impact depends on surrounding block context. |  
| 84 |     pub pdop: f64,        // Position DOP | executable/support line; behavior impact depends on surrounding block context. |  
| 85 |     pub tdop: f64,        // Time DOP | executable/support line; behavior impact depends on surrounding block context.
```

```
| 86 | pub hacc_m: f64, // Horizontal accuracy (metres, 1σ) | executable/support line; behavior impact depends on surrounding block context. |
| 87 | pub vacc_m: f64, // Vertical accuracy | executable/support line; behavior impact depends on surrounding block context. |
| 88 | pub fix_quality: GpsFixQuality, | executable/support line; behavior impact depends on surrounding block context. |
| 89 | (blank) | line is blank; visual separation only. |
| 90 | // ■ Satellites ████████████████████████████████████████████████████████████████████████████ | comment line; no runtime effect. |
| 91 | pub satellites: Vec<Satellite>, | executable/support line; behavior impact depends on surrounding block context. |
| 92 | pub sats_in_view: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 93 | pub sats_used: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 94 | (blank) | line is blank; visual separation only. |
| 95 | // ■ Time ████████████████████████████████████████████████████████████████████████████ | comment line; no runtime effect. |
| 96 | pub utc_time: String, // "HHMMSS.ss" | executable/support line; behavior impact depends on surrounding block context. |
| 97 | pub utc_date: String, // "DDMMYY" | executable/support line; behavior impact depends on surrounding block context. |
| 98 | pub gps_week: u16, | executable/support line; behavior impact depends on surrounding block context. |
| 99 | pub time_of_week_s: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 100 | (blank) | line is blank; visual separation only. |
| 101 | // ■ Dead Reckoning ████████████████████████████████████████████████████████████████████████████ | comment line; no runtime effect. |
| 102 | pub dr_active: bool, // Using dead reckoning (GNSS lost) | executable/support line; behavior impact depends on surrounding block context. |
| 103 | pub dr_time_s: f64, // Seconds since last GNSS fix | executable/support line; behavior impact depends on surrounding block context. |
| 104 | pub dr_error_m: f64, // Accumulated DR error | executable/support line; behavior impact depends on surrounding block context. |
| 105 | (blank) | line is blank; visual separation only. |
| 106 | // ■ Simulation state ████████████████████████████████████████████████████████████████████████████ | comment line; no runtime effect. |
| 107 | pos_x: f64, // Local ENU East (metres from origin) | executable/support line; behavior impact depends on surrounding block context. |
| 108 | pos_y: f64, // Local ENU North | executable/support line; behavior impact depends on surrounding block context. |
| 109 | noise_seed: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 110 | update_timer: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 111 | pub nmea_queue: VecDeque<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 112 | pub update_rate_hz: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 113 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 114 | (blank) | line is blank; visual separation only. |
| 115 | impl Default for GpsModule { | implementation block; method behavior and trait conformance are defined here. |
| 116 | fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 117 |     Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 118 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 119 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 120 | (blank) | line is blank; visual separation only. |
| 121 | impl GpsModule { | implementation block; method behavior and trait conformance are defined here. |
| 122 | pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 123 |     // Start at a typical field location in Brazil (lat -22°, lon -47°) | comment line; documentation intent, no runtime effect. |
| 124 |     let mut gps = GpsModule { | executable/support line; behavior impact depends on surrounding block context. |
| 125 |         latitude_deg: -22.8100, | executable/support line; behavior impact depends on surrounding block context. |
| 126 |         longitude_deg: -47.0620, | executable/support line; behavior impact depends on surrounding block context. |
| 127 |         altitude_msl: 640.0, | executable/support line; behavior impact depends on surrounding block context.
```

```

. |
| 128 |         undulation: -10.5, | executable/support line; behavior impact depends on surrounding block context. |
|
| 129 |         speed_knots: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 130 |         speed_kmh: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 131 |         course_deg: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 132 |         climb_ms: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 133 |         hdop: 0.9, | executable/support line; behavior impact depends on surrounding block context. |
| 134 |         vdop: 1.2, | executable/support line; behavior impact depends on surrounding block context. |
| 135 |         pdop: 1.5, | executable/support line; behavior impact depends on surrounding block context. |
| 136 |         tdop: 0.8, | executable/support line; behavior impact depends on surrounding block context. |
| 137 |         hacc_m: 1.2, | executable/support line; behavior impact depends on surrounding block context. |
| 138 |         vacc_m: 2.0, | executable/support line; behavior impact depends on surrounding block context. |
| 139 |         fix_quality: GpsFixQuality::SpsFix, | executable/support line; behavior impact depends on surrounding block context. |
| 140 |         satellites: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 141 |         sats_in_view: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 142 |         sats_used: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 143 |         utc_time: "120000.00".into(), | executable/support line; behavior impact depends on surrounding block context. |
| 144 |         utc_date: "110826".into(), | executable/support line; behavior impact depends on surrounding block context. |
| 145 |         gps_week: 2326, | executable/support line; behavior impact depends on surrounding block context. |
| 146 |         time_of_week_s: 43200.0, | executable/support line; behavior impact depends on surrounding block context. |
| 147 |         dr_active: false, | executable/support line; behavior impact depends on surrounding block context. |
| 148 |         dr_time_s: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 149 |         dr_error_m: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 150 |         pos_x: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 151 |         pos_y: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 152 |         noise_seed: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 153 |         update_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
|
| 154 |         nmea_queue: VecDeque::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 155 |         update_rate_hz: 10.0, | executable/support line; behavior impact depends on surrounding block context. |
| 156 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
|
| 157 |     gps.init_satellites(); | executable/support line; behavior impact depends on surrounding block context. |
|
| 158 |     gps | executable/support line; behavior impact depends on surrounding block context. |
| 159 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 160 | (blank) | line is blank; visual separation only. |
| 161 | fn init_satellites(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 162 |     // Simulate a realistic satellite constellation visible from Brazil | comment line; documentation intent, no runtime effect. |
| 163 |     let sat_data = [ | executable/support line; behavior impact depends on surrounding block context. |
| 164 |         // PRN, elev, azimuth, SNR, constellation | comment line; documentation intent, no runtime effect. |
| 165 |         (1u8, 55.0, 185.0, 44.0, Constellation::Gps), | executable/support line; behavior impact depends on surrounding block context. |
| 166 |         (3, 32.0, 290.0, 38.0, Constellation::Gps), | executable/support line; behavior impact depends on surrounding block context. |
| 167 |         (6, 18.0, 125.0, 30.0, Constellation::Gps), | executable/support line; behavior impact depends on surrounding block context. |
| 168 |         (7, 72.0, 42.0, 47.0, Constellation::Gps), | executable/support line; behavior impact depends on surrounding block context. |
| 169 |         (11, 44.0, 320.0, 41.0, Constellation::Gps), | executable/support line; behavior impact depends on surrounding block context. |
| 170 |         (14, 28.0, 210.0, 35.0, Constellation::Gps), | executable/support line; behavior impact depends on surrounding block context. |
| 171 |         (17, 61.0, 150.0, 45.0, Constellation::Gps), | executable/support line; behavior impact depends on surrounding block context. |

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```
| 172 |          (19, 12.0, 78.0, 22.0, Constellation::Gps), | executable/support line; behavior impact depends on  
rounding block context. |  
| 173 |          (22, 38.0, 265.0, 39.0, Constellation::Gps), | executable/support line; behavior impact depends o  
urrounding block context. |  
| 174 |          (28, 85.0, 330.0, 48.0, Constellation::Gps), | executable/support line; behavior impact depends o  
urrounding block context. |  
| 175 |          (65, 42.0, 95.0, 37.0, Constellation::Glonass), | executable/support line; behavior impact depend  
n surrounding block context. |  
| 176 |          (66, 29.0, 175.0, 33.0, Constellation::Glonass), | executable/support line; behavior impact dependen  
on surrounding block context. |  
| 177 |          (67, 58.0, 250.0, 43.0, Constellation::Glonass), | executable/support line; behavior impact dependen  
on surrounding block context. |  
| 178 |          (68, 15.0, 310.0, 24.0, Constellation::Glonass), | executable/support line; behavior impact dependen  
on surrounding block context. |  
| 179 |          (11 + 64, 47.0, 60.0, 40.0, Constellation::Galileo), | executable/support line; behavior impact d  
nds on surrounding block context. |  
| 180 |          (12 + 64, 33.0, 220.0, 36.0, Constellation::Galileo), | executable/support line; behavior impact  
ends on surrounding block context. |  
| 181 |          (13 + 64, 68.0, 145.0, 46.0, Constellation::Galileo), | executable/support line; behavior impact  
ends on surrounding block context. |  
| 182 |      ]; | executable/support line; behavior impact depends on surrounding block context. |  
| 183 |      for (prn, elev, azimuth, snr, constellation) in sat_data.iter() { | iteration control; changing bound  
ody affects throughput, timing, and possibly safety guarantees. |  
| 184 |          self.satellites.push(Satellite { | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 185 |              prn: *prn, | executable/support line; behavior impact depends on surrounding block context. |  
| 186 |              elevation: *elev, | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 187 |              azimuth: *azimuth, | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 188 |              snr: *snr, | executable/support line; behavior impact depends on surrounding block context. |  
| 189 |              used: *elev > 10.0, | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 190 |              constellation: *constellation, | executable/support line; behavior impact depends on surround  
block context. |  
| 191 |          }); | executable/support line; behavior impact depends on surrounding block context. |  
| 192 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 193 |      self.sats_in_view = self.satellites.len() as u8; | executable/support line; behavior impact depends o  
urrounding block context. |  
| 194 |      self.sats_used = self.satellites.iter().filter(\|s\| s.used).count() as u8; | executable/support line  
ehavior impact depends on surrounding block context. |  
| 195 |    } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 196 | (blank) | line is blank; visual separation only. |  
| 197 |      /// Main update – call every simulation step. | comment line; documentation intent, no runtime effect. |  
| 198 |      /// `speed_kmh` and `heading_deg` come from vehicle dynamics. | comment line; documentation intent, no ru  
me effect. |  
| 199 |      pub fn update(&mut self, speed_kmh: f64, heading_deg: f64, dt: f64) { | declares executable behavior; sig  
ure/body changes alter API, control flow, and outputs. |  
| 200 |          self.update_timer += dt; | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 201 | (blank) | line is blank; visual separation only. |  
| 202 |          // ■ Update position from vehicle speed ████████████████████████████████████████████ | comment line; docume  
ent, no runtime effect. |  
| 203 |          let speed_ms = speed_kmh / 3.6; | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 204 |          let head_rad = heading_deg * PI / 180.0; | executable/support line; behavior impact depends on surrou  
ng block context. |  
| 205 |          self.pos_x += speed_ms * head_rad.sin() * dt; | executable/support line; behavior impact depends on s  
ounding block context. |  
| 206 |          self.pos_y += speed_ms * head_rad.cos() * dt; | executable/support line; behavior impact depends on s  
ounding block context. |  
| 207 | (blank) | line is blank; visual separation only. |  
| 208 |          // ■ Convert local ENU to lat/lon ████████████████████████████████████████████ | comment line; docu  
ent, no runtime effect. |
```

```
| 209 | // 1° latitude ≈ 111,320 m | comment line; documentation intent, no runtime effect. |  
| 210 | // 1° longitude ≈ 111,320 × cos(lat) m | comment line; documentation intent, no runtime effect. |  
| 211 | let metres_per_deg_lat = 111_320.0; | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 212 | let metres_per_deg_lon = 111_320.0 * (self.latitude_deg * PI / 180.0).cos(); | executable/support lin  
behavior impact depends on surrounding block context. |  
| 213 | self.latitude_deg += (self.pos_y / metres_per_deg_lat) * 0.001 * dt; // slow drift | executable/suppo  
line; behavior impact depends on surrounding block context. |  
| 214 | self.longitude_deg += (self.pos_x / metres_per_deg_lon) * 0.001 * dt; | executable/support line; beha  
r impact depends on surrounding block context. |  
| 215 | // Actual integration – more accurate | comment line; documentation intent, no runtime effect. |  
| 216 | let d_lat = (speed_ms * head_rad.cos() * dt) / metres_per_deg_lat; | executable/support line; behavior  
mpact depends on surrounding block context. |  
| 217 | let d_lon = (speed_ms * head_rad.sin() * dt) / metres_per_deg_lon; | executable/support line; behavior  
mpact depends on surrounding block context. |  
| 218 | self.latitude_deg += d_lat; | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 219 | self.longitude_deg += d_lon; | executable/support line; behavior impact depends on surrounding block  
text. |  
| 220 | (blank) | line is blank; visual separation only. |  
| 221 | self.speed_kmh = speed_kmh; | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 222 | self.speed_knots = speed_kmh * 0.539957; | executable/support line; behavior impact depends on surroun  
ng block context. |  
| 223 | self.course_deg = heading_deg; | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 224 | (blank) | line is blank; visual separation only. |  
| 225 | // ■ Add GPS noise (pseudo-random, position-dependent) ■■■■■■■■■■■■■■■■■■■■■■ | comment line; documentat  
ent, no runtime effect. |  
| 226 | self.noise_seed += dt * 7.3; | executable/support line; behavior impact depends on surrounding block  
text. |  
| 227 | let noise = \|s: f64\| ((s * 127.1 + 311.7).sin() * 43758.5 % 1.0 - 0.5) * 2.0; | executable/support  
e; behavior impact depends on surrounding block context. |  
| 228 | let pos_noise_m = match self.fix_quality { | executable/support line; behavior impact depends on surr  
ding block context. |  
| 229 | GpsFixQuality::RtkFix => 0.02, | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 230 | GpsFixQuality::DgpsFix => 0.5, | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 231 | GpsFixQuality::SpsFix => 2.5, | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 232 | GpsFixQuality::Dead => self.dr_error_m, | executable/support line; behavior impact depends on sur  
nding block context. |  
| 233 | _ => 5.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 234 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 235 | let noise_lat = noise(self.noise_seed) * pos_noise_m / metres_per_deg_lat; | executable/support line;  
havior impact depends on surrounding block context. |  
| 236 | let noise_lon = noise(self.noise_seed + 1.5) * pos_noise_m / metres_per_deg_lon; | executable/support  
ne; behavior impact depends on surrounding block context. |  
| 237 | self.hacc_m = pos_noise_m * 1.5; | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 238 | self.vacc_m = pos_noise_m * 2.5; | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 239 | (blank) | line is blank; visual separation only. |  
| 240 | // ■ Satellite dynamics (elevation/SNR vary slowly) ■■■■■■■■■■■■■■■■■■■■■■ | comment line; documenta  
ent, no runtime effect. |  
| 241 | let t = self.noise_seed; | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 242 | for (i, sat) in self.satellites.iter_mut().enumerate() { | iteration control; changing bounds/body af  
ts throughput, timing, and possibly safety guarantees. |  
| 243 | let drift = (t * 0.01 + i as f64 * 0.7).sin() * 0.3; | executable/support line; behavior impact d  
nds on surrounding block context. |  
| 244 | sat.elevation = (sat.elevation + drift * dt * 0.1).clamp(5.0, 90.0); | executable/support line; b  
vior impact depends on surrounding block context. |
```

```
| 245 |         let snr_noise = noise(t + i as f64 * 3.1) * 0.5; | executable/support line; behavior impact depends on surrounding block context. |
| 246 |             sat.snr = (sat.snr + snr_noise * dt * 5.0).clamp(10.0, 52.0); | executable/support line; behavior impact depends on surrounding block context. |
| 247 |             sat.used = sat.elevation > 10.0 && sat.snr > 18.0; | executable/support line; behavior impact depends on surrounding block context. |
| 248 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope |
| 249 |         self.sats_used = self.satellites.iter().filter(\|s\| s.used).count() as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 250 |         self.sats_in_view = self.satellites.len() as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 251 | (blank) | line is blank; visual separation only. |
| 252 |         // ■ DOP calculation ████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 253 |         // Simplified: better HDOP with more good satellites overhead | comment line; documentation intent, no runtime effect. |
| 254 |         let good_sats = self | executable/support line; behavior impact depends on surrounding block context. |
| 255 |             .satellites | executable/support line; behavior impact depends on surrounding block context. |
| 256 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 257 |             .filter(\|s\| s.used && s.elevation > 30.0) | executable/support line; behavior impact depends on surrounding block context. |
| 258 |                 .count() as f64; | executable/support line; behavior impact depends on surrounding block context. |
| 259 |         self.hdop = (1.5 - good_sats * 0.05).max(0.8) + noise(t + 5.1).abs() * 0.1; | executable/support line; behavior impact depends on surrounding block context. |
| 260 |         self.vdop = self.hdop * 1.35; | executable/support line; behavior impact depends on surrounding block context. |
| 261 |         self.pdop = (self.hdop.powi(2) + self.vdop.powi(2)).sqrt(); | executable/support line; behavior impact depends on surrounding block context. |
| 262 | (blank) | line is blank; visual separation only. |
| 263 |         // ■ Update UTC time ████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 264 |         self.time_of_week_s += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 265 |         let total_s = self.time_of_week_s; | executable/support line; behavior impact depends on surrounding block context. |
| 266 |         let h = ((total_s / 3600.0) as u32) % 24; | executable/support line; behavior impact depends on surrounding block context. |
| 267 |         let m = ((total_s / 60.0) as u32) % 60; | executable/support line; behavior impact depends on surrounding block context. |
| 268 |         let s = total_s % 60.0; | executable/support line; behavior impact depends on surrounding block context. |
| 269 |         self.utc_time = format!("{:02}{:02}:{:05.2}", h, m, s); | executable/support line; behavior impact depends on surrounding block context. |
| 270 | (blank) | line is blank; visual separation only. |
| 271 |         // ■ Generate NMEA sentences at update rate ████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 272 |         if self.update_timer >= 1.0 / self.update_rate_hz { | conditional gate; changing condition shifts runtime path and side effects. |
| 273 |             self.update_timer = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 274 |             let lat_applied = self.latitude_deg + noise_lat; | executable/support line; behavior impact depends on surrounding block context. |
| 275 |             let lon_applied = self.longitude_deg + noise_lon; | executable/support line; behavior impact depends on surrounding block context. |
| 276 |             self.nmea_queue | executable/support line; behavior impact depends on surrounding block context. |
| 277 |                 .push_back(self.build_gga(lat_applied, lon_applied)); | executable/support line; behavior impact depends on surrounding block context. |
| 278 |             self.nmea_queue | executable/support line; behavior impact depends on surrounding block context. |
| 279 |                 .push_back(self.build_rmc(lat_applied, lon_applied)); | executable/support line; behavior impact depends on surrounding block context. |
| 280 |             self.nmea_queue.push_back(self.build_gsa()); | executable/support line; behavior impact depends on surrounding block context. |
| 281 |             if self.nmea_queue.len() > 120 { | conditional gate; changing condition shifts runtime path and side effects. |
```


[illegible]

```

|
| 330 |         .satellites | executable/support line; behavior impact depends on surrounding block context. |
| 331 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 332 |         .filter(\|s\| s.used) | executable/support line; behavior impact depends on surrounding block con
t. |
| 333 |         .take(12) | executable/support line; behavior impact depends on surrounding block context. |
| 334 |         .map(\|s\| format!("{:02}", s.prn)) | executable/support line; behavior impact depends on surroun
g block context. |
| 335 |         .collect::<Vec<_>>() | executable/support line; behavior impact depends on surrounding block cont
. |
| 336 |         .join(","); | executable/support line; behavior impact depends on surrounding block context. |
| 337 |     let body = format!( | executable/support line; behavior impact depends on surrounding block context. |
| 338 |         "GPGSA,A,3,{},{},{:.1},{:.1},{:.1}", | executable/support line; behavior impact depends on surrou
ng block context. |
| 339 |         used, | executable/support line; behavior impact depends on surrounding block context. |
| 340 |         ", ".repeat(12usize.saturating_sub(self.sats_used as usize)), | executable/support line; behavior
act depends on surrounding block context. |
| 341 |         self.pdop, | executable/support line; behavior impact depends on surrounding block context. |
| 342 |         self.hdop, | executable/support line; behavior impact depends on surrounding block context. |
| 343 |         self.vdop | executable/support line; behavior impact depends on surrounding block context. |
| 344 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 345 |     format!("${}*{:02X}\r\n", body, Self::nmea_checksum(&body)) | executable/support line; behavior impac
depends on surrounding block context. |
| 346 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 347 | (blank) | line is blank; visual separation only. |
| 348 | fn deg_to_nmea(deg: f64, pos: char, neg: char) -> (f64, f64, char) { | declares executable behavior; sign
re/body changes alter API, control flow, and outputs. |
| 349 |     let hemi = if deg >= 0.0 { pos } else { neg }; | executable/support line; behavior impact depends on
rounding block context. |
| 350 |     let abs = deg.abs(); | executable/support line; behavior impact depends on surrounding block context. |
| 351 |     let d = abs.floor(); | executable/support line; behavior impact depends on surrounding block context. |
| 352 |     let m = (abs - d) * 60.0; | executable/support line; behavior impact depends on surrounding block con
t. |
| 353 |     (d, d * 100.0 + m, hemi) | executable/support line; behavior impact depends on surrounding block cont
. |
| 354 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 355 | (blank) | line is blank; visual separation only. |
| 356 | fn nmea_checksum(sentence: &str) -> u8 { | declares executable behavior; signature/body changes alter API
ontrol flow, and outputs. |
| 357 |     sentence.bytes().fold(0u8, \|acc, b\| acc ^ b) | executable/support line; behavior impact depends on
rounding block context. |
| 358 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 359 | (blank) | line is blank; visual separation only. |
| 360 | pub fn position_string(&self) -> String { | declares executable behavior; signature/body changes alter AP
ontrol flow, and outputs. |
| 361 |     let lat_d = self.latitude_deg.abs(); | executable/support line; behavior impact depends on surroundi
lock context. |
| 362 |     let lat_h = if self.latitude_deg >= 0.0 { 'N' } else { 'S' }; | executable/support line; behavior imp
depends on surrounding block context. |
| 363 |     let lon_d = self.longitude_deg.abs(); | executable/support line; behavior impact depends on surroundi
block context. |
| 364 |     let lon_h = if self.longitude_deg >= 0.0 { 'E' } else { 'W' }; | executable/support line; behavior im
t depends on surrounding block context. |
| 365 |     format!("{:.6}°{} {:.6}°{}", lat_d, lat_h, lon_d, lon_h) | executable/support line; behavior impact
ends on surrounding block context. |
| 366 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 367 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/implement.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 505 | Non-empty: 442 | Symbols: 36

Symbol Inventory

Line	Kind	Symbol	Change Impact
15	const	HITCH_MAX_RATE	data/structure coupling
17	const	PTO_OVERLOAD_SLIP	data/structure coupling
19	const	PTO_RECOVER_SLIP	data/structure coupling
21	const	AUX_MAX_FLOW	data/structure coupling
23	const	AUX_RELIEF_BAR	data/structure coupling
25	const	HITCH_SENSOR_BW	data/structure coupling
31	enum	PtoMode	data/structure coupling
39	impl	PtoMode	behavior binding and method semantics
41	fn	target_rpm	signature/API coupling
50	fn	required_engine_rpm	signature/API coupling
61	impl	fmt::Display for PtoMode	behavior binding and method semantics
62	fn	fmt	signature/API coupling
77	enum	HitchMode	data/structure coupling
88	impl	fmt::Display for HitchMode	behavior binding and method semantics
89	fn	fmt	signature/API coupling
103	enum	ImplementType	data/structure coupling
114	impl	fmt::Display for ImplementType	behavior binding and method semantics
115	fn	fmt	signature/API coupling
133	enum	ValveDirection	data/structure coupling
139	impl	fmt::Display for ValveDirection	behavior binding and method semantics
140	fn	fmt	signature/API coupling
153	struct	HydraulicBank	data/structure coupling
163	impl	HydraulicBank	behavior binding and method semantics
164	fn	new	signature/API coupling
176	fn	update	signature/API coupling
193	struct	ImplementControl	data/structure coupling
234	impl	ImplementControl	behavior binding and method semantics
235	fn	new	signature/API coupling
278	fn	update	signature/API coupling
286	fn	update_pto	signature/API coupling
353	fn	update_hitch	signature/API coupling
405	fn	update_aux_hydraulics	signature/API coupling
448	fn	pto_torque_demand	signature/API coupling
463	fn	generate_j1939_frames	signature/API coupling
501	impl	Default for ImplementControl	behavior binding and method semantics
502	fn	default	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	/// ISOBUS / ISO 11783-7 Implement Controller	comment line; documentation intent, no runtime effect.
2	///	comment line; documentation intent, no runtime effect.
3	/// Covers: Rear & Front PTO, 3-Point Hitch (position / draft / mixed / float),	comment line; documentation intent, no runtime effect.
4	/// four Auxiliary Hydraulic valve banks, loader hydraulics, and Task Controller	comment line; documentation intent, no runtime effect.
5	/// interface. Emits J1939 PGN 65093 (PTO) and PGN 65091 (Hitch) frames.	comment line; documentation intent, runtime effect.
6	(blank)	line is blank; visual separation only.
7	use crate::j1939::addr;	imports symbols; changing path/name affects compile-time resolution.
8	use crate::j1939::pgn;	imports symbols; changing path/name affects compile-time resolution.
9	use crate::j1939::{Builder, J1939Frame};	imports symbols; changing path/name affects compile-time resolution.
10	use std::fmt;	imports symbols; changing path/name affects compile-time resolution.
11	(blank)	line is blank; visual separation only.
12	// ■■■ Module-level constants	comment line; documentation intent, no runtime effect.
13	(blank)	line is blank; visual separation only.
14	/// Maximum hitch movement rate [%/s]	comment line; documentation intent, no runtime effect.
15	const HITCH_MAX_RATE: f64 = 20.0;	executable/support line; behavior impact depends on surrounding block content.
16	/// PTO slip at which overload protection engages [%]	comment line; documentation intent, no runtime effect.
17	const PTO_OVERLOAD_SLIP: f64 = 15.0;	executable/support line; behavior impact depends on surrounding block content.

```
| ext. |  
| 18 | /// PTO slip at which overload protection clears [%] | comment line; documentation intent, no runtime effect.  
| 19 | const PTO_RECOVER_SLIP: f64 = 5.0; | executable/support line; behavior impact depends on surrounding block context.  
t. |  
| 20 | /// Maximum flow per auxiliary bank [L/min] | comment line; documentation intent, no runtime effect. |  
| 21 | const AUX_MAX_FLOW: f64 = 60.0; | executable/support line; behavior impact depends on surrounding block context  
|  
| 22 | /// Auxiliary circuit relief pressure [bar] | comment line; documentation intent, no runtime effect. |  
| 23 | const AUX_RELIEF_BAR: f64 = 200.0; | executable/support line; behavior impact depends on surrounding block context.  
t. |  
| 24 | /// Hitch draft sensor bandwidth approximation [rad/s] | comment line; documentation intent, no runtime effect.  
| 25 | const HITCH_SENSOR_BW: f64 = 2.0 * std::f64::consts::PI * 5.0; | executable/support line; behavior impact depends  
on surrounding block context. |  
| 26 | (blank) | line is blank; visual separation only. |  
| 27 | // ■■ PtoMode ████████████████████████████████████████████████████████████████████████████████████ | comment  
intent, no runtime effect. |  
| 28 | (blank) | line is blank; visual separation only. |  
| 29 | /// PTO engagement / speed mode | comment line; documentation intent, no runtime effect. |  
| 30 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding block  
context. |  
| 31 | pub enum PtoMode { | declares state variants; adding/removing variants impacts pattern matches and logic branches.  
. |  
| 32 |     Off, | executable/support line; behavior impact depends on surrounding block context. |  
| 33 |     Std540,      // 540 RPM at rated engine speed | executable/support line; behavior impact depends on surrounding  
block context. |  
| 34 |     Std1000,    // 1000 RPM at rated engine speed | executable/support line; behavior impact depends on surround  
ing block context. |  
| 35 |     Economy540, // 540 RPM at reduced engine speed (fuel economy) | executable/support line; behavior impact  
ends on surrounding block context. |  
| 36 |     Economy1000, // 1000 RPM at reduced engine speed | executable/support line; behavior impact depends on surr  
ounding block context. |  
| 37 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 38 | (blank) | line is blank; visual separation only. |  
| 39 | impl PtoMode { | implementation block; method behavior and trait conformance are defined here. |  
| 40 |     /// Target PTO shaft output speed [RPM] | comment line; documentation intent, no runtime effect. |  
| 41 |     pub fn target_rpm(self) -> f64 { | declares executable behavior; signature/body changes alter API, control  
flow, and outputs. |  
| 42 |         match self { | branch dispatch; changing arms alters behavior for variants and input classes. |  
| 43 |             PtoMode::Off => 0.0, | executable/support line; behavior impact depends on surrounding block conte  
nt. |  
| 44 |             PtoMode::Std540 \| PtoMode::Economy540 => 540.0, | executable/support line; behavior impact depend  
n surrounding block context. |  
| 45 |             PtoMode::Std1000 \| PtoMode::Economy1000 => 1000.0, | executable/support line; behavior impact dep  
s on surrounding block context. |  
| 46 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 47 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 48 | (blank) | line is blank; visual separation only. |  
| 49 |     /// Nominal engine RPM required to achieve the target PTO speed | comment line; documentation intent, no r  
ime effect. |  
| 50 |     pub fn required_engine_rpm(self) -> f64 { | declares executable behavior; signature/body changes alter API  
ontrol flow, and outputs. |  
| 51 |         match self { | branch dispatch; changing arms alters behavior for variants and input classes. |  
| 52 |             PtoMode::Off => 800.0, | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 53 |             PtoMode::Std540 => 2100.0, | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 54 |             PtoMode::Economy540 => 1450.0, | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 55 |             PtoMode::Std1000 => 2100.0, | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 56 |             PtoMode::Economy1000 => 1700.0, | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 57 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 58 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 59 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
```

[illegible]

[illegible]

```

193 | pub struct ImplementControl { | declares data layout; field changes may break serialization, tests, and calle
|
| 194 | // ■ PTO System ████████████████████████████████████████████████████████████████████ | comment line; documentation in
e effect. |
| 195 | pub pto_rear_enabled: bool, | executable/support line; behavior impact depends on surrounding block conte
|
| 196 | pub pto_front_enabled: bool, | executable/support line; behavior impact depends on surrounding block conte
. |
| 197 | pub pto_rear_rpm: f64, // actual output shaft speed [RPM] | executable/support line; behavior impact depende
on surrounding block context. |
| 198 | pub pto_front_rpm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 199 | pub pto_target_rpm: f64, // 540 or 1000 per mode | executable/support line; behavior impact depends on su
rounding block context. |
| 200 | pub pto_mode: PtoMode, | executable/support line; behavior impact depends on surrounding block context. |
| 201 | pub pto_shaft_torque_nm: f64, // torque transmitted to implement [Nm] | executable/support line; behavior im
pact depends on surrounding block context. |
| 202 | pub pto_slip_pct: f64, // shaft slip relative to target [%] | executable/support line; behavior im
t depends on surrounding block context. |
| 203 | (blank) | line is blank; visual separation only. |
| 204 | // ■ 3-Point Hitch (rear) ████████████████████████████████████████████████████████ | comment line; documentation inte
me effect. |
| 205 | pub hitch_position_pct: f64, // 0 = fully lowered, 100 = fully raised | executable/support line; behavior im
pact depends on surrounding block context. |
| 206 | pub hitch_target_pct: f64, | executable/support line; behavior impact depends on surrounding block context
|
| 207 | pub hitch_draft_force_kn: f64, // measured draft force [kN] | executable/support line; behavior impact
depends on surrounding block context. |
| 208 | pub hitch_ground_pressure_kpa: f64, // soil contact pressure [kPa] | executable/support line; behavior im
t depends on surrounding block context. |
| 209 | pub hitch_control_mode: HitchMode, | executable/support line; behavior impact depends on surrounding bloc
context. |
| 210 | pub hitch_moving: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 211 | pub hitch_speed_pct_per_s: f64, // current hitch movement rate [%/s] | executable/support line; behavior i
act depends on surrounding block context. |
| 212 | (blank) | line is blank; visual separation only. |
| 213 | // ■ Auxiliary Hydraulic Valves (4 banks) ████████████████████████████████ | comment line; documentation intent, n
e effect. |
| 214 | pub aux_banks: [HydraulicBank; 4], | executable/support line; behavior impact depends on surrounding bloc
context. |
| 215 | (blank) | line is blank; visual separation only. |
| 216 | // ■ Remote Hydraulic (loader / front attachment) ████████████████ | comment line; documentation intent, no
me effect. |
| 217 | pub loader_lift_position_pct: f64, | executable/support line; behavior impact depends on surrounding bloc
context. |
| 218 | pub loader_tilt_position_pct: f64, | executable/support line; behavior impact depends on surrounding bloc
context. |
| 219 | pub loader_pressure_bar: f64, | executable/support line; behavior impact depends on surrounding block con
t. |
| 220 | (blank) | line is blank; visual separation only. |
| 221 | // ■ Attached implement ████████████████████████████████████████████████████████ | comment line; documentation int
me effect. |
| 222 | pub implement_attached: Option<ImplementType>, | executable/support line; behavior impact depends on surr
ding block context. |
| 223 | pub implement_width_m: f64, | executable/support line; behavior impact depends on surrounding block conte
|
| 224 | pub implement_working_depth_cm: f64, | executable/support line; behavior impact depends on surrounding bl
context. |
| 225 | pub isobus_connected: bool, | executable/support line; behavior impact depends on surrounding block conte
|
| 226 | pub task_controller_active: bool, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 227 | (blank) | line is blank; visual separation only. |
| 228 | // ■ Internal state (pub per crate convention) ████████████████████████████████ | comment line; documentation intent, n
me effect. |
| 229 | pub overload_protection_active: bool, | executable/support line; behavior impact depends on surrounding b

```



```

k context. |
| 230 |     pub hitch_draft_target_kn: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 231 |     pub pto_spin_up_timer: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 232 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 233 | (blank) | line is blank; visual separation only. |
| 234 | impl ImplementControl { | implementation block; method behavior and trait conformance are defined here. |
| 235 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 236 |         ImplementControl { | executable/support line; behavior impact depends on surrounding block context. |
| 237 |             pto_rear_enabled: false, | executable/support line; behavior impact depends on surrounding block context. |
| 238 |             pto_front_enabled: false, | executable/support line; behavior impact depends on surrounding block context. |
| 239 |             pto_rear_rpm: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 240 |             pto_front_rpm: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 241 |             pto_target_rpm: 540.0, | executable/support line; behavior impact depends on surrounding block context. |
| 242 |             pto_mode: PtoMode::Off, | executable/support line; behavior impact depends on surrounding block context. |
| 243 |             pto_shaft_torque_nm: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 244 |             pto_slip_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 245 | (blank) | line is blank; visual separation only. |
| 246 |             hitch_position_pct: 100.0, | executable/support line; behavior impact depends on surrounding block context. |
| 247 |             hitch_target_pct: 100.0, | executable/support line; behavior impact depends on surrounding block context. |
| 248 |             hitch_draft_force_kn: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 249 |             hitch_ground_pressure_kpa: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 250 |             hitch_control_mode: HitchMode::Position, | executable/support line; behavior impact depends on surrounding block context. |
| 251 |             hitch_moving: false, | executable/support line; behavior impact depends on surrounding block context. |
| 252 |             hitch_speed_pct_per_s: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 253 | (blank) | line is blank; visual separation only. |
| 254 |             aux_banks: [ | executable/support line; behavior impact depends on surrounding block context. |
| 255 |                 HydraulicBank::new(0), | executable/support line; behavior impact depends on surrounding block context. |
| 256 |                 HydraulicBank::new(1), | executable/support line; behavior impact depends on surrounding block context. |
| 257 |                 HydraulicBank::new(2), | executable/support line; behavior impact depends on surrounding block context. |
| 258 |                 HydraulicBank::new(3), | executable/support line; behavior impact depends on surrounding block context. |
| 259 |             ], | executable/support line; behavior impact depends on surrounding block context. |
| 260 | (blank) | line is blank; visual separation only. |
| 261 |             loader_lift_position_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 262 |             loader_tilt_position_pct: 50.0, | executable/support line; behavior impact depends on surrounding block context. |
| 263 |             loader_pressure_bar: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 264 | (blank) | line is blank; visual separation only. |
| 265 |             implement_attached: None, | executable/support line; behavior impact depends on surrounding block context. |
| 266 |             implement_width_m: 3.0, | executable/support line; behavior impact depends on surrounding block context. |

```

[illegible]

```

| 305 |           Some(ImplementType::Harvester) => 0.80, | executable/support line; behavior impact depends on surrounding block context. |
| 306 |           Some(ImplementType::Plow) => 0.75, | executable/support line; behavior impact depends on surrounding block context. |
| 307 |           Some(ImplementType::Disc) => 0.60, | executable/support line; behavior impact depends on surrounding block context. |
| 308 |           Some(ImplementType::Cultivator) => 0.45, | executable/support line; behavior impact depends on surrounding block context. |
| 309 |           Some(ImplementType::Planter) => 0.30, | executable/support line; behavior impact depends on surrounding block context. |
| 310 |           Some(ImplementType::Sprayer) => 0.12, | executable/support line; behavior impact depends on surrounding block context. |
| 311 |           _ => 0.08, | executable/support line; behavior impact depends on surrounding block context. |
| 312 |       }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 313 |       let desired_torque = 150.0 + load_k * 850.0 * throttle; | executable/support line; behavior impact depends on surrounding block context. |
| 314 | (blank) | line is blank; visual separation only. |
| 315 |       // Overload protection: hysteresis on slip threshold | comment line; documentation intent, no runtime effect. |
| 316 |       if self.pto_slip_pct > PTO_OVERLOAD_SLIP { | conditional gate; changing condition shifts runtime path and side effects. |
| 317 |           self.overload_protection_active = true; | executable/support line; behavior impact depends on surrounding block context. |
| 318 |       } else if self.pto_slip_pct < PTO_RECOVER_SLIP { | executable/support line; behavior impact depends on surrounding block context. |
| 319 |           self.overload_protection_active = false; | executable/support line; behavior impact depends on surrounding block context. |
| 320 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 321 |       let effective_torque = if self.overload_protection_active { | executable/support line; behavior impact depends on surrounding block context. |
| 322 |           desired_torque * 0.25 // severe de-rate to allow slip recovery | executable/support line; behavior impact depends on surrounding block context. |
| 323 |       } else { | executable/support line; behavior impact depends on surrounding block context. |
| 324 |           desired_torque | executable/support line; behavior impact depends on surrounding block context. |
| 325 |       }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 326 | (blank) | line is blank; visual separation only. |
| 327 |       // First-order speed dynamics: acceleration from speed error minus torque drag | comment line; documentation intent, no runtime effect. |
| 328 |       let speed_error = target - self.pto_rear_rpm; | executable/support line; behavior impact depends on surrounding block context. |
| 329 |       let accel = (speed_error * 4.0 - effective_torque * 0.035).clamp(-350.0, 700.0); | executable/support line; behavior impact depends on surrounding block context. |
| 330 |       self.pto_rear_rpm = (self.pto_rear_rpm + accel * dt).clamp(0.0, target * 1.15); | executable/support line; behavior impact depends on surrounding block context. |
| 331 | (blank) | line is blank; visual separation only. |
| 332 |       // Slip is the normalised speed deficit relative to no-load target | comment line; documentation intent, no runtime effect. |
| 333 |       self.pto_slip_pct = if target > 1.0 { | executable/support line; behavior impact depends on surrounding block context. |
| 334 |           ((target - self.pto_rear_rpm) / target * 100.0).clamp(0.0, 100.0) | executable/support line; behavior impact depends on surrounding block context. |
| 335 |       } else { | executable/support line; behavior impact depends on surrounding block context. |
| 336 |           0.0 | executable/support line; behavior impact depends on surrounding block context. |
| 337 |       }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 338 |       // Shaft torque is slightly reduced under slip (clutch modulation) | comment line; documentation intent, no runtime effect. |
| 339 |       self.pto_shaft_torque_nm = effective_torque * (1.0 - self.pto_slip_pct * 0.005); | executable/support line; behavior impact depends on surrounding block context. |
| 340 | (blank) | line is blank; visual separation only. |
| 341 |       // Front PTO: driven from same shaft at a fixed reduction ratio | comment line; documentation intent, no runtime effect. |

```

[illegible]

[illegible]

```
| rounding block context. |  
| 417 |         ValveDirection::Neutral => 0.0, | executable/support line; behavior impact depends on surrounding  
g block context. |  
| 418 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 419 |         self.loader_lift_position_pct = | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 420 |             (self.loader_lift_position_pct + delta).clamp(0.0, 100.0); | executable/support line; behavio  
mpact depends on surrounding block context. |  
| 421 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 422 | (blank) | line is blank; visual separation only. |  
| 423 |     // Bank 1 → loader/bucket tilt cylinder | comment line; documentation intent, no runtime effect. |  
| 424 |     if self.aux_banks[1].engaged { | conditional gate; changing condition shifts runtime path and side ef  
ts. |  
| 425 |         let delta = match self.aux_banks[1].direction { | executable/support line; behavior impact depend  
n surrounding block context. |  
| 426 |             ValveDirection::Extend => 25.0 * dt, | executable/support line; behavior impact depends on su  
unding block context. |  
| 427 |             ValveDirection::Retract => -25.0 * dt, | executable/support line; behavior impact depends on  
rounding block context. |  
| 428 |             ValveDirection::Neutral => 0.0, | executable/support line; behavior impact depends on surround  
g block context. |  
| 429 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 430 |         self.loader_tilt_position_pct = | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 431 |             (self.loader_tilt_position_pct + delta).clamp(0.0, 100.0); | executable/support line; behavio  
mpact depends on surrounding block context. |  
| 432 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 433 | (blank) | line is blank; visual separation only. |  
| 434 |         let active_p = if self.aux_banks[0].engaged \\ \\ self.aux_banks[1].engaged { | executable/support lin  
behavior impact depends on surrounding block context. |  
| 435 |             self.aux_banks[0] | executable/support line; behavior impact depends on surrounding block context  
| 436 |                 .pressure_bar | executable/support line; behavior impact depends on surrounding block context  
| 437 |                 .max(self.aux_banks[1].pressure_bar) | executable/support line; behavior impact depends on su  
unding block context. |  
| 438 |         } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 439 |             0.0 | executable/support line; behavior impact depends on surrounding block context. |  
| 440 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 441 |     // Bleed loader circuit when nothing is active | comment line; documentation intent, no runtime effec  
|  
| 442 |         self.loader_pressure_bar += (active_p - self.loader_pressure_bar) * (5.0 * dt).min(1.0); | executable  
pport line; behavior impact depends on surrounding block context. |  
| 443 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 444 | (blank) | line is blank; visual separation only. |  
| 445 |     // ■■ Queries ████████████████████████████████████████████████████████████████████ | comment  
intent, no runtime effect. |  
| 446 | (blank) | line is blank; visual separation only. |  
| 447 |     /// Torque demand placed on the engine by all active PTO shafts [Nm] | comment line; documentation intent  
o runtime effect. |  
| 448 |     pub fn pto_torque_demand(&self) -> f64 { | declares executable behavior; signature/body changes alter API  
ontrol flow, and outputs. |  
| 449 |         if !self.pto_rear_enabled { | conditional gate; changing condition shifts runtime path and side effec  
|  
| 450 |             return 0.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 451 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 452 |         let front_add = if self.pto_front_enabled { | executable/support line; behavior impact depends on sur  
nding block context. |  
| 453 |             self.pto_shaft_torque_nm * 0.30 | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 454 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
```

[illegible]

[illegible]

```
| tent, no runtime effect. |  
| 59 | pub struct Imu { | declares data layout; field changes may break serialization, tests, and callers. |  
| 60 | // ■ Raw sensor outputs (body frame) ████████████████████████████████████████ | comment line; document  
nt, no runtime effect. |  
| 61 |     pub accel_x: f64, // m/s² (+x = forward) | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 62 |     pub accel_y: f64, // m/s² (+y = left) | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 63 |     pub accel_z: f64, // m/s² (+z = up) | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 64 | (blank) | line is blank; visual separation only. |  
| 65 |     pub gyro_x: f64, // rad/s roll rate | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 66 |     pub gyro_y: f64, // rad/s pitch rate | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 67 |     pub gyro_z: f64, // rad/s yaw rate | executable/support line; behavior impact depends on surrounding block  
nxtent. |  
| 68 | (blank) | line is blank; visual separation only. |  
| 69 |     pub mag_x: f64, // µT (earth field + bias) | executable/support line; behavior impact depends on surroundi  
block context. |  
| 70 |     pub mag_y: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 71 |     pub mag_z: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 72 | (blank) | line is blank; visual separation only. |  
| 73 | // ■ Calibrated values (bias removed) ████████████████████████████████████████ | comment line; docum  
nt, no runtime effect. |  
| 74 |     pub accel_cal: [f64; 3], | executable/support line; behavior impact depends on surrounding block context.  
| 75 |     pub gyro_cal: [f64; 3], | executable/support line; behavior impact depends on surrounding block context. |  
| 76 | (blank) | line is blank; visual separation only. |  
| 77 | // ■ Attitude (Madgwick AHRS) ████████████████████████████████████████ | comment line; do  
tent, no runtime effect. |  
| 78 |     pub q: Quaternion, | executable/support line; behavior impact depends on surrounding block context. |  
| 79 |     pub roll_deg: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 80 |     pub pitch_deg: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 81 |     pub yaw_deg: f64, // heading (0-360, 0=North) | executable/support line; behavior impact depends on surrou  
ng block context. |  
| 82 | (blank) | line is blank; visual separation only. |  
| 83 | // ■ Linear acceleration (gravity removed) ████████████████████████████████████████ | comment line; docum  
tent, no runtime effect. |  
| 84 |     pub lin_accel_x: f64, // m/s² in world frame | executable/support line; behavior impact depends on surroun  
g block context. |  
| 85 |     pub lin_accel_y: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 86 |     pub lin_accel_z: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 87 | (blank) | line is blank; visual separation only. |  
| 88 | // ■ Temperature ████████████████████████████████████████████████████████████ | comment line  
tent, no runtime effect. |  
| 89 |     pub temperature_c: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 90 | (blank) | line is blank; visual separation only. |  
| 91 | // ■ Derived ████████████████████████████████████████████████████████████ | comment line  
tent, no runtime effect. |  
| 92 |     pub lateral_g: f64,      // g-force lateral (positive = right) | executable/support line; behavior impact  
ends on surrounding block context. |  
| 93 |     pub longitudinal_g: f64, // g-force longitudinal (positive = forward) | executable/support line; behavior  
act depends on surrounding block context. |  
| 94 |     pub vertical_g: f64,      // g-force vertical | executable/support line; behavior impact depends on surroun  
g block context. |  
| 95 | (blank) | line is blank; visual separation only. |  
| 96 | // ■ Fault detection ████████████████████████████████████████████████████████████ | comment line  
tent, no runtime effect. |  
| 97 |     pub accel_fault: bool, | executable/support line; behavior impact depends on surrounding block context. |  
| 98 |     pub gyro_fault: bool, | executable/support line; behavior impact depends on surrounding block context. |  
| 99 |     pub mag_fault: bool, | executable/support line; behavior impact depends on surrounding block context. |  
| 100 | (blank) | line is blank; visual separation only. |  
| 101 | // ■ Internal simulation state ████████████████████████████████████████████████████████████ | comment line;  
tent, no runtime effect. |  
| 102 |     accel_bias: [f64; 3], | executable/support line; behavior impact depends on surrounding block context.
```

```

| 103 |     gyro_bias: [f64; 3], | executable/support line; behavior impact depends on surrounding block context. |
| 104 |     noise_t: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 105 |     vibe_amp: f64, // vibration amplitude from engine | executable/support line; behavior impact depends on s
ounding block context. |
| 106 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 107 | (blank) | line is blank; visual separation only. |
| 108 | impl Default for Imu { | implementation block; method behavior and trait conformance are defined here. |
| 109 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, an
utputs. |
| 110 |         Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 111 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 112 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 113 | (blank) | line is blank; visual separation only. |
| 114 | impl Imu { | implementation block; method behavior and trait conformance are defined here. |
| 115 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, an
utputs. |
| 116 |         let mut imu = Imu { | executable/support line; behavior impact depends on surrounding block context.
| 117 |             accel_x: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 118 |             accel_y: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 119 |             accel_z: 9.81, | executable/support line; behavior impact depends on surrounding block context. |
| 120 |             gyro_x: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 121 |             gyro_y: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 122 |             gyro_z: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 123 |             mag_x: 20.0, | executable/support line; behavior impact depends on surrounding block context. |
| 124 |             mag_y: -5.0, | executable/support line; behavior impact depends on surrounding block context. |
| 125 |             mag_z: -40.0, // Brazil magnetic field | executable/support line; behavior impact depends on surr
ding block context. |
| 126 |             accel_cal: [0.0; 3], | executable/support line; behavior impact depends on surrounding block cont
. |
| 127 |             gyro_cal: [0.0; 3], | executable/support line; behavior impact depends on surrounding block conte
|
| 128 |             q: Quaternion::identity(), | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 129 |             roll_deg: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 130 |             pitch_deg: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 131 |             yaw_deg: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 132 |             lin_accel_x: 0.0, | executable/support line; behavior impact depends on surrounding block context
| 133 |             lin_accel_y: 0.0, | executable/support line; behavior impact depends on surrounding block context
| 134 |             lin_accel_z: 0.0, | executable/support line; behavior impact depends on surrounding block context
| 135 |             temperature_c: 25.0, | executable/support line; behavior impact depends on surrounding block cont
. |
| 136 |             lateral_g: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 137 |             longitudinal_g: 0.0, | executable/support line; behavior impact depends on surrounding block cont
. |
| 138 |             vertical_g: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 139 |             accel_fault: false, | executable/support line; behavior impact depends on surrounding block conte
|
| 140 |             gyro_fault: false, | executable/support line; behavior impact depends on surrounding block context
|
| 141 |             mag_fault: false, | executable/support line; behavior impact depends on surrounding block context
| 142 |             accel_bias: [0.002, -0.001, 0.015], | executable/support line; behavior impact depends on surroun
g block context. |
| 143 |             gyro_bias: [0.0003, -0.0002, 0.0001], | executable/support line; behavior impact depends on surro
ing block context. |
| 144 |             noise_t: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 145 |             vibe_amp: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 146 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 147 |         imu.q.normalise(); | executable/support line; behavior impact depends on surrounding block context. |
| 148 |         imu | executable/support line; behavior impact depends on surrounding block context. |
| 149 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 150 | (blank) | line is blank; visual separation only. |
| 151 |     /// Update IMU with vehicle dynamics. | comment line; documentation intent, no runtime effect. |
| 152 |     /// `ax_ms2` = longitudinal acceleration, `yaw_rate` = deg/s, `rpm` = engine RPM | comment line; document
on intent, no runtime effect. |

```

```
| 153 | #[allow(clippy::too_many_arguments)] | executable/support line; behavior impact depends on surrounding block context. |
```

```
| 154 | pub fn update( | declares executable behavior; signature/body changes alter API, control flow, and output
```

```
| 155 | &ampmut self, | executable/support line; behavior impact depends on surrounding block context. |
```

```
| 156 | ax_ms2: f64, | executable/support line; behavior impact depends on surrounding block context. |
```

```
| 157 | lat_acc_ms2: f64, | executable/support line; behavior impact depends on surrounding block context. |
```

```
| 158 | yaw_rate_degs: f64, | executable/support line; behavior impact depends on surrounding block context.
```

```
| 159 | heading_deg: f64, | executable/support line; behavior impact depends on surrounding block context. |
```

```
| 160 | grade_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |
```

```
| 161 | rpm: f64, | executable/support line; behavior impact depends on surrounding block context. |
```

```
| 162 | dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
```

```
| 163 | ) { | executable/support line; behavior impact depends on surrounding block context. |
```

```
| 164 |     self.noise_t += dt; | executable/support line; behavior impact depends on surrounding block context.
```

```
| 165 |     let n = self.noise_t; | executable/support line; behavior impact depends on surrounding block context
```

```
| 166 | (blank) | line is blank; visual separation only. |
```

```
| 167 |     // ■ Engine vibration model (fundamental + harmonics) ██████████ | comment line; document
```

```
tent, no runtime effect. |
```

```
| 168 |     let vibe_freq = rpm / 60.0 * (Imu::ncyl() as f64 / 2.0); // firing frequency Hz | executable/support
```

```
e; behavior impact depends on surrounding block context. |
```

```
| 169 |     self.vibe_amp = (rpm / 2200.0 * 0.15).min(0.3); | executable/support line; behavior impact depends on
```

```
rrounding block context. |
```

```
| 170 |     let vibe = self.vibe_amp | executable/support line; behavior impact depends on surrounding block cont
```

```
. |
```

```
| 171 |         * ((2.0 * PI * vibe_freq * n).sin() + 0.3 * (4.0 * PI * vibe_freq * n).sin()); | executable/suppo
```

```
line; behavior impact depends on surrounding block context. |
```

```
| 172 | (blank) | line is blank; visual separation only. |
```

```
| 173 |     // ■ True accelerations in body frame ██████████ | comment line; doc
```

```
ntent, no runtime effect. |
```

```
| 174 |     let g_component = 9.81 * (grade_pct / 100.0).atan().sin(); // grade | executable/support line; behavi
```

```
impact depends on surrounding block context. |
```

```
| 175 |     let true_ax = ax_ms2 - g_component; | executable/support line; behavior impact depends on surrounding
```

```
ock context. |
```

```
| 176 |     let true_ay = lat_acc_ms2; | executable/support line; behavior impact depends on surrounding block co
```

```
xt. |
```

```
| 177 |     let true_az = -9.81; | executable/support line; behavior impact depends on surrounding block context.
```

```
| 178 | (blank) | line is blank; visual separation only. |
```

```
| 179 |     // ■ Add noise, bias and vibration ██████████ | comment line; do
```

```
ntent, no runtime effect. |
```

```
| 180 |     let an = \|seed: f64, amp: f64\| { | executable/support line; behavior impact depends on surrounding
```

```
ck context. |
```

```
| 181 |         (((seed * 127.1 + 311.7).sin()) * 43758.5).fract() - 0.5) * 2.0 * amp | executable/support line; b
```

```
vior impact depends on surrounding block context. |
```

```
| 182 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
```

```
|
```

```
| 183 |     self.accel_x = true_ax + self.accel_bias[0] + an(n * 1.1, ACCEL_NOISE) + vibe; | executable/support l
```

```
; behavior impact depends on surrounding block context. |
```

```
| 184 |     self.accel_y = true_ay + self.accel_bias[1] + an(n * 2.3, ACCEL_NOISE) + vibe * 0.6; | executable/sup
```

```
t line; behavior impact depends on surrounding block context. |
```

```
| 185 |     self.accel_z = true_az + self.accel_bias[2] + an(n * 3.7, ACCEL_NOISE) + vibe * 0.8; | executable/sup
```

```
t line; behavior impact depends on surrounding block context. |
```

```
| 186 | (blank) | line is blank; visual separation only. |
```

```
| 187 |     let yaw_rate_rad = yaw_rate_degs * PI / 180.0; | executable/support line; behavior impact depends on
```

```
rrounding block context. |
```

```
| 188 |     // Gyro bias drifts slowly | comment line; documentation intent, no runtime effect. |
```

```
| 189 |     self.gyro_bias[2] += an(n * 0.5, GYRO_DRIFT) * dt; | executable/support line; behavior impact depends
```

```
surrounding block context. |
```

```
| 190 |     self.gyro_x = 0.0 + self.gyro_bias[0] + an(n * 4.1, GYRO_NOISE); | executable/support line; behavior
```

```
act depends on surrounding block context. |
```

```
| 191 |     self.gyro_y = 0.0 + self.gyro_bias[1] + an(n * 5.3, GYRO_NOISE); | executable/support line; behavior
```

```
act depends on surrounding block context. |
```

```
| 192 |     self.gyro_z = yaw_rate_rad + self.gyro_bias[2] + an(n * 6.7, GYRO_NOISE); | executable/support line;
```

```
avior impact depends on surrounding block context. |
```

```
| 193 | (blank) | line is blank; visual separation only. |
```

```
| 194 |     // Calibrated values | comment line; documentation intent, no runtime effect. |
```

```
| 195 |         self.accel_cal = [ | executable/support line; behavior impact depends on surrounding block context. |  
| 196 |             self.accel_x - self.accel_bias[0], | executable/support line; behavior impact depends on surrounding  
block context. |  
| 197 |             self.accel_y - self.accel_bias[1], | executable/support line; behavior impact depends on surrounding  
block context. |  
| 198 |             self.accel_z - self.accel_bias[2], | executable/support line; behavior impact depends on surrounding  
block context. |  
| 199 | ]; | executable/support line; behavior impact depends on surrounding block context. |  
| 200 | self.gyro_cal = [ | executable/support line; behavior impact depends on surrounding block context. |  
| 201 |     self.gyro_x - self.gyro_bias[0], | executable/support line; behavior impact depends on surrounding  
lock context. |  
| 202 |     self.gyro_y - self.gyro_bias[1], | executable/support line; behavior impact depends on surrounding  
lock context. |  
| 203 |     self.gyro_z - self.gyro_bias[2], | executable/support line; behavior impact depends on surrounding  
lock context. |  
| 204 | ]; | executable/support line; behavior impact depends on surrounding block context. |  
| 205 | (blank) | line is blank; visual separation only. |  
| 206 | // ■ Magnetometer (earth field + vehicle field distortion) ■■■■■■■■■■■■ | comment line; document  
intent, no runtime effect. |  
| 207 | let yaw_rad = heading_deg * PI / 180.0; | executable/support line; behavior impact depends on surround  
g block context. |  
| 208 | self.mag_x = 20.0 * yaw_rad.cos() + an(n * 7.1, MAG_NOISE); | executable/support line; behavior impac  
depends on surrounding block context. |  
| 209 | self.mag_y = 20.0 * (-yaw_rad).sin() + an(n * 8.3, MAG_NOISE); | executable/support line; behavior im  
t depends on surrounding block context. |  
| 210 | self.mag_z = -40.0 + an(n * 9.5, MAG_NOISE); | executable/support line; behavior impact depends on su  
unding block context. |  
| 211 | (blank) | line is blank; visual separation only. |  
| 212 | // ■ Madgwick AHRS filter ■■■■■■■■■■■■ | comment line  
intent, no runtime effect. |  
| 213 | self.madgwick_update(dt); | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 214 | (blank) | line is blank; visual separation only. |  
| 215 | // ■ Extract Euler angles ■■■■■■■■■■■■ | comment line  
intent, no runtime effect. |  
| 216 | let (r, p, y) = self.q.to_euler_deg(); | executable/support line; behavior impact depends on surround  
block context. |  
| 217 | self.roll_deg = r; | executable/support line; behavior impact depends on surrounding block context. |  
| 218 | self.pitch_deg = p; | executable/support line; behavior impact depends on surrounding block context.  
| 219 | self.yaw_deg = (y + 360.0) % 360.0; | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 220 | (blank) | line is blank; visual separation only. |  
| 221 | // ■ Remove gravity to get linear acceleration ■■■■■■■■■■■■ | comment line; docum  
ntent, no runtime effect. |  
| 222 | let (qw, qx, qy, qz) = (self.q.w, self.q.x, self.q.y, self.q.z); | executable/support line; behavior  
act depends on surrounding block context. |  
| 223 | let gx = 2.0 * (qx * qz - qw * qy); | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 224 | let gy = 2.0 * (qw * qx + qy * qz); | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 225 | let gz = qw * qw - qx * qx - qy * qy + qz * qz; | executable/support line; behavior impact depends on  
rounding block context. |  
| 226 | self.lin_accel_x = self.accel_cal[0] - gx * 9.81; | executable/support line; behavior impact depends  
surrounding block context. |  
| 227 | self.lin_accel_y = self.accel_cal[1] - gy * 9.81; | executable/support line; behavior impact depends  
surrounding block context. |  
| 228 | self.lin_accel_z = self.accel_cal[2] - gz * 9.81; | executable/support line; behavior impact depends  
surrounding block context. |  
| 229 | (blank) | line is blank; visual separation only. |  
| 230 | // ■ G-forces ■■■■■■■■■■■■ | comment l  
intent, no runtime effect. |  
| 231 | self.longitudinal_g = self.lin_accel_x / 9.81; | executable/support line; behavior impact depends on  
rounding block context. |  
| 232 | self.lateral_g = self.lin_accel_y / 9.81; | executable/support line; behavior impact depends on surro  
ing block context.
```

```

233 | self.vertical_g = self.accel_cal[2].abs() / 9.81; | executable/support line; behavior impact depends
surrounding block context. |
| 234 | (blank) | line is blank; visual separation only. |
| 235 | // ■ Temperature ████████████████████████████████████████████████████████████████████████████ | comment lin
ntent, no runtime effect. |
| 236 | self.temperature_c += (35.0 + rpm / 2200.0 * 15.0 - self.temperature_c) * dt * 0.01; | executable/sup
t line; behavior impact depends on surrounding block context. |
| 237 | (blank) | line is blank; visual separation only. |
| 238 | // ■ Fault detection ████████████████████████████████████████████████████████████████████████████ | comment line
ntent, no runtime effect. |
| 239 | let accel_mag = (self.accel_x.powi(2) + self.accel_y.powi(2) + self.accel_z.powi(2)).sqrt(); | execut
e/support line; behavior impact depends on surrounding block context. |
| 240 | self.accel_fault = !(4.0..=25.0).contains(&accel_mag); | executable/support line; behavior impact dep
s on surrounding block context. |
| 241 | self.gyro_fault = self.gyro_z.abs() > 10.0; // >10 rad/s = sensor fault | executable/support line; be
ior impact depends on surrounding block context. |
| 242 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 243 | (blank) | line is blank; visual separation only. |
| 244 | /// Madgwick AHRS algorithm – fuses accel + gyro + mag into quaternion attitude. | comment line; document
on intent, no runtime effect. |
| 245 | fn madgwick_update(&mut self, dt: f64) { | declares executable behavior; signature/body changes alter API
ontrol flow, and outputs. |
| 246 | let [ax, ay, az] = self.accel_cal; | executable/support line; behavior impact depends on surrounding
ck context. |
| 247 | let [gx, gy, gz] = self.gyro_cal; | executable/support line; behavior impact depends on surrounding b
k context. |
| 248 | let (mx, my, mz) = (self.mag_x, self.mag_y, self.mag_z); | executable/support line; behavior impact d
nds on surrounding block context. |
| 249 | (blank) | line is blank; visual separation only. |
| 250 | let q0 = self.q.w; | executable/support line; behavior impact depends on surrounding block context. |
| 251 | let q1 = self.q.x; | executable/support line; behavior impact depends on surrounding block context. |
| 252 | let q2 = self.q.y; | executable/support line; behavior impact depends on surrounding block context. |
| 253 | let q3 = self.q.z; | executable/support line; behavior impact depends on surrounding block context. |
| 254 | (blank) | line is blank; visual separation only. |
| 255 | // Normalise accelerometer | comment line; documentation intent, no runtime effect. |
| 256 | let an = (ax * ax + ay * ay + az * az).sqrt(); | executable/support line; behavior impact depends on
rounding block context. |
| 257 | if an < 1e-10 { | conditional gate; changing condition shifts runtime path and side effects. |
| 258 | return; | executable/support line; behavior impact depends on surrounding block context. |
| 259 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 260 | let (ax, ay, az) = (ax / an, ay / an, az / an); | executable/support line; behavior impact depends on
rounding block context. |
| 261 | (blank) | line is blank; visual separation only. |
| 262 | // Normalise magnetometer | comment line; documentation intent, no runtime effect. |
| 263 | let mn = (mx * mx + my * my + mz * mz).sqrt(); | executable/support line; behavior impact depends on
rounding block context. |
| 264 | if mn < 1e-10 { | conditional gate; changing condition shifts runtime path and side effects. |
| 265 | return; | executable/support line; behavior impact depends on surrounding block context. |
| 266 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 267 | let (mx, my, mz) = (mx / mn, my / mn, mz / mn); | executable/support line; behavior impact depends on
rounding block context. |
| 268 | (blank) | line is blank; visual separation only. |
| 269 | // Reference direction of earth's magnetic field | comment line; documentation intent, no runtime eff
. |
| 270 | let hx = 2.0 * mx * (0.5 - q2 * q2 - q3 * q3) | executable/support line; behavior impact depends on s
ounding block context. |
| 271 | + 2.0 * my * (q1 * q2 - q0 * q3) | executable/support line; behavior impact depends on surroundin
lock context. |
| 272 | + 2.0 * mz * (q1 * q3 + q0 * q2); | executable/support line; behavior impact depends on surroundi
block context. |
| 273 | let hy = 2.0 * mx * (q1 * q2 + q0 * q3) | executable/support line; behavior impact depends on surroun
g block context. |
| 274 | + 2.0 * my * (0.5 - q1 * q1 - q3 * q3) | executable/support line; behavior impact depends on sur

```

```

ding block context. |
| 275 |          + 2.0 * mz * (q2 * q3 - q0 * q1); | executable/support line; behavior impact depends on surrounding
block context. |
| 276 |          let hz = 2.0 * mx * (q1 * q3 - q0 * q2) | executable/support line; behavior impact depends on surroundi
g block context. |
| 277 |          + 2.0 * my * (q2 * q3 + q0 * q1) | executable/support line; behavior impact depends on surrounding
lock context. |
| 278 |          + 2.0 * mz * (0.5 - q1 * q1 - q2 * q2); | executable/support line; behavior impact depends on surr
nding block context. |
| 279 |          let bx = (hx * hx + hy * hy).sqrt(); | executable/support line; behavior impact depends on surrounding
lock context. |
| 280 |          let bz = hz; | executable/support line; behavior impact depends on surrounding block context. |
| 281 | (blank) | line is blank; visual separation only. |
| 282 |          // Gradient descent objective function | comment line; documentation intent, no runtime effect. |
| 283 |          let f1 = 2.0 * (q1 * q3 - q0 * q2) - ax; | executable/support line; behavior impact depends on surroundi
ng block context. |
| 284 |          let f2 = 2.0 * (q0 * q1 + q2 * q3) - ay; | executable/support line; behavior impact depends on surroundi
ng block context. |
| 285 |          let f3 = 1.0 - 2.0 * (q1 * q1 + q2 * q2) - az; | executable/support line; behavior impact depends on
rounding block context. |
| 286 |          let f4 = 2.0 * bx * (0.5 - q2 * q2 - q3 * q3) + 2.0 * bz * (q1 * q3 - q0 * q2) - mx; | executable/sup
t line; behavior impact depends on surrounding block context. |
| 287 |          let f5 = 2.0 * bx * (q1 * q2 - q0 * q3) + 2.0 * bz * (q0 * q1 + q2 * q3) - my; | executable/support l
; behavior impact depends on surrounding block context. |
| 288 |          let f6 = 2.0 * bx * (q0 * q2 + q1 * q3) + 2.0 * bz * (0.5 - q1 * q1 - q2 * q2) - mz; | executable/sup
t line; behavior impact depends on surrounding block context. |
| 289 | (blank) | line is blank; visual separation only. |
| 290 |          // Jacobian and gradient | comment line; documentation intent, no runtime effect. |
| 291 |          let j11 = -2.0 * q2; | executable/support line; behavior impact depends on surrounding block context. |
| 292 |          let j12 = 2.0 * q3; | executable/support line; behavior impact depends on surrounding block context. |
| 293 |          let j13 = -2.0 * q0; | executable/support line; behavior impact depends on surrounding block context. |
| 294 |          let j14 = 2.0 * q1; | executable/support line; behavior impact depends on surrounding block context. |
| 295 |          let j21 = 2.0 * q0; | executable/support line; behavior impact depends on surrounding block context. |
| 296 |          let j22 = 2.0 * q1; | executable/support line; behavior impact depends on surrounding block context. |
| 297 |          let j23 = 2.0 * q2; | executable/support line; behavior impact depends on surrounding block context. |
| 298 |          let j24 = 2.0 * q3; | executable/support line; behavior impact depends on surrounding block context. |
| 299 |          let j31 = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 300 |          let j32 = -4.0 * q1; | executable/support line; behavior impact depends on surrounding block context. |
| 301 |          let j33 = -4.0 * q2; | executable/support line; behavior impact depends on surrounding block context. |
| 302 |          let j34 = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 303 | (blank) | line is blank; visual separation only. |
| 304 |          let mut s0 = j11 * f1 + j21 * f2 + j31 * f3 - 2.0 * bz * q2 * f4 | executable/support line; behavior
act depends on surrounding block context. |
| 305 |          + (-2.0 * bx * q3 + 2.0 * bz * q1) * f5 | executable/support line; behavior impact depends on surr
nding block context. |
| 306 |          + 2.0 * bx * q2 * f6; | executable/support line; behavior impact depends on surrounding block con
t. |
| 307 |          let mut s1 = j12 * f1 | executable/support line; behavior impact depends on surrounding block context
| 308 |          + j22 * f2 | executable/support line; behavior impact depends on surrounding block context. |
| 309 |          + j32 * f3 | executable/support line; behavior impact depends on surrounding block context. |
| 310 |          + 2.0 * bz * q3 * f4 | executable/support line; behavior impact depends on surrounding block context
. |
| 311 |          + (2.0 * bx * q2 + 2.0 * bz * q0) * f5 | executable/support line; behavior impact depends on surr
ding block context. |
| 312 |          + (2.0 * bx * q3 - 4.0 * bz * q1) * f6; | executable/support line; behavior impact depends on surr
nding block context. |
| 313 |          let mut s2 = j13 * f1 | executable/support line; behavior impact depends on surrounding block context
| 314 |          + j23 * f2 | executable/support line; behavior impact depends on surrounding block context. |
| 315 |          + j33 * f3 | executable/support line; behavior impact depends on surrounding block context. |
| 316 |          + (-4.0 * bx * q2 - 2.0 * bz * q0) * f4 | executable/support line; behavior impact depends on surr
nding block context. |
| 317 |          + (2.0 * bx * q1 + 2.0 * bz * q3) * f5 | executable/support line; behavior impact depends on surr
ding block context. |
| 318 |          + (2.0 * bx * q0 - 4.0 * bz * q2) * f6; | executable/support line; behavior impact depends on surr
nding block context. |

```

```

| 319 |         let mut s3 = j14 * f1 | executable/support line; behavior impact depends on surrounding block context. |
| 320 |         + j24 * f2 | executable/support line; behavior impact depends on surrounding block context. |
| 321 |         + j34 * f3 | executable/support line; behavior impact depends on surrounding block context. |
| 322 |         + (-4.0 * bx * q3 + 2.0 * bz * q1) * f4 | executable/support line; behavior impact depends on surrounding block context. |
| 323 |         + (-2.0 * bx * q0 + 2.0 * bz * q2) * f5 | executable/support line; behavior impact depends on surrounding block context. |
| 324 |         + 2.0 * bx * q1 * f6; | executable/support line; behavior impact depends on surrounding block context. |
| 325 |         let sn = (s0 * s0 + s1 * s1 + s2 * s2 + s3 * s3).sqrt().max(1e-10); | executable/support line; behavior impact depends on surrounding block context. |
| 326 |         s0 /= sn; | executable/support line; behavior impact depends on surrounding block context. |
| 327 |         s1 /= sn; | executable/support line; behavior impact depends on surrounding block context. |
| 328 |         s2 /= sn; | executable/support line; behavior impact depends on surrounding block context. |
| 329 |         s3 /= sn; | executable/support line; behavior impact depends on surrounding block context. |
| 330 | (blank) | line is blank; visual separation only. |
| 331 |         // Rate of change of quaternion from gyro | comment line; documentation intent, no runtime effect. |
| 332 |         let qdot0 = 0.5 * (-q1 * gx - q2 * gy - q3 * gz) - BETA * s0; | executable/support line; behavior impact depends on surrounding block context. |
| 333 |         let qdot1 = 0.5 * (q0 * gx + q2 * gz - q3 * gy) - BETA * s1; | executable/support line; behavior impact depends on surrounding block context. |
| 334 |         let qdot2 = 0.5 * (q0 * gy - q1 * gz + q3 * gx) - BETA * s2; | executable/support line; behavior impact depends on surrounding block context. |
| 335 |         let qdot3 = 0.5 * (q0 * gz + q1 * gy - q2 * gx) - BETA * s3; | executable/support line; behavior impact depends on surrounding block context. |
| 336 | (blank) | line is blank; visual separation only. |
| 337 |         self.q.w += qdot0 * dt; | executable/support line; behavior impact depends on surrounding block context. |
| 338 |         self.q.x += qdot1 * dt; | executable/support line; behavior impact depends on surrounding block context. |
| 339 |         self.q.y += qdot2 * dt; | executable/support line; behavior impact depends on surrounding block context. |
| 340 |         self.q.z += qdot3 * dt; | executable/support line; behavior impact depends on surrounding block context. |
| 341 |         self.q.normalise(); | executable/support line; behavior impact depends on surrounding block context. |
| 342 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 343 | (blank) | line is blank; visual separation only. |
| 344 |     fn ncy1() -> u8 { | declares executable behavior; signature/body changes alter API, control flow, and ownership. |
| 345 |         6 | executable/support line; behavior impact depends on surrounding block context. |
| 346 |     } // 6-cylinder | executable/support line; behavior impact depends on surrounding block context. |
| 347 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/io/allowlist.rs

Role: I/O and diagnostics stack: adapters, allowlist, flashing, replay, metrics, production phases.
Lines: 230 | Non-empty: 205 | Symbols: 15

Symbol Inventory

Line	Kind	Symbol	Change Impact
10	enum	RuleType	data/structure coupling
16	struct	AllowlistRule	data/structure coupling
28	struct	Allowlist	data/structure coupling
32	impl	Allowlist	behavior binding and method semantics
33	fn	from_path	signature/API coupling
42	fn	is_allowed	signature/API coupling
46	fn	per_rule_rate_limit	signature/API coupling
53	fn	validate	signature/API coupling
95	fn	rule_matches_frame	signature/API coupling
137	fn	parse_u32_hex_or_dec	signature/API coupling
146	fn	parse_mask	signature/API coupling
150	fn	match_hex_pattern	signature/API coupling
177	module	tests	data/structure coupling


```
| 182 | fn | can_rule_matches_mask | signature/API coupling |
| 209 | fn | serial_rule_matches_pattern | signature/API coupling |
```

Line-by-Line Impact Map

```
| Line | Source | Operational Impact If Changed | | |
|---|---|---|---|---|
| 1 | use std::fs; | imports symbols; changing path/name affects compile-time resolution. |
| 2 | use std::path::Path; | imports symbols; changing path/name affects compile-time resolution. |
| 3 | (blank) | line is blank; visual separation only. |
| 4 | use serde::{Deserialize, Serialize}; | imports symbols; changing path/name affects compile-time resolution. |
| 5 | (blank) | line is blank; visual separation only. |
| 6 | use super::hw::{Frame, HwError}; | imports symbols; changing path/name affects compile-time resolution. |
| 7 | (blank) | line is blank; visual separation only. |
| 8 | #[derive(Debug, Clone, PartialEq, Eq, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 9 | #[serde(rename_all = "lowercase")] | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 10 | pub enum RuleType { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 11 |     Can, | executable/support line; behavior impact depends on surrounding block context. |
| 12 |     Serial, | executable/support line; behavior impact depends on surrounding block context. |
| 13 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 14 | (blank) | line is blank; visual separation only. |
| 15 | #[derive(Debug, Clone, PartialEq, Eq, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 16 | pub struct AllowlistRule { | declares data layout; field changes may break serialization, tests, and callers. |
| 17 |     #[serde(rename = "type")] | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 18 |     pub rule_type: RuleType, | executable/support line; behavior impact depends on surrounding block context. |
| 19 |     pub id: Option<u32>, | executable/support line; behavior impact depends on surrounding block context. |
| 20 |     pub mask: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 21 |     pub allowed_bytes: Option<Vec<[usize; 2]>>, | executable/support line; behavior impact depends on surrounding block context. |
| 22 |     pub max_rate_per_sec: Option<u32>, | executable/support line; behavior impact depends on surrounding block context. |
| 23 |     pub pattern_hex: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 24 |     pub description: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 25 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 26 | (blank) | line is blank; visual separation only. |
| 27 | #[derive(Debug, Clone, Default, PartialEq, Eq)] | executable/support line; behavior impact depends on surrounding block context. |
| 28 | pub struct Allowlist { | declares data layout; field changes may break serialization, tests, and callers. |
| 29 |     pub rules: Vec<AllowlistRule>, | executable/support line; behavior impact depends on surrounding block context. |
| 30 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 31 | (blank) | line is blank; visual separation only. |
| 32 | impl Allowlist { | implementation block; method behavior and trait conformance are defined here. |
| 33 |     pub fn from_path(path: &Path) -> Result<Self, String> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 34 |         let raw = fs::read_to_string(path).map_err(|e| format!("read allowlist failed: {e}")); | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 35 |         let rules: Vec<AllowlistRule> = | executable/support line; behavior impact depends on surrounding block context. |
| 36 |             serde_json::from_str(&raw).map_err(|e| format!("parse allowlist failed: {e}")); | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 37 |         let allow = Self { rules }; | executable/support line; behavior impact depends on surrounding block context. |
| 38 |         allow.validate().map_err(|e| e.to_string()); | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 39 |         Ok(allow) | executable/support line; behavior impact depends on surrounding block context. |
| 40 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 41 | (blank) | line is blank; visual separation only. |
```

```

| 42 |     pub fn is_allowed(&self, frame: &Frame) -> bool { | declares executable behavior; signature/body changes alter API, control flow, and outputs. | | | | |
| 43 |         self.rules.iter().any(|r| rule_matches_frame(r, frame)) | executable/support line; behavior impact depends on surrounding block context. |
| 44 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 45 | (blank) | line is blank; visual separation only. |
| 46 |     pub fn per_rule_rate_limit(&self, frame: &Frame) -> Option<u32> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 47 |         self.rules | executable/support line; behavior impact depends on surrounding block context. |
| 48 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 49 |             .find(|r| rule_matches_frame(r, frame)) | executable/support line; behavior impact depends on surrounding block context. |
| 50 |             .and_then(|r| r.max_rate_per_sec) | executable/support line; behavior impact depends on surrounding block context. |
| 51 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 52 | (blank) | line is blank; visual separation only. |
| 53 |     fn validate(&self) -> Result<(), HwError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 54 |         for (idx, rule) in self.rules.iter().enumerate() { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 55 |             match rule.rule_type { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 56 |                 RuleType::Can => { | executable/support line; behavior impact depends on surrounding block context. |
| 57 |                     if rule.id.is_none() { | conditional gate; changing condition shifts runtime path and side effects. |
| 58 |                         return Err(HwError::Unknown(format!("allowlist rule #{idx} missing CAN id" | executable/support line; behavior impact depends on surrounding block context. |
| 59 |                             "allowlist rule #{idx} missing CAN id" | executable/support line; behavior impact depends on surrounding block context. |
| 60 |                             )); | executable/support line; behavior impact depends on surrounding block context. |
| 61 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 62 |                     if let Some(mask) = &rule.mask { | conditional gate; changing condition shifts runtime path and side effects. |
| 63 |                         if parse_u32_hex_or_dec(mask).is_none() { | conditional gate; changing condition shifts runtime path and side effects. |
| 64 |                             return Err(HwError::Unknown(format!("allowlist rule #{idx} has invalid mask" | executable/support line; behavior impact depends on surrounding block context. |
| 65 |                                 "allowlist rule #{idx} has invalid mask" | executable/support line; behavior impact depends on surrounding block context. |
| 66 |                                 )); | executable/support line; behavior impact depends on surrounding block context. |
| 67 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 68 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 69 |                     if let Some(windows) = &rule.allowed_bytes { | conditional gate; changing condition shifts runtime path and side effects. |
| 70 |                         for win in windows { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 71 |                             if win[1] == 0 || win[0] >= 8 || win[0].saturating_add(win[1]) > 8 { | conditional gate; changing condition shifts runtime path and side effects. |
| 72 |                                 return Err(HwError::Unknown(format!("allowlist rule #{idx} has invalid allowed_bytes window" | executable/support line; behavior impact depends on surrounding block context. |
| 73 |                                     "allowlist rule #{idx} has invalid allowed_bytes window" | executable/support line; behavior impact depends on surrounding block context. |
| 74 |                                     )); | executable/support line; behavior impact depends on surrounding block context. |
| 75 |                             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 76 |                         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 77 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

```

| 78 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 79 |           RuleType::Serial => { | executable/support line; behavior impact depends on surrounding block context. |
| 80 |               if let Some(p) = &rule.pattern_hex { | conditional gate; changing condition shifts runtime path and side effects. |
| 81 |                   let compact: String = p.chars().filter(\c\ !c.is_whitespace()).collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 82 |                   if !compact.len().is_multiple_of(2) { | conditional gate; changing condition shifts runtime path and side effects. |
| 83 |                       return Err(HwError::Unknown(format!( | executable/support line; behavior impact depends on surrounding block context. |
| 84 |                           "allowlist rule #{idx} has invalid serial pattern length" | executable/support line; behavior impact depends on surrounding block context. |
| 85 |                           )); | executable/support line; behavior impact depends on surrounding block context. |
| 86 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 87 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 88 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 89 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 90 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 91 |   Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 92 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 93 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 94 | (blank) | line is blank; visual separation only. |
| 95 | fn rule_matches_frame(rule: &AllowlistRule, frame: &Frame) -> bool { | declares executable behavior; signature changes alter API, control flow, and outputs. |
| 96 |     match (rule.rule_type.clone(), frame) { | branch dispatch; changing arms alters behavior for variants and unit classes. |
| 97 |         (RuleType::Can, Frame::Can(cf)) => { | executable/support line; behavior impact depends on surrounding block context. |
| 98 |             let rule_id = match rule.id { | executable/support line; behavior impact depends on surrounding block context. |
| 99 |                 Some(v) => v, | executable/support line; behavior impact depends on surrounding block context. |
| 100 |                 None => return false, | executable/support line; behavior impact depends on surrounding block context. |
| 101 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 102 |             let mask = rule | executable/support line; behavior impact depends on surrounding block context. |
| 103 |                 .mask | executable/support line; behavior impact depends on surrounding block context. |
| 104 |                 .as_deref() | executable/support line; behavior impact depends on surrounding block context. |
| 105 |                 .and_then(parse_u32_hex_or_dec) | executable/support line; behavior impact depends on surrounding block context. |
| 106 |                 .unwrap_or(0x1FFF_FFFF); | executable/support line; behavior impact depends on surrounding block context. |
| 107 | (blank) | line is blank; visual separation only. |
| 108 |             if (cf.id & mask) != (rule_id & mask) { | conditional gate; changing condition shifts runtime path and side effects. |
| 109 |                 return false; | executable/support line; behavior impact depends on surrounding block context. |
| 110 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 111 | (blank) | line is blank; visual separation only. |
| 112 |             // Restrict all payload bytes outside windows when configured. | comment line; documentation intended to have no runtime effect. |
| 113 |             if let Some(allowed) = &rule.allowed_bytes { | conditional gate; changing condition shifts runtime path and side effects. |
| 114 |                 for i in 0..cf.len.min(8) { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 115 |                     if !allowed | conditional gate; changing condition shifts runtime path and side effects. |
| 116 |                         .iter() | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 117 | .any(\|w\| i >= w[0] && i < w[0].saturating_add(w[1])) | executable/support line; beh
or impact depends on surrounding block context. |
| 118 | && cf.data[i] != 0 | executable/support line; behavior impact depends on surrounding
ck context. |
| 119 | { | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 120 | return false; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 121 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 122 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 123 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 124 | (blank) | line is blank; visual separation only. |
| 125 | true | executable/support line; behavior impact depends on surrounding block context. |
| 126 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 127 | (RuleType::Serial, Frame::Serial(sf)) => { | executable/support line; behavior impact depends on surr
ding block context. |
| 128 | let Some(pattern) = &rule.pattern_hex else { | executable/support line; behavior impact depends o
urrounding block context. |
| 129 | return false; | executable/support line; behavior impact depends on surrounding block context
| 130 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 131 | match_hex_pattern(pattern, &sf.bytes) | executable/support line; behavior impact depends on surro
ing block context. |
| 132 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 133 | _ => false, | executable/support line; behavior impact depends on surrounding block context. |
| 134 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 135 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 136 | (blank) | line is blank; visual separation only. |
| 137 | fn parse_u32_hex_or_dec(s: &str) -> Option<u32> { | declares executable behavior; signature/body changes alte
PI, control flow, and outputs. |
| 138 | let t = s.trim(); | executable/support line; behavior impact depends on surrounding block context. |
| 139 | if let Some(hex) = t.strip_prefix("0x").or_else(\| \| t.strip_prefix("0X")) { | conditional gate; changing
ndition shifts runtime path and side effects. |
| 140 | u32::from_str_radix(hex, 16).ok() | executable/support line; behavior impact depends on surrounding b
k context. |
| 141 | } else { | executable/support line; behavior impact depends on surrounding block context. |
| 142 | t.parse::<u32>().ok() | executable/support line; behavior impact depends on surrounding block context
| 143 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 144 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 145 | (blank) | line is blank; visual separation only. |
| 146 | pub fn parse_mask(mask: &str) -> Result<u32, HwError> { | declares executable behavior; signature/body change
lter API, control flow, and outputs. |
| 147 | parse_u32_hex_or_dec(mask).ok_or_else(\| \| HwError::Unknown("invalid allowlist mask".into())) | executabl
upport line; behavior impact depends on surrounding block context. |
| 148 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 149 | (blank) | line is blank; visual separation only. |
| 150 | fn match_hex_pattern(pattern: &str, bytes: &[u8]) -> bool { | declares executable behavior; signature/body ch
es alter API, control flow, and outputs. |
| 151 | let compact: String = pattern.chars().filter(\|c\| !c.is_whitespace()).collect(); | executable/support li
behavior impact depends on surrounding block context. |
| 152 | if !compact.len().is_multiple_of(2) { | conditional gate; changing condition shifts runtime path and side
fects. |
| 153 | return false; | executable/support line; behavior impact depends on surrounding block context. |
| 154 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 155 | (blank) | line is blank; visual separation only. |
| 156 | let mut idx = 0usize; | executable/support line; behavior impact depends on surrounding block context. |
| 157 | for chunk in compact.as_bytes().chunks(2) { | iteration control; changing bounds/body affects throughput,
ming, and possibly safety guarantees. |
| 158 | if idx >= bytes.len() { | conditional gate; changing condition shifts runtime path and side effects.
| 159 | return false; | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 160 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 161 |         let pair = std::str::from_utf8(chunk).unwrap_or_default(); | executable/support line; behavior impact
depends on surrounding block context. |
| 162 |         if pair != "." { | conditional gate; changing condition shifts runtime path and side effects. |
| 163 |             let Ok(expected) = u8::from_str_radix(pair, 16) else { | executable/support line; behavior impact
depends on surrounding block context. |
| 164 |                 return false; | executable/support line; behavior impact depends on surrounding block context
| 165 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
pes. |
| 166 |             if bytes[idx] != expected { | conditional gate; changing condition shifts runtime path and side e
cts. |
| 167 |                 return false; | executable/support line; behavior impact depends on surrounding block context
| 168 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 169 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 170 |         idx += 1; | executable/support line; behavior impact depends on surrounding block context. |
| 171 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 172 | (blank) | line is blank; visual separation only. |
| 173 |     idx == bytes.len() | executable/support line; behavior impact depends on surrounding block context. |
| 174 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 175 | (blank) | line is blank; visual separation only. |
| 176 | #[cfg(test)] | executable/support line; behavior impact depends on surrounding block context. |
| 177 | mod tests { | declares a module boundary; changing module name/path breaks references. |
| 178 |     use super::*; | imports symbols; changing path/name affects compile-time resolution. |
| 179 |     use crate::io::hw::{CanFrame, Frame, SerialFrame}; | imports symbols; changing path/name affects compile-
e resolution. |
| 180 | (blank) | line is blank; visual separation only. |
| 181 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 182 |     fn can_rule_matches_mask() { | declares executable behavior; signature/body changes alter API, control fl
and outputs. |
| 183 |         let allow = Allowlist { | executable/support line; behavior impact depends on surrounding block conte
|
| 184 |             rules: vec![AllowlistRule { | executable/support line; behavior impact depends on surrounding blo
context. |
| 185 |                 rule_type: RuleType::Can, | executable/support line; behavior impact depends on surrounding b
k context. |
| 186 |                 id: Some(0x18FF50E5), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 187 |                 mask: Some("0x1FFFFFFF".to_string()), | executable/support line; behavior impact depends on s
ounding block context. |
| 188 |                 allowed_bytes: Some(vec![[2, 2]]), | executable/support line; behavior impact depends on surr
ding block context. |
| 189 |                 max_rate_per_sec: None, | executable/support line; behavior impact depends on surrounding blo
context. |
| 190 |                 pattern_hex: None, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 191 |                 description: None, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 192 |             }}, | executable/support line; behavior impact depends on surrounding block context. |
| 193 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 194 | (blank) | line is blank; visual separation only. |
| 195 |         let mut data = [0u8; 8]; | executable/support line; behavior impact depends on surrounding block cont
. |
| 196 |         data[2] = 0xAB; | executable/support line; behavior impact depends on surrounding block context. |
| 197 |         let f = Frame::Can(CanFrame { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 198 |             id: 0x18FF50E5, | executable/support line; behavior impact depends on surrounding block context. |
| 199 |             dlc: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 200 |             data, | executable/support line; behavior impact depends on surrounding block context. |
| 201 |             len: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 202 |             timestamp_ms: None, | executable/support line; behavior impact depends on surrounding block conte
|

```

```

| 203 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 204 | (blank) | line is blank; visual separation only. |
| 205 |         assert!(allow.is_allowed(&f)); | executable/support line; behavior impact depends on surrounding block
context. |
| 206 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 207 | (blank) | line is blank; visual separation only. |
| 208 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 209 |     fn serial_rule_matches_pattern() { | declares executable behavior; signature/body changes alter API, control
flow, and outputs. |
| 210 |         let allow = Allowlist { | executable/support line; behavior impact depends on surrounding block context.
|
| 211 |             rules: vec![AllowlistRule { | executable/support line; behavior impact depends on surrounding block
context. |
| 212 |                 rule_type: RuleType::Serial, | executable/support line; behavior impact depends on surrounding block
context. |
| 213 |                 id: None, | executable/support line; behavior impact depends on surrounding block context. |
| 214 |                 mask: None, | executable/support line; behavior impact depends on surrounding block context. |
| 215 |                 allowed_bytes: None, | executable/support line; behavior impact depends on surrounding block context.
|
| 216 |                 max_rate_per_sec: None, | executable/support line; behavior impact depends on surrounding block context.
|
| 217 |                 pattern_hex: Some("AA55..".to_string()), | executable/support line; behavior impact depends on
surrounding block context. |
| 218 |                 description: None, | executable/support line; behavior impact depends on surrounding block context.
|
| 219 |             }], | executable/support line; behavior impact depends on surrounding block context. |
| 220 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
|
| 221 | (blank) | line is blank; visual separation only. |
| 222 |         let f = Frame::Serial(SerialFrame { | executable/support line; behavior impact depends on surrounding block
context. |
| 223 |             bytes: vec![0xAA, 0x55, 0x01], | executable/support line; behavior impact depends on surrounding block
context. |
| 224 |             protocol_hint: None, | executable/support line; behavior impact depends on surrounding block context.
. |
| 225 |             timestamp_ms: None, | executable/support line; behavior impact depends on surrounding block context.
|
| 226 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 227 | (blank) | line is blank; visual separation only. |
| 228 |         assert!(allow.is_allowed(&f)); | executable/support line; behavior impact depends on surrounding block
context. |
| 229 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 230 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/io/artifact.rs

Role: I/O and diagnostics stack: adapters, allowlist, flashing, replay, metrics, production phases.

Lines: 53 | Non-empty: 46 | Symbols: 2

Symbol Inventory

Line	Kind	Symbol	Change Impact
10	struct	FirmwareArtifactReport	data/structure coupling
18	fn	validate_firmware_artifact	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	use std::fs;	imports symbols; changing path/name affects compile-time resolution.
2	use std::path::Path;	imports symbols; changing path/name affects compile-time resolution.
3	(blank)	line is blank; visual separation only.
4	use serde::{Deserialize, Serialize};	imports symbols; changing path/name affects compile-time resolution.
5	use sha2::{Digest, Sha256};	imports symbols; changing path/name affects compile-time resolution.

```

| 6 | (blank) | line is blank; visual separation only. |
| 7 | use super::hw::HwError; | imports symbols; changing path/name affects compile-time resolution. |
| 8 | (blank) | line is blank; visual separation only. |
| 9 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report com
ibility and tooling. |
| 10 | pub struct FirmwareArtifactReport { | declares data layout; field changes may break serialization, tests, and
lers. |
| 11 |     pub path: String, | executable/support line; behavior impact depends on surrounding block context. |
| 12 |     pub bytes: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 13 |     pub crc32: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 14 |     pub sha256: String, | executable/support line; behavior impact depends on surrounding block context. |
| 15 |     pub valid: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 16 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 17 | (blank) | line is blank; visual separation only. |
| 18 | pub fn validate_firmware_artifact<P: AsRef<Path>>(<P> | declares executable behavior; signature/body changes alte
PI, control flow, and outputs. |
| 19 |     path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 20 |     max_bytes: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 21 | ) -> Result<FirmwareArtifactReport, HwError> { | executable/support line; behavior impact depends on surroundi
block context. |
| 22 |     let p = path.as_ref(); | executable/support line; behavior impact depends on surrounding block context. |
| 23 |     let data = fs::read(p) | executable/support line; behavior impact depends on surrounding block context. |
| 24 |     .map_err(|e| HwError::Unknown(format!("artifact read failed ({}): {}", p.display()))); | error-pro
ating operation; changing it impacts failure semantics and caller recovery paths. |
| 25 | (blank) | line is blank; visual separation only. |
| 26 |     if data.is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 27 |         return Err(HwError::Unknown("artifact payload is empty".to_string())); | executable/support line; beha
r impact depends on surrounding block context. |
| 28 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 29 |     if data.len() > max_bytes { | conditional gate; changing condition shifts runtime path and side effects. |
| 30 |         return Err(HwError::Unknown(format!( | executable/support line; behavior impact depends on surrounding
ock context. |
| 31 |             "artifact payload too large: {} bytes (limit {})", | executable/support line; behavior impact depe
on surrounding block context. |
| 32 |             data.len(), | executable/support line; behavior impact depends on surrounding block context. |
| 33 |             max_bytes | executable/support line; behavior impact depends on surrounding block context. |
| 34 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 35 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 36 | (blank) | line is blank; visual separation only. |
| 37 |     let crc32 = crc32fast::hash(&data); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 38 |     let mut hasher = Sha256::new(); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 39 |     hasher.update(&data); | executable/support line; behavior impact depends on surrounding block context. |
| 40 |     let sha = hasher.finalize(); | executable/support line; behavior impact depends on surrounding block conte
|
| 41 |     let mut sha_hex = String::with_capacity(64); | executable/support line; behavior impact depends on surroundi
g block context. |
| 42 |     for b in sha { | iteration control; changing bounds/body affects throughput, timing, and possibly safety g
antees. |
| 43 |         sha_hex.push_str(&format!("{b:02x}")); | executable/support line; behavior impact depends on surroundi
block context. |
| 44 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 45 | (blank) | line is blank; visual separation only. |
| 46 |     Ok(FirmwareArtifactReport { | executable/support line; behavior impact depends on surrounding block contex
|
| 47 |         path: p.display().to_string(), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 48 |         bytes: data.len(), | executable/support line; behavior impact depends on surrounding block context. |
| 49 |         crc32, | executable/support line; behavior impact depends on surrounding block context. |
| 50 |         sha256: sha_hex, | executable/support line; behavior impact depends on surrounding block context. |
| 51 |         valid: true, | executable/support line; behavior impact depends on surrounding block context. |
| 52 |     }) | executable/support line; behavior impact depends on surrounding block context. |
| 53 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/io/ecm_params.rs

Role: I/O and diagnostics stack: adapters, allowlist, flashing, replay, metrics, production phases.
Lines: 157 | Non-empty: 143 | Symbols: 11

Symbol Inventory

Line	Kind	Symbol	Change Impact
6	const	PGN_EEC1	data/structure coupling
7	const	PGN_ET1	data/structure coupling
8	const	PGN_EFL_P1	data/structure coupling
11	struct	EcmSnapshot	data/structure coupling
23	struct	DecodedParam	data/structure coupling
29	fn	decode_can_frame	signature/API coupling
93	fn	merge_snapshot	signature/API coupling
111	fn	j1939_pgn	signature/API coupling
125	module	tests	data/structure coupling
129	fn	decode_eec1_engine_speed	signature/API coupling
144	fn	decode_et1_coolant	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	use serde::{Deserialize, Serialize};	imports symbols; changing path/name affects compile-time resolution.
2	(blank)	line is blank; visual separation only.
3	use super::hw::CanFrame;	imports symbols; changing path/name affects compile-time resolution.
4	(blank)	line is blank; visual separation only.
5	// J1939 standard PGNs often available on heavy-duty ECM networks, including Caterpillar platforms.	comment L ; documentation intent, no runtime effect.
6	pub const PGN_EEC1: u32 = 61444;	executable/support line; behavior impact depends on surrounding block context.
7	pub const PGN_ET1: u32 = 65262;	executable/support line; behavior impact depends on surrounding block context.
8	pub const PGN_EFL_P1: u32 = 65263;	executable/support line; behavior impact depends on surrounding block context.
9	(blank)	line is blank; visual separation only.
10	#[derive(Debug, Clone, Serialize, Deserialize, Default, PartialEq)]	serialization contract touchpoint; edits can break report compatibility and tooling.
11	pub struct EcmSnapshot {	declares data layout; field changes may break serialization, tests, and callers.
12	pub engine_speed_rpm: Option<f64>,	executable/support line; behavior impact depends on surrounding block context.
13	pub accel_pedal_pct: Option<f64>,	executable/support line; behavior impact depends on surrounding block context.
14	pub coolant_temp_c: Option<f64>,	executable/support line; behavior impact depends on surrounding block context.
15	pub oil_pressure_kpa: Option<f64>,	executable/support line; behavior impact depends on surrounding block context.
16	pub fuel_temp_c: Option<f64>,	executable/support line; behavior impact depends on surrounding block context.
17	pub last_seen_pgn: Option<u32>,	executable/support line; behavior impact depends on surrounding block context.
18	pub source_address: Option<u8>,	executable/support line; behavior impact depends on surrounding block context.
19	pub timestamp_ms: Option<u64>,	executable/support line; behavior impact depends on surrounding block context.
20	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
21	(blank)	line is blank; visual separation only.
22	#[derive(Debug, Clone, Serialize, Deserialize, PartialEq)]	serialization contract touchpoint; edits can break report compatibility and tooling.
23	pub struct DecodedParam {	declares data layout; field changes may break serialization, tests, and callers.
24	pub key: &'static str,	executable/support line; behavior impact depends on surrounding block context.
25	pub value: f64,	executable/support line; behavior impact depends on surrounding block context.
26	pub unit: &'static str,	executable/support line; behavior impact depends on surrounding block context.
27	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.


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| 28 | (blank) | line is blank; visual separation only. |
| 29 | pub fn decode_can_frame(frame: &CanFrame) -> Vec<DecodedParam> { | declares executable behavior; signature/body
changes alter API, control flow, and outputs. |
| 30 |     let pgn = j1939_pgn(frame.id); | executable/support line; behavior impact depends on surrounding block context. |
| 31 |     let d = &frame.data; | executable/support line; behavior impact depends on surrounding block context. |
| 32 | (blank) | line is blank; visual separation only. |
| 33 |     match pgn { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 34 |         PGN_EEC1 => { | executable/support line; behavior impact depends on surrounding block context. |
| 35 |             // SPN 190 Engine Speed: bytes 4-5, little-endian, 0.125 rpm/bit. | comment line; documentation intent, no runtime effect. |
| 36 |             // SPN 91 Accelerator Pedal Position 1: byte 2, 0.4 %/bit. | comment line; documentation intent, no runtime effect. |
| 37 |             let eng_raw = u16::from_le_bytes([d[3], d[4]]); | executable/support line; behavior impact depends on surrounding block context. |
| 38 |             let app1_raw = d[1]; | executable/support line; behavior impact depends on surrounding block context. |
| 39 |             let mut out = Vec::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 40 |             if eng_raw != 0xFFFF { | conditional gate; changing condition shifts runtime path and side effects. |
| 41 |                 out.push(DecodedParam { | executable/support line; behavior impact depends on surrounding block context. |
| 42 |                     key: "engine_speed_rpm", | executable/support line; behavior impact depends on surrounding block context. |
| 43 |                     value: f64::from(eng_raw) * 0.125, | executable/support line; behavior impact depends on surrounding block context. |
| 44 |                     unit: "rpm", | executable/support line; behavior impact depends on surrounding block context. |
| 45 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 46 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 47 |             if app1_raw != 0xFF { | conditional gate; changing condition shifts runtime path and side effects. |
| 48 |                 out.push(DecodedParam { | executable/support line; behavior impact depends on surrounding block context. |
| 49 |                     key: "accel_pedal_pct", | executable/support line; behavior impact depends on surrounding block context. |
| 50 |                     value: f64::from(app1_raw) * 0.4, | executable/support line; behavior impact depends on surrounding block context. |
| 51 |                     unit: "%", | executable/support line; behavior impact depends on surrounding block context. |
| 52 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 53 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 54 |             out | executable/support line; behavior impact depends on surrounding block context. |
| 55 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 56 |         PGN_ET1 => { | executable/support line; behavior impact depends on surrounding block context. |
| 57 |             // SPN 110 Coolant Temp: byte 1, 1 C/bit, offset -40 C. | comment line; documentation intent, no runtime effect. |
| 58 |             // SPN 174 Fuel Temp 1: byte 2, 1 C/bit, offset -40 C. | comment line; documentation intent, no runtime effect. |
| 59 |             let mut out = Vec::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 60 |             if d[0] != 0xFF { | conditional gate; changing condition shifts runtime path and side effects. |
| 61 |                 out.push(DecodedParam { | executable/support line; behavior impact depends on surrounding block context. |
| 62 |                     key: "coolant_temp_c", | executable/support line; behavior impact depends on surrounding block context. |
| 63 |                     value: f64::from(d[0]) - 40.0, | executable/support line; behavior impact depends on surrounding block context. |
| 64 |                     unit: "C", | executable/support line; behavior impact depends on surrounding block context. |
| 65 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 66 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 67 |             if d[1] != 0xFF { | conditional gate; changing condition shifts runtime path and side effects. |
| 68 |                 out.push(DecodedParam { | executable/support line; behavior impact depends on surrounding block context. |

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```

| 69 |           key: "fuel_temp_c", | executable/support line; behavior impact depends on surrounding block
context. |
| 70 |           value: f64::from(d[1]) - 40.0, | executable/support line; behavior impact depends on surro
ing block context. |
| 71 |           unit: "C", | executable/support line; behavior impact depends on surrounding block context
| 72 |       }); | executable/support line; behavior impact depends on surrounding block context. |
| 73 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime sco
s. |
| 74 |   out | executable/support line; behavior impact depends on surrounding block context. |
| 75 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 76 | PGN_EFL_P1 => { | executable/support line; behavior impact depends on surrounding block context. |
| 77 |   // SPN 100 Engine Oil Pressure: byte 4, 4 kPa/bit. | comment line; documentation intent, no runtime
ffect. |
| 78 |   let oil_raw = d[3]; | executable/support line; behavior impact depends on surrounding block context
|
| 79 |   if oil_raw != 0xFF { | conditional gate; changing condition shifts runtime path and side effects.
| 80 |     vec![DecodedParam { | executable/support line; behavior impact depends on surrounding block co
xt. |
| 81 |       key: "oil_pressure_kpa", | executable/support line; behavior impact depends on surrounding
ock context. |
| 82 |       value: f64::from(oil_raw) * 4.0, | executable/support line; behavior impact depends on sur
nding block context. |
| 83 |       unit: "kPa", | executable/support line; behavior impact depends on surrounding block conte
|
| 84 |       }); | executable/support line; behavior impact depends on surrounding block context. |
| 85 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 86 |       Vec::new() | executable/support line; behavior impact depends on surrounding block context. |
| 87 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime sco
s. |
| 88 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 89 |   _ => Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 90 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 91 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 92 | (blank) | line is blank; visual separation only. |
| 93 | pub fn merge_snapshot(mut snap: EcmSnapshot, frame: &CanFrame) -> EcmSnapshot { | declares executable behavior
signature/body changes alter API, control flow, and outputs. |
| 94 |   let pgn = j1939_pgn(frame.id); | executable/support line; behavior impact depends on surrounding block con
t. |
| 95 |   for p in decode_can_frame(frame) { | iteration control; changing bounds/body affects throughput, timing, a
possibly safety guarantees. |
| 96 |     match p.key { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 97 |       "engine_speed_rpm" => snap.engine_speed_rpm = Some(p.value), | executable/support line; behavior i
ct depends on surrounding block context. |
| 98 |       "accel_pedal_pct" => snap.accel_pedal_pct = Some(p.value), | executable/support line; behavior imp
depends on surrounding block context. |
| 99 |       "coolant_temp_c" => snap.coolant_temp_c = Some(p.value), | executable/support line; behavior impac
depends on surrounding block context. |
| 100 |       "fuel_temp_c" => snap.fuel_temp_c = Some(p.value), | executable/support line; behavior impact dep
s on surrounding block context. |
| 101 |       "oil_pressure_kpa" => snap.oil_pressure_kpa = Some(p.value), | executable/support line; behavior
act depends on surrounding block context. |
| 102 |       _ => {} | executable/support line; behavior impact depends on surrounding block context. |
| 103 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 104 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 105 |   snap.last_seen_pgn = Some(pgn); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 106 |   snap.source_address = Some((frame.id & 0xFF) as u8); | executable/support line; behavior impact depends o
urrounding block context. |
| 107 |   snap.timestamp_ms = frame.timestamp_ms; | executable/support line; behavior impact depends on surrounding
ock context. |
| 108 |   snap | executable/support line; behavior impact depends on surrounding block context. |
| 109 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 110 | (blank) | line is blank; visual separation only. |
| 111 | pub fn j1939_pgn(can_id: u32) -> u32 { | declares executable behavior; signature/body changes alter API, cont

```

```

flow, and outputs. |
| 112 |      // 29-bit CAN ID for J1939: priority(3), reserved(1), dp(1), pf(8), ps(8), sa(8). | comment line; document
ion intent, no runtime effect. |
| 113 |      let pf = ((can_id >> 16) & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding
ock context. |
| 114 |      let ps = ((can_id >> 8) & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding
ck context. |
| 115 |      let dp = (can_id >> 24) & 0x01 ; | executable/support line; behavior impact depends on surrounding block
text. |
| 116 | (blank) | line is blank; visual separation only. |
| 117 |      if pf < 240 { | conditional gate; changing condition shifts runtime path and side effects. |
| 118 |          (dp << 16) \ | ((pf as u32) << 8) | executable/support line; behavior impact depends on surrounding bl
context. |
| 119 |      } else { | executable/support line; behavior impact depends on surrounding block context. |
| 120 |          (dp << 16) \ | ((pf as u32) << 8) \ | ps as u32 | executable/support line; behavior impact depends on s
ounding block context. |
| 121 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 122 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 123 | (blank) | line is blank; visual separation only. |
| 124 | #[cfg(test)] | executable/support line; behavior impact depends on surrounding block context. |
| 125 | mod tests { | declares a module boundary; changing module name/path breaks references. |
| 126 |     use super::*; | imports symbols; changing path/name affects compile-time resolution. |
| 127 | (blank) | line is blank; visual separation only. |
| 128 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 129 |     fn decode_eec1_engine_speed() { | declares executable behavior; signature/body changes alter API, control
ow, and outputs. |
| 130 |         let frame = CanFrame { | executable/support line; behavior impact depends on surrounding block context
|
| 131 |             id: 0x0CF00400, // PGN 61444 | executable/support line; behavior impact depends on surrounding bl
context. |
| 132 |             dlc: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 133 |             data: [0x00, 50, 0x00, 0x40, 0x1F, 0, 0, 0], | executable/support line; behavior impact depends o
urrounding block context. |
| 134 |             len: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 135 |             timestamp_ms: Some(1), | executable/support line; behavior impact depends on surrounding block co
xt. |
| 136 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 137 | (blank) | line is blank; visual separation only. |
| 138 |         let vals = decode_can_frame(&frame); | executable/support line; behavior impact depends on surroundin
lock context. |
| 139 |         assert!(vals.iter().any(\ | v \ | v.key == "engine_speed_rpm")); | executable/support line; behavior impa
depends on surrounding block context. |
| 140 |         assert!(vals.iter().any(\ | v \ | v.key == "accel_pedal_pct")); | executable/support line; behavior impac
depends on surrounding block context. |
| 141 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 142 | (blank) | line is blank; visual separation only. |
| 143 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 144 |     fn decode_et1_coolant() { | declares executable behavior; signature/body changes alter API, control flow,
d outputs. |
| 145 |         let frame = CanFrame { | executable/support line; behavior impact depends on surrounding block context
|
| 146 |             id: 0x18FEEE00, // PGN 65262 | executable/support line; behavior impact depends on surrounding bl
context. |
| 147 |             dlc: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 148 |             data: [90, 100, 0, 0, 0, 0, 0, 0], | executable/support line; behavior impact depends on surround
block context. |
| 149 |             len: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 150 |             timestamp_ms: Some(2), | executable/support line; behavior impact depends on surrounding block co
xt. |
| 151 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 152 | (blank) | line is blank; visual separation only. |
| 153 |         let vals = decode_can_frame(&frame); | executable/support line; behavior impact depends on surroundin
lock context. |

```

```
| 154 |      assert!(vals.iter().any(\|v\| v.key == "coolant_temp_c")); | executable/support line; behavior impact
depends on surrounding block context. |
| 155 |      assert!(vals.iter().any(\|v\| v.key == "fuel_temp_c")); | executable/support line; behavior impact de
ds on surrounding block context. |
| 156 |    } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 157 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
```

File: src/io/hw.rs

Role: I/O and diagnostics stack: adapters, allowlist, flashing, replay, metrics, production phases.
Lines: 405 | Non-empty: 372 | Symbols: 29

Symbol Inventory

Line	Kind	Symbol	Change Impact
12	enum	HwMode	data/structure coupling
18	struct	CanFrame	data/structure coupling
27	struct	SerialFrame	data/structure coupling
34	enum	Frame	data/structure coupling
40	struct	HwCapabilities	data/structure coupling
48	impl	HwCapabilities	behavior binding and method semantics
49	fn	missing_for_live_flash	signature/API coupling
62	struct	HwConfig	data/structure coupling
99	impl	Default for HwConfig	behavior binding and method semantics
100	fn	default	signature/API coupling
140	impl	HwConfig	behavior binding and method semantics
141	fn	from_cli_args	signature/API coupling
223	fn	write_intent_enabled	signature/API coupling
227	fn	write_effectively_enabled	signature/API coupling
231	fn	declared_capabilities	signature/API coupling
253	fn	capabilities_summary	signature/API coupling
263	enum	HwError	data/structure coupling
286	trait	HardwareInterface	signature/API coupling
287	fn	init	signature/API coupling
288	fn	read_frame	signature/API coupling
289	fn	try_read_frame	signature/API coupling
290	fn	send_frame	signature/API coupling
291	fn	close	signature/API coupling
293	fn	adapter_info	signature/API coupling
298	fn	open_real_adapter	signature/API coupling
324	fn	probe_live_adapter	signature/API coupling
335	struct	StartupAudit	data/structure coupling
349	fn	write_listen_only_startup_log	signature/API coupling
400	fn	now_ms	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	use std::fs::{self, File, OpenOptions};	imports symbols; changing path/name affects compile-time resolution.
2	use std::io::Write;	imports symbols; changing path/name affects compile-time resolution.
3	use std::path::PathBuf;	imports symbols; changing path/name affects compile-time resolution.
4	use std::time::{SystemTime, UNIX_EPOCH};	imports symbols; changing path/name affects compile-time resolution.
5	(blank)	line is blank; visual separation only.
6	use flate2::write::GzEncoder;	imports symbols; changing path/name affects compile-time resolution.
7	use flate2::Compression;	imports symbols; changing path/name affects compile-time resolution.
8	use serde::{Deserialize, Serialize};	imports symbols; changing path/name affects compile-time resolution.
9	use thiserror::Error;	imports symbols; changing path/name affects compile-time resolution.
10	(blank)	line is blank; visual separation only.
11	#[derive(Debug, Clone, Copy, PartialEq, Eq, Serialize, Deserialize)]	serialization contract touchpoint; editing an break report compatibility and tooling.
12	pub enum HwMode {	declares state variants; adding/removing variants impacts pattern matches and logic branches
13	Sim,	executable/support line; behavior impact depends on surrounding block context.

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| 14 |     Live, | executable/support line; behavior impact depends on surrounding block context. |
| 15 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 16 | (blank) | line is blank; visual separation only. |
| 17 | #[derive(Debug, Clone, PartialEq, Eq, Serialize, Deserialize)] | serialization contract touchpoint; edits can
ak report compatibility and tooling. |
| 18 | pub struct CanFrame { | declares data layout; field changes may break serialization, tests, and callers. |
| 19 |     pub id: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 20 |     pub dlc: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 21 |     pub data: [u8; 8], | executable/support line; behavior impact depends on surrounding block context. |
| 22 |     pub len: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 23 |     pub timestamp_ms: Option<u64>, | executable/support line; behavior impact depends on surrounding block con
t. |
| 24 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 25 | (blank) | line is blank; visual separation only. |
| 26 | #[derive(Debug, Clone, PartialEq, Eq, Serialize, Deserialize)] | serialization contract touchpoint; edits can
ak report compatibility and tooling. |
| 27 | pub struct SerialFrame { | declares data layout; field changes may break serialization, tests, and callers. |
| 28 |     pub bytes: Vec<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 29 |     pub protocol_hint: Option<String>, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 30 |     pub timestamp_ms: Option<u64>, | executable/support line; behavior impact depends on surrounding block con
t. |
| 31 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 32 | (blank) | line is blank; visual separation only. |
| 33 | #[derive(Debug, Clone, PartialEq, Eq, Serialize, Deserialize)] | serialization contract touchpoint; edits can
ak report compatibility and tooling. |
| 34 | pub enum Frame { | declares state variants; adding/removing variants impacts pattern matches and logic branche
|
| 35 |     Can(CanFrame), | executable/support line; behavior impact depends on surrounding block context. |
| 36 |     Serial(SerialFrame), | executable/support line; behavior impact depends on surrounding block context. |
| 37 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 38 | (blank) | line is blank; visual separation only. |
| 39 | #[derive(Debug, Clone, PartialEq, Eq, Serialize, Deserialize, Default)] | serialization contract touchpoint; e
s can break report compatibility and tooling. |
| 40 | pub struct HwCapabilities { | declares data layout; field changes may break serialization, tests, and callers.
| 41 |     pub can_raw: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 42 |     pub isotp: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 43 |     pub j1939: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 44 |     pub uds_flash: bool, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnosti
utcomes. |
| 45 |     pub vendor_bridge: bool, | executable/support line; behavior impact depends on surrounding block context.
| 46 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 47 | (blank) | line is blank; visual separation only. |
| 48 | impl HwCapabilities { | implementation block; method behavior and trait conformance are defined here. |
| 49 |     pub fn missing_for_live_flash(&self) -> Vec<&'static str> { | declares executable behavior; signature/body
anges alter API, control flow, and outputs. |
| 50 |         let mut missing = Vec::new(); | executable/support line; behavior impact depends on surrounding block
text. |
| 51 |         if !self.uds_flash { | conditional gate; changing condition shifts runtime path and side effects. |
| 52 |             missing.push("uds_flash"); | I/O or protocol-critical logic; edits can affect integration, timing,
d diagnostic outcomes. |
| 53 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 54 |         if self.can_raw && !(self.isotp || self.vendor_bridge) { | conditional gate; changing condition shif
runtime path and side effects. |
| 55 |             missing.push("isotp_or_vendor_bridge"); | executable/support line; behavior impact depends on surr
ding block context. |
| 56 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 57 |         missing | executable/support line; behavior impact depends on surrounding block context. |
| 58 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 59 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 60 | (blank) | line is blank; visual separation only. |
| 61 | #[derive(Debug, Clone, PartialEq, Eq, Serialize, Deserialize)] | serialization contract touchpoint; edits can
ak report compatibility and tooling. |
| 62 | pub struct HwConfig { | declares data layout; field changes may break serialization, tests, and callers. |
| 63 |     pub mode: HwMode, | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 64 |     pub vendor_name: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 65 |     pub vendor_template_dir: Option<PathBuf>, | executable/support line; behavior impact depends on surrounding block context. |
| 66 |     pub vendor_bridge_exe: Option<PathBuf>, | executable/support line; behavior impact depends on surrounding block context. |
| 67 |     pub vendor_bridge_timeout_ms: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 68 |     pub serial_port: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 69 |     pub serial_baud: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 70 |     pub serial_parity: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 71 |     pub serial_stopbits: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 72 |     pub can_interface: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 73 |     pub ethernet_probe: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 74 |     pub enable_write: bool, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 75 |     pub allowlist_path: Option<PathBuf>, | executable/support line; behavior impact depends on surrounding block context. |
| 76 |     pub rate_limit_global_per_sec: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 77 |     pub rate_limit_per_id_per_sec: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 78 |     pub write_retry_count: u8, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 79 |     pub write_retry_backoff_ms: u64, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 80 |     pub reconnect_backoff_base_ms: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 81 |     pub reconnect_backoff_max_ms: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 82 |     pub uds_retry_count: u8, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 83 |     pub uds_timeout_p2_ms: u64, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 84 |     pub uds_timeout_p2star_ms: u64, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 85 |     pub uds_st_min_ms: u64, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 86 |     pub uds_block_size: u8, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 87 |     pub j1939_idle_guard_ms: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 88 |     pub keep_channels_alive: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 89 |     pub parser_max_frame_size: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 90 |     pub dry_run: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 91 |     pub use_isolation_hw: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 92 |     pub log_dir: PathBuf, | executable/support line; behavior impact depends on surrounding block context. |
| 93 |     pub metrics_enabled: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 94 |     pub emergency_stop_hook: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 95 |     pub allow_untrusted_ecu: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 96 |     pub noninteractive_approved: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 97 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 98 | (blank) | line is blank; visual separation only. |
| 99 | impl Default for HwConfig { | implementation block; method behavior and trait conformance are defined here. |
| 100 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and

```

```

utputs. |
| 101 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 102 |             mode: HwMode::Sim, | executable/support line; behavior impact depends on surrounding block context. |
|
| 103 |             vendor_name: None, | executable/support line; behavior impact depends on surrounding block context. |
|
| 104 |             vendor_template_dir: None, | executable/support line; behavior impact depends on surrounding block context. |
| 105 |             vendor_bridge_exe: None, | executable/support line; behavior impact depends on surrounding block context. |
| 106 |             vendor_bridge_timeout_ms: 1_000, | executable/support line; behavior impact depends on surrounding block context. |
| 107 |             serial_port: None, | executable/support line; behavior impact depends on surrounding block context. |
|
| 108 |             serial_baud: 115_200, | executable/support line; behavior impact depends on surrounding block context. |
| 109 |             serial_parity: Some("None".to_string()), | executable/support line; behavior impact depends on surrounding block context. |
| 110 |             serial_stopbits: 1, | executable/support line; behavior impact depends on surrounding block context. |
|
| 111 |             can_interface: None, | executable/support line; behavior impact depends on surrounding block context. |
|
| 112 |             ethernet_probe: None, | executable/support line; behavior impact depends on surrounding block context. |
| 113 |             enable_write: false, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 114 |             allowlist_path: None, | executable/support line; behavior impact depends on surrounding block context. |
| 115 |             rate_limit_global_per_sec: 100, | executable/support line; behavior impact depends on surrounding block context. |
| 116 |             rate_limit_per_id_per_sec: 10, | executable/support line; behavior impact depends on surrounding block context. |
| 117 |             write_retry_count: 3, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 118 |             write_retry_backoff_ms: 200, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 119 |             reconnect_backoff_base_ms: 500, | executable/support line; behavior impact depends on surrounding block context. |
| 120 |             reconnect_backoff_max_ms: 30_000, | executable/support line; behavior impact depends on surrounding block context. |
| 121 |             uds_retry_count: 3, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 122 |             uds_timeout_p2_ms: 1_000, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 123 |             uds_timeout_p2star_ms: 5_000, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 124 |             uds_st_min_ms: 5, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 125 |             uds_block_size: 0, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 126 |             j1939_idle_guard_ms: 2_500, | executable/support line; behavior impact depends on surrounding block context. |
| 127 |             keep_channels_alive: true, | executable/support line; behavior impact depends on surrounding block context. |
| 128 |             parser_max_frame_size: 4096, | executable/support line; behavior impact depends on surrounding block context. |
| 129 |             dry_run: true, | executable/support line; behavior impact depends on surrounding block context. |
| 130 |             use_isolation_hw: false, | executable/support line; behavior impact depends on surrounding block context. |
| 131 |             log_dir: PathBuf::from("./artifacts/hw_logs"), | executable/support line; behavior impact depends on surrounding block context. |
| 132 |             metrics_enabled: false, | executable/support line; behavior impact depends on surrounding block context. |
| 133 |             emergency_stop_hook: None, | executable/support line; behavior impact depends on surrounding block context. |

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| 134 |           allow_untrusted_ecu: false, | executable/support line; behavior impact depends on surrounding block
context. |
| 135 |           noninteractive_approved: false, | executable/support line; behavior impact depends on surrounding
block context. |
| 136 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 137 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 138 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 139 | (blank) | line is blank; visual separation only. |
| 140 | impl HwConfig { | implementation block; method behavior and trait conformance are defined here. |
| 141 |     pub fn from_cli_args(args: &[String]) -> Result<Self, HwError> { | declares executable behavior; signature
body changes alter API, control flow, and outputs. |
| 142 |         let mut cfg = HwConfig::default(); | executable/support line; behavior impact depends on surrounding
block context. |
| 143 |         for a in args { | iteration control; changing bounds/body affects throughput, timing, and possibly sa
y guarantees. |
| 144 |             if let Some(v) = a.strip_prefix("--hw-mode=") { | conditional gate; changing condition shifts run
e path and side effects. |
| 145 |                 cfg.mode = match v.to_ascii_lowercase().as_str() { | executable/support line; behavior impact
depends on surrounding block context. |
| 146 |                     "sim" => HwMode::Sim, | executable/support line; behavior impact depends on surrounding b
k context. |
| 147 |                     "live" => HwMode::Live, | executable/support line; behavior impact depends on surrounding
block context. |
| 148 |                     _ => return Err(HwError::Unknown(format!("invalid --hw-mode value: {v}")), | executable/
port line; behavior impact depends on surrounding block context. |
| 149 |                 }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 150 |             } else if a == "--dry-run" { | executable/support line; behavior impact depends on surrounding bl
context. |
| 151 |                 cfg.dry_run = true; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 152 |             } else if a == "--dry-run=false" { | executable/support line; behavior impact depends on surround
block context. |
| 153 |                 cfg.dry_run = false; | executable/support line; behavior impact depends on surrounding block
text. |
| 154 |             } else if a == "--enable-write" { | I/O or protocol-critical logic; edits can affect integration,
ming, and diagnostic outcomes. |
| 155 |                 cfg.enable_write = true; | I/O or protocol-critical logic; edits can affect integration, timi
and diagnostic outcomes. |
| 156 |             } else if a == "--noninteractive-approved" { | executable/support line; behavior impact depends o
urrounding block context. |
| 157 |                 cfg.noninteractive_approved = true; | executable/support line; behavior impact depends on sur
nding block context. |
| 158 |             } else if let Some(v) = a.strip_prefix("--allowlist=") { | executable/support line; behavior impa
depends on surrounding block context. |
| 159 |                 cfg.allowlist_path = Some(PathBuf::from(v)); | executable/support line; behavior impact depen
on surrounding block context. |
| 160 |             } else if let Some(v) = a.strip_prefix("--log-dir=") { | executable/support line; behavior impact
depends on surrounding block context. |
| 161 |                 cfg.log_dir = PathBuf::from(v); | executable/support line; behavior impact depends on surroun
g block context. |
| 162 |             } else if let Some(v) = a.strip_prefix("--serial-port=") { | executable/support line; behavior im
t depends on surrounding block context. |
| 163 |                 cfg.serial_port = Some(v.to_string()); | executable/support line; behavior impact depends on
rounding block context. |
| 164 |             } else if let Some(v) = a.strip_prefix("--vendor-name=") { | executable/support line; behavior im
t depends on surrounding block context. |
| 165 |                 cfg.vendor_name = Some(v.to_string()); | executable/support line; behavior impact depends on
rounding block context. |
| 166 |             } else if let Some(v) = a.strip_prefix("--vendor-template-dir=") { | executable/support line; beh
or impact depends on surrounding block context. |
| 167 |                 cfg.vendor_template_dir = Some(PathBuf::from(v)); | executable/support line; behavior impact
ends on surrounding block context. |
| 168 |             } else if let Some(v) = a.strip_prefix("--vendor-bridge-exe=") { | executable/support line; behav

```



```

    impact depends on surrounding block context. |
| 169 |         cfg.vendor_bridge_exe = Some(PathBuf::from(v)); | executable/support line; behavior impact de
ds on surrounding block context. |
| 170 |     } else if let Some(v) = a.strip_prefix("--vendor-bridge-timeout-ms=") { | executable/support line
ehavior impact depends on surrounding block context. |
| 171 |         cfg.vendor_bridge_timeout_ms = v.parse::

```

```

| 201 |           }); | error-propagating operation; changing it impacts failure semantics and caller recovery
ths. |
| 202 |           } else if let Some(v) = a.strip_prefix("--uds-st-min-ms=") { | I/O or protocol-critical logic; ed
can affect integration, timing, and diagnostic outcomes. |
| 203 |           cfg.uds_st_min_ms = v.parse::<u64>().map_err(|e| { | I/O or protocol-critical logic; edits
affect integration, timing, and diagnostic outcomes. |
| 204 |           HwError::Unknown(format!("invalid --uds-st-min-ms value: {e}")) | I/O or protocol-critical
logic; edits can affect integration, timing, and diagnostic outcomes. |
| 205 |           }); | error-propagating operation; changing it impacts failure semantics and caller recovery
ths. |
| 206 |           } else if let Some(v) = a.strip_prefix("--uds-block-size=") { | I/O or protocol-critical logic; e
s can affect integration, timing, and diagnostic outcomes. |
| 207 |           cfg.uds_block_size = v.parse::<u8>().map_err(|e| { | I/O or protocol-critical logic; edits
affect integration, timing, and diagnostic outcomes. |
| 208 |           HwError::Unknown(format!("invalid --uds-block-size value: {e}")) | I/O or protocol-critical
logic; edits can affect integration, timing, and diagnostic outcomes. |
| 209 |           }); | error-propagating operation; changing it impacts failure semantics and caller recovery
ths. |
| 210 |           } else if let Some(v) = a.strip_prefix("--j1939-idle-guard-ms=") { | executable/support line; beh
or impact depends on surrounding block context. |
| 211 |           cfg.j1939_idle_guard_ms = v.parse::<u64>().map_err(|e| { | executable/support line; behavio
mpact depends on surrounding block context. |
| 212 |           HwError::Unknown(format!("invalid --j1939-idle-guard-ms value: {e}")) | executable/support
ine; behavior impact depends on surrounding block context. |
| 213 |           }); | error-propagating operation; changing it impacts failure semantics and caller recovery
ths. |
| 214 |           } else if a == "--keep-channels-alive" { | executable/support line; behavior impact depends on su
unding block context. |
| 215 |           cfg.keep_channels_alive = true; | executable/support line; behavior impact depends on surround
g block context. |
| 216 |           } else if a == "--keep-channels-alive=false" { | executable/support line; behavior impact depends
surrounding block context. |
| 217 |           cfg.keep_channels_alive = false; | executable/support line; behavior impact depends on surround
ng block context. |
| 218 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 219 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 220 |           Ok(cfg) | executable/support line; behavior impact depends on surrounding block context. |
| 221 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 222 | (blank) | line is blank; visual separation only. |
| 223 |     pub fn write_intent_enabled(&self) -> bool { | declares executable behavior; signature/body changes alter
I, control flow, and outputs. |
| 224 |         self.enable_write && self.allowlist_path.is_some() && self.noninteractive_approved | I/O or protocol-
tical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 225 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 226 | (blank) | line is blank; visual separation only. |
| 227 |     pub fn write_effectively_enabled(&self) -> bool { | declares executable behavior; signature/body changes
er API, control flow, and outputs. |
| 228 |         self.write_intent_enabled() && !self.dry_run | I/O or protocol-critical logic; edits can affect integ
ion, timing, and diagnostic outcomes. |
| 229 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 230 | (blank) | line is blank; visual separation only. |
| 231 |     pub fn declared_capabilities(&self) -> HwCapabilities { | declares executable behavior; signature/body ch
es alter API, control flow, and outputs. |
| 232 |         let vendor_bridge = self | executable/support line; behavior impact depends on surrounding block cont
. |
| 233 |         .vendor_name | executable/support line; behavior impact depends on surrounding block context. |
| 234 |         .as_deref() | executable/support line; behavior impact depends on surrounding block context. |
| 235 |         .map(|v| v.eq_ignore_ascii_case("cat_comm")) | executable/support line; behavior impact depends
surrounding block context. |
| 236 |         .unwrap_or(false) | executable/support line; behavior impact depends on surrounding block context
|
| 237 |         || self.vendor_template_dir.is_some() | executable/support line; behavior impact depends on surr
nding block context. |
| 238 |         || self.vendor_bridge_exe.is_some(); | executable/support line; behavior impact depends on surr

```

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ding block context. |
| 239 |         let can_raw = self.can_interface.is_some(); | executable/support line; behavior impact depends on surr
nding block context. |
| 240 |         let isotp = can_raw \\| vendor_bridge; | executable/support line; behavior impact depends on surroun
g block context. |
| 241 |         let j1939 = can_raw; | executable/support line; behavior impact depends on surrounding block context.
| 242 |         let uds_flash = can_raw \\| self.serial_port.is_some() \\| vendor_bridge; | I/O or protocol-critical
logic; edits can affect integration, timing, and diagnostic outcomes. |
| 243 | (blank) | line is blank; visual separation only. |
| 244 |         HwCapabilities { | executable/support line; behavior impact depends on surrounding block context. |
| 245 |             can_raw, | executable/support line; behavior impact depends on surrounding block context. |
| 246 |             isotp, | executable/support line; behavior impact depends on surrounding block context. |
| 247 |             j1939, | executable/support line; behavior impact depends on surrounding block context. |
| 248 |             uds_flash, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic
tcomes. |
| 249 |             vendor_bridge, | executable/support line; behavior impact depends on surrounding block context. |
| 250 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 251 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 252 | (blank) | line is blank; visual separation only. |
| 253 |     pub fn capabilities_summary(&self) -> String { | declares executable behavior; signature/body changes alt
API, control flow, and outputs. |
| 254 |         let c = self.declared_capabilities(); | executable/support line; behavior impact depends on surroundi
block context. |
| 255 |         format!( | executable/support line; behavior impact depends on surrounding block context. |
| 256 |             "can_raw={} isotp={} j1939={} uds_flash={} vendor_bridge={}", | I/O or protocol-critical logic; e
s can affect integration, timing, and diagnostic outcomes. |
| 257 |             c.can_raw, c.isotp, c.j1939, c.uds_flash, c.vendor_bridge | I/O or protocol-critical logic; edits
n affect integration, timing, and diagnostic outcomes. |
| 258 |         ) | executable/support line; behavior impact depends on surrounding block context. |
| 259 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 260 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 261 | (blank) | line is blank; visual separation only. |
| 262 | #[derive(Debug, Error)] | executable/support line; behavior impact depends on surrounding block context. |
| 263 | pub enum HwError { | declares state variants; adding/removing variants impacts pattern matches and logic bran
s. |
| 264 |     #[error("port not found: {0}")] | executable/support line; behavior impact depends on surrounding block c
ext. |
| 265 |     PortNotFound(String), | executable/support line; behavior impact depends on surrounding block context. |
| 266 |     #[error("permission denied: {0}")] | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 267 |     PermissionDenied(String), | executable/support line; behavior impact depends on surrounding block context
| 268 |     #[error("parse error: {cause}; raw_data={raw_data}")] | executable/support line; behavior impact depends
surrounding block context. |
| 269 |     ParseError { cause: String, raw_data: String }, | executable/support line; behavior impact depends on surr
nding block context. |
| 270 |     #[error("write blocked by allowlist")] | I/O or protocol-critical logic; edits can affect integration, ti
g, and diagnostic outcomes. |
| 271 |     WriteBlockedAllowlist, | executable/support line; behavior impact depends on surrounding block context. |
| 272 |     #[error("rate limited")] | executable/support line; behavior impact depends on surrounding block context.
| 273 |     RateLimited, | executable/support line; behavior impact depends on surrounding block context. |
| 274 |     #[error("bus off")] | executable/support line; behavior impact depends on surrounding block context. |
| 275 |     BusOff, | executable/support line; behavior impact depends on surrounding block context. |
| 276 |     #[error("transceiver error")] | executable/support line; behavior impact depends on surrounding block con
t. |
| 277 |     TransceiverError, | executable/support line; behavior impact depends on surrounding block context. |
| 278 |     #[error("checksum error")] | executable/support line; behavior impact depends on surrounding block context
|
| 279 |     ChecksumError, | executable/support line; behavior impact depends on surrounding block context. |
| 280 |     #[error("timeout")] | executable/support line; behavior impact depends on surrounding block context. |
| 281 |     Timeout, | executable/support line; behavior impact depends on surrounding block context. |
| 282 |     #[error("unknown: {0}")] | executable/support line; behavior impact depends on surrounding block context.
| 283 |     Unknown(String), | executable/support line; behavior impact depends on surrounding block context. |
| 284 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 285 | (blank) | line is blank; visual separation only. |

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```

| 286 | pub trait HardwareInterface { | defines interface contract; signature changes impact all implementers and call
ites. |
| 287 |     fn init(&mut self, config: &HwConfig) -> Result<(), HwError>; | declares executable behavior; signature/body
changes alter API, control flow, and outputs. |
| 288 |     fn read_frame(&mut self) -> Result<Frame, HwError>; | declares executable behavior; signature/body change
alter API, control flow, and outputs. |
| 289 |     fn try_read_frame(&mut self) -> Result<Option<Frame>, HwError>; | declares executable behavior; signature/b
dy changes alter API, control flow, and outputs. |
| 290 |     fn send_frame(&mut self, frame: Frame) -> Result<(), HwError>; | declares executable behavior; signature/b
y changes alter API, control flow, and outputs. |
| 291 |     fn close(&mut self) -> Result<(), HwError>; | declares executable behavior; signature/body changes alter
, control flow, and outputs. |
| 292 | (blank) | line is blank; visual separation only. |
| 293 |     fn adapter_info(&self) -> Option<String> { | declares executable behavior; signature/body changes alter A
control flow, and outputs. |
| 294 |         None | executable/support line; behavior impact depends on surrounding block context. |
| 295 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 296 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 297 | (blank) | line is blank; visual separation only. |
| 298 | pub fn open_real_adapter(cfg: &HwConfig) -> Result<Box<dyn HardwareInterface>, HwError> { | declares executab
behavior; signature/body changes alter API, control flow, and outputs. |
| 299 |     #[cfg(all(target_os = "windows", feature = "vendor-windows"))] | executable/support line; behavior impact
depends on surrounding block context. |
| 300 |     { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 301 |         if let Some(vendor) = &cfg.vendor_name { | conditional gate; changing condition shifts runtime path a
side effects. |
| 302 |             if vendor.eq_ignore_ascii_case("cat_comm") { | conditional gate; changing condition shifts runtim
ath and side effects. |
| 303 |                 return super::vendor_cat_comm::CatCommAdapter::open(cfg) | executable/support line; behavior
act depends on surrounding block context. |
| 304 |                 .map(|ad| Box::new(ad) as Box<dyn HardwareInterface>); | executable/support line; behav
impact depends on surrounding block context. |
| 305 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 306 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 307 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 308 | (blank) | line is blank; visual separation only. |
| 309 |     if let Some(iface) = &cfg.can_interface { | conditional gate; changing condition shifts runtime path and
e effects. |
| 310 |         let ad = super::socketcan_adapter::SocketCanAdapter::open(iface)?; | error-propagating operation; cha
nging it impacts failure semantics and caller recovery paths. |
| 311 |         return Ok(Box::new(ad)); | executable/support line; behavior impact depends on surrounding block cont
. |
| 312 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 313 | (blank) | line is blank; visual separation only. |
| 314 |     if let Some(port) = &cfg.serial_port { | conditional gate; changing condition shifts runtime path and sid
ffects. |
| 315 |         let ad = super::serial_adapter::SerialAdapter::open(port, cfg.serial_baud)?; | error-propagating oper
on; changing it impacts failure semantics and caller recovery paths. |
| 316 |         return Ok(Box::new(ad)); | executable/support line; behavior impact depends on surrounding block cont
. |
| 317 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 318 | (blank) | line is blank; visual separation only. |
| 319 |     Err(HwError::PortNotFound( | executable/support line; behavior impact depends on surrounding block context
|
| 320 |         "no real interface configured: set --vendor-name=cat_comm with bridge/template, or --can-if, or --ser
-port".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 321 |     )) | executable/support line; behavior impact depends on surrounding block context. |
| 322 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 323 | (blank) | line is blank; visual separation only. |
| 324 | pub fn probe_live_adapter(cfg: &HwConfig) -> Result<String, HwError> { | declares executable behavior; signat
/body changes alter API, control flow, and outputs. |
| 325 |     let mut adapter = open_real_adapter(cfg)?; | error-propagating operation; changing it impacts failure sem
ics and caller recovery paths. |

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| 326 |     adapter.init(cfg)?; | error-propagating operation; changing it impacts failure semantics and caller recovery
paths. |
| 327 |     let info = adapter | executable/support line; behavior impact depends on surrounding block context. |
| 328 |         .adapter_info() | executable/support line; behavior impact depends on surrounding block context. |
| 329 |         .unwrap_or_else(|| "adapter_info=unavailable".to_string()); | executable/support line; behavior impact
depends on surrounding block context. |
| 330 |     adapter.close()?; | error-propagating operation; changing it impacts failure semantics and caller recovery
paths. |
| 331 |     Ok(info) | executable/support line; behavior impact depends on surrounding block context. |
| 332 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 333 | (blank) | line is blank; visual separation only. |
| 334 | #[derive(Debug, Serialize)] | serialization contract touchpoint; edits can break report compatibility and too
g. |
| 335 | struct StartupAudit<'a> { | declares data layout; field changes may break serialization, tests, and callers.
| 336 |     timestamp_ms: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 337 |     component: &'a str, | executable/support line; behavior impact depends on surrounding block context. |
| 338 |     event: &'a str, | executable/support line; behavior impact depends on surrounding block context. |
| 339 |     level: &'a str, | executable/support line; behavior impact depends on surrounding block context. |
| 340 |     direction: &'a str, | executable/support line; behavior impact depends on surrounding block context. |
| 341 |     reason: &'a str, | executable/support line; behavior impact depends on surrounding block context. |
| 342 |     mode: &'a str, | executable/support line; behavior impact depends on surrounding block context. |
| 343 |     dry_run: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 344 |     enable_write: bool, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnosti
utcomes. |
| 345 |     allowlist_present: bool, | executable/support line; behavior impact depends on surrounding block context.
| 346 |     write_effectively_enabled: bool, | I/O or protocol-critical logic; edits can affect integration, timing,
diagnostic outcomes. |
| 347 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 348 | (blank) | line is blank; visual separation only. |
| 349 | pub fn write_listen_only_startup_log(cfg: &HwConfig) -> Result<PathBuf, HwError> { | declares executable beha
r; signature/body changes alter API, control flow, and outputs. |
| 350 |     fs::create_dir_all(&cfg.log_dir) | executable/support line; behavior impact depends on surrounding block
text. |
| 351 |         .map_err(|e| HwError::Unknown(format!("create log dir failed: {e}")))?; | error-propagating operati
changing it impacts failure semantics and caller recovery paths. |
| 352 | (blank) | line is blank; visual separation only. |
| 353 |     let ts = now_ms(); | executable/support line; behavior impact depends on surrounding block context. |
| 354 |     let session = format!("session-{:ts}"); | executable/support line; behavior impact depends on surrounding
ck context. |
| 355 |     let path = cfg.log_dir.join(format!("{session}.json.gz")); | executable/support line; behavior impact de
ds on surrounding block context. |
| 356 | (blank) | line is blank; visual separation only. |
| 357 |     let file = File::create(&path) | executable/support line; behavior impact depends on surrounding block co
xt. |
| 358 |         .map_err(|e| HwError::Unknown(format!("create log file failed: {e}")))?; | error-propagating operat
; changing it impacts failure semantics and caller recovery paths. |
| 359 |     let mut gz = GzEncoder::new(file, Compression::default()); | executable/support line; behavior impact dep
s on surrounding block context. |
| 360 | (blank) | line is blank; visual separation only. |
| 361 |     let mode = match cfg.mode { | executable/support line; behavior impact depends on surrounding block conte
|
| 362 |         HwMode::Sim => "sim", | executable/support line; behavior impact depends on surrounding block context
| 363 |         HwMode::Live => "live", | executable/support line; behavior impact depends on surrounding block conte
|
| 364 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 365 | (blank) | line is blank; visual separation only. |
| 366 |     let audit = StartupAudit { | executable/support line; behavior impact depends on surrounding block context
|
| 367 |         timestamp_ms: ts, | executable/support line; behavior impact depends on surrounding block context. |
| 368 |         component: "io:hw", | executable/support line; behavior impact depends on surrounding block context.
| 369 |         event: "startup_policy", | executable/support line; behavior impact depends on surrounding block conte
. |
| 370 |         level: "info", | executable/support line; behavior impact depends on surrounding block context. |
| 371 |         direction: "blocked", | executable/support line; behavior impact depends on surrounding block context
| 372 |         reason: "safe_by_default_listen_only", | executable/support line; behavior impact depends on surround

```

```

block context. |
| 373 |         mode, | executable/support line; behavior impact depends on surrounding block context. |
| 374 |         dry_run: cfg.dry_run, | executable/support line; behavior impact depends on surrounding block context. |
| 375 |         enable_write: cfg.enable_write, | I/O or protocol-critical logic; edits can affect integration, timing,
and diagnostic outcomes. |
| 376 |         allowlist_present: cfg.allowlist_path.is_some(), | executable/support line; behavior impact depends on
surrounding block context. |
| 377 |         write_effectively_enabled: cfg.write_effectively_enabled(), | I/O or protocol-critical logic; edits can
affect integration, timing, and diagnostic outcomes. |
| 378 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 379 | (blank) | line is blank; visual separation only. |
| 380 |     let line = serde_json::to_string(&audit) | serialization contract touchpoint; edits can break report compati-
bility and tooling. |
| 381 |         .map_err(|e| HwError::Unknown(format!("serialize audit failed: {e}")))?; | error-propagating operation;
; changing it impacts failure semantics and caller recovery paths. |
| 382 |         writeln!(gz, "{line}").map_err(|e| HwError::Unknown(format!("write audit failed: {e}")))?; | error-propagat-
ing operation; changing it impacts failure semantics and caller recovery paths. |
| 383 |         gz.finish() | executable/support line; behavior impact depends on surrounding block context. |
| 384 |         .map_err(|e| HwError::Unknown(format!("finalize gzip failed: {e}")))?; | error-propagating operation;
changing it impacts failure semantics and caller recovery paths. |
| 385 | (blank) | line is blank; visual separation only. |
| 386 |         if cfg.write_effectively_enabled() { | conditional gate; changing condition shifts runtime path and side
effects. |
| 387 |             let idx_path = cfg.log_dir.join("write_enabled_sessions.log"); | I/O or protocol-critical logic; edits
can affect integration, timing, and diagnostic outcomes. |
| 388 |             let mut idx = OpenOptions::new() | executable/support line; behavior impact depends on surrounding block
context. |
| 389 |                 .create(true) | executable/support line; behavior impact depends on surrounding block context. |
| 390 |                 .append(true) | executable/support line; behavior impact depends on surrounding block context. |
| 391 |                 .open(idx_path) | executable/support line; behavior impact depends on surrounding block context. |
| 392 |                 .map_err(|e| HwError::Unknown(format!("open write-enabled index failed: {e}")))?; | error-propagat-
ing operation; changing it impacts failure semantics and caller recovery paths. |
| 393 |                 writeln!(idx, "{session}") | I/O or protocol-critical logic; edits can affect integration, timing, and
diagnostic outcomes. |
| 394 |                 .map_err(|e| HwError::Unknown(format!("append write-enabled index failed: {e}")))?; | error-propagat-
ing operation; changing it impacts failure semantics and caller recovery paths. |
| 395 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 396 | (blank) | line is blank; visual separation only. |
| 397 |         Ok(path) | executable/support line; behavior impact depends on surrounding block context. |
| 398 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 399 | (blank) | line is blank; visual separation only. |
| 400 | fn now_ms() -> u64 { | declares executable behavior; signature/body changes alter API, control flow, and output
. |
| 401 |     SystemTime::now() | executable/support line; behavior impact depends on surrounding block context. | | |
| 402 |     .duration_since(UNIX_EPOCH) | executable/support line; behavior impact depends on surrounding block context. |
| 403 |     .map(|d| d.as_millis() as u64) | executable/support line; behavior impact depends on surrounding block context. |
| 404 |     .unwrap_or(0) | executable/support line; behavior impact depends on surrounding block context. |
| 405 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/io/live_runner.rs

Role: I/O and diagnostics stack: adapters, allowlist, flashing, replay, metrics, production phases.

Lines: 1676 | Non-empty: 1534 | Symbols: 65

Symbol Inventory

Line	Kind	Symbol	Change Impact
19	const	DID_VIN	data/structure coupling
20	const	DID_SW_VERSION	data/structure coupling
21	const	DID_HW_VERSION	data/structure coupling
22	const	DID_CALIBRATION_ID	data/structure coupling
23	const	DID_ECU_SERIAL	data/structure coupling

24	const	DID_FINGERPRINT	data/structure coupling
25	const	DID_BATTERY_VOLTAGE	data/structure coupling
26	const	MIN_FLASH_SUPPLY_V	data/structure coupling
31	struct	DetectResult	data/structure coupling
36	struct	LiveFeed	data/structure coupling
47	struct	SnapshotPoint	data/structure coupling
56	impl	LiveFeed	behavior binding and method semantics
57	fn	latest_snapshot	signature/API coupling
64	fn	last_update_ms	signature/API coupling
68	fn	frames_total	signature/API coupling
72	fn	recent_points	signature/API coupling
81	fn	is_alive	signature/API coupling
85	fn	last_error	signature/API coupling
92	fn	export_csv	signature/API coupling
128	fn	stop	signature/API coupling
133	fn	detect_ecms	signature/API coupling
174	fn	connect_ecm	signature/API coupling
250	fn	start_retrieve_data	signature/API coupling
373	fn	now_ms	signature/API coupling
380	fn	optf	signature/API coupling
388	struct	FlashSummary	data/structure coupling
402	struct	FlashTransportDiagnostics	data/structure coupling
417	struct	BridgeDiagAggregate	data/structure coupling
426	struct	BridgeDiagLine	data/structure coupling
436	struct	EcuIdentity	data/structure coupling
446	struct	FlashPreflight	data/structure coupling
452	struct	FlashPreflightReport	data/structure coupling
459	enum	FlashAuditStatus	data/structure coupling
465	struct	FlashAuditEvent	data/structure coupling
476	fn	hex_lower	signature/API coupling
484	fn	append_flash_audit	signature/API coupling
549	fn	ensure_positive_response	signature/API coupling
612	fn	read_did_ascii	signature/API coupling
655	fn	read_did_battery_voltage	signature/API coupling
679	fn	decode_battery_voltage_from_can	signature/API coupling
695	fn	sample_supply_voltage_from_bus	signature/API coupling
720	fn	read_ecu_identity	signature/API coupling
749	fn	run_flash_preflight	signature/API coupling
778	fn	live_read_ecm_identity	signature/API coupling
793	fn	live_preflight_ecm	signature/API coupling
816	fn	j1939_uds_req_id	signature/API coupling
820	fn	is_uds_response_id	signature/API coupling
827	fn	can_payload	signature/API coupling
831	fn	send_frame_retry	signature/API coupling
856	fn	send_can_sf	signature/API coupling
889	fn	wait_for_fc	signature/API coupling
929	fn	send_can_multiframe	signature/API coupling
1021	fn	send_fc_cts	signature/API coupling
1046	fn	recv_can_uds_response	signature/API coupling
1164	fn	send_uds_and_wait	signature/API coupling
1216	fn	check_live_channel_policy	signature/API coupling
1241	fn	live_clean_ecm	signature/API coupling
1286	fn	live_flash_ecm_firmware	signature/API coupling
1590	fn	read_bridge_diag_since	signature/API coupling
1625	module	tests	data/structure coupling
1631	fn	uds_response_id_matches_expected_target	signature/API coupling
1639	fn	uds_response_id_rejects_wrong_pf_ps_or_sa	signature/API coupling
1650	fn	ensure_positive_response_accepts_expected_sid	signature/API coupling
1658	fn	ensure_positive_response_rejects_negative_response_code	signature/API coupling
1668	fn	ensure_positive_response_rejects_unexpected_sid	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
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```

| 1 | use std::collections::BTreeSet; | imports symbols; changing path/name affects compile-time resolution. |
| 2 | use std::collections::VecDeque; | imports symbols; changing path/name affects compile-time resolution. |
| 3 | use std::fs::{self, File}; | imports symbols; changing path/name affects compile-time resolution. |
| 4 | use std::io::Write; | imports symbols; changing path/name affects compile-time resolution. |
| 5 | use std::net::{TcpStream, ToSocketAddrs}; | imports symbols; changing path/name affects compile-time resolution. |
| 6 | use std::path::Path; | imports symbols; changing path/name affects compile-time resolution. |
| 7 | use std::sync::atomic::{AtomicBool, AtomicU64, Ordering}; | imports symbols; changing path/name affects compile-time resolution. |
| 8 | use std::sync::{Arc, Mutex}; | imports symbols; changing path/name affects compile-time resolution. |
| 9 | use std::sync::OnceLock; | imports symbols; changing path/name affects compile-time resolution. |
| 10 | use std::thread; | imports symbols; changing path/name affects compile-time resolution. |
| 11 | use std::time::{Duration, Instant}; | imports symbols; changing path/name affects compile-time resolution. |
| 12 | (blank) | line is blank; visual separation only. |
| 13 | use super::ecm_params::{j1939_pgn, merge_snapshot, EcmSnapshot, PGN_EEC1, PGN_EFL_P1, PGN_ET1}; | imports symbols; changing path/name affects compile-time resolution. |
| 14 | use super::hw::{CanFrame, Frame, HwConfig, HwError, HwMode}; | imports symbols; changing path/name affects compile-time resolution. |
| 15 | use super::replay::{append_record, frame_to_record, JsonlRecord}; | imports symbols; changing path/name affects compile-time resolution. |
| 16 | use serde::{Deserialize, Serialize}; | imports symbols; changing path/name affects compile-time resolution. |
| 17 | use sha2::{Digest, Sha256}; | imports symbols; changing path/name affects compile-time resolution. |
| 18 | (blank) | line is blank; visual separation only. |
| 19 | const DID_VIN: u16 = 0xF190; | executable/support line; behavior impact depends on surrounding block context. |
| 20 | const DID_SW_VERSION: u16 = 0xF189; | executable/support line; behavior impact depends on surrounding block context. |
| 21 | const DID_HW_VERSION: u16 = 0xF191; | executable/support line; behavior impact depends on surrounding block context. |
| 22 | const DID_CALIBRATION_ID: u16 = 0xF180; | executable/support line; behavior impact depends on surrounding block context. |
| 23 | const DID_ECU_SERIAL: u16 = 0xF18C; | executable/support line; behavior impact depends on surrounding block context. |
| 24 | const DID_FINGERPRINT: u16 = 0xF184; | executable/support line; behavior impact depends on surrounding block context. |
| 25 | const DID_BATTERY_VOLTAGE: u16 = 0xDD0E; | executable/support line; behavior impact depends on surrounding block context. |
| 26 | const MIN_FLASH_SUPPLY_V: f64 = 11.8; | executable/support line; behavior impact depends on surrounding block context. |
| 27 | (blank) | line is blank; visual separation only. |
| 28 | static FLASH_AUDIT_CHAIN: OnceLock<Mutex<u8; 32>> = OnceLock::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 29 | (blank) | line is blank; visual separation only. |
| 30 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 31 | pub struct DetectResult { | declares data layout; field changes may break serialization, tests, and callers. |
| 32 |     pub source_addresses: Vec<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 33 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 34 | (blank) | line is blank; visual separation only. |
| 35 | #[derive(Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 36 | pub struct LiveFeed { | declares data layout; field changes may break serialization, tests, and callers. |
| 37 |     stop: Arc<AtomicBool>, | executable/support line; behavior impact depends on surrounding block context. |
| 38 |     alive: Arc<AtomicBool>, | executable/support line; behavior impact depends on surrounding block context. |
| 39 |     latest_snapshot: Arc<Mutex<EcmSnapshot>>, | executable/support line; behavior impact depends on surrounding lock context. |
| 40 |     history: Arc<Mutex<VecDeque<SnapshotPoint>>>, | executable/support line; behavior impact depends on surrounding block context. |
| 41 |     last_error: Arc<Mutex<Option<String>>>, | executable/support line; behavior impact depends on surrounding block context. |
| 42 |     last_update_ms: Arc<AtomicU64>, | executable/support line; behavior impact depends on surrounding block context. |
| 43 |     frames_total: Arc<AtomicU64>, | executable/support line; behavior impact depends on surrounding block context. |
| 44 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 45 | (blank) | line is blank; visual separation only. |
| 46 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |

```



```

| 47 | pub struct SnapshotPoint { | declares data layout; field changes may break serialization, tests, and callers.
| 48 |     pub ts_ms: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 49 |     pub engine_speed_rpm: Option<f64>, | executable/support line; behavior impact depends on surrounding block
context. |
| 50 |     pub accel_pedal_pct: Option<f64>, | executable/support line; behavior impact depends on surrounding block
context. |
| 51 |     pub coolant_temp_c: Option<f64>, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 52 |     pub fuel_temp_c: Option<f64>, | executable/support line; behavior impact depends on surrounding block cont
. |
| 53 |     pub oil_pressure_kpa: Option<f64>, | executable/support line; behavior impact depends on surrounding block
context. |
| 54 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 55 | (blank) | line is blank; visual separation only. |
| 56 | impl LiveFeed { | implementation block; method behavior and trait conformance are defined here. |
| 57 |     pub fn latest_snapshot(&self) -> EcmSnapshot { | declares executable behavior; signature/body changes alter
API, control flow, and outputs. |
| 58 |         match self.latest_snapshot.lock() { | branch dispatch; changing arms alters behavior for variants and
ut classes. |
| 59 |             Ok(s) => s.clone(), | executable/support line; behavior impact depends on surrounding block context
|
| 60 |             Err(poisoned) => poisoned.into_inner().clone(), | executable/support line; behavior impact depends
surrounding block context. |
| 61 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 62 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 63 | (blank) | line is blank; visual separation only. |
| 64 |     pub fn last_update_ms(&self) -> u64 { | declares executable behavior; signature/body changes alter API, co
ol flow, and outputs. |
| 65 |         self.last_update_ms.load(Ordering::Relaxed) | executable/support line; behavior impact depends on surr
ding block context. |
| 66 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 67 | (blank) | line is blank; visual separation only. |
| 68 |     pub fn frames_total(&self) -> u64 { | declares executable behavior; signature/body changes alter API, cont
flow, and outputs. |
| 69 |         self.frames_total.load(Ordering::Relaxed) | executable/support line; behavior impact depends on surrou
ng block context. |
| 70 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 71 | (blank) | line is blank; visual separation only. |
| 72 |     pub fn recent_points(&self, limit: usize) -> Vec<SnapshotPoint> { | declares executable behavior; signatur
ody changes alter API, control flow, and outputs. |
| 73 |         let hist = match self.history.lock() { | executable/support line; behavior impact depends on surroundi
block context. |
| 74 |             Ok(h) => h, | executable/support line; behavior impact depends on surrounding block context. |
| 75 |             Err(poisoned) => poisoned.into_inner(), | executable/support line; behavior impact depends on surr
ding block context. |
| 76 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 77 |         let count = hist.len().min(limit); | executable/support line; behavior impact depends on surrounding b
k context. |
| 78 |         hist.iter().skip(hist.len() - count).cloned().collect() | executable/support line; behavior impact dep
s on surrounding block context. |
| 79 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 80 | (blank) | line is blank; visual separation only. |
| 81 |     pub fn is_alive(&self) -> bool { | declares executable behavior; signature/body changes alter API, control
ow, and outputs. |
| 82 |         self.alive.load(Ordering::Relaxed) | executable/support line; behavior impact depends on surrounding b
k context. |
| 83 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 84 | (blank) | line is blank; visual separation only. |
| 85 |     pub fn last_error(&self) -> Option<String> { | declares executable behavior; signature/body changes alter
, control flow, and outputs. |
| 86 |         match self.last_error.lock() { | branch dispatch; changing arms alters behavior for variants and input
asses. |
| 87 |             Ok(e) => e.clone(), | executable/support line; behavior impact depends on surrounding block context
|

```

```

| 88 |         Err(poisoned) => poisoned.into_inner().clone(), | executable/support line; behavior impact depends
surrounding block context. |
| 89 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 90 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 91 | (blank) | line is blank; visual separation only. |
| 92 | pub fn export_csv<P: AsRef<Path>>(&self, path: P) -> Result<(), HwError> { | declares executable behavior;
signature/body changes alter API, control flow, and outputs. |
| 93 |     let path = path.as_ref(); | executable/support line; behavior impact depends on surrounding block cont
. |
| 94 |     if let Some(parent) = path.parent() { | conditional gate; changing condition shifts runtime path and s
effects. |
| 95 |         fs::create_dir_all(parent) | executable/support line; behavior impact depends on surrounding block
ntext. |
| 96 |         .map_err(|e| HwError::Unknown(format!("create csv dir failed: {e}")))?; | error-propagating
ration; changing it impacts failure semantics and caller recovery paths. |
| 97 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 98 | (blank) | line is blank; visual separation only. |
| 99 |     let hist = self | executable/support line; behavior impact depends on surrounding block context. |
| 100 |     .history | executable/support line; behavior impact depends on surrounding block context. |
| 101 |     .lock() | executable/support line; behavior impact depends on surrounding block context. |
| 102 |     .map_err(|_| HwError::Unknown("history lock poisoned".to_string()))?; | error-propagating opera
n; changing it impacts failure semantics and caller recovery paths. |
| 103 | (blank) | line is blank; visual separation only. |
| 104 |     let mut f = File::create(path) | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 105 |     .map_err(|e| HwError::Unknown(format!("create csv file failed: {e}")))?; | error-propagating op
tion; changing it impacts failure semantics and caller recovery paths. |
| 106 |     writeln!( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outc
s. |
| 107 |         f, | executable/support line; behavior impact depends on surrounding block context. |
| 108 |         "ts_ms,engine_speed_rpm,accel_pedal_pct,coolant_temp_c,fuel_temp_c,oil_pressure_kpa" | executable
pport line; behavior impact depends on surrounding block context. |
| 109 |     ) | executable/support line; behavior impact depends on surrounding block context. |
| 110 |     .map_err(|e| HwError::Unknown(format!("write csv header failed: {e}")))?; | error-propagating opera
n; changing it impacts failure semantics and caller recovery paths. |
| 111 | (blank) | line is blank; visual separation only. |
| 112 |     for p in hist.iter() { | iteration control; changing bounds/body affects throughput, timing, and poss
y safety guarantees. |
| 113 |         writeln!( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic
comes. |
| 114 |             f, | executable/support line; behavior impact depends on surrounding block context. |
| 115 |             "{}",{},{},{},{},{},{},", | executable/support line; behavior impact depends on surrounding block
text. |
| 116 |             p.ts_ms, | executable/support line; behavior impact depends on surrounding block context. |
| 117 |             optf(p.engine_speed_rpm), | executable/support line; behavior impact depends on surrounding b
k context. |
| 118 |             optf(p.accel_pedal_pct), | executable/support line; behavior impact depends on surrounding bl
context. |
| 119 |             optf(p.coolant_temp_c), | executable/support line; behavior impact depends on surrounding blo
context. |
| 120 |             optf(p.fuel_temp_c), | executable/support line; behavior impact depends on surrounding block
text. |
| 121 |             optf(p.oil_pressure_kpa) | executable/support line; behavior impact depends on surrounding bl
context. |
| 122 |         ) | executable/support line; behavior impact depends on surrounding block context. |
| 123 |         .map_err(|e| HwError::Unknown(format!("write csv row failed: {e}")))?; | error-propagating oper
on; changing it impacts failure semantics and caller recovery paths. |
| 124 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 125 |     Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 126 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 127 | (blank) | line is blank; visual separation only. |
| 128 | pub fn stop(&self) { | declares executable behavior; signature/body changes alter API, control flow, and
puts. |
| 129 |     self.stop.store(true, Ordering::Relaxed); | executable/support line; behavior impact depends on surro

```

```

ing block context. |
| 130 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 131 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 132 | (blank) | line is blank; visual separation only. |
| 133 | pub fn detect_ecms(cfg: &HwConfig, timeout: Duration) -> Result<DetectResult, HwError> { | declares executable
behavior; signature/body changes alter API, control flow, and outputs. |
| 134 |     if cfg.mode != HwMode::Live { | conditional gate; changing condition shifts runtime path and side effects
| 135 |         return Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block
text. |
| 136 |         "Detect requires --hw-mode=live".to_string(), | executable/support line; behavior impact depends
surrounding block context. |
| 137 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 138 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 139 |     check_live_channel_policy(cfg)?; | error-propagating operation; changing it impacts failure semantics and
caller recovery paths. |
| 140 | (blank) | line is blank; visual separation only. |
| 141 |     let mut adapter = super::hw::open_real_adapter(cfg)?; | error-propagating operation; changing it impacts
failure semantics and caller recovery paths. |
| 142 |     adapter.init(cfg)?; | error-propagating operation; changing it impacts failure semantics and caller recov
paths. |
| 143 | (blank) | line is blank; visual separation only. |
| 144 |     let start = Instant::now(); | executable/support line; behavior impact depends on surrounding block conte
|
| 145 |     let mut found = BTreeSet::new(); | executable/support line; behavior impact depends on surrounding block
text. |
| 146 | (blank) | line is blank; visual separation only. |
| 147 |     while start.elapsed() < timeout { | iteration control; changing bounds/body affects throughput, timing, a
possibly safety guarantees. |
| 148 |         match adapter.read_frame() { | branch dispatch; changing arms alters behavior for variants and input
sses. |
| 149 |             Ok(Frame::Can(cf)) => { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 150 |                 let pgn = j1939_pgn(cf.id); | executable/support line; behavior impact depends on surrounding
ock context. |
| 151 |                 let sa = (cf.id & 0xFF) as u8; | executable/support line; behavior impact depends on surround
block context. |
| 152 |                 if matches!(pgn, PGN_EEC1 \ | PGN_ET1 \ | PGN_EFL_P1 \ | 60928) { | conditional gate; changing c
ition shifts runtime path and side effects. |
| 153 |                     found.insert(sa); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 154 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 155 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 156 |             Ok(Frame::Serial(_)) => { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 157 |                 // For serial-only links, one logical ECM channel is considered detected. | comment line; doc
ntation intent, no runtime effect. |
| 158 |                 found.insert(0); | executable/support line; behavior impact depends on surrounding block cont
. |
| 159 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 160 |             Err(HwError::Timeout) => {} | executable/support line; behavior impact depends on surrounding blo
context. |
| 161 |             Err(e) => { | executable/support line; behavior impact depends on surrounding block context. |
| 162 |                 adapter.close()?; | error-propagating operation; changing it impacts failure semantics and ca
r recovery paths. |
| 163 |                 return Err(e); | executable/support line; behavior impact depends on surrounding block context
|
| 164 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 165 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 166 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 167 | (blank) | line is blank; visual separation only. |

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| 168 |     adapter.close()?; | error-propagating operation; changing it impacts failure semantics and caller recovery
aths. |
| 169 |     Ok(DetectResult { | executable/support line; behavior impact depends on surrounding block context. |
| 170 |         source_addresses: found.into_iter().collect(), | executable/support line; behavior impact depends on
rounding block context. |
| 171 |     }) | executable/support line; behavior impact depends on surrounding block context. |
| 172 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 173 | (blank) | line is blank; visual separation only. |
| 174 | pub fn connect_ecm(cfg: &HwConfig, target_sa: Option<u8>) -> Result<(), HwError> { | declares executable beha
r; signature/body changes alter API, control flow, and outputs. |
| 175 |     if cfg.mode != HwMode::Live { | conditional gate; changing condition shifts runtime path and side effects
| 176 |         return Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block
text. |
| 177 |         "Connect requires --hw-mode=live".to_string(), | executable/support line; behavior impact depends
surrounding block context. |
| 178 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 179 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 180 |     check_live_channel_policy(cfg)?; | error-propagating operation; changing it impacts failure semantics and
ller recovery paths. |
| 181 | (blank) | line is blank; visual separation only. |
| 182 |     let mut adapter = super::hw::open_real_adapter(cfg)?; | error-propagating operation; changing it impacts
ture semantics and caller recovery paths. |
| 183 |     adapter.init(cfg)?; | error-propagating operation; changing it impacts failure semantics and caller recov
paths. |
| 184 | (blank) | line is blank; visual separation only. |
| 185 |     // For CAN links, send standard J1939 PGN requests as an integration handshake. | comment line; documenta
n intent, no runtime effect. |
| 186 |     if cfg.can_interface.is_some() { | conditional gate; changing condition shifts runtime path and side effe
. |
| 187 |         let dst = target_sa.unwrap_or(0xFF); | executable/support line; behavior impact depends on surroundin
lock context. |
| 188 |         let reqs = [60928_u32, PGN_EEC1, PGN_ET1, PGN_EFL_P1]; | executable/support line; behavior impact dep
s on surrounding block context. |
| 189 |         for pgn in reqs { | iteration control; changing bounds/body affects throughput, timing, and possibly
ety guarantees. |
| 190 |             let mut data = [0u8; 8]; | executable/support line; behavior impact depends on surrounding block
text. |
| 191 |             data[0] = (pgn & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding b
k context. |
| 192 |             data[1] = ((pgn >> 8) & 0xFF) as u8; | executable/support line; behavior impact depends on surrou
ng block context. |
| 193 |             data[2] = ((pgn >> 16) & 0xFF) as u8; | executable/support line; behavior impact depends on surro
ing block context. |
| 194 |             let req_id = 0x18EA0000 \ | ((dst as u32) << 8) \ | 0xF9; // Request PGN from tool SA=0xF9 | execut
e/support line; behavior impact depends on surrounding block context. |
| 195 |             let req = Frame::Can(CanFrame { | executable/support line; behavior impact depends on surrounding
ock context. |
| 196 |                 id: req_id, | executable/support line; behavior impact depends on surrounding block context.
| 197 |                 dlc: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 198 |                 data, | executable/support line; behavior impact depends on surrounding block context. |
| 199 |                 len: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 200 |                 timestamp_ms: Some(now_ms()), | executable/support line; behavior impact depends on surroundi
block context. |
| 201 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 202 |             if let Err(e) = adapter.send_frame(req) { | conditional gate; changing condition shifts runtime p
and side effects. |
| 203 |                 adapter.close()?; | error-propagating operation; changing it impacts failure semantics and ca
r recovery paths. |
| 204 |                 return Err(e); | executable/support line; behavior impact depends on surrounding block context
|
| 205 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 206 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 207 | (blank) | line is blank; visual separation only. |

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| 208 |          // Authentication/identification proxy: require at least one expected PGN | comment line; documentati
intent, no runtime effect. |
| 209 |          // from the requested source (or any source when no target SA is provided). | comment line; documenta
n intent, no runtime effect. |
| 210 |          let deadline = Instant::now() + Duration::from_secs(2); | executable/support line; behavior impact de
ds on surrounding block context. |
| 211 |          let mut got_response = false; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 212 |          while Instant::now() < deadline { | iteration control; changing bounds/body affects throughput, timin
and possibly safety guarantees. |
| 213 |              match adapter.read_frame() { | branch dispatch; changing arms alters behavior for variants and in
classes. |
| 214 |                  Ok(Frame::Can(cf)) => { | executable/support line; behavior impact depends on surrounding blo
context. |
| 215 |                      let pgn = j1939_pgn(cf.id); | executable/support line; behavior impact depends on surroun
g block context. |
| 216 |                      let sa = (cf.id & 0xFF) as u8; | executable/support line; behavior impact depends on surr
ding block context. |
| 217 |                      let valid_pgn = matches!(pgn, 60928 \| PGN_EEC1 \| PGN_ET1 \| PGN_EFL_P1); | executable/s
ort line; behavior impact depends on surrounding block context. |
| 218 |                      let valid_sa = target_sa.is_none() \| \| Some(sa) == target_sa; | executable/support line;
havior impact depends on surrounding block context. |
| 219 |                      if valid_pgn && valid_sa { | conditional gate; changing condition shifts runtime path and
de effects. |
| 220 |                          got_response = true; | executable/support line; behavior impact depends on surroundin
lock context. |
| 221 |                          break; | executable/support line; behavior impact depends on surrounding block context
|
| 222 |                      } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 223 |                  } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 224 |                  Ok(Frame::Serial(_)) => {} | executable/support line; behavior impact depends on surrounding
ck context. |
| 225 |                  Err(HwError::Timeout) => {} | executable/support line; behavior impact depends on surrounding
ock context. |
| 226 |                  Err(e) => { | executable/support line; behavior impact depends on surrounding block context.
| 227 |                      adapter.close()?; | error-propagating operation; changing it impacts failure semantics an
aller recovery paths. |
| 228 |                      return Err(e); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 229 |                  } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 230 |              } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 231 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 232 |          if !got_response { | conditional gate; changing condition shifts runtime path and side effects. |
| 233 |              adapter.close()?; | error-propagating operation; changing it impacts failure semantics and caller
covery paths. |
| 234 |              return Err(HwError::Timeout); | executable/support line; behavior impact depends on surrounding b
k context. |
| 235 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 236 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 237 | (blank) | line is blank; visual separation only. |
| 238 |      let identity = read_ecu_identity(&mut *adapter, cfg, target_sa)?; | error-propagating operation; changing
impacts failure semantics and caller recovery paths. |
| 239 |      if !cfg.allow_untrusted_ecu && identity.fingerprint.is_none() { | conditional gate; changing condition sh
s runtime path and side effects. |
| 240 |          adapter.close()?; | error-propagating operation; changing it impacts failure semantics and caller rec
ry paths. |
| 241 |          return Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block
text. |
| 242 |              "connect rejected: missing fingerprint DID 0xF184 on trusted profile".to_string(), | executable/s

```

```

ort line; behavior impact depends on surrounding block context. |
| 243 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 244 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 245 | (blank) | line is blank; visual separation only. |
| 246 |     adapter.close()?; | error-propagating operation; changing it impacts failure semantics and caller recovery
aths. |
| 247 |     Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 248 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 249 | (blank) | line is blank; visual separation only. |
| 250 | pub fn start_retrieve_data(cfg: HwConfig, target_sa: Option<u8>) -> Result<LiveFeed, HwError> { | declares ex
table behavior; signature/body changes alter API, control flow, and outputs. |
| 251 |     if cfg.mode != HwMode::Live { | conditional gate; changing condition shifts runtime path and side effects
| 252 |         return Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block
text. |
| 253 |         "Retrieve requires --hw-mode=live".to_string(), | executable/support line; behavior impact depend
n surrounding block context. |
| 254 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 255 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 256 |     check_live_channel_policy(&cfg)?; | error-propagating operation; changing it impacts failure semantics and
aller recovery paths. |
| 257 | (blank) | line is blank; visual separation only. |
| 258 |     let stop = Arc::new(AtomicBool::new(false)); | executable/support line; behavior impact depends on surrou
ng block context. |
| 259 |     let alive = Arc::new(AtomicBool::new(true)); | executable/support line; behavior impact depends on surrou
ng block context. |
| 260 |     let latest_snapshot = Arc::new(Mutex::new(EcmSnapshot::default())); | executable/support line; behavior i
ct depends on surrounding block context. |
| 261 |     let history = Arc::new(Mutex::new(VecDeque::<SnapshotPoint>::new())); | executable/support line; behavior
impact depends on surrounding block context. |
| 262 |     let last_error = Arc::new(Mutex::new(None)); | executable/support line; behavior impact depends on surrou
ng block context. |
| 263 |     let last_update_ms = Arc::new(AtomicU64::new(0)); | executable/support line; behavior impact depends on s
ounding block context. |
| 264 |     let frames_total = Arc::new(AtomicU64::new(0)); | executable/support line; behavior impact depends on sur
nding block context. |
| 265 | (blank) | line is blank; visual separation only. |
| 266 |     let feed = LiveFeed { | executable/support line; behavior impact depends on surrounding block context. |
| 267 |         stop: Arc::clone(&stop), | executable/support line; behavior impact depends on surrounding block cont
. |
| 268 |         alive: Arc::clone(&alive), | executable/support line; behavior impact depends on surrounding block co
xt. |
| 269 |         latest_snapshot: Arc::clone(&latest_snapshot), | executable/support line; behavior impact depends on
rounding block context. |
| 270 |         history: Arc::clone(&history), | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 271 |         last_error: Arc::clone(&last_error), | executable/support line; behavior impact depends on surroundin
lock context. |
| 272 |         last_update_ms: Arc::clone(&last_update_ms), | executable/support line; behavior impact depends on su
unding block context. |
| 273 |         frames_total: Arc::clone(&frames_total), | executable/support line; behavior impact depends on surrou
ng block context. |
| 274 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 275 | (blank) | line is blank; visual separation only. |
| 276 |     thread::spawn(move || { | executable/support line; behavior impact depends on surrounding block context
| 277 |         let mut adapter = match super::hw::open_real_adapter(&cfg) { | executable/support line; behavior impa
depends on surrounding block context. |
| 278 |             Ok(a) => a, | executable/support line; behavior impact depends on surrounding block context. |
| 279 |             Err(e) => { | executable/support line; behavior impact depends on surrounding block context. |
| 280 |                 if let Ok(mut err) = last_error.lock() { | conditional gate; changing condition shifts runtim
ath and side effects. |
| 281 |                     *err = Some(format!("open adapter failed: {e}")); | executable/support line; behavior imp
depends on surrounding block context. |
| 282 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 283 |                 alive.store(false, Ordering::Relaxed); | executable/support line; behavior impact depends on

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```

rounding block context. |
| 284 |           eprintln!("[ecm-live] open adapter failed: {e}"); | executable/support line; behavior impact
ends on surrounding block context. |
| 285 |           return; | executable/support line; behavior impact depends on surrounding block context. |
| 286 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 287 |       }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 288 |       if let Err(e) = adapter.init(&cfg) { | conditional gate; changing condition shifts runtime path and s
effects. |
| 289 |           if let Ok(mut err) = last_error.lock() { | conditional gate; changing condition shifts runtime pa
and side effects. |
| 290 |               *err = Some(format!("[init adapter failed: {e}")); | executable/support line; behavior impact
ends on surrounding block context. |
| 291 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 292 |           alive.store(false, Ordering::Relaxed); | executable/support line; behavior impact depends on surr
ding block context. |
| 293 |           eprintln!("[ecm-live] init adapter failed: {e}"); | executable/support line; behavior impact depende
on surrounding block context. |
| 294 |           return; | executable/support line; behavior impact depends on surrounding block context. |
| 295 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 296 | (blank) | line is blank; visual separation only. |
| 297 |           let rx_log = cfg.log_dir.join("ecm_rx.jsonl"); | executable/support line; behavior impact depends on
rounding block context. |
| 298 |           let snapshot_log = cfg.log_dir.join("ecm_snapshot.jsonl"); | executable/support line; behavior impact
depends on surrounding block context. |
| 299 |           let mut snapshot = EcmSnapshot::default(); | executable/support line; behavior impact depends on surr
ding block context. |
| 300 | (blank) | line is blank; visual separation only. |
| 301 |           while !stop.load(Ordering::Relaxed) { | iteration control; changing bounds/body affects throughput, t
ng, and possibly safety guarantees. |
| 302 |               match adapter.read_frame() { | branch dispatch; changing arms alters behavior for variants and in
classes. |
| 303 |                   Ok(frame) => { | executable/support line; behavior impact depends on surrounding block contex
|
| 304 |                       frames_total.fetch_add(1, Ordering::Relaxed); | executable/support line; behavior impact
ends on surrounding block context. |
| 305 |                       let rec = frame_to_record(&frame, "rx", None, Some(cfg.dry_run)); | executable/support li
behavior impact depends on surrounding block context. |
| 306 |                       let _ = append_record(&rx_log, &rec); | executable/support line; behavior impact depends
surrounding block context. |
| 307 | (blank) | line is blank; visual separation only. |
| 308 |                       if let Frame::Can(cf) = &frame { | conditional gate; changing condition shifts runtime pa
and side effects. |
| 309 |                           let sa = (cf.id & 0xFF) as u8; | executable/support line; behavior impact depends on
rounding block context. |
| 310 |                               if target_sa.is_some() && Some(sa) != target_sa { | conditional gate; changing condit
shifts runtime path and side effects. |
| 311 |                                   continue; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 312 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership an
ifetime scopes. |
| 313 | (blank) | line is blank; visual separation only. |
| 314 |                               snapshot = merge_snapshot(snapshot, cf); | executable/support line; behavior impact d
nds on surrounding block context. |
| 315 |                               if let Ok(mut s) = latest_snapshot.lock() { | conditional gate; changing condition sh
s runtime path and side effects. |
| 316 |                                   *s = snapshot.clone(); | executable/support line; behavior impact depends on surr
ding block context. |
| 317 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership an
ifetime scopes. |
| 318 | (blank) | line is blank; visual separation only. |
| 319 |                               if let Ok(mut hist) = history.lock() { | conditional gate; changing condition shifts

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time path and side effects. |
| 320 |                               hist.push_back(SnapshotPoint { | executable/support line; behavior impact depends
surrounding block context. |
| 321 |                               ts_ms: cf.timestamp_ms.unwrap_or_else(now_ms), | executable/support line; beh
or impact depends on surrounding block context. |
| 322 |                               engine_speed_rpm: snapshot.engine_speed_rpm, | executable/support line; behav
impact depends on surrounding block context. |
| 323 |                               accel_pedal_pct: snapshot.accel_pedal_pct, | executable/support line; behavio
mpact depends on surrounding block context. |
| 324 |                               coolant_temp_c: snapshot.coolant_temp_c, | executable/support line; behavior
act depends on surrounding block context. |
| 325 |                               fuel_temp_c: snapshot.fuel_temp_c, | executable/support line; behavior impact
pend on surrounding block context. |
| 326 |                               oil_pressure_kpa: snapshot.oil_pressure_kpa, | executable/support line; behav
impact depends on surrounding block context. |
| 327 |                               }); | executable/support line; behavior impact depends on surrounding block conte
|
| 328 |                               if hist.len() > 5000 { | conditional gate; changing condition shifts runtime path
d side effects. |
| 329 |                               let _ = hist.pop_front(); | executable/support line; behavior impact depends
surrounding block context. |
| 330 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership
nd lifetime scopes. |
| 331 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership an
ifetime scopes. |
| 332 | (blank) | line is blank; visual separation only. |
| 333 |                               let ts = cf.timestamp_ms.unwrap_or_else(now_ms); | executable/support line; behavior
act depends on surrounding block context. |
| 334 |                               last_update_ms.store(ts, Ordering::Relaxed); | executable/support line; behavior impa
depends on surrounding block context. |
| 335 | (blank) | line is blank; visual separation only. |
| 336 |                               let snap_line = JsonlRecord { | executable/support line; behavior impact depends on s
ounding block context. |
| 337 |                               ts, | executable/support line; behavior impact depends on surrounding block conte
|
| 338 |                               transport: "can".to_string(), | executable/support line; behavior impact depends
surrounding block context. |
| 339 |                               dir: "snapshot".to_string(), | executable/support line; behavior impact depends o
urrounding block context. |
| 340 |                               id: Some(cf.id), | executable/support line; behavior impact depends on surroundin
lock context. |
| 341 |                               dlc: Some(cf.dlc), | executable/support line; behavior impact depends on surround
block context. |
| 342 |                               data: Some( | executable/support line; behavior impact depends on surrounding blo
context. |
| 343 |                               serde_json::to_string(&snapshot) | serialization contract touchpoint; edits c
break report compatibility and tooling. |
| 344 |                               .unwrap_or_else(|_| "{ }".to_string()), | executable/support line; behav
impact depends on surrounding block context. |
| 345 |                               ), | executable/support line; behavior impact depends on surrounding block contex
|
| 346 |                               raw_hex: None, | executable/support line; behavior impact depends on surrounding
ck context. |
| 347 |                               allowed: None, | executable/support line; behavior impact depends on surrounding
ck context. |
| 348 |                               dry_run: Some(cfg.dry_run), | executable/support line; behavior impact depends on
rrounding block context. |
| 349 |                               }; | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 350 |                               let _ = append_record(&snapshot_log, &snap_line); | executable/support line; behavior
pact depends on surrounding block context. |
| 351 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 352 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |

```



```

| 353 |           Err(HwError::Timeout) => { | executable/support line; behavior impact depends on surrounding
ck context. |
| 354 |           thread::sleep(Duration::from_millis(20)); | executable/support line; behavior impact depende
on surrounding block context. |
| 355 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 356 |           Err(e) => { | executable/support line; behavior impact depends on surrounding block context.
| 357 |               if let Ok(mut err) = last_error.lock() { | conditional gate; changing condition shifts ru
me path and side effects. |
| 358 |                   *err = Some(format!("read error: {e}")); | executable/support line; behavior impact d
nds on surrounding block context. |
| 359 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 360 |               eprintln!("[ecm-live] read error: {e}"); | executable/support line; behavior impact dependen
on surrounding block context. |
| 361 |               thread::sleep(Duration::from_millis(150)); | executable/support line; behavior impact depen
s on surrounding block context. |
| 362 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 363 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 364 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 365 | (blank) | line is blank; visual separation only. |
| 366 |   alive.store(false, Ordering::Relaxed); | executable/support line; behavior impact depends on surround
block context. |
| 367 |   let _ = adapter.close(); | executable/support line; behavior impact depends on surrounding block cont
. |
| 368 |   }); | executable/support line; behavior impact depends on surrounding block context. |
| 369 | (blank) | line is blank; visual separation only. |
| 370 |   Ok(feed) | executable/support line; behavior impact depends on surrounding block context. |
| 371 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 372 | (blank) | line is blank; visual separation only. |
| 373 | fn now_ms() -> u64 { | declares executable behavior; signature/body changes alter API, control flow, and outp
. |
| 374 |   std::time::SystemTime::now() | executable/support line; behavior impact depends on surrounding block cont
. |
| 375 |   .duration_since(std::time::UNIX_EPOCH) | executable/support line; behavior impact depends on surround
block context. |
| 376 |   .map(|d| d.as_millis() as u64) | executable/support line; behavior impact depends on surrounding bl
context. |
| 377 |   .unwrap_or(0) | executable/support line; behavior impact depends on surrounding block context. |
| 378 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 379 | (blank) | line is blank; visual separation only. |
| 380 | fn optf(v: Option<f64>) -> String { | declares executable behavior; signature/body changes alter API, control
ow, and outputs. |
| 381 |   match v { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 382 |       Some(x) => format!("{x:.4}"), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 383 |       None => String::new(), | executable/support line; behavior impact depends on surrounding block context
|
| 384 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 385 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 386 | (blank) | line is blank; visual separation only. |
| 387 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 388 | pub struct FlashSummary { | declares data layout; field changes may break serialization, tests, and callers.
| 389 |     pub bytes_sent: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 390 |     pub blocks_sent: usize, | executable/support line; behavior impact depends on surrounding block context.
| 391 |     pub crc32: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 392 |     pub supply_voltage_v: f64, | executable/support line; behavior impact depends on surrounding block context
|
| 393 |     pub vin: String, | executable/support line; behavior impact depends on surrounding block context. |
| 394 |     pub sw_version: String, | executable/support line; behavior impact depends on surrounding block context.
| 395 |     pub hw_version: String, | executable/support line; behavior impact depends on surrounding block context.
| 396 |     pub calibration_id: String, | executable/support line; behavior impact depends on surrounding block conte

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```

|
| 397 |     pub ecu_serial: String, | executable/support line; behavior impact depends on surrounding block context.
| 398 |     pub transport_diagnostics: FlashTransportDiagnostics, | executable/support line; behavior impact depends
surrounding block context. |
| 399 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 400 | (blank) | line is blank; visual separation only. |
| 401 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report c
atibility and tooling. |
| 402 | pub struct FlashTransportDiagnostics { | declares data layout; field changes may break serialization, tests,
callers. |
| 403 |     pub security_seed_positive: bool, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 404 |     pub security_unlock_positive: bool, | executable/support line; behavior impact depends on surrounding blo
context. |
| 405 |     pub request_download_positive: bool, | executable/support line; behavior impact depends on surrounding bl
context. |
| 406 |     pub transfer_data_blocks_attempted: usize, | executable/support line; behavior impact depends on surround
block context. |
| 407 |     pub transfer_data_blocks_acked: usize, | executable/support line; behavior impact depends on surrounding
ck context. |
| 408 |     pub request_transfer_exit_positive: bool, | executable/support line; behavior impact depends on surroundi
block context. |
| 409 |     pub fc_blocksize_seen: Option<u8>, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 410 |     pub fc_stmin_seen_ms: Option<u64>, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 411 |     pub wait_frame_count: u32, | executable/support line; behavior impact depends on surrounding block contex
|
| 412 |     pub sequence_error_count: u32, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 413 |     pub flowcontrol_timeout_count: u32, | executable/support line; behavior impact depends on surrounding blo
context. |
| 414 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 415 | (blank) | line is blank; visual separation only. |
| 416 | #[derive(Debug, Clone, Default)] | executable/support line; behavior impact depends on surrounding block cont
. |
| 417 | struct BridgeDiagAggregate { | declares data layout; field changes may break serialization, tests, and caller
|
| 418 |     wait_frame_count: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 419 |     sequence_error_count: u32, | executable/support line; behavior impact depends on surrounding block contex
|
| 420 |     flowcontrol_timeout_count: u32, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 421 |     fc_blocksize_seen: Option<u8>, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 422 |     fc_stmin_seen_ms: Option<u64>, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 423 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 424 | (blank) | line is blank; visual separation only. |
| 425 | #[derive(Debug, Deserialize)] | serialization contract touchpoint; edits can break report compatibility and t
ing. |
| 426 | struct BridgeDiagLine { | declares data layout; field changes may break serialization, tests, and callers. |
| 427 |     ts_ms: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 428 |     wait_frame_count_delta: Option<u32>, | executable/support line; behavior impact depends on surrounding bl
context. |
| 429 |     sequence_error_count_delta: Option<u32>, | executable/support line; behavior impact depends on surroundi
lock context. |
| 430 |     flowcontrol_timeout_count_delta: Option<u32>, | executable/support line; behavior impact depends on surro
ing block context. |
| 431 |     fc_blocksize: Option<u8>, | executable/support line; behavior impact depends on surrounding block context
| 432 |     fc_stmin_ms: Option<u64>, | executable/support line; behavior impact depends on surrounding block context
| 433 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 434 | (blank) | line is blank; visual separation only. |
| 435 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report c
atibility and tooling. |

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| 436 | pub struct EcuIdentity { | declares data layout; field changes may break serialization, tests, and callers. |
| 437 |     pub vin: String, | executable/support line; behavior impact depends on surrounding block context. |
| 438 |     pub sw_version: String, | executable/support line; behavior impact depends on surrounding block context. |
| 439 |     pub hw_version: String, | executable/support line; behavior impact depends on surrounding block context. |
| 440 |     pub calibration_id: String, | executable/support line; behavior impact depends on surrounding block context. |
| 441 |     pub ecu_serial: String, | executable/support line; behavior impact depends on surrounding block context. |
| 442 |     pub fingerprint: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 443 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 444 | (blank) | line is blank; visual separation only. |
| 445 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 446 | struct FlashPreflight { | declares data layout; field changes may break serialization, tests, and callers. |
| 447 |     supply_voltage_v: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 448 |     identity: EcuIdentity, | executable/support line; behavior impact depends on surrounding block context. |
| 449 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 450 | (blank) | line is blank; visual separation only. |
| 451 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 452 | pub struct FlashPreflightReport { | declares data layout; field changes may break serialization, tests, and callers. |
| 453 |     pub supply_voltage_v: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 454 |     pub identity: EcuIdentity, | executable/support line; behavior impact depends on surrounding block context. |
| 455 |     pub trusted_fingerprint: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 456 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 457 | (blank) | line is blank; visual separation only. |
| 458 | #[derive(Debug, Clone, Copy, Serialize)] | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 459 | pub enum FlashAuditStatus { | declares state variants; adding/removing variants impacts pattern matches and local branches. |
| 460 |     Ok, | executable/support line; behavior impact depends on surrounding block context. |
| 461 |     Failed, | executable/support line; behavior impact depends on surrounding block context. |
| 462 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 463 | (blank) | line is blank; visual separation only. |
| 464 | #[derive(Debug, Serialize)] | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 465 | struct FlashAuditEvent<'a> { | declares data layout; field changes may break serialization, tests, and callers. |
| 466 |     ts_ms: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 467 |     stage: &'a str, | executable/support line; behavior impact depends on surrounding block context. |
| 468 |     status: FlashAuditStatus, | executable/support line; behavior impact depends on surrounding block context. |
| 469 |     detail: String, | executable/support line; behavior impact depends on surrounding block context. |
| 470 |     target_sa: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 471 |     dry_run: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 472 |     prev_hash: String, | executable/support line; behavior impact depends on surrounding block context. |
| 473 |     event_hash: String, | executable/support line; behavior impact depends on surrounding block context. |
| 474 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 475 | (blank) | line is blank; visual separation only. |
| 476 | fn hex_lower(bytes: &[u8]) -> String { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 477 |     let mut s = String::with_capacity(bytes.len() * 2); | executable/support line; behavior impact depends on surrounding block context. |
| 478 |     for b in bytes { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 479 |         s.push_str(&format!("{b:02x}")); | executable/support line; behavior impact depends on surrounding block context. |
| 480 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 481 |     s | executable/support line; behavior impact depends on surrounding block context. |
| 482 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 483 | (blank) | line is blank; visual separation only. |
| 484 | fn append_flash_audit( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |

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| 485 |     cfg: &HwConfig, | executable/support line; behavior impact depends on surrounding block context. |
| 486 |     stage: &str, | executable/support line; behavior impact depends on surrounding block context. |
| 487 |     status: FlashAuditStatus, | executable/support line; behavior impact depends on surrounding block context. |
| 488 |     detail: impl Into<String>, | executable/support line; behavior impact depends on surrounding block context. |
|
| 489 |     target_sa: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. | | |
| 490 | ) { | executable/support line; behavior impact depends on surrounding block context. |
| 491 |     let chain = FLASH_AUDIT_CHAIN.get_or_init(|| Mutex::new([0u8; 32])); | executable/support line; behavior impact depends on surrounding block context. |
| 492 |     let mut chain_guard = match chain.lock() { | executable/support line; behavior impact depends on surrounding block context. |
| 493 |         Ok(g) => g, | executable/support line; behavior impact depends on surrounding block context. |
| 494 |         Err(poisoned) => poisoned.into_inner(), | executable/support line; behavior impact depends on surrounding block context. |
| 495 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 496 |     let prev_hash = *chain_guard; | executable/support line; behavior impact depends on surrounding block context. |
| 497 |     let detail_str = detail.into(); | executable/support line; behavior impact depends on surrounding block context. |
| 498 |     let canonical = format!( | executable/support line; behavior impact depends on surrounding block context. |
| 499 |         "{}\|{\|{:?}\|{\|{:?}\|{\|{:?}\|{\|", | error-propagating operation; changing it impacts failure semantics and all recovery paths. |
| 500 |         now_ms(), | executable/support line; behavior impact depends on surrounding block context. |
| 501 |         stage, | executable/support line; behavior impact depends on surrounding block context. |
| 502 |         status, | executable/support line; behavior impact depends on surrounding block context. |
| 503 |         detail_str, | executable/support line; behavior impact depends on surrounding block context. |
| 504 |         target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 505 |         cfg.dry_run | executable/support line; behavior impact depends on surrounding block context. |
| 506 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 507 |     let mut hasher = Sha256::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 508 |     hasher.update(prev_hash); | executable/support line; behavior impact depends on surrounding block context. |
| 509 |     hasher.update(canonical.as_bytes()); | executable/support line; behavior impact depends on surrounding block context. |
| 510 |     let event_hash_arr: [u8; 32] = hasher.finalize().into(); | executable/support line; behavior impact depends on surrounding block context. |
| 511 | (blank) | line is blank; visual separation only. |
| 512 |     let event = FlashAuditEvent { | executable/support line; behavior impact depends on surrounding block context. |
| 513 |         ts_ms: now_ms(), | executable/support line; behavior impact depends on surrounding block context. |
| 514 |         stage, | executable/support line; behavior impact depends on surrounding block context. |
| 515 |         status, | executable/support line; behavior impact depends on surrounding block context. |
| 516 |         detail: detail_str, | executable/support line; behavior impact depends on surrounding block context. |
| 517 |         target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 518 |         dry_run: cfg.dry_run, | executable/support line; behavior impact depends on surrounding block context. |
| 519 |         prev_hash: hex_lower(&prev_hash), | executable/support line; behavior impact depends on surrounding block context. |
| 520 |         event_hash: hex_lower(&event_hash_arr), | executable/support line; behavior impact depends on surrounding block context. |
| 521 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 522 |     let serialized = serde_json::to_string(&event).unwrap_or_else(|_| { | serialization contract touchpoint; misplaced edits can break report compatibility and tooling. |
| 523 |         format!( | executable/support line; behavior impact depends on surrounding block context. |
| 524 |             "{{\"ts_ms\":{},\"stage\":\"{}\", \"status\":\"Failed\", \"detail\":\"audit_serialize_failed\", \"target_sa\":{}, \"dry_run\":{}}}", | executable/support line; behavior impact depends on surrounding block context. |
| 525 |             event.ts_ms, | executable/support line; behavior impact depends on surrounding block context. |
| 526 |             stage, | executable/support line; behavior impact depends on surrounding block context. |
| 527 |             target_sa | executable/support line; behavior impact depends on surrounding block context. |
| 528 |             .map(|sa| sa.to_string()) | executable/support line; behavior impact depends on surrounding block context. |
| 529 |             .unwrap_or_else(|_| "null".to_string()), | executable/support line; behavior impact depends on surrounding block context. |
| 530 |             cfg.dry_run | executable/support line; behavior impact depends on surrounding block context. |
| 531 |         ) | executable/support line; behavior impact depends on surrounding block context. |
| 532 |     }); | executable/support line; behavior impact depends on surrounding block context. |

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| 533 | (blank) | line is blank; visual separation only. |
| 534 |     let rec = JsonlRecord { | executable/support line; behavior impact depends on surrounding block context. |
| 535 |         ts: now_ms(), | executable/support line; behavior impact depends on surrounding block context. |
| 536 |         transport: "audit".to_string(), | executable/support line; behavior impact depends on surrounding block
context. |
| 537 |         dir: "flash-stage".to_string(), | I/O or protocol-critical logic; edits can affect integration, timing,
and diagnostic outcomes. |
| 538 |         id: None, | executable/support line; behavior impact depends on surrounding block context. |
| 539 |         dlc: None, | executable/support line; behavior impact depends on surrounding block context. |
| 540 |         data: Some(serialized), | executable/support line; behavior impact depends on surrounding block context. |
|
| 541 |         raw_hex: None, | executable/support line; behavior impact depends on surrounding block context. |
| 542 |         allowed: Some(cfg.write_effectively_enabled()), | I/O or protocol-critical logic; edits can affect integration,
timing, and diagnostic outcomes. |
| 543 |         dry_run: Some(cfg.dry_run), | executable/support line; behavior impact depends on surrounding block context. |
| 544 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 545 |     let _ = append_record(&cfg.log_dir.join("ecm_flash_audit.jsonl"), &rec); | I/O or protocol-critical logic; edits
can affect integration, timing, and diagnostic outcomes. |
| 546 |     *chain_guard = event_hash_arr; | executable/support line; behavior impact depends on surrounding block context. |
| 547 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 548 | (blank) | line is blank; visual separation only. |
| 549 | fn ensure_positive_response( | declares executable behavior; signature/body changes alter API, control flow,
outputs. |
| 550 |     rsp: &[u8], | executable/support line; behavior impact depends on surrounding block context. |
| 551 |     req_sid: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 552 |     stage: &str, | executable/support line; behavior impact depends on surrounding block context. |
| 553 |     cfg: &HwConfig, | executable/support line; behavior impact depends on surrounding block context. |
| 554 |     target_sa: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 555 | ) -> Result<(), HwError> { | executable/support line; behavior impact depends on surrounding block context. |
| 556 |     if rsp.is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 557 |         append_flash_audit( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic
outcomes. |
| 558 |             cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 559 |             stage, | executable/support line; behavior impact depends on surrounding block context. |
| 560 |             FlashAuditStatus::Failed, | executable/support line; behavior impact depends on surrounding block
context. |
| 561 |             "empty UDS response", | I/O or protocol-critical logic; edits can affect integration, timing, and
agnostic outcomes. |
| 562 |             target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 563 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 564 |         return Err(HwError::Unknown(format!( | executable/support line; behavior impact depends on surrounding
block context. |
| 565 |             "{stage}: empty UDS response for service 0x{req_sid:02X}" | I/O or protocol-critical logic; edits
can affect integration, timing, and diagnostic outcomes. |
| 566 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 567 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 568 | (blank) | line is blank; visual separation only. |
| 569 |     if rsp[0] == 0x7F { | conditional gate; changing condition shifts runtime path and side effects. |
| 570 |         let nrc = rsp.get(2).copied().unwrap_or(0x00); | executable/support line; behavior impact depends on
surrounding block context. |
| 571 |         let sid = rsp.get(1).copied().unwrap_or(0x00); | executable/support line; behavior impact depends on
surrounding block context. |
| 572 |         append_flash_audit( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic
outcomes. |
| 573 |             cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 574 |             stage, | executable/support line; behavior impact depends on surrounding block context. |
| 575 |             FlashAuditStatus::Failed, | executable/support line; behavior impact depends on surrounding block
context. |
| 576 |             format!("negative response SID=0x{sid:02X} NRC=0x{nrc:02X}"), | executable/support line; behavior
impact depends on surrounding block context. |
| 577 |             target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 578 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 579 |         return Err(HwError::Unknown(format!( | executable/support line; behavior impact depends on surrounding

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lock context. |
| 580 |         "{stage}: UDS negative response SID=0x{sid:02X}, NRC=0x{nrc:02X}" | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 581 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 582 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 583 | (blank) | line is blank; visual separation only. |
| 584 |     let expected = req_sid.wrapping_add(0x40); | executable/support line; behavior impact depends on surrounding block context. |
| 585 |     if rsp[0] != expected { | conditional gate; changing condition shifts runtime path and side effects. |
| 586 |         append_flash_audit( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 587 |             cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 588 |             stage, | executable/support line; behavior impact depends on surrounding block context. |
| 589 |             FlashAuditStatus::Failed, | executable/support line; behavior impact depends on surrounding block context. |
| 590 |             format!( | executable/support line; behavior impact depends on surrounding block context. |
| 591 |                 "unexpected positive response SID=0x{:02X}, expected=0x{expected:02X}", | executable/support line; behavior impact depends on surrounding block context. |
| 592 |                 rsp[0] | executable/support line; behavior impact depends on surrounding block context. |
| 593 |             ), | executable/support line; behavior impact depends on surrounding block context. |
| 594 |             target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 595 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 596 |             return Err(HwError::Unknown(format!( | executable/support line; behavior impact depends on surrounding block context. |
| 597 |                 "{stage}: unexpected response SID=0x{:02X}, expected=0x{expected:02X}", | executable/support line; behavior impact depends on surrounding block context. |
| 598 |                 rsp[0] | executable/support line; behavior impact depends on surrounding block context. |
| 599 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 600 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 601 | (blank) | line is blank; visual separation only. |
| 602 |     append_flash_audit( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 603 |         cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 604 |         stage, | executable/support line; behavior impact depends on surrounding block context. |
| 605 |         FlashAuditStatus::Ok, | executable/support line; behavior impact depends on surrounding block context. |
| 606 |         format!("positive response SID=0x{:02X}", rsp[0]), | executable/support line; behavior impact depends on surrounding block context. |
| 607 |         target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 608 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 609 |     Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 610 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 611 | (blank) | line is blank; visual separation only. |
| 612 | fn read_did_ascii( | declares executable behavior; signature/body changes alter API, control flow, and output |
| 613 |     adapter: &mut dyn super::hw::HardwareInterface, | executable/support line; behavior impact depends on surrounding block context. |
| 614 |     cfg: &HwConfig, | executable/support line; behavior impact depends on surrounding block context. |
| 615 |     target_sa: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 616 |     did: u16, | executable/support line; behavior impact depends on surrounding block context. |
| 617 |     stage: &str, | executable/support line; behavior impact depends on surrounding block context. |
| 618 | ) -> Result<String, HwError> { | executable/support line; behavior impact depends on surrounding block context. |
| 619 |     let req = [0x22, (did >> 8) as u8, (did & 0xFF) as u8]; | executable/support line; behavior impact depends on surrounding block context. |
| 620 |     let rsp = send_uds_and_wait( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 621 |         adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 622 |         cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 623 |         target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 624 |         &req, | executable/support line; behavior impact depends on surrounding block context. |
| 625 |         Duration::from_millis(cfg.uds_timeout_p2_ms.max(1)), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 626 |     ); | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 627 |     ensure_positive_response(&rsp, 0x22, stage, cfg, target_sa)?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |

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| 628 |     if rsp.len() < 3 { | conditional gate; changing condition shifts runtime path and side effects. |
| 629 |         return Err(HwError::ParseError { | executable/support line; behavior impact depends on surrounding block
context. |
| 630 |             cause: format!("{stage}: short DID response"), | executable/support line; behavior impact depends
surrounding block context. |
| 631 |             raw_data: format!("len={}", rsp.len()), | executable/support line; behavior impact depends on sur
nding block context. |
| 632 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 633 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 634 |     let rsp_did = ((rsp[1] as u16) << 8) \| rsp[2] as u16; | executable/support line; behavior impact depends
surrounding block context. |
| 635 |     if rsp_did != did { | conditional gate; changing condition shifts runtime path and side effects. |
| 636 |         return Err(HwError::ParseError { | executable/support line; behavior impact depends on surrounding block
context. |
| 637 |             cause: format!("{stage}: DID echo mismatch"), | executable/support line; behavior impact depends
surrounding block context. |
| 638 |             raw_data: format!("expected=0x{did:04X}, got=0x{rsp_did:04X}"), | executable/support line; behavi
impact depends on surrounding block context. |
| 639 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 640 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 641 |     let payload = &rsp[3..]; | executable/support line; behavior impact depends on surrounding block context. |
| 642 |     if payload.is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 643 |         return Err(HwError::ParseError { | executable/support line; behavior impact depends on surrounding block
context. |
| 644 |             cause: format!("{stage}: empty DID payload"), | executable/support line; behavior impact depends
surrounding block context. |
| 645 |             raw_data: format!("did=0x{did:04X}"), | executable/support line; behavior impact depends on surro
ing block context. |
| 646 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 647 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 648 |     let text = String::from_utf8(payload.to_vec()).map_err(\_| HwError::ParseError { | executable/support l
; behavior impact depends on surrounding block context. |
| 649 |         cause: format!("{stage}: non-utf8 DID payload"), | executable/support line; behavior impact depends o
urrounding block context. |
| 650 |         raw_data: format!("did=0x{did:04X}, bytes={}", hex_lower(payload)), | executable/support line; behavi
impact depends on surrounding block context. |
| 651 |     })?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 652 |     Ok(text.trim().to_string()) | executable/support line; behavior impact depends on surrounding block conte
|
| 653 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 654 | (blank) | line is blank; visual separation only. |
| 655 | fn read_did_battery_voltage( | declares executable behavior; signature/body changes alter API, control flow,
outputs. |
| 656 |     adapter: &mut dyn super::hw::HardwareInterface, | executable/support line; behavior impact depends on sur
nding block context. |
| 657 |     cfg: &HwConfig, | executable/support line; behavior impact depends on surrounding block context. |
| 658 |     target_sa: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 659 | ) -> Result<f64, HwError> { | executable/support line; behavior impact depends on surrounding block context. |
| 660 |     let req = [0x22, (DID_BATTERY_VOLTAGE >> 8) as u8, (DID_BATTERY_VOLTAGE & 0xFF) as u8]; | executable/supp
line; behavior impact depends on surrounding block context. |
| 661 |     let rsp = send_uds_and_wait( | I/O or protocol-critical logic; edits can affect integration, timing, and
gnostic outcomes. |
| 662 |         adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 663 |         cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 664 |         target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 665 |         &req, | executable/support line; behavior impact depends on surrounding block context. |
| 666 |         Duration::from_millis(cfg.uds_timeout_p2_ms.max(1)), | I/O or protocol-critical logic; edits can affe
integration, timing, and diagnostic outcomes. |
| 667 |     )?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 668 |     ensure_positive_response(&rsp, 0x22, "preflight::battery_did", cfg, target_sa)?; | error-propagating oper
on; changing it impacts failure semantics and caller recovery paths. |
| 669 |     if rsp.len() < 5 { | conditional gate; changing condition shifts runtime path and side effects. |
| 670 |         return Err(HwError::ParseError { | executable/support line; behavior impact depends on surrounding block
context. |
| 671 |             cause: "preflight::battery_did short payload".to_string(), | executable/support line; behavior im

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t depends on surrounding block context. |
| 672 |         raw_data: format!("{}", rsp.len()), | executable/support line; behavior impact depends on surrounding block context. |
| 673 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 674 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 675 | let raw = u16::from_be_bytes([rsp[3], rsp[4]]); | executable/support line; behavior impact depends on surrounding block context. |
| 676 | Ok((raw as f64) * 0.05) | executable/support line; behavior impact depends on surrounding block context. |
| 677 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 678 | (blank) | line is blank; visual separation only. |
| 679 | fn decode_battery_voltage_from_can(cf: &CanFrame) -> Option<f64> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 680 |     let pgn = j1939_pgn(cf.id); | executable/support line; behavior impact depends on surrounding block context. |
| 681 |     if !matches!(pgn, 65271 \| 65272) { | conditional gate; changing condition shifts runtime path and side effects. |
| 682 |         return None; | executable/support line; behavior impact depends on surrounding block context. |
| 683 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 684 |     let p = can_payload(cf); | executable/support line; behavior impact depends on surrounding block context. |
| 685 |     if p.len() < 2 { | conditional gate; changing condition shifts runtime path and side effects. |
| 686 |         return None; | executable/support line; behavior impact depends on surrounding block context. |
| 687 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 688 |     let raw = u16::from_le_bytes([p[0], p[1]]); | executable/support line; behavior impact depends on surrounding block context. |
| 689 |     if raw == 0xFFFF { | conditional gate; changing condition shifts runtime path and side effects. |
| 690 |         return None; | executable/support line; behavior impact depends on surrounding block context. |
| 691 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 692 |     Some((raw as f64) * 0.05) | executable/support line; behavior impact depends on surrounding block context. |
| 693 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 694 | (blank) | line is blank; visual separation only. |
| 695 | fn sample_supply_voltage_from_bus( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 696 |     adapter: &mut dyn super::hw::HardwareInterface, | executable/support line; behavior impact depends on surrounding block context. |
| 697 |     timeout: Duration, | executable/support line; behavior impact depends on surrounding block context. |
| 698 | ) -> Result<Option<f64>, HwError> { | executable/support line; behavior impact depends on surrounding block context. |
| 699 |     let deadline = Instant::now() + timeout; | executable/support line; behavior impact depends on surrounding block context. |
| 700 |     let mut max_v = f64::NEG_INFINITY; | executable/support line; behavior impact depends on surrounding block context. |
| 701 |     while Instant::now() < deadline { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 702 |         match adapter.read_frame() { | branch dispatch; changing arms alters behavior for variants and input cases. |
| 703 |             Ok(Frame::Can(cf)) => { | executable/support line; behavior impact depends on surrounding block context. |
| 704 |                 if let Some(v) = decode_battery_voltage_from_can(&cf) { | conditional gate; changing condition shifts runtime path and side effects. |
| 705 |                     max_v = max_v.max(v); | executable/support line; behavior impact depends on surrounding block context. |
| 706 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 707 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 708 |             Ok(Frame::Serial(_)) => {} | executable/support line; behavior impact depends on surrounding block context. |
| 709 |             Err(HwError::Timeout) => {} | executable/support line; behavior impact depends on surrounding block context. |
| 710 |             Err(e) => return Err(e), | executable/support line; behavior impact depends on surrounding block context. |
| 711 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 712 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 713 |     if max_v.is_finite() { | conditional gate; changing condition shifts runtime path and side effects. |

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| 714 |         Ok(Some(max_v)) | executable/support line; behavior impact depends on surrounding block context. |
| 715 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 716 |         Ok(None) | executable/support line; behavior impact depends on surrounding block context. |
| 717 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 718 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 719 | (blank) | line is blank; visual separation only. |
| 720 | fn read_ecu_identity( | declares executable behavior; signature/body changes alter API, control flow, and out
s. |
| 721 |     adapter: &mut dyn super::hw::HardwareInterface, | executable/support line; behavior impact depends on surr
nding block context. |
| 722 |     cfg: &HwConfig, | executable/support line; behavior impact depends on surrounding block context. |
| 723 |     target_sa: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 724 | ) -> Result<EcuIdentity, HwError> { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 725 |     let vin = read_did_ascii(adapter, cfg, target_sa, DID_VIN, "id::vin"?; | error-propagating operation; ch
ing it impacts failure semantics and caller recovery paths. |
| 726 |     let sw = read_did_ascii(adapter, cfg, target_sa, DID_SW_VERSION, "id::sw"?; | error-propagating operatio
changing it impacts failure semantics and caller recovery paths. |
| 727 |     let hw = read_did_ascii(adapter, cfg, target_sa, DID_HW_VERSION, "id::hw"?; | error-propagating operatio
changing it impacts failure semantics and caller recovery paths. |
| 728 |     let cal = read_did_ascii(adapter, cfg, target_sa, DID_CALIBRATION_ID, "id::cal"?; | error-propagating op
tion; changing it impacts failure semantics and caller recovery paths. |
| 729 |     let serial = read_did_ascii(adapter, cfg, target_sa, DID_ECU_SERIAL, "id::serial"?; | error-propagating
ration; changing it impacts failure semantics and caller recovery paths. |
| 730 |     let fingerprint = read_did_ascii( | executable/support line; behavior impact depends on surrounding block
ntext. |
| 731 |         adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 732 |         cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 733 |         target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 734 |         DID_FINGERPRINT, | executable/support line; behavior impact depends on surrounding block context. |
| 735 |         "id::fingerprint", | executable/support line; behavior impact depends on surrounding block context. |
| 736 |     ) | executable/support line; behavior impact depends on surrounding block context. |
| 737 |     .ok(); | executable/support line; behavior impact depends on surrounding block context. |
| 738 | (blank) | line is blank; visual separation only. |
| 739 |     Ok(EcuIdentity { | executable/support line; behavior impact depends on surrounding block context. |
| 740 |         vin, | executable/support line; behavior impact depends on surrounding block context. |
| 741 |         sw_version: sw, | executable/support line; behavior impact depends on surrounding block context. |
| 742 |         hw_version: hw, | executable/support line; behavior impact depends on surrounding block context. |
| 743 |         calibration_id: cal, | executable/support line; behavior impact depends on surrounding block context. |
| 744 |         ecu_serial: serial, | executable/support line; behavior impact depends on surrounding block context. |
| 745 |         fingerprint, | executable/support line; behavior impact depends on surrounding block context. |
| 746 |     }) | executable/support line; behavior impact depends on surrounding block context. |
| 747 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 748 | (blank) | line is blank; visual separation only. |
| 749 | fn run_flash_preflight( | declares executable behavior; signature/body changes alter API, control flow, and o
uts. |
| 750 |     adapter: &mut dyn super::hw::HardwareInterface, | executable/support line; behavior impact depends on surr
nding block context. |
| 751 |     cfg: &HwConfig, | executable/support line; behavior impact depends on surrounding block context. |
| 752 |     target_sa: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 753 | ) -> Result<FlashPreflight, HwError> { | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 754 |     let supply_voltage = match sample_supply_voltage_from_bus(adapter, Duration::from_millis(500))? { | error
opagating operation; changing it impacts failure semantics and caller recovery paths. |
| 755 |         Some(v) => v, | executable/support line; behavior impact depends on surrounding block context. |
| 756 |         None => read_did_battery_voltage(adapter, cfg, target_sa)?, | error-propagating operation; changing i
mpacts failure semantics and caller recovery paths. |
| 757 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 758 |     if supply_voltage < MIN_FLASH_SUPPLY_V { | conditional gate; changing condition shifts runtime path and s
effects. |
| 759 |         return Err(HwError::Unknown(format!( | executable/support line; behavior impact depends on surroundin
lock context. |
| 760 |             "preflight failed: supply voltage {:.2}V below minimum {:.2}V", | executable/support line; behavi
impact depends on surrounding block context. |
| 761 |             supply_voltage, MIN_FLASH_SUPPLY_V | executable/support line; behavior impact depends on surround

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block context. |
| 762 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 763 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 764 | (blank) | line is blank; visual separation only. |
| 765 |     let identity = read_ecu_identity(adapter, cfg, target_sa)?; | error-propagating operation; changing it im
ts failure semantics and caller recovery paths. |
| 766 |     if !cfg.allow_untrusted_ecu && identity.fingerprint.is_none() { | conditional gate; changing condition sh
s runtime path and side effects. |
| 767 |         return Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block
text. |
| 768 |         "preflight failed: ECU fingerprint DID 0xF184 not available; use trusted ECU profile or --allow-u
usted-ecu only in controlled bench".to_string(), | executable/support line; behavior impact depends on surrounding bl
context. |
| 769 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 770 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 771 | (blank) | line is blank; visual separation only. |
| 772 |     Ok(FlashPreflight { | executable/support line; behavior impact depends on surrounding block context. |
| 773 |         supply_voltage_v: supply_voltage, | executable/support line; behavior impact depends on surrounding b
k context. |
| 774 |         identity, | executable/support line; behavior impact depends on surrounding block context. |
| 775 |     }) | executable/support line; behavior impact depends on surrounding block context. |
| 776 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 777 | (blank) | line is blank; visual separation only. |
| 778 | pub fn live_read_ecm_identity(cfg: &HwConfig, target_sa: Option<u8>) -> Result<EcuIdentity, HwError> { | decl
s executable behavior; signature/body changes alter API, control flow, and outputs. |
| 779 |     if cfg.mode != HwMode::Live { | conditional gate; changing condition shifts runtime path and side effects
| 780 |         return Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block
text. |
| 781 |         "identity read requires --hw-mode=live".to_string(), | executable/support line; behavior impact d
nds on surrounding block context. |
| 782 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 783 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 784 |     check_live_channel_policy(cfg)?; | error-propagating operation; changing it impacts failure semantics and
ller recovery paths. |
| 785 | (blank) | line is blank; visual separation only. |
| 786 |     let mut adapter = super::hw::open_real_adapter(cfg)?; | error-propagating operation; changing it impacts
lure semantics and caller recovery paths. |
| 787 |     adapter.init(cfg)?; | error-propagating operation; changing it impacts failure semantics and caller recov
paths. |
| 788 |     let identity = read_ecu_identity(&mut *adapter, cfg, target_sa)?; | error-propagating operation; changing
impacts failure semantics and caller recovery paths. |
| 789 |     adapter.close()?; | error-propagating operation; changing it impacts failure semantics and caller recover
aths. |
| 790 |     Ok(identity) | executable/support line; behavior impact depends on surrounding block context. |
| 791 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 792 | (blank) | line is blank; visual separation only. |
| 793 | pub fn live_preflight_ecm( | declares executable behavior; signature/body changes alter API, control flow, an
utputs. |
| 794 |     cfg: &HwConfig, | executable/support line; behavior impact depends on surrounding block context. |
| 795 |     target_sa: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 796 | ) -> Result<FlashPreflightReport, HwError> { | executable/support line; behavior impact depends on surroundin
lock context. |
| 797 |     if cfg.mode != HwMode::Live { | conditional gate; changing condition shifts runtime path and side effects
| 798 |         return Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block
text. |
| 799 |         "preflight requires --hw-mode=live".to_string(), | executable/support line; behavior impact depen
on surrounding block context. |
| 800 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 801 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 802 |     check_live_channel_policy(cfg)?; | error-propagating operation; changing it impacts failure semantics and
ller recovery paths. |
| 803 | (blank) | line is blank; visual separation only. |
| 804 |     let mut adapter = super::hw::open_real_adapter(cfg)?; | error-propagating operation; changing it impacts
lure semantics and caller recovery paths. |
| 805 |     adapter.init(cfg)?; | error-propagating operation; changing it impacts failure semantics and caller recov

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paths. |
| 806 |     let preflight = run_flash_preflight(&mut *adapter, cfg, target_sa)?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. | | |
| 807 |     adapter.close()?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 808 | (blank) | line is blank; visual separation only. |
| 809 |     Ok(FlashPreflightReport { | executable/support line; behavior impact depends on surrounding block context. |
| 810 |         supply_voltage_v: preflight.supply_voltage_v, | executable/support line; behavior impact depends on surrounding block context. |
| 811 |         trusted_fingerprint: preflight.identity.fingerprint.is_some(), | executable/support line; behavior impact depends on surrounding block context. |
| 812 |         identity: preflight.identity, | executable/support line; behavior impact depends on surrounding block context. |
| 813 |     }) | executable/support line; behavior impact depends on surrounding block context. |
| 814 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 815 | (blank) | line is blank; visual separation only. |
| 816 | fn j1939_uds_req_id(dst_sa: u8) -> u32 { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 817 |     0x18DA0000 \ | ((dst_sa as u32) << 8) \ | 0xF9 | executable/support line; behavior impact depends on surrounding block context. |
| 818 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 819 | (blank) | line is blank; visual separation only. |
| 820 | fn is_uds_response_id(target_sa: Option<u8>, id: u32) -> bool { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 821 |     let pf = ((id >> 16) & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 822 |     let ps = ((id >> 8) & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 823 |     let sa = (id & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 824 |     pf == 0xDA && ps == 0xF9 && target_sa.is_none_or(|t| t == sa) | executable/support line; behavior impact depends on surrounding block context. |
| 825 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 826 | (blank) | line is blank; visual separation only. |
| 827 | fn can_payload(cf: &CanFrame) -> &[u8] { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 828 |     &cf.data[..cf.len.min(8)] | executable/support line; behavior impact depends on surrounding block context. |
| 829 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 830 | (blank) | line is blank; visual separation only. |
| 831 | fn send_frame_retry( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 832 |     adapter: &mut dyn super::hw::HardwareInterface, | executable/support line; behavior impact depends on surrounding block context. |
| 833 |     frame: Frame, | executable/support line; behavior impact depends on surrounding block context. |
| 834 |     retry_count: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 835 |     backoff_ms: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 836 | ) -> Result<(), HwError> { | executable/support line; behavior impact depends on surrounding block context. |
| 837 |     let max_tries = retry_count.saturating_add(1); | executable/support line; behavior impact depends on surrounding block context. |
| 838 |     let mut try_idx = 0u8; | executable/support line; behavior impact depends on surrounding block context. |
| 839 |     loop { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 840 |         match adapter.send_frame(frame.clone()) { | branch dispatch; changing arms alters behavior for various and input classes. |
| 841 |             Ok(()) => return Ok(()), | executable/support line; behavior impact depends on surrounding block context. |
| 842 |             Err(HwError::Timeout) | executable/support line; behavior impact depends on surrounding block context. |
| 843 |             \ | Err(HwError::RateLimited) | executable/support line; behavior impact depends on surrounding block context. |
| 844 |             \ | Err(HwError::TransceiverError) | executable/support line; behavior impact depends on surrounding block context. |
| 845 |             \ | Err(HwError::BusOff) | executable/support line; behavior impact depends on surrounding block context. |
| 846 |                 if try_idx + 1 < max_tries => | conditional gate; changing condition shifts runtime path and

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e effects. |
| 847 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 848 |             try_idx = try_idx.saturating_add(1); | executable/support line; behavior impact depends on su
unding block context. |
| 849 |             thread::sleep(Duration::from_millis(backoff_ms.max(1))); | executable/support line; behavior
act depends on surrounding block context. |
| 850 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 851 |         Err(e) => return Err(e), | executable/support line; behavior impact depends on surrounding block
text. |
| 852 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 853 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 854 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 855 | (blank) | line is blank; visual separation only. |
| 856 | fn send_can_sf( | declares executable behavior; signature/body changes alter API, control flow, and outputs.
| 857 |     adapter: &mut dyn super::hw::HardwareInterface, | executable/support line; behavior impact depends on sur
nding block context. |
| 858 |     cfg: &HwConfig, | executable/support line; behavior impact depends on surrounding block context. |
| 859 |     dst_sa: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 860 |     payload: &[u8], | executable/support line; behavior impact depends on surrounding block context. |
| 861 | ) -> Result<(), HwError> { | executable/support line; behavior impact depends on surrounding block context. |
| 862 |     if payload.len() > 7 { | conditional gate; changing condition shifts runtime path and side effects. |
| 863 |         return Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block
text. |
| 864 |             "CAN single-frame payload exceeds 7 bytes".to_string(), | executable/support line; behavior impac
depends on surrounding block context. |
| 865 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 866 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 867 | (blank) | line is blank; visual separation only. |
| 868 |     let mut data = [0u8; 8]; | executable/support line; behavior impact depends on surrounding block context.
| 869 |     data[0] = payload.len() as u8; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 870 |     if !payload.is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 871 |         data[1..(1 + payload.len())].copy_from_slice(payload); | executable/support line; behavior impact dep
s on surrounding block context. |
| 872 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 873 | (blank) | line is blank; visual separation only. |
| 874 |     let frame = Frame::Can(CanFrame { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 875 |         id: j1939_uds_req_id(dst_sa), | I/O or protocol-critical logic; edits can affect integration, timing,
d diagnostic outcomes. |
| 876 |         dlc: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 877 |         data, | executable/support line; behavior impact depends on surrounding block context. |
| 878 |         len: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 879 |         timestamp_ms: Some(now_ms()), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 880 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 881 |     send_frame_retry( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic
comes. |
| 882 |         adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 883 |         frame, | executable/support line; behavior impact depends on surrounding block context. |
| 884 |         cfg.uds_retry_count, | I/O or protocol-critical logic; edits can affect integration, timing, and diag
tic outcomes. |
| 885 |         cfg.write_retry_backoff_ms, | I/O or protocol-critical logic; edits can affect integration, timing, a
diagnostic outcomes. |
| 886 |     ) | executable/support line; behavior impact depends on surrounding block context. |
| 887 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 888 | (blank) | line is blank; visual separation only. |
| 889 | fn wait_for_fc( | declares executable behavior; signature/body changes alter API, control flow, and outputs.
| 890 |     adapter: &mut dyn super::hw::HardwareInterface, | executable/support line; behavior impact depends on sur
nding block context. |
| 891 |     cfg: &HwConfig, | executable/support line; behavior impact depends on surrounding block context. |
| 892 |     target_sa: u8, | executable/support line; behavior impact depends on surrounding block context. |

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| 893 |     timeout: Duration, | executable/support line; behavior impact depends on surrounding block context. |
| 894 | ) -> Result<(u8, u64), HwError> { | executable/support line; behavior impact depends on surrounding block con
t. |
| 895 |     let deadline = Instant::now() + timeout; | executable/support line; behavior impact depends on surroundin
lock context. |
| 896 |     while Instant::now() < deadline { | iteration control; changing bounds/body affects throughput, timing, a
possibly safety guarantees. |
| 897 |         match adapter.read_frame() { | branch dispatch; changing arms alters behavior for variants and input
sses. |
| 898 |             Ok(Frame::Can(cf)) => { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 899 |                 if !is_uds_response_id(Some(target_sa), cf.id) { | conditional gate; changing condition shift
untime path and side effects. |
| 900 |                     continue; | executable/support line; behavior impact depends on surrounding block context
| 901 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 902 |                 let p = can_payload(&cf); | executable/support line; behavior impact depends on surrounding b
k context. |
| 903 |                 if p.len() < 3 { | conditional gate; changing condition shifts runtime path and side effects.
| 904 |                     continue; | executable/support line; behavior impact depends on surrounding block context
| 905 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 906 |                 let pci_type = (p[0] >> 4) & 0x0F; | executable/support line; behavior impact depends on surr
ding block context. |
| 907 |                 if pci_type != 0x3 { | conditional gate; changing condition shifts runtime path and side effe
. |
| 908 |                     continue; | executable/support line; behavior impact depends on surrounding block context
| 909 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 910 |                 let fs = p[0] & 0x0F; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 911 |                 if fs == 0x1 { | conditional gate; changing condition shifts runtime path and side effects. |
| 912 |                     return Err(HwError::RateLimited); | executable/support line; behavior impact depends on s
ounding block context. |
| 913 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 914 |                 if fs == 0x2 { | conditional gate; changing condition shifts runtime path and side effects. |
| 915 |                     return Err(HwError::Timeout); | executable/support line; behavior impact depends on surro
ing block context. |
| 916 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 917 |                 let bs = p[1]; | executable/support line; behavior impact depends on surrounding block context
|
| 918 |                 let st_min = p[2] as u64; | executable/support line; behavior impact depends on surrounding b
k context. |
| 919 |                 return Ok((bs, st_min.max(cfg.uds_st_min_ms))); | I/O or protocol-critical logic; edits can a
ct integration, timing, and diagnostic outcomes. |
| 920 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 921 |             Ok(_) => {} | executable/support line; behavior impact depends on surrounding block context. |
| 922 |             Err(HwError::Timeout) => {} | executable/support line; behavior impact depends on surrounding blo
context. |
| 923 |             Err(e) => return Err(e), | executable/support line; behavior impact depends on surrounding block
text. |
| 924 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 925 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 926 |     Err(HwError::Timeout) | executable/support line; behavior impact depends on surrounding block context. |
| 927 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 928 | (blank) | line is blank; visual separation only. |
| 929 | fn send_can_multiframe( | declares executable behavior; signature/body changes alter API, control flow, and o
uts. |
| 930 |     adapter: &mut dyn super::hw::HardwareInterface, | executable/support line; behavior impact depends on sur
nding block context. |
| 931 |     cfg: &HwConfig, | executable/support line; behavior impact depends on surrounding block context. |

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| 932 |     dst_sa: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 933 |     payload: &[u8], | executable/support line; behavior impact depends on surrounding block context. |
| 934 | ) -> Result<(), HwError> { | executable/support line; behavior impact depends on surrounding block context. |
| 935 |     if payload.len() <= 7 { | conditional gate; changing condition shifts runtime path and side effects. |
| 936 |         return send_can_sf(adapter, cfg, dst_sa, payload); | I/O or protocol-critical logic; edits can affect
integration, timing, and diagnostic outcomes. |
| 937 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 938 |     if payload.len() > 4095 { | conditional gate; changing condition shifts runtime path and side effects. |
| 939 |         return Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block
text. |
| 940 |             "CAN ISO-TP payload too large (>4095 bytes)".to_string(), | executable/support line; behavior imp
depends on surrounding block context. |
| 941 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 942 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 943 | (blank) | line is blank; visual separation only. |
| 944 |     let total_len = payload.len(); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 945 |     let mut ff = [0u8; 8]; | executable/support line; behavior impact depends on surrounding block context. |
| 946 |     ff[0] = 0x10 \ | (((total_len >> 8) & 0x0F) as u8); | executable/support line; behavior impact depends on
rounding block context. |
| 947 |     ff[1] = (total_len & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 948 |     ff[2..8].copy_from_slice(&payload[..6]); | executable/support line; behavior impact depends on surroundin
lock context. |
| 949 | (blank) | line is blank; visual separation only. |
| 950 |     let ff_frame = Frame::Can(CanFrame { | executable/support line; behavior impact depends on surrounding bl
context. |
| 951 |         id: j1939_uds_req_id(dst_sa), | I/O or protocol-critical logic; edits can affect integration, timing,
d diagnostic outcomes. |
| 952 |         dlc: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 953 |         data: ff, | executable/support line; behavior impact depends on surrounding block context. |
| 954 |         len: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 955 |         timestamp_ms: Some(now_ms()), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 956 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 957 |     send_frame_retry( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic
comes. |
| 958 |         adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 959 |         ff_frame, | executable/support line; behavior impact depends on surrounding block context. |
| 960 |         cfg.uds_retry_count, | I/O or protocol-critical logic; edits can affect integration, timing, and diag
tic outcomes. |
| 961 |         cfg.write_retry_backoff_ms, | I/O or protocol-critical logic; edits can affect integration, timing, a
diagnostic outcomes. |
| 962 |     )?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 963 | (blank) | line is blank; visual separation only. |
| 964 |     let (mut bs, mut st_min) = wait_for_fc( | executable/support line; behavior impact depends on surrounding
ock context. |
| 965 |         adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 966 |         cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 967 |         dst_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 968 |         Duration::from_millis(cfg.uds_timeout_p2_ms.max(1)), | I/O or protocol-critical logic; edits can affe
integration, timing, and diagnostic outcomes. |
| 969 |     )?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 970 |     if cfg.uds_block_size > 0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 971 |         bs = cfg.uds_block_size; | I/O or protocol-critical logic; edits can affect integration, timing, and
gnostic outcomes. |
| 972 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 973 |     st_min = st_min.max(cfg.uds_st_min_ms); | I/O or protocol-critical logic; edits can affect integration, t
ng, and diagnostic outcomes. |
| 974 | (blank) | line is blank; visual separation only. |
| 975 |     let mut idx = 6usize; | executable/support line; behavior impact depends on surrounding block context. |
| 976 |     let mut sn = 1u8; | executable/support line; behavior impact depends on surrounding block context. |
| 977 |     let mut sent_in_block = 0u8; | executable/support line; behavior impact depends on surrounding block cont
. |
| 978 | (blank) | line is blank; visual separation only. |

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| 979 |     while idx < payload.len() { | iteration control; changing bounds/body affects throughput, timing, and pos
ly safety guarantees. |
| 980 |         let end = (idx + 7).min(payload.len()); | executable/support line; behavior impact depends on surroun
g block context. |
| 981 |         let mut data = [0u8; 8]; | executable/support line; behavior impact depends on surrounding block cont
. |
| 982 |         data[0] = 0x20 \| (sn & 0x0F); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 983 |         let count = end - idx; | executable/support line; behavior impact depends on surrounding block context
|
| 984 |         data[1..(1 + count)].copy_from_slice(&payload[idx..end]); | executable/support line; behavior impact
ends on surrounding block context. |
| 985 | (blank) | line is blank; visual separation only. |
| 986 |         let cf = Frame::Can(CanFrame { | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 987 |             id: j1939_uds_req_id(dst_sa), | I/O or protocol-critical logic; edits can affect integration, tim
, and diagnostic outcomes. |
| 988 |             dlc: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 989 |             data, | executable/support line; behavior impact depends on surrounding block context. |
| 990 |             len: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 991 |             timestamp_ms: Some(now_ms()), | executable/support line; behavior impact depends on surrounding b
k context. |
| 992 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 993 |         send_frame_retry(adapter, cf, cfg.uds_retry_count, cfg.write_retry_backoff_ms)?; | error-propagating
ration; changing it impacts failure semantics and caller recovery paths. |
| 994 | (blank) | line is blank; visual separation only. |
| 995 |         idx = end; | executable/support line; behavior impact depends on surrounding block context. |
| 996 |         sn = (sn + 1) & 0x0F; | executable/support line; behavior impact depends on surrounding block context
|
| 997 |         sent_in_block = sent_in_block.saturating_add(1); | executable/support line; behavior impact depends o
urrounding block context. |
| 998 | (blank) | line is blank; visual separation only. |
| 999 |         if st_min > 0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 1000 |             thread::sleep(Duration::from_millis(st_min)); | executable/support line; behavior impact depends
surrounding block context. |
| 1001 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1002 | (blank) | line is blank; visual separation only. |
| 1003 |         if bs > 0 && sent_in_block >= bs && idx < payload.len() { | conditional gate; changing condition shi
runtime path and side effects. |
| 1004 |             let (new_bs, new_st) = wait_for_fc( | executable/support line; behavior impact depends on surroun
ng block context. |
| 1005 |                 adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 1006 |                 cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1007 |                 dst_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1008 |                 Duration::from_millis(cfg.uds_timeout_p2star_ms.max(1)), | I/O or protocol-critical logic; e
s can affect integration, timing, and diagnostic outcomes. |
| 1009 |             ); | error-propagating operation; changing it impacts failure semantics and caller recovery pat
|
| 1010 |             sent_in_block = 0; | executable/support line; behavior impact depends on surrounding block conte
|
| 1011 |             if cfg.uds_block_size == 0 { | conditional gate; changing condition shifts runtime path and side
fects. |
| 1012 |                 bs = new_bs; | executable/support line; behavior impact depends on surrounding block context
|
| 1013 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1014 |             st_min = new_st.max(cfg.uds_st_min_ms); | I/O or protocol-critical logic; edits can affect integ
ion, timing, and diagnostic outcomes. |
| 1015 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1016 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
|
| 1017 | (blank) | line is blank; visual separation only. |
| 1018 |     Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 1019 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1020 | (blank) | line is blank; visual separation only. |
| 1021 | fn send_fc_cts( | declares executable behavior; signature/body changes alter API, control flow, and outputs.

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| 1022 |     adapter: &mut dyn super::hw::HardwareInterface, | executable/support line; behavior impact depends on surrounding block context. |
| 1023 |     cfg: &HwConfig, | executable/support line; behavior impact depends on surrounding block context. |
| 1024 |     target_sa: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 1025 | ) -> Result<(), HwError> { | executable/support line; behavior impact depends on surrounding block context. |
| 1026 |     let mut data = [0u8; 8]; | executable/support line; behavior impact depends on surrounding block context. |
| 1027 |     data[0] = 0x30; | executable/support line; behavior impact depends on surrounding block context. |
| 1028 |     data[1] = cfg.uds_block_size; | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1029 |     data[2] = cfg.uds_st_min_ms.min(0x7F) as u8; | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1030 | (blank) | line is blank; visual separation only. |
| 1031 |     let frame = Frame::Can(CanFrame { | executable/support line; behavior impact depends on surrounding block context. |
| 1032 |         id: j1939_uds_req_id(target_sa), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1033 |         dlc: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 1034 |         data, | executable/support line; behavior impact depends on surrounding block context. |
| 1035 |         len: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 1036 |         timestamp_ms: Some(now_ms()), | executable/support line; behavior impact depends on surrounding block context. |
| 1037 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 1038 |     send_frame_retry( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1039 |         adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 1040 |         frame, | executable/support line; behavior impact depends on surrounding block context. |
| 1041 |         cfg.uds_retry_count, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1042 |         cfg.write_retry_backoff_ms, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1043 |     ) | executable/support line; behavior impact depends on surrounding block context. |
| 1044 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1045 | (blank) | line is blank; visual separation only. |
| 1046 | fn recv_can_uds_response( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1047 |     adapter: &mut dyn super::hw::HardwareInterface, | executable/support line; behavior impact depends on surrounding block context. |
| 1048 |     cfg: &HwConfig, | executable/support line; behavior impact depends on surrounding block context. |
| 1049 |     target_sa: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 1050 |     timeout: Duration, | executable/support line; behavior impact depends on surrounding block context. |
| 1051 | ) -> Result<Vec<u8>, HwError> { | executable/support line; behavior impact depends on surrounding block context. |
| 1052 |     let mut deadline = Instant::now() + timeout; | executable/support line; behavior impact depends on surrounding block context. |
| 1053 |     let mut last_keepalive = Instant::now(); | executable/support line; behavior impact depends on surrounding block context. |
| 1054 | (blank) | line is blank; visual separation only. |
| 1055 |     loop { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 1056 |         if Instant::now() >= deadline { | conditional gate; changing condition shifts runtime path and side effects. |
| 1057 |             return Err(HwError::Timeout); | executable/support line; behavior impact depends on surrounding block context. |
| 1058 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1059 | (blank) | line is blank; visual separation only. |
| 1060 |         match adapter.read_frame() { | branch dispatch; changing arms alters behavior for variants and inputs. |
| 1061 |             Ok(Frame::Can(cf)) => { | executable/support line; behavior impact depends on surrounding block context. |
| 1062 |                 if !is_uds_response_id(target_sa, cf.id) { | conditional gate; changing condition shifts runtime path and side effects. |
| 1063 |                     continue; | executable/support line; behavior impact depends on surrounding block context. |
| 1064 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime

```



```

scopes. |
| 1065 |           let p = can_payload(&cf); | executable/support line; behavior impact depends on surrounding
ck context. |
| 1066 |           if p.is_empty() { | conditional gate; changing condition shifts runtime path and side effect
|
| 1067 |           continue; | executable/support line; behavior impact depends on surrounding block context
|
| 1068 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1069 | (blank) | line is blank; visual separation only. |
| 1070 |           let pci_type = (p[0] >> 4) & 0x0F; | executable/support line; behavior impact depends on surr
nding block context. |
| 1071 |           if pci_type == 0x0 { | conditional gate; changing condition shifts runtime path and side eff
s. |
| 1072 |           let l = (p[0] & 0x0F) as usize; | executable/support line; behavior impact depends on su
unding block context. |
| 1073 |           if p.len() < 1 + l { | conditional gate; changing condition shifts runtime path and side
fects. |
| 1074 |           continue; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1075 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 1076 |           let payload = p[1..(1 + l)].to_vec(); | executable/support line; behavior impact depends
surrounding block context. |
| 1077 |           if payload.len() >= 3 && payload[0] == 0x7F && payload[2] == 0x78 { | conditional gate;
nging condition shifts runtime path and side effects. |
| 1078 |           deadline = Instant::now() | executable/support line; behavior impact depends on surr
ding block context. |
| 1079 |           + Duration::from_millis(cfg.uds_timeout_p2star_ms.max(1)); | I/O or protocol-cri
al logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1080 |           continue; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1081 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 1082 |           return Ok(payload); | executable/support line; behavior impact depends on surrounding bl
context. |
| 1083 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1084 | (blank) | line is blank; visual separation only. |
| 1085 |           if pci_type == 0x1 { | conditional gate; changing condition shifts runtime path and side eff
s. |
| 1086 |           if p.len() < 2 { | conditional gate; changing condition shifts runtime path and side eff
s. |
| 1087 |           continue; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1088 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 1089 |           let total_len = (((p[0] as usize) & 0x0F) << 8) \| (p[1] as usize); | executable/support
ne; behavior impact depends on surrounding block context. |
| 1090 |           if total_len == 0 { | conditional gate; changing condition shifts runtime path and side
ects. |
| 1091 |           continue; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1092 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 1093 | (blank) | line is blank; visual separation only. |
| 1094 |           let src_sa = (cf.id & 0xFF) as u8; | executable/support line; behavior impact depends on
rrounding block context. |
| 1095 |           send_fc_cts(adapter, cfg, src_sa)?; | error-propagating operation; changing it impacts f
ure semantics and caller recovery paths. |
| 1096 | (blank) | line is blank; visual separation only. |
| 1097 |           let mut out = Vec::with_capacity(total_len); | executable/support line; behavior impact
ends on surrounding block context. |
| 1098 |           out.extend_from_slice(&p[2..p.len().min(8)]); | executable/support line; behavior impact
ends on surrounding block context. |

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| 1099 |           let mut expected_sn = 1u8; | executable/support line; behavior impact depends on surround
g block context. |
| 1100 | (blank) | line is blank; visual separation only. |
| 1101 |           while out.len() < total_len { | iteration control; changing bounds/body affects throughp
timing, and possibly safety guarantees. |
| 1102 |           match adapter.read_frame() { | branch dispatch; changing arms alters behavior for va
nts and input classes. |
| 1103 |           Ok(Frame::Can(seg)) => { | executable/support line; behavior impact depends on s
ounding block context. |
| 1104 |               if !is_uds_response_id(target_sa, seg.id) { | conditional gate; changing con
ion shifts runtime path and side effects. |
| 1105 |                   continue; | executable/support line; behavior impact depends on surround
block context. |
| 1106 |               } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 1107 |               let sp = can_payload(&seg); | executable/support line; behavior impact dependen
on surrounding block context. |
| 1108 |               if sp.is_empty() { | conditional gate; changing condition shifts runtime pat
nd side effects. |
| 1109 |                   continue; | executable/support line; behavior impact depends on surround
block context. |
| 1110 |               } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 1111 |               let t = (sp[0] >> 4) & 0x0F; | executable/support line; behavior impact depende
on surrounding block context. |
| 1112 |               if t != 0x2 { | conditional gate; changing condition shifts runtime path and
de effects. |
| 1113 |                   continue; | executable/support line; behavior impact depends on surround
block context. |
| 1114 |               } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 1115 |               let sn_cf = sp[0] & 0x0F; | executable/support line; behavior impact depends
surrounding block context. |
| 1116 |               if sn_cf != expected_sn { | conditional gate; changing condition shifts runt
path and side effects. |
| 1117 |                   return Err(HwError::ParseError { | executable/support line; behavior imp
depends on surrounding block context. |
| 1118 |                       cause: "ISO-TP CF sequence mismatch".to_string(), | executable/suppo
line; behavior impact depends on surrounding block context. |
| 1119 |                       raw_data: format!( | executable/support line; behavior impact depend
n surrounding block context. |
| 1120 |                           "expected_sn={expected_sn}, got_sn={sn_cf}" | executable/support
ne; behavior impact depends on surrounding block context. |
| 1121 |                       ), | executable/support line; behavior impact depends on surrounding
ock context. |
| 1122 |                   }); | executable/support line; behavior impact depends on surrounding bl
context. |
| 1123 |               } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 1124 |               expected_sn = (expected_sn + 1) & 0x0F; | executable/support line; behavior
act depends on surrounding block context. |
| 1125 |               out.extend_from_slice(&sp[1..]); | executable/support line; behavior impact
ends on surrounding block context. |
| 1126 |           } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 1127 |           Ok(_) => {} | executable/support line; behavior impact depends on surrounding bl
context. |
| 1128 |           Err(HwError::Timeout) => {} | executable/support line; behavior impact depends o
urrounding block context. |
| 1129 |           Err(e) => return Err(e), | executable/support line; behavior impact depends on s
ounding block context. |
| 1130 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 1131 | (blank) | line is blank; visual separation only. |
| 1132 |           if Instant::now() >= deadline { | conditional gate; changing condition shifts runt

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```

ath and side effects. |
| 1133 |                               return Err(HwError::Timeout); | executable/support line; behavior impact depends
surrounding block context. |
| 1134 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 1135 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 1136 | (blank) | line is blank; visual separation only. |
| 1137 |                               out.truncate(total_len); | executable/support line; behavior impact depends on surroundi
block context. |
| 1138 |                               if out.len() >= 3 && out[0] == 0x7F && out[2] == 0x78 { | conditional gate; changing con
ion shifts runtime path and side effects. |
| 1139 |                               deadline = Instant::now() | executable/support line; behavior impact depends on surr
ding block context. |
| 1140 |                               + Duration::from_millis(cfg.uds_timeout_p2star_ms.max(1)); | I/O or protocol-criti
al logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1141 |                               continue; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1142 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 1143 |                               return Ok(out); | executable/support line; behavior impact depends on surrounding block
text. |
| 1144 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1145 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1146 |                               Ok(Frame::Serial(_)) => {} | executable/support line; behavior impact depends on surrounding blo
context. |
| 1147 |                               Err(HwError::Timeout) => { | executable/support line; behavior impact depends on surrounding blo
context. |
| 1148 |                               if cfg.keep_channels_alive | conditional gate; changing condition shifts runtime path and si
effects. |
| 1149 |                               && cfg.can_interface.is_some() | executable/support line; behavior impact depends on sur
nding block context. |
| 1150 |                               && last_keepalive.elapsed() | executable/support line; behavior impact depends on surrou
ng block context. |
| 1151 |                               >= Duration::from_millis(cfg.j1939_idle_guard_ms.max(250)) | executable/support line
ehavior impact depends on surrounding block context. |
| 1152 |                               { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1153 |                               if let Some(sa) = target_sa { | conditional gate; changing condition shifts runtime path
d side effects. |
| 1154 |                               let _ = send_can_sf(adapter, cfg, sa, &[0x3E, 0x00]); | I/O or protocol-critical log
edits can affect integration, timing, and diagnostic outcomes. |
| 1155 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 1156 |                               last_keepalive = Instant::now(); | executable/support line; behavior impact depends on s
ounding block context. |
| 1157 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1158 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1159 |                               Err(e) => return Err(e), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1160 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1161 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1162 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1163 | (blank) | line is blank; visual separation only. |
| 1164 | fn send_uds_and_wait( | declares executable behavior; signature/body changes alter API, control flow, and ou
ts. |
| 1165 |     adapter: &mut dyn super::hw::HardwareInterface, | executable/support line; behavior impact depends on su
unding block context. |
| 1166 |     cfg: &HwConfig, | executable/support line; behavior impact depends on surrounding block context. |
| 1167 |     target_sa: Option<u8>, | executable/support line; behavior impact depends on surrounding block context.

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| 1168 |     req: &[u8], | executable/support line; behavior impact depends on surrounding block context. |
| 1169 |     timeout: Duration, | executable/support line; behavior impact depends on surrounding block context. |
| 1170 | ) -> Result<Vec<u8>, HwError> { | executable/support line; behavior impact depends on surrounding block context. |
. |
| 1171 |     if cfg.can_interface.is_some() { | conditional gate; changing condition shifts runtime path and side effects. | | |
| 1172 |         let dst_sa = target_sa.unwrap_or(0x00); | executable/support line; behavior impact depends on surrounding block context. |
| 1173 |         send_can_multiframe(adapter, cfg, dst_sa, req)?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1174 |         return recv_can_uds_response(adapter, cfg, target_sa, timeout); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1175 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1176 | (blank) | line is blank; visual separation only. |
| 1177 |     let frame = Frame::Serial(super::hw::SerialFrame { | executable/support line; behavior impact depends on surrounding block context. |
| 1178 |         bytes: req.to_vec(), | executable/support line; behavior impact depends on surrounding block context. |
| 1179 |         protocol_hint: Some("uds-raw".into()), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1180 |         timestamp_ms: Some(now_ms()), | executable/support line; behavior impact depends on surrounding block context. |
| 1181 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 1182 |     send_frame_retry( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1183 |         adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 1184 |         frame, | executable/support line; behavior impact depends on surrounding block context. |
| 1185 |         cfg.uds_retry_count, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1186 |         cfg.write_retry_backoff_ms, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1187 |     ); | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1188 | (blank) | line is blank; visual separation only. |
| 1189 |     let deadline = Instant::now() + timeout; | executable/support line; behavior impact depends on surrounding block context. |
| 1190 |     while Instant::now() < deadline { | iteration control; changing bounds/body affects throughput, timing, possibly safety guarantees. |
| 1191 |         match adapter.read_frame() { | branch dispatch; changing arms alters behavior for variants and input cases. |
| 1192 |             Ok(Frame::Serial(sf)) => { | executable/support line; behavior impact depends on surrounding block context. |
| 1193 |                 if sf | conditional gate; changing condition shifts runtime path and side effects. |
| 1194 |                     .protocol_hint | executable/support line; behavior impact depends on surrounding block context. |
| 1195 |                     .as_deref() | executable/support line; behavior impact depends on surrounding block context. |
| 1196 |                     .is_some_and(|h| !h.eq_ignore_ascii_case("uds-raw")) | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1197 |                 { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1198 |                     continue; | executable/support line; behavior impact depends on surrounding block context. |
| 1199 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1200 |                 if sf.bytes.is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 1201 |                     continue; | executable/support line; behavior impact depends on surrounding block context. |
| 1202 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1203 |                 if sf.bytes.len() >= 3 && sf.bytes[0] == 0x7F && sf.bytes[2] == 0x78 { | conditional gate; changing condition shifts runtime path and side effects. |
| 1204 |                     continue; | executable/support line; behavior impact depends on surrounding block context. |
| 1205 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

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| 1206 |         return Ok(sf.bytes); | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 1207 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 1208 |         Ok(Frame::Can(_)) => {} | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 1209 |         Err(HwError::Timeout) => {} | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 1210 |         Err(e) => return Err(e), | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 1211 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1212 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1213 | Err(HwError::Timeout) | executable/support line; behavior impact depends on surrounding block context. |
| 1214 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1215 | (blank) | line is blank; visual separation only. |
| 1216 | fn check_live_channel_policy(cfg: &HwConfig) -> Result<(), HwError> { | declares executable behavior; signat
/body changes alter API, control flow, and outputs. |
| 1217 |     if cfg.ethernet_probe.is_some() && cfg.can_interface.is_none() { | conditional gate; changing condition
fts runtime path and side effects. |
| 1218 |         return Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 1219 |             "ethernet/internet checks require CAN/J1939 active (--can-if) to keep channel continuity" | execu
ble/support line; behavior impact depends on surrounding block context. |
| 1220 |             .to_string(), | executable/support line; behavior impact depends on surrounding block context
|
| 1221 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 1222 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1223 | (blank) | line is blank; visual separation only. |
| 1224 |     if let Some(target) = &cfg.ethernet_probe { | conditional gate; changing condition shifts runtime path a
side effects. |
| 1225 |         let mut addrs = target | executable/support line; behavior impact depends on surrounding block conte
|
| 1226 |         .to_socket_addrs() | executable/support line; behavior impact depends on surrounding block conte
|
| 1227 |         .map_err(|e| HwError::Unknown(format!("invalid --eth-probe target: {e}")))?; | error-propagati
operation; changing it impacts failure semantics and caller recovery paths. |
| 1228 |         let addr = addrs | executable/support line; behavior impact depends on surrounding block context. |
| 1229 |         .next() | executable/support line; behavior impact depends on surrounding block context. |
| 1230 |         .ok_or_else(|_| HwError::Unknown("--eth-probe resolved no socket address".to_string()))?; | err
propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1231 |         TcpStream::connect_timeout(&addr, Duration::from_millis(350)).map_err(|e| { | executable/support l
; behavior impact depends on surrounding block context. |
| 1232 |         HwError::Unknown(format!( | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 1233 |             "ethernet/internet check failed ({target}): {e}; preserving J1939 stability" | executable/su
rt line; behavior impact depends on surrounding block context. |
| 1234 |         )) | executable/support line; behavior impact depends on surrounding block context. |
| 1235 |         }); | error-propagating operation; changing it impacts failure semantics and caller recovery paths.
| 1236 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1237 | (blank) | line is blank; visual separation only. |
| 1238 |     Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 1239 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1240 | (blank) | line is blank; visual separation only. |
| 1241 | pub fn live_clean_ecm(cfg: &HwConfig, target_sa: Option<u8>) -> Result<(), HwError> { | declares executable
avior; signature/body changes alter API, control flow, and outputs. |
| 1242 |     if cfg.mode != HwMode::Live { | conditional gate; changing condition shifts runtime path and side effect
|
| 1243 |         return Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 1244 |             "clean requires --hw-mode=live".to_string(), | executable/support line; behavior impact depends
surrounding block context. |
| 1245 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 1246 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1247 |     if !cfg.write_effectively_enabled() { | conditional gate; changing condition shifts runtime path and sid

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ffects. |
| 1248 |         return Err(HwError::WriteBlockedAllowlist); | executable/support line; behavior impact depends on su
unding block context. |
| 1249 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1250 |     check_live_channel_policy(cfg)?; | error-propagating operation; changing it impacts failure semantics and
aller recovery paths. |
| 1251 | (blank) | line is blank; visual separation only. |
| 1252 |     let mut adapter = super::hw::open_real_adapter(cfg)?; | error-propagating operation; changing it impacts
ilure semantics and caller recovery paths. |
| 1253 |     adapter.init(cfg)?; | error-propagating operation; changing it impacts failure semantics and caller reco
y paths. |
| 1254 | (blank) | line is blank; visual separation only. |
| 1255 |     let session_rsp = send_uds_and_wait( | I/O or protocol-critical logic; edits can affect integration, tim
, and diagnostic outcomes. |
| 1256 |         &mut *adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 1257 |         cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1258 |         target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1259 |         &[0x10, 0x02], | executable/support line; behavior impact depends on surrounding block context. |
| 1260 |         Duration::from_millis(cfg.uds_timeout_p2_ms.max(1)), | I/O or protocol-critical logic; edits can aff
integration, timing, and diagnostic outcomes. |
| 1261 |     );?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1262 |     ensure_positive_response(&session_rsp, 0x10, "clean::session", cfg, target_sa)?; | error-propagating ope
ion; changing it impacts failure semantics and caller recovery paths. |
| 1263 | (blank) | line is blank; visual separation only. |
| 1264 |     let clear_rsp = send_uds_and_wait( | I/O or protocol-critical logic; edits can affect integration, timin
and diagnostic outcomes. |
| 1265 |         &mut *adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 1266 |         cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1267 |         target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1268 |         &[0x14, 0xFF, 0xFF, 0xFF], | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1269 |         Duration::from_millis(cfg.uds_timeout_p2_ms.max(1)), | I/O or protocol-critical logic; edits can aff
integration, timing, and diagnostic outcomes. |
| 1270 |     );?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1271 |     ensure_positive_response(&clear_rsp, 0x14, "clean::clear_dtc", cfg, target_sa)?; | error-propagating ope
ion; changing it impacts failure semantics and caller recovery paths. |
| 1272 | (blank) | line is blank; visual separation only. |
| 1273 |     let routine_rsp = send_uds_and_wait( | I/O or protocol-critical logic; edits can affect integration, tim
, and diagnostic outcomes. |
| 1274 |         &mut *adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 1275 |         cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1276 |         target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1277 |         &[0x31, 0x01, 0xDF, 0x04], | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1278 |         Duration::from_millis(cfg.uds_timeout_p2_ms.max(1)), | I/O or protocol-critical logic; edits can aff
integration, timing, and diagnostic outcomes. |
| 1279 |     );?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1280 |     ensure_positive_response(&routine_rsp, 0x31, "clean::routine", cfg, target_sa)?; | error-propagating ope
ion; changing it impacts failure semantics and caller recovery paths. |
| 1281 | (blank) | line is blank; visual separation only. |
| 1282 |     adapter.close()?; | error-propagating operation; changing it impacts failure semantics and caller recove
paths. |
| 1283 |     Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 1284 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1285 | (blank) | line is blank; visual separation only. |
| 1286 | pub fn live_flash_ecm_firmware( | declares executable behavior; signature/body changes alter API, control fl
and outputs. |
| 1287 |     cfg: &HwConfig, | executable/support line; behavior impact depends on surrounding block context. |
| 1288 |     target_sa: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 1289 |     firmware: &[u8], | executable/support line; behavior impact depends on surrounding block context. |
| 1290 | ) -> Result<FlashSummary, HwError> { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1291 |     if cfg.mode != HwMode::Live { | conditional gate; changing condition shifts runtime path and side effect
|
| 1292 |         return Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block

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ntext. |
| 1293 |         "flash requires --hw-mode=live".to_string(), | I/O or protocol-critical logic; edits can affect
egration, timing, and diagnostic outcomes. |
| 1294 |     )); | executable/support line; behavior impact depends on surrounding block context. |
| 1295 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1296 | if firmware.is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 1297 |     return Err(HwError::Unknown("firmware payload is empty".to_string())); | executable/support line; be
ior impact depends on surrounding block context. |
| 1298 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1299 | if !cfg.write_effectively_enabled() { | conditional gate; changing condition shifts runtime path and sid
ffects. |
| 1300 |     return Err(HwError::WriteBlockedAllowlist); | executable/support line; behavior impact depends on su
unding block context. |
| 1301 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1302 | check_live_channel_policy(cfg)?; | error-propagating operation; changing it impacts failure semantics and
aller recovery paths. |
| 1303 | let flash_start_ms = now_ms(); | I/O or protocol-critical logic; edits can affect integration, timing, a
diagnostic outcomes. |
| 1304 | (blank) | line is blank; visual separation only. |
| 1305 | append_flash_audit( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnost
outcomes. |
| 1306 |     cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1307 |     "flash::start", | I/O or protocol-critical logic; edits can affect integration, timing, and diagnost
outcomes. |
| 1308 |     FlashAuditStatus::Ok, | executable/support line; behavior impact depends on surrounding block context
|
| 1309 |     format!("payload_bytes={}", firmware.len()), | executable/support line; behavior impact depends on s
ounding block context. |
| 1310 |     target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1311 | ); | executable/support line; behavior impact depends on surrounding block context. |
| 1312 | (blank) | line is blank; visual separation only. |
| 1313 | let mut adapter = super::hw::open_real_adapter(cfg)?; | error-propagating operation; changing it impacts
ilure semantics and caller recovery paths. |
| 1314 | adapter.init(cfg)?; | error-propagating operation; changing it impacts failure semantics and caller reco
y paths. |
| 1315 | let firmware_crc = crc32fast::hash(firmware); | executable/support line; behavior impact depends on surr
ding block context. |
| 1316 | let mut transfer_data_blocks_attempted = 0usize; | executable/support line; behavior impact depends on s
ounding block context. |
| 1317 | let mut transfer_data_blocks_acked = 0usize; | executable/support line; behavior impact depends on surro
ing block context. |
| 1318 | let wait_frame_count = 0u32; | executable/support line; behavior impact depends on surrounding block con
t. |
| 1319 | let sequence_error_count = 0u32; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1320 | let flowcontrol_timeout_count = 0u32; | executable/support line; behavior impact depends on surrounding
ck context. |
| 1321 | (blank) | line is blank; visual separation only. |
| 1322 | let preflight = run_flash_preflight(&mut *adapter, cfg, target_sa)?; | error-propagating operation; chan
g it impacts failure semantics and caller recovery paths. |
| 1323 | append_flash_audit( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnost
outcomes. |
| 1324 |     cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1325 |     "flash::preflight", | I/O or protocol-critical logic; edits can affect integration, timing, and diag
tic outcomes. |
| 1326 |     FlashAuditStatus::Ok, | executable/support line; behavior impact depends on surrounding block context
|
| 1327 |     format!( | executable/support line; behavior impact depends on surrounding block context. |
| 1328 |         "supply_v={:.2}, vin={}, sw={}, hw={}, cal={}, serial={}, fp_present={}", | executable/support l
; behavior impact depends on surrounding block context. |
| 1329 |         preflight.supply_voltage_v, | executable/support line; behavior impact depends on surrounding bl
context. |
| 1330 |         preflight.identity.vin, | executable/support line; behavior impact depends on surrounding block
text. |
| 1331 |         preflight.identity.sw_version, | executable/support line; behavior impact depends on surrounding

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ock context. |
| 1332 |          preflight.identity.hw_version, | executable/support line; behavior impact depends on surrounding
ock context. |
| 1333 |          preflight.identity.calibration_id, | executable/support line; behavior impact depends on surround
g block context. |
| 1334 |          preflight.identity.ecu_serial, | executable/support line; behavior impact depends on surrounding
ock context. |
| 1335 |          preflight.identity.fingerprint.is_some() | executable/support line; behavior impact depends on s
ounding block context. |
| 1336 |      ), | executable/support line; behavior impact depends on surrounding block context. |
| 1337 |      target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1338 |      ); | executable/support line; behavior impact depends on surrounding block context. |
| 1339 | (blank) | line is blank; visual separation only. |
| 1340 |      let default_session_rsp = send_uds_and_wait( | I/O or protocol-critical logic; edits can affect integrat
, timing, and diagnostic outcomes. |
| 1341 |          &mut *adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 1342 |          cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1343 |          target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1344 |          &[0x10, 0x02], | executable/support line; behavior impact depends on surrounding block context. |
| 1345 |          Duration::from_millis(cfg.uds_timeout_p2_ms.max(1)), | I/O or protocol-critical logic; edits can aff
integration, timing, and diagnostic outcomes. |
| 1346 |      )?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1347 |      ensure_positive_response( | executable/support line; behavior impact depends on surrounding block context
|
| 1348 |          &default_session_rsp, | executable/support line; behavior impact depends on surrounding block context
|
| 1349 |          0x10, | executable/support line; behavior impact depends on surrounding block context. |
| 1350 |          "flash::default_session", | I/O or protocol-critical logic; edits can affect integration, timing, and
diagnostic outcomes. |
| 1351 |          cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1352 |          target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1353 |      )?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1354 | (blank) | line is blank; visual separation only. |
| 1355 |      let seed = send_uds_and_wait( | I/O or protocol-critical logic; edits can affect integration, timing, and
diagnostic outcomes. |
| 1356 |          &mut *adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 1357 |          cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1358 |          target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1359 |          &[0x27, 0x05], | executable/support line; behavior impact depends on surrounding block context. |
| 1360 |          Duration::from_millis(cfg.uds_timeout_p2_ms.max(1)), | I/O or protocol-critical logic; edits can aff
integration, timing, and diagnostic outcomes. |
| 1361 |      )?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1362 |      ensure_positive_response(&seed, 0x27, "flash::security_seed", cfg, target_sa)?; | error-propagating oper
on; changing it impacts failure semantics and caller recovery paths. |
| 1363 |      if seed.len() < 6 { | conditional gate; changing condition shifts runtime path and side effects. |
| 1364 |          append_flash_audit( | I/O or protocol-critical logic; edits can affect integration, timing, and diag
tic outcomes. |
| 1365 |              cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1366 |              "flash::security_seed", | I/O or protocol-critical logic; edits can affect integration, timing,
diagnostic outcomes. |
| 1367 |              FlashAuditStatus::Failed, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1368 |              format!("short seed response len={}", seed.len()), | executable/support line; behavior impact de
ds on surrounding block context. |
| 1369 |              target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1370 |          ); | executable/support line; behavior impact depends on surrounding block context. |
| 1371 |          adapter.close()?; | error-propagating operation; changing it impacts failure semantics and caller re
ery paths. |
| 1372 |      return Err(HwError::Unknown("security seed response invalid".to_string())); | executable/support lin
behavior impact depends on surrounding block context. |
| 1373 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1374 |      let seed_u32 = ((seed[2] as u32) << 24) | executable/support line; behavior impact depends on surroundin
lock context. |
| 1375 |          \ | ((seed[3] as u32) << 16) | executable/support line; behavior impact depends on surrounding block
text. |

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| 1376 |      \ | ((seed[4] as u32) << 8) | executable/support line; behavior impact depends on surrounding block context. |
| 1377 |      \ | seed[5] as u32; | executable/support line; behavior impact depends on surrounding block context. |
| 1378 |      let key = seed_u32 ^ 0xDEADBEEF; | executable/support line; behavior impact depends on surrounding block context. |
| 1379 |      let key_req = [ | executable/support line; behavior impact depends on surrounding block context. |
| 1380 |          0x27, | executable/support line; behavior impact depends on surrounding block context. |
| 1381 |          0x06, | executable/support line; behavior impact depends on surrounding block context. |
| 1382 |          ((key >> 24) & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding block context. |
| 1383 |          ((key >> 16) & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding block context. |
| 1384 |          ((key >> 8) & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding block context. |
| 1385 |          (key & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding block context. |
| 1386 |      ]; | executable/support line; behavior impact depends on surrounding block context. |
| 1387 |      let unlock_rsp = send_uds_and_wait( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1388 |          &mut *adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 1389 |          cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1390 |          target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1391 |          &key_req, | executable/support line; behavior impact depends on surrounding block context. |
| 1392 |          Duration::from_millis(cfg.uds_timeout_p2_ms.max(1)), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1393 |      )?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1394 |      ensure_positive_response(&unlock_rsp, 0x27, "flash::security_unlock", cfg, target_sa)?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1395 | (blank) | line is blank; visual separation only. |
| 1396 |      let prog_session_rsp = send_uds_and_wait( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1397 |          &mut *adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 1398 |          cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1399 |          target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1400 |          &[0x10, 0x03], | executable/support line; behavior impact depends on surrounding block context. |
| 1401 |          Duration::from_millis(cfg.uds_timeout_p2_ms.max(1)), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1402 |      )?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1403 |      ensure_positive_response( | executable/support line; behavior impact depends on surrounding block context. |
| 1404 |          &prog_session_rsp, | executable/support line; behavior impact depends on surrounding block context. |
| 1405 |          0x10, | executable/support line; behavior impact depends on surrounding block context. |
| 1406 |          "flash::programming_session", | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1407 |          cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1408 |          target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1409 |      )?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1410 | (blank) | line is blank; visual separation only. |
| 1411 |      let total_len = firmware.len() as u32; | executable/support line; behavior impact depends on surrounding block context. |
| 1412 |      let req_dl = [ | executable/support line; behavior impact depends on surrounding block context. |
| 1413 |          0x34, | executable/support line; behavior impact depends on surrounding block context. |
| 1414 |          0x00, | executable/support line; behavior impact depends on surrounding block context. |
| 1415 |          0x00, | executable/support line; behavior impact depends on surrounding block context. |
| 1416 |          0x00, | executable/support line; behavior impact depends on surrounding block context. |
| 1417 |          0x00, | executable/support line; behavior impact depends on surrounding block context. |
| 1418 |          0x00, | executable/support line; behavior impact depends on surrounding block context. |
| 1419 |          ((total_len >> 24) & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding block context. |
| 1420 |          ((total_len >> 16) & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding block context. |
| 1421 |          ((total_len >> 8) & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding block context. |
| 1422 |          (total_len & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding block context. |
| 1423 |      ]; | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 1424 |         let dl_rsp = send_uds_and_wait( | I/O or protocol-critical logic; edits can affect integration, timing,
diagnostic outcomes. |
| 1425 |         &mut *adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 1426 |         cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1427 |         target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1428 |         &req_dl, | executable/support line; behavior impact depends on surrounding block context. |
| 1429 |         Duration::from_millis(cfg.uds_timeout_p2_ms.max(1)), | I/O or protocol-critical logic; edits can affect
integration, timing, and diagnostic outcomes. |
| 1430 |     ); | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1431 |     ensure_positive_response(&dl_rsp, 0x34, "flash::request_download", cfg, target_sa)?; | error-propagating
operation; changing it impacts failure semantics and caller recovery paths. |
| 1432 | (blank) | line is blank; visual separation only. |
| 1433 |     let mut max_block_payload = 127usize; | executable/support line; behavior impact depends on surrounding
block context. |
| 1434 |     if dl_rsp.len() >= 4 && dl_rsp[0] == 0x74 { | conditional gate; changing condition shifts runtime path and
side effects. |
| 1435 |         let mbl = ((dl_rsp[2] as u16) << 8) \ dl_rsp[3] as u16; | executable/support line; behavior impact
depends on surrounding block context. |
| 1436 |         if mbl > 2 { | conditional gate; changing condition shifts runtime path and side effects. |
| 1437 |             max_block_payload = (mbl as usize).saturating_sub(2).max(1); | executable/support line; behavior
impact depends on surrounding block context. |
| 1438 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
|
| 1439 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1440 |     max_block_payload = max_block_payload.min(2048); | executable/support line; behavior impact depends on
surrounding block context. |
| 1441 | (blank) | line is blank; visual separation only. |
| 1442 |     let mut block = 1u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1443 |     let mut blocks = 0usize; | executable/support line; behavior impact depends on surrounding block context.
|
| 1444 |     for chunk in firmware.chunks(max_block_payload) { | iteration control; changing bounds/body affects timing,
put, timing, and possibly safety guarantees. |
| 1445 |         let mut req = Vec::with_capacity(chunk.len() + 2); | executable/support line; behavior impact depends
on surrounding block context. |
| 1446 |         req.push(0x36); | executable/support line; behavior impact depends on surrounding block context. |
| 1447 |         req.push(block); | executable/support line; behavior impact depends on surrounding block context. |
| 1448 |         req.extend_from_slice(chunk); | executable/support line; behavior impact depends on surrounding block
context. |
| 1449 |         transfer_data_blocks_attempted += 1; | executable/support line; behavior impact depends on surrounding
block context. |
| 1450 |         let ack = match send_uds_and_wait( | I/O or protocol-critical logic; edits can affect integration, timing,
and diagnostic outcomes. |
| 1451 |             &mut *adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 1452 |             cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1453 |             target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1454 |             &req, | executable/support line; behavior impact depends on surrounding block context. |
| 1455 |             Duration::from_millis(cfg.uds_timeout_p2star_ms.max(1)), | I/O or protocol-critical logic; edits can
affect integration, timing, and diagnostic outcomes. |
| 1456 |         ) { | executable/support line; behavior impact depends on surrounding block context. |
| 1457 |             Ok(v) => v, | executable/support line; behavior impact depends on surrounding block context. |
| 1458 |             Err(HwError::Timeout) => { | executable/support line; behavior impact depends on surrounding block
context. |
| 1459 |                 append_flash_audit( | I/O or protocol-critical logic; edits can affect integration, timing,
diagnostic outcomes. |
| 1460 |                     cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1461 |                     "flash::transfer_data", | I/O or protocol-critical logic; edits can affect integration,
timing, and diagnostic outcomes. |
| 1462 |                     FlashAuditStatus::Failed, | executable/support line; behavior impact depends on surrounding
block context. |
| 1463 |                     format!("timeout waiting transfer ack block={block}"), | executable/support line; behavior
impact depends on surrounding block context. |
| 1464 |                     target_sa, | executable/support line; behavior impact depends on surrounding block context.
|
| 1465 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 1466 |                 adapter.close()?; | error-propagating operation; changing it impacts failure semantics and caller
recovery paths. |

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| 1467 |         return Err(HwError::Timeout); | executable/support line; behavior impact depends on surrounding
    block context. |
| 1468 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime sco
pes. |
| 1469 |         Err(e) => { | executable/support line; behavior impact depends on surrounding block context. |
| 1470 |             adapter.close()?; | error-propagating operation; changing it impacts failure semantics and c
aller recovery paths. |
| 1471 |             return Err(e); | executable/support line; behavior impact depends on surrounding block conte
xt. |
| 1472 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime sco
pes. |
| 1473 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime sco
pes. |
| 1474 |     ensure_positive_response(&ack, 0x36, "flash::transfer_data", cfg, target_sa)?; | error-propagating o
peration; changing it impacts failure semantics and caller recovery paths. |
| 1475 |     if ack.len() < 2 { | conditional gate; changing condition shifts runtime path and side effects. |
| 1476 |         append_flash_audit( | I/O or protocol-critical logic; edits can affect integration, timing, and
diagnostic outcomes. |
| 1477 |             cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1478 |             "flash::transfer_data", | I/O or protocol-critical logic; edits can affect integration, timi
ng, and diagnostic outcomes. |
| 1479 |             FlashAuditStatus::Failed, | executable/support line; behavior impact depends on surrounding
block context. |
| 1480 |             format!("ack block mismatch expected={block} got={}", ack.get(1).copied().unwrap_or(0xFF)),
executable/support line; behavior impact depends on surrounding block context. |
| 1481 |             target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1482 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 1483 |         adapter.close()?; | error-propagating operation; changing it impacts failure semantics and calle
r recovery paths. |
| 1484 |         return Err(HwError::Unknown(format!( | executable/support line; behavior impact depends on surro
unding block context. |
| 1485 |             "transfer ack mismatch for block {block}" | executable/support line; behavior impact depends
on surrounding block context. |
| 1486 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 1487 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
es. |
| 1488 |     transfer_data_blocks_acked += 1; | executable/support line; behavior impact depends on surrounding b
lock context. |
| 1489 |     block = block.wrapping_add(1); | executable/support line; behavior impact depends on surrounding blo
ck context. |
| 1490 |     blocks += 1; | executable/support line; behavior impact depends on surrounding block context. |
| 1491 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1492 | (blank) | line is blank; visual separation only. |
| 1493 | let exit_req_crc = [ | executable/support line; behavior impact depends on surrounding block context. |
| 1494 |     0x37, | executable/support line; behavior impact depends on surrounding block context. |
| 1495 |     ((firmware_crc >> 24) & 0xFF) as u8, | executable/support line; behavior impact depends on surroundi
ng block context. |
| 1496 |     ((firmware_crc >> 16) & 0xFF) as u8, | executable/support line; behavior impact depends on surroundi
ng block context. |
| 1497 |     ((firmware_crc >> 8) & 0xFF) as u8, | executable/support line; behavior impact depends on surroundi
ng block context. |
| 1498 |     (firmware_crc & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding block
context. |
| 1499 | ]; | executable/support line; behavior impact depends on surrounding block context. |
| 1500 | let exit = send_uds_and_wait( | I/O or protocol-critical logic; edits can affect integration, timing, and
diagnostic outcomes. |
| 1501 |     &mut *adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 1502 |     cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1503 |     target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1504 |     &exit_req_crc, | executable/support line; behavior impact depends on surrounding block context. |
| 1505 |     Duration::from_millis(cfg.uds_timeout_p2star_ms.max(1)), | I/O or protocol-critical logic; edits can
affect integration, timing, and diagnostic outcomes. |
| 1506 | ); | executable/support line; behavior impact depends on surrounding block context. |
| 1507 | .or_else(|_| { | executable/support line; behavior impact depends on surrounding block context. |

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| 1508 |         send_uds_and_wait( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnosti
ic outcomes. |
| 1509 |             &mut *adapter, | executable/support line; behavior impact depends on surrounding block context.
| 1510 |             cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1511 |             target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1512 |             &[0x37, 0x00], | executable/support line; behavior impact depends on surrounding block context.
| 1513 |             Duration::from_millis(cfg.uds_timeout_p2star_ms.max(1)), | I/O or protocol-critical logic; edits
n affect integration, timing, and diagnostic outcomes. |
| 1514 |         ) | executable/support line; behavior impact depends on surrounding block context. |
| 1515 |     }?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1516 |     ensure_positive_response(&exit, 0x37, "flash::transfer_exit", cfg, target_sa)?; | error-propagating oper
on; changing it impacts failure semantics and caller recovery paths. |
| 1517 |     if exit.first().copied() != Some(0x77) { | conditional gate; changing condition shifts runtime path and
e effects. |
| 1518 |         append_flash_audit( | I/O or protocol-critical logic; edits can affect integration, timing, and diagn
tic outcomes. |
| 1519 |             cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1520 |             "flash::transfer_exit", | I/O or protocol-critical logic; edits can affect integration, timing,
diagnostic outcomes. |
| 1521 |             FlashAuditStatus::Failed, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1522 |             "transfer exit rejected", | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1523 |             target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1524 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 1525 |         adapter.close()?; | error-propagating operation; changing it impacts failure semantics and caller re
ery paths. |
| 1526 |         return Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1527 |             "transfer exit rejected during integrity verification".to_string(), | executable/support line; b
vior impact depends on surrounding block context. |
| 1528 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 1529 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1530 | (blank) | line is blank; visual separation only. |
| 1531 |     let verify_req = [ | executable/support line; behavior impact depends on surrounding block context. |
| 1532 |         0x31, | executable/support line; behavior impact depends on surrounding block context. |
| 1533 |         0x01, | executable/support line; behavior impact depends on surrounding block context. |
| 1534 |         0xF1, | executable/support line; behavior impact depends on surrounding block context. |
| 1535 |         0x90, | executable/support line; behavior impact depends on surrounding block context. |
| 1536 |         ((firmware_crc >> 24) & 0xFF) as u8, | executable/support line; behavior impact depends on surroundi
block context. |
| 1537 |         ((firmware_crc >> 16) & 0xFF) as u8, | executable/support line; behavior impact depends on surroundi
block context. |
| 1538 |         ((firmware_crc >> 8) & 0xFF) as u8, | executable/support line; behavior impact depends on surroundin
lock context. |
| 1539 |         (firmware_crc & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1540 |     ]; | executable/support line; behavior impact depends on surrounding block context. |
| 1541 |     let verify_rsp = send_uds_and_wait( | I/O or protocol-critical logic; edits can affect integration, timi
and diagnostic outcomes. |
| 1542 |         &mut *adapter, | executable/support line; behavior impact depends on surrounding block context. |
| 1543 |         cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1544 |         target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1545 |         &verify_req, | executable/support line; behavior impact depends on surrounding block context. |
| 1546 |         Duration::from_millis(cfg.uds_timeout_p2star_ms.max(1)), | I/O or protocol-critical logic; edits can
fect integration, timing, and diagnostic outcomes. |
| 1547 |     );?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1548 |     ensure_positive_response(&verify_rsp, 0x31, "flash::verify_routine", cfg, target_sa)?; | error-propagati
operation; changing it impacts failure semantics and caller recovery paths. |
| 1549 | (blank) | line is blank; visual separation only. |
| 1550 |     adapter.close()?; | error-propagating operation; changing it impacts failure semantics and caller recove
paths. |
| 1551 | (blank) | line is blank; visual separation only. |
| 1552 |     append_flash_audit( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnost
outcomes. |

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| 1553 |         cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 1554 |         "flash::complete", | I/O or protocol-critical logic; edits can affect integration, timing, and diagnosti
ic outcomes. |
| 1555 |         FlashAuditStatus::Ok, | executable/support line; behavior impact depends on surrounding block context
|
| 1556 |         format!("bytes={} blocks={} crc32=0x{firmware_crc:08X}", firmware.len(), blocks), | executable/suppo
line; behavior impact depends on surrounding block context. |
| 1557 |         target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1558 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 1559 | (blank) | line is blank; visual separation only. |
| 1560 |     let bridge_diag = read_bridge_diag_since(cfg, flash_start_ms); | I/O or protocol-critical logic; edits c
affect integration, timing, and diagnostic outcomes. |
| 1561 | (blank) | line is blank; visual separation only. |
| 1562 |     Ok(FlashSummary { | executable/support line; behavior impact depends on surrounding block context. |
| 1563 |         bytes_sent: firmware.len(), | executable/support line; behavior impact depends on surrounding block
text. |
| 1564 |         blocks_sent: blocks, | executable/support line; behavior impact depends on surrounding block context
|
| 1565 |         crc32: firmware_crc, | executable/support line; behavior impact depends on surrounding block context
|
| 1566 |         supply_voltage_v: preflight.supply_voltage_v, | executable/support line; behavior impact depends on
rounding block context. |
| 1567 |         vin: preflight.identity.vin, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1568 |         sw_version: preflight.identity.sw_version, | executable/support line; behavior impact depends on sur
nding block context. |
| 1569 |         hw_version: preflight.identity.hw_version, | executable/support line; behavior impact depends on sur
nding block context. |
| 1570 |         calibration_id: preflight.identity.calibration_id, | executable/support line; behavior impact depend
n surrounding block context. |
| 1571 |         ecu_serial: preflight.identity.ecu_serial, | executable/support line; behavior impact depends on sur
nding block context. |
| 1572 |         transport_diagnostics: FlashTransportDiagnostics { | executable/support line; behavior impact depend
n surrounding block context. |
| 1573 |             security_seed_positive: true, | executable/support line; behavior impact depends on surrounding
ck context. |
| 1574 |             security_unlock_positive: true, | executable/support line; behavior impact depends on surroundin
lock context. |
| 1575 |             request_download_positive: true, | executable/support line; behavior impact depends on surroundi
block context. |
| 1576 |             transfer_data_blocks_attempted, | executable/support line; behavior impact depends on surroundin
lock context. |
| 1577 |             transfer_data_blocks_acked, | executable/support line; behavior impact depends on surrounding bl
context. |
| 1578 |             request_transfer_exit_positive: true, | executable/support line; behavior impact depends on surr
ding block context. |
| 1579 |             fc_blocksize_seen: bridge_diag.fc_blocksize_seen.or(Some(cfg.uds_block_size)), | I/O or protocol
ritical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1580 |             fc_stmin_seen_ms: bridge_diag.fc_stmin_seen_ms.or(Some(cfg.uds_st_min_ms)), | I/O or protocol-cr
cal logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1581 |             wait_frame_count: wait_frame_count.saturating_add(bridge_diag.wait_frame_count), | executable/su
rt line; behavior impact depends on surrounding block context. |
| 1582 |             sequence_error_count: sequence_error_count | executable/support line; behavior impact depends on
rrounding block context. |
| 1583 |                 .saturating_add(bridge_diag.sequence_error_count), | executable/support line; behavior impac
depends on surrounding block context. |
| 1584 |             flowcontrol_timeout_count: flowcontrol_timeout_count | executable/support line; behavior impact
ends on surrounding block context. |
| 1585 |                 .saturating_add(bridge_diag.flowcontrol_timeout_count), | executable/support line; behavior
act depends on surrounding block context. |
| 1586 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 1587 |     }) | executable/support line; behavior impact depends on surrounding block context. |
| 1588 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1589 | (blank) | line is blank; visual separation only. |
| 1590 | fn read_bridge_diag_since(cfg: &HwConfig, since_ms: u64) -> BridgeDiagAggregate { | declares executable beha
r; signature/body changes alter API, control flow, and outputs. |
| 1591 |     let path = cfg.log_dir.join("cat_comm_bridge_diag.jsonl"); | executable/support line; behavior impact de

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ds on surrounding block context. |
| 1592 |     let content = match fs::read_to_string(path) { | executable/support line; behavior impact depends on surr
nding block context. |
| 1593 |         Ok(v) => v, | executable/support line; behavior impact depends on surrounding block context. |
| 1594 |         Err(_) => return BridgeDiagAggregate::default(), | executable/support line; behavior impact depends
surrounding block context. |
| 1595 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1596 | (blank) | line is blank; visual separation only. |
| 1597 |     let mut agg = BridgeDiagAggregate::default(); | executable/support line; behavior impact depends on surr
ding block context. |
| 1598 |     for line in content.lines() { | iteration control; changing bounds/body affects throughput, timing, and
sibly safety guarantees. |
| 1599 |         let Ok(r) = serde_json::from_str::<BridgeDiagLine>(line) else { | serialization contract touchpoint;
its can break report compatibility and tooling. |
| 1600 |             continue; | executable/support line; behavior impact depends on surrounding block context. |
| 1601 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime sco
. |
| 1602 |             if r.ts_ms < since_ms { | conditional gate; changing condition shifts runtime path and side effects.
| 1603 |                 continue; | executable/support line; behavior impact depends on surrounding block context. |
| 1604 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1605 |             agg.wait_frame_count = agg | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1606 |                 .wait_frame_count | executable/support line; behavior impact depends on surrounding block contex
|
| 1607 |                 .saturating_add(r.wait_frame_count_delta.unwrap_or(0)); | executable/support line; behavior impa
depends on surrounding block context. |
| 1608 |             agg.sequence_error_count = agg | executable/support line; behavior impact depends on surrounding blo
context. |
| 1609 |                 .sequence_error_count | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1610 |                 .saturating_add(r.sequence_error_count_delta.unwrap_or(0)); | executable/support line; behavior
act depends on surrounding block context. |
| 1611 |             agg.flowcontrol_timeout_count = agg | executable/support line; behavior impact depends on surroundin
lock context. |
| 1612 |                 .flowcontrol_timeout_count | executable/support line; behavior impact depends on surrounding blo
context. |
| 1613 |                 .saturating_add(r.flowcontrol_timeout_count_delta.unwrap_or(0)); | executable/support line; beha
r impact depends on surrounding block context. |
| 1614 |             if r.fc_blocksize.is_some() { | conditional gate; changing condition shifts runtime path and side ef
ts. |
| 1615 |                 agg.fc_blocksize_seen = r.fc_blocksize; | executable/support line; behavior impact depends on su
unding block context. |
| 1616 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1617 |             if r.fc_stmin_ms.is_some() { | conditional gate; changing condition shifts runtime path and side eff
s. |
| 1618 |                 agg.fc_stmin_seen_ms = r.fc_stmin_ms; | executable/support line; behavior impact depends on surr
ding block context. |
| 1619 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1620 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1621 |         agg | executable/support line; behavior impact depends on surrounding block context. |
| 1622 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1623 | (blank) | line is blank; visual separation only. |
| 1624 | #[cfg(test)] | executable/support line; behavior impact depends on surrounding block context. |
| 1625 | mod tests { | declares a module boundary; changing module name/path breaks references. |
| 1626 |     use super::ensure_positive_response; | imports symbols; changing path/name affects compile-time resolution
|
| 1627 |     use super::is_uds_response_id; | imports symbols; changing path/name affects compile-time resolution. |
| 1628 |     use crate::io::hw::HwConfig; | imports symbols; changing path/name affects compile-time resolution. |
| 1629 | (blank) | line is blank; visual separation only. |
| 1630 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1631 |     fn uds_response_id_matches_expected_target() { | declares executable behavior; signature/body changes al
API, control flow, and outputs. |

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| 1632 |          // PF=0xDA, PS=0xF9, SA=0x00 | comment line; documentation intent, no runtime effect. |
| 1633 |          let id = 0x18DAF900u32; | executable/support line; behavior impact depends on surrounding block context. |
| 1634 |          assert!(is_uds_response_id(Some(0x00), id)); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1635 |          assert!(is_uds_response_id(None, id)); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1636 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1637 | (blank) | line is blank; visual separation only. |
| 1638 |      #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1639 |      fn uds_response_id_rejects_wrong_pf_ps_or_sa() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1640 |          let wrong_pf = 0x18EAF900u32; | executable/support line; behavior impact depends on surrounding block context. |
| 1641 |          let wrong_ps = 0x18DA1200u32; | executable/support line; behavior impact depends on surrounding block context. |
| 1642 |          let wrong_sa = 0x18DAF901u32; | executable/support line; behavior impact depends on surrounding block context. |
| 1643 | (blank) | line is blank; visual separation only. |
| 1644 |          assert!(!is_uds_response_id(Some(0x00), wrong_pf)); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1645 |          assert!(!is_uds_response_id(Some(0x00), wrong_ps)); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1646 |          assert!(!is_uds_response_id(Some(0x00), wrong_sa)); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1647 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1648 | (blank) | line is blank; visual separation only. |
| 1649 |      #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1650 |      fn ensure_positive_response_accepts_expected_sid() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1651 |          let cfg = HwConfig::default(); | executable/support line; behavior impact depends on surrounding block context. |
| 1652 |          let rsp = [0x50, 0x02, 0x00, 0x00]; | executable/support line; behavior impact depends on surrounding block context. |
| 1653 |          let result = ensure_positive_response(&rsp, 0x10, "test::session", &cfg, Some(0x00)); | executable/support line; behavior impact depends on surrounding block context. |
| 1654 |          assert!(result.is_ok()); | executable/support line; behavior impact depends on surrounding block context. |
| 1655 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1656 | (blank) | line is blank; visual separation only. |
| 1657 |      #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1658 |      fn ensure_positive_response_rejects_negative_response_code() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1659 |          let cfg = HwConfig::default(); | executable/support line; behavior impact depends on surrounding block context. |
| 1660 |          let rsp = [0x7F, 0x10, 0x22]; | executable/support line; behavior impact depends on surrounding block context. |
| 1661 |          let result = ensure_positive_response(&rsp, 0x10, "test::session", &cfg, Some(0x00)); | executable/support line; behavior impact depends on surrounding block context. |
| 1662 |          assert!(result.is_err()); | executable/support line; behavior impact depends on surrounding block context. |
| 1663 |          let msg = format!("{}", result.err().expect("error expected")); | panic-prone operation; changing/replacing affects crash behavior and robustness. |
| 1664 |          assert!(msg.contains("NRC=0x22"), "unexpected error: {msg}"); | executable/support line; behavior impact depends on surrounding block context. |
| 1665 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1666 | (blank) | line is blank; visual separation only. |
| 1667 |      #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1668 |      fn ensure_positive_response_rejects_unexpected_sid() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1669 |          let cfg = HwConfig::default(); | executable/support line; behavior impact depends on surrounding block context. |
| 1670 |          let rsp = [0x62, 0xF1, 0x90]; | executable/support line; behavior impact depends on surrounding block context. |
| 1671 |          let result = ensure_positive_response(&rsp, 0x10, "test::session", &cfg, Some(0x00)); | executable/support line; behavior impact depends on surrounding block context. |

```

```
ort line; behavior impact depends on surrounding block context. |
| 1672 |          assert!(result.is_err()); | executable/support line; behavior impact depends on surrounding block context. |
| 1673 |          let msg = format!("{}", result.err().expect("error expected")); | panic-prone operation; changing/replacing affects crash behavior and robustness. |
| 1674 |          assert!(msg.contains("expected=0x50"), "unexpected error: {msg}"); | executable/support line; behavior impact depends on surrounding block context. |
| 1675 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1676 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
```

File: src/io/metrics.rs

Role: I/O and diagnostics stack: adapters, allowlist, flashing, replay, metrics, production phases.

Lines: 48 | Non-empty: 40 | Symbols: 10

Symbol Inventory

Line	Kind	Symbol	Change Impact
2	struct	HwMetrics	data/structure coupling
14	impl	HwMetrics	behavior binding and method semantics
15	fn	on_rx_can	signature/API coupling
20	fn	on_rx_serial	signature/API coupling
25	fn	on_tx_can_allowed	signature/API coupling
29	fn	on_tx_serial_allowed	signature/API coupling
33	fn	on_tx_blocked	signature/API coupling
37	fn	on_rate_limited	signature/API coupling
41	fn	on_read_error	signature/API coupling
45	fn	on_write_error	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	#[derive(Debug, Default, Clone)]	executable/support line; behavior impact depends on surrounding block context.
2	pub struct HwMetrics {	declares data layout; field changes may break serialization, tests, and callers.
3	pub hw_rx_frames_total_can: u64,	executable/support line; behavior impact depends on surrounding block context.
4	pub hw_rx_frames_total_serial: u64,	executable/support line; behavior impact depends on surrounding block context.
5	pub hw_tx_frames_total_can_allowed: u64,	executable/support line; behavior impact depends on surrounding block context.
6	pub hw_tx_frames_total_serial_allowed: u64,	executable/support line; behavior impact depends on surrounding block context.
7	pub hw_tx_frames_total_blocked: u64,	executable/support line; behavior impact depends on surrounding block context.
8	pub hw_rate_limited_total: u64,	executable/support line; behavior impact depends on surrounding block context.
9	pub hw_read_errors_total: u64,	executable/support line; behavior impact depends on surrounding block context.
10	pub hw_write_errors_total: u64,	I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes.
11	pub hw_last_rx_timestamp_ms: u64,	executable/support line; behavior impact depends on surrounding block context.
12	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
13	(blank)	line is blank; visual separation only.
14	impl HwMetrics {	implementation block; method behavior and trait conformance are defined here.
15	pub fn on_rx_can(&mut self, ts_ms: u64) {	declares executable behavior; signature/body changes alter API control flow, and outputs.
16	self.hw_rx_frames_total_can += 1;	executable/support line; behavior impact depends on surrounding block context.
17	self.hw_last_rx_timestamp_ms = ts_ms;	executable/support line; behavior impact depends on surrounding block context.
18	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.


```

| 19 | (blank) | line is blank; visual separation only. |
| 20 |     pub fn on_rx_serial(&mut self, ts_ms: u64) { | declares executable behavior; signature/body changes alter
, control flow, and outputs. |
| 21 |         self.hw_rx_frames_total_serial += 1; | executable/support line; behavior impact depends on surrounding
ock context. |
| 22 |         self.hw_last_rx_timestamp_ms = ts_ms; | executable/support line; behavior impact depends on surrounding
lock context. |
| 23 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 24 | (blank) | line is blank; visual separation only. |
| 25 |     pub fn on_tx_can_allowed(&mut self) { | declares executable behavior; signature/body changes alter API, co
ol flow, and outputs. |
| 26 |         self.hw_tx_frames_total_can_allowed += 1; | executable/support line; behavior impact depends on surroun
ng block context. |
| 27 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 28 | (blank) | line is blank; visual separation only. |
| 29 |     pub fn on_tx_serial_allowed(&mut self) { | declares executable behavior; signature/body changes alter API,
ntrol flow, and outputs. |
| 30 |         self.hw_tx_frames_total_serial_allowed += 1; | executable/support line; behavior impact depends on surr
nding block context. |
| 31 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 32 | (blank) | line is blank; visual separation only. |
| 33 |     pub fn on_tx_blocked(&mut self) { | declares executable behavior; signature/body changes alter API, contro
low, and outputs. |
| 34 |         self.hw_tx_frames_total_blocked += 1; | executable/support line; behavior impact depends on surroundin
lock context. |
| 35 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 36 | (blank) | line is blank; visual separation only. |
| 37 |     pub fn on_rate_limited(&mut self) { | declares executable behavior; signature/body changes alter API, contr
flow, and outputs. |
| 38 |         self.hw_rate_limited_total += 1; | executable/support line; behavior impact depends on surrounding blo
context. |
| 39 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 40 | (blank) | line is blank; visual separation only. |
| 41 |     pub fn on_read_error(&mut self) { | declares executable behavior; signature/body changes alter API, contro
low, and outputs. |
| 42 |         self.hw_read_errors_total += 1; | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 43 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 44 | (blank) | line is blank; visual separation only. |
| 45 |     pub fn on_write_error(&mut self) { | declares executable behavior; signature/body changes alter API, contr
flow, and outputs. |
| 46 |         self.hw_write_errors_total += 1; | I/O or protocol-critical logic; edits can affect integration, timin
and diagnostic outcomes. |
| 47 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 48 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/io/mock.rs

Role: I/O and diagnostics stack: adapters, allowlist, flashing, replay, metrics, production phases.

Lines: 187 | Non-empty: 160 | Symbols: 13

Symbol Inventory

Line	Kind	Symbol	Change Impact
11	struct	MockAdapter	data/structure coupling
23	impl	MockAdapter	behavior binding and method semantics
24	fn	new	signature/API coupling
28	fn	inject_rx	signature/API coupling
32	fn	inject_disconnect	signature/API coupling
36	fn	clear_disconnect	signature/API coupling
41	impl	HardwareInterface for MockAdapter	behavior binding and method semantics
42	fn	init	signature/API coupling
61	fn	read_frame	signature/API coupling
89	fn	try_read_frame	signature/API coupling

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| 114 | fn | send_frame | signature/API coupling |
| 176 | fn | close | signature/API coupling |
| 182 | fn | now_ms | signature/API coupling |
```

Line-by-Line Impact Map

```
| Line | Source | Operational Impact If Changed |
|---:|---|---|
| 1 | use std::collections::VecDeque; | imports symbols; changing path/name affects compile-time resolution. |
| 2 | use std::path::PathBuf; | imports symbols; changing path/name affects compile-time resolution. |
| 3 | (blank) | line is blank; visual separation only. |
| 4 | use super::allowlist::Allowlist; | imports symbols; changing path/name affects compile-time resolution. |
| 5 | use super::hw::{Frame, HardwareInterface, HwConfig, HwError}; | imports symbols; changing path/name affects compile-time resolution. |
| 6 | use super::metrics::HwMetrics; | imports symbols; changing path/name affects compile-time resolution. |
| 7 | use super::rate_limiter::TwoLevelRateLimiter; | imports symbols; changing path/name affects compile-time resolution. |
| 8 | use super::replay::{append_record, frame_to_record}; | imports symbols; changing path/name affects compile-time resolution. |
| 9 | (blank) | line is blank; visual separation only. |
| 10 | #[derive(Debug, Default)] | executable/support line; behavior impact depends on surrounding block context. |
| 11 | pub struct MockAdapter { | declares data layout; field changes may break serialization, tests, and callers. |
| 12 |     config: Option<HwConfig>, | executable/support line; behavior impact depends on surrounding block context. |
| 13 |     closed: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 14 |     disconnected: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 15 |     pub rx_queue: VecDeque<Frame>, | executable/support line; behavior impact depends on surrounding block context. |
| 16 |     pub tx_queue: VecDeque<Frame>, | executable/support line; behavior impact depends on surrounding block context. |
| 17 |     allowlist: Option<Allowlist>, | executable/support line; behavior impact depends on surrounding block context. |
| 18 |     limiter: Option<TwoLevelRateLimiter>, | executable/support line; behavior impact depends on surrounding block context. |
| 19 |     log_path: Option<PathBuf>, | executable/support line; behavior impact depends on surrounding block context. |
| 20 |     pub metrics: HwMetrics, | executable/support line; behavior impact depends on surrounding block context. |
| 21 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 22 | (blank) | line is blank; visual separation only. |
| 23 | impl MockAdapter { | implementation block; method behavior and trait conformance are defined here. |
| 24 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 25 |         Self::default() | executable/support line; behavior impact depends on surrounding block context. |
| 26 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 27 | (blank) | line is blank; visual separation only. |
| 28 |     pub fn inject_rx(&mut self, frame: Frame) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 29 |         self.rx_queue.push_back(frame); | executable/support line; behavior impact depends on surrounding block context. |
| 30 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 31 | (blank) | line is blank; visual separation only. |
| 32 |     pub fn inject_disconnect(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 33 |         self.disconnected = true; | executable/support line; behavior impact depends on surrounding block context. |
| 34 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 35 | (blank) | line is blank; visual separation only. |
| 36 |     pub fn clear_disconnect(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 37 |         self.disconnected = false; | executable/support line; behavior impact depends on surrounding block context. |
| 38 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 39 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 40 | (blank) | line is blank; visual separation only. |
| 41 | impl HardwareInterface for MockAdapter { | implementation block; method behavior and trait conformance are defined here. |
| 42 |     fn init(&mut self, config: &HwConfig) -> Result<(), HwError> { | declares executable behavior; signature/body
```

```

changes alter API, control flow, and outputs. |
| 43 |         self.config = Some(config.clone()); | executable/support line; behavior impact depends on surrounding
ck context. |
| 44 |         self.closed = false; | executable/support line; behavior impact depends on surrounding block context.
| 45 |         self.allowlist = if let Some(path) = &config.allowlist_path { | executable/support line; behavior impa
depends on surrounding block context. |
| 46 |             Some( | executable/support line; behavior impact depends on surrounding block context. |
| 47 |                 Allowlist::from_path(path) | executable/support line; behavior impact depends on surrounding b
k context. |
| 48 |                 .map_err(\|e\| HwError::Unknown(format!("load allowlist failed: {e}")))?, | error-propagat
operation; changing it impacts failure semantics and caller recovery paths. |
| 49 |             ) | executable/support line; behavior impact depends on surrounding block context. |
| 50 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 51 |             None | executable/support line; behavior impact depends on surrounding block context. |
| 52 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 53 |         self.limiter = Some(TwoLevelRateLimiter::new( | executable/support line; behavior impact depends on su
unding block context. |
| 54 |             config.rate_limit_global_per_sec, | executable/support line; behavior impact depends on surroundin
lock context. |
| 55 |             config.rate_limit_per_id_per_sec, | executable/support line; behavior impact depends on surroundin
lock context. |
| 56 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 57 |         self.log_path = Some(config.log_dir.join("mock-session.jsonl")); | executable/support line; behavior i
ct depends on surrounding block context. |
| 58 |         Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 59 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 60 | (blank) | line is blank; visual separation only. |
| 61 |     fn read_frame(&mut self) -> Result<Frame, HwError> { | declares executable behavior; signature/body change
lter API, control flow, and outputs. |
| 62 |         if self.disconnected { | conditional gate; changing condition shifts runtime path and side effects. |
| 63 |             self.metrics.on_read_error(); | executable/support line; behavior impact depends on surrounding bl
context. |
| 64 |             return Err(HwError::Timeout); | executable/support line; behavior impact depends on surrounding bl
context. |
| 65 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 66 | (blank) | line is blank; visual separation only. |
| 67 |         let frame = self.rx_queue.pop_front().ok_or_else(\|_| { | executable/support line; behavior impact dep
s on surrounding block context. |
| 68 |             self.metrics.on_read_error(); | executable/support line; behavior impact depends on surrounding bl
context. |
| 69 |             HwError::Timeout | executable/support line; behavior impact depends on surrounding block context.
| 70 |         })?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 71 | (blank) | line is blank; visual separation only. |
| 72 |         match &frame { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 73 |             Frame::Can(cf) => self | executable/support line; behavior impact depends on surrounding block con
t. |
| 74 |                 .metrics | executable/support line; behavior impact depends on surrounding block context. |
| 75 |                 .on_rx_can(cf.timestamp_ms.unwrap_or_else(now_ms)), | executable/support line; behavior impact
ends on surrounding block context. |
| 76 |             Frame::Serial(sf) => self | executable/support line; behavior impact depends on surrounding block
text. |
| 77 |                 .metrics | executable/support line; behavior impact depends on surrounding block context. |
| 78 |                 .on_rx_serial(sf.timestamp_ms.unwrap_or_else(now_ms)), | executable/support line; behavior imp
depends on surrounding block context. |
| 79 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 80 | (blank) | line is blank; visual separation only. |
| 81 |         if let Some(path) = &self.log_path { | conditional gate; changing condition shifts runtime path and si
effects. |
| 82 |             let rec = frame_to_record(&frame, "rx", None, None); | executable/support line; behavior impact de
ds on surrounding block context. |
| 83 |             let _ = append_record(path, &rec); | executable/support line; behavior impact depends on surroundi
block context. |
| 84 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 85 | (blank) | line is blank; visual separation only. |

```

```

| 86 |         Ok(frame) | executable/support line; behavior impact depends on surrounding block context. |
| 87 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 88 | (blank) | line is blank; visual separation only. |
| 89 |     fn try_read_frame(&mut self) -> Result<Option<Frame>, HwError> { | declares executable behavior; signature
dy changes alter API, control flow, and outputs. |
| 90 |         if self.disconnected { | conditional gate; changing condition shifts runtime path and side effects. |
| 91 |             self.metrics.on_read_error(); | executable/support line; behavior impact depends on surrounding bl
context. |
| 92 |             return Err(HwError::Timeout); | executable/support line; behavior impact depends on surrounding bl
context. |
| 93 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 94 | (blank) | line is blank; visual separation only. |
| 95 |         let out = self.rx_queue.pop_front(); | executable/support line; behavior impact depends on surrounding
ock context. |
| 96 |         if let Some(frame) = &out { | conditional gate; changing condition shifts runtime path and side effect
|
| 97 |             match frame { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 98 |                 Frame::Can(cf) => self | executable/support line; behavior impact depends on surrounding block
ntext. |
| 99 |                     .metrics | executable/support line; behavior impact depends on surrounding block context.
| 100 |                     .on_rx_can(cf.timestamp_ms.unwrap_or_else(now_ms)), | executable/support line; behavior i
ct depends on surrounding block context. |
| 101 |                 Frame::Serial(sf) => self | executable/support line; behavior impact depends on surrounding b
k context. |
| 102 |                     .metrics | executable/support line; behavior impact depends on surrounding block context.
| 103 |                     .on_rx_serial(sf.timestamp_ms.unwrap_or_else(now_ms)), | executable/support line; behavio
mpact depends on surrounding block context. |
| 104 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 105 |         if let Some(path) = &self.log_path { | conditional gate; changing condition shifts runtime path a
side effects. |
| 106 |             let rec = frame_to_record(frame, "rx", None, None); | executable/support line; behavior impac
depends on surrounding block context. |
| 107 |             let _ = append_record(path, &rec); | executable/support line; behavior impact depends on surr
ding block context. |
| 108 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 109 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 110 | (blank) | line is blank; visual separation only. |
| 111 |     Ok(out) | executable/support line; behavior impact depends on surrounding block context. |
| 112 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 113 | (blank) | line is blank; visual separation only. |
| 114 | fn send_frame(&mut self, frame: Frame) -> Result<(), HwError> { | declares executable behavior; signature
dy changes alter API, control flow, and outputs. |
| 115 |     if self.disconnected { | conditional gate; changing condition shifts runtime path and side effects. |
| 116 |         self.metrics.on_write_error(); | I/O or protocol-critical logic; edits can affect integration, ti
g, and diagnostic outcomes. |
| 117 |         return Err(HwError::TransceiverError); | executable/support line; behavior impact depends on surr
ding block context. |
| 118 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 119 | (blank) | line is blank; visual separation only. |
| 120 |     let cfg = self | executable/support line; behavior impact depends on surrounding block context. |
| 121 |         .config | executable/support line; behavior impact depends on surrounding block context. |
| 122 |         .as_ref() | executable/support line; behavior impact depends on surrounding block context. |
| 123 |         .ok_or_else(|| HwError::Unknown("adapter not initialized".to_string()))?; | error-propagating o
ation; changing it impacts failure semantics and caller recovery paths. |
| 124 | (blank) | line is blank; visual separation only. |
| 125 |     if !cfg.write_intent_enabled() { | conditional gate; changing condition shifts runtime path and side
ects. |
| 126 |         self.metrics.on_tx_blocked(); | executable/support line; behavior impact depends on surrounding b
k context. |
| 127 |         return Err(HwError::WriteBlockedAllowlist); | executable/support line; behavior impact depends on
rrounding block context. |

```

```

| 128 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 129 | (blank) | line is blank; visual separation only. |
| 130 |         let is_allowed = self | executable/support line; behavior impact depends on surrounding block context
| 131 |         .allowlist | executable/support line; behavior impact depends on surrounding block context. |
| 132 |         .as_ref() | executable/support line; behavior impact depends on surrounding block context. |
| 133 |         .is_some_and(\|al\| al.is_allowed(&frame)); | executable/support line; behavior impact depends on
rrounding block context. |
| 134 |         if !is_allowed { | conditional gate; changing condition shifts runtime path and side effects. |
| 135 |             self.metrics.on_tx_blocked(); | executable/support line; behavior impact depends on surrounding b
k context. |
| 136 |             return Err(HwError::WriteBlockedAllowlist); | executable/support line; behavior impact depends on
rrounding block context. |
| 137 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 138 | (blank) | line is blank; visual separation only. |
| 139 |         let is_rate_ok = if let Some(lim) = self.limiter.as_mut() { | executable/support line; behavior impac
depends on surrounding block context. |
| 140 |             match &frame { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 141 |                 Frame::Can(cf) => lim.check_can(cf.id), | executable/support line; behavior impact depends on
rrounding block context. |
| 142 |                 Frame::Serial(_) => lim.check_serial(), | executable/support line; behavior impact depends on
rrounding block context. |
| 143 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 144 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 145 |             true | executable/support line; behavior impact depends on surrounding block context. |
| 146 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 147 | (blank) | line is blank; visual separation only. |
| 148 |         if !is_rate_ok { | conditional gate; changing condition shifts runtime path and side effects. |
| 149 |             self.metrics.on_rate_limited(); | executable/support line; behavior impact depends on surrounding
ock context. |
| 150 |             self.metrics.on_write_error(); | I/O or protocol-critical logic; edits can affect integration, ti
g, and diagnostic outcomes. |
| 151 |             return Err(HwError::RateLimited); | executable/support line; behavior impact depends on surroundi
block context. |
| 152 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 153 | (blank) | line is blank; visual separation only. |
| 154 |         if let Some(path) = &self.log_path { | conditional gate; changing condition shifts runtime path and s
effects. |
| 155 |             let rec = frame_to_record(&frame, "tx", Some(true), Some(cfg.dry_run)); | executable/support line
ehavior impact depends on surrounding block context. |
| 156 |             let _ = append_record(path, &rec); | executable/support line; behavior impact depends on surround
block context. |
| 157 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 158 | (blank) | line is blank; visual separation only. |
| 159 |         if cfg.dry_run { | conditional gate; changing condition shifts runtime path and side effects. |
| 160 |             match &frame { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 161 |                 Frame::Can(_) => self.metrics.on_tx_can_allowed(), | executable/support line; behavior impac
depends on surrounding block context. |
| 162 |                 Frame::Serial(_) => self.metrics.on_tx_serial_allowed(), | executable/support line; behavior
act depends on surrounding block context. |
| 163 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 164 |             return Ok(()); | executable/support line; behavior impact depends on surrounding block context. |
| 165 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 166 | (blank) | line is blank; visual separation only. |
| 167 |         self.tx_queue.push_back(frame.clone()); | executable/support line; behavior impact depends on surround
g block context. |
| 168 |         match frame { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 169 |             Frame::Can(_) => self.metrics.on_tx_can_allowed(), | executable/support line; behavior impact dep

```

```

s on surrounding block context. |
| 170 |         Frame::Serial(_) => self.metrics.on_tx_serial_allowed(), | executable/support line; behavior impact
depends on surrounding block context. |
| 171 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
|
| 172 | (blank) | line is blank; visual separation only. |
| 173 |     Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 174 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 175 | (blank) | line is blank; visual separation only. |
| 176 | fn close(&mut self) -> Result<(), HwError> { | declares executable behavior; signature/body changes alter
I, control flow, and outputs. |
| 177 |     self.closed = true; | executable/support line; behavior impact depends on surrounding block context. |
| 178 |     Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 179 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 180 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 181 | (blank) | line is blank; visual separation only. |
| 182 | fn now_ms() -> u64 { | declares executable behavior; signature/body changes alter API, control flow, and output
. |
| 183 |     std::time::SystemTime::now() | executable/support line; behavior impact depends on surrounding block context
. |
| 184 |         .duration_since(std::time::UNIX_EPOCH) | executable/support line; behavior impact depends on surrounding
block context. |
| 185 |         .map(|d| d.as_millis() as u64) | executable/support line; behavior impact depends on surrounding block
context. |
| 186 |         .unwrap_or(0) | executable/support line; behavior impact depends on surrounding block context. |
| 187 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/io/mod.rs

Role: I/O and diagnostics stack: adapters, allowlist, flashing, replay, metrics, production phases.
Lines: 15 | Non-empty: 15 | Symbols: 13

Symbol Inventory

Line	Kind	Symbol	Change Impact
1	module	allowlist	data/structure coupling
2	module	artifact	data/structure coupling
3	module	ecm_params	data/structure coupling
4	module	hw	data/structure coupling
5	module	live_runner	data/structure coupling
6	module	metrics	data/structure coupling
7	module	mock	data/structure coupling
9	module	production_program	data/structure coupling
10	module	rate_limiter	data/structure coupling
11	module	replay	data/structure coupling
12	module	serial_adapter	data/structure coupling
13	module	socketcan_adapter	data/structure coupling
15	module	vendor_cat_comm	data/structure coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	pub mod allowlist;	declares a module boundary; changing module name/path breaks references.
2	pub mod artifact;	declares a module boundary; changing module name/path breaks references.
3	pub mod ecm_params;	declares a module boundary; changing module name/path breaks references.
4	pub mod hw;	declares a module boundary; changing module name/path breaks references.
5	pub mod live_runner;	declares a module boundary; changing module name/path breaks references.
6	pub mod metrics;	declares a module boundary; changing module name/path breaks references.
7	pub mod mock;	declares a module boundary; changing module name/path breaks references.
8	#[cfg(not(test))]	executable/support line; behavior impact depends on surrounding block context.
9	pub mod production_program;	declares a module boundary; changing module name/path breaks references.
10	pub mod rate_limiter;	declares a module boundary; changing module name/path breaks references.
11	pub mod replay;	declares a module boundary; changing module name/path breaks references.

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| 12 | pub mod serial_adapter; | declares a module boundary; changing module name/path breaks references. |
| 13 | pub mod socketcan_adapter; | declares a module boundary; changing module name/path breaks references. |
| 14 | #[cfg(all(target_os = "windows", feature = "vendor-windows"))] | executable/support line; behavior impact depends on surrounding block context. |
| 15 | pub mod vendor_cat_comm; | declares a module boundary; changing module name/path breaks references. |
```

File: src/io/production_program.rs

Role: I/O and diagnostics stack: adapters, allowlist, flashing, replay, metrics, production phases.

Lines: 797 | Non-empty: 750 | Symbols: 18

Symbol Inventory

Line	Kind	Symbol	Change Impact
18	enum	PhaseStatus	data/structure coupling
25	struct	PhaseResult	data/structure coupling
33	struct	ProgramReport	data/structure coupling
50	struct	ServiceConformanceResult	data/structure coupling
57	struct	ConformanceSummary	data/structure coupling
63	struct	ProgramOptions	data/structure coupling
68	impl	Default for ProgramOptions	behavior binding and method semantics
69	fn	default	signature/API coupling
77	fn	run_all_phases	signature/API coupling
568	fn	run_sim_baseline_check	signature/API coupling
587	fn	run_sim_connectivity_check	signature/API coupling
601	fn	run_sim_identity_check	signature/API coupling
613	fn	build_conformance_summary	signature/API coupling
731	fn	run_sim_preflight_check	signature/API coupling
744	fn	run_uds_runtime_self_test	signature/API coupling
763	fn	run_sim_stress_check	signature/API coupling
784	fn	hex_sha	signature/API coupling
792	fn	now_ms	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	use std::fs;	imports symbols; changing path/name affects compile-time resolution.
2	use std::path::{Path, PathBuf};	imports symbols; changing path/name affects compile-time resolution.
3	use std::time::{Duration, SystemTime, UNIX_EPOCH};	imports symbols; changing path/name affects compile-time resolution.
4	(blank)	line is blank; visual separation only.
5	use serde::{Deserialize, Serialize};	imports symbols; changing path/name affects compile-time resolution.
6	use sha2::{Digest, Sha256};	imports symbols; changing path/name affects compile-time resolution.
7	(blank)	line is blank; visual separation only.
8	use crate::HeavyMachinery;	imports symbols; changing path/name affects compile-time resolution.
9	(blank)	line is blank; visual separation only.
10	use super::artifact::{validate_firmware_artifact, FirmwareArtifactReport};	imports symbols; changing path/name affects compile-time resolution.
11	use super::hw::{HwCapabilities, HwConfig, HwError, HwMode};	imports symbols; changing path/name affects compile-time resolution.
12	use super::live_runner::	imports symbols; changing path/name affects compile-time resolution.
13	connect_ecm, detect_ecms, live_flash_ecm_firmware, live_preflight_ecm, live_read_ecm_identity,	I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes.
14	DetectResult, FlashPreflightReport, FlashSummary,	executable/support line; behavior impact depends on surrounding block context.
15	};	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
16	(blank)	line is blank; visual separation only.
17	#[derive(Debug, Clone, Copy, Serialize, Deserialize, PartialEq, Eq)]	serialization contract touchpoint; edits can break report compatibility and tooling.
18	pub enum PhaseStatus {	declares state variants; adding/removing variants impacts pattern matches and logic branches.
19	Passed,	executable/support line; behavior impact depends on surrounding block context.
20	Failed,	executable/support line; behavior impact depends on surrounding block context.

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| 21 |     Blocked, | executable/support line; behavior impact depends on surrounding block context. |
| 22 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 23 | (blank) | line is blank; visual separation only. |
| 24 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report co
tibility and tooling. |
| 25 | pub struct PhaseResult { | declares data layout; field changes may break serialization, tests, and callers. |
| 26 |     pub phase: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 27 |     pub title: String, | executable/support line; behavior impact depends on surrounding block context. |
| 28 |     pub status: PhaseStatus, | executable/support line; behavior impact depends on surrounding block context. |
| 29 |     pub details: String, | executable/support line; behavior impact depends on surrounding block context. |
| 30 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 31 | (blank) | line is blank; visual separation only. |
| 32 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report co
tibility and tooling. |
| 33 | pub struct ProgramReport { | declares data layout; field changes may break serialization, tests, and callers. |
| 34 |     pub generated_at_ms: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 35 |     pub mode: String, | executable/support line; behavior impact depends on surrounding block context. |
| 36 |     pub target_sa: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 37 |     pub strict_mode: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 38 |     pub execute_flash: bool, | I/O or protocol-critical logic; edits can affect integration, timing, and diagn
ic outcomes. |
| 39 |     pub artifact: Option<FirmwareArtifactReport>, | executable/support line; behavior impact depends on surrou
ng block context. |
| 40 |     pub detect: Option<DetectResult>, | executable/support line; behavior impact depends on surrounding block
text. |
| 41 |     pub preflight: Option<FlashPreflightReport>, | executable/support line; behavior impact depends on surroun
g block context. |
| 42 |     pub identity: Option<super::live_runner::EcuIdentity>, | executable/support line; behavior impact depends
surrounding block context. |
| 43 |     pub conformance_summary: Option<ConformanceSummary>, | executable/support line; behavior impact depends on
rounding block context. |
| 44 |     pub phases: Vec<PhaseResult>, | executable/support line; behavior impact depends on surrounding block cont
. |
| 45 |     pub overall_passed: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 46 |     pub report_sha256: Option<String>, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 47 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 48 | (blank) | line is blank; visual separation only. |
| 49 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report co
tibility and tooling. |
| 50 | pub struct ServiceConformanceResult { | declares data layout; field changes may break serialization, tests, and
allers. |
| 51 |     pub service_sid: String, | executable/support line; behavior impact depends on surrounding block context. |
| 52 |     pub passed: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 53 |     pub evidence: String, | executable/support line; behavior impact depends on surrounding block context. |
| 54 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 55 | (blank) | line is blank; visual separation only. |
| 56 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report co
tibility and tooling. |
| 57 | pub struct ConformanceSummary { | declares data layout; field changes may break serialization, tests, and call
. |
| 58 |     pub passed: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 59 |     pub services: Vec<ServiceConformanceResult>, | executable/support line; behavior impact depends on surroun
g block context. |
| 60 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 61 | (blank) | line is blank; visual separation only. |
| 62 | #[derive(Debug, Clone, Copy)] | executable/support line; behavior impact depends on surrounding block context. |
| 63 | pub struct ProgramOptions { | declares data layout; field changes may break serialization, tests, and callers. |
| 64 |     pub strict_mode: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 65 |     pub execute_flash: bool, | I/O or protocol-critical logic; edits can affect integration, timing, and diagn
ic outcomes. |
| 66 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 67 | (blank) | line is blank; visual separation only. |
| 68 | impl Default for ProgramOptions { | implementation block; method behavior and trait conformance are defined he
|

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```

| 69 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
tputs. |
| 70 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 71 |             strict_mode: true, | executable/support line; behavior impact depends on surrounding block context
| 72 |             execute_flash: false, | I/O or protocol-critical logic; edits can affect integration, timing, and
gnostic outcomes. |
| 73 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 74 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 75 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 76 | (blank) | line is blank; visual separation only. |
| 77 | pub fn run_all_phases( | declares executable behavior; signature/body changes alter API, control flow, and out
s. |
| 78 |     cfg: &HwConfig, | executable/support line; behavior impact depends on surrounding block context. |
| 79 |     target_sa: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 80 |     artifact_path: Option<&Path>, | executable/support line; behavior impact depends on surrounding block cont
. |
| 81 |     out_dir: &Path, | executable/support line; behavior impact depends on surrounding block context. |
| 82 |     options: ProgramOptions, | executable/support line; behavior impact depends on surrounding block context.
| 83 | ) -> Result<PathBuf, HwError> { | executable/support line; behavior impact depends on surrounding block context
|
| 84 |     fs::create_dir_all(out_dir) | executable/support line; behavior impact depends on surrounding block context
|
| 85 |         .map_err(\|e\| HwError::Unknown(format!("create report dir failed: {e}")))?; | error-propagating opera
n; changing it impacts failure semantics and caller recovery paths. |
| 86 | (blank) | line is blank; visual separation only. |
| 87 |     let mut phases = Vec::with_capacity(12); | executable/support line; behavior impact depends on surrounding
ock context. |
| 88 |     let mut artifact = None; | executable/support line; behavior impact depends on surrounding block context.
| 89 |     let mut detect = None; | executable/support line; behavior impact depends on surrounding block context. |
| 90 |     let mut preflight = None; | executable/support line; behavior impact depends on surrounding block context.
| 91 |     let mut identity = None; | executable/support line; behavior impact depends on surrounding block context.
| 92 |     let mut flash_summary: Option<FlashSummary> = None; | I/O or protocol-critical logic; edits can affect int
ation, timing, and diagnostic outcomes. |
| 93 |     let capabilities = cfg.declared_capabilities(); | executable/support line; behavior impact depends on surr
ding block context. |
| 94 |     let missing_live_caps = capabilities.missing_for_live_flash(); | I/O or protocol-critical logic; edits can
fect integration, timing, and diagnostic outcomes. |
| 95 | (blank) | line is blank; visual separation only. |
| 96 |     // Phase 1: Compliance baseline | comment line; documentation intent, no runtime effect. |
| 97 |     let docs_ok = Path::new("docs/ECM-Data.md").exists() | executable/support line; behavior impact depends on
rrounding block context. |
| 98 |         && Path::new("docs/CAT_COMM_TEMPLATE_PROTOCOL.md").exists(); | executable/support line; behavior impac
depends on surrounding block context. |
| 99 |     phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding block context.
| 100 |         phase: 1, | executable/support line; behavior impact depends on surrounding block context. |
| 101 |         title: "Baseline e Compliance".to_string(), | executable/support line; behavior impact depends on sur
nding block context. |
| 102 |         status: if docs_ok { | executable/support line; behavior impact depends on surrounding block context.
| 103 |             PhaseStatus::Passed | executable/support line; behavior impact depends on surrounding block conte
|
| 104 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 105 |             PhaseStatus::Failed | executable/support line; behavior impact depends on surrounding block conte
|
| 106 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 107 |         details: if docs_ok { | executable/support line; behavior impact depends on surrounding block context
| 108 |             "runbook e protocolo presentes".to_string() | executable/support line; behavior impact depends on
rrounding block context. |
| 109 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 110 |             "faltam documentos obrigatorios em docs/".to_string() | executable/support line; behavior impact
ends on surrounding block context. |
| 111 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 112 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 113 | (blank) | line is blank; visual separation only. |
| 114 |     // Phase 2: Hardware bench readiness | comment line; documentation intent, no runtime effect. |
| 115 |     let hw_ready = matches!(cfg.mode, HwMode::Live) && missing_live_caps.is_empty(); | executable/support lin

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behavior impact depends on surrounding block context. |
| 116 |     let sim_baseline_ok = run_sim_baseline_check(); | executable/support line; behavior impact depends on surrounding block context. |
| 117 |     phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding block context. |
| 118 |         phase: 2, | executable/support line; behavior impact depends on surrounding block context. |
| 119 |         title: "Plataforma Fisica Real".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 120 |         status: if hw_ready { | executable/support line; behavior impact depends on surrounding block context. |
| 121 |             PhaseStatus::Passed | executable/support line; behavior impact depends on surrounding block context. |
| 122 |         } else if matches!(cfg.mode, HwMode::Sim) && sim_baseline_ok.0 { | executable/support line; behavior impact depends on surrounding block context. |
| 123 |             PhaseStatus::Passed | executable/support line; behavior impact depends on surrounding block context. |
| 124 |         } else if matches!(cfg.mode, HwMode::Sim) { | executable/support line; behavior impact depends on surrounding block context. |
| 125 |             PhaseStatus::Failed | executable/support line; behavior impact depends on surrounding block context. |
| 126 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 127 |             PhaseStatus::Failed | executable/support line; behavior impact depends on surrounding block context. |
| 128 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 129 |         details: if hw_ready { | executable/support line; behavior impact depends on surrounding block context. |
| 130 |             format!("capabilities: {}", cfg.capabilities_summary()) | executable/support line; behavior impact depends on surrounding block context. |
| 131 |         } else if matches!(cfg.mode, HwMode::Sim) { | executable/support line; behavior impact depends on surrounding block context. |
| 132 |             format!("sim_harness={}: {}", sim_baseline_ok.0, sim_baseline_ok.1) | executable/support line; behavior impact depends on surrounding block context. |
| 133 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 134 |             format!("capabilities insufficients: {}; missing={}", | executable/support line; behavior impact depends on surrounding block context. |
| 135 |                 cfg.capabilities_summary(), | executable/support line; behavior impact depends on surrounding block context. |
| 136 |                 if missing_live_caps.is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 137 |                     "none".to_string() | executable/support line; behavior impact depends on surrounding block context. |
| 138 |                 } else { | executable/support line; behavior impact depends on surrounding block context. |
| 139 |                     missing_live_caps.join(",") | executable/support line; behavior impact depends on surrounding block context. |
| 140 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 141 |             ) | executable/support line; behavior impact depends on surrounding block context. |
| 142 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 143 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 144 | (blank) | line is blank; visual separation only. |
| 145 | // Phase 3: Live detect/connect | comment line; documentation intent, no runtime effect. |
| 146 | if hw_ready { | conditional gate; changing condition shifts runtime path and side effects. |
| 147 |     match detect_ecms(cfg, Duration::from_secs(3)) { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 148 |         Ok(d) => { | executable/support line; behavior impact depends on surrounding block context. |
| 149 |             let found = d.source_addresses.len(); | executable/support line; behavior impact depends on surrounding block context. |
| 150 |             detect = Some(d); | executable/support line; behavior impact depends on surrounding block context. |
| 151 |             phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding block context. |
| 152 |                 phase: 3, | executable/support line; behavior impact depends on surrounding block context. |
| 153 |                 title: "Conexao Confiavel".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 154 |                 status: if found > 0 { | executable/support line; behavior impact depends on surrounding block context. |

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| 156 |           PhaseStatus::Passed | executable/support line; behavior impact depends on surrounding
ock context. |
| 157 |           } else { | executable/support line; behavior impact depends on surrounding block context.
| 158 |           PhaseStatus::Failed | executable/support line; behavior impact depends on surrounding
ock context. |
| 159 |           }, | executable/support line; behavior impact depends on surrounding block context. |
| 160 |           details: format!("ECMS detectados: {found}"), | executable/support line; behavior impact
ends on surrounding block context. |
| 161 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 162 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 163 |       Err(e) => phases.push(PhaseResult { | executable/support line; behavior impact depends on surroun
g block context. |
| 164 |           phase: 3, | executable/support line; behavior impact depends on surrounding block context. |
| 165 |           title: "Conexao Confiavel".to_string(), | executable/support line; behavior impact depends on
rrounding block context. |
| 166 |           status: PhaseStatus::Failed, | executable/support line; behavior impact depends on surroundin
lock context. |
| 167 |           details: format!("falha detect/connect: {e}"), | executable/support line; behavior impact dep
s on surrounding block context. |
| 168 |       }); | executable/support line; behavior impact depends on surrounding block context. |
| 169 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 170 | (blank) | line is blank; visual separation only. |
| 171 |       match connect_ecm(cfg, target_sa) { | branch dispatch; changing arms alters behavior for variants and
put classes. |
| 172 |           Ok(()) => {} | executable/support line; behavior impact depends on surrounding block context. |
| 173 |           Err(e) => phases.push(PhaseResult { | executable/support line; behavior impact depends on surroun
g block context. |
| 174 |               phase: 3, | executable/support line; behavior impact depends on surrounding block context. |
| 175 |               title: "Conexao Confiavel (Connect)".to_string(), | executable/support line; behavior impact
ends on surrounding block context. |
| 176 |               status: PhaseStatus::Failed, | executable/support line; behavior impact depends on surroundin
lock context. |
| 177 |               details: format!("connect falhou: {e}"), | executable/support line; behavior impact depends o
urrounding block context. |
| 178 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 179 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 180 |   } else if matches!(cfg.mode, HwMode::Sim) { | executable/support line; behavior impact depends on surroun
g block context. |
| 181 |       let (ok, detail) = run_sim_connectivity_check(); | executable/support line; behavior impact depends o
urrounding block context. |
| 182 |       phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding block con
t. |
| 183 |           phase: 3, | executable/support line; behavior impact depends on surrounding block context. |
| 184 |           title: "Conexao Confiavel".to_string(), | executable/support line; behavior impact depends on sur
nding block context. |
| 185 |           status: if ok { PhaseStatus::Passed } else { PhaseStatus::Failed }, | executable/support line; be
ior impact depends on surrounding block context. |
| 186 |           details: detail, | executable/support line; behavior impact depends on surrounding block context.
| 187 |       }); | executable/support line; behavior impact depends on surrounding block context. |
| 188 |   } else { | executable/support line; behavior impact depends on surrounding block context. |
| 189 |       phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding block con
t. |
| 190 |           phase: 3, | executable/support line; behavior impact depends on surrounding block context. |
| 191 |           title: "Conexao Confiavel".to_string(), | executable/support line; behavior impact depends on sur
nding block context. |
| 192 |           status: PhaseStatus::Failed, | executable/support line; behavior impact depends on surrounding bl
context. |
| 193 |           details: "hardware live indisponivel".to_string(), | executable/support line; behavior impact dep
s on surrounding block context. |
| 194 |       }); | executable/support line; behavior impact depends on surrounding block context. |
| 195 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 196 | (blank) | line is blank; visual separation only. |

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| 197 |      // Phase 4: ECU identity | comment line; documentation intent, no runtime effect. |
| 198 |      if hw_ready { | conditional gate; changing condition shifts runtime path and side effects. |
| 199 |          match live_read_ecm_identity(cfg, target_sa) { | branch dispatch; changing arms alters behavior for v
ants and input classes. |
| 200 |              Ok(id) => { | executable/support line; behavior impact depends on surrounding block context. |
| 201 |                  let fp = id.fingerprint.is_some(); | executable/support line; behavior impact depends on surr
ding block context. |
| 202 |                  identity = Some(id); | executable/support line; behavior impact depends on surrounding block
text. |
| 203 |                  phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding b
k context. |
| 204 |                      phase: 4, | executable/support line; behavior impact depends on surrounding block context
| 205 |                      title: "Leitura e Identidade ECM".to_string(), | executable/support line; behavior impact
depends on surrounding block context. |
| 206 |                      status: if fp \\|\\| cfg.allow_untrusted_ecu { | executable/support line; behavior impact d
nds on surrounding block context. |
| 207 |                          PhaseStatus::Passed | executable/support line; behavior impact depends on surrounding
ock context. |
| 208 |                          } else { | executable/support line; behavior impact depends on surrounding block context.
| 209 |                          PhaseStatus::Failed | executable/support line; behavior impact depends on surrounding
ock context. |
| 210 |                      }, | executable/support line; behavior impact depends on surrounding block context. |
| 211 |                      details: format!("fingerprint={fp} allow_untrusted_ecu={}", cfg.allow_untrusted_ecu), | e
utable/support line; behavior impact depends on surrounding block context. |
| 212 |                  }); | executable/support line; behavior impact depends on surrounding block context. |
| 213 |              } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 214 |              Err(e) => phases.push(PhaseResult { | executable/support line; behavior impact depends on surroun
g block context. |
| 215 |                  phase: 4, | executable/support line; behavior impact depends on surrounding block context. |
| 216 |                  title: "Leitura e Identidade ECM".to_string(), | executable/support line; behavior impact dep
s on surrounding block context. |
| 217 |                  status: PhaseStatus::Failed, | executable/support line; behavior impact depends on surroundin
lock context. |
| 218 |                  details: format!("falha leitura DID: {e}"), | executable/support line; behavior impact depend
n surrounding block context. |
| 219 |                  }); | executable/support line; behavior impact depends on surrounding block context. |
| 220 |              } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 221 |          } else if matches!(cfg.mode, HwMode::Sim) { | executable/support line; behavior impact depends on surroun
g block context. |
| 222 |              let (ok, detail) = run_sim_identity_check(); | executable/support line; behavior impact depends on su
unding block context. |
| 223 |              phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding block con
t. |
| 224 |                  phase: 4, | executable/support line; behavior impact depends on surrounding block context. |
| 225 |                  title: "Leitura e Identidade ECM".to_string(), | executable/support line; behavior impact depends
surrounding block context. |
| 226 |                  status: if ok { PhaseStatus::Passed } else { PhaseStatus::Failed }, | executable/support line; be
ior impact depends on surrounding block context. |
| 227 |                  details: detail, | executable/support line; behavior impact depends on surrounding block context.
| 228 |                  }); | executable/support line; behavior impact depends on surrounding block context. |
| 229 |          } else { | executable/support line; behavior impact depends on surrounding block context. |
| 230 |              phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding block con
t. |
| 231 |                  phase: 4, | executable/support line; behavior impact depends on surrounding block context. |
| 232 |                  title: "Leitura e Identidade ECM".to_string(), | executable/support line; behavior impact depends
surrounding block context. |
| 233 |                  status: PhaseStatus::Failed, | executable/support line; behavior impact depends on surrounding bl
context. |
| 234 |                  details: "requer modo live".to_string(), | executable/support line; behavior impact depends on su
unding block context. |
| 235 |                  }); | executable/support line; behavior impact depends on surrounding block context. |
| 236 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 237 | (blank) | line is blank; visual separation only. |

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| 238 |      // Phase 5: Artifact validation | comment line; documentation intent, no runtime effect. |
| 239 |      match artifact_path { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 240 |          Some(path) => match validate_firmware_artifact(path, 64 * 1024 * 1024) { | executable/support line; b
vior impact depends on surrounding block context. |
| 241 |              Ok(r) => { | executable/support line; behavior impact depends on surrounding block context. |
| 242 |                  artifact = Some(r.clone()); | executable/support line; behavior impact depends on surrounding
ock context. |
| 243 |                      phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding b
k context. |
| 244 |                          phase: 5, | executable/support line; behavior impact depends on surrounding block context
| 245 |                          title: "Validacao de Artefato".to_string(), | executable/support line; behavior impact de
ds on surrounding block context. |
| 246 |                          status: PhaseStatus::Passed, | executable/support line; behavior impact depends on surrou
ng block context. |
| 247 |                          details: format!("{}", bytes crc32=0x{:08X}", r.bytes, r.crc32), | executable/support line;
havior impact depends on surrounding block context. |
| 248 |                      }); | executable/support line; behavior impact depends on surrounding block context. |
| 249 |              } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 250 |              Err(e) => phases.push(PhaseResult { | executable/support line; behavior impact depends on surroun
g block context. |
| 251 |                  phase: 5, | executable/support line; behavior impact depends on surrounding block context. |
| 252 |                  title: "Validacao de Artefato".to_string(), | executable/support line; behavior impact depend
n surrounding block context. |
| 253 |                  status: PhaseStatus::Failed, | executable/support line; behavior impact depends on surroundin
lock context. |
| 254 |                  details: format!("artefato invalido: {e}"), | executable/support line; behavior impact depend
n surrounding block context. |
| 255 |              }), | executable/support line; behavior impact depends on surrounding block context. |
| 256 |          }, | executable/support line; behavior impact depends on surrounding block context. |
| 257 |          None => phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding b
k context. |
| 258 |              phase: 5, | executable/support line; behavior impact depends on surrounding block context. |
| 259 |              title: "Validacao de Artefato".to_string(), | executable/support line; behavior impact depends on
rounding block context. |
| 260 |              status: if options.strict_mode { | executable/support line; behavior impact depends on surroundin
lock context. |
| 261 |                  PhaseStatus::Failed | executable/support line; behavior impact depends on surrounding block c
ext. |
| 262 |              } else { | executable/support line; behavior impact depends on surrounding block context. |
| 263 |                  PhaseStatus::Blocked | executable/support line; behavior impact depends on surrounding block
text. |
| 264 |              }, | executable/support line; behavior impact depends on surrounding block context. |
| 265 |              details: "forneca --firmware=<arquivo.bin>".to_string(), | executable/support line; behavior impa
depends on surrounding block context. |
| 266 |          }), | executable/support line; behavior impact depends on surrounding block context. |
| 267 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 268 | (blank) | line is blank; visual separation only. |
| 269 |      // Phase 6: Flash preflight gate | comment line; documentation intent, no runtime effect. |
| 270 |      if hw_ready { | conditional gate; changing condition shifts runtime path and side effects. |
| 271 |          match live_preflight_ecm(cfg, target_sa) { | branch dispatch; changing arms alters behavior for varia
and input classes. |
| 272 |              Ok(pf) => { | executable/support line; behavior impact depends on surrounding block context. |
| 273 |                  let v_ok = pf.supply_voltage_v >= 11.8; | executable/support line; behavior impact depends on
rounding block context. |
| 274 |                  preflight = Some(pf.clone()); | executable/support line; behavior impact depends on surroundi
block context. |
| 275 |                  let mut status = if v_ok { | executable/support line; behavior impact depends on surrounding
ck context. |
| 276 |                      PhaseStatus::Passed | executable/support line; behavior impact depends on surrounding blo
context. |
| 277 |                  } else { | executable/support line; behavior impact depends on surrounding block context. |
| 278 |                      PhaseStatus::Failed | executable/support line; behavior impact depends on surrounding blo
context. |
| 279 |                  }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet

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scopes. |
| 280 |         let mut details = format!( | executable/support line; behavior impact depends on surrounding
ck context. |
| 281 |             "supply_v={:.2} trusted_fp={}", | executable/support line; behavior impact depends on sur
nding block context. |
| 282 |             pf.supply_voltage_v, pf.trusted_fingerprint | executable/support line; behavior impact de
ds on surrounding block context. |
| 283 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 284 |             if options.execute_flash { | conditional gate; changing condition shifts runtime path and sid
ffects. |
| 285 |                 if let (true, Some(path), true) = ( | conditional gate; changing condition shifts runtime
th and side effects. |
| 286 |                     status == PhaseStatus::Passed, | executable/support line; behavior impact depends on
rounding block context. |
| 287 |                     artifact_path, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 288 |                     cfg.write_effectively_enabled(), | I/O or protocol-critical logic; edits can affect i
gration, timing, and diagnostic outcomes. |
| 289 |                 ) { | executable/support line; behavior impact depends on surrounding block context. |
| 290 |                     match fs::read(path) { | branch dispatch; changing arms alters behavior for variants
input classes. |
| 291 |                         Ok(payload) => match live_flash_ecm_firmware(cfg, target_sa, &payload) { | I/O or
otocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 292 |                             Ok(sum) => { | executable/support line; behavior impact depends on surroundin
lock context. |
| 293 |                                 flash_summary = Some(sum.clone()); | I/O or protocol-critical logic; edit
an affect integration, timing, and diagnostic outcomes. |
| 294 |                                 details = format!( | executable/support line; behavior impact depends on
rounding block context. |
| 295 |                                     "{} \\\ flash_ok bytes={} blocks={} crc32=0x{:08X} \\\ td_ack={} /{} seq
r={} fc_timeout={}", | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes.
| 296 |                                     details, | executable/support line; behavior impact depends on surrou
ng block context. |
| 297 |                                     sum.bytes_sent, | executable/support line; behavior impact depends on
rounding block context. |
| 298 |                                     sum.blocks_sent, | executable/support line; behavior impact depends o
urrounding block context. |
| 299 |                                     sum.crc32, | executable/support line; behavior impact depends on surr
ding block context. |
| 300 |                                     sum.transport_diagnostics.transfer_data_blocks_acked, | executable/su
rt line; behavior impact depends on surrounding block context. |
| 301 |                                     sum.transport_diagnostics.transfer_data_blocks_attempted, | executabl
upport line; behavior impact depends on surrounding block context. |
| 302 |                                     sum.transport_diagnostics.sequence_error_count, | executable/support
e; behavior impact depends on surrounding block context. |
| 303 |                                     sum.transport_diagnostics.flowcontrol_timeout_count | executable/supp
line; behavior impact depends on surrounding block context. |
| 304 |                                 ); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 305 |                             } | scope delimiter; structure-only line, but misplaced edits can change owne
ip and lifetime scopes. |
| 306 |                             Err(e) => { | executable/support line; behavior impact depends on surrounding
ock context. |
| 307 |                                 status = PhaseStatus::Failed; | executable/support line; behavior impact
ends on surrounding block context. |
| 308 |                                 details = format!("{}", \\\ flash_fail={}, details, e); | I/O or protocol-c
ical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 309 |                             } | scope delimiter; structure-only line, but misplaced edits can change owne
ip and lifetime scopes. |
| 310 |                                     }, | executable/support line; behavior impact depends on surrounding block contex
|
| 311 |                             Err(e) => { | executable/support line; behavior impact depends on surrounding blo
context. |
| 312 |                                 status = PhaseStatus::Failed; | executable/support line; behavior impact depe
on surrounding block context. |

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| 313 |                 details = format!("{}", \ firmware_read_fail={}, details, e); | executable/support
rt line; behavior impact depends on surrounding block context. |
| 314 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership
nd lifetime scopes. |
| 315 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership an
ifetime scopes. |
| 316 |                 } else { | executable/support line; behavior impact depends on surrounding block context.
| 317 |                 status = PhaseStatus::Failed; | executable/support line; behavior impact depends on s
urrounding block context. |
| 318 |                 details = format!( | executable/support line; behavior impact depends on surrounding
ck context. |
| 319 |                 "{} \ flash requested but missing requirements (artifact + write_enabled + prefl
t_ok)", | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 320 |                 details | executable/support line; behavior impact depends on surrounding block c
ext. |
| 321 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 322 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 323 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 324 |                 phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding b
k context. |
| 325 |                     phase: 6, | executable/support line; behavior impact depends on surrounding block context
| 326 |                     title: "Flash Seguro (Preflight)".to_string(), | executable/support line; behavior impact
ends on surrounding block context. |
| 327 |                     status, | executable/support line; behavior impact depends on surrounding block context.
| 328 |                     details, | executable/support line; behavior impact depends on surrounding block context.
| 329 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 330 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 331 |                 Err(e) => phases.push(PhaseResult { | executable/support line; behavior impact depends on surroun
g block context. |
| 332 |                     phase: 6, | executable/support line; behavior impact depends on surrounding block context. |
| 333 |                     title: "Flash Seguro (Preflight)".to_string(), | executable/support line; behavior impact dep
s on surrounding block context. |
| 334 |                     status: PhaseStatus::Failed, | executable/support line; behavior impact depends on surroundin
lock context. |
| 335 |                     details: format!("preflight falhou: {e}"), | executable/support line; behavior impact depends
surrounding block context. |
| 336 |                 }}, | executable/support line; behavior impact depends on surrounding block context. |
| 337 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 338 |             } else if matches!(cfg.mode, HwMode::Sim) { | executable/support line; behavior impact depends on surroun
g block context. |
| 339 |                 let (ok, detail) = run_sim_preflight_check(); | executable/support line; behavior impact depends on s
urrounding block context. |
| 340 |                 phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding block con
t. |
| 341 |                     phase: 6, | executable/support line; behavior impact depends on surrounding block context. |
| 342 |                     title: "Flash Seguro (Preflight)".to_string(), | executable/support line; behavior impact depends
surrounding block context. |
| 343 |                     status: if ok { PhaseStatus::Passed } else { PhaseStatus::Failed }, | executable/support line; be
ior impact depends on surrounding block context. |
| 344 |                     details: detail, | executable/support line; behavior impact depends on surrounding block context.
| 345 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 346 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 347 |                 phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding block con
t. |
| 348 |                     phase: 6, | executable/support line; behavior impact depends on surrounding block context. |
| 349 |                     title: "Flash Seguro (Preflight)".to_string(), | executable/support line; behavior impact depends
surrounding block context. |
| 350 |                     status: PhaseStatus::Failed, | executable/support line; behavior impact depends on surrounding bl
context. |
| 351 |                     details: "requer modo live".to_string(), | executable/support line; behavior impact depends on su
urrounding block context. |

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| 352 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 353 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 354 | (blank) | line is blank; visual separation only. |
| 355 |     // Phase 7: Diagnostics readiness | comment line; documentation intent, no runtime effect. |
| 356 |     let diag_ready_cfg = cfg.uds_timeout_p2_ms > 0 | I/O or protocol-critical logic; edits can affect integra
n, timing, and diagnostic outcomes. |
| 357 |         && cfg.uds_timeout_p2star_ms >= cfg.uds_timeout_p2_ms | I/O or protocol-critical logic; edits can aff
integration, timing, and diagnostic outcomes. |
| 358 |         && cfg.uds_retry_count > 0; | I/O or protocol-critical logic; edits can affect integration, timing, a
diagnostic outcomes. |
| 359 |     let diag_runtime = run_uds_runtime_self_test(); | I/O or protocol-critical logic; edits can affect integ
on, timing, and diagnostic outcomes. |
| 360 |     let diag_ready = diag_ready_cfg && diag_runtime.0; | executable/support line; behavior impact depends on
rounding block context. |
| 361 |     phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding block context
| 362 |         phase: 7, | executable/support line; behavior impact depends on surrounding block context. |
| 363 |         title: "Diagnostic e Debug".to_string(), | executable/support line; behavior impact depends on surro
ing block context. |
| 364 |         status: if diag_ready { | executable/support line; behavior impact depends on surrounding block conte
|
| 365 |             PhaseStatus::Passed | executable/support line; behavior impact depends on surrounding block conte
|
| 366 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 367 |             PhaseStatus::Failed | executable/support line; behavior impact depends on surrounding block conte
|
| 368 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 369 |         details: format!( | executable/support line; behavior impact depends on surrounding block context. |
| 370 |             "uds_retry={} p2={} p2*= {} runtime={}: {}", | I/O or protocol-critical logic; edits can affect i
gration, timing, and diagnostic outcomes. |
| 371 |             cfg.uds_retry_count, | I/O or protocol-critical logic; edits can affect integration, timing, and
gnostic outcomes. |
| 372 |             cfg.uds_timeout_p2_ms, | I/O or protocol-critical logic; edits can affect integration, timing, an
diagnostic outcomes. |
| 373 |             cfg.uds_timeout_p2star_ms, | I/O or protocol-critical logic; edits can affect integration, timing
nd diagnostic outcomes. |
| 374 |             diag_runtime.0, | executable/support line; behavior impact depends on surrounding block context. |
| 375 |             diag_runtime.1 | executable/support line; behavior impact depends on surrounding block context. |
| 376 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 377 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 378 | (blank) | line is blank; visual separation only. |
| 379 |     // Phase 8: Governance/security | comment line; documentation intent, no runtime effect. |
| 380 |     let gov = cfg.allowlist_path.is_some() && cfg.noninteractive_approved; | executable/support line; behavio
mpact depends on surrounding block context. |
| 381 |     let write_policy = if cfg.enable_write { | I/O or protocol-critical logic; edits can affect integration,
ing, and diagnostic outcomes. |
| 382 |         cfg.write_effectively_enabled() | I/O or protocol-critical logic; edits can affect integration, timin
and diagnostic outcomes. |
| 383 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 384 |         true | executable/support line; behavior impact depends on surrounding block context. |
| 385 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 386 |     phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding block context
| 387 |         phase: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 388 |         title: "Governanca e Seguranca".to_string(), | executable/support line; behavior impact depends on su
unding block context. |
| 389 |         status: if gov && write_policy { | I/O or protocol-critical logic; edits can affect integration, timi
and diagnostic outcomes. |
| 390 |             PhaseStatus::Passed | executable/support line; behavior impact depends on surrounding block conte
|
| 391 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 392 |             PhaseStatus::Failed | executable/support line; behavior impact depends on surrounding block conte
|
| 393 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 394 |         details: format!( | executable/support line; behavior impact depends on surrounding block context. |
| 395 |             "allowlist_present={} noninteractive_approved={} write_effective={}", | I/O or protocol-critical
ic; edits can affect integration, timing, and diagnostic outcomes. |

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| 396 |         cfg.allowlist_path.is_some(), | executable/support line; behavior impact depends on surrounding block context. |
| 397 |         cfg.noninteractive_approved, | executable/support line; behavior impact depends on surrounding block context. |
| 398 |         cfg.write_effectively_enabled() | I/O or protocol-critical logic; edits can affect integration, testing, and diagnostic outcomes. |
| 399 |     ), | executable/support line; behavior impact depends on surrounding block context. |
| 400 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 401 | (blank) | line is blank; visual separation only. |
| 402 |     // Phase 9: Reliability/stress readiness | comment line; documentation intent, no runtime effect. |
| 403 |     let reliability_ok = cfg.rate_limit_global_per_sec > 0 | executable/support line; behavior impact depends on surrounding block context. |
| 404 |         && cfg.rate_limit_per_id_per_sec > 0 | executable/support line; behavior impact depends on surrounding block context. |
| 405 |         && cfg.reconnect_backoff_max_ms >= cfg.reconnect_backoff_base_ms | executable/support line; behavior impact depends on surrounding block context. |
| 406 |         && run_sim_stress_check().0; | executable/support line; behavior impact depends on surrounding block context. |
| 407 |     let sim_stress = run_sim_stress_check(); | executable/support line; behavior impact depends on surrounding block context. |
| 408 |     phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding block context. |
| 409 |         phase: 9, | executable/support line; behavior impact depends on surrounding block context. |
| 410 |         title: "Confiabilidade e Stress".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 411 |         status: if reliability_ok { | executable/support line; behavior impact depends on surrounding block context. |
| 412 |             PhaseStatus::Passed | executable/support line; behavior impact depends on surrounding block context. |
| 413 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 414 |             PhaseStatus::Failed | executable/support line; behavior impact depends on surrounding block context. |
| 415 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 416 |         details: format!( | executable/support line; behavior impact depends on surrounding block context. |
| 417 |             "rate_global={} rate_per_id={} reconnect={}..{} ms", | executable/support line; behavior impact depends on surrounding block context. |
| 418 |             cfg.rate_limit_global_per_sec, | executable/support line; behavior impact depends on surrounding block context. |
| 419 |             cfg.rate_limit_per_id_per_sec, | executable/support line; behavior impact depends on surrounding block context. |
| 420 |             cfg.reconnect_backoff_base_ms, | executable/support line; behavior impact depends on surrounding block context. |
| 421 |             cfg.reconnect_backoff_max_ms | executable/support line; behavior impact depends on surrounding block context. |
| 422 |         ) + &format!(" stress={}: {}", sim_stress.0, sim_stress.1), | executable/support line; behavior impact depends on surrounding block context. |
| 423 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 424 | (blank) | line is blank; visual separation only. |
| 425 |     // Phase 10: Industrialization | comment line; documentation intent, no runtime effect. |
| 426 |     let industrial_ok = Path::new("Cargo.toml").exists() | executable/support line; behavior impact depends on surrounding block context. |
| 427 |         && Path::new(".github/workflows/integration-mock.yml").exists() | executable/support line; behavior impact depends on surrounding block context. |
| 428 |         && Path::new("docs/ECM-Data.md").exists(); | executable/support line; behavior impact depends on surrounding block context. |
| 429 |     phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding block context. |
| 430 |         phase: 10, | executable/support line; behavior impact depends on surrounding block context. |
| 431 |         title: "Industrializacao".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 432 |         status: if industrial_ok { | executable/support line; behavior impact depends on surrounding block context. |
| 433 |             PhaseStatus::Passed | executable/support line; behavior impact depends on surrounding block context. |
| 434 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 435 |             PhaseStatus::Failed | executable/support line; behavior impact depends on surrounding block context. |
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| 436 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 437 |         details: if industrial_ok { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 438 |             "pipeline de build/test encontrado".to_string() | executable/support line; behavior impact depend
n surrounding block context. |
| 439 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 440 |             "faltam artefatos de pipeline".to_string() | executable/support line; behavior impact depends on
rounding block context. |
| 441 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 442 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 443 | (blank) | line is blank; visual separation only. |
| 444 |     // Phase 11: Homologation dossier generation (placeholder, finalized after writing artifacts) | comment l
; documentation intent, no runtime effect. |
| 445 |     phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding block context
| 446 |         phase: 11, | executable/support line; behavior impact depends on surrounding block context. |
| 447 |         title: "Homologacao Final".to_string(), | executable/support line; behavior impact depends on surround
g block context. |
| 448 |         status: PhaseStatus::Passed, | executable/support line; behavior impact depends on surrounding block
text. |
| 449 |         details: "gerando dossie json/markdown".to_string(), | executable/support line; behavior impact depen
on surrounding block context. |
| 450 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 451 | (blank) | line is blank; visual separation only. |
| 452 |     // Phase 12: Production gate | comment line; documentation intent, no runtime effect. |
| 453 |     let overall_passed = phases | executable/support line; behavior impact depends on surrounding block conte
|
| 454 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 455 |         .filter(\|p\| p.phase <= 10) | executable/support line; behavior impact depends on surrounding block
text. |
| 456 |         .all(\|p\| match p.status { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 457 |             PhaseStatus::Passed => true, | executable/support line; behavior impact depends on surrounding bl
context. |
| 458 |             PhaseStatus::Blocked => !options.strict_mode, | executable/support line; behavior impact depends
surrounding block context. |
| 459 |             PhaseStatus::Failed => false, | executable/support line; behavior impact depends on surrounding b
k context. |
| 460 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 461 |     phases.push(PhaseResult { | executable/support line; behavior impact depends on surrounding block context
| 462 |         phase: 12, | executable/support line; behavior impact depends on surrounding block context. |
| 463 |         title: "Gate de Producao".to_string(), | executable/support line; behavior impact depends on surround
block context. |
| 464 |         status: if overall_passed { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 465 |             PhaseStatus::Passed | executable/support line; behavior impact depends on surrounding block conte
|
| 466 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 467 |             PhaseStatus::Failed | executable/support line; behavior impact depends on surrounding block conte
|
| 468 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 469 |         details: if overall_passed { | executable/support line; behavior impact depends on surrounding block
text. |
| 470 |             "gates criticos satisfeitos para o contexto atual".to_string() | executable/support line; behavior
mpact depends on surrounding block context. |
| 471 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 472 |             "ha falhas bloqueantes nas fases anteriores".to_string() | executable/support line; behavior impa
depends on surrounding block context. |
| 473 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 474 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 475 | (blank) | line is blank; visual separation only. |
| 476 |     let conformance_summary = Some(build_conformance_summary( | executable/support line; behavior impact depe
on surrounding block context. |
| 477 |     options.execute_flash, | I/O or protocol-critical logic; edits can affect integration, timing, and di
ostic outcomes. |
| 478 |     options.strict_mode, | executable/support line; behavior impact depends on surrounding block context.

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| 479 |         &flash_summary, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic
utcomes. |
| 480 |         &capabilities, | executable/support line; behavior impact depends on surrounding block context. |
| 481 |     )); | executable/support line; behavior impact depends on surrounding block context. |
| 482 | (blank) | line is blank; visual separation only. |
| 483 |     let report = ProgramReport { | executable/support line; behavior impact depends on surrounding block context. |
. |
| 484 |         generated_at_ms: now_ms(), | executable/support line; behavior impact depends on surrounding block context. |
| 485 |         mode: match cfg.mode { | executable/support line; behavior impact depends on surrounding block context. |
|
| 486 |             HwMode::Sim => "sim".to_string(), | executable/support line; behavior impact depends on surrounding
block context. |
| 487 |             HwMode::Live => "live".to_string(), | executable/support line; behavior impact depends on surrounding
g block context. |
| 488 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 489 |         target_sa, | executable/support line; behavior impact depends on surrounding block context. |
| 490 |         strict_mode: options.strict_mode, | executable/support line; behavior impact depends on surrounding block
context. |
| 491 |         execute_flash: options.execute_flash, | I/O or protocol-critical logic; edits can affect integration,
ming, and diagnostic outcomes. |
| 492 |         artifact, | executable/support line; behavior impact depends on surrounding block context. |
| 493 |         detect, | executable/support line; behavior impact depends on surrounding block context. |
| 494 |         preflight, | executable/support line; behavior impact depends on surrounding block context. |
| 495 |         identity, | executable/support line; behavior impact depends on surrounding block context. |
| 496 |         conformance_summary, | executable/support line; behavior impact depends on surrounding block context. |
| 497 |         phases, | executable/support line; behavior impact depends on surrounding block context. |
| 498 |         overall_passed, | executable/support line; behavior impact depends on surrounding block context. |
| 499 |         report_sha256: None, | executable/support line; behavior impact depends on surrounding block context. |
| 500 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 501 | (blank) | line is blank; visual separation only. |
| 502 |     let ts = now_ms(); | executable/support line; behavior impact depends on surrounding block context. |
| 503 |     let json_path = out_dir.join(format!("production_phase_report_{ts}.json")); | executable/support line; behavior
impact depends on surrounding block context. |
| 504 |     let md_path = out_dir.join(format!("production_phase_report_{ts}.md")); | executable/support line; behavior
impact depends on surrounding block context. |
| 505 | (blank) | line is blank; visual separation only. |
| 506 |     let json = serde_json::to_string_pretty(&report) | serialization contract touchpoint; edits can break rep
compatibility and tooling. |
| 507 |         .map_err(|e| HwError::Unknown(format!("serialize report json failed: {e}")))?; | error-propagating
ration; changing it impacts failure semantics and caller recovery paths. |
| 508 |     fs::write(&json_path, json) | I/O or protocol-critical logic; edits can affect integration, timing, and d
nostic outcomes. |
| 509 |         .map_err(|e| HwError::Unknown(format!("write report json failed: {e}")))?; | error-propagating oper
on; changing it impacts failure semantics and caller recovery paths. |
| 510 | (blank) | line is blank; visual separation only. |
| 511 |     let mut md = String::new(); | executable/support line; behavior impact depends on surrounding block context. |
|
| 512 |     md.push_str("# Production Program Report\n\n"); | executable/support line; behavior impact depends on sur
nding block context. |
| 513 |     md.push_str(&format!("- generated_at_ms: {}\n", report.generated_at_ms)); | executable/support line; beha
r impact depends on surrounding block context. |
| 514 |     md.push_str(&format!("- mode: {}\n", report.mode)); | executable/support line; behavior impact depends on
rrounding block context. |
| 515 |     md.push_str(&format!("- target_sa: {?:}\n", report.target_sa)); | error-propagating operation; changing i
mpacts failure semantics and caller recovery paths. |
| 516 |     md.push_str(&format!("- overall_passed: {}\n\n", report.overall_passed)); | executable/support line; beha
r impact depends on surrounding block context. |
| 517 |     if let Some(c) = &report.conformance_summary { | conditional gate; changing condition shifts runtime path
d side effects. |
| 518 |         md.push_str("## UDS Conformance Summary\n\n"); | I/O or protocol-critical logic; edits can affect int
ation, timing, and diagnostic outcomes. |
| 519 |         md.push_str(&format!("- passed: {}\n", c.passed)); | executable/support line; behavior impact depends
surrounding block context. |
| 520 |         for s in &c.services { | iteration control; changing bounds/body affects throughput, timing, and poss

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y safety guarantees. |
| 521 |           md.push_str(&format!( | executable/support line; behavior impact depends on surrounding block con
t. |
| 522 |           "- SID {}: passed={} ({})\n", | executable/support line; behavior impact depends on surroundi
block context. |
| 523 |           s.service_sid, s.passed, s.evidence | executable/support line; behavior impact depends on surro
nding block context. |
| 524 |           )); | executable/support line; behavior impact depends on surrounding block context. |
| 525 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 526 |           md.push_str("\n"); | executable/support line; behavior impact depends on surrounding block context. |
| 527 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 528 |       md.push_str("## Phase Results\n\n"); | executable/support line; behavior impact depends on surrounding bl
context. |
| 529 |       for p in &report.phases { | iteration control; changing bounds/body affects throughput, timing, and possi
safety guarantees. |
| 530 |           md.push_str(&format!( | executable/support line; behavior impact depends on surrounding block context
| 531 |           "- Fase {} - {}: {:?} ({})\n", | error-propagating operation; changing it impacts failure semanti
and caller recovery paths. |
| 532 |           p.phase, p.title, p.status, p.details | executable/support line; behavior impact depends on surro
ing block context. |
| 533 |           )); | executable/support line; behavior impact depends on surrounding block context. |
| 534 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 535 |       fs::write(&md_path, md) | I/O or protocol-critical logic; edits can affect integration, timing, and diagn
ic outcomes. |
| 536 |           .map_err(|e| HwError::Unknown(format!("write report markdown failed: {e}")))?; | error-propagating
ration; changing it impacts failure semantics and caller recovery paths. |
| 537 | (blank) | line is blank; visual separation only. |
| 538 |       let json_bytes = fs::read(&json_path) | executable/support line; behavior impact depends on surrounding b
k context. |
| 539 |           .map_err(|e| HwError::Unknown(format!("read report json failed: {e}")))?; | error-propagating opera
n; changing it impacts failure semantics and caller recovery paths. |
| 540 |       let mut h = Sha256::new(); | executable/support line; behavior impact depends on surrounding block context
|
| 541 |       h.update(&json_bytes); | executable/support line; behavior impact depends on surrounding block context. |
| 542 |       let report_hash = hex_sha(h.finalize().as_slice()); | executable/support line; behavior impact depends on
rounding block context. |
| 543 | (blank) | line is blank; visual separation only. |
| 544 |       let mut report_final = report; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 545 |       if let Some(p11) = report_final.phases.iter_mut().find(|p| p.phase == 11) { | conditional gate; changin
ondition shifts runtime path and side effects. |
| 546 |           p11.status = if Path::new(&json_path).exists() && Path::new(&md_path).exists() { | executable/support
ne; behavior impact depends on surrounding block context. |
| 547 |               PhaseStatus::Passed | executable/support line; behavior impact depends on surrounding block conte
|
| 548 |           } else { | executable/support line; behavior impact depends on surrounding block context. |
| 549 |               PhaseStatus::Failed | executable/support line; behavior impact depends on surrounding block conte
|
| 550 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 551 |           p11.details = format!( | executable/support line; behavior impact depends on surrounding block context
|
| 552 |           "json={} md={} sha256={}", | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 553 |           json_path.display(), | executable/support line; behavior impact depends on surrounding block cont
. |
| 554 |           md_path.display(), | executable/support line; behavior impact depends on surrounding block context
|
| 555 |           report_hash | executable/support line; behavior impact depends on surrounding block context. |
| 556 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 557 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 558 |       report_final.report_sha256 = Some(report_hash); | executable/support line; behavior impact depends on sur
nding block context. |
| 559 | (blank) | line is blank; visual separation only. |

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| 560 |     let json_final = serde_json::to_string_pretty(&report_final) | serialization contract touchpoint; edits c
break report compatibility and tooling. |
| 561 |         .map_err(\|e\| HwError::Unknown(format!("serialize final report json failed: {e}")))?; | error-propagat
ng operation; changing it impacts failure semantics and caller recovery paths. |
| 562 |     fs::write(&json_path, json_final) | I/O or protocol-critical logic; edits can affect integration, timing,
d diagnostic outcomes. |
| 563 |         .map_err(\|e\| HwError::Unknown(format!("write final report json failed: {e}")))?; | error-propagatin
peration; changing it impacts failure semantics and caller recovery paths. |
| 564 | (blank) | line is blank; visual separation only. |
| 565 |     Ok(json_path) | executable/support line; behavior impact depends on surrounding block context. |
| 566 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 567 | (blank) | line is blank; visual separation only. |
| 568 | fn run_sim_baseline_check() -> (bool, String) { | declares executable behavior; signature/body changes alter
, control flow, and outputs. |
| 569 |     let mut sim = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding bl
context. |
| 570 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 571 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 572 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 573 |     sim.throttle_pct = 35.0; | executable/support line; behavior impact depends on surrounding block context. |
| 574 |     for _ in 0..360 { | iteration control; changing bounds/body affects throughput, timing, and possibly safe
guarantees. |
| 575 |         sim.tick(1.0 / 60.0); | executable/support line; behavior impact depends on surrounding block context
| 576 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 577 |     let ok = sim.elapsed > 5.0 && sim.metrics.steps_completed > 0; | executable/support line; behavior impact
depends on surrounding block context. |
| 578 |     ( | executable/support line; behavior impact depends on surrounding block context. |
| 579 |         ok, | executable/support line; behavior impact depends on surrounding block context. |
| 580 |         format!( | executable/support line; behavior impact depends on surrounding block context. |
| 581 |             "elapsed={:.2}s steps={} speed={:.2}kmh", | executable/support line; behavior impact depends on s
ounding block context. |
| 582 |             sim.elapsed, sim.metrics.steps_completed, sim.tcm.ground_speed_kmh | executable/support line; beh
or impact depends on surrounding block context. |
| 583 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 584 |     ) | executable/support line; behavior impact depends on surrounding block context. |
| 585 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 586 | (blank) | line is blank; visual separation only. |
| 587 | fn run_sim_connectivity_check() -> (bool, String) { | declares executable behavior; signature/body changes al
API, control flow, and outputs. |
| 588 |     let mut sim = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding bl
context. |
| 589 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 590 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 591 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 592 |     sim.throttle_pct = 20.0; | executable/support line; behavior impact depends on surrounding block context. |
| 593 |     for _ in 0..240 { | iteration control; changing bounds/body affects throughput, timing, and possibly safe
guarantees. |
| 594 |         sim.tick(1.0 / 60.0); | executable/support line; behavior impact depends on surrounding block context
| 595 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 596 |     let can_health = sim.can_net.network_health_score_01(); | executable/support line; behavior impact depend
n surrounding block context. |
| 597 |     let ok = (0.0..=1.0).contains(&can_health); | executable/support line; behavior impact depends on surroun
g block context. |
| 598 |     (ok, format!("can_health={:.3} errors={}", can_health, sim.can_net.total_errors())) | executable/support
e; behavior impact depends on surrounding block context. |
| 599 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 600 | (blank) | line is blank; visual separation only. |
| 601 | fn run_sim_identity_check() -> (bool, String) { | declares executable behavior; signature/body changes alter
, control flow, and outputs. |
| 602 |     let mut sim = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding bl
context. |
| 603 |     let vin = sim.uds_ecm.process(&[0x22, 0xF1, 0x90], 0.0); | I/O or protocol-critical logic; edits can affe
integration, timing, and diagnostic outcomes. |
| 604 |     let sw = sim.uds_ecm.process(&[0x22, 0xF1, 0x89], 0.1); | I/O or protocol-critical logic; edits can affect
ntegration, timing, and diagnostic outcomes. |

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| 605 |     let vin_ok = vin.first().copied() == Some(0x62) && vin.len() > 6; | executable/support line; behavior impact
depends on surrounding block context. |
| 606 |     let sw_ok = sw.first().copied() == Some(0x62) && sw.len() > 6; | executable/support line; behavior impact
depends on surrounding block context. |
| 607 |     ( | executable/support line; behavior impact depends on surrounding block context. |
| 608 |         vin_ok && sw_ok, | executable/support line; behavior impact depends on surrounding block context. |
| 609 |         format!("vin_ok={} sw_ok={} vin_len={} sw_len={}", vin_ok, sw_ok, vin.len(), sw.len()), | executable/
port line; behavior impact depends on surrounding block context. |
| 610 |     ) | executable/support line; behavior impact depends on surrounding block context. |
| 611 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 612 | (blank) | line is blank; visual separation only. |
| 613 | fn build_conformance_summary( | declares executable behavior; signature/body changes alter API, control flow,
d outputs. |
| 614 |     execute_flash: bool, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnosti
outcomes. |
| 615 |     strict_mode: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 616 |     flash_summary: &Option<FlashSummary>, | I/O or protocol-critical logic; edits can affect integration, tim
, and diagnostic outcomes. |
| 617 |     capabilities: &HwCapabilities, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 618 | ) -> ConformanceSummary { | executable/support line; behavior impact depends on surrounding block context. |
| 619 |     if !execute_flash { | conditional gate; changing condition shifts runtime path and side effects. |
| 620 |         return ConformanceSummary { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 621 |             passed: !strict_mode, | executable/support line; behavior impact depends on surrounding block con
t. |
| 622 |             services: vec![ | executable/support line; behavior impact depends on surrounding block context.
| 623 |                 ServiceConformanceResult { | executable/support line; behavior impact depends on surrounding
ck context. |
| 624 |                     service_sid: "0x27".to_string(), | executable/support line; behavior impact depends on su
unding block context. |
| 625 |                     passed: !strict_mode, | executable/support line; behavior impact depends on surrounding b
k context. |
| 626 |                     evidence: "flash path not executed".to_string(), | I/O or protocol-critical logic; edits
affect integration, timing, and diagnostic outcomes. |
| 627 |                 }, | executable/support line; behavior impact depends on surrounding block context. |
| 628 |                 ServiceConformanceResult { | executable/support line; behavior impact depends on surrounding
ck context. |
| 629 |                     service_sid: "0x34".to_string(), | executable/support line; behavior impact depends on su
unding block context. |
| 630 |                     passed: !strict_mode, | executable/support line; behavior impact depends on surrounding b
k context. |
| 631 |                     evidence: "flash path not executed".to_string(), | I/O or protocol-critical logic; edits
affect integration, timing, and diagnostic outcomes. |
| 632 |                 }, | executable/support line; behavior impact depends on surrounding block context. |
| 633 |                 ServiceConformanceResult { | executable/support line; behavior impact depends on surrounding
ck context. |
| 634 |                     service_sid: "0x36".to_string(), | executable/support line; behavior impact depends on su
unding block context. |
| 635 |                     passed: !strict_mode, | executable/support line; behavior impact depends on surrounding b
k context. |
| 636 |                     evidence: "flash path not executed".to_string(), | I/O or protocol-critical logic; edits
affect integration, timing, and diagnostic outcomes. |
| 637 |                 }, | executable/support line; behavior impact depends on surrounding block context. |
| 638 |                 ServiceConformanceResult { | executable/support line; behavior impact depends on surrounding
ck context. |
| 639 |                     service_sid: "0x37".to_string(), | executable/support line; behavior impact depends on su
unding block context. |
| 640 |                     passed: !strict_mode, | executable/support line; behavior impact depends on surrounding b
k context. |
| 641 |                     evidence: "flash path not executed".to_string(), | I/O or protocol-critical logic; edits
affect integration, timing, and diagnostic outcomes. |
| 642 |                 }, | executable/support line; behavior impact depends on surrounding block context. |
| 643 |             ], | executable/support line; behavior impact depends on surrounding block context. |
| 644 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop

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|
| 645 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 646 | (blank) | line is blank; visual separation only. |
| 647 |     if let Some(sum) = flash_summary { | conditional gate; changing condition shifts runtime path and side ef
ts. |
| 648 |         let d = &sum.transport_diagnostics; | executable/support line; behavior impact depends on surrounding
ock context. |
| 649 |         let sid27_ok = d.security_seed_positive && d.security_unlock_positive; | executable/support line; beh
or impact depends on surrounding block context. |
| 650 |         let sid34_ok = d.request_download_positive; | executable/support line; behavior impact depends on sur
nding block context. |
| 651 |         let sid36_ok = d.transfer_data_blocks_attempted > 0 | executable/support line; behavior impact depend
n surrounding block context. |
| 652 |         && d.transfer_data_blocks_attempted == d.transfer_data_blocks_acked | executable/support line; be
ior impact depends on surrounding block context. |
| 653 |         && d.sequence_error_count == 0 | executable/support line; behavior impact depends on surrounding
ck context. |
| 654 |         && d.flowcontrol_timeout_count == 0; | executable/support line; behavior impact depends on surrou
ng block context. |
| 655 |         let sid37_ok = d.request_transfer_exit_positive; | executable/support line; behavior impact depends o
urrounding block context. |
| 656 | (blank) | line is blank; visual separation only. |
| 657 |         let services = vec![] | executable/support line; behavior impact depends on surrounding block context.
| 658 |         ServiceConformanceResult { | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 659 |             service_sid: "0x27".to_string(), | executable/support line; behavior impact depends on surrou
ng block context. |
| 660 |             passed: sid27_ok, | executable/support line; behavior impact depends on surrounding block con
t. |
| 661 |             evidence: format!( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 662 |                 "seed_ok={} unlock_ok={}", | executable/support line; behavior impact depends on surround
block context. |
| 663 |                 d.security_seed_positive, d.security_unlock_positive | executable/support line; behavior
act depends on surrounding block context. |
| 664 |             ), | executable/support line; behavior impact depends on surrounding block context. |
| 665 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 666 |         ServiceConformanceResult { | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 667 |             service_sid: "0x34".to_string(), | executable/support line; behavior impact depends on surrou
ng block context. |
| 668 |             passed: sid34_ok, | executable/support line; behavior impact depends on surrounding block con
t. |
| 669 |             evidence: format!("request_download_positive={}", d.request_download_positive), | executable/
port line; behavior impact depends on surrounding block context. |
| 670 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 671 |         ServiceConformanceResult { | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 672 |             service_sid: "0x36".to_string(), | executable/support line; behavior impact depends on surrou
ng block context. |
| 673 |             passed: sid36_ok, | executable/support line; behavior impact depends on surrounding block con
t. |
| 674 |             evidence: format!( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 675 |                 "acked={} / {} seq_err={} fc_timeout={} stmin_ms={:?} bs={:?}", | error-propagating operati
changing it impacts failure semantics and caller recovery paths. |
| 676 |             d.transfer_data_blocks_acked, | executable/support line; behavior impact depends on surrou
ing block context. |
| 677 |             d.transfer_data_blocks_attempted, | executable/support line; behavior impact depends on s
ounding block context. |
| 678 |             d.sequence_error_count, | executable/support line; behavior impact depends on surrounding
ock context. |
| 679 |             d.flowcontrol_timeout_count, | executable/support line; behavior impact depends on surrou
ng block context. |
| 680 |             d.fc_stmin_seen_ms, | executable/support line; behavior impact depends on surrounding blo

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context. |
| 681 |           d.fc_blocksize_seen | executable/support line; behavior impact depends on surrounding blo
context. |
| 682 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 683 |       }, | executable/support line; behavior impact depends on surrounding block context. |
| 684 |       ServiceConformanceResult { | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 685 |           service_sid: "0x37".to_string(), | executable/support line; behavior impact depends on surrou
ng block context. |
| 686 |           passed: sid37_ok, | executable/support line; behavior impact depends on surrounding block con
t. |
| 687 |           evidence: format!("request_transfer_exit_positive={}", d.request_transfer_exit_positive), | e
utable/support line; behavior impact depends on surrounding block context. |
| 688 |       }, | executable/support line; behavior impact depends on surrounding block context. |
| 689 |     ]; | executable/support line; behavior impact depends on surrounding block context. |
| 690 | (blank) | line is blank; visual separation only. |
| 691 |     return ConformanceSummary { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 692 |         passed: services.iter().all(|s| s.passed), | executable/support line; behavior impact depends o
urrounding block context. |
| 693 |         services, | executable/support line; behavior impact depends on surrounding block context. |
| 694 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 695 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 696 | (blank) | line is blank; visual separation only. |
| 697 |     ConformanceSummary { | executable/support line; behavior impact depends on surrounding block context. |
| 698 |         passed: false, | executable/support line; behavior impact depends on surrounding block context. |
| 699 |         services: vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 700 |             ServiceConformanceResult { | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 701 |                 service_sid: "0x27".to_string(), | executable/support line; behavior impact depends on surrou
ng block context. |
| 702 |                 passed: false, | executable/support line; behavior impact depends on surrounding block contex
|
| 703 |                 evidence: format!( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 704 |                     "flash path requested without summary; capabilities={}", | I/O or protocol-critical logic
dits can affect integration, timing, and diagnostic outcomes. |
| 705 |                     if capabilities.missing_for_live_flash().is_empty() { | conditional gate; changing condit
shifts runtime path and side effects. |
| 706 |                         "present".to_string() | executable/support line; behavior impact depends on surroundi
block context. |
| 707 |                     } else { | executable/support line; behavior impact depends on surrounding block context.
| 708 |                         capabilities.missing_for_live_flash().join(",") | I/O or protocol-critical logic; edi
can affect integration, timing, and diagnostic outcomes. |
| 709 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 710 |                 }, | executable/support line; behavior impact depends on surrounding block context. |
| 711 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 712 |             ServiceConformanceResult { | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 713 |                 service_sid: "0x34".to_string(), | executable/support line; behavior impact depends on surrou
ng block context. |
| 714 |                 passed: false, | executable/support line; behavior impact depends on surrounding block contex
|
| 715 |                 evidence: "flash summary unavailable".to_string(), | I/O or protocol-critical logic; edits ca
ffect integration, timing, and diagnostic outcomes. |
| 716 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 717 |             ServiceConformanceResult { | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 718 |                 service_sid: "0x36".to_string(), | executable/support line; behavior impact depends on surrou
ng block context. |
| 719 |                 passed: false, | executable/support line; behavior impact depends on surrounding block contex
|
| 720 |                 evidence: "flash summary unavailable".to_string(), | I/O or protocol-critical logic; edits ca

```



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ffect integration, timing, and diagnostic outcomes. |
| 721 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 722 |         ServiceConformanceResult { | executable/support line; behavior impact depends on surrounding block
context. |
| 723 |             service_sid: "0x37".to_string(), | executable/support line; behavior impact depends on surround
ng block context. |
| 724 |             passed: false, | executable/support line; behavior impact depends on surrounding block context
|
| 725 |             evidence: "flash summary unavailable".to_string(), | I/O or protocol-critical logic; edits ca
ffect integration, timing, and diagnostic outcomes. |
| 726 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 727 |     ], | executable/support line; behavior impact depends on surrounding block context. |
| 728 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 729 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 730 | (blank) | line is blank; visual separation only. |
| 731 | fn run_sim_preflight_check() -> (bool, String) { | declares executable behavior; signature/body changes alter
I, control flow, and outputs. |
| 732 |     let mut sim = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding bl
context. |
| 733 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 734 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 735 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 736 |     for _ in 0..120 { | iteration control; changing bounds/body affects throughput, timing, and possibly safe
guarantees. |
| 737 |         sim.tick(1.0 / 60.0); | executable/support line; behavior impact depends on surrounding block context
|
| 738 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 739 |     let v = sim.bcm.battery_voltage; | executable/support line; behavior impact depends on surrounding block
text. |
| 740 |     let ok = v >= 11.8; | executable/support line; behavior impact depends on surrounding block context. |
| 741 |     (ok, format!("battery_v={:.2}", v)) | executable/support line; behavior impact depends on surrounding blo
context. |
| 742 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 743 | (blank) | line is blank; visual separation only. |
| 744 | fn run_uds_runtime_self_test() -> (bool, String) { | declares executable behavior; signature/body changes alt
API, control flow, and outputs. |
| 745 |     let mut sim = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding bl
context. |
| 746 |     let sess = sim.uds_ecm.process(&[0x10, 0x01], 0.0); | I/O or protocol-critical logic; edits can affect in
ration, timing, and diagnostic outcomes. |
| 747 |     let vin = sim.uds_ecm.process(&[0x22, 0xF1, 0x90], 0.1); | I/O or protocol-critical logic; edits can affe
integration, timing, and diagnostic outcomes. |
| 748 |     let tester = sim.uds_ecm.process(&[0x3E, 0x00], 0.2); | I/O or protocol-critical logic; edits can affect
egration, timing, and diagnostic outcomes. |
| 749 |     let ok = sess.first().copied() == Some(0x50) | executable/support line; behavior impact depends on surround
ng block context. |
| 750 |         && vin.first().copied() == Some(0x62) | executable/support line; behavior impact depends on surroundi
block context. |
| 751 |         && tester.first().copied() == Some(0x7E); | executable/support line; behavior impact depends on surrou
ing block context. |
| 752 |     ( | executable/support line; behavior impact depends on surrounding block context. |
| 753 |         ok, | executable/support line; behavior impact depends on surrounding block context. |
| 754 |         format!( | executable/support line; behavior impact depends on surrounding block context. |
| 755 |             "session_sid=0x{:02X} vin_sid=0x{:02X} tester_sid=0x{:02X}", | executable/support line; behavior
act depends on surrounding block context. |
| 756 |             sess.first().copied().unwrap_or(0), | executable/support line; behavior impact depends on surround
g block context. |
| 757 |             vin.first().copied().unwrap_or(0), | executable/support line; behavior impact depends on surround
block context. |
| 758 |             tester.first().copied().unwrap_or(0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 759 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 760 |     ) | executable/support line; behavior impact depends on surrounding block context. |
| 761 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 762 | (blank) | line is blank; visual separation only. |
| 763 | fn run_sim_stress_check() -> (bool, String) { | declares executable behavior; signature/body changes alter AP

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control flow, and outputs. |
| 764 |     let mut sim = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding block context. | | |
| 765 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 766 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 767 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 768 |     let mut max_speed = 0.0_f64; | executable/support line; behavior impact depends on surrounding block context. |
| 769 |     for i in 0..1800 { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 770 |         sim.throttle_pct = if i % 300 < 180 { 55.0 } else { 15.0 }; | executable/support line; behavior impact depends on surrounding block context. |
| 771 |         sim.brake_pct = if i % 450 > 400 { 20.0 } else { 0.0 }; | executable/support line; behavior impact depends on surrounding block context. |
| 772 |         sim.tick(1.0 / 60.0); | executable/support line; behavior impact depends on surrounding block context. |
| 773 |         max_speed = max_speed.max(sim.tcm.ground_speed_kmh); | executable/support line; behavior impact depends on surrounding block context. |
| 774 |         if sim.tcm.ground_speed_kmh.is_nan() { | conditional gate; changing condition shifts runtime path and side effects. |
| 775 |             return (false, "speed NaN detected".to_string()); | executable/support line; behavior impact depends on surrounding block context. |
| 776 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 777 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 778 |     ( | executable/support line; behavior impact depends on surrounding block context. |
| 779 |         true, | executable/support line; behavior impact depends on surrounding block context. |
| 780 |         format!("max_speed={:.2} fuel={:.2}%", max_speed, sim.ecm.fuel_level_pct), | executable/support line; behavior impact depends on surrounding block context. |
| 781 |     ) | executable/support line; behavior impact depends on surrounding block context. |
| 782 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 783 | (blank) | line is blank; visual separation only. |
| 784 | fn hex_sha(bytes: &[u8]) -> String { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 785 |     let mut out = String::with_capacity(bytes.len() * 2); | executable/support line; behavior impact depends on surrounding block context. |
| 786 |     for b in bytes { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 787 |         out.push_str(&format!("{b:02x}")); | executable/support line; behavior impact depends on surrounding block context. |
| 788 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 789 |     out | executable/support line; behavior impact depends on surrounding block context. |
| 790 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 791 | (blank) | line is blank; visual separation only. |
| 792 | fn now_ms() -> u64 { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 793 |     SystemTime::now() | executable/support line; behavior impact depends on surrounding block context. |
| 794 |     .duration_since(UNIX_EPOCH) | executable/support line; behavior impact depends on surrounding block context. |
| 795 |     .map(|d| d.as_millis() as u64) | executable/support line; behavior impact depends on surrounding block context. |
| 796 |     .unwrap_or(0) | executable/support line; behavior impact depends on surrounding block context. |
| 797 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/io/rate_limiter.rs

Role: I/O and diagnostics stack: adapters, allowlist, flashing, replay, metrics, production phases.
Lines: 85 | Non-empty: 73 | Symbols: 12

Symbol Inventory

Line	Kind	Symbol	Change Impact
5	struct	TokenBucket	data/structure coupling
12	impl	TokenBucket	behavior binding and method semantics
13	fn	new	signature/API coupling

```
| 23 | fn | try_take | signature/API coupling |
| 33 | fn | refill | signature/API coupling |
| 42 | struct | TwoLevelRateLimiter | data/structure coupling |
| 48 | impl | TwoLevelRateLimiter | behavior binding and method semantics |
| 49 | fn | new | signature/API coupling |
| 57 | fn | check_can | signature/API coupling |
| 70 | fn | check_serial | signature/API coupling |
| 76 | module | tests | data/structure coupling |
| 80 | fn | limiter_blocks_when_empty | signature/API coupling |
```

Line-by-Line Impact Map

```
| Line | Source | Operational Impact If Changed |
|---:|---|---|
| 1 | use std::collections::HashMap; | imports symbols; changing path/name affects compile-time resolution. |
| 2 | use std::time::Instant; | imports symbols; changing path/name affects compile-time resolution. |
| 3 | (blank) | line is blank; visual separation only. |
| 4 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 5 | struct TokenBucket { | declares data layout; field changes may break serialization, tests, and callers. |
| 6 |     tokens: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 7 |     capacity: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 8 |     refill_per_sec: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 9 |     last_refill: Instant, | executable/support line; behavior impact depends on surrounding block context. |
| 10 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 11 | (blank) | line is blank; visual separation only. |
| 12 | impl TokenBucket { | implementation block; method behavior and trait conformance are defined here. |
| 13 |     fn new(rate_per_sec: u32, burst: u32) -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 14 |         let cap = burst.max(1) as f64; | executable/support line; behavior impact depends on surrounding block context. |
| 15 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 16 |             tokens: cap, | executable/support line; behavior impact depends on surrounding block context. |
| 17 |             capacity: cap, | executable/support line; behavior impact depends on surrounding block context. |
| 18 |             refill_per_sec: rate_per_sec.max(1) as f64, | executable/support line; behavior impact depends on surrounding block context. |
| 19 |             last_refill: Instant::now(), | executable/support line; behavior impact depends on surrounding block context. |
| 20 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 21 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 22 | (blank) | line is blank; visual separation only. |
| 23 |     fn try_take(&mut self, amount: f64) -> bool { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 24 |         self.refill(); | executable/support line; behavior impact depends on surrounding block context. |
| 25 |         if self.tokens >= amount { | conditional gate; changing condition shifts runtime path and side effects. |
| 26 |             self.tokens -= amount; | executable/support line; behavior impact depends on surrounding block context. |
| 27 |             true | executable/support line; behavior impact depends on surrounding block context. |
| 28 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 29 |             false | executable/support line; behavior impact depends on surrounding block context. |
| 30 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 31 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 32 | (blank) | line is blank; visual separation only. |
| 33 |     fn refill(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 34 |         let now = Instant::now(); | executable/support line; behavior impact depends on surrounding block context. |
| 35 |         let elapsed = now.duration_since(self.last_refill).as_secs_f64(); | executable/support line; behavior impact depends on surrounding block context. |
| 36 |         self.last_refill = now; | executable/support line; behavior impact depends on surrounding block context. |
| 37 |         self.tokens = (self.tokens + elapsed * self.refill_per_sec).min(self.capacity); | executable/support line; behavior impact depends on surrounding block context. |
| 38 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 39 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 40 | (blank) | line is blank; visual separation only. |
```

```

| 41 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 42 | pub struct TwoLevelRateLimiter { | declares data layout; field changes may break serialization, tests, and cal
s. |
| 43 |     global: TokenBucket, | executable/support line; behavior impact depends on surrounding block context. |
| 44 |     per_id_rate: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 45 |     per_id: HashMap<u32, TokenBucket>, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 46 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 47 | (blank) | line is blank; visual separation only. |
| 48 | impl TwoLevelRateLimiter { | implementation block; method behavior and trait conformance are defined here. |
| 49 |     pub fn new(global_per_sec: u32, per_id_per_sec: u32) -> Self { | declares executable behavior; signature/b
changes alter API, control flow, and outputs. |
| 50 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 51 |             global: TokenBucket::new(global_per_sec, global_per_sec), | executable/support line; behavior impa
depends on surrounding block context. |
| 52 |             per_id_rate: per_id_per_sec.max(1), | executable/support line; behavior impact depends on surround
block context. |
| 53 |             per_id: HashMap::new(), | executable/support line; behavior impact depends on surrounding block co
xt. |
| 54 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 55 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 56 | (blank) | line is blank; visual separation only. |
| 57 |     pub fn check_can(&mut self, can_id: u32) -> bool { | declares executable behavior; signature/body changes
er API, control flow, and outputs. |
| 58 |         if !self.global.try_take(1.0) { | conditional gate; changing condition shifts runtime path and side ef
ts. |
| 59 |             return false; | executable/support line; behavior impact depends on surrounding block context. |
| 60 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 61 | (blank) | line is blank; visual separation only. |
| 62 |         let per = self | executable/support line; behavior impact depends on surrounding block context. |
| 63 |         .per_id | executable/support line; behavior impact depends on surrounding block context. |
| 64 |         .entry(can_id) | executable/support line; behavior impact depends on surrounding block context. |
| 65 |         .or_insert_with(\|\| TokenBucket::new(self.per_id_rate, self.per_id_rate)); | executable/support l
; behavior impact depends on surrounding block context. |
| 66 | (blank) | line is blank; visual separation only. |
| 67 |         per.try_take(1.0) | executable/support line; behavior impact depends on surrounding block context. |
| 68 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 69 | (blank) | line is blank; visual separation only. |
| 70 |     pub fn check_serial(&mut self) -> bool { | declares executable behavior; signature/body changes alter API,
ntrol flow, and outputs. |
| 71 |         self.global.try_take(1.0) | executable/support line; behavior impact depends on surrounding block cont
. |
| 72 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 73 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 74 | (blank) | line is blank; visual separation only. |
| 75 | #[cfg(test)] | executable/support line; behavior impact depends on surrounding block context. |
| 76 | mod tests { | declares a module boundary; changing module name/path breaks references. |
| 77 |     use super::*; | imports symbols; changing path/name affects compile-time resolution. |
| 78 | (blank) | line is blank; visual separation only. |
| 79 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 80 |     fn limiter_blocks_when_empty() { | declares executable behavior; signature/body changes alter API, control
ow, and outputs. |
| 81 |         let mut lim = TwoLevelRateLimiter::new(1, 1); | executable/support line; behavior impact depends on su
unding block context. |
| 82 |         assert!(lim.check_can(0x100)); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 83 |         assert!(lim.check_can(0x100)); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 84 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 85 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/io/replay.rs

Role: I/O and diagnostics stack: adapters, allowlist, flashing, replay, metrics, production phases.

Lines: 183 | Non-empty: 165 | Symbols: 11

Symbol Inventory

Line	Kind	Symbol	Change Impact
10	struct	JsonlRecord	data/structure coupling
22	fn	append_record	signature/API coupling
39	fn	read_records	signature/API coupling
55	fn	frame_to_record	signature/API coupling
87	fn	record_to_frame	signature/API coupling
121	fn	now_ms	signature/API coupling
128	fn	hex_encode	signature/API coupling
137	fn	hex_decode	signature/API coupling
151	fn	nibble	signature/API coupling
159	module	tests	data/structure coupling
163	fn	can_record_roundtrip	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	use std::fs::{self, OpenOptions};	imports symbols; changing path/name affects compile-time resolution.
2	use std::io::{BufRead, BufReader, Write};	imports symbols; changing path/name affects compile-time resolution.
3	use std::path::Path;	imports symbols; changing path/name affects compile-time resolution.
4	(blank)	line is blank; visual separation only.
5	use serde::{Deserialize, Serialize};	imports symbols; changing path/name affects compile-time resolution.
6	(blank)	line is blank; visual separation only.
7	use super::hw::{CanFrame, Frame, HwError, SerialFrame};	imports symbols; changing path/name affects compile-time resolution.
8	(blank)	line is blank; visual separation only.
9	#[derive(Debug, Clone, Serialize, Deserialize, PartialEq, Eq)]	serialization contract touchpoint; edits can break report compatibility and tooling.
10	pub struct JsonlRecord {	declares data layout; field changes may break serialization, tests, and callers.
11	pub ts: u64,	executable/support line; behavior impact depends on surrounding block context.
12	pub transport: String,	executable/support line; behavior impact depends on surrounding block context.
13	pub dir: String,	executable/support line; behavior impact depends on surrounding block context.
14	pub id: Option<u32>,	executable/support line; behavior impact depends on surrounding block context.
15	pub dlc: Option<u8>,	executable/support line; behavior impact depends on surrounding block context.
16	pub data: Option<String>,	executable/support line; behavior impact depends on surrounding block context.
17	pub raw_hex: Option<String>,	executable/support line; behavior impact depends on surrounding block context.
18	pub allowed: Option<bool>,	executable/support line; behavior impact depends on surrounding block context.
19	pub dry_run: Option<bool>,	executable/support line; behavior impact depends on surrounding block context.
20	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
21	(blank)	line is blank; visual separation only.
22	pub fn append_record(path: &Path, rec: &JsonlRecord) -> Result<(), HwError> {	declares executable behavior; nature/body changes alter API, control flow, and outputs.
23	if let Some(parent) = path.parent() {	conditional gate; changing condition shifts runtime path and side effects.
24	fs::create_dir_all(parent)	executable/support line; behavior impact depends on surrounding block context.
25	.map_err(e HwError::Unknown(format!("create log dir failed: {e}")))?;	error-propagating operation; changing it impacts failure semantics and caller recovery paths.
26	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
27	(blank)	line is blank; visual separation only.
28	let mut f = OpenOptions::new()	executable/support line; behavior impact depends on surrounding block context.
29	.create(true)	executable/support line; behavior impact depends on surrounding block context.
30	.append(true)	executable/support line; behavior impact depends on surrounding block context.
31	.open(path)	executable/support line; behavior impact depends on surrounding block context.
32	.map_err(e HwError::Unknown(format!("open jsonl log failed: {e}")))?;	error-propagating operation; changing it impacts failure semantics and caller recovery paths.
33	(blank)	line is blank; visual separation only.
34	let line = serde_json::to_string(rec)	serialization contract touchpoint; edits can break report compatibility and tooling.

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| 35 |         .map_err(\|e\| HwError::Unknown(format!("json encode failed: {e}")))?; | error-propagating operation;
nging it impacts failure semantics and caller recovery paths. |
| 36 |         writeln!(f, "{line}").map_err(\|e\| HwError::Unknown(format!("write jsonl failed: {e}"))) | I/O or protocol
ritical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 37 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 38 | (blank) | line is blank; visual separation only. |
| 39 | pub fn read_records(path: &Path) -> Result<Vec<JsonlRecord>, HwError> { | declares executable behavior; signature
/body changes alter API, control flow, and outputs. |
| 40 |     let f = OpenOptions::new() | executable/support line; behavior impact depends on surrounding block context. |
| 41 |     .read(true) | executable/support line; behavior impact depends on surrounding block context. |
| 42 |     .open(path) | executable/support line; behavior impact depends on surrounding block context. |
| 43 |     .map_err(\|e\| HwError::Unknown(format!("open replay file failed: {e}")))?; | error-propagating operation;
; changing it impacts failure semantics and caller recovery paths. |
| 44 |     let r = BufReader::new(f); | executable/support line; behavior impact depends on surrounding block context. |
| 45 |     let mut out = Vec::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 46 |     for line in r.lines() { | iteration control; changing bounds/body affects throughput, timing, and possibly saf
fety guarantees. |
| 47 |         let line = line.map_err(\|e\| HwError::Unknown(format!("read line failed: {e}")))?; | error-propagating op
eration; changing it impacts failure semantics and caller recovery paths. |
| 48 |         let rec = serde_json::from_str::<JsonlRecord>(&line) | serialization contract touchpoint; edits can break
report compatibility and tooling. |
| 49 |         .map_err(\|e\| HwError::Unknown(format!("parse line failed: {e}")))?; | error-propagating operation;
changing it impacts failure semantics and caller recovery paths. |
| 50 |         out.push(rec); | executable/support line; behavior impact depends on surrounding block context. |
| 51 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 52 |     Ok(out) | executable/support line; behavior impact depends on surrounding block context. |
| 53 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 54 | (blank) | line is blank; visual separation only. |
| 55 | pub fn frame_to_record( | declares executable behavior; signature/body changes alter API, control flow, and outp
ts. |
| 56 |     frame: &Frame, | executable/support line; behavior impact depends on surrounding block context. |
| 57 |     dir: &str, | executable/support line; behavior impact depends on surrounding block context. |
| 58 |     allowed: Option<bool>, | executable/support line; behavior impact depends on surrounding block context. |
| 59 |     dry_run: Option<bool>, | executable/support line; behavior impact depends on surrounding block context. |
| 60 | ) -> JsonlRecord { | executable/support line; behavior impact depends on surrounding block context. |
| 61 |     match frame { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 62 |         Frame::Can(cf) => JsonlRecord { | executable/support line; behavior impact depends on surrounding block
context. |
| 63 |             ts: cf.timestamp_ms.unwrap_or_else(now_ms), | executable/support line; behavior impact depends on
rounding block context. |
| 64 |             transport: "can".to_string(), | executable/support line; behavior impact depends on surrounding block
context. |
| 65 |             dir: dir.to_string(), | executable/support line; behavior impact depends on surrounding block context.
. |
| 66 |             id: Some(cf.id), | executable/support line; behavior impact depends on surrounding block context. |
| 67 |             dlc: Some(cf.dlc), | executable/support line; behavior impact depends on surrounding block context. |
| 68 |             data: Some(hex_encode(&cf.data[..cf.len.min(8)])), | executable/support line; behavior impact depende
on surrounding block context. |
| 69 |             raw_hex: None, | executable/support line; behavior impact depends on surrounding block context. |
| 70 |             allowed, | executable/support line; behavior impact depends on surrounding block context. |
| 71 |             dry_run, | executable/support line; behavior impact depends on surrounding block context. |
| 72 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 73 |         Frame::Serial(sf) => JsonlRecord { | executable/support line; behavior impact depends on surrounding block
context. |
| 74 |             ts: sf.timestamp_ms.unwrap_or_else(now_ms), | executable/support line; behavior impact depends on
rounding block context. |
| 75 |             transport: "serial".to_string(), | executable/support line; behavior impact depends on surrounding block
context. |
| 76 |             dir: dir.to_string(), | executable/support line; behavior impact depends on surrounding block context.
. |
| 77 |             id: None, | executable/support line; behavior impact depends on surrounding block context. |
| 78 |             dlc: Some((sf.bytes.len().min(255)) as u8), | executable/support line; behavior impact depends on
rounding block context. |
| 79 |             data: Some(hex_encode(&sf.bytes)), | executable/support line; behavior impact depends on surrounding
block context. |

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```

| 80 |         raw_hex: None, | executable/support line; behavior impact depends on surrounding block context. | | |
| 81 |         allowed, | executable/support line; behavior impact depends on surrounding block context. |
| 82 |         dry_run, | executable/support line; behavior impact depends on surrounding block context. |
| 83 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 84 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 85 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 86 | (blank) | line is blank; visual separation only. |
| 87 | pub fn record_to_frame(rec: &JsonlRecord) -> Option<Frame> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 88 |     if rec.dir != "rx" { | conditional gate; changing condition shifts runtime path and side effects. |
| 89 |         return None; | executable/support line; behavior impact depends on surrounding block context. |
| 90 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 91 | (blank) | line is blank; visual separation only. |
| 92 |     if rec.transport == "can" { | conditional gate; changing condition shifts runtime path and side effects. |
| 93 |         let id = rec.id?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 94 |         let data_hex = rec.data.as_deref()?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 95 |         let bytes = hex_decode(data_hex)?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 96 |         let mut data = [0u8; 8]; | executable/support line; behavior impact depends on surrounding block context. |
| 97 |         let len = bytes.len().min(8); | executable/support line; behavior impact depends on surrounding block context. |
| 98 |         data[..len].copy_from_slice(&bytes[..len]); | executable/support line; behavior impact depends on surrounding block context. |
| 99 |         return Some(Frame::Can(CanFrame { | executable/support line; behavior impact depends on surrounding block context. |
| 100 |             id, | executable/support line; behavior impact depends on surrounding block context. |
| 101 |             dlc: len as u8, | executable/support line; behavior impact depends on surrounding block context. |
| 102 |             data, | executable/support line; behavior impact depends on surrounding block context. |
| 103 |             len, | executable/support line; behavior impact depends on surrounding block context. |
| 104 |             timestamp_ms: Some(rec.ts), | executable/support line; behavior impact depends on surrounding block context. |
| 105 |         })); | executable/support line; behavior impact depends on surrounding block context. |
| 106 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 107 | (blank) | line is blank; visual separation only. |
| 108 |     if rec.transport == "serial" { | conditional gate; changing condition shifts runtime path and side effects. |
| 109 |         let data_hex = rec.data.as_deref()?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 110 |         let bytes = hex_decode(data_hex)?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 111 |         return Some(Frame::Serial(SerialFrame { | executable/support line; behavior impact depends on surrounding block context. |
| 112 |             bytes, | executable/support line; behavior impact depends on surrounding block context. |
| 113 |             protocol_hint: None, | executable/support line; behavior impact depends on surrounding block context. |
| 114 |             timestamp_ms: Some(rec.ts), | executable/support line; behavior impact depends on surrounding block context. |
| 115 |         })); | executable/support line; behavior impact depends on surrounding block context. |
| 116 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 117 | (blank) | line is blank; visual separation only. |
| 118 |     None | executable/support line; behavior impact depends on surrounding block context. |
| 119 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 120 | (blank) | line is blank; visual separation only. |
| 121 | fn now_ms() -> u64 { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 122 |     std::time::SystemTime::now() | executable/support line; behavior impact depends on surrounding block context. |
| 123 |     .duration_since(std::time::UNIX_EPOCH) | executable/support line; behavior impact depends on surrounding block context. |
| 124 |     .map(|d| d.as_millis() as u64) | executable/support line; behavior impact depends on surrounding block context. |
| 125 |     .unwrap_or(0) | executable/support line; behavior impact depends on surrounding block context. |

```

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| 126 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 127 | (blank) | line is blank; visual separation only. |
| 128 | fn hex_encode(bytes: &[u8]) -> String { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 129 |     let mut out = String::with_capacity(bytes.len() * 2); | executable/support line; behavior impact depends on surrounding block context. |
| 130 |     for b in bytes { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 131 |         out.push(nibble((*b >> 4) & 0x0F)); | executable/support line; behavior impact depends on surrounding block context. |
| 132 |         out.push(nibble(*b & 0x0F)); | executable/support line; behavior impact depends on surrounding block context. |
| 133 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 134 |     out | executable/support line; behavior impact depends on surrounding block context. |
| 135 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 136 | (blank) | line is blank; visual separation only. |
| 137 | fn hex_decode(hex: &str) -> Option<Vec<u8>> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 138 |     if !hex.len().is_multiple_of(2) { | conditional gate; changing condition shifts runtime path and side effects. |
| 139 |         return None; | executable/support line; behavior impact depends on surrounding block context. |
| 140 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 141 |     let mut out = Vec::with_capacity(hex.len() / 2); | executable/support line; behavior impact depends on surrounding block context. |
| 142 |     let mut chars = hex.chars(); | executable/support line; behavior impact depends on surrounding block context. |
| 143 |     while let (Some(h), Some(l)) = (chars.next(), chars.next()) { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 144 |         let hi = h.to_digit(16)? as u8; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 145 |         let lo = l.to_digit(16)? as u8; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 146 |         out.push((hi << 4) \| lo); | executable/support line; behavior impact depends on surrounding block context. |
| 147 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 148 |     Some(out) | executable/support line; behavior impact depends on surrounding block context. |
| 149 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 150 | (blank) | line is blank; visual separation only. |
| 151 | fn nibble(v: u8) -> char { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 152 |     match v { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 153 |         0..=9 => (b'0' + v) as char, | executable/support line; behavior impact depends on surrounding block context. |
| 154 |         _ => (b'a' + (v - 10)) as char, | executable/support line; behavior impact depends on surrounding block context. |
| 155 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 156 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 157 | (blank) | line is blank; visual separation only. |
| 158 | #[cfg(test)] | executable/support line; behavior impact depends on surrounding block context. |
| 159 | mod tests { | declares a module boundary; changing module name/path breaks references. |
| 160 |     use super::*; | imports symbols; changing path/name affects compile-time resolution. |
| 161 | (blank) | line is blank; visual separation only. |
| 162 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 163 |     fn can_record_roundtrip() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 164 |         let f = Frame::Can(CanFrame { | executable/support line; behavior impact depends on surrounding block context. |
| 165 |             id: 0x123, | executable/support line; behavior impact depends on surrounding block context. |
| 166 |             dlc: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 167 |             data: [1, 2, 3, 4, 5, 6, 7, 8], | executable/support line; behavior impact depends on surrounding block context. |
| 168 |             len: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 169 |             timestamp_ms: Some(12345), | executable/support line; behavior impact depends on surrounding block context. |
| 170 |             }); | executable/support line; behavior impact depends on surrounding block context. |

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| 171 |         let rec = frame_to_record(&f, "rx", None, None); | executable/support line; behavior impact depends on
surrounding block context. |
| 172 |         let got = record_to_frame(&rec).expect("frame from record"); | panic-prone operation; changing/removing
affects crash behavior and robustness. |
| 173 |         assert_eq!(rec.ts, 12345); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 174 |         assert!(matches!(got, Frame::Can(_)), "expected can frame"); | executable/support line; behavior impa
depends on surrounding block context. |
| 175 |         if let Frame::Can(cf) = got { | conditional gate; changing condition shifts runtime path and side eff
s. |
| 176 |             assert_eq!(cf.id, 0x123); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 177 |             assert_eq!(cf.dlc, 8); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 178 |             assert_eq!(cf.len, 8); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 179 |             assert_eq!(cf.data, [1, 2, 3, 4, 5, 6, 7, 8]); | executable/support line; behavior impact depends
surrounding block context. |
| 180 |             assert_eq!(cf.timestamp_ms, Some(12345)); | executable/support line; behavior impact depends on s
ounding block context. |
| 181 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 182 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 183 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/io/serial_adapter.rs

Role: I/O and diagnostics stack: adapters, allowlist, flashing, replay, metrics, production phases.
Lines: 96 | Non-empty: 81 | Symbols: 10

Symbol Inventory

Line	Kind	Symbol	Change Impact
9	struct	SerialAdapter	data/structure coupling
13	impl	SerialAdapter	behavior binding and method semantics
14	fn	open	signature/API coupling
30	impl	HardwareInterface for SerialAdapter	behavior binding and method semantics
31	fn	init	signature/API coupling
35	fn	read_frame	signature/API coupling
61	fn	try_read_frame	signature/API coupling
69	fn	send_frame	signature/API coupling
85	fn	close	signature/API coupling
91	fn	now_ms	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	use std::io::{Read, Write};	imports symbols; changing path/name affects compile-time resolution.
2	use std::time::Duration;	imports symbols; changing path/name affects compile-time resolution.
3	(blank)	line is blank; visual separation only.
4	use serialport::SerialPort;	imports symbols; changing path/name affects compile-time resolution.
5	(blank)	line is blank; visual separation only.
6	use super::hw::{Frame, HardwareInterface, HwConfig, HwError, SerialFrame};	imports symbols; changing path/name affects compile-time resolution.
7	(blank)	line is blank; visual separation only.
8	#[derive(Debug, Default)]	executable/support line; behavior impact depends on surrounding block context.
9	pub struct SerialAdapter {	declares data layout; field changes may break serialization, tests, and callers.
10	port: Option<Box<dyn SerialPort>>,	executable/support line; behavior impact depends on surrounding block context.
11	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
12	(blank)	line is blank; visual separation only.
13	impl SerialAdapter {	implementation block; method behavior and trait conformance are defined here.
14	pub fn open(port_name: &str, baud: u32) -> Result<Self, HwError> {	declares executable behavior; signature

```

body changes alter API, control flow, and outputs. |
| 15 |         let port = serialport::new(port_name, baud) | executable/support line; behavior impact depends on surrounding block context. |
| 16 |         .timeout(Duration::from_millis(250)) | executable/support line; behavior impact depends on surrounding block context. |
| 17 |         .open() | executable/support line; behavior impact depends on surrounding block context. |
| 18 |         .map_err(\|e\| { | executable/support line; behavior impact depends on surrounding block context. |
| 19 |             let msg = e.to_string(); | executable/support line; behavior impact depends on surrounding block context. |
| 20 |             if msg.to_ascii_lowercase().contains("permission") { | conditional gate; changing condition shifts runtime path and side effects. |
| 21 |                 HwError::PermissionDenied(msg) | executable/support line; behavior impact depends on surrounding block context. |
| 22 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 23 |                 HwError::PortNotFound(msg) | executable/support line; behavior impact depends on surrounding block context. |
| 24 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 25 |             }); | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 26 |         Ok(Self { port: Some(port) }) | executable/support line; behavior impact depends on surrounding block context. |
| 27 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 28 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 29 | (blank) | line is blank; visual separation only. |
| 30 | impl HardwareInterface for SerialAdapter { | implementation block; method behavior and trait conformance are defined here. |
| 31 |     fn init(&mut self, _config: &HwConfig) -> Result<(), HwError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 32 |         Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 33 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 34 | (blank) | line is blank; visual separation only. |
| 35 |     fn read_frame(&mut self) -> Result<Frame, HwError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 36 |         let Some(port) = self.port.as_mut() else { | executable/support line; behavior impact depends on surrounding block context. |
| 37 |             return Err(HwError::PortNotFound("serial port not initialized".into())); | executable/support line; behavior impact depends on surrounding block context. |
| 38 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 39 | (blank) | line is blank; visual separation only. |
| 40 |         let mut buf = vec![0u8; 1024]; | executable/support line; behavior impact depends on surrounding block context. |
| 41 |         let n = port.read(&mut buf).map_err(\|e\| { | executable/support line; behavior impact depends on surrounding block context. |
| 42 |             if e.kind() == std::io::ErrorKind::TimedOut { | conditional gate; changing condition shifts runtime path and side effects. |
| 43 |                 HwError::Timeout | executable/support line; behavior impact depends on surrounding block context. |
| 44 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 45 |                 HwError::TransceiverError | executable/support line; behavior impact depends on surrounding block context. |
| 46 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 47 |             }); | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 48 | (blank) | line is blank; visual separation only. |
| 49 |         if n == 0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 50 |             return Err(HwError::Timeout); | executable/support line; behavior impact depends on surrounding block context. |
| 51 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 52 | (blank) | line is blank; visual separation only. |
| 53 |         buf.truncate(n); | executable/support line; behavior impact depends on surrounding block context. |
| 54 |         Ok(Frame::Serial(SerialFrame { | executable/support line; behavior impact depends on surrounding block context. |
| 55 |             bytes: buf, | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 56 |         protocol_hint: Some("raw-serial".into()), | executable/support line; behavior impact depends on surrounding block context. | | |
| 57 |         timestamp_ms: Some(now_ms()), | executable/support line; behavior impact depends on surrounding block context. |
| 58 |     ))) | executable/support line; behavior impact depends on surrounding block context. |
| 59 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 60 | (blank) | line is blank; visual separation only. |
| 61 | fn try_read_frame(&mut self) -> Result<Option<Frame>, HwError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 62 |     match self.read_frame() { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 63 |         Ok(f) => Ok(Some(f)), | executable/support line; behavior impact depends on surrounding block context. |
| 64 |         Err(HwError::Timeout) => Ok(None), | executable/support line; behavior impact depends on surrounding block context. |
| 65 |         Err(e) => Err(e), | executable/support line; behavior impact depends on surrounding block context. |
| 66 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 67 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 68 | (blank) | line is blank; visual separation only. |
| 69 | fn send_frame(&mut self, frame: Frame) -> Result<(), HwError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 70 |     let Some(port) = self.port.as_mut() else { | executable/support line; behavior impact depends on surrounding block context. |
| 71 |         return Err(HwError::PortNotFound("serial port not initialized".into())); | executable/support line; behavior impact depends on surrounding block context. |
| 72 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 73 | (blank) | line is blank; visual separation only. |
| 74 |     let bytes: Vec<u8> = match frame { | executable/support line; behavior impact depends on surrounding block context. |
| 75 |         Frame::Serial(sf) => sf.bytes, | executable/support line; behavior impact depends on surrounding block context. |
| 76 |         Frame::Can(cf) => cf.data[..cf.len.min(8)].to_vec(), | executable/support line; behavior impact depends on surrounding block context. |
| 77 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 78 | (blank) | line is blank; visual separation only. |
| 79 |     port.write_all(&bytes) | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 80 |     .map_err(|_| HwError::TransceiverError)?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 81 |     port.flush().map_err(|_| HwError::TransceiverError)?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 82 |     Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 83 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 84 | (blank) | line is blank; visual separation only. |
| 85 | fn close(&mut self) -> Result<(), HwError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 86 |     self.port = None; | executable/support line; behavior impact depends on surrounding block context. |
| 87 |     Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 88 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 89 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 90 | (blank) | line is blank; visual separation only. |
| 91 | fn now_ms() -> u64 { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 92 |     std::time::SystemTime::now() | executable/support line; behavior impact depends on surrounding block context. |
| 93 |     .duration_since(std::time::UNIX_EPOCH) | executable/support line; behavior impact depends on surrounding block context. |
| 94 |     .map(|d| d.as_millis() as u64) | executable/support line; behavior impact depends on surrounding block context. |
| 95 |     .unwrap_or(0) | executable/support line; behavior impact depends on surrounding block context. |
| 96 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/io/socketcan_adapter.rs

Role: I/O and diagnostics stack: adapters, allowlist, flashing, replay, metrics, production phases.
Lines: 145 | Non-empty: 124 | Symbols: 13

Symbol Inventory

Line	Kind	Symbol	Change Impact	
---: --- --- ---				
10	struct	SocketCanAdapter	data/structure coupling	
15	impl	SocketCanAdapter	behavior binding and method semantics	
17	fn	open	signature/API coupling	
28	fn	open	signature/API coupling	
35	impl	HardwareInterface for SocketCanAdapter	behavior binding and method semantics	
36	fn	init	signature/API coupling	
41	fn	read_frame	signature/API coupling	
74	fn	read_frame	signature/API coupling	
80	fn	try_read_frame	signature/API coupling	
89	fn	send_frame	signature/API coupling	
124	fn	send_frame	signature/API coupling	
130	fn	close	signature/API coupling	
140	fn	now_ms	signature/API coupling	

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed	
---: --- ---			
1	use super::hw::{Frame, HardwareInterface, HwConfig, HwError};	imports symbols; changing path/name affects compile-time resolution.	
2	(blank)	line is blank; visual separation only.	
3	#[cfg(target_os = "linux")]	executable/support line; behavior impact depends on surrounding block context.	
4	use super::hw::CanFrame;	imports symbols; changing path/name affects compile-time resolution.	
5	(blank)	line is blank; visual separation only.	
6	#[cfg(target_os = "linux")]	executable/support line; behavior impact depends on surrounding block context.	
7	use socketcan::{CanFrame as LinuxCanFrame, CanSocket, EmbeddedFrame, ExtendedId, Socket};	imports symbols; changing path/name affects compile-time resolution.	
8	(blank)	line is blank; visual separation only.	
9	#[derive(Debug, Default)]	executable/support line; behavior impact depends on surrounding block context.	
10	pub struct SocketCanAdapter {	declares data layout; field changes may break serialization, tests, and caller	
11	#[cfg(target_os = "linux")]	executable/support line; behavior impact depends on surrounding block context.	
12	socket: Option<CanSocket>,	executable/support line; behavior impact depends on surrounding block context.	
13	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.	
14	(blank)	line is blank; visual separation only.	
15	impl SocketCanAdapter {	implementation block; method behavior and trait conformance are defined here.	
16	#[cfg(target_os = "linux")]	executable/support line; behavior impact depends on surrounding block context.	
17	pub fn open(iface: &str) -> Result<Self, HwError> {	declares executable behavior; signature/body changes	
18	let socket = CanSocket::open(iface).map_err(e HwError::PortNotFound(e.to_string()))?;	error-propagating operation; changing it impacts failure semantics and caller recovery paths.	
19	socket	executable/support line; behavior impact depends on surrounding block context.	
20	.set_read_timeout(std::time::Duration::from_millis(250))	executable/support line; behavior impact depends on surrounding block context.	
21	.map_err(e HwError::TransceiverError)?;	error-propagating operation; changing it impacts failure semantics and caller recovery paths.	
22	Ok(Self {	executable/support line; behavior impact depends on surrounding block context.	
23	socket: Some(socket),	executable/support line; behavior impact depends on surrounding block context.	
24	})	executable/support line; behavior impact depends on surrounding block context.	
25	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.	
26	(blank)	line is blank; visual separation only.	
27	#[cfg(not(target_os = "linux"))]	executable/support line; behavior impact depends on surrounding block context.	
28	pub fn open(_iface: &str) -> Result<Self, HwError> {	declares executable behavior; signature/body changes	

```

lter API, control flow, and outputs. |
| 29 |         Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block context.
| 30 |             "SocketCAN real adapter is supported on Linux only".to_string(), | executable/support line; behavi
impact depends on surrounding block context. |
| 31 |         )) | executable/support line; behavior impact depends on surrounding block context. |
| 32 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 33 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 34 | (blank) | line is blank; visual separation only. |
| 35 | impl HardwareInterface for SocketCanAdapter { | implementation block; method behavior and trait conformance ar
efined here. |
| 36 |     fn init(&mut self, _config: &HwConfig) -> Result<(), HwError> { | declares executable behavior; signature/
y changes alter API, control flow, and outputs. |
| 37 |         Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 38 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 39 | (blank) | line is blank; visual separation only. |
| 40 |     #[cfg(target_os = "linux")] | executable/support line; behavior impact depends on surrounding block context
|
| 41 |     fn read_frame(&mut self) -> Result<Frame, HwError> { | declares executable behavior; signature/body change
lter API, control flow, and outputs. |
| 42 |         let Some(socket) = self.socket.as_ref() else { | executable/support line; behavior impact depends on s
ounding block context. |
| 43 |             return Err(HwError::PortNotFound("CAN socket not initialized".into())); | executable/support line;
havior impact depends on surrounding block context. |
| 44 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 45 | (blank) | line is blank; visual separation only. |
| 46 |         let f = socket.read_frame().map_err(|e| { | executable/support line; behavior impact depends on surr
ding block context. |
| 47 |             if e.kind() == std::io::ErrorKind::TimedOut { | conditional gate; changing condition shifts runtim
ath and side effects. |
| 48 |                 HwError::Timeout | executable/support line; behavior impact depends on surrounding block conte
|
| 49 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 50 |                 HwError::TransceiverError | executable/support line; behavior impact depends on surrounding bl
context. |
| 51 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime sc
s. |
| 52 |         }}?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 53 | (blank) | line is blank; visual separation only. |
| 54 |         let mut data = [0u8; 8]; | executable/support line; behavior impact depends on surrounding block conte
|
| 55 |         let payload = f.data(); | executable/support line; behavior impact depends on surrounding block context
|
| 56 |         let len = payload.len().min(8); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 57 |         data[..len].copy_from_slice(&payload[..len]); | executable/support line; behavior impact depends on su
unding block context. |
| 58 | (blank) | line is blank; visual separation only. |
| 59 |         let id = match f.id() { | executable/support line; behavior impact depends on surrounding block context
|
| 60 |             socketcan::Id::Standard(stdid) => u32::from(stdid.as_raw()), | executable/support line; behavior i
ct depends on surrounding block context. |
| 61 |             socketcan::Id::Extended(extid) => extid.as_raw(), | executable/support line; behavior impact depen
on surrounding block context. |
| 62 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 63 | (blank) | line is blank; visual separation only. |
| 64 |         Ok(Frame::Can(CanFrame { | executable/support line; behavior impact depends on surrounding block conte
|
| 65 |             id, | executable/support line; behavior impact depends on surrounding block context. |
| 66 |             dlc: len as u8, | executable/support line; behavior impact depends on surrounding block context. |
| 67 |             data, | executable/support line; behavior impact depends on surrounding block context. |
| 68 |             len, | executable/support line; behavior impact depends on surrounding block context. |
| 69 |             timestamp_ms: Some(now_ms()), | executable/support line; behavior impact depends on surrounding bl
context. |

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```

| 70 |         )) | executable/support line; behavior impact depends on surrounding block context. |
| 71 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 72 | (blank) | line is blank; visual separation only. |
| 73 |     #[cfg(not(target_os = "linux"))] | executable/support line; behavior impact depends on surrounding block context. |
| 74 |     fn read_frame(&mut self) -> Result<Frame, HwError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 75 |         Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block context. |
| 76 |             "SocketCAN real adapter is supported on Linux only".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 77 |         )) | executable/support line; behavior impact depends on surrounding block context. |
| 78 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 79 | (blank) | line is blank; visual separation only. |
| 80 |     fn try_read_frame(&mut self) -> Result<Option<Frame>, HwError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 81 |         match self.read_frame() { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 82 |             Ok(f) => Ok(Some(f)), | executable/support line; behavior impact depends on surrounding block context. |
| 83 |             Err(HwError::Timeout) => Ok(None), | executable/support line; behavior impact depends on surrounding block context. |
| 84 |             Err(e) => Err(e), | executable/support line; behavior impact depends on surrounding block context. |
| 85 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 86 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 87 | (blank) | line is blank; visual separation only. |
| 88 |     #[cfg(target_os = "linux")] | executable/support line; behavior impact depends on surrounding block context. |
| 89 |     fn send_frame(&mut self, frame: Frame) -> Result<(), HwError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 90 |         let Some(socket) = self.socket.as_ref() else { | executable/support line; behavior impact depends on surrounding block context. |
| 91 |             return Err(HwError::PortNotFound("CAN socket not initialized".into())); | executable/support line; behavior impact depends on surrounding block context. |
| 92 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 93 | (blank) | line is blank; visual separation only. |
| 94 |         let cf = match frame { | executable/support line; behavior impact depends on surrounding block context. |
| 95 |             Frame::Can(c) => c, | executable/support line; behavior impact depends on surrounding block context. |
| 96 |             Frame::Serial(_) => { | executable/support line; behavior impact depends on surrounding block context. |
| 97 |                 return Err(HwError::ParseError { | executable/support line; behavior impact depends on surrounding block context. |
| 98 |                     cause: "cannot send serial frame over SocketCAN".to_string(), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 99 |                     raw_data: "serial-frame".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 100 |                 }) | executable/support line; behavior impact depends on surrounding block context. |
| 101 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 102 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 103 | (blank) | line is blank; visual separation only. |
| 104 |         let Some(ext_id) = ExtendedId::new(cf.id) else { | executable/support line; behavior impact depends on surrounding block context. |
| 105 |             return Err(HwError::ParseError { | executable/support line; behavior impact depends on surrounding block context. |
| 106 |                 cause: "invalid CAN extended id".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 107 |                 raw_data: format!("{}", cf.id), | executable/support line; behavior impact depends on surrounding block context. |
| 108 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 109 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 110 | (blank) | line is blank; visual separation only. |

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| 111 |         let Some(frame) = LinuxCanFrame::new(ext_id, &cf.data[..cf.len.min(8)]) else { | executable/support line; behavior impact depends on surrounding block context. | | |
| 112 |             return Err(HwError::ParseError { | executable/support line; behavior impact depends on surrounding block context. |
| 113 |                 cause: "failed to build CAN frame".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 114 |                 raw_data: format!("id={}", cf.id), | executable/support line; behavior impact depends on surrounding block context. |
| 115 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 116 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 117 | (blank) | line is blank; visual separation only. |
| 118 |         socket | executable/support line; behavior impact depends on surrounding block context. |
| 119 |         .write_frame(&frame) | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 120 |         .map_err(|_| HwError::TransceiverError) | executable/support line; behavior impact depends on surrounding block context. |
| 121 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 122 | (blank) | line is blank; visual separation only. |
| 123 |     #[cfg(not(target_os = "linux"))] | executable/support line; behavior impact depends on surrounding block context. |
| 124 |     fn send_frame(&mut self, _frame: Frame) -> Result<(), HwError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 125 |         Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block context. |
| 126 |             "SocketCAN real adapter is supported on Linux only".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 127 |         )) | executable/support line; behavior impact depends on surrounding block context. |
| 128 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 129 | (blank) | line is blank; visual separation only. |
| 130 |     fn close(&mut self) -> Result<(), HwError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 131 |         #[cfg(target_os = "linux")] | executable/support line; behavior impact depends on surrounding block context. |
| 132 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 133 |             self.socket = None; | executable/support line; behavior impact depends on surrounding block context. |
| 134 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 135 |         Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 136 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 137 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 138 | (blank) | line is blank; visual separation only. |
| 139 | #[cfg(target_os = "linux")] | executable/support line; behavior impact depends on surrounding block context. |
| 140 | fn now_ms() -> u64 { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 141 |     std::time::SystemTime::now() | executable/support line; behavior impact depends on surrounding block context. |
| 142 |     .duration_since(std::time::UNIX_EPOCH) | executable/support line; behavior impact depends on surrounding block context. |
| 143 |     .map(|d| d.as_millis() as u64) | executable/support line; behavior impact depends on surrounding block context. |
| 144 |     .unwrap_or(0) | executable/support line; behavior impact depends on surrounding block context. |
| 145 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/io/vendor_cat_comm.rs

Role: I/O and diagnostics stack: adapters, allowlist, flashing, replay, metrics, production phases.

Lines: 418 | Non-empty: 377 | Symbols: 27

Symbol Inventory

Line	Kind	Symbol	Change Impact
10	const	DEFAULT_TEMPLATE_BRIDGE_EXE	data/structure coupling

13	struct	CatCommAdapter	data/structure coupling
24	struct	CatCommBridgeInfo	data/structure coupling
29	impl	CatCommBridgeInfo	behavior binding and method semantics
30	fn	supports	signature/API coupling
37	enum	BridgeRequest	data/structure coupling
57	struct	BridgeResponse	data/structure coupling
68	struct	BridgeDiagPayload	data/structure coupling
77	struct	BridgeDiagRecord	data/structure coupling
87	impl	CatCommAdapter	behavior binding and method semantics
88	fn	open	signature/API coupling
154	fn	ensure_connected	signature/API coupling
164	fn	send_bridge_request	signature/API coupling
182	fn	read_bridge_response	signature/API coupling
201	fn	append_diag_record	signature/API coupling
232	impl	HardwareInterface for CatCommAdapter	behavior binding and method semantics
233	fn	init	signature/API coupling
246	fn	read_frame	signature/API coupling
261	fn	try_read_frame	signature/API coupling
273	fn	send_frame	signature/API coupling
286	fn	close	signature/API coupling
305	fn	adapter_info	signature/API coupling
318	fn	resolve_bridge_exe	signature/API coupling
347	fn	map_bridge_error	signature/API coupling
368	fn	bridge_info_from_response	signature/API coupling
375	fn	ensure_bridge_compatibility	signature/API coupling
413	fn	now_ms	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	use std::fs::{self, OpenOptions};	imports symbols; changing path/name affects compile-time resolution.
2	use std::io::{BufRead, BufReader, Write};	imports symbols; changing path/name affects compile-time resolution.
3	use std::path::PathBuf;	imports symbols; changing path/name affects compile-time resolution.
4	use std::process::{Child, ChildStdin, Command, Stdio};	imports symbols; changing path/name affects compile-time resolution.
5	(blank)	line is blank; visual separation only.
6	use serde::{Deserialize, Serialize};	imports symbols; changing path/name affects compile-time resolution.
7	(blank)	line is blank; visual separation only.
8	use super::hw::{Frame, HardwareInterface, HwConfig, HwError};	imports symbols; changing path/name affects compile-time resolution.
9	(blank)	line is blank; visual separation only.
10	const DEFAULT_TEMPLATE_BRIDGE_EXE: &str = "cat_comm_bridge.exe";	executable/support line; behavior impact depends on surrounding block context.
11	(blank)	line is blank; visual separation only.
12	#[derive(Debug)]	executable/support line; behavior impact depends on surrounding block context.
13	pub struct CatCommAdapter {	declares data layout; field changes may break serialization, tests, and callers.
14	connected: bool,	executable/support line; behavior impact depends on surrounding block context.
15	child: Child,	executable/support line; behavior impact depends on surrounding block context.
16	child_stdin: ChildStdin,	executable/support line; behavior impact depends on surrounding block context.
17	child_stdout: BufReader<std::process::ChildStdout>,	executable/support line; behavior impact depends on surrounding block context.
18	timeout_ms: u64,	executable/support line; behavior impact depends on surrounding block context.
19	bridge_info: CatCommBridgeInfo,	executable/support line; behavior impact depends on surrounding block context.
20	diag_log_path: PathBuf,	executable/support line; behavior impact depends on surrounding block context.
21	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
22	(blank)	line is blank; visual separation only.
23	#[derive(Debug, Clone)]	executable/support line; behavior impact depends on surrounding block context.
24	struct CatCommBridgeInfo {	declares data layout; field changes may break serialization, tests, and callers.
25	protocol_version: Option<String>,	executable/support line; behavior impact depends on surrounding block context.
26	capabilities: Vec<String>,	executable/support line; behavior impact depends on surrounding block context.
27	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
28	(blank)	line is blank; visual separation only.


```

| 29 | impl CatCommBridgeInfo { | implementation block; method behavior and trait conformance are defined here. |
| 30 |     fn supports(&self, cap: &str) -> bool { | declares executable behavior; signature/body changes alter API,
|    | control flow, and outputs. |
| 31 |         self.capabilities.iter().any(|c| c.eq_ignore_ascii_case(cap)) | executable/support line; behavior im
|    | t depends on surrounding block context. | |
| 32 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 33 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 34 | (blank) | line is blank; visual separation only. |
| 35 | #[derive(Debug, Serialize)] | serialization contract touchpoint; edits can break report compatibility and tool
|    | . |
| 36 | #[serde(tag = "cmd", rename_all = "snake_case")] | serialization contract touchpoint; edits can break report c
|    | atibility and tooling. |
| 37 | enum BridgeRequest { | declares state variants; adding/removing variants impacts pattern matches and logic bra
|    | es. | |
| 38 |     Init { | executable/support line; behavior impact depends on surrounding block context. |
| 39 |         dry_run: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 40 |         enable_write: bool, | I/O or protocol-critical logic; edits can affect integration, timing, and diagno
|    | c outcomes. |
| 41 |         can_interface: Option<String>, | executable/support line; behavior impact depends on surrounding block
|    | ntext. |
| 42 |         serial_port: Option<String>, | executable/support line; behavior impact depends on surrounding block c
|    | ext. |
| 43 |         template_dir: Option<String>, | executable/support line; behavior impact depends on surrounding block
|    | text. | |
| 44 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 45 |     Read { | executable/support line; behavior impact depends on surrounding block context. |
| 46 |         timeout_ms: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 47 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 48 |     TryRead, | executable/support line; behavior impact depends on surrounding block context. |
| 49 |     Send { | executable/support line; behavior impact depends on surrounding block context. |
| 50 |         frame: Frame, | executable/support line; behavior impact depends on surrounding block context. |
| 51 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 52 |     Close, | executable/support line; behavior impact depends on surrounding block context. |
| 53 |     Ping, | executable/support line; behavior impact depends on surrounding block context. |
| 54 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 55 | (blank) | line is blank; visual separation only. |
| 56 | #[derive(Debug, Deserialize)] | serialization contract touchpoint; edits can break report compatibility and to
|    | ng. | |
| 57 | struct BridgeResponse { | declares data layout; field changes may break serialization, tests, and callers. |
| 58 |     ok: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 59 |     frame: Option<Frame>, | executable/support line; behavior impact depends on surrounding block context. |
| 60 |     error: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 61 |     kind: Option<String>, | executable/support line; behavior impact depends on surrounding block context. |
| 62 |     protocol_version: Option<String>, | executable/support line; behavior impact depends on surrounding block
|    | text. |
| 63 |     capabilities: Option<Vec<String>>, | executable/support line; behavior impact depends on surrounding block
|    | ntext. |
| 64 |     diag: Option<BridgeDiagPayload>, | executable/support line; behavior impact depends on surrounding block c
|    | ext. | |
| 65 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 66 | (blank) | line is blank; visual separation only. |
| 67 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report co
|    | tibility and tooling. |
| 68 | struct BridgeDiagPayload { | declares data layout; field changes may break serialization, tests, and callers.
| 69 |     wait_frame_count_delta: Option<u32>, | executable/support line; behavior impact depends on surrounding blo
|    | context. |
| 70 |     sequence_error_count_delta: Option<u32>, | executable/support line; behavior impact depends on surrounding
|    | ock context. |
| 71 |     flowcontrol_timeout_count_delta: Option<u32>, | executable/support line; behavior impact depends on surrou
|    | ng block context. |
| 72 |     fc_blocksize: Option<u8>, | executable/support line; behavior impact depends on surrounding block context.
| 73 |     fc_stmin_ms: Option<u64>, | executable/support line; behavior impact depends on surrounding block context.
| 74 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 75 | (blank) | line is blank; visual separation only. |
| 76 | #[derive(Debug, Serialize)] | serialization contract touchpoint; edits can break report compatibility and tool

```

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. |
| 77 | struct BridgeDiagRecord<'a> { | declares data layout; field changes may break serialization, tests, and caller
|
| 78 |     ts_ms: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 79 |     source: &'a str, | executable/support line; behavior impact depends on surrounding block context. |
| 80 |     wait_frame_count_delta: u32, | executable/support line; behavior impact depends on surrounding block conte
|
| 81 |     sequence_error_count_delta: u32, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 82 |     flowcontrol_timeout_count_delta: u32, | executable/support line; behavior impact depends on surrounding bl
context. |
| 83 |     fc_blocksize: Option<u8>, | executable/support line; behavior impact depends on surrounding block context.
| 84 |     fc_stmin_ms: Option<u64>, | executable/support line; behavior impact depends on surrounding block context.
| 85 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 86 | (blank) | line is blank; visual separation only. |
| 87 | impl CatCommAdapter { | implementation block; method behavior and trait conformance are defined here. |
| 88 |     pub fn open(cfg: &HwConfig) -> Result<Self, HwError> { | declares executable behavior; signature/body chan
alter API, control flow, and outputs. |
| 89 |         let _vendor = cfg | executable/support line; behavior impact depends on surrounding block context. |
| 90 |         .vendor_name | executable/support line; behavior impact depends on surrounding block context. |
| 91 |         .as_deref() | executable/support line; behavior impact depends on surrounding block context. |
| 92 |         .unwrap_or("cat_comm") | executable/support line; behavior impact depends on surrounding block con
t. |
| 93 |         .to_ascii_lowercase(); | executable/support line; behavior impact depends on surrounding block con
t. |
| 94 | (blank) | line is blank; visual separation only. |
| 95 |         let bridge_exe = resolve_bridge_exe(cfg)?; | error-propagating operation; changing it impacts failure
antics and caller recovery paths. |
| 96 |         let mut child = Command::new(&bridge_exe) | executable/support line; behavior impact depends on surrou
ng block context. |
| 97 |             .arg("--mode=stdio-jsonl") | executable/support line; behavior impact depends on surrounding block
ntext. |
| 98 |             .stdin(Stdio::piped()) | executable/support line; behavior impact depends on surrounding block con
t. |
| 99 |             .stdout(Stdio::piped()) | executable/support line; behavior impact depends on surrounding block co
xt. |
| 100 |             .stderr(Stdio::piped()) | executable/support line; behavior impact depends on surrounding block c
ext. |
| 101 |             .spawn() | executable/support line; behavior impact depends on surrounding block context. |
| 102 |             .map_err(|e| { | executable/support line; behavior impact depends on surrounding block context.
| 103 |                 HwError::Unknown(format!( | executable/support line; behavior impact depends on surrounding b
k context. |
| 104 |                 "failed to start Cat Comm bridge at {}: {e}", | executable/support line; behavior impact
ends on surrounding block context. |
| 105 |                 bridge_exe.display() | executable/support line; behavior impact depends on surrounding bl
context. |
| 106 |             )) | executable/support line; behavior impact depends on surrounding block context. |
| 107 |             }?); | error-propagating operation; changing it impacts failure semantics and caller recovery pat
|
| 108 | (blank) | line is blank; visual separation only. |
| 109 |         let child_stdin = child.stdin.take().ok_or_else(|_| { | executable/support line; behavior impact dep
s on surrounding block context. |
| 110 |             HwError::Unknown("failed to capture Cat Comm bridge stdin".to_string()) | executable/support line
ehavior impact depends on surrounding block context. |
| 111 |         }?); | error-propagating operation; changing it impacts failure semantics and caller recovery paths.
| 112 |         let child_stdout = child.stdout.take().ok_or_else(|_| { | executable/support line; behavior impact d
nds on surrounding block context. |
| 113 |             HwError::Unknown("failed to capture Cat Comm bridge stdout".to_string()) | executable/support lin
behavior impact depends on surrounding block context. |
| 114 |         }?); | error-propagating operation; changing it impacts failure semantics and caller recovery paths.
| 115 | (blank) | line is blank; visual separation only. |
| 116 |         let mut adapter = Self { | executable/support line; behavior impact depends on surrounding block cont
. |
| 117 |         connected: false, | executable/support line; behavior impact depends on surrounding block context
| 118 |         child, | executable/support line; behavior impact depends on surrounding block context. |

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| 119 |         child_stdin, | executable/support line; behavior impact depends on surrounding block context. | | |
| 120 |         child_stdout: BufReader::new(child_stdout), | executable/support line; behavior impact depends on surrounding block context. |
| 121 |         timeout_ms: cfg.vendor_bridge_timeout_ms.max(100), | executable/support line; behavior impact depends on surrounding block context. |
| 122 |         bridge_info: CatCommBridgeInfo { | executable/support line; behavior impact depends on surrounding block context. |
| 123 |             protocol_version: None, | executable/support line; behavior impact depends on surrounding block context. |
| 124 |             capabilities: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 125 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 126 |         diag_log_path: cfg.log_dir.join("cat_comm_bridge_diag.jsonl"), | executable/support line; behavior impact depends on surrounding block context. |
| 127 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 128 |     fs::create_dir_all(&cfg.log_dir) | executable/support line; behavior impact depends on surrounding block context. |
| 129 |         .map_err(|e| HwError::Unknown(format!("create bridge log dir failed: {e}")))?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 130 | (blank) | line is blank; visual separation only. |
| 131 |     let init_req = BridgeRequest::Init { | executable/support line; behavior impact depends on surrounding block context. |
| 132 |         dry_run: cfg.dry_run, | executable/support line; behavior impact depends on surrounding block context. |
| 133 |         enable_write: cfg.write_effectively_enabled(), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 134 |         can_interface: cfg.can_interface.clone(), | executable/support line; behavior impact depends on surrounding block context. |
| 135 |         serial_port: cfg.serial_port.clone(), | executable/support line; behavior impact depends on surrounding block context. |
| 136 |         template_dir: cfg | executable/support line; behavior impact depends on surrounding block context. |
| 137 |         .vendor_template_dir | executable/support line; behavior impact depends on surrounding block context. |
| 138 |         .as_ref() | executable/support line; behavior impact depends on surrounding block context. |
| 139 |         .map(|p| p.display().to_string()), | executable/support line; behavior impact depends on surrounding block context. |
| 140 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 141 |     let init_rsp = adapter.send_bridge_request(&init_req)?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 142 |     if !init_rsp.ok { | conditional gate; changing condition shifts runtime path and side effects. |
| 143 |         return Err(map_bridge_error( | executable/support line; behavior impact depends on surrounding block context. |
| 144 |             init_rsp.kind, | executable/support line; behavior impact depends on surrounding block context. |
| 145 |             init_rsp.error.unwrap_or_else(|| "bridge init failed".to_string()), | executable/support line; behavior impact depends on surrounding block context. |
| 146 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 147 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 148 |     adapter.bridge_info = bridge_info_from_response(&init_rsp); | executable/support line; behavior impact depends on surrounding block context. |
| 149 |     ensure_bridge_compatibility(cfg, &adapter.bridge_info)?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 150 | (blank) | line is blank; visual separation only. |
| 151 |     Ok(adapter) | executable/support line; behavior impact depends on surrounding block context. |
| 152 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 153 | (blank) | line is blank; visual separation only. |
| 154 | fn ensure_connected(&self) -> Result<(), HwError> { | declares executable behavior; signature/body change after API, control flow, and outputs. |
| 155 |     if self.connected { | conditional gate; changing condition shifts runtime path and side effects. |
| 156 |         Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 157 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 158 |         Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block context.

```

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t. |
| 159 |         "Cat Comm bridge is not initialized".to_string(), | executable/support line; behavior impact
ends on surrounding block context. |
| 160 |         )) | executable/support line; behavior impact depends on surrounding block context. |
| 161 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 162 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 163 | (blank) | line is blank; visual separation only. |
| 164 |     fn send_bridge_request(&mut self, req: &BridgeRequest) -> Result<BridgeResponse, HwError> { | declares ex
table behavior; signature/body changes alter API, control flow, and outputs. |
| 165 |         let line = serde_json::to_string(req) | serialization contract touchpoint; edits can break report com
ibility and tooling. |
| 166 |         .map_err(|e| HwError::Unknown(format!("bridge request encode failed: {e}")))?; | error-propagat
operation; changing it impacts failure semantics and caller recovery paths. |
| 167 |         self.child_stdin | executable/support line; behavior impact depends on surrounding block context. |
| 168 |         .write_all(line.as_bytes()) | I/O or protocol-critical logic; edits can affect integration, timin
and diagnostic outcomes. |
| 169 |         .map_err(|e| HwError::Unknown(format!("bridge stdin write failed: {e}")))?; | error-propagating
eration; changing it impacts failure semantics and caller recovery paths. |
| 170 |         self.child_stdin | executable/support line; behavior impact depends on surrounding block context. |
| 171 |         .write_all(b"\n") | I/O or protocol-critical logic; edits can affect integration, timing, and dia
stic outcomes. |
| 172 |         .map_err(|e| HwError::Unknown(format!("bridge stdin newline failed: {e}")))?; | error-propagati
operation; changing it impacts failure semantics and caller recovery paths. |
| 173 |         self.child_stdin | executable/support line; behavior impact depends on surrounding block context. |
| 174 |         .flush() | executable/support line; behavior impact depends on surrounding block context. |
| 175 |         .map_err(|e| HwError::Unknown(format!("bridge stdin flush failed: {e}")))?; | error-propagating
eration; changing it impacts failure semantics and caller recovery paths. |
| 176 | (blank) | line is blank; visual separation only. |
| 177 |         let rsp = self.read_bridge_response()?; | error-propagating operation; changing it impacts failure se
tics and caller recovery paths. |
| 178 |         self.append_diag_record(&rsp); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 179 |         Ok(rsp) | executable/support line; behavior impact depends on surrounding block context. |
| 180 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 181 | (blank) | line is blank; visual separation only. |
| 182 |     fn read_bridge_response(&mut self) -> Result<BridgeResponse, HwError> { | declares executable behavior; s
ature/body changes alter API, control flow, and outputs. |
| 183 |         let mut buf = String::new(); | executable/support line; behavior impact depends on surrounding block
text. |
| 184 |         self.child_stdout | executable/support line; behavior impact depends on surrounding block context. |
| 185 |         .read_line(&mut buf) | executable/support line; behavior impact depends on surrounding block cont
. |
| 186 |         .map_err(|e| HwError::Unknown(format!("bridge stdout read failed: {e}")))?; | error-propagating
eration; changing it impacts failure semantics and caller recovery paths. |
| 187 |         if buf.trim().is_empty() { | conditional gate; changing condition shifts runtime path and side effect
|
| 188 |             return Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding bl
context. |
| 189 |             "Cat Comm bridge closed stream or returned empty response".to_string(), | executable/support
e; behavior impact depends on surrounding block context. |
| 190 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 191 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 192 | (blank) | line is blank; visual separation only. |
| 193 |         serde_json::from_str::<BridgeResponse>(buf.trim()).map_err(|e| { | serialization contract touchpoint
edits can break report compatibility and tooling. |
| 194 |             HwError::ParseError { | executable/support line; behavior impact depends on surrounding block con
t. |
| 195 |                 cause: format!("bridge response parse failed: {e}"), | executable/support line; behavior impa
depends on surrounding block context. |
| 196 |                 raw_data: buf, | executable/support line; behavior impact depends on surrounding block contex
|
| 197 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |

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| 198 |         }) | executable/support line; behavior impact depends on surrounding block context. |
| 199 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 200 | (blank) | line is blank; visual separation only. |
| 201 |     fn append_diag_record(&self, rsp: &BridgeResponse) { | declares executable behavior; signature/body change
alter API, control flow, and outputs. |
| 202 |         let Some(diag) = &rsp.diag else { | executable/support line; behavior impact depends on surrounding block
context. |
| 203 |             return; | executable/support line; behavior impact depends on surrounding block context. |
| 204 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 205 | (blank) | line is blank; visual separation only. |
| 206 |         let rec = BridgeDiagRecord { | executable/support line; behavior impact depends on surrounding block context. |
| 207 |             ts_ms: now_ms(), | executable/support line; behavior impact depends on surrounding block context. |
| 208 |             source: "vendor_cat_comm", | executable/support line; behavior impact depends on surrounding block context. |
| 209 |             wait_frame_count_delta: diag.wait_frame_count_delta.unwrap_or(0), | executable/support line; behavior impact depends on surrounding block context. |
| 210 |             sequence_error_count_delta: diag.sequence_error_count_delta.unwrap_or(0), | executable/support line; behavior impact depends on surrounding block context. |
| 211 |             flowcontrol_timeout_count_delta: diag.flowcontrol_timeout_count_delta.unwrap_or(0), | executable/support line; behavior impact depends on surrounding block context. |
| 212 |             fc_blocksize: diag.fc_blocksize, | executable/support line; behavior impact depends on surrounding block context. |
| 213 |             fc_stmin_ms: diag.fc_stmin_ms, | executable/support line; behavior impact depends on surrounding block context. |
| 214 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 215 | (blank) | line is blank; visual separation only. |
| 216 |         let Ok(mut f) = OpenOptions::new() | executable/support line; behavior impact depends on surrounding block context. |
| 217 |             .create(true) | executable/support line; behavior impact depends on surrounding block context. |
| 218 |             .append(true) | executable/support line; behavior impact depends on surrounding block context. |
| 219 |             .open(&self.diag_log_path) | executable/support line; behavior impact depends on surrounding block context. |
| 220 |         else { | executable/support line; behavior impact depends on surrounding block context. |
| 221 |             return; | executable/support line; behavior impact depends on surrounding block context. |
| 222 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 223 | (blank) | line is blank; visual separation only. |
| 224 |         let line = match serde_json::to_string(&rec) { | serialization contract touchpoint; edits can break runtime compatibility and tooling. |
| 225 |             Ok(v) => v, | executable/support line; behavior impact depends on surrounding block context. |
| 226 |             Err(_) => return, | executable/support line; behavior impact depends on surrounding block context. |
| 227 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 228 |         let _ = writeln!(f, "{line}"); | I/O or protocol-critical logic; edits can affect integration, timing and diagnostic outcomes. |
| 229 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 230 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 231 | (blank) | line is blank; visual separation only. |
| 232 | impl HardwareInterface for CatCommAdapter { | implementation block; method behavior and trait conformance are defined here. |
| 233 |     fn init(&mut self, _config: &HwConfig) -> Result<(), HwError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 234 |         let rsp = self.send_bridge_request(&BridgeRequest::Ping)?; | error-propagating operation; changing impacts failure semantics and caller recovery paths. |
| 235 |         if rsp.ok { | conditional gate; changing condition shifts runtime path and side effects. |
| 236 |             self.bridge_info = bridge_info_from_response(&rsp); | executable/support line; behavior impact depends on surrounding block context. |
| 237 |             self.connected = true; | executable/support line; behavior impact depends on surrounding block context. |
| 238 |             return Ok(()); | executable/support line; behavior impact depends on surrounding block context. |
| 239 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

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| 240 |         Err(map_bridge_error( | executable/support line; behavior impact depends on surrounding block context. |
| 241 |             rsp.kind, | executable/support line; behavior impact depends on surrounding block context. |
| 242 |             rsp.error.unwrap_or_else(\|\| "bridge ping failed".to_string()), | executable/support line; behavior impact depends on surrounding block context. |
| 243 |         )) | executable/support line; behavior impact depends on surrounding block context. |
| 244 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 245 | (blank) | line is blank; visual separation only. |
| 246 |     fn read_frame(&mut self) -> Result<Frame, HwError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 247 |         self.ensure_connected()?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 248 |         let rsp = self.send_bridge_request(&BridgeRequest::Read { | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 249 |             timeout_ms: self.timeout_ms, | executable/support line; behavior impact depends on surrounding block context. |
| 250 |         }); | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 251 |         if !rsp.ok { | conditional gate; changing condition shifts runtime path and side effects. |
| 252 |             return Err(map_bridge_error( | executable/support line; behavior impact depends on surrounding block context. |
| 253 |                 rsp.kind, | executable/support line; behavior impact depends on surrounding block context. |
| 254 |                 rsp.error.unwrap_or_else(\|\| "bridge read failed".to_string()), | executable/support line; behavior impact depends on surrounding block context. |
| 255 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 256 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 257 |         rsp.frame | executable/support line; behavior impact depends on surrounding block context. |
| 258 |         .ok_or_else(\|\| HwError::Unknown("bridge read returned no frame".to_string())) | executable/support line; behavior impact depends on surrounding block context. |
| 259 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 260 | (blank) | line is blank; visual separation only. |
| 261 |     fn try_read_frame(&mut self) -> Result<Option<Frame>, HwError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 262 |         self.ensure_connected()?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 263 |         let rsp = self.send_bridge_request(&BridgeRequest::TryRead)?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 264 |         if !rsp.ok { | conditional gate; changing condition shifts runtime path and side effects. |
| 265 |             return Err(map_bridge_error( | executable/support line; behavior impact depends on surrounding block context. |
| 266 |                 rsp.kind, | executable/support line; behavior impact depends on surrounding block context. |
| 267 |                 rsp.error.unwrap_or_else(\|\| "bridge try_read failed".to_string()), | executable/support line; behavior impact depends on surrounding block context. |
| 268 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 269 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 270 |         Ok(rsp.frame) | executable/support line; behavior impact depends on surrounding block context. |
| 271 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 272 | (blank) | line is blank; visual separation only. |
| 273 |     fn send_frame(&mut self, frame: Frame) -> Result<(), HwError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 274 |         self.ensure_connected()?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 275 |         let rsp = self.send_bridge_request(&BridgeRequest::Send { frame }); | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 276 |         if rsp.ok { | conditional gate; changing condition shifts runtime path and side effects. |
| 277 |             Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 278 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 279 |             Err(map_bridge_error( | executable/support line; behavior impact depends on surrounding block context. |
| 280 |                 rsp.kind, | executable/support line; behavior impact depends on surrounding block context. |
| 281 |                 rsp.error.unwrap_or_else(\|\| "bridge send failed".to_string()), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 282 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 283 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

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| 284 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 285 | (blank) | line is blank; visual separation only. |
| 286 |     fn close(&mut self) -> Result<(), HwError> { | declares executable behavior; signature/body changes alter
I, control flow, and outputs. |
| 287 |         let _ = self.send_bridge_request(&BridgeRequest::Close); | I/O or protocol-critical logic; edits can
ect integration, timing, and diagnostic outcomes. |
| 288 |         self.connected = false; | executable/support line; behavior impact depends on surrounding block conte
|
| 289 | (blank) | line is blank; visual separation only. |
| 290 |         if let Err(e) = self.child.kill() { | conditional gate; changing condition shifts runtime path and si
effects. |
| 291 |             let already_exited = self | executable/support line; behavior impact depends on surrounding block
ntext. |
| 292 |                 .child | executable/support line; behavior impact depends on surrounding block context. |
| 293 |                 .try_wait() | executable/support line; behavior impact depends on surrounding block context.
| 294 |                 .ok() | executable/support line; behavior impact depends on surrounding block context. |
| 295 |                 .flatten() | executable/support line; behavior impact depends on surrounding block context. |
| 296 |                 .is_some(); | executable/support line; behavior impact depends on surrounding block context.
| 297 |             if !already_exited { | conditional gate; changing condition shifts runtime path and side effects.
| 298 |                 return Err(HwError::Unknown(format!("failed to stop Cat Comm bridge: {e}"))); | executable/su
rt line; behavior impact depends on surrounding block context. |
| 299 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 300 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 301 |         let _ = self.child.wait(); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 302 |         Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 303 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 304 | (blank) | line is blank; visual separation only. |
| 305 |     fn adapter_info(&self) -> Option<String> { | declares executable behavior; signature/body changes alter A
control flow, and outputs. |
| 306 |         Some(format!( | executable/support line; behavior impact depends on surrounding block context. |
| 307 |             "bridge_protocol={:?} capabilities={}", | error-propagating operation; changing it impacts failur
emantics and caller recovery paths. |
| 308 |             self.bridge_info.protocol_version, | executable/support line; behavior impact depends on surround
block context. |
| 309 |             if self.bridge_info.capabilities.is_empty() { | conditional gate; changing condition shifts runti
path and side effects. |
| 310 |                 "none".to_string() | executable/support line; behavior impact depends on surrounding block co
xt. |
| 311 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 312 |                 self.bridge_info.capabilities.join(",") | executable/support line; behavior impact depends on
rounding block context. |
| 313 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 314 |         )) | executable/support line; behavior impact depends on surrounding block context. |
| 315 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 316 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 317 | (blank) | line is blank; visual separation only. |
| 318 | fn resolve_bridge_exe(cfg: &HwConfig) -> Result<PathBuf, HwError> { | declares executable behavior; signature
dy changes alter API, control flow, and outputs. |
| 319 |     if let Some(explicit) = &cfg.vendor_bridge_exe { | conditional gate; changing condition shifts runtime pa
and side effects. |
| 320 |         if explicit.exists() { | conditional gate; changing condition shifts runtime path and side effects. |
| 321 |             return Ok(explicit.clone()); | executable/support line; behavior impact depends on surrounding bl
context. |
| 322 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 323 |         return Err(HwError::PortNotFound(format!( | executable/support line; behavior impact depends on surro
ing block context. |
| 324 |             "vendor bridge executable not found: {}", | executable/support line; behavior impact depends on s
ounding block context. |
| 325 |             explicit.display() | executable/support line; behavior impact depends on surrounding block contex
|

```

```

| 326 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 327 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 328 | (blank) | line is blank; visual separation only. |
| 329 |     if let Some(template_dir) = &cfg.vendor_template_dir { | conditional gate; changing condition shifts runtime path and side effects. |
| 330 |         let candidate = template_dir.join(DEFAULT_TEMPLATE_BRIDGE_EXE); | executable/support line; behavior impact depends on surrounding block context. |
| 331 |         if candidate.exists() { | conditional gate; changing condition shifts runtime path and side effects. |
| 332 |             return Ok(candidate); | executable/support line; behavior impact depends on surrounding block context. |
| 333 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 334 |         return Err(HwError::PortNotFound(format!( | executable/support line; behavior impact depends on surrounding block context. |
| 335 |             "template bridge not found at {}. Upload your template bridge executable as {}", | executable/support line; behavior impact depends on surrounding block context. |
| 336 |             candidate.display(), | executable/support line; behavior impact depends on surrounding block context. |
| 337 |             DEFAULT_TEMPLATE_BRIDGE_EXE | executable/support line; behavior impact depends on surrounding block context. |
| 338 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 339 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 340 | (blank) | line is blank; visual separation only. |
| 341 |     Err(HwError::PortNotFound( | executable/support line; behavior impact depends on surrounding block context. |
| 342 |         "Cat Comm bridge not configured. Set --vendor-bridge-exe=<path> or --vendor-template-dir=<dir>" | executable/support line; behavior impact depends on surrounding block context. |
| 343 |         .to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 344 |     )) | executable/support line; behavior impact depends on surrounding block context. |
| 345 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 346 | (blank) | line is blank; visual separation only. |
| 347 | fn map_bridge_error(kind: Option<String>, msg: String) -> HwError { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 348 |     match kind | branch dispatch; changing arms alters behavior for variants and input classes. |
| 349 |         .as_deref() | executable/support line; behavior impact depends on surrounding block context. |
| 350 |         .unwrap_or_default() | executable/support line; behavior impact depends on surrounding block context. |
| 351 |         .to_ascii_lowercase() | executable/support line; behavior impact depends on surrounding block context. |
| 352 |         .as_str() | executable/support line; behavior impact depends on surrounding block context. |
| 353 |     { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 354 |         "timeout" => HwError::Timeout, | executable/support line; behavior impact depends on surrounding block context. |
| 355 |         "rate_limited" => HwError::RateLimited, | executable/support line; behavior impact depends on surrounding block context. |
| 356 |         "bus_off" => HwError::BusOff, | executable/support line; behavior impact depends on surrounding block context. |
| 357 |         "transceiver" => HwError::TransceiverError, | executable/support line; behavior impact depends on surrounding block context. |
| 358 |         "permission" => HwError::PermissionDenied(msg), | executable/support line; behavior impact depends on surrounding block context. |
| 359 |         "write_blocked" => HwError::WriteBlockedAllowlist, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 360 |         "parse" => HwError::ParseError { | executable/support line; behavior impact depends on surrounding block context. |
| 361 |             cause: "bridge parse error".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 362 |             raw_data: msg, | executable/support line; behavior impact depends on surrounding block context. |
| 363 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 364 |         _ => HwError::Unknown(msg), | executable/support line; behavior impact depends on surrounding block context. |
| 365 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 366 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 367 | (blank) | line is blank; visual separation only. |
| 368 | fn bridge_info_from_response(rsp: &BridgeResponse) -> CatCommBridgeInfo { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 369 |     CatCommBridgeInfo { | executable/support line; behavior impact depends on surrounding block context. |

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| 370 |         protocol_version: rsp.protocol_version.clone(), | executable/support line; behavior impact depends on surrounding block context. | | |
| 371 |         capabilities: rsp.capabilities.clone().unwrap_or_default(), | executable/support line; behavior impact depends on surrounding block context. |
| 372 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 373 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 374 | (blank) | line is blank; visual separation only. |
| 375 | fn ensure_bridge_compatibility(cfg: &HwConfig, info: &CatCommBridgeInfo) -> Result<(), HwError> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 376 |     if let Some(ver) = &info.protocol_version { | conditional gate; changing condition shifts runtime path and side effects. |
| 377 |         let major = ver | executable/support line; behavior impact depends on surrounding block context. |
| 378 |         .split('.') | executable/support line; behavior impact depends on surrounding block context. |
| 379 |         .next() | executable/support line; behavior impact depends on surrounding block context. |
| 380 |         .and_then(|s| s.parse::<u32>().ok()) | executable/support line; behavior impact depends on surrounding block context. |
| 381 |         .unwrap_or(0); | executable/support line; behavior impact depends on surrounding block context. |
| 382 |         if major > 1 { | conditional gate; changing condition shifts runtime path and side effects. |
| 383 |             return Err(HwError::Unknown(format!( | executable/support line; behavior impact depends on surrounding block context. |
| 384 |                 "unsupported Cat Comm bridge protocol version {ver}; expected major version 1" | executable/support line; behavior impact depends on surrounding block context. |
| 385 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 386 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 387 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 388 | (blank) | line is blank; visual separation only. |
| 389 |     // Capability checks are strict only when the bridge explicitly reports capabilities. | comment line; documentation intent, no runtime effect. |
| 390 |     if !info.capabilities.is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 391 |         if cfg.write_effectively_enabled() && !info.supports("write") { | conditional gate; changing condition shifts runtime path and side effects. |
| 392 |             return Err(HwError::PermissionDenied( | executable/support line; behavior impact depends on surrounding block context. |
| 393 |                 "bridge does not advertise write capability".to_string(), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 394 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 395 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 396 |         if cfg.can_interface.is_some() && !info.supports("can") { | conditional gate; changing condition shifts runtime path and side effects. |
| 397 |             return Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block context. |
| 398 |                 "bridge does not advertise CAN capability for requested --can-if path" | executable/support line; behavior impact depends on surrounding block context. |
| 399 |                 .to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 400 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 401 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 402 |         if cfg.serial_port.is_some() && !info.supports("serial") { | conditional gate; changing condition shifts runtime path and side effects. |
| 403 |             return Err(HwError::Unknown( | executable/support line; behavior impact depends on surrounding block context. |
| 404 |                 "bridge does not advertise serial capability for requested --serial-port path" | executable/support line; behavior impact depends on surrounding block context. |
| 405 |                 .to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 406 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 407 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 408 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 409 | (blank) | line is blank; visual separation only. |
| 410 |     Ok(()) | executable/support line; behavior impact depends on surrounding block context. |

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| 411 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 412 | (blank) | line is blank; visual separation only. |
| 413 | fn now_ms() -> u64 { | declares executable behavior; signature/body changes alter API, control flow, and output
. |
| 414 |     std::time::SystemTime::now() | executable/support line; behavior impact depends on surrounding block context
. |
| 415 |         .duration_since(std::time::UNIX_EPOCH) | executable/support line; behavior impact depends on surrounding
block context. |
| 416 |         .map(|d| d.as_millis() as u64) | executable/support line; behavior impact depends on surrounding block
context. |
| 417 |         .unwrap_or(0) | executable/support line; behavior impact depends on surrounding block context. |
| 418 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/j1939.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 1656 | Non-empty: 1611 | Symbols: 129

Symbol Inventory

Line	Kind	Symbol	Change Impact
14	module	addr	data/structure coupling
15	const	ECM_1	data/structure coupling
16	const	ECM_2	data/structure coupling
17	const	TURBOCHARGER	data/structure coupling
18	const	TRANSMISSION	data/structure coupling
19	const	BRAKES	data/structure coupling
20	const	TASK_CTRL	data/structure coupling
21	const	HEADWAY	data/structure coupling
22	const	IMPLEMENT	data/structure coupling
23	const	ISOBUS_VT	data/structure coupling
24	const	CAB	data/structure coupling
25	const	ARMREST	data/structure coupling
26	const	INSTRUMENT	data/structure coupling
27	const	HITCH	data/structure coupling
28	const	NULL	data/structure coupling
29	const	BROADCAST	data/structure coupling
33	module	pgn	data/structure coupling
35	const	EBC1	data/structure coupling
36	const	ETC1	data/structure coupling
37	const	ETC2	data/structure coupling
38	const	EEC1	data/structure coupling
39	const	EEC2	data/structure coupling
40	const	EEC3	data/structure coupling
41	const	ET1	data/structure coupling
42	const	EFL_P1	data/structure coupling
43	const	IC1	data/structure coupling
44	const	LFE	data/structure coupling
45	const	HOURS	data/structure coupling
46	const	FUEL1	data/structure coupling
47	const	EEC5	data/structure coupling
49	const	CCVS	data/structure coupling
50	const	VD	data/structure coupling
51	const	AMB	data/structure coupling
52	const	FD	data/structure coupling
54	const	PTO	data/structure coupling
55	const	HITCH	data/structure coupling
56	const	WS_MST	data/structure coupling
57	const	WS_MBR	data/structure coupling
58	const	SC	data/structure coupling
59	const	GSC	data/structure coupling
61	const	DM1	data/structure coupling
62	const	DM2	data/structure coupling
63	const	DM3	data/structure coupling

64	const	DM11	data/structure coupling
65	const	DM15	data/structure coupling
66	const	DM16	data/structure coupling
67	const	RQST	data/structure coupling
69	const	TP_CM	data/structure coupling
70	const	TP_DT	data/structure coupling
71	const	ECAN	data/structure coupling
72	const	PROP_A	data/structure coupling
73	const	ADAS1	data/structure coupling
74	const	FUS1	data/structure coupling
75	const	ENG1	data/structure coupling
78	const	TSC1	data/structure coupling
79	const	CM1	data/structure coupling
80	const	TCFG	data/structure coupling
81	const	EC1	data/structure coupling
82	const	HRVD	data/structure coupling
83	const	SHUTDN	data/structure coupling
84	const	SOFT	data/structure coupling
85	const	ECFG	data/structure coupling
86	const	PTODE	data/structure coupling
87	const	TD	data/structure coupling
88	const	TIRE	data/structure coupling
89	const	EI	data/structure coupling
90	const	VEP1	data/structure coupling
91	const	SERV	data/structure coupling
92	const	EBC2	data/structure coupling
94	const	GSC1	data/structure coupling
96	const	NM_ACK	data/structure coupling
97	const	DM13	data/structure coupling
98	const	DM14	data/structure coupling
99	const	DM18	data/structure coupling
100	const	DM19	data/structure coupling
101	const	DM20	data/structure coupling
102	const	DM21	data/structure coupling
103	const	DM25	data/structure coupling
104	const	DM26	data/structure coupling
105	const	DM27	data/structure coupling
107	const	SEC	data/structure coupling
108	const	SECACK	data/structure coupling
114	struct	SpnDef	data/structure coupling
125	struct	PgnDef	data/structure coupling
1112	fn	find_pgn	signature/API coupling
1119	struct	J1939Frame	data/structure coupling
1132	struct	SpnValue	data/structure coupling
1139	impl	J1939Frame	behavior binding and method semantics
1140	fn	from_raw	signature/API coupling
1172	fn	build_id	signature/API coupling
1179	fn	decode	signature/API coupling
1203	fn	pgn_name	signature/API coupling
1207	fn	sa_name	signature/API coupling
1227	fn	data_hex	signature/API coupling
1236	fn	bits	signature/API coupling
1251	struct	Builder	data/structure coupling
1252	impl	Builder	behavior binding and method semantics
1253	fn	eec1	signature/API coupling
1265	fn	eec2	signature/API coupling
1271	fn	et1	signature/API coupling
1280	fn	efl_p1	signature/API coupling
1287	fn	ic1	signature/API coupling
1296	fn	lfe	signature/API coupling
1304	fn	pto	signature/API coupling
1312	fn	ccvs	signature/API coupling
1322	fn	dm1	signature/API coupling
1342	fn	hours	signature/API coupling
1353	fn	adas1	signature/API coupling

[illegible]

[illegible]

[illegible]

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145 |         bit_length: 4, | executable/support line; behavior impact depends on surrounding block context.
146 |         factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
147 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
148 |         unit: "", | executable/support line; behavior impact depends on surrounding block context. |
149 |     }, | executable/support line; behavior impact depends on surrounding block context. |
150 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
151 |         spn: 512, | executable/support line; behavior impact depends on surrounding block context. |
152 |         name: "Driver Demand Torque", | executable/support line; behavior impact depends on surrounding
block context. |
153 |         byte_offset: 1, | executable/support line; behavior impact depends on surrounding block context.
154 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context.
155 |         bit_length: 8, | executable/support line; behavior impact depends on surrounding block context.
156 |         factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
157 |         offset: -125.0, | executable/support line; behavior impact depends on surrounding block context.
158 |         unit: "%", | executable/support line; behavior impact depends on surrounding block context. |
159 |     }, | executable/support line; behavior impact depends on surrounding block context. |
160 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
161 |         spn: 513, | executable/support line; behavior impact depends on surrounding block context. |
162 |         name: "Actual Engine Torque", | executable/support line; behavior impact depends on surrounding
block context. |
163 |         byte_offset: 2, | executable/support line; behavior impact depends on surrounding block context.
164 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context.
165 |         bit_length: 8, | executable/support line; behavior impact depends on surrounding block context.
166 |         factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
167 |         offset: -125.0, | executable/support line; behavior impact depends on surrounding block context.
168 |         unit: "%", | executable/support line; behavior impact depends on surrounding block context. |
169 |     }, | executable/support line; behavior impact depends on surrounding block context. |
170 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
171 |         spn: 190, | executable/support line; behavior impact depends on surrounding block context. |
172 |         name: "Engine Speed", | executable/support line; behavior impact depends on surrounding block
context. |
173 |         byte_offset: 3, | executable/support line; behavior impact depends on surrounding block context.
174 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context.
175 |         bit_length: 16, | executable/support line; behavior impact depends on surrounding block context.
176 |         factor: 0.125, | executable/support line; behavior impact depends on surrounding block context.
177 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
178 |         unit: "rpm", | executable/support line; behavior impact depends on surrounding block context.
179 |     }, | executable/support line; behavior impact depends on surrounding block context. |
180 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
181 |         spn: 1675, | executable/support line; behavior impact depends on surrounding block context. |
182 |         name: "Engine Starter Mode", | executable/support line; behavior impact depends on surrounding
lock context. |
183 |         byte_offset: 6, | executable/support line; behavior impact depends on surrounding block context.
184 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context.
185 |         bit_length: 4, | executable/support line; behavior impact depends on surrounding block context.
186 |         factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
187 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
188 |         unit: "", | executable/support line; behavior impact depends on surrounding block context. |

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| 189 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 190 |     ], | executable/support line; behavior impact depends on surrounding block context. |
| 191 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 192 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 193 |     pgn: 61445, | executable/support line; behavior impact depends on surrounding block context. |
| 194 |     name: "EEC2", | executable/support line; behavior impact depends on surrounding block context. |
| 195 |     desc: "Electronic Engine Control 2", | executable/support line; behavior impact depends on surrounding
lock context. |
| 196 |     rate_ms: 50, | executable/support line; behavior impact depends on surrounding block context. |
| 197 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
| 198 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 199 |             spn: 558, | executable/support line; behavior impact depends on surrounding block context. |
| 200 |             name: "Accel Pedal Pos 1", | executable/support line; behavior impact depends on surrounding
ck context. |
| 201 |                 byte_offset: 1, | executable/support line; behavior impact depends on surrounding block conte
|
| 202 |                 bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 203 |                 bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
| 204 |                 factor: 0.4, | executable/support line; behavior impact depends on surrounding block context.
| 205 |                 offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 206 |                 unit: "%", | executable/support line; behavior impact depends on surrounding block context. |
| 207 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 208 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 209 |             spn: 91, | executable/support line; behavior impact depends on surrounding block context. |
| 210 |             name: "Throttle Position", | executable/support line; behavior impact depends on surrounding
ck context. |
| 211 |                 byte_offset: 1, | executable/support line; behavior impact depends on surrounding block conte
|
| 212 |                 bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 213 |                 bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
| 214 |                 factor: 0.4, | executable/support line; behavior impact depends on surrounding block context.
| 215 |                 offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 216 |                 unit: "%", | executable/support line; behavior impact depends on surrounding block context. |
| 217 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 218 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 219 |             spn: 92, | executable/support line; behavior impact depends on surrounding block context. |
| 220 |             name: "Engine Percent Load", | executable/support line; behavior impact depends on surrounding
lock context. |
| 221 |                 byte_offset: 2, | executable/support line; behavior impact depends on surrounding block conte
|
| 222 |                 bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 223 |                 bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
| 224 |                 factor: 0.4, | executable/support line; behavior impact depends on surrounding block context.
| 225 |                 offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 226 |                 unit: "%", | executable/support line; behavior impact depends on surrounding block context. |
| 227 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 228 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 229 |             spn: 974, | executable/support line; behavior impact depends on surrounding block context. |
| 230 |             name: "Remote Accel Pedal", | executable/support line; behavior impact depends on surrounding
ock context. |
| 231 |                 byte_offset: 3, | executable/support line; behavior impact depends on surrounding block conte
|
| 232 |                 bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 233 |                 bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
| 234 |                 factor: 0.4, | executable/support line; behavior impact depends on surrounding block context.
| 235 |                 offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 236 |                 unit: "%", | executable/support line; behavior impact depends on surrounding block context. |

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| 237 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 238 |     ], | executable/support line; behavior impact depends on surrounding block context. |
| 239 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 240 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 241 |     pgn: 61442, | executable/support line; behavior impact depends on surrounding block context. |
| 242 |     name: "ETC1", | executable/support line; behavior impact depends on surrounding block context. |
| 243 |     desc: "Electronic Transmission Control 1", | executable/support line; behavior impact depends on surrou
ding block context. |
| 244 |     rate_ms: 20, | executable/support line; behavior impact depends on surrounding block context. |
| 245 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
| 246 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 247 |             spn: 560, | executable/support line; behavior impact depends on surrounding block context. |
| 248 |             name: "Trans Driveline Engaged", | executable/support line; behavior impact depends on surrou
ng block context. |
| 249 |             byte_offset: 0, | executable/support line; behavior impact depends on surrounding block conte
|
| 250 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 251 |             bit_length: 2, | executable/support line; behavior impact depends on surrounding block context
|
| 252 |             factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 253 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 254 |             unit: "", | executable/support line; behavior impact depends on surrounding block context. |
| 255 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 256 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 257 |             spn: 573, | executable/support line; behavior impact depends on surrounding block context. |
| 258 |             name: "Torque Conv Lockup", | executable/support line; behavior impact depends on surrounding
ock context. |
| 259 |             byte_offset: 0, | executable/support line; behavior impact depends on surrounding block conte
|
| 260 |             bit_offset: 2, | executable/support line; behavior impact depends on surrounding block context
|
| 261 |             bit_length: 2, | executable/support line; behavior impact depends on surrounding block context
|
| 262 |             factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 263 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 264 |             unit: "", | executable/support line; behavior impact depends on surrounding block context. |
| 265 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 266 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 267 |             spn: 191, | executable/support line; behavior impact depends on surrounding block context. |
| 268 |             name: "Output Shaft Speed", | executable/support line; behavior impact depends on surrounding
ock context. |
| 269 |             byte_offset: 3, | executable/support line; behavior impact depends on surrounding block conte
|
| 270 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 271 |             bit_length: 16, | executable/support line; behavior impact depends on surrounding block conte
|
| 272 |             factor: 0.125, | executable/support line; behavior impact depends on surrounding block context
|
| 273 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 274 |             unit: "rpm", | executable/support line; behavior impact depends on surrounding block context.
| 275 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 276 |     ], | executable/support line; behavior impact depends on surrounding block context. |
| 277 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 278 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 279 |     pgn: 61443, | executable/support line; behavior impact depends on surrounding block context. |
| 280 |     name: "ETC2", | executable/support line; behavior impact depends on surrounding block context. |
| 281 |     desc: "Electronic Transmission Control 2", | executable/support line; behavior impact depends on surrou
ding block context. |
| 282 |     rate_ms: 20, | executable/support line; behavior impact depends on surrounding block context. |
| 283 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
| 284 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 285 |             spn: 524, | executable/support line; behavior impact depends on surrounding block context. |
| 286 |             name: "Selected Gear", | executable/support line; behavior impact depends on surrounding bloc

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ontext. |
| 287 |         byte_offset: 3, | executable/support line; behavior impact depends on surrounding block conte
|
| 288 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block contex
|
| 289 |         bit_length: 8, | executable/support line; behavior impact depends on surrounding block contex
|
| 290 |         factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 291 |         offset: -125.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 292 |         unit: "idx", | executable/support line; behavior impact depends on surrounding block context.
| 293 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 294 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 295 |         spn: 523, | executable/support line; behavior impact depends on surrounding block context. |
| 296 |         name: "Current Gear", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 297 |         byte_offset: 4, | executable/support line; behavior impact depends on surrounding block conte
|
| 298 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block contex
|
| 299 |         bit_length: 8, | executable/support line; behavior impact depends on surrounding block contex
|
| 300 |         factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 301 |         offset: -125.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 302 |         unit: "idx", | executable/support line; behavior impact depends on surrounding block context.
| 303 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 304 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 305 |         spn: 525, | executable/support line; behavior impact depends on surrounding block context. |
| 306 |         name: "Current Range", | executable/support line; behavior impact depends on surrounding block
ontext. |
| 307 |         byte_offset: 7, | executable/support line; behavior impact depends on surrounding block conte
|
| 308 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block contex
|
| 309 |         bit_length: 8, | executable/support line; behavior impact depends on surrounding block contex
|
| 310 |         factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 311 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 312 |         unit: "", | executable/support line; behavior impact depends on surrounding block context. |
| 313 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 314 | ], | executable/support line; behavior impact depends on surrounding block context. |
| 315 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 316 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 317 |     pgn: 61441, | executable/support line; behavior impact depends on surrounding block context. |
| 318 |     name: "EBC1", | executable/support line; behavior impact depends on surrounding block context. |
| 319 |     desc: "Electronic Brake Controller 1", | executable/support line; behavior impact depends on surround
block context. |
| 320 |     rate_ms: 20, | executable/support line; behavior impact depends on surrounding block context. |
| 321 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
| 322 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 323 |             spn: 561, | executable/support line; behavior impact depends on surrounding block context. |
| 324 |             name: "ABS Active", | executable/support line; behavior impact depends on surrounding block c
ext. |
| 325 |         byte_offset: 0, | executable/support line; behavior impact depends on surrounding block conte
|
| 326 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block contex
|
| 327 |         bit_length: 2, | executable/support line; behavior impact depends on surrounding block contex
|
| 328 |         factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 329 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 330 |         unit: "", | executable/support line; behavior impact depends on surrounding block context. |
| 331 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 332 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |

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| 333 |           spn: 562, | executable/support line; behavior impact depends on surrounding block context. |
| 334 |           name: "ABS Offroad Switch", | executable/support line; behavior impact depends on surrounding
ock context. |
| 335 |           byte_offset: 0, | executable/support line; behavior impact depends on surrounding block conte
|
| 336 |           bit_offset: 2, | executable/support line; behavior impact depends on surrounding block context
|
| 337 |           bit_length: 2, | executable/support line; behavior impact depends on surrounding block context
|
| 338 |           factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 339 |           offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 340 |           unit: "", | executable/support line; behavior impact depends on surrounding block context. |
| 341 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 342 | SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 343 |     spn: 563, | executable/support line; behavior impact depends on surrounding block context. |
| 344 |     name: "ASR Brake Control", | executable/support line; behavior impact depends on surrounding
ck context. |
| 345 |     byte_offset: 0, | executable/support line; behavior impact depends on surrounding block conte
|
| 346 |     bit_offset: 4, | executable/support line; behavior impact depends on surrounding block context
|
| 347 |     bit_length: 2, | executable/support line; behavior impact depends on surrounding block context
|
| 348 |     factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 349 |     offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 350 |     unit: "", | executable/support line; behavior impact depends on surrounding block context. |
| 351 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 352 | SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 353 |     spn: 1121, | executable/support line; behavior impact depends on surrounding block context. |
| 354 |     name: "ESP Active", | executable/support line; behavior impact depends on surrounding block c
ext. |
| 355 |     byte_offset: 1, | executable/support line; behavior impact depends on surrounding block conte
|
| 356 |     bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 357 |     bit_length: 2, | executable/support line; behavior impact depends on surrounding block context
|
| 358 |     factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 359 |     offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 360 |     unit: "", | executable/support line; behavior impact depends on surrounding block context. |
| 361 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 362 | SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 363 |     spn: 521, | executable/support line; behavior impact depends on surrounding block context. |
| 364 |     name: "Front Axle Brake Demand", | executable/support line; behavior impact depends on surrou
ng block context. |
| 365 |     byte_offset: 2, | executable/support line; behavior impact depends on surrounding block conte
|
| 366 |     bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 367 |     bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
| 368 |     factor: 0.4, | executable/support line; behavior impact depends on surrounding block context.
| 369 |     offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 370 |     unit: "%", | executable/support line; behavior impact depends on surrounding block context. |
| 371 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 372 | SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 373 |     spn: 522, | executable/support line; behavior impact depends on surrounding block context. |
| 374 |     name: "Rear Axle Brake Demand", | executable/support line; behavior impact depends on surroun
g block context. |
| 375 |     byte_offset: 3, | executable/support line; behavior impact depends on surrounding block conte
|
| 376 |     bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 377 |     bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|

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378 |         factor: 0.4, | executable/support line; behavior impact depends on surrounding block context. |
379 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
380 |         unit: "%", | executable/support line; behavior impact depends on surrounding block context. |
381 |     }, | executable/support line; behavior impact depends on surrounding block context. |
382 | ], | executable/support line; behavior impact depends on surrounding block context. |
383 | }, | executable/support line; behavior impact depends on surrounding block context. |
384 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
385 |     pgn: 65215, | executable/support line; behavior impact depends on surrounding block context. |
386 |     name: "EBC2", | executable/support line; behavior impact depends on surrounding block context. |
387 |     desc: "Electronic Brake Controller 2", | executable/support line; behavior impact depends on surrounding
block context. |
388 |     rate_ms: 100, | executable/support line; behavior impact depends on surrounding block context. |
389 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
390 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
391 |             spn: 904, | executable/support line; behavior impact depends on surrounding block context. |
392 |             name: "Wheel Speed FL", | executable/support line; behavior impact depends on surrounding blo
context. |
393 |             byte_offset: 0, | executable/support line; behavior impact depends on surrounding block conte
|
394 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block contex
|
395 |             bit_length: 16, | executable/support line; behavior impact depends on surrounding block conte
|
396 |             factor: 0.125, | executable/support line; behavior impact depends on surrounding block contex
|
397 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
398 |             unit: "km/h", | executable/support line; behavior impact depends on surrounding block context
399 |         }, | executable/support line; behavior impact depends on surrounding block context. |
400 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
401 |             spn: 905, | executable/support line; behavior impact depends on surrounding block context. |
402 |             name: "Wheel Speed FR", | executable/support line; behavior impact depends on surrounding blo
context. |
403 |             byte_offset: 2, | executable/support line; behavior impact depends on surrounding block conte
|
404 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block contex
|
405 |             bit_length: 16, | executable/support line; behavior impact depends on surrounding block conte
|
406 |             factor: 0.125, | executable/support line; behavior impact depends on surrounding block contex
|
407 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
408 |             unit: "km/h", | executable/support line; behavior impact depends on surrounding block context
409 |         }, | executable/support line; behavior impact depends on surrounding block context. |
410 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
411 |             spn: 906, | executable/support line; behavior impact depends on surrounding block context. |
412 |             name: "Wheel Speed RL", | executable/support line; behavior impact depends on surrounding blo
context. |
413 |             byte_offset: 4, | executable/support line; behavior impact depends on surrounding block conte
|
414 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block contex
|
415 |             bit_length: 16, | executable/support line; behavior impact depends on surrounding block conte
|
416 |             factor: 0.125, | executable/support line; behavior impact depends on surrounding block contex
|
417 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
418 |             unit: "km/h", | executable/support line; behavior impact depends on surrounding block context
419 |         }, | executable/support line; behavior impact depends on surrounding block context. |
420 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
421 |             spn: 907, | executable/support line; behavior impact depends on surrounding block context. |
422 |             name: "Wheel Speed RR", | executable/support line; behavior impact depends on surrounding blo
context. |
423 |             byte_offset: 6, | executable/support line; behavior impact depends on surrounding block conte
|
424 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block contex

```

```

|
| 425 |         bit_length: 16, | executable/support line; behavior impact depends on surrounding block conte
|
| 426 |         factor: 0.125, | executable/support line; behavior impact depends on surrounding block context
|
| 427 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 428 |         unit: "km/h", | executable/support line; behavior impact depends on surrounding block context
| 429 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 430 | ], | executable/support line; behavior impact depends on surrounding block context. |
| 431 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 432 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 433 |     pgn: 65262, | executable/support line; behavior impact depends on surrounding block context. |
| 434 |     name: "ET1", | executable/support line; behavior impact depends on surrounding block context. |
| 435 |     desc: "Engine Temperature 1", | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 436 |     rate_ms: 1000, | executable/support line; behavior impact depends on surrounding block context. |
| 437 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
| 438 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 439 |             spn: 110, | executable/support line; behavior impact depends on surrounding block context. |
| 440 |             name: "Engine Coolant Temp", | executable/support line; behavior impact depends on surroundin
lock context. |
| 441 |             byte_offset: 0, | executable/support line; behavior impact depends on surrounding block conte
|
| 442 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 443 |             bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
| 444 |             factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 445 |             offset: -40.0, | executable/support line; behavior impact depends on surrounding block context
|
| 446 |             unit: "°C", | executable/support line; behavior impact depends on surrounding block context.
| 447 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 448 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 449 |             spn: 174, | executable/support line; behavior impact depends on surrounding block context. |
| 450 |             name: "Fuel Temperature", | executable/support line; behavior impact depends on surrounding b
k context. |
| 451 |             byte_offset: 1, | executable/support line; behavior impact depends on surrounding block conte
|
| 452 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 453 |             bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
| 454 |             factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 455 |             offset: -40.0, | executable/support line; behavior impact depends on surrounding block context
|
| 456 |             unit: "°C", | executable/support line; behavior impact depends on surrounding block context.
| 457 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 458 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 459 |             spn: 175, | executable/support line; behavior impact depends on surrounding block context. |
| 460 |             name: "Engine Oil Temperature", | executable/support line; behavior impact depends on surround
g block context. |
| 461 |             byte_offset: 2, | executable/support line; behavior impact depends on surrounding block conte
|
| 462 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 463 |             bit_length: 16, | executable/support line; behavior impact depends on surrounding block conte
|
| 464 |             factor: 0.03125, | executable/support line; behavior impact depends on surrounding block cont
. |
| 465 |             offset: -273.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 466 |             unit: "°C", | executable/support line; behavior impact depends on surrounding block context.
| 467 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 468 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 469 |             spn: 176, | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 470 |         name: "Turbo Oil Temperature", | executable/support line; behavior impact depends on surrounding
block context. |
| 471 |         byte_offset: 4, | executable/support line; behavior impact depends on surrounding block context. |
|
| 472 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
|
| 473 |         bit_length: 16, | executable/support line; behavior impact depends on surrounding block context. |
|
| 474 |         factor: 0.03125, | executable/support line; behavior impact depends on surrounding block context. |
. |
| 475 |         offset: -273.0, | executable/support line; behavior impact depends on surrounding block context. |
|
| 476 |         unit: "°C", | executable/support line; behavior impact depends on surrounding block context. |
| 477 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 478 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 479 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 480 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 481 |     pgn: 65263, | executable/support line; behavior impact depends on surrounding block context. |
| 482 |     name: "EFL/P1", | executable/support line; behavior impact depends on surrounding block context. |
| 483 |     desc: "Engine Fluid Level/Pressure 1", | executable/support line; behavior impact depends on surrounding
block context. |
| 484 |     rate_ms: 500, | executable/support line; behavior impact depends on surrounding block context. |
| 485 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
| 486 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 487 |             spn: 94, | executable/support line; behavior impact depends on surrounding block context. |
| 488 |             name: "Fuel Delivery Pressure", | executable/support line; behavior impact depends on surrounding
g block context. |
| 489 |             byte_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
|
| 490 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
|
| 491 |             bit_length: 8, | executable/support line; behavior impact depends on surrounding block context. |
|
| 492 |             factor: 4.0, | executable/support line; behavior impact depends on surrounding block context. |
| 493 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 494 |             unit: "kPa", | executable/support line; behavior impact depends on surrounding block context. |
| 495 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 496 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 497 |             spn: 98, | executable/support line; behavior impact depends on surrounding block context. |
| 498 |             name: "Engine Oil Level", | executable/support line; behavior impact depends on surrounding block
k context. |
| 499 |             byte_offset: 3, | executable/support line; behavior impact depends on surrounding block context. |
|
| 500 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
|
| 501 |             bit_length: 8, | executable/support line; behavior impact depends on surrounding block context. |
|
| 502 |             factor: 0.4, | executable/support line; behavior impact depends on surrounding block context. |
| 503 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 504 |             unit: "%", | executable/support line; behavior impact depends on surrounding block context. |
| 505 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 506 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 507 |             spn: 100, | executable/support line; behavior impact depends on surrounding block context. |
| 508 |             name: "Engine Oil Pressure", | executable/support line; behavior impact depends on surrounding
lock context. |
| 509 |             byte_offset: 4, | executable/support line; behavior impact depends on surrounding block context. |
|
| 510 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
|
| 511 |             bit_length: 8, | executable/support line; behavior impact depends on surrounding block context. |
|
| 512 |             factor: 4.0, | executable/support line; behavior impact depends on surrounding block context. |
| 513 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 514 |             unit: "kPa", | executable/support line; behavior impact depends on surrounding block context. |
| 515 |         }, | executable/support line; behavior impact depends on surrounding block context. |

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```

516 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
517 |             spn: 109, | executable/support line; behavior impact depends on surrounding block context. |
518 |             name: "Coolant Pressure", | executable/support line; behavior impact depends on surrounding b
k context. |
519 |             byte_offset: 6, | executable/support line; behavior impact depends on surrounding block conte
|
520 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
521 |             bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
522 |             factor: 2.0, | executable/support line; behavior impact depends on surrounding block context.
523 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
524 |             unit: "kPa", | executable/support line; behavior impact depends on surrounding block context.
525 |         }, | executable/support line; behavior impact depends on surrounding block context. |
526 |     ], | executable/support line; behavior impact depends on surrounding block context. |
527 | }, | executable/support line; behavior impact depends on surrounding block context. |
528 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
529 |     pgn: 65270, | executable/support line; behavior impact depends on surrounding block context. |
530 |     name: "IC1", | executable/support line; behavior impact depends on surrounding block context. |
531 |     desc: "Inlet/Exhaust Conditions 1", | executable/support line; behavior impact depends on surrounding
ock context. |
532 |     rate_ms: 500, | executable/support line; behavior impact depends on surrounding block context. |
533 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
534 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
535 |             spn: 102, | executable/support line; behavior impact depends on surrounding block context. |
536 |             name: "Boost Pressure", | executable/support line; behavior impact depends on surrounding blo
context. |
537 |             byte_offset: 0, | executable/support line; behavior impact depends on surrounding block conte
|
538 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
539 |             bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
540 |             factor: 2.0, | executable/support line; behavior impact depends on surrounding block context.
541 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
542 |             unit: "kPa", | executable/support line; behavior impact depends on surrounding block context.
543 |         }, | executable/support line; behavior impact depends on surrounding block context. |
544 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
545 |             spn: 105, | executable/support line; behavior impact depends on surrounding block context. |
546 |             name: "Intake Manifold Temp", | executable/support line; behavior impact depends on surroundi
block context. |
547 |             byte_offset: 1, | executable/support line; behavior impact depends on surrounding block conte
|
548 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
549 |             bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
550 |             factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
551 |             offset: -40.0, | executable/support line; behavior impact depends on surrounding block context
|
552 |             unit: "°C", | executable/support line; behavior impact depends on surrounding block context.
553 |         }, | executable/support line; behavior impact depends on surrounding block context. |
554 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
555 |             spn: 106, | executable/support line; behavior impact depends on surrounding block context. |
556 |             name: "Air Inlet Pressure", | executable/support line; behavior impact depends on surrounding
ock context. |
557 |             byte_offset: 2, | executable/support line; behavior impact depends on surrounding block conte
|
558 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
559 |             bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
560 |             factor: 2.0, | executable/support line; behavior impact depends on surrounding block context.
561 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
562 |             unit: "kPa", | executable/support line; behavior impact depends on surrounding block context.

```



```

| 563 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 564 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 565 |             spn: 107, | executable/support line; behavior impact depends on surrounding block context. |
| 566 |             name: "Air Filter Diff Pressure", | executable/support line; behavior impact depends on surro
ing block context. |
| 567 |             byte_offset: 3, | executable/support line; behavior impact depends on surrounding block conte
|
| 568 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 569 |             bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
| 570 |             factor: 0.5, | executable/support line; behavior impact depends on surrounding block context.
| 571 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 572 |             unit: "kPa", | executable/support line; behavior impact depends on surrounding block context.
| 573 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 574 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 575 |             spn: 173, | executable/support line; behavior impact depends on surrounding block context. |
| 576 |             name: "Exhaust Gas Temperature", | executable/support line; behavior impact depends on surrou
ng block context. |
| 577 |             byte_offset: 4, | executable/support line; behavior impact depends on surrounding block conte
|
| 578 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 579 |             bit_length: 16, | executable/support line; behavior impact depends on surrounding block conte
|
| 580 |             factor: 0.03125, | executable/support line; behavior impact depends on surrounding block cont
. |
| 581 |             offset: -273.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 582 |             unit: "°C", | executable/support line; behavior impact depends on surrounding block context.
| 583 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 584 |     ], | executable/support line; behavior impact depends on surrounding block context. |
| 585 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 586 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 587 |     pgn: 65266, | executable/support line; behavior impact depends on surrounding block context. |
| 588 |     name: "LFE", | executable/support line; behavior impact depends on surrounding block context. |
| 589 |     desc: "Fuel Economy", | executable/support line; behavior impact depends on surrounding block context
| 590 |     rate_ms: 100, | executable/support line; behavior impact depends on surrounding block context. |
| 591 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
| 592 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 593 |             spn: 183, | executable/support line; behavior impact depends on surrounding block context. |
| 594 |             name: "Fuel Rate", | executable/support line; behavior impact depends on surrounding block co
xt. |
| 595 |             byte_offset: 0, | executable/support line; behavior impact depends on surrounding block conte
|
| 596 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 597 |             bit_length: 16, | executable/support line; behavior impact depends on surrounding block conte
|
| 598 |             factor: 0.05, | executable/support line; behavior impact depends on surrounding block context
| 599 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 600 |             unit: "L/h", | executable/support line; behavior impact depends on surrounding block context.
| 601 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 602 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 603 |             spn: 184, | executable/support line; behavior impact depends on surrounding block context. |
| 604 |             name: "Instant Fuel Economy", | executable/support line; behavior impact depends on surroun
block context. |
| 605 |             byte_offset: 2, | executable/support line; behavior impact depends on surrounding block conte
|
| 606 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 607 |             bit_length: 16, | executable/support line; behavior impact depends on surrounding block conte
|
| 608 |             factor: 1.0 / 512.0, | executable/support line; behavior impact depends on surrounding block
text. |

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```

| 609 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 610 |         unit: "km/L", | executable/support line; behavior impact depends on surrounding block context. |
| 611 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 612 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 613 |         spn: 51, | executable/support line; behavior impact depends on surrounding block context. |
| 614 |         name: "Throttle Position", | executable/support line; behavior impact depends on surrounding
ck context. |
| 615 |         byte_offset: 6, | executable/support line; behavior impact depends on surrounding block conte
|
| 616 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 617 |         bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
| 618 |         factor: 0.4, | executable/support line; behavior impact depends on surrounding block context. |
| 619 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 620 |         unit: "%", | executable/support line; behavior impact depends on surrounding block context. |
| 621 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 622 | ], | executable/support line; behavior impact depends on surrounding block context. |
| 623 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 624 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 625 |     pgn: 65253, | executable/support line; behavior impact depends on surrounding block context. |
| 626 |     name: "HOURS", | executable/support line; behavior impact depends on surrounding block context. |
| 627 |     desc: "Engine Hours/Revolutions", | executable/support line; behavior impact depends on surrounding b
k context. |
| 628 |     rate_ms: 1000, | executable/support line; behavior impact depends on surrounding block context. |
| 629 |     spns: &[SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 630 |         spn: 247, | executable/support line; behavior impact depends on surrounding block context. |
| 631 |         name: "Total Engine Hours", | executable/support line; behavior impact depends on surrounding blo
context. |
| 632 |         byte_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 633 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 634 |         bit_length: 32, | executable/support line; behavior impact depends on surrounding block context. |
| 635 |         factor: 0.05, | executable/support line; behavior impact depends on surrounding block context. |
| 636 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 637 |         unit: "h", | executable/support line; behavior impact depends on surrounding block context. |
| 638 |     }], | executable/support line; behavior impact depends on surrounding block context. |
| 639 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 640 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 641 |     pgn: 65265, | executable/support line; behavior impact depends on surrounding block context. |
| 642 |     name: "CCVS", | executable/support line; behavior impact depends on surrounding block context. |
| 643 |     desc: "Cruise Control/Vehicle Speed", | executable/support line; behavior impact depends on surroundi
block context. |
| 644 |     rate_ms: 100, | executable/support line; behavior impact depends on surrounding block context. |
| 645 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
| 646 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 647 |             spn: 70, | executable/support line; behavior impact depends on surrounding block context. |
| 648 |             name: "Parking Brake Switch", | executable/support line; behavior impact depends on surroundi
block context. |
| 649 |             byte_offset: 0, | executable/support line; behavior impact depends on surrounding block conte
|
| 650 |             bit_offset: 2, | executable/support line; behavior impact depends on surrounding block context
|
| 651 |             bit_length: 2, | executable/support line; behavior impact depends on surrounding block context
|
| 652 |             factor: 1.0, | executable/support line; behavior impact depends on surrounding block context. |
| 653 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 654 |             unit: "", | executable/support line; behavior impact depends on surrounding block context. |
| 655 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 656 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 657 |             spn: 84, | executable/support line; behavior impact depends on surrounding block context. |
| 658 |             name: "Wheel-Based Speed", | executable/support line; behavior impact depends on surrounding
ck context. |
| 659 |             byte_offset: 1, | executable/support line; behavior impact depends on surrounding block conte
|
| 660 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context

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|
| 661 |         bit_length: 16, | executable/support line; behavior impact depends on surrounding block conte
|
| 662 |         factor: 1.0 / 256.0, | executable/support line; behavior impact depends on surrounding block
text. |
| 663 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 664 |         unit: "km/h", | executable/support line; behavior impact depends on surrounding block context
| 665 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 666 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 667 |         spn: 85, | executable/support line; behavior impact depends on surrounding block context. |
| 668 |         name: "Cruise Control Active", | executable/support line; behavior impact depends on surround
block context. |
| 669 |         byte_offset: 3, | executable/support line; behavior impact depends on surrounding block conte
|
| 670 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 671 |         bit_length: 2, | executable/support line; behavior impact depends on surrounding block context
|
| 672 |         factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 673 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 674 |         unit: "", | executable/support line; behavior impact depends on surrounding block context. |
| 675 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 676 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 677 |         spn: 86, | executable/support line; behavior impact depends on surrounding block context. |
| 678 |         name: "CC Set Speed", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 679 |         byte_offset: 4, | executable/support line; behavior impact depends on surrounding block conte
|
| 680 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 681 |         bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
| 682 |         factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 683 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 684 |         unit: "km/h", | executable/support line; behavior impact depends on surrounding block context
| 685 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 686 | ], | executable/support line; behavior impact depends on surrounding block context. |
| 687 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 688 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 689 |     pgn: 65093, | executable/support line; behavior impact depends on surrounding block context. |
| 690 |     name: "PTO", | executable/support line; behavior impact depends on surrounding block context. |
| 691 |     desc: "Power Take-Off Info", | executable/support line; behavior impact depends on surrounding block
text. |
| 692 |     rate_ms: 100, | executable/support line; behavior impact depends on surrounding block context. |
| 693 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
| 694 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 695 |             spn: 1691, | executable/support line; behavior impact depends on surrounding block context. |
| 696 |             name: "Rear PTO State", | executable/support line; behavior impact depends on surrounding blo
context. |
| 697 |             byte_offset: 0, | executable/support line; behavior impact depends on surrounding block conte
|
| 698 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 699 |             bit_length: 5, | executable/support line; behavior impact depends on surrounding block context
|
| 700 |             factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 701 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 702 |             unit: "", | executable/support line; behavior impact depends on surrounding block context. |
| 703 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 704 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 705 |             spn: 900, | executable/support line; behavior impact depends on surrounding block context. |
| 706 |             name: "PTO Engagement Control", | executable/support line; behavior impact depends on surround
g block context. |
| 707 |             byte_offset: 1, | executable/support line; behavior impact depends on surrounding block conte
|

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708 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
709 |         bit_length: 2, | executable/support line; behavior impact depends on surrounding block context. |
710 |         factor: 1.0, | executable/support line; behavior impact depends on surrounding block context. |
711 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
712 |         unit: "", | executable/support line; behavior impact depends on surrounding block context. |
713 |     }, | executable/support line; behavior impact depends on surrounding block context. |
714 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
715 |         spn: 1693, | executable/support line; behavior impact depends on surrounding block context. |
716 |         name: "Rear PTO Output RPM", | executable/support line; behavior impact depends on surrounding block context. |
717 |         byte_offset: 2, | executable/support line; behavior impact depends on surrounding block context. |
718 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
719 |         bit_length: 16, | executable/support line; behavior impact depends on surrounding block context. |
720 |         factor: 0.125, | executable/support line; behavior impact depends on surrounding block context. |
721 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
722 |         unit: "rpm", | executable/support line; behavior impact depends on surrounding block context. |
723 |     }, | executable/support line; behavior impact depends on surrounding block context. |
724 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
725 |         spn: 1694, | executable/support line; behavior impact depends on surrounding block context. |
726 |         name: "Front PTO Output RPM", | executable/support line; behavior impact depends on surrounding block context. |
727 |         byte_offset: 4, | executable/support line; behavior impact depends on surrounding block context. |
728 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
729 |         bit_length: 16, | executable/support line; behavior impact depends on surrounding block context. |
730 |         factor: 0.125, | executable/support line; behavior impact depends on surrounding block context. |
731 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
732 |         unit: "rpm", | executable/support line; behavior impact depends on surrounding block context. |
733 |     }, | executable/support line; behavior impact depends on surrounding block context. |
734 | ], | executable/support line; behavior impact depends on surrounding block context. |
735 | }, | executable/support line; behavior impact depends on surrounding block context. |
736 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
737 |     pgn: 65091, | executable/support line; behavior impact depends on surrounding block context. |
738 |     name: "HITCH", | executable/support line; behavior impact depends on surrounding block context. |
739 |     desc: "Hitch / Hydraulic Status", | executable/support line; behavior impact depends on surrounding block context. |
740 |     rate_ms: 100, | executable/support line; behavior impact depends on surrounding block context. |
741 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
742 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
743 |             spn: 50001, | executable/support line; behavior impact depends on surrounding block context. |
744 |             name: "Hydraulic System Pressure", | executable/support line; behavior impact depends on surrounding block context. |
745 |             byte_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
746 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
747 |             bit_length: 8, | executable/support line; behavior impact depends on surrounding block context. |
748 |             factor: 2.0, | executable/support line; behavior impact depends on surrounding block context. |
749 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
750 |             unit: "bar", | executable/support line; behavior impact depends on surrounding block context. |
751 |         }, | executable/support line; behavior impact depends on surrounding block context. |
752 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
753 |             spn: 50002, | executable/support line; behavior impact depends on surrounding block context. |
754 |             name: "Hydraulic Fluid Temp", | executable/support line; behavior impact depends on surrounding block context. |

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755 |         byte_offset: 1, | executable/support line; behavior impact depends on surrounding block context. |
756 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
757 |         bit_length: 8, | executable/support line; behavior impact depends on surrounding block context. |
758 |         factor: 1.0, | executable/support line; behavior impact depends on surrounding block context. |
759 |         offset: -40.0, | executable/support line; behavior impact depends on surrounding block context. |
760 |         unit: "C", | executable/support line; behavior impact depends on surrounding block context. |
761 |     }, | executable/support line; behavior impact depends on surrounding block context. |
762 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
763 |         spn: 50003, | executable/support line; behavior impact depends on surrounding block context. |
764 |         name: "Pump Flow", | executable/support line; behavior impact depends on surrounding block context. |
765 |         byte_offset: 2, | executable/support line; behavior impact depends on surrounding block context. |
766 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
767 |         bit_length: 8, | executable/support line; behavior impact depends on surrounding block context. |
768 |         factor: 2.0, | executable/support line; behavior impact depends on surrounding block context. |
769 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
770 |         unit: "L/min", | executable/support line; behavior impact depends on surrounding block context. |
771 |     }, | executable/support line; behavior impact depends on surrounding block context. |
772 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
773 |         spn: 50004, | executable/support line; behavior impact depends on surrounding block context. |
774 |         name: "Hydraulic Alarm Flags", | executable/support line; behavior impact depends on surrounding block context. |
775 |         byte_offset: 3, | executable/support line; behavior impact depends on surrounding block context. |
776 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
777 |         bit_length: 8, | executable/support line; behavior impact depends on surrounding block context. |
778 |         factor: 1.0, | executable/support line; behavior impact depends on surrounding block context. |
779 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
780 |         unit: "bits", | executable/support line; behavior impact depends on surrounding block context. |
781 |     }, | executable/support line; behavior impact depends on surrounding block context. |
782 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
783 |         spn: 50005, | executable/support line; behavior impact depends on surrounding block context. |
784 |         name: "Hitch Position", | executable/support line; behavior impact depends on surrounding block context. |
785 |         byte_offset: 4, | executable/support line; behavior impact depends on surrounding block context. |
786 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
787 |         bit_length: 8, | executable/support line; behavior impact depends on surrounding block context. |
788 |         factor: 100.0 / 255.0, | executable/support line; behavior impact depends on surrounding block context. |
789 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
790 |         unit: "%", | executable/support line; behavior impact depends on surrounding block context. |
791 |     }, | executable/support line; behavior impact depends on surrounding block context. |
792 | ], | executable/support line; behavior impact depends on surrounding block context. |
793 | }, | executable/support line; behavior impact depends on surrounding block context. |
794 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
795 |     pgn: 65269, | executable/support line; behavior impact depends on surrounding block context. |
796 |     name: "AMB", | executable/support line; behavior impact depends on surrounding block context. |
797 |     desc: "Ambient Conditions", | executable/support line; behavior impact depends on surrounding block context. |
798 |     rate_ms: 1000, | executable/support line; behavior impact depends on surrounding block context. |
799 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
800 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |

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| 801 |           spn: 171, | executable/support line; behavior impact depends on surrounding block context. |
| 802 |           name: "Ambient Air Temp", | executable/support line; behavior impact depends on surrounding b
k context. |
| 803 |           byte_offset: 0, | executable/support line; behavior impact depends on surrounding block conte
|
| 804 |           bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 805 |           bit_length: 16, | executable/support line; behavior impact depends on surrounding block conte
|
| 806 |           factor: 0.03125, | executable/support line; behavior impact depends on surrounding block cont
. |
| 807 |           offset: -273.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 808 |           unit: "C", | executable/support line; behavior impact depends on surrounding block context. |
| 809 |       }, | executable/support line; behavior impact depends on surrounding block context. |
| 810 |       SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 811 |           spn: 108, | executable/support line; behavior impact depends on surrounding block context. |
| 812 |           name: "Barometric Pressure", | executable/support line; behavior impact depends on surroundin
lock context. |
| 813 |           byte_offset: 2, | executable/support line; behavior impact depends on surrounding block conte
|
| 814 |           bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 815 |           bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
| 816 |           factor: 0.5, | executable/support line; behavior impact depends on surrounding block context.
| 817 |           offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 818 |           unit: "kPa", | executable/support line; behavior impact depends on surrounding block context.
| 819 |       }, | executable/support line; behavior impact depends on surrounding block context. |
| 820 |       SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 821 |           spn: 174, | executable/support line; behavior impact depends on surrounding block context. |
| 822 |           name: "Cabin / Inlet Temp", | executable/support line; behavior impact depends on surrounding
ock context. |
| 823 |           byte_offset: 3, | executable/support line; behavior impact depends on surrounding block conte
|
| 824 |           bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 825 |           bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
| 826 |           factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 827 |           offset: -40.0, | executable/support line; behavior impact depends on surrounding block context
|
| 828 |           unit: "C", | executable/support line; behavior impact depends on surrounding block context. |
| 829 |       }, | executable/support line; behavior impact depends on surrounding block context. |
| 830 |   ], | executable/support line; behavior impact depends on surrounding block context. |
| 831 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 832 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 833 |     pgn: 65271, | executable/support line; behavior impact depends on surrounding block context. |
| 834 |     name: "VEP1", | executable/support line; behavior impact depends on surrounding block context. |
| 835 |     desc: "Vehicle Electrical Power 1", | executable/support line; behavior impact depends on surrounding
ock context. |
| 836 |     rate_ms: 500, | executable/support line; behavior impact depends on surrounding block context. |
| 837 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
| 838 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 839 |             spn: 168, | executable/support line; behavior impact depends on surrounding block context. |
| 840 |             name: "Battery Potential", | executable/support line; behavior impact depends on surrounding
ck context. |
| 841 |             byte_offset: 0, | executable/support line; behavior impact depends on surrounding block conte
|
| 842 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 843 |             bit_length: 16, | executable/support line; behavior impact depends on surrounding block conte
|
| 844 |             factor: 0.05, | executable/support line; behavior impact depends on surrounding block context
| 845 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.

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| 846 |         unit: "V", | executable/support line; behavior impact depends on surrounding block context. |
| 847 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 848 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 849 |         spn: 158, | executable/support line; behavior impact depends on surrounding block context. |
| 850 |         name: "Alternator Current", | executable/support line; behavior impact depends on surrounding
ock context. |
| 851 |         byte_offset: 2, | executable/support line; behavior impact depends on surrounding block conte
|
| 852 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 853 |         bit_length: 16, | executable/support line; behavior impact depends on surrounding block conte
|
| 854 |         factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 855 |         offset: -32000.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 856 |         unit: "A", | executable/support line; behavior impact depends on surrounding block context. |
| 857 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 858 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 859 |         spn: 167, | executable/support line; behavior impact depends on surrounding block context. |
| 860 |         name: "Charging Voltage", | executable/support line; behavior impact depends on surrounding b
k context. |
| 861 |         byte_offset: 4, | executable/support line; behavior impact depends on surrounding block conte
|
| 862 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 863 |         bit_length: 16, | executable/support line; behavior impact depends on surrounding block conte
|
| 864 |         factor: 0.05, | executable/support line; behavior impact depends on surrounding block context
| 865 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 866 |         unit: "V", | executable/support line; behavior impact depends on surrounding block context. |
| 867 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 868 |     ], | executable/support line; behavior impact depends on surrounding block context. |
| 869 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 870 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 871 |     pgn: 65280, | executable/support line; behavior impact depends on surrounding block context. |
| 872 |     name: "ADAS1", | executable/support line; behavior impact depends on surrounding block context. |
| 873 |     desc: "ADAS status and interventions", | executable/support line; behavior impact depends on surround
block context. |
| 874 |     rate_ms: 100, | executable/support line; behavior impact depends on surrounding block context. |
| 875 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
| 876 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 877 |             spn: 52001, | executable/support line; behavior impact depends on surrounding block context.
| 878 |             name: "LKA Active", | executable/support line; behavior impact depends on surrounding block c
ext. |
| 879 |             byte_offset: 0, | executable/support line; behavior impact depends on surrounding block conte
|
| 880 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 881 |             bit_length: 2, | executable/support line; behavior impact depends on surrounding block context
|
| 882 |             factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 883 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 884 |             unit: "", | executable/support line; behavior impact depends on surrounding block context. |
| 885 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 886 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 887 |             spn: 52002, | executable/support line; behavior impact depends on surrounding block context.
| 888 |             name: "ACC Active", | executable/support line; behavior impact depends on surrounding block c
ext. |
| 889 |             byte_offset: 0, | executable/support line; behavior impact depends on surrounding block conte
|
| 890 |             bit_offset: 2, | executable/support line; behavior impact depends on surrounding block context
|
| 891 |             bit_length: 2, | executable/support line; behavior impact depends on surrounding block context
|
| 892 |             factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.

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| 893 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 894 |         unit: "", | executable/support line; behavior impact depends on surrounding block context. |
| 895 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 896 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 897 |         spn: 52003, | executable/support line; behavior impact depends on surrounding block context.
| 898 |         name: "AEB Active", | executable/support line; behavior impact depends on surrounding block c
ext. |
| 899 |         byte_offset: 0, | executable/support line; behavior impact depends on surrounding block conte
|
| 900 |         bit_offset: 4, | executable/support line; behavior impact depends on surrounding block context
|
| 901 |         bit_length: 2, | executable/support line; behavior impact depends on surrounding block context
|
| 902 |         factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 903 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 904 |         unit: "", | executable/support line; behavior impact depends on surrounding block context. |
| 905 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 906 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 907 |         spn: 52004, | executable/support line; behavior impact depends on surrounding block context.
| 908 |         name: "Lead Distance", | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 909 |         byte_offset: 1, | executable/support line; behavior impact depends on surrounding block conte
|
| 910 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 911 |         bit_length: 16, | executable/support line; behavior impact depends on surrounding block conte
|
| 912 |         factor: 0.1, | executable/support line; behavior impact depends on surrounding block context.
| 913 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 914 |         unit: "m", | executable/support line; behavior impact depends on surrounding block context. |
| 915 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 916 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 917 |         spn: 52005, | executable/support line; behavior impact depends on surrounding block context.
| 918 |         name: "TTC", | executable/support line; behavior impact depends on surrounding block context.
| 919 |         byte_offset: 3, | executable/support line; behavior impact depends on surrounding block conte
|
| 920 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 921 |         bit_length: 16, | executable/support line; behavior impact depends on surrounding block conte
|
| 922 |         factor: 0.01, | executable/support line; behavior impact depends on surrounding block context
| 923 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 924 |         unit: "s", | executable/support line; behavior impact depends on surrounding block context. |
| 925 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 926 |     SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 927 |         spn: 52006, | executable/support line; behavior impact depends on surrounding block context.
| 928 |         name: "Speed Limit", | executable/support line; behavior impact depends on surrounding block
text. |
| 929 |         byte_offset: 5, | executable/support line; behavior impact depends on surrounding block conte
|
| 930 |         bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context
|
| 931 |         bit_length: 8, | executable/support line; behavior impact depends on surrounding block context
|
| 932 |         factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 933 |         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 934 |         unit: "km/h", | executable/support line; behavior impact depends on surrounding block context
| 935 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 936 | ], | executable/support line; behavior impact depends on surrounding block context. |
| 937 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 938 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 939 |     pgn: 65281, | executable/support line; behavior impact depends on surrounding block context. |
| 940 |     name: "FUS1", | executable/support line; behavior impact depends on surrounding block context. |
| 941 |     desc: "Sensor fusion hazard status", | executable/support line; behavior impact depends on surroundin
lock context. |

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| 942 |         rate_ms: 100, | executable/support line; behavior impact depends on surrounding block context. |
| 943 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
| 944 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 945 |             spn: 52101, | executable/support line; behavior impact depends on surrounding block context. |
| 946 |             name: "Fused Objects", | executable/support line; behavior impact depends on surrounding block
context. |
| 947 |             byte_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 948 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 949 |             bit_length: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 950 |             factor: 1.0, | executable/support line; behavior impact depends on surrounding block context. |
| 951 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 952 |             unit: "count", | executable/support line; behavior impact depends on surrounding block context. |
| 953 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 954 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 955 |             spn: 52102, | executable/support line; behavior impact depends on surrounding block context. |
| 956 |             name: "Critical TTC", | executable/support line; behavior impact depends on surrounding block
context. |
| 957 |             byte_offset: 1, | executable/support line; behavior impact depends on surrounding block context. |
| 958 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 959 |             bit_length: 16, | executable/support line; behavior impact depends on surrounding block context. |
| 960 |             factor: 0.01, | executable/support line; behavior impact depends on surrounding block context. |
| 961 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 962 |             unit: "s", | executable/support line; behavior impact depends on surrounding block context. |
| 963 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 964 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 965 |             spn: 52103, | executable/support line; behavior impact depends on surrounding block context. |
| 966 |             name: "Fusion Confidence", | executable/support line; behavior impact depends on surrounding
block context. |
| 967 |             byte_offset: 3, | executable/support line; behavior impact depends on surrounding block context. |
| 968 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 969 |             bit_length: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 970 |             factor: 0.4, | executable/support line; behavior impact depends on surrounding block context. |
| 971 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 972 |             unit: "%", | executable/support line; behavior impact depends on surrounding block context. |
| 973 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 974 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 975 |             spn: 52104, | executable/support line; behavior impact depends on surrounding block context. |
| 976 |             name: "In-Path Hazard", | executable/support line; behavior impact depends on surrounding block
context. |
| 977 |             byte_offset: 4, | executable/support line; behavior impact depends on surrounding block context. |
| 978 |             bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 979 |             bit_length: 1, | executable/support line; behavior impact depends on surrounding block context. |
| 980 |             factor: 1.0, | executable/support line; behavior impact depends on surrounding block context. |
| 981 |             offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 982 |             unit: "", | executable/support line; behavior impact depends on surrounding block context. |
| 983 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 984 |     ], | executable/support line; behavior impact depends on surrounding block context. |
| 985 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 986 | PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 987 |     pgn: 65282, | executable/support line; behavior impact depends on surrounding block context. |
| 988 |     name: "ENGY1", | executable/support line; behavior impact depends on surrounding block context. |
| 989 |     desc: "Vehicle energy balance", | executable/support line; behavior impact depends on surrounding block context. |

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context. |
| 990 |         rate_ms: 100, | executable/support line; behavior impact depends on surrounding block context. |
| 991 |         spns: &[ | executable/support line; behavior impact depends on surrounding block context. |
| 992 |             SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 993 |                 spn: 52201, | executable/support line; behavior impact depends on surrounding block context.
| 994 |                 name: "Battery Voltage", | executable/support line; behavior impact depends on surrounding block
context. |
| 995 |                 byte_offset: 0, | executable/support line; behavior impact depends on surrounding block context.
|
| 996 |                 bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context.
|
| 997 |                 bit_length: 16, | executable/support line; behavior impact depends on surrounding block context.
|
| 998 |                 factor: 0.05, | executable/support line; behavior impact depends on surrounding block context.
| 999 |                 offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 1000 |                 unit: "V", | executable/support line; behavior impact depends on surrounding block context.
| 1001 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 1002 |             SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 1003 |                 spn: 52202, | executable/support line; behavior impact depends on surrounding block context.
| 1004 |                 name: "Alternator Current", | executable/support line; behavior impact depends on surrounding block
lock context. |
| 1005 |                 byte_offset: 2, | executable/support line; behavior impact depends on surrounding block context.
|
| 1006 |                 bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context.
|
| 1007 |                 bit_length: 16, | executable/support line; behavior impact depends on surrounding block context.
|
| 1008 |                 factor: 1.0, | executable/support line; behavior impact depends on surrounding block context.
| 1009 |                 offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 1010 |                 unit: "A", | executable/support line; behavior impact depends on surrounding block context.
| 1011 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 1012 |             SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 1013 |                 spn: 52203, | executable/support line; behavior impact depends on surrounding block context.
| 1014 |                 name: "Total Electrical Load", | executable/support line; behavior impact depends on surrounding block
g block context. |
| 1015 |                 byte_offset: 4, | executable/support line; behavior impact depends on surrounding block context.
|
| 1016 |                 bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context.
|
| 1017 |                 bit_length: 16, | executable/support line; behavior impact depends on surrounding block context.
|
| 1018 |                 factor: 0.01, | executable/support line; behavior impact depends on surrounding block context.
|
| 1019 |                 offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 1020 |                 unit: "kW", | executable/support line; behavior impact depends on surrounding block context.
| 1021 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 1022 |             SpnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 1023 |                 spn: 52204, | executable/support line; behavior impact depends on surrounding block context.
| 1024 |                 name: "Hydraulic Power", | executable/support line; behavior impact depends on surrounding block
k context. |
| 1025 |                 byte_offset: 6, | executable/support line; behavior impact depends on surrounding block context.
|
| 1026 |                 bit_offset: 0, | executable/support line; behavior impact depends on surrounding block context.
|
| 1027 |                 bit_length: 16, | executable/support line; behavior impact depends on surrounding block context.
|
| 1028 |                 factor: 0.01, | executable/support line; behavior impact depends on surrounding block context.
|
| 1029 |                 offset: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 1030 |                 unit: "kW", | executable/support line; behavior impact depends on surrounding block context.
| 1031 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 1032 |         ], | executable/support line; behavior impact depends on surrounding block context. |
| 1033 |     }, | executable/support line; behavior impact depends on surrounding block context. |
| 1034 |     PgnDef { | executable/support line; behavior impact depends on surrounding block context. |
| 1035 |         pgn: 65226, | executable/support line; behavior impact depends on surrounding block context. |

```

[illegible]

```
| 1084 |         name: "FUEL1",| executable/support line; behavior impact depends on surrounding block context. |  
| 1085 | desc: "Fuel Consumption (cumulative)", | executable/support line; behavior impact depends on surround  
g block context. |  
| 1086 |     rate_ms: 1000, | executable/support line; behavior impact depends on surrounding block context. |  
| 1087 |     spns: &[ | executable/support line; behavior impact depends on surrounding block context. |  
| 1088 |         SpnDef { | executable/support line; behavior impact depends on surrounding block context. |  
| 1089 |             spn: 182, | executable/support line; behavior impact depends on surrounding block context. |  
| 1090 |             name: "Trip Fuel", | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 1091 |                 byte_offset: 0, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 1092 |                 bit_offset: 0, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 1093 |                 bit_length: 32, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 1094 |                     factor: 0.5, | executable/support line; behavior impact depends on surrounding block context  
| 1095 |                     offset: 0.0, | executable/support line; behavior impact depends on surrounding block context  
| 1096 |                     unit: "L", | executable/support line; behavior impact depends on surrounding block context.  
| 1097 |                 }, | executable/support line; behavior impact depends on surrounding block context. |  
| 1098 |             SpnDef { | executable/support line; behavior impact depends on surrounding block context. |  
| 1099 |                 spn: 250, | executable/support line; behavior impact depends on surrounding block context. |  
| 1100 |                 name: "Total Fuel Used", | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 1101 |                     byte_offset: 4, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 1102 |                     bit_offset: 0, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 1103 |                     bit_length: 32, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 1104 |                         factor: 0.5, | executable/support line; behavior impact depends on surrounding block context  
| 1105 |                         offset: 0.0, | executable/support line; behavior impact depends on surrounding block context  
| 1106 |                         unit: "L", | executable/support line; behavior impact depends on surrounding block context.  
| 1107 |                     }, | executable/support line; behavior impact depends on surrounding block context. |  
| 1108 |                 ], | executable/support line; behavior impact depends on surrounding block context. |  
| 1109 |             }, | executable/support line; behavior impact depends on surrounding block context. |  
| 1110 | ]; | executable/support line; behavior impact depends on surrounding block context. |  
| 1111 | (blank) | line is blank; visual separation only. |  
| 1112 | pub fn find_pgn(pgn: u32) -> Option<&'static PgnDef> { | declares executable behavior; signature/body change  
lter API, control flow, and outputs. |  
| 1113 |     REGISTRY.iter().find(|p| p.pgn == pgnd)| executable/support line; behavior impact depends on surroundi  
block context. |  
| 1114 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 1115 | (blank) | line is blank; visual separation only. |  
| 1116 | // ■■■ J1939 Frame ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ | comment l  
ntent, no runtime effect. |  
| 1117 | (blank) | line is blank; visual separation only. |  
| 1118 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |  
| 1119 | pub struct J1939Frame { | declares data layout; field changes may break serialization, tests, and callers. |  
| 1120 |     pub timestamp: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 1121 |     pub raw_id: u32, | executable/support line; behavior impact depends on surrounding block context. |  
| 1122 |     pub priority: u8, | executable/support line; behavior impact depends on surrounding block context. |  
| 1123 |     pub pgn: u32, | executable/support line; behavior impact depends on surrounding block context. |  
| 1124 |     pub sa: u8, | executable/support line; behavior impact depends on surrounding block context. |  
| 1125 |     pub da: u8, // 0xFF = broadcast | executable/support line; behavior impact depends on surrounding block  
text. |  
| 1126 |         pub data: [u8; 8], | executable/support line; behavior impact depends on surrounding block context. |  
| 1127 |         pub dlc: u8, | executable/support line; behavior impact depends on surrounding block context. |  
| 1128 |         pub decoded: Vec<SpnValue>, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 1129 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 1130 | (blank) | line is blank; visual separation only. |  
| 1131 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |  
| 1132 | pub struct SpnValue { | declares data layout; field changes may break serialization, tests, and callers. |  
| 1133 |     pub spn: u32, | executable/support line; behavior impact depends on surrounding block context. |  
| 1134 |     pub name: &'static str, | executable/support line; behavior impact depends on surrounding block context.
```

```

| 1135 |     pub physical: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 1136 |     pub unit: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 1137 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1138 | (blank) | line is blank; visual separation only. |
| 1139 | impl J1939Frame { | implementation block; method behavior and trait conformance are defined here. |
| 1140 |     pub fn from_raw(ts: f64, raw_id: u32, data: &[u8]) -> Self { | declares executable behavior; signature/b
changes alter API, control flow, and outputs. |
| 1141 |         let priority = ((raw_id >> 26) & 0x7) as u8; | executable/support line; behavior impact depends on s
urrounding block context. |
| 1142 |         let dp = (raw_id >> 24) & 0x1; | executable/support line; behavior impact depends on surrounding blo
context. |
| 1143 |         let pf = ((raw_id >> 16) & 0xFF) as u8; | executable/support line; behavior impact depends on surrou
ng block context. |
| 1144 |         let ps = ((raw_id >> 8) & 0xFF) as u8; | executable/support line; behavior impact depends on surroun
g block context. |
| 1145 |         let sa = (raw_id & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding bl
context. |
| 1146 | (blank) | line is blank; visual separation only. |
| 1147 |         let (pgn, da) = if pf < 0xF0 { | executable/support line; behavior impact depends on surrounding blo
context. |
| 1148 |             (((dp << 17) \ | ((pf as u32) << 8)), ps) | executable/support line; behavior impact depends on s
urrounding block context. |
| 1149 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 1150 |             (((dp << 17) \ | ((pf as u32) << 8) \ | (ps as u32)), 0xFF) | executable/support line; behavior im
t depends on surrounding block context. |
| 1151 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime sco
. |
| 1152 | (blank) | line is blank; visual separation only. |
| 1153 |         let mut arr = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1154 |         let len = data.len().min(8); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1155 |         arr[..len].copy_from_slice(&data[..len]); | executable/support line; behavior impact depends on surr
ding block context. |
| 1156 | (blank) | line is blank; visual separation only. |
| 1157 |         let mut f = J1939Frame { | executable/support line; behavior impact depends on surrounding block con
t. |
| 1158 |             timestamp: ts, | executable/support line; behavior impact depends on surrounding block context. |
| 1159 |             raw_id, | executable/support line; behavior impact depends on surrounding block context. |
| 1160 |             priority, | executable/support line; behavior impact depends on surrounding block context. |
| 1161 |             pgn, | executable/support line; behavior impact depends on surrounding block context. |
| 1162 |             sa, | executable/support line; behavior impact depends on surrounding block context. |
| 1163 |             da, | executable/support line; behavior impact depends on surrounding block context. |
| 1164 |             data: arr, | executable/support line; behavior impact depends on surrounding block context. |
| 1165 |             dlc: len as u8, | executable/support line; behavior impact depends on surrounding block context. |
| 1166 |             decoded: Vec::new(), | executable/support line; behavior impact depends on surrounding block con
t. |
| 1167 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime sco
. |
| 1168 |         f.decode(); | executable/support line; behavior impact depends on surrounding block context. |
| 1169 |         f | executable/support line; behavior impact depends on surrounding block context. |
| 1170 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1171 | (blank) | line is blank; visual separation only. |
| 1172 |     pub fn build_id(priority: u8, pgn: u32, sa: u8, da: u8) -> u32 { | declares executable behavior; signatu
body changes alter API, control flow, and outputs. |
| 1173 |         let dp = (pgn >> 17) & 0x1; | executable/support line; behavior impact depends on surrounding block
text. |
| 1174 |         let pf = (pgn >> 8) & 0xFF; | executable/support line; behavior impact depends on surrounding block
text. |
| 1175 |         let ps = if pf < 0xF0 { da as u32 } else { pgn & 0xFF }; | executable/support line; behavior impact
ends on surrounding block context. |
| 1176 |         ((priority as u32 & 7) << 26) \ | (dp << 24) \ | (pf << 16) \ | (ps << 8) \ | (sa as u32) | executable/s
ort line; behavior impact depends on surrounding block context. |
| 1177 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1178 | (blank) | line is blank; visual separation only. |

```

```

| 1179 |     fn decode(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and
outputs. |
| 1180 |         let def = match find_pgn(self.pgn) { | executable/support line; behavior impact depends on surrounding
block context. |
| 1181 |             Some(d) => d, | executable/support line; behavior impact depends on surrounding block context. |
| 1182 |             None => return, | executable/support line; behavior impact depends on surrounding block context. |
| 1183 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
. |
| 1184 |         for spn in def.spns { | iteration control; changing bounds/body affects throughput, timing, and possib
y safety guarantees. |
| 1185 |             let end_byte = | executable/support line; behavior impact depends on surrounding block context. |
| 1186 |                 spn.byte_offset + (spn.bit_offset as usize + spn.bit_length as usize).div_ceil(8); | executab
/support line; behavior impact depends on surrounding block context. |
| 1187 |             if end_byte > self.dlc as usize { | conditional gate; changing condition shifts runtime path and s
de effects. |
| 1188 |                 continue; | executable/support line; behavior impact depends on surrounding block context. |
| 1189 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime sc
pes. |
| 1190 |             let raw = bits(&self.data, spn.byte_offset, spn.bit_offset, spn.bit_length); | executable/support
ine; behavior impact depends on surrounding block context. |
| 1191 |             if raw == (1u64 << spn.bit_length) - 1 { | conditional gate; changing condition shifts runtime p
and side effects. |
| 1192 |                 continue; | executable/support line; behavior impact depends on surrounding block context. |
| 1193 |             } // all-ones = not available | executable/support line; behavior impact depends on surrounding
ck context. |
| 1194 |             self.decoded.push(SpnValue { | executable/support line; behavior impact depends on surrounding b
k context. |
| 1195 |                 spn: spn.spn, | executable/support line; behavior impact depends on surrounding block context
|
| 1196 |                 name: spn.name, | executable/support line; behavior impact depends on surrounding block cont
. |
| 1197 |                 physical: raw as f64 * spn.factor + spn.offset, | executable/support line; behavior impact d
nds on surrounding block context. |
| 1198 |                 unit: spn.unit, | executable/support line; behavior impact depends on surrounding block cont
. |
| 1199 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 1200 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1201 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1202 | (blank) | line is blank; visual separation only. |
| 1203 |     pub fn pgn_name(&self) -> &'static str { | declares executable behavior; signature/body changes alter AP
control flow, and outputs. |
| 1204 |         find_pgn(self.pgn).map(|p| p.name).unwrap_or("???") | error-propagating operation; changing it imp
s failure semantics and caller recovery paths. |
| 1205 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1206 | (blank) | line is blank; visual separation only. |
| 1207 |     pub fn sa_name(&self) -> &'static str { | declares executable behavior; signature/body changes alter API
ontrol flow, and outputs. |
| 1208 |         match self.sa { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 1209 |             0x00 => "ECM1 ", | executable/support line; behavior impact depends on surrounding block context
| 1210 |             0x01 => "ECM2 ", | executable/support line; behavior impact depends on surrounding block context
| 1211 |             0x02 => "TURBO", | executable/support line; behavior impact depends on surrounding block context
| 1212 |             0x03 => "TRANS", | executable/support line; behavior impact depends on surrounding block context
| 1213 |             0x0B => "ABS  ", | executable/support line; behavior impact depends on surrounding block context
| 1214 |             0x0D => "TC   ", | executable/support line; behavior impact depends on surrounding block context
| 1215 |             0x1C => "DASH ", | executable/support line; behavior impact depends on surrounding block context
| 1216 |             0x25 => "IMPL ", | executable/support line; behavior impact depends on surrounding block context
| 1217 |             0x26 => "VT   ", | executable/support line; behavior impact depends on surrounding block context
| 1218 |             0x27 => "CAB  ", | executable/support line; behavior impact depends on surrounding block context
| 1219 |             0x28 => "ARST ", | executable/support line; behavior impact depends on surrounding block context
| 1220 |             0x1E => "HITCH", | executable/support line; behavior impact depends on surrounding block context
| 1221 |             0xFE => "NULL ", | executable/support line; behavior impact depends on surrounding block context
| 1222 |             0xFF => "BCAST", | executable/support line; behavior impact depends on surrounding block context
| 1223 |             _ => "UNK  ", | executable/support line; behavior impact depends on surrounding block context. |
| 1224 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop

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```
| 1225 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 1226 | (blank) | line is blank; visual separation only. |  
| 1227 |     pub fn data_hex(&self) -> String { | declares executable behavior; signature/body changes alter API, control  
flow, and outputs. |  
| 1228 |         self.data[..self.dlc as usize] | executable/support line; behavior impact depends on surrounding block  
context. |  
| 1229 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |  
| 1230 |             .map(|b| format!("{:02X}", b)) | executable/support line; behavior impact depends on surrounding  
block context. |  
| 1231 |                 .collect:::<Vec<>>>() | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 1232 |                     .join(" ") | executable/support line; behavior impact depends on surrounding block context. |  
| 1233 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 1234 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 1235 | (blank) | line is blank; visual separation only. |  
| 1236 | fn bits(data: &[u8], byte_off: usize, bit_off: u8, len: u8) -> u64 { | declares executable behavior; signatu  
body changes alter API, control flow, and outputs. |  
| 1237 |     let mut v = 0u64; | executable/support line; behavior impact depends on surrounding block context. |  
| 1238 |     for i in 0..len as usize { | iteration control; changing bounds/body affects throughput, timing, and pos  
ly safety guarantees. |  
| 1239 |         let abs = byte_off * 8 + bit_off as usize + i; | executable/support line; behavior impact depends on  
rounding block context. |  
| 1240 |         let b = abs / 8; | executable/support line; behavior impact depends on surrounding block context. |  
| 1241 |         let k = abs % 8; | executable/support line; behavior impact depends on surrounding block context. |  
| 1242 |         if b < data.len() && (data[b] >> k) & 1 == 1 { | conditional gate; changing condition shifts runtime  
th and side effects. |  
| 1243 |             v |= 1 << i; | executable/support line; behavior impact depends on surrounding block context. |  
| 1244 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 1245 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 1246 |     v | executable/support line; behavior impact depends on surrounding block context. |  
| 1247 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 1248 | (blank) | line is blank; visual separation only. |  
| 1249 | // ■ J1939 Frame Builders ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ | comment line  
nment, no runtime effect. |  
| 1250 | (blank) | line is blank; visual separation only. |  
| 1251 | pub struct Builder; | declares data layout; field changes may break serialization, tests, and callers. |  
| 1252 | impl Builder { | implementation block; method behavior and trait conformance are defined here. |  
| 1253 |     pub fn eec1(ts: f64, rpm: f64, actual_pct: f64, demand_pct: f64, sa: u8) -> J1939Frame { | declares execu  
ble behavior; signature/body changes alter API, control flow, and outputs. |  
| 1254 |         let mut d = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 1255 |         d[0] = 0xF0; | executable/support line; behavior impact depends on surrounding block context. |  
| 1256 |         d[1] = (demand_pct + 125.0).clamp(0.0, 250.0) as u8; | executable/support line; behavior impact depende  
on surrounding block context. |  
| 1257 |         d[2] = (actual_pct + 125.0).clamp(0.0, 250.0) as u8; | executable/support line; behavior impact depende  
on surrounding block context. |  
| 1258 |         let s = (rpm / 0.125) as u16; | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 1259 |         d[3] = (s & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 1260 |         d[4] = (s >> 8) as u8; | executable/support line; behavior impact depends on surrounding block conte  
|  
| 1261 |         d[5] = sa; | executable/support line; behavior impact depends on surrounding block context. |  
| 1262 |         d[6] = 0xF0; | executable/support line; behavior impact depends on surrounding block context. |  
| 1263 |         J1939Frame::from_raw(ts, J1939Frame::build_id(3, pgn::EEC1, sa, 0xFF), &d) | executable/support line  
ehavior impact depends on surrounding block context. |  
| 1264 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 1265 |     pub fn eec2(ts: f64, throttle: f64, load: f64, sa: u8) -> J1939Frame { | declares executable behavior; si  
ature/body changes alter API, control flow, and outputs. |  
| 1266 |         let mut d = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 1267 |         d[1] = (throttle / 0.4).clamp(0.0, 250.0) as u8; | executable/support line; behavior impact depends  
surrounding block context. |
```

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| 1268 |         d[2] = (load / 0.4).clamp(0.0, 250.0) as u8; | executable/support line; behavior impact depends on s
ounding block context. |
| 1269 |         J1939Frame::from_raw(ts, J1939Frame::build_id(3, pgn::EEC2, sa, 0xFF), &d) | executable/support line
ehavior impact depends on surrounding block context. |
| 1270 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1271 |     pub fn et1(ts: f64, coolant: f64, fuel_t: f64, oil_t: f64, sa: u8) -> J1939Frame { | declares executable
havior; signature/body changes alter API, control flow, and outputs. |
| 1272 |         let mut d = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block con
t. |
| 1273 |         d[0] = (coolant + 40.0).clamp(0.0, 250.0) as u8; | executable/support line; behavior impact depends
surrounding block context. |
| 1274 |         d[1] = (fuel_t + 40.0).clamp(0.0, 250.0) as u8; | executable/support line; behavior impact depends o
urrounding block context. |
| 1275 |         let o = ((oil_t + 273.0) / 0.03125) as u16; | executable/support line; behavior impact depends on su
unding block context. |
| 1276 |         d[2] = (o & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block con
t. |
| 1277 |         d[3] = (o >> 8) as u8; | executable/support line; behavior impact depends on surrounding block conte
|
| 1278 |         J1939Frame::from_raw(ts, J1939Frame::build_id(6, pgn::ET1, sa, 0xFF), &d) | executable/support line;
havior impact depends on surrounding block context. |
| 1279 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1280 |     pub fn efl_p1(ts: f64, fuel_kpa: f64, oil_lvl: f64, oil_kpa: f64, sa: u8) -> J1939Frame { | declares exe
able behavior; signature/body changes alter API, control flow, and outputs. |
| 1281 |         let mut d = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block con
t. |
| 1282 |         d[0] = (fuel_kpa / 4.0).clamp(0.0, 250.0) as u8; | executable/support line; behavior impact depends
surrounding block context. |
| 1283 |         d[3] = (oil_lvl / 0.4).clamp(0.0, 250.0) as u8; | executable/support line; behavior impact depends o
urrounding block context. |
| 1284 |         d[4] = (oil_kpa / 4.0).clamp(0.0, 250.0) as u8; | executable/support line; behavior impact depends o
urrounding block context. |
| 1285 |         J1939Frame::from_raw(ts, J1939Frame::build_id(6, pgn::EFL_P1, sa, 0xFF), &d) | executable/support li
behavior impact depends on surrounding block context. |
| 1286 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1287 |     pub fn ic1(ts: f64, boost_kpa: f64, intake_t: f64, exh_t: f64, sa: u8) -> J1939Frame { | declares execut
e behavior; signature/body changes alter API, control flow, and outputs. |
| 1288 |         let mut d = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block con
t. |
| 1289 |         d[0] = (boost_kpa / 2.0).clamp(0.0, 250.0) as u8; | executable/support line; behavior impact depends
surrounding block context. |
| 1290 |         d[1] = (intake_t + 40.0).clamp(0.0, 250.0) as u8; | executable/support line; behavior impact depends
surrounding block context. |
| 1291 |         let e = ((exh_t + 273.0) / 0.03125) as u16; | executable/support line; behavior impact depends on su
unding block context. |
| 1292 |         d[4] = (e & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block con
t. |
| 1293 |         d[5] = (e >> 8) as u8; | executable/support line; behavior impact depends on surrounding block conte
|
| 1294 |         J1939Frame::from_raw(ts, J1939Frame::build_id(6, pgn::IC1, sa, 0xFF), &d) | executable/support line;
havior impact depends on surrounding block context. |
| 1295 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1296 |     pub fn lfe(ts: f64, fuel_lph: f64, throttle: f64, sa: u8) -> J1939Frame { | declares executable behavior
ignature/body changes alter API, control flow, and outputs. |
| 1297 |         let mut d = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block con
t. |
| 1298 |         let f = (fuel_lph / 0.05) as u16; | executable/support line; behavior impact depends on surrounding
ck context. |
| 1299 |         d[0] = (f & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block con
t. |
| 1300 |         d[1] = (f >> 8) as u8; | executable/support line; behavior impact depends on surrounding block conte
|
| 1301 |         d[6] = (throttle / 0.4).clamp(0.0, 250.0) as u8; | executable/support line; behavior impact depends
surrounding block context. |
| 1302 |         J1939Frame::from_raw(ts, J1939Frame::build_id(6, pgn::LFE, sa, 0xFF), &d) | executable/support line;

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havior impact depends on surrounding block context. |
| 1303 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1304 |     pub fn pto(ts: f64, rpm: f64, enabled: bool, sa: u8) -> J1939Frame { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1305 |         let mut d = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block context. |
| 1306 |         d[0] = if enabled { 0x01 } else { 0x00 }; | executable/support line; behavior impact depends on surrounding block context. |
| 1307 |         let r = (rpm / 0.125) as u16; | executable/support line; behavior impact depends on surrounding block context. |
| 1308 |         d[2] = (r & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1309 |         d[3] = (r >> 8) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1310 |         J1939Frame::from_raw(ts, J1939Frame::build_id(6, pgn::PTO, sa, 0xFF), &d) | executable/support line; behavior impact depends on surrounding block context. |
| 1311 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1312 |     pub fn ccvs(ts: f64, speed: f64, cc_active: bool, cc_speed: f64, sa: u8) -> J1939Frame { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1313 |         let mut d = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block context. |
| 1314 |         d[0] = 0b11111100; | executable/support line; behavior impact depends on surrounding block context. |
| 1315 |         let s = (speed * 256.0) as u16; | executable/support line; behavior impact depends on surrounding block context. |
| 1316 |         d[1] = (s & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1317 |         d[2] = (s >> 8) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1318 |         d[3] = if cc_active { 0x01 } else { 0x00 }; | executable/support line; behavior impact depends on surrounding block context. |
| 1319 |         d[4] = cc_speed.clamp(0.0, 250.0) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1320 |         J1939Frame::from_raw(ts, J1939Frame::build_id(6, pgn::CCVS, sa, 0xFF), &d) | executable/support line; behavior impact depends on surrounding block context. |
| 1321 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1322 |     pub fn dm1( | declares executable behavior; signature/body changes alter API, control flow, and outputs. | |
| 1323 |         ts: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 1324 |         amber: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 1325 |         red: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 1326 |         mil: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 1327 |         dtc_spn: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 1328 |         fmi: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 1329 |         sa: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 1330 |     ) -> J1939Frame { | executable/support line; behavior impact depends on surrounding block context. |
| 1331 |         let mut d = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block context. |
| 1332 |         d[0] = (if mil { 0x40 } else { 0 }) | executable/support line; behavior impact depends on surrounding block context. |
| 1333 |         \ | (if red { 0x10 } else { 0 }) | executable/support line; behavior impact depends on surrounding block context. |
| 1334 |         \ | (if amber { 0x04 } else { 0 }); | executable/support line; behavior impact depends on surrounding block context. |
| 1335 |         d[1] = 0xFF; | executable/support line; behavior impact depends on surrounding block context. |
| 1336 |         d[2] = (dtc_spn & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1337 |         d[3] = ((dtc_spn >> 8) & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1338 |         d[4] = (((dtc_spn >> 16) & 0x7) as u8) \ | ((fmi & 0x1F) << 3); | executable/support line; behavior impact depends on surrounding block context. |
| 1339 |         d[5] = 1; // occurrence count | executable/support line; behavior impact depends on surrounding block context. |
| 1340 |         J1939Frame::from_raw(ts, J1939Frame::build_id(6, pgn::DM1, sa, 0xFF), &d) | executable/support line; behavior impact depends on surrounding block context. |
| 1341 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1342 |     pub fn hours(ts: f64, total_h: f64, sa: u8) -> J1939Frame { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |

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changes alter API, control flow, and outputs. |
| 1343 |         let mut d = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block context. | |
| 1344 |         let h = (total_h / 0.05) as u32; | executable/support line; behavior impact depends on surrounding block context. |
| 1345 |         d[0] = (h & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1346 |         d[1] = ((h >> 8) & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1347 |         d[2] = ((h >> 16) & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1348 |         d[3] = ((h >> 24) & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1349 |         J1939Frame::from_raw(ts, J1939Frame::build_id(6, pgn::HOURS, sa, 0xFF), &d) | executable/support line; behavior impact depends on surrounding block context. |
| 1350 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1351 | (blank) | line is blank; visual separation only. |
| 1352 |     #[allow(clippy::too_many_arguments)] | executable/support line; behavior impact depends on surrounding block context. |
| 1353 |     pub fn adas1( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1354 |         ts: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 1355 |         lka_on: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 1356 |         acc_on: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 1357 |         aeb_on: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 1358 |         lead_distance_m: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 1359 |         ttc_s: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 1360 |         speed_limit_kmh: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 1361 |         sa: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 1362 |     ) -> J1939Frame { | executable/support line; behavior impact depends on surrounding block context. |
| 1363 |         let mut d = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block context. |
| 1364 |         d[0] = (if lka_on { 0x01 } else { 0x00 }) | executable/support line; behavior impact depends on surrounding block context. |
| 1365 |         \ | (if acc_on { 0x01 } else { 0x00 }) << 2 | executable/support line; behavior impact depends on surrounding block context. |
| 1366 |         \ | (if aeb_on { 0x01 } else { 0x00 }) << 4; | executable/support line; behavior impact depends on surrounding block context. |
| 1367 |         let lead = (lead_distance_m.clamp(0.0, 6553.0) / 0.1) as u16; | executable/support line; behavior impact depends on surrounding block context. |
| 1368 |         d[1] = (lead & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1369 |         d[2] = (lead >> 8) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1370 |         let ttc = (ttc_s.clamp(0.0, 655.35) / 0.01) as u16; | executable/support line; behavior impact depends on surrounding block context. |
| 1371 |         d[3] = (ttc & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1372 |         d[4] = (ttc >> 8) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1373 |         d[5] = speed_limit_kmh.clamp(0.0, 250.0) as u8; | executable/support line; behavior impact depends on surrounding block context. |
| 1374 |         J1939Frame::from_raw(ts, J1939Frame::build_id(6, pgn::ADAS1, sa, 0xFF), &d) | executable/support line; behavior impact depends on surrounding block context. |
| 1375 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1376 | (blank) | line is blank; visual separation only. |
| 1377 |     pub fn fus1( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1378 |         ts: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 1379 |         fused_objects: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 1380 |         critical_ttc_s: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 1381 |         confidence_01: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 1382 |         in_path_hazard: bool, | executable/support line; behavior impact depends on surrounding block context. |

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[illegible]

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| 1423 | pub struct Dtc { | declares data layout; field changes may break serialization, tests, and callers. |
| 1424 |     pub spn: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 1425 |     pub fmi: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 1426 |     pub count: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 1427 |     pub active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 1428 |     pub desc: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 1429 |     pub severity: DtcSeverity, | executable/support line; behavior impact depends on surrounding block context. |
| 1430 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1431 | (blank) | line is blank; visual separation only. |
| 1432 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding block context. |
| 1433 | pub enum DtcSeverity { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 1434 |     Protect, | executable/support line; behavior impact depends on surrounding block context. |
| 1435 |     Amber, | executable/support line; behavior impact depends on surrounding block context. |
| 1436 |     Red, | executable/support line; behavior impact depends on surrounding block context. |
| 1437 |     Mil, | executable/support line; behavior impact depends on surrounding block context. |
| 1438 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1439 | (blank) | line is blank; visual separation only. |
| 1440 | impl std::fmt::Display for DtcSeverity { | implementation block; method behavior and trait conformance are defined here. |
| 1441 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1442 |         match self { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 1443 |             DtcSeverity::Protect => write!(f, "PROT"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1444 |             DtcSeverity::Amber => write!(f, "AMBR"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1445 |             DtcSeverity::Red => write!(f, "RED!"), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1446 |             DtcSeverity::Mil => write!(f, "MIL "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1447 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1448 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1449 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1450 | (blank) | line is blank; visual separation only. |
| 1451 | pub fn fmi_name(fmi: u8) -> &'static str { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1452 |     match fmi { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 1453 |         0 => "Above Normal – Most Severe", | executable/support line; behavior impact depends on surrounding block context. |
| 1454 |         1 => "Below Normal – Most Severe", | executable/support line; behavior impact depends on surrounding block context. |
| 1455 |         2 => "Erratic / Intermittent", | executable/support line; behavior impact depends on surrounding block context. |
| 1456 |         3 => "Voltage High / Short to High", | executable/support line; behavior impact depends on surrounding block context. |
| 1457 |         4 => "Voltage Low / Short to Low", | executable/support line; behavior impact depends on surrounding block context. |
| 1458 |         5 => "Current Low / Open Circuit", | executable/support line; behavior impact depends on surrounding block context. |
| 1459 |         6 => "Current High / Grounded", | executable/support line; behavior impact depends on surrounding block context. |
| 1460 |         7 => "Mechanical System Not Responding", | executable/support line; behavior impact depends on surrounding block context. |
| 1461 |         8 => "Abnormal Frequency / Pulse Width", | executable/support line; behavior impact depends on surrounding block context. |
| 1462 |         9 => "Abnormal Update Rate", | executable/support line; behavior impact depends on surrounding block context. |
| 1463 |         10 => "Abnormal Rate of Change", | executable/support line; behavior impact depends on surrounding block context. |
| 1464 |         11 => "Root Cause Not Known", | executable/support line; behavior impact depends on surrounding block context. |

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[illegible]

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| 1510 | (blank) | line is blank; visual separation only. |
| 1511 |     pub fn push(&mut self, frame: J1939Frame) { | declares executable behavior; signature/body changes alter
I, control flow, and outputs. |
| 1512 |         self.frames.push_front(frame); | executable/support line; behavior impact depends on surrounding blo
context. |
| 1513 |         if self.frames.len() > 200 { | conditional gate; changing condition shifts runtime path and side eff
s. |
| 1514 |             self.frames.pop_back(); | executable/support line; behavior impact depends on surrounding block
text. |
| 1515 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1516 |         self.total_frames += 1; | executable/support line; behavior impact depends on surrounding block cont
. |
| 1517 |         self.frame_count += 1; | executable/support line; behavior impact depends on surrounding block conte
|
| 1518 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1519 | (blank) | line is blank; visual separation only. |
| 1520 |     pub fn tick(&mut self, dt: f64) { | declares executable behavior; signature/body changes alter API, cont
flow, and outputs. |
| 1521 |         self.frame_timer += dt; | executable/support line; behavior impact depends on surrounding block cont
. |
| 1522 |         if self.frame_timer >= 1.0 { | conditional gate; changing condition shifts runtime path and side eff
s. |
| 1523 |             self.fps = self.frame_count; | executable/support line; behavior impact depends on surrounding b
k context. |
| 1524 |             // 500 kbps, min frame ~55 bits → ~9090 frames/s max | comment line; documentation intent, no r
me effect. |
| 1525 |             self.bus_load_pct = (self.frame_count as f64 / 9090.0 * 100.0).min(100.0); | executable/support
e; behavior impact depends on surrounding block context. |
| 1526 |             self.frame_count = 0; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1527 |             self.frame_timer = 0.0; | executable/support line; behavior impact depends on surrounding block
text. |
| 1528 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1529 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1530 | (blank) | line is blank; visual separation only. |
| 1531 |     pub fn visible_frames(&self) -> impl Iterator<Item = &J1939Frame> { | declares executable behavior; sign
re/body changes alter API, control flow, and outputs. |
| 1532 |         self.frames.iter().filter(\f\| { | executable/support line; behavior impact depends on surrounding
ck context. |
| 1533 |             self.filter_pgn.is_none_or(\p\| f.pgn == p) | executable/support line; behavior impact depends
surrounding block context. |
| 1534 |             && self.filter_sa.is_none_or(\s\| f.sa == s) | executable/support line; behavior impact dep
s on surrounding block context. |
| 1535 |         }) | executable/support line; behavior impact depends on surrounding block context. |
| 1536 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1537 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1538 | (blank) | line is blank; visual separation only. |
| 1539 | #[cfg(test)] | executable/support line; behavior impact depends on surrounding block context. |
| 1540 | mod tests { | declares a module boundary; changing module name/path breaks references. |
| 1541 |     use super::*; | imports symbols; changing path/name affects compile-time resolution. |
| 1542 | (blank) | line is blank; visual separation only. |
| 1543 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1544 |     fn pdu1_id_roundtrip_preserves_destination() { | declares executable behavior; signature/body changes al
API, control flow, and outputs. |
| 1545 |         let id = J1939Frame::build_id(3, pgn::TSC1, addr::ECM_1, addr::TRANSMISSION); | executable/support l
; behavior impact depends on surrounding block context. |
| 1546 |         let frame = J1939Frame::from_raw(0.0, id, &[0xFF; 8]); | executable/support line; behavior impact de
ds on surrounding block context. |
| 1547 |         assert_eq!(frame.pgn, pgn::TSC1); | executable/support line; behavior impact depends on surrounding
ck context. |
| 1548 |         assert_eq!(frame.da, addr::TRANSMISSION); | executable/support line; behavior impact depends on surr
ding block context. |
| 1549 |         assert_eq!(frame.sa, addr::ECM_1); | executable/support line; behavior impact depends on surrounding

```

```

ock context. |
| 1550 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1551 | (blank) | line is blank; visual separation only. |
| 1552 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1553 |     fn ebc2_decodes_all_wheel_speeds() { | declares executable behavior; signature/body changes alter API, c
rol flow, and outputs. |
| 1554 |         let mut data = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block
text. |
| 1555 |         let fl = (48.0 / 0.125) as u16; | executable/support line; behavior impact depends on surrounding bl
context. |
| 1556 |         let fr = (50.0 / 0.125) as u16; | executable/support line; behavior impact depends on surrounding bl
context. |
| 1557 |         let rl = (47.0 / 0.125) as u16; | executable/support line; behavior impact depends on surrounding bl
context. |
| 1558 |         let rr = (49.0 / 0.125) as u16; | executable/support line; behavior impact depends on surrounding bl
context. |
| 1559 |         data[0] = (fl & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1560 |         data[1] = (fl >> 8) as u8; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1561 |         data[2] = (fr & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1562 |         data[3] = (fr >> 8) as u8; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1563 |         data[4] = (rl & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1564 |         data[5] = (rl >> 8) as u8; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1565 |         data[6] = (rr & 0xFF) as u8; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1566 |         data[7] = (rr >> 8) as u8; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1567 |         let id = J1939Frame::build_id(6, pgn::EBC2, addr::BRAKES, 0xFF); | executable/support line; behavior
pact depends on surrounding block context. |
| 1568 |         let frame = J1939Frame::from_raw(0.0, id, &data); | executable/support line; behavior impact depends
surrounding block context. |
| 1569 |         assert_eq!(frame.pgn_name(), "EBC2"); | executable/support line; behavior impact depends on surround
block context. |
| 1570 |         assert!(frame | executable/support line; behavior impact depends on surrounding block context. |
| 1571 |             .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 1572 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1573 |             .any(|d| d.dspn == 904 && (d.physical - 48.0).abs() < 0.2)); | executable/support line; behavior
mpact depends on surrounding block context. |
| 1574 |         assert!(frame | executable/support line; behavior impact depends on surrounding block context. |
| 1575 |             .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 1576 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1577 |             .any(|d| d.dspn == 905 && (d.physical - 50.0).abs() < 0.2)); | executable/support line; behavior
mpact depends on surrounding block context. |
| 1578 |         assert!(frame | executable/support line; behavior impact depends on surrounding block context. |
| 1579 |             .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 1580 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1581 |             .any(|d| d.dspn == 906 && (d.physical - 47.0).abs() < 0.2)); | executable/support line; behavior
mpact depends on surrounding block context. |
| 1582 |         assert!(frame | executable/support line; behavior impact depends on surrounding block context. |
| 1583 |             .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 1584 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1585 |             .any(|d| d.dspn == 907 && (d.physical - 49.0).abs() < 0.2)); | executable/support line; behavior
mpact depends on surrounding block context. |
| 1586 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1587 | (blank) | line is blank; visual separation only. |
| 1588 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1589 |     fn etc2_gear_decode_matches_encoded_value() { | declares executable behavior; signature/body changes alt
API, control flow, and outputs. |
| 1590 |         let mut data = [0xFFu8; 8]; | executable/support line; behavior impact depends on surrounding block
text. |

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| 1591 |         data[3] = 132; // selected gear = 7 after offset -125 | executable/support line; behavior impact depends on surrounding block context. | | |
| 1592 |         data[4] = 132; // current gear = 7 | executable/support line; behavior impact depends on surrounding block context. |
| 1593 |         data[7] = 3; // range C | executable/support line; behavior impact depends on surrounding block context. |
| 1594 |         let id = J1939Frame::build_id(3, pgn::ETC2, addr::TRANSMISSION, 0xFF); | executable/support line; behavior impact depends on surrounding block context. |
| 1595 |         let frame = J1939Frame::from_raw(0.0, id, &data); | executable/support line; behavior impact depends on surrounding block context. |
| 1596 |         assert_eq!(frame.pgn_name(), "ETC2"); | executable/support line; behavior impact depends on surrounding block context. |
| 1597 |         assert!(frame | executable/support line; behavior impact depends on surrounding block context. |
| 1598 |             .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 1599 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1600 |             .any(|d| d.dspn == 524 && (d.physical - 7.0).abs() < 0.1)); | executable/support line; behavior impact depends on surrounding block context. |
| 1601 |         assert!(frame | executable/support line; behavior impact depends on surrounding block context. |
| 1602 |             .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 1603 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1604 |             .any(|d| d.dspn == 523 && (d.physical - 7.0).abs() < 0.1)); | executable/support line; behavior impact depends on surrounding block context. |
| 1605 |         assert!(frame | executable/support line; behavior impact depends on surrounding block context. |
| 1606 |             .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 1607 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1608 |             .any(|d| d.dspn == 525 && (d.physical - 3.0).abs() < 0.1)); | executable/support line; behavior impact depends on surrounding block context. |
| 1609 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1610 | (blank) | line is blank; visual separation only. | | |
| 1611 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1612 |     fn adas_builder_roundtrip_exposes_core_spns() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1613 |         let frame = Builder::adas1(0.0, true, true, false, 23.4, 1.85, 80.0, addr::HEADWAY); | executable/support line; behavior impact depends on surrounding block context. |
| 1614 |         assert_eq!(frame.pgn, pgn::ADAS1); | executable/support line; behavior impact depends on surrounding block context. |
| 1615 |         assert_eq!(frame.pgn_name(), "ADAS1"); | executable/support line; behavior impact depends on surrounding block context. |
| 1616 |         assert!(frame | executable/support line; behavior impact depends on surrounding block context. |
| 1617 |             .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 1618 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1619 |             .any(|d| d.dspn == 52001 && d.physical >= 1.0)); | executable/support line; behavior impact depends on surrounding block context. |
| 1620 |         assert!(frame | executable/support line; behavior impact depends on surrounding block context. |
| 1621 |             .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 1622 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1623 |             .any(|d| d.dspn == 52002 && d.physical >= 1.0)); | executable/support line; behavior impact depends on surrounding block context. |
| 1624 |         assert!(frame | executable/support line; behavior impact depends on surrounding block context. |
| 1625 |             .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 1626 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1627 |             .any(|d| d.dspn == 52004 && (d.physical - 23.4).abs() < 0.2)); | executable/support line; behavior impact depends on surrounding block context. |
| 1628 |         assert!(frame | executable/support line; behavior impact depends on surrounding block context. |
| 1629 |             .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 1630 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1631 |             .any(|d| d.dspn == 52005 && (d.physical - 1.85).abs() < 0.05)); | executable/support line; behavior impact depends on surrounding block context. |
| 1632 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1633 | (blank) | line is blank; visual separation only. |
| 1634 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1635 |     fn energy_builder_roundtrip_exposes_power_values() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1636 |         let frame = Builder::engy1(0.0, 27.4, 84.0, 2.30, 19.75, addr::HEADWAY); | executable/support line; behavior impact depends on surrounding block context. |

```



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| 1637 |          assert_eq!(frame.pgn, pgn::ENGY1); | executable/support line; behavior impact depends on surrounding
ock context. |
| 1638 |          assert_eq!(frame.pgn_name(), "ENGY1"); | executable/support line; behavior impact depends on surround
g block context. |
| 1639 |          assert!(frame | executable/support line; behavior impact depends on surrounding block context. |
| 1640 |              .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 1641 |              .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1642 |              .any(|d| d.spn == 52201 && (d.physical - 27.4).abs() < 0.2)); | executable/support line; behav
impact depends on surrounding block context. |
| 1643 |          assert!(frame | executable/support line; behavior impact depends on surrounding block context. |
| 1644 |              .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 1645 |              .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1646 |              .any(|d| d.spn == 52202 && (d.physical - 84.0).abs() < 1.0)); | executable/support line; behav
impact depends on surrounding block context. |
| 1647 |          assert!(frame | executable/support line; behavior impact depends on surrounding block context. |
| 1648 |              .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 1649 |              .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1650 |              .any(|d| d.spn == 52203 && (d.physical - 2.30).abs() < 0.05)); | executable/support line; beha
r impact depends on surrounding block context. |
| 1651 |          assert!(frame | executable/support line; behavior impact depends on surrounding block context. |
| 1652 |              .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 1653 |              .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1654 |              .any(|d| d.spn == 52204 && (d.physical - 19.75).abs() < 0.05)); | executable/support line; beh
or impact depends on surrounding block context. |
| 1655 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1656 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/leak_physics.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 2151 | Non-empty: 1994 | Symbols: 95

Symbol Inventory

Line	Kind	Symbol	Change Impact
8	struct	LeakModelCoefficients	data/structure coupling
16	impl	Default for LeakModelCoefficients	behavior binding and method semantics
17	fn	default	signature/API coupling
29	struct	CalibrationCsvSample	data/structure coupling
43	struct	CalibrationCircuitReport	data/structure coupling
56	struct	CalibrationGroupReport	data/structure coupling
65	struct	CalibrationReport	data/structure coupling
73	fn	aggregate_group_report	signature/API coupling
107	enum	LeakAlertLevel	data/structure coupling
116	enum	PressureBand	data/structure coupling
125	impl	std::fmt::Display for PressureBand	behavior binding and method semantics
126	fn	fmt	signature/API coupling
140	enum	CircuitComponent	data/structure coupling
147	enum	OilType	data/structure coupling
195	struct	OilRuntimeProfile	data/structure coupling
203	const	OIL_TYPE_VARIANTS	data/structure coupling
248	impl	OilType	behavior binding and method semantics
249	fn	all	signature/API coupling
253	fn	name	signature/API coupling
257	fn	runtime_profile	signature/API coupling
307	impl	OilType	behavior binding and method semantics
308	fn	density_kg_m3	signature/API coupling
312	fn	viscosity_index	signature/API coupling
317	impl	std::fmt::Display for LeakAlertLevel	behavior binding and method semantics
318	fn	fmt	signature/API coupling
331	enum	OringMaterial	data/structure coupling
351	struct	MaterialRuntimeProfile	data/structure coupling
358	const	ORING_MATERIAL_VARIANTS	data/structure coupling
373	impl	OringMaterial	behavior binding and method semantics

374	fn	all	signature/API coupling
378	fn	name	signature/API coupling
382	fn	runtime_profile	signature/API coupling
403	fn	temp_window_c	signature/API coupling
408	fn	gas_permeability_factor	signature/API coupling
414	struct	MaterialCatalogEntry	data/structure coupling
435	struct	OilCatalogEntry	data/structure coupling
458	fn	engineering_material_catalog	signature/API coupling
496	fn	engineering_oil_catalog	signature/API coupling
540	struct	PressureEnvelope	data/structure coupling
549	struct	OringSpec	data/structure coupling
563	struct	CircuitInput	data/structure coupling
573	struct	CircuitResult	data/structure coupling
599	struct	ManualCircuitParams	data/structure coupling
615	struct	ScenarioPrediction	data/structure coupling
626	struct	LeakAlert	data/structure coupling
634	struct	PressureStats	data/structure coupling
640	impl	PressureStats	behavior binding and method semantics
641	fn	new	signature/API coupling
649	fn	update	signature/API coupling
657	struct	SealState	data/structure coupling
671	impl	SealState	behavior binding and method semantics
672	fn	new	signature/API coupling
690	struct	LeakCircuit	data/structure coupling
707	impl	LeakCircuit	behavior binding and method semantics
709	fn	new	signature/API coupling
740	fn	step	signature/API coupling
1011	fn	reset	signature/API coupling
1016	fn	apply_manual_params	signature/API coupling
1062	struct	LeakPhysicsRig	data/structure coupling
1069	impl	Default for LeakPhysicsRig	behavior binding and method semantics
1070	fn	default	signature/API coupling
1075	impl	LeakPhysicsRig	behavior binding and method semantics
1076	fn	new	signature/API coupling
1085	fn	with_default_machine_presets	signature/API coupling
1184	fn	add_circuit	signature/API coupling
1188	fn	circuit_mut	signature/API coupling
1192	fn	oil_density_for	signature/API coupling
1200	fn	set_runtime_pressure_state	signature/API coupling
1212	fn	apply_manual_params	signature/API coupling
1221	fn	step	signature/API coupling
1246	fn	reset	signature/API coupling
1255	fn	export_last_results_json	signature/API coupling
1267	fn	export_last_results_csv	signature/API coupling
1316	fn	export_predictions_json	signature/API coupling
1332	fn	export_predictions_csv	signature/API coupling
1362	fn	export_material_catalog_json	signature/API coupling
1374	fn	export_material_catalog_csv	signature/API coupling
1408	fn	export_oil_catalog_json	signature/API coupling
1420	fn	export_oil_catalog_csv	signature/API coupling
1457	fn	calibrate_from_csv	signature/API coupling
1462	struct	CsvRow	data/structure coupling
1654	fn	export_calibration_report_json	signature/API coupling
1670	fn	export_calibration_report_csv	signature/API coupling
1725	fn	run_monte_carlo	signature/API coupling
1816	fn	predict_scenarios	signature/API coupling
1908	module	tests	data/structure coupling
1912	fn	high_pressure_eventually_ruptures	signature/API coupling
1944	fn	manual_params_are_applied	signature/API coupling
1975	fn	scenario_prediction_returns_ranked_data	signature/API coupling
1983	fn	monte_carlo_produces_samples	signature/API coupling
1990	fn	exports_json_and_csv_files	signature/API coupling
2030	fn	report_captures_pressure_band_and_safe_window	signature/API coupling
2055	fn	rupture_event_records_pressure_and_time	signature/API coupling
2089	fn	engineering_catalogs_are_large_and_coherent	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	/// O-ring leak / rupture physics for hydraulic, oil and refrigerant circuits.	comment line; documentation in t, no runtime effect.
2	(blank)	line is blank; visual separation only.
3	use serde::{Deserialize, Serialize};	imports symbols; changing path/name affects compile-time resolution.
4	use std::fs;	imports symbols; changing path/name affects compile-time resolution.
5	use std::path::Path;	imports symbols; changing path/name affects compile-time resolution.
6	(blank)	line is blank; visual separation only.
7	#[derive(Debug, Clone, Copy, Serialize, Deserialize)]	serialization contract touchpoint; edits can break report compatibility and tooling.
8	pub struct LeakModelCoefficients {	declares data layout; field changes may break serialization, tests, and ca rs.
9	pub damage_rate_scale: f64,	executable/support line; behavior impact depends on surrounding block context.
10	pub extrusion_rate_scale: f64,	executable/support line; behavior impact depends on surrounding block context.
11	pub thermal_rate_scale: f64,	executable/support line; behavior impact depends on surrounding block context.
12	pub flow_rate_scale: f64,	executable/support line; behavior impact depends on surrounding block context.
13	pub rupture_area_scale: f64,	executable/support line; behavior impact depends on surrounding block context.
14	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
15	(blank)	line is blank; visual separation only.
16	impl Default for LeakModelCoefficients {	implementation block; method behavior and trait conformance are defined here.
17	fn default() -> Self {	declares executable behavior; signature/body changes alter API, control flow, and tputs.
18	Self {	executable/support line; behavior impact depends on surrounding block context.
19	damage_rate_scale: 1.0,	executable/support line; behavior impact depends on surrounding block context.
20	extrusion_rate_scale: 1.0,	executable/support line; behavior impact depends on surrounding block context.
21	thermal_rate_scale: 1.0,	executable/support line; behavior impact depends on surrounding block context.
22	flow_rate_scale: 1.0,	executable/support line; behavior impact depends on surrounding block context.
23	rupture_area_scale: 1.0,	executable/support line; behavior impact depends on surrounding block context.
24	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
25	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
26	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
27	(blank)	line is blank; visual separation only.
28	#[derive(Debug, Clone, Serialize, Deserialize)]	serialization contract touchpoint; edits can break report compatibility and tooling.
29	pub struct CalibrationCsvSample {	declares data layout; field changes may break serialization, tests, and ca rs.
30	pub timestamp_s: Option<f64>,	executable/support line; behavior impact depends on surrounding block context.
31	pub circuit_name: String,	executable/support line; behavior impact depends on surrounding block context.
32	pub pressure_bar: f64,	executable/support line; behavior impact depends on surrounding block context.
33	pub delta_p_bar: f64,	executable/support line; behavior impact depends on surrounding block context.
34	pub temp_c: f64,	executable/support line; behavior impact depends on surrounding block context.
35	pub cycles_per_s: f64,	executable/support line; behavior impact depends on surrounding block context.
36	pub duty_01: f64,	executable/support line; behavior impact depends on surrounding block context.
37	pub fluid_density_kg_m3: f64,	executable/support line; behavior impact depends on surrounding block context.
38	pub measured_leak_lpm: f64,	executable/support line; behavior impact depends on surrounding block context.
39	pub observed_rupture: Option<bool>,	executable/support line; behavior impact depends on surrounding block context.
40	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.

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| 41 | (blank) | line is blank; visual separation only. |
| 42 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report co
tibility and tooling. |
| 43 | pub struct CalibrationCircuitReport { | declares data layout; field changes may break serialization, tests, and
allers. |
| 44 |     pub circuit_name: String, | executable/support line; behavior impact depends on surrounding block context.
| 45 |     pub material: String, | executable/support line; behavior impact depends on surrounding block context. |
| 46 |     pub oil: String, | executable/support line; behavior impact depends on surrounding block context. |
| 47 |     pub sample_count: usize, | executable/support line; behavior impact depends on surrounding block context.
| 48 |     pub rmse_leak_lpm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 49 |     pub mape_leak_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 50 |     pub max_abs_error_lpm: f64, | executable/support line; behavior impact depends on surrounding block context.
|
| 51 |     pub rupture_accuracy_pct: f64, | executable/support line; behavior impact depends on surrounding block con
t. |
| 52 |     pub fitted: LeakModelCoefficients, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 53 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 54 | (blank) | line is blank; visual separation only. |
| 55 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report co
tibility and tooling. |
| 56 | pub struct CalibrationGroupReport { | declares data layout; field changes may break serialization, tests, and
lers. |
| 57 |     pub group: String, | executable/support line; behavior impact depends on surrounding block context. |
| 58 |     pub sample_count: usize, | executable/support line; behavior impact depends on surrounding block context.
| 59 |     pub mean_rmse_lpm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 60 |     pub mean_mape_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 61 |     pub mean_rupture_accuracy_pct: f64, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 62 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 63 | (blank) | line is blank; visual separation only. |
| 64 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report co
tibility and tooling. |
| 65 | pub struct CalibrationReport { | declares data layout; field changes may break serialization, tests, and calle
|
| 66 |     pub total_samples: usize, | executable/support line; behavior impact depends on surrounding block context.
| 67 |     pub calibrated_circuits: usize, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 68 |     pub circuit_reports: Vec<CalibrationCircuitReport>, | executable/support line; behavior impact depends on
rounding block context. |
| 69 |     pub by_material: Vec<CalibrationGroupReport>, | executable/support line; behavior impact depends on surrou
ng block context. |
| 70 |     pub by_oil: Vec<CalibrationGroupReport>, | executable/support line; behavior impact depends on surrounding
ock context. |
| 71 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 72 | (blank) | line is blank; visual separation only. |
| 73 | fn aggregate_group_report<F>( | declares executable behavior; signature/body changes alter API, control flow,
outputs. |
| 74 |     items: &[CalibrationCircuitReport], | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 75 |     mut key_fn: F, | executable/support line; behavior impact depends on surrounding block context. |
| 76 | ) -> Vec<CalibrationGroupReport> | executable/support line; behavior impact depends on surrounding block conte
|
| 77 | where | executable/support line; behavior impact depends on surrounding block context. |
| 78 |     F: FnMut(&CalibrationCircuitReport) -> String, | executable/support line; behavior impact depends on surro
ing block context. |
| 79 | { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 80 |     use std::collections::BTreeMap; | imports symbols; changing path/name affects compile-time resolution. |
| 81 |     let mut acc: BTreeMap<String, (usize, f64, f64, f64, usize)> = BTreeMap::new(); | executable/support line;
havior impact depends on surrounding block context. |
| 82 |     for r in items { | iteration control; changing bounds/body affects throughput, timing, and possibly safety
arantees. |
| 83 |         let key = key_fn(r); | executable/support line; behavior impact depends on surrounding block context.
| 84 |         let entry = acc.entry(key).or_insert((0, 0.0, 0.0, 0.0, 0)); | executable/support line; behavior impac
depends on surrounding block context. |

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| 85 |         entry.0 += 1; | executable/support line; behavior impact depends on surrounding block context. |
| 86 |         entry.1 += r.rmse_leak_lpm; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 87 |         entry.2 += r.mape_leak_pct; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 88 |         entry.3 += r.rupture_accuracy_pct; | executable/support line; behavior impact depends on surrounding b
k context. |
| 89 |         entry.4 += r.sample_count; | executable/support line; behavior impact depends on surrounding block con
t. |
| 90 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 91 | (blank) | line is blank; visual separation only. |
| 92 |     let mut out = Vec::with_capacity(acc.len()); | executable/support line; behavior impact depends on surroun
g block context. |
| 93 |     for (group, (count, rmse, mape, rupture_acc, sample_count)) in acc { | iteration control; changing bounds/
y affects throughput, timing, and possibly safety guarantees. |
| 94 |         let n = count.max(1) as f64; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 95 |         out.push(CalibrationGroupReport { | executable/support line; behavior impact depends on surrounding bl
context. |
| 96 |             group, | executable/support line; behavior impact depends on surrounding block context. |
| 97 |             sample_count, | executable/support line; behavior impact depends on surrounding block context. |
| 98 |             mean_rmse_lpm: rmse / n, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 99 |             mean_mape_pct: mape / n, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 100 |             mean_rupture_accuracy_pct: rupture_acc / n, | executable/support line; behavior impact depends on
rounding block context. |
| 101 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 102 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 103 |     out | executable/support line; behavior impact depends on surrounding block context. |
| 104 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 105 | (blank) | line is blank; visual separation only. |
| 106 | #[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Serialize, Deserialize)] | serialization contrac
touchpoint; edits can break report compatibility and tooling. |
| 107 | pub enum LeakAlertLevel { | declares state variants; adding/removing variants impacts pattern matches and log
branches. |
| 108 |     Normal, | executable/support line; behavior impact depends on surrounding block context. |
| 109 |     Watch, | executable/support line; behavior impact depends on surrounding block context. |
| 110 |     Warning, | executable/support line; behavior impact depends on surrounding block context. |
| 111 |     Critical, | executable/support line; behavior impact depends on surrounding block context. |
| 112 |     Ruptured, | executable/support line; behavior impact depends on surrounding block context. |
| 113 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 114 | (blank) | line is blank; visual separation only. |
| 115 | #[derive(Debug, Clone, Copy, PartialEq, Eq, Serialize, Deserialize)] | serialization contract touchpoint; edi
can break report compatibility and tooling. |
| 116 | pub enum PressureBand { | declares state variants; adding/removing variants impacts pattern matches and logic
anches. |
| 117 |     BelowMinimum, | executable/support line; behavior impact depends on surrounding block context. |
| 118 |     LowBand, | executable/support line; behavior impact depends on surrounding block context. |
| 119 |     MidBand, | executable/support line; behavior impact depends on surrounding block context. |
| 120 |     HighBand, | executable/support line; behavior impact depends on surrounding block context. |
| 121 |     OverMaximum, | executable/support line; behavior impact depends on surrounding block context. |
| 122 |     AtRupture, | executable/support line; behavior impact depends on surrounding block context. |
| 123 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 124 | (blank) | line is blank; visual separation only. |
| 125 | impl std::fmt::Display for PressureBand { | implementation block; method behavior and trait conformance are d
ned here. |
| 126 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; sign
re/body changes alter API, control flow, and outputs. |
| 127 |         let s = match self { | executable/support line; behavior impact depends on surrounding block context.
| 128 |             PressureBand::BelowMinimum => "below_min", | executable/support line; behavior impact depends on
rounding block context. |
| 129 |             PressureBand::LowBand => "low", | executable/support line; behavior impact depends on surrounding
ock context. |
| 130 |             PressureBand::MidBand => "mid", | executable/support line; behavior impact depends on surrounding

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ock context. |
| 131 |           PressureBand::HighBand => "high", | executable/support line; behavior impact depends on surrounding block context. |
| 132 |           PressureBand::OverMaximum => "over_max", | executable/support line; behavior impact depends on surrounding block context. |
| 133 |           PressureBand::AtRupture => "at_rupture", | executable/support line; behavior impact depends on surrounding block context. |
| 134 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 135 |           write!(f, "{}", s) | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 136 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 137 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 138 | (blank) | line is blank; visual separation only. |
| 139 | #[derive(Debug, Clone, Copy, PartialEq, Eq, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 140 | pub enum CircuitComponent { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 141 |     Oring, | executable/support line; behavior impact depends on surrounding block context. |
| 142 |     Seal, | executable/support line; behavior impact depends on surrounding block context. |
| 143 |     AchHose, | executable/support line; behavior impact depends on surrounding block context. |
| 144 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 145 | (blank) | line is blank; visual separation only. |
| 146 | #[derive(Debug, Clone, Copy, PartialEq, Eq, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 147 | pub enum OilType { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 148 |     HydraulicIso22, | executable/support line; behavior impact depends on surrounding block context. |
| 149 |     HydraulicIso32, | executable/support line; behavior impact depends on surrounding block context. |
| 150 |     HydraulicIso46, | executable/support line; behavior impact depends on surrounding block context. |
| 151 |     HydraulicIso68, | executable/support line; behavior impact depends on surrounding block context. |
| 152 |     HydraulicIso100, | executable/support line; behavior impact depends on surrounding block context. |
| 153 |     HydraulicIso150, | executable/support line; behavior impact depends on surrounding block context. |
| 154 |     HydraulicIso220, | executable/support line; behavior impact depends on surrounding block context. |
| 155 |     Aw32, | executable/support line; behavior impact depends on surrounding block context. |
| 156 |     Aw46, | executable/support line; behavior impact depends on surrounding block context. |
| 157 |     Aw68, | executable/support line; behavior impact depends on surrounding block context. |
| 158 |     Utto10w30, | executable/support line; behavior impact depends on surrounding block context. |
| 159 |     Utto80w, | executable/support line; behavior impact depends on surrounding block context. |
| 160 |     Gear75w90, | executable/support line; behavior impact depends on surrounding block context. |
| 161 |     Gear80w90, | executable/support line; behavior impact depends on surrounding block context. |
| 162 |     AtfDexronIii, | executable/support line; behavior impact depends on surrounding block context. |
| 163 |     AtfDexronVi, | executable/support line; behavior impact depends on surrounding block context. |
| 164 |     Engine0w20, | executable/support line; behavior impact depends on surrounding block context. |
| 165 |     Engine5w30, | executable/support line; behavior impact depends on surrounding block context. |
| 166 |     Engine5w40, | executable/support line; behavior impact depends on surrounding block context. |
| 167 |     Engine10w30, | executable/support line; behavior impact depends on surrounding block context. |
| 168 |     Engine15w40, | executable/support line; behavior impact depends on surrounding block context. |
| 169 |     Engine20w50, | executable/support line; behavior impact depends on surrounding block context. |
| 170 |     TransformerInhibited, | executable/support line; behavior impact depends on surrounding block context. |
| 171 |     Turbine32, | executable/support line; behavior impact depends on surrounding block context. |
| 172 |     Turbine46, | executable/support line; behavior impact depends on surrounding block context. |
| 173 |     Compressor46, | executable/support line; behavior impact depends on surrounding block context. |
| 174 |     Compressor68, | executable/support line; behavior impact depends on surrounding block context. |
| 175 |     BiodegradableHees46, | executable/support line; behavior impact depends on surrounding block context. |
| 176 |     BiodegradableHepr46, | executable/support line; behavior impact depends on surrounding block context. |
| 177 |     FireResistantHfdu46, | executable/support line; behavior impact depends on surrounding block context. |
| 178 |     BrakeDot3, | executable/support line; behavior impact depends on surrounding block context. |
| 179 |     BrakeDot4, | executable/support line; behavior impact depends on surrounding block context. |
| 180 |     BrakeDot51, | executable/support line; behavior impact depends on surrounding block context. |
| 181 |     SyntheticEsters100, | executable/support line; behavior impact depends on surrounding block context. |
| 182 |     RefrigerationMineral3gs, | executable/support line; behavior impact depends on surrounding block context. |
| 183 |     Pag46, | executable/support line; behavior impact depends on surrounding block context. |
| 184 |     Pag100, | executable/support line; behavior impact depends on surrounding block context. |
| 185 |     Pag150, | executable/support line; behavior impact depends on surrounding block context. |

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| 186 |     Poe32, | executable/support line; behavior impact depends on surrounding block context. |
| 187 |     Poe46, | executable/support line; behavior impact depends on surrounding block context. |
| 188 |     Poe68, | executable/support line; behavior impact depends on surrounding block context. |
| 189 |     Poe100, | executable/support line; behavior impact depends on surrounding block context. |
| 190 |     VacuumPump100, | executable/support line; behavior impact depends on surrounding block context. |
| 191 |     Custom, | executable/support line; behavior impact depends on surrounding block context. |
| 192 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 193 | (blank) | line is blank; visual separation only. |
| 194 | #[derive(Debug, Clone, Copy, Serialize, Deserialize)] | serialization contract touchpoint; edits can break re
t compatibility and tooling. |
| 195 | pub struct OilRuntimeProfile { | declares data layout; field changes may break serialization, tests, and call
. |
| 196 |     pub label: &'static str, | executable/support line; behavior impact depends on surrounding block context.
| 197 |     pub density_kg_m3: f64, | executable/support line; behavior impact depends on surrounding block context.
| 198 |     pub viscosity_index: f64, | executable/support line; behavior impact depends on surrounding block context
| 199 |     pub min_oper_temp_c: f64, | executable/support line; behavior impact depends on surrounding block context
| 200 |     pub max_oper_temp_c: f64, | executable/support line; behavior impact depends on surrounding block context
| 201 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 202 | (blank) | line is blank; visual separation only. |
| 203 | const OIL_TYPE_VARIANTS: [OilType; 42] = [ | executable/support line; behavior impact depends on surrounding
ck context. |
| 204 |     OilType::HydraulicIso22, | executable/support line; behavior impact depends on surrounding block context.
| 205 |     OilType::HydraulicIso32, | executable/support line; behavior impact depends on surrounding block context.
| 206 |     OilType::HydraulicIso46, | executable/support line; behavior impact depends on surrounding block context.
| 207 |     OilType::HydraulicIso68, | executable/support line; behavior impact depends on surrounding block context.
| 208 |     OilType::HydraulicIso100, | executable/support line; behavior impact depends on surrounding block context
| 209 |     OilType::HydraulicIso150, | executable/support line; behavior impact depends on surrounding block context
| 210 |     OilType::HydraulicIso220, | executable/support line; behavior impact depends on surrounding block context
| 211 |     OilType::Aw32, | executable/support line; behavior impact depends on surrounding block context. |
| 212 |     OilType::Aw46, | executable/support line; behavior impact depends on surrounding block context. |
| 213 |     OilType::Aw68, | executable/support line; behavior impact depends on surrounding block context. |
| 214 |     OilType::Utto10w30, | executable/support line; behavior impact depends on surrounding block context. |
| 215 |     OilType::Utto80w, | executable/support line; behavior impact depends on surrounding block context. |
| 216 |     OilType::Gear75w90, | executable/support line; behavior impact depends on surrounding block context. |
| 217 |     OilType::Gear80w90, | executable/support line; behavior impact depends on surrounding block context. |
| 218 |     OilType::AtfDexronIii, | executable/support line; behavior impact depends on surrounding block context. |
| 219 |     OilType::AtfDexronVi, | executable/support line; behavior impact depends on surrounding block context. |
| 220 |     OilType::Engine0w20, | executable/support line; behavior impact depends on surrounding block context. |
| 221 |     OilType::Engine5w30, | executable/support line; behavior impact depends on surrounding block context. |
| 222 |     OilType::Engine5w40, | executable/support line; behavior impact depends on surrounding block context. |
| 223 |     OilType::Engine10w30, | executable/support line; behavior impact depends on surrounding block context. |
| 224 |     OilType::Engine15w40, | executable/support line; behavior impact depends on surrounding block context. |
| 225 |     OilType::Engine20w50, | executable/support line; behavior impact depends on surrounding block context. |
| 226 |     OilType::TransformerInhibited, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 227 |     OilType::Turbine32, | executable/support line; behavior impact depends on surrounding block context. |
| 228 |     OilType::Turbine46, | executable/support line; behavior impact depends on surrounding block context. |
| 229 |     OilType::Compressor46, | executable/support line; behavior impact depends on surrounding block context. |
| 230 |     OilType::Compressor68, | executable/support line; behavior impact depends on surrounding block context. |
| 231 |     OilType::BiodegradableHees46, | executable/support line; behavior impact depends on surrounding block con
t. |
| 232 |     OilType::BiodegradableHepr46, | executable/support line; behavior impact depends on surrounding block con
t. |
| 233 |     OilType::FireResistantHfdu46, | executable/support line; behavior impact depends on surrounding block con
t. |
| 234 |     OilType::BrakeDot3, | executable/support line; behavior impact depends on surrounding block context. |
| 235 |     OilType::BrakeDot4, | executable/support line; behavior impact depends on surrounding block context. |
| 236 |     OilType::BrakeDot51, | executable/support line; behavior impact depends on surrounding block context. |
| 237 |     OilType::SyntheticEsters100, | executable/support line; behavior impact depends on surrounding block cont
. |
| 238 |     OilType::RefrigerationMineral3gs, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 239 |     OilType::Pag46, | executable/support line; behavior impact depends on surrounding block context. |
| 240 |     OilType::Pag100, | executable/support line; behavior impact depends on surrounding block context. |
| 241 |     OilType::Pag150, | executable/support line; behavior impact depends on surrounding block context. |

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| 242 | OilType::Poe32, | executable/support line; behavior impact depends on surrounding block context. |
| 243 | OilType::Poe46, | executable/support line; behavior impact depends on surrounding block context. |
| 244 | OilType::Poe68, | executable/support line; behavior impact depends on surrounding block context. |
| 245 | OilType::Poe100, | executable/support line; behavior impact depends on surrounding block context. |
| 246 | ]; | executable/support line; behavior impact depends on surrounding block context. |
| 247 | (blank) | line is blank; visual separation only. |
| 248 | impl OilType { | implementation block; method behavior and trait conformance are defined here. |
| 249 |     pub fn all() -> &'static [OilType] { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 250 |         &OIL_TYPE_VARIANTS | executable/support line; behavior impact depends on surrounding block context. |
| 251 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 252 | (blank) | line is blank; visual separation only. |
| 253 |     pub fn name(self) -> &'static str { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 254 |         self.runtime_profile().label | executable/support line; behavior impact depends on surrounding block context. |
| 255 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 256 | (blank) | line is blank; visual separation only. |
| 257 |     pub fn runtime_profile(self) -> OilRuntimeProfile { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 258 |         match self { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 259 |             OilType::HydraulicIso22 => OilRuntimeProfile { label: "Hydraulic ISO VG 22", density_kg_m3: 852.0, viscosity_index: 108.0, min_oper_temp_c: -20.0, max_oper_temp_c: 85.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 260 |             OilType::HydraulicIso32 => OilRuntimeProfile { label: "Hydraulic ISO VG 32", density_kg_m3: 858.0, viscosity_index: 106.0, min_oper_temp_c: -18.0, max_oper_temp_c: 90.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 261 |             OilType::HydraulicIso46 => OilRuntimeProfile { label: "Hydraulic ISO VG 46", density_kg_m3: 865.0, viscosity_index: 105.0, min_oper_temp_c: -15.0, max_oper_temp_c: 95.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 262 |             OilType::HydraulicIso68 => OilRuntimeProfile { label: "Hydraulic ISO VG 68", density_kg_m3: 875.0, viscosity_index: 102.0, min_oper_temp_c: -10.0, max_oper_temp_c: 100.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 263 |             OilType::HydraulicIso100 => OilRuntimeProfile { label: "Hydraulic ISO VG 100", density_kg_m3: 882.0, viscosity_index: 100.0, min_oper_temp_c: -8.0, max_oper_temp_c: 105.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 264 |             OilType::HydraulicIso150 => OilRuntimeProfile { label: "Hydraulic ISO VG 150", density_kg_m3: 890.0, viscosity_index: 98.0, min_oper_temp_c: -5.0, max_oper_temp_c: 110.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 265 |             OilType::HydraulicIso220 => OilRuntimeProfile { label: "Hydraulic ISO VG 220", density_kg_m3: 896.0, viscosity_index: 97.0, min_oper_temp_c: 0.0, max_oper_temp_c: 115.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 266 |             OilType::Aw32 => OilRuntimeProfile { label: "AW Hydraulic 32", density_kg_m3: 860.0, viscosity_index: 110.0, min_oper_temp_c: -20.0, max_oper_temp_c: 95.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 267 |             OilType::Aw46 => OilRuntimeProfile { label: "AW Hydraulic 46", density_kg_m3: 868.0, viscosity_index: 108.0, min_oper_temp_c: -18.0, max_oper_temp_c: 100.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 268 |             OilType::Aw68 => OilRuntimeProfile { label: "AW Hydraulic 68", density_kg_m3: 878.0, viscosity_index: 105.0, min_oper_temp_c: -12.0, max_oper_temp_c: 105.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 269 |             OilType::Utto10w30 => OilRuntimeProfile { label: "UTTO 10W-30", density_kg_m3: 872.0, viscosity_index: 142.0, min_oper_temp_c: -30.0, max_oper_temp_c: 115.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 270 |             OilType::Utto80w => OilRuntimeProfile { label: "UTTO 80W", density_kg_m3: 885.0, viscosity_index: 138.0, min_oper_temp_c: -20.0, max_oper_temp_c: 120.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 271 |             OilType::Gear75w90 => OilRuntimeProfile { label: "Gear Oil 75W-90", density_kg_m3: 875.0, viscosity_index: 155.0, min_oper_temp_c: -40.0, max_oper_temp_c: 140.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 272 |             OilType::Gear80w90 => OilRuntimeProfile { label: "Gear Oil 80W-90", density_kg_m3: 890.0, viscosity_index: 135.0, min_oper_temp_c: -26.0, max_oper_temp_c: 140.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 273 |             OilType::AtfDexronIii => OilRuntimeProfile { label: "ATF Dexron III", density_kg_m3: 850.0, viscosity_index: 180.0, min_oper_temp_c: -40.0, max_oper_temp_c: 130.0 }, | executable/support line; behavior impact depends

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surrounding block context. |
| 274 |           OilType::AtfDexronVi => OilRuntimeProfile { label: "ATF Dexron VI", density_kg_m3: 846.0, viscosity_index: 165.0, min_oper_temp_c: -45.0, max_oper_temp_c: 135.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 275 |           OilType::Engine0w20 => OilRuntimeProfile { label: "Engine 0W-20", density_kg_m3: 845.0, viscosity_index: 170.0, min_oper_temp_c: -40.0, max_oper_temp_c: 125.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 276 |           OilType::Engine5w30 => OilRuntimeProfile { label: "Engine 5W-30", density_kg_m3: 855.0, viscosity_index: 165.0, min_oper_temp_c: -35.0, max_oper_temp_c: 130.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 277 |           OilType::Engine5w40 => OilRuntimeProfile { label: "Engine 5W-40", density_kg_m3: 862.0, viscosity_index: 162.0, min_oper_temp_c: -35.0, max_oper_temp_c: 135.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 278 |           OilType::Engine10w30 => OilRuntimeProfile { label: "Engine 10W-30", density_kg_m3: 870.0, viscosity_index: 150.0, min_oper_temp_c: -30.0, max_oper_temp_c: 130.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 279 |           OilType::Engine15w40 => OilRuntimeProfile { label: "Engine 15W-40", density_kg_m3: 880.0, viscosity_index: 140.0, min_oper_temp_c: -25.0, max_oper_temp_c: 135.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 280 |           OilType::Engine20w50 => OilRuntimeProfile { label: "Engine 20W-50", density_kg_m3: 892.0, viscosity_index: 132.0, min_oper_temp_c: -15.0, max_oper_temp_c: 140.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 281 |           OilType::TransformerInhibited => OilRuntimeProfile { label: "Transformer Oil Inhibited", density_kg_m3: 885.0, viscosity_index: 95.0, min_oper_temp_c: -30.0, max_oper_temp_c: 110.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 282 |           OilType::Turbine32 => OilRuntimeProfile { label: "Turbine Oil 32", density_kg_m3: 855.0, viscosity_index: 102.0, min_oper_temp_c: -10.0, max_oper_temp_c: 120.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 283 |           OilType::Turbine46 => OilRuntimeProfile { label: "Turbine Oil 46", density_kg_m3: 865.0, viscosity_index: 100.0, min_oper_temp_c: -8.0, max_oper_temp_c: 125.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 284 |           OilType::Compressor46 => OilRuntimeProfile { label: "Compressor Oil 46", density_kg_m3: 862.0, viscosity_index: 108.0, min_oper_temp_c: -15.0, max_oper_temp_c: 150.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 285 |           OilType::Compressor68 => OilRuntimeProfile { label: "Compressor Oil 68", density_kg_m3: 872.0, viscosity_index: 105.0, min_oper_temp_c: -10.0, max_oper_temp_c: 155.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 286 |           OilType::BiodegradableHees46 => OilRuntimeProfile { label: "HEES Biodegradable 46", density_kg_m3: 820.0, viscosity_index: 190.0, min_oper_temp_c: -35.0, max_oper_temp_c: 90.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 287 |           OilType::BiodegradableHepr46 => OilRuntimeProfile { label: "HEPR Biodegradable 46", density_kg_m3: 800.0, viscosity_index: 165.0, min_oper_temp_c: -30.0, max_oper_temp_c: 105.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 288 |           OilType::FireResistantHfdu46 => OilRuntimeProfile { label: "HF DU Fire Resistant 46", density_kg_m3: 930.0, viscosity_index: 150.0, min_oper_temp_c: -20.0, max_oper_temp_c: 160.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 289 |           OilType::BrakeDot3 => OilRuntimeProfile { label: "Brake Fluid DOT3", density_kg_m3: 1035.0, viscosity_index: 220.0, min_oper_temp_c: -40.0, max_oper_temp_c: 205.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 290 |           OilType::BrakeDot4 => OilRuntimeProfile { label: "Brake Fluid DOT4", density_kg_m3: 1060.0, viscosity_index: 230.0, min_oper_temp_c: -45.0, max_oper_temp_c: 230.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 291 |           OilType::BrakeDot51 => OilRuntimeProfile { label: "Brake Fluid DOT5.1", density_kg_m3: 1045.0, viscosity_index: 240.0, min_oper_temp_c: -50.0, max_oper_temp_c: 240.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 292 |           OilType::SyntheticEsters100 => OilRuntimeProfile { label: "Synthetic Ester 100", density_kg_m3: 900.0, viscosity_index: 175.0, min_oper_temp_c: -35.0, max_oper_temp_c: 180.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 293 |           OilType::RefrigerationMineral3gs => OilRuntimeProfile { label: "Refrigeration Mineral 3GS", density_kg_m3: 895.0, viscosity_index: 85.0, min_oper_temp_c: -40.0, max_oper_temp_c: 80.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 294 |           OilType::Pag46 => OilRuntimeProfile { label: "PAG 46", density_kg_m3: 995.0, viscosity_index: 130.0, min_oper_temp_c: -40.0, max_oper_temp_c: 170.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 295 |           OilType::Pag100 => OilRuntimeProfile { label: "PAG 100", density_kg_m3: 1002.0, viscosity_index:
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.0, min_oper_temp_c: -35.0, max_oper_temp_c: 175.0 }, | executable/support line; behavior impact depends on surrounding
block context. |
| 296 |           OilType::Pag150 => OilRuntimeProfile { label: "PAG 150", density_kg_m3: 1010.0, viscosity_index:
.0, min_oper_temp_c: -30.0, max_oper_temp_c: 180.0 }, | executable/support line; behavior impact depends on surrounding
block context. |
| 297 |           OilType::Poe32 => OilRuntimeProfile { label: "POE 32", density_kg_m3: 970.0, viscosity_index: 140
min_oper_temp_c: -45.0, max_oper_temp_c: 170.0 }, | executable/support line; behavior impact depends on surrounding
ck context. |
| 298 |           OilType::Poe46 => OilRuntimeProfile { label: "POE 46", density_kg_m3: 975.0, viscosity_index: 143
min_oper_temp_c: -40.0, max_oper_temp_c: 175.0 }, | executable/support line; behavior impact depends on surrounding
ck context. |
| 299 |           OilType::Poe68 => OilRuntimeProfile { label: "POE 68", density_kg_m3: 980.0, viscosity_index: 145
min_oper_temp_c: -35.0, max_oper_temp_c: 180.0 }, | executable/support line; behavior impact depends on surrounding
ck context. |
| 300 |           OilType::Poe100 => OilRuntimeProfile { label: "POE 100", density_kg_m3: 988.0, viscosity_index: 1
0, min_oper_temp_c: -30.0, max_oper_temp_c: 185.0 }, | executable/support line; behavior impact depends on surrounding
lock context. |
| 301 |           OilType::VacuumPump100 => OilRuntimeProfile { label: "Vacuum Pump 100", density_kg_m3: 885.0, vis
ity_index: 98.0, min_oper_temp_c: -10.0, max_oper_temp_c: 135.0 }, | executable/support line; behavior impact depends
surrounding block context. |
| 302 |           OilType::Custom => OilRuntimeProfile { label: "Custom", density_kg_m3: 900.0, viscosity_index: 12
, min_oper_temp_c: -20.0, max_oper_temp_c: 120.0 }, | executable/support line; behavior impact depends on surrounding
ock context. |
| 303 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 304 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 305 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 306 | (blank) | line is blank; visual separation only. |
| 307 | impl OilType { | implementation block; method behavior and trait conformance are defined here. |
| 308 |     pub fn density_kg_m3(self) -> f64 { | declares executable behavior; signature/body changes alter API, con
l flow, and outputs. |
| 309 |         self.runtime_profile().density_kg_m3 | executable/support line; behavior impact depends on surrounding
lock context. |
| 310 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 311 | (blank) | line is blank; visual separation only. |
| 312 |     pub fn viscosity_index(self) -> f64 { | declares executable behavior; signature/body changes alter API, c
rol flow, and outputs. |
| 313 |         self.runtime_profile().viscosity_index | executable/support line; behavior impact depends on surround
block context. |
| 314 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 315 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 316 | (blank) | line is blank; visual separation only. |
| 317 | impl std::fmt::Display for LeakAlertLevel { | implementation block; method behavior and trait conformance are
fined here. |
| 318 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; sign
re/body changes alter API, control flow, and outputs. |
| 319 |         let s = match self { | executable/support line; behavior impact depends on surrounding block context.
| 320 |             LeakAlertLevel::Normal => "NORMAL", | executable/support line; behavior impact depends on surround
g block context. |
| 321 |             LeakAlertLevel::Watch => "WATCH", | executable/support line; behavior impact depends on surroundi
block context. |
| 322 |             LeakAlertLevel::Warning => "WARNING", | executable/support line; behavior impact depends on surro
ing block context. |
| 323 |             LeakAlertLevel::Critical => "CRITICAL", | executable/support line; behavior impact depends on sur
nding block context. |
| 324 |             LeakAlertLevel::Ruptured => "RUPTURED", | executable/support line; behavior impact depends on sur
nding block context. |
| 325 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 326 |         write!(f, "{}", s) | I/O or protocol-critical logic; edits can affect integration, timing, and diagno
c outcomes. |
| 327 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 328 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 329 | (blank) | line is blank; visual separation only. |
| 330 | #[derive(Debug, Clone, Copy, PartialEq, Eq, Serialize, Deserialize)] | serialization contract touchpoint; edi

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```

can break report compatibility and tooling. |
| 331 | pub enum OringMaterial { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 332 |     Nbr, | executable/support line; behavior impact depends on surrounding block context. |
| 333 |     Hnbr, | executable/support line; behavior impact depends on surrounding block context. |
| 334 |     Fkm, | executable/support line; behavior impact depends on surrounding block context. |
| 335 |     Epdm, | executable/support line; behavior impact depends on surrounding block context. |
| 336 |     Silicone, | executable/support line; behavior impact depends on surrounding block context. |
| 337 |     Fluorosilicone, | executable/support line; behavior impact depends on surrounding block context. |
| 338 |     Aflas, | executable/support line; behavior impact depends on surrounding block context. |
| 339 |     Ffkm, | executable/support line; behavior impact depends on surrounding block context. |
| 340 |     Ptfе, | executable/support line; behavior impact depends on surrounding block context. |
| 341 |     Polyurethane, | executable/support line; behavior impact depends on surrounding block context. |
| 342 |     Acм, | executable/support line; behavior impact depends on surrounding block context. |
| 343 |     Cr, | executable/support line; behavior impact depends on surrounding block context. |
| 344 |     Vmq, | executable/support line; behavior impact depends on surrounding block context. |
| 345 |     Fvmq, | executable/support line; behavior impact depends on surrounding block context. |
| 346 |     Sbr, | executable/support line; behavior impact depends on surrounding block context. |
| 347 |     Ecor, | executable/support line; behavior impact depends on surrounding block context. |
| 348 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 349 | (blank) | line is blank; visual separation only. |
| 350 | #[derive(Debug, Clone, Copy, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 351 | pub struct MaterialRuntimeProfile { | declares data layout; field changes may break serialization, tests, and toolers. |
| 352 |     pub label: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 353 |     pub min_temp_c: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 354 |     pub max_temp_c: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 355 |     pub gas_permeability_factor: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 356 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 357 | (blank) | line is blank; visual separation only. |
| 358 | const ORING_MATERIAL_VARIANTS: [OringMaterial; 12] = [ | executable/support line; behavior impact depends on surrounding block context. |
| 359 |     OringMaterial::Nbr, | executable/support line; behavior impact depends on surrounding block context. |
| 360 |     OringMaterial::Hnbr, | executable/support line; behavior impact depends on surrounding block context. |
| 361 |     OringMaterial::Fkm, | executable/support line; behavior impact depends on surrounding block context. |
| 362 |     OringMaterial::Epdm, | executable/support line; behavior impact depends on surrounding block context. |
| 363 |     OringMaterial::Silicone, | executable/support line; behavior impact depends on surrounding block context. |
| 364 |     OringMaterial::Fluorosilicone, | executable/support line; behavior impact depends on surrounding block context. |
| 365 |     OringMaterial::Aflas, | executable/support line; behavior impact depends on surrounding block context. |
| 366 |     OringMaterial::Ffkm, | executable/support line; behavior impact depends on surrounding block context. |
| 367 |     OringMaterial::Ptfе, | executable/support line; behavior impact depends on surrounding block context. |
| 368 |     OringMaterial::Polyurethane, | executable/support line; behavior impact depends on surrounding block context. |
| 369 |     OringMaterial::Acм, | executable/support line; behavior impact depends on surrounding block context. |
| 370 |     OringMaterial::Cr, | executable/support line; behavior impact depends on surrounding block context. |
| 371 | ]; | executable/support line; behavior impact depends on surrounding block context. |
| 372 | (blank) | line is blank; visual separation only. |
| 373 | impl OringMaterial { | implementation block; method behavior and trait conformance are defined here. |
| 374 |     pub fn all() -> &'static [OringMaterial] { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 375 |         &ORING_MATERIAL_VARIANTS | executable/support line; behavior impact depends on surrounding block context. |
| 376 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 377 | (blank) | line is blank; visual separation only. |
| 378 |     pub fn name(self) -> &'static str { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 379 |         self.runtime_profile().label | executable/support line; behavior impact depends on surrounding block context. |
| 380 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 381 | (blank) | line is blank; visual separation only. |
| 382 |     pub fn runtime_profile(self) -> MaterialRuntimeProfile { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |

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| 383 |         match self { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 384 |             OringMaterial::Nbr => MaterialRuntimeProfile { label: "NBR", min_temp_c: -35.0, max_temp_c: 110.0,
as_permeability_factor: 1.00 }, | executable/support line; behavior impact depends on surrounding block context. |
| 385 |             OringMaterial::Hnbr => MaterialRuntimeProfile { label: "HNBR", min_temp_c: -40.0, max_temp_c: 150.0,
gas_permeability_factor: 0.92 }, | executable/support line; behavior impact depends on surrounding block context. |
| 386 |             OringMaterial::Fkm => MaterialRuntimeProfile { label: "FKM", min_temp_c: -20.0, max_temp_c: 200.0,
as_permeability_factor: 0.72 }, | executable/support line; behavior impact depends on surrounding block context. |
| 387 |             OringMaterial::Epdm => MaterialRuntimeProfile { label: "EPDM", min_temp_c: -45.0, max_temp_c: 140.0,
gas_permeability_factor: 1.10 }, | executable/support line; behavior impact depends on surrounding block context. |
| 388 |             OringMaterial::Silicone => MaterialRuntimeProfile { label: "VMQ Silicone", min_temp_c: -60.0, max_
mp_c: 210.0, gas_permeability_factor: 1.35 }, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 389 |             OringMaterial::Fluorosilicone => MaterialRuntimeProfile { label: "FVMQ Fluorosilicone", min_temp_
-58.0, max_temp_c: 190.0, gas_permeability_factor: 1.18 }, | executable/support line; behavior impact depends on surr
ding block context. |
| 390 |             OringMaterial::Aflas => MaterialRuntimeProfile { label: "AFLAS", min_temp_c: -10.0, max_temp_c: 2
0, gas_permeability_factor: 0.80 }, | executable/support line; behavior impact depends on surrounding block context.
| 391 |             OringMaterial::Ffkm => MaterialRuntimeProfile { label: "FFKM", min_temp_c: -15.0, max_temp_c: 320
gas_permeability_factor: 0.52 }, | executable/support line; behavior impact depends on surrounding block context |
| 392 |             OringMaterial::Ptfе => MaterialRuntimeProfile { label: "PTFE", min_temp_c: -80.0, max_temp_c: 260
gas_permeability_factor: 0.58 }, | executable/support line; behavior impact depends on surrounding block context. |
| 393 |             OringMaterial::Polyurethane => MaterialRuntimeProfile { label: "PU", min_temp_c: -35.0, max_temp_
105.0, gas_permeability_factor: 0.88 }, | executable/support line; behavior impact depends on surrounding block conte
|
| 394 |             OringMaterial::Acm => MaterialRuntimeProfile { label: "ACM", min_temp_c: -25.0, max_temp_c: 165.0,
as_permeability_factor: 0.86 }, | executable/support line; behavior impact depends on surrounding block context. |
| 395 |             OringMaterial::Cr => MaterialRuntimeProfile { label: "CR", min_temp_c: -40.0, max_temp_c: 115.0,
_permeability_factor: 1.08 }, | executable/support line; behavior impact depends on surrounding block context. |
| 396 |             OringMaterial::Vmq => MaterialRuntimeProfile { label: "VMQ", min_temp_c: -60.0, max_temp_c: 205.0,
as_permeability_factor: 1.35 }, | executable/support line; behavior impact depends on surrounding block context. |
| 397 |             OringMaterial::Fvmq => MaterialRuntimeProfile { label: "FVMQ", min_temp_c: -58.0, max_temp_c: 190
gas_permeability_factor: 1.18 }, | executable/support line; behavior impact depends on surrounding block context. |
| 398 |             OringMaterial::Sbr => MaterialRuntimeProfile { label: "SBR", min_temp_c: -40.0, max_temp_c: 100.0,
as_permeability_factor: 1.22 }, | executable/support line; behavior impact depends on surrounding block context. |
| 399 |             OringMaterial::Ecor => MaterialRuntimeProfile { label: "ECO", min_temp_c: -35.0, max_temp_c: 140.
gas_permeability_factor: 0.95 }, | executable/support line; behavior impact depends on surrounding block context. |
| 400 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 401 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 402 | (blank) | line is blank; visual separation only. |
| 403 |     pub fn temp_window_c(self) -> (f64, f64) { | declares executable behavior; signature/body changes alter A
control flow, and outputs. |
| 404 |         let p = self.runtime_profile(); | executable/support line; behavior impact depends on surrounding blo
context. |
| 405 |         (p.min_temp_c, p.max_temp_c) | executable/support line; behavior impact depends on surrounding block
text. |
| 406 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 407 | (blank) | line is blank; visual separation only. |
| 408 |     pub fn gas_permeability_factor(self) -> f64 { | declares executable behavior; signature/body changes alte
PI, control flow, and outputs. |
| 409 |         self.runtime_profile().gas_permeability_factor | executable/support line; behavior impact depends on
rounding block context. |
| 410 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 411 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 412 | (blank) | line is blank; visual separation only. |
| 413 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report c
atibility and tooling. |
| 414 | pub struct MaterialCatalogEntry { | declares data layout; field changes may break serialization, tests, and c
ers. |
| 415 |     pub id: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 416 |     pub family: &'static str, | executable/support line; behavior impact depends on surrounding block context
| 417 |     pub designation: &'static str, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 418 |     pub hardness_shore: f64, | executable/support line; behavior impact depends on surrounding block context.
| 419 |     pub tensile_strength_mpa: f64, | executable/support line; behavior impact depends on surrounding block co

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xt. |
| 420 | pub elongation_pct: f64, | executable/support line; behavior impact depends on surrounding block context.
| 421 | pub compression_set_pct_70h: f64, | executable/support line; behavior impact depends on surrounding block context.
ntext. |
| 422 | pub tear_strength_kn_m: f64, | executable/support line; behavior impact depends on surrounding block context.
. |
| 423 | pub density_g_cm3: f64, | executable/support line; behavior impact depends on surrounding block context.
| 424 | pub temp_min_c: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 425 | pub temp_max_c: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 426 | pub abrasion_index: f64, | executable/support line; behavior impact depends on surrounding block context.
| 427 | pub corrosion_resistance_index: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 428 | pub hydraulic_oil_compat: f64, | executable/support line; behavior impact depends on surrounding block context.
xt. |
| 429 | pub engine_oil_compat: f64, | executable/support line; behavior impact depends on surrounding block context.
|
| 430 | pub refrigerant_oil_compat: f64, | executable/support line; behavior impact depends on surrounding block context.
|
| 431 | pub notes: &'static str, | executable/support line; behavior impact depends on surrounding block context.
| 432 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 433 | (blank) | line is blank; visual separation only. |
| 434 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report card
atibility and tooling. |
| 435 | pub struct OilCatalogEntry { | declares data layout; field changes may break serialization, tests, and caller
|
| 436 | pub id: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 437 | pub family: &'static str, | executable/support line; behavior impact depends on surrounding block context.
| 438 | pub grade: &'static str, | executable/support line; behavior impact depends on surrounding block context.
| 439 | pub base_stock: &'static str, | executable/support line; behavior impact depends on surrounding block context.
t. |
| 440 | pub density_15c_kg_m3: f64, | executable/support line; behavior impact depends on surrounding block context.
|
| 441 | pub viscosity_40c_cst: f64, | executable/support line; behavior impact depends on surrounding block context.
|
| 442 | pub viscosity_100c_cst: f64, | executable/support line; behavior impact depends on surrounding block context.
. |
| 443 | pub viscosity_index: f64, | executable/support line; behavior impact depends on surrounding block context.
| 444 | pub pour_point_c: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 445 | pub flash_point_c: f64, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic
ic outcomes. |
| 446 | pub tan_mg_koh_g: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 447 | pub water_content_ppm_limit: f64, | executable/support line; behavior impact depends on surrounding block context.
ntext. |
| 448 | pub demulsibility_min: f64, | executable/support line; behavior impact depends on surrounding block context.
|
| 449 | pub copper_corrosion_level: f64, | executable/support line; behavior impact depends on surrounding block context.
text. |
| 450 | pub oxidation_stability_h: f64, | executable/support line; behavior impact depends on surrounding block context.
ext. |
| 451 | pub recommended_min_temp_c: f64, | executable/support line; behavior impact depends on surrounding block context.
text. |
| 452 | pub recommended_max_temp_c: f64, | executable/support line; behavior impact depends on surrounding block context.
text. |
| 453 | pub anti_wear_index: f64, | executable/support line; behavior impact depends on surrounding block context.
| 454 | pub corrosion_inhibition_index: f64, | executable/support line; behavior impact depends on surrounding block context.
context. |
| 455 | pub notes: &'static str, | executable/support line; behavior impact depends on surrounding block context.
| 456 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 457 | (blank) | line is blank; visual separation only. |
| 458 | pub fn engineering_material_catalog() -> Vec<MaterialCatalogEntry> { | declares executable behavior; signature
ody changes alter API, control flow, and outputs. |
| 459 |     vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 460 |         MaterialCatalogEntry { id: "OR-NBR-70", family: "O-RING", designation: "NBR 70", hardness_shore: 70.0,
ensile_strength_mpa: 12.0, elongation_pct: 280.0, compression_set_pct_70h: 18.0, tear_strength_kn_m: 28.0, density_g_cm3:
: 1.20, temp_min_c: -35.0, temp_max_c: 110.0, abrasion_index: 0.72, corrosion_resistance_index: 0.64, hydraulic_oil_compat:

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at: 0.92, engine_oil_compat: 0.86, refrigerant_oil_compat: 0.41, notes: "General hydraulic service" }, | executable/support line; behavior impact depends on surrounding block context. |

| 461 | MaterialCatalogEntry { id: "OR-NBR-90", family: "O-RING", designation: "NBR 90", hardness_shore: 90.0, tensile_strength_mpa: 14.0, elongation_pct: 220.0, compression_set_pct_70h: 16.0, tear_strength_kn_m: 34.0, density_g_cm3: 1.22, temp_min_c: -30.0, temp_max_c: 110.0, abrasion_index: 0.84, corrosion_resistance_index: 0.66, hydraulic_oil_compat: 0.95, engine_oil_compat: 0.88, refrigerant_oil_compat: 0.38, notes: "High-pressure extrusion resistance" }, | executable/support line; behavior impact depends on surrounding block context. |

| 462 | MaterialCatalogEntry { id: "OR-HNBR-80", family: "O-RING", designation: "HNBR 80", hardness_shore: 80.0, tensile_strength_mpa: 17.0, elongation_pct: 260.0, compression_set_pct_70h: 13.0, tear_strength_kn_m: 36.0, density_g_cm3: 1.18, temp_min_c: -40.0, temp_max_c: 150.0, abrasion_index: 0.82, corrosion_resistance_index: 0.71, hydraulic_oil_compat: 0.94, engine_oil_compat: 0.91, refrigerant_oil_compat: 0.65, notes: "Preferred for thermal cycling" }, | executable/support line; behavior impact depends on surrounding block context. |

| 463 | MaterialCatalogEntry { id: "OR-FKM-75", family: "O-RING", designation: "FKM 75", hardness_shore: 75.0, tensile_strength_mpa: 13.0, elongation_pct: 220.0, compression_set_pct_70h: 12.0, tear_strength_kn_m: 24.0, density_g_cm3: 1.85, temp_min_c: -20.0, temp_max_c: 200.0, abrasion_index: 0.63, corrosion_resistance_index: 0.88, hydraulic_oil_compat: 0.86, engine_oil_compat: 0.98, refrigerant_oil_compat: 0.89, notes: "Strong hydrocarbon and heat resistance" }, | executable/support line; behavior impact depends on surrounding block context. |

| 464 | MaterialCatalogEntry { id: "OR-EPDM-70", family: "O-RING", designation: "EPDM 70", hardness_shore: 70.0, tensile_strength_mpa: 11.0, elongation_pct: 320.0, compression_set_pct_70h: 20.0, tear_strength_kn_m: 26.0, density_g_cm3: 1.14, temp_min_c: -45.0, temp_max_c: 140.0, abrasion_index: 0.58, corrosion_resistance_index: 0.79, hydraulic_oil_compat: 0.30, engine_oil_compat: 0.25, refrigerant_oil_compat: 0.78, notes: "Steam/water dominant service" }, | executable/support line; behavior impact depends on surrounding block context. |

| 465 | MaterialCatalogEntry { id: "OR-AFLAS-80", family: "O-RING", designation: "AFLAS 80", hardness_shore: 80.0, tensile_strength_mpa: 14.0, elongation_pct: 200.0, compression_set_pct_70h: 15.0, tear_strength_kn_m: 27.0, density_g_cm3: 1.62, temp_min_c: -10.0, temp_max_c: 230.0, abrasion_index: 0.66, corrosion_resistance_index: 0.91, hydraulic_oil_compat: 0.88, engine_oil_compat: 0.94, refrigerant_oil_compat: 0.80, notes: "Amines/sour gas tolerant" }, | executable/support line; behavior impact depends on surrounding block context. |

| 466 | MaterialCatalogEntry { id: "OR-FFKM-75", family: "O-RING", designation: "FFKM 75", hardness_shore: 75.0, tensile_strength_mpa: 16.0, elongation_pct: 170.0, compression_set_pct_70h: 10.0, tear_strength_kn_m: 20.0, density_g_cm3: 1.95, temp_min_c: -15.0, temp_max_c: 320.0, abrasion_index: 0.55, corrosion_resistance_index: 0.97, hydraulic_oil_compat: 0.96, engine_oil_compat: 0.99, refrigerant_oil_compat: 0.94, notes: "Extreme high-temp chemical duty" }, | executable/support line; behavior impact depends on surrounding block context. |

| 467 | MaterialCatalogEntry { id: "OR-SIL-70", family: "O-RING", designation: "VMQ 70", hardness_shore: 70.0, tensile_strength_mpa: 9.0, elongation_pct: 350.0, compression_set_pct_70h: 24.0, tear_strength_kn_m: 18.0, density_g_cm3: 1.10, temp_min_c: -60.0, temp_max_c: 210.0, abrasion_index: 0.42, corrosion_resistance_index: 0.76, hydraulic_oil_compat: 0.48, engine_oil_compat: 0.38, refrigerant_oil_compat: 0.72, notes: "Low-temp flexibility priority" }, | executable/support line; behavior impact depends on surrounding block context. |

| 468 | MaterialCatalogEntry { id: "OR-FSIL-70", family: "O-RING", designation: "FVMQ 70", hardness_shore: 70.0, tensile_strength_mpa: 10.0, elongation_pct: 280.0, compression_set_pct_70h: 20.0, tear_strength_kn_m: 19.0, density_g_cm3: 1.34, temp_min_c: -58.0, temp_max_c: 190.0, abrasion_index: 0.45, corrosion_resistance_index: 0.80, hydraulic_oil_compat: 0.70, engine_oil_compat: 0.74, refrigerant_oil_compat: 0.88, notes: "Aviation fuel and oil compatibility" }, | executable/support line; behavior impact depends on surrounding block context. |

| 469 | MaterialCatalogEntry { id: "OR-PTFE", family: "O-RING", designation: "PTFE Virgin", hardness_shore: 60.0, tensile_strength_mpa: 25.0, elongation_pct: 220.0, compression_set_pct_70h: 6.0, tear_strength_kn_m: 14.0, density_g_cm3: 2.16, temp_min_c: -80.0, temp_max_c: 260.0, abrasion_index: 0.39, corrosion_resistance_index: 0.96, hydraulic_oil_compat: 0.98, engine_oil_compat: 0.98, refrigerant_oil_compat: 0.95, notes: "Low friction, needs backup ring" }, | executable/support line; behavior impact depends on surrounding block context. |

| 470 | MaterialCatalogEntry { id: "OR-PU-93", family: "O-RING", designation: "PU 93A", hardness_shore: 93.0, tensile_strength_mpa: 42.0, elongation_pct: 430.0, compression_set_pct_70h: 22.0, tear_strength_kn_m: 75.0, density_g_cm3: 1.21, temp_min_c: -35.0, temp_max_c: 105.0, abrasion_index: 0.95, corrosion_resistance_index: 0.65, hydraulic_oil_compat: 0.90, engine_oil_compat: 0.80, refrigerant_oil_compat: 0.52, notes: "Excellent abrasion and extrusion resistance" }, | executable/support line; behavior impact depends on surrounding block context. |

| 471 | MaterialCatalogEntry { id: "OR-ACM-70", family: "O-RING", designation: "ACM 70", hardness_shore: 70.0, tensile_strength_mpa: 11.5, elongation_pct: 210.0, compression_set_pct_70h: 14.0, tear_strength_kn_m: 21.0, density_g_cm3: 1.24, temp_min_c: -25.0, temp_max_c: 165.0, abrasion_index: 0.60, corrosion_resistance_index: 0.82, hydraulic_oil_compat: 0.81, engine_oil_compat: 0.93, refrigerant_oil_compat: 0.59, notes: "Hot engine oil applications" }, | executable/support line; behavior impact depends on surrounding block context. |

| 472 | MaterialCatalogEntry { id: "HS-NBR-TUBE", family: "HOSE_INNER_TUBE", designation: "NBR Tube", hardness_shore: 75.0, tensile_strength_mpa: 14.0, elongation_pct: 250.0, compression_set_pct_70h: 18.0, tear_strength_kn_m: 29.0, density_g_cm3: 1.18, temp_min_c: -30.0, temp_max_c: 120.0, abrasion_index: 0.74, corrosion_resistance_index: 0.68, hydraulic_oil_compat: 0.94, engine_oil_compat: 0.89, refrigerant_oil_compat: 0.40, notes: "General hydraulic hoses" }, | executable/support line; behavior impact depends on surrounding block context. |

| 473 | MaterialCatalogEntry { id: "HS-HNBR-TUBE", family: "HOSE_INNER_TUBE", designation: "HNBR Tube", hardness_shore: 78.0, tensile_strength_mpa: 17.0, elongation_pct: 260.0, compression_set_pct_70h: 14.0, tear_strength_kn_m: 30.0, density_g_cm3: 1.20, temp_min_c: -40.0, temp_max_c: 150.0, abrasion_index: 0.81, corrosion_resistance_index: 0.74, notes: "High-pressure hydraulic hoses" }, | executable/support line; behavior impact depends on surrounding block context. |

raulic_oil_compat: 0.95, engine_oil_compat: 0.92, refrigerant_oil_compat: 0.62, notes: "High-temp pressure hose" }, |
executable/support line; behavior impact depends on surrounding block context. |
474	MaterialCatalogEntry { id: "HS-CSM-COVER", family: "HOSE_COVER", designation: "CSM Cover", hardness_shore: 68.0, tensile_strength_mpa: 12.0, elongation_pct: 300.0, compression_set_pct_70h: 22.0, tear_strength_kn_m: 23.0, density_g_cm3: 1.30, temp_min_c: -25.0, temp_max_c: 125.0, abrasion_index: 0.70, corrosion_resistance_index: 0.77, hydraulic_oil_compat: 0.76, engine_oil_compat: 0.71, refrigerant_oil_compat: 0.63, notes: "Weather/ozone resistance" },	executable/support line; behavior impact depends on surrounding block context.
475	MaterialCatalogEntry { id: "HS-PTFE-TUBE", family: "HOSE_INNER_TUBE", designation: "PTFE Hose Tube", hardness_shore: 62.0, tensile_strength_mpa: 24.0, elongation_pct: 220.0, compression_set_pct_70h: 7.0, tear_strength_kn_m: 16.0, density_g_cm3: 2.15, temp_min_c: -60.0, temp_max_c: 260.0, abrasion_index: 0.40, corrosion_resistance_index: 0.77, hydraulic_oil_compat: 0.99, engine_oil_compat: 0.99, refrigerant_oil_compat: 0.95, notes: "Chemical extreme hose assemblies" },	executable/support line; behavior impact depends on surrounding block context.
476	MaterialCatalogEntry { id: "HS-ARAMID-BRAID", family: "HOSE_REINFORCEMENT", designation: "Aramid Braided Hose", hardness_shore: 0.0, tensile_strength_mpa: 3000.0, elongation_pct: 3.0, compression_set_pct_70h: 0.0, tear_strength_kn_m: 0.0, density_g_cm3: 1.44, temp_min_c: -40.0, temp_max_c: 180.0, abrasion_index: 0.88, corrosion_resistance_index: 0.77, hydraulic_oil_compat: 0.95, engine_oil_compat: 0.95, refrigerant_oil_compat: 0.95, notes: "High burst-pressure reinforcement" },	executable/support line; behavior impact depends on surrounding block context.
477	MaterialCatalogEntry { id: "HS-STEEL-WIRE", family: "HOSE_REINFORCEMENT", designation: "High-tensile Steel Wire", hardness_shore: 0.0, tensile_strength_mpa: 2500.0, elongation_pct: 2.0, compression_set_pct_70h: 0.0, tear_strength_kn_m: 0.0, density_g_cm3: 7.85, temp_min_c: -50.0, temp_max_c: 200.0, abrasion_index: 0.94, corrosion_resistance_index: 0.74, hydraulic_oil_compat: 0.98, engine_oil_compat: 0.98, refrigerant_oil_compat: 0.98, notes: "Spiral-wire hose reinforcement" },	executable/support line; behavior impact depends on surrounding block context.
478	MaterialCatalogEntry { id: "SL-PTFE-LIP", family: "DYNAMIC_SEAL", designation: "PTFE Lip Seal", hardness_shore: 63.0, tensile_strength_mpa: 22.0, elongation_pct: 210.0, compression_set_pct_70h: 8.0, tear_strength_kn_m: 15.0, density_g_cm3: 2.12, temp_min_c: -70.0, temp_max_c: 250.0, abrasion_index: 0.44, corrosion_resistance_index: 0.95, hydraulic_oil_compat: 0.99, engine_oil_compat: 0.99, refrigerant_oil_compat: 0.92, notes: "Low-friction dynamic shaft seals" },	executable/support line; behavior impact depends on surrounding block context.
479	MaterialCatalogEntry { id: "SL-PU-LIP", family: "DYNAMIC_SEAL", designation: "PU Lip Seal", hardness_shore: 92.0, tensile_strength_mpa: 40.0, elongation_pct: 410.0, compression_set_pct_70h: 20.0, tear_strength_kn_m: 72.0, density_g_cm3: 1.20, temp_min_c: -35.0, temp_max_c: 110.0, abrasion_index: 0.96, corrosion_resistance_index: 0.66, hydraulic_oil_compat: 0.92, engine_oil_compat: 0.84, refrigerant_oil_compat: 0.50, notes: "Heavy-duty rod/piston seals" },	executable/support line; behavior impact depends on surrounding block context.
480	MaterialCatalogEntry { id: "SL-NBR-LIP", family: "DYNAMIC_SEAL", designation: "NBR Lip", hardness_shore: 75.0, tensile_strength_mpa: 13.0, elongation_pct: 260.0, compression_set_pct_70h: 17.0, tear_strength_kn_m: 29.0, density_g_cm3: 1.19, temp_min_c: -35.0, temp_max_c: 120.0, abrasion_index: 0.76, corrosion_resistance_index: 0.66, hydraulic_oil_compat: 0.93, engine_oil_compat: 0.87, refrigerant_oil_compat: 0.42, notes: "Cost-effective rotating shaft seal" },	executable/support line; behavior impact depends on surrounding block context.
481	MaterialCatalogEntry { id: "BK-PTFE-GLASS", family: "BACKUP_RING", designation: "PTFE 25% Glass", hardness_shore: 64.0, tensile_strength_mpa: 28.0, elongation_pct: 120.0, compression_set_pct_70h: 6.0, tear_strength_kn_m: 10.0, density_g_cm3: 2.22, temp_min_c: -70.0, temp_max_c: 260.0, abrasion_index: 0.52, corrosion_resistance_index: 0.96, hydraulic_oil_compat: 0.99, engine_oil_compat: 0.99, refrigerant_oil_compat: 0.94, notes: "Anti-extrusion backup rings" },	executable/support line; behavior impact depends on surrounding block context.
482	MaterialCatalogEntry { id: "BK-PA66", family: "BACKUP_RING", designation: "PA66", hardness_shore: 80.0, tensile_strength_mpa: 78.0, elongation_pct: 60.0, compression_set_pct_70h: 10.0, tear_strength_kn_m: 0.0, density_g_cm3: 1.14, temp_min_c: -30.0, temp_max_c: 120.0, abrasion_index: 0.78, corrosion_resistance_index: 0.72, hydraulic_oil_compat: 0.90, engine_oil_compat: 0.80, refrigerant_oil_compat: 0.57, notes: "Split backup rings for hydraulics" },	executable/support line; behavior impact depends on surrounding block context.
483	MaterialCatalogEntry { id: "BK-PEEK", family: "BACKUP_RING", designation: "PEEK", hardness_shore: 85.0, tensile_strength_mpa: 95.0, elongation_pct: 45.0, compression_set_pct_70h: 5.0, tear_strength_kn_m: 0.0, density_g_cm3: 1.32, temp_min_c: -50.0, temp_max_c: 250.0, abrasion_index: 0.84, corrosion_resistance_index: 0.94, hydraulic_oil_compat: 0.98, engine_oil_compat: 0.97, refrigerant_oil_compat: 0.89, notes: "High-pressure/high-temp anti-extrusion" },	executable/support line; behavior impact depends on surrounding block context.
484	MaterialCatalogEntry { id: "MS-CARBON", family: "MECH_SEAL_FACE", designation: "Carbon Graphite", hardness_shore: 0.0, tensile_strength_mpa: 70.0, elongation_pct: 0.5, compression_set_pct_70h: 0.0, tear_strength_kn_m: 0.0, density_g_cm3: 1.75, temp_min_c: -50.0, temp_max_c: 260.0, abrasion_index: 0.66, corrosion_resistance_index: 0.84, hydraulic_oil_compat: 0.97, engine_oil_compat: 0.95, refrigerant_oil_compat: 0.88, notes: "Mechanical seal rotating face" },	executable/support line; behavior impact depends on surrounding block context.
485	MaterialCatalogEntry { id: "MS-SIC", family: "MECH_SEAL_FACE", designation: "Silicon Carbide", hardness_shore: 0.0, tensile_strength_mpa: 380.0, elongation_pct: 0.1, compression_set_pct_70h: 0.0, tear_strength_kn_m: 0.0, density_g_cm3: 3.10, temp_min_c: -80.0, temp_max_c: 300.0, abrasion_index: 0.98, corrosion_resistance_index: 0.93, hydraulic_oil_compat: 0.99, engine_oil_compat: 0.99, refrigerant_oil_compat: 0.96, notes: "Extreme wear mechanical seal face" },	executable/support line; behavior impact depends on surrounding block context.
486	MaterialCatalogEntry { id: "MS-TC", family: "MECH_SEAL_FACE", designation: "Tungsten Carbide", hardness_shore: 0.0, tensile_strength_mpa: 500.0, elongation_pct: 0.1, compression_set_pct_70h: 0.0, tear_strength_kn_m: 0.0, density_g_cm3: 14.5, temp_min_c: -80.0, temp_max_c: 400.0, abrasion_index: 0.99, corrosion_resistance_index: 0.89, hydraulic_oil_compat: 0.99, engine_oil_compat: 0.99, refrigerant_oil_compat: 0.99, notes: "Extreme wear mechanical seal face" },	executable/support line; behavior impact depends on surrounding block context.

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c_oil_compat: 0.99, engine_oil_compat: 0.99, refrigerant_oil_compat: 0.95, notes: "High-load slurry-capable face" },
executable/support line; behavior impact depends on surrounding block context. |
| 487 |         MaterialCatalogEntry { id: "WR-POM", family: "WEAR_RING", designation: "POM Wear Ring", hardness_shore:
83.0, tensile_strength_mpa: 65.0, elongation_pct: 20.0, compression_set_pct_70h: 9.0, tear_strength_kn_m: 0.0, density_g_cm3: 1.42, temp_min_c: -40.0, temp_max_c: 110.0, abrasion_index: 0.80, corrosion_resistance_index: 0.70, hydraulic_oil_compat: 0.92, engine_oil_compat: 0.84, refrigerant_oil_compat: 0.58, notes: "Hydraulic cylinder guidance" }, | executable/support line; behavior impact depends on surrounding block context. |
| 488 |         MaterialCatalogEntry { id: "WR-PTFE-BRONZE", family: "WEAR_RING", designation: "PTFE Bronze Filled", hardness_shore: 65.0, tensile_strength_mpa: 26.0, elongation_pct: 130.0, compression_set_pct_70h: 7.0, tear_strength_kn_m: 0.0, density_g_cm3: 2.35, temp_min_c: -70.0, temp_max_c: 250.0, abrasion_index: 0.72, corrosion_resistance_index: 0.9, hydraulic_oil_compat: 0.99, engine_oil_compat: 0.98, refrigerant_oil_compat: 0.90, notes: "Low friction wear ring" }, | executable/support line; behavior impact depends on surrounding block context. |
| 489 |         MaterialCatalogEntry { id: "WR-PHENOLIC", family: "WEAR_RING", designation: "Fabric Phenolic", hardness_shore: 88.0, tensile_strength_mpa: 120.0, elongation_pct: 2.0, compression_set_pct_70h: 8.0, tear_strength_kn_m: 0.0, density_g_cm3: 1.35, temp_min_c: -40.0, temp_max_c: 135.0, abrasion_index: 0.86, corrosion_resistance_index: 0.75, hydraulic_oil_compat: 0.94, engine_oil_compat: 0.90, refrigerant_oil_compat: 0.62, notes: "High load bearing guidance" }, | executable/support line; behavior impact depends on surrounding block context. |
| 490 |         MaterialCatalogEntry { id: "OR-ECO-80", family: "O-RING", designation: "ECO 80", hardness_shore: 80.0, tensile_strength_mpa: 12.5, elongation_pct: 240.0, compression_set_pct_70h: 16.0, tear_strength_kn_m: 24.0, density_g_cm3: 1.28, temp_min_c: -35.0, temp_max_c: 140.0, abrasion_index: 0.68, corrosion_resistance_index: 0.78, hydraulic_oil_compat: 0.88, engine_oil_compat: 0.90, refrigerant_oil_compat: 0.66, notes: "Fuel/oil resistant elastomer" }, | executable/support line; behavior impact depends on surrounding block context. |
| 491 |         MaterialCatalogEntry { id: "OR-CR-70", family: "O-RING", designation: "CR 70", hardness_shore: 70.0, tensile_strength_mpa: 11.0, elongation_pct: 300.0, compression_set_pct_70h: 21.0, tear_strength_kn_m: 22.0, density_g_cm3: 1.24, temp_min_c: -40.0, temp_max_c: 115.0, abrasion_index: 0.60, corrosion_resistance_index: 0.73, hydraulic_oil_compat: 0.76, engine_oil_compat: 0.72, refrigerant_oil_compat: 0.70, notes: "Weather and moderate oil service" }, | executable/support line; behavior impact depends on surrounding block context. |
| 492 |         MaterialCatalogEntry { id: "OR-VMQ-50", family: "O-RING", designation: "VMQ 50", hardness_shore: 50.0, tensile_strength_mpa: 8.0, elongation_pct: 420.0, compression_set_pct_70h: 28.0, tear_strength_kn_m: 14.0, density_g_cm3: 1.09, temp_min_c: -65.0, temp_max_c: 200.0, abrasion_index: 0.36, corrosion_resistance_index: 0.74, hydraulic_oil_compat: 0.44, engine_oil_compat: 0.36, refrigerant_oil_compat: 0.70, notes: "High flexibility, low pressure sealing" }, | executable/support line; behavior impact depends on surrounding block context. |
| 493 |     ] | executable/support line; behavior impact depends on surrounding block context. |
| 494 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 495 | (blank) | line is blank; visual separation only. |
| 496 | pub fn engineering_oil_catalog() -> Vec<OilCatalogEntry> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 497 |     vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 498 |         OilCatalogEntry { id: "HYD-ISO-22", family: "HYDRAULIC", grade: "ISO VG 22", base_stock: "Mineral", density_15c_kg_m3: 852.0, viscosity_40c_cst: 22.0, viscosity_100c_cst: 4.5, viscosity_index: 108.0, pour_point_c: -30.0, flash_point_c: 205.0, tan_mg_koh_g: 0.7, water_content_ppm_limit: 300.0, demulsibility_min: 20.0, copper_corrosion_level_1h_h: 0.0, oxidation_stability_h: 1800.0, recommended_min_temp_c: -20.0, recommended_max_temp_c: 85.0, anti_wear_index: 0.78, corrosion_inhibition_index: 0.74, notes: "Fine servo hydraulic systems" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 499 |         OilCatalogEntry { id: "HYD-ISO-32", family: "HYDRAULIC", grade: "ISO VG 32", base_stock: "Mineral", density_15c_kg_m3: 858.0, viscosity_40c_cst: 32.0, viscosity_100c_cst: 5.4, viscosity_index: 106.0, pour_point_c: -27.0, flash_point_c: 215.0, tan_mg_koh_g: 0.7, water_content_ppm_limit: 300.0, demulsibility_min: 20.0, copper_corrosion_level_1h_h: 0.0, oxidation_stability_h: 1900.0, recommended_min_temp_c: -18.0, recommended_max_temp_c: 90.0, anti_wear_index: 0.80, corrosion_inhibition_index: 0.76, notes: "General mobile hydraulics" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 500 |         OilCatalogEntry { id: "HYD-ISO-46", family: "HYDRAULIC", grade: "ISO VG 46", base_stock: "Mineral", density_15c_kg_m3: 865.0, viscosity_40c_cst: 46.0, viscosity_100c_cst: 6.8, viscosity_index: 105.0, pour_point_c: -24.0, flash_point_c: 225.0, tan_mg_koh_g: 0.8, water_content_ppm_limit: 300.0, demulsibility_min: 25.0, copper_corrosion_level_1h_h: 0.0, oxidation_stability_h: 2200.0, recommended_min_temp_c: -15.0, recommended_max_temp_c: 95.0, anti_wear_index: 0.82, corrosion_inhibition_index: 0.78, notes: "Standard heavy equipment hydraulic oil" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 501 |         OilCatalogEntry { id: "HYD-ISO-68", family: "HYDRAULIC", grade: "ISO VG 68", base_stock: "Mineral", density_15c_kg_m3: 875.0, viscosity_40c_cst: 68.0, viscosity_100c_cst: 8.7, viscosity_index: 102.0, pour_point_c: -21.0, flash_point_c: 235.0, tan_mg_koh_g: 0.8, water_content_ppm_limit: 300.0, demulsibility_min: 25.0, copper_corrosion_level_1h_h: 0.0, oxidation_stability_h: 2300.0, recommended_min_temp_c: -10.0, recommended_max_temp_c: 100.0, anti_wear_index: 0.8, corrosion_inhibition_index: 0.79, notes: "High-load hydraulic circuits" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 502 |         OilCatalogEntry { id: "HYD-ISO-100", family: "HYDRAULIC", grade: "ISO VG 100", base_stock: "Mineral", density_15c_kg_m3: 882.0, viscosity_40c_cst: 100.0, viscosity_100c_cst: 11.0, viscosity_index: 100.0, pour_point_c: -15.0, flash_point_c: 240.0, tan_mg_koh_g: 0.9, water_content_ppm_limit: 300.0, demulsibility_min: 30.0, copper_corrosion_level_1h_h: 0.0, oxidation_stability_h: 2400.0, recommended_min_temp_c: -5.0, recommended_max_temp_c: 105.0, anti_wear_index: 0.85, corrosion_inhibition_index: 0.80, notes: "Ultra-high pressure hydraulic systems" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

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l: 1.0, oxidation_stability_h: 2400.0, recommended_min_temp_c: -8.0, recommended_max_temp_c: 105.0, anti_wear_index: 4, corrosion_inhibition_index: 0.80, notes: "Slow-speed high pressure actuators" }, | I/O or protocol-critical logic; its can affect integration, timing, and diagnostic outcomes. |

| 503 | OilCatalogEntry { id: "AW-46", family: "HYDRAULIC_AW", grade: "AW 46", base_stock: "Mineral", density_15c_kg_m3: 868.0, viscosity_40c_cst: 46.0, viscosity_100c_cst: 6.9, viscosity_index: 108.0, pour_point_c: -27.0, flash_point_c: 228.0, tan_mg_koh_g: 0.75, water_content_ppm_limit: 250.0, demulsibility_min: 20.0, copper_corrosion_level: 1.0, oxidation_stability_h: 2600.0, recommended_min_temp_c: -18.0, recommended_max_temp_c: 100.0, anti_wear_index: 0.90, corrosion_inhibition_index: 0.82, notes: "ZDDP anti-wear package" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 504 | OilCatalogEntry { id: "HEES-46", family: "BIODEGRADABLE", grade: "HEES 46", base_stock: "Synthetic Ester", density_15c_kg_m3: 920.0, viscosity_40c_cst: 46.0, viscosity_100c_cst: 9.3, viscosity_index: 190.0, pour_point_c: -48.0, flash_point_c: 300.0, tan_mg_koh_g: 1.2, water_content_ppm_limit: 150.0, demulsibility_min: 35.0, copper_corrosion_level: 1.0, oxidation_stability_h: 1500.0, recommended_min_temp_c: -35.0, recommended_max_temp_c: 90.0, anti_wear_index: 0.88, corrosion_inhibition_index: 0.84, notes: "Environmentally sensitive sites" }, | I/O or protocol-critical logic; its can affect integration, timing, and diagnostic outcomes. |

| 505 | OilCatalogEntry { id: "HEPR-46", family: "BIODEGRADABLE", grade: "HEPR 46", base_stock: "PAO", density_15c_kg_m3: 900.0, viscosity_40c_cst: 46.0, viscosity_100c_cst: 8.6, viscosity_index: 165.0, pour_point_c: -48.0, flash_point_c: 260.0, tan_mg_koh_g: 0.8, water_content_ppm_limit: 200.0, demulsibility_min: 30.0, copper_corrosion_level: 1.0, oxidation_stability_h: 2200.0, recommended_min_temp_c: -30.0, recommended_max_temp_c: 105.0, anti_wear_index: 0.86, corrosion_inhibition_index: 0.83, notes: "Longer life bio-hydraulic oil" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 506 | OilCatalogEntry { id: "HFDU-46", family: "FIRE_RESISTANT", grade: "HFDU 46", base_stock: "Synthetic Ester", density_15c_kg_m3: 930.0, viscosity_40c_cst: 46.0, viscosity_100c_cst: 8.5, viscosity_index: 150.0, pour_point_c: -48.0, flash_point_c: 320.0, tan_mg_koh_g: 1.5, water_content_ppm_limit: 150.0, demulsibility_min: 40.0, copper_corrosion_level: 1.0, oxidation_stability_h: 1200.0, recommended_min_temp_c: -20.0, recommended_max_temp_c: 160.0, anti_wear_index: 0.84, corrosion_inhibition_index: 0.88, notes: "Steel plant/high fire risk" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 507 | OilCatalogEntry { id: "ENG-0W20", family: "ENGINE", grade: "0W-20", base_stock: "Group III/PAO", density_15c_kg_m3: 845.0, viscosity_40c_cst: 43.0, viscosity_100c_cst: 8.5, viscosity_index: 170.0, pour_point_c: -45.0, flash_point_c: 228.0, tan_mg_koh_g: 1.5, water_content_ppm_limit: 200.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 400.0, recommended_min_temp_c: -40.0, recommended_max_temp_c: 125.0, anti_wear_index: 0.82, corrosion_inhibition_index: 0.78, notes: "Fuel economy modern engines" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 508 | OilCatalogEntry { id: "ENG-5W30", family: "ENGINE", grade: "5W-30", base_stock: "Group II/III", density_15c_kg_m3: 855.0, viscosity_40c_cst: 61.0, viscosity_100c_cst: 10.5, viscosity_index: 165.0, pour_point_c: -39.0, flash_point_c: 232.0, tan_mg_koh_g: 1.6, water_content_ppm_limit: 200.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 380.0, recommended_min_temp_c: -35.0, recommended_max_temp_c: 130.0, anti_wear_index: 0.84, corrosion_inhibition_index: 0.79, notes: "Broad fleet use" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 509 | OilCatalogEntry { id: "ENG-10W30", family: "ENGINE", grade: "10W-30", base_stock: "Group II", density_15c_kg_m3: 870.0, viscosity_40c_cst: 72.0, viscosity_100c_cst: 11.5, viscosity_index: 150.0, pour_point_c: -33.0, flash_point_c: 235.0, tan_mg_koh_g: 1.7, water_content_ppm_limit: 250.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 350.0, recommended_min_temp_c: -30.0, recommended_max_temp_c: 130.0, anti_wear_index: 0.86, corrosion_inhibition_index: 0.80, notes: "Standard diesel engine oil" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 510 | OilCatalogEntry { id: "ENG-15W40", family: "ENGINE", grade: "15W-40", base_stock: "Group II", density_15c_kg_m3: 880.0, viscosity_40c_cst: 108.0, viscosity_100c_cst: 14.5, viscosity_index: 140.0, pour_point_c: -27.0, flash_point_c: 238.0, tan_mg_koh_g: 1.8, water_content_ppm_limit: 250.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 330.0, recommended_min_temp_c: -25.0, recommended_max_temp_c: 135.0, anti_wear_index: 0.88, corrosion_inhibition_index: 0.82, notes: "Heavy duty diesel baseline" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 511 | OilCatalogEntry { id: "ENG-20W50", family: "ENGINE", grade: "20W-50", base_stock: "Group II", density_15c_kg_m3: 892.0, viscosity_40c_cst: 155.0, viscosity_100c_cst: 18.0, viscosity_index: 132.0, pour_point_c: -21.0, flash_point_c: 245.0, tan_mg_koh_g: 1.9, water_content_ppm_limit: 250.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 320.0, recommended_min_temp_c: -15.0, recommended_max_temp_c: 140.0, anti_wear_index: 0.90, corrosion_inhibition_index: 0.82, notes: "High temp/worn engine support" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 512 | OilCatalogEntry { id: "ATF-DIII", family: "TRANSMISSION", grade: "Dexron III", base_stock: "Mineral/Synthetic", density_15c_kg_m3: 850.0, viscosity_40c_cst: 35.0, viscosity_100c_cst: 7.2, viscosity_index: 180.0, pour_point_c: -48.0, flash_point_c: 210.0, tan_mg_koh_g: 1.0, water_content_ppm_limit: 200.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 500.0, recommended_min_temp_c: -40.0, recommended_max_temp_c: 130.0, anti_wear_index: 0.80, corrosion_inhibition_index: 0.78, notes: "Legacy automatic transmissions" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 513 | OilCatalogEntry { id: "ATF-DVI", family: "TRANSMISSION", grade: "Dexron VI", base_stock: "Synthetic", density_15c_kg_m3: 846.0, viscosity_40c_cst: 30.0, viscosity_100c_cst: 6.0, viscosity_index: 165.0, pour_point_c: -51.0

lash_point_c: 220.0, tan_mg_koh_g: 0.9, water_content_ppm_limit: 200.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 600.0, recommended_min_temp_c: -45.0, recommended_max_temp_c: 135.0, anti_wear_index: 0.8, corrosion_inhibition_index: 0.80, notes: "Lower viscosity high-efficiency ATF" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 514 | OilCatalogEntry { id: "GEAR-75W90", family: "GEAR", grade: "75W-90", base_stock: "PA0", density_15c_kg_m3: 875.0, viscosity_40c_cst: 95.0, viscosity_100c_cst: 15.0, viscosity_index: 155.0, pour_point_c: -45.0, flash_point_c: 210.0, tan_mg_koh_g: 1.4, water_content_ppm_limit: 250.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 450.0, recommended_min_temp_c: -40.0, recommended_max_temp_c: 140.0, anti_wear_index: 0.93, corrosion_inhibition_index: 0.84, notes: "Final drive and gearboxes" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 515 | OilCatalogEntry { id: "GEAR-80W90", family: "GEAR", grade: "80W-90", base_stock: "Mineral", density_15c_kg_m3: 890.0, viscosity_40c_cst: 135.0, viscosity_100c_cst: 14.5, viscosity_index: 135.0, pour_point_c: -30.0, flash_point_c: 215.0, tan_mg_koh_g: 1.5, water_content_ppm_limit: 250.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 380.0, recommended_min_temp_c: -26.0, recommended_max_temp_c: 140.0, anti_wear_index: 0.94, corrosion_inhibition_index: 0.84, notes: "Heavy axle service" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 516 | OilCatalogEntry { id: "PAG-46", family: "REFRIGERATION", grade: "PAG 46", base_stock: "Polyalkylene Glycol", density_15c_kg_m3: 995.0, viscosity_40c_cst: 46.0, viscosity_100c_cst: 8.3, viscosity_index: 130.0, pour_point_c: -48.0, flash_point_c: 210.0, tan_mg_koh_g: 0.05, water_content_ppm_limit: 50.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 700.0, recommended_min_temp_c: -40.0, recommended_max_temp_c: 170.0, anti_wear_index: 0.62, corrosion_inhibition_index: 0.70, notes: "R134a compressor lubricant" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 517 | OilCatalogEntry { id: "PAG-100", family: "REFRIGERATION", grade: "PAG 100", base_stock: "Polyalkylene Glycol", density_15c_kg_m3: 1002.0, viscosity_40c_cst: 100.0, viscosity_100c_cst: 14.0, viscosity_index: 135.0, pour_point_c: -45.0, flash_point_c: 220.0, tan_mg_koh_g: 0.06, water_content_ppm_limit: 50.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 760.0, recommended_min_temp_c: -35.0, recommended_max_temp_c: 175.0, anti_wear_index: 0.64, corrosion_inhibition_index: 0.72, notes: "Higher viscosity compressor oil" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 518 | OilCatalogEntry { id: "PAG-150", family: "REFRIGERATION", grade: "PAG 150", base_stock: "Polyalkylene Glycol", density_15c_kg_m3: 1010.0, viscosity_40c_cst: 150.0, viscosity_100c_cst: 18.0, viscosity_index: 140.0, pour_point_c: -40.0, flash_point_c: 225.0, tan_mg_koh_g: 0.07, water_content_ppm_limit: 50.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 780.0, recommended_min_temp_c: -30.0, recommended_max_temp_c: 180.0, anti_wear_index: 0.66, corrosion_inhibition_index: 0.73, notes: "High load compressor application" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 519 | OilCatalogEntry { id: "POE-32", family: "REFRIGERATION", grade: "POE 32", base_stock: "Polyol Ester", density_15c_kg_m3: 970.0, viscosity_40c_cst: 32.0, viscosity_100c_cst: 5.9, viscosity_index: 140.0, pour_point_c: -54.0, flash_point_c: 245.0, tan_mg_koh_g: 0.05, water_content_ppm_limit: 35.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 900.0, recommended_min_temp_c: -45.0, recommended_max_temp_c: 170.0, anti_wear_index: 0.6, corrosion_inhibition_index: 0.76, notes: "HFC/POE refrigeration systems" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 520 | OilCatalogEntry { id: "POE-46", family: "REFRIGERATION", grade: "POE 46", base_stock: "Polyol Ester", density_15c_kg_m3: 975.0, viscosity_40c_cst: 46.0, viscosity_100c_cst: 7.4, viscosity_index: 143.0, pour_point_c: -50.0, flash_point_c: 250.0, tan_mg_koh_g: 0.05, water_content_ppm_limit: 35.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 950.0, recommended_min_temp_c: -40.0, recommended_max_temp_c: 175.0, anti_wear_index: 0.6, corrosion_inhibition_index: 0.77, notes: "Common automotive POE" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 521 | OilCatalogEntry { id: "POE-68", family: "REFRIGERATION", grade: "POE 68", base_stock: "Polyol Ester", density_15c_kg_m3: 980.0, viscosity_40c_cst: 68.0, viscosity_100c_cst: 9.2, viscosity_index: 145.0, pour_point_c: -45.0, flash_point_c: 255.0, tan_mg_koh_g: 0.06, water_content_ppm_limit: 35.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 980.0, recommended_min_temp_c: -35.0, recommended_max_temp_c: 180.0, anti_wear_index: 0.7, corrosion_inhibition_index: 0.78, notes: "High-temp compressor operation" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 522 | OilCatalogEntry { id: "POE-100", family: "REFRIGERATION", grade: "POE 100", base_stock: "Polyol Ester", density_15c_kg_m3: 988.0, viscosity_40c_cst: 100.0, viscosity_100c_cst: 11.5, viscosity_index: 148.0, pour_point_c: -40.0, flash_point_c: 260.0, tan_mg_koh_g: 0.06, water_content_ppm_limit: 35.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 1000.0, recommended_min_temp_c: -30.0, recommended_max_temp_c: 185.0, anti_wear_index: 0.71, corrosion_inhibition_index: 0.78, notes: "Very high compressor loading" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 523 | OilCatalogEntry { id: "TRF-INH", family: "ELECTRICAL", grade: "Transformer Inhibited", base_stock: "Mineral", density_15c_kg_m3: 885.0, viscosity_40c_cst: 9.0, viscosity_100c_cst: 2.4, viscosity_index: 95.0, pour_point_c: -5.0, flash_point_c: 150.0, tan_mg_koh_g: 0.03, water_content_ppm_limit: 20.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 164.0, recommended_min_temp_c: -30.0, recommended_max_temp_c: 110.0, anti_wear_index: 0.20, corrosion_inhibition_index: 0.88, notes: "Insulating fluid, not hydraulic" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 524 | OilCatalogEntry { id: "TURB-32", family: "TURBINE", grade: "ISO 32", base_stock: "Mineral", density_15c_kg_m3: 885.0, viscosity_40c_cst: 32.0, viscosity_100c_cst: 5.9, viscosity_index: 140.0, pour_point_c: -54.0, flash_point_c: 245.0, tan_mg_koh_g: 0.05, water_content_ppm_limit: 35.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 900.0, recommended_min_temp_c: -45.0, recommended_max_temp_c: 170.0, anti_wear_index: 0.6, corrosion_inhibition_index: 0.76, notes: "HFC/POE refrigeration systems" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

kg_m3: 855.0, viscosity_40c_cst: 32.0, viscosity_100c_cst: 5.3, viscosity_index: 102.0, pour_point_c: -15.0, flash_point_c: 220.0, tan_mg_koh_g: 0.2, water_content_ppm_limit: 200.0, demulsibility_min: 15.0, copper_corrosion_level: 1.0, oxidation_stability_h: 4000.0, recommended_min_temp_c: -10.0, recommended_max_temp_c: 120.0, anti_wear_index: 0.50, corrosion_inhibition_index: 0.86, notes: "Long-life oxidation resistance" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 525 | OilCatalogEntry { id: "TURB-46", family: "TURBINE", grade: "ISO 46", base_stock: "Mineral", density_15c_kg_m3: 865.0, viscosity_40c_cst: 46.0, viscosity_100c_cst: 6.8, viscosity_index: 100.0, pour_point_c: -12.0, flash_point_c: 225.0, tan_mg_koh_g: 0.2, water_content_ppm_limit: 200.0, demulsibility_min: 15.0, copper_corrosion_level: 1.0, oxidation_stability_h: 4200.0, recommended_min_temp_c: -8.0, recommended_max_temp_c: 125.0, anti_wear_index: 0.52, corrosion_inhibition_index: 0.86, notes: "Steam/gas turbine lubrication" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 526 | OilCatalogEntry { id: "COMP-46", family: "COMPRESSOR", grade: "ISO 46", base_stock: "Mineral", density_15c_kg_m3: 862.0, viscosity_40c_cst: 46.0, viscosity_100c_cst: 7.2, viscosity_index: 108.0, pour_point_c: -30.0, flash_point_c: 230.0, tan_mg_koh_g: 0.5, water_content_ppm_limit: 250.0, demulsibility_min: 25.0, copper_corrosion_level: 1.0, oxidation_stability_h: 2500.0, recommended_min_temp_c: -15.0, recommended_max_temp_c: 150.0, anti_wear_index: 0.74, corrosion_inhibition_index: 0.80, notes: "Rotary screw compressor" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 527 | OilCatalogEntry { id: "COMP-68", family: "COMPRESSOR", grade: "ISO 68", base_stock: "Mineral", density_15c_kg_m3: 872.0, viscosity_40c_cst: 68.0, viscosity_100c_cst: 9.2, viscosity_index: 105.0, pour_point_c: -24.0, flash_point_c: 235.0, tan_mg_koh_g: 0.5, water_content_ppm_limit: 250.0, demulsibility_min: 25.0, copper_corrosion_level: 1.0, oxidation_stability_h: 2600.0, recommended_min_temp_c: -10.0, recommended_max_temp_c: 155.0, anti_wear_index: 0.76, corrosion_inhibition_index: 0.81, notes: "High discharge temperature compressors" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 528 | OilCatalogEntry { id: "DOT3", family: "BRAKE", grade: "DOT 3", base_stock: "Glycol Ether", density_15c_kg_m3: 1035.0, viscosity_40c_cst: 5.0, viscosity_100c_cst: 2.0, viscosity_index: 220.0, pour_point_c: -60.0, flash_point_c: 120.0, tan_mg_koh_g: 1.0, water_content_ppm_limit: 500.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 0.0, recommended_min_temp_c: -40.0, recommended_max_temp_c: 205.0, anti_wear_index: 0.35, corrosion_inhibition_index: 0.75, notes: "Dry boiling around 205C" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 529 | OilCatalogEntry { id: "DOT4", family: "BRAKE", grade: "DOT 4", base_stock: "Borate Ester", density_15c_kg_m3: 1060.0, viscosity_40c_cst: 6.0, viscosity_100c_cst: 2.2, viscosity_index: 230.0, pour_point_c: -60.0, flash_point_c: 150.0, tan_mg_koh_g: 1.0, water_content_ppm_limit: 500.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 0.0, recommended_min_temp_c: -45.0, recommended_max_temp_c: 230.0, anti_wear_index: 0.37, corrosion_inhibition_index: 0.78, notes: "Higher boiling than DOT3" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 530 | OilCatalogEntry { id: "DOT5.1", family: "BRAKE", grade: "DOT 5.1", base_stock: "Borate Ester", density_15c_kg_m3: 1045.0, viscosity_40c_cst: 6.2, viscosity_100c_cst: 2.3, viscosity_index: 240.0, pour_point_c: -65.0, flash_point_c: 155.0, tan_mg_koh_g: 1.0, water_content_ppm_limit: 500.0, demulsibility_min: 0.0, copper_corrosion_level: 1.0, oxidation_stability_h: 0.0, recommended_min_temp_c: -50.0, recommended_max_temp_c: 240.0, anti_wear_index: 0.38, corrosion_inhibition_index: 0.79, notes: "Low-temp ABS response" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 531 | OilCatalogEntry { id: "UTTO-10W30", family: "TRACTOR", grade: "UTTO 10W-30", base_stock: "Mineral", density_15c_kg_m3: 872.0, viscosity_40c_cst: 64.0, viscosity_100c_cst: 10.5, viscosity_index: 142.0, pour_point_c: -39.0, flash_point_c: 220.0, tan_mg_koh_g: 1.2, water_content_ppm_limit: 300.0, demulsibility_min: 20.0, copper_corrosion_level: 1.0, oxidation_stability_h: 900.0, recommended_min_temp_c: -30.0, recommended_max_temp_c: 115.0, anti_wear_index: 0.8, corrosion_inhibition_index: 0.80, notes: "Combined hydraulic/transmission tractors" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 532 | OilCatalogEntry { id: "UTTO-80W", family: "TRACTOR", grade: "UTTO 80W", base_stock: "Mineral", density_15c_kg_m3: 885.0, viscosity_40c_cst: 90.0, viscosity_100c_cst: 12.0, viscosity_index: 128.0, pour_point_c: -30.0, flash_point_c: 225.0, tan_mg_koh_g: 1.3, water_content_ppm_limit: 300.0, demulsibility_min: 20.0, copper_corrosion_level: 1.0, oxidation_stability_h: 850.0, recommended_min_temp_c: -20.0, recommended_max_temp_c: 120.0, anti_wear_index: 0.86, corrosion_inhibition_index: 0.80, notes: "Older tractor drivetrain systems" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 533 | OilCatalogEntry { id: "VAC-100", family: "VACUUM", grade: "ISO 100", base_stock: "Mineral", density_15c_kg_m3: 885.0, viscosity_40c_cst: 100.0, viscosity_100c_cst: 11.2, viscosity_index: 98.0, pour_point_c: -18.0, flash_point_c: 240.0, tan_mg_koh_g: 0.4, water_content_ppm_limit: 200.0, demulsibility_min: 30.0, copper_corrosion_level: 1.0, oxidation_stability_h: 1400.0, recommended_min_temp_c: -10.0, recommended_max_temp_c: 135.0, anti_wear_index: 0.60, corrosion_inhibition_index: 0.82, notes: "Rotary vane vacuum pumps" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

| 534 | OilCatalogEntry { id: "ESTER-100", family: "SYNTHETIC", grade: "ISO 100", base_stock: "Synthetic Ester", density_15c_kg_m3: 915.0, viscosity_40c_cst: 100.0, viscosity_100c_cst: 14.5, viscosity_index: 175.0, pour_point_c: -35.0, flash_point_c: 285.0, tan_mg_koh_g: 0.9, water_content_ppm_limit: 150.0, demulsibility_min: 35.0, copper_corrosion_level: 1.0, oxidation_stability_h: 2000.0, recommended_min_temp_c: -35.0, recommended_max_temp_c: 180.0, anti_wear_index: 0.88, corrosion_inhibition_index: 0.87, notes: "High temp synthetic chain lubrication" }, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

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| 535 |         OilCatalogEntry { id: "REF-MIN-3GS", family: "REFRIGERATION", grade: "3GS", base_stock: "Mineral Naph
nic", density_15c_kg_m3: 895.0, viscosity_40c_cst: 32.0, viscosity_100c_cst: 4.8, viscosity_index: 85.0, pour_point_c
42.0, flash_point_c: 180.0, tan_mg_koh_g: 0.05, water_content_ppm_limit: 50.0, demulsibility_min: 0.0, copper_corrosi
level: 1.0, oxidation_stability_h: 600.0, recommended_min_temp_c: -40.0, recommended_max_temp_c: 80.0, anti_wear_inde
0.50, corrosion_inhibition_index: 0.68, notes: "Legacy ammonia/CFC systems" }, | I/O or protocol-critical logic; edit
an affect integration, timing, and diagnostic outcomes. |
| 536 |     ] | executable/support line; behavior impact depends on surrounding block context. |
| 537 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 538 | (blank) | line is blank; visual separation only. |
| 539 | #[derive(Debug, Clone, Copy, Serialize, Deserialize)] | serialization contract touchpoint; edits can break re
t compatibility and tooling. |
| 540 | pub struct PressureEnvelope { | declares data layout; field changes may break serialization, tests, and calle
|
| 541 |     pub min_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 542 |     pub mean_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 543 |     pub ideal_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 544 |     pub max_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 545 |     pub rupture_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 546 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 547 | (blank) | line is blank; visual separation only. |
| 548 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report c
atibility and tooling. |
| 549 | pub struct OringSpec { | declares data layout; field changes may break serialization, tests, and callers. |
| 550 |     pub id_tag: String, | executable/support line; behavior impact depends on surrounding block context. |
| 551 |     pub material: OringMaterial, | executable/support line; behavior impact depends on surrounding block cont
. |
| 552 |     pub shore_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 553 |     pub cross_section_mm: f64, | executable/support line; behavior impact depends on surrounding block context
|
| 554 |     pub squeeze_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 555 |     pub extrusion_gap_mm: f64, | executable/support line; behavior impact depends on surrounding block context
|
| 556 |     pub compression_set_pct: f64, | executable/support line; behavior impact depends on surrounding block con
t. |
| 557 |     pub design_life_hours: f64, | executable/support line; behavior impact depends on surrounding block conte
|
| 558 |     pub base_leak_area_mm2: f64, | executable/support line; behavior impact depends on surrounding block cont
. |
| 559 |     pub discharge_coeff: f64, | executable/support line; behavior impact depends on surrounding block context
| 560 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 561 | (blank) | line is blank; visual separation only. |
| 562 | #[derive(Debug, Clone, Copy, Serialize, Deserialize)] | serialization contract touchpoint; edits can break re
t compatibility and tooling. |
| 563 | pub struct CircuitInput { | declares data layout; field changes may break serialization, tests, and callers. |
| 564 |     pub pressure_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 565 |     pub delta_p_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 566 |     pub temp_c: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 567 |     pub cycles_per_s: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 568 |     pub duty_01: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 569 |     pub fluid_density_kg_m3: f64, | executable/support line; behavior impact depends on surrounding block con
t. |
| 570 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 571 | (blank) | line is blank; visual separation only. |
| 572 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report c
atibility and tooling. |
| 573 | pub struct CircuitResult { | declares data layout; field changes may break serialization, tests, and callers. |
| 574 |     pub name: String, | executable/support line; behavior impact depends on surrounding block context. |
| 575 |     pub alert: LeakAlertLevel, | executable/support line; behavior impact depends on surrounding block context
|
| 576 |     pub pressure_band: PressureBand, | executable/support line; behavior impact depends on surrounding block
text. |
| 577 |     pub current_pressure_bar: f64, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 578 |     pub rupture_probability_pct: f64, | executable/support line; behavior impact depends on surrounding block
ntext. |

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| 579 |     pub predicted_hours_to_rupture: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 580 |     pub rupture_pressure_bar: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 581 |     pub rupture_elapsed_h: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 582 |     pub leak_area_mm2: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 583 |     pub leak_lpm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 584 |     pub pressure_min_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 585 |     pub pressure_mean_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 586 |     pub pressure_max_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 587 |     pub recommended_hold_min_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 588 |     pub recommended_hold_target_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 589 |     pub recommended_hold_max_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 590 |     pub margin_to_max_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 591 |     pub margin_to_rupture_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 592 |     pub rca_hint: String, | executable/support line; behavior impact depends on surrounding block context. |
| 593 |     pub pca_recommendation: String, | executable/support line; behavior impact depends on surrounding block context. |
| 594 |     pub rupture_confirmed: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 595 |     pub warning_text: String, | executable/support line; behavior impact depends on surrounding block context. |
| 596 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 597 | (blank) | line is blank; visual separation only. |
| 598 | #[derive(Debug, Clone, Copy, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 599 | pub struct ManualCircuitParams { | declares data layout; field changes may break serialization, tests, and calculations. |
| 600 |     pub oil_type: Option<OilType>, | executable/support line; behavior impact depends on surrounding block context. |
| 601 |     pub piston_pressure_bar: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 602 |     pub operation_pressure_bar: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 603 |     pub pressure_min_bar: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 604 |     pub pressure_mean_bar: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 605 |     pub pressure_ideal_bar: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 606 |     pub pressure_max_bar: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 607 |     pub pressure_rupture_bar: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 608 |     pub oring_squeeze_pct: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 609 |     pub compression_set_pct: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 610 |     pub base_leak_area_mm2: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 611 |     pub max_supported_temp_c: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 612 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 613 | (blank) | line is blank; visual separation only. |
| 614 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 615 | pub struct ScenarioPrediction { | declares data layout; field changes may break serialization, tests, and calculations.

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s. |
| 616 |     pub circuit_name: String, | executable/support line; behavior impact depends on surrounding block context. |
| 617 |     pub scenario_name: String, | executable/support line; behavior impact depends on surrounding block context. |
|
| 618 |     pub peak_pressure_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
|
| 619 |     pub final_alert: LeakAlertLevel, | executable/support line; behavior impact depends on surrounding block context. |
| 620 |     pub final_rupture_probability_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 621 |     pub hours_to_rupture: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 622 |     pub likely_failure_mode: String, | executable/support line; behavior impact depends on surrounding block context. |
| 623 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 624 | (blank) | line is blank; visual separation only. |
| 625 | #[derive(Debug, Clone, Serialize, Deserialize)] | serialization contract touchpoint; edits can break report cardinality and tooling. |
| 626 | pub struct LeakAlert { | declares data layout; field changes may break serialization, tests, and callers. |
| 627 |     pub circuit_name: String, | executable/support line; behavior impact depends on surrounding block context. |
| 628 |     pub level: LeakAlertLevel, | executable/support line; behavior impact depends on surrounding block context. |
|
| 629 |     pub message: String, | executable/support line; behavior impact depends on surrounding block context. |
| 630 |     pub predicted_hours_to_rupture: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 631 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 632 | (blank) | line is blank; visual separation only. |
| 633 | #[derive(Debug, Clone, Copy)] | executable/support line; behavior impact depends on surrounding block context. |
| 634 | struct PressureStats { | declares data layout; field changes may break serialization, tests, and callers. |
| 635 |     min_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 636 |     max_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 637 |     mean_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 638 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 639 | (blank) | line is blank; visual separation only. |
| 640 | impl PressureStats { | implementation block; method behavior and trait conformance are defined here. |
| 641 |     fn new(seed_bar: f64) -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 642 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 643 |             min_bar: seed_bar, | executable/support line; behavior impact depends on surrounding block context. |
|
| 644 |             max_bar: seed_bar, | executable/support line; behavior impact depends on surrounding block context. |
|
| 645 |             mean_bar: seed_bar, | executable/support line; behavior impact depends on surrounding block context. |
|
| 646 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
|
| 647 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 648 | (blank) | line is blank; visual separation only. |
| 649 |     fn update(&mut self, p_bar: f64) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 650 |         self.min_bar = self.min_bar.min(p_bar); | executable/support line; behavior impact depends on surrounding block context. |
| 651 |         self.max_bar = self.max_bar.max(p_bar); | executable/support line; behavior impact depends on surrounding block context. |
| 652 |         self.mean_bar = self.mean_bar * 0.992 + p_bar * 0.008; | executable/support line; behavior impact depends on surrounding block context. |
| 653 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 654 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 655 | (blank) | line is blank; visual separation only. |
| 656 | #[derive(Debug, Clone, Copy)] | executable/support line; behavior impact depends on surrounding block context. |
| 657 | struct SealState { | declares data layout; field changes may break serialization, tests, and callers. |
| 658 |     damage_index: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 659 |     extrusion_index: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 660 |     thermal_damage: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 661 |     cumulative_overpressure_s: f64, | executable/support line; behavior impact depends on surrounding block context. |

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ext. |
| 662 |     total_operating_h: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 663 |     leak_area_mm2: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 664 |     leak_lpm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 665 |     rupture_confirmed: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 666 |     rupture_pressure_bar: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 667 |     rupture_elapsed_h: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 668 |     rupture_probability_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 669 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 670 | (blank) | line is blank; visual separation only. |
| 671 | impl SealState { | implementation block; method behavior and trait conformance are defined here. |
| 672 |     fn new(base_area_mm2: f64) -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 673 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 674 |             damage_index: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 675 |             extrusion_index: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 676 |             thermal_damage: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 677 |             cumulative_overpressure_s: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 678 |             total_operating_h: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 679 |             leak_area_mm2: base_area_mm2, | executable/support line; behavior impact depends on surrounding block context. |
| 680 |             leak_lpm: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 681 |             rupture_confirmed: false, | executable/support line; behavior impact depends on surrounding block context. |
| 682 |             rupture_pressure_bar: None, | executable/support line; behavior impact depends on surrounding block context. |
| 683 |             rupture_elapsed_h: None, | executable/support line; behavior impact depends on surrounding block context. |
| 684 |             rupture_probability_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 685 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 686 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 687 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 688 | (blank) | line is blank; visual separation only. |
| 689 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 690 | pub struct LeakCircuit { | declares data layout; field changes may break serialization, tests, and callers. |
| 691 |     pub name: String, | executable/support line; behavior impact depends on surrounding block context. |
| 692 |     pub application: String, | executable/support line; behavior impact depends on surrounding block context. |
| 693 |     pub component: CircuitComponent, | executable/support line; behavior impact depends on surrounding block context. |
| 694 |     pub oil_type: OilType, | executable/support line; behavior impact depends on surrounding block context. |
| 695 |     pub seal_count: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 696 |     pub piston_pressure_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 697 |     pub operation_pressure_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 698 |     pub pressure: PressureEnvelope, | executable/support line; behavior impact depends on surrounding block context. |
| 699 |     pub spec: OringSpec, | executable/support line; behavior impact depends on surrounding block context. |
| 700 |     pub coeffs: LeakModelCoefficients, | executable/support line; behavior impact depends on surrounding block context. |
| 701 |     pub reservoir_volume_l: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 702 |     pub pressure_support_lpm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 703 |     stats: PressureStats, | executable/support line; behavior impact depends on surrounding block context. |

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| 704 |     state: SealState, | executable/support line; behavior impact depends on surrounding block context. |
| 705 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 706 | (blank) | line is blank; visual separation only. |
| 707 | impl LeakCircuit { | implementation block; method behavior and trait conformance are defined here. |
| 708 |     #[allow(clippy::too_many_arguments)] | executable/support line; behavior impact depends on surrounding block
context. |
| 709 |     pub fn new( | declares executable behavior; signature/body changes alter API, control flow, and outputs.
| 710 |         name: &str, | executable/support line; behavior impact depends on surrounding block context. |
| 711 |         application: &str, | executable/support line; behavior impact depends on surrounding block context. |
| 712 |         component: CircuitComponent, | executable/support line; behavior impact depends on surrounding block
context. |
| 713 |         oil_type: OilType, | executable/support line; behavior impact depends on surrounding block context. |
| 714 |         seal_count: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 715 |         piston_pressure_bar: f64, | executable/support line; behavior impact depends on surrounding block con
text. |
| 716 |         operation_pressure_bar: f64, | executable/support line; behavior impact depends on surrounding block
context. |
| 717 |         pressure: PressureEnvelope, | executable/support line; behavior impact depends on surrounding block c
ontext. |
| 718 |         spec: OringSpec, | executable/support line; behavior impact depends on surrounding block context. |
| 719 |         reservoir_volume_l: f64, | executable/support line; behavior impact depends on surrounding block cont
ext. |
| 720 |         pressure_support_lpm: f64, | executable/support line; behavior impact depends on surrounding block co
ntext. |
| 721 |     ) -> Self { | executable/support line; behavior impact depends on surrounding block context. |
| 722 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 723 |             name: name.to_string(), | executable/support line; behavior impact depends on surrounding block c
ontext. |
| 724 |             application: application.to_string(), | executable/support line; behavior impact depends on surro
unding block context. |
| 725 |             component, | executable/support line; behavior impact depends on surrounding block context. |
| 726 |             oil_type, | executable/support line; behavior impact depends on surrounding block context. |
| 727 |             seal_count, | executable/support line; behavior impact depends on surrounding block context. |
| 728 |             piston_pressure_bar, | executable/support line; behavior impact depends on surrounding block cont
ext. |
| 729 |             operation_pressure_bar, | executable/support line; behavior impact depends on surrounding block c
ontext. |
| 730 |             pressure, | executable/support line; behavior impact depends on surrounding block context. |
| 731 |             coeffs: LeakModelCoefficients::default(), | executable/support line; behavior impact depends on s
urrounding block context. |
| 732 |             stats: PressureStats::new(pressure.mean_bar), | executable/support line; behavior impact depends
on surrounding block context. |
| 733 |             state: SealState::new(spec.base_leak_area_mm2), | executable/support line; behavior impact depend
s on surrounding block context. |
| 734 |             spec, | executable/support line; behavior impact depends on surrounding block context. |
| 735 |             reservoir_volume_l, | executable/support line; behavior impact depends on surrounding block conte
xt. |
| 736 |             pressure_support_lpm, | executable/support line; behavior impact depends on surrounding block con
text. |
| 737 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
s. |
| 738 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 739 | (blank) | line is blank; visual separation only. |
| 740 |     pub fn step(&mut self, input: CircuitInput, dt: f64) -> CircuitResult { | declares executable behavior; s
ignature/body changes alter API, control flow, and outputs. |
| 741 |         let dt = dt.max(1e-6); | executable/support line; behavior impact depends on surrounding block context
. |
| 742 |         self.stats.update(input.pressure_bar.max(0.0)); | executable/support line; behavior impact depends on
surrounding block context. |
| 743 |         self.state.total_operating_h += dt / 3600.0; | executable/support line; behavior impact depends on su
rrounding block context. |
| 744 | (blank) | line is blank; visual separation only. |
| 745 |         let (t_min, t_max) = self.spec.material.temp_window_c(); | executable/support line; behavior impact d
epends on surrounding block context. |
| 746 |         let temp_excess = if input.temp_c > t_max { | executable/support line; behavior impact depends on sur

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nding block context. |
| 747 |           (input.temp_c - t_max) / 20.0 | executable/support line; behavior impact depends on surrounding b
k context. |
| 748 |       } else if input.temp_c < t_min { | executable/support line; behavior impact depends on surrounding bl
context. |
| 749 |           (t_min - input.temp_c) / 25.0 | executable/support line; behavior impact depends on surrounding b
k context. |
| 750 |       } else { | executable/support line; behavior impact depends on surrounding block context. |
| 751 |           0.0 | executable/support line; behavior impact depends on surrounding block context. |
| 752 |       }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 753 | (blank) | line is blank; visual separation only. |
| 754 |       let pressure_ratio_max = input.pressure_bar / self.pressure.max_bar.max(1e-3); | executable/support l
; behavior impact depends on surrounding block context. |
| 755 |       let pressure_ratio_rupt = input.pressure_bar / self.pressure.rupture_bar.max(1e-3); | executable/supp
line; behavior impact depends on surrounding block context. |
| 756 | (blank) | line is blank; visual separation only. |
| 757 |       if input.pressure_bar > self.pressure.max_bar { | conditional gate; changing condition shifts runtime
th and side effects. |
| 758 |           self.state.cumulative_overpressure_s += dt; | executable/support line; behavior impact depends on
rrounding block context. |
| 759 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 760 | (blank) | line is blank; visual separation only. |
| 761 |       let squeeze_target = 0.20; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 762 |       let squeeze = (self.spec.squeeze_pct / 100.0).clamp(0.05, 0.45); | executable/support line; behavior
act depends on surrounding block context. |
| 763 |       let squeeze_penalty = ((squeeze - squeeze_target).abs() / squeeze_target).clamp(0.0, 2.0); | executab
support line; behavior impact depends on surrounding block context. |
| 764 | (blank) | line is blank; visual separation only. |
| 765 |       let hardness_penalty = ((80.0 - self.spec.shore_a).max(0.0) / 40.0).clamp(0.0, 1.0); | executable/sup
t line; behavior impact depends on surrounding block context. |
| 766 |       let gap_penalty = (self.spec.extrusion_gap_mm / 0.25).clamp(0.2, 3.0); | executable/support line; beh
or impact depends on surrounding block context. |
| 767 |       let age_ratio = | executable/support line; behavior impact depends on surrounding block context. |
| 768 |           (self.state.total_operating_h / self.spec.design_life_hours.max(1.0)).clamp(0.0, 4.0); | executab
support line; behavior impact depends on surrounding block context. |
| 769 |       let cycle_factor = (input.cycles_per_s / 2.0).clamp(0.0, 4.0); | executable/support line; behavior im
t depends on surrounding block context. |
| 770 |       let duty = input.duty_01.clamp(0.0, 1.0); | executable/support line; behavior impact depends on surro
ing block context. |
| 771 |       let oil_vi = self.oil_type.viscosity_index(); | executable/support line; behavior impact depends on s
ounding block context. |
| 772 |       let vi_factor = (130.0 / oil_vi.max(60.0)).clamp(0.7, 1.8); | executable/support line; behavior impac
depends on surrounding block context. |
| 773 |       let piston_overload = | executable/support line; behavior impact depends on surrounding block context
| 774 |           (self.piston_pressure_bar / self.pressure.ideal_bar.max(1e-3)).clamp(0.4, 2.2); | executable/supp
line; behavior impact depends on surrounding block context. |
| 775 |       let operation_stress = | executable/support line; behavior impact depends on surrounding block context
|
| 776 |           (self.operation_pressure_bar / self.pressure.ideal_bar.max(1e-3)).clamp(0.4, 2.5); | executable/s
ort line; behavior impact depends on surrounding block context. |
| 777 | (blank) | line is blank; visual separation only. |
| 778 |       let mut damage_rate = 2.2e-6 * self.coeffs.damage_rate_scale.max(0.05); | executable/support line; be
ior impact depends on surrounding block context. |
| 779 |       damage_rate *= 1.0 + 1.9 * (pressure_ratio_max - 0.85).max(0.0).powi(2); | executable/support line; b
vior impact depends on surrounding block context. |
| 780 |       damage_rate *= 1.0 + 0.75 * temp_excess.max(0.0); | executable/support line; behavior impact depends
surrounding block context. |
| 781 |       damage_rate *= 1.0 + 0.45 * squeeze_penalty; | executable/support line; behavior impact depends on su
unding block context. |
| 782 |       damage_rate *= 1.0 + 0.35 * cycle_factor; | executable/support line; behavior impact depends on surro
ing block context. |
| 783 |       damage_rate *= 1.0 + 0.28 * age_ratio; | executable/support line; behavior impact depends on surround

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block context. |
| 784 |         damage_rate *= 0.8 + 0.6 * duty; | executable/support line; behavior impact depends on surrounding block context. | | | | |
| 785 |         damage_rate *= vi_factor; | executable/support line; behavior impact depends on surrounding block context. |
| 786 |         damage_rate *= 0.75 + 0.25 * piston_overload; | executable/support line; behavior impact depends on surrounding block context. |
| 787 |         damage_rate *= 0.75 + 0.25 * operation_stress; | executable/support line; behavior impact depends on surrounding block context. |
| 788 | (blank) | line is blank; visual separation only. |
| 789 |         self.state.damage_index = (self.state.damage_index + damage_rate * dt).clamp(0.0, 1.6); | executable/support line; behavior impact depends on surrounding block context. |
| 790 | (blank) | line is blank; visual separation only. |
| 791 |         let extrusion_drive = | executable/support line; behavior impact depends on surrounding block context. |
| 792 |             (pressure_ratio_max - 0.9).max(0.0) * (1.0 + hardness_penalty) * gap_penalty; | executable/support line; behavior impact depends on surrounding block context. |
| 793 |         self.state.extrusion_index = (self.state.extrusion_index | executable/support line; behavior impact depends on surrounding block context. |
| 794 |             + extrusion_drive * dt * 1.8e-4 * self.coeffs.extrusion_rate_scale.max(0.05)) | executable/support line; behavior impact depends on surrounding block context. |
| 795 |             .clamp(0.0, 1.5); | executable/support line; behavior impact depends on surrounding block context. |
| 796 | (blank) | line is blank; visual separation only. |
| 797 |         let thermal_rate = (temp_excess.max(0.0) * (0.9 + 0.4 * duty)) | executable/support line; behavior impact depends on surrounding block context. |
| 798 |             * 2.5e-5 | executable/support line; behavior impact depends on surrounding block context. |
| 799 |             * self.coeffs.thermal_rate_scale.max(0.05); | executable/support line; behavior impact depends on surrounding block context. |
| 800 |         self.state.thermal_damage = (self.state.thermal_damage + thermal_rate * dt).clamp(0.0, 1.2); | executable/support line; behavior impact depends on surrounding block context. |
| 801 | (blank) | line is blank; visual separation only. |
| 802 |         if matches!(self.component, CircuitComponent::AcHose) { | conditional gate; changing condition shifts runtime path and side effects. |
| 803 |             self.state.extrusion_index = (self.state.extrusion_index | executable/support line; behavior impact depends on surrounding block context. |
| 804 |                 + (pressure_ratio_max - 0.75).max(0.0) | executable/support line; behavior impact depends on surrounding block context. |
| 805 |                     * dt | executable/support line; behavior impact depends on surrounding block context. |
| 806 |                     * 2.2e-4 | executable/support line; behavior impact depends on surrounding block context. |
| 807 |                     * self.coeffs.extrusion_rate_scale.max(0.05)) | executable/support line; behavior impact depends on surrounding block context. |
| 808 |                 .clamp(0.0, 1.8); | executable/support line; behavior impact depends on surrounding block context. |
| 809 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
| 810 | (blank) | line is blank; visual separation only. |
| 811 |         if !self.state.ruption_confirmed { | conditional gate; changing condition shifts runtime path and side effects. |
| 812 |             let immediate_burst = input.pressure_bar >= self.pressure.ruption_bar; | executable/support line; behavior impact depends on surrounding block context. |
| 813 |             let fatigue_burst = self.state.damage_index > 1.0 && self.state.extrusion_index > 0.8; | executable/support line; behavior impact depends on surrounding block context. |
| 814 |             let thermal_burst = self.state.thermal_damage > 1.0 && pressure_ratio_max > 0.95; | executable/support line; behavior impact depends on surrounding block context. |
| 815 |             if immediate_burst || fatigue_burst || thermal_burst { | conditional gate; changing condition shifts runtime path and side effects. |
| 816 |                 self.state.ruption_confirmed = true; | executable/support line; behavior impact depends on surrounding block context. |
| 817 |                 self.state.ruption_pressure_bar = Some(input.pressure_bar); | executable/support line; behavior impact depends on surrounding block context. |
| 818 |                 self.state.ruption_elapsed_h = Some(self.state.total_operating_h); | executable/support line; behavior impact depends on surrounding block context. |
| 819 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
| 820 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |

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| 821 | (blank) | line is blank; visual separation only. |
| 822 |         let micro_area = self.spec.base_leak_area_mm2 | executable/support line; behavior impact depends on s
ounding block context. |
| 823 |         * (1.0 + 0.65 * squeeze_penalty + 0.012 * self.spec.compression_set_pct) | executable/support lin
behavior impact depends on surrounding block context. |
| 824 |         * self.spec.material.gas_permeability_factor(); | executable/support line; behavior impact depend
n surrounding block context. |
| 825 |         let fatigue_area = self.spec.cross_section_mm * self.state.damage_index.powi(2) * 0.003; | executable
pport line; behavior impact depends on surrounding block context. |
| 826 |         let extrusion_area = self.spec.cross_section_mm * self.state.extrusion_index * 0.0025; | executable/s
ort line; behavior impact depends on surrounding block context. |
| 827 |         let thermal_area = self.spec.cross_section_mm * self.state.thermal_damage * 0.0018; | executable/supp
line; behavior impact depends on surrounding block context. |
| 828 |         let rupture_area = if self.state.rupture_confirmed { | executable/support line; behavior impact dependen
on surrounding block context. |
| 829 |             if matches!(self.component, CircuitComponent::ACHose) { | conditional gate; changing condition sh
s runtime path and side effects. |
| 830 |                 self.spec.cross_section_mm | executable/support line; behavior impact depends on surrounding
ck context. |
| 831 |                 * self.spec.cross_section_mm | executable/support line; behavior impact depends on surround
ng block context. |
| 832 |                 * 0.14 | executable/support line; behavior impact depends on surrounding block context. |
| 833 |                 * self.coeffs.rupture_area_scale.max(0.05) | executable/support line; behavior impact depen
s on surrounding block context. |
| 834 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 835 |                 self.spec.cross_section_mm | executable/support line; behavior impact depends on surrounding
ck context. |
| 836 |                 * self.spec.cross_section_mm | executable/support line; behavior impact depends on surround
ng block context. |
| 837 |                 * 0.06 | executable/support line; behavior impact depends on surrounding block context. |
| 838 |                 * self.coeffs.rupture_area_scale.max(0.05) | executable/support line; behavior impact depen
s on surrounding block context. |
| 839 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 840 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 841 |             0.0 | executable/support line; behavior impact depends on surrounding block context. |
| 842 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 843 | (blank) | line is blank; visual separation only. |
| 844 |         self.state.leak_area_mm2 = | executable/support line; behavior impact depends on surrounding block co
xt. |
| 845 |         (micro_area + fatigue_area + extrusion_area + thermal_area + rupture_area) | executable/support l
; behavior impact depends on surrounding block context. |
| 846 |         * self.seal_count.max(1) as f64; | executable/support line; behavior impact depends on surround
ng block context. |
| 847 | (blank) | line is blank; visual separation only. |
| 848 |         let dp_pa = (input.delta_p_bar.max(input.pressure_bar).max(0.0)) * 1.0e5; | executable/support line;
avior impact depends on surrounding block context. |
| 849 |         let rho = input.fluid_density_kg_m3.max(120.0); | executable/support line; behavior impact depends on
rrounding block context. |
| 850 |         let area_m2 = self.state.leak_area_mm2 * 1.0e-6; | executable/support line; behavior impact depends o
urrounding block context. |
| 851 |         let q_m3_s = self.spec.discharge_coeff.clamp(0.05, 1.0) | executable/support line; behavior impact de
ds on surrounding block context. |
| 852 |         * self.coeffs.flow_rate_scale.max(0.05) | executable/support line; behavior impact depends on surr
nding block context. |
| 853 |         * area_m2 | executable/support line; behavior impact depends on surrounding block context. |
| 854 |         * (2.0 * dp_pa / rho).sqrt(); | executable/support line; behavior impact depends on surrounding b
k context. |
| 855 |         self.state.leak_lpm = q_m3_s * 60000.0; | executable/support line; behavior impact depends on surroun
g block context. |
| 856 | (blank) | line is blank; visual separation only. |
| 857 |         let risk = 34.0 * (pressure_ratio_max - 0.8).max(0.0) | executable/support line; behavior impact depe
on surrounding block context. |
| 858 |         + 26.0 * (pressure_ratio_rupt - 0.7).max(0.0) | executable/support line; behavior impact depends

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```

surrounding block context. |
| 859 |           + 18.0 * self.state.damage_index | executable/support line; behavior impact depends on surrounding
lock context. |
| 860 |           + 12.0 * self.state.extrusion_index | executable/support line; behavior impact depends on surroundi
g block context. |
| 861 |           + 10.0 * self.state.thermal_damage; | executable/support line; behavior impact depends on surroundi
g block context. |
| 862 |           self.state.rupture_probability_pct = risk.clamp(0.0, 100.0); | executable/support line; behavior impa
depends on surrounding block context. |
| 863 | (blank) | line is blank; visual separation only. |
| 864 |           let rate_to_failure = damage_rate | executable/support line; behavior impact depends on surrounding b
k context. |
| 865 |           + extrusion_drive * 1.8e-4 * self.coeffs.extrusion_rate_scale.max(0.05) | executable/support line
ehavior impact depends on surrounding block context. |
| 866 |           + thermal_rate; | executable/support line; behavior impact depends on surrounding block context.
| 867 |           let pred_h = if self.state.rupture_confirmed { | executable/support line; behavior impact depends on
rounding block context. |
| 868 |               Some(0.0) | executable/support line; behavior impact depends on surrounding block context. |
| 869 |           } else if rate_to_failure > 1e-8 { | executable/support line; behavior impact depends on surrounding
ck context. |
| 870 |               Some(((1.0 - self.state.damage_index).max(0.0) / rate_to_failure) / 3600.0) | executable/support
e; behavior impact depends on surrounding block context. |
| 871 |           } else { | executable/support line; behavior impact depends on surrounding block context. |
| 872 |               None | executable/support line; behavior impact depends on surrounding block context. |
| 873 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 874 | (blank) | line is blank; visual separation only. |
| 875 |           let alert = if self.state.rupture_confirmed { | executable/support line; behavior impact depends on s
ounding block context. |
| 876 |               LeakAlertLevel::Ruptured | executable/support line; behavior impact depends on surrounding block
text. |
| 877 |           } else if pressure_ratio_rupt > 0.9 \\| self.state.rupture_probability_pct > 85.0 { | executable/sup
t line; behavior impact depends on surrounding block context. |
| 878 |               LeakAlertLevel::Critical | executable/support line; behavior impact depends on surrounding block
text. |
| 879 |           } else if pressure_ratio_max > 1.0 \\| self.state.rupture_probability_pct > 65.0 { | executable/supp
line; behavior impact depends on surrounding block context. |
| 880 |               LeakAlertLevel::Warning | executable/support line; behavior impact depends on surrounding block c
ext. |
| 881 |           } else if pressure_ratio_max > 0.85 \\| self.state.rupture_probability_pct > 40.0 { | executable/sup
t line; behavior impact depends on surrounding block context. |
| 882 |               LeakAlertLevel::Watch | executable/support line; behavior impact depends on surrounding block con
t. |
| 883 |           } else { | executable/support line; behavior impact depends on surrounding block context. |
| 884 |               LeakAlertLevel::Normal | executable/support line; behavior impact depends on surrounding block co
xt. |
| 885 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 886 | (blank) | line is blank; visual separation only. |
| 887 |           let pressure_band = if input.pressure_bar >= self.pressure.rupture_bar { | executable/support line; b
vior impact depends on surrounding block context. |
| 888 |               PressureBand::AtRupture | executable/support line; behavior impact depends on surrounding block c
ext. |
| 889 |           } else if input.pressure_bar > self.pressure.max_bar { | executable/support line; behavior impact dep
s on surrounding block context. |
| 890 |               PressureBand::OverMaximum | executable/support line; behavior impact depends on surrounding block
ntext. |
| 891 |           } else if input.pressure_bar < self.pressure.min_bar { | executable/support line; behavior impact dep
s on surrounding block context. |
| 892 |               PressureBand::BelowMinimum | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 893 |           } else { | executable/support line; behavior impact depends on surrounding block context. |
| 894 |               let span = (self.pressure.max_bar - self.pressure.min_bar).max(1e-6); | executable/support line;
avior impact depends on surrounding block context. |
| 895 |               let ratio = (input.pressure_bar - self.pressure.min_bar) / span; | executable/support line; behav

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```

impact depends on surrounding block context. |
| 896 |         if ratio < 0.33 { | conditional gate; changing condition shifts runtime path and side effects. |
| 897 |             PressureBand::LowBand | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 898 |         } else if ratio < 0.66 { | executable/support line; behavior impact depends on surrounding block
text. |
| 899 |             PressureBand::MidBand | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 900 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 901 |             PressureBand::HighBand | executable/support line; behavior impact depends on surrounding block
oncontext. |
| 902 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 903 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 904 | (blank) | line is blank; visual separation only. |
| 905 |     let recommended_hold_min_bar = (self.pressure.ideal_bar * 0.65).max(self.pressure.min_bar); | execut
/support line; behavior impact depends on surrounding block context. |
| 906 |     let recommended_hold_target_bar = self.pressure.ideal_bar; | executable/support line; behavior impact
depends on surrounding block context. |
| 907 |     let recommended_hold_max_bar = (self.pressure.max_bar * 0.88).min(self.pressure.rupture_bar * 0.78);
executable/support line; behavior impact depends on surrounding block context. |
| 908 |     let margin_to_max_bar = self.pressure.max_bar - input.pressure_bar; | executable/support line; behavi
impact depends on surrounding block context. |
| 909 |     let margin_to_rupture_bar = self.pressure.rupture_bar - input.pressure_bar; | executable/support line
ehavior impact depends on surrounding block context. |
| 910 | (blank) | line is blank; visual separation only. |
| 911 |     let rca_hint = if self.state.rupture_confirmed { | executable/support line; behavior impact depends o
urrounding block context. |
| 912 |         format!( | executable/support line; behavior impact depends on surrounding block context. |
| 913 |             "ruptured_at={:.2}bar; damage={:.3}; extrusion={:.3}; thermal={:.3}", | executable/support li
behavior impact depends on surrounding block context. |
| 914 |             self.state.rupture_pressure_bar.unwrap_or(input.pressure_bar), | executable/support line; beh
or impact depends on surrounding block context. |
| 915 |             self.state.damage_index, | executable/support line; behavior impact depends on surrounding bl
context. |
| 916 |             self.state.extrusion_index, | executable/support line; behavior impact depends on surrounding
ock context. |
| 917 |             self.state.thermal_damage | executable/support line; behavior impact depends on surrounding b
k context. |
| 918 |         ) | executable/support line; behavior impact depends on surrounding block context. |
| 919 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 920 |         format!( | executable/support line; behavior impact depends on surrounding block context. |
| 921 |             "band={}; margin_to_max={:.2}bar; margin_to_rupture={:.2}bar", | executable/support line; beh
or impact depends on surrounding block context. |
| 922 |             pressure_band, margin_to_max_bar, margin_to_rupture_bar | executable/support line; behavior i
ct depends on surrounding block context. |
| 923 |         ) | executable/support line; behavior impact depends on surrounding block context. |
| 924 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 925 | (blank) | line is blank; visual separation only. |
| 926 |     let pca_recommendation = if self.state.rupture_confirmed { | executable/support line; behavior impact
depends on surrounding block context. |
| 927 |         format!( | executable/support line; behavior impact depends on surrounding block context. |
| 928 |             "keep_between_{:.1}_and_{:.1}_bar; reduce_cycles_or_temp; review_material={}", | executable/s
ort line; behavior impact depends on surrounding block context. |
| 929 |             recommended_hold_min_bar, | executable/support line; behavior impact depends on surrounding b
k context. |
| 930 |             recommended_hold_max_bar, | executable/support line; behavior impact depends on surrounding b
k context. |
| 931 |             self.spec.material.name() | executable/support line; behavior impact depends on surrounding b
k context. |
| 932 |         ) | executable/support line; behavior impact depends on surrounding block context. |
| 933 |     } else if matches!(pressure_band, PressureBand::OverMaximum \| PressureBand::AtRupture) { | executabl
upport line; behavior impact depends on surrounding block context. |

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| 934 |         format!( | executable/support line; behavior impact depends on surrounding block context. |
| 935 |             "immediate_derate_to_{:.1}_bar_target_{:.1}_bar", | executable/support line; behavior impact
ends on surrounding block context. |
| 936 |             recommended_hold_max_bar, recommended_hold_target_bar | executable/support line; behavior imp
depends on surrounding block context. |
| 937 |         ) | executable/support line; behavior impact depends on surrounding block context. |
| 938 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 939 |         format!( | executable/support line; behavior impact depends on surrounding block context. |
| 940 |             "hold_{:.1}..{:.1}_bar_target_{:.1}_bar", | executable/support line; behavior impact depends
surrounding block context. |
| 941 |             recommended_hold_min_bar, recommended_hold_max_bar, recommended_hold_target_bar | executable/s
port line; behavior impact depends on surrounding block context. |
| 942 |         ) | executable/support line; behavior impact depends on surrounding block context. |
| 943 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 944 | (blank) | line is blank; visual separation only. |
| 945 |     let warning_text = match alert { | executable/support line; behavior impact depends on surrounding bl
context. |
| 946 |         LeakAlertLevel::Ruptured => format!( | executable/support line; behavior impact depends on surrou
ng block context. |
| 947 |             "{:} ruptura detectada em {:.2} bar ({:.3} h); vazamento {:.2} L/min; manter {:.1}-{:.1} bar"
executable/support line; behavior impact depends on surrounding block context. |
| 948 |             self.application, | executable/support line; behavior impact depends on surrounding block con
t. |
| 949 |             self.state.rupture_pressure_bar.unwrap_or(input.pressure_bar), | executable/support line; beh
or impact depends on surrounding block context. |
| 950 |             self.state.rupture_elapsed_h.unwrap_or(self.state.total_operating_h), | executable/support li
behavior impact depends on surrounding block context. |
| 951 |             self.state.leak_lpm, | executable/support line; behavior impact depends on surrounding block
text. |
| 952 |             recommended_hold_min_bar, | executable/support line; behavior impact depends on surrounding b
k context. |
| 953 |             recommended_hold_max_bar | executable/support line; behavior impact depends on surrounding bl
context. |
| 954 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 955 |         LeakAlertLevel::Critical => format!( | executable/support line; behavior impact depends on surrou
ng block context. |
| 956 |             "{:} risco critico ({:.0}%); ponto={:} p={:.1}bar; margem ruptura {:.1}bar", | executable/supp
line; behavior impact depends on surrounding block context. |
| 957 |             self.application, | executable/support line; behavior impact depends on surrounding block con
t. |
| 958 |             self.state.rupture_probability_pct, | executable/support line; behavior impact depends on sur
nding block context. |
| 959 |             pressure_band, | executable/support line; behavior impact depends on surrounding block context
|
| 960 |             input.pressure_bar, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 961 |             margin_to_rupture_bar | executable/support line; behavior impact depends on surrounding block
ntext. |
| 962 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 963 |         LeakAlertLevel::Warning => format!( | executable/support line; behavior impact depends on surrou
g block context. |
| 964 |             "{:} sobrecarga progressiva; ponto={:} p={:.1}bar; manter <= {:.1}bar", | executable/support l
; behavior impact depends on surrounding block context. |
| 965 |             self.application, | executable/support line; behavior impact depends on surrounding block con
t. |
| 966 |             pressure_band, | executable/support line; behavior impact depends on surrounding block context
|
| 967 |             input.pressure_bar, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 968 |             recommended_hold_max_bar | executable/support line; behavior impact depends on surrounding bl
context. |
| 969 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 970 |         LeakAlertLevel::Watch => format!( | executable/support line; behavior impact depends on surroundi
block context. |

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```

| 971 |         "{: degradacao progressiva; ponto={} p={:.1}bar; alvo {:.1}bar", | executable/support line;
avior impact depends on surrounding block context. |
| 972 |         self.application, | executable/support line; behavior impact depends on surrounding block con
t. |
| 973 |         pressure_band, | executable/support line; behavior impact depends on surrounding block context
|
| 974 |         input.pressure_bar, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 975 |         recommended_hold_target_bar | executable/support line; behavior impact depends on surrounding
ock context. |
| 976 |     ), | executable/support line; behavior impact depends on surrounding block context. |
| 977 |     LeakAlertLevel::Normal => format!( | executable/support line; behavior impact depends on surround
block context. |
| 978 |         "{: nominal; ponto={} p={:.1}bar", | executable/support line; behavior impact depends on sur
nding block context. |
| 979 |         self.application, | executable/support line; behavior impact depends on surrounding block con
t. |
| 980 |         pressure_band, | executable/support line; behavior impact depends on surrounding block context
|
| 981 |         input.pressure_bar | executable/support line; behavior impact depends on surrounding block co
xt. |
| 982 |     ), | executable/support line; behavior impact depends on surrounding block context. |
| 983 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 984 | (blank) | line is blank; visual separation only. |
| 985 |     CircuitResult { | executable/support line; behavior impact depends on surrounding block context. |
| 986 |         name: self.name.clone(), | executable/support line; behavior impact depends on surrounding block
text. |
| 987 |         alert, | executable/support line; behavior impact depends on surrounding block context. |
| 988 |         pressure_band, | executable/support line; behavior impact depends on surrounding block context. |
| 989 |         current_pressure_bar: input.pressure_bar, | executable/support line; behavior impact depends on s
urrounding block context. |
| 990 |         rupture_probability_pct: self.state.rupture_probability_pct, | executable/support line; behavior
act depends on surrounding block context. |
| 991 |         predicted_hours_to_rupture: pred_h, | executable/support line; behavior impact depends on surroun
g block context. |
| 992 |         rupture_pressure_bar: self.state.rupture_pressure_bar, | executable/support line; behavior impact
ends on surrounding block context. |
| 993 |         rupture_elapsed_h: self.state.rupture_elapsed_h, | executable/support line; behavior impact depen
on surrounding block context. |
| 994 |         leak_area_mm2: self.state.leak_area_mm2, | executable/support line; behavior impact depends on su
urrounding block context. |
| 995 |         leak_lpm: self.state.leak_lpm, | executable/support line; behavior impact depends on surrounding
ck context. |
| 996 |         pressure_min_bar: self.stats.min_bar, | executable/support line; behavior impact depends on surro
ing block context. |
| 997 |         pressure_mean_bar: self.stats.mean_bar, | executable/support line; behavior impact depends on sur
nding block context. |
| 998 |         pressure_max_bar: self.stats.max_bar, | executable/support line; behavior impact depends on surro
ing block context. |
| 999 |         recommended_hold_min_bar, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1000 |         recommended_hold_target_bar, | executable/support line; behavior impact depends on surrounding b
k context. |
| 1001 |         recommended_hold_max_bar, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1002 |         margin_to_max_bar, | executable/support line; behavior impact depends on surrounding block conte
|
| 1003 |         margin_to_rupture_bar, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1004 |         rca_hint, | executable/support line; behavior impact depends on surrounding block context. |
| 1005 |         pca_recommendation, | executable/support line; behavior impact depends on surrounding block cont
. |
| 1006 |         rupture_confirmed: self.state.rupture_confirmed, | executable/support line; behavior impact depe
on surrounding block context. |

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| 1007 |           warning_text, | executable/support line; behavior impact depends on surrounding block context. |
| 1008 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1009 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1010 | (blank) | line is blank; visual separation only. |
| 1011 |   pub fn reset(&mut self) { | declares executable behavior; signature/body changes alter API, control flow
nd outputs. |
| 1012 |       self.stats = PressureStats::new(self.pressure.mean_bar); | executable/support line; behavior impact
ends on surrounding block context. |
| 1013 |       self.state = SealState::new(self.spec.base_leak_area_mm2); | executable/support line; behavior impac
depends on surrounding block context. |
| 1014 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1015 | (blank) | line is blank; visual separation only. |
| 1016 |   pub fn apply_manual_params(&mut self, p: ManualCircuitParams) { | declares executable behavior; signatur
ody changes alter API, control flow, and outputs. |
| 1017 |       if let Some(v) = p.oil_type { | conditional gate; changing condition shifts runtime path and side ef
ts. |
| 1018 |           self.oil_type = v; | executable/support line; behavior impact depends on surrounding block conte
|
| 1019 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1020 |       if let Some(v) = p.piston_pressure_bar { | conditional gate; changing condition shifts runtime path
side effects. |
| 1021 |           self.piston_pressure_bar = v.max(0.0); | executable/support line; behavior impact depends on surr
nding block context. |
| 1022 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1023 |       if let Some(v) = p.operation_pressure_bar { | conditional gate; changing condition shifts runtime pa
and side effects. |
| 1024 |           self.operation_pressure_bar = v.max(0.0); | executable/support line; behavior impact depends on
rounding block context. |
| 1025 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1026 |       if let Some(v) = p.pressure_min_bar { | conditional gate; changing condition shifts runtime path and
de effects. |
| 1027 |           self.pressure.min_bar = v.max(0.0); | executable/support line; behavior impact depends on surrou
ng block context. |
| 1028 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1029 |       if let Some(v) = p.pressure_mean_bar { | conditional gate; changing condition shifts runtime path an
ide effects. |
| 1030 |           self.pressure.mean_bar = v.max(0.0); | executable/support line; behavior impact depends on surro
ing block context. |
| 1031 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1032 |       if let Some(v) = p.pressure_ideal_bar { | conditional gate; changing condition shifts runtime path a
side effects. |
| 1033 |           self.pressure.ideal_bar = v.max(0.0); | executable/support line; behavior impact depends on surr
ding block context. |
| 1034 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1035 |       if let Some(v) = p.pressure_max_bar { | conditional gate; changing condition shifts runtime path and
de effects. |
| 1036 |           self.pressure.max_bar = v.max(0.0); | executable/support line; behavior impact depends on surrou
ng block context. |
| 1037 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1038 |       if let Some(v) = p.pressure_rupture_bar { | conditional gate; changing condition shifts runtime path
d side effects. |
| 1039 |           self.pressure.rupture_bar = v.max(self.pressure.max_bar + 0.1); | executable/support line; behav
impact depends on surrounding block context. |
| 1040 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1041 |       if let Some(v) = p.orring_squeeze_pct { | conditional gate; changing condition shifts runtime path an
ide effects. |

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| 1042 |         self.spec.squeeze_pct = v.clamp(5.0, 45.0); | executable/support line; behavior impact depends on
surrounding block context. |
| 1043 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1044 |         if let Some(v) = p.compression_set_pct { | conditional gate; changing condition shifts runtime path
side effects. |
| 1045 |             self.spec.compression_set_pct = v.clamp(0.0, 80.0); | executable/support line; behavior impact d
nds on surrounding block context. |
| 1046 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1047 |         if let Some(v) = p.base_leak_area_mm2 { | conditional gate; changing condition shifts runtime path a
side effects. |
| 1048 |             self.spec.base_leak_area_mm2 = v.max(0.0); | executable/support line; behavior impact depends on
rrounding block context. |
| 1049 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1050 |         if let Some(v) = p.max_supported_temp_c { | conditional gate; changing condition shifts runtime path
d side effects. |
| 1051 |             let (tmin, _) = self.spec.material.temp_window_c(); | executable/support line; behavior impact d
nds on surrounding block context. |
| 1052 |             let _ = tmin; | executable/support line; behavior impact depends on surrounding block context. |
| 1053 |             // user-specific temp ceiling shifts modeled via compression set drift | comment line; documenta
n intent, no runtime effect. |
| 1054 |             let over = (self.operation_pressure_bar * 0.0 + 1.0) * (v / 180.0).clamp(0.4, 1.4); | executable
pport line; behavior impact depends on surrounding block context. |
| 1055 |             self.spec.compression_set_pct = | executable/support line; behavior impact depends on surroundin
lock context. |
| 1056 |                 (self.spec.compression_set_pct * (2.0 - over)).clamp(0.0, 80.0); | executable/support line;
avior impact depends on surrounding block context. |
| 1057 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1058 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1059 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1060 | (blank) | line is blank; visual separation only. |
| 1061 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 1062 | pub struct LeakPhysicsRig { | declares data layout; field changes may break serialization, tests, and caller
|
| 1063 |     pub circuits: Vec<LeakCircuit>, | executable/support line; behavior impact depends on surrounding block
text. |
| 1064 |     pub alerts: Vec<LeakAlert>, | executable/support line; behavior impact depends on surrounding block cont
. |
| 1065 |     pub total_leak_lpm: f64, | executable/support line; behavior impact depends on surrounding block context
| 1066 |     pub last_results: Vec<CircuitResult>, | executable/support line; behavior impact depends on surrounding
ck context. |
| 1067 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1068 | (blank) | line is blank; visual separation only. |
| 1069 | impl Default for LeakPhysicsRig { | implementation block; method behavior and trait conformance are defined
e. |
| 1070 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, a
outputs. |
| 1071 |         Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 1072 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1073 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1074 | (blank) | line is blank; visual separation only. |
| 1075 | impl LeakPhysicsRig { | implementation block; method behavior and trait conformance are defined here. |
| 1076 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, a
outputs. |
| 1077 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 1078 |             circuits: Vec::new(), | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1079 |             alerts: Vec::new(), | executable/support line; behavior impact depends on surrounding block cont
. |
| 1080 |             total_leak_lpm: 0.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 1081 |             last_results: Vec::new(), | executable/support line; behavior impact depends on surrounding bloc

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ontext. |
| 1082 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1083 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1084 | (blank) | line is blank; visual separation only. |
| 1085 |     pub fn with_default_machine_presets() -> Self { | declares executable behavior; signature/body changes a
r API, control flow, and outputs. |
| 1086 |         let mut rig = Self::new(); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1087 | (blank) | line is blank; visual separation only. |
| 1088 |         rig.circuits.push(LeakCircuit::new( | executable/support line; behavior impact depends on surroundin
lock context. |
| 1089 |             "HYD_MAIN", | executable/support line; behavior impact depends on surrounding block context. |
| 1090 |             "O-ring circuito hidraulico principal", | executable/support line; behavior impact depends on su
unding block context. |
| 1091 |             CircuitComponent::Oring, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1092 |             OilType::HydraulicIso46, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1093 |             8, | executable/support line; behavior impact depends on surrounding block context. |
| 1094 |             170.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1095 |             140.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1096 |             PressureEnvelope { | executable/support line; behavior impact depends on surrounding block conte
|
| 1097 |                 min_bar: 45.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 1098 |                 mean_bar: 135.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 1099 |                 ideal_bar: 160.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1100 |                 max_bar: 230.0, | executable/support line; behavior impact depends on surrounding block cont
. |
| 1101 |                 rupture_bar: 285.0, | executable/support line; behavior impact depends on surrounding block
text. |
| 1102 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 1103 |             OringSpec { | executable/support line; behavior impact depends on surrounding block context. |
| 1104 |                 id_tag: "AS568-228 NBR90".to_string(), | executable/support line; behavior impact depends on
rounding block context. |
| 1105 |                 material: OringMaterial::Nbr, | executable/support line; behavior impact depends on surround
block context. |
| 1106 |                 shore_a: 90.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 1107 |                 cross_section_mm: 5.33, | executable/support line; behavior impact depends on surrounding bl
context. |
| 1108 |                 squeeze_pct: 18.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1109 |                 extrusion_gap_mm: 0.20, | executable/support line; behavior impact depends on surrounding bl
context. |
| 1110 |                 compression_set_pct: 12.0, | executable/support line; behavior impact depends on surrounding
ock context. |
| 1111 |                 design_life_hours: 12000.0, | executable/support line; behavior impact depends on surroundin
lock context. |
| 1112 |                 base_leak_area_mm2: 0.0007, | executable/support line; behavior impact depends on surroundin
lock context. |
| 1113 |                 discharge_coeff: 0.62, | executable/support line; behavior impact depends on surrounding blo
context. |
| 1114 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 1115 |             90.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1116 |             180.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1117 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 1118 | (blank) | line is blank; visual separation only. |
| 1119 |         rig.circuits.push(LeakCircuit::new( | executable/support line; behavior impact depends on surroundin
lock context. |
| 1120 |             "ENG_OIL", | executable/support line; behavior impact depends on surrounding block context. |
| 1121 |             "O-ring pressao de oleo do motor", | executable/support line; behavior impact depends on surround

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g block context. |
| 1122 |      CircuitComponent::Seal, | executable/support line; behavior impact depends on surrounding block
text. |
| 1123 |      OilType::Engine15w40, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1124 |      3, | executable/support line; behavior impact depends on surrounding block context. |
| 1125 |      4.4, | executable/support line; behavior impact depends on surrounding block context. |
| 1126 |      3.4, | executable/support line; behavior impact depends on surrounding block context. |
| 1127 |      PressureEnvelope { | executable/support line; behavior impact depends on surrounding block conte
|
| 1128 |          min_bar: 1.2, | executable/support line; behavior impact depends on surrounding block context
|
| 1129 |          mean_bar: 3.2, | executable/support line; behavior impact depends on surrounding block conte
|
| 1130 |          ideal_bar: 4.0, | executable/support line; behavior impact depends on surrounding block cont
. |
| 1131 |          max_bar: 5.2, | executable/support line; behavior impact depends on surrounding block context
|
| 1132 |          rupture_bar: 7.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1133 |      }, | executable/support line; behavior impact depends on surrounding block context. |
| 1134 |      OringSpec { | executable/support line; behavior impact depends on surrounding block context. |
| 1135 |          id_tag: "AS568-214 FKM75".to_string(), | executable/support line; behavior impact depends on
rounding block context. |
| 1136 |          material: OringMaterial::Fkm, | executable/support line; behavior impact depends on surround
block context. |
| 1137 |          shore_a: 75.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 1138 |          cross_section_mm: 3.53, | executable/support line; behavior impact depends on surrounding bl
context. |
| 1139 |          squeeze_pct: 20.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1140 |          extrusion_gap_mm: 0.08, | executable/support line; behavior impact depends on surrounding bl
context. |
| 1141 |          compression_set_pct: 10.0, | executable/support line; behavior impact depends on surrounding
ock context. |
| 1142 |          design_life_hours: 10000.0, | executable/support line; behavior impact depends on surroundin
lock context. |
| 1143 |          base_leak_area_mm2: 0.0003, | executable/support line; behavior impact depends on surroundin
lock context. |
| 1144 |          discharge_coeff: 0.58, | executable/support line; behavior impact depends on surrounding blo
context. |
| 1145 |      }, | executable/support line; behavior impact depends on surrounding block context. |
| 1146 |      28.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1147 |      45.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1148 |      )); | executable/support line; behavior impact depends on surrounding block context. |
| 1149 | (blank) | line is blank; visual separation only. |
| 1150 |      rig.circuits.push(LeakCircuit::new( | executable/support line; behavior impact depends on surroundin
lock context. |
| 1151 |          "AC_HIGH", | executable/support line; behavior impact depends on surrounding block context. |
| 1152 |          "O-ring alta do ar condicionado", | executable/support line; behavior impact depends on surround
block context. |
| 1153 |      CircuitComponent::ACHose, | executable/support line; behavior impact depends on surrounding blo
ontext. |
| 1154 |      OilType::Pag46, | executable/support line; behavior impact depends on surrounding block context.
|
| 1155 |      6, | executable/support line; behavior impact depends on surrounding block context. |
| 1156 |      16.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1157 |      12.5, | executable/support line; behavior impact depends on surrounding block context. |
| 1158 |      PressureEnvelope { | executable/support line; behavior impact depends on surrounding block conte
|
| 1159 |          min_bar: 7.0, | executable/support line; behavior impact depends on surrounding block context
|
| 1160 |          mean_bar: 12.0, | executable/support line; behavior impact depends on surrounding block cont
. |
| 1161 |          ideal_bar: 14.0, | executable/support line; behavior impact depends on surrounding block con

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t. |
| 1162 |           max_bar: 20.0, | executable/support line; behavior impact depends on surrounding block context. | | |
| 1163 |           rupture_bar: 28.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1164 |       }, | executable/support line; behavior impact depends on surrounding block context. |
| 1165 |       OringSpec { | executable/support line; behavior impact depends on surrounding block context. |
| 1166 |           id_tag: "AS568-210 HNBR80".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 1167 |           material: OringMaterial::Hnbr, | executable/support line; behavior impact depends on surrounding block context. |
| 1168 |           shore_a: 80.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1169 |           cross_section_mm: 3.53, | executable/support line; behavior impact depends on surrounding block context. |
| 1170 |           squeeze_pct: 17.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1171 |           extrusion_gap_mm: 0.10, | executable/support line; behavior impact depends on surrounding block context. |
| 1172 |           compression_set_pct: 14.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1173 |           design_life_hours: 9000.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1174 |           base_leak_area_mm2: 0.0005, | executable/support line; behavior impact depends on surrounding block context. |
| 1175 |           discharge_coeff: 0.55, | executable/support line; behavior impact depends on surrounding block context. |
| 1176 |       }, | executable/support line; behavior impact depends on surrounding block context. |
| 1177 |       1.1, | executable/support line; behavior impact depends on surrounding block context. |
| 1178 |       12.0, | executable/support line; behavior impact depends on surrounding block context. |
| 1179 |     )); | executable/support line; behavior impact depends on surrounding block context. |
| 1180 | (blank) | line is blank; visual separation only. |
| 1181 |     rig | executable/support line; behavior impact depends on surrounding block context. |
| 1182 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1183 | (blank) | line is blank; visual separation only. |
| 1184 |     pub fn add_circuit(&mut self, circuit: LeakCircuit) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1185 |         self.circuits.push(circuit); | executable/support line; behavior impact depends on surrounding block context. |
| 1186 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1187 | (blank) | line is blank; visual separation only. |
| 1188 |     pub fn circuit_mut(&mut self, name: &str) -> Option<&mut LeakCircuit> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1189 |         self.circuits.iter_mut().find(|c| c.name == name) | executable/support line; behavior impact depends on surrounding block context. |
| 1190 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1191 | (blank) | line is blank; visual separation only. |
| 1192 |     pub fn oil_density_for(&self, name: &str) -> f64 { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1193 |         self.circuits | executable/support line; behavior impact depends on surrounding block context. |
| 1194 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1195 |             .find(|c| c.name == name) | executable/support line; behavior impact depends on surrounding block context. |
| 1196 |             .map(|c| c.oil_type.density_kg_m3()) | executable/support line; behavior impact depends on surrounding block context. |
| 1197 |             .unwrap_or(900.0) | executable/support line; behavior impact depends on surrounding block context. |
| 1198 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1199 | (blank) | line is blank; visual separation only. |
| 1200 |     pub fn set_runtime_pressure_state( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1201 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 1202 |         name: &str, | executable/support line; behavior impact depends on surrounding block context. |
| 1203 |         piston_pressure_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
xt. |

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| 1204 |           operation_pressure_bar: f64, | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 1205 |       ) { | executable/support line; behavior impact depends on surrounding block context. |
| 1206 |           if let Some(c) = self.circuit_mut(name) { | conditional gate; changing condition shifts runtime path
d side effects. |
| 1207 |               c.piston_pressure_bar = piston_pressure_bar.max(0.0); | executable/support line; behavior impact
depends on surrounding block context. |
| 1208 |               c.operation_pressure_bar = operation_pressure_bar.max(0.0); | executable/support line; behavior
act depends on surrounding block context. |
| 1209 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1210 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1211 | (blank) | line is blank; visual separation only. |
| 1212 |       pub fn apply_manual_params(&mut self, name: &str, params: ManualCircuitParams) -> bool { | declares exec
ble behavior; signature/body changes alter API, control flow, and outputs. |
| 1213 |           if let Some(c) = self.circuit_mut(name) { | conditional gate; changing condition shifts runtime path
d side effects. |
| 1214 |               c.apply_manual_params(params); | executable/support line; behavior impact depends on surrounding
ock context. |
| 1215 |               true | executable/support line; behavior impact depends on surrounding block context. |
| 1216 |           } else { | executable/support line; behavior impact depends on surrounding block context. |
| 1217 |               false | executable/support line; behavior impact depends on surrounding block context. |
| 1218 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1219 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1220 | (blank) | line is blank; visual separation only. |
| 1221 |       pub fn step(&mut self, inputs: &[(&str, CircuitInput)], dt: f64) -> Vec<CircuitResult> { | declares exec
ble behavior; signature/body changes alter API, control flow, and outputs. |
| 1222 |           self.alerts.clear(); | executable/support line; behavior impact depends on surrounding block context
| 1223 |           self.total_leak_lpm = 0.0; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1224 | (blank) | line is blank; visual separation only. |
| 1225 |           let mut results = Vec::new(); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1226 |           for (name, input) in inputs { | iteration control; changing bounds/body affects throughput, timing,
possibly safety guarantees. |
| 1227 |               if let Some(circuit) = self.circuits.iter_mut().find(|c| c.name == *name) { | conditional gate
changing condition shifts runtime path and side effects. |
| 1228 |                   let out = circuit.step(*input, dt); | executable/support line; behavior impact depends on su
unding block context. |
| 1229 |                   self.total_leak_lpm += out.leak_lpm; | executable/support line; behavior impact depends on s
ounding block context. |
| 1230 |                   if out.alert >= LeakAlertLevel::Watch { | conditional gate; changing condition shifts runtim
ath and side effects. |
| 1231 |                       self.alerts.push(LeakAlert { | executable/support line; behavior impact depends on surro
ing block context. |
| 1232 |                           circuit_name: out.name.clone(), | executable/support line; behavior impact depends o
urrounding block context. |
| 1233 |                           level: out.alert, | executable/support line; behavior impact depends on surrounding
ck context. |
| 1234 |                           message: out.warning_text.clone(), | executable/support line; behavior impact depend
n surrounding block context. |
| 1235 |                           predicted_hours_to_rupture: out.predicted_hours_to_rupture, | executable/support lin
behavior impact depends on surrounding block context. |
| 1236 |                       }); | executable/support line; behavior impact depends on surrounding block context. |
| 1237 |                   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1238 |                   results.push(out); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1239 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1240 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1241 | (blank) | line is blank; visual separation only. |
| 1242 |           self.last_results = results.clone(); | executable/support line; behavior impact depends on surroundi

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block context. |
| 1243 |         results | executable/support line; behavior impact depends on surrounding block context. |
| 1244 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1245 | (blank) | line is blank; visual separation only. |
| 1246 |     pub fn reset(&mut self) { | declares executable behavior; signature/body changes alter API, control flow
and outputs. |
| 1247 |         self.total_leak_lpm = 0.0; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1248 |         self.alerts.clear(); | executable/support line; behavior impact depends on surrounding block context
| 1249 |         self.last_results.clear(); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1250 |         for c in &mut self.circuits { | iteration control; changing bounds/body affects throughput, timing,
possibly safety guarantees. |
| 1251 |             c.reset(); | executable/support line; behavior impact depends on surrounding block context. |
| 1252 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1253 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1254 | (blank) | line is blank; visual separation only. |
| 1255 |     pub fn export_last_results_json<P: AsRef<Path>>(&self, path: P) -> std::io::Result<()> { | declares execu
ble behavior; signature/body changes alter API, control flow, and outputs. |
| 1256 |         let path_ref = path.as_ref(); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1257 |         if let Some(parent) = path_ref.parent() { | conditional gate; changing condition shifts runtime path
d side effects. |
| 1258 |             if !parent.as_os_str().is_empty() { | conditional gate; changing condition shifts runtime path a
side effects. |
| 1259 |                 fs::create_dir_all(parent)?; | error-propagating operation; changing it impacts failure sema
cs and caller recovery paths. |
| 1260 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1261 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1262 |         let payload = serde_json::to_string_pretty(&self.last_results) | serialization contract touchpoint;
ts can break report compatibility and tooling. |
| 1263 |         .map_err(|e| std::io::Error::other(format!("json serialization failed: {e}")))?; | error-propa
ing operation; changing it impacts failure semantics and caller recovery paths. |
| 1264 |         fs::write(path_ref, payload) | I/O or protocol-critical logic; edits can affect integration, timing,
d diagnostic outcomes. |
| 1265 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1266 | (blank) | line is blank; visual separation only. |
| 1267 |     pub fn export_last_results_csv<P: AsRef<Path>>(&self, path: P) -> std::io::Result<()> { | declares execu
le behavior; signature/body changes alter API, control flow, and outputs. |
| 1268 |         let path_ref = path.as_ref(); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1269 |         if let Some(parent) = path_ref.parent() { | conditional gate; changing condition shifts runtime path
d side effects. |
| 1270 |             if !parent.as_os_str().is_empty() { | conditional gate; changing condition shifts runtime path a
side effects. |
| 1271 |                 fs::create_dir_all(parent)?; | error-propagating operation; changing it impacts failure sema
cs and caller recovery paths. |
| 1272 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1273 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1274 |         let mut out = String::from( | executable/support line; behavior impact depends on surrounding block
text. |
| 1275 |             "name,alert,pressure_band,current_pressure_bar,rupture_probability_pct,predicted_hours_to_ruptur
upture_pressure_bar,rupture_elapsed_h,leak_area_mm2,leak_lpm,pressure_min_bar,pressure_mean_bar,pressure_max_bar,reco
nded_hold_min_bar,recommended_hold_target_bar,recommended_hold_max_bar,margin_to_max_bar,margin_to_rupture_bar,rca_hi
pca_recommendation,rupture_confirmed,warning_text\n", | executable/support line; behavior impact depends on surroundi
block context. |
| 1276 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 1277 |         for r in &self.last_results { | iteration control; changing bounds/body affects throughput, timing,
possibly safety guarantees. |
| 1278 |             let warn = r.warning_text.replace("'", "'").replace(',', ","); | executable/support line; behav

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    impact depends on surrounding block context. |
| 1279 |         let rca = r.rca_hint.replace('','').replace(',',' '); | executable/support line; behavior impact depends on surrounding block context. | | |
| 1280 |         let pca = r.pca_recommendation.replace('','').replace(',',' '); | executable/support line; behavior impact depends on surrounding block context. |
| 1281 |         out.push_str(&format!( | executable/support line; behavior impact depends on surrounding block context. |
| 1282 |             "{},{},{:.6},{:.6},{},{},{},{:.9},{:.9},{:.6},{:.6},{:.6},{:.6},{:.6},{:.6},{:.6},{:.6},\n",\n","{\}\n","{\}\n","{\}\n", | executable/support line; behavior impact depends on surrounding block context. |
| 1283 |             r.name, | executable/support line; behavior impact depends on surrounding block context. |
| 1284 |             r.alert, | executable/support line; behavior impact depends on surrounding block context. |
| 1285 |             r.pressure_band, | executable/support line; behavior impact depends on surrounding block context. |
| 1286 |             r.current_pressure_bar, | executable/support line; behavior impact depends on surrounding block context. |
| 1287 |             r.rupture_probability_pct, | executable/support line; behavior impact depends on surrounding block context. |
| 1288 |             r.predicted_hours_to_rupture | executable/support line; behavior impact depends on surrounding block context. |
| 1289 |             .map(|v| format!("{v:.6}")) | executable/support line; behavior impact depends on surrounding block context. |
| 1290 |             .unwrap_or_default(), | executable/support line; behavior impact depends on surrounding block context. |
| 1291 |             r.rupture_pressure_bar | executable/support line; behavior impact depends on surrounding block context. |
| 1292 |             .map(|v| format!("{v:.6}")) | executable/support line; behavior impact depends on surrounding block context. |
| 1293 |             .unwrap_or_default(), | executable/support line; behavior impact depends on surrounding block context. |
| 1294 |             r.rupture_elapsed_h | executable/support line; behavior impact depends on surrounding block context. |
| 1295 |             .map(|v| format!("{v:.6}")) | executable/support line; behavior impact depends on surrounding block context. |
| 1296 |             .unwrap_or_default(), | executable/support line; behavior impact depends on surrounding block context. |
| 1297 |             r.leak_area_mm2, | executable/support line; behavior impact depends on surrounding block context. |
| 1298 |             r.leak_lpm, | executable/support line; behavior impact depends on surrounding block context. |
| 1299 |             r.pressure_min_bar, | executable/support line; behavior impact depends on surrounding block context. |
| 1300 |             r.pressure_mean_bar, | executable/support line; behavior impact depends on surrounding block context. |
| 1301 |             r.pressure_max_bar, | executable/support line; behavior impact depends on surrounding block context. |
| 1302 |             r.recommended_hold_min_bar, | executable/support line; behavior impact depends on surrounding block context. |
| 1303 |             r.recommended_hold_target_bar, | executable/support line; behavior impact depends on surrounding block context. |
| 1304 |             r.recommended_hold_max_bar, | executable/support line; behavior impact depends on surrounding block context. |
| 1305 |             r.margin_to_max_bar, | executable/support line; behavior impact depends on surrounding block context. |
| 1306 |             r.margin_to_rupture_bar, | executable/support line; behavior impact depends on surrounding block context. |
| 1307 |             rca, | executable/support line; behavior impact depends on surrounding block context. |
| 1308 |             pca, | executable/support line; behavior impact depends on surrounding block context. |
| 1309 |             r.rupture_confirmed, | executable/support line; behavior impact depends on surrounding block context. |
| 1310 |             warn | executable/support line; behavior impact depends on surrounding block context. |
| 1311 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 1312 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1313 |     fs::write(path_ref, out) | I/O or protocol-critical logic; edits can affect integration, timing, and agnostic outcomes. |
| 1314 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.

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| 1315 | (blank) | line is blank; visual separation only. |
| 1316 |     pub fn export_predictions_json<P: AsRef<Path>>() { | declares executable behavior; signature/body changes al
r API, control flow, and outputs. |
| 1317 |         &self, | executable/support line; behavior impact depends on surrounding block context. |
| 1318 |         path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1319 |         predictions: &[ScenarioPrediction], | executable/support line; behavior impact depends on surroundin
lock context. |
| 1320 |     ) -> std::io::Result<()> { | executable/support line; behavior impact depends on surrounding block conte
|
| 1321 |         let path_ref = path.as_ref(); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1322 |         if let Some(parent) = path_ref.parent() { | conditional gate; changing condition shifts runtime path
d side effects. |
| 1323 |             if !parent.as_os_str().is_empty() { | conditional gate; changing condition shifts runtime path a
side effects. |
| 1324 |                 fs::create_dir_all(parent)?; | error-propagating operation; changing it impacts failure sema
cs and caller recovery paths. |
| 1325 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1326 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1327 |         let payload = serde_json::to_string_pretty(predictions) | serialization contract touchpoint; edits c
break report compatibility and tooling. |
| 1328 |         .map_err(|e| std::io::Error::other(format!("json serialization failed: {e}")))?; | error-propa
ing operation; changing it impacts failure semantics and caller recovery paths. |
| 1329 |         fs::write(path_ref, payload) | I/O or protocol-critical logic; edits can affect integration, timing,
d diagnostic outcomes. |
| 1330 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1331 | (blank) | line is blank; visual separation only. |
| 1332 |     pub fn export_predictions_csv<P: AsRef<Path>>() { | declares executable behavior; signature/body changes al
API, control flow, and outputs. |
| 1333 |         &self, | executable/support line; behavior impact depends on surrounding block context. |
| 1334 |         path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1335 |         predictions: &[ScenarioPrediction], | executable/support line; behavior impact depends on surroundin
lock context. |
| 1336 |     ) -> std::io::Result<()> { | executable/support line; behavior impact depends on surrounding block conte
|
| 1337 |         let path_ref = path.as_ref(); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1338 |         if let Some(parent) = path_ref.parent() { | conditional gate; changing condition shifts runtime path
d side effects. |
| 1339 |             if !parent.as_os_str().is_empty() { | conditional gate; changing condition shifts runtime path a
side effects. |
| 1340 |                 fs::create_dir_all(parent)?; | error-propagating operation; changing it impacts failure sema
cs and caller recovery paths. |
| 1341 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1342 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1343 |         let mut out = String::from("circuit_name,scenario_name,peak_pressure_bar,final_alert,final_rupture_p
ability_pct,hours_to_rupture,likely_failure_mode\n"); | executable/support line; behavior impact depends on surroundi
block context. |
| 1344 |         for p in predictions { | iteration control; changing bounds/body affects throughput, timing, and pos
ly safety guarantees. |
| 1345 |             let mode = p.likely_failure_mode.replace("'", "").replace(',', ";"); | executable/support line
ehavior impact depends on surrounding block context. |
| 1346 |             out.push_str(&format!( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1347 |                 "{},{},{:.6},{},{:.6},{},{ }\n", | executable/support line; behavior impact depends on su
unding block context. |
| 1348 |                 p.circuit_name, | executable/support line; behavior impact depends on surrounding block cont
. |
| 1349 |                 p.scenario_name, | executable/support line; behavior impact depends on surrounding block con
t. |
| 1350 |                 p.peak_pressure_bar, | executable/support line; behavior impact depends on surrounding block

```



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ntext. |
| 1351 |           p.final_alert, | executable/support line; behavior impact depends on surrounding block context. |
| 1352 |           p.final_rupture_probability_pct, | executable/support line; behavior impact depends on surrounding block context. |
| 1353 |           p.hours_to_rupture | executable/support line; behavior impact depends on surrounding block context. |
| 1354 |           .map(\|v\| format!("{v:.6}")) | executable/support line; behavior impact depends on surrounding block context. |
| 1355 |           .unwrap_or_default(), | executable/support line; behavior impact depends on surrounding block context. |
| 1356 |           mode | executable/support line; behavior impact depends on surrounding block context. |
| 1357 |       )); | executable/support line; behavior impact depends on surrounding block context. |
| 1358 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1359 |       fs::write(path_ref, out) | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1360 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1361 | (blank) | line is blank; visual separation only. |
| 1362 |   pub fn export_material_catalog_json<P: AsRef<Path>>(&self, path: P) -> std::io::Result<()> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1363 |       let path_ref = path.as_ref(); | executable/support line; behavior impact depends on surrounding block context. |
| 1364 |       if let Some(parent) = path_ref.parent() { | conditional gate; changing condition shifts runtime path and side effects. |
| 1365 |           if !parent.as_os_str().is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 1366 |               fs::create_dir_all(parent)?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1367 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1368 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1369 |       let payload = serde_json::to_string_pretty(&engineering_material_catalog()) | serialization contract touchpoint; edits can break report compatibility and tooling. |
| 1370 |       .map_err(\|e\| std::io::Error::other(format!("json serialization failed: {e}")))?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1371 |       fs::write(path_ref, payload) | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1372 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1373 | (blank) | line is blank; visual separation only. |
| 1374 |   pub fn export_material_catalog_csv<P: AsRef<Path>>(&self, path: P) -> std::io::Result<()> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1375 |       let path_ref = path.as_ref(); | executable/support line; behavior impact depends on surrounding block context. |
| 1376 |       if let Some(parent) = path_ref.parent() { | conditional gate; changing condition shifts runtime path and side effects. |
| 1377 |           if !parent.as_os_str().is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 1378 |               fs::create_dir_all(parent)?; | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1379 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1380 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1381 |       let mut out = String::from("id,family,designation,hardness_shore,tensile_strength_mpa,elongation_pct,compression_set_pct_70h,tear_strength_kn_m,density_g_cm3,temp_min_c,temp_max_c,abrasion_index,corrosion_resistance_index,hydraulic_oil_compat,engine_oil_compat,refrigerant_oil_compat,notes\n"); | executable/support line; behavior impact depends on surrounding block context. |
| 1382 |       for m in engineering_material_catalog() { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 1383 |           let notes = m.notes.replace("'", "").replace(',', ";"); | executable/support line; behavior impact depends on surrounding block context. |
| 1384 |           out.push_str(&format!("{}", m)); | executable/support line; behavior impact depends on surrounding block context. |

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```
| 1385 | " {}, { }, { : . 3 } , { : . 3 } , { : . 3 } , { : . 3 } , { : . 3 } , { : . 3 } , { : . 3 } , { : . 3 } , { : . 3 } , \"{}|  
n\", | executable/support line; behavior impact depends on surrounding block context. |  
| 1386 | m.id, | executable/support line; behavior impact depends on surrounding block context. |  
| 1387 | m.family, | executable/support line; behavior impact depends on surrounding block context. |  
| 1388 | m.designation, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 1389 | m.hardness_shore, | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 1390 | m.tensile_strength_mpa, | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 1391 | m.elongation_pct, | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 1392 | m.compression_set_pct_70h, | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 1393 | m.tear_strength_kn_m, | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 1394 | m.density_g_cm3, | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 1395 | m.temp_min_c, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 1396 | m.temp_max_c, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 1397 | m.abrasion_index, | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 1398 | m.corrosion_resistance_index, | executable/support line; behavior impact depends on surround  
block context. |  
| 1399 | m.hydraulic_oil_compat, | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 1400 | m.engine_oil_compat, | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 1401 | m.refrigerant_oil_compat, | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 1402 | notes, | executable/support line; behavior impact depends on surrounding block context. |  
| 1403 | )); | executable/support line; behavior impact depends on surrounding block context. |  
| 1404 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 1405 | fs::write(path_ref, out) | I/O or protocol-critical logic; edits can affect integration, timing, and  
agnostic outcomes. |  
| 1406 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 1407 | (blank) | line is blank; visual separation only. |  
| 1408 | pub fn export_oil_catalog_json<P: AsRef<Path>>(&self, path: P) -> std::io::Result<()> { | declares execu  
le behavior; signature/body changes alter API, control flow, and outputs. |  
| 1409 | let path_ref = path.as_ref(); | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 1410 | if let Some(parent) = path_ref.parent() { | conditional gate; changing condition shifts runtime path  
d side effects. |  
| 1411 | if !parent.as_os_str().is_empty() { | conditional gate; changing condition shifts runtime path a  
side effects. |  
| 1412 | fs::create_dir_all(parent)?; | error-propagating operation; changing it impacts failure sema  
cs and caller recovery paths. |  
| 1413 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 1414 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 1415 | let payload = serde_json::to_string_pretty(&engineering_oil_catalog()) | serialization contract touc  
int; edits can break report compatibility and tooling. |  
| 1416 | .map_err(|e| std::io::Error::other(format!(\"json serialization failed: {e}\")))?; | error-propa  
ing operation; changing it impacts failure semantics and caller recovery paths. |  
| 1417 | fs::write(path_ref, payload) | I/O or protocol-critical logic; edits can affect integration, timing,  
d diagnostic outcomes. |  
| 1418 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 1419 | (blank) | line is blank; visual separation only. |  
| 1420 | pub fn export_oil_catalog_csv<P: AsRef<Path>>(&self, path: P) -> std::io::Result<()> { | declares execut  
e behavior; signature/body changes alter API, control flow, and outputs. |  
| 1421 | let path_ref = path.as_ref(); | executable/support line; behavior impact depends on surrounding blo
```

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|
| context. |
| 1422 |         if let Some(parent) = path_ref.parent() { | conditional gate; changing condition shifts runtime path
d side effects. |
| 1423 |             if !parent.as_os_str().is_empty() { | conditional gate; changing condition shifts runtime path a
side effects. |
| 1424 |                 fs::create_dir_all(parent)?; | error-propagating operation; changing it impacts failure sema
cs and caller recovery paths. |
| 1425 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1426 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1427 |         let mut out = String::from("id,family,grade,base_stock,density_15c_kg_m3,viscosity_40c_cst,viscosity
0c_cst,viscosity_index,pour_point_c,flash_point_c,tan_mg_koh_g,water_content_ppm_limit,demulsibility_min,copper_corro
n_level,oxidation_stability_h,recommended_min_temp_c,recommended_max_temp_c,anti_wear_index,corrosion_inhibition_inde
otes\n"); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1428 |         for o in engineering_oil_catalog() { | iteration control; changing bounds/body affects throughput, t
ng, and possibly safety guarantees. |
| 1429 |             let notes = o.notes.replace('\"', '\"').replace(',', ', '); | executable/support line; behavior im
t depends on surrounding block context. |
| 1430 |             out.push_str(&format!( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1431 |                 "{},{},{},{},{:.3},{:.3},{:.3},{:.3},{:.3},{:.3},{:.3},{:.3},{:.3},{:.3},{:.3},{:.3},{{
}}, {:.3}, \"{}\\n\", | executable/support line; behavior impact depends on surrounding block context. |
| 1432 |                 o.id, | executable/support line; behavior impact depends on surrounding block context. |
| 1433 |                 o.family, | executable/support line; behavior impact depends on surrounding block context. |
| 1434 |                 o.grade, | executable/support line; behavior impact depends on surrounding block context. |
| 1435 |                 o.base_stock, | executable/support line; behavior impact depends on surrounding block contex
|
| 1436 |                 o.density_15c_kg_m3, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1437 |                 o.viscosity_40c_cst, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1438 |                 o.viscosity_100c_cst, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1439 |                 o.viscosity_index, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1440 |                 o.pour_point_c, | executable/support line; behavior impact depends on surrounding block cont
. |
| 1441 |                 o.flash_point_c, | I/O or protocol-critical logic; edits can affect integration, timing, and
agnostic outcomes. |
| 1442 |                 o.tan_mg_koh_g, | executable/support line; behavior impact depends on surrounding block cont
. |
| 1443 |                 o.water_content_ppm_limit, | executable/support line; behavior impact depends on surrounding
ock context. |
| 1444 |                 o.demulsibility_min, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1445 |                 o.copper_corrosion_level, | executable/support line; behavior impact depends on surrounding
ck context. |
| 1446 |                 o.oxidation_stability_h, | executable/support line; behavior impact depends on surrounding b
k context. |
| 1447 |                 o.recommended_min_temp_c, | executable/support line; behavior impact depends on surrounding
ck context. |
| 1448 |                 o.recommended_max_temp_c, | executable/support line; behavior impact depends on surrounding
ck context. |
| 1449 |                 o.anti_wear_index, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1450 |                 o.corrosion_inhibition_index, | executable/support line; behavior impact depends on surround
block context. |
| 1451 |                 notes, | executable/support line; behavior impact depends on surrounding block context. |
| 1452 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 1453 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1454 |         fs::write(path_ref, out) | I/O or protocol-critical logic; edits can affect integration, timing, and
agnostic outcomes.
| 1455 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
```

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| 1456 | (blank) | line is blank; visual separation only. |
| 1457 |     pub fn calibrate_from_csv<P: AsRef<Path>>() | declares executable behavior; signature/body changes alter
, control flow, and outputs. |
| 1458 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 1459 |         csv_path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1460 |     ) -> std::io::Result<CalibrationReport> { | executable/support line; behavior impact depends on surround
block context. |
| 1461 |         #[derive(Debug, Deserialize)] | serialization contract touchpoint; edits can break report compatibil
and tooling. |
| 1462 |         struct CsvRow { | declares data layout; field changes may break serialization, tests, and callers. |
| 1463 |             timestamp_s: Option<f64>, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1464 |             circuit_name: String, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1465 |             pressure_bar: f64, | executable/support line; behavior impact depends on surrounding block conte
|
| 1466 |             delta_p_bar: Option<f64>, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1467 |             temp_c: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 1468 |             cycles_per_s: f64, | executable/support line; behavior impact depends on surrounding block conte
|
| 1469 |             duty_01: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 1470 |             fluid_density_kg_m3: Option<f64>, | executable/support line; behavior impact depends on surround
block context. |
| 1471 |             measured_leak_lpm: f64, | executable/support line; behavior impact depends on surrounding block
text. |
| 1472 |             observed_rupture: Option<bool>, | executable/support line; behavior impact depends on surroundin
lock context. |
| 1473 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1474 | (blank) | line is blank; visual separation only. |
| 1475 |         let mut rdr = csv::ReaderBuilder::new() | executable/support line; behavior impact depends on surrou
ng block context. |
| 1476 |             .has_headers(true) | executable/support line; behavior impact depends on surrounding block conte
|
| 1477 |             .flexible(true) | executable/support line; behavior impact depends on surrounding block context.
|
| 1478 |             .trim(csv::Trim::All) | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1479 |             .from_path(csv_path.as_ref()) | executable/support line; behavior impact depends on surrounding
ck context. |
| 1480 |             .map_err(\|e\| std::io::Error::other(format!("open csv failed: {e}")))?; | error-propagating ope
ion; changing it impacts failure semantics and caller recovery paths. |
| 1481 | (blank) | line is blank; visual separation only. |
| 1482 |         let mut samples_by_circuit: std::collections::BTreeMap<String, Vec<CalibrationCsvSample>> = | execut
e/support line; behavior impact depends on surrounding block context. |
| 1483 |             std::collections::BTreeMap::new(); | executable/support line; behavior impact depends on surround
g block context. |
| 1484 | (blank) | line is blank; visual separation only. |
| 1485 |         for rec in rdr.deserialize::<CsvRow>() { | iteration control; changing bounds/body affects throughpu
timing, and possibly safety guarantees. |
| 1486 |             let row = rec.map_err(\|e\| std::io::Error::other(format!("csv parse failed: {e}")))?; | error-p
agating operation; changing it impacts failure semantics and caller recovery paths. |
| 1487 |             let circuit = row.circuit_name.trim().to_string(); | executable/support line; behavior impact de
ds on surrounding block context. |
| 1488 |             let density = row.fluid_density_kg_m3.unwrap_or(900.0); | executable/support line; behavior impa
depends on surrounding block context. |
| 1489 |             samples_by_circuit | executable/support line; behavior impact depends on surrounding block conte
|
| 1490 |                 .entry(circuit.clone()) | executable/support line; behavior impact depends on surrounding bl
context. |
| 1491 |                 .or_default() | executable/support line; behavior impact depends on surrounding block contex
|
| 1492 |                 .push(CalibrationCsvSample { | executable/support line; behavior impact depends on surroundi
block context. |
| 1493 |                     timestamp_s: row.timestamp_s, | executable/support line; behavior impact depends on surr

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ding block context. |
| 1494 |          circuit_name: circuit, | executable/support line; behavior impact depends on surrounding
ock context. |
| 1495 |          pressure_bar: row.pressure_bar, | executable/support line; behavior impact depends on su
unding block context. |
| 1496 |          delta_p_bar: row.delta_p_bar.unwrap_or(row.pressure_bar), | executable/support line; beh
or impact depends on surrounding block context. |
| 1497 |          temp_c: row.temp_c, | executable/support line; behavior impact depends on surrounding bl
context. |
| 1498 |          cycles_per_s: row.cycles_per_s, | executable/support line; behavior impact depends on su
unding block context. |
| 1499 |          duty_01: row.duty_01, | executable/support line; behavior impact depends on surrounding
ck context. |
| 1500 |          fluid_density_kg_m3: density, | executable/support line; behavior impact depends on surr
ding block context. |
| 1501 |          measured_leak_lpm: row.measured_leak_lpm, | executable/support line; behavior impact dep
s on surrounding block context. |
| 1502 |          observed_rupture: row.observed_rupture, | executable/support line; behavior impact depen
on surrounding block context. |
| 1503 |          }); | executable/support line; behavior impact depends on surrounding block context. |
| 1504 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1505 | (blank) | line is blank; visual separation only. |
| 1506 |      let mut circuit_reports = Vec::new(); | executable/support line; behavior impact depends on surround
block context. |
| 1507 | (blank) | line is blank; visual separation only. |
| 1508 |      for circuit in &mut self.circuits { | iteration control; changing bounds/body affects throughput, ti
g, and possibly safety guarantees. |
| 1509 |          let Some(samples) = samples_by_circuit.get(&circuit.name) else { | executable/support line; beha
r impact depends on surrounding block context. |
| 1510 |              continue; | executable/support line; behavior impact depends on surrounding block context. |
| 1511 |          }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
opes. |
| 1512 |          if samples.is_empty() { | conditional gate; changing condition shifts runtime path and side effe
. |
| 1513 |              continue; | executable/support line; behavior impact depends on surrounding block context. |
| 1514 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1515 | (blank) | line is blank; visual separation only. |
| 1516 |          let mut best = circuit.coeffs; | executable/support line; behavior impact depends on surrounding
ock context. |
| 1517 |          let mut best_score = f64::INFINITY; | executable/support line; behavior impact depends on surrou
ng block context. |
| 1518 | (blank) | line is blank; visual separation only. |
| 1519 |          let grid_damage = [0.6, 0.8, 1.0, 1.2, 1.4]; | executable/support line; behavior impact depends
surrounding block context. |
| 1520 |          let grid_extr = [0.6, 0.85, 1.0, 1.2, 1.4]; | executable/support line; behavior impact depends o
urrounding block context. |
| 1521 |          let grid_thermal = [0.6, 0.85, 1.0, 1.2, 1.4]; | executable/support line; behavior impact depend
n surrounding block context. |
| 1522 |          let grid_flow = [0.7, 0.85, 1.0, 1.15, 1.3]; | executable/support line; behavior impact depends
surrounding block context. |
| 1523 | (blank) | line is blank; visual separation only. |
| 1524 |          for ds in grid_damage { | iteration control; changing bounds/body affects throughput, timing, an
possibly safety guarantees. |
| 1525 |              for es in grid_extr { | iteration control; changing bounds/body affects throughput, timing,
possibly safety guarantees. |
| 1526 |                  for ts in grid_thermal { | iteration control; changing bounds/body affects throughput, t
ng, and possibly safety guarantees. |
| 1527 |                      for fs in grid_flow { | iteration control; changing bounds/body affects throughput,
ing, and possibly safety guarantees. |
| 1528 |                          let mut sim = circuit.clone(); | executable/support line; behavior impact depend
n surrounding block context. |
| 1529 |                          sim.reset(); | executable/support line; behavior impact depends on surrounding b
k context. |

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| 1530 |                 sim.coeffs = LeakModelCoefficients { | executable/support line; behavior impact
ends on surrounding block context. |
| 1531 |                 damage_rate_scale: ds, | executable/support line; behavior impact depends on
rrounding block context. |
| 1532 |                 extrusion_rate_scale: es, | executable/support line; behavior impact depends
surrounding block context. |
| 1533 |                 thermal_rate_scale: ts, | executable/support line; behavior impact depends o
urrounding block context. |
| 1534 |                 flow_rate_scale: fs, | executable/support line; behavior impact depends on s
ounding block context. |
| 1535 |                 rupture_area_scale: 1.0, | executable/support line; behavior impact depends
surrounding block context. |
| 1536 |                 }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 1537 | (blank) | line is blank; visual separation only. |
| 1538 |                 let mut score = 0.0; | executable/support line; behavior impact depends on surro
ing block context. |
| 1539 |                 let mut prev_t: Option<f64> = None; | executable/support line; behavior impact d
nds on surrounding block context. |
| 1540 |                 for s in samples { | iteration control; changing bounds/body affects throughput,
ming, and possibly safety guarantees. |
| 1541 |                     let dt = match (prev_t, s.timestamp_s) { | executable/support line; behavior
pact depends on surrounding block context. |
| 1542 |                         (Some(p), Some(t)) => (t - p).max(0.01), | executable/support line; beha
r impact depends on surrounding block context. |
| 1543 |                         _ => 0.05, | executable/support line; behavior impact depends on surroun
g block context. |
| 1544 |                     }; | scope delimiter; structure-only line, but misplaced edits can change ow
ship and lifetime scopes. |
| 1545 |                     prev_t = s.timestamp_s; | executable/support line; behavior impact depends o
urrounding block context. |
| 1546 | (blank) | line is blank; visual separation only. |
| 1547 |                     let out = sim.step( | executable/support line; behavior impact depends on su
unding block context. |
| 1548 |                     CircuitInput { | executable/support line; behavior impact depends on sur
nding block context. |
| 1549 |                         pressure_bar: s.pressure_bar, | executable/support line; behavior im
t depends on surrounding block context. |
| 1550 |                         delta_p_bar: s.delta_p_bar, | executable/support line; behavior impa
depends on surrounding block context. |
| 1551 |                         temp_c: s.temp_c, | executable/support line; behavior impact depends
surrounding block context. |
| 1552 |                         cycles_per_s: s.cycles_per_s, | executable/support line; behavior im
t depends on surrounding block context. |
| 1553 |                         duty_01: s.duty_01, | executable/support line; behavior impact depen
on surrounding block context. |
| 1554 |                         fluid_density_kg_m3: s.fluid_density_kg_m3, | executable/support lin
behavior impact depends on surrounding block context. |
| 1555 |                         }, | executable/support line; behavior impact depends on surrounding blo
context. |
| 1556 |                         dt, | executable/support line; behavior impact depends on surrounding bl
context. |
| 1557 |                     ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1558 | (blank) | line is blank; visual separation only. |
| 1559 |                     let abs_err = (out.leak_lpm - s.measured_leak_lpm).abs(); | executable/suppo
line; behavior impact depends on surrounding block context. |
| 1560 |                     let rel = abs_err / s.measured_leak_lpm.abs().max(0.05); | executable/suppor
ine; behavior impact depends on surrounding block context. |
| 1561 |                     score += rel; | executable/support line; behavior impact depends on surround
block context. |
| 1562 |                     if let Some(obs_rupt) = s.observed_rupture { | conditional gate; changing co
tion shifts runtime path and side effects. |
| 1563 |                         if obs_rupt != out.rupture_confirmed { | conditional gate; changing cond
on shifts runtime path and side effects. |

```

```

| 1564 |                                     score += 2.5; | executable/support line; behavior impact depends on
rounding block context. |
| 1565 |                                     } | scope delimiter; structure-only line, but misplaced edits can change
nership and lifetime scopes. |
| 1566 |                                     } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 1567 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 1568 | (blank) | line is blank; visual separation only. |
| 1569 |                                     if score < best_score { | conditional gate; changing condition shifts runtime pa
and side effects. |
| 1570 |                                     best_score = score; | executable/support line; behavior impact depends on su
unding block context. |
| 1571 |                                     best = sim.coeffs; | executable/support line; behavior impact depends on sur
nding block context. |
| 1572 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 1573 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 1574 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 1575 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1576 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1577 | (blank) | line is blank; visual separation only. |
| 1578 |                                     circuit.coeffs = best; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1579 | (blank) | line is blank; visual separation only. |
| 1580 |                                     let mut eval = circuit.clone(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 1581 |                                     eval.reset(); | executable/support line; behavior impact depends on surrounding block context. |
| 1582 |                                     let mut prev_t: Option<f64> = None; | executable/support line; behavior impact depends on surrou
ng block context. |
| 1583 |                                     let mut se = 0.0; | executable/support line; behavior impact depends on surrounding block contex
|
| 1584 |                                     let mut ape = 0.0; | executable/support line; behavior impact depends on surrounding block conte
|
| 1585 |                                     let mut max_abs: f64 = 0.0; | executable/support line; behavior impact depends on surrounding bl
context. |
| 1586 |                                     let mut rupture_hits = 0usize; | executable/support line; behavior impact depends on surrounding
ock context. |
| 1587 |                                     let mut rupture_total = 0usize; | executable/support line; behavior impact depends on surroundin
lock context. |
| 1588 |                                     for s in samples { | iteration control; changing bounds/body affects throughput, timing, and pos
ly safety guarantees. |
| 1589 |                                     let dt = match (prev_t, s.timestamp_s) { | executable/support line; behavior impact depends
surrounding block context. |
| 1590 |                                     (Some(p), Some(t)) => (t - p).max(0.01), | executable/support line; behavior impact depe
on surrounding block context. |
| 1591 |                                     _ => 0.05, | executable/support line; behavior impact depends on surrounding block conte
|
| 1592 |                                     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and life
e scopes. |
| 1593 |                                     prev_t = s.timestamp_s; | executable/support line; behavior impact depends on surrounding bl
context. |
| 1594 | (blank) | line is blank; visual separation only. |
| 1595 |                                     let out = eval.step( | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1596 |                                     CircuitInput { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1597 |                                     pressure_bar: s.pressure_bar, | executable/support line; behavior impact depends on
rounding block context. |
| 1598 |                                     delta_p_bar: s.delta_p_bar, | executable/support line; behavior impact depends on su
unding block context. |

```

```

| 1599 |           temp_c: s.temp_c, | executable/support line; behavior impact depends on surrounding
ck context. |
| 1600 |           cycles_per_s: s.cycles_per_s, | executable/support line; behavior impact depends on
rounding block context. |
| 1601 |           duty_01: s.duty_01, | executable/support line; behavior impact depends on surrounding
lock context. |
| 1602 |           fluid_density_kg_m3: s.fluid_density_kg_m3, | executable/support line; behavior impact
depends on surrounding block context. |
| 1603 |           }, | executable/support line; behavior impact depends on surrounding block context. |
| 1604 |           dt, | executable/support line; behavior impact depends on surrounding block context. |
| 1605 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 1606 |           let err = out.leak_lpm - s.measured_leak_lpm; | executable/support line; behavior impact dep
s on surrounding block context. |
| 1607 |           let abs_err = err.abs(); | executable/support line; behavior impact depends on surrounding b
k context. |
| 1608 |           se += err * err; | executable/support line; behavior impact depends on surrounding block con
t. |
| 1609 |           ape += abs_err / s.measured_leak_lpm.abs().max(0.05); | executable/support line; behavior im
t depends on surrounding block context. |
| 1610 |           max_abs = max_abs.max(abs_err); | executable/support line; behavior impact depends on surrou
ng block context. |
| 1611 |           if let Some(obs) = s.observed_rupture { | conditional gate; changing condition shifts runtime
ath and side effects. |
| 1612 |               rupture_total += 1; | executable/support line; behavior impact depends on surrounding bl
context. |
| 1613 |               if obs == out.rupture_confirmed { | conditional gate; changing condition shifts runtime
h and side effects. |
| 1614 |                   rupture_hits += 1; | executable/support line; behavior impact depends on surrounding
ock context. |
| 1615 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 1616 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1617 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1618 | (blank) | line is blank; visual separation only. |
| 1619 |           let n = samples.len() as f64; | executable/support line; behavior impact depends on surrounding
ck context. |
| 1620 |           let rmse = (se / n.max(1.0)).sqrt(); | executable/support line; behavior impact depends on surro
ing block context. |
| 1621 |           let mape = (ape / n.max(1.0)) * 100.0; | executable/support line; behavior impact depends on sur
nding block context. |
| 1622 |           let rupture_acc = if rupture_total > 0 { | executable/support line; behavior impact depends on s
ounding block context. |
| 1623 |               rupture_hits as f64 / rupture_total as f64 * 100.0 | executable/support line; behavior impac
depends on surrounding block context. |
| 1624 |           } else { | executable/support line; behavior impact depends on surrounding block context. |
| 1625 |               100.0 | executable/support line; behavior impact depends on surrounding block context. |
| 1626 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
opes. |
| 1627 | (blank) | line is blank; visual separation only. |
| 1628 |           circuit_reports.push(CalibrationCircuitReport { | executable/support line; behavior impact depen
on surrounding block context. |
| 1629 |               circuit_name: circuit.name.clone(), | executable/support line; behavior impact depends on su
unding block context. |
| 1630 |               material: circuit.spec.material.name().to_string(), | executable/support line; behavior impa
depends on surrounding block context. |
| 1631 |               oil: circuit.oil_type.name().to_string(), | executable/support line; behavior impact depends
surrounding block context. |
| 1632 |               sample_count: samples.len(), | executable/support line; behavior impact depends on surroundi
block context. |
| 1633 |               rmse_leak_lpm: rmse, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1634 |               mape_leak_pct: mape, | executable/support line; behavior impact depends on surrounding block
ntext. |

```



```

| 1635 |           max_abs_error_lpm: max_abs, | executable/support line; behavior impact depends on surrounding
lock context. |
| 1636 |           rupture_accuracy_pct: rupture_acc, | executable/support line; behavior impact depends on sur
nding block context. |
| 1637 |           fitted: best, | executable/support line; behavior impact depends on surrounding block context
|
| 1638 |       }); | executable/support line; behavior impact depends on surrounding block context. |
| 1639 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1640 | (blank) | line is blank; visual separation only. |
| 1641 |   let by_material = aggregate_group_report(&circuit_reports, \|r\| r.material.clone()); | executable/s
ort line; behavior impact depends on surrounding block context. |
| 1642 |   let by_oil = aggregate_group_report(&circuit_reports, \|r\| r.oil.clone()); | executable/support lin
behavior impact depends on surrounding block context. |
| 1643 |   let total_samples = circuit_reports.iter().map(\|r\| r.sample_count).sum(); | executable/support lin
behavior impact depends on surrounding block context. |
| 1644 | (blank) | line is blank; visual separation only. |
| 1645 |   Ok(CalibrationReport { | executable/support line; behavior impact depends on surrounding block conte
|
| 1646 |       total_samples, | executable/support line; behavior impact depends on surrounding block context.
| 1647 |       calibrated_circuits: circuit_reports.len(), | executable/support line; behavior impact depends o
urrounding block context. |
| 1648 |       circuit_reports, | executable/support line; behavior impact depends on surrounding block context
| 1649 |       by_material, | executable/support line; behavior impact depends on surrounding block context. |
| 1650 |       by_oil, | executable/support line; behavior impact depends on surrounding block context. |
| 1651 |   }) | executable/support line; behavior impact depends on surrounding block context. |
| 1652 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1653 | (blank) | line is blank; visual separation only. |
| 1654 | pub fn export_calibration_report_json<P: AsRef<Path>>(< | declares executable behavior; signature/body ch
es alter API, control flow, and outputs. |
| 1655 |     &self, | executable/support line; behavior impact depends on surrounding block context. |
| 1656 |     path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1657 |     report: &CalibrationReport, | executable/support line; behavior impact depends on surrounding block
text. |
| 1658 | ) -> std::io::Result<()> { | executable/support line; behavior impact depends on surrounding block conte
|
| 1659 |     let path_ref = path.as_ref(); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1660 |     if let Some(parent) = path_ref.parent() { | conditional gate; changing condition shifts runtime path
d side effects. |
| 1661 |         if !parent.as_os_str().is_empty() { | conditional gate; changing condition shifts runtime path a
side effects. |
| 1662 |             fs::create_dir_all(parent)?; | error-propagating operation; changing it impacts failure sema
cs and caller recovery paths. |
| 1663 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1664 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1665 |     let payload = serde_json::to_string_pretty(report) | serialization contract touchpoint; edits can br
report compatibility and tooling. |
| 1666 |     .map_err(\|e\| std::io::Error::other(format!("json serialization failed: {e}")))?; | error-propa
ing operation; changing it impacts failure semantics and caller recovery paths. |
| 1667 |     fs::write(path_ref, payload) | I/O or protocol-critical logic; edits can affect integration, timing,
d diagnostic outcomes. |
| 1668 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1669 | (blank) | line is blank; visual separation only. |
| 1670 | pub fn export_calibration_report_csv<P: AsRef<Path>>(< | declares executable behavior; signature/body cha
s alter API, control flow, and outputs. |
| 1671 |     &self, | executable/support line; behavior impact depends on surrounding block context. |
| 1672 |     path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1673 |     report: &CalibrationReport, | executable/support line; behavior impact depends on surrounding block
text. |
| 1674 | ) -> std::io::Result<()> { | executable/support line; behavior impact depends on surrounding block conte
|
| 1675 |     let path_ref = path.as_ref(); | executable/support line; behavior impact depends on surrounding bloc

```

```

ontext. |
| 1676 |         if let Some(parent) = path_ref.parent() { | conditional gate; changing condition shifts runtime path
d side effects. |
| 1677 |             if !parent.as_os_str().is_empty() { | conditional gate; changing condition shifts runtime path a
side effects. |
| 1678 |                 fs::create_dir_all(parent)?; | error-propagating operation; changing it impacts failure sema
cs and caller recovery paths. |
| 1679 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1680 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1681 | (blank) | line is blank; visual separation only. |
| 1682 |         let mut out = String::from("section,key,sample_count,rmse_leak_lpm,mape_leak_pct,max_abs_error_lpm,r
ure_accuracy_pct,damage_rate_scale,extrusion_rate_scale,thermal_rate_scale,flow_rate_scale,rupture_area_scale\n"); |
cutable/support line; behavior impact depends on surrounding block context. |
| 1683 | (blank) | line is blank; visual separation only. |
| 1684 |         for r in &report.circuit_reports { | iteration control; changing bounds/body affects throughput, tim
, and possibly safety guarantees. |
| 1685 |             out.push_str(&format!( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1686 |                 "circuit,{},{},{:.6},{:.6},{:.6},{:.6},{:.6},{:.6},{:.6},{:.6},{:.6}\n", | executable/support
ine; behavior impact depends on surrounding block context. |
| 1687 |                 r.circuit_name, | executable/support line; behavior impact depends on surrounding block cont
. |
| 1688 |                 r.sample_count, | executable/support line; behavior impact depends on surrounding block cont
. |
| 1689 |                 r.rmse_leak_lpm, | executable/support line; behavior impact depends on surrounding block con
t. |
| 1690 |                 r.mape_leak_pct, | executable/support line; behavior impact depends on surrounding block con
t. |
| 1691 |                 r.max_abs_error_lpm, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1692 |                 r.rupture_accuracy_pct, | executable/support line; behavior impact depends on surrounding bl
context. |
| 1693 |                 r.fitted.damage_rate_scale, | executable/support line; behavior impact depends on surroundin
lock context. |
| 1694 |                 r.fitted.extrusion_rate_scale, | executable/support line; behavior impact depends on surround
g block context. |
| 1695 |                 r.fitted.thermal_rate_scale, | executable/support line; behavior impact depends on surroundi
block context. |
| 1696 |                 r.fitted.flow_rate_scale, | executable/support line; behavior impact depends on surrounding
ck context. |
| 1697 |                 r.fitted.rupture_area_scale, | executable/support line; behavior impact depends on surroundi
block context. |
| 1698 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 1699 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1700 | (blank) | line is blank; visual separation only. |
| 1701 |         for g in &report.by_material { | iteration control; changing bounds/body affects throughput, timing,
d possibly safety guarantees. |
| 1702 |             out.push_str(&format!( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1703 |                 "material,{},{},{:.6},{:.6},{:.6},,,{:.6},,,,,\n", | executable/support line; behavior impact depen
on surrounding block context. |
| 1704 |                 g.group, | executable/support line; behavior impact depends on surrounding block context. |
| 1705 |                 g.sample_count, | executable/support line; behavior impact depends on surrounding block cont
. |
| 1706 |                 g.mean_rmse_lpm, | executable/support line; behavior impact depends on surrounding block con
t. |
| 1707 |                 g.mean_mape_pct, | executable/support line; behavior impact depends on surrounding block con
t. |
| 1708 |                 g.mean_rupture_accuracy_pct, | executable/support line; behavior impact depends on surroundi
block context. |
| 1709 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 1710 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop

```

```

|
| 1711 | (blank) | line is blank; visual separation only. |
| 1712 |         for g in &report.by_oil { | iteration control; changing bounds/body affects throughput, timing, and
sibly safety guarantees. |
| 1713 |             out.push_str(&format!( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1714 |                 "oil,{},{},{:.6},{:.6},{:.6},,,,,\n", | executable/support line; behavior impact depends on
rrounding block context. |
| 1715 |                 g.group, | executable/support line; behavior impact depends on surrounding block context. |
| 1716 |                 g.sample_count, | executable/support line; behavior impact depends on surrounding block cont
. |
| 1717 |                 g.mean_rmse_lpm, | executable/support line; behavior impact depends on surrounding block con
t. |
| 1718 |                 g.mean_mape_pct, | executable/support line; behavior impact depends on surrounding block con
t. |
| 1719 |                 g.mean_ruption_accuracy_pct, | executable/support line; behavior impact depends on surroundi
block context. |
| 1720 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 1721 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1722 |         fs::write(path_ref, out) | I/O or protocol-critical logic; edits can affect integration, timing, and
agnostic outcomes. |
| 1723 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1724 | (blank) | line is blank; visual separation only. |
| 1725 |     pub fn run_monte_carlo( | declares executable behavior; signature/body changes alter API, control flow,
outputs. |
| 1726 |         &self, | executable/support line; behavior impact depends on surrounding block context. |
| 1727 |         runs: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 1728 |         horizon_s: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 1729 |         dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 1730 |         variation_pct: f64, | executable/support line; behavior impact depends on surrounding block context.
| 1731 |     ) -> Vec<ScenarioPrediction> { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1732 |         let mut out = Vec::new(); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1733 |         let n = runs.max(1); | executable/support line; behavior impact depends on surrounding block context
| 1734 |         let spread = (variation_pct / 100.0).clamp(0.0, 0.95); | executable/support line; behavior impact de
ds on surrounding block context. |
| 1735 |         for base in &self.circuits { | iteration control; changing bounds/body affects throughput, timing, a
possibly safety guarantees. |
| 1736 |             for i in 0..n { | iteration control; changing bounds/body affects throughput, timing, and possib
safety guarantees. |
| 1737 |                 let mut c = base.clone(); | executable/support line; behavior impact depends on surrounding
ck context. |
| 1738 |                 c.reset(); | executable/support line; behavior impact depends on surrounding block context.
| 1739 |                 let phi = ((i as f64 * 1.618_033_988_75).fract() - 0.5) * 2.0; | executable/support line; be
ior impact depends on surrounding block context. |
| 1740 |                 let factor = 1.0 + phi * spread; | executable/support line; behavior impact depends on surro
ing block context. |
| 1741 |                 c.operation_pressure_bar *= factor; | executable/support line; behavior impact depends on su
unding block context. |
| 1742 |                 c.piston_pressure_bar *= 1.0 + phi * spread * 0.7; | executable/support line; behavior impac
depends on surrounding block context. |
| 1743 |                 c.spec.squeeze_pct = | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1744 |                     (c.spec.squeeze_pct * (1.0 - phi * spread * 0.2)).clamp(5.0, 45.0); | executable/support
ne; behavior impact depends on surrounding block context. |
| 1745 |                 c.spec.compression_set_pct = (c.spec.compression_set_pct | executable/support line; behavior
pact depends on surrounding block context. |
| 1746 |                     * (1.0 + phi.abs() * spread * 0.3)) | executable/support line; behavior impact depends o
urrounding block context. |
| 1747 |                     .clamp(0.0, 80.0); | executable/support line; behavior impact depends on surrounding blo
context. |
| 1748 | (blank) | line is blank; visual separation only. |
| 1749 |         let mut t = 0.0; | executable/support line; behavior impact depends on surrounding block con

```

```

t. |
| 1750 |           let mut last = CircuitResult { | executable/support line; behavior impact depends on surround
g block context. |
| 1751 |           name: c.name.clone(), | executable/support line; behavior impact depends on surrounding
ck context. |
| 1752 |           alert: LeakAlertLevel::Normal, | executable/support line; behavior impact depends on surr
nding block context. |
| 1753 |           pressure_band: PressureBand::MidBand, | executable/support line; behavior impact depends
surrounding block context. |
| 1754 |           current_pressure_bar: c.pressure.mean_bar, | executable/support line; behavior impact de
ds on surrounding block context. |
| 1755 |           rupture_probability_pct: 0.0, | executable/support line; behavior impact depends on surr
ding block context. |
| 1756 |           predicted_hours_to_rupture: None, | executable/support line; behavior impact depends on
rounding block context. |
| 1757 |           rupture_pressure_bar: None, | executable/support line; behavior impact depends on surrou
ng block context. |
| 1758 |           rupture_elapsed_h: None, | executable/support line; behavior impact depends on surroundi
block context. |
| 1759 |           leak_area_mm2: 0.0, | executable/support line; behavior impact depends on surrounding bl
context. |
| 1760 |           leak_lpm: 0.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1761 |           pressure_min_bar: c.pressure.min_bar, | executable/support line; behavior impact depends
surrounding block context. |
| 1762 |           pressure_mean_bar: c.pressure.mean_bar, | executable/support line; behavior impact depen
on surrounding block context. |
| 1763 |           pressure_max_bar: c.pressure.max_bar, | executable/support line; behavior impact depends
surrounding block context. |
| 1764 |           recommended_hold_min_bar: c.pressure.ideal_bar * 0.65, | executable/support line; behavi
impact depends on surrounding block context. |
| 1765 |           recommended_hold_target_bar: c.pressure.ideal_bar, | executable/support line; behavior i
ct depends on surrounding block context. |
| 1766 |           recommended_hold_max_bar: c.pressure.max_bar * 0.88, | executable/support line; behavior
pact depends on surrounding block context. |
| 1767 |           margin_to_max_bar: c.pressure.max_bar - c.pressure.mean_bar, | executable/support line;
avior impact depends on surrounding block context. |
| 1768 |           margin_to_rupture_bar: c.pressure.rupture_bar - c.pressure.mean_bar, | executable/suppor
ine; behavior impact depends on surrounding block context. |
| 1769 |           rca_hint: String::new(), | executable/support line; behavior impact depends on surroundi
block context. |
| 1770 |           pca_recommendation: String::new(), | executable/support line; behavior impact depends on
rounding block context. |
| 1771 |           rupture_confirmed: false, | executable/support line; behavior impact depends on surround
block context. |
| 1772 |           warning_text: String::new(), | executable/support line; behavior impact depends on surro
ing block context. |
| 1773 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and life
e scopes. |
| 1774 | (blank) | line is blank; visual separation only. |
| 1775 |           while t < horizon_s { | iteration control; changing bounds/body affects throughput, timing,
possibly safety guarantees. |
| 1776 |               let p = c.pressure.mean_bar * (1.0 + (t * 0.27).sin() * 0.1) * factor; | executable/supp
line; behavior impact depends on surrounding block context. |
| 1777 |               last = c.step( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1778 |               CircuitInput { | executable/support line; behavior impact depends on surrounding blo
context. |
| 1779 |                   pressure_bar: p, | executable/support line; behavior impact depends on surroundi
block context. |
| 1780 |                   delta_p_bar: (p - c.pressure.min_bar).max(0.0), | executable/support line; behav
impact depends on surrounding block context. |
| 1781 |                   temp_c: 40.0 + (t * 0.05).sin() * 8.0, | executable/support line; behavior impac
depends on surrounding block context. |
| 1782 |                   cycles_per_s: 1.2 + (t * 0.11).sin().abs() * 1.8, | executable/support line; beh

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or impact depends on surrounding block context. |
| 1783 |           duty_01: 0.55 + (t * 0.07).sin().abs() * 0.35, | executable/support line; behavior
impact depends on surrounding block context. |
| 1784 |           fluid_density_kg_m3: c.oil_type.density_kg_m3(), | executable/support line; beha
r impact depends on surrounding block context. |
| 1785 |           }, | executable/support line; behavior impact depends on surrounding block context.
| 1786 |           dt.max(0.005), | executable/support line; behavior impact depends on surrounding blo
context. |
| 1787 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 1788 |           t += dt.max(0.005); | executable/support line; behavior impact depends on surrounding bl
context. |
| 1789 |           if last.alert == LeakAlertLevel::Ruptured { | conditional gate; changing condition shift
untime path and side effects. |
| 1790 |               break; | executable/support line; behavior impact depends on surrounding block conte
|
| 1791 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 1792 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1793 | (blank) | line is blank; visual separation only. |
| 1794 |       out.push(ScenarioPrediction { | executable/support line; behavior impact depends on surround
block context. |
| 1795 |           circuit_name: c.name.clone(), | executable/support line; behavior impact depends on surr
ding block context. |
| 1796 |           scenario_name: format!("MC_RUN_{:04}", i), | executable/support line; behavior impact de
ds on surrounding block context. |
| 1797 |           peak_pressure_bar: c.stats.max_bar, | executable/support line; behavior impact depends o
urrounding block context. |
| 1798 |           final_alert: last.alert, | executable/support line; behavior impact depends on surroundi
block context. |
| 1799 |           final_rupture_probability_pct: last.rupture_probability_pct, | executable/support line;
avior impact depends on surrounding block context. |
| 1800 |           hours_to_rupture: last.predicted_hours_to_rupture, | executable/support line; behavior i
ct depends on surrounding block context. |
| 1801 |           likely_failure_mode: if last.rupture_confirmed { | executable/support line; behavior imp
depends on surrounding block context. |
| 1802 |               "MonteCarlo rupture".into() | executable/support line; behavior impact depends on su
unding block context. |
| 1803 |           } else { | executable/support line; behavior impact depends on surrounding block context
| 1804 |               "MonteCarlo degradation".into() | executable/support line; behavior impact depends o
urrounding block context. |
| 1805 |           }, | executable/support line; behavior impact depends on surrounding block context. |
| 1806 |       }); | executable/support line; behavior impact depends on surrounding block context. |
| 1807 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1808 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1809 |       out.sort_by(\|a, b\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 1810 |           b.final_rupture_probability_pct | executable/support line; behavior impact depends on surroundin
lock context. |
| 1811 |           .total_cmp(&a.final_rupture_probability_pct) | executable/support line; behavior impact depe
on surrounding block context. |
| 1812 |       }); | executable/support line; behavior impact depends on surrounding block context. |
| 1813 |       out | executable/support line; behavior impact depends on surrounding block context. |
| 1814 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1815 | (blank) | line is blank; visual separation only. |
| 1816 |   pub fn predict_scenarios(&self, horizon_s: f64, dt: f64) -> Vec<ScenarioPrediction> { | declares execut
behavior; signature/body changes alter API, control flow, and outputs. |
| 1817 |       let scenarios: [&str, f64, f64, f64, f64; 4] = [ | executable/support line; behavior impact depend
n surrounding block context. |
| 1818 |           ("Nominal", 1.00, 1.00, 1.00, 1.00), | executable/support line; behavior impact depends on surro
ing block context. |
| 1819 |           ("Sobrepessao Ciclica", 1.25, 1.15, 1.20, 1.10), | executable/support line; behavior impact dep
s on surrounding block context. |

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| 1820 |         ("Choque Termico", 1.05, 1.35, 1.30, 1.10), | executable/support line; behavior impact depends on
surrounding block context. |
| 1821 |         ("Pico + Vibracao", 1.35, 1.10, 1.50, 1.20), | executable/support line; behavior impact depends
surrounding block context. |
| 1822 |     ]; | executable/support line; behavior impact depends on surrounding block context. |
| 1823 | (blank) | line is blank; visual separation only. |
| 1824 |     let mut out = Vec::new(); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1825 |     for base in &self.circuits { | iteration control; changing bounds/body affects throughput, timing, a
possibly safety guarantees. |
| 1826 |         for (name, p_mul, t_mul, cyc_mul, duty_mul) in scenarios { | iteration control; changing bounds/
y affects throughput, timing, and possibly safety guarantees. |
| 1827 |             let mut c = base.clone(); | executable/support line; behavior impact depends on surrounding
ck context. |
| 1828 |             c.reset(); | executable/support line; behavior impact depends on surrounding block context.
| 1829 |             let mut t = 0.0; | executable/support line; behavior impact depends on surrounding block con
t. |
| 1830 |             let mut last = CircuitResult { | executable/support line; behavior impact depends on surroun
g block context. |
| 1831 |                 name: c.name.clone(), | executable/support line; behavior impact depends on surrounding
ck context. |
| 1832 |                 alert: LeakAlertLevel::Normal, | executable/support line; behavior impact depends on sur
nding block context. |
| 1833 |                 pressure_band: PressureBand::MidBand, | executable/support line; behavior impact depends
surrounding block context. |
| 1834 |                 current_pressure_bar: c.pressure.mean_bar, | executable/support line; behavior impact de
ds on surrounding block context. |
| 1835 |                 rupture_probability_pct: 0.0, | executable/support line; behavior impact depends on surr
ding block context. |
| 1836 |                 predicted_hours_to_rupture: None, | executable/support line; behavior impact depends on
rounding block context. |
| 1837 |                 rupture_pressure_bar: None, | executable/support line; behavior impact depends on surrou
ng block context. |
| 1838 |                 rupture_elapsed_h: None, | executable/support line; behavior impact depends on surroundi
block context. |
| 1839 |                 leak_area_mm2: 0.0, | executable/support line; behavior impact depends on surrounding bl
context. |
| 1840 |                 leak_lpm: 0.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1841 |                 pressure_min_bar: c.pressure.min_bar, | executable/support line; behavior impact depends
surrounding block context. |
| 1842 |                 pressure_mean_bar: c.pressure.mean_bar, | executable/support line; behavior impact depen
on surrounding block context. |
| 1843 |                 pressure_max_bar: c.pressure.max_bar, | executable/support line; behavior impact depends
surrounding block context. |
| 1844 |                 recommended_hold_min_bar: c.pressure.ideal_bar * 0.65, | executable/support line; behavi
impact depends on surrounding block context. |
| 1845 |                 recommended_hold_target_bar: c.pressure.ideal_bar, | executable/support line; behavior i
ct depends on surrounding block context. |
| 1846 |                 recommended_hold_max_bar: c.pressure.max_bar * 0.88, | executable/support line; behavior
pact depends on surrounding block context. |
| 1847 |                 margin_to_max_bar: c.pressure.max_bar - c.pressure.mean_bar, | executable/support line;
avior impact depends on surrounding block context. |
| 1848 |                 margin_to_rupture_bar: c.pressure.rupture_bar - c.pressure.mean_bar, | executable/suppor
ine; behavior impact depends on surrounding block context. |
| 1849 |                 rca_hint: String::new(), | executable/support line; behavior impact depends on surroundi
block context. |
| 1850 |                 pca_recommendation: String::new(), | executable/support line; behavior impact depends on
rrounding block context. |
| 1851 |                 rupture_confirmed: false, | executable/support line; behavior impact depends on surround
block context. |
| 1852 |                 warning_text: String::new(), | executable/support line; behavior impact depends on surro
ing block context. |
| 1853 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and life
e scopes. |

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| 1854 | (blank) | line is blank; visual separation only. |
| 1855 |             while t < horizon_s { | iteration control; changing bounds/body affects throughput, timing,
possibly safety guarantees. |
| 1856 |                 let phase = (t / 4.5).sin() * 0.08 + 1.0; | executable/support line; behavior impact dep
s on surrounding block context. |
| 1857 |                 let p = c.pressure.mean_bar * p_mul * phase; | executable/support line; behavior impact
ends on surrounding block context. |
| 1858 |                 let input = CircuitInput { | executable/support line; behavior impact depends on surround
g block context. |
| 1859 |                     pressure_bar: p, | executable/support line; behavior impact depends on surrounding b
k context. |
| 1860 |                     delta_p_bar: (p - c.pressure.min_bar).max(0.0), | executable/support line; behavior
act depends on surrounding block context. |
| 1861 |                     temp_c: 42.0 * t_mul, | executable/support line; behavior impact depends on surround
block context. |
| 1862 |                     cycles_per_s: 1.6 * cyc_mul, | executable/support line; behavior impact depends on s
ounding block context. |
| 1863 |                     duty_01: (0.62 * duty_mul).clamp(0.0, 1.0), | executable/support line; behavior impa
depends on surrounding block context. |
| 1864 |                     fluid_density_kg_m3: c.oil_type.density_kg_m3(), | executable/support line; behavior
pact depends on surrounding block context. |
| 1865 |                 }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 1866 |                 last = c.step(input, dt); | executable/support line; behavior impact depends on surround
block context. |
| 1867 |                 t += dt; | executable/support line; behavior impact depends on surrounding block context
| 1868 |                 if last.alert == LeakAlertLevel::Ruptured { | conditional gate; changing condition shift
untime path and side effects. |
| 1869 |                     break; | executable/support line; behavior impact depends on surrounding block conte
|
| 1870 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 1871 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1872 | (blank) | line is blank; visual separation only. |
| 1873 |             let mode = if last.rupture_confirmed { | executable/support line; behavior impact depends on
rounding block context. |
| 1874 |                 if matches!(c.component, CircuitComponent::AcHose) { | conditional gate; changing condit
shifts runtime path and side effects. |
| 1875 |                     "Burst de mangueira A/C" | executable/support line; behavior impact depends on surro
ing block context. |
| 1876 |                 } else if c.state.extrusion_index > c.state.thermal_damage { | executable/support line;
avior impact depends on surrounding block context. |
| 1877 |                     "Extrusao de vedacao" | executable/support line; behavior impact depends on surround
block context. |
| 1878 |                 } else { | executable/support line; behavior impact depends on surrounding block context
| 1879 |                     "Ruptura por fadiga termica" | executable/support line; behavior impact depends on s
ounding block context. |
| 1880 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 1881 |                 } else if last.alert >= LeakAlertLevel::Warning { | executable/support line; behavior impact
pends on surrounding block context. |
| 1882 |                     "Falha progressiva de vedacao" | executable/support line; behavior impact depends on sur
nding block context. |
| 1883 |                 } else { | executable/support line; behavior impact depends on surrounding block context. |
| 1884 |                     "Sem ruptura no horizonte" | executable/support line; behavior impact depends on surroun
g block context. |
| 1885 |                 }; | scope delimiter; structure-only line, but misplaced edits can change ownership and life
e scopes. |
| 1886 | (blank) | line is blank; visual separation only. |
| 1887 |                 out.push(ScenarioPrediction { | executable/support line; behavior impact depends on surround
block context. |
| 1888 |                     circuit_name: c.name.clone(), | executable/support line; behavior impact depends on surr
ding block context. |
| 1889 |                     scenario_name: name.to_string(), | executable/support line; behavior impact depends on s

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ounding block context. |
| 1890 |             peak_pressure_bar: c.stats.max_bar, | executable/support line; behavior impact depends on
urrounding block context. |
| 1891 |             final_alert: last.alert, | executable/support line; behavior impact depends on surrounding
block context. |
| 1892 |             final_rupture_probability_pct: last.rupture_probability_pct, | executable/support line;
avior impact depends on surrounding block context. |
| 1893 |             hours_to_rupture: last.predicted_hours_to_rupture, | executable/support line; behavior i
ct depends on surrounding block context. |
| 1894 |             likely_failure_mode: mode.to_string(), | executable/support line; behavior impact depend
n surrounding block context. |
| 1895 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 1896 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1897 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1898 | (blank) | line is blank; visual separation only. |
| 1899 |         out.sort_by(\|a, b\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 1900 |             b.final_rupture_probability_pct | executable/support line; behavior impact depends on surrounding
lock context. |
| 1901 |             .total_cmp(&a.final_rupture_probability_pct) | executable/support line; behavior impact depende
on surrounding block context. |
| 1902 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 1903 |         out | executable/support line; behavior impact depends on surrounding block context. |
| 1904 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1905 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1906 | (blank) | line is blank; visual separation only. |
| 1907 | #[cfg(test)] | executable/support line; behavior impact depends on surrounding block context. |
| 1908 | mod tests { | declares a module boundary; changing module name/path breaks references. |
| 1909 |     use super::*; | imports symbols; changing path/name affects compile-time resolution. |
| 1910 | (blank) | line is blank; visual separation only. |
| 1911 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1912 |     fn high_pressure_eventually_ruptures() { | declares executable behavior; signature/body changes alter AP
control flow, and outputs. |
| 1913 |         let mut rig = LeakPhysicsRig::with_default_machine_presets(); | executable/support line; behavior im
t depends on surrounding block context. |
| 1914 |         let mut ruptured = false; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1915 |         for _ in 0..4000 { | iteration control; changing bounds/body affects throughput, timing, and possibl
afety guarantees. |
| 1916 |             let results = rig.step( | executable/support line; behavior impact depends on surrounding block
text. |
| 1917 |             &[( | executable/support line; behavior impact depends on surrounding block context. |
| 1918 |                 "HYD_MAIN", | executable/support line; behavior impact depends on surrounding block conte
. |
| 1919 |                 CircuitInput { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1920 |                     pressure_bar: 320.0, | executable/support line; behavior impact depends on surrounding
block context. |
| 1921 |                     delta_p_bar: 300.0, | executable/support line; behavior impact depends on surrounding
lock context. |
| 1922 |                     temp_c: 105.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 1923 |                     cycles_per_s: 3.0, | executable/support line; behavior impact depends on surrounding
ock context. |
| 1924 |                     duty_01: 0.95, | executable/support line; behavior impact depends on surrounding blo
context. |
| 1925 |                     fluid_density_kg_m3: 860.0, | executable/support line; behavior impact depends on su
unding block context. |
| 1926 |                     }, | executable/support line; behavior impact depends on surrounding block context. |
| 1927 |                 )], | executable/support line; behavior impact depends on surrounding block context. |
| 1928 |                 0.02, | executable/support line; behavior impact depends on surrounding block context. |
| 1929 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 1930 |             if let Some(r) = results.first() { | conditional gate; changing condition shifts runtime path an

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ide effects. |
| 1931 |           if r.rupture_confirmed { | conditional gate; changing condition shifts runtime path and side
fects. |
| 1932 |               ruptured = true; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1933 |               break; | executable/support line; behavior impact depends on surrounding block context.
| 1934 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1935 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1936 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1937 |       assert!( | executable/support line; behavior impact depends on surrounding block context. |
| 1938 |           ruptured, | executable/support line; behavior impact depends on surrounding block context. |
| 1939 |           "expected rupture under sustained extreme overpressure" | executable/support line; behavior impa
depends on surrounding block context. |
| 1940 |       ); | executable/support line; behavior impact depends on surrounding block context. |
| 1941 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1942 | (blank) | line is blank; visual separation only. |
| 1943 |   #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1944 |   fn manual_params_are_applied() { | declares executable behavior; signature/body changes alter API, contr
flow, and outputs. |
| 1945 |       let mut rig = LeakPhysicsRig::with_default_machine_presets(); | executable/support line; behavior im
t depends on surrounding block context. |
| 1946 |       let ok = rig.apply_manual_params( | executable/support line; behavior impact depends on surrounding
ck context. |
| 1947 |           "ENG_OIL", | executable/support line; behavior impact depends on surrounding block context. |
| 1948 |           ManualCircuitParams { | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1949 |               oil_type: Some(OilType::Engine10w30), | executable/support line; behavior impact depends on
rounding block context. |
| 1950 |               piston_pressure_bar: Some(4.8), | executable/support line; behavior impact depends on surrou
ng block context. |
| 1951 |               operation_pressure_bar: Some(3.9), | executable/support line; behavior impact depends on sur
nding block context. |
| 1952 |               pressure_min_bar: Some(1.1), | executable/support line; behavior impact depends on surroundi
block context. |
| 1953 |               pressure_mean_bar: Some(3.5), | executable/support line; behavior impact depends on surround
block context. |
| 1954 |               pressure_ideal_bar: Some(4.4), | executable/support line; behavior impact depends on surround
g block context. |
| 1955 |               pressure_max_bar: Some(5.9), | executable/support line; behavior impact depends on surroundi
block context. |
| 1956 |               pressure_rupture_bar: Some(7.2), | executable/support line; behavior impact depends on surro
ing block context. |
| 1957 |               oring_squeeze_pct: Some(22.0), | executable/support line; behavior impact depends on surround
g block context. |
| 1958 |               compression_set_pct: Some(15.0), | executable/support line; behavior impact depends on surro
ing block context. |
| 1959 |               base_leak_area_mm2: Some(0.0004), | executable/support line; behavior impact depends on surr
ding block context. |
| 1960 |               max_supported_temp_c: None, | executable/support line; behavior impact depends on surroundi
lock context. |
| 1961 |           }, | executable/support line; behavior impact depends on surrounding block context. |
| 1962 |       ); | executable/support line; behavior impact depends on surrounding block context. |
| 1963 |       assert!(ok); | executable/support line; behavior impact depends on surrounding block context. |
| 1964 |       let c = rig | executable/support line; behavior impact depends on surrounding block context. |
| 1965 |           .circuits | executable/support line; behavior impact depends on surrounding block context. |
| 1966 |           .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1967 |           .find(|c| c.name == "ENG_OIL") | executable/support line; behavior impact depends on surroundi
block context. |
| 1968 |           .expect("circuit exists"); | panic-prone operation; changing/removing affects crash behavior and
bustness. |
| 1969 |       assert_eq!(c.oil_type, OilType::Engine10w30); | executable/support line; behavior impact depends on
rounding block context. |

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| 1970 |         assert!(c.pressure.max_bar > 5.8); | executable/support line; behavior impact depends on surrounding
ock context. |
| 1971 |         assert!(c.spec.squeeze_pct > 21.0); | executable/support line; behavior impact depends on surrounding
lock context. |
| 1972 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1973 | (blank) | line is blank; visual separation only. |
| 1974 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1975 |     fn scenario_prediction_returns_ranked_data() { | declares executable behavior; signature/body changes al
API, control flow, and outputs. |
| 1976 |         let rig = LeakPhysicsRig::with_default_machine_presets(); | executable/support line; behavior impact
ends on surrounding block context. |
| 1977 |         let pred = rig.predict_scenarios(180.0, 0.05); | executable/support line; behavior impact depends on
rrounding block context. |
| 1978 |         assert!(!pred.is_empty()); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1979 |         assert!(pred.len() >= rig.circuits.len() * 4); | executable/support line; behavior impact depends on
rrounding block context. |
| 1980 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1981 | (blank) | line is blank; visual separation only. |
| 1982 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1983 |     fn monte_carlo_produces_samples() { | declares executable behavior; signature/body changes alter API, co
ol flow, and outputs. |
| 1984 |         let rig = LeakPhysicsRig::with_default_machine_presets(); | executable/support line; behavior impact
ends on surrounding block context. |
| 1985 |         let pred = rig.run_monte_carlo(20, 120.0, 0.05, 20.0); | executable/support line; behavior impact de
ds on surrounding block context. |
| 1986 |         assert!(pred.len() >= rig.circuits.len() * 20); | executable/support line; behavior impact depends o
urrounding block context. |
| 1987 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1988 | (blank) | line is blank; visual separation only. |
| 1989 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 1990 |     fn exports_json_and_csv_files() { | declares executable behavior; signature/body changes alter API, cont
flow, and outputs. |
| 1991 |         let mut rig = LeakPhysicsRig::with_default_machine_presets(); | executable/support line; behavior im
t depends on surrounding block context. |
| 1992 |         let _ = rig.step( | executable/support line; behavior impact depends on surrounding block context. |
| 1993 |             &[( | executable/support line; behavior impact depends on surrounding block context. |
| 1994 |                 "AC_HIGH", | executable/support line; behavior impact depends on surrounding block context.
| 1995 |                 CircuitInput { | executable/support line; behavior impact depends on surrounding block conte
|
| 1996 |                 pressure_bar: 16.0, | executable/support line; behavior impact depends on surrounding bl
context. |
| 1997 |                 delta_p_bar: 12.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 1998 |                 temp_c: 55.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1999 |                 cycles_per_s: 2.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 2000 |                 duty_01: 0.7, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 2001 |                 fluid_density_kg_m3: 995.0, | executable/support line; behavior impact depends on surroun
ng block context. |
| 2002 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 2003 |         )], | executable/support line; behavior impact depends on surrounding block context. |
| 2004 |         0.1, | executable/support line; behavior impact depends on surrounding block context. |
| 2005 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 2006 |         let pred = rig.predict_scenarios(60.0, 0.05); | executable/support line; behavior impact depends on
rounding block context. |
| 2007 | (blank) | line is blank; visual separation only. |
| 2008 |         let root = std::env::temp_dir().join("autobreaking_leak_tests"); | executable/support line; behavior
pact depends on surrounding block context. |
| 2009 |         let runtime_json = root.join("runtime.json"); | executable/support line; behavior impact depends on
rounding block context. |
| 2010 |         let runtime_csv = root.join("runtime.csv"); | executable/support line; behavior impact depends on su
unding block context. |

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| 2011 |          let pred_json = root.join("pred.json"); | executable/support line; behavior impact depends on surrounding block context. |
| 2012 |          let pred_csv = root.join("pred.csv"); | executable/support line; behavior impact depends on surrounding block context. |
| 2013 | (blank) | line is blank; visual separation only. |
| 2014 |          rig.export_last_results_json(&runtime_json) | executable/support line; behavior impact depends on surrounding block context. |
| 2015 |          .expect("runtime json"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 2016 |          rig.export_last_results_csv(&runtime_csv) | executable/support line; behavior impact depends on surrounding block context. |
| 2017 |          .expect("runtime csv"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 2018 |          rig.export_predictions_json(&pred_json, &pred) | executable/support line; behavior impact depends on surrounding block context. |
| 2019 |          .expect("pred json"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 2020 |          rig.export_predictions_csv(&pred_csv, &pred) | executable/support line; behavior impact depends on surrounding block context. |
| 2021 |          .expect("pred csv"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 2022 | (blank) | line is blank; visual separation only. |
| 2023 |          assert!(runtime_json.exists()); | executable/support line; behavior impact depends on surrounding block context. |
| 2024 |          assert!(runtime_csv.exists()); | executable/support line; behavior impact depends on surrounding block context. |
| 2025 |          assert!(pred_json.exists()); | executable/support line; behavior impact depends on surrounding block context. |
| 2026 |          assert!(pred_csv.exists()); | executable/support line; behavior impact depends on surrounding block context. |
| 2027 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2028 | (blank) | line is blank; visual separation only. |
| 2029 |      #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 2030 |      fn report_captures_pressure_band_and_safe_window() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 2031 |          let mut rig = LeakPhysicsRig::with_default_machine_presets(); | executable/support line; behavior impact depends on surrounding block context. |
| 2032 |          let out = rig.step( | executable/support line; behavior impact depends on surrounding block context. |
| 2033 |              &[( | executable/support line; behavior impact depends on surrounding block context. |
| 2034 |                  "HYD_MAIN", | executable/support line; behavior impact depends on surrounding block context. |
| 2035 |                  CircuitInput { | executable/support line; behavior impact depends on surrounding block context. |
| 2036 |                      pressure_bar: 150.0, | executable/support line; behavior impact depends on surrounding block context. |
| 2037 |                      delta_p_bar: 110.0, | executable/support line; behavior impact depends on surrounding block context. |
| 2038 |                      temp_c: 60.0, | executable/support line; behavior impact depends on surrounding block context. |
| 2039 |                      cycles_per_s: 1.8, | executable/support line; behavior impact depends on surrounding block context. |
| 2040 |                      duty_01: 0.65, | executable/support line; behavior impact depends on surrounding block context. |
| 2041 |                      fluid_density_kg_m3: 865.0, | executable/support line; behavior impact depends on surrounding block context. |
| 2042 |                      }, | executable/support line; behavior impact depends on surrounding block context. |
| 2043 |                  )], | executable/support line; behavior impact depends on surrounding block context. |
| 2044 |                  0.05, | executable/support line; behavior impact depends on surrounding block context. |
| 2045 |              ); | executable/support line; behavior impact depends on surrounding block context. |
| 2046 | (blank) | line is blank; visual separation only. |
| 2047 |          let r = out.first().expect("result exists"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 2048 |          assert!(matches!(r.pressure_band, PressureBand::LowBand \| PressureBand::MidBand \| PressureBand::HighBand)); | executable/support line; behavior impact depends on surrounding block context. |
| 2049 |          assert!(r.recommended_hold_min_bar < r.recommended_hold_target_bar); | executable/support line; behavior impact depends on surrounding block context. |

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| 2050 |         assert!(r.recommended_hold_target_bar < r.recommended_hold_max_bar); | executable/support line; beha
r impact depends on surrounding block context. |
| 2051 |         assert!(r.margin_to_rupture_bar > 0.0); | executable/support line; behavior impact depends on surrou
ng block context. |
| 2052 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 2053 | (blank) | line is blank; visual separation only. |
| 2054 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 2055 |     fn rupture_event_records_pressure_and_time() { | declares executable behavior; signature/body changes al
API, control flow, and outputs. |
| 2056 |         let mut rig = LeakPhysicsRig::with_default_machine_presets(); | executable/support line; behavior im
t depends on surrounding block context. |
| 2057 |         let mut rupture: Option<CircuitResult> = None; | executable/support line; behavior impact depends on
rrounding block context. |
| 2058 |         for _ in 0..6000 { | iteration control; changing bounds/body affects throughput, timing, and possibl
afety guarantees. |
| 2059 |             let out = rig.step( | executable/support line; behavior impact depends on surrounding block cont
. |
| 2060 |                 &[( | executable/support line; behavior impact depends on surrounding block context. |
| 2061 |                     "HYD_MAIN", | executable/support line; behavior impact depends on surrounding block cont
. |
| 2062 |                     CircuitInput { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 2063 |                         pressure_bar: 330.0, | executable/support line; behavior impact depends on surroundi
block context. |
| 2064 |                         delta_p_bar: 320.0, | executable/support line; behavior impact depends on surroundin
lock context. |
| 2065 |                         temp_c: 120.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 2066 |                         cycles_per_s: 3.8, | executable/support line; behavior impact depends on surrounding
ock context. |
| 2067 |                         duty_01: 0.98, | executable/support line; behavior impact depends on surrounding blo
context. |
| 2068 |                         fluid_density_kg_m3: 860.0, | executable/support line; behavior impact depends on su
unding block context. |
| 2069 |                     }, | executable/support line; behavior impact depends on surrounding block context. |
| 2070 |                 )], | executable/support line; behavior impact depends on surrounding block context. |
| 2071 |                 0.02, | executable/support line; behavior impact depends on surrounding block context. |
| 2072 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 2073 |             if let Some(r) = out.first() { | conditional gate; changing condition shifts runtime path and si
effects. |
| 2074 |                 if r.rupture_confirmed { | conditional gate; changing condition shifts runtime path and side
fects. |
| 2075 |                     rupture = Some(r.clone()); | executable/support line; behavior impact depends on surroun
g block context. |
| 2076 |                     break; | executable/support line; behavior impact depends on surrounding block context.
| 2077 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 2078 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2079 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 2080 | (blank) | line is blank; visual separation only. |
| 2081 |         let r = rupture.expect("rupture expected in deterministic stress profile"); | panic-prone operation;
anging/removing affects crash behavior and robustness. |
| 2082 |         assert!(r.rupture_pressure_bar.is_some()); | executable/support line; behavior impact depends on sur
nding block context. |
| 2083 |         assert!(r.rupture_elapsed_h.is_some()); | executable/support line; behavior impact depends on surrou
ng block context. |
| 2084 |         assert!(r.rupture_pressure_bar.expect("has rupture pressure") >= 280.0); | panic-prone operation; ch
ing/removing affects crash behavior and robustness. |
| 2085 |         assert!(r.warning_text.contains("ruptura")); | executable/support line; behavior impact depends on s
ounding block context. |
| 2086 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 2087 | (blank) | line is blank; visual separation only. |
| 2088 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 2089 |     fn engineering_catalogs_are_large_and_coherent() { | declares executable behavior; signature/body change
alter API, control flow, and outputs. |
| 2090 |         let mats = engineering_material_catalog(); | executable/support line; behavior impact depends on surr
nding block context. |
| 2091 |         let oils = engineering_oil_catalog(); | executable/support line; behavior impact depends on surround
block context. |
| 2092 |         assert!(mats.len() >= 30, "material catalog should be large"); | executable/support line; behavior i
ct depends on surrounding block context. |
| 2093 |         assert!(oils.len() >= 30, "oil catalog should be large"); | executable/support line; behavior impact
ends on surrounding block context. |
| 2094 | (blank) | line is blank; visual separation only. |
| 2095 |         let mut mat_ids: Vec<&str> = mats.iter().map(|m| m.id).collect(); | executable/support line; behav
impact depends on surrounding block context. |
| 2096 |         mat_ids.sort_unstable(); | executable/support line; behavior impact depends on surrounding block con
t. |
| 2097 |         mat_ids.dedup(); | executable/support line; behavior impact depends on surrounding block context. |
| 2098 |         assert_eq!(mat_ids.len(), mats.len(), "material IDs must be unique"); | executable/support line; beh
or impact depends on surrounding block context. |
| 2099 | (blank) | line is blank; visual separation only. |
| 2100 |         let mut oil_ids: Vec<&str> = oils.iter().map(|o| o.id).collect(); | executable/support line; behav
impact depends on surrounding block context. |
| 2101 |         oil_ids.sort_unstable(); | executable/support line; behavior impact depends on surrounding block con
t. |
| 2102 |         oil_ids.dedup(); | executable/support line; behavior impact depends on surrounding block context. |
| 2103 |         assert_eq!(oil_ids.len(), oils.len(), "oil IDs must be unique"); | executable/support line; behavior
pact depends on surrounding block context. |
| 2104 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 2105 | (blank) | line is blank; visual separation only. |
| 2106 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 2107 |     fn calibration_mode_ingests_csv_and_exports_reports() { | declares executable behavior; signature/body c
ges alter API, control flow, and outputs. |
| 2108 |         let mut rig = LeakPhysicsRig::with_default_machine_presets(); | executable/support line; behavior im
t depends on surrounding block context. |
| 2109 |         let root = std::env::temp_dir().join("autobreaking_calibration_tests"); | executable/support line; b
vior impact depends on surrounding block context. |
| 2110 |         std::fs::create_dir_all(&root).expect("create calibration temp dir"); | panic-prone operation; chang
/removing affects crash behavior and robustness. |
| 2111 |         let csv_path = root.join("bench_data.csv"); | executable/support line; behavior impact depends on su
unding block context. |
| 2112 | (blank) | line is blank; visual separation only. |
| 2113 |         let mut csv = String::from( | executable/support line; behavior impact depends on surrounding block
text. |
| 2114 |             "timestamp_s,circuit_name,pressure_bar,delta_p_bar,temp_c,cycles_per_s,duty_01,fluid_density_kg_
measured_leak_lpm,observed_rupture\n", | executable/support line; behavior impact depends on surrounding block context
|
| 2115 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 2116 |         for i in 0..40 { | iteration control; changing bounds/body affects throughput, timing, and possibly
ety guarantees. |
| 2117 |             let t = i as f64 * 0.05; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 2118 |             let p = 120.0 + (i as f64 * 0.5); | executable/support line; behavior impact depends on surround
block context. |
| 2119 |             let dp = 95.0 + (i as f64 * 0.4); | executable/support line; behavior impact depends on surround
block context. |
| 2120 |             let temp = 65.0 + (i as f64 * 0.08); | executable/support line; behavior impact depends on surro
ing block context. |
| 2121 |             let measured_hyd = 0.11 + (i as f64 * 0.0012); | executable/support line; behavior impact depend
n surrounding block context. |
| 2122 |             csv.push_str(&format!( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 2123 |                 "{t:.3},HYD_MAIN,{p:.4},{dp:.4},{temp:.3},1.60,0.62,860.0,{measured_hyd:.6},false\n" | execu
le/support line; behavior impact depends on surrounding block context. |
| 2124 |             )); | executable/support line; behavior impact depends on surrounding block context. |
| 2125 | (blank) | line is blank; visual separation only. |
| 2126 |             let p_ac = 14.0 + (i as f64 * 0.07); | executable/support line; behavior impact depends on surro

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ing block context. |
| 2127 |           let dp_ac = 10.0 + (i as f64 * 0.05); | executable/support line; behavior impact depends on surr
ding block context. |
| 2128 |           let measured_ac = 0.018 + (i as f64 * 0.00035); | executable/support line; behavior impact depen
on surrounding block context. |
| 2129 |           csv.push_str(&format!( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 2130 |               "{t:.3},AC_HIGH,{p_ac:.4},{dp_ac:.4},48.0,1.20,0.55,995.0,{measured_ac:.6},false\n" | execut
e/support line; behavior impact depends on surrounding block context. |
| 2131 |           )); | executable/support line; behavior impact depends on surrounding block context. |
| 2132 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 2133 |           std::fs::write(&csv_path, csv).expect("write bench csv"); | panic-prone operation; changing/removing
fects crash behavior and robustness. |
| 2134 | (blank) | line is blank; visual separation only. |
| 2135 |           let report = rig.calibrate_from_csv(&csv_path).expect("run calibration"); | panic-prone operation; c
ging/removing affects crash behavior and robustness. |
| 2136 |           assert!(report.calibrated_circuits >= 2); | executable/support line; behavior impact depends on surr
ding block context. |
| 2137 |           assert!(report.total_samples >= 80); | executable/support line; behavior impact depends on surroun
block context. |
| 2138 |           assert!(report.by_material.iter().any(|g| g.sample_count > 0)); | executable/support line; behavio
mpact depends on surrounding block context. |
| 2139 |           assert!(report.by_oil.iter().any(|g| g.sample_count > 0)); | executable/support line; behavior imp
depends on surrounding block context. |
| 2140 | (blank) | line is blank; visual separation only. |
| 2141 |           let csv_report = root.join("calibration_report.csv"); | executable/support line; behavior impact dep
s on surrounding block context. |
| 2142 |           let json_report = root.join("calibration_report.json"); | executable/support line; behavior impact d
nds on surrounding block context. |
| 2143 |           rig.export_calibration_report_csv(&csv_report, &report) | executable/support line; behavior impact d
nds on surrounding block context. |
| 2144 |               .expect("export calibration csv"); | panic-prone operation; changing/removing affects crash beha
r and robustness. |
| 2145 |           rig.export_calibration_report_json(&json_report, &report) | executable/support line; behavior impact
ends on surrounding block context. |
| 2146 |               .expect("export calibration json"); | panic-prone operation; changing/removing affects crash beh
or and robustness. |
| 2147 | (blank) | line is blank; visual separation only. |
| 2148 |           assert!(csv_report.exists()); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 2149 |           assert!(json_report.exists()); | executable/support line; behavior impact depends on surrounding blo
context. |
| 2150 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 2151 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/lib.rs

Role: Library root and module graph for the simulation and protocol ecosystem.

Lines: 1169 | Non-empty: 1081 | Symbols: 77

Symbol Inventory

Line	Kind	Symbol	Change Impact
8	module	boot_sequence	data/structure coupling
9	module	can_gateway	data/structure coupling
10	module	ecu_abs	data/structure coupling
11	module	ecu_bcm	data/structure coupling
12	module	ecu_ecm	data/structure coupling
13	module	ecu_hcm	data/structure coupling
14	module	ecu_icm	data/structure coupling
15	module	ecu_tcm	data/structure coupling
16	module	electrical	data/structure coupling
17	module	implement	data/structure coupling

18	module	j1939	data/structure coupling
19	module	network_mgmt	data/structure coupling
20	module	nvm	data/structure coupling
21	module	uds	data/structure coupling
22	module	io	data/structure coupling
25	module	autonomous	data/structure coupling
26	module	camera	data/structure coupling
27	module	can_network	data/structure coupling
28	module	ecu_vcm	data/structure coupling
29	module	gps	data/structure coupling
30	module	imu	data/structure coupling
31	module	leak_physics	data/structure coupling
32	module	lidar	data/structure coupling
33	module	observability	data/structure coupling
34	module	radar	data/structure coupling
35	module	sim_core	data/structure coupling
36	module	v2x_telematics	data/structure coupling
40	module	adas	data/structure coupling
42	module	can_bus	data/structure coupling
44	module	chassis	data/structure coupling
45	module	ecm_heavy	data/structure coupling
47	module	engine	data/structure coupling
49	module	sensors	data/structure coupling
51	module	transmission	data/structure coupling
53	module	vehicle	data/structure coupling
103	enum	FaultType	data/structure coupling
136	impl	std::fmt::Display for FaultType	behavior binding and method semantics
137	fn	fmt	signature/API coupling
168	const	FAULT_TYPES	data/structure coupling
197	struct	HeavyMachinery	data/structure coupling
254	impl	Default for HeavyMachinery	behavior binding and method semantics
255	fn	default	signature/API coupling
260	impl	HeavyMachinery	behavior binding and method semantics
261	fn	new	signature/API coupling
315	fn	tick	signature/API coupling
768	fn	vehicle_heading	signature/API coupling
771	fn	vehicle_accel_ms2	signature/API coupling
775	fn	run_leak_physics	signature/API coupling
877	fn	build_radar_traffic	signature/API coupling
881	fn	implement_traffic	signature/API coupling
913	fn	build_lidar_world	signature/API coupling
924	fn	apply_fault	signature/API coupling
1006	fn	inject_fault	signature/API coupling
1011	fn	clear_faults	signature/API coupling
1021	fn	key_advance	signature/API coupling
1026	fn	key_off	signature/API coupling
1033	fn	reset	signature/API coupling
1037	fn	apply_leak_manual_params	signature/API coupling
1045	fn	add_custom_leak_circuit	signature/API coupling
1049	fn	predict_leak_scenarios	signature/API coupling
1053	fn	export_leak_report_json	signature/API coupling
1060	fn	export_leak_report_csv	signature/API coupling
1067	fn	export_leak_predictions_json	signature/API coupling
1075	fn	export_leak_predictions_csv	signature/API coupling
1083	fn	export_leak_material_catalog_json	signature/API coupling
1090	fn	export_leak_material_catalog_csv	signature/API coupling
1097	fn	export_leak_oil_catalog_json	signature/API coupling
1104	fn	export_leak_oil_catalog_csv	signature/API coupling
1111	fn	calibrate_leak_model_from_csv	signature/API coupling
1118	fn	export_leak_calibration_report_json	signature/API coupling
1126	fn	export_leak_calibration_report_csv	signature/API coupling
1134	fn	monte_carlo_leak_predictions	signature/API coupling
1145	fn	export_can_snapshot_json	signature/API coupling
1152	fn	export_can_snapshot_csv	signature/API coupling
1160	fn	ignition	signature/API coupling

1163	fn	engine_running	signature/API coupling
1166	fn	all_ecus_online	signature/API coupling

[illegible]

[illegible]

[illegible]

```

k context. |
| 128 |      // CAN faults | comment line; documentation intent, no runtime effect. |
| 129 |      BusOffInjection, // force bus-off condition | executable/support line; behavior impact depends on surround
g block context. |
| 130 |      KillEcm,          // power off ECM node | executable/support line; behavior impact depends on surrounding
ck context. |
| 131 |      KillTcm,          // power off TCM node | executable/support line; behavior impact depends on surrounding
ck context. |
| 132 |      // None selected | comment line; documentation intent, no runtime effect. |
| 133 |      None, | executable/support line; behavior impact depends on surrounding block context. |
| 134 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 135 | (blank) | line is blank; visual separation only. |
| 136 | impl std::fmt::Display for FaultType { | implementation block; method behavior and trait conformance are defini
here. |
| 137 |      fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signat
re/body changes alter API, control flow, and outputs. |
| 138 |          let s = match self { | executable/support line; behavior impact depends on surrounding block context.
| 139 |              FaultType::HighCoolantTemp => "High Coolant Temp (SPN110 FMI0)", | executable/support line; behav
impact depends on surrounding block context. |
| 140 |              FaultType::LowOilPressure => "Low Oil Pressure (SPN100 FMI1)", | executable/support line; behavio
mpact depends on surrounding block context. |
| 141 |              FaultType::LowFuelPressure => "Low Fuel Pressure (SPN94 FMI1)", | executable/support line; behavi
impact depends on surrounding block context. |
| 142 |              FaultType::CriticalDefLevel => "Critical DEF Level (SPN3361 FMI1)", | executable/support line; be
ior impact depends on surrounding block context. |
| 143 |              FaultType::HighDpfSoot => "High DPF Soot (SPN3251 FMI16)", | executable/support line; behavior im
t depends on surrounding block context. |
| 144 |              FaultType::HighBoostPressure => "High Boost Pressure (SPN102 FMI0)", | executable/support line; b
vior impact depends on surrounding block context. |
| 145 |              FaultType::LowBoostPressure => "Low Boost Pressure (SPN102 FMI1)", | executable/support line; beh
or impact depends on surrounding block context. |
| 146 |              FaultType::LowCoolantPressure => "Low Coolant Pressure (SPN111 FMI1)", | executable/support line;
havior impact depends on surrounding block context. |
| 147 |              FaultType::HighIntakeTemp => "High Intake Temp (SPN105 FMI15)", | executable/support line; behavi
impact depends on surrounding block context. |
| 148 |              FaultType::HighExhaustTemp => "High Exhaust Temp (SPN173 FMI0)", | executable/support line; behav
impact depends on surrounding block context. |
| 149 |              FaultType::PoorDefQuality => "Poor DEF Quality (SPN1761 FMI3)", | executable/support line; behavi
impact depends on surrounding block context. |
| 150 |              FaultType::HighNoxTailpipe => "High Tailpipe NOx (SPN3226 FMI16)", | executable/support line; beh
or impact depends on surrounding block context. |
| 151 |              FaultType::LowFuelRailPressure => "Low Fuel Rail Pressure (SPN157 FMI1)", | executable/support li
behavior impact depends on surrounding block context. |
| 152 |              FaultType::TransClutchOverheat => "Trans Clutch Overheat", | executable/support line; behavior im
t depends on surrounding block context. |
| 153 |              FaultType::TransNeutralFailure => "Trans Neutral Failure", | executable/support line; behavior im
t depends on surrounding block context. |
| 154 |              FaultType::LowBatteryVoltage => "Low Battery Voltage", | executable/support line; behavior impact
ends on surrounding block context. |
| 155 |              FaultType::AlternatorFault => "Alternator Fault", | executable/support line; behavior impact depe
on surrounding block context. |
| 156 |              FaultType::HighHydTemp => "Hydraulic Fluid High Temp", | executable/support line; behavior impact
ends on surrounding block context. |
| 157 |              FaultType::LowHydLevel => "Hydraulic Level Low", | executable/support line; behavior impact depen
on surrounding block context. |
| 158 |              FaultType::HydFilterBypass => "Hyd Filter Bypass Open", | executable/support line; behavior impac
depends on surrounding block context. |
| 159 |              FaultType::BusOffInjection => "CAN Bus-Off Injection", | executable/support line; behavior impact
ends on surrounding block context. |
| 160 |              FaultType::KillEcm => "Kill ECM Node", | executable/support line; behavior impact depends on surr
ding block context. |
| 161 |              FaultType::KillTcm => "Kill TCM Node", | executable/support line; behavior impact depends on surr
ding block context. |
| 162 |              FaultType::None => "(no fault selected)", | executable/support line; behavior impact depends on s
ounding block context. |

```

```
| 163 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 164 | write!(f, "{}", s) | I/O or protocol-critical logic; edits can affect integration, timing, and diagno  
c outcomes. |  
| 165 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 166 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 167 | (blank) | line is blank; visual separation only. |  
| 168 | pub const FAULT_TYPES: &[FaultType] = &[ | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 169 | FaultType::None, | executable/support line; behavior impact depends on surrounding block context. |  
| 170 | FaultType::HighCoolantTemp, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 171 | FaultType::LowOilPressure, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 172 | FaultType::LowFuelPressure, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 173 | FaultType::CriticalDefLevel, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 174 | FaultType::HighDpfSoot, | executable/support line; behavior impact depends on surrounding block context.  
| 175 | FaultType::HighBoostPressure, | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 176 | FaultType::LowBoostPressure, | executable/support line; behavior impact depends on surrounding block conti  
. |  
| 177 | FaultType::LowCoolantPressure, | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 178 | FaultType::HighIntakeTemp, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 179 | FaultType::HighExhaustTemp, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 180 | FaultType::PoorDefQuality, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 181 | FaultType::HighNoxTailpipe, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 182 | FaultType::LowFuelRailPressure, | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 183 | FaultType::TransClutchOverheat, | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 184 | FaultType::TransNeutralFailure, | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 185 | FaultType::LowBatteryVoltage, | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 186 | FaultType::AlternatorFault, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 187 | FaultType::HighHydTemp, | executable/support line; behavior impact depends on surrounding block context.  
| 188 | FaultType::LowHydLevel, | executable/support line; behavior impact depends on surrounding block context.  
| 189 | FaultType::HydFilterBypass, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 190 | FaultType::BusOffInjection, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 191 | FaultType::KillEcm, | executable/support line; behavior impact depends on surrounding block context. |  
| 192 | FaultType::KillTcm, | executable/support line; behavior impact depends on surrounding block context. |  
| 193 | ]; | executable/support line; behavior impact depends on surrounding block context. |  
| 194 | (blank) | line is blank; visual separation only. |  
| 195 | // ■■ Top-level Heavy Machinery Bench ██████████ | comment line; do  
tent, no runtime effect. |  
| 196 | (blank) | line is blank; visual separation only. |  
| 197 | pub struct HeavyMachinery { | declares data layout; field changes may break serialization, tests, and callers  
| 198 | // ■ CAN infrastructure ██████████ | comment line; do  
nt, no runtime effect. |  
| 199 | pub gateway: CanGateway, | executable/support line; behavior impact depends on surrounding block context.  
| 200 | pub boot: BootSequence, | executable/support line; behavior impact depends on surrounding block context.  
| 201 | pub net_mgmt: NetworkManager, // OSEK NM – coordinated sleep/wake | executable/support line; behavior imp  
depends on surrounding block context. |  
| 202 | (blank) | line is blank; visual separation only. |  
| 203 | // ■ ECU modules (each in its own .rs – mirrors real architecture) ████████ | comment line; documentation
```

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nt, no runtime effect. |
| 204 | pub ecm: EcuEcm, // Engine Control Module | executable/support line; behavior impact depends on surrounding block context. |
| 205 | pub tcm: EcuTcm, // Transmission Control Module | executable/support line; behavior impact depends on surrounding block context. |
| 206 | pub bcm: EcuBcm, // Body Control Module | executable/support line; behavior impact depends on surrounding block context. |
| 207 | pub icm: EcuIcm, // Instrument Cluster | executable/support line; behavior impact depends on surrounding block context. |
| 208 | pub hcm: EcuHcm, // Hydraulic Control Module | executable/support line; behavior impact depends on surrounding block context. |
| 209 | pub abs: EcuAbs, // ABS / ESP / TCS | executable/support line; behavior impact depends on surrounding block context. |
| 210 | pub electrical: ElectricalSystem, | executable/support line; behavior impact depends on surrounding block context. |
| 211 | (blank) | line is blank; visual separation only. |
| 212 | // ■ Diagnostic services (one UDS server per ECU in real life) ■■■■■■■■■■ | comment line; documentation, no runtime effect. |
| 213 | pub uds_ecm: UdsServer, // ECM diagnostic endpoint | I/O or protocol-critical logic; edits can affect initialization, timing, and diagnostic outcomes. |
| 214 | pub uds_tcm: UdsServer, // TCM diagnostic endpoint | I/O or protocol-critical logic; edits can affect initialization, timing, and diagnostic outcomes. |
| 215 | (blank) | line is blank; visual separation only. |
| 216 | // ■ Non-volatile memory (shared EEPROM simulation for ECM) ■■■■■■■■■■ | comment line; documentation, no runtime effect. |
| 217 | pub nvm: NvmStore, | executable/support line; behavior impact depends on surrounding block context. |
| 218 | (blank) | line is blank; visual separation only. |
| 219 | // ■ Physical / implement systems ■■■■■■■■■■ | comment line; documentation, no runtime effect. |
| 220 | pub implement: ImplementControl, | executable/support line; behavior impact depends on surrounding block context. |
| 221 | (blank) | line is blank; visual separation only. |
| 222 | // ■ Simulation time ■■■■■■■■■■ | comment line; documentation, no runtime effect. |
| 223 | pub elapsed: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 224 | (blank) | line is blank; visual separation only. |
| 225 | // ■ Driver inputs ■■■■■■■■■■ | comment line; documentation, no runtime effect. |
| 226 | pub throttle_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 227 | pub brake_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 228 | pub hitch_joystick: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 229 | pub loader_lift_cmd: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 230 | pub loader_tilt_cmd: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 231 | (blank) | line is blank; visual separation only. |
| 232 | // ■ Fault simulation ■■■■■■■■■■ | comment line; documentation, no runtime effect. |
| 233 | pub selected_fault: FaultType, | executable/support line; behavior impact depends on surrounding block context. |
| 234 | pub fault_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 235 | (blank) | line is blank; visual separation only. |
| 236 | // ■■ Sensor suite (autonomous / ADAS) ■■■■■■■■■■ | comment line; documentation, no runtime effect. |
| 237 | pub gps: GpsModule, | executable/support line; behavior impact depends on surrounding block context. |
| 238 | pub imu: Imu, | executable/support line; behavior impact depends on surrounding block context. |
| 239 | pub radar: RadarSuite, | executable/support line; behavior impact depends on surrounding block context. |
| 240 | pub lidar: LidarSensor, | executable/support line; behavior impact depends on surrounding block context. |
| 241 | pub ad: AutonomousController, | executable/support line; behavior impact depends on surrounding block context. |
| 242 | pub v2x: V2xModule, | executable/support line; behavior impact depends on surrounding block context. |
| 243 | pub telematics: TelematicsModule, | executable/support line; behavior impact depends on surrounding block context. |
| 244 | /// Multi-bus CAN network (HS-CAN powertrain/chassis, MS-CAN body, ISOBUS) | comment line; documentation, no runtime effect. |
| 245 | pub can_net: CanNetwork, | executable/support line; behavior impact depends on surrounding block context. |
| 246 | pub vcm: EcuVcm, | executable/support line; behavior impact depends on surrounding block context. |

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| 247 |     pub camera: CameraSystem, | executable/support line; behavior impact depends on surrounding block context. |
| 248 |     pub fusion: SensorFusion, | executable/support line; behavior impact depends on surrounding block context. |
| 249 |     pub leak_rig: LeakPhysicsRig, | executable/support line; behavior impact depends on surrounding block context. |
| 250 |     pub leak_reports: Vec<CircuitResult>, | executable/support line; behavior impact depends on surrounding block context. |
| 251 |     pub metrics: SimMetrics, | executable/support line; behavior impact depends on surrounding block context. |
| 252 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 253 | (blank) | line is blank; visual separation only. |
| 254 | impl Default for HeavyMachinery { | implementation block; method behavior and trait conformance are defined here. |
| 255 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 256 |         Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 257 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 258 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 259 | (blank) | line is blank; visual separation only. |
| 260 | impl HeavyMachinery { | implementation block; method behavior and trait conformance are defined here. |
| 261 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 262 |         let mut gw = CanGateway::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 263 |         gw.register_node(j1939::addr::ECM_1, "ECM #1"); | executable/support line; behavior impact depends on surrounding block context. |
| 264 |         gw.register_node(j1939::addr::TRANSMISSION, "TCM"); | executable/support line; behavior impact depends on surrounding block context. |
| 265 |         gw.register_node(j1939::addr::CAB, "BCM/CAB"); | executable/support line; behavior impact depends on surrounding block context. |
| 266 |         gw.register_node(j1939::addr::INSTRUMENT, "ICM/DASH"); | executable/support line; behavior impact depends on surrounding block context. |
| 267 |         gw.register_node(j1939::addr::HITCH, "HCM/HITCH"); | executable/support line; behavior impact depends on surrounding block context. |
| 268 |         gw.register_node(j1939::addr::BRAKES, "ABS/ESP"); | executable/support line; behavior impact depends on surrounding block context. |
| 269 |         gw.register_node(j1939::addr::ISOBUS_VT, "ISOBUS-VT"); | executable/support line; behavior impact depends on surrounding block context. |
| 270 |         gw.register_node(j1939::addr::TASK_CTRL, "ISOBUS-TC"); | executable/support line; behavior impact depends on surrounding block context. |
| 271 |         gw.register_node(j1939::addr::IMPLEMENT, "IMPLEMENT"); | executable/support line; behavior impact depends on surrounding block context. |
| 272 | (blank) | line is blank; visual separation only. |
| 273 |     HeavyMachinery { | executable/support line; behavior impact depends on surrounding block context. |
| 274 |         gateway: gw, | executable/support line; behavior impact depends on surrounding block context. |
| 275 |         boot: BootSequence::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 276 |         net_mgmt: NetworkManager::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 277 |         ecm: EcuEcm::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 278 |         tcm: EcuTcm::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 279 |         bcm: EcuBcm::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 280 |         icm: EcuIcm::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 281 |         hcm: EcuHcm::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 282 |         abs: EcuAbs::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 283 |         electrical: ElectricalSystem::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 284 |         uds_ecm: UdsServer::new("ECM #1", j1939::addr::ECM_1), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 285 |         uds_tcm: UdsServer::new("TCM", j1939::addr::TRANSMISSION), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

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```
| 286 |         nvm: NvmStore::new(), | executable/support line; behavior impact depends on surrounding block context. |  
t. |  
| 287 |         implement: ImplementControl::new(), | executable/support line; behavior impact depends on surrounding block context. |  
g block context. |  
| 288 |         elapsed: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 289 |         throttle_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
|  
| 290 |         brake_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 291 |         hitch_joyystick: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
. |  
| 292 |         loader_lift_cmd: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
t. |  
| 293 |         loader_tilt_cmd: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
t. |  
| 294 |         selected_fault: FaultType::None, | executable/support line; behavior impact depends on surrounding block context. |  
lock context. |  
| 295 |         fault_active: false, | executable/support line; behavior impact depends on surrounding block context. |  
. |  
| 296 |         gps: GpsModule::new(), | executable/support line; behavior impact depends on surrounding block context. |  
xt. |  
| 297 |         imu: Imu::new(), | executable/support line; behavior impact depends on surrounding block context. |  
| 298 |         radar: RadarSuite::new(), | executable/support line; behavior impact depends on surrounding block context. |  
ntext. |  
| 299 |         lidar: LidarSensor::new_vlp16(), | executable/support line; behavior impact depends on surrounding block context. |  
lock context. |  
| 300 |         ad: AutonomousController::new(), | executable/support line; behavior impact depends on surrounding block context. |  
lock context. |  
| 301 |         v2x: V2xModule::new(), | executable/support line; behavior impact depends on surrounding block context. |  
xt. |  
| 302 |         telematics: TelematicsModule::new(), | executable/support line; behavior impact depends on surrounding block context. |  
ng block context. |  
| 303 |         can_net: CanNetwork::new(), | executable/support line; behavior impact depends on surrounding block context. |  
context. |  
| 304 |         vcm: EcuVcm::new(), | executable/support line; behavior impact depends on surrounding block context. |  
|  
| 305 |         camera: CameraSystem::new(), | executable/support line; behavior impact depends on surrounding block context. |  
context. |  
| 306 |         fusion: SensorFusion::new(), | executable/support line; behavior impact depends on surrounding block context. |  
context. |  
| 307 |         leak_rig: LeakPhysicsRig::with_default_machine_presets(), | executable/support line; behavior impact depends on surrounding block context. |  
depends on surrounding block context. |  
| 308 |         leak_reports: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |  
ntext. |  
| 309 |         metrics: SimMetrics::default(), | executable/support line; behavior impact depends on surrounding block context. |  
ock context. |  
| 310 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
|  
| 311 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 312 | (blank) | line is blank; visual separation only. |  
| 313 | // ████████████████████████████████████████████████████████████████████████████████████████████████████████ | comment  
intent, no runtime effect. |  
| 314 | /// Master simulation tick – called every 16 ms (~ 60 Hz). | comment line; documentation intent, no runtime effect. |  
effect. |  
| 315 | pub fn tick(&ampmut self, dt: f64) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
flow, and outputs. |  
| 316 |     let step_started = Instant::now(); | executable/support line; behavior impact depends on surrounding block context. |  
ck context. |  
| 317 |     let body_load_requested_a = { | executable/support line; behavior impact depends on surrounding block context. |  
ntext. |  
| 318 |         let mut load = 0.0; | executable/support line; behavior impact depends on surrounding block context. |  
|  
| 319 |         if self.bcm.work_lights_front { | conditional gate; changing condition shifts runtime path and side effects. |  
effects. |  
| 320 |             load += 18.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 321 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
```

```

es. |
| 322 |         if self.bcm.work_lights_rear { | conditional gate; changing condition shifts runtime path and side effects. |
| 323 |             load += 14.0; | executable/support line; behavior impact depends on surrounding block context. |
| 324 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
es. |
| 325 |         if self.bcm.road_lights { | conditional gate; changing condition shifts runtime path and side effects. |
| 326 |             load += 8.0; | executable/support line; behavior impact depends on surrounding block context. |
| 327 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
es. |
| 328 |         if self.bcm.beacon_light { | conditional gate; changing condition shifts runtime path and side effects. |
| 329 |             load += 4.0; | executable/support line; behavior impact depends on surrounding block context. |
| 330 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
es. |
| 331 |         if self.bcm.hazard_lights { | conditional gate; changing condition shifts runtime path and side effects. |
| 332 |             load += 6.0; | executable/support line; behavior impact depends on surrounding block context. |
| 333 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
es. |
| 334 |         if self.bcm.cab_interior_light { | conditional gate; changing condition shifts runtime path and side effects. |
| 335 |             load += 2.0; | executable/support line; behavior impact depends on surrounding block context. |
| 336 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
es. |
| 337 |         if self.bcm.hvac_on { | conditional gate; changing condition shifts runtime path and side effects. |
| 338 |             load += 15.0; | executable/support line; behavior impact depends on surrounding block context. |
| 339 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
es. |
| 340 |         if self.bcm.horn_active { | conditional gate; changing condition shifts runtime path and side effects. |
| 341 |             load += 8.0; | executable/support line; behavior impact depends on surrounding block context. |
| 342 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
es. |
| 343 |         load | executable/support line; behavior impact depends on surrounding block context. |
| 344 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
|
| 345 |         let electrical_loads = ElectricalLoadRequest { | executable/support line; behavior impact depends on surrounding block context. |
| 346 |             bcm_supply_a: 5.0 + body_load_requested_a, | executable/support line; behavior impact depends on surrounding block context. |
| 347 |             vcm_supply_a: 4.0, | executable/support line; behavior impact depends on surrounding block context. |
| 348 |             ecm_supply_a: if matches!(self.boot.ignition, IgnitionState::On \| IgnitionState::Cranking \| IgnitionState::Running) { 6.5 } else { 0.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 349 |             tcm_supply_a: if matches!(self.boot.ignition, IgnitionState::On \| IgnitionState::Cranking \| IgnitionState::Running) { 3.2 } else { 0.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 350 |             abs_supply_a: if matches!(self.boot.ignition, IgnitionState::On \| IgnitionState::Cranking \| IgnitionState::Running) { 7.5 } else { 0.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 351 |             hcm_supply_a: if matches!(self.boot.ignition, IgnitionState::On \| IgnitionState::Cranking \| IgnitionState::Running) { 4.8 } else { 0.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 352 |             icm_supply_a: if matches!(self.boot.ignition, IgnitionState::Accessory \| IgnitionState::On \| IgnitionState::Cranking \| IgnitionState::Running) { 1.8 } else { 0.0 }, | executable/support line; behavior impact depends on surrounding block context. |
| 353 |             body_load_a: body_load_requested_a, | executable/support line; behavior impact depends on surrounding block context. |
| 354 |             hs_can_a: 0.15, | executable/support line; behavior impact depends on surrounding block context. |
| 355 |             ms_can_a: 0.08, | executable/support line; behavior impact depends on surrounding block context. |
| 356 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
|

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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```

ext. |
| 590 |         self.radar.ttc_front, | executable/support line; behavior impact depends on surrounding block context. |
t. |
| 591 |         self.elapsed, | executable/support line; behavior impact depends on surrounding block context. |
| 592 |         dt, | executable/support line; behavior impact depends on surrounding block context. |
| 593 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 594 |     self.fusion | executable/support line; behavior impact depends on surrounding block context. |
| 595 |         .fuse(&self.camera, &self.radar.front_center.targets, spd_ms, dt); | executable/support line; behavior impact depends on surrounding block context. |
| 596 | (blank) | line is blank; visual separation only. |
| 597 |         // AD controller | comment line; documentation intent, no runtime effect. |
| 598 |         let lane_off = self.imu.lateral_g * 0.2; // rough lane offset from lateral G | executable/support line; behavior impact depends on surrounding block context. |
| 599 |         let lane_conf = if self.tcm.ground_speed_kmh > 5.0 { | executable/support line; behavior impact depends on surrounding block context. |
| 600 |             0.92 | executable/support line; behavior impact depends on surrounding block context. |
| 601 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 602 |             0.0 | executable/support line; behavior impact depends on surrounding block context. |
| 603 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
| 604 |         let fused_ttc = self.fusion.ttc_critical.min(self.radar.ttc_front); | executable/support line; behavior impact depends on surrounding block context. |
| 605 |         let fused_lead = self.fusion.lead_dist_m.min(self.radar.closest_front_m); | executable/support line; behavior impact depends on surrounding block context. |
| 606 |         let ttc = fused_ttc; | executable/support line; behavior impact depends on surrounding block context. |
| 607 |         let lead_spd = if self.radar.front_center.closest_inpath().is_some() { | executable/support line; behavior impact depends on surrounding block context. |
| 608 |             (self.tcm.ground_speed_kmh - 5.0).max(0.0) / 3.6 | executable/support line; behavior impact depends on surrounding block context. |
| 609 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 610 |             0.0 | executable/support line; behavior impact depends on surrounding block context. |
| 611 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
| 612 |         let (ad_thr, ad_brk, ad_str) = self.ad.tick( | executable/support line; behavior impact depends on surrounding block context. |
| 613 |             self.tcm.ground_speed_kmh, | executable/support line; behavior impact depends on surrounding block context. |
| 614 |             self.throttle_pct / 100.0, | executable/support line; behavior impact depends on surrounding block context. |
| 615 |             self.brake_pct / 100.0, | executable/support line; behavior impact depends on surrounding block context. |
| 616 |             fused_lead, | executable/support line; behavior impact depends on surrounding block context. |
| 617 |             lead_spd, | executable/support line; behavior impact depends on surrounding block context. |
| 618 |             lane_off, | executable/support line; behavior impact depends on surrounding block context. |
| 619 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 620 |             lane_conf, | executable/support line; behavior impact depends on surrounding block context. |
| 621 |             ttc, | executable/support line; behavior impact depends on surrounding block context. |
| 622 |             self.radar.bsm_left, | executable/support line; behavior impact depends on surrounding block context. |
| 623 |             self.radar.bsm_right, | executable/support line; behavior impact depends on surrounding block context. |
| 624 |             dt, | executable/support line; behavior impact depends on surrounding block context. |
| 625 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 626 |         // AD overrides apply back to vehicle (if engaged) | comment line; documentation intent, no runtime effect. |
| 627 |         if self.ad.engaged { | conditional gate; changing condition shifts runtime path and side effects. |
| 628 |             if let Some(v) = ad_thr { | conditional gate; changing condition shifts runtime path and side effects. |
| 629 |                 self.throttle_pct = (v * 100.0).clamp(0.0, 100.0); | executable/support line; behavior impact depends on surrounding block context. |
| 630 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
| 631 |             if let Some(v) = ad_brk { | conditional gate; changing condition shifts runtime path and side effects. |
| 632 |                 self.brake_pct = (v * 100.0).clamp(0.0, 100.0); | executable/support line; behavior impact depends on surrounding block context. |

```

```

ds on surrounding block context. |
| 633 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 634 |         if let Some(v) = ad_str { | conditional gate; changing condition shifts runtime path and side eff
s. |
| 635 |             self.ad.steer_cmd_deg = v; | executable/support line; behavior impact depends on surrounding
ck context. |
| 636 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 637 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 638 | (blank) | line is blank; visual separation only. |
| 639 |     // V2X | comment line; documentation intent, no runtime effect. |
| 640 |     self.v2x.update( | executable/support line; behavior impact depends on surrounding block context. |
| 641 |         self.gps.latitude_deg, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 642 |         self.gps.longitude_deg, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 643 |         spd_ms, | executable/support line; behavior impact depends on surrounding block context. |
| 644 |         self.vehicle_heading(), | executable/support line; behavior impact depends on surrounding block c
ext. |
| 645 |         self.imu.longitudinal_g * 9.81, | executable/support line; behavior impact depends on surrounding
ock context. |
| 646 |         self.imu.lateral_g * 9.81, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 647 |         self.abs.abs_system_active, | executable/support line; behavior impact depends on surrounding blo
context. |
| 648 |         self.elapsed, | executable/support line; behavior impact depends on surrounding block context. |
| 649 |         dt, | executable/support line; behavior impact depends on surrounding block context. |
| 650 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 651 | (blank) | line is blank; visual separation only. |
| 652 |     // Telematics | comment line; documentation intent, no runtime effect. |
| 653 |     self.telematics.update( | executable/support line; behavior impact depends on surrounding block conte
|
| 654 |         self.gps.latitude_deg, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 655 |         self.gps.longitude_deg, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 656 |         self.tcm.ground_speed_kmh, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 657 |         self.ecm.engine_hours, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 658 |         self.ecm.active_dtcs.len(), | executable/support line; behavior impact depends on surrounding blo
context. |
| 659 |         self.elapsed, | executable/support line; behavior impact depends on surrounding block context. |
| 660 |         dt, | executable/support line; behavior impact depends on surrounding block context. |
| 661 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 662 | (blank) | line is blank; visual separation only. |
| 663 |     // VCM | comment line; documentation intent, no runtime effect. |
| 664 |     let abs_torque_limit_nm = if self.abs.abs_system_active || self.abs.tcs_system_active { | executabl
upport line; behavior impact depends on surrounding block context. |
| 665 |         260.0 | executable/support line; behavior impact depends on surrounding block context. |
| 666 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 667 |         1050.0 | executable/support line; behavior impact depends on surrounding block context. |
| 668 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 669 |     let ad_torque_limit_nm = if fused_ttc < 1.2 { | executable/support line; behavior impact depends on s
ounding block context. |
| 670 |         0.0 | executable/support line; behavior impact depends on surrounding block context. |
| 671 |     } else if fused_ttc < 2.0 { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 672 |         220.0 | executable/support line; behavior impact depends on surrounding block context. |
| 673 |     } else if fused_ttc < 3.5 { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 674 |         520.0 | executable/support line; behavior impact depends on surrounding block context. |

```



```

| 675 |         } else if fused_lead < 12.0 { | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 676 |             680.0 | executable/support line; behavior impact depends on surrounding block context. |
| 677 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 678 |             1050.0 | executable/support line; behavior impact depends on surrounding block context. |
| 679 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 680 |     let vcm_frames = self.vcm.tick( | executable/support line; behavior impact depends on surrounding blo
context. |
| 681 |         self.elapsed, | executable/support line; behavior impact depends on surrounding block context. |
| 682 |         self.ecm.rpm, | executable/support line; behavior impact depends on surrounding block context. |
| 683 |         self.tcm.ground_speed_kmh, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 684 |         self.ecm.actual_torque_nm, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 685 |         self.hcm.hydraulic_power_kw, | executable/support line; behavior impact depends on surrounding bl
context. |
| 686 |         self.implement.pto_torque_demand(), | executable/support line; behavior impact depends on surroun
g block context. |
| 687 |         self.bcm.total_load_amps, | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 688 |         abs_torque_limit_nm, | executable/support line; behavior impact depends on surrounding block cont
. |
| 689 |         ad_torque_limit_nm, | executable/support line; behavior impact depends on surrounding block conte
|
| 690 |         self.ecm.active_dtcs.len() as u32, | executable/support line; behavior impact depends on surround
block context. |
| 691 |         &self.gateway.dispatched.clone(), | executable/support line; behavior impact depends on surroundi
block context. |
| 692 |         dt, | executable/support line; behavior impact depends on surrounding block context. |
| 693 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 694 |     for f in vcm_frames { | iteration control; changing bounds/body affects throughput, timing, and possi
safety guarantees. |
| 695 |         self.gateway.transmit(f); | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 696 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 697 | (blank) | line is blank; visual separation only. |
| 698 |     // ADAS/fusion/energy status frames on the powertrain network via VCM SA. | comment line; documentati
intent, no runtime effect. |
| 699 |     let lead_distance = self.fusion.lead_dist_m.min(self.radar.closest_front_m); | executable/support lin
behavior impact depends on surrounding block context. |
| 700 |     let critical_ttc = self.fusion.ttc_critical.min(self.radar.ttc_front); | executable/support line; beh
or impact depends on surrounding block context. |
| 701 |     let in_path_hazard = self | executable/support line; behavior impact depends on surrounding block con
t. |
| 702 |         .fusion | executable/support line; behavior impact depends on surrounding block context. |
| 703 |         .objects | executable/support line; behavior impact depends on surrounding block context. |
| 704 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 705 |         .any(\|o\| o.in_path && o.ttc_s < 3.0); | executable/support line; behavior impact depends on sur
nding block context. |
| 706 |     let speed_limit = self.camera.active_speed_limit.unwrap_or(0) as f64; | executable/support line; beha
r impact depends on surrounding block context. |
| 707 |     let adas = j1939::Builder::adas1( | executable/support line; behavior impact depends on surrounding b
k context. |
| 708 |         self.elapsed, | executable/support line; behavior impact depends on surrounding block context. |
| 709 |         self.ad.lka_state == FeatureState::Active, | executable/support line; behavior impact depends on
rounding block context. |
| 710 |         self.ad.acc_state == FeatureState::Active, | executable/support line; behavior impact depends on
rounding block context. |
| 711 |         self.ad.aeb_state == FeatureState::Active, | executable/support line; behavior impact depends on
rounding block context. |
| 712 |         lead_distance, | executable/support line; behavior impact depends on surrounding block context. |
| 713 |         critical_ttc, | executable/support line; behavior impact depends on surrounding block context. |
| 714 |         speed_limit, | executable/support line; behavior impact depends on surrounding block context. |

```

[illegible]

```

on surrounding block context. |
| 756 |         event: "tick_metrics", | executable/support line; behavior impact depends on surrounding block
context. |
| 757 |         details: format!( | executable/support line; behavior impact depends on surrounding block con
t. |
| 758 |             "loop_duration_ms={:.3},steps_completed={},error_count={},replay_failures={}", | executab
support line; behavior impact depends on surrounding block context. |
| 759 |             self.metrics.loop_duration_ms, | executable/support line; behavior impact depends on surr
ding block context. |
| 760 |             self.metrics.steps_completed, | executable/support line; behavior impact depends on surro
ing block context. |
| 761 |             self.metrics.error_count, | executable/support line; behavior impact depends on surroundi
block context. |
| 762 |             self.metrics.replay_failures | executable/support line; behavior impact depends on surrou
ng block context. |
| 763 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 764 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 765 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 766 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 767 | (blank) | line is blank; visual separation only. |
| 768 | fn vehicle_heading(&self) -> f64 { | declares executable behavior; signature/body changes alter API, cont
flow, and outputs. |
| 769 |     self.gps.course_deg | executable/support line; behavior impact depends on surrounding block context.
| 770 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 771 | fn vehicle_accel_ms2(&self) -> f64 { | declares executable behavior; signature/body changes alter API, co
ol flow, and outputs. |
| 772 |     self.ecm.acceleration_ms2() | executable/support line; behavior impact depends on surrounding block c
ext. |
| 773 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 774 | (blank) | line is blank; visual separation only. |
| 775 | fn run_leak_physics(&mut self, dt: f64) { | declares executable behavior; signature/body changes alter AP
control flow, and outputs. |
| 776 |     let hvac_duty = if self.bcm.hvac_on { 0.82 } else { 0.18 }; | executable/support line; behavior impac
depends on surrounding block context. |
| 777 |     let ac_high_bar = if self.bcm.hvac_on { | executable/support line; behavior impact depends on surroun
g block context. |
| 778 |         (6.5 + (self.ecm.rpm / 2200.0).clamp(0.0, 1.4) * 9.5 | executable/support line; behavior impact d
nds on surrounding block context. |
| 779 |         + (self.ecm.ambient_temp_c - 20.0).max(0.0) * 0.12) | executable/support line; behavior impac
depends on surrounding block context. |
| 780 |         .clamp(4.0, 32.0) | executable/support line; behavior impact depends on surrounding block con
t. |
| 781 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 782 |         5.5 | executable/support line; behavior impact depends on surrounding block context. |
| 783 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 784 | (blank) | line is blank; visual separation only. |
| 785 |     self.leak_rig.set_runtime_pressure_state( | executable/support line; behavior impact depends on surro
ing block context. |
| 786 |         "HYD_MAIN", | executable/support line; behavior impact depends on surrounding block context. |
| 787 |         self.hcm.system_pressure_bar, | executable/support line; behavior impact depends on surrounding b
k context. |
| 788 |         self.hcm.system_pressure_bar, | executable/support line; behavior impact depends on surrounding b
k context. |
| 789 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 790 |     self.leak_rig.set_runtime_pressure_state( | executable/support line; behavior impact depends on surro
ing block context. |
| 791 |         "ENG_OIL", | executable/support line; behavior impact depends on surrounding block context. |
| 792 |         (self.ecm.oil_pressure_kpa / 100.0).max(0.0), | executable/support line; behavior impact depends
surrounding block context. |
| 793 |         (self.ecm.oil_pressure_kpa / 100.0).max(0.0), | executable/support line; behavior impact depends
surrounding block context. |
| 794 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 795 |     self.leak_rig | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 796 |         .set_runtime_pressure_state("AC_HIGH", ac_high_bar, ac_high_bar); | executable/support line; beha
r impact depends on surrounding block context. |
| 797 | (blank) | line is blank; visual separation only. |
| 798 |         let hyd_rho = self.leak_rig.oil_density_for("HYD_MAIN"); | executable/support line; behavior impact d
nds on surrounding block context. |
| 799 |         let eng_rho = self.leak_rig.oil_density_for("ENG_OIL"); | executable/support line; behavior impact de
ds on surrounding block context. |
| 800 |         let ac_rho = self.leak_rig.oil_density_for("AC_HIGH"); | executable/support line; behavior impact dep
s on surrounding block context. |
| 801 | (blank) | line is blank; visual separation only. |
| 802 |         let inputs = [ | executable/support line; behavior impact depends on surrounding block context. |
| 803 |             ( | executable/support line; behavior impact depends on surrounding block context. |
| 804 |                 "HYD_MAIN", | executable/support line; behavior impact depends on surrounding block context.
| 805 |                 CircuitInput { | executable/support line; behavior impact depends on surrounding block contex
|
| 806 |                 pressure_bar: self.hcm.system_pressure_bar, | executable/support line; behavior impact de
ds on surrounding block context. |
| 807 |                 delta_p_bar: (self.hcm.system_pressure_bar - self.hcm.return_pres_bar).max(0.0), | execut
e/support line; behavior impact depends on surrounding block context. |
| 808 |                 temp_c: self.hcm.fluid_temp_c, | executable/support line; behavior impact depends on surr
ding block context. |
| 809 |                 cycles_per_s: (self.hcm.total_flow_demand_lpm / 45.0).clamp(0.0, 5.0), | executable/suppo
line; behavior impact depends on surrounding block context. |
| 810 |                 duty_01: (self.hcm.total_flow_demand_lpm / self.hcm.pump_flow_lpm.max(1.0)) | executable/
port line; behavior impact depends on surrounding block context. |
| 811 |                 .clamp(0.0, 1.0), | executable/support line; behavior impact depends on surrounding b
k context. |
| 812 |                 fluid_density_kg_m3: hyd_rho, | executable/support line; behavior impact depends on surro
ing block context. |
| 813 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 814 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 815 |         ( | executable/support line; behavior impact depends on surrounding block context. |
| 816 |             "ENG_OIL", | executable/support line; behavior impact depends on surrounding block context. |
| 817 |             CircuitInput { | executable/support line; behavior impact depends on surrounding block contex
|
| 818 |             pressure_bar: (self.ecm.oil_pressure_kpa / 100.0).max(0.0), | executable/support line; be
ior impact depends on surrounding block context. |
| 819 |             delta_p_bar: (self.ecm.oil_pressure_kpa / 100.0).max(0.0), | executable/support line; beh
or impact depends on surrounding block context. |
| 820 |             temp_c: self.ecm.oil_temp_c, | executable/support line; behavior impact depends on surrou
ng block context. |
| 821 |             cycles_per_s: (self.ecm.rpm / 60.0).clamp(0.0, 60.0), | executable/support line; behavior
pact depends on surrounding block context. |
| 822 |             duty_01: (self.ecm.active_throttle / 100.0).clamp(0.0, 1.0), | executable/support line; b
vior impact depends on surrounding block context. |
| 823 |             fluid_density_kg_m3: eng_rho, | executable/support line; behavior impact depends on surro
ing block context. |
| 824 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 825 |     ), | executable/support line; behavior impact depends on surrounding block context. |
| 826 |     ( | executable/support line; behavior impact depends on surrounding block context. |
| 827 |         "AC_HIGH", | executable/support line; behavior impact depends on surrounding block context. |
| 828 |         CircuitInput { | executable/support line; behavior impact depends on surrounding block contex
|
| 829 |         pressure_bar: ac_high_bar, | executable/support line; behavior impact depends on surround
block context. |
| 830 |         delta_p_bar: (ac_high_bar - 2.0).max(0.0), | executable/support line; behavior impact dep
s on surrounding block context. |
| 831 |         temp_c: self.bcm.cab_temp_c + 8.0, | executable/support line; behavior impact depends on
rounding block context. |
| 832 |         cycles_per_s: (self.ecm.rpm / 120.0).clamp(0.0, 30.0), | executable/support line; behavior
mpact depends on surrounding block context. |
| 833 |         duty_01: hvac_duty, | executable/support line; behavior impact depends on surrounding blo
context. |
| 834 |         fluid_density_kg_m3: ac_rho, | executable/support line; behavior impact depends on surrou
ng block context. |

```

```

| 835 |           }, | executable/support line; behavior impact depends on surrounding block context. |
| 836 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 837 |       ]; | executable/support line; behavior impact depends on surrounding block context. |
| 838 | (blank) | line is blank; visual separation only. |
| 839 |       let reports = self.leak_rig.step(&inputs, dt); | executable/support line; behavior impact depends on
rounding block context. |
| 840 |       self.leak_reports = reports.clone(); | executable/support line; behavior impact depends on surrounding
lock context. |
| 841 | (blank) | line is blank; visual separation only. |
| 842 |       for rep in &reports { | iteration control; changing bounds/body affects throughput, timing, and possi
safety guarantees. |
| 843 |           let leaked_l = rep.leak_lpm * dt / 60.0; | executable/support line; behavior impact depends on su
unding block context. |
| 844 |           match rep.name.as_str() { | branch dispatch; changing arms alters behavior for variants and input
asses. |
| 845 |               "HYD_MAIN" => { | executable/support line; behavior impact depends on surrounding block conte
|
| 846 |                   let pressure_drop_bar = leaked_l * 85.0; | executable/support line; behavior impact dependen
on surrounding block context. |
| 847 |                   self.hcm.system_pressure_bar = | executable/support line; behavior impact depends on surr
ding block context. |
| 848 |                   (self.hcm.system_pressure_bar - pressure_drop_bar).max(0.0); | executable/support lin
behavior impact depends on surrounding block context. |
| 849 |                   self.hcm.fluid_level_pct = | executable/support line; behavior impact depends on surround
block context. |
| 850 |                   (self.hcm.fluid_level_pct - leaked_l / 90.0 * 100.0).clamp(0.0, 100.0); | executable/su
port line; behavior impact depends on surrounding block context. |
| 851 |                   if rep.alert >= LeakAlertLevel::Warning { | conditional gate; changing condition shifts r
ime path and side effects. |
| 852 |                       self.hcm.alarm_low_pressure = true; | executable/support line; behavior impact depend
n surrounding block context. |
| 853 |                   } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 854 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 855 |               "ENG_OIL" => { | executable/support line; behavior impact depends on surrounding block context
|
| 856 |                   let pressure_scale = (1.0 - (rep.leak_lpm / 12.0)).clamp(0.12, 1.0); | executable/support
ne; behavior impact depends on surrounding block context. |
| 857 |                   self.ecm.oil_pressure_kpa *= pressure_scale; | executable/support line; behavior impact d
nds on surrounding block context. |
| 858 |                   self.ecm.oil_level_pct = | executable/support line; behavior impact depends on surroundin
lock context. |
| 859 |                   (self.ecm.oil_level_pct - leaked_l / 28.0 * 100.0).clamp(0.0, 100.0); | executable/su
rt line; behavior impact depends on surrounding block context. |
| 860 |                   if rep.alert >= LeakAlertLevel::Warning && self.ecm.rpm > 600.0 { | conditional gate; cha
ng condition shifts runtime path and side effects. |
| 861 |                       self.ecm.amber_lamp = true; | executable/support line; behavior impact depends on surr
nding block context. |
| 862 |                   } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 863 |                       if rep.alert >= LeakAlertLevel::Critical && self.ecm.rpm > 600.0 { | conditional gate; ch
ing condition shifts runtime path and side effects. |
| 864 |                           self.ecm.red_lamp = true; | executable/support line; behavior impact depends on surro
ing block context. |
| 865 |                       } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 866 |                   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 867 |               "AC_HIGH" | executable/support line; behavior impact depends on surrounding block context. |
| 868 |               if self.bcm.hvac_on => { | conditional gate; changing condition shifts runtime path and s
effects. |
| 869 |                   let cooling_loss = (rep.leak_lpm / 1.5).clamp(0.0, 1.0); | executable/support line; b
vior impact depends on surrounding block context. |
| 870 |                   self.bcm.cab_temp_c += (0.22 + cooling_loss * 0.55) * dt; | executable/support line;

```

```

avior impact depends on surrounding block context. |
| 871 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 872 |         _ => {} | executable/support line; behavior impact depends on surrounding block context. |
| 873 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 874 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 875 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 876 | (blank) | line is blank; visual separation only. |
| 877 |     fn build_radar_traffic(&self) -> Vec<SimTrafficObj> { | declares executable behavior; signature/body chan
alter API, control flow, and outputs. |
| 878 |         self.implement_traffic() | executable/support line; behavior impact depends on surrounding block cont
. |
| 879 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 880 | (blank) | line is blank; visual separation only. |
| 881 |     fn implement_traffic(&self) -> Vec<SimTrafficObj> { | declares executable behavior; signature/body change
lter API, control flow, and outputs. |
| 882 |         use radar::RadarObjectClass; | imports symbols; changing path/name affects compile-time resolution. |
| 883 |         let t = self.elapsed; | executable/support line; behavior impact depends on surrounding block context
| 884 |         let lead_dist = 50.0 + (t * 0.1).sin() * 30.0; | executable/support line; behavior impact depends on
rounding block context. |
| 885 |         vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 886 |             SimTrafficObj { | executable/support line; behavior impact depends on surrounding block context.
| 887 |                 id: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 888 |                 distance_m: lead_dist.max(5.0), | executable/support line; behavior impact depends on surroun
g block context. |
| 889 |                 lateral_offset_m: 0.0, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 890 |                 speed_ms: 10.0 + (t * 0.05).sin() * 5.0, | executable/support line; behavior impact depends o
urrounding block context. |
| 891 |                 rcs_dbzm: 12.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 892 |                 object_type: RadarObjectClass::Car, | executable/support line; behavior impact depends on sur
nding block context. |
| 893 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 894 |             SimTrafficObj { | executable/support line; behavior impact depends on surrounding block context.
| 895 |                 id: 1, | executable/support line; behavior impact depends on surrounding block context. |
| 896 |                 distance_m: 25.0 + (t * 0.07).sin() * 10.0, | executable/support line; behavior impact depend
n surrounding block context. |
| 897 |                 lateral_offset_m: 3.5, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 898 |                 speed_ms: 8.0, | executable/support line; behavior impact depends on surrounding block context
|
| 899 |                 rcs_dbzm: 8.0, | executable/support line; behavior impact depends on surrounding block context
|
| 900 |                 object_type: RadarObjectClass::Car, | executable/support line; behavior impact depends on sur
nding block context. |
| 901 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 902 |             SimTrafficObj { | executable/support line; behavior impact depends on surrounding block context.
| 903 |                 id: 2, | executable/support line; behavior impact depends on surrounding block context. |
| 904 |                 distance_m: 80.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 905 |                 lateral_offset_m: -0.5, | executable/support line; behavior impact depends on surrounding blo
context. |
| 906 |                 speed_ms: 15.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 907 |                 rcs_dbzm: 22.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 908 |                 object_type: RadarObjectClass::Truck, | executable/support line; behavior impact depends on s
ounding block context. |
| 909 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 910 |         ] | executable/support line; behavior impact depends on surrounding block context. |
| 911 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 912 | (blank) | line is blank; visual separation only. |

```

```
| 913 | fn build_lidar_world(&self) -> Vec<WorldObstacle> { | declares executable behavior; signature/body change  
alter API, control flow, and outputs. |  
| 914 | let t = self.elapsed; | executable/support line; behavior impact depends on surrounding block context.  
| 915 | vec![] | executable/support line; behavior impact depends on surrounding block context. |  
| 916 | WorldObstacle::new_vehicle(50.0 + (t * 0.1).sin() * 20.0, 0.0, 10.0), | executable/support line;  
avior impact depends on surrounding block context. |  
| 917 | WorldObstacle::new_vehicle(25.0, 3.5, 8.0), | executable/support line; behavior impact depends on  
rounding block context. |  
| 918 | WorldObstacle::new_barrier(30.0, -8.0, 10.0), | executable/support line; behavior impact depends  
surrounding block context. |  
| 919 | WorldObstacle::new_pedestrian(12.0, 2.0), | executable/support line; behavior impact depends on s  
ounding block context. |  
| 920 | ] | executable/support line; behavior impact depends on surrounding block context. |  
| 921 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 922 | (blank) | line is blank; visual separation only. |  
| 923 | // ████████████████████████████████████████████████████████████████████████████████████████████████ | comment  
ntent, no runtime effect. |  
| 924 | fn apply_fault(&mut self) { | declares executable behavior; signature/body changes alter API, control flo  
and outputs. |  
| 925 | match self.selected_fault { | branch dispatch; changing arms alters behavior for variants and input c  
ses. |  
| 926 | FaultType::HighCoolantTemp => { | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 927 | self.ecm.coolant_temp_c = 115.0; | executable/support line; behavior impact depends on surrou  
ng block context. |  
| 928 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 929 | FaultType::LowOilPressure => { | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 930 | self.ecm.oil_pressure_kpa = 40.0; | executable/support line; behavior impact depends on surro  
ing block context. |  
| 931 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 932 | FaultType::LowFuelPressure => { | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 933 | self.ecm.fuel_pressure_kpa = 50.0; | executable/support line; behavior impact depends on surr  
ding block context. |  
| 934 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 935 | FaultType::CriticalDefLevel => { | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 936 | self.ecm.def_level_pct = 1.0; | executable/support line; behavior impact depends on surroundi  
block context. |  
| 937 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 938 | FaultType::HighDpfSoot => { | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 939 | self.ecm.dpf_soot_pct = 95.0; | executable/support line; behavior impact depends on surroundi  
block context. |  
| 940 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 941 | FaultType::HighBoostPressure => { | executable/support line; behavior impact depends on surroundi  
block context. |  
| 942 | self.ecm.boost_pressure_kpa = 340.0; | executable/support line; behavior impact depends on su  
unding block context. |  
| 943 | self.ecm.intake_manifold_kpa = 340.0; | executable/support line; behavior impact depends on s  
ounding block context. |  
| 944 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 945 | FaultType::LowBoostPressure => { | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 946 | self.ecm.boost_pressure_kpa = 80.0; | executable/support line; behavior impact depends on sur  
nding block context. |  
| 947 | self.ecm.intake_manifold_kpa = 82.0; | executable/support line; behavior impact depends on su  
unding block context. |
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| 948 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 949 |           FaultType::LowCoolantPressure => { | executable/support line; behavior impact depends on surround
    block context. |
| 950 |               self.ecm.coolant_pres_kpa = 40.0; | executable/support line; behavior impact depends on surro
ing block context. |
| 951 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 952 |           FaultType::HighIntakeTemp => { | executable/support line; behavior impact depends on surrounding
ck context. |
| 953 |               self.ecm.intake_temp_c = 115.0; | executable/support line; behavior impact depends on surround
g block context. |
| 954 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 955 |           FaultType::HighExhaustTemp => { | executable/support line; behavior impact depends on surrounding
ock context. |
| 956 |               self.ecm.exhaust_temp_c = 860.0; | executable/support line; behavior impact depends on surround
ng block context. |
| 957 |               self.ecm.dpf_temp_c = 780.0; | executable/support line; behavior impact depends on surroundin
lock context. |
| 958 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 959 |           FaultType::PoorDefQuality => { | executable/support line; behavior impact depends on surrounding
ck context. |
| 960 |               self.ecm.def_quality_pct = 55.0; | executable/support line; behavior impact depends on surround
ng block context. |
| 961 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 962 |           FaultType::HighNoxTailpipe => { | executable/support line; behavior impact depends on surrounding
ock context. |
| 963 |               self.ecm.nox_tailpipe_ppm = 1100.0; | executable/support line; behavior impact depends on sur
nding block context. |
| 964 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 965 |           FaultType::LowFuelRailPressure => { | executable/support line; behavior impact depends on surround
g block context. |
| 966 |               self.ecm.fuel_rail_pressure_mpa = 45.0; | executable/support line; behavior impact depends on
rounding block context. |
| 967 |               self.ecm.fuel_pressure_kpa = 120.0; | executable/support line; behavior impact depends on sur
nding block context. |
| 968 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 969 |           FaultType::TransClutchOverheat => { | executable/support line; behavior impact depends on surround
g block context. |
| 970 |               self.tcm.clutch_temp_c = 250.0; | executable/support line; behavior impact depends on surround
g block context. |
| 971 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 972 |           FaultType::TransNeutralFailure => { /* TCM cannot leave neutral */ } | executable/support line; b
vior impact depends on surrounding block context. |
| 973 |           FaultType::LowBatteryVoltage => { | executable/support line; behavior impact depends on surroundi
block context. |
| 974 |               self.bcm.battery_voltage = 10.8; | executable/support line; behavior impact depends on surround
ng block context. |
| 975 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 976 |           FaultType::AlternatorFault => { | executable/support line; behavior impact depends on surrounding
ock context. |
| 977 |               self.ecm.alternator_v = 11.0; | executable/support line; behavior impact depends on surroundi
block context. |
| 978 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 979 |           FaultType::HighHydTemp => { | executable/support line; behavior impact depends on surrounding blo
context. |
| 980 |               self.hcm.fluid_temp_c = 115.0; | executable/support line; behavior impact depends on surround

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| block context. |
| 981 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 982 |         FaultType::LowHydLevel => { | executable/support line; behavior impact depends on surrounding blo
context. |
| 983 |             self.hcm.fluid_level_pct = 10.0; | executable/support line; behavior impact depends on surrou
ng block context. |
| 984 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 985 |         FaultType::HydFilterBypass => { | executable/support line; behavior impact depends on surrounding
ock context. |
| 986 |             self.hcm.filter_dp_bar = 8.0; | executable/support line; behavior impact depends on surroundi
block context. |
| 987 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 988 |         FaultType::BusOffInjection => { | executable/support line; behavior impact depends on surrounding
ock context. |
| 989 |             self.gateway.inject_bus_off = true; | executable/support line; behavior impact depends on sur
nding block context. |
| 990 |             self.fault_active = false; | executable/support line; behavior impact depends on surrounding
ck context. |
| 991 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 992 |         FaultType::KillEcm => { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 993 |             self.gateway.kill_node(j1939::addr::ECM_1); | executable/support line; behavior impact depend
n surrounding block context. |
| 994 |             self.fault_active = false; | executable/support line; behavior impact depends on surrounding
ck context. |
| 995 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 996 |         FaultType::KillTcm => { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 997 |             self.gateway.kill_node(j1939::addr::TRANSMISSION); | executable/support line; behavior impact
ends on surrounding block context. |
| 998 |             self.fault_active = false; | executable/support line; behavior impact depends on surrounding
ck context. |
| 999 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 1000 |         FaultType::None => {} | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1001 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1002 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1003 | (blank) | line is blank; visual separation only. |
| 1004 |     // ████████████████████████████████████████████████████████████████████████████████████████████ | commen
intent, no runtime effect. |
| 1005 |     /// Activate the selected fault | comment line; documentation intent, no runtime effect. |
| 1006 |     pub fn inject_fault(&mut self) { | declares executable behavior; signature/body changes alter API, contr
flow, and outputs. |
| 1007 |         self.fault_active = true; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1008 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1009 | (blank) | line is blank; visual separation only. |
| 1010 |     /// Clear all faults and restore modules | comment line; documentation intent, no runtime effect. |
| 1011 |     pub fn clear_faults(&mut self) { | declares executable behavior; signature/body changes alter API, contr
flow, and outputs. |
| 1012 |         self.fault_active = false; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1013 |         self.ecm.clear_dtcs(); | executable/support line; behavior impact depends on surrounding block conte
|
| 1014 |         self.gateway.revive_node(j1939::addr::ECM_1); | executable/support line; behavior impact depends on
rounding block context. |
| 1015 |         self.gateway.revive_node(j1939::addr::TRANSMISSION); | executable/support line; behavior impact depe
on surrounding block context. |
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| 1016 |         self.leak_rig.reset(); | executable/support line; behavior impact depends on surrounding block context. |
| 1017 |         self.leak_reports.clear(); | executable/support line; behavior impact depends on surrounding block context. |
| 1018 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1019 | (blank) | line is blank; visual separation only. |
| 1020 |     /// Advance ignition key one position | comment line; documentation intent, no runtime effect. |
| 1021 |     pub fn key_advance(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1022 |         self.boot.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 1023 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1024 | (blank) | line is blank; visual separation only. |
| 1025 |     /// Turn key fully off – powers down all ECUs | comment line; documentation intent, no runtime effect. |
| 1026 |     pub fn key_off(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1027 |         self.boot.key_off(); | executable/support line; behavior impact depends on surrounding block context. |
| 1028 |         self.throttle_pct = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 1029 |         self.brake_pct = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 1030 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1031 | (blank) | line is blank; visual separation only. |
| 1032 |     /// Hard reset – rebuilds entire bench from scratch | comment line; documentation intent, no runtime effect. |
| 1033 |     pub fn reset(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1034 |         *self = Self::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 1035 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1036 | (blank) | line is blank; visual separation only. |
| 1037 |     pub fn apply_leak_manual_params( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1038 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 1039 |         circuit_name: &str, | executable/support line; behavior impact depends on surrounding block context. |
| 1040 |         params: ManualCircuitParams, | executable/support line; behavior impact depends on surrounding block context. |
| 1041 |     ) -> bool { | executable/support line; behavior impact depends on surrounding block context. |
| 1042 |         self.leak_rig.apply_manual_params(circuit_name, params) | executable/support line; behavior impact depends on surrounding block context. |
| 1043 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1044 | (blank) | line is blank; visual separation only. |
| 1045 |     pub fn add_custom_leak_circuit(&mut self, circuit: LeakCircuit) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1046 |         self.leak_rig.add_circuit(circuit); | executable/support line; behavior impact depends on surrounding block context. |
| 1047 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1048 | (blank) | line is blank; visual separation only. |
| 1049 |     pub fn predict_leak_scenarios(&self, horizon_s: f64, dt: f64) -> Vec<ScenarioPrediction> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1050 |         self.leak_rig.predict_scenarios(horizon_s, dt) | executable/support line; behavior impact depends on surrounding block context. |
| 1051 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1052 | (blank) | line is blank; visual separation only. |
| 1053 |     pub fn export_leak_report_json<P: AsRef<std::path::Path>>(&mut self, | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1054 |         &self, | executable/support line; behavior impact depends on surrounding block context. |
| 1055 |         path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1056 |     ) -> std::io::Result<> { | executable/support line; behavior impact depends on surrounding block context. |
| 1057 |         self.leak_rig.export_last_results_json(path) | executable/support line; behavior impact depends on surrounding block context. |
| 1058 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1059 | (blank) | line is blank; visual separation only. |
| 1060 |     pub fn export_leak_report_csv<P: AsRef<std::path::Path>>(&mut self, | declares executable behavior; signature/body changes alter API, control flow, and outputs. |

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| 1061 |         &self, | executable/support line; behavior impact depends on surrounding block context. |
| 1062 |         path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1063 |     ) -> std::io::Result<()> { | executable/support line; behavior impact depends on surrounding block conte
|
| 1064 |         self.leak_rig.export_last_results_csv(path) | executable/support line; behavior impact depends on su
unding block context. |
| 1065 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1066 | (blank) | line is blank; visual separation only. |
| 1067 |     pub fn export_leak_predictions_json<P: AsRef<std::path::Path>>(< | declares executable behavior; signatur
ody changes alter API, control flow, and outputs. |
| 1068 |         &self, | executable/support line; behavior impact depends on surrounding block context. |
| 1069 |         path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1070 |         predictions: &[ScenarioPrediction], | executable/support line; behavior impact depends on surroundin
lock context. |
| 1071 |     ) -> std::io::Result<()> { | executable/support line; behavior impact depends on surrounding block conte
|
| 1072 |         self.leak_rig.export_predictions_json(path, predictions) | executable/support line; behavior impact
ends on surrounding block context. |
| 1073 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1074 | (blank) | line is blank; visual separation only. |
| 1075 |     pub fn export_leak_predictions_csv<P: AsRef<std::path::Path>>(< | declares executable behavior; signature
dy changes alter API, control flow, and outputs. |
| 1076 |         &self, | executable/support line; behavior impact depends on surrounding block context. |
| 1077 |         path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1078 |         predictions: &[ScenarioPrediction], | executable/support line; behavior impact depends on surroundin
lock context. |
| 1079 |     ) -> std::io::Result<()> { | executable/support line; behavior impact depends on surrounding block conte
|
| 1080 |         self.leak_rig.export_predictions_csv(path, predictions) | executable/support line; behavior impact d
nds on surrounding block context. |
| 1081 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1082 | (blank) | line is blank; visual separation only. |
| 1083 |     pub fn export_leak_material_catalog_json<P: AsRef<std::path::Path>>(< | declares executable behavior; sig
ure/body changes alter API, control flow, and outputs. |
| 1084 |         &self, | executable/support line; behavior impact depends on surrounding block context. |
| 1085 |         path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1086 |     ) -> std::io::Result<()> { | executable/support line; behavior impact depends on surrounding block conte
|
| 1087 |         self.leak_rig.export_material_catalog_json(path) | executable/support line; behavior impact depends
surrounding block context. |
| 1088 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1089 | (blank) | line is blank; visual separation only. |
| 1090 |     pub fn export_leak_material_catalog_csv<P: AsRef<std::path::Path>>(< | declares executable behavior; sign
re/body changes alter API, control flow, and outputs. |
| 1091 |         &self, | executable/support line; behavior impact depends on surrounding block context. |
| 1092 |         path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1093 |     ) -> std::io::Result<()> { | executable/support line; behavior impact depends on surrounding block conte
|
| 1094 |         self.leak_rig.export_material_catalog_csv(path) | executable/support line; behavior impact depends o
urrounding block context. |
| 1095 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1096 | (blank) | line is blank; visual separation only. |
| 1097 |     pub fn export_leak_oil_catalog_json<P: AsRef<std::path::Path>>(< | declares executable behavior; signatur
ody changes alter API, control flow, and outputs. |
| 1098 |         &self, | executable/support line; behavior impact depends on surrounding block context. |
| 1099 |         path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1100 |     ) -> std::io::Result<()> { | executable/support line; behavior impact depends on surrounding block conte
|
| 1101 |         self.leak_rig.export_oil_catalog_json(path) | executable/support line; behavior impact depends on su
unding block context. |
| 1102 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1103 | (blank) | line is blank; visual separation only. |
| 1104 |     pub fn export_leak_oil_catalog_csv<P: AsRef<std::path::Path>>(< | declares executable behavior; signature
dy changes alter API, control flow, and outputs. |
| 1105 |         &self, | executable/support line; behavior impact depends on surrounding block context. |

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| 1106 |         path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1107 |     ) -> std::io::Result<()> { | executable/support line; behavior impact depends on surrounding block conte
|
| 1108 |         self.leak_rig.export_oil_catalog_csv(path) | executable/support line; behavior impact depends on surr
nding block context. |
| 1109 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1110 | (blank) | line is blank; visual separation only. |
| 1111 |     pub fn calibrate_leak_model_from_csv<P: AsRef<std::path::Path>>(< | declares executable behavior; signatu
body changes alter API, control flow, and outputs. |
| 1112 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 1113 |         csv_path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1114 |     ) -> std::io::Result<CalibrationReport> { | executable/support line; behavior impact depends on surround
block context. |
| 1115 |         self.leak_rig.calibrate_from_csv(csv_path) | executable/support line; behavior impact depends on surr
nding block context. |
| 1116 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1117 | (blank) | line is blank; visual separation only. |
| 1118 |     pub fn export_leak_calibration_report_json<P: AsRef<std::path::Path>>(< | declares executable behavior; s
ature/body changes alter API, control flow, and outputs. |
| 1119 |         &self, | executable/support line; behavior impact depends on surrounding block context. |
| 1120 |         path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1121 |         report: &CalibrationReport, | executable/support line; behavior impact depends on surrounding block
text. |
| 1122 |     ) -> std::io::Result<()> { | executable/support line; behavior impact depends on surrounding block conte
|
| 1123 |         self.leak_rig.export_calibration_report_json(path, report) | executable/support line; behavior impac
depends on surrounding block context. |
| 1124 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1125 | (blank) | line is blank; visual separation only. |
| 1126 |     pub fn export_leak_calibration_report_csv<P: AsRef<std::path::Path>>(< | declares executable behavior; si
ture/body changes alter API, control flow, and outputs. |
| 1127 |         &self, | executable/support line; behavior impact depends on surrounding block context. |
| 1128 |         path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1129 |         report: &CalibrationReport, | executable/support line; behavior impact depends on surrounding block
text. |
| 1130 |     ) -> std::io::Result<()> { | executable/support line; behavior impact depends on surrounding block conte
|
| 1131 |         self.leak_rig.export_calibration_report_csv(path, report) | executable/support line; behavior impact
depends on surrounding block context. |
| 1132 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1133 | (blank) | line is blank; visual separation only. |
| 1134 |     pub fn monte_carlo_leak_predictions(< | declares executable behavior; signature/body changes alter API, c
rol flow, and outputs. |
| 1135 |         &self, | executable/support line; behavior impact depends on surrounding block context. |
| 1136 |         runs: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 1137 |         horizon_s: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 1138 |         dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 1139 |         variation_pct: f64, | executable/support line; behavior impact depends on surrounding block context.
| 1140 |     ) -> Vec<ScenarioPrediction> { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1141 |         self.leak_rig | executable/support line; behavior impact depends on surrounding block context. |
| 1142 |         .run_monte_carlo(runs, horizon_s, dt, variation_pct) | executable/support line; behavior impact
ends on surrounding block context. |
| 1143 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1144 | (blank) | line is blank; visual separation only. |
| 1145 |     pub fn export_can_snapshot_json<P: AsRef<std::path::Path>>(< | declares executable behavior; signature/bo
changes alter API, control flow, and outputs. |
| 1146 |         &self, | executable/support line; behavior impact depends on surrounding block context. |
| 1147 |         path: P, | executable/support line; behavior impact depends on surrounding block context. |
| 1148 |     ) -> std::io::Result<()> { | executable/support line; behavior impact depends on surrounding block conte
|
| 1149 |         self.can_net.export_snapshot_json(path) | executable/support line; behavior impact depends on surrou
ng block context. |
| 1150 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1151 | (blank) | line is blank; visual separation only. |

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[illegible]

[illegible]

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103 | // Returns distance to ray intersection, or None if no hit | comment line; documentation intent, no runtime effect. |
| 104 | pub fn ray_intersect( | declares executable behavior; signature/body changes alter API, control flow, and outputs. | | |
| 105 |     &self, | executable/support line; behavior impact depends on surrounding block context. |
| 106 |     origin_x: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 107 |     origin_y: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 108 |     dir_x: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 109 |     dir_y: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 110 |     z: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 111 | ) -> Option<f64> { | executable/support line; behavior impact depends on surrounding block context. |
| 112 |     if z < self.z_min - 0.1 || z > self.z_max + 0.1 { | conditional gate; changing condition shifts runtime path and side effects. |
| 113 |         return None; | executable/support line; behavior impact depends on surrounding block context. |
| 114 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 115 |     // AABB intersection | comment line; documentation intent, no runtime effect. |
| 116 |     let inv_dx = if dir_x.abs() > 1e-10 { | executable/support line; behavior impact depends on surrounding block context. |
| 117 |         1.0 / dir_x | executable/support line; behavior impact depends on surrounding block context. |
| 118 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 119 |         1e10 | executable/support line; behavior impact depends on surrounding block context. |
| 120 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 121 |     let inv_dy = if dir_y.abs() > 1e-10 { | executable/support line; behavior impact depends on surrounding block context. |
| 122 |         1.0 / dir_y | executable/support line; behavior impact depends on surrounding block context. |
| 123 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 124 |         1e10 | executable/support line; behavior impact depends on surrounding block context. |
| 125 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 126 |     let t1x = ((self.x - self.half_l) - origin_x) * inv_dx; | executable/support line; behavior impact depends on surrounding block context. |
| 127 |     let t2x = ((self.x + self.half_l) - origin_x) * inv_dx; | executable/support line; behavior impact depends on surrounding block context. |
| 128 |     let t1y = ((self.y - self.half_w) - origin_y) * inv_dy; | executable/support line; behavior impact depends on surrounding block context. |
| 129 |     let t2y = ((self.y + self.half_w) - origin_y) * inv_dy; | executable/support line; behavior impact depends on surrounding block context. |
| 130 |     let tmin = t1x.min(t2x).max(t1y.min(t2y)); | executable/support line; behavior impact depends on surrounding block context. |
| 131 |     let tmax = t1x.max(t2x).min(t1y.max(t2y)); | executable/support line; behavior impact depends on surrounding block context. |
| 132 |     if tmax >= tmin && tmin > 0.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 133 |         Some(tmin) | executable/support line; behavior impact depends on surrounding block context. |
| 134 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 135 |         None | executable/support line; behavior impact depends on surrounding block context. |
| 136 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 137 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 138 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 139 | (blank) | line is blank; visual separation only. |
| 140 | // ■■ LIDAR sensor ████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 141 | pub struct LidarSensor { | declares data layout; field changes may break serialization, tests, and callers. |
| 142 |     // ■ Configuration ████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 143 |     pub channels: u8, // 16, 32, or 64 | executable/support line; behavior impact depends on surrounding block context. |
| 144 |     pub rotation_hz: f64, // 10 or 20 Hz | executable/support line; behavior impact depends on surrounding block context. |
| 145 |     pub h_resolution: f64, // horizontal resolution degrees (0.2° for VLP-16) | executable/support line; behavior impact depends on surrounding block context. |
| 146 |     pub max_range_m: f64, // maximum range (100m VLP-16, 200m Ouster) | executable/support line; behavior impact depends on surrounding block context. |

```



```
| depends on surrounding block context. |
| 147 | pub min_range_m: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 148 | pub active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 149 | (blank) | line is blank; visual separation only. |
| 150 | // ■ Output ████████████████████████████████████████████████████████████████████████████████ | comment l
tent, no runtime effect. |
| 151 | pub points: Vec<LidarPoint>, | executable/support line; behavior impact depends on surrounding block cont
. |
| 152 | pub clusters: Vec<LidarCluster>, | executable/support line; behavior impact depends on surrounding block
text. |
| 153 | pub points_per_scan: u32, | executable/support line; behavior impact depends on surrounding block context
| 154 | pub scan_time_ms: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 155 | (blank) | line is blank; visual separation only. |
| 156 | // ■ Occupancy grid (simplified) ████████████████████████████████████████████████████████████████ | comment line;
tent, no runtime effect. |
| 157 | pub occupancy_grid: Vec<Vec<u8>>, // 100×100 grid, 1m cells, centre=ego | executable/support line; behavi
impact depends on surrounding block context. |
| 158 | pub grid_size: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 159 | (blank) | line is blank; visual separation only. |
| 160 | // ■ Stats ████████████████████████████████████████████████████████████████████████████████ | comment
tent, no runtime effect. |
| 161 | pub total_scans: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 162 | pub obstacles_seen: usize, | executable/support line; behavior impact depends on surrounding block contex
|
| 163 | (blank) | line is blank; visual separation only. |
| 164 | // ■ Ground model ████████████████████████████████████████████████████████████████████████████████ | comment li
tent, no runtime effect. |
| 165 | pub ground_level: f64, // metres (accounts for hills) | executable/support line; behavior impact depends
surrounding block context. |
| 166 | (blank) | line is blank; visual separation only. |
| 167 | // ■ Internal ████████████████████████████████████████████████████████████████████████████████ | comment l
tent, no runtime effect. |
| 168 | #[allow(dead_code)] | executable/support line; behavior impact depends on surrounding block context. |
| 169 | rotation_angle: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 170 | scan_timer: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 171 | cluster_id_counter: u32, | executable/support line; behavior impact depends on surrounding block context.
| 172 | noise_t: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 173 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 174 | (blank) | line is blank; visual separation only. |
| 175 | impl LidarSensor { | implementation block; method behavior and trait conformance are defined here. |
| 176 | pub fn new_vlp16() -> Self { | declares executable behavior; signature/body changes alter API, control fl
and outputs. |
| 177 | LidarSensor { | executable/support line; behavior impact depends on surrounding block context. |
| 178 | channels: 16, | executable/support line; behavior impact depends on surrounding block context. |
| 179 | rotation_hz: 10.0, | executable/support line; behavior impact depends on surrounding block context
|
| 180 | h_resolution: 0.2, | executable/support line; behavior impact depends on surrounding block context
|
| 181 | max_range_m: 100.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 182 | min_range_m: 0.3, | executable/support line; behavior impact depends on surrounding block context
| 183 | active: true, | executable/support line; behavior impact depends on surrounding block context. |
| 184 | points: Vec::new(), | executable/support line; behavior impact depends on surrounding block conte
|
| 185 | clusters: Vec::new(), | executable/support line; behavior impact depends on surrounding block con
t. |
| 186 | points_per_scan: 0, | executable/support line; behavior impact depends on surrounding block conte
|
| 187 | scan_time_ms: 0.0, | executable/support line; behavior impact depends on surrounding block context
|
| 188 | occupancy_grid: vec![vec![@u8; 100]; 100], | executable/support line; behavior impact depends on
rounding block context. |
| 189 | grid_size: 100, | executable/support line; behavior impact depends on surrounding block context.
| 190 | total_scans: 0, | executable/support line; behavior impact depends on surrounding block context.
| 191 | obstacles_seen: 0, | executable/support line; behavior impact depends on surrounding block context
```

```

|
| 192 |         ground_level: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
|
| 193 |         rotation_angle: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
|
| 194 |         scan_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 195 |         cluster_id_counter: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 196 |         noise_t: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 197 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
|
| 198 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 199 | (blank) | line is blank; visual separation only. |
| 200 |     /// Update LIDAR – generates point cloud from world obstacles. | comment line; documentation intent, no runtime effect. |
| 201 |     pub fn update(&mut self, world: &[WorldObstacle], vehicle_pitch: f64, dt: f64) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 202 |         self.noise_t += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 203 |         self.scan_timer += dt; | executable/support line; behavior impact depends on surrounding block context. |
|
| 204 |         if self.scan_timer < 1.0 / self.rotation_hz { | conditional gate; changing condition shifts runtime path and side effects. |
| 205 |             return; | executable/support line; behavior impact depends on surrounding block context. |
| 206 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
|
| 207 |         self.scan_timer = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
|
| 208 | (blank) | line is blank; visual separation only. | | |
| 209 |         let t0 = std::time::Instant::now(); | executable/support line; behavior impact depends on surrounding block context. |
| 210 |         self.points.clear(); | executable/support line; behavior impact depends on surrounding block context. |
| 211 |         self.clear_grid(); | executable/support line; behavior impact depends on surrounding block context. |
| 212 | (blank) | line is blank; visual separation only. |
| 213 |         // VLP-16 vertical angles: -15° to +15° in 2° steps | comment line; documentation intent, no runtime effect. |
| 214 |         let vert_angles: Vec<f64> = (0..self.channels as i32) | executable/support line; behavior impact depends on surrounding block context. |
| 215 |             .map(|i| -15.0 + i as f64 * 2.0) // degrees | executable/support line; behavior impact depends on surrounding block context. |
| 216 |             .collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 217 | (blank) | line is blank; visual separation only. |
| 218 |         let n = \|s: f64\| ((s * 127.1 + 311.7).sin() * 43758.5 % 1.0 - 0.5) * 2.0; | executable/support line; behavior impact depends on surrounding block context. |
| 219 |         let nt = self.noise_t; | executable/support line; behavior impact depends on surrounding block context. |
|
| 220 | (blank) | line is blank; visual separation only. |
| 221 |         // Full 360° scan | comment line; documentation intent, no runtime effect. |
| 222 |         let mut az = 0.0_f64; | executable/support line; behavior impact depends on surrounding block context. |
| 223 |         while az < 360.0 { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 224 |             let az_rad = az * PI / 180.0; | executable/support line; behavior impact depends on surrounding block context. |
| 225 |             let dir_x = az_rad.cos(); | executable/support line; behavior impact depends on surrounding block context. |
| 226 |             let dir_y = az_rad.sin(); | executable/support line; behavior impact depends on surrounding block context. |
| 227 | (blank) | line is blank; visual separation only. |
| 228 |             for (ring, vert_deg) in vert_angles.iter().enumerate() { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 229 |                 let vert_rad = (vert_deg + vehicle_pitch) * PI / 180.0; | executable/support line; behavior impact depends on surrounding block context. |
| 230 |                 let h_factor = vert_rad.cos(); | executable/support line; behavior impact depends on surrounding block context. |
| 231 |                 let z_per_m = vert_rad.sin(); | executable/support line; behavior impact depends on surrounding block context. |

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| 232 | (blank) | line is blank; visual separation only. |
| 233 |           // Find closest intersection | comment line; documentation intent, no runtime effect. |
| 234 |           let mut min_dist = self.max_range_m; | executable/support line; behavior impact depends on su
unding block context. |
| 235 |           let mut hit_intensity = 0u8; | executable/support line; behavior impact depends on surroundin
lock context. |
| 236 | (blank) | line is blank; visual separation only. |
| 237 |           // Ground return | comment line; documentation intent, no runtime effect. |
| 238 |           if *vert_deg < 0.0 { | conditional gate; changing condition shifts runtime path and side effe
. |
| 239 |           let ground_dist = (self.ground_level - 1.8) / (-vert_rad.sin()).max(1e-6); | executable/s
ort line; behavior impact depends on surrounding block context. |
| 240 |           if ground_dist < min_dist && ground_dist > self.min_range_m { | conditional gate; changin
ondition shifts runtime path and side effects. |
| 241 |           min_dist = ground_dist; | executable/support line; behavior impact depends on surround
g block context. |
| 242 |           hit_intensity = 30; | executable/support line; behavior impact depends on surrounding
ock context. |
| 243 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 244 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 245 | (blank) | line is blank; visual separation only. |
| 246 |           // Obstacle returns | comment line; documentation intent, no runtime effect. |
| 247 |           for obs in world { | iteration control; changing bounds/body affects throughput, timing, and
sibly safety guarantees. |
| 248 |           let z_at_range = \|r: f64\| self.ground_level + z_per_m * r + 1.8; // sensor height 1.8m
xecutable/support line; behavior impact depends on surrounding block context. |
| 249 |                                           // Check at expected obs
le position | comment line; documentation intent, no runtime effect. |
| 250 |           if let Some(dist) = obs.ray_intersect( | conditional gate; changing condition shifts runt
path and side effects. |
| 251 |           0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 252 |           0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 253 |           dir_x * h_factor, | executable/support line; behavior impact depends on surrounding b
k context. |
| 254 |           dir_y * h_factor, | executable/support line; behavior impact depends on surrounding b
k context. |
| 255 |           z_at_range(obs.x.hypot(obs.y)), | executable/support line; behavior impact depends on
rounding block context. |
| 256 |           ) { | executable/support line; behavior impact depends on surrounding block context. |
| 257 |           if dist < min_dist { | conditional gate; changing condition shifts runtime path and s
effects. |
| 258 |           min_dist = dist; | executable/support line; behavior impact depends on surrounding
lock context. |
| 259 |           let snr = obs.reflectivity * (1.0 - dist / self.max_range_m); | executable/support
ine; behavior impact depends on surrounding block context. |
| 260 |           hit_intensity = (snr * 200.0 + 20.0).clamp(0.0, 255.0) as u8; | executable/support
ine; behavior impact depends on surrounding block context. |
| 261 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership an
ifetime scopes. |
| 262 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 263 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 264 | (blank) | line is blank; visual separation only. |
| 265 |           if min_dist < self.max_range_m && min_dist > self.min_range_m { | conditional gate; changing
dition shifts runtime path and side effects. |
| 266 |           let noise_m = n(nt + az * 0.01 + ring as f64 * 0.1) * 0.02; // ±2cm noise | executable/su
rt line; behavior impact depends on surrounding block context. |
| 267 |           let d = (min_dist + noise_m) as f32; | executable/support line; behavior impact depends o
urrounding block context. |
| 268 |           let px = d * dir_x as f32 * vert_rad.cos() as f32; | executable/support line; behavior im
t depends on surrounding block context. |
| 269 |           let py = d * dir_y as f32 * vert_rad.cos() as f32; | executable/support line; behavior im

```

```

t depends on surrounding block context. |
| 270 |         let pz = (self.ground_level + z_per_m * min_dist) as f32; | executable/support line; beha
r impact depends on surrounding block context. |
| 271 | (blank) | line is blank; visual separation only. |
| 272 |         self.points.push(LidarPoint { | executable/support line; behavior impact depends on surro
ing block context. |
| 273 |             x: px, | executable/support line; behavior impact depends on surrounding block context
|
| 274 |             y: py, | executable/support line; behavior impact depends on surrounding block context
|
| 275 |             z: pz, | executable/support line; behavior impact depends on surrounding block context
|
| 276 |             intensity: hit_intensity, | executable/support line; behavior impact depends on surro
ing block context. |
| 277 |             ring: ring as u8, | executable/support line; behavior impact depends on surrounding b
k context. |
| 278 |             azimuth: az as f32, | executable/support line; behavior impact depends on surrounding
ock context. |
| 279 |             distance: d, | executable/support line; behavior impact depends on surrounding block
text. |
| 280 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 281 | (blank) | line is blank; visual separation only. |
| 282 |         // Update occupancy grid | comment line; documentation intent, no runtime effect. |
| 283 |         let gx = (50.0 + px as f64) as usize; | executable/support line; behavior impact depends
surrounding block context. |
| 284 |         let gy = (50.0 + py as f64) as usize; | executable/support line; behavior impact depends
surrounding block context. |
| 285 |         if gx < self.grid_size && gy < self.grid_size && pz > 0.2 { | conditional gate; changing
dition shifts runtime path and side effects. |
| 286 |             self.occupancy_grid[gy][gx] = | executable/support line; behavior impact depends on s
ounding block context. |
| 287 |             self.occupancy_grid[gy][gx].saturating_add(10); | executable/support line; behavi
impact depends on surrounding block context. |
| 288 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 289 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 290 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 291 |         az += self.h_resolution; | executable/support line; behavior impact depends on surrounding block
text. |
| 292 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 293 | (blank) | line is blank; visual separation only. |
| 294 |     self.points_per_scan = self.points.len() as u32; | executable/support line; behavior impact depends o
urrounding block context. |
| 295 |     self.scan_time_ms = t0.elapsed().as_secs_f64() * 1000.0; | executable/support line; behavior impact d
nds on surrounding block context. |
| 296 |     self.total_scans += 1; | executable/support line; behavior impact depends on surrounding block context
|
| 297 | (blank) | line is blank; visual separation only. |
| 298 |     // Simple clustering: group nearby occupied grid cells | comment line; documentation intent, no runti
effect. |
| 299 |     self.cluster_obstacles(world); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 300 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 301 | (blank) | line is blank; visual separation only. |
| 302 |     fn cluster_obstacles(&mut self, world: &[WorldObstacle]) { | declares executable behavior; signature/body
anges alter API, control flow, and outputs. |
| 303 |         self.clusters.clear(); | executable/support line; behavior impact depends on surrounding block context
|
| 304 |         for obs in world.iter() { | iteration control; changing bounds/body affects throughput, timing, and p
ibly safety guarantees. |
| 305 |             let dist = (obs.x.powi(2) + obs.y.powi(2)).sqrt(); | executable/support line; behavior impact dep
s on surrounding block context. |

```

```

| 306 |         if dist > self.max_range_m { | conditional gate; changing condition shifts runtime path and side
ects. |
| 307 |             continue; | executable/support line; behavior impact depends on surrounding block context. |
| 308 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 309 |             let obj_type = if obs.z_max > 1.2 && obs.half_w > 0.5 { | executable/support line; behavior impac
depends on surrounding block context. |
| 310 |                 LidarObjectType::Vehicle | executable/support line; behavior impact depends on surrounding bl
context. |
| 311 |             } else if obs.z_max > 1.5 { | executable/support line; behavior impact depends on surrounding blo
context. |
| 312 |                 LidarObjectType::Pedestrian | executable/support line; behavior impact depends on surrounding
ock context. |
| 313 |             } else if obs.z_max < 0.9 { | executable/support line; behavior impact depends on surrounding blo
context. |
| 314 |                 LidarObjectType::Barrier | executable/support line; behavior impact depends on surrounding bl
context. |
| 315 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 316 |                 LidarObjectType::Unknown | executable/support line; behavior impact depends on surrounding bl
context. |
| 317 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 318 |         let pts = self | executable/support line; behavior impact depends on surrounding block context. |
| 319 |             .points | executable/support line; behavior impact depends on surrounding block context. |
| 320 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 321 |             .filter(\|p\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 322 |                 ((p.x as f64 - obs.x).powi(2) + (p.y as f64 - obs.y).powi(2)).sqrt() < 3.0 | executable/s
ort line; behavior impact depends on surrounding block context. |
| 323 |             }) | executable/support line; behavior impact depends on surrounding block context. |
| 324 |             .count(); | executable/support line; behavior impact depends on surrounding block context. |
| 325 |         if pts > 0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 326 |             self.clusters.push(LidarCluster { | executable/support line; behavior impact depends on surrou
ing block context. |
| 327 |                 id: self.cluster_id_counter, | executable/support line; behavior impact depends on surrou
ng block context. |
| 328 |                 centroid_x: obs.x, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 329 |                 centroid_y: obs.y, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 330 |                 centroid_z: obs.z_max / 2.0, | executable/support line; behavior impact depends on surrou
ng block context. |
| 331 |                 length_m: obs.half_l * 2.0, | executable/support line; behavior impact depends on surroun
g block context. |
| 332 |                 width_m: obs.half_w * 2.0, | executable/support line; behavior impact depends on surroun
block context. |
| 333 |                 height_m: obs.z_max, | executable/support line; behavior impact depends on surrounding bl
context. |
| 334 |                 distance_m: dist, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 335 |                 object_type: obj_type, | executable/support line; behavior impact depends on surrounding
ck context. |
| 336 |                 point_count: pts, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 337 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 338 |             self.cluster_id_counter = self.cluster_id_counter.wrapping_add(1); | executable/support line;
havior impact depends on surrounding block context. |
| 339 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 340 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 341 |     self.obstacles_seen = self.clusters.len(); | executable/support line; behavior impact depends on surr
ding block context. |
| 342 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 343 | (blank) | line is blank; visual separation only. |

```

```

| 344 |     fn clear_grid(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 345 |         for row in &mut self.occupancy_grid { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 346 |             for cell in row.iter_mut() { | iteration control; changing bounds/body affects throughput, timing and possibly safety guarantees. |
| 347 |                 *cell = (*cell).saturating_sub(5); | executable/support line; behavior impact depends on surrounding block context. |
| 348 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 349 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 350 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 351 | (blank) | line is blank; visual separation only. |
| 352 |     /// Get occupancy as percentage for a given grid cell | comment line; documentation intent, no runtime effect. |
| 353 |     pub fn cell_occupied(&self, row: usize, col: usize) -> bool { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 354 |         row < self.grid_size && col < self.grid_size && self.occupancy_grid[row][col] > 50 | executable/support line; behavior impact depends on surrounding block context. |
| 355 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 356 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/main.rs

Role: Desktop GUI runtime orchestration and presentation layer.

Lines: 9466 | Non-empty: 9212 | Symbols: 113

Symbol Inventory

Line	Kind	Symbol	Change Impact
8	module	io	data/structure coupling
9	module	ecm_mock_provider	data/structure coupling
10	module	widgets	data/structure coupling
37	const	DT	data/structure coupling
38	const	PLOT_WINDOW	data/structure coupling
39	const	EVENT_MAX	data/structure coupling
46	enum	Cmd	data/structure coupling
145	enum	Tab	data/structure coupling
167	struct	LeakManualUi	data/structure coupling
181	impl	Default for LeakManualUi	behavior binding and method semantics
182	fn	default	signature/API coupling
200	struct	LeakCustomUi	data/structure coupling
226	impl	Default for LeakCustomUi	behavior binding and method semantics
227	fn	default	signature/API coupling
257	enum	CanMode	data/structure coupling
264	enum	EventLevel	data/structure coupling
273	struct	AppEvent	data/structure coupling
280	struct	Signal	data/structure coupling
294	struct	App	data/structure coupling
426	impl	App	behavior binding and method semantics
427	fn	new	signature/API coupling
542	fn	pick_save_path	signature/API coupling
553	fn	pick_open_path	signature/API coupling
560	fn	refresh_mock_ecm_live_from_sim	signature/API coupling
572	fn	ecm_treeview	signature/API coupling
603	fn	oil_type_from_idx	signature/API coupling
607	fn	component_from_idx	signature/API coupling
615	fn	material_from_idx	signature/API coupling
622	fn	idx_from_oil_type	signature/API coupling
626	fn	sync_leak_manual_from_selected	signature/API coupling
642	fn	sanitize_leak_manual	signature/API coupling
667	fn	sanitize_leak_custom	signature/API coupling
690	fn	validate_leak_manual	signature/API coupling

704	fn	validate_leak_custom	signature/API coupling
721	fn	build_manual_leak_params	signature/API coupling
738	fn	apply_manual_to_selected_circuit	signature/API coupling
754	fn	can_bus_from_idx	signature/API coupling
764	fn	parse_u32_filter	signature/API coupling
773	fn	can_filter_match_signal	signature/API coupling
799	fn	can_filter_match_trace	signature/API coupling
819	fn	ema	signature/API coupling
823	fn	update_v2x_kpis	signature/API coupling
829	fn	approach_f32	signature/API coupling
839	fn	shape_pedal	signature/API coupling
846	fn	flush_cmds	signature/API coupling
1297	fn	sync_params_from_bench	signature/API coupling
1309	fn	update_signals	signature/API coupling
1377	fn	detect_events	signature/API coupling
1561	fn	push_plots	signature/API coupling
1581	fn	auto_sequence	signature/API coupling
1591	fn	now_ms	signature/API coupling
1601	fn	main	signature/API coupling
1651	impl	eFrame::App	for App behavior binding and method semantics
1652	fn	update	signature/API coupling
1769	impl	App	behavior binding and method semantics
1770	fn	tab_bar	signature/API coupling
1847	fn	ad_readiness_issues	signature/API coupling
1867	fn	ux_alerts	signature/API coupling
1935	fn	ux_context_items	signature/API coupling
2204	fn	render_ux_context_strip	signature/API coupling
2222	fn	statusbar	signature/API coupling
2283	fn	render_ux_guide	signature/API coupling
2413	fn	render_ux_quick_actions	signature/API coupling
2560	fn	electrical_wire_mm2	signature/API coupling
2580	fn	electrical_cap_nf_per_m	signature/API coupling
2594	fn	electrical_node_online	signature/API coupling
2601	fn	tab_ecm_live_data	signature/API coupling
2890	fn	toolbar	signature/API coupling
3116	impl	App	behavior binding and method semantics
3117	fn	tab_help	signature/API coupling
3199	fn	tab_cluster	signature/API coupling
3673	impl	App	behavior binding and method semantics
3674	fn	tab_can	signature/API coupling
3794	fn	can_network_overview	signature/API coupling
3819	const	BUS_NAMES	data/structure coupling
3995	fn	can_signals	signature/API coupling
4114	fn	can_trace	signature/API coupling
4284	impl	App	behavior binding and method semantics
4285	fn	tab_events	signature/API coupling
4437	impl	App	behavior binding and method semantics
4438	fn	tab_ecu_net	signature/API coupling
4621	impl	App	behavior binding and method semantics
4622	fn	tab_engine	signature/API coupling
4974	impl	App	behavior binding and method semantics
4975	fn	dtc_fault_storage_addr	signature/API coupling
4980	fn	dtc_dm1_payload_hex	signature/API coupling
4997	fn	fault_selection_hint	signature/API coupling
5018	fn	tab_faults	signature/API coupling
5327	impl	App	behavior binding and method semantics
5328	fn	tab_boot	signature/API coupling
5460	impl	App	behavior binding and method semantics
5461	fn	tab_implements	signature/API coupling
5918	impl	App	behavior binding and method semantics
5919	fn	tab_params	signature/API coupling
6235	impl	App	behavior binding and method semantics
6236	fn	uds_process_sa	signature/API coupling
6246	fn	uds_send_and_log	signature/API coupling
6278	fn	run_uds_flash_pipeline	signature/API coupling

```

| 6360 | fn | leak_ascii_cad | signature/API coupling |
| 6364 | const | SECTORS | data/structure coupling |
| 6468 | fn | tab_leak_lab | signature/API coupling |
| 7022 | const | COMP_NAMES | data/structure coupling |
| 7327 | fn | tab_electrical | signature/API coupling |
| 7676 | impl | App | behavior binding and method semantics |
| 7677 | fn | tab_sensors | signature/API coupling |
| 8067 | impl | App | behavior binding and method semantics |
| 8068 | fn | tab_autonomous | signature/API coupling |
| 8623 | impl | App | behavior binding and method semantics |
| 8624 | fn | tab_v2x | signature/API coupling |
| 9039 | impl | App | behavior binding and method semantics |
| 9040 | fn | tab_uds | signature/API coupling |
| 9357 | impl | App | behavior binding and method semantics |
| 9358 | fn | tab_plots | signature/API coupling |

```

Line-by-Line Impact Map

```

| Line | Source | Operational Impact If Changed |
|---:|---|---|
| 1 | //! Heavy Machinery ECU Bench – GUI v3.0 | comment line; documentation intent, no runtime effect. |
| 2 | //! Command-queue architecture: all UI→simulation mutations go through Cmd enum, | comment line; documentation
ent, no runtime effect. |
| 3 | //! executed AFTER the UI frame, eliminating all borrow-checker closure conflicts. | comment line; documentatio
ntent, no runtime effect. |
| 4 | (blank) | line is blank; visual separation only. |
| 5 | #![cfg_attr(not(debug_assertions), windows_subsystem = "windows")] | executable/support line; behavior impact d
nds on surrounding block context. |
| 6 | #![allow(float_literal_f32_fallback)] | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7 | (blank) | line is blank; visual separation only. |
| 8 | pub mod io; | declares a module boundary; changing module name/path breaks references. |
| 9 | mod ecm_mock_provider; | declares a module boundary; changing module name/path breaks references. |
| 10 | mod widgets; | declares a module boundary; changing module name/path breaks references. |
| 11 | (blank) | line is blank; visual separation only. |
| 12 | use auto_breaking::{ | imports symbols; changing path/name affects compile-time resolution. |
| 13 |     autonomous::FeatureState, | executable/support line; behavior impact depends on surrounding block context. |
| 14 |     boot_sequence::EcuBootStage, | executable/support line; behavior impact depends on surrounding block conte
|
| 15 |     can_gateway::BusState, | executable/support line; behavior impact depends on surrounding block context. |
| 16 |     CalibrationReport, | executable/support line; behavior impact depends on surrounding block context. |
| 17 |     ElectricalControlPoint, ElectricalFaultMode, | executable/support line; behavior impact depends on surround
g block context. |
| 18 |     ecu_abs::EspCondition, | executable/support line; behavior impact depends on surrounding block context. |
| 19 |     ecu_ecm::AftertreatmentState, | executable/support line; behavior impact depends on surrounding block cont
. |
| 20 |     ecu_tcm::{AutoShiftMode, ClutchState, Direction}, | executable/support line; behavior impact depends on su
unding block context. |
| 21 |     implement::PtoMode, | executable/support line; behavior impact depends on surrounding block context. |
| 22 |     j1939::DtcSeverity, | executable/support line; behavior impact depends on surrounding block context. |
| 23 |     v2x_telematics::ConnectState, | executable/support line; behavior impact depends on surrounding block cont
. |
| 24 |     CircuitComponent, HeavyMachinery, IgnitionState, LeakCircuit, ManualCircuitParams, OilType, | executable/s
ort line; behavior impact depends on surrounding block context. |
| 25 |     OringMaterial, OringSpec, PressureEnvelope, ScenarioPrediction, FAULT_TYPES, | executable/support line; be
ior impact depends on surrounding block context. |
| 26 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 27 | use chrono::Local; | imports symbols; changing path/name affects compile-time resolution. |
| 28 | use eframe::{egui, NativeOptions}; | imports symbols; changing path/name affects compile-time resolution. |
| 29 | use egui::*; | imports symbols; changing path/name affects compile-time resolution. |
| 30 | use egui_plot::{Line, Plot, PlotPoints}; | imports symbols; changing path/name affects compile-time resolution
| 31 | use rfd::FileDialog; | imports symbols; changing path/name affects compile-time resolution. |
| 32 | use std::collections::{BTreeMap, HashMap, VecDeque}; | imports symbols; changing path/name affects compile-tim
esolution. |
| 33 | use std::path::PathBuf; | imports symbols; changing path/name affects compile-time resolution. |

```



```
| 34 | use std::time::Duration; | imports symbols; changing path/name affects compile-time resolution. |  
| 35 | use widgets::{arc_gauge, bar_gauge, digital_readout, direction_selector, warning_lamp}; | imports symbols; cha  
ng path/name affects compile-time resolution. |  
| 36 | (blank) | line is blank; visual separation only. |  
| 37 | const DT: f64 = 1.0 / 60.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 38 | const PLOT_WINDOW: usize = 600; | executable/support line; behavior impact depends on surrounding block contex  
|  
| 39 | const EVENT_MAX: usize = 500; | executable/support line; behavior impact depends on surrounding block context.  
| 40 | type TraceRow = (f64, u32, u8, u8, String, String, String); | executable/support line; behavior impact depends  
surrounding block context. |  
| 41 | (blank) | line is blank; visual separation only. |  
| 42 | // ████████████████████████████████████████████████████████████████████████████████ | comment  
tent, no runtime effect. |  
| 43 | // Command queue – ALL mutations to bench go through here, no closures needed | comment line; documentation in  
t, no runtime effect. |  
| 44 | // ████████████████████████████████████████████████████████████████████████████████ | commen  
tent, no runtime effect. |  
| 45 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |  
| 46 | enum Cmd { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |  
| 47 |     // Ignition | comment line; documentation intent, no runtime effect. |  
| 48 |     KeyAdvance, | executable/support line; behavior impact depends on surrounding block context. |  
| 49 |     KeyOff, | executable/support line; behavior impact depends on surrounding block context. |  
| 50 |     // Throttle/brake (raw value 0-100) | comment line; documentation intent, no runtime effect. |  
| 51 |     SetThrottle(f32), | executable/support line; behavior impact depends on surrounding block context. |  
| 52 |     SetBrake(f32), | executable/support line; behavior impact depends on surrounding block context. |  
| 53 |     // Direction | comment line; documentation intent, no runtime effect. |  
| 54 |     SetDirection(Direction), | executable/support line; behavior impact depends on surrounding block context.  
| 55 |     SetNeutral, | executable/support line; behavior impact depends on surrounding block context. |  
| 56 |     // Gearbox | comment line; documentation intent, no runtime effect. |  
| 57 |     ToggleAutoShift, | executable/support line; behavior impact depends on surrounding block context. |  
| 58 |     ToggleCreeper, | executable/support line; behavior impact depends on surrounding block context. |  
| 59 |     ManualUp, | executable/support line; behavior impact depends on surrounding block context. |  
| 60 |     ManualDn, | executable/support line; behavior impact depends on surrounding block context. |  
| 61 |     // ECM parameter overrides | comment line; documentation intent, no runtime effect. |  
| 62 |     SetFuelLevel(f64), | executable/support line; behavior impact depends on surrounding block context. |  
| 63 |     SetDefLevel(f64), | executable/support line; behavior impact depends on surrounding block context. |  
| 64 |     SetCoolantTemp(f64), | executable/support line; behavior impact depends on surrounding block context. |  
| 65 |     SetOilPressure(f64), | executable/support line; behavior impact depends on surrounding block context. |  
| 66 |     SetBoostPressure(f64), | executable/support line; behavior impact depends on surrounding block context. |  
| 67 |     SetEngineHours(f64), | executable/support line; behavior impact depends on surrounding block context. |  
| 68 |     ClearDtcs, | executable/support line; behavior impact depends on surrounding block context. |  
| 69 |     // BCM | comment line; documentation intent, no runtime effect. |  
| 70 |     ToggleWorkLights, | executable/support line; behavior impact depends on surrounding block context. |  
| 71 |     ToggleRoadLights, | executable/support line; behavior impact depends on surrounding block context. |  
| 72 |     ToggleBeacon, | executable/support line; behavior impact depends on surrounding block context. |  
| 73 |     Honk, | executable/support line; behavior impact depends on surrounding block context. |  
| 74 |     CycleWiper, | executable/support line; behavior impact depends on surrounding block context. |  
| 75 |     ToggleHvac, | executable/support line; behavior impact depends on surrounding block context. |  
| 76 |     ResetBcmFuses, | executable/support line; behavior impact depends on surrounding block context. |  
| 77 |     ElectricalSetBranchFault(String, ElectricalFaultMode), | executable/support line; behavior impact depends  
surrounding block context. |  
| 78 |     ElectricalSetControlFault(ElectricalControlPoint, ElectricalFaultMode), | executable/support line; behavio  
mpact depends on surrounding block context. |  
| 79 |     ElectricalResetFaults, | executable/support line; behavior impact depends on surrounding block context. |  
| 80 |     // Implements | comment line; documentation intent, no runtime effect. |  
| 81 |     SetPtoMode(PtoMode), | executable/support line; behavior impact depends on surrounding block context. |  
| 82 |     SetHitchTarget(f64), | executable/support line; behavior impact depends on surrounding block context. |  
| 83 |     SetAuxValve(usize, f64), | executable/support line; behavior impact depends on surrounding block context.  
| 84 |     SetLoaderLift(f64), | executable/support line; behavior impact depends on surrounding block context. |  
| 85 |     SetLoaderTilt(f64), | executable/support line; behavior impact depends on surrounding block context. |  
| 86 |     // Hydraulics | comment line; documentation intent, no runtime effect. |  
| 87 |     SetHitchJoystick(f64), | executable/support line; behavior impact depends on surrounding block context. |  
| 88 |     // Autonomous | comment line; documentation intent, no runtime effect. |  
| 89 |     EngageAD, | executable/support line; behavior impact depends on surrounding block context. |  
| 90 |     DisengageAD, | executable/support line; behavior impact depends on surrounding block context. |
```

[illegible]

```
| intent, no runtime effect. |  
| 142 | // Tab enum | comment line; documentation intent, no runtime effect. |  
| 143 | // ████████████████████████████████████████████████████████████████████████ | commen  
ntent, no runtime effect. |  
| 144 | #[derive(PartialEq, Clone, Copy, Debug)] | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 145 | enum Tab { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |  
| 146 |     Help, | executable/support line; behavior impact depends on surrounding block context. |  
| 147 |     Cluster, | executable/support line; behavior impact depends on surrounding block context. |  
| 148 |     Electrical, | executable/support line; behavior impact depends on surrounding block context. |  
| 149 |     CanBus, | executable/support line; behavior impact depends on surrounding block context. |  
| 150 |     Events, | executable/support line; behavior impact depends on surrounding block context. |  
| 151 |     EcuNet, | executable/support line; behavior impact depends on surrounding block context. |  
| 152 |     Engine, | executable/support line; behavior impact depends on surrounding block context. |  
| 153 |     Faults, | executable/support line; behavior impact depends on surrounding block context. |  
| 154 |     Boot, | executable/support line; behavior impact depends on surrounding block context. |  
| 155 |     Implements, | executable/support line; behavior impact depends on surrounding block context. |  
| 156 |     Params, | executable/support line; behavior impact depends on surrounding block context. |  
| 157 |     Sensors, | executable/support line; behavior impact depends on surrounding block context. |  
| 158 |     Autonomous, | executable/support line; behavior impact depends on surrounding block context. |  
| 159 |     V2X, | executable/support line; behavior impact depends on surrounding block context. |  
| 160 |     Uds, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |  
| 161 |     EcmliveData, | executable/support line; behavior impact depends on surrounding block context. |  
| 162 |     LeakLab, | executable/support line; behavior impact depends on surrounding block context. |  
| 163 |     Plots, | executable/support line; behavior impact depends on surrounding block context. |  
| 164 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 165 | (blank) | line is blank; visual separation only. |  
| 166 | #[derive(Clone)] | executable/support line; behavior impact depends on surrounding block context. |  
| 167 | struct LeakManualUi { | declares data layout; field changes may break serialization, tests, and callers. |  
| 168 |     oil_type_idx: usize, | executable/support line; behavior impact depends on surrounding block context. |  
| 169 |     piston_pressure_bar: f64, | executable/support line; behavior impact depends on surrounding block context.  
| 170 |     operation_pressure_bar: f64, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 171 |     pressure_min_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 172 |     pressure_mean_bar: f64, | executable/support line; behavior impact depends on surrounding block context.  
| 173 |     pressure_ideal_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 174 |     pressure_max_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 175 |     pressure_rupture_bar: f64, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 176 |     squeeze_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 177 |     compression_set_pct: f64, | executable/support line; behavior impact depends on surrounding block context.  
| 178 |     base_leak_area_mm2: f64, | executable/support line; behavior impact depends on surrounding block context.  
| 179 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 180 | (blank) | line is blank; visual separation only. |  
| 181 | impl Default for LeakManualUi { | implementation block; method behavior and trait conformance are defined her  
|  
| 182 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, an  
utputs. |  
| 183 |         Self { | executable/support line; behavior impact depends on surrounding block context. |  
| 184 |             oil_type_idx: 0, | executable/support line; behavior impact depends on surrounding block context.  
| 185 |             piston_pressure_bar: 100.0, | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 186 |             operation_pressure_bar: 80.0, | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 187 |             pressure_min_bar: 20.0, | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 188 |             pressure_mean_bar: 60.0, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 189 |             pressure_ideal_bar: 80.0, | executable/support line; behavior impact depends on surrounding block  
ncontext. |  
| 190 |             pressure_max_bar: 120.0, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 191 |             pressure_rupture_bar: 160.0, | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 192 |             squeeze_pct: 18.0, | executable/support line; behavior impact depends on surrounding block conte
```

```

|
| 193 |           compression_set_pct: 12.0, | executable/support line; behavior impact depends on surrounding block
context. |
| 194 |           base_leak_area_mm2: 0.0005, | executable/support line; behavior impact depends on surrounding block
context. |
| 195 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
|
| 196 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 197 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 198 | (blank) | line is blank; visual separation only. |
| 199 | #[derive(Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 200 | struct LeakCustomUi { | declares data layout; field changes may break serialization, tests, and callers. |
| 201 |     name: String, | executable/support line; behavior impact depends on surrounding block context. |
| 202 |     application: String, | executable/support line; behavior impact depends on surrounding block context. |
| 203 |     component_idx: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 204 |     oil_type_idx: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 205 |     seal_count: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 206 |     piston_pressure_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 207 |     operation_pressure_bar: f64, | executable/support line; behavior impact depends on surrounding block context.
. |
| 208 |     min_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 209 |     mean_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 210 |     ideal_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 211 |     max_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 212 |     rupture_bar: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 213 |     material_idx: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 214 |     shore_a: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 215 |     cross_section_mm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 216 |     squeeze_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 217 |     extrusion_gap_mm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 218 |     compression_set_pct: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 219 |     design_life_hours: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 220 |     base_leak_area_mm2: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 221 |     discharge_coeff: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 222 |     reservoir_volume_l: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 223 |     support_lpm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 224 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 225 | (blank) | line is blank; visual separation only. |
| 226 | impl Default for LeakCustomUi { | implementation block; method behavior and trait conformance are defined here
|
| 227 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
outputs. |
| 228 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 229 |             name: "CUSTOM_01".into(), | executable/support line; behavior impact depends on surrounding block
context. |
| 230 |             application: "Circuito custom de producao".into(), | executable/support line; behavior impact depends
on surrounding block context. |
| 231 |             component_idx: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 232 |             oil_type_idx: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 233 |             seal_count: 4, | executable/support line; behavior impact depends on surrounding block context. |
| 234 |             piston_pressure_bar: 120.0, | executable/support line; behavior impact depends on surrounding block
context. |
| 235 |             operation_pressure_bar: 95.0, | executable/support line; behavior impact depends on surrounding block
context. |
| 236 |             min_bar: 20.0, | executable/support line; behavior impact depends on surrounding block context. |
| 237 |             mean_bar: 70.0, | executable/support line; behavior impact depends on surrounding block context. |
| 238 |             ideal_bar: 95.0, | executable/support line; behavior impact depends on surrounding block context. |
| 239 |             max_bar: 130.0, | executable/support line; behavior impact depends on surrounding block context. |
| 240 |             rupture_bar: 180.0, | executable/support line; behavior impact depends on surrounding block context.
|
| 241 |             material_idx: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 242 |             shore_a: 80.0, | executable/support line; behavior impact depends on surrounding block context. |
| 243 |             cross_section_mm: 3.53, | executable/support line; behavior impact depends on surrounding block context.
|
| 244 |             squeeze_pct: 18.0, | executable/support line; behavior impact depends on surrounding block context.

```

```
| 245 |           extrusion_gap_mm: 0.12, | executable/support line; behavior impact depends on surrounding block context. |
| 246 |           compression_set_pct: 10.0, | executable/support line; behavior impact depends on surrounding block context. |
| 247 |           design_life_hours: 8000.0, | executable/support line; behavior impact depends on surrounding block context. |
| 248 |           base_leak_area_mm2: 0.0005, | executable/support line; behavior impact depends on surrounding block context. |
| 249 |           discharge_coeff: 0.58, | executable/support line; behavior impact depends on surrounding block context. |
| 250 |           reservoir_volume_l: 20.0, | executable/support line; behavior impact depends on surrounding block context. |
| 251 |           support_lpm: 40.0, | executable/support line; behavior impact depends on surrounding block context. |
| 252 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 253 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 254 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 255 | (blank) | line is blank; visual separation only. |
| 256 | #[derive(PartialEq, Clone, Copy)] | executable/support line; behavior impact depends on surrounding block context. |
| 257 | enum CanMode { | declares state variants; adding/removing variants impacts pattern matches and logic branches |
| 258 |     Signals, | executable/support line; behavior impact depends on surrounding block context. |
| 259 |     Trace, | executable/support line; behavior impact depends on surrounding block context. |
| 260 |     Network, | executable/support line; behavior impact depends on surrounding block context. |
| 261 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 262 | (blank) | line is blank; visual separation only. |
| 263 | #[derive(Clone, Copy, PartialEq, Debug)] | executable/support line; behavior impact depends on surrounding block context. |
| 264 | pub enum EventLevel { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 265 |     Debug, | executable/support line; behavior impact depends on surrounding block context. |
| 266 |     Info, | executable/support line; behavior impact depends on surrounding block context. |
| 267 |     Ok, | executable/support line; behavior impact depends on surrounding block context. |
| 268 |     Warn, | executable/support line; behavior impact depends on surrounding block context. |
| 269 |     Critical, | executable/support line; behavior impact depends on surrounding block context. |
| 270 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 271 | (blank) | line is blank; visual separation only. |
| 272 | #[derive(Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 273 | struct AppEvent { | declares data layout; field changes may break serialization, tests, and callers. |
| 274 |     ts: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 275 |     source: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 276 |     msg: String, | executable/support line; behavior impact depends on surrounding block context. |
| 277 |     lvl: EventLevel, | executable/support line; behavior impact depends on surrounding block context. |
| 278 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 279 | (blank) | line is blank; visual separation only. |
| 280 | struct Signal { | declares data layout; field changes may break serialization, tests, and callers. |
| 281 |     pgn_name: &'static str, | executable/support line; behavior impact depends on surrounding block context. |
| 282 |     sa_name: String, | executable/support line; behavior impact depends on surrounding block context. |
| 283 |     raw_id: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 284 |     last_ts: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 285 |     count: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 286 |     period_ms: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 287 |     decoded: Vec<(String, f64, String)>, | executable/support line; behavior impact depends on surrounding block context. |
| 288 |     fresh: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 289 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 290 | (blank) | line is blank; visual separation only. |
| 291 | // ████████████████████████████████████████████████████████████████████████████ | comment intent, no runtime effect. |
| 292 | // Application state | comment line; documentation intent, no runtime effect. |
| 293 | // ████████████████████████████████████████████████████████████████████████████ | comment intent, no runtime effect. |
| 294 | struct App { | declares data layout; field changes may break serialization, tests, and callers. |
```

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| 295 |     bench: HeavyMachinery, | executable/support line; behavior impact depends on surrounding block context. |
| 296 |     hw_cfg: io::hw::HwConfig, | executable/support line; behavior impact depends on surrounding block context. |
| 297 |     tab: Tab, | executable/support line; behavior impact depends on surrounding block context. |
| 298 |     cmds: Vec<Cmd>, // deferred command queue | executable/support line; behavior impact depends on surrounding
block context. |
| 299 | (blank) | line is blank; visual separation only. |
| 300 |     // ECM live operations state | comment line; documentation intent, no runtime effect. |
| 301 |     ecm_detected_sas: Vec<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 302 |     ecm_selected_idx: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 303 |     ecm_connected: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 304 |     ecm_live_feed: Option<io::live_runner::LiveFeed>, | executable/support line; behavior impact depends on s
urrounding block context. |
| 305 |     ecm_live_status: String, | executable/support line; behavior impact depends on surrounding block context. |
| 306 |     ecm_live_snapshot: io::ecm_params::EcmSnapshot, | executable/support line; behavior impact depends on surr
ounding block context. |
| 307 |     ecm_live_last_update_ms: u64, | executable/support line; behavior impact depends on surrounding block con
t. |
| 308 |     ecm_live_frames_total: u64, | executable/support line; behavior impact depends on surrounding block conte
|
| 309 | (blank) | line is blank; visual separation only. |
| 310 |     // Inputs mirrored here (bench mutations happen via Cmd) | comment line; documentation intent, no runtime
fect. |
| 311 |     throttle: f32, | executable/support line; behavior impact depends on surrounding block context. |
| 312 |     brake: f32, | executable/support line; behavior impact depends on surrounding block context. |
| 313 |     throttle_target: f32, | executable/support line; behavior impact depends on surrounding block context. |
| 314 |     brake_target: f32, | executable/support line; behavior impact depends on surrounding block context. |
| 315 |     throttle_cmd_last: f32, | executable/support line; behavior impact depends on surrounding block context. |
| 316 |     brake_cmd_last: f32, | executable/support line; behavior impact depends on surrounding block context. |
| 317 |     throttle_cmd_last_tick: u64, | executable/support line; behavior impact depends on surrounding block cont
. |
| 318 |     brake_cmd_last_tick: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 319 | (blank) | line is blank; visual separation only. |
| 320 |     // CAN monitor | comment line; documentation intent, no runtime effect. |
| 321 |     can_mode: CanMode, | executable/support line; behavior impact depends on surrounding block context. |
| 322 |     can_freeze: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 323 |     can_filter: String, | executable/support line; behavior impact depends on surrounding block context. |
| 324 |     can_bus_idx: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 325 |     can_note: String, | executable/support line; behavior impact depends on surrounding block context. |
| 326 |     can_trace_tree: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 327 |     can_trace_limit: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 328 |     can_signal_limit: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 329 |     can_hide_stale: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 330 |     can_trace_update_every_ticks: u64, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 331 |     can_trace_tick_counter: u64, | executable/support line; behavior impact depends on surrounding block cont
. |
| 332 |     can_trace_pause_on_scroll: bool, | executable/support line; behavior impact depends on surrounding block
text. |
| 333 |     can_trace_pause_ticks_left: u32, | executable/support line; behavior impact depends on surrounding block
text. |
| 334 |     sig_map: HashMap<(u32, u8), Signal>, | executable/support line; behavior impact depends on surrounding bl
context. |
| 335 |     // (ts, raw_id, sa, dlc, hex_data, pgn[sa], decoded_str) | comment line; documentation intent, no runtime
fect. |
| 336 |     trace_snap: Vec<(f64, u32, u8, u8, String, String, String)>, | executable/support line; behavior impact d
nds on surrounding block context. |
| 337 | (blank) | line is blank; visual separation only. |
| 338 |     // Events | comment line; documentation intent, no runtime effect. |
| 339 |     events: Vec<AppEvent>, | executable/support line; behavior impact depends on surrounding block context. |
| 340 |     ev_pause: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 341 |     ev_filter: String, | executable/support line; behavior impact depends on surrounding block context. |
| 342 |     ev_min_level: EventLevel, | executable/support line; behavior impact depends on surrounding block context. |
| 343 | (blank) | line is blank; visual separation only. |
| 344 |     // Previous state for event detection | comment line; documentation intent, no runtime effect. |

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| 345 |     prev_rpm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 346 |     prev_gear: String, | executable/support line; behavior impact depends on surrounding block context. |
| 347 |     prev_dtcs: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 348 |     prev_abs: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 349 |     prev_esp: EspCondition, | executable/support line; behavior impact depends on surrounding block context. |
| 350 |     prev_ign: IgnitionState, | executable/support line; behavior impact depends on surrounding block context. |
| 351 |     prev_regen: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 352 |     prev_mode: String, | executable/support line; behavior impact depends on surrounding block context. |
| 353 | (blank) | line is blank; visual separation only. |
| 354 |     // Parameter editing – user-editable simulation params | comment line; documentation intent, no runtime e
ct. |
| 355 |     p_fuel: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 356 |     p_def: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 357 |     p_coolant: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 358 |     p_oil_prs: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 359 |     p_boost: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 360 |     p_engine_h: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 361 |     params_dirty: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 362 | (blank) | line is blank; visual separation only. |
| 363 |     // Implements controls | comment line; documentation intent, no runtime effect. |
| 364 |     #[allow(dead_code)] | executable/support line; behavior impact depends on surrounding block context. |
| 365 |     pto_target_rpm: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 366 |     hitch_target: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 367 |     aux_cmds: [f64; 4], | executable/support line; behavior impact depends on surrounding block context. |
| 368 | (blank) | line is blank; visual separation only. |
| 369 |     // AD path planning | comment line; documentation intent, no runtime effect. |
| 370 |     ad_waypoints: Vec<[f64; 2]>, // local ENU metres | executable/support line; behavior impact depends on su
rounding block context. |
| 371 |     #[allow(dead_code)] | executable/support line; behavior impact depends on surrounding block context. |
| 372 |     ad_canvas_pos: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 373 | (blank) | line is blank; visual separation only. |
| 374 |     // Fault panel | comment line; documentation intent, no runtime effect. |
| 375 |     fault_idx: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 376 |     fault_feedback: String, | executable/support line; behavior impact depends on surrounding block context. |
| 377 |     fault_last_dtc_count: usize, | executable/support line; behavior impact depends on surrounding block cont
. |
| 378 | (blank) | line is blank; visual separation only. |
| 379 |     // Implements operator feedback | comment line; documentation intent, no runtime effect. |
| 380 |     implement_feedback: String, | executable/support line; behavior impact depends on surrounding block conte
|
| 381 | (blank) | line is blank; visual separation only. |
| 382 |     // V2X smoothed KPIs | comment line; documentation intent, no runtime effect. |
| 383 |     v2x_range_ema: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 384 |     v2x_loss_ema: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 385 |     v2x_lat_ema: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 386 | (blank) | line is blank; visual separation only. |
| 387 |     // UDS console | comment line; documentation intent, no runtime effect. |
| 388 |     uds_input: String, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic
tcomes. |
| 389 |     uds_sa: u8, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcom
|
| 390 |     uds_flash_path: String, | I/O or protocol-critical logic; edits can affect integration, timing, and diagn
ic outcomes. |
| 391 |     uds_log: Vec<(bool, String)>, | I/O or protocol-critical logic; edits can affect integration, timing, and
agnostic outcomes. |
| 392 | (blank) | line is blank; visual separation only. |
| 393 |     // Leak physics lab | comment line; documentation intent, no runtime effect. |
| 394 |     leak_sel_idx: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 395 |     leak_manual: LeakManualUi, | executable/support line; behavior impact depends on surrounding block context
|
| 396 |     leak_custom: LeakCustomUi, | executable/support line; behavior impact depends on surrounding block context
|
| 397 |     leak_horizon_s: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 398 |     leak_scenario_dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 399 |     leak_predictions: Vec<ScenarioPrediction>, | executable/support line; behavior impact depends on surround

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block context. |
| 400 |     leak_temporal_trace: VecDeque<(f64, f64, f64, f64)>, // (t, pressure, risk, leak) | executable/support line; behavior impact depends on surrounding block context. |
| 401 |     leak_calibration_csv_path: String, | executable/support line; behavior impact depends on surrounding block context. |
| 402 |     leak_calibration_report: Option<CalibrationReport>, | executable/support line; behavior impact depends on surrounding block context. |
| 403 |     leak_note: String, | executable/support line; behavior impact depends on surrounding block context. |
| 404 |     leak_view_yaw_deg: f32, | executable/support line; behavior impact depends on surrounding block context. |
| 405 |     leak_view_pitch_deg: f32, | executable/support line; behavior impact depends on surrounding block context. |
| 406 |     leak_view_zoom: f32, | executable/support line; behavior impact depends on surrounding block context. |
| 407 |     leak_timeback_window_s: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 408 |     leak_timeback_latest_first: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 409 |     leak_timeback_stride: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 410 | (blank) | line is blank; visual separation only. |
| 411 |     // Plots | comment line; documentation intent, no runtime effect. |
| 412 |     pl_rpm: Vec<[f64; 2]>, | executable/support line; behavior impact depends on surrounding block context. |
| 413 |     pl_spd: Vec<[f64; 2]>, | executable/support line; behavior impact depends on surrounding block context. |
| 414 |     pl_torque: Vec<[f64; 2]>, | executable/support line; behavior impact depends on surrounding block context. |
| 415 |     pl_fuel: Vec<[f64; 2]>, | executable/support line; behavior impact depends on surrounding block context. |
| 416 |     pl_coolant: Vec<[f64; 2]>, | executable/support line; behavior impact depends on surrounding block context. |
| 417 |     pl_dpf: Vec<[f64; 2]>, | executable/support line; behavior impact depends on surrounding block context. |
| 418 |     pl_boost: Vec<[f64; 2]>, | executable/support line; behavior impact depends on surrounding block context. |
| 419 |     pl_hyd: Vec<[f64; 2]>, | executable/support line; behavior impact depends on surrounding block context. |
| 420 | (blank) | line is blank; visual separation only. |
| 421 |     ticks: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 422 |     ux_show_guide: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 423 |     ux_compact_mode: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 424 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 425 | (blank) | line is blank; visual separation only. |
| 426 | impl App { | implementation block; method behavior and trait conformance are defined here. |
| 427 |     fn new(cc: &frame::CreationContext, hw_cfg: io::hw::HwConfig) -> Self { | declares executable behavior; nature/body changes alter API, control flow, and outputs. |
| 428 |         let mut vis = Visuals::dark(); | executable/support line; behavior impact depends on surrounding block context. |
| 429 |         vis.panel_fill = Color32::from_gray(14); | executable/support line; behavior impact depends on surrounding block context. |
| 430 |         cc.egui_ctx.set_visuals(vis); | executable/support line; behavior impact depends on surrounding block context. |
| 431 | (blank) | line is blank; visual separation only. |
| 432 |         let mut bench = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 433 |         bench.tcm.set_direction(Direction::Forward); | executable/support line; behavior impact depends on surrounding block context. |
| 434 |         let fuel = bench.ecm.fuel_level_pct; | executable/support line; behavior impact depends on surrounding block context. |
| 435 |         let def = bench.ecm.def_level_pct; | executable/support line; behavior impact depends on surrounding block context. |
| 436 |         let cool = bench.ecm.coolant_temp_c; | executable/support line; behavior impact depends on surrounding block context. |
| 437 |         let oil = bench.ecm.oil_pressure_kpa; | executable/support line; behavior impact depends on surrounding block context. |
| 438 |         let boost = bench.ecm.boost_pressure_kpa; | executable/support line; behavior impact depends on surrounding block context. |
| 439 |         let hrs = bench.ecm.engine_hours; | executable/support line; behavior impact depends on surrounding block context. |
| 440 | (blank) | line is blank; visual separation only. |
| 441 |         App { | executable/support line; behavior impact depends on surrounding block context. |
| 442 |             bench, | executable/support line; behavior impact depends on surrounding block context. |
| 443 |             hw_cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 444 |             tab: Tab::Cluster, | executable/support line; behavior impact depends on surrounding block context. |

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|
| 445 |         cmds: Vec::new(), | executable/support line; behavior impact depends on surrounding block context.
| 446 |         ecm_detected_sas: Vec::new(), | executable/support line; behavior impact depends on surrounding block context.
| 447 |         ecm_selected_idx: 0, | executable/support line; behavior impact depends on surrounding block context.
| 448 |         ecm_connected: false, | executable/support line; behavior impact depends on surrounding block context.
| 449 |         ecm_live_feed: None, | executable/support line; behavior impact depends on surrounding block context.
| 450 |         ecm_live_status: "Idle. Press Detect to scan ECMS on the network.".into(), | executable/support line; behavior impact depends on surrounding block context.
| 451 |         ecm_live_snapshot: io::ecm_params::EcmSnapshot::default(), | executable/support line; behavior impact depends on surrounding block context.
| 452 |         ecm_live_last_update_ms: 0, | executable/support line; behavior impact depends on surrounding block context.
| 453 |         ecm_live_frames_total: 0, | executable/support line; behavior impact depends on surrounding block context.
| 454 |         throttle: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 455 |         brake: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 456 |         throttle_target: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 457 |         brake_target: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 458 |         throttle_cmd_last: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 459 |         brake_cmd_last: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 460 |         throttle_cmd_last_tick: 0, | executable/support line; behavior impact depends on surrounding block context.
| 461 |         brake_cmd_last_tick: 0, | executable/support line; behavior impact depends on surrounding block context.
| 462 |         can_mode: CanMode::Signals, | executable/support line; behavior impact depends on surrounding block context.
| 463 |         can_freeze: false, | executable/support line; behavior impact depends on surrounding block context.
| 464 |         can_filter: String::new(), | executable/support line; behavior impact depends on surrounding block context.
| 465 |         sig_map: HashMap::new(), | executable/support line; behavior impact depends on surrounding block context.
| 466 |         trace_snap: Vec::new(), | executable/support line; behavior impact depends on surrounding block context.
| 467 |         can_bus_idx: 0, | executable/support line; behavior impact depends on surrounding block context.
| 468 |         can_note: String::new(), | executable/support line; behavior impact depends on surrounding block context.
| 469 |         can_trace_tree: true, | executable/support line; behavior impact depends on surrounding block context.
| 470 |         can_trace_limit: 500, | executable/support line; behavior impact depends on surrounding block context.
| 471 |         can_signal_limit: 300, | executable/support line; behavior impact depends on surrounding block context.
| 472 |         can_hide_stale: false, | executable/support line; behavior impact depends on surrounding block context.
| 473 |         can_trace_update_every_ticks: 1, | executable/support line; behavior impact depends on surrounding block context.
| 474 |         can_trace_tick_counter: 0, | executable/support line; behavior impact depends on surrounding block context.
| 475 |         can_trace_pause_on_scroll: true, | executable/support line; behavior impact depends on surrounding block context.
| 476 |         can_trace_pause_ticks_left: 0, | executable/support line; behavior impact depends on surrounding block context.
| 477 |         events: Vec::new(), | executable/support line; behavior impact depends on surrounding block context.
| 478 |         ev_pause: false, | executable/support line; behavior impact depends on surrounding block context.
| 479 |         ev_filter: String::new(), | executable/support line; behavior impact depends on surrounding block context.

```

```

ntext. |
| 480 |           ev_min_level: EventLevel::Info, | executable/support line; behavior impact depends on surrounding
ock context. |
| 481 |           prev_rpm: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 482 |           prev_gear: "N".into(), | executable/support line; behavior impact depends on surrounding block co
xt. |
| 483 |           prev_dtcs: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 484 |           prev_abs: false, | executable/support line; behavior impact depends on surrounding block context.
| 485 |           prev_esp: EspCondition::Neutral, | executable/support line; behavior impact depends on surroundin
lock context. |
| 486 |           prev_ign: IgnitionState::Off, | executable/support line; behavior impact depends on surrounding b
k context. |
| 487 |           prev_regen: false, | executable/support line; behavior impact depends on surrounding block contex
|
| 488 |           prev_mode: String::new(), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 489 |           p_fuel: fuel, | executable/support line; behavior impact depends on surrounding block context. |
| 490 |           p_def: def, | executable/support line; behavior impact depends on surrounding block context. |
| 491 |           p_coolant: cool, | executable/support line; behavior impact depends on surrounding block context.
| 492 |           p_oil_prs: oil, | executable/support line; behavior impact depends on surrounding block context.
| 493 |           p_boost: boost, | executable/support line; behavior impact depends on surrounding block context.
| 494 |           p_engine_h: hrs, | executable/support line; behavior impact depends on surrounding block context.
| 495 |           params_dirty: false, | executable/support line; behavior impact depends on surrounding block cont
. |
| 496 |           pto_target_rpm: 540.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 497 |           hitch_target: 100.0, | executable/support line; behavior impact depends on surrounding block cont
. |
| 498 |           aux_cmds: [0.0; 4], | executable/support line; behavior impact depends on surrounding block conte
|
| 499 |           ad_waypoints: Vec::new(), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 500 |           ad_canvas_pos: 0.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 501 |           fault_idx: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 502 |           fault_feedback: String::new(), | executable/support line; behavior impact depends on surrounding
ck context. |
| 503 |           fault_last_dtc_count: 0, | executable/support line; behavior impact depends on surrounding block
text. |
| 504 |           implement_feedback: String::new(), | executable/support line; behavior impact depends on surround
block context. |
| 505 |           v2x_range_ema: 300.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 506 |           v2x_loss_ema: 2.0, | executable/support line; behavior impact depends on surrounding block context
|
| 507 |           v2x_lat_ema: 15.0, | executable/support line; behavior impact depends on surrounding block context
|
| 508 |           uds_input: "10 02".into(), | I/O or protocol-critical logic; edits can affect integration, timing
nd diagnostic outcomes. |
| 509 |           uds_sa: 0x00, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnos
outcomes. |
| 510 |           uds_flash_path: String::new(), | I/O or protocol-critical logic; edits can affect integration, ti
g, and diagnostic outcomes. |
| 511 |           uds_log: Vec::new(), | I/O or protocol-critical logic; edits can affect integration, timing, and
gnostic outcomes. |
| 512 |           leak_sel_idx: 0, | executable/support line; behavior impact depends on surrounding block context.
| 513 |           leak_manual: LeakManualUi::default(), | executable/support line; behavior impact depends on surro
ing block context. |
| 514 |           leak_custom: LeakCustomUi::default(), | executable/support line; behavior impact depends on surro
ing block context. |
| 515 |           leak_horizon_s: 600.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 516 |           leak_scenario_dt: 0.05, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 517 |           leak_predictions: Vec::new(), | executable/support line; behavior impact depends on surrounding b

```

```

k context. |
| 518 |          leak_temporal_trace: VecDeque::new(), | executable/support line; behavior impact depends on surro
ing block context. |
| 519 |          leak_calibration_csv_path: String::new(), | executable/support line; behavior impact depends on s
ounding block context. |
| 520 |          leak_calibration_report: None, | executable/support line; behavior impact depends on surrounding
ck context. |
| 521 |          leak_note: String::new(), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 522 |          leak_view_yaw_deg: 25.0, | executable/support line; behavior impact depends on surrounding block
text. |
| 523 |          leak_view_pitch_deg: 20.0, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 524 |          leak_view_zoom: 1.0, | executable/support line; behavior impact depends on surrounding block cont
. |
| 525 |          leak_timeback_window_s: 120.0, | executable/support line; behavior impact depends on surrounding
ck context. |
| 526 |          leak_timeback_latest_first: false, | executable/support line; behavior impact depends on surround
block context. |
| 527 |          leak_timeback_stride: 1, | executable/support line; behavior impact depends on surrounding block
text. |
| 528 |          pl_rpm: vec![], | executable/support line; behavior impact depends on surrounding block context.
| 529 |          pl_spd: vec![], | executable/support line; behavior impact depends on surrounding block context.
| 530 |          pl_torque: vec![], | executable/support line; behavior impact depends on surrounding block contex
|
| 531 |          pl_fuel: vec![], | executable/support line; behavior impact depends on surrounding block context.
| 532 |          pl_coolant: vec![], | executable/support line; behavior impact depends on surrounding block conte
|
| 533 |          pl_dpfp: vec![], | executable/support line; behavior impact depends on surrounding block context.
| 534 |          pl_boost: vec![], | executable/support line; behavior impact depends on surrounding block context
| 535 |          pl_hyd: vec![], | executable/support line; behavior impact depends on surrounding block context.
| 536 |          ticks: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 537 |          ux_show_guide: true, | executable/support line; behavior impact depends on surrounding block cont
. |
| 538 |          ux_compact_mode: false, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 539 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 540 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 541 | (blank) | line is blank; visual separation only. |
| 542 |      fn pick_save_path(default_name: &str, ext: &str) -> Option<String> { | declares executable behavior; sign
re/body changes alter API, control flow, and outputs. |
| 543 |          let default_dir = PathBuf::from("reports"); | executable/support line; behavior impact depends on sur
nding block context. |
| 544 |          let _ = std::fs::create_dir_all(&default_dir); | executable/support line; behavior impact depends on
rounding block context. |
| 545 |          FileDialog::new() | executable/support line; behavior impact depends on surrounding block context. |
| 546 |          .set_directory(default_dir) | executable/support line; behavior impact depends on surrounding blo
context. |
| 547 |          .set_file_name(default_name) | executable/support line; behavior impact depends on surrounding bl
context. |
| 548 |          .add_filter(ext, &[ext]) | executable/support line; behavior impact depends on surrounding block
text. |
| 549 |          .save_file() | executable/support line; behavior impact depends on surrounding block context. |
| 550 |          .map(|p| p.display().to_string()) | executable/support line; behavior impact depends on surround
g block context. |
| 551 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 552 | (blank) | line is blank; visual separation only. |
| 553 |      fn pick_open_path(ext: &str) -> Option<String> { | declares executable behavior; signature/body changes a
r API, control flow, and outputs. |
| 554 |          FileDialog::new() | executable/support line; behavior impact depends on surrounding block context. |
| 555 |          .add_filter(ext, &[ext]) | executable/support line; behavior impact depends on surrounding block
text. |
| 556 |          .pick_file() | executable/support line; behavior impact depends on surrounding block context. |
| 557 |          .map(|p| p.display().to_string()) | executable/support line; behavior impact depends on surround

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g block context. |
| 558 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 559 | (blank) | line is blank; visual separation only. |
| 560 |     fn refresh_mock_ecm_live_from_sim(&mut self) { | declares executable behavior; signature/body changes alt
API, control flow, and outputs. |
| 561 |         let e = &self.bench.ecm; | executable/support line; behavior impact depends on surrounding block context. |
| 562 |         self.ecm_live_snapshot.engine_speed_rpm = Some(e.rpm); | executable/support line; behavior impact depends on surrounding block context. |
| 563 |         self.ecm_live_snapshot.accel_pedal_pct = Some(e.active_throttle); | executable/support line; behavior impact depends on surrounding block context. |
| 564 |         self.ecm_live_snapshot.coolant_temp_c = Some(e.coolant_temp_c); | executable/support line; behavior impact depends on surrounding block context. |
| 565 |         self.ecm_live_snapshot.fuel_temp_c = Some(e.fuel_temp_c); | executable/support line; behavior impact depends on surrounding block context. |
| 566 |         self.ecm_live_snapshot.oil_pressure_kpa = Some(e.oil_pressure_kpa); | executable/support line; behavior impact depends on surrounding block context. |
| 567 |         self.ecm_live_snapshot.last_seen_pgn = Some(61444); | executable/support line; behavior impact depends on surrounding block context. |
| 568 |         self.ecm_live_snapshot.source_address = Some(0x00); | executable/support line; behavior impact depends on surrounding block context. |
| 569 |         self.ecm_live_last_update_ms = now_ms(); | executable/support line; behavior impact depends on surrounding block context. |
| 570 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 571 | (blank) | line is blank; visual separation only. |
| 572 |     fn ecm_treeview(&self, ui: &mut Ui, mock_mode: bool) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 573 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 574 |         ui.label(if mock_mode { | executable/support line; behavior impact depends on surrounding block context. |
|
| 575 |             "ECM Explorer (Mock/CAT ET style)" | executable/support line; behavior impact depends on surrounding block context. |
| 576 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 577 |             "ECM Explorer (CAT ET style)" | executable/support line; behavior impact depends on surrounding block context. |
| 578 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 579 | (blank) | line is blank; visual separation only. |
| 580 |         let nodes = ecm_mock_provider::build_mock_ecm_tree(&self.bench, &self.ecm_live_snapshot); | executable/support line; behavior impact depends on surrounding block context. |
| 581 |         for node in nodes { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 582 |             CollapsingHeader::new(format!("{}", (SA 0x{:02X})", node.name, node.source_address)) | executable/support line; behavior impact depends on surrounding block context. |
| 583 |                 .default_open(node.source_address == 0x00) | executable/support line; behavior impact depends on surrounding block context. |
| 584 |                 .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
|
| 585 |                     for f in node.functions { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 586 |                         CollapsingHeader::new(f.name).default_open(false).show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
| 587 |                             Grid::new(format!("ecm_tree_{}_{}", node.source_address, f.name)) | executable/support line; behavior impact depends on surrounding block context. |
| 588 |                                 .num_columns(3) | executable/support line; behavior impact depends on surrounding block context. |
| 589 |                             .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
|
| 590 |                                 for p in f.params { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 591 |                                     ui.label(p.key); | executable/support line; behavior impact depends on surrounding block context. |
| 592 |                                     ui.label(format!("{:.3}", p.value)); | executable/support line; behavior impact depends on surrounding block context. |
| 593 |                                     ui.label(p.unit); | executable/support line; behavior impact depends on surrounding block context. |

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| 594 |                                     ui.end_row(); | executable/support line; behavior impact depends on s
ounding block context. |
| 595 |                                     } | scope delimiter; structure-only line, but misplaced edits can change
ership and lifetime scopes. |
| 596 |                                     }); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 597 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 598 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 599 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 600 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 601 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 602 | (blank) | line is blank; visual separation only. |
| 603 |     fn oil_type_from_idx(idx: usize) -> OilType { | declares executable behavior; signature/body changes altere
PI, control flow, and outputs. |
| 604 |         OilType::all().get(idx).copied().unwrap_or(OilType::Custom) | executable/support line; behavior impac
depends on surrounding block context. |
| 605 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 606 | (blank) | line is blank; visual separation only. |
| 607 |     fn component_from_idx(idx: usize) -> CircuitComponent { | declares executable behavior; signature/body ch
es alter API, control flow, and outputs. |
| 608 |         match idx { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 609 |             0 => CircuitComponent::Oring, | executable/support line; behavior impact depends on surrounding b
k context. |
| 610 |             1 => CircuitComponent::Seal, | executable/support line; behavior impact depends on surrounding bl
context. |
| 611 |             _ => CircuitComponent::AcHose, | executable/support line; behavior impact depends on surrounding
ck context. |
| 612 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 613 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 614 | (blank) | line is blank; visual separation only. |
| 615 |     fn material_from_idx(idx: usize) -> OringMaterial { | declares executable behavior; signature/body change
lter API, control flow, and outputs. |
| 616 |         OringMaterial::all() | executable/support line; behavior impact depends on surrounding block context. |
| 617 |             .get(idx) | executable/support line; behavior impact depends on surrounding block context. |
| 618 |             .copied() | executable/support line; behavior impact depends on surrounding block context. |
| 619 |             .unwrap_or(OringMaterial::Nbr) | executable/support line; behavior impact depends on surrounding
ck context. |
| 620 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 621 | (blank) | line is blank; visual separation only. |
| 622 |     fn idx_from_oil_type(o: OilType) -> usize { | declares executable behavior; signature/body changes alter
, control flow, and outputs. |
| 623 |         OilType::all().iter().position(|x| *x == o).unwrap_or(0) | executable/support line; behavior impact
depends on surrounding block context. |
| 624 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 625 | (blank) | line is blank; visual separation only. |
| 626 |     fn sync_leak_manual_from_selected(&mut self) { | declares executable behavior; signature/body changes alt
API, control flow, and outputs. |
| 627 |         if let Some(c) = self.bench.leak_rig.circuits.get(self.leak_sel_idx) { | conditional gate; changing c
ition shifts runtime path and side effects. |
| 628 |             self.leak_manual.oil_type_idx = Self::idx_from_oil_type(c.oil_type); | executable/support line; b
vior impact depends on surrounding block context. |
| 629 |             self.leak_manual.piston_pressure_bar = c.piston_pressure_bar; | executable/support line; behavior
pact depends on surrounding block context. |
| 630 |             self.leak_manual.operation_pressure_bar = c.operation_pressure_bar; | executable/support line; be
ior impact depends on surrounding block context. |
| 631 |             self.leak_manual.pressure_min_bar = c.pressure.min_bar; | executable/support line; behavior impac
depends on surrounding block context. |
| 632 |             self.leak_manual.pressure_mean_bar = c.pressure.mean_bar; | executable/support line; behavior imp
depends on surrounding block context. |
| 633 |             self.leak_manual.pressure_ideal_bar = c.pressure.ideal_bar; | executable/support line; behavior i
ct depends on surrounding block context. |
| 634 |             self.leak_manual.pressure_max_bar = c.pressure.max_bar; | executable/support line; behavior impac

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depends on surrounding block context. |
| 635 |         self.leak_manual.pressure_rupture_bar = c.pressure.rupture_bar; | executable/support line; behavior
impact depends on surrounding block context. |
| 636 |         self.leak_manual.squeeze_pct = c.spec.squeeze_pct; | executable/support line; behavior impact dep
s on surrounding block context. |
| 637 |         self.leak_manual.compression_set_pct = c.spec.compression_set_pct; | executable/support line; beh
or impact depends on surrounding block context. |
| 638 |         self.leak_manual.base_leak_area_mm2 = c.spec.base_leak_area_mm2; | executable/support line; behav
impact depends on surrounding block context. |
| 639 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 640 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 641 | (blank) | line is blank; visual separation only. |
| 642 |     fn sanitize_leak_manual(&mut self) { | declares executable behavior; signature/body changes alter API, co
ol flow, and outputs. |
| 643 |         self.leak_manual.piston_pressure_bar = self.leak_manual.piston_pressure_bar.clamp(0.0, 1000.0); | exe
able/support line; behavior impact depends on surrounding block context. |
| 644 |         self.leak_manual.operation_pressure_bar = self.leak_manual.operation_pressure_bar.clamp(0.0, 1000.0);
executable/support line; behavior impact depends on surrounding block context. |
| 645 |         self.leak_manual.pressure_min_bar = self.leak_manual.pressure_min_bar.max(0.0); | executable/support
e; behavior impact depends on surrounding block context. |
| 646 |         self.leak_manual.pressure_mean_bar = self | executable/support line; behavior impact depends on surro
ing block context. |
| 647 |             .leak_manual | executable/support line; behavior impact depends on surrounding block context. |
| 648 |             .pressure_mean_bar | executable/support line; behavior impact depends on surrounding block contex
|
| 649 |             .max(self.leak_manual.pressure_min_bar); | executable/support line; behavior impact depends on su
unding block context. |
| 650 |         self.leak_manual.pressure_ideal_bar = self | executable/support line; behavior impact depends on surr
ding block context. |
| 651 |             .leak_manual | executable/support line; behavior impact depends on surrounding block context. |
| 652 |             .pressure_ideal_bar | executable/support line; behavior impact depends on surrounding block conte
|
| 653 |             .max(self.leak_manual.pressure_mean_bar); | executable/support line; behavior impact depends on s
ounding block context. |
| 654 |         self.leak_manual.pressure_max_bar = self | executable/support line; behavior impact depends on surrou
ng block context. |
| 655 |             .leak_manual | executable/support line; behavior impact depends on surrounding block context. |
| 656 |             .pressure_max_bar | executable/support line; behavior impact depends on surrounding block context
|
| 657 |             .max(self.leak_manual.pressure_ideal_bar); | executable/support line; behavior impact depends on
rounding block context. |
| 658 |         self.leak_manual.pressure_rupture_bar = self | executable/support line; behavior impact depends on su
unding block context. |
| 659 |             .leak_manual | executable/support line; behavior impact depends on surrounding block context. |
| 660 |             .pressure_rupture_bar | executable/support line; behavior impact depends on surrounding block con
t. |
| 661 |             .max(self.leak_manual.pressure_max_bar + 0.1); | executable/support line; behavior impact depends
surrounding block context. |
| 662 |         self.leak_manual.squeeze_pct = self.leak_manual.squeeze_pct.clamp(5.0, 45.0); | executable/support li
behavior impact depends on surrounding block context. |
| 663 |         self.leak_manual.compression_set_pct = self.leak_manual.compression_set_pct.clamp(0.0, 80.0); | execu
le/support line; behavior impact depends on surrounding block context. |
| 664 |         self.leak_manual.base_leak_area_mm2 = self.leak_manual.base_leak_area_mm2.clamp(0.0, 0.1); | executab
support line; behavior impact depends on surrounding block context. |
| 665 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 666 | (blank) | line is blank; visual separation only. |
| 667 |     fn sanitize_leak_custom(&mut self) { | declares executable behavior; signature/body changes alter API, co
ol flow, and outputs. |
| 668 |         self.leak_custom.name = self.leak_custom.name.trim().to_string(); | executable/support line; behavior
pact depends on surrounding block context. |
| 669 |         self.leak_custom.application = self.leak_custom.application.trim().to_string(); | executable/support
e; behavior impact depends on surrounding block context. |
| 670 |         self.leak_custom.seal_count = self.leak_custom.seal_count.clamp(1, 128); | executable/support line; b
vior impact depends on surrounding block context. |
| 671 |         self.leak_custom.shore_a = self.leak_custom.shore_a.clamp(40.0, 95.0); | executable/support line; beh

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or impact depends on surrounding block context. |
| 672 |         self.leak_custom.piston_pressure_bar = self.leak_custom.piston_pressure_bar.clamp(0.0, 1000.0); | executable/support line; behavior impact depends on surrounding block context. |
| 673 |         self.leak_custom.operation_pressure_bar = self.leak_custom.operation_pressure_bar.clamp(0.0, 1000.0); | executable/support line; behavior impact depends on surrounding block context. |
| 674 |         self.leak_custom.min_bar = self.leak_custom.min_bar.max(0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 675 |         self.leak_custom.mean_bar = self.leak_custom.mean_bar.max(self.leak_custom.min_bar); | executable/support line; behavior impact depends on surrounding block context. |
| 676 |         self.leak_custom.ideal_bar = self.leak_custom.ideal_bar.max(self.leak_custom.mean_bar); | executable/support line; behavior impact depends on surrounding block context. |
| 677 |         self.leak_custom.max_bar = self.leak_custom.max_bar.max(self.leak_custom.ideal_bar); | executable/support line; behavior impact depends on surrounding block context. |
| 678 |         self.leak_custom.rupture_bar = self.leak_custom.rupture_bar.max(self.leak_custom.max_bar + 0.1); | executable/support line; behavior impact depends on surrounding block context. |
| 679 |         self.leak_custom.cross_section_mm = self.leak_custom.cross_section_mm.clamp(0.5, 20.0); | executable/support line; behavior impact depends on surrounding block context. |
| 680 |         self.leak_custom.squeeze_pct = self.leak_custom.squeeze_pct.clamp(5.0, 45.0); | executable/support line; behavior impact depends on surrounding block context. |
| 681 |         self.leak_custom.extrusion_gap_mm = self.leak_custom.extrusion_gap_mm.clamp(0.01, 3.0); | executable/support line; behavior impact depends on surrounding block context. |
| 682 |         self.leak_custom.compression_set_pct = self.leak_custom.compression_set_pct.clamp(0.0, 80.0); | executable/support line; behavior impact depends on surrounding block context. |
| 683 |         self.leak_custom.design_life_hours = self.leak_custom.design_life_hours.clamp(1.0, 200000.0); | executable/support line; behavior impact depends on surrounding block context. |
| 684 |         self.leak_custom.base_leak_area_mm2 = self.leak_custom.base_leak_area_mm2.clamp(0.0, 0.1); | executable/support line; behavior impact depends on surrounding block context. |
| 685 |         self.leak_custom.discharge_coeff = self.leak_custom.discharge_coeff.clamp(0.05, 1.0); | executable/support line; behavior impact depends on surrounding block context. |
| 686 |         self.leak_custom.reservoir_volume_l = self.leak_custom.reservoir_volume_l.clamp(0.1, 2000.0); | executable/support line; behavior impact depends on surrounding block context. |
| 687 |         self.leak_custom.support_lpm = self.leak_custom.support_lpm.clamp(0.0, 5000.0); | executable/support line; behavior impact depends on surrounding block context. |
| 688 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 689 | (blank) | line is blank; visual separation only. |
| 690 |     fn validate_leak_manual(&self) -> Result<(), String> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 691 |         if !(self.leak_manual.pressure_min_bar <= self.leak_manual.pressure_mean_bar | conditional gate; changing condition shifts runtime path and side effects. |
| 692 |             && self.leak_manual.pressure_mean_bar <= self.leak_manual.pressure_ideal_bar | executable/support line; behavior impact depends on surrounding block context. |
| 693 |             && self.leak_manual.pressure_ideal_bar <= self.leak_manual.pressure_max_bar | executable/support line; behavior impact depends on surrounding block context. |
| 694 |             && self.leak_manual.pressure_max_bar < self.leak_manual.pressure_rupture_bar) | executable/support line; behavior impact depends on surrounding block context. |
| 695 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 696 |             return Err("Envelope invalido: requer min <= mean <= ideal <= max < rupture".into()); | executable/support line; behavior impact depends on surrounding block context. |
| 697 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 698 |         if self.leak_manual.base_leak_area_mm2 <= 0.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 699 |             return Err("Base leak deve ser > 0".into()); | executable/support line; behavior impact depends on surrounding block context. |
| 700 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 701 |         Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 702 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 703 | (blank) | line is blank; visual separation only. |
| 704 |     fn validate_leak_custom(&self) -> Result<(), String> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 705 |         if self.leak_custom.name.is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 706 |             return Err("Nome do circuito custom nao pode ser vazio".into()); | executable/support line; behavior impact depends on surrounding block context. |

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impact depends on surrounding block context. |
| 707 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 708 |         if !(self.leak_custom.min_bar <= self.leak_custom.mean_bar | conditional gate; changing condition shi
runtime path and side effects. |
| 709 |             && self.leak_custom.mean_bar <= self.leak_custom.ideal_bar | executable/support line; behavior im
t depends on surrounding block context. |
| 710 |             && self.leak_custom.ideal_bar <= self.leak_custom.max_bar | executable/support line; behavior imp
depends on surrounding block context. |
| 711 |             && self.leak_custom.max_bar < self.leak_custom.rupture_bar) | executable/support line; behavior i
ct depends on surrounding block context. |
| 712 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 713 |             return Err("Envelope custom invalido: requer min <= mean <= ideal <= max < rupture".into()); | ex
table/support line; behavior impact depends on surrounding block context. |
| 714 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 715 |         if self.leak_custom.base_leak_area_mm2 <= 0.0 { | conditional gate; changing condition shifts runtime
th and side effects. |
| 716 |             return Err("Base leak custom deve ser > 0".into()); | executable/support line; behavior impact de
ds on surrounding block context. |
| 717 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 718 |         Ok(()) | executable/support line; behavior impact depends on surrounding block context. |
| 719 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 720 | (blank) | line is blank; visual separation only. |
| 721 |     fn build_manual_leak_params(&self) -> ManualCircuitParams { | declares executable behavior; signature/bod
changes alter API, control flow, and outputs. |
| 722 |         ManualCircuitParams { | executable/support line; behavior impact depends on surrounding block context
| 723 |             oil_type: Some(Self::oil_type_from_idx(self.leak_manual.oil_type_idx)), | executable/support line
ehavior impact depends on surrounding block context. |
| 724 |             piston_pressure_bar: Some(self.leak_manual.piston_pressure_bar), | executable/support line; behav
impact depends on surrounding block context. |
| 725 |             operation_pressure_bar: Some(self.leak_manual.operation_pressure_bar), | executable/support line;
havior impact depends on surrounding block context. |
| 726 |             pressure_min_bar: Some(self.leak_manual.pressure_min_bar), | executable/support line; behavior im
t depends on surrounding block context. |
| 727 |             pressure_mean_bar: Some(self.leak_manual.pressure_mean_bar), | executable/support line; behavior
act depends on surrounding block context. |
| 728 |             pressure_ideal_bar: Some(self.leak_manual.pressure_ideal_bar), | executable/support line; behavio
mpact depends on surrounding block context. |
| 729 |             pressure_max_bar: Some(self.leak_manual.pressure_max_bar), | executable/support line; behavior im
t depends on surrounding block context. |
| 730 |             pressure_rupture_bar: Some(self.leak_manual.pressure_rupture_bar), | executable/support line; beh
or impact depends on surrounding block context. |
| 731 |             oring_squeeze_pct: Some(self.leak_manual.squeeze_pct), | executable/support line; behavior impact
ends on surrounding block context. |
| 732 |             compression_set_pct: Some(self.leak_manual.compression_set_pct), | executable/support line; behav
impact depends on surrounding block context. |
| 733 |             base_leak_area_mm2: Some(self.leak_manual.base_leak_area_mm2), | executable/support line; behavio
mpact depends on surrounding block context. |
| 734 |             max_supported_temp_c: None, | executable/support line; behavior impact depends on surrounding blo
context. |
| 735 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 736 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 737 | (blank) | line is blank; visual separation only. |
| 738 |     fn apply_manual_to_selected_circuit(&mut self) -> Result<String, String> { | declares executable behavior
signature/body changes alter API, control flow, and outputs. |
| 739 |         self.validate_leak_manual()?; | error-propagating operation; changing it impacts failure semantics an
aller recovery paths. |
| 740 |         let Some(c) = self.bench.leak_rig.circuits.get(self.leak_sel_idx) else { | executable/support line; b
vior impact depends on surrounding block context. |
| 741 |             return Err("Nenhum circuito selecionado".into()); | executable/support line; behavior impact depe
on surrounding block context. |

```



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| 742 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 743 |         let name = c.name.clone(); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 744 |         let ok = self | executable/support line; behavior impact depends on surrounding block context. |
| 745 |             .bench | executable/support line; behavior impact depends on surrounding block context. |
| 746 |             .apply_leak_manual_params(&name, self.build_manual_leak_params()); | executable/support line; beh
or impact depends on surrounding block context. |
| 747 |             if ok { | conditional gate; changing condition shifts runtime path and side effects. |
| 748 |                 Ok(name) | executable/support line; behavior impact depends on surrounding block context. |
| 749 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 750 |                 Err("Falha no backend ao aplicar parametros".into()) | executable/support line; behavior impact d
nds on surrounding block context. |
| 751 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 752 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 753 | (blank) | line is blank; visual separation only. |
| 754 |     fn can_bus_from_idx(idx: usize) -> auto_breaking::BusId { | declares executable behavior; signature/body
nges alter API, control flow, and outputs. |
| 755 |         match idx { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 756 |             0 => auto_breaking::BusId::PowertrainHs, | executable/support line; behavior impact depends on su
unding block context. |
| 757 |             1 => auto_breaking::BusId::ChassisHs, | executable/support line; behavior impact depends on surro
ing block context. |
| 758 |             2 => auto_breaking::BusId::BodyMs, | executable/support line; behavior impact depends on surround
block context. |
| 759 |             3 => auto_breaking::BusId::IsoBus, | executable/support line; behavior impact depends on surround
block context. |
| 760 |             _ => auto_breaking::BusId::Diagnostic, | executable/support line; behavior impact depends on surr
ding block context. |
| 761 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 762 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 763 | (blank) | line is blank; visual separation only. |
| 764 |     fn parse_u32_filter(input: &str) -> Option<u32> { | declares executable behavior; signature/body changes
er API, control flow, and outputs. |
| 765 |         let s = input.trim().to_lowercase(); | executable/support line; behavior impact depends on surroundi
lock context. |
| 766 |         if let Some(hex) = s.strip_prefix("0x") { | conditional gate; changing condition shifts runtime path
side effects. |
| 767 |             u32::from_str_radix(hex, 16).ok() | executable/support line; behavior impact depends on surroundi
block context. |
| 768 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 769 |             s.parse::<u32>().ok() | executable/support line; behavior impact depends on surrounding block con
t. |
| 770 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 771 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 772 | (blank) | line is blank; visual separation only. |
| 773 |     fn can_filter_match_signal(filter: &str, pgn: u32, sa: u8, sig: &Signal) -> bool { | declares executable
avior; signature/body changes alter API, control flow, and outputs. |
| 774 |         let filt = filter.trim().to_lowercase(); | executable/support line; behavior impact depends on surrou
ng block context. |
| 775 |         if filt.is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 776 |             return true; | executable/support line; behavior impact depends on surrounding block context. |
| 777 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 778 |         if let Some(v) = filt.strip_prefix("sa:") { | conditional gate; changing condition shifts runtime pat
nd side effects. |
| 779 |             return Self::parse_u32_filter(v) == Some(sa as u32); | executable/support line; behavior impact d
nds on surrounding block context. |
| 780 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 781 |         if let Some(v) = filt.strip_prefix("pgn:") { | conditional gate; changing condition shifts runtime pa
and side effects. |

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| 782 |         return Self::parse_u32_filter(v) == Some(pgn); | executable/support line; behavior impact depends
surrounding block context. |
| 783 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 784 |         if let Some(v) = filt.strip_prefix("id:") { | conditional gate; changing condition shifts runtime pat
nd side effects. |
| 785 |             return Self::parse_u32_filter(v) == Some(sig.raw_id); | executable/support line; behavior impact
ends on surrounding block context. |
| 786 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 787 |         sig.pgn_name.to_lowercase().contains(&filt) | executable/support line; behavior impact depends on surr
nding block context. |
| 788 |         \\| sig.sa_name.to_lowercase().contains(&filt) | executable/support line; behavior impact depend
n surrounding block context. |
| 789 |         \\| format!("{:02x}", sa).contains(&filt) | executable/support line; behavior impact depends on
rounding block context. |
| 790 |         \\| format!("{:0x{:02x}", sa).contains(&filt) | executable/support line; behavior impact depends o
urrounding block context. |
| 791 |         \\| format!("{:05}", pgn).contains(&filt) | executable/support line; behavior impact depends on
rounding block context. |
| 792 |         \\| format!("{:08x}", sig.raw_id).contains(&filt) | executable/support line; behavior impact dep
s on surrounding block context. |
| 793 |         \\| sig | executable/support line; behavior impact depends on surrounding block context. |
| 794 |         .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 795 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 796 |         .any(\\| (name, _, _) \\| name.to_lowercase().contains(&filt)) | executable/support line; behavio
mpact depends on surrounding block context. |
| 797 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 798 | (blank) | line is blank; visual separation only. |
| 799 |     fn can_filter_match_trace(filter: &str, row: &TraceRow) -> bool { | declares executable behavior; signatu
body changes alter API, control flow, and outputs. |
| 800 |         let filt = filter.trim().to_lowercase(); | executable/support line; behavior impact depends on surrou
ng block context. |
| 801 |         if filt.is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 802 |             return true; | executable/support line; behavior impact depends on surrounding block context. |
| 803 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 804 |         let (_ts, raw_id, sa, _dlc, hex, pgn_sa, decoded) = row; | executable/support line; behavior impact d
nds on surrounding block context. |
| 805 |         if let Some(v) = filt.strip_prefix("sa:") { | conditional gate; changing condition shifts runtime pat
nd side effects. |
| 806 |             return Self::parse_u32_filter(v) == Some(*sa as u32); | executable/support line; behavior impact
ends on surrounding block context. |
| 807 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 808 |         if let Some(v) = filt.strip_prefix("id:") { | conditional gate; changing condition shifts runtime pat
nd side effects. |
| 809 |             return Self::parse_u32_filter(v) == Some(*raw_id); | executable/support line; behavior impact dep
s on surrounding block context. |
| 810 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 811 |         pgn_sa.to_lowercase().contains(&filt) | executable/support line; behavior impact depends on surroundi
block context. |
| 812 |         \\| decoded.to_lowercase().contains(&filt) | executable/support line; behavior impact depends on
rrounding block context. |
| 813 |         \\| hex.to_lowercase().contains(&filt) | executable/support line; behavior impact depends on surr
nding block context. |
| 814 |         \\| format!("{:02x}", sa).contains(&filt) | executable/support line; behavior impact depends on
rounding block context. |
| 815 |         \\| format!("{:0x{:02x}", sa).contains(&filt) | executable/support line; behavior impact depends o
urrounding block context. |
| 816 |         \\| format!("{:08x}", raw_id).contains(&filt) | executable/support line; behavior impact depends
surrounding block context. |
| 817 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 818 | (blank) | line is blank; visual separation only. |

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| 819 |         fn ema(prev: f64, value: f64, alpha: f64) -> f64 { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 820 |             prev + alpha * (value - prev) | executable/support line; behavior impact depends on surrounding block context.  
| 821 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 822 | (blank) | line is blank; visual separation only. |  
| 823 |     fn update_v2x_kpis(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 824 |         self.v2x_range_ema = Self::ema(self.v2x_range_ema, self.bench.v2x.range_m, 0.10); | executable/support line; behavior impact depends on surrounding block context. |  
| 825 |         self.v2x_loss_ema = Self::ema(self.v2x_loss_ema, self.bench.v2x.packet_loss_pct, 0.12); | executable/support line; behavior impact depends on surrounding block context. |  
| 826 |         self.v2x_lat_ema = Self::ema(self.v2x_lat_ema, self.bench.v2x.latency_ms, 0.12); | executable/support line; behavior impact depends on surrounding block context. |  
| 827 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 828 | (blank) | line is blank; visual separation only. |  
| 829 |     fn approach_f32(current: f32, target: f32, max_step: f32) -> f32 { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 830 |         if (target - current).abs() <= max_step { | conditional gate; changing condition shifts runtime path and side effects. |  
| 831 |             target | executable/support line; behavior impact depends on surrounding block context. |  
| 832 |         } else if target > current { | executable/support line; behavior impact depends on surrounding block context. |  
| 833 |             current + max_step | executable/support line; behavior impact depends on surrounding block context. |  
| 834 |         } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 835 |             current - max_step | executable/support line; behavior impact depends on surrounding block context. |  
| 836 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 837 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 838 | (blank) | line is blank; visual separation only. |  
| 839 |     fn shape_pedal(x: f32, deadzone: f32) -> f64 { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 840 |         let n = ((x - deadzone) / (1.0 - deadzone)).clamp(0.0, 1.0); | executable/support line; behavior impact depends on surrounding block context. |  
| 841 |         let shaped = n * n * (3.0 - 2.0 * n); // smoothstep | executable/support line; behavior impact depends on surrounding block context. |  
| 842 |         shaped as f64 | executable/support line; behavior impact depends on surrounding block context. |  
| 843 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 844 | (blank) | line is blank; visual separation only. |  
| 845 |         // ■ Execute all queued commands ████████████████████████████████████████████████████ | comment line; documentation, no runtime effect. |  
| 846 |     fn flush_cmds(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 847 |         let cmds: Vec<Cmd> = self.cmds.drain(..).collect(); | executable/support line; behavior impact depends on surrounding block context. |  
| 848 |         for cmd in cmds { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |  
| 849 |             match cmd { | branch dispatch; changing arms alters behavior for variants and input classes. |  
| 850 |                 Cmd::KeyAdvance => self.bench.key_advance(), | executable/support line; behavior impact depends on surrounding block context. |  
| 851 |                 Cmd::KeyOff => { | executable/support line; behavior impact depends on surrounding block context. |  
| 852 |                     self.bench.key_off(); | executable/support line; behavior impact depends on surrounding block context. |  
| 853 |                     self.throttle = 0.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 854 |                     self.brake = 0.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 855 |                     self.throttle_target = 0.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 856 |                     self.brake_target = 0.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 857 |                     self.throttle_cmd_last = 0.0; | executable/support line; behavior impact depends on surrounding block context.
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ing block context. |
| 858 |           self.brake_cmd_last = 0.0; | executable/support line; behavior impact depends on surrounding
block context. |
| 859 |           self.throttle_cmd_last_tick = self.ticks; | executable/support line; behavior impact depends
on surrounding block context. |
| 860 |           self.brake_cmd_last_tick = self.ticks; | executable/support line; behavior impact depends
surrounding block context. |
| 861 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 862 |       Cmd::SetThrottle(v) => { | executable/support line; behavior impact depends on surrounding bl
context. |
| 863 |           self.throttle_target = v.clamp(0.0, 1.0); | executable/support line; behavior impact depende
on surrounding block context. |
| 864 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 865 |       Cmd::SetBrake(v) => { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 866 |           self.brake_target = v.clamp(0.0, 1.0); | executable/support line; behavior impact depends
surrounding block context. |
| 867 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 868 |       Cmd::SetDirection(d) => self.bench.tcm.set_direction(d), | executable/support line; behavior
act depends on surrounding block context. |
| 869 |       Cmd::SetNeutral => self.bench.tcm.set_neutral(), | executable/support line; behavior impact d
nds on surrounding block context. |
| 870 |       Cmd::ToggleAutoShift => self.bench.tcm.toggle_auto(), | executable/support line; behavior imp
depends on surrounding block context. |
| 871 |       Cmd::ToggleCreeper => self.bench.tcm.toggle_creeper(), | executable/support line; behavior im
t depends on surrounding block context. |
| 872 |       Cmd::ManualUp => self.bench.tcm.manual_upshift(), | executable/support line; behavior impact
ends on surrounding block context. |
| 873 |       Cmd::ManualDn => self.bench.tcm.manual_downshift(), | executable/support line; behavior impac
depends on surrounding block context. |
| 874 |       // ECM overrides | comment line; documentation intent, no runtime effect. |
| 875 |       Cmd::SetFuelLevel(v) => { | executable/support line; behavior impact depends on surrounding b
k context. |
| 876 |           self.bench.ecm.fuel_level_pct = v.clamp(0.0, 100.0); | executable/support line; behavior
act depends on surrounding block context. |
| 877 |           self.p_fuel = v; | executable/support line; behavior impact depends on surrounding block
text. |
| 878 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 879 |       Cmd::SetDefLevel(v) => { | executable/support line; behavior impact depends on surrounding bl
context. |
| 880 |           self.bench.ecm.def_level_pct = v.clamp(0.0, 100.0); | executable/support line; behavior i
ct depends on surrounding block context. |
| 881 |           self.p_def = v; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 882 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 883 |       Cmd::SetCoolantTemp(v) => { | executable/support line; behavior impact depends on surrounding
ock context. |
| 884 |           self.bench.ecm.coolant_temp_c = v.clamp(-40.0, 130.0); | executable/support line; behavior
mpact depends on surrounding block context. |
| 885 |           self.p_coolant = v; | executable/support line; behavior impact depends on surrounding blo
context. |
| 886 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 887 |       Cmd::SetOilPressure(v) => { | executable/support line; behavior impact depends on surrounding
ock context. |
| 888 |           self.bench.ecm.oil_pressure_kpa = v.clamp(0.0, 800.0); | executable/support line; behavior
mpact depends on surrounding block context. |
| 889 |           self.p_oil_prs = v; | executable/support line; behavior impact depends on surrounding blo
context. |
| 890 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti

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scopes. |
| 891 |           Cmd::SetBoostPressure(v) => { | executable/support line; behavior impact depends on surrounding
block context. |
| 892 |           self.bench.ecm.boost_pressure_kpa = v.clamp(0.0, 350.0); | executable/support line; behavior
impact depends on surrounding block context. |
| 893 |           self.p_boost = v; | executable/support line; behavior impact depends on surrounding block
context. |
| 894 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 895 |           Cmd::SetEngineHours(v) => { | executable/support line; behavior impact depends on surrounding
block context. |
| 896 |           self.bench.ecm.engine_hours = v.max(0.0); | executable/support line; behavior impact depende
on surrounding block context. |
| 897 |           self.p_engine_h = v; | executable/support line; behavior impact depends on surrounding block
context. |
| 898 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 899 |           Cmd::ClearDtcs => self.bench.clear_faults(), | executable/support line; behavior impact depende
on surrounding block context. |
| 900 |           // BCM | comment line; documentation intent, no runtime effect. |
| 901 |           Cmd::ToggleWorkLights => self.bench.bcm.toggle_work_lights(), | executable/support line; beha
r impact depends on surrounding block context. |
| 902 |           Cmd::ToggleRoadLights => self.bench.bcm.toggle_road_lights(), | executable/support line; beha
r impact depends on surrounding block context. |
| 903 |           Cmd::ToggleBeacon => self.bench.bcm.toggle_beacon(), | executable/support line; behavior impa
depends on surrounding block context. |
| 904 |           Cmd::Honk => self.bench.bcm.honk(), | executable/support line; behavior impact depends on surr
nding block context. |
| 905 |           Cmd::CycleWiper => self.bench.bcm.cycle_wiper(), | executable/support line; behavior impact d
nds on surrounding block context. |
| 906 |           Cmd::ToggleHvac => self.bench.bcm.toggle_hvac(), | executable/support line; behavior impact d
nds on surrounding block context. |
| 907 |           Cmd::ResetBcmFuses => self.bench.bcm.reset_all_fuses(), | executable/support line; behavior i
ct depends on surrounding block context. |
| 908 |           Cmd::ElectricalSetBranchFault(id, fault) => { | executable/support line; behavior impact depende
on surrounding block context. |
| 909 |           self.bench.electrical.set_branch_fault(&id, fault) | executable/support line; behavior im
t depends on surrounding block context. |
| 910 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 911 |           Cmd::ElectricalSetControlFault(point, fault) => { | executable/support line; behavior impact
ends on surrounding block context. |
| 912 |           self.bench.electrical.set_control_fault(point, fault) | executable/support line; behavior
pact depends on surrounding block context. |
| 913 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 914 |           Cmd::ElectricalResetFaults => self.bench.electrical.reset_faults(), | executable/support line
ehavior impact depends on surrounding block context. |
| 915 |           // Implements | comment line; documentation intent, no runtime effect. |
| 916 |           Cmd::SetPtoMode(m) => { | executable/support line; behavior impact depends on surrounding blo
context. |
| 917 |           self.bench.implement.pto_rear_enabled = m != PtoMode::Off; | executable/support line; beh
or impact depends on surrounding block context. |
| 918 |           self.bench.implement.pto_mode = m; | executable/support line; behavior impact depends on
rounding block context. |
| 919 |           self.implement_feedback = format!("Rear PTO mode -> {}", m); | executable/support line; b
vior impact depends on surrounding block context. |
| 920 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 921 |           Cmd::SetHitchTarget(v) => { | executable/support line; behavior impact depends on surrounding
block context. |
| 922 |           self.bench.implement.hitch_target_pct = v.clamp(0.0, 100.0); | executable/support line; b
vior impact depends on surrounding block context. |
| 923 |           self.hitch_target = v; | executable/support line; behavior impact depends on surrounding
ck context. |

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| 924 |                 self.implement_feedback = | executable/support line; behavior impact depends on surrounding
block context. |
| 925 |                 format!("Hitch target set -> {:.1}%", self.hitch_target); | executable/support line;
avior impact depends on surrounding block context. |
| 926 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 927 |                 Cmd::SetAuxValve(i, v) => { | executable/support line; behavior impact depends on surrounding
ock context. |
| 928 |                 if i < 4 { | conditional gate; changing condition shifts runtime path and side effects. |
| 929 |                 self.bench.hcm.set_aux_cmd(i, v); | executable/support line; behavior impact depends
surrounding block context. |
| 930 |                 self.bench.implement.aux_banks[i].direction = if v > 0.1 { | executable/support line;
havior impact depends on surrounding block context. |
| 931 |                 auto_breaking::implement::ValveDirection::Extend | executable/support line; behav
impact depends on surrounding block context. |
| 932 |                 } else if v < -0.1 { | executable/support line; behavior impact depends on surroundin
lock context. |
| 933 |                 auto_breaking::implement::ValveDirection::Retract | executable/support line; beha
r impact depends on surrounding block context. |
| 934 |                 } else { | executable/support line; behavior impact depends on surrounding block cont
. |
| 935 |                 auto_breaking::implement::ValveDirection::Neutral | executable/support line; beha
r impact depends on surrounding block context. |
| 936 |                 }; | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 937 |                 self.bench.implement.aux_banks[i].engaged = v.abs() > 0.05; | executable/support line
ehavior impact depends on surrounding block context. |
| 938 |                 self.aux_cmds[i] = v; | executable/support line; behavior impact depends on surroundi
block context. |
| 939 |                 self.implement_feedback = | executable/support line; behavior impact depends on surro
ing block context. |
| 940 |                 format!("Aux bank {} command -> {:+.1}", i, self.aux_cmds[i]); | executable/suppo
line; behavior impact depends on surrounding block context. |
| 941 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 942 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 943 |                 Cmd::SetLoaderLift(v) => { | executable/support line; behavior impact depends on surrounding
ck context. |
| 944 |                 self.bench.loader_lift_cmd = v; | executable/support line; behavior impact depends on sur
nding block context. |
| 945 |                 self.implement_feedback = format!("Loader lift cmd -> {:+.2}", v); | executable/support l
; behavior impact depends on surrounding block context. |
| 946 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 947 |                 Cmd::SetLoaderTilt(v) => { | executable/support line; behavior impact depends on surrounding
ck context. |
| 948 |                 self.bench.loader_tilt_cmd = v; | executable/support line; behavior impact depends on sur
nding block context. |
| 949 |                 self.implement_feedback = format!("Loader tilt cmd -> {:+.2}", v); | executable/support l
; behavior impact depends on surrounding block context. |
| 950 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 951 |                 Cmd::SetHitchJoystick(v) => { | executable/support line; behavior impact depends on surroundi
block context. |
| 952 |                 self.bench.hitch_joystick = v; | executable/support line; behavior impact depends on surr
ding block context. |
| 953 |                 self.implement_feedback = format!("Hitch joystick -> {:+.1}", v); | executable/support li
behavior impact depends on surrounding block context. |
| 954 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 955 |                 // AD | comment line; documentation intent, no runtime effect. |
| 956 |                 Cmd::EngageAD => { | executable/support line; behavior impact depends on surrounding block co
xt. |
| 957 |                 self.bench.ad.engage(self.bench.tcm.ground_speed_kmh); | executable/support line; behavior

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mpact depends on surrounding block context. |
| 958 |             if self.bench.tcm.direction == Direction::Neutral | conditional gate; changing condition
fts runtime path and side effects. |
| 959 |             \\| self.bench.tcm.direction == Direction::Park | executable/support line; behavior
act depends on surrounding block context. |
| 960 |             { | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 961 |                 self.bench.tcm.set_direction(Direction::Forward); | executable/support line; behavior
pact depends on surrounding block context. |
| 962 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 963 |             if self.bench.tcm.auto_mode != AutoShiftMode::Auto { | conditional gate; changing conditi
shifts runtime path and side effects. |
| 964 |                 self.bench.tcm.toggle_auto(); | executable/support line; behavior impact depends on s
ounding block context. |
| 965 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 966 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 967 |         Cmd::DisengageAD => self.bench.ad.disengage(), | executable/support line; behavior impact dep
s on surrounding block context. |
| 968 |         Cmd::ToggleLka => self.bench.ad.toggle_lka(), | executable/support line; behavior impact depe
on surrounding block context. |
| 969 |         Cmd::AccSpeedSet(v) => self.bench.ad.set_acc_speed(v), | executable/support line; behavior im
t depends on surrounding block context. |
| 970 |         Cmd::AccHeadwaySet(v) => self.bench.ad.set_headway(v), | executable/support line; behavior im
t depends on surrounding block context. |
| 971 |         Cmd::AddWaypoint(x, y) => self.ad_waypoints.push([x, y]), | executable/support line; behavior
pact depends on surrounding block context. |
| 972 |         Cmd::ClearWaypoints => self.ad_waypoints.clear(), | executable/support line; behavior impact
ends on surrounding block context. |
| 973 |         // Faults | comment line; documentation intent, no runtime effect. |
| 974 |         Cmd::SelectFault(i) => { | executable/support line; behavior impact depends on surrounding bl
context. |
| 975 |             self.fault_idx = i.min(FAULT_TYPES.len().saturating_sub(1)); | executable/support line; b
vior impact depends on surrounding block context. |
| 976 |             self.bench.selected_fault = FAULT_TYPES[self.fault_idx]; | executable/support line; behav
impact depends on surrounding block context. |
| 977 |             self.fault_feedback = | executable/support line; behavior impact depends on surrounding b
k context. |
| 978 |             format!("Selected fault: {}", self.bench.selected_fault); | executable/support line;
avior impact depends on surrounding block context. |
| 979 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 980 |         Cmd::InjectFault => { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 981 |             self.bench.inject_fault(); | executable/support line; behavior impact depends on surround
block context. |
| 982 |             self.fault_last_dtc_count = self.bench.ecm.active_dtcs.len(); | executable/support line;
avior impact depends on surrounding block context. |
| 983 |             self.fault_feedback = format!( | executable/support line; behavior impact depends on surr
ding block context. |
| 984 |                 "Injected {}. Waiting ECM/DM1 propagation...", | executable/support line; behavior im
t depends on surrounding block context. |
| 985 |                 self.bench.selected_fault | executable/support line; behavior impact depends on surro
ing block context. |
| 986 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 987 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 988 |         Cmd::ClearFaults => { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 989 |             self.bench.clear_faults(); | executable/support line; behavior impact depends on surround
block context. |
| 990 |             self.fault_last_dtc_count = self.bench.ecm.active_dtcs.len(); | executable/support line;
avior impact depends on surrounding block context. |

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| 991 |             self.fault_feedback = "All injected faults cleared".into(); | executable/support line; behavior impact depends on surrounding block context. |
| 992 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 993 |         // Telematics | comment line; documentation intent, no runtime effect. |
| 994 |         Cmd::StartOta => self.bench.telematics.start_ota(), | executable/support line; behavior impact depends on surrounding block context. |
| 995 |         // Leak Lab | comment line; documentation intent, no runtime effect. |
| 996 |         Cmd::LeakSelectCircuit(i) => { | executable/support line; behavior impact depends on surrounding block context. |
| 997 |             self.leak_sel_idx = i.min(self.bench.leak_rig.circuits.len().saturating_sub(1)); | executable/support line; behavior impact depends on surrounding block context. |
| 998 |             self.sync_leak_manual_from_selected(); | executable/support line; behavior impact depends on surrounding block context. |
| 999 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1000 |         Cmd::LeakApplyManual => { | executable/support line; behavior impact depends on surrounding block context. |
| 1001 |             self.leak_note = match self.apply_manual_to_selected_circuit() { | executable/support line; behavior impact depends on surrounding block context. |
| 1002 |                 Ok(name) => format!("Parametros aplicados com sucesso em {}", name), | executable/support line; behavior impact depends on surrounding block context. |
| 1003 |                 Err(e) => format!("Falha ao aplicar parametros: {}", e), | executable/support line; behavior impact depends on surrounding block context. |
| 1004 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1005 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1006 |         Cmd::LeakPredictScenarios => { | executable/support line; behavior impact depends on surrounding block context. |
| 1007 |             self.leak_predictions = self.bench.predict_leak_scenarios( | executable/support line; behavior impact depends on surrounding block context. |
| 1008 |                 self.leak_horizon_s, | executable/support line; behavior impact depends on surrounding block context. |
| 1009 |                 self.leak_scenario_dt.max(0.01), | executable/support line; behavior impact depends on surrounding block context. |
| 1010 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 1011 |             self.leak_note = format!("{}", self.leak_predictions.len()); | executable/support line; behavior impact depends on surrounding block context. |
| 1012 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1013 |         Cmd::LeakAddCustomCircuit => { | executable/support line; behavior impact depends on surrounding block context. |
| 1014 |             if let Err(e) = self.validate_leak_custom() { | conditional gate; changing condition shifts runtime path and side effects. |
| 1015 |                 self.leak_note = format!("Falha ao adicionar circuito custom: {}", e); | executable/support line; behavior impact depends on surrounding block context. |
| 1016 |                 continue; | executable/support line; behavior impact depends on surrounding block context. |
| 1017 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1018 |             let c = LeakCircuit::new( | executable/support line; behavior impact depends on surrounding block context. |
| 1019 |                 &self.leak_custom.name, | executable/support line; behavior impact depends on surrounding block context. |
| 1020 |                 &self.leak_custom.application, | executable/support line; behavior impact depends on surrounding block context. |
| 1021 |                 Self::component_from_idx(self.leak_custom.component_idx), | executable/support line; behavior impact depends on surrounding block context. |
| 1022 |                 Self::oil_type_from_idx(self.leak_custom.oil_type_idx), | executable/support line; behavior impact depends on surrounding block context. |
| 1023 |                 self.leak_custom.seal_count.max(1), | executable/support line; behavior impact depends on surrounding block context. |
| 1024 |                 self.leak_custom.piston_pressure_bar, | executable/support line; behavior impact depends on surrounding block context. |

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| 1025 | self.leak_custom.operation_pressure_bar, | executable/support line; behavior impact
ends on surrounding block context. |
| 1026 | PressureEnvelope { | executable/support line; behavior impact depends on surrounding
ock context. |
| 1027 | min_bar: self.leak_custom.min_bar, | executable/support line; behavior impact de
ds on surrounding block context. |
| 1028 | mean_bar: self.leak_custom.mean_bar, | executable/support line; behavior impact
ends on surrounding block context. |
| 1029 | ideal_bar: self.leak_custom.ideal_bar, | executable/support line; behavior impac
depends on surrounding block context. |
| 1030 | max_bar: self.leak_custom.max_bar, | executable/support line; behavior impact de
ds on surrounding block context. |
| 1031 | rupture_bar: self | executable/support line; behavior impact depends on surround
block context. |
| 1032 | .leak_custom | executable/support line; behavior impact depends on surroundi
block context. |
| 1033 | .rupture_bar | executable/support line; behavior impact depends on surroundi
block context. |
| 1034 | .max(self.leak_custom.max_bar + 0.1), | executable/support line; behavior im
t depends on surrounding block context. |
| 1035 | }, | executable/support line; behavior impact depends on surrounding block context.
| 1036 | OringSpec { | executable/support line; behavior impact depends on surrounding block
text. |
| 1037 | id_tag: format!("{}", self.leak_custom.name), | executable/support line;
avior impact depends on surrounding block context. |
| 1038 | material: Self::material_from_idx(self.leak_custom.material_idx), | executable/s
ort line; behavior impact depends on surrounding block context. |
| 1039 | shore_a: self.leak_custom.shore_a, | executable/support line; behavior impact de
ds on surrounding block context. |
| 1040 | cross_section_mm: self.leak_custom.cross_section_mm, | executable/support line;
avior impact depends on surrounding block context. |
| 1041 | squeeze_pct: self.leak_custom.squeeze_pct, | executable/support line; behavior i
ct depends on surrounding block context. |
| 1042 | extrusion_gap_mm: self.leak_custom.extrusion_gap_mm, | executable/support line;
avior impact depends on surrounding block context. |
| 1043 | compression_set_pct: self.leak_custom.compression_set_pct, | executable/support
e; behavior impact depends on surrounding block context. |
| 1044 | design_life_hours: self.leak_custom.design_life_hours, | executable/support line
ehavior impact depends on surrounding block context. |
| 1045 | base_leak_area_mm2: self.leak_custom.base_leak_area_mm2, | executable/support li
behavior impact depends on surrounding block context. |
| 1046 | discharge_coeff: self.leak_custom.discharge_coeff, | executable/support line; be
ior impact depends on surrounding block context. |
| 1047 | }, | executable/support line; behavior impact depends on surrounding block context.
| 1048 | self.leak_custom.reservoir_volume_l, | executable/support line; behavior impact depe
on surrounding block context. |
| 1049 | self.leak_custom.support_lpm, | executable/support line; behavior impact depends on
rounding block context. |
| 1050 | ); | executable/support line; behavior impact depends on surrounding block context. |
| 1051 | self.bench.add_custom_leak_circuit(c); | executable/support line; behavior impact depend
n surrounding block context. |
| 1052 | self.leak_sel_idx = self.bench.leak_rig.circuits.len().saturating_sub(1); | executable/s
ort line; behavior impact depends on surrounding block context. |
| 1053 | self.sync_leak_manual_from_selected(); | executable/support line; behavior impact depend
n surrounding block context. |
| 1054 | self.leak_note = "Circuito custom adicionado".into(); | executable/support line; behavio
mpact depends on surrounding block context. |
| 1055 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1056 | Cmd::LeakExportReportCsv(path) => { | executable/support line; behavior impact depends on su
unding block context. |
| 1057 | self.leak_note = match self.bench.export_leak_report_csv(&path) { | executable/support l
; behavior impact depends on surrounding block context. |
| 1058 | Ok(_) => format!("CSV salvo em {}", path), | executable/support line; behavior impac
depends on surrounding block context. |

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| 1059 | Err(e) => format!("Falha export CSV: {}", e), | executable/support line; behavior im
t depends on surrounding block context. |
| 1060 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 1061 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1062 | Cmd::LeakExportReportJson(path) => { | executable/support line; behavior impact depends on s
ounding block context. |
| 1063 | self.leak_note = match self.bench.export_leak_report_json(&path) { | executable/support
e; behavior impact depends on surrounding block context. |
| 1064 | Ok(_) => format!("JSON salvo em {}", path), | executable/support line; behavior impa
depends on surrounding block context. |
| 1065 | Err(e) => format!("Falha export JSON: {}", e), | executable/support line; behavior i
ct depends on surrounding block context. |
| 1066 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 1067 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1068 | Cmd::LeakExportPredCsv(path) => { | executable/support line; behavior impact depends on surr
ding block context. |
| 1069 | self.leak_note = match self | executable/support line; behavior impact depends on surrou
ng block context. |
| 1070 | .bench | executable/support line; behavior impact depends on surrounding block conte
|
| 1071 | .export_leak_predictions_csv(&path, &self.leak_predictions) | executable/support lin
behavior impact depends on surrounding block context. |
| 1072 | { | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 1073 | Ok(_) => format!("Pred CSV salvo em {}", path), | executable/support line; behavior
act depends on surrounding block context. |
| 1074 | Err(e) => format!("Falha export pred CSV: {}", e), | executable/support line; behavi
impact depends on surrounding block context. |
| 1075 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 1076 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1077 | Cmd::LeakExportPredJson(path) => { | executable/support line; behavior impact depends on sur
nding block context. |
| 1078 | self.leak_note = match self | executable/support line; behavior impact depends on surrou
ng block context. |
| 1079 | .bench | executable/support line; behavior impact depends on surrounding block conte
|
| 1080 | .export_leak_predictions_json(&path, &self.leak_predictions) | executable/support li
behavior impact depends on surrounding block context. |
| 1081 | { | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 1082 | Ok(_) => format!("Pred JSON salvo em {}", path), | executable/support line; behavior
pact depends on surrounding block context. |
| 1083 | Err(e) => format!("Falha export pred JSON: {}", e), | executable/support line; behav
impact depends on surrounding block context. |
| 1084 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 1085 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1086 | Cmd::LeakExportCatalogMaterialsCsv(path) => { | executable/support line; behavior impact dep
s on surrounding block context. |
| 1087 | self.leak_note = match self.bench.leak_rig.export_material_catalog_csv(&path) { | execut
e/support line; behavior impact depends on surrounding block context. |
| 1088 | Ok(_) => format!("Catalogo materiais CSV salvo em {}", path), | executable/support l
; behavior impact depends on surrounding block context. |
| 1089 | Err(e) => format!("Falha export catalogo materiais CSV: {}", e), | executable/suppor
ine; behavior impact depends on surrounding block context. |
| 1090 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 1091 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet

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scopes. |
| 1092 | Cmd::LeakExportCatalogMaterialsJson(path) => { | executable/support line; behavior impact de
ds on surrounding block context. |
| 1093 | self.leak_note = | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1094 | match self.bench.leak_rig.export_material_catalog_json(&path) { | branch dispatch; c
ging arms alters behavior for variants and input classes. |
| 1095 | Ok(_) => format!("Catalogo materiais JSON salvo em {}", path), | executable/supp
line; behavior impact depends on surrounding block context. |
| 1096 | Err(e) => format!("Falha export catalogo materiais JSON: {}", e), | executable/s
ort line; behavior impact depends on surrounding block context. |
| 1097 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |
| 1098 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1099 | Cmd::LeakExportCatalogOilsCsv(path) => { | executable/support line; behavior impact depends
surrounding block context. |
| 1100 | self.leak_note = match self.bench.leak_rig.export_oil_catalog_csv(&path) { | executable/
port line; behavior impact depends on surrounding block context. |
| 1101 | Ok(_) => format!("Catalogo oleos CSV salvo em {}", path), | executable/support line;
havior impact depends on surrounding block context. |
| 1102 | Err(e) => format!("Falha export catalogo oleos CSV: {}", e), | executable/support li
behavior impact depends on surrounding block context. |
| 1103 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 1104 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1105 | Cmd::LeakExportCatalogOilsJson(path) => { | executable/support line; behavior impact depends
surrounding block context. |
| 1106 | self.leak_note = match self.bench.leak_rig.export_oil_catalog_json(&path) { | executable
pport line; behavior impact depends on surrounding block context. |
| 1107 | Ok(_) => format!("Catalogo oleos JSON salvo em {}", path), | executable/support line
ehavior impact depends on surrounding block context. |
| 1108 | Err(e) => format!("Falha export catalogo oleos JSON: {}", e), | executable/support l
; behavior impact depends on surrounding block context. |
| 1109 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 1110 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1111 | Cmd::LeakCalibrateFromCsv(path) => { | executable/support line; behavior impact depends on s
ounding block context. |
| 1112 | self.leak_note = match self.bench.calibrate_leak_model_from_csv(&path) { | executable/su
rt line; behavior impact depends on surrounding block context. |
| 1113 | Ok(report) => { | executable/support line; behavior impact depends on surrounding bl
context. |
| 1114 | let circuits = report.calibrated_circuits; | executable/support line; behavior i
ct depends on surrounding block context. |
| 1115 | let samples = report.total_samples; | executable/support line; behavior impact d
nds on surrounding block context. |
| 1116 | self.leak_calibration_report = Some(report); | executable/support line; behavior
pact depends on surrounding block context. |
| 1117 | self.leak_calibration_csv_path = path.clone(); | executable/support line; behavi
impact depends on surrounding block context. |
| 1118 | format!( | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1119 | "Calibracao concluida: {} circuitos, {} amostras", | executable/support line
ehavior impact depends on surrounding block context. |
| 1120 | circuits, samples | executable/support line; behavior impact depends on surr
ding block context. |
| 1121 | ) | executable/support line; behavior impact depends on surrounding block context
|
| 1122 | } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 1123 | Err(e) => format!("Falha calibracao CSV: {}", e), | executable/support line; behavio
mpact depends on surrounding block context. |

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| 1124 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 1125 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1126 |           Cmd::LeakExportCalibrationCsv(path) => { | executable/support line; behavior impact depends
surrounding block context. |
| 1127 |               self.leak_note = match self.leak_calibration_report.as_ref() { | executable/support line
ehavior impact depends on surrounding block context. |
| 1128 |                   Some(report) => match self | executable/support line; behavior impact depends on surr
nding block context. |
| 1129 |                       .bench | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1130 |                       .export_leak_calibration_report_csv(&path, report) | executable/support line; be
ior impact depends on surrounding block context. |
| 1131 |                   { | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 1132 |                       Ok(_) => format!("Calibracao CSV salva em {}", path), | executable/support line;
havior impact depends on surrounding block context. |
| 1133 |                       Err(e) => format!("Falha export calibracao CSV: {}", e), | executable/support li
behavior impact depends on surrounding block context. |
| 1134 |                   }, | executable/support line; behavior impact depends on surrounding block context.
| 1135 |                   None => "Execute calibracao antes de exportar".into(), | executable/support line; be
ior impact depends on surrounding block context. |
| 1136 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 1137 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1138 |       Cmd::LeakExportCalibrationJson(path) => { | executable/support line; behavior impact depends
surrounding block context. |
| 1139 |           self.leak_note = match self.leak_calibration_report.as_ref() { | executable/support line
ehavior impact depends on surrounding block context. |
| 1140 |               Some(report) => match self | executable/support line; behavior impact depends on surr
nding block context. |
| 1141 |                   .bench | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1142 |                   .export_leak_calibration_report_json(&path, report) | executable/support line; b
vior impact depends on surrounding block context. |
| 1143 |               { | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 1144 |                   Ok(_) => format!("Calibracao JSON salva em {}", path), | executable/support line
ehavior impact depends on surrounding block context. |
| 1145 |                   Err(e) => format!("Falha export calibracao JSON: {}", e), | executable/support l
; behavior impact depends on surrounding block context. |
| 1146 |               }, | executable/support line; behavior impact depends on surrounding block context.
| 1147 |               None => "Execute calibracao antes de exportar".into(), | executable/support line; be
ior impact depends on surrounding block context. |
| 1148 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 1149 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1150 |       Cmd::LeakRunMonteCarlo => { | executable/support line; behavior impact depends on surroundin
lock context. |
| 1151 |           self.leak_predictions = self.bench.monte_carlo_leak_predictions( | executable/support li
behavior impact depends on surrounding block context. |
| 1152 |               120, | executable/support line; behavior impact depends on surrounding block context
| 1153 |               self.leak_horizon_s, | executable/support line; behavior impact depends on surroundi
block context. |
| 1154 |               self.leak_scenario_dt.max(0.01), | executable/support line; behavior impact depends
surrounding block context. |
| 1155 |               25.0, | executable/support line; behavior impact depends on surrounding block context
|
| 1156 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 1157 |           self.leak_note = format!( | executable/support line; behavior impact depends on surround
block context. |
| 1158 |               "Monte Carlo concluido: {} amostras", | executable/support line; behavior impact dep

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s on surrounding block context. |
| 1159 |                 self.leak_predictions.len() | executable/support line; behavior impact depends on su
unding block context. |
| 1160 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 1161 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1162 |             Cmd::LeakClearScenarioOutputs => { | executable/support line; behavior impact depends on sur
unding block context. |
| 1163 |                 self.leak_predictions.clear(); | executable/support line; behavior impact depends on sur
unding block context. |
| 1164 |                 self.leak_temporal_trace.clear(); | executable/support line; behavior impact depends on
rounding block context. |
| 1165 |                 self.leak_note = "Saida de simulacao limpa. Pronto para nova rodada.".into(); | executab
support line; behavior impact depends on surrounding block context. |
| 1166 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1167 |             Cmd::LeakResetSimulation => { | executable/support line; behavior impact depends on surround
block context. |
| 1168 |                 self.bench.leak_rig.reset(); | executable/support line; behavior impact depends on surro
ing block context. |
| 1169 |                 self.bench.leak_reports.clear(); | executable/support line; behavior impact depends on s
ounding block context. |
| 1170 |                 self.leak_predictions.clear(); | executable/support line; behavior impact depends on sur
unding block context. |
| 1171 |                 self.leak_temporal_trace.clear(); | executable/support line; behavior impact depends on
rounding block context. |
| 1172 |                 self.leak_calibration_report = None; | executable/support line; behavior impact depends
surrounding block context. |
| 1173 |                 self.leak_note = | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1174 |                 "Leak Lab resetado: estado fisico, alertas, historico e cenarios limpos." | executab
support line; behavior impact depends on surrounding block context. |
| 1175 |                 .into(); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1176 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1177 |             Cmd::LeakApplyAndPredict => { | executable/support line; behavior impact depends on surround
block context. |
| 1178 |                 match self.apply_manual_to_selected_circuit() { | branch dispatch; changing arms alters
avior for variants and input classes. |
| 1179 |                     Ok(_) => { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1180 |                         self.leak_temporal_trace.clear(); | executable/support line; behavior impact dep
s on surrounding block context. |
| 1181 |                         self.leak_predictions = self.bench.predict_leak_scenarios( | executable/support
e; behavior impact depends on surrounding block context. |
| 1182 |                         self.leak_horizon_s, | executable/support line; behavior impact depends on s
ounding block context. |
| 1183 |                         self.leak_scenario_dt.max(0.01), | executable/support line; behavior impact
ends on surrounding block context. |
| 1184 |                         ); | executable/support line; behavior impact depends on surrounding block conte
|
| 1185 |                         self.leak_note = | executable/support line; behavior impact depends on surroundi
block context. |
| 1186 |                         format!("Nova rodada: parametros aplicados + {} cenarios", self.leak_predict
s.len()); | executable/support line; behavior impact depends on surrounding block context. |
| 1187 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 1188 |                     Err(e) => { | executable/support line; behavior impact depends on surrounding block
text. |
| 1189 |                         self.leak_note = format!("Nova rodada: falha ao aplicar parametros: {}", e); | e
utable/support line; behavior impact depends on surrounding block context. |
| 1190 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 1191 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and l

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time scopes. |
| 1192 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 1193 |           Cmd::LeakApplyAndMonteCarlo => { | executable/support line; behavior impact depends on surrounding
ing block context. |
| 1194 |           match self.apply_manual_to_selected_circuit() { | branch dispatch; changing arms alters behavior
avior for variants and input classes. |
| 1195 |           Ok(_) => { | executable/support line; behavior impact depends on surrounding block context. |
ext. |
| 1196 |           self.leak_temporal_trace.clear(); | executable/support line; behavior impact depends on surrounding
s on surrounding block context. |
| 1197 |           self.leak_predictions = self.bench.monte_carlo_leak_predictions( | executable/support line; behavior
rt line; behavior impact depends on surrounding block context. |
| 1198 |           120, | executable/support line; behavior impact depends on surrounding block context. |
ntext. |
| 1199 |           self.leak_horizon_s, | executable/support line; behavior impact depends on surrounding block context.
ounding block context. |
| 1200 |           self.leak_scenario_dt.max(0.01), | executable/support line; behavior impact depends on surrounding block
ends on surrounding block context. |
| 1201 |           25.0, | executable/support line; behavior impact depends on surrounding block context. |
ontext. |
| 1202 |           ); | executable/support line; behavior impact depends on surrounding block context. |
|
| 1203 |           self.leak_note = | executable/support line; behavior impact depends on surrounding block context. |
block context. |
| 1204 |           format!("Nova rodada MC: parametros aplicados + {} amostras", self.leak_predictions.len()); | executable/
support line; behavior impact depends on surrounding block context. |
| 1205 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 1206 |           Err(e) => { | executable/support line; behavior impact depends on surrounding block context. |
text. |
| 1207 |           self.leak_note = format!("Nova rodada MC: falha ao aplicar parametros: {}", e); | executable/
support line; behavior impact depends on surrounding block context. |
| 1208 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 1209 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
time scopes. |
| 1210 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 1211 |           Cmd::CanInjectBitError(idx) => { | executable/support line; behavior impact depends on surrounding
ing block context. |
| 1212 |           let bus = Self::can_bus_from_idx(idx); | executable/support line; behavior impact depends on surrounding
n surrounding block context. |
| 1213 |           self.bench | executable/support line; behavior impact depends on surrounding block context. |
|
| 1214 |           .can_net | executable/support line; behavior impact depends on surrounding block context. |
t. |
| 1215 |           .inject_error_once(bus, auto_breaking::CanErrorKind::Bit); | executable/support line; behavior impact
depends on surrounding block context. |
| 1216 |           self.can_note = format!("Bit error armado em {}", bus); | executable/support line; behavior impact
depends on surrounding block context. |
| 1217 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 1218 |           Cmd::CanInjectAckError(idx) => { | executable/support line; behavior impact depends on surrounding
ing block context. |
| 1219 |           let bus = Self::can_bus_from_idx(idx); | executable/support line; behavior impact depends on surrounding
n surrounding block context. |
| 1220 |           self.bench | executable/support line; behavior impact depends on surrounding block context. |
|
| 1221 |           .can_net | executable/support line; behavior impact depends on surrounding block context. |
t. |
| 1222 |           .inject_error_once(bus, auto_breaking::CanErrorKind::Ack); | executable/support line; behavior impact
depends on surrounding block context. |
| 1223 |           self.can_note = format!("ACK error armado em {}", bus); | executable/support line; behavior impact
depends on surrounding block context. |

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| 1224 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1225 |           Cmd::CanInjectBusOff(idx) => { | executable/support line; behavior impact depends on surroun
g block context. |
| 1226 |           let bus = Self::can_bus_from_idx(idx); | executable/support line; behavior impact depend
n surrounding block context. |
| 1227 |           self.bench.can_net.inject_bus_off_once(bus); | executable/support line; behavior impact
ends on surrounding block context. |
| 1228 |           self.can_note = format!("Bus-Off armado em {}", bus); | executable/support line; behavio
mpact depends on surrounding block context. |
| 1229 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1230 |           Cmd::CanInjectBabbling(idx) => { | executable/support line; behavior impact depends on surro
ing block context. |
| 1231 |           let bus = Self::can_bus_from_idx(idx); | executable/support line; behavior impact depend
n surrounding block context. |
| 1232 |           self.bench | executable/support line; behavior impact depends on surrounding block conte
|
| 1233 |           .can_net | executable/support line; behavior impact depends on surrounding block con
t. |
| 1234 |           .inject_error_once(bus, auto_breaking::CanErrorKind::BabblingIdiot); | executable/sup
t line; behavior impact depends on surrounding block context. |
| 1235 |           self.can_note = format!("Babbling armado em {}", bus); | executable/support line; behavi
impact depends on surrounding block context. |
| 1236 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1237 |           Cmd::CanClearBusInjections(idx) => { | executable/support line; behavior impact depends on s
ounding block context. |
| 1238 |           let bus = Self::can_bus_from_idx(idx); | executable/support line; behavior impact depend
n surrounding block context. |
| 1239 |           self.bench.can_net.clear_injections(Some(bus)); | executable/support line; behavior impa
depends on surrounding block context. |
| 1240 |           self.can_note = format!("Injecoes limpas em {}", bus); | executable/support line; behavi
impact depends on surrounding block context. |
| 1241 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1242 |           Cmd::CanClearAllInjections => { | executable/support line; behavior impact depends on surrou
ng block context. |
| 1243 |           self.bench.can_net.clear_injections(None); | executable/support line; behavior impact de
ds on surrounding block context. |
| 1244 |           self.can_note = "Injecoes limpas em todos os barramentos".into(); | executable/support l
; behavior impact depends on surrounding block context. |
| 1245 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1246 |           Cmd::CanExportSnapshotCsv(path) => { | executable/support line; behavior impact depends on s
ounding block context. |
| 1247 |           self.can_note = match self.bench.export_can_snapshot_csv(&path) { | executable/support l
; behavior impact depends on surrounding block context. |
| 1248 |           Ok(_) => format!("Snapshot CSV salvo em {}", path), | executable/support line; behav
impact depends on surrounding block context. |
| 1249 |           Err(e) => format!("Falha snapshot CSV: {}", e), | executable/support line; behavior
act depends on surrounding block context. |
| 1250 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 1251 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1252 |           Cmd::CanExportSnapshotJson(path) => { | executable/support line; behavior impact depends on
rounding block context. |
| 1253 |           self.can_note = match self.bench.export_can_snapshot_json(&path) { | executable/support
e; behavior impact depends on surrounding block context. |
| 1254 |           Ok(_) => format!("Snapshot JSON salvo em {}", path), | executable/support line; beha
r impact depends on surrounding block context. |
| 1255 |           Err(e) => format!("Falha snapshot JSON: {}", e), | executable/support line; behavior
pact depends on surrounding block context. |
| 1256 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and

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etime scopes. |
| 1257 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1258 |           // Simulation | comment line; documentation intent, no runtime effect. |
| 1259 |           Cmd::Reset => { | executable/support line; behavior impact depends on surrounding block cont
. |
| 1260 |               self.bench.reset(); | executable/support line; behavior impact depends on surrounding bl
context. |
| 1261 |               self.bench.tcm.set_direction(Direction::Forward); | executable/support line; behavior im
t depends on surrounding block context. |
| 1262 |               self.throttle = 0.0; | executable/support line; behavior impact depends on surrounding b
k context. |
| 1263 |               self.brake = 0.0; | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1264 |               self.throttle_target = 0.0; | executable/support line; behavior impact depends on surrou
ng block context. |
| 1265 |               self.brake_target = 0.0; | executable/support line; behavior impact depends on surroundi
block context. |
| 1266 |               self.throttle_cmd_last = 0.0; | executable/support line; behavior impact depends on surr
ding block context. |
| 1267 |               self.brake_cmd_last = 0.0; | executable/support line; behavior impact depends on surroun
g block context. |
| 1268 |               self.throttle_cmd_last_tick = self.ticks; | executable/support line; behavior impact dep
s on surrounding block context. |
| 1269 |               self.brake_cmd_last_tick = self.ticks; | executable/support line; behavior impact depend
n surrounding block context. |
| 1270 |               self.sig_map.clear(); | executable/support line; behavior impact depends on surrounding
ck context. |
| 1271 |               self.events.clear(); | executable/support line; behavior impact depends on surrounding b
k context. |
| 1272 |               self.ticks = 0; | executable/support line; behavior impact depends on surrounding block
text. |
| 1273 |               for v in [ | iteration control; changing bounds/body affects throughput, timing, and pos
ly safety guarantees. |
| 1274 |                   &mut self.pl_rpm, | executable/support line; behavior impact depends on surrounding
ck context. |
| 1275 |                   &mut self.pl_spd, | executable/support line; behavior impact depends on surrounding
ck context. |
| 1276 |                   &mut self.pl_torque, | executable/support line; behavior impact depends on surroundi
block context. |
| 1277 |                   &mut self.pl_fuel, | executable/support line; behavior impact depends on surrounding
ock context. |
| 1278 |                   &mut self.pl_coolant, | executable/support line; behavior impact depends on surround
block context. |
| 1279 |                   &mut self.pl_dpfr, | executable/support line; behavior impact depends on surrounding
ck context. |
| 1280 |                   &mut self.pl_boost, | executable/support line; behavior impact depends on surroundin
lock context. |
| 1281 |                   &mut self.pl_hyd, | executable/support line; behavior impact depends on surrounding
ck context. |
| 1282 |               ] { | executable/support line; behavior impact depends on surrounding block context. |
| 1283 |                   v.clear(); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1284 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 1285 |           self.uds_log.clear(); | I/O or protocol-critical logic; edits can affect integration, ti
g, and diagnostic outcomes. |
| 1286 |           self.ad_waypoints.clear(); | executable/support line; behavior impact depends on surroun
g block context. |
| 1287 |           self.leak_predictions.clear(); | executable/support line; behavior impact depends on sur
nding block context. |
| 1288 |           self.leak_note.clear(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 1289 |           self.can_note.clear(); | executable/support line; behavior impact depends on surrounding
ock context. |

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| 1290 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. | |
| 1291 |             Cmd::Pause \ | Cmd::Resume => {} // UI-only, no bench mutation needed | executable/support line; behavior impact depends on surrounding block context. |  
| 1292 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 1293 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 1294 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 1295 | (blank) | line is blank; visual separation only. |  
| 1296 | // ■ Sync parameter panel from simulation (when not dirty) ██████████ | comment line; documentation, no runtime effect. |  
| 1297 | fn sync_params_from_bench(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 1298 |     if !self.params_dirty { | conditional gate; changing condition shifts runtime path and side effects. |  
| 1299 |         self.p_fuel = self.bench.ecm.fuel_level_pct; | executable/support line; behavior impact depends on surrounding block context. |  
| 1300 |         self.p_def = self.bench.ecm.def_level_pct; | executable/support line; behavior impact depends on surrounding block context. |  
| 1301 |         self.p_coolant = self.bench.ecm.coolant_temp_c; | executable/support line; behavior impact depends on surrounding block context. |  
| 1302 |         self.p_oil_prs = self.bench.ecm.oil_pressure_kpa; | executable/support line; behavior impact depends on surrounding block context. |  
| 1303 |         self.p_boost = self.bench.ecm.boost_pressure_kpa; | executable/support line; behavior impact depends on surrounding block context. |  
| 1304 |         self.p_engine_h = self.bench.ecm.engine_hours; | executable/support line; behavior impact depends on surrounding block context. |  
| 1305 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 1306 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 1307 | (blank) | line is blank; visual separation only. |  
| 1308 | // ■ Update CAN signal map ████████████████████████████████████████████ | comment line; documentation, no runtime effect. |  
| 1309 | fn update_signals(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 1310 |     let t = self.bench.elapsed(); | executable/support line; behavior impact depends on surrounding block context. |  
| 1311 |     self.can_trace_tick_counter = self.can_trace_tick_counter.saturating_add(1); | executable/support line; behavior impact depends on surrounding block context. |  
| 1312 |     if !self.can_freeze { | conditional gate; changing condition shifts runtime path and side effects. |  
| 1313 |         for frame in self.bench.gateway.dispatched.clone() { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |  
| 1314 |             let e = self.sig_map.entry((frame.pgn, frame.sa)).or_insert(Signal { | executable/support line; behavior impact depends on surrounding block context. |  
| 1315 |                 pgn_name: frame.pgn_name(), | executable/support line; behavior impact depends on surrounding block context. |  
| 1316 |                 sa_name: frame.sa_name().into(), | executable/support line; behavior impact depends on surrounding block context. |  
| 1317 |                 raw_id: frame.raw_id, | executable/support line; behavior impact depends on surrounding block context. |  
| 1318 |                 last_ts: t, | executable/support line; behavior impact depends on surrounding block context. |  
| 1319 |                 count: 0, | executable/support line; behavior impact depends on surrounding block context. |  
| 1320 |                 period_ms: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 1321 |                 decoded: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |  
| 1322 |                 fresh: true, | executable/support line; behavior impact depends on surrounding block context. |  
| 1323 |             }); | executable/support line; behavior impact depends on surrounding block context. |  
| 1324 |             let dt_ms = (t - e.last_ts) * 1000.0; | executable/support line; behavior impact depends on surrounding block context. |  
| 1325 |             if e.count > 0 { | conditional gate; changing condition shifts runtime path and side effects. |  
| 1326 |                 e.period_ms = e.period_ms * 0.85 + dt_ms * 0.15; | executable/support line; behavior im
```

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depends on surrounding block context. |
| 1327 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 1328 |         e.last_ts = t; | executable/support line; behavior impact depends on surrounding block context. |
|
| 1329 |         e.count += 1; | executable/support line; behavior impact depends on surrounding block context. |
|
| 1330 |         e.fresh = true; | executable/support line; behavior impact depends on surrounding block context. |
. |
| 1331 |         e.decoded = frame | executable/support line; behavior impact depends on surrounding block context. |
xt. |
| 1332 |             .decoded | executable/support line; behavior impact depends on surrounding block context. |
| 1333 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1334 |             .map(\|v\| (v.name.to_string(), v.physical, v.unit.to_string())) | executable/support line; behavior impact depends on surrounding block context. |
behavior impact depends on surrounding block context. |
| 1335 |             .collect(); | executable/support line; behavior impact depends on surrounding block context. |
. |
| 1336 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 1337 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
|
| 1338 |     for sig in self.sig_map.values_mut() { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 1339 |         if t - sig.last_ts > 0.25 { | conditional gate; changing condition shifts runtime path and side effects. |
| 1340 |             sig.fresh = false; | executable/support line; behavior impact depends on surrounding block context. |
ext. |
| 1341 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 1342 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
|
| 1343 |     let trace_due = self | executable/support line; behavior impact depends on surrounding block context. |
| 1344 |         .can_trace_tick_counter | executable/support line; behavior impact depends on surrounding block context. |
text. |
| 1345 |         .is_multiple_of(self.can_trace_update_every_ticks.max(1)); | executable/support line; behavior impact depends on surrounding block context. |
|
| 1346 |         if self.can_trace_pause_ticks_left > 0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 1347 |             self.can_trace_pause_ticks_left = self.can_trace_pause_ticks_left.saturating_sub(1); | executable/support line; behavior impact depends on surrounding block context. |
support line; behavior impact depends on surrounding block context. |
| 1348 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
|
| 1349 |         if !self.can_freeze && trace_due && self.can_trace_pause_ticks_left == 0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 1350 |             self.trace_snap = self | executable/support line; behavior impact depends on surrounding block context. |
ext. |
| 1351 |             .bench | executable/support line; behavior impact depends on surrounding block context. |
| 1352 |             .gateway | executable/support line; behavior impact depends on surrounding block context. |
| 1353 |             .bus | executable/support line; behavior impact depends on surrounding block context. |
| 1354 |             .frames | executable/support line; behavior impact depends on surrounding block context. |
| 1355 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 1356 |             .map(\|f\| { | executable/support line; behavior impact depends on surrounding block context. |
| 1357 |                 ( | executable/support line; behavior impact depends on surrounding block context. |
| 1358 |                     f.timestamp, | executable/support line; behavior impact depends on surrounding block context. |
ntext. |
| 1359 |                     f.raw_id, | executable/support line; behavior impact depends on surrounding block context. |
xt. |
| 1360 |                     f.sa, | executable/support line; behavior impact depends on surrounding block context. |
|
| 1361 |                     f.dlc, | executable/support line; behavior impact depends on surrounding block context. |
|
| 1362 |                     f.data_hex(), | executable/support line; behavior impact depends on surrounding block context. |
ontext. |
| 1363 |                     format!("{}", f.pgn_name(), f.sa_name()), | executable/support line; behavior impact depends on surrounding block context. |
t depends on surrounding block context. |

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| 1364 |         f.decoded | executable/support line; behavior impact depends on surrounding block context. |
|      | xt. |
| 1365 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
|      | text. |
| 1366 |             .take(3) | executable/support line; behavior impact depends on surrounding block context. |
|      | ntext. |
| 1367 |             .map(\v\ format!("{:.1}{}", v.physical, v.unit)) | executable/support line; behavior impact depends on surrounding block context. |
| 1368 |             .collect::<Vec<_>>() | executable/support line; behavior impact depends on surrounding block context. |
| 1369 |             .join(" "), | executable/support line; behavior impact depends on surrounding block context. |
| 1370 |                 ) | executable/support line; behavior impact depends on surrounding block context. |
| 1371 |             }) | executable/support line; behavior impact depends on surrounding block context. |
| 1372 |             .collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 1373 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1374 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1375 | (blank) | line is blank; visual separation only. |
| 1376 |     // ■ Generate meaningful events ████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 1377 |     fn detect_events(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1378 |         if self.ev_pause { | conditional gate; changing condition shifts runtime path and side effects. |
| 1379 |             return; | executable/support line; behavior impact depends on surrounding block context. |
| 1380 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1381 |         let t = self.bench.elapsed; | executable/support line; behavior impact depends on surrounding block context. |
| 1382 |         let ecm = &self.bench.ecm; | executable/support line; behavior impact depends on surrounding block context. |
| 1383 |         let abs = &self.bench.abs; | executable/support line; behavior impact depends on surrounding block context. |
| 1384 |         let ign = self.bench.ignition(); | executable/support line; behavior impact depends on surrounding block context. |
| 1385 | (blank) | line is blank; visual separation only. |
| 1386 |         macro_rules! ev { | executable/support line; behavior impact depends on surrounding block context. |
| 1387 |             ($src:expr,$msg:expr,$lvl:expr) => { | executable/support line; behavior impact depends on surrounding block context. |
| 1388 |                 if self.events.len() < EVENT_MAX { | conditional gate; changing condition shifts runtime path and side effects. |
| 1389 |                     self.events.push(AppEvent { | executable/support line; behavior impact depends on surrounding block context. |
| 1390 |                         ts: t, | executable/support line; behavior impact depends on surrounding block context. |
| 1391 |                         source: $src, | executable/support line; behavior impact depends on surrounding block context. |
| 1392 |                         msg: $msg.into(), | executable/support line; behavior impact depends on surrounding block context. |
| 1393 |                         lvl: $lvl, | executable/support line; behavior impact depends on surrounding block context. |
| 1394 |                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 1395 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1396 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1397 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1398 | (blank) | line is blank; visual separation only. |
| 1399 |         // Ignition state | comment line; documentation intent, no runtime effect. |
| 1400 |         if ign != self.prev_ign { | conditional gate; changing condition shifts runtime path and side effects. |
| 1401 |             ev!( | executable/support line; behavior impact depends on surrounding block context. |
| 1402 |                 "IGN", | executable/support line; behavior impact depends on surrounding block context. |
| 1403 |                 format!("Ignition state changed from {} to {}", self.prev_ign, ign), | executable/support line; behavior impact depends on surrounding block context.
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| 1404 |           "Key → {:?} [{}]", | error-propagating operation; changing it impacts failure semantic
d caller recovery paths. |
| 1405 |           ign, | executable/support line; behavior impact depends on surrounding block context. |
| 1406 |           if self.bench | conditional gate; changing condition shifts runtime path and side effect
|
| 1407 |           .boot | executable/support line; behavior impact depends on surrounding block context
|
| 1408 |           .crank_inhibited { "INHIBITED" } else { "OK" } | executable/support line; behavior i
ct depends on surrounding block context. |
| 1409 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 1410 |           EventLevel::Info | executable/support line; behavior impact depends on surrounding block con
t. |
| 1411 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 1412 |           self.prev_ign = ign; | executable/support line; behavior impact depends on surrounding block con
t. |
| 1413 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1414 |       // Engine start/stop | comment line; documentation intent, no runtime effect. |
| 1415 |       if ecm.rpm > 400.0 && self.prev_rpm <= 400.0 { | conditional gate; changing condition shifts runtime
th and side effects. |
| 1416 |           ev!( | executable/support line; behavior impact depends on surrounding block context. |
| 1417 |               "ECM", | executable/support line; behavior impact depends on surrounding block context. |
| 1418 |               format!("ENGINE STARTED RPM={:.0} T={:.1}s", ecm.rpm, t), | executable/support line; behav
impact depends on surrounding block context. |
| 1419 |               EventLevel::Ok | executable/support line; behavior impact depends on surrounding block conte
|
| 1420 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 1421 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1422 |       if ecm.rpm <= 400.0 && self.prev_rpm > 400.0 { | conditional gate; changing condition shifts runtime
th and side effects. |
| 1423 |           ev!("ECM", "ENGINE STOPPED", EventLevel::Warn); | executable/support line; behavior impact dependen
on surrounding block context. |
| 1424 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1425 |       self.prev_rpm = ecm.rpm; | executable/support line; behavior impact depends on surrounding block con
t. |
| 1426 |       // Gear change | comment line; documentation intent, no runtime effect. |
| 1427 |       let gear = self.bench.tcm.gear_label.clone(); | executable/support line; behavior impact depends on
rounding block context. |
| 1428 |       if gear != self.prev_gear { | conditional gate; changing condition shifts runtime path and side effe
. |
| 1429 |           ev!( | executable/support line; behavior impact depends on surrounding block context. |
| 1430 |               "TCM", | executable/support line; behavior impact depends on surrounding block context. |
| 1431 |               format!( | executable/support line; behavior impact depends on surrounding block context. |
| 1432 |                   "GEAR {} → {} (RPM {:.0} Spd {:.1}km/h)", | executable/support line; behavior impact
ends on surrounding block context. |
| 1433 |                   self.prev_gear, gear, ecm.rpm, self.bench.tcm.ground_speed_kmh | executable/support line
ehavior impact depends on surrounding block context. |
| 1434 |               ), | executable/support line; behavior impact depends on surrounding block context. |
| 1435 |               EventLevel::Info | executable/support line; behavior impact depends on surrounding block con
t. |
| 1436 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 1437 |           self.prev_gear = gear; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1438 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1439 |       // DTC set/clear | comment line; documentation intent, no runtime effect. |
| 1440 |       let dtcs = ecm.active_dtcs.len(); | executable/support line; behavior impact depends on surrounding
ck context. |
| 1441 |       if dtcs > self.prev_dtcs { | conditional gate; changing condition shifts runtime path and side effec
|
| 1442 |           if let Some(d) = ecm.active_dtcs.last() { | conditional gate; changing condition shifts runtime
h and side effects. |
| 1443 |               let severity = format!("{}", d.severity); | executable/support line; behavior impact depends

```

```

surrounding block context. |
| 1444 |         ev!( | executable/support line; behavior impact depends on surrounding block context. |
| 1445 |             "ECM", | executable/support line; behavior impact depends on surrounding block context. |
| 1446 |             format!( | executable/support line; behavior impact depends on surrounding block context. |
| 1447 |                 "DTC SET SPN:{:5} FMI:{:2} [{:4}] {}", | executable/support line; behavior impact
depends on surrounding block context. |
| 1448 |                 d.spn, d.fmi, severity, d.desc | executable/support line; behavior impact depends on
rrounding block context. |
| 1449 |             ), | executable/support line; behavior impact depends on surrounding block context. |
| 1450 |             EventLevel::Critical | executable/support line; behavior impact depends on surrounding b
k context. |
| 1451 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 1452 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1453 | } else if dtcs < self.prev_dtcs { | executable/support line; behavior impact depends on surrounding
ck context. |
| 1454 |     ev!( | executable/support line; behavior impact depends on surrounding block context. |
| 1455 |         "ECM", | executable/support line; behavior impact depends on surrounding block context. |
| 1456 |         format!("DTCs CLEARED ({} were active)", self.prev_dtcs), | executable/support line; behavio
mpact depends on surrounding block context. |
| 1457 |         EventLevel::Ok | executable/support line; behavior impact depends on surrounding block conte
|
| 1458 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 1459 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1460 | self.prev_dtcs = dtcs; | executable/support line; behavior impact depends on surrounding block conte
|
| 1461 | // Warning lamps | comment line; documentation intent, no runtime effect. |
| 1462 | if ecm.red_lamp && self.prev_dtcs == 0 { | conditional gate; changing condition shifts runtime path
side effects. |
| 1463 |     ev!("ECM", "■ RED STOP LAMP ON", EventLevel::Critical); | executable/support line; behavior impa
depends on surrounding block context. |
| 1464 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1465 | if ecm.amber_lamp && t > 1.0 { /* only on transition */ } | conditional gate; changing condition shi
runtime path and side effects. |
| 1466 | // ABS | comment line; documentation intent, no runtime effect. |
| 1467 | if abs.abs_system_active && !self.prev_abs { | conditional gate; changing condition shifts runtime p
and side effects. |
| 1468 |     ev!( | executable/support line; behavior impact depends on surrounding block context. |
| 1469 |         "ABS", | executable/support line; behavior impact depends on surrounding block context. |
| 1470 |         format!( | executable/support line; behavior impact depends on surrounding block context. |
| 1471 |             "ABS ACTIVATED Speed={:.1}km/h Brake={:.0}%", | executable/support line; behavior impa
depends on surrounding block context. |
| 1472 |             self.bench.tcm.ground_speed_kmh, self.bench.brake_pct | executable/support line; behavio
mpact depends on surrounding block context. |
| 1473 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 1474 |         EventLevel::Warn | executable/support line; behavior impact depends on surrounding block con
t. |
| 1475 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 1476 | } else if !abs.abs_system_active && self.prev_abs { | executable/support line; behavior impact depen
on surrounding block context. |
| 1477 |     ev!("ABS", "ABS DEACTIVATED – slip recovered", EventLevel::Ok); | executable/support line; behav
impact depends on surrounding block context. |
| 1478 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1479 | self.prev_abs = abs.abs_system_active; | executable/support line; behavior impact depends on surround
g block context. |
| 1480 | // ESP | comment line; documentation intent, no runtime effect. |
| 1481 | if abs.esp_condition != self.prev_esp { | conditional gate; changing condition shifts runtime path a
side effects. |
| 1482 |     if abs.esp_condition != EspCondition::Neutral { | conditional gate; changing condition shifts ru
me path and side effects. |
| 1483 |         ev!( | executable/support line; behavior impact depends on surrounding block context. |
| 1484 |             "ESP", | executable/support line; behavior impact depends on surrounding block context.

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| 1485 |             format!( | executable/support line; behavior impact depends on surrounding block context. |
| 1486 |                 "ESP INTERVENTION: {:?} Yaw-err={:.1}°/s", | error-propagating operation; changing
impacts failure semantics and caller recovery paths. |
| 1487 |                 abs.esp_condition, abs.esp_yaw_error | executable/support line; behavior impact depends
on surrounding block context. |
| 1488 |             ), | executable/support line; behavior impact depends on surrounding block context. |
| 1489 |             EventLevel::Warn | executable/support line; behavior impact depends on surrounding block co
ntext. |
| 1490 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 1491 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 1492 |         ev!("ESP", "ESP STABLE", EventLevel::Ok); | executable/support line; behavior impact depends
surrounding block context. |
| 1493 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1494 |         self.prev_esp = abs.esp_condition; | executable/support line; behavior impact depends on surround
g block context. |
| 1495 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1496 |         // DPF regen | comment line; documentation intent, no runtime effect. |
| 1497 |         let regen = self.bench.ecm.regen_requested; | executable/support line; behavior impact depends on su
unding block context. |
| 1498 |         if regen && !self.prev_regen { | conditional gate; changing condition shifts runtime path and side e
cts. |
| 1499 |             ev!( | executable/support line; behavior impact depends on surrounding block context. |
| 1500 |                 "DPF", | executable/support line; behavior impact depends on surrounding block context. |
| 1501 |                 format!( | executable/support line; behavior impact depends on surrounding block context. |
| 1502 |                     "DPF REGEN STARTED Soot={:.0}% Exhaust={:.0}°C", | executable/support line; behavior i
ct depends on surrounding block context. |
| 1503 |                     ecm.dpf_soot_pct, ecm.exhaust_temp_c | executable/support line; behavior impact depends
surrounding block context. |
| 1504 |                 ), | executable/support line; behavior impact depends on surrounding block context. |
| 1505 |                 EventLevel::Warn | executable/support line; behavior impact depends on surrounding block co
t. |
| 1506 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 1507 |         } else if !regen && self.prev_regen { | executable/support line; behavior impact depends on surround
block context. |
| 1508 |             ev!( | executable/support line; behavior impact depends on surrounding block context. |
| 1509 |                 "DPF", | executable/support line; behavior impact depends on surrounding block context. |
| 1510 |                 format!("DPF REGEN COMPLETE Soot={:.0}%", ecm.dpf_soot_pct), | executable/support line; beh
or impact depends on surrounding block context. |
| 1511 |                 EventLevel::Ok | executable/support line; behavior impact depends on surrounding block conte
|
| 1512 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 1513 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1514 |         self.prev_regen = regen; | executable/support line; behavior impact depends on surrounding block con
t. |
| 1515 |         // Coolant overheat | comment line; documentation intent, no runtime effect. |
| 1516 |         if ecm.coolant_temp_c > 102.0 && self.prev_rpm > 400.0 | conditional gate; changing condition shifts
runtime path and side effects. |
| 1517 |             && (t as u32).is_multiple_of(5) { | executable/support line; behavior impact depends on surround
block context. |
| 1518 |                 // throttle event rate | comment line; documentation intent, no runtime effect. |
| 1519 |                 ev!( | executable/support line; behavior impact depends on surrounding block context. |
| 1520 |                     "ECM", | executable/support line; behavior impact depends on surrounding block context. |
| 1521 |                     format!("HIGH COOLANT TEMP {:.1}°C", ecm.coolant_temp_c), | executable/support line; beh
or impact depends on surrounding block context. |
| 1522 |                     EventLevel::Critical | executable/support line; behavior impact depends on surrounding b
k context. |
| 1523 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 1524 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1525 |             // DEF warning | comment line; documentation intent, no runtime effect. |
| 1526 |             if ecm.def_level_pct < 10.0 && ecm.def_level_pct > 0.0 | conditional gate; changing condition shifts
runtime path and side effects. |

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```

| 1527 |         && (t as u32).is_multiple_of(30) { | executable/support line; behavior impact depends on surround
g block context. |
| 1528 |             ev!( | executable/support line; behavior impact depends on surrounding block context. |
| 1529 |                 "SCR", | executable/support line; behavior impact depends on surrounding block context.
| 1530 |                 format!( | executable/support line; behavior impact depends on surrounding block context
| 1531 |                 "DEF LOW {:.1}% NOx={:.0}ppm", | executable/support line; behavior impact depends o
urrounding block context. |
| 1532 |                 ecm.def_level_pct, ecm.nox_tailpipe_ppm | executable/support line; behavior impact d
nds on surrounding block context. |
| 1533 |             ), | executable/support line; behavior impact depends on surrounding block context. |
| 1534 |             EventLevel::Warn | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1535 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 1536 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1537 |     // Boot events | comment line; documentation intent, no runtime effect. |
| 1538 |     let mode = format!("{}", self.bench.vcm.mode); | executable/support line; behavior impact depends on
rrounding block context. |
| 1539 |     if mode != self.prev_mode { | conditional gate; changing condition shifts runtime path and side effe
. |
| 1540 |         ev!("VCM", format!("Vehicle mode → {}", mode), EventLevel::Info); | executable/support line; beh
or impact depends on surrounding block context. |
| 1541 |         self.prev_mode = mode; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1542 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1543 |     // NM state | comment line; documentation intent, no runtime effect. |
| 1544 |     let nm_state = format!("{}", self.bench.net_mgmt.bus_state); | executable/support line; behavior imp
depends on surrounding block context. |
| 1545 |     let _ = nm_state; // could add NM events | executable/support line; behavior impact depends on surro
ing block context. |
| 1546 |         // TCS | comment line; documentation intent, no runtime effect. |
| 1547 |     if abs.tcs_system_active | conditional gate; changing condition shifts runtime path and side effects
| 1548 |         && (t as u32).is_multiple_of(2) { | executable/support line; behavior impact depends on surround
block context. |
| 1549 |             ev!( | executable/support line; behavior impact depends on surrounding block context. |
| 1550 |                 "TCS", | executable/support line; behavior impact depends on surrounding block context.
| 1551 |                 format!( | executable/support line; behavior impact depends on surrounding block context
| 1552 |                 "TCS cut={:.0}% Speed={:.1}km/h", | executable/support line; behavior impact depend
n surrounding block context. |
| 1553 |                 abs.tcs_throttle_cut * 100.0, | executable/support line; behavior impact depends on
rounding block context. |
| 1554 |                 self.bench.tcm.ground_speed_kmh | executable/support line; behavior impact depends o
urrounding block context. |
| 1555 |             ), | executable/support line; behavior impact depends on surrounding block context. |
| 1556 |             EventLevel::Warn | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1557 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 1558 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1559 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1560 | (blank) | line is blank; visual separation only. |
| 1561 | fn push_plots(&mut self) { | declares executable behavior; signature/body changes alter API, control flo
and outputs. |
| 1562 |     let t = self.bench.elapsed; | executable/support line; behavior impact depends on surrounding block
text. |
| 1563 |     macro_rules! p { | executable/support line; behavior impact depends on surrounding block context. |
| 1564 |         ($v:expr, $x:expr) => { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 1565 |             $v.push([t, $x]); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1566 |             if $v.len() > PLOT_WINDOW { | conditional gate; changing condition shifts runtime path and s
effects. |
| 1567 |                 $v.remove(0); | executable/support line; behavior impact depends on surrounding block co
xt. |

```

```
| 1568 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
scopes. |  
| 1569 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
opes. |  
| 1570 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 1571 |         p!(self.pl_rpm, self.bench.ecm.rpm); | executable/support line; behavior impact depends on surroundi  
block context. |  
| 1572 |         p!(self.pl_spd, self.bench.tcm.ground_speed_kmh); | executable/support line; behavior impact depends  
surrounding block context. |  
| 1573 |         p!(self.pl_torque, self.bench.ecm.actual_torque_nm); | executable/support line; behavior impact depe  
on surrounding block context. |  
| 1574 |         p!(self.pl_fuel, self.bench.ecm.fuel_rate_lph); | executable/support line; behavior impact depends o  
urrounding block context. |  
| 1575 |         p!(self.pl_coolant, self.bench.ecm.coolant_temp_c); | executable/support line; behavior impact depen  
on surrounding block context. |  
| 1576 |         p!(self.pl_dpfp, self.bench.ecm.dpfp_soot_pct); | executable/support line; behavior impact depends on  
rounding block context. |  
| 1577 |         p!(self.pl_boost, self.bench.ecm.boost_pressure_kpa); | executable/support line; behavior impact dep  
s on surrounding block context. |  
| 1578 |         p!(self.pl_hyd, self.bench.hcm.system_pressure_bar); | executable/support line; behavior impact depe  
on surrounding block context. |  
| 1579 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 1580 | (blank) | line is blank; visual separation only. |  
| 1581 |     fn auto_sequence(&mut self) { | declares executable behavior; signature/body changes alter API, control  
w, and outputs. |  
| 1582 |         match self.ticks { | branch dispatch; changing arms alters behavior for variants and input classes.  
| 1583 |             60 => self.cmds.push(Cmd::KeyAdvance), | executable/support line; behavior impact depends on surr  
nding block context. |  
| 1584 |             120 => self.cmds.push(Cmd::KeyAdvance), | executable/support line; behavior impact depends on su  
unding block context. |  
| 1585 |             180 => self.cmds.push(Cmd::KeyAdvance), | executable/support line; behavior impact depends on su  
unding block context. |  
| 1586 |             _ => {} | executable/support line; behavior impact depends on surrounding block context. |  
| 1587 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 1588 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 1589 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 1590 | (blank) | line is blank; visual separation only. |  
| 1591 | fn now_ms() -> u64 { | declares executable behavior; signature/body changes alter API, control flow, and out  
s. |  
| 1592 |     std::time::SystemTime::now() | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 1593 |     .duration_since(std::time::UNIX_EPOCH) | executable/support line; behavior impact depends on surround  
g block context. |  
| 1594 |     .map(|\d| d.as_millis() as u64) | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 1595 |     .unwrap_or(0) | executable/support line; behavior impact depends on surrounding block context. |  
| 1596 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 1597 | (blank) | line is blank; visual separation only. |  
| 1598 | // ████████████████████████████████████████████████████████████████████████████████████████████████ | comm  
intent, no runtime effect. |  
| 1599 | // Entry point | comment line; documentation intent, no runtime effect. |  
| 1600 | // ████████████████████████████████████████████████████████████████████████████████████████████████ | comm  
intent, no runtime effect. |  
| 1601 | fn main() -> eframe::Result<> { | declares executable behavior; signature/body changes alter API, control  
w, and outputs. |  
| 1602 |     let args: Vec<String> = std::env::args().collect(); | executable/support line; behavior impact depends o  
urrounding block context. |  
| 1603 |     let hw_cfg = match io::hw::HwConfig::from_cli_args(&args) { | executable/support line; behavior impact d  
nds on surrounding block context. |  
| 1604 |         Ok(cfg) => cfg, | executable/support line; behavior impact depends on surrounding block context. |  
| 1605 |         Err(e) => { | executable/support line; behavior impact depends on surrounding block context. |  
| 1606 |             eprintln!("[hw-config] invalid arguments, using defaults: {}", e); | executable/support line; be  
ior impact depends on surrounding block context. |
```



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| 1607 |         io::hw::HWConfig::default() | executable/support line; behavior impact depends on surrounding block context. |
| 1608 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1609 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1610 | (blank) | line is blank; visual separation only. |
| 1611 |     match io::hw::write_listen_only_startup_log(&hw_cfg) { | branch dispatch; changing arms alters behavior variants and input classes. |
| 1612 |         Ok(path) => { | executable/support line; behavior impact depends on surrounding block context. |
| 1613 |             println!( | executable/support line; behavior impact depends on surrounding block context. |
| 1614 |                 "[hw-audit] mode={:?} dry_run={} allowlist_present={} write_enabled={} log={}", | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |
| 1615 |                 hw_cfg.mode, | executable/support line; behavior impact depends on surrounding block context. |
| 1616 |                 hw_cfg.dry_run, | executable/support line; behavior impact depends on surrounding block context. |
| 1617 |                 hw_cfg.allowlist_path.is_some(), | executable/support line; behavior impact depends on surrounding block context. |
| 1618 |                 hw_cfg.write_effectively_enabled(), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1619 |                 path.display() | executable/support line; behavior impact depends on surrounding block context. |
| 1620 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 1621 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1622 |         Err(e) => { | executable/support line; behavior impact depends on surrounding block context. |
| 1623 |             eprintln!("[hw-audit] failed to write startup audit: {}", e); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1624 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1625 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1626 | (blank) | line is blank; visual separation only. |
| 1627 |     #[cfg(feature = "advanced_observability")] | executable/support line; behavior impact depends on surrounding block context. |
| 1628 |     { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1629 |         let _ = tracing_subscriber::fmt() | executable/support line; behavior impact depends on surrounding block context. |
| 1630 |             .json() | executable/support line; behavior impact depends on surrounding block context. |
| 1631 |             .with_current_span(false) | executable/support line; behavior impact depends on surrounding block context. |
| 1632 |             .with_target(false) | executable/support line; behavior impact depends on surrounding block context. |
| 1633 |             .try_init(); | executable/support line; behavior impact depends on surrounding block context. |
| 1634 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1635 | (blank) | line is blank; visual separation only. |
| 1636 |     eframe::run_native( | executable/support line; behavior impact depends on surrounding block context. |
| 1637 |         "■ Heavy Machinery ECU Bench v3.0", | executable/support line; behavior impact depends on surrounding block context. |
| 1638 |         NativeOptions { | executable/support line; behavior impact depends on surrounding block context. |
| 1639 |             viewport: ViewportBuilder::default() | executable/support line; behavior impact depends on surrounding block context. |
| 1640 |                 .with_inner_size([1700.0, 1020.0]) | executable/support line; behavior impact depends on surrounding block context. |
| 1641 |                 .with_min_inner_size([1200.0, 700.0]), | executable/support line; behavior impact depends on surrounding block context. |
| 1642 |                 ..Default::default() | executable/support line; behavior impact depends on surrounding block context. |
| 1643 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 1644 |             Box::new(move \|cc\| Ok(Box::new(App::new(cc, hw_cfg.clone())))), | executable/support line; behavior impact depends on surrounding block context. |
| 1645 |         ) | executable/support line; behavior impact depends on surrounding block context. |
| 1646 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1647 | (blank) | line is blank; visual separation only. |
| 1648 | // ████████████████████████████████████████████████████████████████████████████████████████████████████████████████ | comment intent, no runtime effect. |
| 1649 | // Main loop | comment line; documentation intent, no runtime effect. |

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```

1650 | // ████████████████████████████████████████████████████████████████████████████ | comment
intent, no runtime effect. |
| 1651 | impl eframe::App for App { | implementation block; method behavior and trait conformance are defined here. |
| 1652 |     fn update(&ampmut self, ctx: &Context, _: &ampmut eframe::Frame) { | declares executable behavior; signature/b
changes alter API, control flow, and outputs. |
| 1653 |         // ███ Execute deferred commands ████████████████████████████████████████████████████████████████████████████ | comment line; do
tent, no runtime effect. |
| 1654 |         self.flush_cmds(); | executable/support line; behavior impact depends on surrounding block context.
| 1655 | (blank) | line is blank; visual separation only. |
| 1656 |         // ███ Simulation tick ████████████████████████████████████████████████████████████████████████████ | comment line;
tent, no runtime effect. |
| 1657 |         if !self.bench.ad.engaged { | conditional gate; changing condition shifts runtime path and side effe
. |
| 1658 |             let thr_step = (DT as f32) * 1.8; // ~0.56s 0->100% | executable/support line; behavior impact d
nds on surrounding block context. |
| 1659 |             let brk_step = (DT as f32) * 3.0; // brakes react faster | executable/support line; behavior imp
depends on surrounding block context. |
| 1660 |             self.throttle = Self::approach_f32(self.throttle, self.throttle_target, thr_step); | executable/s
port line; behavior impact depends on surrounding block context. |
| 1661 |             self.brake = Self::approach_f32(self.brake, self.brake_target, brk_step); | executable/support l
; behavior impact depends on surrounding block context. |
| 1662 | (blank) | line is blank; visual separation only. |
| 1663 |             let thr_eff = Self::shape_pedal(self.throttle, 0.03); | executable/support line; behavior impact
ends on surrounding block context. |
| 1664 |             let brk_eff = Self::shape_pedal(self.brake, 0.02); | executable/support line; behavior impact de
ds on surrounding block context. |
| 1665 |             self.bench.throttle_pct = (thr_eff * 100.0).clamp(0.0, 100.0); | executable/support line; behavi
impact depends on surrounding block context. |
| 1666 |             self.bench.brake_pct = (brk_eff * 100.0).clamp(0.0, 100.0); | executable/support line; behavior
act depends on surrounding block context. |
| 1667 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1668 |         self.auto_sequence(); | executable/support line; behavior impact depends on surrounding block contex
|
| 1669 |         self.bench.tick(DT); | executable/support line; behavior impact depends on surrounding block context
| 1670 |         if self.bench.ad.engaged { | conditional gate; changing condition shifts runtime path and side effec
|
| 1671 |             // Mirror AD control outputs back into UI sliders while engaged. | comment line; documentation i
nt, no runtime effect. |
| 1672 |             self.throttle = (self.bench.throttle_pct / 100.0) as f32; | executable/support line; behavior im
t depends on surrounding block context. |
| 1673 |             self.brake = (self.bench.brake_pct / 100.0) as f32; | executable/support line; behavior impact d
nds on surrounding block context. |
| 1674 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1675 |         if matches!(self.tab, Tab::CanBus) | conditional gate; changing condition shifts runtime path and si
effects. |
| 1676 |             && matches!(self.can_mode, CanMode::Trace) | executable/support line; behavior impact depends on
rounding block context. |
| 1677 |             && self.can_trace_pause_on_scroll | executable/support line; behavior impact depends on surround
block context. |
| 1678 |             && !self.can_freeze | executable/support line; behavior impact depends on surrounding block cont
. |
| 1679 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1680 |             let scroll_delta = ctx.input(|i| i.raw_scroll_delta.length()); | executable/support line; beha
r impact depends on surrounding block context. |
| 1681 |             if scroll_delta > 0.0 { | conditional gate; changing condition shifts runtime path and side effe
. |
| 1682 |                 let hold = (self.can_trace_update_every_ticks.max(1) * 4) as u32; | executable/support line;
avior impact depends on surrounding block context. |
| 1683 |                 self.can_trace_pause_ticks_left = self.can_trace_pause_ticks_left.max(hold); | executable/su
rt line; behavior impact depends on surrounding block context. |
| 1684 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |

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| 1685 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1686 |         if let Some(sel) = self.bench.leak_rig.circuits.get(self.leak_sel_idx) { | conditional gate; changin
ondition shifts runtime path and side effects. |
| 1687 |             if let Some(rep) = self.bench.leak_reports.iter().find(|r| r.name == sel.name) { | conditional
te; changing condition shifts runtime path and side effects. |
| 1688 |                 self.leak_temporal_trace.push_back(( | executable/support line; behavior impact depends on s
ounding block context. |
| 1689 |                     self.bench.elapsed, | executable/support line; behavior impact depends on surrounding bl
context. |
| 1690 |                     rep.current_pressure_bar, | executable/support line; behavior impact depends on surround
block context. |
| 1691 |                     rep.rupture_probability_pct, | executable/support line; behavior impact depends on surro
ing block context. |
| 1692 |                     rep.leak_lpm, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1693 |                 )); | executable/support line; behavior impact depends on surrounding block context. |
| 1694 |                 if self.leak_temporal_trace.len() > 7200 { | conditional gate; changing condition shifts run
e path and side effects. |
| 1695 |                     self.leak_temporal_trace.pop_front(); | executable/support line; behavior impact depends
surrounding block context. |
| 1696 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1697 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1698 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1699 |         self.ticks += 1; | executable/support line; behavior impact depends on surrounding block context. |
| 1700 |         self.update_v2x_kpis(); | executable/support line; behavior impact depends on surrounding block conte
. |
| 1701 |         self.update_signals(); | executable/support line; behavior impact depends on surrounding block conte
|
| 1702 |         self.detect_events(); | executable/support line; behavior impact depends on surrounding block context
|
| 1703 |         self.push_plots(); | executable/support line; behavior impact depends on surrounding block context.
| 1704 |         self.sync_params_from_bench(); | executable/support line; behavior impact depends on surrounding blo
context. |
| 1705 |         let mut clear_live_feed = false; | executable/support line; behavior impact depends on surrounding b
k context. |
| 1706 |         if let Some(feed) = &self.ecm_live_feed { | conditional gate; changing condition shifts runtime path
d side effects. |
| 1707 |             self.ecm_live_snapshot = feed.latest_snapshot(); | executable/support line; behavior impact depe
on surrounding block context. |
| 1708 |             self.ecm_live_last_update_ms = feed.last_update_ms(); | executable/support line; behavior impact
ends on surrounding block context. |
| 1709 |             self.ecm_live_frames_total = feed.frames_total(); | executable/support line; behavior impact dep
s on surrounding block context. |
| 1710 |             if !feed.is_alive() { | conditional gate; changing condition shifts runtime path and side effect
|
| 1711 |                 if let Some(err) = feed.last_error() { | conditional gate; changing condition shifts runtime
th and side effects. |
| 1712 |                     self.ecm_live_status = format!("Retrieve stopped with error: {}", err); | executable/sup
t line; behavior impact depends on surrounding block context. |
| 1713 |                 } else { | executable/support line; behavior impact depends on surrounding block context. |
| 1714 |                     self.ecm_live_status = "Retrieve stopped.".into(); | executable/support line; behavior i
ct depends on surrounding block context. |
| 1715 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1716 |                 clear_live_feed = true; | executable/support line; behavior impact depends on surrounding bl
context. |
| 1717 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1718 |         } else if matches!(self.hw_cfg.mode, io::hw::HwMode::Sim) { | executable/support line; behavior impa
depends on surrounding block context. |
| 1719 |             self.refresh_mock_ecm_live_from_sim(); | executable/support line; behavior impact depends on sur

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ending block context. |
| 1720 |         self.ecm_live_frames_total = self.ecm_live_frames_total.saturating_add(1); | executable/support
e; behavior impact depends on surrounding block context. |
| 1721 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1722 |     if clear_live_feed { | conditional gate; changing condition shifts runtime path and side effects. |
| 1723 |         self.ecm_live_feed = None; | executable/support line; behavior impact depends on surrounding blo
context. |
| 1724 |         self.ecm_connected = false; | executable/support line; behavior impact depends on surrounding bl
context. |
| 1725 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1726 |     ctx.request_repaint(); | executable/support line; behavior impact depends on surrounding block conte
|
| 1727 | (blank) | line is blank; visual separation only. |
| 1728 |     // ■■ Panels ████████████████████████████████████████████████████████████████████████████████ | comment li
ntent, no runtime effect. |
| 1729 |     TopBottomPanel::top("tb") | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1730 |         .min_height(56.0) | executable/support line; behavior impact depends on surrounding block contex
|
| 1731 |         .show(ctx, \|ui\| self.toolbar(ui)); | executable/support line; behavior impact depends on surro
ing block context. |
| 1732 |     TopBottomPanel::top("tabs") | executable/support line; behavior impact depends on surrounding block
text. |
| 1733 |         .exact_height(58.0) | executable/support line; behavior impact depends on surrounding block cont
. |
| 1734 |         .show(ctx, \|ui\| self.tab_bar(ui)); | executable/support line; behavior impact depends on surro
ing block context. |
| 1735 |     TopBottomPanel::bottom("sb") | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1736 |         .exact_height(22.0) | executable/support line; behavior impact depends on surrounding block cont
. |
| 1737 |         .show(ctx, \|ui\| self.statusbar(ui)); | executable/support line; behavior impact depends on sur
nding block context. |
| 1738 |     CentralPanel::default().show(ctx, \|ui\| { | executable/support line; behavior impact depends on sur
nding block context. |
| 1739 |         self.render_ux_guide(ui); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1740 |         if self.ux_compact_mode { | conditional gate; changing condition shifts runtime path and side ef
ts. |
| 1741 |             ui.spacing_mut().item_spacing = Vec2::new(6.0, 5.0); | executable/support line; behavior imp
depends on surrounding block context. |
| 1742 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1743 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 1744 |         match self.tab { | branch dispatch; changing arms alters behavior for variants and input classes
| 1745 |             Tab::Help => ui.push_id("tab::help", \|ui\| self.tab_help(ui)), | executable/support line; b
vior impact depends on surrounding block context. |
| 1746 |             Tab::Cluster => ui.push_id("tab::cluster", \|ui\| self.tab_cluster(ui)), | executable/suppor
ine; behavior impact depends on surrounding block context. |
| 1747 |             Tab::Electrical => ui.push_id("tab::electrical", \|ui\| self.tab_electrical(ui)), | executab
support line; behavior impact depends on surrounding block context. |
| 1748 |             Tab::CanBus => ui.push_id("tab::can", \|ui\| self.tab_can(ui)), | executable/support line; b
vior impact depends on surrounding block context. |
| 1749 |             Tab::Events => ui.push_id("tab::events", \|ui\| self.tab_events(ui)), | executable/support l
; behavior impact depends on surrounding block context. |
| 1750 |             Tab::EcuNet => ui.push_id("tab::ecu_net", \|ui\| self.tab_ecu_net(ui)), | executable/support
ne; behavior impact depends on surrounding block context. |
| 1751 |             Tab::Engine => ui.push_id("tab::engine", \|ui\| self.tab_engine(ui)), | executable/support l
; behavior impact depends on surrounding block context. |
| 1752 |             Tab::Faults => ui.push_id("tab::faults", \|ui\| self.tab_faults(ui)), | executable/support l
; behavior impact depends on surrounding block context. |
| 1753 |             Tab::Boot => ui.push_id("tab::boot", \|ui\| self.tab_boot(ui)), | executable/support line; b
vior impact depends on surrounding block context. |

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[illegible]

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| 1792 | (Tab::EcuNet, "■ ECU NET"), | executable/support line; behavior impact depends on su
unding block context. |
| 1793 | (Tab::Faults, "■ FAULTS"), | executable/support line; behavior impact depends on sur
nding block context. |
| 1794 | (Tab::Boot, "■ BOOT"), | executable/support line; behavior impact depends on surroun
g block context. |
| 1795 | (Tab::Params, "■ PARAMS"), | executable/support line; behavior impact depends on sur
nding block context. |
| 1796 | (Tab::Sensors, "■ SENSORS"), | executable/support line; behavior impact depends on s
ounding block context. |
| 1797 | (Tab::V2X, "■ V2X"), | executable/support line; behavior impact depends on surroundi
block context. |
| 1798 | (Tab::Uds, "■ UDS"), | I/O or protocol-critical logic; edits can affect integration,
ming, and diagnostic outcomes. |
| 1799 | (Tab::EcmLiveData, "■ ECM LIVE"), | executable/support line; behavior impact depends
surrounding block context. |
| 1800 | ], | executable/support line; behavior impact depends on surrounding block context. |
| 1801 | ), | executable/support line; behavior impact depends on surrounding block context. |
| 1802 | ]; | executable/support line; behavior impact depends on surrounding block context. |
| 1803 | (blank) | line is blank; visual separation only. |
| 1804 | for (section_name, tabs) in nav_sections { | iteration control; changing bounds/body affects thro
hput, timing, and possibly safety guarantees. |
| 1805 | ui.horizontal_wrapped(ui\| { | executable/support line; behavior impact depends on surround
g block context. |
| 1806 | ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 1807 | RichText::new(section_name) | executable/support line; behavior impact depends on su
unding block context. |
| 1808 | .size(9.6) | executable/support line; behavior impact depends on surrounding blo
context. |
| 1809 | .color(Color32::from_gray(145)) | executable/support line; behavior impact dependen
on surrounding block context. |
| 1810 | .strong(), | executable/support line; behavior impact depends on surrounding blo
context. |
| 1811 | ); | executable/support line; behavior impact depends on surrounding block context. |
| 1812 | ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 1813 | for (t, lbl) in tabs { | iteration control; changing bounds/body affects throughput, tim
, and possibly safety guarantees. |
| 1814 | let sel = self.tab == *t; | executable/support line; behavior impact depends on surr
ding block context. |
| 1815 | let fill = if sel { | executable/support line; behavior impact depends on surroundin
lock context. |
| 1816 | Color32::from_rgb(45, 105, 210) | executable/support line; behavior impact dependen
on surrounding block context. |
| 1817 | } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 1818 | Color32::from_gray(30) | executable/support line; behavior impact depends on sur
nding block context. |
| 1819 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |
| 1820 | let label = if *t == Tab::Faults && dtcs > 0 { | executable/support line; behavior i
ct depends on surrounding block context. |
| 1821 | format!("■ FAULTS({})", dtcs) | executable/support line; behavior impact depends
surrounding block context. |
| 1822 | } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 1823 | (*lbl).into() | executable/support line; behavior impact depends on surrounding
ck context. |
| 1824 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |
| 1825 | let col = if *t == Tab::Faults && dtcs > 0 { | executable/support line; behavior imp
depends on surrounding block context. |
| 1826 | Color32::RED | executable/support line; behavior impact depends on surrounding b
k context. |

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| 1827 |           } else if sel { | executable/support line; behavior impact depends on surrounding block context. |
| 1828 |           Color32::WHITE | executable/support line; behavior impact depends on surrounding block context. |
| 1829 |           } else { | executable/support line; behavior impact depends on surrounding block context. |
| 1830 |           Color32::from_gray(160) | executable/support line; behavior impact depends on surrounding block context. |
| 1831 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1832 |           if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 1833 |             .add(Button::new(RichText::new(label).size(10.8).color(col)).fill(fill)) | executable/support line; behavior impact depends on surrounding block context. |
| 1834 |             .clicked() | executable/support line; behavior impact depends on surrounding block context. |
| 1835 |           { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1836 |             self.tab = *t; | executable/support line; behavior impact depends on surrounding block context. |
| 1837 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1838 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1839 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 1840 |         if section_name == "OPERACAO" { | conditional gate; changing condition shifts runtime path and side effects. |
| 1841 |           ui.add_space(2.0); | executable/support line; behavior impact depends on surrounding block context. |
| 1842 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1843 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1844 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 1845 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1846 | (blank) | line is blank; visual separation only. |
| 1847 |   fn ad_readiness_issues(&self) -> Vec<String> { | declares executable behavior; signature/body changes API, control flow, and outputs. |
| 1848 |     let mut issues = Vec::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 1849 |     if !self.bench.engine_running() { | conditional gate; changing condition shifts runtime path and side effects. |
| 1850 |       issues.push("Motor desligado: o AD nao consegue acelerar nem sustentar velocidade.".to_string()); | executable/support line; behavior impact depends on surrounding block context. |
| 1851 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1852 |     if matches!(self.bench.tcm.direction, Direction::Park) { | conditional gate; changing condition shifts runtime path and side effects. |
| 1853 |       issues.push("Transmissao em Park: selecione Forward para permitir deslocamento.".to_string()); | executable/support line; behavior impact depends on surrounding block context. |
| 1854 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1855 |     if self.bench.tcm.auto_mode != AutoShiftMode::Auto { | conditional gate; changing condition shifts runtime path and side effects. |
| 1856 |       issues.push("Transmissao fora de AUTO: o AD perde previsibilidade de troca.".to_string()); | executable/support line; behavior impact depends on surrounding block context. |
| 1857 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1858 |     if self.bench.ad.lane.confidence <= 0.4 { | conditional gate; changing condition shifts runtime path and side effects. |
| 1859 |       issues.push("Lane confidence baixa: o LKA permanece em espera abaixo de 40%.".to_string()); | executable/support line; behavior impact depends on surrounding block context. |
| 1860 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 1861 |     if self.bench.tcm.ground_speed_kmh < 5.0 { | conditional gate; changing condition shifts runtime path and side effects. |

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| 1862 |         issues.push("Abaixo de 5 km/h o LKA ainda nao atua; o ACC pode acelerar a maquina.".to_string())
executable/support line; behavior impact depends on surrounding block context. |
| 1863 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1864 |         issues | executable/support line; behavior impact depends on surrounding block context. |
| 1865 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 1866 | (blank) | line is blank; visual separation only. |
| 1867 |     fn ux_alerts(&self) -> Vec<(Color32, String)> { | declares executable behavior; signature/body changes a
r API, control flow, and outputs. |
| 1868 |         let mut alerts = Vec::new(); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1869 |         match self.tab { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 1870 |             Tab::Cluster \| Tab::Electrical \| Tab::Engine \| Tab::Implements => { | executable/support line
ehavior impact depends on surrounding block context. |
| 1871 |                 if !self.bench.engine_running() { | conditional gate; changing condition shifts runtime path
d side effects. |
| 1872 |                     alerts.push(( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1873 |                         Color32::YELLOW, | executable/support line; behavior impact depends on surrounding b
k context. |
| 1874 |                         "Motor desligado: esta aba so reflete dinamica real depois de RUN.".to_string(), | e
utable/support line; behavior impact depends on surrounding block context. |
| 1875 |                     )); | executable/support line; behavior impact depends on surrounding block context. |
| 1876 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1877 |                 if self.bench.ecm.red_lamp { | conditional gate; changing condition shifts runtime path and
e effects. |
| 1878 |                     alerts.push(( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1879 |                         Color32::RED, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1880 |                         "RED STOP ativo: existe falha critica impactando a operacao.".to_string(), | executa
/support line; behavior impact depends on surrounding block context. |
| 1881 |                     )); | executable/support line; behavior impact depends on surrounding block context. |
| 1882 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1883 |                 if matches!(self.tab, Tab::Electrical) | conditional gate; changing condition shifts runtime
th and side effects. |
| 1884 |                     && self.bench.bcm.fuses.iter().any(\|f\| f.blown) | executable/support line; behavior im
t depends on surrounding block context. |
| 1885 |                 { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1886 |                     alerts.push(( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1887 |                         Color32::RED, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1888 |                         "Existe ao menos um fusivel aberto: rearme antes de validar o ramo afetado.".to_stri
), | executable/support line; behavior impact depends on surrounding block context. |
| 1889 |                     )); | executable/support line; behavior impact depends on surrounding block context. |
| 1890 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1891 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1892 |             Tab::Autonomous => { | executable/support line; behavior impact depends on surrounding block con
t. |
| 1893 |                 if !self.bench.engine_running() { | conditional gate; changing condition shifts runtime path
d side effects. |
| 1894 |                     alerts.push(( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1895 |                         Color32::YELLOW, | executable/support line; behavior impact depends on surrounding b
k context. |
| 1896 |                         "Motor desligado: o AD nao consegue gerar tracao nem validar longitudinal.".to_strin
, | executable/support line; behavior impact depends on surrounding block context. |
| 1897 |                     )); | executable/support line; behavior impact depends on surrounding block context. |
| 1898 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet

```



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scopes. |
| 1899 |           if self.bench.ad.engaged { | conditional gate; changing condition shifts runtime path and si
effects. |
| 1900 |           let msg = if let Some(reason) = self.bench.ad.degrade_reason { | executable/support line
ehavior impact depends on surrounding block context. |
| 1901 |           format!("AD engatado em modo degradado: {}.", reason) | executable/support line; beh
or impact depends on surrounding block context. |
| 1902 |           } else { | executable/support line; behavior impact depends on surrounding block context
| 1903 |           "AD engatado: comandos manuais principais ficam travados para evitar disputa.".to_st
g() | executable/support line; behavior impact depends on surrounding block context. |
| 1904 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 1905 |           alerts.push((Color32::from_rgb(90, 210, 130), msg)); | executable/support line; behavior
pact depends on surrounding block context. |
| 1906 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1907 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1908 |           Tab::CanBus \| Tab::Events \| Tab::EcuNet \| Tab::Faults \| Tab::Uds => { | I/O or protocol-crit
l logic; edits can affect integration, timing, and diagnostic outcomes. |
| 1909 |           if self.bench.gateway.bus_load_pct > 85.0 { | conditional gate; changing condition shifts ru
me path and side effects. |
| 1910 |           alerts.push(( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1911 |           Color32::YELLOW, | executable/support line; behavior impact depends on surrounding b
k context. |
| 1912 |           format!("Carga CAN alta ({:.0}%): diagnostico pode ficar ruidoso nesta aba.", self.b
h.gateway.bus_load_pct), | executable/support line; behavior impact depends on surrounding block context. |
| 1913 |           )); | executable/support line; behavior impact depends on surrounding block context. |
| 1914 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1915 |           if matches!(self.tab, Tab::Faults \| Tab::Uds) && self.bench.ecm.red_lamp { | conditional ga
changing condition shifts runtime path and side effects. |
| 1916 |           alerts.push(( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1917 |           Color32::RED, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 1918 |           "Falha critica ativa: compare Faults, Events e Engine antes de continuar.".to_string
| executable/support line; behavior impact depends on surrounding block context. |
| 1919 |           )); | executable/support line; behavior impact depends on surrounding block context. |
| 1920 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1921 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1922 |           Tab::EcmLiveData => { | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1923 |           if self.ecm_live_feed.as_ref().is_some_and(\|feed\| !feed.is_alive()) { | conditional gate;
nging condition shifts runtime path and side effects. |
| 1924 |           alerts.push(( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 1925 |           Color32::YELLOW, | executable/support line; behavior impact depends on surrounding b
k context. |
| 1926 |           "ECM Live sem stream valido: reconecte ou reinicie Retrieve Data.".to_string(), | ex
table/support line; behavior impact depends on surrounding block context. |
| 1927 |           )); | executable/support line; behavior impact depends on surrounding block context. |
| 1928 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 1929 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 1930 |           Tab::Help \| Tab::Boot \| Tab::Params \| Tab::Sensors \| Tab::V2X \| Tab::LeakLab \| Tab::Plots
{} | executable/support line; behavior impact depends on surrounding block context. |
| 1931 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 1932 |           alerts | executable/support line; behavior impact depends on surrounding block context. |
| 1933 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.

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| 1934 | (blank) | line is blank; visual separation only. |
| 1935 |     fn ux_context_items(&self) -> Vec<(String, Color32)> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 1936 |         match self.tab { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 1937 |             Tab::Help => vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 1938 |                 ( | executable/support line; behavior impact depends on surrounding block context. |
| 1939 |                     format!("Workspace {:.1}s", self.bench.elapsed), | executable/support line; behavior impact depends on surrounding block context. |
| 1940 |                         Color32::from_gray(180), | executable/support line; behavior impact depends on surrounding block context. |
| 1941 |                             ), | executable/support line; behavior impact depends on surrounding block context. |
| 1942 |                             ( | executable/support line; behavior impact depends on surrounding block context. |
| 1943 |                                 format!("{}", abas operacionais", 17), | executable/support line; behavior impact depends on surrounding block context. |
| 1944 |                                     Color32::from_rgb(110, 180, 255), | executable/support line; behavior impact depends on surrounding block context. |
| 1945 |                                         ), | executable/support line; behavior impact depends on surrounding block context. |
| 1946 |                                         ( | executable/support line; behavior impact depends on surrounding block context. |
| 1947 |                                             format!("{}", DTC ativos", self.bench.ecm.active_dtcs.len()), | executable/support line; behavior impact depends on surrounding block context. |
| 1948 |                                             if self.bench.ecm.active_dtcs.is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 1949 |                                                 Color32::GREEN | executable/support line; behavior impact depends on surrounding block context. |
| 1950 |                                                     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 1951 |                                                         Color32::YELLOW | executable/support line; behavior impact depends on surrounding block context. |
| 1952 |                                                             }, | executable/support line; behavior impact depends on surrounding block context. |
| 1953 |                                                         ), | executable/support line; behavior impact depends on surrounding block context. |
| 1954 |                             ], | executable/support line; behavior impact depends on surrounding block context. |
| 1955 |             Tab::Cluster => vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 1956 |                 ( | executable/support line; behavior impact depends on surrounding block context. |
| 1957 |                     format!("RPM {:.0}", self.bench.ecm.rpm), | executable/support line; behavior impact depends on surrounding block context. |
| 1958 |                         Color32::GREEN, | executable/support line; behavior impact depends on surrounding block context. |
| 1959 |                             ), | executable/support line; behavior impact depends on surrounding block context. |
| 1960 |                             ( | executable/support line; behavior impact depends on surrounding block context. |
| 1961 |                                 format!("SPD {:.1} km/h", self.bench.tcm.ground_speed_kmh), | executable/support line; behavior impact depends on surrounding block context. |
| 1962 |                                     Color32::LIGHT_BLUE, | executable/support line; behavior impact depends on surrounding block context. |
| 1963 |                                         ), | executable/support line; behavior impact depends on surrounding block context. |
| 1964 |                                         ( | executable/support line; behavior impact depends on surrounding block context. |
| 1965 |                                             format!("Gear {}", self.bench.tcm.gear_label), | executable/support line; behavior impact depends on surrounding block context. |
| 1966 |                                                 Color32::YELLOW, | executable/support line; behavior impact depends on surrounding block context. |
| 1967 |                                                     ), | executable/support line; behavior impact depends on surrounding block context. |
| 1968 |                                                         ], | executable/support line; behavior impact depends on surrounding block context. |
| 1969 |             Tab::Electrical => vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 1970 |                 ( | executable/support line; behavior impact depends on surrounding block context. |
| 1971 |                     format!("Batt {:.1} V", self.bench.bcm.battery_voltage), | executable/support line; behavior impact depends on surrounding block context. |
| 1972 |                         if self.bench.bcm.battery_voltage < 11.8 { | conditional gate; changing condition shifts runtime path and side effects. |
| 1973 |                             Color32::RED | executable/support line; behavior impact depends on surrounding block context. |
| 1974 |                                 } else { | executable/support line; behavior impact depends on surrounding block context. |
| 1975 |                                     Color32::GREEN | executable/support line; behavior impact depends on surrounding block context. |
| 1976 |                                         }, | executable/support line; behavior impact depends on surrounding block context. |

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| 1977 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 1978 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 1979 |             format!("Load {:.0} A", self.bench.bcm.total_load_amps), | executable/support line; beha
r impact depends on surrounding block context. |
| 1980 |             Color32::YELLOW, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1981 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 1982 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 1983 |             format!("Load shed {}", if self.bench.vcm.power.load_shed_active { "on" } else { "off" }
| executable/support line; behavior impact depends on surrounding block context. |
| 1984 |             if self.bench.vcm.power.load_shed_active { | conditional gate; changing condition shifts
untime path and side effects. |
| 1985 |             Color32::RED | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1986 |             } else { | executable/support line; behavior impact depends on surrounding block context
| 1987 |             Color32::from_gray(180) | executable/support line; behavior impact depends on surrou
ng block context. |
| 1988 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 1989 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 1990 |         ], | executable/support line; behavior impact depends on surrounding block context. |
| 1991 |         Tab::CanBus => vec![ | executable/support line; behavior impact depends on surrounding block con
t. |
| 1992 |         ( | executable/support line; behavior impact depends on surrounding block context. |
| 1993 |           format!("Bus load {:.0}%", self.bench.gateway.bus_load_pct), | executable/support line;
avior impact depends on surrounding block context. |
| 1994 |           if self.bench.gateway.bus_load_pct > 85.0 { | conditional gate; changing condition shift
untime path and side effects. |
| 1995 |           Color32::RED | executable/support line; behavior impact depends on surrounding block
ntext. |
| 1996 |           } else { | executable/support line; behavior impact depends on surrounding block context
| 1997 |           Color32::GOLD | executable/support line; behavior impact depends on surrounding block
ontext. |
| 1998 |           }, | executable/support line; behavior impact depends on surrounding block context. |
| 1999 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 2000 |         ( | executable/support line; behavior impact depends on surrounding block context. |
| 2001 |           format!("Signals {}", self.sig_map.len()), | executable/support line; behavior impact de
ds on surrounding block context. |
| 2002 |           Color32::from_rgb(110, 180, 255), | executable/support line; behavior impact depends on
rounding block context. |
| 2003 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2004 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2005 |             format!("Trace {}", self.trace_snap.len()), | executable/support line; behavior impact d
nds on surrounding block context. |
| 2006 |             Color32::from_gray(180), | executable/support line; behavior impact depends on surroundi
block context. |
| 2007 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2008 |         ], | executable/support line; behavior impact depends on surrounding block context. |
| 2009 |         Tab::Events => vec![ | executable/support line; behavior impact depends on surrounding block con
t. |
| 2010 |         ( | executable/support line; behavior impact depends on surrounding block context. |
| 2011 |           format!("Eventos {}", self.events.len()), | executable/support line; behavior impact dep
s on surrounding block context. |
| 2012 |           Color32::from_rgb(110, 180, 255), | executable/support line; behavior impact depends on
rounding block context. |
| 2013 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2014 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2015 |             format!("Filtro {}", if self.ev_filter.is_empty() { "livre" } else { "ativo" })), | execu
le/support line; behavior impact depends on surrounding block context. |
| 2016 |             if self.ev_filter.is_empty() { | conditional gate; changing condition shifts runtime pat
nd side effects. |
| 2017 |             Color32::from_gray(180) | executable/support line; behavior impact depends on surrou
ng block context. |
| 2018 |             } else { | executable/support line; behavior impact depends on surrounding block context
| 2019 |             Color32::YELLOW | executable/support line; behavior impact depends on surrounding bl
context. |

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| 2020 |           }, | executable/support line; behavior impact depends on surrounding block context. |
| 2021 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2022 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2023 |           format!("Pause {}", if self.ev_pause { "on" } else { "off" })), | executable/support line;
behavior impact depends on surrounding block context. |
| 2024 |           if self.ev_pause { Color32::YELLOW } else { Color32::GREEN }, | conditional gate; changing
condition shifts runtime path and side effects. |
| 2025 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2026 |           ], | executable/support line; behavior impact depends on surrounding block context. |
| 2027 |           Tab::EcuNet => vec![ | executable/support line; behavior impact depends on surrounding block con
t. |
| 2028 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2029 |           format!( | executable/support line; behavior impact depends on surrounding block context
| 2030 |           "Online {}/{}", | executable/support line; behavior impact depends on surrounding bl
context. |
| 2031 |           self.bench.boot.ecus.iter().filter(|e| e.is_online()).count(), | executable/support
ine; behavior impact depends on surrounding block context. |
| 2032 |           self.bench.boot.ecus.len() | executable/support line; behavior impact depends on surr
nding block context. |
| 2033 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2034 |           Color32::from_rgb(110, 180, 255), | executable/support line; behavior impact depends on
rounding block context. |
| 2035 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2036 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2037 |           format!("NM {}", self.bench.net_mgmt.bus_state), | executable/support line; behavior imp
depends on surrounding block context. |
| 2038 |           Color32::from_gray(180), | executable/support line; behavior impact depends on surroundi
block context. |
| 2039 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2040 |           ], | executable/support line; behavior impact depends on surrounding block context. |
| 2041 |           Tab::Engine => vec![ | executable/support line; behavior impact depends on surrounding block con
t. |
| 2042 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2043 |           format!("Load {:.0}%", self.bench.ecm.percent_load), | executable/support line; behavior
pact depends on surrounding block context. |
| 2044 |           Color32::YELLOW, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 2045 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2046 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2047 |           format!("Torque {:.0} Nm", self.bench.ecm.actual_torque_nm), | executable/support line;
avior impact depends on surrounding block context. |
| 2048 |           Color32::from_rgb(80, 180, 255), | executable/support line; behavior impact depends on s
ounding block context. |
| 2049 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2050 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2051 |           format!("Coolant {:.0} C", self.bench.ecm.coolant_temp_c), | executable/support line; be
ior impact depends on surrounding block context. |
| 2052 |           if self.bench.ecm.coolant_temp_c > 105.0 { | conditional gate; changing condition shifts
runtime path and side effects. |
| 2053 |           Color32::RED | executable/support line; behavior impact depends on surrounding block
ntext. |
| 2054 |           } else { | executable/support line; behavior impact depends on surrounding block context
| 2055 |           Color32::from_rgb(255, 130, 30) | executable/support line; behavior impact depends o
urrounding block context. |
| 2056 |           }, | executable/support line; behavior impact depends on surrounding block context. |
| 2057 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2058 |           ], | executable/support line; behavior impact depends on surrounding block context. |
| 2059 |           Tab::Faults => vec![ | executable/support line; behavior impact depends on surrounding block con
t. |
| 2060 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2061 |           format!("Active DTC {}", self.bench.ecm.active_dtcs.len()), | executable/support line; b
vior impact depends on surrounding block context. |
| 2062 |           if self.bench.ecm.active_dtcs.is_empty() { | conditional gate; changing condition shifts
runtime path and side effects. |
| 2063 |           Color32::GREEN | executable/support line; behavior impact depends on surrounding blo

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context. |
| 2064 |           } else { | executable/support line; behavior impact depends on surrounding block context. |
| 2065 |           Color32::RED | executable/support line; behavior impact depends on surrounding block context. |
| 2066 |           }, | executable/support line; behavior impact depends on surrounding block context. |
| 2067 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2068 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2069 |           format!("Selected {}", self.bench.selected_fault), | executable/support line; behavior impact depends on surrounding block context. |
| 2070 |           Color32::from_rgb(110, 180, 255), | executable/support line; behavior impact depends on surrounding block context. |
| 2071 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2072 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2073 |           format!("Fault inject {}", if self.bench.fault_active { "armed" } else { "idle" })), | executable/support line; behavior impact depends on surrounding block context. |
| 2074 |           if self.bench.fault_active { Color32::YELLOW } else { Color32::GREEN }, | conditional gate; changing condition shifts runtime path and side effects. |
| 2075 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2076 |           ], | executable/support line; behavior impact depends on surrounding block context. |
| 2077 |           Tab::Boot => vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 2078 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2079 |           format!("Ignition {}", self.bench.ignition()), | executable/support line; behavior impact depends on surrounding block context. |
| 2080 |           Color32::from_rgb(110, 180, 255), | executable/support line; behavior impact depends on surrounding block context. |
| 2081 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2082 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2083 |           format!("Engine {}", if self.bench.engine_running() { "RUN" } else { "OFF" })), | executable/support line; behavior impact depends on surrounding block context. |
| 2084 |           if self.bench.engine_running() { Color32::GREEN } else { Color32::YELLOW }, | conditional gate; changing condition shifts runtime path and side effects. |
| 2085 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2086 |           ], | executable/support line; behavior impact depends on surrounding block context. |
| 2087 |           Tab::Implements => vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 2088 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2089 |           format!("PTO {}", if self.bench.implement.pto_rear_enabled { "on" } else { "off" })), | executable/support line; behavior impact depends on surrounding block context. |
| 2090 |           if self.bench.implement.pto_rear_enabled { Color32::GREEN } else { Color32::from_gray(170) }, | conditional gate; changing condition shifts runtime path and side effects. |
| 2091 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2092 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2093 |           format!("Hyd {:.0} bar", self.bench.hcm.system_pressure_bar), | executable/support line; behavior impact depends on surrounding block context. |
| 2094 |           Color32::from_rgb(0, 210, 210), | executable/support line; behavior impact depends on surrounding block context. |
| 2095 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2096 |           ], | executable/support line; behavior impact depends on surrounding block context. |
| 2097 |           Tab::Params => vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 2098 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2099 |           "Live tuning panel".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 2100 |           Color32::from_rgb(110, 180, 255), | executable/support line; behavior impact depends on surrounding block context. |
| 2101 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2102 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 2103 |           format!("Mode {}", if self.ux_compact_mode { "compact" } else { "full" })), | executable/support line; behavior impact depends on surrounding block context. |
| 2104 |           Color32::from_gray(180), | executable/support line; behavior impact depends on surrounding block context. |
| 2105 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2106 |           ], | executable/support line; behavior impact depends on surrounding block context. |
| 2107 |           Tab::Sensors => vec![ | executable/support line; behavior impact depends on surrounding block context.

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xt. |
| 2108 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2109 |          format!("GPS {:.1} km/h", self.bench.gps.speed_kmh), | executable/support line; behavior
pact depends on surrounding block context. |
| 2110 |          Color32::LIGHT_BLUE, | executable/support line; behavior impact depends on surrounding b
k context. |
| 2111 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2112 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2113 |          format!("Radar TTC {:.1}s", self.bench.radar.ttc_front.min(99.9)), | executable/support
e; behavior impact depends on surrounding block context. |
| 2114 |          if self.bench.radar.ttc_front < 2.0 { Color32::RED } else { Color32::YELLOW }, | conditi
l gate; changing condition shifts runtime path and side effects. |
| 2115 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2116 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2117 |          format!("Lane conf {:.0}%", self.bench.ad.lane.confidence * 100.0), | executable/support
ne; behavior impact depends on surrounding block context. |
| 2118 |          if self.bench.ad.lane.confidence > 0.7 { Color32::GREEN } else { Color32::YELLOW }, | co
tional gate; changing condition shifts runtime path and side effects. |
| 2119 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2120 |          ], | executable/support line; behavior impact depends on surrounding block context. |
| 2121 |          Tab::Autonomous => vec![ | executable/support line; behavior impact depends on surrounding block
ntext. |
| 2122 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2123 |          format!("AD {} ", if self.bench.ad.engaged { "engaged" } else { "standby" })), | executabl
upport line; behavior impact depends on surrounding block context. |
| 2124 |          if self.bench.ad.engaged { Color32::GREEN } else { Color32::from_gray(180) }, | conditio
gate; changing condition shifts runtime path and side effects. |
| 2125 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2126 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2127 |          format!("ACC set {:.0} km/h", self.bench.ad.acc_set_speed_kmh), | executable/support lin
behavior impact depends on surrounding block context. |
| 2128 |          Color32::LIGHT_BLUE, | executable/support line; behavior impact depends on surrounding b
k context. |
| 2129 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2130 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2131 |          format!("Lane {:.0}%", self.bench.ad.lane.confidence * 100.0), | executable/support line
ehavior impact depends on surrounding block context. |
| 2132 |          if self.bench.ad.lane.confidence > 0.7 { Color32::GREEN } else { Color32::YELLOW }, | co
tional gate; changing condition shifts runtime path and side effects. |
| 2133 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2134 |          ], | executable/support line; behavior impact depends on surrounding block context. |
| 2135 |          Tab::V2X => vec![ | executable/support line; behavior impact depends on surrounding block contex
|
| 2136 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2137 |          format!("Link {:.0}%", self.v2x_range_ema.mul_add(0.0, (100.0 - (self.v2x_loss_ema * 2.0
lamp(0.0, 60.0) - ((self.v2x_lat_ema - 10.0) * 1.6).clamp(0.0, 40.0)).clamp(0.0, 100.0))), | executable/support line;
havior impact depends on surrounding block context. |
| 2138 |          Color32::from_rgb(110, 180, 255), | executable/support line; behavior impact depends on
rounding block context. |
| 2139 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2140 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2141 |          format!("Nearby {} ", self.bench.v2x.nearby_vehicles.len()), | executable/support line; b
vior impact depends on surrounding block context. |
| 2142 |          Color32::from_gray(180), | executable/support line; behavior impact depends on surroundi
block context. |
| 2143 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2144 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2145 |          format!("OTA {} ", if self.bench.telematics.ota_downloading { "downloading" } else if sel
ench.telematics.ota_available { "available" } else { "idle" })), | executable/support line; behavior impact depends on
rrounding block context. |
| 2146 |          if self.bench.telematics.ota_downloading { Color32::YELLOW } else { Color32::GREEN }, |
ditional gate; changing condition shifts runtime path and side effects. |
| 2147 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2148 |          ], | executable/support line; behavior impact depends on surrounding block context. |
| 2149 |          Tab::Uds => vec![ | I/O or protocol-critical logic; edits can affect integration, timing, and di

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ostic outcomes. |
| 2150 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2151 |          format!("Session {}", self.bench.uds_ecm.session), | I/O or protocol-critical logic; edi
can affect integration, timing, and diagnostic outcomes. |
| 2152 |          Color32::from_rgb(110, 180, 255), | executable/support line; behavior impact depends on
rounding block context. |
| 2153 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2154 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2155 |          format!("Security {:?}", self.bench.uds_ecm.security), | error-propagating operation; ch
ing it impacts failure semantics and caller recovery paths. |
| 2156 |          Color32::YELLOW, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 2157 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2158 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2159 |          format!("UDS log {}", self.uds_log.len()), | I/O or protocol-critical logic; edits can a
ct integration, timing, and diagnostic outcomes. |
| 2160 |          Color32::from_gray(180), | executable/support line; behavior impact depends on surroundi
block context. |
| 2161 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2162 |          ], | executable/support line; behavior impact depends on surrounding block context. |
| 2163 |          Tab::EcmLiveData => vec![ | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 2164 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2165 |          format!("Mode {:?}", self.hw_cfg.mode), | error-propagating operation; changing it impac
failure semantics and caller recovery paths. |
| 2166 |          Color32::from_rgb(110, 180, 255), | executable/support line; behavior impact depends on
rounding block context. |
| 2167 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2168 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2169 |          format!("Detected {}", self.ecm_detected_sas.len()), | executable/support line; behavior
pact depends on surrounding block context. |
| 2170 |          Color32::from_gray(180), | executable/support line; behavior impact depends on surroundi
block context. |
| 2171 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2172 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2173 |          format!("Feed {}", if self.ecm_live_feed.is_some() { "active" } else { "idle" })), | exec
ble/support line; behavior impact depends on surrounding block context. |
| 2174 |          if self.ecm_live_feed.is_some() { Color32::GREEN } else { Color32::YELLOW }, | condition
gate; changing condition shifts runtime path and side effects. |
| 2175 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2176 |          ], | executable/support line; behavior impact depends on surrounding block context. |
| 2177 |          Tab::LeakLab => vec![ | executable/support line; behavior impact depends on surrounding block co
xt. |
| 2178 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2179 |          format!("Circuitos {}", self.bench.leak_rig.circuits.len()), | executable/support line;
avior impact depends on surrounding block context. |
| 2180 |          Color32::from_rgb(110, 180, 255), | executable/support line; behavior impact depends on
rounding block context. |
| 2181 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2182 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2183 |          format!("Predicoes {}", self.leak_predictions.len()), | executable/support line; behavior
mpact depends on surrounding block context. |
| 2184 |          Color32::YELLOW, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 2185 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2186 |          ( | executable/support line; behavior impact depends on surrounding block context. |
| 2187 |          format!("Calibrado {}", if self.leak_calibration_report.is_some() { "sim" } else { "nao"
, | executable/support line; behavior impact depends on surrounding block context. |
| 2188 |          if self.leak_calibration_report.is_some() { Color32::GREEN } else { Color32::from_gray(1
}, | conditional gate; changing condition shifts runtime path and side effects. |
| 2189 |          ), | executable/support line; behavior impact depends on surrounding block context. |
| 2190 |          ], | executable/support line; behavior impact depends on surrounding block context. |
| 2191 |          Tab::Plots => vec![ | executable/support line; behavior impact depends on surrounding block cont
. |
| 2192 |          ( | executable/support line; behavior impact depends on surrounding block context. |

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| 2193 |          format!("{}", self.pl_rpm.len()), | executable/support line; behavior impact depends on surrounding block context. |
| 2194 |          Color32::from_rgb(110, 180, 255), | executable/support line; behavior impact depends on surrounding block context. |
| 2195 |      ), | executable/support line; behavior impact depends on surrounding block context. |
| 2196 |      ( | executable/support line; behavior impact depends on surrounding block context. |
| 2197 |          format!("{}", self.bench.tcm.ground_speed_kmh), | executable/support line; behavior impact depends on surrounding block context. |
| 2198 |          Color32::LIGHT_BLUE, | executable/support line; behavior impact depends on surrounding block context. |
| 2199 |      ), | executable/support line; behavior impact depends on surrounding block context. |
| 2200 |      ], | executable/support line; behavior impact depends on surrounding block context. |
| 2201 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2202 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2203 | (blank) | line is blank; visual separation only. |
| 2204 |      fn render_ux_context_strip(&self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 2205 |          let items = self.ux_context_items(); | executable/support line; behavior impact depends on surrounding block context. |
| 2206 |          if items.is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 2207 |              return; | executable/support line; behavior impact depends on surrounding block context. |
| 2208 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2209 |          ui.add_space(4.0); | executable/support line; behavior impact depends on surrounding block context. |
| 2210 |          ui.horizontal_wrapped(ui) { | executable/support line; behavior impact depends on surrounding block context. |
| 2211 |              for (text, color) in items { | iteration control; changing bounds/body affects throughput, timing and possibly safety guarantees. |
| 2212 |                  egui::Frame::group(ui.style()) | executable/support line; behavior impact depends on surrounding block context. |
| 2213 |                      .fill(Color32::from_gray(20)) | executable/support line; behavior impact depends on surrounding block context. |
| 2214 |                      .inner_margin(Margin::symmetric(8.0, 4.0)) | executable/support line; behavior impact depends on surrounding block context. |
| 2215 |                      .show(ui, ui) { | executable/support line; behavior impact depends on surrounding block context. |
| 2216 |                          ui.label(RichText::new(text).size(10.0).color(color)); | executable/support line; behavior impact depends on surrounding block context. |
| 2217 |                      }); | executable/support line; behavior impact depends on surrounding block context. |
| 2218 |                  } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2219 |              }); | executable/support line; behavior impact depends on surrounding block context. |
| 2220 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2221 | (blank) | line is blank; visual separation only. |
| 2222 |      fn statusbar(&self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 2223 |          let t = self.bench.elapsed(); | executable/support line; behavior impact depends on surrounding block context. |
| 2224 |          let rpm = self.bench.ecm.rpm; | executable/support line; behavior impact depends on surrounding block context. |
| 2225 |          let spd = self.bench.tcm.ground_speed_kmh; | executable/support line; behavior impact depends on surrounding block context. |
| 2226 |          let gear = &self.bench.tcm.gear_label; | executable/support line; behavior impact depends on surrounding block context. |
| 2227 |          let can = self.bench.gateway.bus_load_pct; | executable/support line; behavior impact depends on surrounding block context. |
| 2228 |          let dtcs = self.bench.ecm.active_dtcs.len(); | executable/support line; behavior impact depends on surrounding block context. |
| 2229 |          let nm = format!("{}", self.bench.net_mgmt.bus_state); | executable/support line; behavior impact depends on surrounding block context. |
| 2230 |          let red = self.bench.ecm.red_lamp; | executable/support line; behavior impact depends on surrounding block context. |
| 2231 |          let amber = self.bench.ecm.amber_lamp; | executable/support line; behavior impact depends on surrounding block context. |

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| 2232 |         let abs_on = self.bench.abs.abs_system_active; | executable/support line; behavior impact depends on
rounding block context. |
| 2233 |         let esp_on = self.bench.abs.esp_system_active; | executable/support line; behavior impact depends on
rounding block context. |
| 2234 |         ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 2235 |             macro_rules! s { | executable/support line; behavior impact depends on surrounding block context
| 2236 |                 ($v:expr,$c:expr) => { | executable/support line; behavior impact depends on surrounding blo
context. |
| 2237 |                 ui.label(RichText::new($v).size(11.0).color($c)); | executable/support line; behavior im
t depends on surrounding block context. |
| 2238 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and life
e scopes. |
| 2239 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2240 |             s!(format!("t={:.1}s", t), Color32::from_gray(120)); | executable/support line; behavior impact
ends on surrounding block context. |
| 2241 |             ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 2242 |             s!(format!("RPM {:.0}", rpm), Color32::GREEN); | executable/support line; behavior impact depend
n surrounding block context. |
| 2243 |             ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 2244 |             s!(format!("{:.1}km/h", spd), Color32::LIGHT_BLUE); | executable/support line; behavior impact d
nds on surrounding block context. |
| 2245 |             ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 2246 |             s!(format!("Gear {}", gear), Color32::YELLOW); | executable/support line; behavior impact depend
n surrounding block context. |
| 2247 |             ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 2248 |             s!(format!("CAN {:.0}%", can), Color32::GOLD); | executable/support line; behavior impact depend
n surrounding block context. |
| 2249 |             ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 2250 |             s!( | executable/support line; behavior impact depends on surrounding block context. |
| 2251 |                 format!("{}", DTC", dtcs), | executable/support line; behavior impact depends on surrounding b
k context. |
| 2252 |                 if dtcs == 0 { | conditional gate; changing condition shifts runtime path and side effects.
| 2253 |                     Color32::from_gray(100) | executable/support line; behavior impact depends on surroundin
lock context. |
| 2254 |                     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 2255 |                     Color32::RED | executable/support line; behavior impact depends on surrounding block con
t. |
| 2256 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 2257 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 2258 |             ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 2259 |             s!(format!("NM:{})", nm), Color32::from_gray(120)); | executable/support line; behavior impact de
ds on surrounding block context. |
| 2260 |             if red { | conditional gate; changing condition shifts runtime path and side effects. |
| 2261 |                 ui.separator(); | executable/support line; behavior impact depends on surrounding block cont
. |
| 2262 |                 s!("■ RED STOP", Color32::RED); | executable/support line; behavior impact depends on surrou
ng block context. |
| 2263 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2264 |             if amber { | conditional gate; changing condition shifts runtime path and side effects. |
| 2265 |                 ui.separator(); | executable/support line; behavior impact depends on surrounding block cont
. |
| 2266 |                 s!("■ AMBER", Color32::GOLD); | executable/support line; behavior impact depends on surround
block context. |
| 2267 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2268 |             if abs_on { | conditional gate; changing condition shifts runtime path and side effects. |
| 2269 |                 ui.separator(); | executable/support line; behavior impact depends on surrounding block cont
. |
| 2270 |                 s!("■ ABS", Color32::WHITE); | executable/support line; behavior impact depends on surroundi
block context. |
| 2271 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime

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pes. |
| 2272 |         if esp_on { | conditional gate; changing condition shifts runtime path and side effects. |
| 2273 |             ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
. |
| 2274 |             s!("\u25a0 ESP", Color32::LIGHT_BLUE); | executable/support line; behavior impact depends on surrounding block context. |
| 2275 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2276 |         if self.bench.ad.engaged { | conditional gate; changing condition shifts runtime path and side effects. |
| 2277 |             ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
. |
| 2278 |             s!("\u25a0 AD ON", Color32::GREEN); | executable/support line; behavior impact depends on surrounding block context. |
| 2279 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2280 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 2281 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2282 | (blank) | line is blank; visual separation only. |
| 2283 | fn render_ux_guide(&mut self, ui: &mut Ui) { | declares executable behavior; signature/body changes algorithm, control flow, and outputs. |
| 2284 |     let (title, guide, legend) = match self.tab { | executable/support line; behavior impact depends on surrounding block context. |
| 2285 |         Tab::Help => ( | executable/support line; behavior impact depends on surrounding block context. |
| 2286 |             "Help", | executable/support line; behavior impact depends on surrounding block context. |
| 2287 |             "Use esta aba para fluxo rapido de operacao, diagnostico e leitura da Leak Lab.", | executable/support line; behavior impact depends on surrounding block context. |
| 2288 |             "Comece por Help -> Cluster -> Engine -> Leaks e depois avanca para CAN/UDS.", | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 2289 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 2290 |         Tab::Cluster => ( | executable/support line; behavior impact depends on surrounding block context. |
| 2291 |             "Cluster", | executable/support line; behavior impact depends on surrounding block context. |
| 2292 |             "Use ignition -> throttle/brake -> direction to validate baseline vehicle response.", | executable/support line; behavior impact depends on surrounding block context. |
| 2293 |             "Lamps: RED=critical stop, AMBER=warning, ABS/ESP=assist active.", | executable/support line; behavior impact depends on surrounding block context. |
| 2294 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 2295 |         Tab::Electrical => ( | executable/support line; behavior impact depends on surrounding block context. |
| 2296 |             "Electrical", | executable/support line; behavior impact depends on surrounding block context. |
| 2297 |             "Use esta aba para enxergar alimentacao, fusiveis, ramos ECU e referencias de chicote do modulo atual.", | executable/support line; behavior impact depends on surrounding block context. |
| 2298 |             "Valores de bitola/capacitancia sao baseline de simulacao e devem ser substituidos por dados M quando houver harness real.", | executable/support line; behavior impact depends on surrounding block context. |
| 2299 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 2300 |         Tab::CanBus => ( | executable/support line; behavior impact depends on surrounding block context. |
| 2301 |             "CAN Bus", | executable/support line; behavior impact depends on surrounding block context. |
| 2302 |             "Start in Signals, then Trace, then Network for RCA. Freeze before exporting evidence.", | executable/support line; behavior impact depends on surrounding block context. |
| 2303 |             "State colors: green=active, yellow=passive, red=bus-off/high error.", | executable/support line; behavior impact depends on surrounding block context. |
| 2304 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 2305 |         Tab::Events => ( | executable/support line; behavior impact depends on surrounding block context. |
| 2306 |             "Events", | executable/support line; behavior impact depends on surrounding block context. |
| 2307 |             "Filter by severity and subsystem, then correlate timestamp with Faults/UDS actions.", | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 2308 |             "Icon color maps to severity; pause freezes feed for reading.", | executable/support line; behavior impact depends on surrounding block context. |
| 2309 |         ), | executable/support line; behavior impact depends on surrounding block context. |
| 2310 |         Tab::EcuNet => ( | executable/support line; behavior impact depends on surrounding block context. |
| 2311 |             "ECU Net", | executable/support line; behavior impact depends on surrounding block context. |
| 2312 |             "Check node online state and boot stage before deep diagnostics.", | executable/support line; behavior impact depends on surrounding block context. |

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| 2313 |           "Degraded/offline nodes usually correlate with CAN or power faults.", | executable/support line; behavior impact depends on surrounding block context. |
| 2314 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2315 |       Tab::Engine => ( | executable/support line; behavior impact depends on surrounding block context. |
| 2316 |           "Engine", | executable/support line; behavior impact depends on surrounding block context. |
| 2317 |           "Validate thermal, pressure and torque limits under controlled load changes.", | executable/support line; behavior impact depends on surrounding block context. |
| 2318 |           "Watch red/yellow status first, then inspect numeric channels.", | executable/support line; behavior impact depends on surrounding block context. |
| 2319 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2320 |       Tab::Faults => ( | executable/support line; behavior impact depends on surrounding block context. |
| 2321 |           "Faults", | executable/support line; behavior impact depends on surrounding block context. |
| 2322 |           "Inject one fault at a time; observe Events/CAN/Engine and clear to compare recovery.", | executable/support line; behavior impact depends on surrounding block context. |
| 2323 |           "Use CLEAR after each test case to isolate cause/effect.", | executable/support line; behavior impact depends on surrounding block context. |
| 2324 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2325 |       Tab::Boot => ( | executable/support line; behavior impact depends on surrounding block context. |
| 2326 |           "Boot", | executable/support line; behavior impact depends on surrounding block context. |
| 2327 |           "Confirm startup sequence and timing before runtime validation.", | executable/support line; behavior impact depends on surrounding block context. |
| 2328 |           "Blocked stage means readiness is not complete for higher-level tests.", | executable/support line; behavior impact depends on surrounding block context. |
| 2329 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2330 |       Tab::Implements => ( | executable/support line; behavior impact depends on surrounding block context. |
| 2331 |           "Implements", | executable/support line; behavior impact depends on surrounding block context. |
| 2332 |           "Operate PTO/hitch/aux valves incrementally and monitor hydraulic limits.", | executable/support line; behavior impact depends on surrounding block context. |
| 2333 |           "Large jumps in commands can mask root cause of instability.", | executable/support line; behavior impact depends on surrounding block context. |
| 2334 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2335 |       Tab::Params => ( | executable/support line; behavior impact depends on surrounding block context. |
| 2336 |           "Params", | executable/support line; behavior impact depends on surrounding block context. |
| 2337 |           "Edit one family of parameters at a time; sync back before changing scenario.", | executable/support line; behavior impact depends on surrounding block context. |
| 2338 |           "Use Sync from simulation to avoid stale values.", | executable/support line; behavior impact depends on surrounding block context. |
| 2339 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2340 |       Tab::Sensors => ( | executable/support line; behavior impact depends on surrounding block context. |
| 2341 |           "Sensors", | executable/support line; behavior impact depends on surrounding block context. |
| 2342 |           "Validate GNSS/IMU/radar/lidar/camera consistency before AD validation.", | executable/support line; behavior impact depends on surrounding block context. |
| 2343 |           "Prioritize confidence/quality metrics before control metrics.", | executable/support line; behavior impact depends on surrounding block context. |
| 2344 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2345 |       Tab::Autonomous => ( | executable/support line; behavior impact depends on surrounding block context. |
| 2346 |           "AD", | executable/support line; behavior impact depends on surrounding block context. |
| 2347 |           "Engage only after sensors and base dynamics are stable; tune ACC/LKA gradually.", | executable/support line; behavior impact depends on surrounding block context. |
| 2348 |           "TTC/THW and lane confidence are primary safety indicators.", | executable/support line; behavior impact depends on surrounding block context. |
| 2349 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2350 |       Tab::V2X => ( | executable/support line; behavior impact depends on surrounding block context. |
| 2351 |           "V2X", | executable/support line; behavior impact depends on surrounding block context. |
| 2352 |           "Evaluate latency/loss first, then cooperative alerts impact on control.", | executable/support line; behavior impact depends on surrounding block context. |
| 2353 |           "Higher packet loss can invalidate AD cooperative assumptions.", | executable/support line; behavior impact depends on surrounding block context. |
| 2354 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2355 |       Tab::Uds => ( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |

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| 2356 |           "UDS", | I/O or protocol-critical logic; edits can affect integration, timing, and diagnosti
utcomes. |
| 2357 |           "Follow sequence: session -> security -> read/write/routine -> transfer exit.", | I/O or pro
ol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 2358 |           "Negative response 0x7F indicates rejected service/subfunction/conditions.", | executable/su
rt line; behavior impact depends on surrounding block context. |
| 2359 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2360 |           Tab::EcmLiveData => ( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 2361 |           "ECM Live", | executable/support line; behavior impact depends on surrounding block context.
| 2362 |           "Detect -> Connect -> Retrieve -> Export. Keep mode and SA aligned.", | executable/support l
; behavior impact depends on surrounding block context. |
| 2363 |           "In SIM mode, panels are mock; in LIVE mode, policy gates apply.", | executable/support line
ehavior impact depends on surrounding block context. |
| 2364 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2365 |           Tab::LeakLab => ( | executable/support line; behavior impact depends on surrounding block contex
|
| 2366 |           "Leak Lab", | executable/support line; behavior impact depends on surrounding block context.
| 2367 |           "Select circuit, apply parameters, then run Scenario or Monte Carlo and review ranking.", |
cutable/support line; behavior impact depends on surrounding block context. |
| 2368 |           "Use ASCII legend and temporal map to interpret pressure/risk/flow quickly.", | executable/s
ort line; behavior impact depends on surrounding block context. |
| 2369 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2370 |           Tab::Plots => ( | executable/support line; behavior impact depends on surrounding block context.
| 2371 |           "Plots", | executable/support line; behavior impact depends on surrounding block context. |
| 2372 |           "Use for before/after comparison around faults, AD engagement or UDS actions.", | I/O or pro
ol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 2373 |           "Trend shape matters more than single-point values.", | executable/support line; behavior im
t depends on surrounding block context. |
| 2374 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 2375 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime sco
. |
| 2376 | (blank) | line is blank; visual separation only. |
| 2377 |           egui::Frame::group(ui.style()) | executable/support line; behavior impact depends on surrounding blo
context. |
| 2378 |           .fill(Color32::from_gray(16)) | executable/support line; behavior impact depends on surrounding
ck context. |
| 2379 |           .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 2380 |           ui.horizontal_wrapped(\|ui\| { | executable/support line; behavior impact depends on surround
g block context. |
| 2381 |           ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 2382 |           RichText::new(format!("UX Guide - {} ", title)) | executable/support line; behavior i
ct depends on surrounding block context. |
| 2383 |           .size(11.6) | executable/support line; behavior impact depends on surrounding bl
context. |
| 2384 |           .color(Color32::from_rgb(110, 180, 255)) | executable/support line; behavior imp
depends on surrounding block context. |
| 2385 |           .strong(), | executable/support line; behavior impact depends on surrounding blo
context. |
| 2386 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 2387 |           ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 2388 |           ui.checkbox(&mut self.ux_show_guide, "Show details"); | executable/support line; behavio
mpact depends on surrounding block context. |
| 2389 |           ui.checkbox(&mut self.ux_compact_mode, "Compact mode"); | executable/support line; behav
impact depends on surrounding block context. |
| 2390 |           ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 2391 |           self.render_ux_quick_actions(ui); | executable/support line; behavior impact depends on
rounding block context. |
| 2392 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 2393 | (blank) | line is blank; visual separation only. |
| 2394 |           for (color, msg) in self.ux_alerts() { | iteration control; changing bounds/body affects thr

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hput, timing, and possibly safety guarantees. |
| 2395 |           ui.add_space(4.0); | executable/support line; behavior impact depends on surrounding block
context. |
| 2396 |           ui.label(RichText::new(format!("{}", msg)).size(10.3).color(color)); | executable/support
line; behavior impact depends on surrounding block context. |
| 2397 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2398 | (blank) | line is blank; visual separation only. |
| 2399 |           self.render_ux_context_strip(ui); | executable/support line; behavior impact depends on surrounding
block context. |
| 2400 | (blank) | line is blank; visual separation only. |
| 2401 |           if self.ux_show_guide { | conditional gate; changing condition shifts runtime path and side
effects. |
| 2402 |           ui.add_space(4.0); | executable/support line; behavior impact depends on surrounding block
context. |
| 2403 |           ui.label(RichText::new(format!("How to use: {}", guide)).size(10.2)); | executable/support
line; behavior impact depends on surrounding block context. |
| 2404 |           ui.label( | executable/support line; behavior impact depends on surrounding block context.
|
| 2405 |           RichText::new(format!("Legend/Interpretation: {}", legend)) | executable/support line;
behavior impact depends on surrounding block context. |
| 2406 |           .size(10.0) | executable/support line; behavior impact depends on surrounding block
context. |
| 2407 |           .color(Color32::from_gray(175)), | executable/support line; behavior impact depends
on surrounding block context. |
| 2408 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 2409 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2410 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 2411 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2412 | (blank) | line is blank; visual separation only. |
| 2413 |     fn render_ux_quick_actions(&mut self, ui: &mut Ui) { | declares executable behavior; signature/body changes
alter API, control flow, and outputs. |
| 2414 |         match self.tab { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 2415 |             Tab::Help => { | executable/support line; behavior impact depends on surrounding block context.
|
| 2416 |                 if ui.button("Reset simulation").clicked() { | conditional gate; changing condition shifts runtime
path and side effects. |
| 2417 |                     self.cmds.push(Cmd::Reset); | executable/support line; behavior impact depends on surrounding
block context. |
| 2418 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2419 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2420 |             Tab::Cluster => { | executable/support line; behavior impact depends on surrounding block context.
|
| 2421 |                 if ui.button("Key OFF").clicked() { | conditional gate; changing condition shifts runtime path
and side effects. |
| 2422 |                     self.cmds.push(Cmd::KeyOff); | executable/support line; behavior impact depends on surrounding
block context. |
| 2423 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2424 |                 if ui.button("Reset simulation").clicked() { | conditional gate; changing condition shifts runtime
path and side effects. |
| 2425 |                     self.cmds.push(Cmd::Reset); | executable/support line; behavior impact depends on surrounding
block context. |
| 2426 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2427 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2428 |             Tab::Electrical => { | executable/support line; behavior impact depends on surrounding block context.
|
| 2429 |                 if ui.button("Toggle HVAC").clicked() { | conditional gate; changing condition shifts runtime
path and side effects. |
| 2430 |                     self.cmds.push(Cmd::ToggleHvac); | executable/support line; behavior impact depends on surrounding
block context. |

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| 2431 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2432 |           if ui.button("Reset fuses").clicked() { | conditional gate; changing condition shifts runtime
ath and side effects. |
| 2433 |               self.cmds.push(Cmd::ResetBcmFuses); | executable/support line; behavior impact depends on
urrounding block context. |
| 2434 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2435 |           if ui.button("Toggle work lights").clicked() { | conditional gate; changing condition shifts
runtime path and side effects. |
| 2436 |               self.cmds.push(Cmd::ToggleWorkLights); | executable/support line; behavior impact depends
n surrounding block context. |
| 2437 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2438 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2439 |       Tab::Events => { | executable/support line; behavior impact depends on surrounding block context
| 2440 |           if ui.button("Clear events").clicked() { | conditional gate; changing condition shifts runtime
path and side effects. |
| 2441 |               self.events.clear(); | executable/support line; behavior impact depends on surrounding b
k context. |
| 2442 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2443 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2444 |       Tab::CanBus => { | executable/support line; behavior impact depends on surrounding block context
| 2445 |           if ui.button("Clear CAN views").clicked() { | conditional gate; changing condition shifts ru
me path and side effects. |
| 2446 |               self.sig_map.clear(); | executable/support line; behavior impact depends on surrounding
ck context. |
| 2447 |               self.trace_snap.clear(); | executable/support line; behavior impact depends on surroundi
block context. |
| 2448 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2449 |           if ui.button("Clear CAN injections").clicked() { | conditional gate; changing condition shif
runtime path and side effects. |
| 2450 |               self.cmds.push(Cmd::CanClearAllInjections); | executable/support line; behavior impact d
nds on surrounding block context. |
| 2451 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2452 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2453 |       Tab::EcuNet => { | executable/support line; behavior impact depends on surrounding block context
| 2454 |           if ui.button("Key OFF").clicked() { | conditional gate; changing condition shifts runtime pa
and side effects. |
| 2455 |               self.cmds.push(Cmd::KeyOff); | executable/support line; behavior impact depends on surro
ing block context. |
| 2456 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2457 |           if ui.button("Reset simulation").clicked() { | conditional gate; changing condition shifts r
ime path and side effects. |
| 2458 |               self.cmds.push(Cmd::Reset); | executable/support line; behavior impact depends on surrou
ng block context. |
| 2459 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2460 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2461 |       Tab::Engine => { | executable/support line; behavior impact depends on surrounding block context
| 2462 |           if ui.button("Clear DTCs").clicked() { | conditional gate; changing condition shifts runtime
th and side effects. |
| 2463 |               self.cmds.push(Cmd::ClearDtcs); | executable/support line; behavior impact depends on su
unding block context. |
| 2464 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2465 |           if ui.button("Reset simulation").clicked() { | conditional gate; changing condition shifts r

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ime path and side effects. |
| 2466 |           self.cmds.push(Cmd::Reset); | executable/support line; behavior impact depends on surround
ng block context. |
| 2467 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 2468 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2469 |       Tab::Faults => { | executable/support line; behavior impact depends on surrounding block context
| 2470 |           if ui.button("Inject selected").clicked() { | conditional gate; changing condition shifts ru
ime path and side effects. |
| 2471 |           self.cmds.push(Cmd::InjectFault); | executable/support line; behavior impact depends on
rounding block context. |
| 2472 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 2473 |           if ui.button("Clear faults").clicked() { | conditional gate; changing condition shifts runti
path and side effects. |
| 2474 |           self.cmds.push(Cmd::ClearFaults); | executable/support line; behavior impact depends on
rounding block context. |
| 2475 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 2476 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2477 |       Tab::Boot => { | executable/support line; behavior impact depends on surrounding block context.
| 2478 |           if ui.button("Key advance").clicked() { | conditional gate; changing condition shifts runtime
ath and side effects. |
| 2479 |           self.cmds.push(Cmd::KeyAdvance); | executable/support line; behavior impact depends on s
ounding block context. |
| 2480 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 2481 |           if ui.button("Key OFF").clicked() { | conditional gate; changing condition shifts runtime pa
and side effects. |
| 2482 |           self.cmds.push(Cmd::KeyOff); | executable/support line; behavior impact depends on surro
ing block context. |
| 2483 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 2484 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2485 |       Tab::Implements => { | executable/support line; behavior impact depends on surrounding block con
t. |
| 2486 |           if ui.button("PTO OFF").clicked() { | conditional gate; changing condition shifts runtime pa
and side effects. |
| 2487 |           self.cmds.push(Cmd::SetPtoMode(PtoMode::Off)); | executable/support line; behavior impac
depends on surrounding block context. |
| 2488 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 2489 |           if ui.button("Reset simulation").clicked() { | conditional gate; changing condition shifts r
ime path and side effects. |
| 2490 |           self.cmds.push(Cmd::Reset); | executable/support line; behavior impact depends on surround
ng block context. |
| 2491 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 2492 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2493 |       Tab::Params => { | executable/support line; behavior impact depends on surrounding block context
| 2494 |           if ui.button("Reset simulation").clicked() { | conditional gate; changing condition shifts r
ime path and side effects. |
| 2495 |           self.cmds.push(Cmd::Reset); | executable/support line; behavior impact depends on surround
ng block context. |
| 2496 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 2497 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2498 |       Tab::Sensors => { | executable/support line; behavior impact depends on surrounding block contex
|
| 2499 |           if ui.button("Reset simulation").clicked() { | conditional gate; changing condition shifts r

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ime path and side effects. |
| 2500 |           self.cmds.push(Cmd::Reset); | executable/support line; behavior impact depends on surround
ng block context. |
| 2501 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 2502 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2503 |       Tab::Autonomous => { | executable/support line; behavior impact depends on surrounding block con
t. |
| 2504 |           if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 2505 |               .button(if self.bench.ad.engaged { "Disengage AD" } else { "Engage AD" }) | executable/s
ort line; behavior impact depends on surrounding block context. |
| 2506 |               .clicked() | executable/support line; behavior impact depends on surrounding block conte
|
| 2507 |           { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 2508 |               self.cmds.push(if self.bench.ad.engaged { | executable/support line; behavior impact dep
s on surrounding block context. |
| 2509 |                   Cmd::DisengageAD | executable/support line; behavior impact depends on surrounding b
k context. |
| 2510 |                   } else { | executable/support line; behavior impact depends on surrounding block context
| 2511 |                       Cmd::EngageAD | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 2512 |                   }); | executable/support line; behavior impact depends on surrounding block context. |
| 2513 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 2514 |           if ui.button("Clear path").clicked() { | conditional gate; changing condition shifts runtime
th and side effects. |
| 2515 |               self.cmds.push(Cmd::ClearWaypoints); | executable/support line; behavior impact depends
surrounding block context. |
| 2516 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 2517 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2518 |       Tab::V2X => { | executable/support line; behavior impact depends on surrounding block context. |
| 2519 |           if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 2520 |               .add_enabled(self.bench.telematics.ota_available, Button::new("Start OTA")) | executable
pport line; behavior impact depends on surrounding block context. |
| 2521 |               .clicked() | executable/support line; behavior impact depends on surrounding block conte
|
| 2522 |           { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 2523 |               self.cmds.push(Cmd::StartOta); | executable/support line; behavior impact depends on surr
nding block context. |
| 2524 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 2525 |           if ui.button("Reset simulation").clicked() { | conditional gate; changing condition shifts r
ime path and side effects. |
| 2526 |               self.cmds.push(Cmd::Reset); | executable/support line; behavior impact depends on surround
ng block context. |
| 2527 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 2528 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2529 |       Tab::Uds => { | I/O or protocol-critical logic; edits can affect integration, timing, and diagno
c outcomes. |
| 2530 |           if ui.button("Clear UDS log").clicked() { | conditional gate; changing condition shifts runt
path and side effects. |
| 2531 |               self.uds_log.clear(); | I/O or protocol-critical logic; edits can affect integration, ti
g, and diagnostic outcomes. |
| 2532 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 2533 |           if ui.button("Default Session").clicked() { | conditional gate; changing condition shifts ru
me path and side effects. |
| 2534 |               self.uds_input = "10 01".into(); | I/O or protocol-critical logic; edits can affect inte

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tion, timing, and diagnostic outcomes. |
| 2535 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2536 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2537 |       Tab::EcmLiveData => { | executable/support line; behavior impact depends on surrounding block context. |
| 2538 |           if ui.button("Clear live snapshot").clicked() { | conditional gate; changing condition shifts runtime
path and side effects. |
| 2539 |               self.ecm_live_snapshot = io::ecm_params::EcmSnapshot::default(); | executable/support line; behavior
impact depends on surrounding block context. |
| 2540 |               self.ecm_live_frames_total = 0; | executable/support line; behavior impact depends on surrounding
block context. |
| 2541 |               self.ecm_live_last_update_ms = 0; | executable/support line; behavior impact depends on surrounding
block context. |
| 2542 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2543 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2544 |       Tab::LeakLab => { | executable/support line; behavior impact depends on surrounding block context. |
| 2545 |           if ui.button("Clear leak outputs").clicked() { | conditional gate; changing condition shifts runtime
path and side effects. |
| 2546 |               self.cmds.push(Cmd::LeakClearScenarioOutputs); | executable/support line; behavior impact
depends on surrounding block context. |
| 2547 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2548 |           if ui.button("Reset leak sim").clicked() { | conditional gate; changing condition shifts runtime
path and side effects. |
| 2549 |               self.cmds.push(Cmd::LeakResetSimulation); | executable/support line; behavior impact depends
on surrounding block context. |
| 2550 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2551 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2552 |       Tab::Plots => { | executable/support line; behavior impact depends on surrounding block context. |
| 2553 |           if ui.button("Reset simulation").clicked() { | conditional gate; changing condition shifts runtime
path and side effects. |
| 2554 |               self.cmds.push(Cmd::Reset); | executable/support line; behavior impact depends on surrounding
block context. |
| 2555 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2556 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2557 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2558 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2559 | (blank) | line is blank; visual separation only. |
| 2560 | fn electrical_wire_mm2(current_a: f64) -> f64 { | declares executable behavior; signature/body changes a
r API, control flow, and outputs. |
| 2561 |     if current_a <= 5.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 2562 |         0.75 | executable/support line; behavior impact depends on surrounding block context. |
| 2563 |     } else if current_a <= 10.0 { | executable/support line; behavior impact depends on surrounding block
context. |
| 2564 |         1.0 | executable/support line; behavior impact depends on surrounding block context. |
| 2565 |     } else if current_a <= 15.0 { | executable/support line; behavior impact depends on surrounding block
context. |
| 2566 |         1.5 | executable/support line; behavior impact depends on surrounding block context. |
| 2567 |     } else if current_a <= 25.0 { | executable/support line; behavior impact depends on surrounding block
context. |
| 2568 |         2.5 | executable/support line; behavior impact depends on surrounding block context. |
| 2569 |     } else if current_a <= 40.0 { | executable/support line; behavior impact depends on surrounding block
context. |
| 2570 |         4.0 | executable/support line; behavior impact depends on surrounding block context. |
| 2571 |     } else if current_a <= 60.0 { | executable/support line; behavior impact depends on surrounding block

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ontext. |
| 2572 |         6.0 | executable/support line; behavior impact depends on surrounding block context. |
| 2573 |     } else if current_a <= 100.0 { | executable/support line; behavior impact depends on surrounding block
context. |
| 2574 |         16.0 | executable/support line; behavior impact depends on surrounding block context. |
| 2575 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 2576 |         25.0 | executable/support line; behavior impact depends on surrounding block context. |
| 2577 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 2578 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 2579 | (blank) | line is blank; visual separation only. |
| 2580 | fn electrical_cap_nf_per_m(gauge_mm2: f64, twisted_pair: bool) -> f64 { | declares executable behavior;
nature/body changes alter API, control flow, and outputs. |
| 2581 |     if twisted_pair { | conditional gate; changing condition shifts runtime path and side effects. |
| 2582 |         0.05 | executable/support line; behavior impact depends on surrounding block context. |
| 2583 |     } else if gauge_mm2 <= 1.0 { | executable/support line; behavior impact depends on surrounding block
context. |
| 2584 |         0.09 | executable/support line; behavior impact depends on surrounding block context. |
| 2585 |     } else if gauge_mm2 <= 2.5 { | executable/support line; behavior impact depends on surrounding block
context. |
| 2586 |         0.11 | executable/support line; behavior impact depends on surrounding block context. |
| 2587 |     } else if gauge_mm2 <= 6.0 { | executable/support line; behavior impact depends on surrounding block
context. |
| 2588 |         0.13 | executable/support line; behavior impact depends on surrounding block context. |
| 2589 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 2590 |         0.16 | executable/support line; behavior impact depends on surrounding block context. |
| 2591 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 2592 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 2593 | (blank) | line is blank; visual separation only. |
| 2594 | fn electrical_node_online(&self, sa: u8) -> bool { | declares executable behavior; signature/body change
alter API, control flow, and outputs. |
| 2595 |     self.bench | executable/support line; behavior impact depends on surrounding block context. |
| 2596 |     .boot | executable/support line; behavior impact depends on surrounding block context. |
| 2597 |     .ecu_by_sa(sa) | executable/support line; behavior impact depends on surrounding block context.
| 2598 |     .is_some_and(\|ecu\| ecu.is_online()) | executable/support line; behavior impact depends on surr
rounding block context. |
| 2599 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 2600 | (blank) | line is blank; visual separation only. |
| 2601 | fn tab_ecm_live_data(&mut self, ui: &mut Ui) { | declares executable behavior; signature/body changes al
API, control flow, and outputs. |
| 2602 |     ui.heading("ECM-Live Data"); | executable/support line; behavior impact depends on surrounding block
context. |
| 2603 |     ui.add_space(6.0); | executable/support line; behavior impact depends on surrounding block context.
| 2604 |     let live_mode = matches!(self.hw_cfg.mode, io::hw::HwMode::Live); | executable/support line; behavior
impact depends on surrounding block context. |
| 2605 | (blank) | line is blank; visual separation only. |
| 2606 |     ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block conte
|
| 2607 |         .auto_shrink([false, false]) | executable/support line; behavior impact depends on surrounding b
k context. |
| 2608 |         .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 2609 | (blank) | line is blank; visual separation only. |
| 2610 |             if !live_mode { | conditional gate; changing condition shifts runtime path and side effects.
| 2611 |                 ui.colored_label( | executable/support line; behavior impact depends on surrounding block
context. |
| 2612 |                     Color32::YELLOW, | executable/support line; behavior impact depends on surrounding b
k context. |
| 2613 |                     "SIM mode active: showing mock ECM view. Switch to --hw-mode=live for hardware Detect
connect/Retrieve.", | executable/support line; behavior impact depends on surrounding block context. |
| 2614 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 2615 |                 if self.ecm_detected_sas.is_empty() { | conditional gate; changing condition shifts runt
path and side effects. |
| 2616 |                     self.ecm_detected_sas = vec![0x00]; | executable/support line; behavior impact depen

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on surrounding block context. |
| 2617 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2618 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2619 | (blank) | line is blank; visual separation only. |
| 2620 |     ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
| 2621 |         if ui.add_enabled(live_mode, Button::new("Detect")).clicked() { | conditional gate; changing condition shifts runtime path and side effects. |
| 2622 |             match io::live_runner::detect_ecms(&self.hw_cfg, Duration::from_secs(3)) { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 2623 |                 Ok(res) => { | executable/support line; behavior impact depends on surrounding block context. |
| 2624 |                     self.ecm_detected_sas = res.source_addresses; | executable/support line; behavior impact depends on surrounding block context. |
| 2625 |                     self.ecm_selected_idx = 0; | executable/support line; behavior impact depends on surrounding block context. |
| 2626 |                     self.ecm_connected = false; | executable/support line; behavior impact depends on surrounding block context. |
| 2627 |                     if self.ecm_detected_sas.is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 2628 |                         self.ecm_live_status = | executable/support line; behavior impact depends on surrounding block context. |
| 2629 |                         "Detect completed: no ECM found in timeout window.".into(); | executable/support line; behavior impact depends on surrounding block context. |
| 2630 |                     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 2631 |                         self.ecm_live_status = format!( | executable/support line; behavior impact depends on surrounding block context. |
| 2632 |                             "Detect completed: {} ECM source address(es) found.", | executable/support line; behavior impact depends on surrounding block context. |
| 2633 |                             self.ecm_detected_sas.len() | executable/support line; behavior impact depends on surrounding block context. |
| 2634 |                         ); | executable/support line; behavior impact depends on surrounding block context. |
| 2635 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2636 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2637 |                 Err(e) => { | executable/support line; behavior impact depends on surrounding block context. |
| 2638 |                     self.ecm_live_status = format!("Detect failed: {e}"); | executable/support line; behavior impact depends on surrounding block context. |
| 2639 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2640 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2641 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2642 | (blank) | line is blank; visual separation only. |
| 2643 |         let can_connect = live_mode && !self.ecm_detected_sas.is_empty(); | executable/support line; behavior impact depends on surrounding block context. |
| 2644 |         if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 2645 |             .add_enabled(can_connect, Button::new("Connect")) | executable/support line; behavior impact depends on surrounding block context. |
| 2646 |             .clicked() | executable/support line; behavior impact depends on surrounding block context. |
| 2647 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2648 |             let selected_sa = self.ecm_detected_sas.get(self.ecm_selected_idx).copied(); | executable/support line; behavior impact depends on surrounding block context. |
| 2649 |             let target_sa = if self.hw_cfg.can_interface.is_some() { | executable/support line; behavior impact depends on surrounding block context. |
| 2650 |                 selected_sa | executable/support line; behavior impact depends on surrounding block context. |
| 2651 |             }

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| 2651 |           } else { | executable/support line; behavior impact depends on surrounding block context. |
| 2652 |               None | executable/support line; behavior impact depends on surrounding block context. |
| 2653 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2654 | (blank) | line is blank; visual separation only. |
| 2655 |           match io::live_runner::connect_ecm(&self.hw_cfg, target_sa) { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 2656 |               Ok(()) => { | executable/support line; behavior impact depends on surrounding block context. |
| 2657 |                   self.ecm_connected = true; | executable/support line; behavior impact depends on surrounding block context. |
| 2658 |                   self.ecm_live_status = "Connect successful. Ready to Retrieve Data.".into(); | executable/support line; behavior impact depends on surrounding block context. |
| 2659 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2660 |               Err(e) => { | executable/support line; behavior impact depends on surrounding block context. |
| 2661 |                   self.ecm_connected = false; | executable/support line; behavior impact depends on surrounding block context. |
| 2662 |                   self.ecm_live_status = format!("Connect failed: {e}"); | executable/support line; behavior impact depends on surrounding block context. |
| 2663 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2664 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2665 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2666 | (blank) | line is blank; visual separation only. |
| 2667 |       let can_retrieve = live_mode && self.ecm_connected && self.ecm_live_feed.is_none(); | executable/support line; behavior impact depends on surrounding block context. |
| 2668 |       if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 2669 |           .add_enabled(can_retrieve, Button::new("Retrieve Data")) | executable/support line; behavior impact depends on surrounding block context. |
| 2670 |           .clicked() | executable/support line; behavior impact depends on surrounding block context. |
| 2671 |       { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2672 |           let selected_sa = self.ecm_detected_sas.get(self.ecm_selected_idx).copied(); | executable/support line; behavior impact depends on surrounding block context. |
| 2673 |           let target_sa = if self.hw_cfg.can_interface.is_some() { | executable/support line; behavior impact depends on surrounding block context. |
| 2674 |               selected_sa | executable/support line; behavior impact depends on surrounding block context. |
| 2675 |           } else { | executable/support line; behavior impact depends on surrounding block context. |
| 2676 |               None | executable/support line; behavior impact depends on surrounding block context. |
| 2677 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2678 | (blank) | line is blank; visual separation only. |
| 2679 |           match io::live_runner::start_retrieve_data(self.hw_cfg.clone(), target_sa) { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 2680 |               Ok(feed) => { | executable/support line; behavior impact depends on surrounding block context. |
| 2681 |                   self.ecm_live_feed = Some(feed); | executable/support line; behavior impact depends on surrounding block context. |
| 2682 |                   self.ecm_live_status = | executable/support line; behavior impact depends on surrounding block context. |
| 2683 |                       "Retrieve started. Real-time parameter updates are active.".into(); | executable/support line; behavior impact depends on surrounding block context. |
| 2684 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 2685 |               Err(e) => { | executable/support line; behavior impact depends on surrounding block context. |
| 2686 |                   self.ecm_live_status = format!("Retrieve failed: {e}"); | executable/support line; behavior impact depends on surrounding block context. |
| 2687 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

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| 2688 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2689 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2690 | (blank) | line is blank; visual separation only. |
| 2691 |         if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 2692 |             .add_enabled(self.ecm_live_feed.is_some(), Button::new("Stop")) | executable/support line; behavior
impact depends on surrounding block context. |
| 2693 |             .clicked() | executable/support line; behavior impact depends on surrounding block context. |
| 2694 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2695 |             if let Some(feed) = &self.ecm_live_feed { | conditional gate; changing condition shifts runtime
path and side effects. |
| 2696 |                 feed.stop(); | executable/support line; behavior impact depends on surrounding block context. |
| 2697 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2698 |             self.ecm_live_feed = None; | executable/support line; behavior impact depends on surrounding block
context. |
| 2699 |             self.ecm_live_status = "Retrieve stopped by operator.".into(); | executable/support line; behavior
impact depends on surrounding block context. |
| 2700 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2701 | (blank) | line is blank; visual separation only. |
| 2702 |         if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 2703 |             .add_enabled(self.ecm_live_feed.is_some(), Button::new("Export CSV")) | executable/support line;
behavior impact depends on surrounding block context. |
| 2704 |             .clicked() | executable/support line; behavior impact depends on surrounding block context. |
| 2705 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2706 |             if let Some(feed) = &self.ecm_live_feed { | conditional gate; changing condition shifts runtime
path and side effects. |
| 2707 |                 let ts = chrono::Local::now().format("%Y%m%d_%H%M%S").to_string(); | executable/support line;
behavior impact depends on surrounding block context. |
| 2708 |                 let name = format!("ecm_live_{}.csv", ts); | executable/support line; behavior impact depends
on surrounding block context. |
| 2709 |                 if let Some(path) = Self::pick_save_path(&name, "csv") { | conditional gate; changing condition
shifts runtime path and side effects. |
| 2710 |                     match feed.export_csv(&path) { | branch dispatch; changing arms alters behavior for
variants and input classes. |
| 2711 |                         Ok(()) => { | executable/support line; behavior impact depends on surrounding block
context. |
| 2712 |                             self.ecm_live_status = format!("CSV exported to {}", path); | executable/support
line; behavior impact depends on surrounding block context. |
| 2713 |                         } | scope delimiter; structure-only line, but misplaced edits can change ownership
and lifetime scopes. |
| 2714 |                     Err(e) => { | executable/support line; behavior impact depends on surrounding block
context. |
| 2715 |                         self.ecm_live_status = format!("CSV export failed: {e}"); | executable/support
line; behavior impact depends on surrounding block context. |
| 2716 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership
and lifetime scopes. |
| 2717 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and
lifetime scopes. |
| 2718 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and
lifetime scopes. |
| 2719 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2720 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 2721 |     }); | executable/support line; behavior impact depends on surrounding block context. | | |
| 2722 | (blank) | line is blank; visual separation only. |
| 2723 |     ui.add_space(8.0); | executable/support line; behavior impact depends on surrounding block context. |
| 2724 |     ui.horizontal(|ui| { | executable/support line; behavior impact depends on surrounding block context. |
|

```

```

| 2725 |         ui.label("Detected ECM SA:"); | executable/support line; behavior impact depends on surrounding
ck context. |
| 2726 |         if self.ecm_detected_sas.is_empty() { | conditional gate; changing condition shifts runtime path
d side effects. |
| 2727 |             ui.label("(none)"); | executable/support line; behavior impact depends on surrounding block
text. |
| 2728 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 2729 |             let mut selected = self.ecm_selected_idx.min(self.ecm_detected_sas.len() - 1); | executable/su
port line; behavior impact depends on surrounding block context. |
| 2730 |             ComboBox::from_id_salt("ecm_live::sa_combo") | executable/support line; behavior impact depe
on surrounding block context. |
| 2731 |                 .selected_text(format!("{}", self.ecm_detected_sas[selected])) | executable/support
ine; behavior impact depends on surrounding block context. |
| 2732 |                 .show_ui(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding
ck context. |
| 2733 |                     for (i, sa) in self.ecm_detected_sas.iter().enumerate() { | iteration control; chang
bounds/body affects throughput, timing, and possibly safety guarantees. |
| 2734 |                         ui.selectable_value(&mut selected, i, format!("{}", sa)); | executable/sup
t line; behavior impact depends on surrounding block context. |
| 2735 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 2736 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 2737 |                 self.ecm_selected_idx = selected; | executable/support line; behavior impact depends on surr
ding block context. |
| 2738 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2739 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 2740 | (blank) | line is blank; visual separation only. |
| 2741 |         ui.add_space(8.0); | executable/support line; behavior impact depends on surrounding block context.
|
| 2742 |         ui.label(format!("Status: {}", self.ecm_live_status)); | executable/support line; behavior impact de
ds on surrounding block context. |
| 2743 |         ui.label(format!( | executable/support line; behavior impact depends on surrounding block context. |
| 2744 |             "Connected: {} \| Streaming: {} \| Frames: {} \| Last update(ms): {}", | executable/support line
ehavior impact depends on surrounding block context. |
| 2745 |             self.ecm_connected, | executable/support line; behavior impact depends on surrounding block cont
. |
| 2746 |             self.ecm_live_feed.is_some(), | executable/support line; behavior impact depends on surrounding
ck context. |
| 2747 |             self.ecm_live_frames_total, | executable/support line; behavior impact depends on surrounding bl
context. |
| 2748 |             self.ecm_live_last_update_ms | executable/support line; behavior impact depends on surrounding b
k context. |
| 2749 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 2750 | (blank) | line is blank; visual separation only. |
| 2751 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 2752 |         if let Some(feed) = &self.ecm_live_feed { | conditional gate; changing condition shifts runtime path
d side effects. |
| 2753 |             let points = feed.recent_points(300); | executable/support line; behavior impact depends on surr
ding block context. |
| 2754 |             if !points.is_empty() { | conditional gate; changing condition shifts runtime path and side effe
. |
| 2755 |                 let mut rpm_min = f64::INFINITY; | executable/support line; behavior impact depends on surro
ing block context. |
| 2756 |                 let mut rpm_max = f64::NEG_INFINITY; | executable/support line; behavior impact depends on s
ounding block context. |
| 2757 |                 let mut rpm_sum = 0.0; | executable/support line; behavior impact depends on surrounding blo
context. |
| 2758 |                 let mut rpm_count = 0usize; | executable/support line; behavior impact depends on surroundin
lock context. |
| 2759 |                 let mut cool_max = f64::NEG_INFINITY; | executable/support line; behavior impact depends on
rounding block context. |
| 2760 |                 let mut oil_min = f64::INFINITY; | executable/support line; behavior impact depends on surro
ing block context. |
| 2761 | (blank) | line is blank; visual separation only. |
| 2762 |                 for p in &points { | iteration control; changing bounds/body affects throughput, timing, and

```

```

ssibly safety guarantees. |
| 2763 |         if let Some(r) = p.engine_speed_rpm { | conditional gate; changing condition shifts runtime
    path and side effects. |
| 2764 |             rpm_min = rpm_min.min(r); | executable/support line; behavior impact depends on surrounding
    ding block context. |
| 2765 |             rpm_max = rpm_max.max(r); | executable/support line; behavior impact depends on surrounding
    ding block context. |
| 2766 |             rpm_sum += r; | executable/support line; behavior impact depends on surrounding block
    ontext. |
| 2767 |             rpm_count += 1; | executable/support line; behavior impact depends on surrounding block
    context. |
| 2768 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
    time scopes. |
| 2769 |         if let Some(c) = p.coolant_temp_c { | conditional gate; changing condition shifts runtime
    ath and side effects. |
| 2770 |             cool_max = cool_max.max(c); | executable/support line; behavior impact depends on surrounding
    unding block context. |
| 2771 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
    time scopes. |
| 2772 |         if let Some(o) = p.oil_pressure_kpa { | conditional gate; changing condition shifts runtime
    path and side effects. |
| 2773 |             oil_min = oil_min.min(o); | executable/support line; behavior impact depends on surrounding
    ding block context. |
| 2774 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
    time scopes. |
| 2775 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
    scopes. |
| 2776 | (blank) | line is blank; visual separation only. |
| 2777 |         ui.label("Post-analysis dashboard (rolling window)"); | executable/support line; behavior impact
    t depends on surrounding block context. |
| 2778 |         Grid::new("ecm_live::dashboard_grid").show(ui, \ui\ { | executable/support line; behavior
    act depends on surrounding block context. |
| 2779 |             ui.label("Samples"); | executable/support line; behavior impact depends on surrounding block
    k context. |
| 2780 |             ui.label(format!("{}", points.len())); | executable/support line; behavior impact depends
    n surrounding block context. |
| 2781 |             ui.end_row(); | executable/support line; behavior impact depends on surrounding block co
    xt. |
| 2782 | (blank) | line is blank; visual separation only. |
| 2783 |             ui.label("RPM Min"); | executable/support line; behavior impact depends on surrounding block
    k context. |
| 2784 |             ui.label(if rpm_count > 0 { | executable/support line; behavior impact depends on surrounding
    ng block context. |
| 2785 |                 format!("{:.1}", rpm_min) | executable/support line; behavior impact depends on surrounding
    ding block context. |
| 2786 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 2787 |                 "-".to_string() | executable/support line; behavior impact depends on surrounding block
    context. |
| 2788 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 2789 |             ui.end_row(); | executable/support line; behavior impact depends on surrounding block co
    xt. |
| 2790 | (blank) | line is blank; visual separation only. |
| 2791 |             ui.label("RPM Avg"); | executable/support line; behavior impact depends on surrounding block
    k context. |
| 2792 |             ui.label(if rpm_count > 0 { | executable/support line; behavior impact depends on surrounding
    ng block context. |
| 2793 |                 format!("{:.1}", rpm_sum / rpm_count as f64) | executable/support line; behavior imp
    depends on surrounding block context. |
| 2794 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 2795 |                 "-".to_string() | executable/support line; behavior impact depends on surrounding block
    context. |
| 2796 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 2797 |             ui.end_row(); | executable/support line; behavior impact depends on surrounding block co
    xt. |
| 2798 | (blank) | line is blank; visual separation only. |

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| 2799 |                ui.label("RPM Max"); | executable/support line; behavior impact depends on surrounding block context. |
| 2800 |                ui.label(if rpm_count > 0 { | executable/support line; behavior impact depends on surrounding block context. |
| 2801 |                    format!("{:.1}", rpm_max) | executable/support line; behavior impact depends on surrounding block context. |
| 2802 |                } else { | executable/support line; behavior impact depends on surrounding block context. |
| 2803 |                    "-".to_string() | executable/support line; behavior impact depends on surrounding block context. |
| 2804 |                }); | executable/support line; behavior impact depends on surrounding block context. |
| 2805 |                ui.end_row(); | executable/support line; behavior impact depends on surrounding block context. |
| 2806 | (blank) | line is blank; visual separation only. |
| 2807 |                ui.label("Coolant Peak C"); | executable/support line; behavior impact depends on surrounding block context. |
| 2808 |                ui.label(if cool_max.is_finite() { | executable/support line; behavior impact depends on surrounding block context. |
| 2809 |                    format!("{:.1}", cool_max) | executable/support line; behavior impact depends on surrounding block context. |
| 2810 |                } else { | executable/support line; behavior impact depends on surrounding block context. |
| 2811 |                    "-".to_string() | executable/support line; behavior impact depends on surrounding block context. |
| 2812 |                }); | executable/support line; behavior impact depends on surrounding block context. |
| 2813 |                ui.end_row(); | executable/support line; behavior impact depends on surrounding block context. |
| 2814 | (blank) | line is blank; visual separation only. |
| 2815 |                ui.label("Oil Pressure Min kPa"); | executable/support line; behavior impact depends on surrounding block context. |
| 2816 |                ui.label(if oil_min.is_finite() { | executable/support line; behavior impact depends on surrounding block context. |
| 2817 |                    format!("{:.1}", oil_min) | executable/support line; behavior impact depends on surrounding block context. |
| 2818 |                } else { | executable/support line; behavior impact depends on surrounding block context. |
| 2819 |                    "-".to_string() | executable/support line; behavior impact depends on surrounding block context. |
| 2820 |                }); | executable/support line; behavior impact depends on surrounding block context. |
| 2821 |                ui.end_row(); | executable/support line; behavior impact depends on surrounding block context. |
| 2822 |                }); | executable/support line; behavior impact depends on surrounding block context. |
| 2823 |                ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 2824 |                } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime. |
| 2825 |            } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
| 2826 | (blank) | line is blank; visual separation only. |
| 2827 |                ui.label("Real-time ECM Snapshot:"); | executable/support line; behavior impact depends on surrounding block context. |
| 2828 |                Grid::new("ecm_live::snapshot_grid").show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
| 2829 |                    ui.label("Engine Speed"); | executable/support line; behavior impact depends on surrounding block context. |
| 2830 |                    ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 2831 |                        self.ecm_live_snapshot | executable/support line; behavior impact depends on surrounding block context. |
| 2832 |                    .engine_speed_rpm | executable/support line; behavior impact depends on surrounding block context. |
| 2833 |                    .map_or("-".into(), \|v\| format!("{v:.1} rpm")); | executable/support line; behavior impact depends on surrounding block context. |
| 2834 |                }); | executable/support line; behavior impact depends on surrounding block context. |
| 2835 |                ui.end_row(); | executable/support line; behavior impact depends on surrounding block context. |
| 2836 | (blank) | line is blank; visual separation only. |
| 2837 |                ui.label("Accel Pedal"); | executable/support line; behavior impact depends on surrounding block context. |
| 2838 |                ui.label( | executable/support line; behavior impact depends on surrounding block context. |

```



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| 2839 |         self.ecm_live_snapshot | executable/support line; behavior impact depends on surrounding block
context. |
| 2840 |         .accel_pedal_pct | executable/support line; behavior impact depends on surrounding block
context. |
| 2841 |         .map_or("-".into(), \|v\| format!("{v:.1} %")), | executable/support line; behavior impact
depends on surrounding block context. |
| 2842 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 2843 |         ui.end_row(); | executable/support line; behavior impact depends on surrounding block context. |
| 2844 | (blank) | line is blank; visual separation only. |
| 2845 |         ui.label("Coolant Temp"); | executable/support line; behavior impact depends on surrounding block
context. |
| 2846 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 2847 |         self.ecm_live_snapshot | executable/support line; behavior impact depends on surrounding block
context. |
| 2848 |         .coolant_temp_c | executable/support line; behavior impact depends on surrounding block
context. |
| 2849 |         .map_or("-".into(), \|v\| format!("{v:.1} C")), | executable/support line; behavior impact
depends on surrounding block context. |
| 2850 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 2851 |         ui.end_row(); | executable/support line; behavior impact depends on surrounding block context. |
| 2852 | (blank) | line is blank; visual separation only. |
| 2853 |         ui.label("Fuel Temp"); | executable/support line; behavior impact depends on surrounding block
context. |
| 2854 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 2855 |         self.ecm_live_snapshot | executable/support line; behavior impact depends on surrounding block
context. |
| 2856 |         .fuel_temp_c | executable/support line; behavior impact depends on surrounding block
context. |
| 2857 |         .map_or("-".into(), \|v\| format!("{v:.1} C")), | executable/support line; behavior impact
depends on surrounding block context. |
| 2858 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 2859 |         ui.end_row(); | executable/support line; behavior impact depends on surrounding block context. |
| 2860 | (blank) | line is blank; visual separation only. |
| 2861 |         ui.label("Oil Pressure"); | executable/support line; behavior impact depends on surrounding block
context. |
| 2862 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 2863 |         self.ecm_live_snapshot | executable/support line; behavior impact depends on surrounding block
context. |
| 2864 |         .oil_pressure_kpa | executable/support line; behavior impact depends on surrounding block
context. |
| 2865 |         .map_or("-".into(), \|v\| format!("{v:.1} kPa")), | executable/support line; behavior impact
depends on surrounding block context. |
| 2866 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 2867 |         ui.end_row(); | executable/support line; behavior impact depends on surrounding block context. |
| 2868 | (blank) | line is blank; visual separation only. |
| 2869 |         ui.label("Last PGN"); | executable/support line; behavior impact depends on surrounding block
context. |
| 2870 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 2871 |         self.ecm_live_snapshot | executable/support line; behavior impact depends on surrounding block
context. |
| 2872 |         .last_seen_pgn | executable/support line; behavior impact depends on surrounding block
context. |
| 2873 |         .map_or("-".into(), \|v\| format!("{v}")), | executable/support line; behavior impact
depends on surrounding block context. |
| 2874 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 2875 |         ui.end_row(); | executable/support line; behavior impact depends on surrounding block context. |
| 2876 | (blank) | line is blank; visual separation only. |
| 2877 |         ui.label("Source Address"); | executable/support line; behavior impact depends on surrounding block
context. |
| 2878 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 2879 |         self.ecm_live_snapshot | executable/support line; behavior impact depends on surrounding block
context. |
| 2880 |         .source_address | executable/support line; behavior impact depends on surrounding block
context. |
| 2881 |         .map_or("-".into(), \|v\| format!("{v:02X}")), | executable/support line; behavior impact

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depends on surrounding block context. |
| 2882 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 2883 |         ui.end_row(); | executable/support line; behavior impact depends on surrounding block context. |
| 2884 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 2885 | (blank) | line is blank; visual separation only. |
| 2886 |         self.ecm_treeview(ui, !live_mode); | executable/support line; behavior impact depends on sur
nding block context. |
| 2887 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 2888 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 2889 | (blank) | line is blank; visual separation only. |
| 2890 | fn toolbar(&mut self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter API, c
rol flow, and outputs. |
| 2891 |     ui.push_id("ui::toolbar", \|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 2892 |         ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 2893 |             ui.add_space(6.0); | executable/support line; behavior impact depends on surrounding block conte
|
| 2894 |             ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 2895 |                 RichText::new("■ ECU BENCH v3.0") | executable/support line; behavior impact depends on surr
ding block context. |
| 2896 |                 .size(17.0) | executable/support line; behavior impact depends on surrounding block cont
. |
| 2897 |                 .color(Color32::from_rgb(80, 155, 255)) | executable/support line; behavior impact depen
on surrounding block context. |
| 2898 |                 .strong(), | executable/support line; behavior impact depends on surrounding block conte
|
| 2899 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 2900 |             ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 2901 |             // IGN key | comment line; documentation intent, no runtime effect. |
| 2902 |             ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 2903 |                 RichText::new("IGN") | executable/support line; behavior impact depends on surrounding block
ntext. |
| 2904 |                 .size(11.0) | executable/support line; behavior impact depends on surrounding block cont
. |
| 2905 |                 .color(Color32::from_gray(150)), | executable/support line; behavior impact depends on s
ounding block context. |
| 2906 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 2907 |             let ign = self.bench.ignition(); | executable/support line; behavior impact depends on surroundi
block context. |
| 2908 |             for (lbl, target) in [ | iteration control; changing bounds/body affects throughput, timing, and
ssibly safety guarantees. |
| 2909 |                 ("OFF", IgnitionState::Off), | executable/support line; behavior impact depends on surroundi
block context. |
| 2910 |                 ("ACC", IgnitionState::Accessory), | executable/support line; behavior impact depends on sur
nding block context. |
| 2911 |                 ("ON", IgnitionState::On), | executable/support line; behavior impact depends on surrounding
ock context. |
| 2912 |                 ("START", IgnitionState::Cranking), | executable/support line; behavior impact depends on su
unding block context. |
| 2913 |             ] { | executable/support line; behavior impact depends on surrounding block context. |
| 2914 |                 let cur = ign == target; | executable/support line; behavior impact depends on surrounding b
k context. |
| 2915 |                 let col = if cur { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 2916 |                     match target { | branch dispatch; changing arms alters behavior for variants and input c
ses. |
| 2917 |                         IgnitionState::Off => Color32::from_gray(110), | executable/support line; behavior i
ct depends on surrounding block context. |
| 2918 |                         IgnitionState::Accessory => Color32::YELLOW, | executable/support line; behavior imp
depends on surrounding block context. |
| 2919 |                         IgnitionState::On => Color32::LIGHT_BLUE, | executable/support line; behavior impact
ends on surrounding block context. |
| 2920 |                         IgnitionState::Cranking => Color32::from_rgb(255, 130, 40), | executable/support lin
behavior impact depends on surrounding block context. |

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| 2921 | IgnitionState::Running => Color32::GREEN, | executable/support line; behavior impact
depends on surrounding block context. |
| 2922 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 2923 | } else { | executable/support line; behavior impact depends on surrounding block context. |
| 2924 | Color32::from_gray(40) | executable/support line; behavior impact depends on surrounding
ock context. |
| 2925 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and life
e scopes. |
| 2926 | let btn = Button::new(RichText::new(lbl).size(11.0).color(if cur { | executable/support line
ehavior impact depends on surrounding block context. |
| 2927 | Color32::BLACK | executable/support line; behavior impact depends on surrounding block c
ext. |
| 2928 | } else { | executable/support line; behavior impact depends on surrounding block context. |
| 2929 | Color32::from_gray(155) | executable/support line; behavior impact depends on surroundin
lock context. |
| 2930 | ))) | executable/support line; behavior impact depends on surrounding block context. |
| 2931 | .fill(col) | executable/support line; behavior impact depends on surrounding block context.
| 2932 | .min_size(Vec2::new(48.0, 26.0)); | executable/support line; behavior impact depends on surr
ding block context. |
| 2933 | if ui.add(btn).clicked() { | conditional gate; changing condition shifts runtime path and si
effects. |
| 2934 | match target { | branch dispatch; changing arms alters behavior for variants and input c
ses. |
| 2935 | IgnitionState::Off => self.cmds.push(Cmd::KeyOff), | executable/support line; behavi
impact depends on surrounding block context. |
| 2936 | _ => { | executable/support line; behavior impact depends on surrounding block conte
|
| 2937 | for _ in 0..5 { | iteration control; changing bounds/body affects throughput, ti
g, and possibly safety guarantees. |
| 2938 | if self.bench.ignition() == target { | conditional gate; changing condition
fts runtime path and side effects. |
| 2939 | break; | executable/support line; behavior impact depends on surrounding
ock context. |
| 2940 | } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 2941 | self.cmds.push(Cmd::KeyAdvance); | executable/support line; behavior impact
ends on surrounding block context. |
| 2942 | } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 2943 | } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 2944 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 2945 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 2946 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2947 | let run = self.bench.engine_running(); | executable/support line; behavior impact depends on sur
nding block context. |
| 2948 | let ad_locked = self.bench.ad.engaged; | executable/support line; behavior impact depends on sur
nding block context. |
| 2949 | ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 2950 | RichText::new(if run { "● RUN" } else { "■ OFF" }) | executable/support line; behavior impac
depends on surrounding block context. |
| 2951 | .size(11.0) | executable/support line; behavior impact depends on surrounding block cont
. |
| 2952 | .color(if run { | executable/support line; behavior impact depends on surrounding block
text. |
| 2953 | Color32::GREEN | executable/support line; behavior impact depends on surrounding blo
context. |
| 2954 | } else { | executable/support line; behavior impact depends on surrounding block context
| 2955 | Color32::from_gray(80) | executable/support line; behavior impact depends on surroun
g block context. |
| 2956 | }), | executable/support line; behavior impact depends on surrounding block context. |

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| 2957 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 2958 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 2959 |         // Throttle | comment line; documentation intent, no runtime effect. |
| 2960 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 2961 |             RichText::new("THROT") | executable/support line; behavior impact depends on surrounding blo
context. |
| 2962 |             .size(11.0) | executable/support line; behavior impact depends on surrounding block cont
. |
| 2963 |             .color(Color32::from_gray(145)), | executable/support line; behavior impact depends on s
urrounding block context. |
| 2964 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 2965 |         let mut thr = self.throttle_target; | executable/support line; behavior impact depends on surrou
ng block context. |
| 2966 |         let thr_resp = ui.add_enabled( | executable/support line; behavior impact depends on surrounding
ock context. |
| 2967 |             !ad_locked, | executable/support line; behavior impact depends on surrounding block context. |
| 2968 |             Slider::new(&mut thr, 0.0..=1.0) | executable/support line; behavior impact depends on surro
ing block context. |
| 2969 |             .show_value(false) | executable/support line; behavior impact depends on surrounding blo
context. |
| 2970 |             .trailing_fill(true), | executable/support line; behavior impact depends on surrounding
ck context. |
| 2971 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 2972 |         if ad_locked { | conditional gate; changing condition shifts runtime path and side effects. |
| 2973 |             let _ = thr_resp.on_disabled_hover_text("Throttle manual bloqueado enquanto o AD estiver ati
"); | executable/support line; behavior impact depends on surrounding block context. |
| 2974 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 2975 |             let thr_changed = (thr - self.throttle_cmd_last).abs() > 0.01; | executable/support line; be
ior impact depends on surrounding block context. |
| 2976 |             let thr_due = self.ticks.saturating_sub(self.throttle_cmd_last_tick) >= 2; | executable/supp
line; behavior impact depends on surrounding block context. |
| 2977 |             if thr != self.throttle_target && (thr_changed || thr_due) { | conditional gate; changing
dition shifts runtime path and side effects. |
| 2978 |                 self.cmds.push(Cmd::SetThrottle(thr)); | executable/support line; behavior impact depend
n surrounding block context. |
| 2979 |                 self.throttle_cmd_last = thr; | executable/support line; behavior impact depends on surr
ding block context. |
| 2980 |                 self.throttle_cmd_last_tick = self.ticks; | executable/support line; behavior impact dep
s on surrounding block context. |
| 2981 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 2982 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 2983 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 2984 |             RichText::new(format!("{}", "cmd {:3.0}% / act {:3.0}%", self.throttle_target * 100.0, self.thrott
* 100.0)) | executable/support line; behavior impact depends on surrounding block context. |
| 2985 |             .size(11.0) | executable/support line; behavior impact depends on surrounding block cont
. |
| 2986 |             .monospace() | executable/support line; behavior impact depends on surrounding block con
t. |
| 2987 |             .color(Color32::YELLOW), | executable/support line; behavior impact depends on surroundi
block context. |
| 2988 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 2989 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 2990 |         // Brake | comment line; documentation intent, no runtime effect. |
| 2991 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 2992 |             RichText::new("BRAKE") | executable/support line; behavior impact depends on surrounding blo
context. |
| 2993 |             .size(11.0) | executable/support line; behavior impact depends on surrounding block cont
. |
| 2994 |             .color(Color32::from_gray(145)), | executable/support line; behavior impact depends on s
urrounding block context. |
| 2995 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 2996 |         let mut brk = self.brake_target; | executable/support line; behavior impact depends on surroundi
block context. |

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| 2997 |         let brk_resp = ui.add_enabled( | executable/support line; behavior impact depends on surrounding
ock context. |
| 2998 |             !ad_locked, | executable/support line; behavior impact depends on surrounding block context.
| 2999 |             Slider::new(&mut brk, 0.0..=1.0) | executable/support line; behavior impact depends on surro
ing block context. |
| 3000 |                 .show_value(false) | executable/support line; behavior impact depends on surrounding blo
context. |
| 3001 |                 .trailing_fill(true), | executable/support line; behavior impact depends on surrounding
ck context. |
| 3002 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 3003 |             if ad_locked { | conditional gate; changing condition shifts runtime path and side effects. |
| 3004 |                 let _ = brk_resp.on_disabled_hover_text("Freio manual bloqueado enquanto o AD estiver ativo.
| executable/support line; behavior impact depends on surrounding block context. |
| 3005 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 3006 |                 let brk_changed = (brk - self.brake_cmd_last).abs() > 0.01; | executable/support line; behav
impact depends on surrounding block context. |
| 3007 |                 let brk_due = self.ticks.saturating_sub(self.brake_cmd_last_tick) >= 2; | executable/support
ne; behavior impact depends on surrounding block context. |
| 3008 |                 if brk != self.brake_target && (brk_changed || brk_due) { | conditional gate; changing con
ion shifts runtime path and side effects. |
| 3009 |                     self.cmds.push(Cmd::SetBrake(brk)); | executable/support line; behavior impact depends o
urrounding block context. |
| 3010 |                     self.brake_cmd_last = brk; | executable/support line; behavior impact depends on surroun
g block context. |
| 3011 |                     self.brake_cmd_last_tick = self.ticks; | executable/support line; behavior impact depend
n surrounding block context. |
| 3012 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 3013 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3014 |             ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 3015 |                 RichText::new(format!("{}", self.brake_target * 100.0, self.brake * 1
0)) | executable/support line; behavior impact depends on surrounding block context. |
| 3016 |                 .size(11.0) | executable/support line; behavior impact depends on surrounding block cont
. |
| 3017 |                 .monospace() | executable/support line; behavior impact depends on surrounding block con
t. |
| 3018 |                 .color(Color32::RED), | executable/support line; behavior impact depends on surrounding
ck context. |
| 3019 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 3020 |             ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 3021 |             // Direction | comment line; documentation intent, no runtime effect. |
| 3022 |             let dir_s = match self.bench.tcm.direction { | executable/support line; behavior impact depends
surrounding block context. |
| 3023 |                 Direction::Forward => "F", | executable/support line; behavior impact depends on surrounding
ock context. |
| 3024 |                 Direction::Reverse => "R", | executable/support line; behavior impact depends on surrounding
ock context. |
| 3025 |                 Direction::Neutral => "N", | executable/support line; behavior impact depends on surrounding
ock context. |
| 3026 |                 Direction::Park => "P", | executable/support line; behavior impact depends on surrounding bl
context. |
| 3027 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
opes. |
| 3028 |             let dir_resp = ui.add_enabled_ui(!ad_locked, \|ui\| direction_selector(ui, dir_s)); | executable
pport line; behavior impact depends on surrounding block context. |
| 3029 |             if ad_locked { | conditional gate; changing condition shifts runtime path and side effects. |
| 3030 |                 let _ = dir_resp | executable/support line; behavior impact depends on surrounding block con
t. |
| 3031 |                 .response | executable/support line; behavior impact depends on surrounding block context
|
| 3032 |                 .on_disabled_hover_text("Direcao bloqueada enquanto o AD estiver ativo."); | executable/
port line; behavior impact depends on surrounding block context. |
| 3033 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |

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| 3034 |         if let Some(k) = dir_resp.inner { | conditional gate; changing condition shifts runtime path and
de effects. |
| 3035 |             match k { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 3036 |                 'F' => self.cmds.push(Cmd::SetDirection(Direction::Forward)), | executable/support line;
havior impact depends on surrounding block context. |
| 3037 |                 'R' => self.cmds.push(Cmd::SetDirection(Direction::Reverse)), | executable/support line;
havior impact depends on surrounding block context. |
| 3038 |                 'N' => self.cmds.push(Cmd::SetNeutral), | executable/support line; behavior impact depen
on surrounding block context. |
| 3039 |                 _ => {} | executable/support line; behavior impact depends on surrounding block context.
| 3040 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 3041 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3042 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 3043 |         // Auto-shift | comment line; documentation intent, no runtime effect. |
| 3044 |         let auto = self.bench.tcm.auto_mode == AutoShiftMode::Auto; | executable/support line; behavior
act depends on surrounding block context. |
| 3045 |         let auto_btn = Button::new(RichText::new(if auto { "AUTO✓" } else { "MANUAL" })).size(11.0)) | ex
table/support line; behavior impact depends on surrounding block context. |
| 3046 |         .fill(if auto { | executable/support line; behavior impact depends on surrounding block cont
. |
| 3047 |             Color32::from_rgb(20, 85, 35) | executable/support line; behavior impact depends on surr
ding block context. |
| 3048 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 3049 |             Color32::from_gray(35) | executable/support line; behavior impact depends on surrounding
ock context. |
| 3050 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 3051 |         let auto_resp = ui.add_enabled(!ad_locked, auto_btn); | executable/support line; behavior impact
ends on surrounding block context. |
| 3052 |         if ad_locked { | conditional gate; changing condition shifts runtime path and side effects. |
| 3053 |             let _ = auto_resp.on_disabled_hover_text("Modo da transmissao bloqueado enquanto o AD estive
tivo."); | executable/support line; behavior impact depends on surrounding block context. |
| 3054 |         } else if auto_resp.clicked() { | executable/support line; behavior impact depends on surroundin
lock context. |
| 3055 |             self.cmds.push(Cmd::ToggleAutoShift); | executable/support line; behavior impact depends on
rounding block context. |
| 3056 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3057 |         // PTO | comment line; documentation intent, no runtime effect. |
| 3058 |         let pto = self.bench.implement.pto_rear_enabled; | executable/support line; behavior impact depe
on surrounding block context. |
| 3059 |         if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 3060 |             .add( | executable/support line; behavior impact depends on surrounding block context. |
| 3061 |                 Button::new(RichText::new(if pto { "PTO ON" } else { "PTO OFF" })).size(11.0)) | executab
support line; behavior impact depends on surrounding block context. |
| 3062 |             .fill(if pto { | executable/support line; behavior impact depends on surrounding blo
context. |
| 3063 |                 Color32::from_rgb(10, 95, 10) | executable/support line; behavior impact depends
surrounding block context. |
| 3064 |             } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 3065 |                 Color32::from_gray(35) | executable/support line; behavior impact depends on surr
nding block context. |
| 3066 |             }), | executable/support line; behavior impact depends on surrounding block context.
| 3067 |         ) | executable/support line; behavior impact depends on surrounding block context. |
| 3068 |         .clicked() | executable/support line; behavior impact depends on surrounding block context.
| 3069 |     { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3070 |         self.cmds.push(if pto { | executable/support line; behavior impact depends on surrounding bl
context. |
| 3071 |             Cmd::SetPtoMode(PtoMode::Off) | executable/support line; behavior impact depends on surr
ding block context. |
| 3072 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 3073 |             Cmd::SetPtoMode(PtoMode::Std540) | executable/support line; behavior impact depends on s

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| surrounding block context. |  
| 3074 |         }); | executable/support line; behavior impact depends on surrounding block context. |  
| 3075 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 3076 |     ui.separator(); | executable/support line; behavior impact depends on surrounding block context.  
| 3077 | // Lights | comment line; documentation intent, no runtime effect. |  
| 3078 |     let wl = self.bench.bcm.work_lights_front; | executable/support line; behavior impact depends on  
rounding block context. |  
| 3079 |     if ui | conditional gate; changing condition shifts runtime path and side effects. |  
| 3080 |         .add( | executable/support line; behavior impact depends on surrounding block context. |  
| 3081 |             Button::new(RichText::new(if wl { "■ WRK" } else { "■ off" })).size(11.0)) | executabl  
ort line; behavior impact depends on surrounding block context. |  
| 3082 |                 .fill(if wl { | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 3083 |                     Color32::from_rgb(85, 65, 8) | executable/support line; behavior impact depends  
surrounding block context. |  
| 3084 |                         ) else { | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 3085 |                             Color32::from_gray(32) | executable/support line; behavior impact depends on surr  
nding block context. |  
| 3086 |                                 }), | executable/support line; behavior impact depends on surrounding block context.  
| 3087 |                             ) | executable/support line; behavior impact depends on surrounding block context. |  
| 3088 |                             .clicked() | executable/support line; behavior impact depends on surrounding block context.  
| 3089 |     { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 3090 |         self.cmds.push(Cmd::ToggleWorkLights); | executable/support line; behavior impact depends on  
rounding block context. |  
| 3091 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 3092 |     ui.separator(); | executable/support line; behavior impact depends on surrounding block context.  
| 3093 | // Pause / Reset | comment line; documentation intent, no runtime effect. |  
| 3094 |     if ui | conditional gate; changing condition shifts runtime path and side effects. |  
| 3095 |         .add(Button::new(RichText::new("■ PAUSE")).size(12.0)).fill(Color32::from_gray(42))) | execut  
e/support line; behavior impact depends on surrounding block context. |  
| 3096 |             .clicked() | executable/support line; behavior impact depends on surrounding block context.  
| 3097 |             { /* toggle handled via bench.paused would need lib support; just use ticks */ | executable/supp  
line; behavior impact depends on surrounding block context. |  
| 3098 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 3099 |         if ui | conditional gate; changing condition shifts runtime path and side effects. |  
| 3100 |             .add( | executable/support line; behavior impact depends on surrounding block context. |  
| 3101 |                 Button::new(RichText::new("■ RESET").size(12.0)) | executable/support line; behavior imp  
depends on surrounding block context. |  
| 3102 |                     .fill(Color32::from_rgb(70, 22, 22)), | executable/support line; behavior impact dep  
s on surrounding block context. |  
| 3103 |                         ) | executable/support line; behavior impact depends on surrounding block context. |  
| 3104 |                         .clicked() | executable/support line; behavior impact depends on surrounding block context.  
| 3105 |             { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 3106 |                 self.cmds.push(Cmd::Reset); | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 3107 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 3108 |         }); | executable/support line; behavior impact depends on surrounding block context. |  
| 3109 |         }); | executable/support line; behavior impact depends on surrounding block context. |  
| 3110 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 3111 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 3112 | (blank) | line is blank; visual separation only. |  
| 3113 | // ████████████████████████████████████████████████████████████████ | comm  
intent, no runtime effect. |  
| 3114 | // TAB CLUSTER | comment line; documentation intent, no runtime effect. |  
| 3115 | // ████████████████████████████████████████████████████████████████ | comm  
intent, no runtime effect. |  
| 3116 | impl App { | implementation block; method behavior and trait conformance are defined here. |  
| 3117 |     fn tab_help(&mut self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter API,
```

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trol flow, and outputs. |
| 3118 |         ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block conte
|
| 3119 |         .auto_shrink([false, false]) | executable/support line; behavior impact depends on surrounding b
k context. |
| 3120 |         .show(ui, \ui\ { | executable/support line; behavior impact depends on surrounding block conte
|
| 3121 |             ui.heading("Help - Operacao Rapida e Guia de Diagnostico"); | executable/support line; behav
impact depends on surrounding block context. |
| 3122 |             ui.add_space(6.0); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3123 | (blank) | line is blank; visual separation only. |
| 3124 |             ui.group(\ui\ { | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3125 |                 ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 3126 |                     RichText::new("Fluxo recomendado (conciso)") | executable/support line; behavior imp
depends on surrounding block context. |
| 3127 |                     .size(12.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 3128 |                     .color(Color32::from_rgb(95, 170, 255)) | executable/support line; behavior impa
depends on surrounding block context. |
| 3129 |                     .strong(), | executable/support line; behavior impact depends on surrounding blo
context. |
| 3130 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 3131 |                 ui.label("1) BOOT: confirmar ignicao e readiness."); | executable/support line; behavior
impact depends on surrounding block context. |
| 3132 |                 ui.label("2) CLUSTER/ENGINE: validar rpm, pressao, temperatura e marcha."); | executable
support line; behavior impact depends on surrounding block context. |
| 3133 |                 ui.label("3) LEAK LAB: aplicar parametros, rodar cenarios/MC e interpretar risco."); | e
utable/support line; behavior impact depends on surrounding block context. |
| 3134 |                 ui.label("4) CAN/EVENTS/UDS: fazer RCA e exportar evidencia."); | I/O or protocol-critic
logic; edits can affect integration, timing, and diagnostic outcomes. |
| 3135 |                 ui.label("5) PLOTS: comparar antes/depois das mudancas."); | executable/support line; be
ior impact depends on surrounding block context. |
| 3136 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 3137 | (blank) | line is blank; visual separation only. |
| 3138 |             ui.add_space(8.0); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3139 |             ui.group(\ui\ { | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3140 |                 ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 3141 |                     RichText::new("Leak Lab - O que significa cada bloco") | executable/support line; be
ior impact depends on surrounding block context. |
| 3142 |                     .size(12.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 3143 |                     .color(Color32::from_rgb(95, 170, 255)) | executable/support line; behavior impa
depends on surrounding block context. |
| 3144 |                     .strong(), | executable/support line; behavior impact depends on surrounding blo
context. |
| 3145 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 3146 |                 ui.label("Runtime Circuits: estado atual de cada circuito com alertas e leak L/min."); |
executable/support line; behavior impact depends on surrounding block context. |
| 3147 |                 ui.label("Manual Engineering Input: parametros de pressao e vedacao para calibracao diri
a."); | executable/support line; behavior impact depends on surrounding block context. |
| 3148 |                 ui.label("Custom Circuit Builder: cria circuito novo para bancada/engenharia."); | execu
le/support line; behavior impact depends on surrounding block context. |
| 3149 |                 ui.label("Scenario Ranking: ordena cenarios por risco final e pressao de pico."); | exec
ble/support line; behavior impact depends on surrounding block context. |
| 3150 |                 ui.label("Timeback/ASCII/Plot: historico temporal de pressao, risco e vazao."); | execut
e/support line; behavior impact depends on surrounding block context. |
| 3151 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 3152 | (blank) | line is blank; visual separation only. |
| 3153 |             ui.add_space(8.0); | executable/support line; behavior impact depends on surrounding block c

```



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ext. |
| 3154 |           ui.group(\|ui\| { | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3155 |           ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 3156 |           RichText::new("Leak Lab - Envelope correto") | executable/support line; behavior imp
depends on surrounding block context. |
| 3157 |           .size(12.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 3158 |           .color(Color32::from_rgb(95, 170, 255)) | executable/support line; behavior impa
depends on surrounding block context. |
| 3159 |           .strong(), | executable/support line; behavior impact depends on surrounding blo
context. |
| 3160 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 3161 |           ui.label("Regra obrigatoria: min <= mean <= ideal <= max < rupture"); | executable/suppo
line; behavior impact depends on surrounding block context. |
| 3162 |           ui.label("Squeeze: tipico 10%-30%, Compression Set: menor tende a maior vida util."); |
cutable/support line; behavior impact depends on surrounding block context. |
| 3163 |           ui.label("Base leak area: influencia vazao de fuga inicial (mm²)." ); | executable/suppo
line; behavior impact depends on surrounding block context. |
| 3164 |           ui.label("Rupture bar define limite de falha catastrofica e o risco de burst."); | execu
le/support line; behavior impact depends on surrounding block context. |
| 3165 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 3166 | (blank) | line is blank; visual separation only. |
| 3167 |           ui.add_space(8.0); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3168 |           ui.group(\|ui\| { | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3169 |           ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 3170 |           RichText::new("Leak Lab - Como operar sem erro") | executable/support line; behavior
pact depends on surrounding block context. |
| 3171 |           .size(12.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 3172 |           .color(Color32::from_rgb(95, 170, 255)) | executable/support line; behavior impa
depends on surrounding block context. |
| 3173 |           .strong(), | executable/support line; behavior impact depends on surrounding blo
context. |
| 3174 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 3175 |           ui.label("1) Selecione circuito."); | executable/support line; behavior impact depends o
urrounding block context. |
| 3176 |           ui.label("2) Ajuste parametros e valide (a UI bloqueia aplicacao invalida)."); | executa
/support line; behavior impact depends on surrounding block context. |
| 3177 |           ui.label("3) APPLY MANUAL PARAMETERS."); | executable/support line; behavior impact depe
on surrounding block context. |
| 3178 |           ui.label("4) RUN SCENARIO PREDICTION ou RUN MONTE CARLO."); | executable/support line; b
vior impact depends on surrounding block context. |
| 3179 |           ui.label("5) Leia ranking + timeline + plot + alertas."); | executable/support line; beh
or impact depends on surrounding block context. |
| 3180 |           ui.label("6) Exporte CSV/JSON para rastreabilidade."); | executable/support line; behavi
impact depends on surrounding block context. |
| 3181 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 3182 | (blank) | line is blank; visual separation only. |
| 3183 |           ui.add_space(8.0); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3184 |           ui.group(\|ui\| { | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3185 |           ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 3186 |           RichText::new("Atalhos de diagnostico") | executable/support line; behavior impact d
nds on surrounding block context. |
| 3187 |           .size(12.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 3188 |           .color(Color32::from_rgb(95, 170, 255)) | executable/support line; behavior impa
depends on surrounding block context. |

```

```

| 3189 |             .strong(), | executable/support line; behavior impact depends on surrounding block
context. |
| 3190 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 3191 |             ui.label("- EVENTS: filtre por severidade e subsystem."); | executable/support line; beh
or impact depends on surrounding block context. |
| 3192 |             ui.label("- CAN: use Freeze antes de exportar snapshot."); | executable/support line; be
ior impact depends on surrounding block context. |
| 3193 |             ui.label("- UDS: siga Session -> Security -> Service -> Transfer Exit."); | I/O or proto
-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 3194 |             ui.label("- ECM LIVE: Detect -> Connect -> Retrieve -> Export."); | executable/support l
; behavior impact depends on surrounding block context. |
| 3195 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 3196 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 3197 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 3198 | (blank) | line is blank; visual separation only. |
| 3199 |         fn tab_cluster(&self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter API, c
rol flow, and outputs. |
| 3200 |             let ecm = &self.bench.ecm; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3201 |             let tcm = &self.bench.tcm; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3202 |             let abs = &self.bench.abs; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3203 |             ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 3204 |                 // Left: big gauges + warning lamps | comment line; documentation intent, no runtime effect. |
| 3205 |                 ui.vertical(\|ui\| { | executable/support line; behavior impact depends on surrounding block con
t. |
| 3206 |                     ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding blo
context. |
| 3207 |                         egui::Frame::dark_canvas(ui.style()).show(ui, \|ui\| { | executable/support line; behavi
impact depends on surrounding block context. |
| 3208 |                             arc_gauge( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3209 |                                 ui, ecm.rpm, 0.0, 3000.0, "ENGINE", "RPM", 155.0, 2000.0, 2400.0, None, | execut
e/support line; behavior impact depends on surrounding block context. |
| 3210 |                                     ) | executable/support line; behavior impact depends on surrounding block context. |
| 3211 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 3212 |                             egui::Frame::dark_canvas(ui.style()).show(ui, \|ui\| { | executable/support line; behavi
impact depends on surrounding block context. |
| 3213 |                                 arc_gauge( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3214 |                                     ui, | executable/support line; behavior impact depends on surrounding block cont
. |
| 3215 |                                     tcm.ground_speed_kmh, | executable/support line; behavior impact depends on surr
ding block context. |
| 3216 |                                     0.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 3217 |                                     50.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3218 |                                     "SPEED", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 3219 |                                     "km/h", | executable/support line; behavior impact depends on surrounding block
text. |
| 3220 |                                     155.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3221 |                                     40.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3222 |                                     48.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3223 |                                     None, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3224 |                                 ) | executable/support line; behavior impact depends on surrounding block context. |
| 3225 |                             }); | executable/support line; behavior impact depends on surrounding block context. |
| 3226 |                             egui::Frame::dark_canvas(ui.style()).show(ui, \|ui\| { | executable/support line; behavi

```

```

impact depends on surrounding block context. |
| 3227 |         arc_gauge( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3228 |             ui, | executable/support line; behavior impact depends on surrounding block cont
. |
| 3229 |             ecm.coolant_temp_c, | executable/support line; behavior impact depends on surrou
ng block context. |
| 3230 |             0.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 3231 |             130.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3232 |             "COOLANT", | executable/support line; behavior impact depends on surrounding blo
context. |
| 3233 |             "°C", | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3234 |             125.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3235 |             100.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3236 |             108.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3237 |             None, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3238 |         ) | executable/support line; behavior impact depends on surrounding block context. |
| 3239 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 3240 |     egui::Frame::dark_canvas(ui.style()).show(ui, \|ui\| { | executable/support line; behavi
impact depends on surrounding block context. |
| 3241 |         arc_gauge( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3242 |             ui, | executable/support line; behavior impact depends on surrounding block cont
. |
| 3243 |             ecm.oil_pressure_kpa, | executable/support line; behavior impact depends on surr
ding block context. |
| 3244 |             0.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 3245 |             700.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3246 |             "OIL PRESS", | executable/support line; behavior impact depends on surrounding b
k context. |
| 3247 |             "kPa", | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3248 |             125.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3249 |             0.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 3250 |             0.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 3251 |             Some(80.0), | executable/support line; behavior impact depends on surrounding bl
context. |
| 3252 |         ) | executable/support line; behavior impact depends on surrounding block context. |
| 3253 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 3254 |     egui::Frame::dark_canvas(ui.style()).show(ui, \|ui\| { | executable/support line; behavi
impact depends on surrounding block context. |
| 3255 |         arc_gauge( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3256 |             ui, | executable/support line; behavior impact depends on surrounding block cont
. |
| 3257 |             ecm.boost_pressure_kpa, | executable/support line; behavior impact depends on su
unding block context. |
| 3258 |             0.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 3259 |             300.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3260 |             "BOOST", | executable/support line; behavior impact depends on surrounding block
ntext. |

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| 3261 | "kPa", | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3262 | 110.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3263 | 250.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3264 | 280.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3265 | None, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3266 | ) | executable/support line; behavior impact depends on surrounding block context. |
| 3267 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 3268 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 3269 | egui::Frame::group(ui.style()) | executable/support line; behavior impact depends on surroun
g block context. |
| 3270 | .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on surr
ding block context. |
| 3271 | .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding blo
context. |
| 3272 | ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3273 | RichText::new(" ■ WARNING CLUSTER") | executable/support line; behavior impact o
nds on surrounding block context. |
| 3274 | .size(11.0) | executable/support line; behavior impact depends on surroundin
lock context. |
| 3275 | .color(Color32::from_gray(130)), | executable/support line; behavior impact
ends on surrounding block context. |
| 3276 | ); | executable/support line; behavior impact depends on surrounding block context.
| 3277 | ui.horizontal_wrapped(\|ui\| { | executable/support line; behavior impact depends on
rounding block context. |
| 3278 | warning_lamp(ui, ecm.red_lamp, "RED STOP", Color32::RED); | executable/support l
; behavior impact depends on surrounding block context. |
| 3279 | warning_lamp(ui, ecm.amber_lamp, "AMBER", Color32::GOLD); | executable/support l
; behavior impact depends on surrounding block context. |
| 3280 | warning_lamp(ui, ecm.mil_active, "MIL", Color32::LIGHT_BLUE); | executable/suppo
line; behavior impact depends on surrounding block context. |
| 3281 | warning_lamp( | executable/support line; behavior impact depends on surrounding
ck context. |
| 3282 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 3283 | ecm.protect_lamp, | executable/support line; behavior impact depends on surr
ding block context. |
| 3284 | "PROTECT", | executable/support line; behavior impact depends on surrounding
ock context. |
| 3285 | Color32::from_rgb(0, 210, 210), | executable/support line; behavior impact d
nds on surrounding block context. |
| 3286 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 3287 | warning_lamp( | executable/support line; behavior impact depends on surrounding
ck context. |
| 3288 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 3289 | ecm.oil_pressure_kpa < 100.0 && ecm.rpm > 500.0, | executable/support line;
avior impact depends on surrounding block context. |
| 3290 | "OIL PRESS", | executable/support line; behavior impact depends on surroundi
block context. |
| 3291 | Color32::RED, | executable/support line; behavior impact depends on surround
block context. |
| 3292 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 3293 | warning_lamp( | executable/support line; behavior impact depends on surrounding
ck context. |
| 3294 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 3295 | ecm.coolant_temp_c > 105.0, | executable/support line; behavior impact depen

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```

on surrounding block context. |
| 3296 |           "COOLANT HI", | executable/support line; behavior impact depends on surround
block context. |
| 3297 |           Color32::RED, | executable/support line; behavior impact depends on surround
block context. |
| 3298 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 3299 |           warning_lamp(ui, ecm.def_level_pct < 10.0, "DEF LOW", Color32::YELLOW); | execut
e/support line; behavior impact depends on surrounding block context. |
| 3300 |           warning_lamp( | executable/support line; behavior impact depends on surrounding
ck context. |
| 3301 |           ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 3302 |           ecm.dpf_soot_pct > 75.0, | executable/support line; behavior impact depends
surrounding block context. |
| 3303 |           "DPF REGEN", | executable/support line; behavior impact depends on surroundi
block context. |
| 3304 |           Color32::from_rgb(200, 80, 200), | executable/support line; behavior impact
ends on surrounding block context. |
| 3305 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 3306 |           warning_lamp( | executable/support line; behavior impact depends on surrounding
ck context. |
| 3307 |           ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 3308 |           ecm.fuel_level_pct < 10.0, | executable/support line; behavior impact depend
n surrounding block context. |
| 3309 |           "FUEL LOW", | executable/support line; behavior impact depends on surroundin
lock context. |
| 3310 |           Color32::YELLOW, | executable/support line; behavior impact depends on surro
ing block context. |
| 3311 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 3312 |           warning_lamp(ui, abs.abs_system_active, "ABS", Color32::WHITE); | executable/sup
t line; behavior impact depends on surrounding block context. |
| 3313 |           warning_lamp(ui, abs.esp_system_active, "ESP", Color32::LIGHT_BLUE); | executabl
upport line; behavior impact depends on surrounding block context. |
| 3314 |           warning_lamp( | executable/support line; behavior impact depends on surrounding
ck context. |
| 3315 |           ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 3316 |           abs.tcs_system_active, | executable/support line; behavior impact depends on
rrounding block context. |
| 3317 |           "TCS", | executable/support line; behavior impact depends on surrounding blo
context. |
| 3318 |           Color32::from_rgb(80, 200, 255), | executable/support line; behavior impact
ends on surrounding block context. |
| 3319 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 3320 |           warning_lamp( | executable/support line; behavior impact depends on surrounding
ck context. |
| 3321 |           ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 3322 |           self.bench.bcm.work_lights_front, | executable/support line; behavior impact
ends on surrounding block context. |
| 3323 |           "WORK LT", | executable/support line; behavior impact depends on surrounding
ock context. |
| 3324 |           Color32::from_rgb(255, 215, 0), | executable/support line; behavior impact d
nds on surrounding block context. |
| 3325 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 3326 |           warning_lamp( | executable/support line; behavior impact depends on surrounding
ck context. |
| 3327 |           ui, | executable/support line; behavior impact depends on surrounding block
text. |

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| 3328 | self.bench.implement.pto_rear_enabled, | executable/support line; behavior i
ct depends on surrounding block context. |
| 3329 | "PT0", | executable/support line; behavior impact depends on surrounding blo
context. |
| 3330 | Color32::GREEN, | executable/support line; behavior impact depends on surrou
ng block context. |
| 3331 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 3332 | warning_lamp(ui, self.bench.ad.engaged, "AD ON", Color32::GREEN); | executable/s
ort line; behavior impact depends on surrounding block context. |
| 3333 | warning_lamp( | executable/support line; behavior impact depends on surrounding
ck context. |
| 3334 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 3335 | self.bench.hcm.alarm_high_temp, | executable/support line; behavior impact d
nds on surrounding block context. |
| 3336 | "HYD TEMP", | executable/support line; behavior impact depends on surroundin
lock context. |
| 3337 | Color32::RED, | executable/support line; behavior impact depends on surround
block context. |
| 3338 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 3339 | warning_lamp( | executable/support line; behavior impact depends on surrounding
ck context. |
| 3340 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 3341 | self.bench.hcm.filter_bypass_open, | executable/support line; behavior impac
depends on surrounding block context. |
| 3342 | "HYD FILT", | executable/support line; behavior impact depends on surroundin
lock context. |
| 3343 | Color32::YELLOW, | executable/support line; behavior impact depends on surro
ing block context. |
| 3344 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 3345 | }); | executable/support line; behavior impact depends on surrounding block context.
| 3346 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 3347 | egui::Frame::group(ui.style()) | executable/support line; behavior impact depends on surroun
g block context. |
| 3348 | .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on surr
ding block context. |
| 3349 | .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding blo
context. |
| 3350 | ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3351 | RichText::new("ENGINE") | executable/support line; behavior impact depends on su
unding block context. |
| 3352 | .size(11.0) | executable/support line; behavior impact depends on surrounding
lock context. |
| 3353 | .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior
pact depends on surrounding block context. |
| 3354 | ); | executable/support line; behavior impact depends on surrounding block context.
| 3355 | let w = 130.0; | executable/support line; behavior impact depends on surrounding blo
context. |
| 3356 | bar_gauge(ui, "Load", ecm.percent_load, 150.0, "%", w, Color32::YELLOW); | executabl
upport line; behavior impact depends on surrounding block context. |
| 3357 | bar_gauge( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3358 | ui, | executable/support line; behavior impact depends on surrounding block cont
. |
| 3359 | "Torque", | executable/support line; behavior impact depends on surrounding blo
ontext. |
| 3360 | ecm.actual_torque_nm, | executable/support line; behavior impact depends on surr
ding block context. |
| 3361 | 1200.0, | executable/support line; behavior impact depends on surrounding block
text. |

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```
| 3362 | "Nm", | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3363 | w, | executable/support line; behavior impact depends on surrounding block conte
|
| 3364 | Color32::from_rgb(80, 180, 255), | executable/support line; behavior impact depe
on surrounding block context. |
| 3365 | ); | executable/support line; behavior impact depends on surrounding block context.
| 3366 | bar_gauge( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3367 | ui, | executable/support line; behavior impact depends on surrounding block cont
. |
| 3368 | "Power", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 3369 | ecm.power_kw(), | executable/support line; behavior impact depends on surroundin
lock context. |
| 3370 | 260.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3371 | "kw", | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3372 | w, | executable/support line; behavior impact depends on surrounding block conte
|
| 3373 | Color32::from_rgb(100, 220, 100), | executable/support line; behavior impact dep
s on surrounding block context. |
| 3374 | ); | executable/support line; behavior impact depends on surrounding block context.
| 3375 | bar_gauge( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3376 | ui, | executable/support line; behavior impact depends on surrounding block cont
. |
| 3377 | "Rail P", | executable/support line; behavior impact depends on surrounding blo
ontext. |
| 3378 | ecm.fuel_rail_pressure_mpa * 10.0, | executable/support line; behavior impact de
ds on surrounding block context. |
| 3379 | 2000.0, | executable/support line; behavior impact depends on surrounding block
text. |
| 3380 | "bar", | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3381 | w, | executable/support line; behavior impact depends on surrounding block conte
|
| 3382 | Color32::from_rgb(0, 210, 210), | executable/support line; behavior impact depen
on surrounding block context. |
| 3383 | ); | executable/support line; behavior impact depends on surrounding block context.
| 3384 | bar_gauge( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3385 | ui, | executable/support line; behavior impact depends on surrounding block cont
. |
| 3386 | "Exhaust", | executable/support line; behavior impact depends on surrounding blo
context. |
| 3387 | ecm.exhaust_temp_c, | executable/support line; behavior impact depends on surrou
ng block context. |
| 3388 | 900.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3389 | "°C", | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3390 | w, | executable/support line; behavior impact depends on surrounding block conte
|
| 3391 | Color32::from_rgb(255, 120, 30), | executable/support line; behavior impact depe
on surrounding block context. |
| 3392 | ); | executable/support line; behavior impact depends on surrounding block context.
| 3393 | bar_gauge( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3394 | ui, | executable/support line; behavior impact depends on surrounding block cont
. |
| 3395 | "Fuel/h", | executable/support line; behavior impact depends on surrounding blo
ontext. |
| 3396 | ecm.fuel_rate_lph, | executable/support line; behavior impact depends on surroun
```

```

g block context. |
| 3397 | 30.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3398 | "L/h", | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3399 | w, | executable/support line; behavior impact depends on surrounding block conte
|
| 3400 | Color32::YELLOW, | executable/support line; behavior impact depends on surroundi
block context. |
| 3401 | ); | executable/support line; behavior impact depends on surrounding block context.
| 3402 | bar_gauge( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3403 | ui, | executable/support line; behavior impact depends on surrounding block cont
. |
| 3404 | "DPF", | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3405 | ecm.dpf_soot_pct, | executable/support line; behavior impact depends on surround
block context. |
| 3406 | 100.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3407 | "%", | executable/support line; behavior impact depends on surrounding block con
t. |
| 3408 | w, | executable/support line; behavior impact depends on surrounding block conte
|
| 3409 | if ecm.dpf_soot_pct > 75.0 { | conditional gate; changing condition shifts runti
path and side effects. |
| 3410 | Color32::RED | executable/support line; behavior impact depends on surroundi
block context. |
| 3411 | } else { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 3412 | Color32::from_rgb(180, 100, 220) | executable/support line; behavior impact
ends on surrounding block context. |
| 3413 | }, | executable/support line; behavior impact depends on surrounding block conte
|
| 3414 | ); | executable/support line; behavior impact depends on surrounding block context.
| 3415 | bar_gauge( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3416 | ui, | executable/support line; behavior impact depends on surrounding block cont
. |
| 3417 | "DEF", | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3418 | ecm.def_level_pct, | executable/support line; behavior impact depends on surround
g block context. |
| 3419 | 100.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3420 | "%", | executable/support line; behavior impact depends on surrounding block con
t. |
| 3421 | w, | executable/support line; behavior impact depends on surrounding block conte
|
| 3422 | if ecm.def_level_pct < 10.0 { | conditional gate; changing condition shifts runt
path and side effects. |
| 3423 | Color32::RED | executable/support line; behavior impact depends on surroundi
block context. |
| 3424 | } else { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 3425 | Color32::from_rgb(0, 210, 210) | executable/support line; behavior impact de
ds on surrounding block context. |
| 3426 | }, | executable/support line; behavior impact depends on surrounding block conte
|
| 3427 | ); | executable/support line; behavior impact depends on surrounding block context.
| 3428 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 3429 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 3430 | ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 3431 | // Centre: transmission | comment line; documentation intent, no runtime effect. |
| 3432 | ui.vertical(\|ui\| { | executable/support line; behavior impact depends on surrounding block con

```



```

t. |
| 3433 |           egui::Frame::group(ui.style()) | executable/support line; behavior impact depends on surrounding block context. |
| 3434 |           .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on surrounding block context. |
| 3435 |           .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
| 3436 |               ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 3437 |                   RichText::new("TRANSMISSION") | executable/support line; behavior impact depends on surrounding block context. |
| 3438 |                   .size(11.0) | executable/support line; behavior impact depends on surrounding block context. |
| 3439 |                   .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior impact depends on surrounding block context. |
| 3440 |               ); | executable/support line; behavior impact depends on surrounding block context. |
| 3441 |               let dc = match tcm.direction { | executable/support line; behavior impact depends on surrounding block context. |
| 3442 |                   Direction::Forward => Color32::GREEN, | executable/support line; behavior impact depends on surrounding block context. |
| 3443 |                   Direction::Reverse => Color32::RED, | executable/support line; behavior impact depends on surrounding block context. |
| 3444 |                   Direction::Neutral => Color32::YELLOW, | executable/support line; behavior impact depends on surrounding block context. |
| 3445 |                   Direction::Park => Color32::from_gray(180), | executable/support line; behavior impact depends on surrounding block context. |
| 3446 |               }; | scope delimiter; structure-only line, but misplaced edits can change ownership of lifetime scopes. |
| 3447 |               let dl = match tcm.direction { | executable/support line; behavior impact depends on surrounding block context. |
| 3448 |                   Direction::Forward => "■ FWD", | executable/support line; behavior impact depends on surrounding block context. |
| 3449 |                   Direction::Reverse => "■ REV", | executable/support line; behavior impact depends on surrounding block context. |
| 3450 |                   Direction::Neutral => "■ NEU", | executable/support line; behavior impact depends on surrounding block context. |
| 3451 |                   Direction::Park => "P", | executable/support line; behavior impact depends on surrounding block context. |
| 3452 |               }; | scope delimiter; structure-only line, but misplaced edits can change ownership of lifetime scopes. |
| 3453 |               ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
| 3454 |                   ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 3455 |                       RichText::new(&tcm.gear_label) | executable/support line; behavior impact depends on surrounding block context. |
| 3456 |                       .size(38.0) | executable/support line; behavior impact depends on surrounding block context. |
| 3457 |                       .color(Color32::from_rgb(80, 220, 80)) | executable/support line; behavior impact depends on surrounding block context. |
| 3458 |                       .strong(), | executable/support line; behavior impact depends on surrounding block context. |
| 3459 |                   ); | executable/support line; behavior impact depends on surrounding block context. |
| 3460 |                   ui.add_space(8.0); | executable/support line; behavior impact depends on surrounding block context. |
| 3461 |                   ui.label(RichText::new(dl).size(22.0).color(dc).strong()); | executable/support line; behavior impact depends on surrounding block context. |
| 3462 |               }); | executable/support line; behavior impact depends on surrounding block context. |
| 3463 |               let am = match tcm.auto_mode { | executable/support line; behavior impact depends on surrounding block context. |
| 3464 |                   AutoShiftMode::Auto => "AUTO", | executable/support line; behavior impact depends on surrounding block context. |
| 3465 |                   AutoShiftMode::Manual => "MANUAL", | executable/support line; behavior impact depends on surrounding block context. |

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```

| 3466 |           AutoShiftMode::Hold => "HOLD", | executable/support line; behavior impact depend
n surrounding block context. |
| 3467 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |
| 3468 |           ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surroundi
g block context. |
| 3469 |           ui.label(RichText::new(am).size(12.0).color(Color32::YELLOW)); | executable/supp
line; behavior impact depends on surrounding block context. |
| 3470 |           ui.separator(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 3471 |           ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 3472 |           RichText::new(format!("Rng {} ", tcm.range)) | executable/support line; behav
impact depends on surrounding block context. |
| 3473 |           .size(12.0) | executable/support line; behavior impact depends on surroundi
ng block context. |
| 3474 |           .color(Color32::LIGHT_BLUE), | executable/support line; behavior impact
ends on surrounding block context. |
| 3475 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 3476 |           if tcm.creeper_engaged { | conditional gate; changing condition shifts runtime p
and side effects. |
| 3477 |           ui.separator(); | executable/support line; behavior impact depends on surroundi
ng block context. |
| 3478 |           ui.label(RichText::new("CREEP").size(11.0).color(Color32::YELLOW)); | execut
e/support line; behavior impact depends on surrounding block context. |
| 3479 |           } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 3480 |           }); | executable/support line; behavior impact depends on surrounding block context.
| 3481 |           let cc = match tcm.clutch_state { | executable/support line; behavior impact depends
surrounding block context. |
| 3482 |           ClutchState::Locked => Color32::GREEN, | executable/support line; behavior impac
depends on surrounding block context. |
| 3483 |           ClutchState::Modulating => Color32::YELLOW, | executable/support line; behavior
act depends on surrounding block context. |
| 3484 |           ClutchState::Filling => Color32::from_rgb(0, 220, 220), | executable/support lin
behavior impact depends on surrounding block context. |
| 3485 |           ClutchState::Open => Color32::from_gray(70), | executable/support line; behavior
pact depends on surrounding block context. |
| 3486 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |
| 3487 |           digital_readout(ui, "Clutch", &format!("{}", tcm.clutch_state), cc); | executable/su
rt line; behavior impact depends on surrounding block context. |
| 3488 |           bar_gauge( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3489 |           ui, | executable/support line; behavior impact depends on surrounding block cont
. |
| 3490 |           "Slip", | executable/support line; behavior impact depends on surrounding block
text. |
| 3491 |           tcm.clutch_slip_pct, | executable/support line; behavior impact depends on surrou
ing block context. |
| 3492 |           100.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3493 |           "%", | executable/support line; behavior impact depends on surrounding block con
t. |
| 3494 |           120.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3495 |           if tcm.clutch_slip_pct > 20.0 { | conditional gate; changing condition shifts ru
me path and side effects. |
| 3496 |           Color32::YELLOW | executable/support line; behavior impact depends on surrou
ng block context. |
| 3497 |           } else { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 3498 |           Color32::GREEN | executable/support line; behavior impact depends on surroun
g block context. |

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```

| 3499 |           }, | executable/support line; behavior impact depends on surrounding block context. |
|      |
| 3500 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 3501 |           if tcm.is_shifting { | conditional gate; changing condition shifts runtime path and
e effects. |
| 3502 |               ui.label( | executable/support line; behavior impact depends on surrounding block
ontext. |
| 3503 |                   RichText::new("■■ SHIFTING...") | executable/support line; behavior impact dep
s on surrounding block context. |
| 3504 |                       .size(11.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 3505 |                       .color(Color32::YELLOW), | executable/support line; behavior impact depe
on surrounding block context. |
| 3506 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 3507 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 3508 |       digital_readout( | executable/support line; behavior impact depends on surrounding b
k context. |
| 3509 |           ui, | executable/support line; behavior impact depends on surrounding block cont
. |
| 3510 |           "Speed", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 3511 |           &format!("{:.2} km/h", tcm.ground_speed_kmh), | executable/support line; behavio
mpact depends on surrounding block context. |
| 3512 |           Color32::LIGHT_BLUE, | executable/support line; behavior impact depends on surro
ing block context. |
| 3513 |       ); | executable/support line; behavior impact depends on surrounding block context. |
| 3514 |       digital_readout( | executable/support line; behavior impact depends on surrounding b
k context. |
| 3515 |           ui, | executable/support line; behavior impact depends on surrounding block cont
. |
| 3516 |           "Out RPM", | executable/support line; behavior impact depends on surrounding blo
context. |
| 3517 |           &format!("{:.0}", tcm.output_shaft_rpm), | executable/support line; behavior imp
depends on surrounding block context. |
| 3518 |           Color32::from_gray(200), | executable/support line; behavior impact depends on s
ounding block context. |
| 3519 |       ); | executable/support line; behavior impact depends on surrounding block context. |
| 3520 |       digital_readout( | executable/support line; behavior impact depends on surrounding b
k context. |
| 3521 |           ui, | executable/support line; behavior impact depends on surrounding block cont
. |
| 3522 |           "Shifts", | executable/support line; behavior impact depends on surrounding blo
ontext. |
| 3523 |           &format!("{}", tcm.total_shifts), | executable/support line; behavior impact dep
s on surrounding block context. |
| 3524 |           Color32::from_gray(160), | executable/support line; behavior impact depends on s
ounding block context. |
| 3525 |       ); | executable/support line; behavior impact depends on surrounding block context. |
| 3526 |       }); | executable/support line; behavior impact depends on surrounding block context. |
| 3527 |       egui::Frame::group(ui.style()) | executable/support line; behavior impact depends on surroun
g block context. |
| 3528 |           .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on surr
ding block context. |
| 3529 |           .show(ui, \ui\ { | executable/support line; behavior impact depends on surrounding blo
context. |
| 3530 |               ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3531 |                   RichText::new("ABS / WHEEL SPEEDS") | executable/support line; behavior impact d
nds on surrounding block context. |
| 3532 |                       .size(11.0) | executable/support line; behavior impact depends on surroundin
lock context. |
| 3533 |                       .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior
pact depends on surrounding block context. |

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```

| 3534 |           ); | executable/support line; behavior impact depends on surrounding block context.
| 3535 |           for (i, w) in abs.wheels.iter().enumerate() { | iteration control; changing bounds/b
affects throughput, timing, and possibly safety guarantees. |
| 3536 |               let n = ["FL", "FR", "RL", "RR"][i]; | executable/support line; behavior impact
ends on surrounding block context. |
| 3537 |               let c = if w.abs_active { | executable/support line; behavior impact depends on
rounding block context. |
| 3538 |                   Color32::RED | executable/support line; behavior impact depends on surroundi
block context. |
| 3539 |               } else if w.tcs_active { | executable/support line; behavior impact depends on s
ounding block context. |
| 3540 |                   Color32::from_rgb(80, 200, 255) | executable/support line; behavior impact d
nds on surrounding block context. |
| 3541 |               } else { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 3542 |                   Color32::GREEN | executable/support line; behavior impact depends on surroun
g block context. |
| 3543 |               }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 3544 |               ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on sur
nding block context. |
| 3545 |                   ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 3546 |                       RichText::new(format!( | executable/support line; behavior impact depend
n surrounding block context. |
| 3547 |                           "{:5.1}km/h {}", | executable/support line; behavior impact depe
on surrounding block context. |
| 3548 |                           n, w.speed, w.valve_state | executable/support line; behavior impact
ends on surrounding block context. |
| 3549 |                       )) | executable/support line; behavior impact depends on surrounding blo
context. |
| 3550 |                           .size(10.5) | executable/support line; behavior impact depends on surrou
ng block context. |
| 3551 |                           .monospace() | executable/support line; behavior impact depends on surro
ing block context. |
| 3552 |                           .color(c), | executable/support line; behavior impact depends on surroun
g block context. |
| 3553 |                   ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3554 |                   if w.slip_ratio > 0.1 { | conditional gate; changing condition shifts runtim
ath and side effects. |
| 3555 |                       ui.label( | executable/support line; behavior impact depends on surround
block context. |
| 3556 |                           RichText::new(format!("slip: {:.0}%", w.slip_ratio * 100.0)) | execut
e/support line; behavior impact depends on surrounding block context. |
| 3557 |                           .size(10.0) | executable/support line; behavior impact depends o
urrounding block context. |
| 3558 |                           .color(Color32::YELLOW), | executable/support line; behavior imp
depends on surrounding block context. |
| 3559 |                       ); | executable/support line; behavior impact depends on surrounding blo
context. |
| 3560 |                   } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 3561 |               }); | executable/support line; behavior impact depends on surrounding block cont
. |
| 3562 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 3563 |               let ec = match abs.esp_condition { | executable/support line; behavior impact depend
n surrounding block context. |
| 3564 |                   EspCondition::Neutral => Color32::GREEN, | executable/support line; behavior imp
depends on surrounding block context. |
| 3565 |                   EspCondition::Understeer => Color32::YELLOW, | executable/support line; behavior
pact depends on surrounding block context. |
| 3566 |                   _ => Color32::RED, | executable/support line; behavior impact depends on surroun
g block context. |

```

```

| 3567 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |
| 3568 |           ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surround
g block context. |
| 3569 |               ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 3570 |                   RichText::new("ESP:") | executable/support line; behavior impact depends on
rounding block context. |
| 3571 |                       .size(11.0) | executable/support line; behavior impact depends on surround
ng block context. |
| 3572 |                       .color(Color32::from_gray(140)), | executable/support line; behavior imp
depends on surrounding block context. |
| 3573 |                   ); | executable/support line; behavior impact depends on surrounding block conte
|
| 3574 |               ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 3575 |                   RichText::new(format!("{}", abs.esp_condition)) | executable/support line; b
vior impact depends on surrounding block context. |
| 3576 |                       .size(11.0) | executable/support line; behavior impact depends on surround
ng block context. |
| 3577 |                       .color(ec) | executable/support line; behavior impact depends on surround
g block context. |
| 3578 |                       .strong(), | executable/support line; behavior impact depends on surround
g block context. |
| 3579 |                   ); | executable/support line; behavior impact depends on surrounding block conte
|
| 3580 |               }); | executable/support line; behavior impact depends on surrounding block context.
| 3581 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 3582 |       }); | executable/support line; behavior impact depends on surrounding block context. |
| 3583 |       ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 3584 |       // Right: fuel/def/battery + hitch | comment line; documentation intent, no runtime effect. |
| 3585 |       ui.vertical(\|ui\| { | executable/support line; behavior impact depends on surrounding block con
t. |
| 3586 |           egui::Frame::dark_canvas(ui.style()).show(ui, \|ui\| { | executable/support line; behavior i
ct depends on surrounding block context. |
| 3587 |               arc_gauge( | executable/support line; behavior impact depends on surrounding block conte
|
| 3588 |                   ui, | executable/support line; behavior impact depends on surrounding block context.
| 3589 |                   ecm.fuel_level_pct, | executable/support line; behavior impact depends on surroundin
lock context. |
| 3590 |                       0.0, | executable/support line; behavior impact depends on surrounding block context
| 3591 |                       100.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 3592 |                       "FUEL", | executable/support line; behavior impact depends on surrounding block cont
. |
| 3593 |                       "%", | executable/support line; behavior impact depends on surrounding block context
| 3594 |                       125.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 3595 |                       0.0, | executable/support line; behavior impact depends on surrounding block context
| 3596 |                       0.0, | executable/support line; behavior impact depends on surrounding block context
| 3597 |                       Some(12.0), | executable/support line; behavior impact depends on surrounding block
text. |
| 3598 |                   ) | executable/support line; behavior impact depends on surrounding block context. |
| 3599 |               }); | executable/support line; behavior impact depends on surrounding block context. |
| 3600 |           egui::Frame::dark_canvas(ui.style()).show(ui, \|ui\| { | executable/support line; behavior i
ct depends on surrounding block context. |
| 3601 |               arc_gauge( | executable/support line; behavior impact depends on surrounding block conte
|
| 3602 |                   ui, | executable/support line; behavior impact depends on surrounding block context.
| 3603 |                   ecm.def_level_pct, | executable/support line; behavior impact depends on surrounding
ock context. |
| 3604 |                       0.0, | executable/support line; behavior impact depends on surrounding block context
| 3605 |                       100.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 3606 |                       "DEF/AdBlue", | executable/support line; behavior impact depends on surrounding bloc

```

```

ontext. |
| 3607 |          "%", | executable/support line; behavior impact depends on surrounding block context. |
| 3608 |          125.0, | executable/support line; behavior impact depends on surrounding block context. |
|      |
| 3609 |          0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 3610 |          0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 3611 |          Some(10.0), | executable/support line; behavior impact depends on surrounding block context. |
| 3612 |          ) | executable/support line; behavior impact depends on surrounding block context. |
| 3613 |      }); | executable/support line; behavior impact depends on surrounding block context. |
| 3614 |      egui::Frame::dark_canvas(ui.style()).show(ui, \ui\ { | executable/support line; behavior impact depends on surrounding block context. |
| 3615 |          arc_gauge( | executable/support line; behavior impact depends on surrounding block context. |
|      |
| 3616 |          ui, | executable/support line; behavior impact depends on surrounding block context. |
| 3617 |          self.bench.bcm.battery_voltage, | executable/support line; behavior impact depends on surrounding block context. |
| 3618 |          9.0, | executable/support line; behavior impact depends on surrounding block context. |
| 3619 |          16.0, | executable/support line; behavior impact depends on surrounding block context. |
|      |
| 3620 |          "BATTERY", | executable/support line; behavior impact depends on surrounding block context. |
| 3621 |          "V", | executable/support line; behavior impact depends on surrounding block context. |
| 3622 |          105.0, | executable/support line; behavior impact depends on surrounding block context. |
|      |
| 3623 |          0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 3624 |          0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 3625 |          Some(11.5), | executable/support line; behavior impact depends on surrounding block context. |
| 3626 |          ) | executable/support line; behavior impact depends on surrounding block context. |
| 3627 |      }); | executable/support line; behavior impact depends on surrounding block context. |
| 3628 |      egui::Frame::group(ui.style()) | executable/support line; behavior impact depends on surrounding block context. |
| 3629 |          .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on surrounding block context. |
| 3630 |          .show(ui, \ui\ { | executable/support line; behavior impact depends on surrounding block context. |
| 3631 |              ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 3632 |                  RichText::new("SERVICE") | executable/support line; behavior impact depends on surrounding block context. |
| 3633 |                  .size(11.0) | executable/support line; behavior impact depends on surrounding block context. |
| 3634 |                  .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior impact depends on surrounding block context. |
| 3635 |                  ); | executable/support line; behavior impact depends on surrounding block context. |
| 3636 |                  digital_readout( | executable/support line; behavior impact depends on surrounding block context. |
| 3637 |                      ui, | executable/support line; behavior impact depends on surrounding block context. |
|      |
| 3638 |                      "Hours", | executable/support line; behavior impact depends on surrounding block context. |
| 3639 |                      &format!("{:.1} h", ecm.engine_hours), | executable/support line; behavior impact depends on surrounding block context. |
| 3640 |                      Color32::from_gray(200), | executable/support line; behavior impact depends on surrounding block context. |
| 3641 |                      ); | executable/support line; behavior impact depends on surrounding block context. |
| 3642 |                      digital_readout( | executable/support line; behavior impact depends on surrounding block context. |
| 3643 |                          ui, | executable/support line; behavior impact depends on surrounding block context. |
|      |
| 3644 |                          "Svc due", | executable/support line; behavior impact depends on surrounding block context. |
| 3645 |                          &format!("{:.1} h", ecm.service_hours_left), | executable/support line; behavior impact depends on surrounding block context. |

```

```
| 3646 | if ecm.service_hours_left < 50.0 { | conditional gate; changing condition shifts  
ntime path and side effects. |  
| 3647 | Color32::RED | executable/support line; behavior impact depends on surroundi  
block context. |  
| 3648 | } else if ecm.service_hours_left < 100.0 { | executable/support line; behavior i  
ct depends on surrounding block context. |  
| 3649 | Color32::YELLOW | executable/support line; behavior impact depends on surrou  
ng block context. |  
| 3650 | } else { | executable/support line; behavior impact depends on surrounding block  
ncontext. |  
| 3651 | Color32::GREEN | executable/support line; behavior impact depends on surround  
g block context. |  
| 3652 | }, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 3653 | ); | executable/support line; behavior impact depends on surrounding block context.  
| 3654 | digital_readout( | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 3655 | ui, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 3656 | "NOx tail", | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 3657 | &format!("{:.0} ppm", ecm.nox_tailpipe_ppm), | executable/support line; behavior  
impact depends on surrounding block context. |  
| 3658 | if ecm.nox_tailpipe_ppm > 500.0 { | conditional gate; changing condition shifts  
time path and side effects. |  
| 3659 | Color32::RED | executable/support line; behavior impact depends on surroundi  
block context. |  
| 3660 | } else { | executable/support line; behavior impact depends on surrounding block  
ncontext. |  
| 3661 | Color32::from_gray(180) | executable/support line; behavior impact depends o  
urrounding block context. |  
| 3662 | }, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 3663 | ); | executable/support line; behavior impact depends on surrounding block context.  
| 3664 | }); | executable/support line; behavior impact depends on surrounding block context. |  
| 3665 | }); | executable/support line; behavior impact depends on surrounding block context. |  
| 3666 | }); | executable/support line; behavior impact depends on surrounding block context. |  
| 3667 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 3668 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 3669 | (blank) | line is blank; visual separation only. |  
| 3670 | // ████████████████████████████████████████████████████████████████████████ | comm  
intent, no runtime effect. |  
| 3671 | // TAB CAN BUS | comment line; documentation intent, no runtime effect. |  
| 3672 | // ████████████████████████████████████████████████████████████████████████ | comm  
intent, no runtime effect. |  
| 3673 | impl App { | implementation block; method behavior and trait conformance are defined here. |  
| 3674 | fn tab_can(&mut self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter API, c  
rol flow, and outputs. |  
| 3675 | // Controls bar | comment line; documentation intent, no runtime effect. |  
| 3676 | ui.horizontal(\ui\ { | executable/support line; behavior impact depends on surrounding block conte  
|  
| 3677 | let mc = if self.can_mode == CanMode::Signals { | executable/support line; behavior impact depen  
on surrounding block context. |  
| 3678 | Color32::from_rgb(45, 105, 210) | executable/support line; behavior impact depends on surrou  
ng block context. |  
| 3679 | } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 3680 | Color32::from_gray(35) | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 3681 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
opes. |  
| 3682 | let tc = if self.can_mode == CanMode::Trace { | executable/support line; behavior impact depends  
surrounding block context. |  
| 3683 | Color32::from_rgb(45, 105, 210) | executable/support line; behavior impact depends on surrou  
ng block context. |  
| 3684 | } else { | executable/support line; behavior impact depends on surrounding block context. |
```

```

| 3685 |           Color32::from_gray(35) | executable/support line; behavior impact depends on surrounding block
context. |
| 3686 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
opes. |
| 3687 |           let nc = if self.can_mode == CanMode::Network { | executable/support line; behavior impact depend
on surrounding block context. |
| 3688 |               Color32::from_rgb(45, 105, 210) | executable/support line; behavior impact depends on surround
ng block context. |
| 3689 |           } else { | executable/support line; behavior impact depends on surrounding block context. |
| 3690 |               Color32::from_gray(35) | executable/support line; behavior impact depends on surrounding blo
context. |
| 3691 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
opes. |
| 3692 |           if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 3693 |               .add(Button::new(RichText::new("■ Signals").size(12.0)).fill(mc)) | executable/support line;
havior impact depends on surrounding block context. |
| 3694 |               .clicked() | executable/support line; behavior impact depends on surrounding block context.
| 3695 |           { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3696 |               self.can_mode = CanMode::Signals; | executable/support line; behavior impact depends on surr
ding block context. |
| 3697 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3698 |           if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 3699 |               .add(Button::new(RichText::new("■ Trace").size(12.0)).fill(tc)) | executable/support line; b
vior impact depends on surrounding block context. |
| 3700 |               .clicked() | executable/support line; behavior impact depends on surrounding block context.
| 3701 |           { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3702 |               self.can_mode = CanMode::Trace; | executable/support line; behavior impact depends on surrou
ng block context. |
| 3703 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3704 |           if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 3705 |               .add(Button::new(RichText::new("■ Network").size(12.0)).fill(nc)) | executable/support line;
havior impact depends on surrounding block context. |
| 3706 |               .clicked() | executable/support line; behavior impact depends on surrounding block context.
| 3707 |           { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3708 |               self.can_mode = CanMode::Network; | executable/support line; behavior impact depends on surr
ding block context. |
| 3709 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3710 |           ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 3711 |           let fc = if self.can_freeze { | executable/support line; behavior impact depends on surrounding
ck context. |
| 3712 |               Color32::RED | executable/support line; behavior impact depends on surrounding block context
| 3713 |           } else { | executable/support line; behavior impact depends on surrounding block context. |
| 3714 |               Color32::from_gray(38) | executable/support line; behavior impact depends on surrounding blo
context. |
| 3715 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
opes. |
| 3716 |           if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 3717 |               .add( | executable/support line; behavior impact depends on surrounding block context. |
| 3718 |                   Button::new( | executable/support line; behavior impact depends on surrounding block con
t. |
| 3719 |                       RichText::new(if self.can_freeze { | executable/support line; behavior impact depend
n surrounding block context. |
| 3720 |                           "■ FROZEN" | executable/support line; behavior impact depends on surrounding blo
context. |
| 3721 |                       } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 3722 |                           "■ Freeze" | executable/support line; behavior impact depends on surrounding blo
context. |
| 3723 |                       }) | executable/support line; behavior impact depends on surrounding block context.

```



```

| 3724 |         .size(12.0), | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 3725 |         ) | executable/support line; behavior impact depends on surrounding block context. |
| 3726 |         .fill(fc), | executable/support line; behavior impact depends on surrounding block conte
|
| 3727 |         ) | executable/support line; behavior impact depends on surrounding block context. |
| 3728 |         .clicked() | executable/support line; behavior impact depends on surrounding block context.
| 3729 |     { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3730 |         self.can_freeze = !self.can_freeze; | executable/support line; behavior impact depends on su
unding block context. |
| 3731 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3732 |         if ui.button("■ Clear").clicked() { | conditional gate; changing condition shifts runtime path a
side effects. |
| 3733 |             self.sig_map.clear(); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 3734 |             self.trace_snap.clear(); | executable/support line; behavior impact depends on surrounding b
k context. |
| 3735 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3736 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 3737 |         ui.label(RichText::new("Filter:").size(11.0).color(Color32::GRAY)); | executable/support line; b
vior impact depends on surrounding block context. |
| 3738 |         ui.add_sized( | executable/support line; behavior impact depends on surrounding block context. |
| 3739 |             [220.0, 20.0], | executable/support line; behavior impact depends on surrounding block conte
|
| 3740 |             TextEdit::singleline(&mut self.can_filter) | executable/support line; behavior impact depend
n surrounding block context. |
| 3741 |             .hint_text("text \\\ sa:00 \\\ pgn:61444 \\\ id:18F00400"), | executable/support line; beha
r impact depends on surrounding block context. |
| 3742 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 3743 |         ui.checkbox(&mut self.can_hide_stale, "Hide stale"); | executable/support line; behavior impact
ends on surrounding block context. |
| 3744 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 3745 |         ui.label(RichText::new("Rows").size(10.8).color(Color32::from_gray(145))); | executable/support
e; behavior impact depends on surrounding block context. |
| 3746 |         ui.add(Slider::new(&mut self.can_signal_limit, 50..=1200).text("sig")); | executable/support lin
behavior impact depends on surrounding block context. |
| 3747 |         ui.add(Slider::new(&mut self.can_trace_limit, 100..=2000).text("trace")); | executable/support l
; behavior impact depends on surrounding block context. |
| 3748 |         if matches!(self.can_mode, CanMode::Trace) { | conditional gate; changing condition shifts runti
path and side effects. |
| 3749 |             ui.separator(); | executable/support line; behavior impact depends on surrounding block cont
. |
| 3750 |             ui.label(RichText::new("Trace tick").size(10.8).color(Color32::from_gray(145))); | executabl
upport line; behavior impact depends on surrounding block context. |
| 3751 |             ui.add( | executable/support line; behavior impact depends on surrounding block context. |
| 3752 |                 Slider::new(&mut self.can_trace_update_every_ticks, 1..=30) | executable/support line; b
vior impact depends on surrounding block context. |
| 3753 |                 .text("every") | executable/support line; behavior impact depends on surrounding blo
context. |
| 3754 |                 .suffix("t"), | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 3755 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 3756 |             ui.checkbox(&mut self.can_trace_pause_on_scroll, "Pause on scroll"); | executable/support li
behavior impact depends on surrounding block context. |
| 3757 |             if self.can_trace_pause_ticks_left > 0 { | conditional gate; changing condition shifts runti
path and side effects. |
| 3758 |                 ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 3759 |                 RichText::new(format!("{}", self.can_trace_pause_ticks_left)) | executable/s
ort line; behavior impact depends on surrounding block context. |
| 3760 |                 .size(10.8) | executable/support line; behavior impact depends on surrounding bl
context. |

```

```

| 3761 |                                     .color(Color32::YELLOW), | executable/support line; behavior impact depends on s
ounding block context. |
| 3762 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 3763 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 3764 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3765 |                                     ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 3766 |                                     let gw = &self.bench.gateway; | executable/support line; behavior impact depends on surrounding
ck context. |
| 3767 |                                     let state_col = match gw.bus_state { | executable/support line; behavior impact depends on surro
ing block context. |
| 3768 |                                     BusState::ErrorActive => Color32::GREEN, | executable/support line; behavior impact depends
surrounding block context. |
| 3769 |                                     BusState::ErrorPassive => Color32::YELLOW, | executable/support line; behavior impact depend
n surrounding block context. |
| 3770 |                                     BusState::BusOff => Color32::RED, | executable/support line; behavior impact depends on surr
ding block context. |
| 3771 |                                     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
opes. |
| 3772 |                                     ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 3773 |                                     RichText::new(format!( | executable/support line; behavior impact depends on surrounding blo
context. |
| 3774 |                                     "Bus:{{}} Load:{{:.0}}% Frames:{{}} {{}}/s Err:{{}}", | executable/support line; behavior impact d
nds on surrounding block context. |
| 3775 |                                     gw.bus_state, gw.bus_load_pct, gw.total_tx, gw.bus.fps, gw.total_errors | executable/sup
t line; behavior impact depends on surrounding block context. |
| 3776 |                                     )) | executable/support line; behavior impact depends on surrounding block context. |
| 3777 |                                     .size(11.0) | executable/support line; behavior impact depends on surrounding block context. |
| 3778 |                                     .color(state_col), | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3779 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 3780 |                                     )); | executable/support line; behavior impact depends on surrounding block context. |
| 3781 |                                     ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 3782 |                                     RichText::new("Tip: use sa:XX / pgm:NNNN / id:XXXXXXXX for deterministic filtering") | executabl
upport line; behavior impact depends on surrounding block context. |
| 3783 |                                     .size(9.8) | executable/support line; behavior impact depends on surrounding block context. |
| 3784 |                                     .color(Color32::from_gray(125)), | executable/support line; behavior impact depends on surro
ing block context. |
| 3785 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 3786 |                                     ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 3787 |                                     match self.can_mode { | branch dispatch; changing arms alters behavior for variants and input classe
|
| 3788 |                                     CanMode::Signals => self.can_signals(ui), | executable/support line; behavior impact depends on
rounding block context. |
| 3789 |                                     CanMode::Trace => self.can_trace(ui), | executable/support line; behavior impact depends on surr
ding block context. |
| 3790 |                                     CanMode::Network => self.can_network_overview(ui), | executable/support line; behavior impact de
ds on surrounding block context. |
| 3791 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 3792 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 3793 | (blank) | line is blank; visual separation only. |
| 3794 |         fn can_network_overview(&mut self, ui: &mut Ui) { | declares executable behavior; signature/body changes
ter API, control flow, and outputs. |
| 3795 |             let net = &self.bench.can_net; | executable/support line; behavior impact depends on surrounding blo
context. |
| 3796 |             let health = (net.network_health_score_01() * 100.0).round(); | executable/support line; behavior im
t depends on surrounding block context. |
| 3797 |             let hc = if health >= 85.0 { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 3798 |                 Color32::GREEN | executable/support line; behavior impact depends on surrounding block context. |
| 3799 |             } else if health >= 65.0 { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 3800 |                 Color32::YELLOW | executable/support line; behavior impact depends on surrounding block context.

```

```

| 3801 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 3802 |             Color32::RED | executable/support line; behavior impact depends on surrounding block context. |
| 3803 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime sco
. |
| 3804 |         ui.label(RichText::new(format!( | executable/support line; behavior impact depends on surrounding bl
context. |
| 3805 |             "Multi-bus overview \ Health: {:.0}% \ Frames(all buses): {} \ Nodes online: {} \ Errors(tot
: {} ", | executable/support line; behavior impact depends on surrounding block context. |
| 3806 |             health, | executable/support line; behavior impact depends on surrounding block context. |
| 3807 |             net.total_frames_all_buses, | executable/support line; behavior impact depends on surrounding bl
context. |
| 3808 |             net.online_count(), | executable/support line; behavior impact depends on surrounding block cont
. |
| 3809 |             net.total_errors() | executable/support line; behavior impact depends on surrounding block conte
|
| 3810 |         )).size(11.0).color(hc)); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3811 |         if !self.can_note.is_empty() { | conditional gate; changing condition shifts runtime path and side e
cts. |
| 3812 |             ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 3813 |                 RichText::new(&self.can_note) | executable/support line; behavior impact depends on surround
block context. |
| 3814 |                     .size(10.4) | executable/support line; behavior impact depends on surrounding block cont
. |
| 3815 |                     .color(Color32::LIGHT_BLUE), | executable/support line; behavior impact depends on surro
ing block context. |
| 3816 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 3817 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 3818 |         ui.horizontal(\ui\ { | executable/support line; behavior impact depends on surrounding block conte
|
| 3819 |             const BUS_NAMES: [&str; 5] = ["Powertrain", "Chassis", "Body", "ISOBUS", "Diagnostic"]; | execut
e/support line; behavior impact depends on surrounding block context. |
| 3820 |             ComboBox::from_id_salt("can::target_bus") | executable/support line; behavior impact depends on
rounding block context. |
| 3821 |                 .selected_text(BUS_NAMES[self.can_bus_idx.min(4)]) | executable/support line; behavior impac
depends on surrounding block context. |
| 3822 |                 .show_ui(ui, \ui\ { | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 3823 |                     for (i, n) in BUS_NAMES.iter().enumerate() { | iteration control; changing bounds/body a
cts throughput, timing, and possibly safety guarantees. |
| 3824 |                         ui.selectable_value(&mut self.can_bus_idx, i, *n); | executable/support line; behavi
impact depends on surrounding block context. |
| 3825 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 3826 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 3827 | (blank) | line is blank; visual separation only. |
| 3828 |                 if ui.button("Inject Bit").clicked() { | conditional gate; changing condition shifts runtime pat
nd side effects. |
| 3829 |                     self.cmds.push(Cmd::CanInjectBitError(self.can_bus_idx)); | executable/support line; behavio
mpact depends on surrounding block context. |
| 3830 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3831 |                 if ui.button("Inject Ack").clicked() { | conditional gate; changing condition shifts runtime pat
nd side effects. |
| 3832 |                     self.cmds.push(Cmd::CanInjectAckError(self.can_bus_idx)); | executable/support line; behavio
mpact depends on surrounding block context. |
| 3833 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3834 |                 if ui.button("Inject BusOff").clicked() { | conditional gate; changing condition shifts runtime
h and side effects. |
| 3835 |                     self.cmds.push(Cmd::CanInjectBusOff(self.can_bus_idx)); | executable/support line; behavior
act depends on surrounding block context. |
| 3836 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |

```

```

| 3837 |         if ui.button("Inject Babbling").clicked() { | conditional gate; changing condition shifts runtime
ath and side effects. |
| 3838 |             self.cmds.push(Cmd::CanInjectBabbling(self.can_bus_idx)); | executable/support line; behavior
mpact depends on surrounding block context. |
| 3839 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3840 |         if ui.button("Clear Bus").clicked() { | conditional gate; changing condition shifts runtime path
d side effects. |
| 3841 |             self.cmds.push(Cmd::CanClearBusInjections(self.can_bus_idx)); | executable/support line; beh
or impact depends on surrounding block context. |
| 3842 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3843 |         if ui.button("Clear All").clicked() { | conditional gate; changing condition shifts runtime path
d side effects. |
| 3844 |             self.cmds.push(Cmd::CanClearAllInjections); | executable/support line; behavior impact depen
on surrounding block context. |
| 3845 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3846 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 3847 |         if ui.button("Export CSV").clicked() { | conditional gate; changing condition shifts runtime pat
nd side effects. |
| 3848 |             let ts = Local::now().format("%Y%m%d_%H%M%S"); | executable/support line; behavior impact de
ds on surrounding block context. |
| 3849 |             let name = format!("can_network_snapshot_{}.csv", ts); | executable/support line; behavior i
ct depends on surrounding block context. |
| 3850 |             if let Some(path) = Self::pick_save_path(&name, "csv") { | conditional gate; changing condit
shifts runtime path and side effects. |
| 3851 |                 self.cmds.push(Cmd::CanExportSnapshotCsv(path)); | executable/support line; behavior imp
depends on surrounding block context. |
| 3852 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 3853 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3854 |         if ui.button("Export JSON").clicked() { | conditional gate; changing condition shifts runtime pa
and side effects. |
| 3855 |             let ts = Local::now().format("%Y%m%d_%H%M%S"); | executable/support line; behavior impact de
ds on surrounding block context. |
| 3856 |             let name = format!("can_network_snapshot_{}.json", ts); | executable/support line; behavior
act depends on surrounding block context. |
| 3857 |             if let Some(path) = Self::pick_save_path(&name, "json") { | conditional gate; changing condi
n shifts runtime path and side effects. |
| 3858 |                 self.cmds.push(Cmd::CanExportSnapshotJson(path)); | executable/support line; behavior im
t depends on surrounding block context. |
| 3859 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 3860 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 3861 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 3862 |     ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 3863 | (blank) | line is blank; visual separation only. |
| 3864 |     let bus_cards = [ | executable/support line; behavior impact depends on surrounding block context. |
| 3865 |         ("HS-CAN Powertrain", &net.powertrain), | executable/support line; behavior impact depends on su
unding block context. |
| 3866 |         ("HS-CAN Chassis", &net.chassis), | executable/support line; behavior impact depends on surround
block context. |
| 3867 |         ("MS-CAN Body", &net.body), | executable/support line; behavior impact depends on surrounding bl
context. |
| 3868 |         ("ISOBUS", &net.isobus), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 3869 |         ("Diag", &net.diagnostic), | executable/support line; behavior impact depends on surrounding blo
context. |
| 3870 |     ]; | executable/support line; behavior impact depends on surrounding block context. |
| 3871 | (blank) | line is blank; visual separation only. |
| 3872 |     egui::Grid::new("can::net_buses") | executable/support line; behavior impact depends on surrounding
ck context. |

```

```

| 3873 |         .num_columns(6) | executable/support line; behavior impact depends on surrounding block context.
| 3874 |         .striped(true) | executable/support line; behavior impact depends on surrounding block context.
| 3875 |         .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 3876 |             for h in ["BUS", "STATE", "LOAD", "TX", "LOG", "ERRORS"] { | iteration control; changing bou
/body affects throughput, timing, and possibly safety guarantees. |
| 3877 |                 ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 3878 |                     RichText::new(h) | executable/support line; behavior impact depends on surrounding b
k context. |
| 3879 |                         .size(10.8) | executable/support line; behavior impact depends on surrounding bl
context. |
| 3880 |                         .color(Color32::from_gray(160)) | executable/support line; behavior impact depen
on surrounding block context. |
| 3881 |                         .strong(), | executable/support line; behavior impact depends on surrounding blo
context. |
| 3882 |                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 3883 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 3884 |             ui.end_row(); | executable/support line; behavior impact depends on surrounding block context
|
| 3885 | (blank) | line is blank; visual separation only. |
| 3886 |             for (name, bus) in bus_cards { | iteration control; changing bounds/body affects throughput,
ming, and possibly safety guarantees. |
| 3887 |                 let state_col = match bus.state { | executable/support line; behavior impact depends on
rounding block context. |
| 3888 |                     auto_breaking::can_network::BusState::ErrorActive => Color32::GREEN, | executable/su
rt line; behavior impact depends on surrounding block context. |
| 3889 |                     auto_breaking::can_network::BusState::ErrorPassive => Color32::YELLOW, | executable/
port line; behavior impact depends on surrounding block context. |
| 3890 |                     auto_breaking::can_network::BusState::BusOff => Color32::RED, | executable/support l
; behavior impact depends on surrounding block context. |
| 3891 |                 }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 3892 |                 ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 3893 |                     RichText::new(name) | executable/support line; behavior impact depends on surroundin
lock context. |
| 3894 |                         .size(10.8) | executable/support line; behavior impact depends on surrounding bl
context. |
| 3895 |                         .color(Color32::from_gray(220)), | executable/support line; behavior impact depe
on surrounding block context. |
| 3896 |                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 3897 |                 ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 3898 |                     RichText::new(format!("{}", bus.state)) | executable/support line; behavior impact d
nds on surrounding block context. |
| 3899 |                         .size(10.8) | executable/support line; behavior impact depends on surrounding bl
context. |
| 3900 |                         .monospace() | executable/support line; behavior impact depends on surrounding b
k context. |
| 3901 |                         .color(state_col), | executable/support line; behavior impact depends on surroun
g block context. |
| 3902 |                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 3903 |                 ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 3904 |                     RichText::new(format!("{:.1}%", bus.bus_load_pct)) | executable/support line; behavi
impact depends on surrounding block context. |
| 3905 |                         .size(10.8) | executable/support line; behavior impact depends on surrounding bl
context. |
| 3906 |                         .monospace() | executable/support line; behavior impact depends on surrounding b
k context. |
| 3907 |                         .color(Color32::from_rgb(100, 205, 255)), | executable/support line; behavior im
t depends on surrounding block context. |
| 3908 |                     ); | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 3909 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 3910 |         RichText::new(format!("{}", bus.total_tx)) | executable/support line; behavior impact depends on surrounding block context. |
| 3911 |         .size(10.8) | executable/support line; behavior impact depends on surrounding block context. |
| 3912 |         .monospace() | executable/support line; behavior impact depends on surrounding block context. |
| 3913 |         .color(Color32::from_gray(190)), | executable/support line; behavior impact depends on surrounding block context. |
| 3914 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 3915 |     ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 3916 |     RichText::new(format!("{}", bus.frame_log.len())) | executable/support line; behavior impact depends on surrounding block context. |
| 3917 |     .size(10.8) | executable/support line; behavior impact depends on surrounding block context. |
| 3918 |     .monospace() | executable/support line; behavior impact depends on surrounding block context. |
| 3919 |     .color(Color32::from_gray(160)), | executable/support line; behavior impact depends on surrounding block context. |
| 3920 | ); | executable/support line; behavior impact depends on surrounding block context. |
| 3921 | ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 3922 | RichText::new(format!("{}", bus.total_errors)) | executable/support line; behavior impact depends on surrounding block context. |
| 3923 | .size(10.8) | executable/support line; behavior impact depends on surrounding block context. |
| 3924 | .monospace() | executable/support line; behavior impact depends on surrounding block context. |
| 3925 | .color(if bus.total_errors > 0 { | executable/support line; behavior impact depends on surrounding block context. |
| 3926 |     Color32::YELLOW | executable/support line; behavior impact depends on surrounding block context. |
| 3927 | } else { | executable/support line; behavior impact depends on surrounding block context. |
| 3928 |     Color32::GREEN | executable/support line; behavior impact depends on surrounding block context. |
| 3929 | })), | executable/support line; behavior impact depends on surrounding block context. |
| 3930 | ); | executable/support line; behavior impact depends on surrounding block context. |
| 3931 | ui.end_row(); | executable/support line; behavior impact depends on surrounding block context. |
| 3932 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 3933 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 3934 | (blank) | line is blank; visual separation only. |
| 3935 | ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 3936 | ui.columns(2, \cols\ { | executable/support line; behavior impact depends on surrounding block context. |
| 3937 |     egui::Frame::group(cols[0].style()) | executable/support line; behavior impact depends on surrounding block context. |
| 3938 |     .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on surrounding block context. |
| 3939 |     .show(&mut cols[0], \ui\ { | executable/support line; behavior impact depends on surrounding block context. |
| 3940 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 3941 |         RichText::new("VCM ROUTING TABLE") | executable/support line; behavior impact depends on surrounding block context. |
| 3942 |         .size(11.0) | executable/support line; behavior impact depends on surrounding block context. |
| 3943 |         .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior impact depends on surrounding block context. |
| 3944 |     ); | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 3945 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 3946 |         for (pgn_id, src, dst) in &net.routing { | iteration control; changing bounds/body affects
throughput, timing, and possibly safety guarantees. |
| 3947 |             let pgn_name = auto_breaking::find_pgn(*pgn_id) | executable/support line; behavior
act depends on surrounding block context. |
| 3948 |                 .map(\|p\| p.name) | executable/support line; behavior impact depends on surround
g block context. |
| 3949 |                 .unwrap_or("???"); | error-propagating operation; changing it impacts failure se
tics and caller recovery paths. |
| 3950 |         ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3951 |             RichText::new(format!( | executable/support line; behavior impact depends on surr
nding block context. |
| 3952 |                 "PGN {:5} {:8} {} -> {}", | executable/support line; behavior impact depend
n surrounding block context. |
| 3953 |                 pgn_id, pgn_name, src, dst | executable/support line; behavior impact depend
n surrounding block context. |
| 3954 |             )) | executable/support line; behavior impact depends on surrounding block conte
|
| 3955 |                 .size(10.2) | executable/support line; behavior impact depends on surrounding bl
context. |
| 3956 |                 .monospace() | executable/support line; behavior impact depends on surrounding b
k context. |
| 3957 |                 .color(Color32::from_gray(200)), | executable/support line; behavior impact depe
on surrounding block context. |
| 3958 |             ); | executable/support line; behavior impact depends on surrounding block context.
| 3959 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 3960 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 3961 | (blank) | line is blank; visual separation only. |
| 3962 |     egui::Frame::group(cols[1].style()) | executable/support line; behavior impact depends on surrou
ng block context. |
| 3963 |         .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on surround
block context. |
| 3964 |         .show(&mut cols[1], \|ui\| { | executable/support line; behavior impact depends on surroundi
block context. |
| 3965 |             ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 3966 |                 RichText::new("LATEST NETWORK ERRORS") | executable/support line; behavior impact de
ds on surrounding block context. |
| 3967 |                 .size(11.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 3968 |                 .color(Color32::from_rgb(255, 150, 80)), | executable/support line; behavior imp
depends on surrounding block context. |
| 3969 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 3970 |             ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 3971 |             for err in net.all_errors().take(14) { | iteration control; changing bounds/body affects
roughput, timing, and possibly safety guarantees. |
| 3972 |                 let sa = err.source_sa.map_or("--".into(), \|v\| format!("{:02X}", v)); | executable
pport line; behavior impact depends on surrounding block context. |
| 3973 |                 let id = err | executable/support line; behavior impact depends on surrounding block
ntext. |
| 3974 |                 .raw_id | executable/support line; behavior impact depends on surrounding block
text. |
| 3975 |                 .map_or("-----".into(), \|v\| format!("{:08X}", v)); | executable/support lin
behavior impact depends on surrounding block context. |
| 3976 |             ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3977 |                 RichText::new(format!( | executable/support line; behavior impact depends on surr
nding block context. |
| 3978 |                 "{:.2}s [{}] SA {} ID {} {:?}", | error-propagating operation; changing it i
cts failure semantics and caller recovery paths. |
| 3979 |                 err.timestamp, err.bus, sa, id, err.kind | executable/support line; behavior

```

```

pact depends on surrounding block context. |
| 3980 |             )) | executable/support line; behavior impact depends on surrounding block conte
|
| 3981 |             .size(9.9) | executable/support line; behavior impact depends on surrounding blo
context. |
| 3982 |             .monospace() | executable/support line; behavior impact depends on surrounding b
k context. |
| 3983 |             .color(Color32::from_rgb(255, 170, 110)), | executable/support line; behavior im
t depends on surrounding block context. |
| 3984 |             ); | executable/support line; behavior impact depends on surrounding block context.
| 3985 |             ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 3986 |             RichText::new(err.description) | executable/support line; behavior impact depend
n surrounding block context. |
| 3987 |             .size(9.6) | executable/support line; behavior impact depends on surrounding
ock context. |
| 3988 |             .color(Color32::from_gray(150)), | executable/support line; behavior impact
ends on surrounding block context. |
| 3989 |             ); | executable/support line; behavior impact depends on surrounding block context.
| 3990 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 3991 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 3992 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 3993 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 3994 | (blank) | line is blank; visual separation only. |
| 3995 |     fn can_signals(&self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter API, c
rol flow, and outputs. |
| 3996 |         let filt = self.can_filter.clone(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 3997 |         // Table headers | comment line; documentation intent, no runtime effect. |
| 3998 |         egui::Grid::new("can::sig_hdr") | executable/support line; behavior impact depends on surrounding bl
context. |
| 3999 |             .num_columns(6) | executable/support line; behavior impact depends on surrounding block context.
| 4000 |             .min_col_width(60.0) | executable/support line; behavior impact depends on surrounding block con
t. |
| 4001 |             .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 4002 |                 for h in ["AGE(s)", "PGN", "SA", "PERIOD", "COUNT", "DECODED VALUES"] { | iteration control;
anging bounds/body affects throughput, timing, and possibly safety guarantees. |
| 4003 |                     ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 4004 |                         RichText::new(h) | executable/support line; behavior impact depends on surrounding b
k context. |
| 4005 |                         .size(10.5) | executable/support line; behavior impact depends on surrounding bl
context. |
| 4006 |                         .color(Color32::from_gray(155)) | executable/support line; behavior impact dependen
on surrounding block context. |
| 4007 |                         .strong(), | executable/support line; behavior impact depends on surrounding blo
context. |
| 4008 |                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 4009 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 4010 |                 ui.end_row(); | executable/support line; behavior impact depends on surrounding block context
|
| 4011 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 4012 |             ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 4013 |             let t = self.bench.elapsed; | executable/support line; behavior impact depends on surrounding block
text. |
| 4014 |             let mut sigs: Vec<(&u32, u8), &Signal> = self | executable/support line; behavior impact depends o
urrounding block context. |
| 4015 |                 .sig_map | executable/support line; behavior impact depends on surrounding block context. |
| 4016 |                 .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 4017 |                 .filter(\|((pgn, sa), s)\| Self::can_filter_match_signal(&filt, *pgn, *sa, s)) | executable/supp
line; behavior impact depends on surrounding block context. |
| 4018 |                 .filter(\|(_, s)\| !self.can_hide_stale \| s.fresh) | executable/support line; behavior impact

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depends on surrounding block context. |
| 4019 |         .collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 4020 |         sigs.sort_by_key(\|((pgn, sa), _)\| (*sa, *pgn)); | executable/support line; behavior impact depends
surrounding block context. |
| 4021 | (blank) | line is blank; visual separation only. |
| 4022 |         let total = sigs.len(); | executable/support line; behavior impact depends on surrounding block cont
. |
| 4023 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 4024 |             RichText::new(format!( | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4025 |                 "Showing {} / {} signals", | executable/support line; behavior impact depends on surrounding
ock context. |
| 4026 |                 total.min(self.can_signal_limit), | executable/support line; behavior impact depends on surr
ding block context. |
| 4027 |                 total | executable/support line; behavior impact depends on surrounding block context. |
| 4028 |             )) | executable/support line; behavior impact depends on surrounding block context. |
| 4029 |             .size(10.2) | executable/support line; behavior impact depends on surrounding block context. |
| 4030 |             .color(Color32::from_gray(130)), | executable/support line; behavior impact depends on surroundi
block context. |
| 4031 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 4032 | (blank) | line is blank; visual separation only. |
| 4033 |         ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block conte
|
| 4034 |             .auto_shrink([false, false]) | executable/support line; behavior impact depends on surrounding b
k context. |
| 4035 |             .max_height(ui.available_height()) | executable/support line; behavior impact depends on surroun
g block context. |
| 4036 |             .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 4037 |             for ((pgn, sa), sig) in sigs.iter().take(self.can_signal_limit) { | iteration control; chang
bounds/body affects throughput, timing, and possibly safety guarantees. |
| 4038 |                 let age = t - sig.last_ts; | executable/support line; behavior impact depends on surroun
g block context. |
| 4039 |                 let ac = if age < 0.05 { | executable/support line; behavior impact depends on surroundi
block context. |
| 4040 |                     Color32::GREEN | executable/support line; behavior impact depends on surrounding blo
context. |
| 4041 |                     } else if age < 0.5 { | executable/support line; behavior impact depends on surrounding
ck context. |
| 4042 |                     Color32::from_gray(220) | executable/support line; behavior impact depends on surrou
ng block context. |
| 4043 |                     } else if age < 2.0 { | executable/support line; behavior impact depends on surrounding
ck context. |
| 4044 |                     Color32::from_gray(150) | executable/support line; behavior impact depends on surrou
ng block context. |
| 4045 |                     } else { | executable/support line; behavior impact depends on surrounding block context
|
| 4046 |                     Color32::from_gray(80) | executable/support line; behavior impact depends on surroun
g block context. |
| 4047 |                 }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 4048 |                 let vals = sig | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4049 |                     .decoded | executable/support line; behavior impact depends on surrounding block con
t. |
| 4050 |                     .iter() | executable/support line; behavior impact depends on surrounding block cont
. |
| 4051 |                     .take(4) | executable/support line; behavior impact depends on surrounding block con
t. |
| 4052 |                     .map(\|(n, v, u)\| { | executable/support line; behavior impact depends on surroundi
block context. |
| 4053 |                         format!( | executable/support line; behavior impact depends on surrounding block
ntext. |
| 4054 |                             "{:}:{:.1}{}", | executable/support line; behavior impact depends on surround
block context. |
| 4055 |                             n.split_whitespace().next().unwrap_or("?"), | error-propagating operation; c

```

```

ging it impacts failure semantics and caller recovery paths. |
| 4056 |           v, | executable/support line; behavior impact depends on surrounding block co
ext. |
| 4057 |           u | executable/support line; behavior impact depends on surrounding block co
xt. |
| 4058 |           ) | executable/support line; behavior impact depends on surrounding block contex
|
| 4059 |       }) | executable/support line; behavior impact depends on surrounding block context.
| 4060 |       .collect::<Vec<_>>() | executable/support line; behavior impact depends on surroundi
block context. |
| 4061 |       .join(" "); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 4062 |       ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 4063 |           ui.add_sized( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 4064 |               [56.0, 18.0], | executable/support line; behavior impact depends on surrounding
ck context. |
| 4065 |               Label::new( | executable/support line; behavior impact depends on surrounding bl
context. |
| 4066 |                   RichText::new(format!("{:>5.2}", age)) | executable/support line; behavior i
ct depends on surrounding block context. |
| 4067 |                   .size(10.2) | executable/support line; behavior impact depends on surrou
ng block context. |
| 4068 |                   .monospace() | executable/support line; behavior impact depends on surro
ing block context. |
| 4069 |                   .color(ac), | executable/support line; behavior impact depends on surrou
ng block context. |
| 4070 |               ), | executable/support line; behavior impact depends on surrounding block conte
|
| 4071 |           ); | executable/support line; behavior impact depends on surrounding block context.
| 4072 |       ui.add_sized( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 4073 |               [190.0, 18.0], | executable/support line; behavior impact depends on surrounding
ock context. |
| 4074 |               Label::new( | executable/support line; behavior impact depends on surrounding bl
context. |
| 4075 |                   RichText::new(format!("{:>5} {}", pgn, sig.pgn_name)) | executable/support l
; behavior impact depends on surrounding block context. |
| 4076 |                   .size(10.2) | executable/support line; behavior impact depends on surrou
ng block context. |
| 4077 |                   .monospace() | executable/support line; behavior impact depends on surro
ing block context. |
| 4078 |                   .color(Color32::GOLD), | executable/support line; behavior impact depend
n surrounding block context. |
| 4079 |               ), | executable/support line; behavior impact depends on surrounding block conte
|
| 4080 |           ); | executable/support line; behavior impact depends on surrounding block context.
| 4081 |       ui.add_sized( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 4082 |               [125.0, 18.0], | executable/support line; behavior impact depends on surrounding
ock context. |
| 4083 |               Label::new( | executable/support line; behavior impact depends on surrounding bl
context. |
| 4084 |                   RichText::new(format!("0x{:02X} {}", sa, sig.sa_name)) | executable/support
e; behavior impact depends on surrounding block context. |
| 4085 |                   .size(10.2) | executable/support line; behavior impact depends on surrou
ng block context. |
| 4086 |                   .monospace() | executable/support line; behavior impact depends on surro
ing block context. |
| 4087 |                   .color(Color32::from_rgb(120, 200, 255)), | executable/support line; beh
or impact depends on surrounding block context. |
| 4088 |               ), | executable/support line; behavior impact depends on surrounding block conte
|
| 4089 |           ); | executable/support line; behavior impact depends on surrounding block context.

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| 4090 |                 ui.add_sized( | executable/support line; behavior impact depends on surrounding block
ontext. |
| 4091 |                 [76.0, 18.0], | executable/support line; behavior impact depends on surrounding
ck context. |
| 4092 |                 Label::new( | executable/support line; behavior impact depends on surrounding bl
context. |
| 4093 |                 RichText::new(format!("{:>5.0}ms", sig.period_ms)) | executable/support line
ehavior impact depends on surrounding block context. |
| 4094 |                 .size(10.2) | executable/support line; behavior impact depends on surrou
ng block context. |
| 4095 |                 .monospace() | executable/support line; behavior impact depends on surro
ing block context. |
| 4096 |                 .color(Color32::from_gray(150)), | executable/support line; behavior imp
depends on surrounding block context. |
| 4097 |                 ), | executable/support line; behavior impact depends on surrounding block conte
|
| 4098 |                 ); | executable/support line; behavior impact depends on surrounding block context.
| 4099 |                 ui.add_sized( | executable/support line; behavior impact depends on surrounding block
ontext. |
| 4100 |                 [74.0, 18.0], | executable/support line; behavior impact depends on surrounding
ck context. |
| 4101 |                 Label::new( | executable/support line; behavior impact depends on surrounding bl
context. |
| 4102 |                 RichText::new(format!("{:>7}", sig.count)) | executable/support line; behavi
impact depends on surrounding block context. |
| 4103 |                 .size(10.2) | executable/support line; behavior impact depends on surrou
ng block context. |
| 4104 |                 .monospace() | executable/support line; behavior impact depends on surro
ing block context. |
| 4105 |                 .color(Color32::from_gray(130)), | executable/support line; behavior imp
depends on surrounding block context. |
| 4106 |                 ), | executable/support line; behavior impact depends on surrounding block conte
|
| 4107 |                 ); | executable/support line; behavior impact depends on surrounding block context.
| 4108 |                 ui.label(RichText::new(&vals).size(10.2).color(ac)); | executable/support line; beha
r impact depends on surrounding block context. |
| 4109 |                 )); | executable/support line; behavior impact depends on surrounding block context. |
| 4110 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 4111 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 4112 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 4113 | (blank) | line is blank; visual separation only. |
| 4114 |             fn can_trace(&mut self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter API,
ontrol flow, and outputs. |
| 4115 |                 ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 4116 |                 ui.label(RichText::new("Trace View:").size(10.8).color(Color32::GRAY)); | executable/support lin
behavior impact depends on surrounding block context. |
| 4117 |                 ui.selectable_value(&mut self.can_trace_tree, false, "Table"); | executable/support line; behavi
impact depends on surrounding block context. |
| 4118 |                 ui.selectable_value(&mut self.can_trace_tree, true, "Tree"); | executable/support line; behavior
impact depends on surrounding block context. |
| 4119 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 4120 |                 ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 4121 | (blank) | line is blank; visual separation only. |
| 4122 |                 let filtered_rows: Vec<&TraceRow> = self | executable/support line; behavior impact depends on surro
ing block context. |
| 4123 |                 .trace_snap | executable/support line; behavior impact depends on surrounding block context. |
| 4124 |                 .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 4125 |                 .filter(\|r\| Self::can_filter_match_trace(&self.can_filter, r)) | executable/support line; beha
r impact depends on surrounding block context. |
| 4126 |                 .collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 4127 |                 ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 4128 |                 RichText::new(format!( | executable/support line; behavior impact depends on surrounding block c
ext. |

```

```

| 4129 |         "Rows after filter: {} (render limit {})", | executable/support line; behavior impact depends
n surrounding block context. |
| 4130 |         filtered_rows.len(), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 4131 |         self.can_trace_limit | executable/support line; behavior impact depends on surrounding block
ntext. |
| 4132 |     )) | executable/support line; behavior impact depends on surrounding block context. |
| 4133 |     .size(10.2) | executable/support line; behavior impact depends on surrounding block context. |
| 4134 |     .color(Color32::from_gray(130)), | executable/support line; behavior impact depends on surroundi
block context. |
| 4135 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 4136 | (blank) | line is blank; visual separation only. |
| 4137 |     if self.can_trace_tree { | conditional gate; changing condition shifts runtime path and side effects
| 4138 |         let mut by_sa: BTreeMap<u8, Vec<TraceRow>> = BTreeMap::new(); | executable/support line; behavio
mpact depends on surrounding block context. |
| 4139 |         for row in filtered_rows.iter().rev().take(self.can_trace_limit) { | iteration control; changing
unds/body affects throughput, timing, and possibly safety guarantees. |
| 4140 |             by_sa.entry(row.2).or_default().push((*row).clone()); | executable/support line; behavior im
t depends on surrounding block context. |
| 4141 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 4142 | (blank) | line is blank; visual separation only. |
| 4143 |         ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4144 |             .auto_shrink([false, false]) | executable/support line; behavior impact depends on surroundi
block context. |
| 4145 |             .max_height(ui.available_height()) | executable/support line; behavior impact depends on surr
nding block context. |
| 4146 |             .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4147 |                 for (sa, rows) in by_sa { | iteration control; changing bounds/body affects throughput,
ing, and possibly safety guarantees. |
| 4148 |                     CollapsingHeader::new(format!("Node SA 0x{:02X} ({} frames)", sa, rows.len())) | exe
able/support line; behavior impact depends on surrounding block context. |
| 4149 |                         .default_open(sa == 0x00 \| \| sa == 0x03) | executable/support line; behavior im
t depends on surrounding block context. |
| 4150 |                         .show(ui, \|ui\| { | executable/support line; behavior impact depends on surroun
g block context. |
| 4151 |                             let mut by_pgn: BTreeMap<String, Vec<TraceRow>> = BTreeMap::new(); | executab
/support line; behavior impact depends on surrounding block context. |
| 4152 |                             for r in rows { | iteration control; changing bounds/body affects throughput
iming, and possibly safety guarantees. |
| 4153 |                                 by_pgn.entry(r.5.clone()).or_default().push(r); | executable/support lin
behavior impact depends on surrounding block context. |
| 4154 |                             } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 4155 |                             for (pgn, mut frames) in by_pgn { | iteration control; changing bounds/body
ects throughput, timing, and possibly safety guarantees. |
| 4156 |                                 frames.sort_by(\|a, b\| a.0.partial_cmp(&b.0).unwrap_or(std::cmp::Orderi
:Equal)); | executable/support line; behavior impact depends on surrounding block context. |
| 4157 |                                 CollapsingHeader::new(format!("{}", pgn, frames.len())) | executabl
upport line; behavior impact depends on surrounding block context. |
| 4158 |                                     .default_open(false) | executable/support line; behavior impact depe
on surrounding block context. |
| 4159 |                                     .show(ui, \|ui\| { | executable/support line; behavior impact depend
n surrounding block context. |
| 4160 |                                         for (ts, raw_id, _sa, dlc, hex, _pgn_sa, decoded) in | iteration
ntrol; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 4161 |                                             frames.iter().rev().take(12).rev() | executable/support line
ehavior impact depends on surrounding block context. |
| 4162 |                                             { | scope delimiter; structure-only line, but misplaced edits ca
hange ownership and lifetime scopes. |
| 4163 |                                                 ui.horizontal_wrapped(\|ui\| { | executable/support line; be
ior impact depends on surrounding block context. |
| 4164 |                                                     ui.label( | executable/support line; behavior impact dep

```

```

s on surrounding block context. |
| 4165 |                                RichText::new(format!("{}", ts)) | executable
pport line; behavior impact depends on surrounding block context. |
| 4166 |                                .monospace() | executable/support line; behavior
pact depends on surrounding block context. |
| 4167 |                                .size(10.0) | executable/support line; behavior
act depends on surrounding block context. |
| 4168 |                                .color(Color32::from_gray(130)), | executable/su
rt line; behavior impact depends on surrounding block context. |
| 4169 |                                ); | executable/support line; behavior impact depends on
rrounding block context. |
| 4170 |                                ui.label( | executable/support line; behavior impact dep
s on surrounding block context. |
| 4171 |                                RichText::new(format!("{}", raw_id)) | execut
e/support line; behavior impact depends on surrounding block context. |
| 4172 |                                .monospace() | executable/support line; behavior
pact depends on surrounding block context. |
| 4173 |                                .size(10.0) | executable/support line; behavior
act depends on surrounding block context. |
| 4174 |                                .color(Color32::LIGHT_BLUE), | executable/suppor
ine; behavior impact depends on surrounding block context. |
| 4175 |                                ); | executable/support line; behavior impact depends on
rrounding block context. |
| 4176 |                                ui.label( | executable/support line; behavior impact dep
s on surrounding block context. |
| 4177 |                                RichText::new(format!("{}", dlc)) | executable/s
ort line; behavior impact depends on surrounding block context. |
| 4178 |                                .monospace() | executable/support line; behavior
pact depends on surrounding block context. |
| 4179 |                                .size(10.0) | executable/support line; behavior
act depends on surrounding block context. |
| 4180 |                                .color(Color32::from_gray(140)), | executable/su
rt line; behavior impact depends on surrounding block context. |
| 4181 |                                ); | executable/support line; behavior impact depends on
rrounding block context. |
| 4182 |                                ui.label( | executable/support line; behavior impact dep
s on surrounding block context. |
| 4183 |                                RichText::new(hex) | executable/support line; behavi
impact depends on surrounding block context. |
| 4184 |                                .monospace() | executable/support line; behavior
pact depends on surrounding block context. |
| 4185 |                                .size(10.0) | executable/support line; behavior
act depends on surrounding block context. |
| 4186 |                                .color(Color32::from_gray(200)), | executable/su
rt line; behavior impact depends on surrounding block context. |
| 4187 |                                ); | executable/support line; behavior impact depends on
rrounding block context. |
| 4188 |                                if !decoded.is_empty() { | conditional gate; changing co
tion shifts runtime path and side effects. |
| 4189 |                                ui.label( | executable/support line; behavior impact
pends on surrounding block context. |
| 4190 |                                RichText::new(decoded) | executable/support line
ehavior impact depends on surrounding block context. |
| 4191 |                                .size(10.0) | executable/support line; behav
impact depends on surrounding block context. |
| 4192 |                                .color(Color32::from_gray(190)), | executabl
upport line; behavior impact depends on surrounding block context. |
| 4193 |                                ); | executable/support line; behavior impact depend
n surrounding block context. |
| 4194 |                                } | scope delimiter; structure-only line, but misplaced
ts can change ownership and lifetime scopes. |
| 4195 |                                }); | executable/support line; behavior impact depends on su
unding block context. |
| 4196 |                                } | scope delimiter; structure-only line, but misplaced edits ca
hange ownership and lifetime scopes. |

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```

| 4197 |                                     }); | executable/support line; behavior impact depends on surrounding
lock context. |
| 4198 |                                     } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 4199 |                                     }); | executable/support line; behavior impact depends on surrounding block cont
. |
| 4200 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 4201 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 4202 |                                     return; | executable/support line; behavior impact depends on surrounding block context. |
| 4203 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 4204 | (blank) | line is blank; visual separation only. |
| 4205 |         egui::Grid::new("can::trace_hdr") | executable/support line; behavior impact depends on surrounding
ck context. |
| 4206 |             .num_columns(6) | executable/support line; behavior impact depends on surrounding block context.
| 4207 |             .min_col_width(50.0) | executable/support line; behavior impact depends on surrounding block con
t. |
| 4208 |             .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 4209 |                 for h in ["TIME(s)", "CAN-ID", "DLC", "HEX DATA", "PGN", "DECODED"] { | iteration control; c
ging bounds/body affects throughput, timing, and possibly safety guarantees. |
| 4210 |                     ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 4211 |                         RichText::new(h) | executable/support line; behavior impact depends on surrounding b
k context. |
| 4212 |                             .size(10.5) | executable/support line; behavior impact depends on surrounding bl
context. |
| 4213 |                             .color(Color32::from_gray(155)) | executable/support line; behavior impact depen
on surrounding block context. |
| 4214 |                             .strong(), | executable/support line; behavior impact depends on surrounding blo
context. |
| 4215 |                                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 4216 |                                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 4217 |                                 ui.end_row(); | executable/support line; behavior impact depends on surrounding block context
|
| 4218 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 4219 |                                     ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 4220 |                                     // stick_to_bottom only when live; when frozen the user can scroll freely | comment line; documentat
intent, no runtime effect. |
| 4221 |                                     ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block conte
|
| 4222 |                                     .auto_shrink([false, false]) | executable/support line; behavior impact depends on surrounding b
k context. |
| 4223 |                                     .max_height(ui.available_height()) | executable/support line; behavior impact depends on surroun
g block context. |
| 4224 |                                     .stick_to_bottom(!self.can_freeze) | executable/support line; behavior impact depends on surroun
g block context. |
| 4225 |                                     .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 4226 |                                         for row in filtered_rows.iter().rev().take(self.can_trace_limit).rev() { | iteration control
hanging bounds/body affects throughput, timing, and possibly safety guarantees. |
| 4227 |                                             let (ts, raw_id, sa, dlc, hex, pgn_sa, decoded) = *row; | executable/support line; behav
impact depends on surrounding block context. |
| 4228 |                                             let sc = match *sa { | executable/support line; behavior impact depends on surrounding b
k context. |
| 4229 |                                                 0x00 => Color32::from_rgb(80, 220, 80), | executable/support line; behavior impact d
nds on surrounding block context. |
| 4230 |                                                 0x03 => Color32::from_rgb(0, 210, 210), | executable/support line; behavior impact d
nds on surrounding block context. |
| 4231 |                                                 0x27 => Color32::LIGHT_BLUE, | executable/support line; behavior impact depends on s
ounding block context. |
| 4232 |                                                 0x1E => Color32::YELLOW, | executable/support line; behavior impact depends on surro
ing block context. |

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| 4233 |                                0x0B => Color32::WHITE, | executable/support line; behavior impact depends on surrounding block context. |
| 4234 |                                _ => Color32::from_gray(175), | executable/support line; behavior impact depends on surrounding block context. |
| 4235 |                                }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 4236 |                                ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
| 4237 |                                ui.add_sized( | executable/support line; behavior impact depends on surrounding block context. |
| 4238 |                                [68.0, 18.0], | executable/support line; behavior impact depends on surrounding block context. |
| 4239 |                                Label::new( | executable/support line; behavior impact depends on surrounding block context. |
| 4240 |                                RichText::new(format!("{:8.3}", ts)) | executable/support line; behavior impact depends on surrounding block context. |
| 4241 |                                .size(10.0) | executable/support line; behavior impact depends on surrounding block context. |
| 4242 |                                .monospace() | executable/support line; behavior impact depends on surrounding block context. |
| 4243 |                                .color(Color32::from_gray(110)), | executable/support line; behavior impact depends on surrounding block context. |
| 4244 |                                ), | executable/support line; behavior impact depends on surrounding block context. |
| 4245 |                                ); | executable/support line; behavior impact depends on surrounding block context. |
| 4246 |                                ui.add_sized( | executable/support line; behavior impact depends on surrounding block context. |
| 4247 |                                [84.0, 18.0], | executable/support line; behavior impact depends on surrounding block context. |
| 4248 |                                Label::new( | executable/support line; behavior impact depends on surrounding block context. |
| 4249 |                                RichText::new(format!("{:08X}", raw_id)) | executable/support line; behavior impact depends on surrounding block context. |
| 4250 |                                .size(10.0) | executable/support line; behavior impact depends on surrounding block context. |
| 4251 |                                .monospace() | executable/support line; behavior impact depends on surrounding block context. |
| 4252 |                                .color(Color32::LIGHT_BLUE), | executable/support line; behavior impact depends on surrounding block context. |
| 4253 |                                ), | executable/support line; behavior impact depends on surrounding block context. |
| 4254 |                                ); | executable/support line; behavior impact depends on surrounding block context. |
| 4255 |                                ui.add_sized( | executable/support line; behavior impact depends on surrounding block context. |
| 4256 |                                [32.0, 18.0], | executable/support line; behavior impact depends on surrounding block context. |
| 4257 |                                Label::new( | executable/support line; behavior impact depends on surrounding block context. |
| 4258 |                                RichText::new(format!("{}", dlc)) | executable/support line; behavior impact depends on surrounding block context. |
| 4259 |                                .size(10.0) | executable/support line; behavior impact depends on surrounding block context. |
| 4260 |                                .monospace() | executable/support line; behavior impact depends on surrounding block context. |
| 4261 |                                .color(Color32::from_gray(130)), | executable/support line; behavior impact depends on surrounding block context. |
| 4262 |                                ), | executable/support line; behavior impact depends on surrounding block context. |
| 4263 |                                ); | executable/support line; behavior impact depends on surrounding block context. |
| 4264 |                                ui.add_sized( | executable/support line; behavior impact depends on surrounding block context. |
| 4265 |                                [205.0, 18.0], | executable/support line; behavior impact depends on surrounding block context. |
| 4266 |                                Label::new( | executable/support line; behavior impact depends on surrounding block context. |

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| 4267 | RichText::new(hex) | executable/support line; behavior impact depends on surroun
inding block context. |
| 4268 | .size(10.0) | executable/support line; behavior impact depends on surroun
ng block context. |
| 4269 | .monospace() | executable/support line; behavior impact depends on surro
ing block context. |
| 4270 | .color(Color32::from_gray(195)), | executable/support line; behavior imp
depends on surrounding block context. |
| 4271 | ), | executable/support line; behavior impact depends on surrounding block conte
|
| 4272 | ); | executable/support line; behavior impact depends on surrounding block context.
| 4273 | ui.add_sized([180.0, 18.0], Label::new(RichText::new(pgn_sa).size(10.0).color(sc))),
executable/support line; behavior impact depends on surrounding block context. |
| 4274 | ui.label(RichText::new(decoded).size(10.0).color(Color32::from_gray(200))); | execut
e/support line; behavior impact depends on surrounding block context. |
| 4275 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 4276 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 4277 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 4278 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 4279 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 4280 | (blank) | line is blank; visual separation only. |
| 4281 | // ████████████████████████████████████████████████████████████████████████████ | comm
intent, no runtime effect. |
| 4282 | // TAB EVENTS | comment line; documentation intent, no runtime effect. |
| 4283 | // ████████████████████████████████████████████████████████████████████████████ | comm
intent, no runtime effect. |
| 4284 | impl App { | implementation block; method behavior and trait conformance are defined here. |
| 4285 |     fn tab_events(&mut self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter API
ontrol flow, and outputs. |
| 4286 |         ui.horizontal(|ui| { | executable/support line; behavior impact depends on surrounding block conte
|
| 4287 |             let pc = if self.ev_pause { | executable/support line; behavior impact depends on surrounding bl
context. |
| 4288 |                 Color32::RED | executable/support line; behavior impact depends on surrounding block context
| 4289 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 4290 |                 Color32::from_gray(38) | executable/support line; behavior impact depends on surrounding blo
context. |
| 4291 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
opes. |
| 4292 |             if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 4293 |                 .add( | executable/support line; behavior impact depends on surrounding block context. |
| 4294 |                     Button::new( | executable/support line; behavior impact depends on surrounding block con
t. |
| 4295 |                         RichText::new(if self.ev_pause { | executable/support line; behavior impact depends
surrounding block context. |
| 4296 |                             "■ FROZEN" | executable/support line; behavior impact depends on surrounding blo
context. |
| 4297 |                         } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 4298 |                             "■ Pause" | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 4299 |                         }) | executable/support line; behavior impact depends on surrounding block context.
| 4300 |                         .size(12.0), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 4301 |                     ) | executable/support line; behavior impact depends on surrounding block context. |
| 4302 |                     .fill(pc), | executable/support line; behavior impact depends on surrounding block conte
|
| 4303 |                 ) | executable/support line; behavior impact depends on surrounding block context. |
| 4304 |                 .clicked() | executable/support line; behavior impact depends on surrounding block context.
| 4305 |             { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 4306 |                 self.ev_pause = !self.ev_pause; | executable/support line; behavior impact depends on surrou
ng block context. |
| 4307 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime

```



```

pes. |
| 4308 |         if ui.button("■ Clear").clicked() { | conditional gate; changing condition shifts runtime path and
side effects. |
| 4309 |             self.events.clear(); | executable/support line; behavior impact depends on surrounding block
context. |
| 4310 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 4311 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 4312 |         ui.label(RichText::new("Filter:").size(11.0).color(Color32::GRAY)); | executable/support line; b
behavior impact depends on surrounding block context. |
| 4313 |         ui.add_sized( | executable/support line; behavior impact depends on surrounding block context. |
| 4314 |             [150.0, 20.0], | executable/support line; behavior impact depends on surrounding block conte
|
| 4315 |             TextEdit::singleline(&mut self.ev_filter).hint_text("text filter..."), | executable/support li
behavior impact depends on surrounding block context. |
| 4316 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 4317 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 4318 |         // Level filter buttons | comment line; documentation intent, no runtime effect. |
| 4319 |         for (lvl, lbl, col) in [ | iteration control; changing bounds/body affects throughput, timing, a
possibly safety guarantees. |
| 4320 |             (EventLevel::Debug, "DBG", Color32::from_gray(140)), | executable/support line; behavior imp
depends on surrounding block context. |
| 4321 |             (EventLevel::Info, "INFO", Color32::LIGHT_BLUE), | executable/support line; behavior impact
ends on surrounding block context. |
| 4322 |             (EventLevel::Ok, "OK", Color32::GREEN), | executable/support line; behavior impact depends o
urrounding block context. |
| 4323 |             (EventLevel::Warn, "WARN", Color32::YELLOW), | executable/support line; behavior impact depe
on surrounding block context. |
| 4324 |             (EventLevel::Critical, "CRIT", Color32::RED), | executable/support line; behavior impact dep
s on surrounding block context. |
| 4325 |         ] { | executable/support line; behavior impact depends on surrounding block context. |
| 4326 |             let sel = self.ev_min_level == lvl; | executable/support line; behavior impact depends on su
unding block context. |
| 4327 |             let fill = if sel { | executable/support line; behavior impact depends on surrounding block
text. |
| 4328 |                 Color32::from_gray(60) | executable/support line; behavior impact depends on surrounding
ock context. |
| 4329 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 4330 |                 Color32::from_gray(30) | executable/support line; behavior impact depends on surrounding
ock context. |
| 4331 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and life
e scopes. |
| 4332 |             if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 4333 |                 .add(Button::new(RichText::new(lbl).size(11.0).color(col)).fill(fill)) | executable/supp
line; behavior impact depends on surrounding block context. |
| 4334 |                 .clicked() | executable/support line; behavior impact depends on surrounding block conte
|
| 4335 |             { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 4336 |                 self.ev_min_level = lvl; | executable/support line; behavior impact depends on surroundi
block context. |
| 4337 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 4338 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 4339 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 4340 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 4341 |             RichText::new(format!("{}", events", self.events.len())) | executable/support line; behavior i
ct depends on surrounding block context. |
| 4342 |             .size(11.0) | executable/support line; behavior impact depends on surrounding block cont
. |
| 4343 |             .color(Color32::from_gray(140)), | executable/support line; behavior impact depends on s
ounding block context. |
| 4344 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 4345 |     }); | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 4346 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 4347 | (blank) | line is blank; visual separation only. |
| 4348 |         // Column headers | comment line; documentation intent, no runtime effect. |
| 4349 |         egui::Grid::new("events::hdr") | executable/support line; behavior impact depends on surrounding block
context. |
| 4350 |             .num_columns(4) | executable/support line; behavior impact depends on surrounding block context.
| 4351 |             .min_col_width(40.0) | executable/support line; behavior impact depends on surrounding block con
t. |
| 4352 |             .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 4353 |                 for h in ["TIME(s)", "LVL", "SOURCE", "EVENT"] { | iteration control; changing bounds/body a
cts throughput, timing, and possibly safety guarantees. |
| 4354 |                     ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 4355 |                         RichText::new(h) | executable/support line; behavior impact depends on surrounding b
k context. |
| 4356 |                             .size(10.5) | executable/support line; behavior impact depends on surrounding bl
context. |
| 4357 |                             .color(Color32::from_gray(150)) | executable/support line; behavior impact depen
on surrounding block context. |
| 4358 |                             .strong(), | executable/support line; behavior impact depends on surrounding blo
context. |
| 4359 |                         ); | executable/support line; behavior impact depends on surrounding block context. |
| 4360 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 4361 |                 ui.end_row(); | executable/support line; behavior impact depends on surrounding block context
|
| 4362 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 4363 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 4364 | (blank) | line is blank; visual separation only. |
| 4365 |         let filt = self.ev_filter.to_lowercase(); | executable/support line; behavior impact depends on surr
ding block context. |
| 4366 |         let min_lvl = self.ev_min_level; | executable/support line; behavior impact depends on surrounding b
k context. |
| 4367 |         let level_order = \|l: &EventLevel\| -> u8 { | executable/support line; behavior impact depends on s
ounding block context. |
| 4368 |             match l { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 4369 |                 EventLevel::Debug => 0, | executable/support line; behavior impact depends on surrounding bl
context. |
| 4370 |                 EventLevel::Info => 1, | executable/support line; behavior impact depends on surrounding blo
context. |
| 4371 |                 EventLevel::Ok => 2, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 4372 |                 EventLevel::Warn => 3, | executable/support line; behavior impact depends on surrounding blo
context. |
| 4373 |                 EventLevel::Critical => 4, | executable/support line; behavior impact depends on surrounding
ock context. |
| 4374 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 4375 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime sco
. |
| 4376 |         let min_ord = level_order(&min_lvl); | executable/support line; behavior impact depends on surroundi
block context. |
| 4377 | (blank) | line is blank; visual separation only. |
| 4378 |         ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block conte
|
| 4379 |             .auto_shrink([false, false]) | executable/support line; behavior impact depends on surrounding b
k context. |
| 4380 |             .stick_to_bottom(!self.ev_pause) | executable/support line; behavior impact depends on surroundi
block context. |
| 4381 |             .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 4382 |                 for (idx, ev) in self | iteration control; changing bounds/body affects throughput, timing,
possibly safety guarantees. |
| 4383 |                     .events | executable/support line; behavior impact depends on surrounding block context.

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```

| 4384 |         .iter() | executable/support line; behavior impact depends on surrounding block context.
| 4385 |         .rev() | executable/support line; behavior impact depends on surrounding block context.
| 4386 |         .filter(\|e\| level_order(&e.lvl) >= min_ord) | executable/support line; behavior impact
depends on surrounding block context. |
| 4387 |         .filter(\|e\| { | executable/support line; behavior impact depends on surrounding block
text. |
| 4388 |             filt.is_empty() | executable/support line; behavior impact depends on surrounding bl
context. |
| 4389 |             \| \| e.msg.to_lowercase().contains(&filt) | executable/support line; behavior im
t depends on surrounding block context. |
| 4390 |             \| \| e.source.to_lowercase().contains(&filt) | executable/support line; behavior
pact depends on surrounding block context. |
| 4391 |         }) | executable/support line; behavior impact depends on surrounding block context. |
| 4392 |         .take(300) | executable/support line; behavior impact depends on surrounding block conte
|
| 4393 |         .enumerate() | executable/support line; behavior impact depends on surrounding block con
t. |
| 4394 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 4395 |             let (icon, col) = match ev.lvl { | executable/support line; behavior impact depends on s
ounding block context. |
| 4396 |                 EventLevel::Debug => (".", Color32::from_gray(100)), | executable/support line; beha
r impact depends on surrounding block context. |
| 4397 |                 EventLevel::Info => ("■", Color32::LIGHT_BLUE), | executable/support line; behavior
act depends on surrounding block context. |
| 4398 |                 EventLevel::Ok => ("✓", Color32::GREEN), | executable/support line; behavior impact
ends on surrounding block context. |
| 4399 |                 EventLevel::Warn => ("■", Color32::YELLOW), | executable/support line; behavior impa
depends on surrounding block context. |
| 4400 |                 EventLevel::Critical => ("✗", Color32::RED), | executable/support line; behavior imp
depends on surrounding block context. |
| 4401 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 4402 |             ui.push_id(format!("events::row_{}", idx), \|ui\| { | executable/support line; behavior
act depends on surrounding block context. |
| 4403 |                 ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surroun
g block context. |
| 4404 |                     ui.add_sized( | executable/support line; behavior impact depends on surrounding
ck context. |
| 4405 |                         [56.0, 18.0], | executable/support line; behavior impact depends on surroun
block context. |
| 4406 |                         Label::new( | executable/support line; behavior impact depends on surroundin
lock context. |
| 4407 |                             RichText::new(format!("{:7.2}", ev.ts)) | executable/support line; behav
impact depends on surrounding block context. |
| 4408 |                             .size(10.5) | executable/support line; behavior impact depends on su
unding block context. |
| 4409 |                             .monospace() | executable/support line; behavior impact depends on s
ounding block context. |
| 4410 |                             .color(Color32::from_gray(115)), | executable/support line; behavior
pact depends on surrounding block context. |
| 4411 |                             ), | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4412 |                         ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4413 |                     ui.add_sized( | executable/support line; behavior impact depends on surrounding
ck context. |
| 4414 |                         [20.0, 18.0], | executable/support line; behavior impact depends on surroun
block context. |
| 4415 |                         Label::new(RichText::new(icon.to_string()).size(11.0).color(col)), | executa
/support line; behavior impact depends on surrounding block context. |
| 4416 |                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4417 |                 ui.add_sized( | executable/support line; behavior impact depends on surrounding
ck context. |

```

```
[78.0, 18.0], | executable/support line; behavior impact depends on surrounding block context. |  
| 4419 | Label::new( | executable/support line; behavior impact depends on surrounding block context. |  
lock context. |  
| 4420 | RichText::new(format!("{:<5}", ev.source)) | executable/support line;  
avior impact depends on surrounding block context. |  
| 4421 | .size(10.5) | executable/support line; behavior impact depends on surrounding block context. |  
| 4422 | .monospace() | executable/support line; behavior impact depends on surrounding block context. |  
ounding block context. |  
| 4423 | .color(Color32::from_gray(165)), | executable/support line; behavior impact depends on surrounding block context. |  
ext. |  
| 4424 | ), | executable/support line; behavior impact depends on surrounding block context. |  
| 4425 | ); | executable/support line; behavior impact depends on surrounding block context. |  
| 4426 | ui.label(RichText::new(&ev.msg).size(10.5).color(col)); | executable/support line;  
behavior impact depends on surrounding block context. |  
| 4427 | }); | executable/support line; behavior impact depends on surrounding block context. |  
| 4428 | }); | executable/support line; behavior impact depends on surrounding block context. |  
| 4429 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 4430 | }); | executable/support line; behavior impact depends on surrounding block context. |  
| 4431 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 4432 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 4433 | (blank) | line is blank; visual separation only. |  
| 4434 | // ████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |  
intent, no runtime effect. |  
| 4435 | // TAB ECU NETWORK | comment line; documentation intent, no runtime effect. |  
| 4436 | // ████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |  
intent, no runtime effect. |  
| 4437 | impl App { | implementation block; method behavior and trait conformance are defined here. |  
| 4438 | fn tab_ecu_net(&self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 4439 | let gw = &self.bench.gateway; | executable/support line; behavior impact depends on surrounding block context. |  
context. |  
| 4440 | let online = self | executable/support line; behavior impact depends on surrounding block context. | | |
| 4441 | .bench | executable/support line; behavior impact depends on surrounding block context. |  
| 4442 | .boot | executable/support line; behavior impact depends on surrounding block context. |  
| 4443 | .ecus | executable/support line; behavior impact depends on surrounding block context. |  
| 4444 | .iter() | executable/support line; behavior impact depends on surrounding block context. |  
| 4445 | .filter(|e| e.is_online()) | executable/support line; behavior impact depends on surrounding block context. |  
k context. |  
| 4446 | .count(); | executable/support line; behavior impact depends on surrounding block context. |  
| 4447 | let total = self.bench.boot.ecus.len(); | executable/support line; behavior impact depends on surrounding block context. |  
ng block context. |  
| 4448 | let unhealthy_nodes = gw | executable/support line; behavior impact depends on surrounding block context. |  
t. |  
| 4449 | .nodes | executable/support line; behavior impact depends on surrounding block context. | | | | | | |
| 4450 | .iter() | executable/support line; behavior impact depends on surrounding block context. |  
| 4451 | .filter(|n| n.tec > 0 || n.rec > 0 || !matches!(n.state, BusState::ErrorActive)) | executable/support line; behavior impact depends on surrounding block context. |  
/support line; behavior impact depends on surrounding block context. |  
| 4452 | .count(); | executable/support line; behavior impact depends on surrounding block context. |  
| 4453 | let net_health = if matches!(gw.bus_state, BusState::BusOff) { | executable/support line; behavior impact depends on surrounding block context. |  
ct depends on surrounding block context. |  
| 4454 | "CRITICAL" | executable/support line; behavior impact depends on surrounding block context. |  
| 4455 | } else if unhealthy_nodes > 0 || gw.total_errors > 0 { | executable/support line; behavior impact depends on surrounding block context. |  
ends on surrounding block context. |  
| 4456 | "DEGRADED" | executable/support line; behavior impact depends on surrounding block context. |  
| 4457 | } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 4458 | "HEALTHY" | executable/support line; behavior impact depends on surrounding block context. |  
| 4459 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
. |  
| 4460 | let net_col = match net_health { | executable/support line; behavior impact depends on surrounding block context. |
```

```

| 4461 |         "CRITICAL" => Color32::RED, | executable/support line; behavior impact depends on surrounding block context. |
| 4462 |         "DEGRADED" => Color32::YELLOW, | executable/support line; behavior impact depends on surrounding block context. |
| 4463 |         _ => Color32::GREEN, | executable/support line; behavior impact depends on surrounding block context. |
| 4464 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope. |
| 4465 |     ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 4466 |         RichText::new(format!( | executable/support line; behavior impact depends on surrounding block context. |
| 4467 |             "J1939 HS-CAN 500kbps \\\ Bus:{{}} \\\ Nodes:{{}}/{{}} \\\ TotalFrames:{{}}", | executable/support line; behavior impact depends on surrounding block context. |
| 4468 |             gw.bus_state, | executable/support line; behavior impact depends on surrounding block context. |
| 4469 |             online, | executable/support line; behavior impact depends on surrounding block context. |
| 4470 |             total, | executable/support line; behavior impact depends on surrounding block context. |
| 4471 |             gw.total_tx | executable/support line; behavior impact depends on surrounding block context. |
| 4472 |         )) | executable/support line; behavior impact depends on surrounding block context. |
| 4473 |         .size(12.0) | executable/support line; behavior impact depends on surrounding block context. |
| 4474 |         .color(Color32::from_gray(185)), | executable/support line; behavior impact depends on surrounding block context. |
| 4475 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 4476 |     ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 4477 |         RichText::new(format!( | executable/support line; behavior impact depends on surrounding block context. |
| 4478 |             "Network health: {{}} \\\ Unhealthy nodes: {{}} \\\ Total errors: {{}}", | executable/support line; behavior impact depends on surrounding block context. |
| 4479 |             net_health, unhealthy_nodes, gw.total_errors | executable/support line; behavior impact depends on surrounding block context. |
| 4480 |         )) | executable/support line; behavior impact depends on surrounding block context. |
| 4481 |         .size(11.0) | executable/support line; behavior impact depends on surrounding block context. |
| 4482 |         .color(net_col), | executable/support line; behavior impact depends on surrounding block context. |
| 4483 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 4484 |     ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 4485 | (blank) | line is blank; visual separation only. |
| 4486 |     ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block context. |
| 4487 |         .auto_shrink([false, false]) | executable/support line; behavior impact depends on surrounding block context. |
| 4488 |         .show(ui, \\\ui\\ { | executable/support line; behavior impact depends on surrounding block context. |
| 4489 |             egui::Grid::new("ecu_net::grid") | executable/support line; behavior impact depends on surrounding block context. |
| 4490 |                 .num_columns(7) | executable/support line; behavior impact depends on surrounding block context. |
| 4491 |                 .striped(true) | executable/support line; behavior impact depends on surrounding block context. |
| 4492 |                 .show(ui, \\\ui\\ { | executable/support line; behavior impact depends on surrounding block context. |
| 4493 |                     for h in [ | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 4494 |                         "MODULE", | executable/support line; behavior impact depends on surrounding block context. |
| 4495 |                         "SA", | executable/support line; behavior impact depends on surrounding block context. |
| 4496 |                         "STAGE", | executable/support line; behavior impact depends on surrounding block context. |
| 4497 |                         "ONLINE AT", | executable/support line; behavior impact depends on surrounding block context. |
| 4498 |                         "TEC", | executable/support line; behavior impact depends on surrounding block context. |
| 4499 |                         "REC", | executable/support line; behavior impact depends on surrounding block context. |
| 4500 |                         "BUS STATE", | executable/support line; behavior impact depends on surrounding block context. |

```

```

k context. |
| 4501 |           ] { | executable/support line; behavior impact depends on surrounding block context.
| 4502 |           ui.label(RichText::new(h).size(11.0).color(Color32::from_gray(150))); | executable
support line; behavior impact depends on surrounding block context. |
| 4503 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 4504 |           ui.end_row(); | executable/support line; behavior impact depends on surrounding block
context. |
| 4505 |           for ecu in &self.bench.boot.ecus { | iteration control; changing bounds/body affects
roughput, timing, and possibly safety guarantees. |
| 4506 |           let sc = match ecu.stage { | executable/support line; behavior impact depends on
surrounding block context. |
| 4507 |           EcuBootStage::Running => Color32::GREEN, | executable/support line; behavior
impact depends on surrounding block context. |
| 4508 |           EcuBootStage::Fault => Color32::RED, | executable/support line; behavior imp
depends on surrounding block context. |
| 4509 |           EcuBootStage::Unpowered => Color32::from_gray(55), | executable/support line
behavior impact depends on surrounding block context. |
| 4510 |           EcuBootStage::AddressClaiming => Color32::YELLOW, | executable/support line;
havior impact depends on surrounding block context. |
| 4511 |           _ => Color32::from_gray(185), | executable/support line; behavior impact dep
s on surrounding block context. |
| 4512 |           }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 4513 |           let node = gw.nodes.iter().find(|n| n.source_addr == ecu.sa); | executable/sup
t line; behavior impact depends on surrounding block context. |
| 4514 |           let (tec, rec, bs) = node.map_or((0u16, 0u16, "N/A"), |n| { | executable/suppo
line; behavior impact depends on surrounding block context. |
| 4515 |           ( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 4516 |           n.tec, | executable/support line; behavior impact depends on surrounding
ock context. |
| 4517 |           n.rec, | executable/support line; behavior impact depends on surrounding
ock context. |
| 4518 |           match n.state { | branch dispatch; changing arms alters behavior for var
ts and input classes. |
| 4519 |           BusState::ErrorActive => "EA", | executable/support line; behavior i
ct depends on surrounding block context. |
| 4520 |           BusState::ErrorPassive => "EP", | executable/support line; behavior
act depends on surrounding block context. |
| 4521 |           BusState::BusOff => "B0", | executable/support line; behavior impact
ends on surrounding block context. |
| 4522 |           }, | executable/support line; behavior impact depends on surrounding blo
context. |
| 4523 |           ) | executable/support line; behavior impact depends on surrounding block co
xt. |
| 4524 |           }); | executable/support line; behavior impact depends on surrounding block cont
. |
| 4525 |           ui.label( | executable/support line; behavior impact depends on surrounding bloc
context. |
| 4526 |           RichText::new(ecu.name) | executable/support line; behavior impact depends o
urrounding block context. |
| 4527 |           .size(11.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 4528 |           .color(Color32::from_gray(220)), | executable/support line; behavior imp
depends on surrounding block context. |
| 4529 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4530 |           ui.label( | executable/support line; behavior impact depends on surrounding bloc
context. |
| 4531 |           RichText::new(format!("{}", ecu.sa)) | executable/support line; behavi
impact depends on surrounding block context. |
| 4532 |           .size(11.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 4533 |           .monospace() | executable/support line; behavior impact depends on surro

```

```

ing block context. |
| 4534 |                                     .color(Color32::YELLOW), | executable/support line; behavior impact depends
on surrounding block context. |
| 4535 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
|
| 4536 |                                     ui.label(RichText::new(format!("{}", ecu.stage)).size(11.0).color(sc)); | executable/support line; behavior impact depends on surrounding block context. |
| 4537 |                                     ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 4538 |                                     RichText::new( | executable/support line; behavior impact depends on surrounding block context. |
| 4539 |                                     ecu.online_at.map_or("-",).into(), \|t\| format!("{:.2}s", t)), | executable/support line; behavior impact depends on surrounding block context. |
| 4540 |                                     ) | executable/support line; behavior impact depends on surrounding block context. |
| 4541 |                                     .size(11.0) | executable/support line; behavior impact depends on surrounding block context. |
| 4542 |                                     .color(Color32::from_gray(150)), | executable/support line; behavior impact depends on surrounding block context. |
| 4543 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
|
| 4544 |                                     ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 4545 |                                     RichText::new(format!("{}", tec)) | executable/support line; behavior impact depends on surrounding block context. |
| 4546 |                                     .size(11.0) | executable/support line; behavior impact depends on surrounding block context. |
| 4547 |                                     .monospace() | executable/support line; behavior impact depends on surrounding block context. |
| 4548 |                                     .color(if tec > 0 { | executable/support line; behavior impact depends on surrounding block context. |
| 4549 |                                     Color32::YELLOW | executable/support line; behavior impact depends on surrounding block context. |
| 4550 |                                     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 4551 |                                     Color32::GREEN | executable/support line; behavior impact depends on surrounding block context. |
| 4552 |                                     }), | executable/support line; behavior impact depends on surrounding block context. |
| 4553 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
|
| 4554 |                                     ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 4555 |                                     RichText::new(format!("{}", rec)) | executable/support line; behavior impact depends on surrounding block context. |
| 4556 |                                     .size(11.0) | executable/support line; behavior impact depends on surrounding block context. |
| 4557 |                                     .monospace() | executable/support line; behavior impact depends on surrounding block context. |
| 4558 |                                     .color(if rec > 0 { | executable/support line; behavior impact depends on surrounding block context. |
| 4559 |                                     Color32::YELLOW | executable/support line; behavior impact depends on surrounding block context. |
| 4560 |                                     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 4561 |                                     Color32::GREEN | executable/support line; behavior impact depends on surrounding block context. |
| 4562 |                                     }), | executable/support line; behavior impact depends on surrounding block context. |
| 4563 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
|
| 4564 |                                     ui.label(RichText::new(bs).size(11.0).color(match bs { | executable/support line; behavior impact depends on surrounding block context. |
| 4565 |                                     "B0" => Color32::RED, | executable/support line; behavior impact depends on surrounding block context. |

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| 4566 |                                     "EP" => Color32::YELLOW, | executable/support line; behavior impact depends
surrounding block context. |
| 4567 |                                     _ => Color32::GREEN, | executable/support line; behavior impact depends on s
ounding block context. |
| 4568 |                                     })); | executable/support line; behavior impact depends on surrounding block con
t. |
| 4569 |                                     ui.end_row(); | executable/support line; behavior impact depends on surrounding
ck context. |
| 4570 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 4571 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 4572 |                                     ui.separator(); | executable/support line; behavior impact depends on surrounding block cont
. |
| 4573 |                                     ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 4574 |                                     RichText::new("CRITICAL NODES (TEC/REC or non-EA state):") | executable/support line; be
ior impact depends on surrounding block context. |
| 4575 |                                     .size(11.0) | executable/support line; behavior impact depends on surrounding block
text. |
| 4576 |                                     .color(Color32::from_gray(140)), | executable/support line; behavior impact depends
surrounding block context. |
| 4577 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 4578 |                                     for n in gw | iteration control; changing bounds/body affects throughput, timing, and possib
safety guarantees. |
| 4579 |                                     .nodes | executable/support line; behavior impact depends on surrounding block context.
| 4580 |                                     .iter() | executable/support line; behavior impact depends on surrounding block context.
| 4581 |                                     .filter(\|n\| n.tec > 0 \| \| n.rec > 0 \| \| !matches!(n.state, BusState::ErrorActive)) |
executable/support line; behavior impact depends on surrounding block context. |
| 4582 |                                     .take(8) | executable/support line; behavior impact depends on surrounding block context
| 4583 |                                     { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 4584 |                                     ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 4585 |                                     RichText::new(format!( | executable/support line; behavior impact depends on surround
g block context. |
| 4586 |                                     "SA 0x{:02X} TEC:{} REC:{} State:{})", | executable/support line; behavior impa
depends on surrounding block context. |
| 4587 |                                     n.source_addr, n.tec, n.rec, n.state | executable/support line; behavior impact
ends on surrounding block context. |
| 4588 |                                     )) | executable/support line; behavior impact depends on surrounding block context.
| 4589 |                                     .size(10.5) | executable/support line; behavior impact depends on surrounding block
text. |
| 4590 |                                     .monospace() | executable/support line; behavior impact depends on surrounding block
ntext. |
| 4591 |                                     .color(Color32::YELLOW), | executable/support line; behavior impact depends on surro
ing block context. |
| 4592 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 4593 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 4594 |                                     ui.separator(); | executable/support line; behavior impact depends on surrounding block cont
. |
| 4595 |                                     ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 4596 |                                     RichText::new("BUS ERROR LOG:") | executable/support line; behavior impact depends on su
unding block context. |
| 4597 |                                     .size(11.0) | executable/support line; behavior impact depends on surrounding block
text. |
| 4598 |                                     .color(Color32::from_gray(140)), | executable/support line; behavior impact depends
surrounding block context. |
| 4599 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 4600 |                                     for err in gw.error_log.iter().rev().take(40) { | iteration control; changing bounds/body af
ts throughput, timing, and possibly safety guarantees. |
| 4601 |                                     let sa = err | executable/support line; behavior impact depends on surrounding block con
t. |
| 4602 |                                     .source_sa | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4603 |                                     .map_or("----".to_string(), \|s\| format!("{}", s)); | executable/support line

```



```
| behavior impact depends on surrounding block context. |  
| 4604 |                ui.label( | executable/support line; behavior impact depends on surrounding block context. |  
|      |  
| 4605 |                RichText::new(format!( | executable/support line; behavior impact depends on surrounding  
g block context. |  
| 4606 |                "{:6.2}s SA:{ } {?:} {} ", | error-propagating operation; changing it impacts fa  
re semantics and caller recovery paths. |  
| 4607 |                err.timestamp, sa, err.kind, err.description | executable/support line; behavior  
impact depends on surrounding block context. |  
| 4608 |                )) | executable/support line; behavior impact depends on surrounding block context.  
| 4609 |                .size(10.5) | executable/support line; behavior impact depends on surrounding block  
text. |  
| 4610 |                .monospace() | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 4611 |                .color(Color32::RED), | executable/support line; behavior impact depends on surround  
block context. |  
| 4612 |                ); | executable/support line; behavior impact depends on surrounding block context. |  
| 4613 |            } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet  
scopes. |  
| 4614 |        }); | executable/support line; behavior impact depends on surrounding block context. |  
| 4615 |    } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 4616 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 4617 | (blank) | line is blank; visual separation only. |  
| 4618 | // ████████████████████████████████████████████████████████████████████████████████████████████████████ | comm  
intent, no runtime effect. |  
| 4619 | // TAB ENGINE (detailed ECM) | comment line; documentation intent, no runtime effect. |  
| 4620 | // ████████████████████████████████████████████████████████████████████████████████████████████████████ | comm  
intent, no runtime effect. |  
| 4621 | impl App { | implementation block; method behavior and trait conformance are defined here. |  
| 4622 |     fn tab_engine(&self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter API, co  
ol flow, and outputs. |  
| 4623 |         let e = &self.bench.ecm; | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 4624 |         ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block conte  
|  
| 4625 |             .auto_shrink([false, false]) | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 4626 |             .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte  
|  
| 4627 |                 ui.columns(3, \|cols\| { | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 4628 |                     // Col 0: Engine core | comment line; documentation intent, no runtime effect. |  
| 4629 |                     egui::Frame::group(cols[0].style()) | executable/support line; behavior impact depends o  
urrounding block context. |  
| 4630 |                         .fill(Color32::from_gray(16)) | executable/support line; behavior impact depends on  
rounding block context. |  
| 4631 |                         .show(&mut cols[0], \|ui\| { | executable/support line; behavior impact depends on s  
ounding block context. |  
| 4632 |                             ui.label( | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 4633 |                                 RichText::new("ENGINE CORE") | executable/support line; behavior impact depe  
on surrounding block context. |  
| 4634 |                                     .size(12.0) | executable/support line; behavior impact depends on surroun  
ng block context. |  
| 4635 |                                         .color(Color32::from_rgb(80, 155, 255)), | executable/support line; beha  
r impact depends on surrounding block context. |  
| 4636 |                                             ); | executable/support line; behavior impact depends on surrounding block conte  
|  
| 4637 |                                     let w = 140.0; | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 4638 |                                     digital_readout( | executable/support line; behavior impact depends on surroundi  
block context. |  
| 4639 |                                         ui, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 4640 |                                             "Governor", | executable/support line; behavior impact depends on surroundin
```

```

lock context. |
| 4641 |                                     &format!("{}", e.governor_mode), | executable/support line; behavior impact
ends on surrounding block context. |
| 4642 |                                     Color32::YELLOW, | executable/support line; behavior impact depends on surro
ing block context. |
| 4643 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4644 |                                     digital_readout( | executable/support line; behavior impact depends on surrounding
block context. |
| 4645 |                                     ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4646 |                                     "Fuel Map", | executable/support line; behavior impact depends on surrounding
lock context. |
| 4647 |                                     &format!("{}", e.fuel_map), | executable/support line; behavior impact dependen
on surrounding block context. |
| 4648 |                                     Color32::YELLOW, | executable/support line; behavior impact depends on surro
ing block context. |
| 4649 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4650 |                                     bar_gauge(ui, "RPM", e.rpm, 3000.0, "rpm", w, Color32::GREEN); | executable/supp
line; behavior impact depends on surrounding block context. |
| 4651 |                                     bar_gauge(ui, "Load", e.percent_load, 150.0, "%", w, Color32::YELLOW); | execut
/support line; behavior impact depends on surrounding block context. |
| 4652 |                                     bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 4653 |                                     ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4654 |                                     "Torque", | executable/support line; behavior impact depends on surrounding
ck context. |
| 4655 |                                     e.actual_torque_nm, | executable/support line; behavior impact depends on su
unding block context. |
| 4656 |                                     1200.0, | executable/support line; behavior impact depends on surrounding bl
context. |
| 4657 |                                     "Nm", | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 4658 |                                     w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4659 |                                     Color32::from_rgb(80, 180, 255), | executable/support line; behavior impact
ends on surrounding block context. |
| 4660 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4661 |                                     bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 4662 |                                     ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4663 |                                     "Power", | executable/support line; behavior impact depends on surrounding b
k context. |
| 4664 |                                     e.power_kw(), | executable/support line; behavior impact depends on surround
block context. |
| 4665 |                                     260.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 4666 |                                     "kW", | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 4667 |                                     w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4668 |                                     Color32::from_rgb(100, 220, 100), | executable/support line; behavior impact
ends on surrounding block context. |
| 4669 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4670 |                                     bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 4671 |                                     ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4672 |                                     "Throttle", | executable/support line; behavior impact depends on surroundin
lock context. |

```

```
| 4673 | e.active_throttle, | executable/support line; behavior impact depends on sur
inding block context. |
| 4674 | 100.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 4675 | "%", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 4676 | w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4677 | Color32::YELLOW, | executable/support line; behavior impact depends on surro
ing block context. |
| 4678 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4679 | ui.separator(); | executable/support line; behavior impact depends on surrounding
lock context. |
| 4680 | bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 4681 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4682 | "Rail P", | executable/support line; behavior impact depends on surrounding
ck context. |
| 4683 | e.fuel_rail_pressure_mpa * 10.0, | executable/support line; behavior impact
ends on surrounding block context. |
| 4684 | 2000.0, | executable/support line; behavior impact depends on surrounding bl
context. |
| 4685 | "bar", | executable/support line; behavior impact depends on surrounding blo
context. |
| 4686 | w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4687 | Color32::from_rgb(0, 210, 210), | executable/support line; behavior impact d
nds on surrounding block context. |
| 4688 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4689 | bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 4690 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4691 | "Fuel Prs", | executable/support line; behavior impact depends on surroundin
lock context. |
| 4692 | e.fuel_pressure_kpa, | executable/support line; behavior impact depends on s
ounding block context. |
| 4693 | 600.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 4694 | "kPa", | executable/support line; behavior impact depends on surrounding blo
context. |
| 4695 | w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4696 | Color32::from_rgb(0, 210, 210), | executable/support line; behavior impact d
nds on surrounding block context. |
| 4697 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4698 | bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 4699 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4700 | "Fuel/h", | executable/support line; behavior impact depends on surrounding
ck context. |
| 4701 | e.fuel_rate_lph, | executable/support line; behavior impact depends on surro
ing block context. |
| 4702 | 30.0, | executable/support line; behavior impact depends on surrounding blo
ontext. |
| 4703 | "L/h", | executable/support line; behavior impact depends on surrounding blo
context. |
| 4704 | w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4705 | Color32::YELLOW, | executable/support line; behavior impact depends on surro
```

```

ing block context. |
| 4706 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4707 |           bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 4708 |               ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4709 |               "Boost", | executable/support line; behavior impact depends on surrounding b
k context. |
| 4710 |               e.boost_pressure_kpa, | executable/support line; behavior impact depends on
rounding block context. |
| 4711 |               300.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 4712 |               "kPa", | executable/support line; behavior impact depends on surrounding blo
context. |
| 4713 |               w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4714 |               Color32::from_rgb(180, 80, 255), | executable/support line; behavior impact
ends on surrounding block context. |
| 4715 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4716 |           bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 4717 |               ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4718 |               "VGT", | executable/support line; behavior impact depends on surrounding blo
context. |
| 4719 |               e.vgt_position_pct, | executable/support line; behavior impact depends on su
unding block context. |
| 4720 |               100.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 4721 |               "%", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 4722 |               w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4723 |               Color32::from_gray(200), | executable/support line; behavior impact depends
surrounding block context. |
| 4724 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4725 |           bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 4726 |               ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4727 |               "EGR", | executable/support line; behavior impact depends on surrounding blo
context. |
| 4728 |               e.egr_valve_pct, | executable/support line; behavior impact depends on surro
ing block context. |
| 4729 |               100.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 4730 |               "%", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 4731 |               w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4732 |               Color32::from_gray(200), | executable/support line; behavior impact depends
surrounding block context. |
| 4733 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4734 |           }); | executable/support line; behavior impact depends on surrounding block context.
| 4735 |           // Col 1: Temperatures & pressures | comment line; documentation intent, no runtime effe
|
| 4736 |           egui::Frame::group(cols[1].style()) | executable/support line; behavior impact depends o
urrounding block context. |
| 4737 |           .fill(Color32::from_gray(16)) | executable/support line; behavior impact depends on
rounding block context. |
| 4738 |           .show(&mut cols[1], \|ui\| { | executable/support line; behavior impact depends on s

```

```

ounding block context. |
| 4739 |                 ui.label( | executable/support line; behavior impact depends on surrounding block
ontext. |
| 4740 |                 RichText::new("THERMAL & PRESSURE") | executable/support line; behavior impact
depends on surrounding block context. |
| 4741 |                 .size(12.0) | executable/support line; behavior impact depends on surrounding block
ng block context. |
| 4742 |                 .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior impact
r impact depends on surrounding block context. |
| 4743 |             ); | executable/support line; behavior impact depends on surrounding block context. |
|
| 4744 |             let w = 140.0; | executable/support line; behavior impact depends on surrounding block
ock context. |
| 4745 |             bar_gauge( | executable/support line; behavior impact depends on surrounding block context. |
context. |
| 4746 |                 ui, | executable/support line; behavior impact depends on surrounding block context. |
text. |
| 4747 |                 "Coolant", | executable/support line; behavior impact depends on surrounding block context. |
ock context. |
| 4748 |                 e.coolant_temp_c, | executable/support line; behavior impact depends on surrounding block context. |
ding block context. |
| 4749 |                 130.0, | executable/support line; behavior impact depends on surrounding block context. |
context. |
| 4750 |                 "°C", | executable/support line; behavior impact depends on surrounding block context. |
ontext. |
| 4751 |                 w, | executable/support line; behavior impact depends on surrounding block context. |
ext. |
| 4752 |                 if e.coolant_temp_c > 100.0 { | conditional gate; changing condition shifts behavior impact
time path and side effects. |
| 4753 |                     Color32::RED | executable/support line; behavior impact depends on surrounding block context. |
ing block context. |
| 4754 |                 } else { | executable/support line; behavior impact depends on surrounding block context. |
k context. |
| 4755 |                     Color32::GREEN | executable/support line; behavior impact depends on surrounding block context. |
nding block context. |
| 4756 |                 }, | executable/support line; behavior impact depends on surrounding block context. |
ext. |
| 4757 |             ); | executable/support line; behavior impact depends on surrounding block context. |
|
| 4758 |             bar_gauge( | executable/support line; behavior impact depends on surrounding block context. |
context. |
| 4759 |                 ui, | executable/support line; behavior impact depends on surrounding block context. |
text. |
| 4760 |                 "Oil", | executable/support line; behavior impact depends on surrounding block context. |
context. |
| 4761 |                 e.oil_temp_c, | executable/support line; behavior impact depends on surrounding block context. |
block context. |
| 4762 |                 160.0, | executable/support line; behavior impact depends on surrounding block context. |
context. |
| 4763 |                 "°C", | executable/support line; behavior impact depends on surrounding block context. |
ontext. |
| 4764 |                 w, | executable/support line; behavior impact depends on surrounding block context. |
ext. |
| 4765 |                 Color32::from_rgb(200, 150, 50), | executable/support line; behavior impact depends on surrounding block context. |
ends on surrounding block context. |
| 4766 |             ); | executable/support line; behavior impact depends on surrounding block context. |
|
| 4767 |             bar_gauge( | executable/support line; behavior impact depends on surrounding block context. |
context. |
| 4768 |                 ui, | executable/support line; behavior impact depends on surrounding block context. |
text. |
| 4769 |                 "Exhaust", | executable/support line; behavior impact depends on surrounding block context. |
ock context. |
| 4770 |                 e.exhaust_temp_c, | executable/support line; behavior impact depends on surrounding block context. |
ding block context. |

```

```
| 4771 |          900.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 4772 |          "°C", | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 4773 |          w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4774 |          Color32::from_rgb(255, 120, 30), | executable/support line; behavior impact
ends on surrounding block context. |
| 4775 |      ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4776 |      bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 4777 |          ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4778 |          "Intake", | executable/support line; behavior impact depends on surrounding
ck context. |
| 4779 |          e.intake_temp_c, | executable/support line; behavior impact depends on surro
ing block context. |
| 4780 |          80.0, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 4781 |          "°C", | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 4782 |          w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4783 |          Color32::from_gray(200), | executable/support line; behavior impact depends
surrounding block context. |
| 4784 |      ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4785 |      bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 4786 |          ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4787 |          "Fuel T", | executable/support line; behavior impact depends on surrounding
ck context. |
| 4788 |          e.fuel_temp_c, | executable/support line; behavior impact depends on surroun
g block context. |
| 4789 |          80.0, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 4790 |          "°C", | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 4791 |          w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4792 |          Color32::from_gray(200), | executable/support line; behavior impact depends
surrounding block context. |
| 4793 |      ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4794 |      ui.separator(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 4795 |      bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 4796 |          ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4797 |          "Oil Prs", | executable/support line; behavior impact depends on surrounding
ock context. |
| 4798 |          e.oil_pressure_kpa, | executable/support line; behavior impact depends on su
unding block context. |
| 4799 |          700.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 4800 |          "kPa", | executable/support line; behavior impact depends on surrounding blo
context. |
| 4801 |          w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4802 |          if e.oil_pressure_kpa < 100.0 { | conditional gate; changing condition shift
untime path and side effects. |
| 4803 |              Color32::RED | executable/support line; behavior impact depends on surro
```

```

ing block context. |
| 4804 |           } else { | executable/support line; behavior impact depends on surrounding b
k context. |
| 4805 |           Color32::GREEN | executable/support line; behavior impact depends on sur
nding block context. |
| 4806 |           }, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4807 |       ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4808 |       bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 4809 |           ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4810 |           "Coolant P", | executable/support line; behavior impact depends on surroundi
block context. |
| 4811 |           e.coolant_pres_kpa, | executable/support line; behavior impact depends on su
nding block context. |
| 4812 |           150.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 4813 |           "kPa", | executable/support line; behavior impact depends on surrounding blo
context. |
| 4814 |           w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4815 |           Color32::from_gray(200), | executable/support line; behavior impact depends
surrounding block context. |
| 4816 |       ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4817 |       bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 4818 |           ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4819 |           "Air Filter", | executable/support line; behavior impact depends on surround
block context. |
| 4820 |           e.air_filter_dp_kpa, | executable/support line; behavior impact depends on s
ounding block context. |
| 4821 |           10.0, | executable/support line; behavior impact depends on surrounding blo
ontext. |
| 4822 |           "kPa", | executable/support line; behavior impact depends on surrounding blo
context. |
| 4823 |           w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4824 |           if e.air_filter_dp_kpa > 5.0 { | conditional gate; changing condition shifts
ntime path and side effects. |
| 4825 |           Color32::YELLOW | executable/support line; behavior impact depends on su
ounding block context. |
| 4826 |           } else { | executable/support line; behavior impact depends on surrounding b
k context. |
| 4827 |           Color32::GREEN | executable/support line; behavior impact depends on sur
nding block context. |
| 4828 |           }, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4829 |       ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4830 |       ui.separator(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 4831 |       digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 4832 |           ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4833 |           "Alt V", | executable/support line; behavior impact depends on surrounding b
k context. |
| 4834 |           &format!("{:.1}V", e.alternator_v), | executable/support line; behavior impa
depends on surrounding block context. |
| 4835 |           Color32::from_gray(200), | executable/support line; behavior impact depends
surrounding block context. |

```

```

| 4836 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4837 |                                     digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 4838 |                                     ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4839 |                                     "Batt V", | executable/support line; behavior impact depends on surrounding
ck context. |
| 4840 |                                     &format!("{:.1}V", self.bench.bcm.battery_voltage), | executable/support lin
behavior impact depends on surrounding block context. |
| 4841 |                                     Color32::from_gray(200), | executable/support line; behavior impact depends
surrounding block context. |
| 4842 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4843 |                                     digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 4844 |                                     ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4845 |                                     "Engine H", | executable/support line; behavior impact depends on surroundin
lock context. |
| 4846 |                                     &format!("{:.1}h", e.engine_hours), | executable/support line; behavior impa
depends on surrounding block context. |
| 4847 |                                     Color32::from_gray(200), | executable/support line; behavior impact depends
surrounding block context. |
| 4848 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4849 |                                     digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 4850 |                                     ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 4851 |                                     "Svc due", | executable/support line; behavior impact depends on surrounding
ock context. |
| 4852 |                                     &format!("{:.1}h", e.service_hours_left), | executable/support line; behavior
mpact depends on surrounding block context. |
| 4853 |                                     if e.service_hours_left < 50.0 { | conditional gate; changing condition shif
runtime path and side effects. |
| 4854 |                                     Color32::RED | executable/support line; behavior impact depends on surro
ing block context. |
| 4855 |                                     } else { | executable/support line; behavior impact depends on surrounding b
k context. |
| 4856 |                                     Color32::GREEN | executable/support line; behavior impact depends on sur
nding block context. |
| 4857 |                                     }, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4858 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4859 |                                     }); | executable/support line; behavior impact depends on surrounding block context.
| 4860 |                                     // Col 2: Aftertreatment + DTCs | comment line; documentation intent, no runtime effect.
| 4861 |                                     equi::Frame::group(cols[2].style()) | executable/support line; behavior impact depends o
urrounding block context. |
| 4862 |                                     .fill(Color32::from_gray(16)) | executable/support line; behavior impact depends on
rounding block context. |
| 4863 |                                     .show(&mut cols[2], \|ui\| { | executable/support line; behavior impact depends on s
ounding block context. |
| 4864 |                                     ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 4865 |                                     RichText::new("AFTERTREATMENT & DTCS") | executable/support line; behavior i
ct depends on surrounding block context. |
| 4866 |                                     .size(12.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 4867 |                                     .color(Color32::from_rgb(80, 155, 255)), | executable/support line; beha
r impact depends on surrounding block context. |
| 4868 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4869 |                                     let w = 130.0; | executable/support line; behavior impact depends on surrounding

```



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ock context. |
| 4870 |           let ac = match e.aftertreatment { | executable/support line; behavior impact depends on surrounding block context. |
| 4871 |               AftertreatmentState::Normal => Color32::GREEN, | executable/support line; behavior impact depends on surrounding block context. |
| 4872 |               AftertreatmentState::DpfRegen => Color32::from_rgb(200, 80, 200), | executable/support line; behavior impact depends on surrounding block context. |
| 4873 |               AftertreatmentState::ScrDegraded => Color32::YELLOW, | executable/support line; behavior impact depends on surrounding block context. |
| 4874 |               _ => Color32::RED, | executable/support line; behavior impact depends on surrounding block context. |
| 4875 |           }; | scope delimiter; structure-only line, but misplaced edits can change owners and lifetime scopes. |
| 4876 |           digital_readout(ui, "State", &format!("{}", e.aftertreatment), ac); | executable/support line; behavior impact depends on surrounding block context. |
| 4877 |           bar_gauge( | executable/support line; behavior impact depends on surrounding block context. |
| 4878 |               ui, | executable/support line; behavior impact depends on surrounding block context. |
| 4879 |               "DPF Soot", | executable/support line; behavior impact depends on surrounding block context. |
| 4880 |               e.dpf_soot_pct, | executable/support line; behavior impact depends on surrounding block context. |
| 4881 |               100.0, | executable/support line; behavior impact depends on surrounding block context. |
| 4882 |               "%", | executable/support line; behavior impact depends on surrounding block context. |
| 4883 |               w, | executable/support line; behavior impact depends on surrounding block context. |
| 4884 |               if e.dpf_soot_pct > 75.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 4885 |                   Color32::RED | executable/support line; behavior impact depends on surrounding block context. |
| 4886 |               } else { | executable/support line; behavior impact depends on surrounding block context. |
| 4887 |                   Color32::from_rgb(180, 100, 220) | executable/support line; behavior impact depends on surrounding block context. |
| 4888 |               }, | executable/support line; behavior impact depends on surrounding block context. |
| 4889 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 4890 |           bar_gauge( | executable/support line; behavior impact depends on surrounding block context. |
| 4891 |               ui, | executable/support line; behavior impact depends on surrounding block context. |
| 4892 |               "DPF Temp", | executable/support line; behavior impact depends on surrounding block context. |
| 4893 |               e.dpf_temp_c, | executable/support line; behavior impact depends on surrounding block context. |
| 4894 |               700.0, | executable/support line; behavior impact depends on surrounding block context. |
| 4895 |               "°C", | executable/support line; behavior impact depends on surrounding block context. |
| 4896 |               w, | executable/support line; behavior impact depends on surrounding block context. |
| 4897 |               Color32::from_rgb(255, 130, 30), | executable/support line; behavior impact depends on surrounding block context. |
| 4898 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 4899 |           bar_gauge( | executable/support line; behavior impact depends on surrounding block context. |
| 4900 |               ui, | executable/support line; behavior impact depends on surrounding block context. |
| 4901 |               "DEF Level", | executable/support line; behavior impact depends on surrounding block context. |

```

4902	e.def_level_pct, executable/support line; behavior impact depends on surro
ing block context.	
4903	100.0, executable/support line; behavior impact depends on surrounding blo
context.	
4904	"%", executable/support line; behavior impact depends on surrounding block
ntext.	
4905	w, executable/support line; behavior impact depends on surrounding block c
ext.	
4906	if e.def_level_pct < 10.0 { conditional gate; changing condition shifts ru
me path and side effects.	
4907	Color32::RED executable/support line; behavior impact depends on surro
ing block context.	
4908	} else { executable/support line; behavior impact depends on surrounding b
k context.	
4909	Color32::from_rgb(0, 210, 210) executable/support line; behavior impac
depends on surrounding block context.	
4910	}, executable/support line; behavior impact depends on surrounding block c
ext.	
4911	); executable/support line; behavior impact depends on surrounding block conte
4912	bar_gauge(executable/support line; behavior impact depends on surrounding blo
context.	
4913	ui, executable/support line; behavior impact depends on surrounding block
text.	
4914	"SCR Eff", executable/support line; behavior impact depends on surrounding
ock context.	
4915	e.scr_efficiency_pct, executable/support line; behavior impact depends on
rounding block context.	
4916	100.0, executable/support line; behavior impact depends on surrounding blo
context.	
4917	"%", executable/support line; behavior impact depends on surrounding block
ntext.	
4918	w, executable/support line; behavior impact depends on surrounding block c
ext.	
4919	if e.scr_efficiency_pct < 80.0 { conditional gate; changing condition shif
runtime path and side effects.	
4920	Color32::YELLOW executable/support line; behavior impact depends on su
unding block context.	
4921	} else { executable/support line; behavior impact depends on surrounding b
k context.	
4922	Color32::GREEN executable/support line; behavior impact depends on sur
nding block context.	
4923	}, executable/support line; behavior impact depends on surrounding block c
ext.	
4924	); executable/support line; behavior impact depends on surrounding block conte
4925	digital_readout(executable/support line; behavior impact depends on surroundi
block context.	
4926	ui, executable/support line; behavior impact depends on surrounding block
text.	
4927	"NOx raw", executable/support line; behavior impact depends on surrounding
ock context.	
4928	&format!("{:.0} ppm", e.nox_raw_ppm), executable/support line; behavior im
t depends on surrounding block context.	
4929	Color32::from_gray(180), executable/support line; behavior impact depends
surrounding block context.	
4930	); executable/support line; behavior impact depends on surrounding block conte
4931	digital_readout(executable/support line; behavior impact depends on surroundi
block context.	
4932	ui, executable/support line; behavior impact depends on surrounding block
text.	
4933	"NOx tail", executable/support line; behavior impact depends on surroundin
lock context.	
4934	&format!("{:.0} ppm", e.nox_tailpipe_ppm), executable/support line; behavi

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impact depends on surrounding block context. |
| 4935 |             if e.nox_tailpipe_ppm > 500.0 { | conditional gate; changing condition shift
untime path and side effects. |
| 4936 |                 Color32::RED | executable/support line; behavior impact depends on surro
ing block context. |
| 4937 |                 } else { | executable/support line; behavior impact depends on surrounding b
k context. |
| 4938 |                 Color32::from_gray(180) | executable/support line; behavior impact depen
on surrounding block context. |
| 4939 |                 }, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4940 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4941 |             ui.separator(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 4942 |             ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 4943 |                 RichText::new("ACTIVE DTCs:") | executable/support line; behavior impact dep
s on surrounding block context. |
| 4944 |                 .size(11.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 4945 |                 .color(Color32::from_gray(140)), | executable/support line; behavior imp
depends on surrounding block context. |
| 4946 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 4947 |             if e.active_dtc.is_empty() { | conditional gate; changing condition shifts runt
path and side effects. |
| 4948 |                 ui.label(RichText::new("✓ None").size(11.0).color(Color32::GREEN)); | execut
e/support line; behavior impact depends on surrounding block context. |
| 4949 |             } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 4950 |             for dtc in &e.active_dtc { | iteration control; changing bounds/body affects th
ghput, timing, and possibly safety guarantees. |
| 4951 |                 let c = match dtc.severity { | executable/support line; behavior impact depen
on surrounding block context. |
| 4952 |                     DtcSeverity::Red => Color32::RED, | executable/support line; behavior im
t depends on surrounding block context. |
| 4953 |                     DtcSeverity::Amber => Color32::GOLD, | executable/support line; behavior
pact depends on surrounding block context. |
| 4954 |                     _ => Color32::LIGHT_BLUE, | executable/support line; behavior impact dep
s on surrounding block context. |
| 4955 |                 }; | scope delimiter; structure-only line, but misplaced edits can change ow
ship and lifetime scopes. |
| 4956 |                 ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 4957 |                     RichText::new(format!( | executable/support line; behavior impact depend
n surrounding block context. |
| 4958 |                         "SPN:{ } FMI:{ } ×{ } { }", | executable/support line; behavior impact
ends on surrounding block context. |
| 4959 |                         dtc.spn, dtc.fmi, dtc.count, dtc.desc | executable/support line; beh
or impact depends on surrounding block context. |
| 4960 |                     )) | executable/support line; behavior impact depends on surrounding blo
context. |
| 4961 |                     .size(10.5) | executable/support line; behavior impact depends on surrou
ng block context. |
| 4962 |                     .color(c), | executable/support line; behavior impact depends on surroun
g block context. |
| 4963 |                 ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 4964 |             } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 4965 |             }); | executable/support line; behavior impact depends on surrounding block context.
| 4966 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 4967 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 4968 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.

```

```
| 4969 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 4970 | (blank) | line is blank; visual separation only. |  
| 4971 | // ████████████████████████████████████████████████████████████████████████████████████████████████ | comment  
intent, no runtime effect. |  
| 4972 | // TAB FAULTS | comment line; documentation intent, no runtime effect. |  
| 4973 | // ████████████████████████████████████████████████████████████████████████████████████████████████ | comm  
intent, no runtime effect. |  
| 4974 | impl App { | implementation block; method behavior and trait conformance are defined here. |  
| 4975 |     fn dtc_fault_storage_addr(spn: u32, fmi: u8) -> u32 { | declares executable behavior; signature/body cha  
s alter API, control flow, and outputs. |  
| 4976 |         // Virtualized NVM map for fault records in ECM region. | comment line; documentation intent, no run  
e effect. |  
| 4977 |             0x0802_0000u32 + (spn & 0xFFFF) * 0x20 + ((fmi as u32) * 0x2) | executable/support line; behavior im  
t depends on surrounding block context. |  
| 4978 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 4979 | (blank) | line is blank; visual separation only. |  
| 4980 |     fn dtc_dm1_payload_hex(amber: bool, red: bool, mil: bool, spn: u32, fmi: u8) -> String { | declares exec  
ble behavior; signature/body changes alter API, control flow, and outputs. |  
| 4981 |         let dm1 = auto_breaking::j1939::Builder::dm1( | executable/support line; behavior impact depends on  
rounding block context. |  
| 4982 |             0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 4983 |             amber, | executable/support line; behavior impact depends on surrounding block context. |  
| 4984 |             red, | executable/support line; behavior impact depends on surrounding block context. |  
| 4985 |             mil, | executable/support line; behavior impact depends on surrounding block context. |  
| 4986 |             spn, | executable/support line; behavior impact depends on surrounding block context. |  
| 4987 |             fmi, | executable/support line; behavior impact depends on surrounding block context. |  
| 4988 |             auto_breaking::j1939::addr::ECM_1, | executable/support line; behavior impact depends on surroun  
g block context. |  
| 4989 |         ); | executable/support line; behavior impact depends on surrounding block context. |  
| 4990 |         dm1.data | executable/support line; behavior impact depends on surrounding block context. |  
| 4991 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |  
| 4992 |             .map(|b| format!("{:02X}", b)) | executable/support line; behavior impact depends on surroundi  
block context. |  
| 4993 |                 .collect::() | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 4994 |                     .join(" ") | executable/support line; behavior impact depends on surrounding block context. |  
| 4995 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 4996 | (blank) | line is blank; visual separation only. |  
| 4997 |     fn fault_selection_hint(selected: auto_breaking::FaultType) -> Option<(u32, u8, &'static str)> { | decla  
executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 4998 |         match selected { | branch dispatch; changing arms alters behavior for variants and input classes. |  
| 4999 |             auto_breaking::FaultType::HighCoolantTemp => { | executable/support line; behavior impact depend  
n surrounding block context. |  
| 5000 |                 Some((110, 0, "Coolant temp above severe threshold")) | executable/support line; behavior im  
t depends on surrounding block context. |  
| 5001 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 5002 |             auto_breaking::FaultType::LowOilPressure => { | executable/support line; behavior impact depends  
surrounding block context. |  
| 5003 |                 Some((100, 1, "Oil pressure below severe threshold")) | executable/support line; behavior im  
t depends on surrounding block context. |  
| 5004 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 5005 |             auto_breaking::FaultType::LowFuelPressure => { | executable/support line; behavior impact depend  
n surrounding block context. |  
| 5006 |                 Some((94, 1, "Fuel delivery pressure below threshold")) | executable/support line; behavior  
act depends on surrounding block context. |  
| 5007 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 5008 |             auto_breaking::FaultType::CriticalDefLevel => { | executable/support line; behavior impact dependen  
on surrounding block context. |  
| 5009 |                 Some((3361, 1, "DEF critically low (derate trigger)") | executable/support line; behavior i  
ct depends on surrounding block context. |  
| 5010 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |
```

```

| 5011 |         auto_breaking::FaultType::HighDpfSoot => { | executable/support line; behavior impact depends on
rounding block context. |
| 5012 |             Some((3251, 16, "DPF soot above high threshold")) | executable/support line; behavior impact
depends on surrounding block context. |
| 5013 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
scopes. |
| 5014 |         _ => None, | executable/support line; behavior impact depends on surrounding block context. |
| 5015 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 5016 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 5017 | (blank) | line is blank; visual separation only. |
| 5018 |     fn tab_faults(&mut self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter API
ontrol flow, and outputs. |
| 5019 |         let dtc_now = self.bench.ecm.active_dtcs.len(); | executable/support line; behavior impact depends o
urrounding block context. |
| 5020 |         let dtc_delta = dtc_now as isize - self.fault_last_dtc_count as isize; | executable/support line; be
ior impact depends on surrounding block context. |
| 5021 |         if dtc_delta != 0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 5022 |             self.fault_feedback = format!( | executable/support line; behavior impact depends on surrounding
ock context. |
| 5023 |                 "ECM DTC update observed: {} -> {} ({}:~)", | executable/support line; behavior impact depen
on surrounding block context. |
| 5024 |                 self.fault_last_dtc_count, dtc_now, dtc_delta | executable/support line; behavior impact dep
s on surrounding block context. |
| 5025 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 5026 |             self.fault_last_dtc_count = dtc_now; | executable/support line; behavior impact depends on surro
ing block context. |
| 5027 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 5028 |         ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block conte
|
| 5029 |             .id_salt("faults::page_scroll") | executable/support line; behavior impact depends on surroundin
lock context. |
| 5030 |             .auto_shrink([false, false]) | executable/support line; behavior impact depends on surrounding b
k context. |
| 5031 |             .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 5032 |         ui.columns(2, \|cols\| { | executable/support line; behavior impact depends on surrounding block con
t. |
| 5033 |             // Left: active DTCs | comment line; documentation intent, no runtime effect. |
| 5034 |             let ui = &mut cols[0]; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 5035 |             egui::Frame::group(ui.style()) | executable/support line; behavior impact depends on surrounding
ock context. |
| 5036 |                 .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on surround
block context. |
| 5037 |                 .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 5038 |                     ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 5039 |                         RichText::new("ACTIVE DTCs (DM1)") | executable/support line; behavior impact depend
n surrounding block context. |
| 5040 |                             .size(12.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 5041 |                             .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior imp
depends on surrounding block context. |
| 5042 |                             ); | executable/support line; behavior impact depends on surrounding block context. |
| 5043 |                         ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 5044 |                     let dtcs_empty = self.bench.ecm.active_dtcs.is_empty(); | executable/support line; behav
impact depends on surrounding block context. |
| 5045 |                     if dtcs_empty { | conditional gate; changing condition shifts runtime path and side effe
. |
| 5046 |                         ui.add_space(10.0); | executable/support line; behavior impact depends on surroundin
lock context. |

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| 5047 |                ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5048 |                RichText::new("✓ No active fault codes") | executable/support line; behavior im
t depends on surrounding block context. |
| 5049 |                .size(13.0) | executable/support line; behavior impact depends on surroundin
lock context. |
| 5050 |                .color(Color32::GREEN), | executable/support line; behavior impact depends o
urrounding block context. |
| 5051 |                ); | executable/support line; behavior impact depends on surrounding block context.
| 5052 |                } else { | executable/support line; behavior impact depends on surrounding block context
| 5053 |                ScrollArea::vertical() | executable/support line; behavior impact depends on surround
g block context. |
| 5054 |                .id_salt("faults::active_dtcs_scroll") | executable/support line; behavior impac
depends on surrounding block context. |
| 5055 |                .max_height(250.0) | executable/support line; behavior impact depends on surround
g block context. |
| 5056 |                .auto_shrink([false, false]) | executable/support line; behavior impact depends
surrounding block context. |
| 5057 |                .show(ui, \|ui\| { | executable/support line; behavior impact depends on surround
g block context. |
| 5058 |                for dtc in &self.bench.ecm.active_dtcs { | iteration control; changing bound
ody affects throughput, timing, and possibly safety guarantees. |
| 5059 |                let c = match dtc.severity { | executable/support line; behavior impact
ends on surrounding block context. |
| 5060 |                DtcSeverity::Red => Color32::RED, | executable/support line; behavior
mpact depends on surrounding block context. |
| 5061 |                DtcSeverity::Amber => Color32::GOLD, | executable/support line; beha
r impact depends on surrounding block context. |
| 5062 |                DtcSeverity::Mil => Color32::LIGHT_BLUE, | executable/support line;
avior impact depends on surrounding block context. |
| 5063 |                DtcSeverity::Protect => Color32::from_rgb(0, 210, 210), | executable
pport line; behavior impact depends on surrounding block context. |
| 5064 |                }; | scope delimiter; structure-only line, but misplaced edits can chang
wnership and lifetime scopes. |
| 5065 |                egui::Frame::group(ui.style()) | executable/support line; behavior impac
depends on surrounding block context. |
| 5066 |                .fill(Color32::from_gray(20)) | executable/support line; behavior im
t depends on surrounding block context. |
| 5067 |                .show(ui, \|ui\| { | executable/support line; behavior impact depend
n surrounding block context. |
| 5068 |                ui.horizontal(\|ui\| { | executable/support line; behavior impac
depends on surrounding block context. |
| 5069 |                ui.label( | executable/support line; behavior impact depends
surrounding block context. |
| 5070 |                RichText::new(format!("{}", dtc.spn)) | executable
pport line; behavior impact depends on surrounding block context. |
| 5071 |                .size(13.0) | executable/support line; behavior impa
depends on surrounding block context. |
| 5072 |                .color(c) | executable/support line; behavior impact
pends on surrounding block context. |
| 5073 |                .strong(), | executable/support line; behavior impac
depends on surrounding block context. |
| 5074 |                ); | executable/support line; behavior impact depends on sur
nding block context. |
| 5075 |                ui.label( | executable/support line; behavior impact depends
surrounding block context. |
| 5076 |                RichText::new(format!("{}", dtc.fmi)) | executable
pport line; behavior impact depends on surrounding block context. |
| 5077 |                .size(12.0) | executable/support line; behavior impa
depends on surrounding block context. |
| 5078 |                .color(Color32::GOLD), | executable/support line; be
ior impact depends on surrounding block context. |
| 5079 |                ); | executable/support line; behavior impact depends on sur
nding block context. |
| 5080 |                ui.label( | executable/support line; behavior impact depends

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surrounding block context. |
| 5081 |                                RichText::new(format!("{}", dtc.count)) | executable/su
rt line; behavior impact depends on surrounding block context. |
| 5082 |                                .size(11.0) | executable/support line; behavior impa
depends on surrounding block context. |
| 5083 |                                .color(Color32::GRAY), | executable/support line; be
ior impact depends on surrounding block context. |
| 5084 |                                ); | executable/support line; behavior impact depends on sur
nding block context. |
| 5085 |                                ui.label( | executable/support line; behavior impact depends
surrounding block context. |
| 5086 |                                RichText::new(format!("{}", dtc.severity)) | executabl
upport line; behavior impact depends on surrounding block context. |
| 5087 |                                .size(11.0) | executable/support line; behavior impa
depends on surrounding block context. |
| 5088 |                                .color(c), | executable/support line; behavior impac
depends on surrounding block context. |
| 5089 |                                ); | executable/support line; behavior impact depends on sur
nding block context. |
| 5090 |                                }); | executable/support line; behavior impact depends on surrou
ng block context. |
| 5091 |                                ui.label( | executable/support line; behavior impact depends on
rounding block context. |
| 5092 |                                RichText::new(dtc.desc) | executable/support line; behavior
act depends on surrounding block context. |
| 5093 |                                .size(11.0) | executable/support line; behavior impact d
nds on surrounding block context. |
| 5094 |                                .color(Color32::from_gray(200)), | executable/support li
behavior impact depends on surrounding block context. |
| 5095 |                                ); | executable/support line; behavior impact depends on surroun
g block context. |
| 5096 |                                let mem_addr = Self::dtc_fault_storage_addr(dtc.spn, dtc.fmi); |
executable/support line; behavior impact depends on surrounding block context. |
| 5097 |                                let dm1_hex = Self::dtc_dm1_payload_hex( | executable/support li
behavior impact depends on surrounding block context. |
| 5098 |                                matches!(dtc.severity, DtcSeverity::Amber), | executable/sup
t line; behavior impact depends on surrounding block context. |
| 5099 |                                matches!(dtc.severity, DtcSeverity::Red), | executable/suppo
line; behavior impact depends on surrounding block context. |
| 5100 |                                matches!(dtc.severity, DtcSeverity::Mil), | executable/suppo
line; behavior impact depends on surrounding block context. |
| 5101 |                                dtc.spn, | executable/support line; behavior impact depends
surrounding block context. |
| 5102 |                                dtc.fmi, | executable/support line; behavior impact depends
surrounding block context. |
| 5103 |                                ); | executable/support line; behavior impact depends on surroun
g block context. |
| 5104 |                                ui.label( | executable/support line; behavior impact depends on
rounding block context. |
| 5105 |                                RichText::new(format!( | executable/support line; behavior i
ct depends on surrounding block context. |
| 5106 |                                "ECM fault storage addr: 0x{mem_addr:08X} (NVM.FaultStor
)" | executable/support line; behavior impact depends on surrounding block context. |
| 5107 |                                )) | executable/support line; behavior impact depends on sur
nding block context. |
| 5108 |                                .size(9.7) | executable/support line; behavior impact depend
n surrounding block context. |
| 5109 |                                .monospace() | executable/support line; behavior impact depe
on surrounding block context. |
| 5110 |                                .color(Color32::from_rgb(140, 210, 255)), | executable/suppo
line; behavior impact depends on surrounding block context. |
| 5111 |                                ); | executable/support line; behavior impact depends on surroun
g block context. |
| 5112 |                                ui.label( | executable/support line; behavior impact depends on
rounding block context. |

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| 5113 |                                RichText::new( | executable/support line; behavior impact de
ds on surrounding block context. |
| 5114 |                                "Diagnostic request -> ECM SA 0x00: UDS 19 02 FF (ReadDT
y status mask)" | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 5115 |                                .to_string(), | executable/support line; behavior im
t depends on surrounding block context. |
| 5116 |                                ) | executable/support line; behavior impact depends on surr
ding block context. |
| 5117 |                                .size(9.6) | executable/support line; behavior impact depend
n surrounding block context. |
| 5118 |                                .monospace() | executable/support line; behavior impact depe
on surrounding block context. |
| 5119 |                                .color(Color32::from_gray(170)), | executable/support line;
avior impact depends on surrounding block context. |
| 5120 |                                ); | executable/support line; behavior impact depends on surroun
g block context. |
| 5121 |                                ui.label( | executable/support line; behavior impact depends on
rounding block context. |
| 5122 |                                RichText::new(format!( | executable/support line; behavior i
ct depends on surrounding block context. |
| 5123 |                                "ECM recognition broadcast -> J1939 DM1 PGN 65226 DATA [
", | executable/support line; behavior impact depends on surrounding block context. |
| 5124 |                                dm1_hex | executable/support line; behavior impact depen
on surrounding block context. |
| 5125 |                                )) | executable/support line; behavior impact depends on surr
nding block context. |
| 5126 |                                .size(9.6) | executable/support line; behavior impact depend
n surrounding block context. |
| 5127 |                                .monospace() | executable/support line; behavior impact depe
on surrounding block context. |
| 5128 |                                .color(Color32::from_gray(170)), | executable/support line;
avior impact depends on surrounding block context. |
| 5129 |                                ); | executable/support line; behavior impact depends on surroun
g block context. |
| 5130 |                                }); | executable/support line; behavior impact depends on surroun
lock context. |
| 5131 |                                } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 5132 |                                }); | executable/support line; behavior impact depends on surrounding block cont
. |
| 5133 |                                } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 5134 |                                ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 5135 |                                ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 5136 |                                RichText::new(format!( | executable/support line; behavior impact depends on surroun
g block context. |
| 5137 |                                "Fault pipeline: selected -> inject -> ECM evaluate -> DM1 broadcast \ | DTC now:
({:+})", | executable/support line; behavior impact depends on surrounding block context. |
| 5138 |                                dtc_now, dtc_delta | executable/support line; behavior impact depends on surroun
g block context. |
| 5139 |                                )) | executable/support line; behavior impact depends on surrounding block context.
| 5140 |                                .size(10.0) | executable/support line; behavior impact depends on surrounding block
text. |
| 5141 |                                .color(Color32::from_gray(150)), | executable/support line; behavior impact depends
surrounding block context. |
| 5142 |                                ); | executable/support line; behavior impact depends on surrounding block context. |
| 5143 |                                if !self.fault_feedback.is_empty() { | conditional gate; changing condition shifts runti
path and side effects. |
| 5144 |                                ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5145 |                                RichText::new(format!("Last action: {}", self.fault_feedback)) | executable/supp
line; behavior impact depends on surrounding block context. |
| 5146 |                                .size(10.2) | executable/support line; behavior impact depends on surroundin

```



```

lock context. |
| 5147 |                                     .color(Color32::LIGHT_BLUE), | executable/support line; behavior impact depends
on surrounding block context. |
| 5148 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 5149 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 5150 |                                     ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 5151 |                                     // Warning lamps | comment line; documentation intent, no runtime effect. |
| 5152 |                                     ui.horizontal(\ui\ { | executable/support line; behavior impact depends on surrounding
ock context. |
| 5153 |                                     let e = &self.bench.ecm; | executable/support line; behavior impact depends on surro
ing block context. |
| 5154 |                                     for (lbl, on, c) in [ | iteration control; changing bounds/body affects throughput,
ing, and possibly safety guarantees. |
| 5155 |                                     ("■ RED", e.red_lamp, Color32::RED), | executable/support line; behavior impact
ends on surrounding block context. |
| 5156 |                                     ("■ AMB", e.amber_lamp, Color32::GOLD), | executable/support line; behavior impa
depends on surrounding block context. |
| 5157 |                                     ("■ MIL", e.mil_active, Color32::LIGHT_BLUE), | executable/support line; behavio
mpact depends on surrounding block context. |
| 5158 |                                     ("■ PROT", e.protect_lamp, Color32::from_rgb(0, 210, 210)), | executable/support
ne; behavior impact depends on surrounding block context. |
| 5159 |                                     ] { | executable/support line; behavior impact depends on surrounding block context. |
| 5160 |                                     ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 5161 |                                     RichText::new(lbl) | executable/support line; behavior impact depends on surr
nding block context. |
| 5162 |                                     .size(12.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 5163 |                                     .color(if on { c } else { Color32::from_gray(55) }) | executable/support
ne; behavior impact depends on surrounding block context. |
| 5164 |                                     .strong(), | executable/support line; behavior impact depends on surroun
g block context. |
| 5165 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 5166 |                                     ui.add_space(4.0); | executable/support line; behavior impact depends on surroun
g block context. |
| 5167 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 5168 |                                     )); | executable/support line; behavior impact depends on surrounding block context. |
| 5169 |                                     ui.add_space(6.0); | executable/support line; behavior impact depends on surrounding blo
context. |
| 5170 |                                     if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 5171 |                                     .add( | executable/support line; behavior impact depends on surrounding block context
|
| 5172 |                                     Button::new(RichText::new("■ CLEAR ALL DTCs").size(12.0)) | executable/support l
; behavior impact depends on surrounding block context. |
| 5173 |                                     .fill(Color32::from_rgb(50, 22, 22)), | executable/support line; behavior im
t depends on surrounding block context. |
| 5174 |                                     ) | executable/support line; behavior impact depends on surrounding block context. |
| 5175 |                                     .clicked() | executable/support line; behavior impact depends on surrounding block c
ext. |
| 5176 |                                     { | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 5177 |                                     self.cmds.push(Cmd::ClearDtcs); | executable/support line; behavior impact depends o
urrounding block context. |
| 5178 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 5179 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 5180 | (blank) | line is blank; visual separation only. |
| 5181 |                                     // Right: fault injection | comment line; documentation intent, no runtime effect. |
| 5182 |                                     let ui = &mut cols[1]; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 5183 |                                     equi::Frame::group(ui.style()) | executable/support line; behavior impact depends on surrounding

```

```

ock context. |
| 5184 |         .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on surrounding
    block context. |
| 5185 |         .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
ext. |
| 5186 |             ui.label( | executable/support line; behavior impact depends on surrounding block context. |
|
| 5187 |                 RichText::new("FAULT INJECTION PANEL") | executable/support line; behavior impact depends
ds on surrounding block context. |
| 5188 |                 .size(12.0) | executable/support line; behavior impact depends on surrounding block context. |
    context. |
| 5189 |                 .color(Color32::from_rgb(200, 80, 80)), | executable/support line; behavior impact
depends on surrounding block context. |
| 5190 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 5191 |             ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
text. |
| 5192 |             ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block context. |
ock context. |
| 5193 |                 .id_salt("faults::selector_scroll") | executable/support line; behavior impact depends
on surrounding block context. |
| 5194 |                 .max_height(340.0) | executable/support line; behavior impact depends on surrounding block context. |
ock context. |
| 5195 |                 .auto_shrink([false, false]) | executable/support line; behavior impact depends on surrounding block context. |
ounding block context. |
| 5196 |                 .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
ock context. |
| 5197 |                     for (i, ft) in FAULT_TYPES.iter().enumerate() { | iteration control; changing body
s/body affects throughput, timing, and possibly safety guarantees. |
| 5198 |                         ui.push_id(format!("fault::selector::{}", i), \|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
e; behavior impact depends on surrounding block context. |
| 5199 |                             let sel = i == self.fault_idx; | executable/support line; behavior impact depends on surrounding block context. |
epends on surrounding block context. |
| 5200 |                             let act = | executable/support line; behavior impact depends on surrounding block context. |
    block context. |
| 5201 |                                 self.bench.fault_active && self.bench.selected_fault == *ft; | executable/support line; behavior impact depends on surrounding block context. |
le/support line; behavior impact depends on surrounding block context. |
| 5202 |                                 let fill = if act { | executable/support line; behavior impact depends on surrounding block context. |
urrounding block context. |
| 5203 |                                     Color32::from_rgb(70, 15, 15) | executable/support line; behavior impact depends on surrounding block context. |
t depends on surrounding block context. |
| 5204 |                                     } else if sel { | executable/support line; behavior impact depends on surrounding block context. |
unding block context. |
| 5205 |                                     Color32::from_rgb(25, 45, 75) | executable/support line; behavior impact depends on surrounding block context. |
t depends on surrounding block context. |
| 5206 |                                     } else { | executable/support line; behavior impact depends on surrounding block context. |
block context. |
| 5207 |                                     Color32::TRANSPARENT | executable/support line; behavior impact depends on surrounding block context. |
on surrounding block context. |
| 5208 |                                     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
wnership and lifetime scopes. |
| 5209 |                                     let col = if act { | executable/support line; behavior impact depends on surrounding block context. |
rrounding block context. |
| 5210 |                                     Color32::RED | executable/support line; behavior impact depends on surrounding block context. |
ounding block context. |
| 5211 |                                     } else if sel { | executable/support line; behavior impact depends on surrounding block context. |
unding block context. |
| 5212 |                                     Color32::from_rgb(180, 220, 255) | executable/support line; behavior impact depends on surrounding block context. |
pact depends on surrounding block context. |
| 5213 |                                     } else { | executable/support line; behavior impact depends on surrounding block context. |
block context. |
| 5214 |                                     Color32::from_gray(155) | executable/support line; behavior impact depends on surrounding block context. |
nds on surrounding block context. |
| 5215 |                                     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
wnership and lifetime scopes. |
| 5216 |                                     let btn = Button::new( | executable/support line; behavior impact depends

```

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n surrounding block context. |
| 5217 |             RichText::new(format!("{}", ft)).size(11.0).color(col), | executable
pport line; behavior impact depends on surrounding block context. |
| 5218 |             ) | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 5219 |             .fill(fill) | executable/support line; behavior impact depends on surrou
ng block context. |
| 5220 |             .min_size(Vec2::new(ui.available_width(), 22.0)); | executable/support l
; behavior impact depends on surrounding block context. |
| 5221 |             if ui.add(btn).clicked() { | conditional gate; changing condition shifts
runtime path and side effects. |
| 5222 |                 self.cmds.push(Cmd::SelectFault(i)); | executable/support line; beha
r impact depends on surrounding block context. |
| 5223 |             } | scope delimiter; structure-only line, but misplaced edits can change
nership and lifetime scopes. |
| 5224 |             }); | executable/support line; behavior impact depends on surrounding block
text. |
| 5225 |             } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 5226 |             }); | executable/support line; behavior impact depends on surrounding block context.
| 5227 |             ui.add_space(6.0); | executable/support line; behavior impact depends on surrounding blo
context. |
| 5228 |             ui.horizontal(\ui\ { | executable/support line; behavior impact depends on surrounding
ock context. |
| 5229 |                 let ic = if self.bench.fault_active { | executable/support line; behavior impact dep
s on surrounding block context. |
| 5230 |                     Color32::RED | executable/support line; behavior impact depends on surrounding b
k context. |
| 5231 |                 } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 5232 |                     Color32::from_rgb(160, 35, 35) | executable/support line; behavior impact depend
n surrounding block context. |
| 5233 |                 }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |
| 5234 |                 let lbl = if self.bench.fault_active { | executable/support line; behavior impact de
ds on surrounding block context. |
| 5235 |                     "■ ACTIVE – Inject Again" | executable/support line; behavior impact depends on
rounding block context. |
| 5236 |                 } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 5237 |                     "■ INJECT FAULT" | executable/support line; behavior impact depends on surroundi
block context. |
| 5238 |                 }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |
| 5239 |                 if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 5240 |                     .add( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5241 |                         Button::new(RichText::new(lbl).size(13.0)) | executable/support line; behavi
impact depends on surrounding block context. |
| 5242 |                         .fill(ic) | executable/support line; behavior impact depends on surround
block context. |
| 5243 |                         .min_size(Vec2::new(185.0, 36.0)), | executable/support line; behavior i
ct depends on surrounding block context. |
| 5244 |                     ) | executable/support line; behavior impact depends on surrounding block contex
|
| 5245 |                     .clicked() | executable/support line; behavior impact depends on surrounding blo
context. |
| 5246 |                 { | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 5247 |                     self.cmds.push(Cmd::InjectFault); | executable/support line; behavior impact dep
s on surrounding block context. |
| 5248 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 5249 |                 if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 5250 |                     .add( | executable/support line; behavior impact depends on surrounding block co

```

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xt. |
| 5251 | Button::new(RichText::new("■ CLEAR").size(13.0)) | executable/support line;
avior impact depends on surrounding block context. |
| 5252 | .fill(Color32::from_rgb(22, 68, 32)) | executable/support line; behavior
pact depends on surrounding block context. |
| 5253 | .min_size(Vec2::new(90.0, 36.0)), | executable/support line; behavior im
t depends on surrounding block context. |
| 5254 | ) | executable/support line; behavior impact depends on surrounding block context.
|
| 5255 | .clicked() | executable/support line; behavior impact depends on surrounding blo
context. |
| 5256 | { | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 5257 | self.cmds.push(Cmd::ClearFaults); | executable/support line; behavior impact dep
s on surrounding block context. |
| 5258 | } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 5259 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 5260 | if self.bench.fault_active { | conditional gate; changing condition shifts runtime path
side effects. |
| 5261 | ui.add_space(4.0); | executable/support line; behavior impact depends on surrounding
ock context. |
| 5262 | ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5263 | RichText::new(format!("■ ACTIVE FAULT: {}", self.bench.selected_fault)) | execut
e/support line; behavior impact depends on surrounding block context. |
| 5264 | .size(11.5) | executable/support line; behavior impact depends on surroundin
lock context. |
| 5265 | .color(Color32::RED) | executable/support line; behavior impact depends on s
ounding block context. |
| 5266 | .strong(), | executable/support line; behavior impact depends on surrounding
ock context. |
| 5267 | ); | executable/support line; behavior impact depends on surrounding block context.
| 5268 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 5269 | ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 5270 | ui.label( | executable/support line; behavior impact depends on surrounding block context.
|
| 5271 | RichText::new("ECM DTC RECOGNITION GUIDE") | executable/support line; behavior impac
depends on surrounding block context. |
| 5272 | .size(11.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 5273 | .color(Color32::from_rgb(110, 180, 255)) | executable/support line; behavior imp
depends on surrounding block context. |
| 5274 | .strong(), | executable/support line; behavior impact depends on surrounding blo
context. |
| 5275 | ); | executable/support line; behavior impact depends on surrounding block context. |
| 5276 | if let Some((spn, fmi, msg)) = | conditional gate; changing condition shifts runtime pat
nd side effects. |
| 5277 | Self::fault_selection_hint(self.bench.selected_fault) | executable/support line; beh
or impact depends on surrounding block context. |
| 5278 | { | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 5279 | let addr = Self::dtc_fault_storage_addr(spn, fmi); | executable/support line; behavi
impact depends on surrounding block context. |
| 5280 | let dm1_hex = Self::dtc_dm1_payload_hex(true, false, false, spn, fmi); | executable/
port line; behavior impact depends on surrounding block context. |
| 5281 | ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5282 | RichText::new(format!( | executable/support line; behavior impact depends on sur
nding block context. |
| 5283 | "Selected fault maps to SPN {spn} FMI {fmi} -> addr 0x{addr:08X}" | executab
support line; behavior impact depends on surrounding block context. |
| 5284 | )) | executable/support line; behavior impact depends on surrounding block conte

```

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|
| 5285 |                                     .size(10.2) | executable/support line; behavior impact depends on surrounding block
context. |
| 5286 |                                     .monospace() | executable/support line; behavior impact depends on surrounding block
context. |
| 5287 |                                     .color(Color32::from_rgb(140, 210, 255)), | executable/support line; behavior impact
t depends on surrounding block context. |
| 5288 |                                     ); | executable/support line; behavior impact depends on surrounding block context.
| 5289 |                                     ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5290 |                                     RichText::new(format!("Trigger condition: {msg}")) | executable/support line; be
ior impact depends on surrounding block context. |
| 5291 |                                     .size(10.0) | executable/support line; behavior impact depends on surrounding
lock context. |
| 5292 |                                     .color(Color32::from_gray(180)), | executable/support line; behavior impact
ends on surrounding block context. |
| 5293 |                                     ); | executable/support line; behavior impact depends on surrounding block context.
| 5294 |                                     ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5295 |                                     RichText::new("Message to query/ack in diagnostics: UDS 19 02 FF -> ECM SA 0x00"
I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 5296 |                                     .size(9.8) | executable/support line; behavior impact depends on surrounding
ock context. |
| 5297 |                                     .monospace() | executable/support line; behavior impact depends on surroundi
block context. |
| 5298 |                                     .color(Color32::from_gray(170)), | executable/support line; behavior impact
ends on surrounding block context. |
| 5299 |                                     ); | executable/support line; behavior impact depends on surrounding block context.
| 5300 |                                     ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5301 |                                     RichText::new(format!( | executable/support line; behavior impact depends on surr
nding block context. |
| 5302 |                                     "Expected ECM DM1 payload for this DTC: [{}]", | executable/support line; be
ior impact depends on surrounding block context. |
| 5303 |                                     dm1_hex | executable/support line; behavior impact depends on surrounding block
context. |
| 5304 |                                     )) | executable/support line; behavior impact depends on surrounding block conte
|
| 5305 |                                     .size(9.8) | executable/support line; behavior impact depends on surrounding block
context. |
| 5306 |                                     .monospace() | executable/support line; behavior impact depends on surrounding b
k context. |
| 5307 |                                     .color(Color32::from_gray(170)), | executable/support line; behavior impact depende
on surrounding block context. |
| 5308 |                                     ); | executable/support line; behavior impact depends on surrounding block context.
| 5309 |                                     } else { | executable/support line; behavior impact depends on surrounding block context
| 5310 |                                     ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5311 |                                     RichText::new( | executable/support line; behavior impact depends on surrounding
ock context. |
| 5312 |                                     "This selected fault is network/power domain and may not map to an ECM SPN/F
DTC record.", | executable/support line; behavior impact depends on surrounding block context. |
| 5313 |                                     ) | executable/support line; behavior impact depends on surrounding block context
|
| 5314 |                                     .size(9.8) | executable/support line; behavior impact depends on surrounding block
context. |
| 5315 |                                     .color(Color32::from_gray(155)), | executable/support line; behavior impact depende
on surrounding block context. |
| 5316 |                                     ); | executable/support line; behavior impact depends on surrounding block context.
| 5317 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
ifetime scopes. |
| 5318 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 5319 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 5320 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 5321 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.

```

```
| 5322 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 5323 | (blank) | line is blank; visual separation only. |  
| 5324 | // ██████████ | comment  
intent, no runtime effect. |  
| 5325 | // TAB BOOT | comment line; documentation intent, no runtime effect. |  
| 5326 | // ██████████ | comm  
intent, no runtime effect. |  
| 5327 | impl App { | implementation block; method behavior and trait conformance are defined here. |  
| 5328 |     fn tab_boot(&self, ui: &ampmut Ui) { | declares executable behavior; signature/body changes alter API, cont  
flow, and outputs. |  
| 5329 |         let ign = self.bench.ignition(); | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 5330 |         ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding block conte  
|  
| 5331 |             for (lbl, s) in [ | iteration control; changing bounds/body affects throughput, timing, and poss  
y safety guarantees. |  
| 5332 |                 ("OFF", IgnitionState::Off), | executable/support line; behavior impact depends on surroundi  
block context. |  
| 5333 |                 ("ACC", IgnitionState::Accessory), | executable/support line; behavior impact depends on surr  
nding block context. |  
| 5334 |                 ("ON", IgnitionState::On), | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 5335 |                 ("CRANK", IgnitionState::Cranking), | executable/support line; behavior impact depends on su  
unding block context. |  
| 5336 |                 ("RUN", IgnitionState::Running), | executable/support line; behavior impact depends on surro  
ing block context. |  
| 5337 |             ] { | executable/support line; behavior impact depends on surrounding block context. |  
| 5338 |                 let cur = ign == s; | executable/support line; behavior impact depends on surrounding block  
text. |  
| 5339 |                 let c = if cur { | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 5340 |                     Color32::GREEN | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 5341 |                 } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 5342 |                     Color32::from_gray(80) | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 5343 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and life  
e scopes. |  
| 5344 |             ui.label( | executable/support line; behavior impact depends on surrounding block context. |  
| 5345 |                 RichText::new(format!("{}", lbl)) | executable/support line; behavior impact depends o  
urrounding block context. |  
| 5346 |                 .size(13.0) | executable/support line; behavior impact depends on surrounding block  
text. |  
| 5347 |                 .color(c) | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 5348 |                 .strong(), | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 5349 |             ); | executable/support line; behavior impact depends on surrounding block context. |  
| 5350 |             if s != IgnitionState::Running { | conditional gate; changing condition shifts runtime path  
side effects. |  
| 5351 |                 ui.label(RichText::new("→").size(11.0).color(Color32::from_gray(80))); | executable/supp  
line; behavior impact depends on surrounding block context. |  
| 5352 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet  
scopes. |  
| 5353 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime  
pes. |  
| 5354 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context.  
| 5355 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |  
| 5356 |             RichText::new(format!( | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 5357 |                 "t={:.1}s Online:{}{}/{}", | executable/support line; behavior impact depends on surround  
block context. |  
| 5358 |                 self.bench.elapsed, | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 5359 |                 self.bench | executable/support line; behavior impact depends on surrounding block cont
```

```

|
| 5360 |             .boot | executable/support line; behavior impact depends on surrounding block context. |
|
| 5361 |             .ecus | executable/support line; behavior impact depends on surrounding block context. |
|
| 5362 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| . |
| 5363 |             .filter(\|e\| e.is_online()) | executable/support line; behavior impact depends on surrounding block context. |
| 5364 |             .count(), | executable/support line; behavior impact depends on surrounding block context. |
| 5365 |             self.bench.boot.ecus.len() | executable/support line; behavior impact depends on surrounding block context. |
| 5366 |             )) | executable/support line; behavior impact depends on surrounding block context. |
| 5367 |             .size(11.0) | executable/support line; behavior impact depends on surrounding block context. |
| 5368 |             .color(Color32::from_gray(170)), | executable/support line; behavior impact depends on surrounding block context. |
| 5369 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 5370 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 5371 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 5372 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 5373 |             RichText::new("SAFETY INTERLOCKS:") | executable/support line; behavior impact depends on surrounding block context. |
| 5374 |             .size(11.0) | executable/support line; behavior impact depends on surrounding block context. |
| 5375 |             .color(Color32::from_gray(140)), | executable/support line; behavior impact depends on surrounding block context. |
| 5376 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 5377 |         ui.columns(3, \|cols\| { | executable/support line; behavior impact depends on surrounding block context. |
| 5378 |             let ils = &self.bench.boot.safety_interlocks; | executable/support line; behavior impact depends on surrounding block context. |
| 5379 |             let per = ils.len().div_ceil(3); | executable/support line; behavior impact depends on surrounding block context. |
| 5380 |             for (ci, col) in cols.iter_mut().enumerate() { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 5381 |                 for il in ils.iter().skip(ci * per).take(per) { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 5382 |                     let (sym, c) = if il.satisfied { | executable/support line; behavior impact depends on surrounding block context. |
| 5383 |                         ("✓", Color32::GREEN) | executable/support line; behavior impact depends on surrounding block context. |
| 5384 |                     } else if il.blocks_start { | executable/support line; behavior impact depends on surrounding block context. |
| 5385 |                         ("✗", Color32::RED) | executable/support line; behavior impact depends on surrounding block context. |
| 5386 |                     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 5387 |                         ("■", Color32::YELLOW) | executable/support line; behavior impact depends on surrounding block context. |
| 5388 |                     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 5389 |                     col.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
| 5390 |                         ui.label(RichText::new(sym).size(13.0).color(c)); | executable/support line; behavior impact depends on surrounding block context. |
| 5391 |                         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 5392 |                             RichText::new(format!("{}", il.id, il.description)) | executable/support line; behavior impact depends on surrounding block context. |
| 5393 |                             .size(10.5) | executable/support line; behavior impact depends on surrounding block context. |
| 5394 |                             .color(Color32::from_gray(185)), | executable/support line; behavior impact depends on surrounding block context. |
| 5395 |                             ); | executable/support line; behavior impact depends on surrounding block context. |
| 5396 |                             if il.blocks_start && !il.satisfied { | conditional gate; changing condition shifts time path and side effects. |

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| 5397 |                 ui.label(RichText::new("[BLOCKS]").size(10.0).color(Color32::RED)); | executable/
support line; behavior impact depends on surrounding block context. |
| 5398 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 5399 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 5400 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 5401 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 5402 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 5403 |     if self.bench.boot.crank_inhibited { | conditional gate; changing condition shifts runtime path and
e effects. |
| 5404 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 5405 |             RichText::new(format!( | executable/support line; behavior impact depends on surrounding blo
context. |
| 5406 |                 "■ CRANK INHIBITED: {}", | executable/support line; behavior impact depends on surroundi
block context. |
| 5407 |                 self.bench.boot.crank_inhibit_reason.unwrap_or("?") | error-propagating operation; chang
it impacts failure semantics and caller recovery paths. |
| 5408 |             )) | executable/support line; behavior impact depends on surrounding block context. |
| 5409 |             .size(12.0) | executable/support line; behavior impact depends on surrounding block context.
| 5410 |             .color(Color32::RED), | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 5411 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 5412 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 5413 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 5414 |             RichText::new("✓ CRANK CLEAR") | executable/support line; behavior impact depends on surround
g block context. |
| 5415 |             .size(12.0) | executable/support line; behavior impact depends on surrounding block cont
. |
| 5416 |             .color(Color32::GREEN), | executable/support line; behavior impact depends on surroundin
lock context. |
| 5417 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 5418 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 5419 |     ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 5420 |     ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 5421 |         RichText::new("BOOT EVENT LOG:") | executable/support line; behavior impact depends on surroundi
block context. |
| 5422 |         .size(11.0) | executable/support line; behavior impact depends on surrounding block context.
| 5423 |         .color(Color32::from_gray(140)), | executable/support line; behavior impact depends on surro
ing block context. |
| 5424 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 5425 |     ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block conte
|
| 5426 |         .auto_shrink([false, false]) | executable/support line; behavior impact depends on surrounding b
k context. |
| 5427 |         .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 5428 |             for ev in self.bench.boot.event_log.iter().rev().take(80) { | iteration control; changing bo
s/body affects throughput, timing, and possibly safety guarantees. |
| 5429 |                 let c = match ev.event { | executable/support line; behavior impact depends on surroundi
block context. |
| 5430 |                     auto_breaking::boot_sequence::BootEventKind::Running => Color32::GREEN, | executable
support line; behavior impact depends on surrounding block context. |
| 5431 |                     auto_breaking::boot_sequence::BootEventKind::Fault => Color32::RED, | executable/sup
t line; behavior impact depends on surrounding block context. |
| 5432 |                     auto_breaking::boot_sequence::BootEventKind::AddressClaimed => { | executable/suppor
ine; behavior impact depends on surrounding block context. |
| 5433 |                         Color32::from_rgb(0, 210, 210) | executable/support line; behavior impact depend
n surrounding block context. |
| 5434 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 5435 |                 _ => Color32::from_gray(195), | executable/support line; behavior impact depends on
rounding block context. |

```



```

5436 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 5437 |             ui.horizontal(\ui\ { | executable/support line; behavior impact depends on surrounding
ock context. |
| 5438 |                 ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5439 |                     RichText::new(format!("{:7.3}s", ev.timestamp)) | executable/support line; behav
impact depends on surrounding block context. |
| 5440 |                         .size(10.5) | executable/support line; behavior impact depends on surroundin
lock context. |
| 5441 |                             .monospace() | executable/support line; behavior impact depends on surroundi
block context. |
| 5442 |                                 .color(Color32::from_gray(115)), | executable/support line; behavior impact
ends on surrounding block context. |
| 5443 |                                     ); | executable/support line; behavior impact depends on surrounding block context.
| 5444 |                                         ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5445 |                                             RichText::new(format!("[{:<12}]", ev.ecu_name)) | executable/support line; behav
impact depends on surrounding block context. |
| 5446 |                                                 .size(10.5) | executable/support line; behavior impact depends on surroundin
lock context. |
| 5447 |                                                     .monospace() | executable/support line; behavior impact depends on surroundi
block context. |
| 5448 |                                                         .color(Color32::from_gray(165)), | executable/support line; behavior impact
ends on surrounding block context. |
| 5449 |                                                             ); | executable/support line; behavior impact depends on surrounding block context.
| 5450 |                                                                 ui.label(RichText::new(&ev.description).size(10.5).color(c)); | executable/support l
; behavior impact depends on surrounding block context. |
| 5451 |                                                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 5452 |                                                                         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 5453 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 5454 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 5455 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 5456 | (blank) | line is blank; visual separation only. |
| 5457 | // ████████████████████████████████████████████████████████████████████████████████████████████████████████████████ | commm
intent, no runtime effect. |
| 5458 | // TAB IMPLEMENTS – fully interactive | comment line; documentation intent, no runtime effect. |
| 5459 | // ████████████████████████████████████████████████████████████████████████████████████████████████████████████████ | commm
intent, no runtime effect. |
| 5460 | impl App { | implementation block; method behavior and trait conformance are defined here. |
| 5461 |     fn tab_implements(&mut self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter
I, control flow, and outputs. |
| 5462 |         // Snapshot display values | comment line; documentation intent, no runtime effect. |
| 5463 |         let pto_rpm = self.bench.implement.pto_rear_rpm; | executable/support line; behavior impact depends
surrounding block context. |
| 5464 |         let pto_mode_s = format!("{}", self.bench.implement.pto_mode); | executable/support line; behavior i
ct depends on surrounding block context. |
| 5465 |         let pto_target = self.bench.implement.pto_target_rpm; | executable/support line; behavior impact dep
s on surrounding block context. |
| 5466 |         let pto_slip = self.bench.implement.pto_slip_pct; | executable/support line; behavior impact depends
surrounding block context. |
| 5467 |         let pto_torq = self.bench.implement.pto_shaft_torque_nm; | executable/support line; behavior impact
ends on surrounding block context. |
| 5468 |         let pto_over = self.bench.implement.overload_protection_active; | executable/support line; behavior
act depends on surrounding block context. |
| 5469 |         let _pto_en = self.bench.implement.pto_rear_enabled; | executable/support line; behavior impact depe
on surrounding block context. |
| 5470 |         let hitch_pos = self.bench.implement.hitch_position_pct; | executable/support line; behavior impact
ends on surrounding block context. |
| 5471 |         let hitch_draft = self.bench.implement.hitch_draft_force_kn; | executable/support line; behavior imp
depends on surrounding block context. |
| 5472 |         let hitch_mode = format!("{}", self.bench.implement.hitch_control_mode); | executable/support line;
avior impact depends on surrounding block context. |
| 5473 |         let hitch_gp = self.bench.implement.hitch_ground_pressure kpa; |executable/support line; behavior i

```

[illegible]

```

impact depends on surrounding block context. |
| 5515 |         arc_gauge( | executable/support line; behavior impact depends on surrounding block co
ext. |
| 5516 |             ui, pto_rpm, 0.0, 1100.0, "PTO REAR", "rpm", 140.0, 900.0, 1050.0, None, | execu
le/support line; behavior impact depends on surrounding block context. |
| 5517 |             ); | executable/support line; behavior impact depends on surrounding block context.
| 5518 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 5519 |         digital_readout(ui, "Mode", &pto_mode_s, Color32::YELLOW); | executable/support line; be
ior impact depends on surrounding block context. |
| 5520 |         digital_readout( | executable/support line; behavior impact depends on surrounding block
ntext. |
| 5521 |             ui, | executable/support line; behavior impact depends on surrounding block context.
| 5522 |             "Target", | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5523 |             &format!("{:.0} rpm", pto_target), | executable/support line; behavior impact depend
n surrounding block context. |
| 5524 |             Color32::from_gray(185), | executable/support line; behavior impact depends on surro
ing block context. |
| 5525 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 5526 |         digital_readout( | executable/support line; behavior impact depends on surrounding block
ntext. |
| 5527 |             ui, | executable/support line; behavior impact depends on surrounding block context.
| 5528 |             "Shaft Tq", | executable/support line; behavior impact depends on surrounding block
text. |
| 5529 |             &format!("{:.0} Nm", pto_torq), | executable/support line; behavior impact depends o
urrounding block context. |
| 5530 |             Color32::from_rgb(0, 210, 210), | executable/support line; behavior impact depends o
urrounding block context. |
| 5531 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 5532 |         bar_gauge( | executable/support line; behavior impact depends on surrounding block conte
|
| 5533 |             ui, | executable/support line; behavior impact depends on surrounding block context.
| 5534 |             "Slip", | executable/support line; behavior impact depends on surrounding block cont
. |
| 5535 |             pto_slip, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5536 |             100.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 5537 |             "%", | executable/support line; behavior impact depends on surrounding block context
| 5538 |             120.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 5539 |             if pto_slip > 10.0 { | conditional gate; changing condition shifts runtime path and
e effects. |
| 5540 |                 Color32::YELLOW | executable/support line; behavior impact depends on surroundin
lock context. |
| 5541 |                 } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 5542 |                 Color32::GREEN | executable/support line; behavior impact depends on surrounding
ock context. |
| 5543 |                 }, | executable/support line; behavior impact depends on surrounding block context.
| 5544 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 5545 |             if pto_over { | conditional gate; changing condition shifts runtime path and side effect
|
| 5546 |             ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5547 |                 RichText::new("■ OVERLOAD!") | executable/support line; behavior impact depends
surrounding block context. |
| 5548 |                 .size(12.0) | executable/support line; behavior impact depends on surrounding
lock context. |
| 5549 |                 .color(Color32::RED) | executable/support line; behavior impact depends on s
ounding block context. |
| 5550 |                 .strong(), | executable/support line; behavior impact depends on surrounding
ock context. |
| 5551 |             ); | executable/support line; behavior impact depends on surrounding block context.
| 5552 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and L

```

[illegible]

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|
| 5588 |           RichText::new("3-POINT HITCH") | executable/support line; behavior impact depends on
rounding block context. |
| 5589 |           .size(12.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 5590 |           .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior imp
depends on surrounding block context. |
| 5591 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 5592 |           // Visual position bar | comment line; documentation intent, no runtime effect. |
| 5593 |           let (rect, _) = ui | executable/support line; behavior impact depends on surrounding blo
context. |
| 5594 |           .allocate_exact_size(Vec2::new(ui.available_width(), 70.0), Sense::hover()); | execu
le/support line; behavior impact depends on surrounding block context. |
| 5595 |           let p = ui.painter_at(rect); | executable/support line; behavior impact depends on surro
ing block context. |
| 5596 |           p.rect_filled(rect, 4.0, Color32::from_gray(20)); | executable/support line; behavior im
t depends on surrounding block context. |
| 5597 |           let frac = 1.0 - hitch_pos as f32 / 100.0; | executable/support line; behavior impact de
ds on surrounding block context. |
| 5598 |           let fill_h = rect.height() * frac; | executable/support line; behavior impact depends on
rounding block context. |
| 5599 |           let fill_r = Rect::from_min_size( | executable/support line; behavior impact depends on
rounding block context. |
| 5600 |           Pos2::new(rect.min.x, rect.min.y + fill_h), | executable/support line; behavior impa
depends on surrounding block context. |
| 5601 |           Vec2::new(rect.width(), rect.height() - fill_h), | executable/support line; behavior
pact depends on surrounding block context. |
| 5602 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 5603 |           p.rect_filled(fill_r, 4.0, Color32::from_rgb(30, 140, 200)); | executable/support line;
avior impact depends on surrounding block context. |
| 5604 |           p.text( | executable/support line; behavior impact depends on surrounding block context.
| 5605 |           rect.center(), | executable/support line; behavior impact depends on surrounding blo
context. |
| 5606 |           Align2::CENTER_CENTER, | executable/support line; behavior impact depends on surroun
g block context. |
| 5607 |           format!("{:.1}%", hitch_pos), | executable/support line; behavior impact depends on
rounding block context. |
| 5608 |           FontId::proportional(18.0), | executable/support line; behavior impact depends on su
unding block context. |
| 5609 |           Color32::WHITE, | executable/support line; behavior impact depends on surrounding bl
context. |
| 5610 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 5611 | (blank) | line is blank; visual separation only. |
| 5612 |           bar_gauge(ui, "Draft", hitch_draft, 50.0, "kN", 120.0, Color32::YELLOW); | executable/su
rt line; behavior impact depends on surrounding block context. |
| 5613 |           digital_readout(ui, "Mode", &hitch_mode, Color32::from_gray(200)); | executable/support
e; behavior impact depends on surrounding block context. |
| 5614 |           digital_readout( | executable/support line; behavior impact depends on surrounding block
ntext. |
| 5615 |           ui, | executable/support line; behavior impact depends on surrounding block context.
| 5616 |           "Gnd P", | executable/support line; behavior impact depends on surrounding block con
t. |
| 5617 |           &format!("{:.1} kPa", hitch_gp), | executable/support line; behavior impact depends
surrounding block context. |
| 5618 |           Color32::from_gray(180), | executable/support line; behavior impact depends on surro
ing block context. |
| 5619 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 5620 |           ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 5621 |           // Target slider | comment line; documentation intent, no runtime effect. |
| 5622 |           ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 5623 |           RichText::new("Target position:") | executable/support line; behavior impact depends
surrounding block context. |
| 5624 |           .size(11.0) | executable/support line; behavior impact depends on surrounding bl

```

[illegible]

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| 5660 | bar_gauge( | executable/support line; behavior impact depends on surrounding block context. |
| |
| 5661 | ui, | executable/support line; behavior impact depends on surrounding block context. |
| 5662 | "Flow", | executable/support line; behavior impact depends on surrounding block context. |
. |
| 5663 | hcm_flow, | executable/support line; behavior impact depends on surrounding block context. |
| 5664 | 300.0, | executable/support line; behavior impact depends on surrounding block context. |
| |
| 5665 | "L/m", | executable/support line; behavior impact depends on surrounding block context. |
| |
| 5666 | w, | executable/support line; behavior impact depends on surrounding block context. |
| 5667 | Color32::from_rgb(0, 210, 210), | executable/support line; behavior impact depends on surrounding block context. |
| 5668 | ); | executable/support line; behavior impact depends on surrounding block context. |
| 5669 | bar_gauge( | executable/support line; behavior impact depends on surrounding block context. |
| |
| 5670 | ui, | executable/support line; behavior impact depends on surrounding block context. |
| 5671 | "Temp", | executable/support line; behavior impact depends on surrounding block context. |
. |
| 5672 | hcm_temp, | executable/support line; behavior impact depends on surrounding block context. |
| 5673 | 120.0, | executable/support line; behavior impact depends on surrounding block context. |
| |
| 5674 | "°C", | executable/support line; behavior impact depends on surrounding block context. |
| |
| 5675 | w, | executable/support line; behavior impact depends on surrounding block context. |
| 5676 | if hcm_temp > 90.0 { | conditional gate; changing condition shifts runtime path and behavior effects. |
| 5677 | Color32::RED | executable/support line; behavior impact depends on surrounding block context. |
| 5678 | } else { | executable/support line; behavior impact depends on surrounding block context. |
| 5679 | Color32::from_rgb(80, 180, 255) | executable/support line; behavior impact depends on surrounding block context. |
| 5680 | }, | executable/support line; behavior impact depends on surrounding block context. |
| 5681 | ); | executable/support line; behavior impact depends on surrounding block context. |
| 5682 | digital_readout(ui, "Mode", &hcm_mode, Color32::from_gray(200)); | executable/support line; behavior impact depends on surrounding block context. |
| 5683 | digital_readout(ui, "Power", &format!("{:.1} kW", hcm_pwr), Color32::YELLOW); | executable/support line; behavior impact depends on surrounding block context. |
| 5684 | ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 5685 | ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| |
| 5686 | RichText::new("LOADER:") | executable/support line; behavior impact depends on surrounding block context. |
| 5687 | .size(11.0) | executable/support line; behavior impact depends on surrounding block context. |
| 5688 | .color(Color32::from_gray(140)), | executable/support line; behavior impact depends on surrounding block context. |
| 5689 | ); | executable/support line; behavior impact depends on surrounding block context. |
| 5690 | // Loader lift slider | comment line; documentation intent, no runtime effect. |
| 5691 | ui.horizontal(\ui\ { | executable/support line; behavior impact depends on surrounding block context. |
| 5692 | ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| |
| 5693 | RichText::new("Lift") | executable/support line; behavior impact depends on surrounding block context. |
| 5694 | .size(11.0) | executable/support line; behavior impact depends on surrounding block context. |
| 5695 | .color(Color32::from_gray(150)), | executable/support line; behavior impact depends on surrounding block context. |
| 5696 | ); | executable/support line; behavior impact depends on surrounding block context. |
| 5697 | let fill_w = (loader_lift as f32 / 100.0 * 80.0).max(2.0); | executable/support line

```

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ehavior impact depends on surrounding block context. |
| 5698 |           let (bar, _) = | executable/support line; behavior impact depends on surrounding blo
context. |
| 5699 |           ui.allocate_exact_size(Vec2::new(80.0, 14.0), Sense::hover()); | executable/supp
line; behavior impact depends on surrounding block context. |
| 5700 |           let p = ui.painter_at(bar); | executable/support line; behavior impact depends on su
unding block context. |
| 5701 |           p.rect_filled(bar, 3.0, Color32::from_gray(22)); | executable/support line; behavior
pact depends on surrounding block context. |
| 5702 |           p.rect_filled( | executable/support line; behavior impact depends on surrounding blo
context. |
| 5703 |           Rect::from_min_size(bar.min, Vec2::new(fill_w, bar.height())), | executable/supp
line; behavior impact depends on surrounding block context. |
| 5704 |           3.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 5705 |           Color32::from_rgb(80, 180, 255), | executable/support line; behavior impact depe
on surrounding block context. |
| 5706 |           ); | executable/support line; behavior impact depends on surrounding block context.
| 5707 |           ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5708 |           RichText::new(format!("{:.0}%", loader_lift)) | executable/support line; behavio
mpact depends on surrounding block context. |
| 5709 |           .size(10.5) | executable/support line; behavior impact depends on surroundin
lock context. |
| 5710 |           .color(Color32::LIGHT_BLUE), | executable/support line; behavior impact depe
on surrounding block context. |
| 5711 |           ); | executable/support line; behavior impact depends on surrounding block context.
| 5712 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 5713 |           let mut ll = self.bench.loader_lift_cmd as f32; | executable/support line; behavior impa
depends on surrounding block context. |
| 5714 |           ui.add_sized( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5715 |           [ui.available_width() - 10.0, 20.0], | executable/support line; behavior impact depe
on surrounding block context. |
| 5716 |           Slider::new(&mut ll, -1.0..=1.0).text("Lift cmd"), | executable/support line; behavi
impact depends on surrounding block context. |
| 5717 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 5718 |           if ll as f64 != self.bench.loader_lift_cmd { | conditional gate; changing condition shif
runtime path and side effects. |
| 5719 |           self.cmds.push(Cmd::SetLoaderLift(ll as f64)); | executable/support line; behavior i
ct depends on surrounding block context. |
| 5720 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 5721 |           let mut lt = self.bench.loader_tilt_cmd as f32; | executable/support line; behavior impa
depends on surrounding block context. |
| 5722 |           ui.add_sized( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5723 |           [ui.available_width() - 10.0, 20.0], | executable/support line; behavior impact depe
on surrounding block context. |
| 5724 |           Slider::new(&mut lt, -1.0..=1.0).text("Tilt cmd"), | executable/support line; behavi
impact depends on surrounding block context. |
| 5725 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 5726 |           if lt as f64 != self.bench.loader_tilt_cmd { | conditional gate; changing condition shif
runtime path and side effects. |
| 5727 |           self.cmds.push(Cmd::SetLoaderTilt(lt as f64)); | executable/support line; behavior i
ct depends on surrounding block context. |
| 5728 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 5729 |           ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 5730 |           ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 5731 |           RichText::new("AUX HYDRAULIC VALVES:") | executable/support line; behavior impact de
ds on surrounding block context. |
| 5732 |           .size(11.0) | executable/support line; behavior impact depends on surrounding bl

```



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context. |
| 5733 |         .color(Color32::from_gray(140)), | executable/support line; behavior impact depends
on surrounding block context. |
| 5734 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 5735 |         for i in 0..4 { | iteration control; changing bounds/body affects throughput, timing, an
possibly safety guarantees. |
| 5736 |             let cmd_val = self.aux_cmds[i]; | executable/support line; behavior impact depends on
urrounding block context. |
| 5737 |             ui.push_id(format!("impl::aux::{}", i), \|ui\| { | executable/support line; behavior
pact depends on surrounding block context. |
| 5738 |                 ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surr
nding block context. |
| 5739 |                     ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 5740 |                         RichText::new(format!("Bank{}:", i)) | executable/support line; behavior
pact depends on surrounding block context. |
| 5741 |                             .size(11.0) | executable/support line; behavior impact depends on su
unding block context. |
| 5742 |                             .color(Color32::from_gray(155)), | executable/support line; behavior
pact depends on surrounding block context. |
| 5743 |                             ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 5744 |                             // - button | comment line; documentation intent, no runtime effect. |
| 5745 |                             if ui.small_button("■").clicked() { | conditional gate; changing condition s
ts runtime path and side effects. |
| 5746 |                                 self.cmds.push(Cmd::SetAuxValve(i, -1.0)); | executable/support line; be
ior impact depends on surrounding block context. |
| 5747 |                             } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 5748 |                             if ui.small_button("■").clicked() { | conditional gate; changing condition s
ts runtime path and side effects. |
| 5749 |                                 self.cmds.push(Cmd::SetAuxValve(i, 0.0)); | executable/support line; beh
or impact depends on surrounding block context. |
| 5750 |                             } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 5751 |                             if ui.small_button("■").clicked() { | conditional gate; changing condition s
ts runtime path and side effects. |
| 5752 |                                 self.cmds.push(Cmd::SetAuxValve(i, 1.0)); | executable/support line; beh
or impact depends on surrounding block context. |
| 5753 |                             } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 5754 |                             let c = if cmd_val.abs() > 0.05 { | executable/support line; behavior impact
pends on surrounding block context. |
| 5755 |                                 Color32::GREEN | executable/support line; behavior impact depends on surr
nding block context. |
| 5756 |                             } else { | executable/support line; behavior impact depends on surrounding b
k context. |
| 5757 |                                 Color32::from_gray(80) | executable/support line; behavior impact depend
n surrounding block context. |
| 5758 |                             }; | scope delimiter; structure-only line, but misplaced edits can change ow
ship and lifetime scopes. |
| 5759 |                             ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 5760 |                                 RichText::new(format!( | executable/support line; behavior impact depend
n surrounding block context. |
| 5761 |                                     "{} {:.0}L/m {:.0}bar", | executable/support line; behavior impact d
nds on surrounding block context. |
| 5762 |                                     aux_dirs[i], aux_flows[i], aux_pres[i] | executable/support line; be
ior impact depends on surrounding block context. |
| 5763 |                                     )) | executable/support line; behavior impact depends on surrounding blo
context. |
| 5764 |                                     .size(10.5) | executable/support line; behavior impact depends on surrou
ng block context. |
| 5765 |                                     .monospace() | executable/support line; behavior impact depends on surro
ing block context. |

```

[illegible]

```

| 5803 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 5804 |         digital_readout( | executable/support line; behavior impact depends on surrounding block
ntext. |
| 5805 |             ui, | executable/support line; behavior impact depends on surrounding block context. |
| 5806 |             "ISOBUS", | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5807 |             if imp.isobus_connected { | conditional gate; changing condition shifts runtime path
d side effects. |
| 5808 |                 "CONNECTED" | executable/support line; behavior impact depends on surrounding bl
context. |
| 5809 |             } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 5810 |                 "disconnected" | executable/support line; behavior impact depends on surrounding
ock context. |
| 5811 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 5812 |             if imp.isobus_connected { | conditional gate; changing condition shifts runtime path
d side effects. |
| 5813 |                 Color32::GREEN | executable/support line; behavior impact depends on surrounding
ock context. |
| 5814 |             } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 5815 |                 Color32::from_gray(100) | executable/support line; behavior impact depends on su
unding block context. |
| 5816 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 5817 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 5818 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 5819 |         ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 5820 |             RichText::new("FRONT PTO:") | executable/support line; behavior impact depends on su
unding block context. |
| 5821 |             .size(11.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 5822 |             .color(Color32::from_gray(140)), | executable/support line; behavior impact depe
on surrounding block context. |
| 5823 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 5824 |         digital_readout( | executable/support line; behavior impact depends on surrounding block
ntext. |
| 5825 |             ui, | executable/support line; behavior impact depends on surrounding block context. |
| 5826 |             "Front RPM", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 5827 |             &format!("{:.0}", imp.pto_front_rpm), | executable/support line; behavior impact dep
s on surrounding block context. |
| 5828 |             if imp.pto_front_enabled { | conditional gate; changing condition shifts runtime pat
nd side effects. |
| 5829 |                 Color32::GREEN | executable/support line; behavior impact depends on surrounding
ock context. |
| 5830 |             } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 5831 |                 Color32::from_gray(100) | executable/support line; behavior impact depends on su
unding block context. |
| 5832 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 5833 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 5834 |         let fen = imp.pto_front_enabled; | executable/support line; behavior impact depends on s
ounding block context. |
| 5835 |         if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 5836 |             .add( | executable/support line; behavior impact depends on surrounding block contex
|
| 5837 |             Button::new( | executable/support line; behavior impact depends on surrounding b
k context. |
| 5838 |                 RichText::new(if fen { "Front PTO ON" } else { "Front PTO OFF" }) | executab
support line; behavior impact depends on surrounding block context. |
| 5839 |                 .size(11.0), | executable/support line; behavior impact depends on surro
ing block context. |
| 5840 |             ) | executable/support line; behavior impact depends on surrounding block contex

```

```

|
| 5841 |             .fill(if fen { | executable/support line; behavior impact depends on surrounding
ock context. |
| 5842 |             Color32::from_rgb(10, 80, 10) | executable/support line; behavior impact dep
s on surrounding block context. |
| 5843 |             } else { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 5844 |             Color32::from_gray(35) | executable/support line; behavior impact depends on
rrounding block context. |
| 5845 |             }}, | executable/support line; behavior impact depends on surrounding block cont
. |
| 5846 |             ) | executable/support line; behavior impact depends on surrounding block context. |
| 5847 |             .clicked() | executable/support line; behavior impact depends on surrounding block c
ext. |
| 5848 |             { | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 5849 |             self.bench.implement.pto_front_enabled = !fen; | executable/support line; behavior i
ct depends on surrounding block context. |
| 5850 |             self.implement_feedback = if fen { | executable/support line; behavior impact depend
n surrounding block context. |
| 5851 |             "Front PTO -> OFF".into() | executable/support line; behavior impact depends on
rounding block context. |
| 5852 |             } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 5853 |             "Front PTO -> ON".into() | executable/support line; behavior impact depends on s
ounding block context. |
| 5854 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |
| 5855 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 5856 |             ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 5857 |             // Hitch control mode | comment line; documentation intent, no runtime effect. |
| 5858 |             ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 5859 |             RichText::new("HITCH CONTROL MODE:") | executable/support line; behavior impact depende
on surrounding block context. |
| 5860 |             .size(11.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 5861 |             .color(Color32::from_gray(140)), | executable/support line; behavior impact depende
on surrounding block context. |
| 5862 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 5863 |             for (mode_s, mode) in [ | iteration control; changing bounds/body affects throughput, ti
g, and possibly safety guarantees. |
| 5864 |             ("Position", &auto_breaking::implement::HitchMode::Position), | executable/support l
; behavior impact depends on surrounding block context. |
| 5865 |             ("Draft", &auto_breaking::implement::HitchMode::Draft), | executable/support line; b
vior impact depends on surrounding block context. |
| 5866 |             ("Mixed", &auto_breaking::implement::HitchMode::Mixed), | executable/support line; b
vior impact depends on surrounding block context. |
| 5867 |             ("Float", &auto_breaking::implement::HitchMode::Float), | executable/support line; b
vior impact depends on surrounding block context. |
| 5868 |             ] { | executable/support line; behavior impact depends on surrounding block context. |
| 5869 |             let cur = hitch_mode.trim() == format!("{}", mode).trim(); | executable/support line
ehavior impact depends on surrounding block context. |
| 5870 |             let fill = if cur { | executable/support line; behavior impact depends on surroundin
lock context. |
| 5871 |             Color32::from_rgb(25, 75, 25) | executable/support line; behavior impact depends
surrounding block context. |
| 5872 |             } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 5873 |             Color32::from_gray(35) | executable/support line; behavior impact depends on sur
nding block context. |
| 5874 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |

```

```

| 5875 |         if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 5876 |             .add( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5877 |                 Button::new(RichText::new(mode_s).size(11.0)) | executable/support line; beh
or impact depends on surrounding block context. |
| 5878 |                 .fill(fill) | executable/support line; behavior impact depends on surrou
ng block context. |
| 5879 |                 .min_size(Vec2::new(90.0, 22.0)), | executable/support line; behavior im
t depends on surrounding block context. |
| 5880 |             ) | executable/support line; behavior impact depends on surrounding block contex
|
| 5881 |             .clicked() | executable/support line; behavior impact depends on surrounding blo
context. |
| 5882 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 5883 |             self.bench.implement.hitch_control_mode = *mode; | executable/support line; beha
r impact depends on surrounding block context. |
| 5884 |             self.implement_feedback = | executable/support line; behavior impact depends on
rounding block context. |
| 5885 |             format!("Hitch control mode -> {}", mode_s); | executable/support line; beha
r impact depends on surrounding block context. |
| 5886 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 5887 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 5888 |     ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 5889 |     let hitch_err = (self.hitch_target - hitch_pos).abs(); | executable/support line; behavi
impact depends on surrounding block context. |
| 5890 |     let hitch_col = if hitch_err > 12.0 { | executable/support line; behavior impact depends
surrounding block context. |
| 5891 |         Color32::YELLOW | executable/support line; behavior impact depends on surrounding bl
context. |
| 5892 |     } else { | executable/support line; behavior impact depends on surrounding block context
|
| 5893 |         Color32::GREEN | executable/support line; behavior impact depends on surrounding blo
context. |
| 5894 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 5895 |     ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 5896 |         RichText::new(format!( | executable/support line; behavior impact depends on surroun
g block context. |
| 5897 |             "Control tracking: Hitch target {:.1}% / actual {:.1}% (err {:.1}%)", | executab
support line; behavior impact depends on surrounding block context. |
| 5898 |             self.hitch_target, hitch_pos, hitch_err | executable/support line; behavior impa
depends on surrounding block context. |
| 5899 |         )) | executable/support line; behavior impact depends on surrounding block context.
|
| 5900 |         .size(10.2) | executable/support line; behavior impact depends on surrounding block
text. |
| 5901 |         .color(hitch_col), | executable/support line; behavior impact depends on surrounding
ock context. |
| 5902 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 5903 |     if !self.implement_feedback.is_empty() { | conditional gate; changing condition shifts r
ime path and side effects. |
| 5904 |         ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5905 |             RichText::new(format!("Last command: {}", self.implement_feedback)) | executable
pport line; behavior impact depends on surrounding block context. |
| 5906 |             .size(10.2) | executable/support line; behavior impact depends on surrounding
lock context. |
| 5907 |             .color(Color32::LIGHT_BLUE), | executable/support line; behavior impact depe
on surrounding block context. |
| 5908 |         ); | executable/support line; behavior impact depends on surrounding block context.
|
| 5909 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |

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```

| 5950 |                                     ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 5951 |                                     RichText::new("Fuel Level %") | executable/support line; behavior impact
depends on surrounding block context. |
| 5952 |                                     .size(11.0) | executable/support line; behavior impact depends on su
unding block context. |
| 5953 |                                     .color(Color32::YELLOW), | executable/support line; behavior impact
ends on surrounding block context. |
| 5954 |                                     ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 5955 |                                     }); | executable/support line; behavior impact depends on surrounding block cont
. |
| 5956 |                                     if ui | conditional gate; changing condition shifts runtime path and side effect
|
| 5957 |                                     .add_sized( | executable/support line; behavior impact depends on surrounding
lock context. |
| 5958 |                                     [ui.available_width() - 10.0, 22.0], | executable/support line; behavior
pact depends on surrounding block context. |
| 5959 |                                     Slider::new(&mut v, 0.0..=100.0).suffix(" %"), | executable/support line
ehavior impact depends on surrounding block context. |
| 5960 |                                     ) | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5961 |                                     .changed() | executable/support line; behavior impact depends on surrounding
ock context. |
| 5962 |                                     { | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 5963 |                                     self.p_fuel = v as f64; | executable/support line; behavior impact depends o
urrounding block context. |
| 5964 |                                     changed = true; | executable/support line; behavior impact depends on surrou
ng block context. |
| 5965 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 5966 |                                     if changed { | conditional gate; changing condition shifts runtime path and side
fects. |
| 5967 |                                     self.cmds.push(Cmd::SetFuelLevel(self.p_fuel)); | executable/support line; b
vior impact depends on surrounding block context. |
| 5968 |                                     self.params_dirty = true; | executable/support line; behavior impact depends
surrounding block context. |
| 5969 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 5970 | (blank) | line is blank; visual separation only. |
| 5971 |                                     let (mut v, mut changed) = (self.p_def as f32, false); | executable/support line
ehavior impact depends on surrounding block context. |
| 5972 |                                     ui.horizontal(\ui\ { | executable/support line; behavior impact depends on sur
nding block context. |
| 5973 |                                     ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 5974 |                                     RichText::new("DEF / AdBlue %") | executable/support line; behavior impa
depends on surrounding block context. |
| 5975 |                                     .size(11.0) | executable/support line; behavior impact depends on su
unding block context. |
| 5976 |                                     .color(Color32::from_rgb(0, 210, 210)), | executable/support line; b
vior impact depends on surrounding block context. |
| 5977 |                                     ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 5978 |                                     }); | executable/support line; behavior impact depends on surrounding block cont
. |
| 5979 |                                     if ui | conditional gate; changing condition shifts runtime path and side effect
|
| 5980 |                                     .add_sized( | executable/support line; behavior impact depends on surrounding
lock context. |
| 5981 |                                     [ui.available_width() - 10.0, 22.0], | executable/support line; behavior
pact depends on surrounding block context. |
| 5982 |                                     Slider::new(&mut v, 0.0..=100.0).suffix(" %"), | executable/support line
ehavior impact depends on surrounding block context. |

```

```

| 5983 |                                     ) | executable/support line; behavior impact depends on surrounding block co
xt. |
| 5984 |                                     .changed() | executable/support line; behavior impact depends on surrounding
ock context. |
| 5985 |                                     { | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 5986 |                                     self.p_def = v as f64; | executable/support line; behavior impact depends on
rrounding block context. |
| 5987 |                                     changed = true; | executable/support line; behavior impact depends on surrou
ng block context. |
| 5988 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 5989 |                                     if changed { | conditional gate; changing condition shifts runtime path and side
fects. |
| 5990 |                                     self.cmds.push(Cmd::SetDefLevel(self.p_def)); | executable/support line; beh
or impact depends on surrounding block context. |
| 5991 |                                     self.params_dirty = true; | executable/support line; behavior impact depends
surrounding block context. |
| 5992 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 5993 | (blank) | line is blank; visual separation only. |
| 5994 |                                     let (mut v, mut changed) = (self.p_engine_h as f32, false); | executable/support
ne; behavior impact depends on surrounding block context. |
| 5995 |                                     ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on sur
nding block context. |
| 5996 |                                     ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 5997 |                                     RichText::new("Engine Hours") | executable/support line; behavior impact
ends on surrounding block context. |
| 5998 |                                     .size(11.0) | executable/support line; behavior impact depends on su
unding block context. |
| 5999 |                                     .color(Color32::from_gray(200)), | executable/support line; behavior
pact depends on surrounding block context. |
| 6000 |                                     ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 6001 |                                     }); | executable/support line; behavior impact depends on surrounding block cont
. |
| 6002 |                                     if ui | conditional gate; changing condition shifts runtime path and side effect
|
| 6003 |                                     .add_sized( | executable/support line; behavior impact depends on surroundin
lock context. |
| 6004 |                                     [ui.available_width() - 10.0, 22.0], | executable/support line; behavior
pact depends on surrounding block context. |
| 6005 |                                     Slider::new(&mut v, 0.0..=100000.0).suffix(" h"), | executable/support l
; behavior impact depends on surrounding block context. |
| 6006 |                                     ) | executable/support line; behavior impact depends on surrounding block co
xt. |
| 6007 |                                     .changed() | executable/support line; behavior impact depends on surrounding
ock context. |
| 6008 |                                     { | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 6009 |                                     self.p_engine_h = v as f64; | executable/support line; behavior impact depen
on surrounding block context. |
| 6010 |                                     changed = true; | executable/support line; behavior impact depends on surrou
ng block context. |
| 6011 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 6012 |                                     if changed { | conditional gate; changing condition shifts runtime path and side
fects. |
| 6013 |                                     self.cmds.push(Cmd::SetEngineHours(self.p_engine_h)); | executable/support l
; behavior impact depends on surrounding block context. |
| 6014 |                                     self.params_dirty = true; | executable/support line; behavior impact depends
surrounding block context. |
| 6015 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |

```


[illegible]

```

xt. |
| 6051 | .changed() | executable/support line; behavior impact depends on surrounding
ock context. |
| 6052 | { | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 6053 | self.p_coolant = v as f64; | executable/support line; behavior impact depend
n surrounding block context. |
| 6054 | changed = true; | executable/support line; behavior impact depends on surrou
ng block context. |
| 6055 | } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 6056 | if changed { | conditional gate; changing condition shifts runtime path and side
fects. |
| 6057 | self.cmds.push(Cmd::SetCoolantTemp(self.p_coolant)); | executable/support li
behavior impact depends on surrounding block context. |
| 6058 | self.params_dirty = true; | executable/support line; behavior impact depends
surrounding block context. |
| 6059 | } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 6060 | (blank) | line is blank; visual separation only. |
| 6061 | let (mut v, mut changed) = (self.p_oil_prs as f32, false); | executable/support
e; behavior impact depends on surrounding block context. |
| 6062 | ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 6063 | RichText::new("Oil Pressure kPa") | executable/support line; behavior impact
ends on surrounding block context. |
| 6064 | .size(11.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 6065 | .color(Color32::GREEN), | executable/support line; behavior impact dependen
on surrounding block context. |
| 6066 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 6067 | if ui | conditional gate; changing condition shifts runtime path and side effect
|
| 6068 | .add_sized( | executable/support line; behavior impact depends on surroundin
lock context. |
| 6069 | [ui.available_width() - 10.0, 22.0], | executable/support line; behavior
pact depends on surrounding block context. |
| 6070 | Slider::new(&mut v, 0.0..=800.0).suffix(" kPa"), | executable/support li
behavior impact depends on surrounding block context. |
| 6071 | ) | executable/support line; behavior impact depends on surrounding block co
xt. |
| 6072 | .changed() | executable/support line; behavior impact depends on surrounding
ock context. |
| 6073 | { | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 6074 | self.p_oil_prs = v as f64; | executable/support line; behavior impact depend
n surrounding block context. |
| 6075 | changed = true; | executable/support line; behavior impact depends on surrou
ng block context. |
| 6076 | } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 6077 | if changed { | conditional gate; changing condition shifts runtime path and side
fects. |
| 6078 | self.cmds.push(Cmd::SetOilPressure(self.p_oil_prs)); | executable/support li
behavior impact depends on surrounding block context. |
| 6079 | self.params_dirty = true; | executable/support line; behavior impact depends
surrounding block context. |
| 6080 | } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 6081 | (blank) | line is blank; visual separation only. |
| 6082 | let (mut v, mut changed) = (self.p_boost as f32, false); | executable/support li
behavior impact depends on surrounding block context. |
| 6083 | ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |

```

[illegible]

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| 6118 |                 ui.label( | executable/support line; behavior impact depends on surrounding block
ontext. |
| 6119 |                 RichText::new("TRANSMISSION CONTROLS") | executable/support line; behavior i
ct depends on surrounding block context. |
| 6120 |                 .size(12.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 6121 |                 .color(Color32::from_rgb(80, 155, 255)), | executable/support line; beha
r impact depends on surrounding block context. |
| 6122 |                 ); | executable/support line; behavior impact depends on surrounding block conte
|
| 6123 |                 ui.separator(); | executable/support line; behavior impact depends on surrounding
lock context. |
| 6124 |                 ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on sur
nding block context. |
| 6125 |                     if ui.button("■ DN").clicked() { | conditional gate; changing condition shift
runtime path and side effects. |
| 6126 |                         self.cmds.push(Cmd::ManualDn); | executable/support line; behavior impac
depends on surrounding block context. |
| 6127 |                     } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 6128 |                     if ui.button("■ UP").clicked() { | conditional gate; changing condition shift
runtime path and side effects. |
| 6129 |                         self.cmds.push(Cmd::ManualUp); | executable/support line; behavior impac
depends on surrounding block context. |
| 6130 |                     } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 6131 |                     if ui | conditional gate; changing condition shifts runtime path and side ef
ts. |
| 6132 |                         .add(Button::new(RichText::new("AUTO").size(11.0)).fill( | executable/su
rt line; behavior impact depends on surrounding block context. |
| 6133 |                         if self.bench.tcm.auto_mode == AutoShiftMode::Auto { | conditional g
; changing condition shifts runtime path and side effects. |
| 6134 |                             Color32::from_rgb(20, 80, 20) | executable/support line; behavior
mpact depends on surrounding block context. |
| 6135 |                         } else { | executable/support line; behavior impact depends on surro
ing block context. |
| 6136 |                             Color32::from_gray(35) | executable/support line; behavior impac
depends on surrounding block context. |
| 6137 |                         }, | executable/support line; behavior impact depends on surrounding
ock context. |
| 6138 |                         )) | executable/support line; behavior impact depends on surrounding blo
context. |
| 6139 |                         .clicked() | executable/support line; behavior impact depends on surroun
g block context. |
| 6140 |                     { | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 6141 |                         self.cmds.push(Cmd::ToggleAutoShift); | executable/support line; behavio
mpact depends on surrounding block context. |
| 6142 |                     } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 6143 |                     if ui | conditional gate; changing condition shifts runtime path and side ef
ts. |
| 6144 |                         .add(Button::new(RichText::new("CREEP").size(11.0)).fill( | executable/s
ort line; behavior impact depends on surrounding block context. |
| 6145 |                         if self.bench.tcm.creeper_engaged { | conditional gate; changing con
ion shifts runtime path and side effects. |
| 6146 |                             Color32::from_rgb(80, 50, 10) | executable/support line; behavior
mpact depends on surrounding block context. |
| 6147 |                         } else { | executable/support line; behavior impact depends on surro
ing block context. |
| 6148 |                             Color32::from_gray(35) | executable/support line; behavior impac
depends on surrounding block context. |
| 6149 |                         }, | executable/support line; behavior impact depends on surrounding
ock context. |
| 6150 |                         )) | executable/support line; behavior impact depends on surrounding blo

```

```

context. |
| 6151 | .clicked() | executable/support line; behavior impact depends on surround
g block context. |
| 6152 | { | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 6153 | self.cmds.push(Cmd::ToggleCreeper); | executable/support line; behavior
act depends on surrounding block context. |
| 6154 | } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 6155 | }); | executable/support line; behavior impact depends on surrounding block cont
. |
| 6156 | ui.separator(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 6157 | ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 6158 | RichText::new("BCM CONTROLS") | executable/support line; behavior impact dep
s on surrounding block context. |
| 6159 | .size(12.0) | executable/support line; behavior impact depends on surround
ng block context. |
| 6160 | .color(Color32::from_rgb(80, 155, 255)), | executable/support line; beha
r impact depends on surrounding block context. |
| 6161 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 6162 | ui.horizontal_wrapped(\|ui\| { | executable/support line; behavior impact depend
n surrounding block context. |
| 6163 | if ui.button("Work Lights").clicked() { | conditional gate; changing conditi
shifts runtime path and side effects. |
| 6164 | self.cmds.push(Cmd::ToggleWorkLights); | executable/support line; behavi
impact depends on surrounding block context. |
| 6165 | } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 6166 | if ui.button("Road Lights").clicked() { | conditional gate; changing conditi
shifts runtime path and side effects. |
| 6167 | self.cmds.push(Cmd::ToggleRoadLights); | executable/support line; behavi
impact depends on surrounding block context. |
| 6168 | } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 6169 | if ui.button("Beacon").clicked() { | conditional gate; changing condition sh
s runtime path and side effects. |
| 6170 | self.cmds.push(Cmd::ToggleBeacon); | executable/support line; behavior i
ct depends on surrounding block context. |
| 6171 | } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 6172 | if ui.button("Honk ■").clicked() { | conditional gate; changing condition sh
s runtime path and side effects. |
| 6173 | self.cmds.push(Cmd::Honk); | executable/support line; behavior impact de
ds on surrounding block context. |
| 6174 | } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 6175 | if ui.button("Wiper").clicked() { | conditional gate; changing condition shi
runtime path and side effects. |
| 6176 | self.cmds.push(Cmd::CycleWiper); | executable/support line; behavior imp
depends on surrounding block context. |
| 6177 | } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 6178 | }); | executable/support line; behavior impact depends on surrounding block cont
. |
| 6179 | ui.separator(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 6180 | ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 6181 | RichText::new("BCM STATUS:") | executable/support line; behavior impact depe
on surrounding block context. |
| 6182 | .size(11.0) | executable/support line; behavior impact depends on surround
ng block context. |

```

```

| 6183 |                                     .color(Color32::from_gray(140)), | executable/support line; behavior imp
depends on surrounding block context. |
| 6184 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 6185 |                                     let b = &self.bench.bcm; | executable/support line; behavior impact depends on s
ounding block context. |
| 6186 |                                     bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 6187 |                                     ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 6188 |                                     "Battery", | executable/support line; behavior impact depends on surrounding
ock context. |
| 6189 |                                     b.battery_voltage - 9.0, | executable/support line; behavior impact depends
surrounding block context. |
| 6190 |                                     7.0, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 6191 |                                     "V", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 6192 |                                     110.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 6193 |                                     if b.battery_voltage < 11.5 { | conditional gate; changing condition shifts
time path and side effects. |
| 6194 |                                     Color32::RED | executable/support line; behavior impact depends on surro
ing block context. |
| 6195 |                                     } else { | executable/support line; behavior impact depends on surrounding b
k context. |
| 6196 |                                     Color32::GREEN | executable/support line; behavior impact depends on sur
nding block context. |
| 6197 |                                     }, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 6198 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 6199 |                                     digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 6200 |                                     ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 6201 |                                     "SoC", | executable/support line; behavior impact depends on surrounding blo
context. |
| 6202 |                                     &format!("{:.0}%", b.battery_soc_pct), | executable/support line; behavior i
ct depends on surrounding block context. |
| 6203 |                                     Color32::from_gray(200), | executable/support line; behavior impact depends
surrounding block context. |
| 6204 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 6205 |                                     digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 6206 |                                     ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 6207 |                                     "Load", | executable/support line; behavior impact depends on surrounding bl
context. |
| 6208 |                                     &format!("{:.0} A", b.total_load_amps), | executable/support line; behavior
act depends on surrounding block context. |
| 6209 |                                     Color32::from_gray(180), | executable/support line; behavior impact depends
surrounding block context. |
| 6210 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 6211 |                                     digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 6212 |                                     ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 6213 |                                     "Wiper", | executable/support line; behavior impact depends on surrounding b
k context. |
| 6214 |                                     &format!("{:?}", b.wiper_speed), | error-propagating operation; changing it
acts failure semantics and caller recovery paths. |
| 6215 |                                     Color32::from_gray(180), | executable/support line; behavior impact depends

```

```
| surrounding block context. |  
| 6216 |                                     ); | executable/support line; behavior impact depends on surrounding block conte  
|  
| 6217 |                                     for fuse in &b.fuses { | iteration control; changing bounds/body affects through  
, timing, and possibly safety guarantees. |  
| 6218 |                                     if fuse.blown { | conditional gate; changing condition shifts runtime path a  
side effects. |  
| 6219 |                                     ui.label( | executable/support line; behavior impact depends on surround  
block context. |  
| 6220 |                                     RichText::new(format!("■ FUSE BLOWN: {}", fuse.id)) | executable/sup  
t line; behavior impact depends on surrounding block context. |  
| 6221 |                                     .size(10.5) | executable/support line; behavior impact depends o  
urrounding block context. |  
| 6222 |                                     .color(Color32::RED), | executable/support line; behavior impact  
depends on surrounding block context. |  
| 6223 |                                     ); | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 6224 |                                     } | scope delimiter; structure-only line, but misplaced edits can change own  
hip and lifetime scopes. |  
| 6225 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownersh  
and lifetime scopes. |  
| 6226 |                                     }); | executable/support line; behavior impact depends on surrounding block context.  
| 6227 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |  
| 6228 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |  
| 6229 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 6230 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 6231 | (blank) | line is blank; visual separation only. |  
| 6232 | // ████████████████████████████████████████████████████████████████████████████████████████████████████████████████ | comm  
intent, no runtime effect. |  
| 6233 | // TAB LEAK LAB | comment line; documentation intent, no runtime effect. |  
| 6234 | // ████████████████████████████████████████████████████████████████████████████████████████████████████████████████ | comm  
intent, no runtime effect. |  
| 6235 | impl App { | implementation block; method behavior and trait conformance are defined here. |  
| 6236 |     fn uds_process_sa(&mut self, sa: u8, req: &[u8], ts: f64) -> Vec<u8> { | declares executable behavior; s  
signature/body changes alter API, control flow, and outputs. |  
| 6237 |         if sa == 0x00 { | conditional gate; changing condition shifts runtime path and side effects. |  
| 6238 |             self.bench.uds_ecm.process(req, ts) | I/O or protocol-critical logic; edits can affect integrati  
timing, and diagnostic outcomes. |  
| 6239 |         } else if sa == 0x03 { | executable/support line; behavior impact depends on surrounding block conte  
|  
| 6240 |             self.bench.uds_tcm.process(req, ts) | I/O or protocol-critical logic; edits can affect integrati  
timing, and diagnostic outcomes. |  
| 6241 |         } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 6242 |             vec![0x7F, req.first().copied().unwrap_or(0x00), 0x11] | executable/support line; behavior impac  
depends on surrounding block context. |  
| 6243 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 6244 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 6245 | (blank) | line is blank; visual separation only. |  
| 6246 |     fn uds_send_and_log(&mut self, sa: u8, req: Vec<u8>, ts: f64) -> Vec<u8> { | declares executable behavio  
signature/body changes alter API, control flow, and outputs. |  
| 6247 |         let req_hex = req | executable/support line; behavior impact depends on surrounding block context. |  
| 6248 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |  
| 6249 |         .map(|b| format!("{:02X}", b)) | executable/support line; behavior impact depends on surroundi  
block context. |  
| 6250 |         .collect::<Vec<_>>() | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 6251 |         .join(" "); | executable/support line; behavior impact depends on surrounding block context. |  
| 6252 |         let svc = auto_breaking::uds::UdsServer::service_name(req[0]); | I/O or protocol-critical logic; edi  
can affect integration, timing, and diagnostic outcomes. |  
| 6253 |         let resp = self.uds_process_sa(sa, &req, ts); | I/O or protocol-critical logic; edits can affect int  
ation, timing, and diagnostic outcomes. |  
| 6254 |         let resp_hex = resp | executable/support line; behavior impact depends on surrounding block context.  
| 6255 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |  
| 6256 |         .map(|b| format!("{:02X}", b)) | executable/support line; behavior impact depends on surroundi
```

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block context. |
| 6257 |         .collect::<Vec<_>>() | executable/support line; behavior impact depends on surrounding block context. | | |
| 6258 |         .join(" "); | executable/support line; behavior impact depends on surrounding block context. |
| 6259 |         let nrc = resp.first().copied() == Some(0x7F); | executable/support line; behavior impact depends on surrounding block context. |
| 6260 |         self.uds_log.push(( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 6261 |             true, | executable/support line; behavior impact depends on surrounding block context. |
| 6262 |             format!("{:.2}s] 0x{:02X} {} -> {}", ts, sa, svc, req_hex), | executable/support line; behavior impact depends on surrounding block context. |
| 6263 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 6264 |         self.uds_log.push(( | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 6265 |             false, | executable/support line; behavior impact depends on surrounding block context. |
| 6266 |             format!( | executable/support line; behavior impact depends on surrounding block context. |
| 6267 |                 "                RESP {}{}", | executable/support line; behavior impact depends on surrounding block context. |
| 6268 |                 resp_hex, | executable/support line; behavior impact depends on surrounding block context. |
| 6269 |                 if nrc { " FAIL" } else { " OK" } | conditional gate; changing condition shifts runtime path and side effects. |
| 6270 |             ), | executable/support line; behavior impact depends on surrounding block context. |
| 6271 |         )); | executable/support line; behavior impact depends on surrounding block context. |
| 6272 |         if self.uds_log.len() > 240 { | conditional gate; changing condition shifts runtime path and side effects. |
| 6273 |             self.uds_log.drain(0..60); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 6274 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 6275 |         resp | executable/support line; behavior impact depends on surrounding block context. |
| 6276 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 6277 | (blank) | line is blank; visual separation only. |
| 6278 |     fn run_uds_flash_pipeline(&mut self, sa: u8, ts: f64) -> Result<String, String> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 6279 |         if self.uds_flash_path.trim().is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 6280 |             return Err("selecione um arquivo .bin para flash".into()); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 6281 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 6282 |         let fw = std::fs::read(&self.uds_flash_path) | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 6283 |         .map_err(|e| format!("falha ao ler firmware: {}", e))?; | error-propagating operation; changing condition impacts failure semantics and caller recovery paths. |
| 6284 |         if fw.is_empty() { | conditional gate; changing condition shifts runtime path and side effects. |
| 6285 |             return Err("arquivo de firmware vazio".into()); | executable/support line; behavior impact depends on surrounding block context. |
| 6286 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 6287 | (blank) | line is blank; visual separation only. |
| 6288 |         // Session + security unlock for programming level. | comment line; documentation intent, no runtime effect. |
| 6289 |         let _ = self.uds_send_and_log(sa, vec![0x10, 0x02], ts); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 6290 |         let seed_resp = self.uds_send_and_log(sa, vec![0x27, 0x05], ts); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 6291 |         if seed_resp.len() < 6 || seed_resp[0] != 0x67 { | conditional gate; changing condition shifts runtime path and side effects. |
| 6292 |             return Err("seed de segurança não recebido".into()); | executable/support line; behavior impact depends on surrounding block context. |
| 6293 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 6294 |         let seed = ((seed_resp[2] as u32) << 24) | executable/support line; behavior impact depends on surrounding block context. |
| 6295 |             || ((seed_resp[3] as u32) << 16) | executable/support line; behavior impact depends on surrounding block context. |

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block context. |
| 6296 |          \ | ((seed_resp[4] as u32) << 8) | executable/support line; behavior impact depends on surrounding
lock context. |
| 6297 |          \ | seed_resp[5] as u32; | executable/support line; behavior impact depends on surrounding block
text. |
| 6298 |          let key = seed ^ 0xDEADBEEF; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 6299 |          let unlock_resp = self.uds_send_and_log( | I/O or protocol-critical logic; edits can affect integrat
, timing, and diagnostic outcomes. |
| 6300 |          sa, | executable/support line; behavior impact depends on surrounding block context. |
| 6301 |          vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 6302 |              0x27, | executable/support line; behavior impact depends on surrounding block context. |
| 6303 |              0x06, | executable/support line; behavior impact depends on surrounding block context. |
| 6304 |              ((key >> 24) & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding
lock context. |
| 6305 |              ((key >> 16) & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding
lock context. |
| 6306 |              ((key >> 8) & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding
ock context. |
| 6307 |              (key & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding block
text. |
| 6308 |          ], | executable/support line; behavior impact depends on surrounding block context. |
| 6309 |          ts, | executable/support line; behavior impact depends on surrounding block context. |
| 6310 |          ); | executable/support line; behavior impact depends on surrounding block context. |
| 6311 |          if unlock_resp.first().copied() != Some(0x67) { | conditional gate; changing condition shifts runtime
ath and side effects. |
| 6312 |              return Err("unlock de segurança falhou".into()); | executable/support line; behavior impact depende
on surrounding block context. |
| 6313 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 6314 | (blank) | line is blank; visual separation only. |
| 6315 |          let _ = self.uds_send_and_log(sa, vec![0x10, 0x03], ts); | I/O or protocol-critical logic; edits can
fect integration, timing, and diagnostic outcomes. |
| 6316 | (blank) | line is blank; visual separation only. |
| 6317 |          let total_len = fw.len() as u32; | executable/support line; behavior impact depends on surrounding b
k context. |
| 6318 |          let req_dl = vec![ | executable/support line; behavior impact depends on surrounding block context.
| 6319 |              0x34, | executable/support line; behavior impact depends on surrounding block context. |
| 6320 |              0x00, | executable/support line; behavior impact depends on surrounding block context. |
| 6321 |              0x00, | executable/support line; behavior impact depends on surrounding block context. |
| 6322 |              0x00, | executable/support line; behavior impact depends on surrounding block context. |
| 6323 |              0x00, | executable/support line; behavior impact depends on surrounding block context. |
| 6324 |              0x00, | executable/support line; behavior impact depends on surrounding block context. |
| 6325 |              ((total_len >> 24) & 0xFF) as u8, | executable/support line; behavior impact depends on surround
block context. |
| 6326 |              ((total_len >> 16) & 0xFF) as u8, | executable/support line; behavior impact depends on surround
block context. |
| 6327 |              ((total_len >> 8) & 0xFF) as u8, | executable/support line; behavior impact depends on surroundi
block context. |
| 6328 |              (total_len & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 6329 |          ]; | executable/support line; behavior impact depends on surrounding block context. |
| 6330 |          let dl_resp = self.uds_send_and_log(sa, req_dl, ts); | I/O or protocol-critical logic; edits can aff
integration, timing, and diagnostic outcomes. |
| 6331 |          if dl_resp.first().copied() != Some(0x74) { | conditional gate; changing condition shifts runtime pa
and side effects. |
| 6332 |              return Err("request download rejeitado".into()); | executable/support line; behavior impact depende
on surrounding block context. |
| 6333 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 6334 | (blank) | line is blank; visual separation only. |
| 6335 |          let mut block: u8 = 1; | executable/support line; behavior impact depends on surrounding block conte
|
| 6336 |          for chunk in fw.chunks(120) { | iteration control; changing bounds/body affects throughput, timing,
possibly safety guarantees. |

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```

| 6337 |         let mut req = Vec::with_capacity(chunk.len() + 2); | executable/support line; behavior impact de
ds on surrounding block context. |
| 6338 |         req.push(0x36); | executable/support line; behavior impact depends on surrounding block context.
| 6339 |         req.push(block); | executable/support line; behavior impact depends on surrounding block context
| 6340 |         req.extend_from_slice(chunk); | executable/support line; behavior impact depends on surrounding
ck context. |
| 6341 |         let tr = self.uds_send_and_log(sa, req, ts); | I/O or protocol-critical logic; edits can affect
egration, timing, and diagnostic outcomes. |
| 6342 |         if tr.first().copied() != Some(0x76) { | conditional gate; changing condition shifts runtime pat
nd side effects. |
| 6343 |             return Err(format!("transfer data falhou no bloco {} ", block)); | executable/support line; b
vior impact depends on surrounding block context. |
| 6344 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 6345 |         block = block.wrapping_add(1); | executable/support line; behavior impact depends on surrounding
ock context. |
| 6346 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 6347 | (blank) | line is blank; visual separation only. |
| 6348 |     let end = self.uds_send_and_log(sa, vec![0x37, 0x00], ts); | I/O or protocol-critical logic; edits c
affect integration, timing, and diagnostic outcomes. |
| 6349 |     if end.first().copied() != Some(0x77) { | conditional gate; changing condition shifts runtime path a
side effects. |
| 6350 |         return Err("request transfer exit falhou".into()); | executable/support line; behavior impact de
ds on surrounding block context. |
| 6351 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 6352 | (blank) | line is blank; visual separation only. |
| 6353 |     Ok(format!( | executable/support line; behavior impact depends on surrounding block context. |
| 6354 |         "flash concluido (simulado): {} bytes enviados para SA 0x{:02X}", | I/O or protocol-critical log
edits can affect integration, timing, and diagnostic outcomes. |
| 6355 |         fw.len(), | executable/support line; behavior impact depends on surrounding block context. |
| 6356 |         sa | executable/support line; behavior impact depends on surrounding block context. |
| 6357 |     )) | executable/support line; behavior impact depends on surrounding block context. |
| 6358 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 6359 | (blank) | line is blank; visual separation only. |
| 6360 | fn leak_ascii_cad( | declares executable behavior; signature/body changes alter API, control flow, and o
uts. |
| 6361 |     circuit: &LeakCircuit, | executable/support line; behavior impact depends on surrounding block conte
|
| 6362 |     report: &auto_breaking::CircuitResult, | executable/support line; behavior impact depends on surroun
g block context. |
| 6363 | ) -> (String, Vec<String>) { | executable/support line; behavior impact depends on surrounding block con
t. |
| 6364 |     const SECTORS: usize = 16; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 6365 |     let mut local_pressure = [0.0_f64; SECTORS]; | executable/support line; behavior impact depends on s
ounding block context. |
| 6366 |     let mut local_frag = [0.0_f64; SECTORS]; | executable/support line; behavior impact depends on surro
ing block context. |
| 6367 |     let mut local_flow = [0.0_f64; SECTORS]; | executable/support line; behavior impact depends on surro
ing block context. |
| 6368 | (blank) | line is blank; visual separation only. |
| 6369 |     for i in 0..SECTORS { | iteration control; changing bounds/body affects throughput, timing, and poss
y safety guarantees. |
| 6370 |         let theta = i as f64 * std::f64::consts::TAU / SECTORS as f64; | executable/support line; behavi
impact depends on surrounding block context. |
| 6371 |         let harmonic = (2.0 * theta).cos() * 0.15 + (theta + 0.4).sin() * 0.07; | executable/support lin
behavior impact depends on surrounding block context. |
| 6372 |         let pressure = (report.current_pressure_bar * (1.0 + harmonic)).max(0.0); | executable/support l
; behavior impact depends on surrounding block context. |
| 6373 |         let pressure_ratio = pressure / circuit.pressure.rupture_bar.max(1e-6); | executable/support lin
behavior impact depends on surrounding block context. |
| 6374 |         let frag = (0.62 * pressure_ratio + 0.38 * (report.rupture_probability_pct / 100.0)) | executabl
upport line; behavior impact depends on surrounding block context. |

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| 6375 |           .clamp(0.0, 1.0); | executable/support line; behavior impact depends on surrounding block context. |
| 6376 |           let flow = report.leak_lpm * (0.65 + 0.35 * harmonic.max(-0.9)); | executable/support line; behavior impact depends on surrounding block context. |
| 6377 | (blank) | line is blank; visual separation only. |
| 6378 |           local_pressure[i] = pressure; | executable/support line; behavior impact depends on surrounding block context. |
| 6379 |           local_frag[i] = frag; | executable/support line; behavior impact depends on surrounding block context. |
| 6380 |           local_flow[i] = flow.max(0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 6381 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 6382 | (blank) | line is blank; visual separation only. |
| 6383 |       let mut max_pressure_idx = 0usize; | executable/support line; behavior impact depends on surrounding block context. |
| 6384 |       let mut max_frag_idx = 0usize; | executable/support line; behavior impact depends on surrounding block context. |
| 6385 |       for i in 1..SECTORS { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 6386 |           if local_pressure[i] > local_pressure[max_pressure_idx] { | conditional gate; changing condition shifts runtime path and side effects. |
| 6387 |               max_pressure_idx = i; | executable/support line; behavior impact depends on surrounding block context. |
| 6388 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 6389 |           if local_frag[i] > local_frag[max_frag_idx] { | conditional gate; changing condition shifts runtime path and side effects. |
| 6390 |               max_frag_idx = i; | executable/support line; behavior impact depends on surrounding block context. |
| 6391 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 6392 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 6393 | (blank) | line is blank; visual separation only. |
| 6394 |       let width = 37usize; | executable/support line; behavior impact depends on surrounding block context. |
| 6395 |       let height = 17usize; | executable/support line; behavior impact depends on surrounding block context. |
| 6396 |       let mut grid = vec![vec![' '; width]; height]; | executable/support line; behavior impact depends on surrounding block context. |
| 6397 |       for (y, row) in grid.iter_mut().enumerate().take(height) { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 6398 |           for (x, cell) in row.iter_mut().enumerate().take(width) { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 6399 |               let nx = (x as f64 / (width - 1) as f64 - 0.5) * 2.0; | executable/support line; behavior impact depends on surrounding block context. |
| 6400 |               let ny = (y as f64 / (height - 1) as f64 - 0.5) * 2.0; | executable/support line; behavior impact depends on surrounding block context. |
| 6401 |               let rr = ((nx * 1.1).powi(2) + (ny * 0.9).powi(2)).sqrt(); | executable/support line; behavior impact depends on surrounding block context. |
| 6402 |               if (0.45..=0.95).contains(&rr) { | conditional gate; changing condition shifts runtime path and side effects. |
| 6403 |                   let ang = ny.atan2(nx); | executable/support line; behavior impact depends on surrounding block context. |
| 6404 |                   let mut idx = ((ang + std::f64::consts::PI) / std::f64::consts::TAU | executable/support line; behavior impact depends on surrounding block context. |
| 6405 |                       * SECTORS as f64) as usize; | executable/support line; behavior impact depends on surrounding block context. |
| 6406 |                   if idx >= SECTORS { | conditional gate; changing condition shifts runtime path and side effects. |
| 6407 |                       idx = SECTORS - 1; | executable/support line; behavior impact depends on surrounding block context. |
| 6408 |                   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 6409 |                   let frag = local_frag[idx]; | executable/support line; behavior impact depends on surrounding block context. |

```

```

ng block context. |
| 6410 |           let mut ch = if frag > 0.85 { | executable/support line; behavior impact depends on surr
ding block context. |
| 6411 |               'X' | executable/support line; behavior impact depends on surrounding block context.
| 6412 |           } else if frag > 0.70 { | executable/support line; behavior impact depends on surrounding
lock context. |
| 6413 |               '!' | executable/support line; behavior impact depends on surrounding block context.
| 6414 |           } else if frag > 0.50 { | executable/support line; behavior impact depends on surrounding
lock context. |
| 6415 |               '*' | executable/support line; behavior impact depends on surrounding block context.
| 6416 |           } else if frag > 0.30 { | executable/support line; behavior impact depends on surrounding
lock context. |
| 6417 |               ':' | executable/support line; behavior impact depends on surrounding block context.
| 6418 |           } else { | executable/support line; behavior impact depends on surrounding block context.
| 6419 |               '.' | executable/support line; behavior impact depends on surrounding block context.
| 6420 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 6421 |           if idx == max_pressure_idx { | conditional gate; changing condition shifts runtime path
side effects. |
| 6422 |               ch = 'P'; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 6423 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and L
time scopes. |
| 6424 |           if idx == max_frag_idx { | conditional gate; changing condition shifts runtime path and
e effects. |
| 6425 |               ch = 'T'; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 6426 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and L
time scopes. |
| 6427 |           *cell = ch; | executable/support line; behavior impact depends on surrounding block cont
. |
| 6428 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 6429 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 6430 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 6431 | (blank) | line is blank; visual separation only. |
| 6432 |           let mut cad = String::new(); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 6433 |           cad.push_str("ENGINEERING ASCII CAD - O-RING / SEAL CROSS-SECTION\n"); | executable/support li
behavior impact depends on surrounding block context. |
| 6434 |           cad.push_str(" legend: T=tear-up point, P=max pressure point, X=fragility critical\n"); | executabl
upport line; behavior impact depends on surrounding block context. |
| 6435 |           cad.push_str(" !=high fragility, *=moderate, :=watch, .=low\n\n"); | executable/support lin
behavior impact depends on surrounding block context. |
| 6436 |           for row in &grid { | iteration control; changing bounds/body affects throughput, timing, and possibl
afety guarantees. |
| 6437 |               let line: String = row.iter().collect(); | executable/support line; behavior impact depends on s
ounding block context. |
| 6438 |               cad.push_str(&line); | executable/support line; behavior impact depends on surrounding block con
t. |
| 6439 |               cad.push('\n'); | executable/support line; behavior impact depends on surrounding block context.
| 6440 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 6441 | (blank) | line is blank; visual separation only. |
| 6442 |           let mut lines = Vec::new(); | executable/support line; behavior impact depends on surrounding block
text. |
| 6443 |           lines.push(format!( | executable/support line; behavior impact depends on surrounding block context.
| 6444 |               "pressure point (max): sector {} \ {:.2} bar", | executable/support line; behavior impact depen
on surrounding block context. |
| 6445 |               max_pressure_idx, local_pressure[max_pressure_idx] | executable/support line; behavior impact de
ds on surrounding block context. |
| 6446 |           )); | executable/support line; behavior impact depends on surrounding block context. |
| 6447 |           lines.push(format!( | executable/support line; behavior impact depends on surrounding block context.

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| 6448 |         "tear-up point (max fragility): sector {} \ | fragility {:.3}", | executable/support line; behavior
impact depends on surrounding block context. |
| 6449 |         max_frag_idx, local_frag[max_frag_idx] | executable/support line; behavior impact depends on sur
nding block context. |
| 6450 |     )); | executable/support line; behavior impact depends on surrounding block context. |
| 6451 |     lines.push(format!( | executable/support line; behavior impact depends on surrounding block context.
| 6452 |         "global leak flow: {:.4} L/min \ | rupture risk {:.1}% \ | band {}", | executable/support line; be
ior impact depends on surrounding block context. |
| 6453 |         report.leak_lpm, report.rupture_probability_pct, report.pressure_band | executable/support line;
havior impact depends on surrounding block context. |
| 6454 |     )); | executable/support line; behavior impact depends on surrounding block context. |
| 6455 | (blank) | line is blank; visual separation only. |
| 6456 |     let mut ranked: Vec<(usize, f64)> = (0..SECTORS).map(|i| (i, local_frag[i])).collect(); | executab
support line; behavior impact depends on surrounding block context. |
| 6457 |     ranked.sort_by(|a, b| b.1.total_cmp(&a.1)); | executable/support line; behavior impact depends on
rounding block context. |
| 6458 |     for (idx, frag) in ranked.into_iter().take(6) { | iteration control; changing bounds/body affects th
ghput, timing, and possibly safety guarantees. |
| 6459 |         lines.push(format!( | executable/support line; behavior impact depends on surrounding block cont
. |
| 6460 |         "fragility point s{:02}: pressure {:.2} bar \ | fragility {:.3} \ | flow {:.4} L/min", | execu
le/support line; behavior impact depends on surrounding block context. |
| 6461 |         idx, local_pressure[idx], frag, local_flow[idx] | executable/support line; behavior impact d
nds on surrounding block context. |
| 6462 |     )); | executable/support line; behavior impact depends on surrounding block context. |
| 6463 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 6464 | (blank) | line is blank; visual separation only. |
| 6465 |     (cad, lines) | executable/support line; behavior impact depends on surrounding block context. |
| 6466 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 6467 | (blank) | line is blank; visual separation only. |
| 6468 | fn tab_leak_lab(&mut self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter A
control flow, and outputs. |
| 6469 |     self.sanitize_leak_manual(); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 6470 |     self.sanitize_leak_custom(); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 6471 |     let manual_validation = self.validate_leak_manual(); | executable/support line; behavior impact depe
on surrounding block context. |
| 6472 |     let custom_validation = self.validate_leak_custom(); | executable/support line; behavior impact depe
on surrounding block context. |
| 6473 |     let manual_ok = manual_validation.is_ok(); | executable/support line; behavior impact depends on sur
nding block context. |
| 6474 |     let custom_ok = custom_validation.is_ok(); | executable/support line; behavior impact depends on sur
nding block context. |
| 6475 | (blank) | line is blank; visual separation only. |
| 6476 |     let circuits: Vec<(usize, String, String, String, String)> = self | executable/support line; behavio
mpact depends on surrounding block context. |
| 6477 |         .bench | executable/support line; behavior impact depends on surrounding block context. |
| 6478 |         .leak_rig | executable/support line; behavior impact depends on surrounding block context. |
| 6479 |         .circuits | executable/support line; behavior impact depends on surrounding block context. |
| 6480 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 6481 |         .enumerate() | executable/support line; behavior impact depends on surrounding block context. |
| 6482 |         .map(|(i, c)| { | executable/support line; behavior impact depends on surrounding block context
|
| 6483 |         ( | executable/support line; behavior impact depends on surrounding block context. |
| 6484 |         i, | executable/support line; behavior impact depends on surrounding block context. |
| 6485 |         c.name.clone(), | executable/support line; behavior impact depends on surrounding block
text. |
| 6486 |         c.application.clone(), | executable/support line; behavior impact depends on surrounding
ock context. |
| 6487 |         format!("{:?}", c.component), | error-propagating operation; changing it impacts failure
mantics and caller recovery paths. |
| 6488 |         format!("{:?}", c.oil_type), | error-propagating operation; changing it impacts failure
antics and caller recovery paths. |

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| 6489 |         ) | executable/support line; behavior impact depends on surrounding block context. |
| 6490 |     }) | executable/support line; behavior impact depends on surrounding block context. |
| 6491 |     .collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 6492 |     let reports = self.bench.leak_reports.clone(); | executable/support line; behavior impact depends on
rounding block context. |
| 6493 |     let alerts = self.bench.leak_rig.alerts.clone(); | executable/support line; behavior impact depends
surrounding block context. |
| 6494 |     let predictions = self.leak_predictions.clone(); | executable/support line; behavior impact depends
surrounding block context. |
| 6495 |     let calibration_report = self.leak_calibration_report.clone(); | executable/support line; behavior i
ct depends on surrounding block context. |
| 6496 |     let selected_circuit = self.bench.leak_rig.circuits.get(self.leak_sel_idx).cloned(); | executable/su
rt line; behavior impact depends on surrounding block context. |
| 6497 |     let selected_report = selected_circuit.as_ref().and_then(|c| { | executable/support line; behavior
pact depends on surrounding block context. |
| 6498 |         reports | executable/support line; behavior impact depends on surrounding block context. |
| 6499 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 6500 |         .find(|r| r.name == c.name) | executable/support line; behavior impact depends on surround
block context. |
| 6501 |         .cloned() | executable/support line; behavior impact depends on surrounding block context. |
| 6502 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 6503 | (blank) | line is blank; visual separation only. |
| 6504 |     ui.horizontal(|ui| { | executable/support line; behavior impact depends on surrounding block conte
|
| 6505 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 6506 |         RichText::new("LEAK PHYSICS LAB – industrial manual + automatic workflow") | executable/supp
line; behavior impact depends on surrounding block context. |
| 6507 |         .size(12.0) | executable/support line; behavior impact depends on surrounding block conte
. |
| 6508 |         .color(Color32::from_rgb(80, 155, 255)) | executable/support line; behavior impact dependen
on surrounding block context. |
| 6509 |         .strong(), | executable/support line; behavior impact depends on surrounding block conte
|
| 6510 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 6511 |     ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 6512 |     ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 6513 |         RichText::new(format!("{}", circuits.len())) | executable/support line; behavior im
t depends on surrounding block context. |
| 6514 |         .size(10.5) | executable/support line; behavior impact depends on surrounding block conte
. |
| 6515 |         .color(Color32::from_gray(150)), | executable/support line; behavior impact depends on s
ounding block context. |
| 6516 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 6517 |     ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 6518 |     ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 6519 |         RichText::new(format!( | executable/support line; behavior impact depends on surrounding blo
context. |
| 6520 |         "Total leak {:.3} L/min", | executable/support line; behavior impact depends on surround
block context. |
| 6521 |         self.bench.leak_rig.total_leak_lpm | executable/support line; behavior impact depends on
rounding block context. |
| 6522 |     )) | executable/support line; behavior impact depends on surrounding block context. |
| 6523 |     .size(10.5) | executable/support line; behavior impact depends on surrounding block context. |
| 6524 |     .color(Color32::YELLOW), | executable/support line; behavior impact depends on surrounding b
k context. |
| 6525 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 6526 |     if !self.leak_note.is_empty() { | conditional gate; changing condition shifts runtime path and s
effects. |
| 6527 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block conte
. |
| 6528 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 6529 |         RichText::new(&self.leak_note) | executable/support line; behavior impact depends on sur
nding block context. |
| 6530 |         .size(10.5) | executable/support line; behavior impact depends on surrounding block
text. |

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| 6531 |                                     .color(Color32::LIGHT_BLUE), | executable/support line; behavior impact depends on s
ounding block context. |
| 6532 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 6533 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 6534 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 6535 |                                     ui.group(\|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
| 6536 |                                     ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 6537 |                                     RichText::new("Como ler esta aba (direto ao ponto)") | executable/support line; behavior imp
depends on surrounding block context. |
| 6538 |                                     .size(10.8) | executable/support line; behavior impact depends on surrounding block conte
. |
| 6539 |                                     .color(Color32::from_rgb(110, 180, 255)) | executable/support line; behavior impact depe
on surrounding block context. |
| 6540 |                                     .strong(), | executable/support line; behavior impact depends on surrounding block conte
|
| 6541 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 6542 |                                     ui.label("1) Runtime Circuits mostra estado atual (pressao, leak, risco, alerta."); | executabl
upport line; behavior impact depends on surrounding block context. |
| 6543 |                                     ui.label("2) Manual Input ajusta engenharia (envelope/oring/oil). A aplicacao invalida fica bloq
da."); | executable/support line; behavior impact depends on surrounding block context. |
| 6544 |                                     ui.label("3) Scenario Ranking ordena pior caso por risco e pico de pressao."); | executable/supp
line; behavior impact depends on surrounding block context. |
| 6545 |                                     ui.label("4) Timeback ASCII/Plot mostra evolucao temporal de pressao-risco-vazao."); | executabl
upport line; behavior impact depends on surrounding block context. |
| 6546 |                                     ui.label("5) Use export CSV/JSON para auditoria e rastreabilidade."); | executable/support line;
havior impact depends on surrounding block context. |
| 6547 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 6548 |                                     ui.horizontal_wrapped(\|ui\| { | executable/support line; behavior impact depends on surrounding blo
context. |
| 6549 |                                     if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 6550 |                                     .add(Button::new("Nova rodada (limpar saidas)").fill(Color32::from_rgb(40, 50, 70))) | execu
le/support line; behavior impact depends on surrounding block context. |
| 6551 |                                     .clicked() | executable/support line; behavior impact depends on surrounding block context.
| 6552 |                                     { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 6553 |                                     self.cmds.push(Cmd::LeakClearScenarioOutputs); | executable/support line; behavior impact de
ds on surrounding block context. |
| 6554 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 6555 |                                     if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 6556 |                                     .add(Button::new("RESET Leak Sim (estado fisico)").fill(Color32::from_rgb(85, 35, 25))) | ex
table/support line; behavior impact depends on surrounding block context. |
| 6557 |                                     .clicked() | executable/support line; behavior impact depends on surrounding block context.
| 6558 |                                     { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 6559 |                                     self.cmds.push(Cmd::LeakResetSimulation); | executable/support line; behavior impact depends
surrounding block context. |
| 6560 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 6561 |                                     if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 6562 |                                     .add_enabled( | executable/support line; behavior impact depends on surrounding block contex
|
| 6563 |                                     manual_ok, | executable/support line; behavior impact depends on surrounding block conte
|
| 6564 |                                     Button::new("Aplicar + Rodar Cenario").fill(Color32::from_rgb(20, 70, 100)), | executabl
upport line; behavior impact depends on surrounding block context. |
| 6565 |                                     ) | executable/support line; behavior impact depends on surrounding block context. |
| 6566 |                                     .clicked() | executable/support line; behavior impact depends on surrounding block context.
| 6567 |                                     { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 6568 |                                     self.cmds.push(Cmd::LeakApplyAndPredict); | executable/support line; behavior impact depends
surrounding block context. |
| 6569 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |

```

```

| 6570 |         if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 6571 |             .add_enabled( | executable/support line; behavior impact depends on surrounding block context. |
|
| 6572 |                 manual_ok, | executable/support line; behavior impact depends on surrounding block context. |
|
| 6573 |                 Button::new("Aplicar + Rodar Monte Carlo").fill(Color32::from_rgb(70, 35, 90)), | executable/support line; behavior impact depends on surrounding block context. |
| 6574 |             ) | executable/support line; behavior impact depends on surrounding block context. |
| 6575 |             .clicked() | executable/support line; behavior impact depends on surrounding block context. |
| 6576 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime. |
pes. |
| 6577 |             self.cmds.push(Cmd::LeakApplyAndMonteCarlo); | executable/support line; behavior impact depends on surrounding block context. |
| 6578 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime. |
pes. |
| 6579 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 6580 |     ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 6581 |     ui.group(\ui\ { | executable/support line; behavior impact depends on surrounding block context. |
| 6582 |         ui.horizontal_wrapped(\ui\ { | executable/support line; behavior impact depends on surrounding block context. |
|
| 6583 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 6584 |             RichText::new("TIMEBACK VIEW") | executable/support line; behavior impact depends on surrounding block context. |
|
| 6585 |                 .size(10.8) | executable/support line; behavior impact depends on surrounding block context. |
|
| 6586 |                 .color(Color32::from_rgb(110, 180, 255)) | executable/support line; behavior impact depends on surrounding block context. |
|
| 6587 |                 .strong(), | executable/support line; behavior impact depends on surrounding block context. |
|
| 6588 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 6589 |             ui.label("janela(s)"); | executable/support line; behavior impact depends on surrounding block context. |
|
| 6590 |             ui.add(DragValue::new(&mut self.leak_timeback_window_s).speed(5.0).range(10.0..=7200.0)); | executable/support line; behavior impact depends on surrounding block context. |
| 6591 |             ui.label("stride"); | executable/support line; behavior impact depends on surrounding block context. |
| 6592 |             ui.add(DragValue::new(&mut self.leak_timeback_stride).speed(1.0).range(1..=60)); | executable/support line; behavior impact depends on surrounding block context. |
| 6593 |             ui.checkbox(&mut self.leak_timeback_latest_first, "latest first"); | executable/support line; behavior impact depends on surrounding block context. |
| 6594 |             ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 6595 |                 RichText::new(format!("hist: {} pontos", self.leak_temporal_trace.len())) | executable/support line; behavior impact depends on surrounding block context. |
| 6596 |                 .size(9.8) | executable/support line; behavior impact depends on surrounding block context. |
|
| 6597 |                 .color(Color32::from_gray(160)), | executable/support line; behavior impact depends on surrounding block context. |
|
| 6598 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 6599 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 6600 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 6601 |             RichText::new("ASCII timeline ponto a ponto: [pressao][risco][vazao] com janela temporal controlavel") | executable/support line; behavior impact depends on surrounding block context. |
| 6602 |             .size(9.6) | executable/support line; behavior impact depends on surrounding block context. |
|
| 6603 |             .color(Color32::from_gray(155)), | executable/support line; behavior impact depends on surrounding block context. |
|
| 6604 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 6605 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 6606 |     ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 6607 | (blank) | line is blank; visual separation only. |
| 6608 |     let mut do_apply = false; | executable/support line; behavior impact depends on surrounding block context. |
|
| 6609 |     let mut do_predict = false; | executable/support line; behavior impact depends on surrounding block context. |
|
| 6610 |     let mut do_add_custom = false; | executable/support line; behavior impact depends on surrounding block context. |

```



```

context. |
| 6611 |          let mut do_export_report_csv = false; | executable/support line; behavior impact depends on surrounding block context. |
| 6612 |          let mut do_export_report_json = false; | executable/support line; behavior impact depends on surrounding block context. |
| 6613 |          let mut do_export_pred_csv = false; | executable/support line; behavior impact depends on surrounding block context. |
| 6614 |          let mut do_export_pred_json = false; | executable/support line; behavior impact depends on surrounding block context. |
| 6615 |          let mut do_export_catalog_mat_csv = false; | executable/support line; behavior impact depends on surrounding block context. |
| 6616 |          let mut do_export_catalog_mat_json = false; | executable/support line; behavior impact depends on surrounding block context. |
| 6617 |          let mut do_export_catalog_oil_csv = false; | executable/support line; behavior impact depends on surrounding block context. |
| 6618 |          let mut do_export_catalog_oil_json = false; | executable/support line; behavior impact depends on surrounding block context. |
| 6619 |          let mut do_pick_calibration_csv = false; | executable/support line; behavior impact depends on surrounding block context. |
| 6620 |          let mut do_run_calibration = false; | executable/support line; behavior impact depends on surrounding block context. |
| 6621 |          let mut do_export_calibration_csv = false; | executable/support line; behavior impact depends on surrounding block context. |
| 6622 |          let mut do_export_calibration_json = false; | executable/support line; behavior impact depends on surrounding block context. |
| 6623 |          let mut do_monte_carlo = false; | executable/support line; behavior impact depends on surrounding block context. |
| 6624 |          let mut pending_select: Option<usize> = None; | executable/support line; behavior impact depends on surrounding block context. |
| 6625 | (blank) | line is blank; visual separation only. |
| 6626 |          ScrollArea::vertical().id_salt("leak::outer_scroll").auto_shrink([false, false]).max_height(ui.available_height()).show(ui, \ui\ { | executable/support line; behavior impact depends on surrounding block context. |
| 6627 |              ui.columns(3, \cols\ { | executable/support line; behavior impact depends on surrounding block context. |
| 6628 |                  // Left: circuit list + runtime status | comment line; documentation intent, no runtime effect
|
| 6629 |                  egui::Frame::group(cols[0].style()).fill(Color32::from_gray(15)).show(&mut cols[0], \ui\ { | executable/support line; behavior impact depends on surrounding block context. |
| 6630 |                      ui.label(RichText::new("RUNTIME CIRCUITS").size(11.5).color(Color32::from_rgb(80,155,255))); | executable/support line; behavior impact depends on surrounding block context. |
| 6631 |                      ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 6632 |                      for (i, name, app, comp, oil) in &circuits { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 6633 |                          ui.push_id(format!("leak::circuit_row_{i}", i), \ui\ { | executable/support line; behavior impact depends on surrounding block context. |
| 6634 |                              let sel = *i == self.leak_sel_idx; | executable/support line; behavior impact depends on surrounding block context. |
| 6635 |                              let fill = if sel { Color32::from_rgb(30,60,100) } else { Color32::from_gray(25) }; | executable/support line; behavior impact depends on surrounding block context. |
| 6636 |                              if ui.add(Button::new(RichText::new(format!("{i} [{i} / {i}]", name, comp, oil)).size(10.5))).fill(fill).min_size(Vec2::new(ui.available_width()-8.0,22.0))).clicked() { | conditional gate; changing conditions shifts runtime path and side effects. |
| 6637 |                                  pending_select = Some(*i); | executable/support line; behavior impact depends on surrounding block context. |
| 6638 |                              } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 6639 |                              }); | executable/support line; behavior impact depends on surrounding block context. |
| 6640 |                              ui.label(RichText::new(app).size(9.8).color(Color32::from_gray(155))); | executable/support line; behavior impact depends on surrounding block context. |
| 6641 |                              } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 6642 |                              ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 6643 |                              ui.label(RichText::new("REAL-TIME REPORT").size(11.0).color(Color32::from_rgb(80,155,255)));

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```

; | executable/support line; behavior impact depends on surrounding block context. |
| 6644 |           for r in &reports { | iteration control; changing bounds/body affects throughput, timing
and possibly safety guarantees. |
| 6645 |           let c = match r.alert { | executable/support line; behavior impact depends on surroundi
ng block context. |
| 6646 |               auto_breaking::LeakAlertLevel::Normal => Color32::GREEN, | executable/support li
behavior impact depends on surrounding block context. |
| 6647 |               auto_breaking::LeakAlertLevel::Watch => Color32::YELLOW, | executable/support li
behavior impact depends on surrounding block context. |
| 6648 |               auto_breaking::LeakAlertLevel::Warning => Color32::from_rgb(255,170,60), | execu
le/support line; behavior impact depends on surrounding block context. |
| 6649 |               auto_breaking::LeakAlertLevel::Critical => Color32::RED, | executable/support li
behavior impact depends on surrounding block context. |
| 6650 |               auto_breaking::LeakAlertLevel::Ruptured => Color32::from_rgb(255,0,0), | executa
/support line; behavior impact depends on surrounding block context. |
| 6651 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |
| 6652 |           ui.label(RichText::new(format!( | executable/support line; behavior impact depends o
urrounding block context. |
| 6653 |               "{} \\\| {} \\\| p={:.1}bar ({} ) \\\| leak {:.3} L/min \\\| risk {:.0}%", | executable/s
ort line; behavior impact depends on surrounding block context. |
| 6654 |               r.name, | executable/support line; behavior impact depends on surrounding block
text. |
| 6655 |               r.alert, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 6656 |               r.current_pressure_bar, | executable/support line; behavior impact depends on su
unding block context. |
| 6657 |               r.pressure_band, | executable/support line; behavior impact depends on surroundi
block context. |
| 6658 |               r.leak_lpm, | executable/support line; behavior impact depends on surrounding bl
context. |
| 6659 |               r.rupture_probability_pct | executable/support line; behavior impact depends on
rounding block context. |
| 6660 |           )),).size(10.0).color(c)); | executable/support line; behavior impact depends on surro
ing block context. |
| 6661 |           ui.label(RichText::new(format!( | executable/support line; behavior impact depends o
urrounding block context. |
| 6662 |               "safe {:.1}-{:.1}bar target {:.1}bar \\\| rupture at {} bar / {} h", | executable/
port line; behavior impact depends on surrounding block context. |
| 6663 |               r.recommended_hold_min_bar, | executable/support line; behavior impact depends o
urrounding block context. |
| 6664 |               r.recommended_hold_max_bar, | executable/support line; behavior impact depends o
urrounding block context. |
| 6665 |               r.recommended_hold_target_bar, | executable/support line; behavior impact depend
n surrounding block context. |
| 6666 |               r.rupture_pressure_bar.map(\\|v\\| format!("{v:.2}")).unwrap_or_else(\\|\\| "n/a".in
)), | executable/support line; behavior impact depends on surrounding block context. |
| 6667 |               r.rupture_elapsed_h.map(\\|v\\| format!("{v:.3}")).unwrap_or_else(\\|\\| "n/a".into(
| executable/support line; behavior impact depends on surrounding block context. |
| 6668 |           )),).size(9.5).color(Color32::from_gray(160))); | executable/support line; behavior im
t depends on surrounding block context. |
| 6669 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 6670 |       if !alerts.is_empty() { | conditional gate; changing condition shifts runtime path and s
effects. |
| 6671 |           ui.separator(); | executable/support line; behavior impact depends on surrounding bl
context. |
| 6672 |           ui.label(RichText::new("ACTIVE ALERTS").size(11.0).color(Color32::RED)); | executabl
upport line; behavior impact depends on surrounding block context. |
| 6673 |           for a in &alerts { | iteration control; changing bounds/body affects throughput, tim
, and possibly safety guarantees. |
| 6674 |               ui.label(RichText::new(format!("{:} {}", a.circuit_name, a.message)).size(10.0).
or(Color32::RED)); | executable/support line; behavior impact depends on surrounding block context. |
| 6675 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |

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| 6676 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 6677 | (blank) | line is blank; visual separation only. |
| 6678 |           ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 6679 |           ui.label(RichText::new("3D PHYSICS/LEAK VIEW").size(11.0).color(Color32::from_rgb(80,155
5))); | executable/support line; behavior impact depends on surrounding block context. |
| 6680 |           ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 6681 |               ui.label("Yaw"); | executable/support line; behavior impact depends on surrounding b
k context. |
| 6682 |               ui.add(Slider::new(&mut self.leak_view_yaw_deg, -180.0..=180.0).show_value(true)); |
executable/support line; behavior impact depends on surrounding block context. |
| 6683 |               ui.label("Pitch"); | executable/support line; behavior impact depends on surrounding
ock context. |
| 6684 |               ui.add(Slider::new(&mut self.leak_view_pitch_deg, -80.0..=80.0).show_value(true)); |
executable/support line; behavior impact depends on surrounding block context. |
| 6685 |               ui.label("Zoom"); | executable/support line; behavior impact depends on surrounding
ck context. |
| 6686 |               ui.add(Slider::new(&mut self.leak_view_zoom, 0.4..=2.5).show_value(true)); | execut
/support line; behavior impact depends on surrounding block context. |
| 6687 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 6688 |           let (rect, _resp) = ui.allocate_exact_size( | executable/support line; behavior impact d
nds on surrounding block context. |
| 6689 |               Vec2::new(ui.available_width() - 4.0, 180.0), | executable/support line; behavior im
t depends on surrounding block context. |
| 6690 |               Sense::hover(), | executable/support line; behavior impact depends on surrounding bl
context. |
| 6691 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 6692 |           let p = ui.painter_at(rect); | executable/support line; behavior impact depends on surro
ing block context. |
| 6693 |           p.rect_filled(rect, 4.0, Color32::from_gray(10)); | executable/support line; behavior im
t depends on surrounding block context. |
| 6694 |           p.rect_stroke(rect, 4.0, Stroke::new(1.0, Color32::from_gray(60))); | executable/support
ne; behavior impact depends on surrounding block context. |
| 6695 | (blank) | line is blank; visual separation only. |
| 6696 |           let yaw = self.leak_view_yaw_deg.to_radians(); | executable/support line; behavior impac
depends on surrounding block context. |
| 6697 |           let pitch = self.leak_view_pitch_deg.to_radians(); | executable/support line; behavior i
ct depends on surrounding block context. |
| 6698 |           let zoom = self.leak_view_zoom; | executable/support line; behavior impact depends on su
unding block context. |
| 6699 |           let center = rect.center(); | executable/support line; behavior impact depends on surrou
ng block context. |
| 6700 |           let scale = 70.0 * zoom; | executable/support line; behavior impact depends on surroundi
block context. |
| 6701 | (blank) | line is blank; visual separation only. |
| 6702 |           let project = \|x: f32, y: f32, z: f32\| -> Pos2 { | executable/support line; behavior i
ct depends on surrounding block context. |
| 6703 |               let cy = yaw.cos(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 6704 |               let sy = yaw.sin(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 6705 |               let cp = pitch.cos(); | executable/support line; behavior impact depends on surround
block context. |
| 6706 |               let sp = pitch.sin(); | executable/support line; behavior impact depends on surround
block context. |
| 6707 | (blank) | line is blank; visual separation only. |
| 6708 |               let xr = x * cy + z * sy; | executable/support line; behavior impact depends on surr
ding block context. |
| 6709 |               let zr = -x * sy + z * cy; | executable/support line; behavior impact depends on sur
nding block context. |
| 6710 |               let yr = y * cp - zr * sp; | executable/support line; behavior impact depends on sur
nding block context. |
| 6711 |               let zr2 = y * sp + zr * cp + 8.0; | executable/support line; behavior impact depends

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surrounding block context. |
| 6712 |           let f = scale / zr2.max(0.3); | executable/support line; behavior impact depends on
rounding block context. |
| 6713 |           Pos2::new(center.x + xr * f, center.y - yr * f) | executable/support line; behavior
act depends on surrounding block context. |
| 6714 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 6715 | (blank) | line is blank; visual separation only. |
| 6716 |           let axis = [ | executable/support line; behavior impact depends on surrounding block con
t. |
| 6717 |           ([-2.2, 0.0, 0.0], [2.2, 0.0, 0.0], Color32::from_rgb(220, 90, 90)), | executable/su
rt line; behavior impact depends on surrounding block context. |
| 6718 |           ([0.0, 0.0, -2.2], [0.0, 0.0, 2.2], Color32::from_rgb(90, 220, 90)), | executable/su
rt line; behavior impact depends on surrounding block context. |
| 6719 |           ([0.0, 0.0, 0.0], [0.0, 2.2, 0.0], Color32::from_rgb(90, 140, 255)), | executable/su
rt line; behavior impact depends on surrounding block context. |
| 6720 |           ]; | executable/support line; behavior impact depends on surrounding block context. |
| 6721 |           for (a, b, c) in axis { | iteration control; changing bounds/body affects throughput, ti
g, and possibly safety guarantees. |
| 6722 |           p.line_segment( | executable/support line; behavior impact depends on surrounding bl
context. |
| 6723 |           [project(a[0], a[1], a[2]), project(b[0], b[1], b[2])], | executable/support lin
behavior impact depends on surrounding block context. |
| 6724 |           Stroke::new(1.2, c), | executable/support line; behavior impact depends on surro
ing block context. |
| 6725 |           ); | executable/support line; behavior impact depends on surrounding block context.
| 6726 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 6727 | (blank) | line is blank; visual separation only. |
| 6728 |           for (i, r) in reports.iter().take(8).enumerate() { | iteration control; changing bounds/
y affects throughput, timing, and possibly safety guarantees. |
| 6729 |           let x = -1.8 + i as f32 * 0.5; | executable/support line; behavior impact depends on
rounding block context. |
| 6730 |           let h = (r.leak_lpm as f32 * 0.9).clamp(0.08, 1.8); | executable/support line; behav
impact depends on surrounding block context. |
| 6731 |           let risk = (r.rupture_probability_pct as f32 / 100.0).clamp(0.0, 1.0); | executable/
port line; behavior impact depends on surrounding block context. |
| 6732 |           let col = Color32::from_rgb( | executable/support line; behavior impact depends on s
ounding block context. |
| 6733 |           (90.0 + 150.0 * risk) as u8, | executable/support line; behavior impact depends
surrounding block context. |
| 6734 |           (210.0 - 120.0 * risk) as u8, | executable/support line; behavior impact depends
surrounding block context. |
| 6735 |           90, | executable/support line; behavior impact depends on surrounding block cont
. |
| 6736 |           ); | executable/support line; behavior impact depends on surrounding block context.
| 6737 | (blank) | line is blank; visual separation only. |
| 6738 |           let y0 = 0.0f32; | executable/support line; behavior impact depends on surrounding b
k context. |
| 6739 |           let y1 = h; | executable/support line; behavior impact depends on surrounding block
text. |
| 6740 |           let z0 = -0.15f32; | executable/support line; behavior impact depends on surrounding
ock context. |
| 6741 |           let z1 = 0.15f32; | executable/support line; behavior impact depends on surrounding
ck context. |
| 6742 |           let x0 = x; | executable/support line; behavior impact depends on surrounding block
text. |
| 6743 |           let x1 = x + 0.3; | executable/support line; behavior impact depends on surrounding
ck context. |
| 6744 | (blank) | line is blank; visual separation only. |
| 6745 |           let corners = [ | executable/support line; behavior impact depends on surrounding bl
context. |
| 6746 |           project(x0, y0, z0), | executable/support line; behavior impact depends on surro
ing block context. |
| 6747 |           project(x1, y0, z0), | executable/support line; behavior impact depends on surro

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ing block context. |
| 6748 | project(x1, y1, z0), | executable/support line; behavior impact depends on surro
ing block context. |
| 6749 | project(x0, y1, z0), | executable/support line; behavior impact depends on surro
ing block context. |
| 6750 | project(x0, y0, z1), | executable/support line; behavior impact depends on surro
ing block context. |
| 6751 | project(x1, y0, z1), | executable/support line; behavior impact depends on surro
ing block context. |
| 6752 | project(x1, y1, z1), | executable/support line; behavior impact depends on surro
ing block context. |
| 6753 | project(x0, y1, z1), | executable/support line; behavior impact depends on surro
ing block context. |
| 6754 | ]; | executable/support line; behavior impact depends on surrounding block context.
| 6755 | (blank) | line is blank; visual separation only. |
| 6756 | let edges = [ | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 6757 | (0, 1), (1, 2), (2, 3), (3, 0), | executable/support line; behavior impact dependen
on surrounding block context. |
| 6758 | (4, 5), (5, 6), (6, 7), (7, 4), | executable/support line; behavior impact dependen
on surrounding block context. |
| 6759 | (0, 4), (1, 5), (2, 6), (3, 7), | executable/support line; behavior impact dependen
on surrounding block context. |
| 6760 | ]; | executable/support line; behavior impact depends on surrounding block context.
| 6761 | for (a, b) in edges { | iteration control; changing bounds/body affects throughput,
ing, and possibly safety guarantees. |
| 6762 | p.line_segment([corners[a], corners[b]], Stroke::new(1.0, col)); | executable/su
rt line; behavior impact depends on surrounding block context. |
| 6763 | } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 6764 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 6765 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 6766 | (blank) | line is blank; visual separation only. |
| 6767 | // Middle: manual params | comment line; documentation intent, no runtime effect. |
| 6768 | egui::Frame::group(cols[1].style()).fill(Color32::from_gray(15)).show(&mut cols[1], \|ui\| {
executable/support line; behavior impact depends on surrounding block context. |
| 6769 | ui.label(RichText::new("MANUAL ENGINEERING INPUT").size(11.5).color(Color32::from_rgb(80
5,255))); | executable/support line; behavior impact depends on surrounding block context. |
| 6770 | ui.label(RichText::new("Use this for production-calibrated pressure/oring/oil settings")
ze(9.8).color(Color32::from_gray(140))); | executable/support line; behavior impact depends on surrounding block cont
. |
| 6771 | ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 6772 | (blank) | line is blank; visual separation only. |
| 6773 | let oil_types = OilType::all(); | executable/support line; behavior impact depends on su
unding block context. |
| 6774 | ComboBox::from_id_salt("leak::manual_oil_type") | executable/support line; behavior impa
depends on surrounding block context. |
| 6775 | .width(180.0) | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 6776 | .selected_text( | executable/support line; behavior impact depends on surrounding bl
context. |
| 6777 | oil_types | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 6778 | .get(self.leak_manual.oil_type_idx) | executable/support line; behavior impa
depends on surrounding block context. |
| 6779 | .copied() | executable/support line; behavior impact depends on surrounding
ck context. |
| 6780 | .unwrap_or(OilType::Custom) | executable/support line; behavior impact dependen
on surrounding block context. |
| 6781 | .name(), | executable/support line; behavior impact depends on surrounding b
k context. |
| 6782 | ) | executable/support line; behavior impact depends on surrounding block context. |
| 6783 | .show_ui(ui, \|ui\| { | executable/support line; behavior impact depends on surround

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    block context. |
| 6784 |           for (i, o) in oil_types.iter().enumerate() { | iteration control; changing bound
ody affects throughput, timing, and possibly safety guarantees. |
| 6785 |               ui.selectable_value(&mut self.leak_manual.oil_type_idx, i, o.name()); | execu
ble/support line; behavior impact depends on surrounding block context. |
| 6786 |           } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 6787 |               }); | executable/support line; behavior impact depends on surrounding block context.
| 6788 |               ui.label(RichText::new("Oil Type").size(9.6).color(Color32::from_gray(150))); | executab
support line; behavior impact depends on surrounding block context. |
| 6789 | (blank) | line is blank; visual separation only. |
| 6790 |               ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 6791 |                   ui.label("Piston bar"); | executable/support line; behavior impact depends on surrou
ng block context. |
| 6792 |                   ui.add(DragValue::new(&mut self.leak_manual.piston_pressure_bar).speed(0.2).range(0.
=1000.0)); | executable/support line; behavior impact depends on surrounding block context. |
| 6793 |                   ui.label("Op bar"); | executable/support line; behavior impact depends on surroundin
lock context. |
| 6794 |                   ui.add(DragValue::new(&mut self.leak_manual.operation_pressure_bar).speed(0.2).range
0..=1000.0)); | executable/support line; behavior impact depends on surrounding block context. |
| 6795 |                   }); | executable/support line; behavior impact depends on surrounding block context. |
| 6796 |                   ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 6797 |                       ui.label("Min"); ui.add(DragValue::new(&mut self.leak_manual.pressure_min_bar).speed
2)); | executable/support line; behavior impact depends on surrounding block context. |
| 6798 |                       ui.label("Mean"); ui.add(DragValue::new(&mut self.leak_manual.pressure_mean_bar).spe
0.2)); | executable/support line; behavior impact depends on surrounding block context. |
| 6799 |                       ui.label("Ideal"); ui.add(DragValue::new(&mut self.leak_manual.pressure_ideal_bar).s
d(0.2)); | executable/support line; behavior impact depends on surrounding block context. |
| 6800 |                   }); | executable/support line; behavior impact depends on surrounding block context. |
| 6801 |                   ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 6802 |                       ui.label("Max"); ui.add(DragValue::new(&mut self.leak_manual.pressure_max_bar).speed
2)); | executable/support line; behavior impact depends on surrounding block context. |
| 6803 |                       ui.label("Rupture"); ui.add(DragValue::new(&mut self.leak_manual.pressure_rupture_ba
speed(0.2)); | executable/support line; behavior impact depends on surrounding block context. |
| 6804 |                       }); | executable/support line; behavior impact depends on surrounding block context. |
| 6805 |                       ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 6806 |                           ui.label("Squeeze %"); ui.add(DragValue::new(&mut self.leak_manual.squeeze_pct).spee
.1).range(5.0..=45.0)); | executable/support line; behavior impact depends on surrounding block context. |
| 6807 |                           ui.label("Comp set %"); ui.add(DragValue::new(&mut self.leak_manual.compression_set_
).speed(0.2).range(0.0..=80.0)); | executable/support line; behavior impact depends on surrounding block context. |
| 6808 |                           }); | executable/support line; behavior impact depends on surrounding block context. |
| 6809 |                           ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 6810 |                               ui.label("Base leak mm2"); | executable/support line; behavior impact depends on sur
nding block context. |
| 6811 |                               ui.add(DragValue::new(&mut self.leak_manual.base_leak_area_mm2).speed(0.0001).range(
..=0.1).max_decimals(6)); | executable/support line; behavior impact depends on surrounding block context. |
| 6812 |                               }); | executable/support line; behavior impact depends on surrounding block context. |
| 6813 | (blank) | line is blank; visual separation only. |
| 6814 |                               if let Err(msg) = &manual_validation { | conditional gate; changing condition shifts run
e path and side effects. |
| 6815 |                                   ui.label(RichText::new(format!("{}", msg)).size(9.8).color(Color32::YELLOW)); | ex
table/support line; behavior impact depends on surrounding block context. |
| 6816 |                               } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 6817 | (blank) | line is blank; visual separation only. |
| 6818 |                               if ui.add_enabled( | conditional gate; changing condition shifts runtime path and side e
cts. |
| 6819 |                                   manual_ok, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 6820 |                                   Button::new(RichText::new("APPLY MANUAL PARAMETERS").size(11.0)).fill(Color32::from_

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(20,80,30)).min_size(Vec2::new(ui.available_width()-8.0,28.0)), | executable/support line; behavior impact depends on
rounding block context. |
| 6821 |         ).clicked() { | executable/support line; behavior impact depends on surrounding block co
xt. |
| 6822 |             do_apply = true; | executable/support line; behavior impact depends on surrounding b
k context. |
| 6823 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 6824 | (blank) | line is blank; visual separation only. |
| 6825 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 6826 |         ui.label(RichText::new("AUTO PREDICTION").size(11.0).color(Color32::from_rgb(80,155,255)
| executable/support line; behavior impact depends on surrounding block context. |
| 6827 |         ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 6828 |             ui.label("Horizon s"); | executable/support line; behavior impact depends on surroun
g block context. |
| 6829 |             ui.add(DragValue::new(&mut self.leak_horizon_s).speed(5.0).range(60.0..=86400.0)); |
executable/support line; behavior impact depends on surrounding block context. |
| 6830 |             ui.label("dt s"); | executable/support line; behavior impact depends on surrounding
ck context. |
| 6831 |             ui.add(DragValue::new(&mut self.leak_scenario_dt).speed(0.01).range(0.01..=1.0)); |
cutable/support line; behavior impact depends on surrounding block context. |
| 6832 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 6833 |         if ui.add(Button::new(RichText::new("RUN SCENARIO PREDICTION").size(11.0)).fill(Color32:
om_rgb(20,50,90)).min_size(Vec2::new(ui.available_width()-8.0,28.0))).clicked() { | conditional gate; changing condit
shifts runtime path and side effects. |
| 6834 |             do_predict = true; | executable/support line; behavior impact depends on surrounding
ock context. |
| 6835 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 6836 |         if ui.add(Button::new(RichText::new("RUN MONTE CARLO (120 runs)").size(11.0)).fill(Color
:from_rgb(70,35,90)).min_size(Vec2::new(ui.available_width()-8.0,26.0))).clicked() { | conditional gate; changing con
ion shifts runtime path and side effects. |
| 6837 |             do_monte_carlo = true; | executable/support line; behavior impact depends on surroun
g block context. |
| 6838 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 6839 | (blank) | line is blank; visual separation only. |
| 6840 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 6841 |         ui.label(RichText::new("O-RING / SEAL ENGINEERING RENDER (ASCII CAD)").size(11.0).color(
or32::from_rgb(80,155,255))); | executable/support line; behavior impact depends on surrounding block context. |
| 6842 |         equi::CollapsingHeader::new("ASCII legend guide") | executable/support line; behavior im
t depends on surrounding block context. |
| 6843 |             .default_open(false) | executable/support line; behavior impact depends on surroundi
block context. |
| 6844 |             .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 6845 |                 ui.label(RichText::new("CAD symbols:").size(9.4).color(Color32::from_gray(180)))
executable/support line; behavior impact depends on surrounding block context. |
| 6846 |                 ui.label(RichText::new("P = ponto de maior pressao local").size(9.2).color(Color
:from_gray(170))); | executable/support line; behavior impact depends on surrounding block context. |
| 6847 |                 ui.label(RichText::new("T = ponto de maior fragilidade (tear-up)").size(9.2).col
Color32::from_gray(170))); | executable/support line; behavior impact depends on surrounding block context. |
| 6848 |                 ui.label(RichText::new("X/!/*/./. = criticidade decrescente da fragilidade").siz
.2).color(Color32::from_gray(170))); | executable/support line; behavior impact depends on surrounding block context. |
| 6849 |                 ui.separator(); | executable/support line; behavior impact depends on surroundi
lock context. |
| 6850 |                 ui.label(RichText::new("Timeline symbols:").size(9.4).color(Color32::from_gray(1
))); | executable/support line; behavior impact depends on surrounding block context. |
| 6851 |                 ui.label(RichText::new("1o char = pressao: ^ alto, : medio, . baixo").size(9.2).
or(Color32::from_gray(170))); | executable/support line; behavior impact depends on surrounding block context. |
| 6852 |                 ui.label(RichText::new("2o char = risco: ! alto, * medio, . baixo").size(9.2).co
(Color32::from_gray(170))); | executable/support line; behavior impact depends on surrounding block context. |

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| 6853 |                 ui.label(RichText::new("3o char = vazao: ~ alta, - media, . baixa").size(9.2).color(
(Color32::from_gray(170)))); | executable/support line; behavior impact depends on surrounding block context. |
| 6854 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 6855 |                 if let (Some(c), Some(r)) = (&selected_circuit, &selected_report) { | conditional gate;
changing condition shifts runtime path and side effects. |
| 6856 |                 let (cad, points) = Self::leak_ascii_cad(c, r); | executable/support line; behavior
act depends on surrounding block context. |
| 6857 |                 ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 6858 |                 RichText::new("Use scroll horizontal para ver o desenho ASCII completo sem corte
| executable/support line; behavior impact depends on surrounding block context. |
| 6859 |                 .size(9.2) | executable/support line; behavior impact depends on surrounding
ock context. |
| 6860 |                 .color(Color32::from_gray(150)), | executable/support line; behavior impact
ends on surrounding block context. |
| 6861 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 6862 |                 ScrollArea::both() | executable/support line; behavior impact depends on surrounding
ock context. |
| 6863 |                 .id_salt("leak::ascii_cad_scroll") | executable/support line; behavior impact de
ds on surrounding block context. |
| 6864 |                 .max_height(245.0) | executable/support line; behavior impact depends on surroun
g block context. |
| 6865 |                 .auto_shrink([false, false]) | executable/support line; behavior impact depends
surrounding block context. |
| 6866 |                 .show(ui, \|ui\| { | executable/support line; behavior impact depends on surroun
g block context. |
| 6867 |                 ui.set_min_width(360.0); | executable/support line; behavior impact depends
surrounding block context. |
| 6868 |                 ui.label(RichText::new(cad).size(8.8).monospace().color(Color32::from_gray(1
))); | executable/support line; behavior impact depends on surrounding block context. |
| 6869 |                 }); | executable/support line; behavior impact depends on surrounding block cont
. |
| 6870 |                 for p in points { | iteration control; changing bounds/body affects throughput, timi
and possibly safety guarantees. |
| 6871 |                 ui.label(RichText::new(p).size(9.2).color(Color32::from_gray(170))); | executabl
upport line; behavior impact depends on surrounding block context. |
| 6872 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 6873 |                 ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 6874 |                 RichText::new(format!( | executable/support line; behavior impact depends on sur
nding block context. |
| 6875 |                 "{ } \| material { } \| cs {:.2}mm \| squeeze {:.1}% \| gap {:.3}mm \| comp se
:.1}% \| shore {:.0}A \| life {:.0}h", | executable/support line; behavior impact depends on surrounding block context
|
| 6876 |                 c.name, | executable/support line; behavior impact depends on surrounding bl
context. |
| 6877 |                 c.spec.material.name(), | executable/support line; behavior impact depends o
urrounding block context. |
| 6878 |                 c.spec.cross_section_mm, | executable/support line; behavior impact depends
surrounding block context. |
| 6879 |                 c.spec.squeeze_pct, | executable/support line; behavior impact depends on su
unding block context. |
| 6880 |                 c.spec.extrusion_gap_mm, | executable/support line; behavior impact depends
surrounding block context. |
| 6881 |                 c.spec.compression_set_pct, | executable/support line; behavior impact depen
on surrounding block context. |
| 6882 |                 c.spec.shore_a, | executable/support line; behavior impact depends on surrou
ng block context. |
| 6883 |                 c.spec.design_life_hours | executable/support line; behavior impact depends
surrounding block context. |
| 6884 |                 )) | executable/support line; behavior impact depends on surrounding block conte
|
| 6885 |                 .size(9.4) | executable/support line; behavior impact depends on surrounding blo
context. |

```



```

| 6886 |                                     .color(Color32::from_gray(170)), | executable/support line; behavior impact depends
on surrounding block context. |
| 6887 |                                     ); | executable/support line; behavior impact depends on surrounding block context.
| 6888 | (blank) | line is blank; visual separation only. |
| 6889 |                                     ui.separator(); | executable/support line; behavior impact depends on surrounding block
context. |
| 6890 |                                     ui.label( | executable/support line; behavior impact depends on surrounding block context.
xt. |
| 6891 |                                     RichText::new("TEMPORAL ASCII EVOLUTION (pressure/risk/flow)") | executable/support
line; behavior impact depends on surrounding block context. |
| 6892 |                                     .size(10.6) | executable/support line; behavior impact depends on surrounding block
lock context. |
| 6893 |                                     .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior impact
depends on surrounding block context. |
| 6894 |                                     ); | executable/support line; behavior impact depends on surrounding block context.
| 6895 |                                     let t_now = self.bench.elapsed(); | executable/support line; behavior impact depends on
surrounding block context. |
| 6896 |                                     let t_min = (t_now - self.leak_timeback_window_s).max(0.0); | executable/support line;
behavior impact depends on surrounding block context. |
| 6897 |                                     let stride = self.leak_timeback_stride.max(1); | executable/support line; behavior impact
depends on surrounding block context. |
| 6898 |                                     let mut tb: Vec<(f64, f64, f64, f64)> = self | executable/support line; behavior impact
depends on surrounding block context. |
| 6899 |                                     .leak_temporal_trace | executable/support line; behavior impact depends on surrounding
block context. |
| 6900 |                                     .iter() | executable/support line; behavior impact depends on surrounding block context.
| 6901 |                                     .copied() | executable/support line; behavior impact depends on surrounding block context.
| 6902 |                                     .filter(\|(t, _, _, _)\| *t >= t_min) | executable/support line; behavior impact
depends on surrounding block context. |
| 6903 |                                     .step_by(stride) | executable/support line; behavior impact depends on surrounding block
context. |
| 6904 |                                     .collect(); | executable/support line; behavior impact depends on surrounding block context.
| 6905 |                                     if self.leak_timeback_latest_first { | conditional gate; changing condition shifts runtime
path and side effects. |
| 6906 |                                     tb.reverse(); | executable/support line; behavior impact depends on surrounding block
context. |
| 6907 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and
lifetime scopes. |
| 6908 | (blank) | line is blank; visual separation only. |
| 6909 |                                     ui.label( | executable/support line; behavior impact depends on surrounding block context.
xt. |
| 6910 |                                     RichText::new(format!( | executable/support line; behavior impact depends on surrounding
block context. |
| 6911 |                                     "window {:.0}s \\\| samples {} \\\| order {}", | executable/support line; behavior impact
depends on surrounding block context. |
| 6912 |                                     self.leak_timeback_window_s, | executable/support line; behavior impact depends on
surrounding block context. |
| 6913 |                                     tb.len(), | executable/support line; behavior impact depends on surrounding block context.
| 6914 |                                     if self.leak_timeback_latest_first { | conditional gate; changing condition shifts
runtime path and side effects. |
| 6915 |                                     "new->old" | executable/support line; behavior impact depends on surrounding block context.
| 6916 |                                     } else { | executable/support line; behavior impact depends on surrounding block context.
| 6917 |                                     "old->new" | executable/support line; behavior impact depends on surrounding block context.
| 6918 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and
lifetime scopes. |
| 6919 |                                     )) | executable/support line; behavior impact depends on surrounding block context.
| 6920 |                                     .size(9.6) | executable/support line; behavior impact depends on surrounding block context.

```

```

context. |
| 6921 | .color(Color32::from_gray(160)), | executable/support line; behavior impact depends
on surrounding block context. |
| 6922 | ); | executable/support line; behavior impact depends on surrounding block context. |
| 6923 | (blank) | line is blank; visual separation only. |
| 6924 | let mut timeline = String::new(); | executable/support line; behavior impact depends
surrounding block context. |
| 6925 | timeline.push_str("legend [P][R][F]: P(^ high, : mid, . low) \\\ R(! high, * mid, . l
\\ F(~ high, - mid, . low)\n"); | executable/support line; behavior impact depends on surrounding block context. |
| 6926 | timeline.push_str("idx \\\ time(s) \\\ ascii\n"); | executable/support line; behavior
act depends on surrounding block context. |
| 6927 | for (i, (tt, p, risk, leak)) in tb.iter().enumerate() { | iteration control; changin
ounds/body affects throughput, timing, and possibly safety guarantees. |
| 6928 | let p_ratio = (p / c.pressure.rupture_bar.max(1e-6)).clamp(0.0, 1.0); | executab
support line; behavior impact depends on surrounding block context. |
| 6929 | let r_ratio = (risk / 100.0).clamp(0.0, 1.0); | executable/support line; behavior
mpact depends on surrounding block context. |
| 6930 | let f_ratio = (leak / 1.5).clamp(0.0, 1.0); | executable/support line; behavior
act depends on surrounding block context. |
| 6931 | let pch = if p_ratio > 0.85 { | executable/support line; behavior impact depends
surrounding block context. |
| 6932 | '^' | executable/support line; behavior impact depends on surrounding block
text. |
| 6933 | } else if p_ratio > 0.55 { | executable/support line; behavior impact depends on
rounding block context. |
| 6934 | ':' | executable/support line; behavior impact depends on surrounding block
text. |
| 6935 | } else { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 6936 | '.' | executable/support line; behavior impact depends on surrounding block
text. |
| 6937 | }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 6938 | let rch = if r_ratio > 0.80 { | executable/support line; behavior impact depends
surrounding block context. |
| 6939 | '!' | executable/support line; behavior impact depends on surrounding block
text. |
| 6940 | } else if r_ratio > 0.45 { | executable/support line; behavior impact depends on
rounding block context. |
| 6941 | '*' | executable/support line; behavior impact depends on surrounding block
text. |
| 6942 | } else { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 6943 | '.' | executable/support line; behavior impact depends on surrounding block
text. |
| 6944 | }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 6945 | let fch = if f_ratio > 0.80 { | executable/support line; behavior impact depends
surrounding block context. |
| 6946 | '~' | executable/support line; behavior impact depends on surrounding block
text. |
| 6947 | } else if f_ratio > 0.45 { | executable/support line; behavior impact depends on
rounding block context. |
| 6948 | '-' | executable/support line; behavior impact depends on surrounding block
text. |
| 6949 | } else { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 6950 | '.' | executable/support line; behavior impact depends on surrounding block
text. |
| 6951 | }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 6952 | timeline.push_str(&format!("{:03} \\\ {:7.2} \\\ {{{}}}\n", i, tt, pch, rch, fch));
executable/support line; behavior impact depends on surrounding block context. |
| 6953 | } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |

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```

| 6954 | ScrollArea::both() | executable/support line; behavior impact depends on surrounding
ock context. |
| 6955 | .id_salt("leak::timeback_ascii_scroll") | executable/support line; behavior impa
depends on surrounding block context. |
| 6956 | .max_height(155.0) | executable/support line; behavior impact depends on surroun
g block context. |
| 6957 | .auto_shrink([false, false]) | executable/support line; behavior impact depends
surrounding block context. |
| 6958 | .show(ui, \|ui\| { | executable/support line; behavior impact depends on surroun
g block context. |
| 6959 | ui.set_min_width(340.0); | executable/support line; behavior impact depends
surrounding block context. |
| 6960 | ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 6961 | RichText::new(timeline) | executable/support line; behavior impact depen
on surrounding block context. |
| 6962 | .size(8.9) | executable/support line; behavior impact depends on sur
nding block context. |
| 6963 | .monospace() | executable/support line; behavior impact depends on s
ounding block context. |
| 6964 | .color(Color32::from_gray(178)), | executable/support line; behavior
pact depends on surrounding block context. |
| 6965 | ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 6966 | }); | executable/support line; behavior impact depends on surrounding block cont
. |
| 6967 | (blank) | line is blank; visual separation only. |
| 6968 | let p_points: Vec<f64; 2> = tb.iter().map(\|(t, p, _, _)\| [*t, *p]).collect(); |
cutable/support line; behavior impact depends on surrounding block context. |
| 6969 | let r_points: Vec<f64; 2> = tb.iter().map(\|(t, _, r, _)\| [*t, *r]).collect(); |
cutable/support line; behavior impact depends on surrounding block context. |
| 6970 | let f_points: Vec<f64; 2> = tb.iter().map(\|(t, _, _, f)\| [*t, *f]).collect(); |
cutable/support line; behavior impact depends on surrounding block context. |
| 6971 | ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 6972 | RichText::new("TIMEBACK VISUAL AID (P/R/F)") | executable/support line; behavior
pact depends on surrounding block context. |
| 6973 | .size(10.4) | executable/support line; behavior impact depends on surroundin
lock context. |
| 6974 | .color(Color32::from_rgb(110, 180, 255)), | executable/support line; behavio
mpact depends on surrounding block context. |
| 6975 | ); | executable/support line; behavior impact depends on surrounding block context.
| 6976 | Plot::new("leak::timeback_plot") | executable/support line; behavior impact depends
surrounding block context. |
| 6977 | .height(150.0) | executable/support line; behavior impact depends on surrounding
ock context. |
| 6978 | .allow_drag(false) | executable/support line; behavior impact depends on surroun
g block context. |
| 6979 | .allow_zoom(false) | executable/support line; behavior impact depends on surroun
g block context. |
| 6980 | .allow_scroll(false) | executable/support line; behavior impact depends on surro
ing block context. |
| 6981 | .show_axes([true, true]) | executable/support line; behavior impact depends on s
ounding block context. |
| 6982 | .show(ui, \|plot_ui\| { | executable/support line; behavior impact depends on su
unding block context. |
| 6983 | let lp = PlotPoints::from_iter(p_points.iter().copied()); | executable/suppo
line; behavior impact depends on surrounding block context. |
| 6984 | let lr = PlotPoints::from_iter(r_points.iter().copied()); | executable/suppo
line; behavior impact depends on surrounding block context. |
| 6985 | let lf = PlotPoints::from_iter(f_points.iter().copied()); | executable/suppo
line; behavior impact depends on surrounding block context. |
| 6986 | plot_ui.line( | executable/support line; behavior impact depends on surround
block context. |
| 6987 | Line::new(lp) | executable/support line; behavior impact depends on surr

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ding block context. |
| 6988 |                                     .color(Color32::from_rgb(255, 170, 60)) | executable/support line; behavior impact depends on surrounding block context. |
| 6989 |                                     .name("Pressure bar"), | executable/support line; behavior impact depends on surrounding block context. |
| 6990 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 6991 |                                     plot_ui.line( | executable/support line; behavior impact depends on surrounding block context. |
| 6992 |                                     Line::new(lr) | executable/support line; behavior impact depends on surrounding block context. |
| 6993 |                                     .color(Color32::from_rgb(255, 80, 80)) | executable/support line; behavior impact depends on surrounding block context. |
| 6994 |                                     .name("Risk %"), | executable/support line; behavior impact depends on surrounding block context. |
| 6995 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 6996 |                                     plot_ui.line( | executable/support line; behavior impact depends on surrounding block context. |
| 6997 |                                     Line::new(lf) | executable/support line; behavior impact depends on surrounding block context. |
| 6998 |                                     .color(Color32::from_rgb(90, 220, 180)) | executable/support line; behavior impact depends on surrounding block context. |
| 6999 |                                     .name("Leak LPM"), | executable/support line; behavior impact depends on surrounding block context. |
| 7000 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 7001 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 7002 |                                     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 7003 |                                     ui.label(RichText::new("Select a circuit with runtime data to render O-ring CAD").size(9.8).color(Color32::from_gray(145))); | executable/support line; behavior impact depends on surrounding block context. |
| 7004 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 7005 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 7006 | (blank) | line is blank; visual separation only. |
| 7007 |                                     // Right: custom circuit + scenario table | comment line; documentation intent, no runtime effect. |
| 7008 |                                     egui::Frame::group(cols[2].style()).fill(Color32::from_gray(15)).show(&mut cols[2], \|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
| 7009 |                                     ui.label(RichText::new("CUSTOM CIRCUIT BUILDER").size(11.5).color(Color32::from_rgb(80, 170, 255))); | executable/support line; behavior impact depends on surrounding block context. |
| 7010 |                                     ui.push_id("leak::custom_name_row", \|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
| 7011 |                                     ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
| 7012 |                                     ui.label("Name"); | executable/support line; behavior impact depends on surrounding block context. |
| 7013 |                                     ui.add_sized([120.0, 20.0], TextEdit::singleline(&mut self.leak_custom.name)); | executable/support line; behavior impact depends on surrounding block context. |
| 7014 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 7015 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 7016 |                                     ui.push_id("leak::custom_app_row", \|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
| 7017 |                                     ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
| 7018 |                                     ui.label("App"); | executable/support line; behavior impact depends on surrounding block context. |
| 7019 |                                     ui.add_sized([180.0, 20.0], TextEdit::singleline(&mut self.leak_custom.application)); | executable/support line; behavior impact depends on surrounding block context. |
| 7020 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 7021 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 7022 |                                     const COMP_NAMES: [&str; 3] = ["O-ring", "Seal", "A/C Hose"]; | executable/support line; behavior impact depends on surrounding block context. |
| 7023 |                                     let oil_types = OilType::all(); | executable/support line; behavior impact depends on surrounding block context. |

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unding block context. |
| 7024 |           let materials = OringMaterial::all(); | executable/support line; behavior impact depends
surrounding block context. |
| 7025 |           ComboBox::from_id_salt("leak::custom_component").selected_text(COMP_NAMES[self.leak_cust
component_idx.min(2)]).show_ui(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7026 |           for (i, n) in COMP_NAMES.iter().enumerate() { ui.selectable_value(&mut self.leak_cust
.component_idx, i, *n); } | iteration control; changing bounds/body affects throughput, timing, and possibly safety g
antees. |
| 7027 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 7028 |           ComboBox::from_id_salt("leak::custom_oil").selected_text( | executable/support line; beh
or impact depends on surrounding block context. |
| 7029 |           oil_types | executable/support line; behavior impact depends on surrounding block co
xt. |
| 7030 |           .get(self.leak_custom.oil_type_idx) | executable/support line; behavior impact d
nds on surrounding block context. |
| 7031 |           .copied() | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 7032 |           .unwrap_or(OilType::Custom) | executable/support line; behavior impact depends o
urrounding block context. |
| 7033 |           .name(), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 7034 |           ).show_ui(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 7035 |           for (i, o) in oil_types.iter().enumerate() { ui.selectable_value(&mut self.leak_cust
oil_type_idx, i, o.name()); } | iteration control; changing bounds/body affects throughput, timing, and possibly safe
guarantees. |
| 7036 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 7037 |           ComboBox::from_id_salt("leak::custom_material").selected_text( | executable/support line
ehavior impact depends on surrounding block context. |
| 7038 |           materials | executable/support line; behavior impact depends on surrounding block co
xt. |
| 7039 |           .get(self.leak_custom.material_idx) | executable/support line; behavior impact d
nds on surrounding block context. |
| 7040 |           .copied() | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 7041 |           .unwrap_or(OringMaterial::Nbr) | executable/support line; behavior impact depend
n surrounding block context. |
| 7042 |           .name(), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 7043 |           ).show_ui(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 7044 |           for (i, m) in materials.iter().enumerate() { ui.selectable_value(&mut self.leak_cust
material_idx, i, m.name()); } | iteration control; changing bounds/body affects throughput, timing, and possibly safe
guarantees. |
| 7045 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 7046 |           ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 7047 |           ui.label("Seals"); ui.add(DragValue::new(&mut self.leak_custom.seal_count).range(1..
8)); | executable/support line; behavior impact depends on surrounding block context. |
| 7048 |           ui.label("ShoreA"); ui.add(DragValue::new(&mut self.leak_custom.shore_a).range(40.0.
5.0)); | executable/support line; behavior impact depends on surrounding block context. |
| 7049 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 7050 |           ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 7051 |           ui.label("Piston"); ui.add(DragValue::new(&mut self.leak_custom.piston_pressure_bar)
eed(0.2)); | executable/support line; behavior impact depends on surrounding block context. |
| 7052 |           ui.label("Op"); ui.add(DragValue::new(&mut self.leak_custom.operation_pressure_bar).
ed(0.2)); | executable/support line; behavior impact depends on surrounding block context. |
| 7053 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 7054 |           ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 7055 |           ui.label("Min"); ui.add(DragValue::new(&mut self.leak_custom.min_bar).speed(0.2)); |
executable/support line; behavior impact depends on surrounding block context. |
| 7056 |           ui.label("Mean"); ui.add(DragValue::new(&mut self.leak_custom.mean_bar).speed(0.2));

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executable/support line; behavior impact depends on surrounding block context. |
| 7057 |                }); | executable/support line; behavior impact depends on surrounding block context. |
| 7058 |                ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 7059 |                ui.label("Ideal"); ui.add(DragValue::new(&mut self.leak_custom.ideal_bar).speed(0.2));
| executable/support line; behavior impact depends on surrounding block context. |
| 7060 |                ui.label("Max"); ui.add(DragValue::new(&mut self.leak_custom.max_bar).speed(0.2)); |
executable/support line; behavior impact depends on surrounding block context. |
| 7061 |                ui.label("Rupt"); ui.add(DragValue::new(&mut self.leak_custom.rupture_bar).speed(0.2));
| executable/support line; behavior impact depends on surrounding block context. |
| 7062 |                }); | executable/support line; behavior impact depends on surrounding block context. |
| 7063 |                ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 7064 |                ui.label("cs mm"); ui.add(DragValue::new(&mut self.leak_custom.cross_section_mm).spee
0.01)); | executable/support line; behavior impact depends on surrounding block context. |
| 7065 |                ui.label("sq %"); ui.add(DragValue::new(&mut self.leak_custom.squeeze_pct).speed(0.1));
| executable/support line; behavior impact depends on surrounding block context. |
| 7066 |                ui.label("gap"); ui.add(DragValue::new(&mut self.leak_custom.extrusion_gap_mm).speed(0.1));
| executable/support line; behavior impact depends on surrounding block context. |
| 7067 |                }); | executable/support line; behavior impact depends on surrounding block context. |
| 7068 |                ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 7069 |                ui.label("comp set"); ui.add(DragValue::new(&mut self.leak_custom.compression_set_pc
speed(0.1)); | executable/support line; behavior impact depends on surrounding block context. |
| 7070 |                ui.label("life h"); ui.add(DragValue::new(&mut self.leak_custom.design_life_hours).s
d(50.0)); | executable/support line; behavior impact depends on surrounding block context. |
| 7071 |                }); | executable/support line; behavior impact depends on surrounding block context. |
| 7072 |                ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 7073 |                ui.label("base leak"); ui.add(DragValue::new(&mut self.leak_custom.base_leak_area_mm
speed(0.0001).max_decimals(6)); | executable/support line; behavior impact depends on surrounding block context. |
| 7074 |                ui.label("Cd"); ui.add(DragValue::new(&mut self.leak_custom.discharge_coeff).speed(0.1));
).range(0.05..=1.0)); | executable/support line; behavior impact depends on surrounding block context. |
| 7075 |                }); | executable/support line; behavior impact depends on surrounding block context. |
| 7076 |                ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 7077 |                ui.label("Reservoir L"); ui.add(DragValue::new(&mut self.leak_custom.reservoir_volum
).speed(0.5)); | executable/support line; behavior impact depends on surrounding block context. |
| 7078 |                ui.label("Support LPM"); ui.add(DragValue::new(&mut self.leak_custom.support_lpm).sp
(0.5)); | executable/support line; behavior impact depends on surrounding block context. |
| 7079 |                }); | executable/support line; behavior impact depends on surrounding block context. |
| 7080 | (blank) | line is blank; visual separation only. |
| 7081 |                if let Err(msg) = &custom_validation { | conditional gate; changing condition shifts run
e path and side effects. |
| 7082 |                ui.label(RichText::new(format!("{}", msg)).size(9.8).color(Color32::YELLOW)); | ex
table/support line; behavior impact depends on surrounding block context. |
| 7083 |                } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 7084 | (blank) | line is blank; visual separation only. |
| 7085 |                if ui.add_enabled( | conditional gate; changing condition shifts runtime path and side e
cts. |
| 7086 |                custom_ok, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7087 |                Button::new(RichText::new("ADD CUSTOM CIRCUIT").size(11.0)).fill(Color32::from_rgb(6
0,10)).min_size(Vec2::new(ui.available_width()-8.0,26.0)), | executable/support line; behavior impact depends on surr
ding block context. |
| 7088 |                ).clicked() { | executable/support line; behavior impact depends on surrounding block co
xt. |
| 7089 |                do_add_custom = true; | executable/support line; behavior impact depends on surround
block context. |
| 7090 |                } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 7091 | (blank) | line is blank; visual separation only. |
| 7092 |                ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |

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| 7093 |             ui.label(RichText::new("SCENARIO RANKING").size(11.0).color(Color32::from_rgb(80,155,255)
; | executable/support line; behavior impact depends on surrounding block context. |
| 7094 |             ScrollArea::vertical().id_salt("leak::scenario_ranking_scroll").max_height(190.0).auto_s
nk([false, false]).show(ui, \ui\ { | executable/support line; behavior impact depends on surrounding block context.
| 7095 |                 for p in predictions.iter().take(50) { | iteration control; changing bounds/body aff
s throughput, timing, and possibly safety guarantees. |
| 7096 |                     let c = match p.final_alert { | executable/support line; behavior impact depends
surrounding block context. |
| 7097 |                         auto_breaking::LeakAlertLevel::Normal => Color32::GREEN, | executable/support
ine; behavior impact depends on surrounding block context. |
| 7098 |                         auto_breaking::LeakAlertLevel::Watch => Color32::YELLOW, | executable/support
ine; behavior impact depends on surrounding block context. |
| 7099 |                         auto_breaking::LeakAlertLevel::Warning => Color32::from_rgb(255,170,60), | e
utable/support line; behavior impact depends on surrounding block context. |
| 7100 |                         auto_breaking::LeakAlertLevel::Critical => Color32::RED, | executable/support
ine; behavior impact depends on surrounding block context. |
| 7101 |                         auto_breaking::LeakAlertLevel::Ruptured => Color32::from_rgb(255,0,0), | exe
able/support line; behavior impact depends on surrounding block context. |
| 7102 |                     }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 7103 |             ui.label(RichText::new(format!("{}", \{ } \{ } \{ } \{ } risk {:.0}% \{ } pmax {:.1}bar",
circuit_name, p.scenario_name, p.final_alert, p.final_rupture_probability_pct, p.peak_pressure_bar)).size(9.8).color(
; | executable/support line; behavior impact depends on surrounding block context. |
| 7104 |             ui.label(RichText::new(format!("mode: {} \{ } ttf: {}", p.likely_failure_mode, p.h
s_to_rupture.map(\h\ format!("{:.1}h", h)).unwrap_or_else(\|\| "n/a".into()))).size(9.4).color(Color32::from_gray(1
)); | executable/support line; behavior impact depends on surrounding block context. |
| 7105 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 7106 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 7107 |             ui.label(RichText::new("3D SCENARIO SIMULATION VIEW").size(10.6).color(Color32::from_rgb
,155,255)); | executable/support line; behavior impact depends on surrounding block context. |
| 7108 |             let (srect, _sresp) = ui.allocate_exact_size( | executable/support line; behavior impact
depends on surrounding block context. |
| 7109 |                 Vec2::new(ui.available_width() - 4.0, 170.0), | executable/support line; behavior im
t depends on surrounding block context. |
| 7110 |                 Sense::hover(), | executable/support line; behavior impact depends on surrounding bl
context. |
| 7111 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 7112 |             let sp = ui.painter_at(srect); | executable/support line; behavior impact depends on surr
nding block context. |
| 7113 |             sp.rect_filled(srect, 4.0, Color32::from_gray(10)); | executable/support line; behavior
act depends on surrounding block context. |
| 7114 |             sp.rect_stroke(srect, 4.0, Stroke::new(1.0, Color32::from_gray(60))); | executable/suppo
line; behavior impact depends on surrounding block context. |
| 7115 | (blank) | line is blank; visual separation only. |
| 7116 |             let center = srect.center(); | executable/support line; behavior impact depends on surro
ing block context. |
| 7117 |             let yaw = 28.0f32.to_radians(); | executable/support line; behavior impact depends on su
unding block context. |
| 7118 |             let pitch = 20.0f32.to_radians(); | executable/support line; behavior impact depends on
rounding block context. |
| 7119 |             let scale = 65.0f32; | executable/support line; behavior impact depends on surrounding b
k context. |
| 7120 |             let project = \|x: f32, y: f32, z: f32\| -> Pos2 { | executable/support line; behavior i
ct depends on surrounding block context. |
| 7121 |                 let cy = yaw.cos(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 7122 |                 let sy = yaw.sin(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 7123 |                 let cp = pitch.cos(); | executable/support line; behavior impact depends on surround
block context. |
| 7124 |                 let spv = pitch.sin(); | executable/support line; behavior impact depends on surroun
g block context. |
| 7125 | (blank) | line is blank; visual separation only. |
| 7126 |                 let xr = x * cy + z * sy; | executable/support line; behavior impact depends on surr

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ding block context. |
| 7127 | let zr = -x * sy + z * cy; | executable/support line; behavior impact depends on sur
nding block context. |
| 7128 | let yr = y * cp - zr * spv; | executable/support line; behavior impact depends on su
unding block context. |
| 7129 | let zr2 = y * spv + zr * cp + 8.5; | executable/support line; behavior impact depend
n surrounding block context. |
| 7130 | let f = scale / zr2.max(0.35); | executable/support line; behavior impact depends on
rrounding block context. |
| 7131 | Pos2::new(center.x + xr * f, center.y - yr * f) | executable/support line; behavior
act depends on surrounding block context. |
| 7132 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 7133 | (blank) | line is blank; visual separation only. |
| 7134 | let axis = [ | executable/support line; behavior impact depends on surrounding block con
t. |
| 7135 | ([-2.4, 0.0, 0.0], [2.4, 0.0, 0.0], Color32::from_rgb(220, 90, 90)), | executable/su
rt line; behavior impact depends on surrounding block context. |
| 7136 | ([0.0, 0.0, -2.4], [0.0, 0.0, 2.4], Color32::from_rgb(90, 220, 90)), | executable/su
rt line; behavior impact depends on surrounding block context. |
| 7137 | ([0.0, 0.0, 0.0], [0.0, 2.4, 0.0], Color32::from_rgb(90, 140, 255)), | executable/su
rt line; behavior impact depends on surrounding block context. |
| 7138 | ]; | executable/support line; behavior impact depends on surrounding block context. |
| 7139 | for (a, b, c) in axis { | iteration control; changing bounds/body affects throughput, ti
g, and possibly safety guarantees. |
| 7140 | sp.line_segment( | executable/support line; behavior impact depends on surrounding b
k context. |
| 7141 | [project(a[0], a[1], a[2]), project(b[0], b[1], b[2])], | executable/support lin
behavior impact depends on surrounding block context. |
| 7142 | Stroke::new(1.1, c), | executable/support line; behavior impact depends on surro
ing block context. |
| 7143 | ); | executable/support line; behavior impact depends on surrounding block context.
| 7144 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 7145 | (blank) | line is blank; visual separation only. |
| 7146 | for (i, scen) in predictions.iter().take(36).enumerate() { | iteration control; changing
unds/body affects throughput, timing, and possibly safety guarantees. |
| 7147 | let x = -2.0 + (i % 12) as f32 * 0.36; | executable/support line; behavior impact de
ds on surrounding block context. |
| 7148 | let z = -1.6 + (i / 12) as f32 * 1.0; | executable/support line; behavior impact dep
s on surrounding block context. |
| 7149 | let risk = (scen.final_rupture_probability_pct as f32 / 100.0).clamp(0.0, 1.0); | ex
table/support line; behavior impact depends on surrounding block context. |
| 7150 | let peak = (scen.peak_pressure_bar as f32 / 320.0).clamp(0.0, 1.0); | executable/sup
t line; behavior impact depends on surrounding block context. |
| 7151 | let h = (0.15 + 1.9 * (0.6 * risk + 0.4 * peak)).clamp(0.1, 2.2); | executable/suppo
line; behavior impact depends on surrounding block context. |
| 7152 | let col = Color32::from_rgb( | executable/support line; behavior impact depends on s
ounding block context. |
| 7153 | (80.0 + 170.0 * risk) as u8, | executable/support line; behavior impact depends
surrounding block context. |
| 7154 | (210.0 - 140.0 * risk) as u8, | executable/support line; behavior impact depends
surrounding block context. |
| 7155 | (90.0 + 90.0 * (1.0 - peak)) as u8, | executable/support line; behavior impact d
nds on surrounding block context. |
| 7156 | ); | executable/support line; behavior impact depends on surrounding block context.
| 7157 | (blank) | line is blank; visual separation only. |
| 7158 | let p0 = project(x, 0.0, z); | executable/support line; behavior impact depends on s
ounding block context. |
| 7159 | let p1 = project(x, h, z); | executable/support line; behavior impact depends on sur
nding block context. |
| 7160 | sp.line_segment([p0, p1], Stroke::new(2.0, col)); | executable/support line; behavio
mpact depends on surrounding block context. |
| 7161 | sp.circle_filled(p1, 2.5, col); | executable/support line; behavior impact depends o
urrounding block context. |

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| 7162 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 7163 |         ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 7164 |             RichText::new("3D axes: X=scenario index, Y=risk/pressure amplitude, Z=scenario grou
| executable/support line; behavior impact depends on surrounding block context. |
| 7165 |             .size(9.0) | executable/support line; behavior impact depends on surrounding blo
context. |
| 7166 |             .color(Color32::from_gray(145)), | executable/support line; behavior impact depe
on surrounding block context. |
| 7167 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 7168 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 7169 |         ui.label(RichText::new("EXPORT REPORTS").size(11.0).color(Color32::from_rgb(80,155,255))
| executable/support line; behavior impact depends on surrounding block context. |
| 7170 |         ui.push_id("leak::export_reports_row", \ui\ { | executable/support line; behavior impa
depends on surrounding block context. |
| 7171 |             ui.horizontal_wrapped(\ui\ { | executable/support line; behavior impact depends on sur
nding block context. |
| 7172 |                 if ui.button("CSV Runtime").clicked() { do_export_report_csv = true; } | conditiona
te; changing condition shifts runtime path and side effects. |
| 7173 |                 if ui.button("JSON Runtime").clicked() { do_export_report_json = true; } | conditiona
gate; changing condition shifts runtime path and side effects. |
| 7174 |                 if ui.button("CSV Prediction").clicked() { do_export_pred_csv = true; } | conditiona
ate; changing condition shifts runtime path and side effects. |
| 7175 |                 if ui.button("JSON Prediction").clicked() { do_export_pred_json = true; } | conditiona
gate; changing condition shifts runtime path and side effects. |
| 7176 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 7177 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 7178 |             ui.label(RichText::new("ENGINEERING CATALOG EXPORT").size(11.0).color(Color32::from_rgb(
155,255))); | executable/support line; behavior impact depends on surrounding block context. |
| 7179 |             ui.push_id("leak::export_catalog_row", \ui\ { | executable/support line; behavior impa
depends on surrounding block context. |
| 7180 |                 ui.horizontal_wrapped(\ui\ { | executable/support line; behavior impact depends on sur
nding block context. |
| 7181 |                     if ui.button("CSV Materials").clicked() { do_export_catalog_mat_csv = true; } | cond
onal gate; changing condition shifts runtime path and side effects. |
| 7182 |                     if ui.button("JSON Materials").clicked() { do_export_catalog_mat_json = true; } | co
ditional gate; changing condition shifts runtime path and side effects. |
| 7183 |                     if ui.button("CSV Oils").clicked() { do_export_catalog_oil_csv = true; } | conditiona
gate; changing condition shifts runtime path and side effects. |
| 7184 |                     if ui.button("JSON Oils").clicked() { do_export_catalog_oil_json = true; } | conditi
l gate; changing condition shifts runtime path and side effects. |
| 7185 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 7186 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 7187 | (blank) | line is blank; visual separation only. |
| 7188 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 7189 |         ui.label(RichText::new("CALIBRATION MODE (BENCH CSV)").size(11.0).color(Color32::from_rg
0,155,255))); | executable/support line; behavior impact depends on surrounding block context. |
| 7190 |         ui.push_id("leak::calib_controls", \ui\ { | executable/support line; behavior impact d
nds on surrounding block context. |
| 7191 |             ui.horizontal(\ui\ { | executable/support line; behavior impact depends on surrounding
ock context. |
| 7192 |                 if ui.button("Select CSV").clicked() { do_pick_calibration_csv = true; } | conditiona
gate; changing condition shifts runtime path and side effects. |
| 7193 |                 let mut run_cal = ui.add_enabled(!self.leak_calibration_csv_path.is_empty(), Button:
w("Run Auto Calibration")); | executable/support line; behavior impact depends on surrounding block context. |
| 7194 |                 if self.leak_calibration_csv_path.is_empty() { | conditional gate; changing conditio
hifts runtime path and side effects. |
| 7195 |                     run_cal = run_cal.on_disabled_hover_text("Selecione um CSV de bancada antes de c
brar"); | executable/support line; behavior impact depends on surrounding block context. |
| 7196 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 7197 |                 if run_cal.clicked() { | conditional gate; changing condition shifts runtime path an

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ide effects. |
| 7198 | do_run_calibration = true; | executable/support line; behavior impact depends on
rounding block context. |
| 7199 | } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 7200 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 7201 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 7202 | if self.leak_calibration_csv_path.is_empty() { | conditional gate; changing condition sh
s runtime path and side effects. |
| 7203 | ui.label(RichText::new("No CSV selected").size(9.8).color(Color32::from_gray(145)));
executable/support line; behavior impact depends on surrounding block context. |
| 7204 | } else { | executable/support line; behavior impact depends on surrounding block context
| 7205 | ui.label(RichText::new(format!("CSV: {}", self.leak_calibration_csv_path)).size(9.6)
lor(Color32::LIGHT_BLUE)); | executable/support line; behavior impact depends on surrounding block context. |
| 7206 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 7207 | if let Some(rep) = &calibration_report { | conditional gate; changing condition shifts r
ime path and side effects. |
| 7208 | ui.label(RichText::new(format!( | executable/support line; behavior impact depends o
urrounding block context. |
| 7209 | "Calibrated circuits: {} \ | samples: {}", | executable/support line; behavior im
t depends on surrounding block context. |
| 7210 | rep.calibrated_circuits, rep.total_samples | executable/support line; behavior i
ct depends on surrounding block context. |
| 7211 | }).size(9.8).color(Color32::YELLOW)); | executable/support line; behavior impact dep
s on surrounding block context. |
| 7212 | for r in rep.circuit_reports.iter().take(5) { | iteration control; changing bounds/b
affects throughput, timing, and possibly safety guarantees. |
| 7213 | ui.label(RichText::new(format!( | executable/support line; behavior impact depen
on surrounding block context. |
| 7214 | "{} \ | RMSE {:.4} LPM \ | MAPE {:.2}% \ | rupture acc {:.1}%", | executable/su
rt line; behavior impact depends on surrounding block context. |
| 7215 | r.circuit_name, r.rmse_leak_lpm, r.mape_leak_pct, r.rupture_accuracy_pct | e
utable/support line; behavior impact depends on surrounding block context. |
| 7216 | }).size(9.4).color(Color32::from_gray(165))); | executable/support line; behavio
mpact depends on surrounding block context. |
| 7217 | } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 7218 | ui.push_id("leak::export_calibration_row", \ | ui\ | { | executable/support line; behav
impact depends on surrounding block context. |
| 7219 | ui.horizontal_wrapped(\ | ui\ | { | executable/support line; behavior impact depends on
rounding block context. |
| 7220 | if ui.button("CSV Calibration Report").clicked() { do_export_calibration_csv = t
; } | conditional gate; changing condition shifts runtime path and side effects. |
| 7221 | if ui.button("JSON Calibration Report").clicked() { do_export_calibration_json =
ue; } | conditional gate; changing condition shifts runtime path and side effects. |
| 7222 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 7223 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 7224 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 7225 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 7226 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 7227 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 7228 | (blank) | line is blank; visual separation only. |
| 7229 | if let Some(i) = pending_select { | conditional gate; changing condition shifts runtime path and sid
ffects. |
| 7230 | self.cmds.push(Cmd::LeakSelectCircuit(i)); | executable/support line; behavior impact depends on
rounding block context. |
| 7231 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 7232 | if do_apply && manual_ok { | conditional gate; changing condition shifts runtime path and side effec
|
| 7233 | self.cmds.push(Cmd::LeakApplyManual); | executable/support line; behavior impact depends on surr
ding block context. |
| 7234 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop

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|
| 7235 |         if do_predict { | conditional gate; changing condition shifts runtime path and side effects. |
| 7236 |             self.cmds.push(Cmd::LeakPredictScenarios); | executable/support line; behavior impact depends on
rounding block context. |
| 7237 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 7238 |         if do_add_custom && custom_ok { | conditional gate; changing condition shifts runtime path and side
ects. |
| 7239 |             self.cmds.push(Cmd::LeakAddCustomCircuit); | executable/support line; behavior impact depends on
rounding block context. |
| 7240 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 7241 |         if do_export_report_csv { | conditional gate; changing condition shifts runtime path and side effect
|
| 7242 |             let ts = Local::now().format("%Y%m%d_%H%M%S"); | executable/support line; behavior impact depend
n surrounding block context. |
| 7243 |             let name = format!("leak_report_{}.csv", ts); | executable/support line; behavior impact depends
surrounding block context. |
| 7244 |             if let Some(path) = Self::pick_save_path(&name, "csv") { | conditional gate; changing condition
fts runtime path and side effects. |
| 7245 |                 self.cmds.push(Cmd::LeakExportReportCsv(path)); | executable/support line; behavior impact d
nds on surrounding block context. |
| 7246 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 7247 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 7248 |         if do_export_report_json { | conditional gate; changing condition shifts runtime path and side effect
|
| 7249 |             let ts = Local::now().format("%Y%m%d_%H%M%S"); | executable/support line; behavior impact depend
n surrounding block context. |
| 7250 |             let name = format!("leak_report_{}.json", ts); | executable/support line; behavior impact depend
n surrounding block context. |
| 7251 |             if let Some(path) = Self::pick_save_path(&name, "json") { | conditional gate; changing condition
ifts runtime path and side effects. |
| 7252 |                 self.cmds.push(Cmd::LeakExportReportJson(path)); | executable/support line; behavior impact
ends on surrounding block context. |
| 7253 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 7254 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 7255 |         if do_export_pred_csv { | conditional gate; changing condition shifts runtime path and side effects.
| 7256 |             let ts = Local::now().format("%Y%m%d_%H%M%S"); | executable/support line; behavior impact depend
n surrounding block context. |
| 7257 |             let name = format!("leak_predictions_{}.csv", ts); | executable/support line; behavior impact de
ds on surrounding block context. |
| 7258 |             if let Some(path) = Self::pick_save_path(&name, "csv") { | conditional gate; changing condition
fts runtime path and side effects. |
| 7259 |                 self.cmds.push(Cmd::LeakExportPredCsv(path)); | executable/support line; behavior impact dep
s on surrounding block context. |
| 7260 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 7261 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 7262 |         if do_export_pred_json { | conditional gate; changing condition shifts runtime path and side effects
| 7263 |             let ts = Local::now().format("%Y%m%d_%H%M%S"); | executable/support line; behavior impact depend
n surrounding block context. |
| 7264 |             let name = format!("leak_predictions_{}.json", ts); | executable/support line; behavior impact d
nds on surrounding block context. |
| 7265 |             if let Some(path) = Self::pick_save_path(&name, "json") { | conditional gate; changing condition
ifts runtime path and side effects. |
| 7266 |                 self.cmds.push(Cmd::LeakExportPredJson(path)); | executable/support line; behavior impact de
ds on surrounding block context. |
| 7267 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 7268 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop

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|
| 7269 |         if do_monte_carlo { | conditional gate; changing condition shifts runtime path and side effects. |
| 7270 |             self.cmds.push(Cmd::LeakRunMonteCarlo); | executable/support line; behavior impact depends on su
unding block context. |
| 7271 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 7272 |         if do_export_catalog_mat_csv { | conditional gate; changing condition shifts runtime path and side e
cts. |
| 7273 |             let ts = Local::now().format("%Y%m%d_%H%M%S"); | executable/support line; behavior impact depend
n surrounding block context. |
| 7274 |             let name = format!("materials_catalog_{}.csv", ts); | executable/support line; behavior impact d
nds on surrounding block context. |
| 7275 |             if let Some(path) = Self::pick_save_path(&name, "csv") { | conditional gate; changing condition
fts runtime path and side effects. |
| 7276 |                 self.cmds.push(Cmd::LeakExportCatalogMaterialsCsv(path)); | executable/support line; behavio
mpact depends on surrounding block context. |
| 7277 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 7278 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 7279 |         if do_export_catalog_mat_json { | conditional gate; changing condition shifts runtime path and side
ects. |
| 7280 |             let ts = Local::now().format("%Y%m%d_%H%M%S"); | executable/support line; behavior impact depend
n surrounding block context. |
| 7281 |             let name = format!("materials_catalog_{}.json", ts); | executable/support line; behavior impact
ends on surrounding block context. |
| 7282 |             if let Some(path) = Self::pick_save_path(&name, "json") { | conditional gate; changing condition
ifts runtime path and side effects. |
| 7283 |                 self.cmds.push(Cmd::LeakExportCatalogMaterialsJson(path)); | executable/support line; behavi
impact depends on surrounding block context. |
| 7284 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 7285 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 7286 |         if do_export_catalog_oil_csv { | conditional gate; changing condition shifts runtime path and side e
cts. |
| 7287 |             let ts = Local::now().format("%Y%m%d_%H%M%S"); | executable/support line; behavior impact depend
n surrounding block context. |
| 7288 |             let name = format!("oils_catalog_{}.csv", ts); | executable/support line; behavior impact depend
n surrounding block context. |
| 7289 |             if let Some(path) = Self::pick_save_path(&name, "csv") { | conditional gate; changing condition
fts runtime path and side effects. |
| 7290 |                 self.cmds.push(Cmd::LeakExportCatalogOilsCsv(path)); | executable/support line; behavior imp
depends on surrounding block context. |
| 7291 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 7292 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 7293 |         if do_export_catalog_oil_json { | conditional gate; changing condition shifts runtime path and side
ects. |
| 7294 |             let ts = Local::now().format("%Y%m%d_%H%M%S"); | executable/support line; behavior impact depend
n surrounding block context. |
| 7295 |             let name = format!("oils_catalog_{}.json", ts); | executable/support line; behavior impact dependen
on surrounding block context. |
| 7296 |             if let Some(path) = Self::pick_save_path(&name, "json") { | conditional gate; changing condition
ifts runtime path and side effects. |
| 7297 |                 self.cmds.push(Cmd::LeakExportCatalogOilsJson(path)); | executable/support line; behavior im
t depends on surrounding block context. |
| 7298 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 7299 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 7300 |         if do_pick_calibration_csv { | conditional gate; changing condition shifts runtime path and side eff
s. |
| 7301 |             if let Some(path) = Self::pick_open_path("csv") { | conditional gate; changing condition shifts

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time path and side effects. |
| 7302 |           self.leak_calibration_csv_path = path; | executable/support line; behavior impact depends on
rounding block context. |
| 7303 |           self.leak_note = "CSV de bancada selecionado".into(); | executable/support line; behavior im
t depends on surrounding block context. |
| 7304 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 7305 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 7306 |           if do_run_calibration { | conditional gate; changing condition shifts runtime path and side effects.
| 7307 |           self.cmds.push(Cmd::LeakCalibrateFromCsv( | executable/support line; behavior impact depends on
rounding block context. |
| 7308 |           self.leak_calibration_csv_path.clone(), | executable/support line; behavior impact depends o
urrounding block context. |
| 7309 |           )); | executable/support line; behavior impact depends on surrounding block context. |
| 7310 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 7311 |           if do_export_calibration_csv { | conditional gate; changing condition shifts runtime path and side e
cts. |
| 7312 |           let ts = Local::now().format("%Y%m%d_%H%M%S"); | executable/support line; behavior impact depend
n surrounding block context. |
| 7313 |           let name = format!("leak_calibration_report_{}.csv", ts); | executable/support line; behavior im
t depends on surrounding block context. |
| 7314 |           if let Some(path) = Self::pick_save_path(&name, "csv") { | conditional gate; changing condition
fts runtime path and side effects. |
| 7315 |           self.cmds.push(Cmd::LeakExportCalibrationCsv(path)); | executable/support line; behavior imp
depends on surrounding block context. |
| 7316 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 7317 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 7318 |           if do_export_calibration_json { | conditional gate; changing condition shifts runtime path and side
ects. |
| 7319 |           let ts = Local::now().format("%Y%m%d_%H%M%S"); | executable/support line; behavior impact depend
n surrounding block context. |
| 7320 |           let name = format!("leak_calibration_report_{}.json", ts); | executable/support line; behavior i
ct depends on surrounding block context. |
| 7321 |           if let Some(path) = Self::pick_save_path(&name, "json") { | conditional gate; changing condition
ifts runtime path and side effects. |
| 7322 |           self.cmds.push(Cmd::LeakExportCalibrationJson(path)); | executable/support line; behavior im
t depends on surrounding block context. |
| 7323 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 7324 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 7325 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 7326 | (blank) | line is blank; visual separation only. |
| 7327 |     fn tab_electrical(&mut self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter
I, control flow, and outputs. |
| 7328 |         let bcm = &self.bench.bcm; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7329 |         let vcm = &self.bench.vcm; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7330 |         let electrical = self.bench.electrical.clone(); | executable/support line; behavior impact depends o
urrounding block context. |
| 7331 |         let snapshot = electrical.snapshot.clone(); | executable/support line; behavior impact depends on su
unding block context. |
| 7332 |         let branches = electrical.branches.clone(); | executable/support line; behavior impact depends on su
unding block context. |
| 7333 |         let ecm_on = self.electrical_node_online(auto_breaking::j1939::addr::ECM_1); | executable/support li
behavior impact depends on surrounding block context. |
| 7334 |         let tcm_on = self.electrical_node_online(auto_breaking::j1939::addr::TRANSMISSION); | executable/sup
t line; behavior impact depends on surrounding block context. |
| 7335 |         let abs_on = self.electrical_node_online(auto_breaking::j1939::addr::BRAKES); | executable/support l
; behavior impact depends on surrounding block context. |

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| 7336 |         let hcm_on = self.electrical_node_online(auto_breaking::j1939::addr::HITCH); | executable/support line
behavior impact depends on surrounding block context. |
| 7337 |         let bcm_on = self.electrical_node_online(auto_breaking::j1939::addr::CAB); | executable/support line
ehavior impact depends on surrounding block context. |
| 7338 |         let icm_on = self.electrical_node_online(auto_breaking::j1939::addr::INSTRUMENT); | executable/suppo
line; behavior impact depends on surrounding block context. |
| 7339 |         let body_load_a = branches | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7340 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 7341 |         .find(\|b\| b.id == "BODY-LOAD") | executable/support line; behavior impact depends on surroundi
block context. |
| 7342 |         .map(\|b\| b.actual_current_a) | executable/support line; behavior impact depends on surrounding
ock context. |
| 7343 |         .unwrap_or(0.0); | executable/support line; behavior impact depends on surrounding block context
| 7344 |         let battery_low = bcm.battery_voltage < 11.8; | executable/support line; behavior impact depends on
rounding block context. |
| 7345 |         let any_blowen = bcm.fuses.iter().any(\|f\| f.blowen) \| \| branches.iter().any(\|b\| b.blowen); | execu
le/support line; behavior impact depends on surrounding block context. |
| 7346 | (blank) | line is blank; visual separation only. |
| 7347 |         ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block conte
|
| 7348 |         .id_salt("electrical::page_scroll") | executable/support line; behavior impact depends on surround
ng block context. |
| 7349 |         .auto_shrink([false, false]) | executable/support line; behavior impact depends on surrounding b
k context. |
| 7350 |         .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 7351 |             ui.group(\|ui\| { | executable/support line; behavior impact depends on surrounding block co
xt. |
| 7352 |                 ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 7353 |                     RichText::new("ELECTRICAL SCHEME – live topology + harness baseline") | executable/s
ort line; behavior impact depends on surrounding block context. |
| 7354 |                     .size(12.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 7355 |                     .color(Color32::from_rgb(110, 180, 255)) | executable/support line; behavior imp
depends on surrounding block context. |
| 7356 |                     .strong(), | executable/support line; behavior impact depends on surrounding blo
context. |
| 7357 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 7358 |                 ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 7359 |                     RichText::new( | executable/support line; behavior impact depends on surrounding blo
context. |
| 7360 |                         "Live values come from BCM/VCM/boot state. Wire gauge and capacitance are engine
ng baseline assumptions for the current simulation, not OEM-certified harness data.", | executable/support line; beha
r impact depends on surrounding block context. |
| 7361 |                     ) | executable/support line; behavior impact depends on surrounding block context. |
| 7362 |                     .size(9.8) | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7363 |                     .color(Color32::from_gray(170)), | executable/support line; behavior impact depends
surrounding block context. |
| 7364 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 7365 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 7366 | (blank) | line is blank; visual separation only. |
| 7367 |             ui.add_space(6.0); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7368 |             ui.horizontal_wrapped(\|ui\| { | executable/support line; behavior impact depends on surround
g block context. |
| 7369 |                 if ui.button("Work Lights").clicked() { | conditional gate; changing condition shifts ru
me path and side effects. |
| 7370 |                     self.cmds.push(Cmd::ToggleWorkLights); | executable/support line; behavior impact de
ds on surrounding block context. |
| 7371 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |

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| 7372 |         if ui.button("Road Lights").clicked() { | conditional gate; changing condition shifts runtime path and side effects. |
| 7373 |             self.cmds.push(Cmd::ToggleRoadLights); | executable/support line; behavior impact depends on surrounding block context. |
| 7374 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 7375 |         if ui.button("Beacon").clicked() { | conditional gate; changing condition shifts runtime path and side effects. |
| 7376 |             self.cmds.push(Cmd::ToggleBeacon); | executable/support line; behavior impact depends on surrounding block context. |
| 7377 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 7378 |         if ui.button("HVAC").clicked() { | conditional gate; changing condition shifts runtime path and side effects. |
| 7379 |             self.cmds.push(Cmd::ToggleHvac); | executable/support line; behavior impact depends on surrounding block context. |
| 7380 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 7381 |         if ui.button("Horn").clicked() { | conditional gate; changing condition shifts runtime path and side effects. |
| 7382 |             self.cmds.push(Cmd::Honk); | executable/support line; behavior impact depends on surrounding block context. |
| 7383 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 7384 |         if ui.button("Wiper").clicked() { | conditional gate; changing condition shifts runtime path and side effects. |
| 7385 |             self.cmds.push(Cmd::CycleWiper); | executable/support line; behavior impact depends on surrounding block context. |
| 7386 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 7387 |         if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 7388 |             .add_enabled(any_blown, Button::new("Reset all fuses")) | executable/support line; behavior impact depends on surrounding block context. |
| 7389 |             .clicked() | executable/support line; behavior impact depends on surrounding block context. |
| 7390 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 7391 |             self.cmds.push(Cmd::ResetBcmFuses); | executable/support line; behavior impact depends on surrounding block context. |
| 7392 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 7393 |         if ui.button("Reset electrical faults").clicked() { | conditional gate; changing condition shifts runtime path and side effects. |
| 7394 |             self.cmds.push(Cmd::ElectricalResetFaults); | executable/support line; behavior impact depends on surrounding block context. |
| 7395 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 7396 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 7397 | (blank) | line is blank; visual separation only. |
| 7398 |     ui.add_space(6.0); | executable/support line; behavior impact depends on surrounding block context. |
| 7399 |     egui::Frame::group(ui.style()) | executable/support line; behavior impact depends on surrounding block context. |
| 7400 |         .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on surrounding block context. |
| 7401 |         .show(ui, \ui\ { | executable/support line; behavior impact depends on surrounding block context. |
| 7402 |             ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 7403 |                 RichText::new("30 / 15 / 31 + RELAY / GROUND CONTROL") | executable/support line; behavior impact depends on surrounding block context. |
| 7404 |                 .size(11.0) | executable/support line; behavior impact depends on surrounding block context. |
| 7405 |                 .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior impact depends on surrounding block context. |

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| 7406 |           ); | executable/support line; behavior impact depends on surrounding block context.
| 7407 |           ui.horizontal_wrapped(\|ui\| { | executable/support line; behavior impact depends on
rrounding block context. |
| 7408 |           ui.label(RichText::new(format!("Line30 {:.2}V", snapshot.line30_v)).color(Color3
YELLOW)); | executable/support line; behavior impact depends on surrounding block context. |
| 7409 |           ui.label(RichText::new(format!("Line15-ACC {:.2}V", snapshot.line15_acc_v)).colo
olor32::LIGHT_BLUE)); | executable/support line; behavior impact depends on surrounding block context. |
| 7410 |           ui.label(RichText::new(format!("Line15-IGN {:.2}V", snapshot.line15_ign_v)).colo
olor32::LIGHT_BLUE)); | executable/support line; behavior impact depends on surrounding block context. |
| 7411 |           ui.label(RichText::new(format!("Line31 drop {:.2}V", snapshot.ground_drop_v)).co
(Color32::from_rgb(255, 140, 80))); | executable/support line; behavior impact depends on surrounding block context. |
| 7412 |           ui.label(RichText::new(format!("Total I {:.1}A", snapshot.total_current_a)).colo
olor32::from_gray(190))); | executable/support line; behavior impact depends on surrounding block context. |
| 7413 |           }); | executable/support line; behavior impact depends on surrounding block context.
| 7414 |           for (point, title, fault) in [ | iteration control; changing bounds/body affects thr
hput, timing, and possibly safety guarantees. |
| 7415 |           (ElectricalControlPoint::MainGround, "Main ground", electrical.main_ground_fault
| executable/support line; behavior impact depends on surrounding block context. |
| 7416 |           (ElectricalControlPoint::AccessoryRelay, "Accessory relay", electrical.accessory
lay_fault), | executable/support line; behavior impact depends on surrounding block context. |
| 7417 |           (ElectricalControlPoint::IgnitionRelay, "Ignition relay", electrical.ignition_re
_fault), | executable/support line; behavior impact depends on surrounding block context. |
| 7418 |           ] { | executable/support line; behavior impact depends on surrounding block context.
| 7419 |           ui.horizontal_wrapped(\|ui\| { | executable/support line; behavior impact depend
n surrounding block context. |
| 7420 |           let c = if fault == ElectricalFaultMode::None { Color32::GREEN } else { Colo
::RED }; | executable/support line; behavior impact depends on surrounding block context. |
| 7421 |           ui.label(RichText::new(format!("{}", title, fault)).color(c).strong());
executable/support line; behavior impact depends on surrounding block context. |
| 7422 |           for (label, mode) in [ | iteration control; changing bounds/body affects thr
hput, timing, and possibly safety guarantees. |
| 7423 |           ("OK", ElectricalFaultMode::None), | executable/support line; behavior i
ct depends on surrounding block context. |
| 7424 |           ("OPEN", ElectricalFaultMode::OpenCircuit), | executable/support line; b
vior impact depends on surrounding block context. |
| 7425 |           ("HI-R", ElectricalFaultMode::HighResistance), | executable/support line
ehavior impact depends on surrounding block context. |
| 7426 |           ] { | executable/support line; behavior impact depends on surrounding block
text. |
| 7427 |           if ui.small_button(label).clicked() { | conditional gate; changing condi
n shifts runtime path and side effects. |
| 7428 |           self.cmds.push(Cmd::ElectricalSetControlFault(point, mode)); | execu
le/support line; behavior impact depends on surrounding block context. |
| 7429 |           } | scope delimiter; structure-only line, but misplaced edits can change
nership and lifetime scopes. |
| 7430 |           } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 7431 |           }); | executable/support line; behavior impact depends on surrounding block cont
. |
| 7432 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 7433 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 7434 | (blank) | line is blank; visual separation only. |
| 7435 |           ui.columns(2, \|cols\| { | executable/support line; behavior impact depends on surrounding b
k context. |
| 7436 |           equi::Frame::group(cols[0].style()) | executable/support line; behavior impact depends o
urrounding block context. |
| 7437 |           .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on
rounding block context. |
| 7438 |           .show(&mut cols[0], \|ui\| { | executable/support line; behavior impact depends on s
ounding block context. |
| 7439 |           ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 7440 |           RichText::new("POWER OVERVIEW") | executable/support line; behavior impact d
nds on surrounding block context. |

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| 7441 |                                     .size(11.0) | executable/support line; behavior impact depends on surround
ng block context. |
| 7442 |                                     .color(Color32::from_rgb(80, 155, 255)), | executable/support line; beha
r impact depends on surrounding block context. |
| 7443 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7444 |                                     let batt_col = if battery_low { | executable/support line; behavior impact dependen
on surrounding block context. |
| 7445 |                                     Color32::RED | executable/support line; behavior impact depends on surrounding
block context. |
| 7446 |                                     } else { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 7447 |                                     Color32::GREEN | executable/support line; behavior impact depends on surround
g block context. |
| 7448 |                                     }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 7449 |                                     bar_gauge(ui, "Battery", bcm.battery_voltage - 9.0, 7.0, "V", 135.0, batt_col);
xecutable/support line; behavior impact depends on surrounding block context. |
| 7450 |                                     digital_readout(ui, "Batt current", &format!("{:.1} A", bcm.battery_current_a),
lor32::from_gray(190)); | executable/support line; behavior impact depends on surrounding block context. |
| 7451 |                                     digital_readout(ui, "Alternator", &format!("{:.0} A", bcm.alternator_amps), Colo
::from_rgb(110, 210, 110)); | executable/support line; behavior impact depends on surrounding block context. |
| 7452 |                                     digital_readout(ui, "ECM branch", if ecm_on { "energized" } else { "off" }, if e
on { Color32::GREEN } else { Color32::RED }); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 7453 |                                     digital_readout(ui, "TCM branch", if tcm_on { "energized" } else { "off" }, if t
on { Color32::GREEN } else { Color32::RED }); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 7454 |                                     digital_readout(ui, "Electrical", &format!("{:.2} kW", vcm.power.electrical_dema
kW), Color32::YELLOW); | executable/support line; behavior impact depends on surrounding block context. |
| 7455 |                                     digital_readout(ui, "Total load", &format!("{:.1} kW", vcm.power.total_load_kW),
vcm.power.overload { Color32::RED } else { Color32::from_gray(190) }); | executable/support line; behavior impact de
ds on surrounding block context. |
| 7456 |                                     digital_readout(ui, "Load shed", if vcm.power.load_shed_active { vcm.power.load_
d_reason.unwrap_or("active") } else { "off" }, if vcm.power.load_shed_active { Color32::RED } else { Color32::GREEN }
| executable/support line; behavior impact depends on surrounding block context. |
| 7457 |                                     ui.separator(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 7458 |                                     ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 7459 |                                     RichText::new("FUSE MATRIX") | executable/support line; behavior impact depende
on surrounding block context. |
| 7460 |                                     .size(11.0) | executable/support line; behavior impact depends on surround
ng block context. |
| 7461 |                                     .color(Color32::from_rgb(80, 155, 255)), | executable/support line; beha
r impact depends on surrounding block context. |
| 7462 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7463 |                                     ScrollArea::vertical() | executable/support line; behavior impact depends on sur
nding block context. |
| 7464 |                                     .id_salt("electrical::fuse_scroll") | executable/support line; behavior impa
depends on surrounding block context. |
| 7465 |                                     .max_height(260.0) | executable/support line; behavior impact depends on sur
nding block context. |
| 7466 |                                     .auto_shrink([false, false]) | executable/support line; behavior impact depende
on surrounding block context. |
| 7467 |                                     .show(ui, \|ui\| { | executable/support line; behavior impact depends on sur
nding block context. |
| 7468 |                                     for fuse in &bcm.fuses { | iteration control; changing bounds/body affect
throughput, timing, and possibly safety guarantees. |
| 7469 |                                     let gauge = Self::electrical_wire_mm2(fuse.rating_a); | executable/s
ort line; behavior impact depends on surrounding block context. |
| 7470 |                                     let cap_total = 3.0 * Self::electrical_cap_nf_per_m(gauge, false); |
ecutable/support line; behavior impact depends on surrounding block context. |
| 7471 |                                     let ratio = if fuse.rating_a > 0.0 { | executable/support line; beha

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r impact depends on surrounding block context. |
| 7472 | (fuse.load_a / fuse.rating_a).clamp(0.0, 1.4) | executable/support
line; behavior impact depends on surrounding block context. |
| 7473 | } else { | executable/support line; behavior impact depends on surro
ing block context. |
| 7474 | 0.0 | executable/support line; behavior impact depends on surrou
ng block context. |
| 7475 | }; | scope delimiter; structure-only line, but misplaced edits can c
ge ownership and lifetime scopes. |
| 7476 | let row_col = if fuse.blown { | executable/support line; behavior im
t depends on surrounding block context. |
| 7477 | Color32::RED | executable/support line; behavior impact depends
surrounding block context. |
| 7478 | } else if ratio > 0.9 { | executable/support line; behavior impact d
nds on surrounding block context. |
| 7479 | Color32::YELLOW | executable/support line; behavior impact dependen
on surrounding block context. |
| 7480 | } else { | executable/support line; behavior impact depends on surro
ing block context. |
| 7481 | Color32::GREEN | executable/support line; behavior impact depend
n surrounding block context. |
| 7482 | }; | scope delimiter; structure-only line, but misplaced edits can c
ge ownership and lifetime scopes. |
| 7483 | egui::Frame::group(ui.style()) | executable/support line; behavior i
ct depends on surrounding block context. |
| 7484 | .fill(Color32::from_gray(20)) | executable/support line; behavior
mpact depends on surrounding block context. |
| 7485 | .show(ui, \ui\ { | executable/support line; behavior impact de
ds on surrounding block context. |
| 7486 | ui.horizontal(\ui\ { | executable/support line; behavior i
ct depends on surrounding block context. |
| 7487 | ui.label(RichText::new(fuse.id).size(10.6).color(row_col
trong())); | executable/support line; behavior impact depends on surrounding block context. |
| 7488 | ui.separator(); | executable/support line; behavior impa
depends on surrounding block context. |
| 7489 | ui.label(RichText::new(format!("{:.1}/{:.1} A", fuse loa
, fuse.rating_a)).size(10.0).monospace().color(Color32::from_gray(195))); | executable/support line; behavior impact
ends on surrounding block context. |
| 7490 | ui.separator(); | executable/support line; behavior impa
depends on surrounding block context. |
| 7491 | ui.label(RichText::new(format!("{:.2} mm2", gauge)).size
8).color(Color32::from_rgb(140, 210, 255))); | executable/support line; behavior impact depends on surrounding block
text. |
| 7492 | ui.separator(); | executable/support line; behavior impa
depends on surrounding block context. |
| 7493 | ui.label(RichText::new(format!("C~{:.2} nF", cap_total))
ze(9.8).color(Color32::from_gray(165))); | executable/support line; behavior impact depends on surrounding block cont
. |
| 7494 | if fuse.blown { | conditional gate; changing condition s
ts runtime path and side effects. |
| 7495 | ui.separator(); | executable/support line; behavior
act depends on surrounding block context. |
| 7496 | ui.label(RichText::new("OPEN").size(9.8).color(Color
:RED).strong()); | executable/support line; behavior impact depends on surrounding block context. |
| 7497 | } | scope delimiter; structure-only line, but misplaced
ts can change ownership and lifetime scopes. |
| 7498 | }); | executable/support line; behavior impact depends on su
unding block context. |
| 7499 | ui.add(ProgressBar::new((ratio / 1.15).clamp(0.0, 1.0) as f3
text("fuse utilization")); | executable/support line; behavior impact depends on surrounding block context. |
| 7500 | }); | executable/support line; behavior impact depends on surrou
ng block context. |
| 7501 | } | scope delimiter; structure-only line, but misplaced edits can change
nership and lifetime scopes. |
| 7502 | }); | executable/support line; behavior impact depends on surrounding block

```

```

text. |
| 7503 |           }); | executable/support line; behavior impact depends on surrounding block context.
| 7504 | (blank) | line is blank; visual separation only. |
| 7505 |           QtGui::Frame::group(cols[1].style()) | executable/support line; behavior impact depends on
surrounding block context. |
| 7506 |           .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on
surrounding block context. |
| 7507 |           .show(&mut cols[1], \\ui\\ { | executable/support line; behavior impact depends on s
urrounding block context. |
| 7508 |           ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 7509 |           RichText::new("LIVE ELECTRICAL TOPOLOGY") | executable/support line; behavior
mpact depends on surrounding block context. |
| 7510 |           .size(11.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 7511 |           .color(Color32::from_rgb(80, 155, 255)), | executable/support line; beha
r impact depends on surrounding block context. |
| 7512 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7513 |           let (rect, _) = ui.allocate_exact_size( | executable/support line; behavior impa
depends on surrounding block context. |
| 7514 |           Vec2::new(ui.available_width(), 340.0), | executable/support line; behavior
act depends on surrounding block context. |
| 7515 |           Sense::hover(), | executable/support line; behavior impact depends on surrou
ng block context. |
| 7516 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7517 |           let p = ui.painter_at(rect); | executable/support line; behavior impact depends
surrounding block context. |
| 7518 |           p.rect_filled(rect, 6.0, Color32::from_gray(10)); | executable/support line; beh
or impact depends on surrounding block context. |
| 7519 |           p.rect_stroke(rect, 6.0, Stroke::new(1.0, Color32::from_gray(55))); | executable
pport line; behavior impact depends on surrounding block context. |
| 7520 | (blank) | line is blank; visual separation only. |
| 7521 |           let map = \\x: f32, y: f32\\ -> Pos2 { | executable/support line; behavior impac
depends on surrounding block context. |
| 7522 |           Pos2::new(rect.left() + x * rect.width(), rect.top() + y * rect.height()) |
cutable/support line; behavior impact depends on surrounding block context. |
| 7523 |           }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 7524 |           let battery = map(0.12, 0.20); | executable/support line; behavior impact depend
n surrounding block context. |
| 7525 |           let alternator = map(0.12, 0.62); | executable/support line; behavior impact dep
s on surrounding block context. |
| 7526 |           let fuse_box = map(0.35, 0.20); | executable/support line; behavior impact depen
on surrounding block context. |
| 7527 |           let bcm_p = map(0.56, 0.10); | executable/support line; behavior impact depends
surrounding block context. |
| 7528 |           let vcm_p = map(0.56, 0.33); | executable/support line; behavior impact depends
surrounding block context. |
| 7529 |           let hcm_p = map(0.56, 0.58); | executable/support line; behavior impact depends
surrounding block context. |
| 7530 |           let icm_p = map(0.56, 0.82); | executable/support line; behavior impact depends
surrounding block context. |
| 7531 |           let ecm_p = map(0.83, 0.10); | executable/support line; behavior impact depends
surrounding block context. |
| 7532 |           let tcm_p = map(0.83, 0.33); | executable/support line; behavior impact depends
surrounding block context. |
| 7533 |           let abs_p = map(0.83, 0.58); | executable/support line; behavior impact depends
surrounding block context. |
| 7534 |           let body_p = map(0.35, 0.82); | executable/support line; behavior impact depends
surrounding block context. |
| 7535 | (blank) | line is blank; visual separation only. |
| 7536 |           let draw_link = \\p: &Painter, a: Pos2, b: Pos2, energized: bool, can_link: bool
{ | executable/support line; behavior impact depends on surrounding block context. |

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| 7537 | let col = if can_link { | executable/support line; behavior impact depends on
surrounding block context. |
| 7538 | if energized { Color32::from_rgb(90, 220, 220) } else { Color32::from_gr
60) } | conditional gate; changing condition shifts runtime path and side effects. |
| 7539 | } else if energized { | executable/support line; behavior impact depends on
rounding block context. |
| 7540 | Color32::from_rgb(255, 210, 90) | executable/support line; behavior impa
depends on surrounding block context. |
| 7541 | } else { | executable/support line; behavior impact depends on surrounding b
k context. |
| 7542 | Color32::from_gray(65) | executable/support line; behavior impact depend
n surrounding block context. |
| 7543 | }; | scope delimiter; structure-only line, but misplaced edits can change ow
ship and lifetime scopes. |
| 7544 | p.line_segment([a, b], Stroke::new(if can_link { 2.0 } else { 2.8 }, col));
executable/support line; behavior impact depends on surrounding block context. |
| 7545 | }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 7546 | (blank) | line is blank; visual separation only. |
| 7547 | let draw_node = \|p: &Painter, pos: Pos2, title: &str, detail: &str, active: boo
fill: Color32\| { | executable/support line; behavior impact depends on surrounding block context. |
| 7548 | let r = Rect::from_center_size(pos, Vec2::new(96.0, 34.0)); | executable/sup
t line; behavior impact depends on surrounding block context. |
| 7549 | p.rect_filled(r, 5.0, if active { fill } else { Color32::from_gray(24) }); |
executable/support line; behavior impact depends on surrounding block context. |
| 7550 | p.rect_stroke(r, 5.0, Stroke::new(1.0, if active { Color32::WHITE } else { C
r32::from_gray(75) })); | executable/support line; behavior impact depends on surrounding block context. |
| 7551 | p.text(r.center_top() + Vec2::new(0.0, 4.0), Align2::CENTER_TOP, title, Font
:proportional(11.0), Color32::WHITE); | executable/support line; behavior impact depends on surrounding block context. |
| 7552 | p.text(r.center_bottom() - Vec2::new(0.0, 4.0), Align2::CENTER_BOTTOM, detail
FontId::proportional(9.0), Color32::from_gray(225)); | executable/support line; behavior impact depends on surroundin
lock context. |
| 7553 | }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 7554 | (blank) | line is blank; visual separation only. |
| 7555 | draw_link(&p, battery, fuse_box, branches.iter().any(\|b\| b.id == "BAT-MAIN" &&
energized), false); | executable/support line; behavior impact depends on surrounding block context. |
| 7556 | draw_link(&p, alternator, battery, branches.iter().any(\|b\| b.id == "ALT-B+" &&
energized), false); | executable/support line; behavior impact depends on surrounding block context. |
| 7557 | draw_link(&p, fuse_box, bcm_p, branches.iter().any(\|b\| b.id == "BCM-PWR" && b.
rgized), false); | executable/support line; behavior impact depends on surrounding block context. |
| 7558 | draw_link(&p, fuse_box, vcm_p, branches.iter().any(\|b\| b.id == "VCM-PWR" && b.
rgized), false); | executable/support line; behavior impact depends on surrounding block context. |
| 7559 | draw_link(&p, fuse_box, hcm_p, branches.iter().any(\|b\| b.id == "HCM-PWR" && b.
rgized), false); | executable/support line; behavior impact depends on surrounding block context. |
| 7560 | draw_link(&p, fuse_box, body_p, branches.iter().any(\|b\| b.id == "BODY-LOAD" &&
energized), false); | executable/support line; behavior impact depends on surrounding block context. |
| 7561 | draw_link(&p, bcm_p, icm_p, branches.iter().any(\|b\| b.id == "ICM-PWR" && b.ene
zed), false); | executable/support line; behavior impact depends on surrounding block context. |
| 7562 | draw_link(&p, vcm_p, ecm_p, branches.iter().any(\|b\| b.id == "ECM-PWR" && b.ene
zed), false); | executable/support line; behavior impact depends on surrounding block context. |
| 7563 | draw_link(&p, vcm_p, tcm_p, branches.iter().any(\|b\| b.id == "TCM-PWR" && b.ene
zed), false); | executable/support line; behavior impact depends on surrounding block context. |
| 7564 | draw_link(&p, vcm_p, abs_p, branches.iter().any(\|b\| b.id == "ABS-PWR" && b.ene
zed), false); | executable/support line; behavior impact depends on surrounding block context. |
| 7565 | draw_link(&p, bcm_p, vcm_p, branches.iter().any(\|b\| b.id == "MS-CAN-BODY" && b.
energized), true); | executable/support line; behavior impact depends on surrounding block context. |
| 7566 | draw_link(&p, vcm_p, ecm_p, branches.iter().any(\|b\| b.id == "HS-CAN-1" && b.ene
ized), true); | executable/support line; behavior impact depends on surrounding block context. |
| 7567 | draw_link(&p, vcm_p, tcm_p, branches.iter().any(\|b\| b.id == "HS-CAN-1" && b.ene
ized), true); | executable/support line; behavior impact depends on surrounding block context. |
| 7568 | draw_link(&p, vcm_p, abs_p, branches.iter().any(\|b\| b.id == "HS-CAN-1" && b.ene
ized), true); | executable/support line; behavior impact depends on surrounding block context. |
| 7569 | draw_link(&p, vcm_p, hcm_p, branches.iter().any(\|b\| b.id == "HS-CAN-1" && b.ene
ized), true); | executable/support line; behavior impact depends on surrounding block context. |

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| 7570 | draw_link(&p, bcm_p, icm_p, branches.iter().any(|b| b.id == "MS-CAN-BODY" && b
ergized), true); | executable/support line; behavior impact depends on surrounding block context. |
| 7571 | (blank) | line is blank; visual separation only. |
| 7572 | draw_node(&p, battery, "BATTERY", &format!("{:.1} V / {:.0}%", bcm.battery_volta
bcm.battery_soc_pct), true, if battery_low { Color32::from_rgb(110, 25, 25) } else { Color32::from_rgb(25, 95, 35) }
| executable/support line; behavior impact depends on surrounding block context. |
| 7573 | draw_node(&p, alternator, "ALTERNATOR", &format!("{:.0} A", bcm.alternator_amps)
cm.alternator_amps > 0.5, Color32::from_rgb(80, 120, 30)); | executable/support line; behavior impact depends on surr
ding block context. |
| 7574 | draw_node(&p, fuse_box, "MAIN FUSE BOX", if any_blowed { "open / short detected"
lse { "all branches closed" }, true, if any_blowed { Color32::from_rgb(100, 35, 20) } else { Color32::from_rgb(80, 80
) }); | executable/support line; behavior impact depends on surrounding block context. |
| 7575 | draw_node(&p, bcm_p, "BCM", &format!("{:.0} A load", bcm.total_load_amps), bcm_o
Color32::from_rgb(30, 70, 105)); | executable/support line; behavior impact depends on surrounding block context. |
| 7576 | draw_node(&p, vcm_p, "VCM", if vcm.power.load_shed_active { "load shed" } else {
ower coord" }, true, Color32::from_rgb(55, 80, 120)); | executable/support line; behavior impact depends on surroundi
block context. |
| 7577 | draw_node(&p, hcm_p, "HCM", if hcm_on { "hyd online" } else { "offline" }, hcm_o
Color32::from_rgb(45, 95, 110)); | executable/support line; behavior impact depends on surrounding block context. |
| 7578 | draw_node(&p, icm_p, "ICM", if icm_on { "cluster online" } else { "offline" }, i
on, Color32::from_rgb(75, 70, 110)); | executable/support line; behavior impact depends on surrounding block context.
| 7579 | draw_node(&p, ecm_p, "ECM", if ecm_on { "engine online" } else { "offline" }, ec
n, Color32::from_rgb(35, 105, 70)); | executable/support line; behavior impact depends on surrounding block context.
| 7580 | draw_node(&p, tcm_p, "TCM", if tcm_on { "trans online" } else { "offline" }, tcm
, Color32::from_rgb(90, 90, 40)); | executable/support line; behavior impact depends on surrounding block context. |
| 7581 | draw_node(&p, abs_p, "ABS/ESP", if abs_on { "brake online" } else { "offline" },
s_on, Color32::from_rgb(110, 55, 45)); | executable/support line; behavior impact depends on surrounding block context
|
| 7582 | draw_node(&p, body_p, "BODY LOADS", &format!("{:.0} A", body_load_a), body_load_
0.1, Color32::from_rgb(70, 55, 100)); | executable/support line; behavior impact depends on surrounding block context
|
| 7583 | (blank) | line is blank; visual separation only. |
| 7584 | p.text(map(0.52, 0.03), Align2::CENTER_TOP, "yellow = power branch \ | cyan = CAN
ckbone", FontId::proportional(9.0), Color32::from_gray(170)); | executable/support line; behavior impact depends on s
ounding block context. |
| 7585 | }); | executable/support line; behavior impact depends on surrounding block context.
| 7586 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 7587 | (blank) | line is blank; visual separation only. |
| 7588 | ui.add_space(8.0); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7589 | egui::Frame::group(ui.style()) | executable/support line; behavior impact depends on surround
g block context. |
| 7590 | .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on surr
ding block context. |
| 7591 | .show(ui, \ | ui \ { | executable/support line; behavior impact depends on surrounding blo
context. |
| 7592 | ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 7593 | RichText::new("HARNESS / ELECTRICAL BRANCH REFERENCE") | executable/support line
ehavior impact depends on surrounding block context. |
| 7594 | .size(11.0) | executable/support line; behavior impact depends on surrounding
lock context. |
| 7595 | .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior
pact depends on surrounding block context. |
| 7596 | ); | executable/support line; behavior impact depends on surrounding block context.
| 7597 | ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 7598 | RichText::new("Gauge, resistance, capacitance and drop below are computed from t
current branch model. Fault buttons act on the live branch and affect ECU power in the simulation.") | executable/sup
t line; behavior impact depends on surrounding block context. |
| 7599 | .size(9.6) | executable/support line; behavior impact depends on surrounding
ock context. |
| 7600 | .color(Color32::from_gray(160)), | executable/support line; behavior impact
ends on surrounding block context. |
| 7601 | ); | executable/support line; behavior impact depends on surrounding block context.

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| 7602 | ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding
g block context. |
| 7603 | .id_salt("electrical::branch_scroll") | executable/support line; behavior impact
depends on surrounding block context. |
| 7604 | .max_height(290.0) | executable/support line; behavior impact depends on surrounding
g block context. |
| 7605 | .auto_shrink([false, false]) | executable/support line; behavior impact depends
surrounding block context. |
| 7606 | .show(ui, \ui\ { | executable/support line; behavior impact depends on surrounding
g block context. |
| 7607 | for branch in &branches { | iteration control; changing bounds/body affects
oughput, timing, and possibly safety guarantees. |
| 7608 | let cap_total = branch.capacitance_nf; | executable/support line; behavior
impact depends on surrounding block context. |
| 7609 | let branch_col = if branch.energized { | executable/support line; behavior
impact depends on surrounding block context. |
| 7610 | if branch.line == auto_breaking::ElectricalLine::HsCan \|\| branch.l
== auto_breaking::ElectricalLine::MsCan { | conditional gate; changing condition shifts runtime path and side effect
|
| 7611 | Color32::from_rgb(90, 220, 220) | executable/support line; behavior
impact depends on surrounding block context. |
| 7612 | } else { | executable/support line; behavior impact depends on surrounding
ing block context. |
| 7613 | Color32::from_rgb(255, 210, 90) | executable/support line; behavior
impact depends on surrounding block context. |
| 7614 | } | scope delimiter; structure-only line, but misplaced edits can change
ownership and lifetime scopes. |
| 7615 | } else { | executable/support line; behavior impact depends on surrounding
block context. |
| 7616 | Color32::from_gray(90) | executable/support line; behavior impact de
ds on surrounding block context. |
| 7617 | }; | scope delimiter; structure-only line, but misplaced edits can change
wnership and lifetime scopes. |
| 7618 | ui.push_id(branch.id, \ui\ { | executable/support line; behavior impact
depends on surrounding block context. |
| 7619 | egui::Frame::group(ui.style()) | executable/support line; behavior impact
ct depends on surrounding block context. |
| 7620 | .fill(Color32::from_gray(20)) | executable/support line; behavior impact
mpact depends on surrounding block context. |
| 7621 | .show(ui, \ui\ { | executable/support line; behavior impact de
ds on surrounding block context. |
| 7622 | ui.horizontal_wrapped(\ui\ { | executable/support line; behavior impact
ior impact depends on surrounding block context. |
| 7623 | ui.label(RichText::new(branch.id).size(10.5).color(branch
ol).strong()); | executable/support line; behavior impact depends on surrounding block context. |
| 7624 | ui.separator(); | executable/support line; behavior impact
depends on surrounding block context. |
| 7625 | ui.label(RichText::new(format!("{}", branch.from,
nch.to)).size(10.0).color(Color32::from_gray(205))); | executable/support line; behavior impact depends on surrounding
lock context. |
| 7626 | ui.separator(); | executable/support line; behavior impact
depends on surrounding block context. |
| 7627 | ui.label(RichText::new(format!("Line {}", branch.line)).
e(9.8).color(Color32::from_gray(170))); | executable/support line; behavior impact depends on surrounding block conte
|
| 7628 | ui.separator(); | executable/support line; behavior impact
depends on surrounding block context. |
| 7629 | ui.label(RichText::new(format!("Fuse {}", branch.fuse_id
size(9.8).color(Color32::from_gray(170))); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 7630 | ui.separator(); | executable/support line; behavior impact
depends on surrounding block context. |
| 7631 | ui.label(RichText::new(format!("{}", branch.fault)).size
8).color(if branch.fault == ElectricalFaultMode::None { Color32::GREEN } else { Color32::RED })); | executable/support
ine; behavior impact depends on surrounding block context. |

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| 7632 |                                     }); | executable/support line; behavior impact depends on su
unding block context. |
| 7633 |                                     ui.label( | executable/support line; behavior impact depends
surrounding block context. |
| 7634 |                                     RichText::new(format!( | executable/support line; behavi
impact depends on surrounding block context. |
| 7635 |                                     "Ireq={:.2} A \ | Iact={:.2} A \ | V={:.2} V \ | R={:.4
hm \ | bitola {:.2} mm2 \ | comprimento {:.1} m \ | C {:.2} nF \ | {})", | executable/support line; behavior impact depend
n surrounding block context. |
| 7636 |                                     branch.requested_current_a, | executable/support lin
behavior impact depends on surrounding block context. |
| 7637 |                                     branch.actual_current_a, | executable/support line;
avior impact depends on surrounding block context. |
| 7638 |                                     branch.voltage_v, | executable/support line; behavior
mpact depends on surrounding block context. |
| 7639 |                                     branch.resistance_ohm, | executable/support line; be
ior impact depends on surrounding block context. |
| 7640 |                                     branch.gauge_mm2, | executable/support line; behavior
mpact depends on surrounding block context. |
| 7641 |                                     branch.length_m, | executable/support line; behavior
pact depends on surrounding block context. |
| 7642 |                                     cap_total, | executable/support line; behavior impac
ends on surrounding block context. |
| 7643 |                                     branch.note | executable/support line; behavior impa
depends on surrounding block context. |
| 7644 |                                     )) | executable/support line; behavior impact depends on
rounding block context. |
| 7645 |                                     .size(9.6) | executable/support line; behavior impact de
ds on surrounding block context. |
| 7646 |                                     .monospace() | executable/support line; behavior impact
ends on surrounding block context. |
| 7647 |                                     .color(Color32::from_gray(180)), | executable/support li
behavior impact depends on surrounding block context. |
| 7648 |                                     ); | executable/support line; behavior impact depends on sur
nding block context. |
| 7649 |                                     ui.horizontal_wrapped(\ | ui\ | { | executable/support line; be
ior impact depends on surrounding block context. |
| 7650 |                                     for (label, mode) in [ | iteration control; changing bou
/body affects throughput, timing, and possibly safety guarantees. |
| 7651 |                                     ("OK", ElectricalFaultMode::None), | executable/supp
line; behavior impact depends on surrounding block context. |
| 7652 |                                     ("OPEN", ElectricalFaultMode::OpenCircuit), | execut
e/support line; behavior impact depends on surrounding block context. |
| 7653 |                                     ("SHORT", ElectricalFaultMode::ShortToGround), | exe
able/support line; behavior impact depends on surrounding block context. |
| 7654 |                                     ("HI-R", ElectricalFaultMode::HighResistance), | exe
able/support line; behavior impact depends on surrounding block context. |
| 7655 |                                     ] { | executable/support line; behavior impact depends o
urrounding block context. |
| 7656 |                                     if ui.small_button(label).clicked() { | conditional
e; changing condition shifts runtime path and side effects. |
| 7657 |                                     self.cmds.push(Cmd::ElectricalSetBranchFault(bra
.id.to_string(), mode)); | executable/support line; behavior impact depends on surrounding block context. |
| 7658 |                                     } | scope delimiter; structure-only line, but mispla
edits can change ownership and lifetime scopes. |
| 7659 |                                     } | scope delimiter; structure-only line, but misplaced
ts can change ownership and lifetime scopes. |
| 7660 |                                     if branch.blown { | conditional gate; changing condition
ifts runtime path and side effects. |
| 7661 |                                     ui.label(RichText::new("branch fuse blown").size(9.6
olor(Color32::RED)); | executable/support line; behavior impact depends on surrounding block context. |
| 7662 |                                     } | scope delimiter; structure-only line, but misplaced
ts can change ownership and lifetime scopes. |
| 7663 |                                     }); | executable/support line; behavior impact depends on su
unding block context. |

```

```
|
7664 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 7665 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 7666 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
hip and lifetime scopes. |
| 7667 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
. |
| 7668 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 7669 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 7670 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 7671 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 7672 | (blank) | line is blank; visual separation only. |
| 7673 | // ████████████████████████████████████████████████████████████████████████████████████████████████████████ | comment intent, no runtime effect. |
| 7674 | // TAB SENSORS | comment line; documentation intent, no runtime effect. |
| 7675 | // ████████████████████████████████████████████████████████████████████████████████████████████████████████ | comment intent, no runtime effect. |
| 7676 | impl App { | implementation block; method behavior and trait conformance are defined here. |
| 7677 |     fn tab_sensors(&self, ui: &mud Ui) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 7678 |         let gps_ok = !matches!( | executable/support line; behavior impact depends on surrounding block context. |
. |
| 7679 |             self.bench.gps.fix_quality, | executable/support line; behavior impact depends on surrounding block context. | | |
| 7680 |             auto_breaking::gps::GpsFixQuality::NoFix | executable/support line; behavior impact depends on surrounding block context. |
| 7681 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 7682 |         let imu_ok = !self.bench.imu.accel_fault && !self.bench.imu.gyro_fault; | executable/support line; behavior impact depends on surrounding block context. |
| 7683 |         let radar_ok = self.bench.radar.total_targets() > 0; | executable/support line; behavior impact depends on surrounding block context. |
| 7684 |         let lidar_ok = self.bench.lidar.points_per_scan > 0; | executable/support line; behavior impact depends on surrounding block context. |
| 7685 |         let healthy_count = [gps_ok, imu_ok, radar_ok, lidar_ok] | executable/support line; behavior impact ends on surrounding block context. |
| 7686 |             .into_iter() | executable/support line; behavior impact depends on surrounding block context. |
| 7687 |             .filter(|v| *v) | executable/support line; behavior impact depends on surrounding block context. |
| 7688 |             .count(); | executable/support line; behavior impact depends on surrounding block context. |
| 7689 |         let sys_col = if healthy_count >= 4 { | executable/support line; behavior impact depends on surrounding block context. |
| 7690 |             Color32::GREEN | executable/support line; behavior impact depends on surrounding block context. |
| 7691 |         } else if healthy_count >= 3 { | executable/support line; behavior impact depends on surrounding block context. |
context. |
| 7692 |             Color32::YELLOW | executable/support line; behavior impact depends on surrounding block context. |
| 7693 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 7694 |             Color32::RED | executable/support line; behavior impact depends on surrounding block context. |
| 7695 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
. |
| 7696 |         ui.horizontal_wrapped(ui) { | executable/support line; behavior impact depends on surrounding block context. |
| 7697 |             ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 7698 |                 RichText::new(format!("Sensor health {}/4", healthy_count)) | executable/support line; behavior impact depends on surrounding block context. |
| 7699 |                     .size(12.0) | executable/support line; behavior impact depends on surrounding block context. |
. |
| 7700 |                     .color(sys_col) | executable/support line; behavior impact depends on surrounding block context. |
text. |
| 7701 |                     .strong(), | executable/support line; behavior impact depends on surrounding block context. |
|
| 7702 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 7703 |         for (name, ok) in [ | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 7704 |             ("GPS", gps_ok), | executable/support line; behavior impact depends on surrounding block context.
```



```

t. |
| 7705 |         ("IMU", imu_ok), | executable/support line; behavior impact depends on surrounding block con
t. |
| 7706 |         ("RADAR", radar_ok), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 7707 |         ("LIDAR", lidar_ok), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 7708 |     ] { | executable/support line; behavior impact depends on surrounding block context. |
| 7709 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 7710 |             RichText::new(format!( | executable/support line; behavior impact depends on surrounding
ock context. |
| 7711 |                 "{}:{}", | executable/support line; behavior impact depends on surrounding block con
t. |
| 7712 |                 name, | executable/support line; behavior impact depends on surrounding block contex
|
| 7713 |                 if ok { "OK" } else { "CHECK" } | conditional gate; changing condition shifts runtime
ath and side effects. |
| 7714 |             )) | executable/support line; behavior impact depends on surrounding block context. |
| 7715 |             .size(10.8) | executable/support line; behavior impact depends on surrounding block conte
. |
| 7716 |             .color(if ok { Color32::GREEN } else { Color32::YELLOW }), | executable/support line; be
ior impact depends on surrounding block context. |
| 7717 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 7718 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 7719 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 7720 |     ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 7721 |     ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block conte
|
| 7722 |         .auto_shrink([false, false]) | executable/support line; behavior impact depends on surrounding b
k context. |
| 7723 |         .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 7724 |             ui.columns(2, \|cols\| { | executable/support line; behavior impact depends on surrounding b
k context. |
| 7725 |                 // GPS | comment line; documentation intent, no runtime effect. |
| 7726 |                 egui::Frame::group(cols[0].style()) | executable/support line; behavior impact depends o
urrounding block context. |
| 7727 |                     .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on
rounding block context. |
| 7728 |                     .show(&mut cols[0], \|ui\| { | executable/support line; behavior impact depends on s
ounding block context. |
| 7729 |                         let g = &self.bench.gps; | executable/support line; behavior impact depends on s
ounding block context. |
| 7730 |                         ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 7731 |                             RichText::new("■ GPS/GNSS – u-blox ZED-F9P") | executable/support line; beha
r impact depends on surrounding block context. |
| 7732 |                             .size(12.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 7733 |                             .color(Color32::from_rgb(80, 155, 255)), | executable/support line; beha
r impact depends on surrounding block context. |
| 7734 |                             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7735 |                         let fc = match g.fix_quality { | executable/support line; behavior impact depend
n surrounding block context. |
| 7736 |                             auto_breaking::gps::GpsFixQuality::RtkFix => Color32::GREEN, | executable/su
rt line; behavior impact depends on surrounding block context. |
| 7737 |                             auto_breaking::gps::GpsFixQuality::DgpsFix => { | executable/support line; b
vior impact depends on surrounding block context. |
| 7738 |                                 Color32::from_rgb(100, 220, 100) | executable/support line; behavior imp
depends on surrounding block context. |
| 7739 |                             } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 7740 |                             auto_breaking::gps::GpsFixQuality::SpsFix => Color32::YELLOW, | executable/s

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ort line; behavior impact depends on surrounding block context. |
| 7741 |             _ => Color32::RED, | executable/support line; behavior impact depends on sur
nding block context. |
| 7742 |             }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 7743 |             ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 7744 |             RichText::new(format!("Fix: {} ", g.fix_quality)) | executable/support line;
avior impact depends on surrounding block context. |
| 7745 |             .size(12.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 7746 |             .color(fc) | executable/support line; behavior impact depends on surroun
g block context. |
| 7747 |             .strong(), | executable/support line; behavior impact depends on surroun
g block context. |
| 7748 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7749 |             digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 7750 |             ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 7751 |             "Position", | executable/support line; behavior impact depends on surroundin
lock context. |
| 7752 |             &g.position_string(), | executable/support line; behavior impact depends on
rounding block context. |
| 7753 |             Color32::LIGHT_BLUE, | executable/support line; behavior impact depends on s
ounding block context. |
| 7754 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7755 |             digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 7756 |             ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 7757 |             "Altitude", | executable/support line; behavior impact depends on surroundin
lock context. |
| 7758 |             &format!("{:.1} m MSL", g.altitude_msl), | executable/support line; behavior
pact depends on surrounding block context. |
| 7759 |             Color32::from_gray(200), | executable/support line; behavior impact depends
surrounding block context. |
| 7760 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7761 |             digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 7762 |             ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 7763 |             "Speed", | executable/support line; behavior impact depends on surrounding b
k context. |
| 7764 |             &format!("{:.2} km/h {:.1}°", g.speed_kmh, g.course_deg), | executable/supp
line; behavior impact depends on surrounding block context. |
| 7765 |             Color32::GREEN, | executable/support line; behavior impact depends on surrou
ng block context. |
| 7766 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7767 |             digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 7768 |             ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 7769 |             "HDOP", | executable/support line; behavior impact depends on surrounding bl
context. |
| 7770 |             &format!("{:.2} PDOP:{:.2}", g.hdop, g.pdop), | executable/support line; be
ior impact depends on surrounding block context. |
| 7771 |             Color32::from_gray(180), | executable/support line; behavior impact depends
surrounding block context. |
| 7772 |             ); | executable/support line; behavior impact depends on surrounding block conte
|

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| 7773 | digital_readout( | executable/support line; behavior impact depends on surrounding
block context. |
| 7774 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 7775 | "Sats", | executable/support line; behavior impact depends on surrounding bl
context. |
| 7776 | &format!("{}", used / {} view", g.sats_used, g.sats_in_view), | executable/sup
t line; behavior impact depends on surrounding block context. |
| 7777 | Color32::YELLOW, | executable/support line; behavior impact depends on surro
ing block context. |
| 7778 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7779 | digital_readout( | executable/support line; behavior impact depends on surrounding
block context. |
| 7780 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 7781 | "H-Acc", | executable/support line; behavior impact depends on surrounding b
k context. |
| 7782 | &format!("{}", m (1σ)", g.hacc_m), | executable/support line; behavior impa
depends on surrounding block context. |
| 7783 | Color32::from_rgb(100, 200, 100), | executable/support line; behavior impact
ends on surrounding block context. |
| 7784 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7785 | digital_readout(ui, "UTC", &g.utc_time, Color32::from_gray(155)); | executable/s
ort line; behavior impact depends on surrounding block context. |
| 7786 | ui.separator(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 7787 | ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 7788 | RichText::new("LATEST NMEA:") | executable/support line; behavior impact dep
s on surrounding block context. |
| 7789 | .size(10.5) | executable/support line; behavior impact depends on surrou
ng block context. |
| 7790 | .color(Color32::from_gray(130)), | executable/support line; behavior imp
depends on surrounding block context. |
| 7791 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7792 | for nmea in g.nmea_queue.iter().take(4) { | iteration control; changing bounds/b
affects throughput, timing, and possibly safety guarantees. |
| 7793 | ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 7794 | RichText::new(nmea.trim()) | executable/support line; behavior impact de
ds on surrounding block context. |
| 7795 | .size(9.5) | executable/support line; behavior impact depends on sur
nding block context. |
| 7796 | .monospace() | executable/support line; behavior impact depends on s
ounding block context. |
| 7797 | .color(Color32::from_gray(180)), | executable/support line; behavior
pact depends on surrounding block context. |
| 7798 | ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7799 | } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 7800 | ui.separator(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 7801 | ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 7802 | RichText::new("SATELLITES:") | executable/support line; behavior impact depe
on surrounding block context. |
| 7803 | .size(10.5) | executable/support line; behavior impact depends on surrou
ng block context. |
| 7804 | .color(Color32::from_gray(130)), | executable/support line; behavior imp
depends on surrounding block context. |
| 7805 | ); | executable/support line; behavior impact depends on surrounding block conte

```

```

|
| 7806 |                 for sat in &g.satellites { | iteration control; changing bounds/body affects thr
hput, timing, and possibly safety guarantees. |
| 7807 |                 let c = if !sat.used { | executable/support line; behavior impact depends on
rrounding block context. |
| 7808 |                     Color32::from_gray(55) | executable/support line; behavior impact depend
n surrounding block context. |
| 7809 |                 } else if sat.snr > 35.0 { | executable/support line; behavior impact depend
n surrounding block context. |
| 7810 |                     Color32::GREEN | executable/support line; behavior impact depends on surr
nding block context. |
| 7811 |                 } else { | executable/support line; behavior impact depends on surrounding b
k context. |
| 7812 |                     Color32::YELLOW | executable/support line; behavior impact depends on su
unding block context. |
| 7813 |                 }; | scope delimiter; structure-only line, but misplaced edits can change ow
ship and lifetime scopes. |
| 7814 |                 let bar: String = | executable/support line; behavior impact depends on surr
ding block context. |
| 7815 |                     "■".repeat(((sat.snr / 52.0 * 12.0) as usize).min(12)); | executable/sup
t line; behavior impact depends on surrounding block context. |
| 7816 |                 ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 7817 |                     RichText::new(format!( | executable/support line; behavior impact depend
n surrounding block context. |
| 7818 |                     "PRN{:02} {} El:{:2.0}° Az:{:3.0}° SNR:{:4.1} {} {}", | executable/s
ort line; behavior impact depends on surrounding block context. |
| 7819 |                     sat.prn, | executable/support line; behavior impact depends on surro
ing block context. |
| 7820 |                     sat.constellation, | executable/support line; behavior impact depend
n surrounding block context. |
| 7821 |                     sat.elevation, | executable/support line; behavior impact depends on
rrounding block context. |
| 7822 |                     sat.azimuth, | executable/support line; behavior impact depends on s
ounding block context. |
| 7823 |                     sat.snr, | executable/support line; behavior impact depends on surro
ing block context. |
| 7824 |                     bar, | executable/support line; behavior impact depends on surroundi
block context. |
| 7825 |                     if sat.used { "✓" } else { "." } | conditional gate; changing condit
shifts runtime path and side effects. |
| 7826 |                 )) | executable/support line; behavior impact depends on surrounding blo
context. |
| 7827 |                 .size(9.5) | executable/support line; behavior impact depends on surroun
g block context. |
| 7828 |                 .monospace() | executable/support line; behavior impact depends on surro
ing block context. |
| 7829 |                 .color(c), | executable/support line; behavior impact depends on surroun
g block context. |
| 7830 |                 ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7831 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 7832 |                 }); | executable/support line; behavior impact depends on surrounding block context.
| 7833 |                 // IMU | comment line; documentation intent, no runtime effect. |
| 7834 |                 egui::Frame::group(cols[1].style()) | executable/support line; behavior impact depends o
urrounding block context. |
| 7835 |                 .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on
rounding block context. |
| 7836 |                 .show(&mut cols[1], \|ui\| { | executable/support line; behavior impact depends on s
ounding block context. |
| 7837 |                 let imu = &self.bench.imu; | executable/support line; behavior impact depends on
rrounding block context. |
| 7838 |                 ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |

```

```

| 7839 |                                RichText::new("■ IMU – Madgwick AHRS 9-DOF") | executable/support line; beha
r impact depends on surrounding block context. |
| 7840 |                                .size(12.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 7841 |                                .color(Color32::from_rgb(80, 155, 255)), | executable/support line; beha
r impact depends on surrounding block context. |
| 7842 |                                ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7843 |                                ui.columns(3, \|c\| { | executable/support line; behavior impact depends on surr
ding block context. |
| 7844 |                                egui::Frame::dark_canvas(c[0].style()).show(&mut c[0], \|ui\| { | executable
pport line; behavior impact depends on surrounding block context. |
| 7845 |                                arc_gauge( | executable/support line; behavior impact depends on surroun
g block context. |
| 7846 |                                ui, | executable/support line; behavior impact depends on surroundin
lock context. |
| 7847 |                                imu.roll_deg, | executable/support line; behavior impact depends on
rounding block context. |
| 7848 |                                -180.0, | executable/support line; behavior impact depends on surrou
ng block context. |
| 7849 |                                180.0, | executable/support line; behavior impact depends on surroun
g block context. |
| 7850 |                                "ROLL", | executable/support line; behavior impact depends on surrou
ng block context. |
| 7851 |                                "°", | executable/support line; behavior impact depends on surroundi
block context. |
| 7852 |                                95.0, | executable/support line; behavior impact depends on surround
block context. |
| 7853 |                                25.0, | executable/support line; behavior impact depends on surround
block context. |
| 7854 |                                45.0, | executable/support line; behavior impact depends on surround
block context. |
| 7855 |                                Some(-25.0), | executable/support line; behavior impact depends on s
ounding block context. |
| 7856 |                                ) | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 7857 |                                }); | executable/support line; behavior impact depends on surrounding block
text. |
| 7858 |                                egui::Frame::dark_canvas(c[1].style()).show(&mut c[1], \|ui\| { | executable
pport line; behavior impact depends on surrounding block context. |
| 7859 |                                arc_gauge( | executable/support line; behavior impact depends on surroun
g block context. |
| 7860 |                                ui, | executable/support line; behavior impact depends on surroundin
lock context. |
| 7861 |                                imu.pitch_deg, | executable/support line; behavior impact depends on
rrounding block context. |
| 7862 |                                -90.0, | executable/support line; behavior impact depends on surroun
g block context. |
| 7863 |                                90.0, | executable/support line; behavior impact depends on surround
block context. |
| 7864 |                                "PITCH", | executable/support line; behavior impact depends on surro
ing block context. |
| 7865 |                                "°", | executable/support line; behavior impact depends on surroundi
block context. |
| 7866 |                                95.0, | executable/support line; behavior impact depends on surround
block context. |
| 7867 |                                15.0, | executable/support line; behavior impact depends on surround
block context. |
| 7868 |                                30.0, | executable/support line; behavior impact depends on surround
block context. |
| 7869 |                                Some(-15.0), | executable/support line; behavior impact depends on s
ounding block context. |
| 7870 |                                ) | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 7871 |                                }); | executable/support line; behavior impact depends on surrounding block

```

```

text. |
| 7872 |             egui::Frame::dark_canvas(c[2].style()).show(&mut c[2], \|ui\| { | executable/s
pport line; behavior impact depends on surrounding block context. |
| 7873 |             arc_gauge( | executable/support line; behavior impact depends on surroundi
g block context. |
| 7874 |             ui, | executable/support line; behavior impact depends on surrounding blo
lock context. |
| 7875 |             imu.yaw_deg, | executable/support line; behavior impact depends on surroundi
ounding block context. |
| 7876 |             0.0, | executable/support line; behavior impact depends on surrounding blo
block context. |
| 7877 |             360.0, | executable/support line; behavior impact depends on surrounding blo
g block context. |
| 7878 |             "YAW/HDG", | executable/support line; behavior impact depends on surroundi
nding block context. |
| 7879 |             "", | executable/support line; behavior impact depends on surrounding blo
block context. |
| 7880 |             95.0, | executable/support line; behavior impact depends on surrounding blo
block context. |
| 7881 |             0.0, | executable/support line; behavior impact depends on surrounding blo
block context. |
| 7882 |             0.0, | executable/support line; behavior impact depends on surrounding blo
block context. |
| 7883 |             None, | executable/support line; behavior impact depends on surrounding blo
block context. |
| 7884 |             ) | executable/support line; behavior impact depends on surrounding block co
ontext. |
| 7885 |             }); | executable/support line; behavior impact depends on surrounding block co
text. |
| 7886 |             }); | executable/support line; behavior impact depends on surrounding block co
. |
| 7887 |             let w = 110.0; | executable/support line; behavior impact depends on surrounding
ock context. |
| 7888 |             ui.label( | executable/support line; behavior impact depends on surrounding blo
ontext. |
| 7889 |             RichText::new("ACCELEROMETER m/s²:") | executable/support line; behavior imp
depends on surrounding block context. |
| 7890 |             .size(10.5) | executable/support line; behavior impact depends on surroundi
ng block context. |
| 7891 |             .color(Color32::from_gray(130)), | executable/support line; behavior imp
depends on surrounding block context. |
| 7892 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7893 |             bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 7894 |             ui, | executable/support line; behavior impact depends on surrounding block co
text. |
| 7895 |             "Ax (fwd)", | executable/support line; behavior impact depends on surrounding
lock context. |
| 7896 |             imu.accel_x, | executable/support line; behavior impact depends on surroundi
block context. |
| 7897 |             20.0, | executable/support line; behavior impact depends on surrounding blo
ontext. |
| 7898 |             "m/s²", | executable/support line; behavior impact depends on surrounding blo
context. |
| 7899 |             w, | executable/support line; behavior impact depends on surrounding block co
ext. |
| 7900 |             Color32::from_rgb(80, 180, 255), | executable/support line; behavior impact
ends on surrounding block context. |
| 7901 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7902 |             bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 7903 |             ui, | executable/support line; behavior impact depends on surrounding block co
text. |

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```

| 7904 | "Ay (lat)", | executable/support line; behavior impact depends on surrounding
lock context. |
| 7905 | imu.accel_y.abs(), | executable/support line; behavior impact depends on sur
nding block context. |
| 7906 | 20.0, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 7907 | "m/s²", | executable/support line; behavior impact depends on surrounding bl
context. |
| 7908 | w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7909 | Color32::from_rgb(80, 180, 255), | executable/support line; behavior impact
ends on surrounding block context. |
| 7910 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7911 | bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 7912 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 7913 | "Az (up)", | executable/support line; behavior impact depends on surrounding
ock context. |
| 7914 | imu.accel_z.abs(), | executable/support line; behavior impact depends on sur
nding block context. |
| 7915 | 15.0, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 7916 | "m/s²", | executable/support line; behavior impact depends on surrounding bl
context. |
| 7917 | w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7918 | Color32::from_rgb(80, 180, 255), | executable/support line; behavior impact
ends on surrounding block context. |
| 7919 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7920 | ui.label( | executable/support line; behavior impact depends on surrounding blo
ontext. |
| 7921 | RichText::new("G-FORCES:") | executable/support line; behavior impact depend
n surrounding block context. |
| 7922 | .size(10.5) | executable/support line; behavior impact depends on surrou
ng block context. |
| 7923 | .color(Color32::from_gray(130)), | executable/support line; behavior imp
depends on surrounding block context. |
| 7924 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7925 | bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 7926 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 7927 | "Long G", | executable/support line; behavior impact depends on surrounding
ck context. |
| 7928 | imu.longitudinal_g.abs(), | executable/support line; behavior impact depends
surrounding block context. |
| 7929 | 1.5, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 7930 | "g", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 7931 | w, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7932 | Color32::RED, | executable/support line; behavior impact depends on surround
block context. |
| 7933 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7934 | bar_gauge(ui, "Lat G", imu.lateral_g.abs(), 1.5, "g", w, Color32::RED); | execut
e/support line; behavior impact depends on surrounding block context. |
| 7935 | digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 7936 | ui, | executable/support line; behavior impact depends on surrounding block

```

```

text. |
| 7937 |             "IMU Temp", | executable/support line; behavior impact depends on surrounding
lock context. |
| 7938 |             &format!("{:.1}°C", imu.temperature_c), | executable/support line; behavior
act depends on surrounding block context. |
| 7939 |             Color32::from_gray(180), | executable/support line; behavior impact depends
surrounding block context. |
| 7940 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7941 |             if imu.accel_fault { | conditional gate; changing condition shifts runtime path
side effects. |
| 7942 |             ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 7943 |             RichText::new("■ ACCEL FAULT") | executable/support line; behavior impac
depends on surrounding block context. |
| 7944 |             .size(11.0) | executable/support line; behavior impact depends on su
unding block context. |
| 7945 |             .color(Color32::RED), | executable/support line; behavior impact dep
s on surrounding block context. |
| 7946 |             ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7947 |             } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 7948 |             if imu.gyro_fault { | conditional gate; changing condition shifts runtime path a
side effects. |
| 7949 |             ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 7950 |             RichText::new("■ GYRO FAULT").size(11.0).color(Color32::RED), | executab
support line; behavior impact depends on surrounding block context. |
| 7951 |             ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7952 |             } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 7953 |             ui.separator(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 7954 |             // RADAR summary | comment line; documentation intent, no runtime effect. |
| 7955 |             let r = &self.bench.radar; | executable/support line; behavior impact depends on
rounding block context. |
| 7956 |             ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 7957 |             RichText::new("■ RADAR 77GHz") | executable/support line; behavior impact de
ds on surrounding block context. |
| 7958 |             .size(12.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 7959 |             .color(Color32::from_rgb(80, 155, 255)), | executable/support line; beha
r impact depends on surrounding block context. |
| 7960 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7961 |             let ttc_col = if r.ttc_front < 2.0 { | executable/support line; behavior impact
ends on surrounding block context. |
| 7962 |             Color32::RED | executable/support line; behavior impact depends on surroundi
block context. |
| 7963 |             } else if r.ttc_front < 4.0 { | executable/support line; behavior impact depends
surrounding block context. |
| 7964 |             Color32::YELLOW | executable/support line; behavior impact depends on surrou
ng block context. |
| 7965 |             } else { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 7966 |             Color32::GREEN | executable/support line; behavior impact depends on surroun
g block context. |
| 7967 |             }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 7968 |             digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 7969 |             ui, | executable/support line; behavior impact depends on surrounding block

```



```

text. |
| 7970 | "Front TTC", | executable/support line; behavior impact depends on surrounding
block context. |
| 7971 | &format!("{:.1}s", r.ttc_front.min(99.9)), | executable/support line; behavior
impact depends on surrounding block context. |
| 7972 | ttc_col, | executable/support line; behavior impact depends on surrounding b
k context. |
| 7973 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7974 | digital_readout( | executable/support line; behavior impact depends on surrounding
block context. |
| 7975 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 7976 | "Lead dist", | executable/support line; behavior impact depends on surrounding
block context. |
| 7977 | &format!("{:.1}m", r.closest_front_m.min(999.0)), | executable/support line;
havior impact depends on surrounding block context. |
| 7978 | Color32::LIGHT_BLUE, | executable/support line; behavior impact depends on s
ounding block context. |
| 7979 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7980 | digital_readout( | executable/support line; behavior impact depends on surrounding
block context. |
| 7981 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 7982 | "BSM L/R", | executable/support line; behavior impact depends on surrounding
ock context. |
| 7983 | &format!( | executable/support line; behavior impact depends on surrounding
ck context. |
| 7984 | "{}/{}", | executable/support line; behavior impact depends on surrounding
block context. |
| 7985 | if r.bsm_left { "■" } else { "✓" }, | conditional gate; changing condit
shifts runtime path and side effects. |
| 7986 | if r.bsm_right { "■" } else { "✓" } | conditional gate; changing condit
shifts runtime path and side effects. |
| 7987 | ), | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7988 | if r.bsm_left \|\| r.bsm_right { | conditional gate; changing condition shif
runtime path and side effects. |
| 7989 | Color32::YELLOW | executable/support line; behavior impact depends on su
ounding block context. |
| 7990 | } else { | executable/support line; behavior impact depends on surrounding b
k context. |
| 7991 | Color32::GREEN | executable/support line; behavior impact depends on sur
nding block context. |
| 7992 | }, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 7993 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 7994 | digital_readout( | executable/support line; behavior impact depends on surrounding
block context. |
| 7995 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 7996 | "Targets", | executable/support line; behavior impact depends on surrounding
ock context. |
| 7997 | &format!("{}", r.total_targets()), | executable/support line; behavior impac
depends on surrounding block context. |
| 7998 | Color32::from_gray(180), | executable/support line; behavior impact depends
surrounding block context. |
| 7999 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8000 | for t in r.front_center.targets.iter().take(5) { | iteration control; changing b
ds/body affects throughput, timing, and possibly safety guarantees. |
| 8001 | let c = if t.ttc_s < 2.0 { | executable/support line; behavior impact depend
n surrounding block context. |

```

```

| 8002 |                                     Color32::RED | executable/support line; behavior impact depends on surro
ing block context. |
| 8003 |                                     } else if t.ttc_s < 4.0 { | executable/support line; behavior impact depends
surrounding block context. |
| 8004 |                                     Color32::YELLOW | executable/support line; behavior impact depends on su
unding block context. |
| 8005 |                                     } else { | executable/support line; behavior impact depends on surrounding b
k context. |
| 8006 |                                     Color32::GREEN | executable/support line; behavior impact depends on sur
nding block context. |
| 8007 |                                     }; | scope delimiter; structure-only line, but misplaced edits can change ow
ship and lifetime scopes. |
| 8008 |                                     ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 8009 |                                     RichText::new(format!( | executable/support line; behavior impact depend
n surrounding block context. |
| 8010 |                                     "ID{:02} {:5.1}m Az:{:+.1}° {:.1}m/s TTC:{:.1}s {}", | executable/su
rt line; behavior impact depends on surrounding block context. |
| 8011 |                                     t.id, | executable/support line; behavior impact depends on surround
block context. |
| 8012 |                                     t.range_m, | executable/support line; behavior impact depends on sur
nding block context. |
| 8013 |                                     t.azimuth_deg, | executable/support line; behavior impact depends on
rrounding block context. |
| 8014 |                                     -t.range_rate_ms, | executable/support line; behavior impact depends
surrounding block context. |
| 8015 |                                     t.ttc_s.min(99.9), | executable/support line; behavior impact depend
n surrounding block context. |
| 8016 |                                     t.object_class | executable/support line; behavior impact depends on
rrounding block context. |
| 8017 |                                     )) | executable/support line; behavior impact depends on surrounding blo
context. |
| 8018 |                                     .size(10.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 8019 |                                     .monospace() | executable/support line; behavior impact depends on surro
ing block context. |
| 8020 |                                     .color(c), | executable/support line; behavior impact depends on surroun
g block context. |
| 8021 |                                     ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8022 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 8023 |                                     ui.separator(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 8024 |                                     let l = &self.bench.lidar; | executable/support line; behavior impact depends on
rrounding block context. |
| 8025 |                                     ui.label( | executable/support line; behavior impact depends on surrounding blo
ontext. |
| 8026 |                                     RichText::new("■ LIDAR VLP-16") | executable/support line; behavior impact c
nds on surrounding block context. |
| 8027 |                                     .size(12.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 8028 |                                     .color(Color32::from_rgb(80, 155, 255)), | executable/support line; beha
r impact depends on surrounding block context. |
| 8029 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8030 |                                     digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 8031 |                                     ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 8032 |                                     "Pts/scan", | executable/support line; behavior impact depends on surroundin
lock context. |
| 8033 |                                     &format!("{}", l.points_per_scan), | executable/support line; behavior impac
depends on surrounding block context. |
| 8034 |                                     Color32::from_gray(200), | executable/support line; behavior impact depends

```

```
| surrounding block context. |  
| 8035 | ); | executable/support line; behavior impact depends on surrounding block conte  
|  
| 8036 | digital_readout( | executable/support line; behavior impact depends on surroundi  
block context. |  
| 8037 | ui, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 8038 | "Clusters", | executable/support line; behavior impact depends on surrounding  
lock context. |  
| 8039 | &format!("{}", l.obstacles_seen), | executable/support line; behavior impact  
depends on surrounding block context. |  
| 8040 | Color32::YELLOW, | executable/support line; behavior impact depends on surro  
ing block context. |  
| 8041 | ); | executable/support line; behavior impact depends on surrounding block conte  
|  
| 8042 | for cl in l.clusters.iter().take(4) { | iteration control; changing bounds/body  
ects throughput, timing, and possibly safety guarantees. |  
| 8043 | ui.label( | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 8044 | RichText::new(format!( | executable/support line; behavior impact depend  
n surrounding block context. |  
| 8045 | "ID{:03} {} {:.1}m {:.1}x{:.1}m {}pts", | executable/support line;  
havior impact depends on surrounding block context. |  
| 8046 | cl.id, | executable/support line; behavior impact depends on surround  
g block context. |  
| 8047 | cl.object_type, | executable/support line; behavior impact depends o  
urrounding block context. |  
| 8048 | cl.distance_m, | executable/support line; behavior impact depends on  
rrounding block context. |  
| 8049 | cl.length_m, | executable/support line; behavior impact depends on s  
ounding block context. |  
| 8050 | cl.width_m, | executable/support line; behavior impact depends on su  
unding block context. |  
| 8051 | cl.point_count | executable/support line; behavior impact depends on  
rrounding block context. |  
| 8052 | )) | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 8053 | .size(10.0) | executable/support line; behavior impact depends on surrou  
ng block context. |  
| 8054 | .monospace() | executable/support line; behavior impact depends on surro  
ing block context. |  
| 8055 | .color(Color32::from_gray(190)), | executable/support line; behavior imp  
depends on surrounding block context. |  
| 8056 | ); | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 8057 | } | scope delimiter; structure-only line, but misplaced edits can change ownersh  
and lifetime scopes. |  
| 8058 | }); | executable/support line; behavior impact depends on surrounding block context.  
| 8059 | }); | executable/support line; behavior impact depends on surrounding block context. |  
| 8060 | }); | executable/support line; behavior impact depends on surrounding block context. |  
| 8061 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 8062 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 8063 | (blank) | line is blank; visual separation only. |  
| 8064 | // ████████████████████████████████████████████████████████████████████████████ | comm  
intent, no runtime effect. |  
| 8065 | // TAB AUTONOMOUS | comment line; documentation intent, no runtime effect. |  
| 8066 | // ████████████████████████████████████████████████████████████████████████████ | comm  
intent, no runtime effect. |  
| 8067 | impl App { | implementation block; method behavior and trait conformance are defined here. |  
| 8068 | fn tab_autonomous(&mut self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter  
I, control flow, and outputs. |  
| 8069 | // All display snapshots | comment line; documentation intent, no runtime effect. |  
| 8070 | let eng = self.bench.ad.engaged; | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 8071 | let lvl_s = format!("{}", self.bench.ad.sae_level); | executable/support line; behavior impact dependen
```

```

on surrounding block context. |
| 8072 |         let acc_s = self.bench.ad.acc_state; | executable/support line; behavior impact depends on surrounding
block context. |
| 8073 |         let acc_spd = self.bench.ad.acc_set_speed_kmh; | executable/support line; behavior impact depends on
surrounding block context. |
| 8074 |         let acc_hdw = self.bench.ad.acc_headway_s; | executable/support line; behavior impact depends on surr
ounding block context. |
| 8075 |         let aeb_s = self.bench.ad.aeb_state; | executable/support line; behavior impact depends on surrounding
block context. |
| 8076 |         let aeb_br = self.bench.ad.aeb_brake_cmd; | executable/support line; behavior impact depends on surr
ounding block context. |
| 8077 |         let aeb_pc = self.bench.ad.aeb_pre_charge; | executable/support line; behavior impact depends on surr
ounding block context. |
| 8078 |         let lka_s = self.bench.ad.lka_state; | executable/support line; behavior impact depends on surrounding
block context. |
| 8079 |         let lka_st = self.bench.ad.lka_steer_cmd; | executable/support line; behavior impact depends on surr
ounding block context. |
| 8080 |         let tja_s = self.bench.ad.tja_state; | executable/support line; behavior impact depends on surrounding
block context. |
| 8081 |         let ttc = self.bench.ad.ttc_s; | executable/support line; behavior impact depends on surrounding blo
ck context. |
| 8082 |         let thw = self.bench.ad.thw_s; | executable/support line; behavior impact depends on surrounding blo
ck context. |
| 8083 |         let lead = self.bench.ad.lead_range_m; | executable/support line; behavior impact depends on surroun
g block context. |
| 8084 |         let lead_v = self.bench.ad.lead_speed_ms; | executable/support line; behavior impact depends on surr
ounding block context. |
| 8085 |         let bsml = self.bench.ad.bsm_left; | executable/support line; behavior impact depends on surrounding
ock context. |
| 8086 |         let bsmr = self.bench.ad.bsm_right; | executable/support line; behavior impact depends on surroundin
lock context. |
| 8087 |         let thr_c = self.bench.ad.throttle_cmd; | executable/support line; behavior impact depends on surrou
ng block context. |
| 8088 |         let brk_c = self.bench.ad.brake_cmd; | executable/support line; behavior impact depends on surroundi
block context. |
| 8089 |         let str_c = self.bench.ad.steer_cmd_deg; | executable/support line; behavior impact depends on surro
ing block context. |
| 8090 |         let det_s = self.bench.ad.lane.detected; | executable/support line; behavior impact depends on surro
ing block context. |
| 8091 |         let conf_s = self.bench.ad.lane.confidence; | executable/support line; behavior impact depends on su
unding block context. |
| 8092 |         let off_s = self.bench.ad.lane.offset_m; | executable/support line; behavior impact depends on surro
ing block context. |
| 8093 |         let w_s = self.bench.ad.lane.lane_width_m; | executable/support line; behavior impact depends on surr
ounding block context. |
| 8094 |         let lt_s = format!("{}", self.bench.ad.lane.left_type); | executable/support line; behavior impact d
nds on surrounding block context. |
| 8095 |         let rt_s = format!("{}", self.bench.ad.lane.right_type); | executable/support line; behavior impact
ends on surrounding block context. |
| 8096 |         let safe_s = format!("{}", self.bench.ad.safety_status); | executable/support line; behavior impact
ends on surrounding block context. |
| 8097 |         let deg_r = self.bench.ad.degrade_reason; | executable/support line; behavior impact depends on surr
ounding block context. |
| 8098 |         let drows = self.bench.ad.drowsiness_pct; | executable/support line; behavior impact depends on surr
ounding block context. |
| 8099 |         let hands = self.bench.ad.hands_off_s; | executable/support line; behavior impact depends on surroun
g block context. |
| 8100 |         let alert = self.bench.ad.driver_alert; | executable/support line; behavior impact depends on surrou
ng block context. |
| 8101 |         let _ego_spd = self.bench.tcm.ground_speed_kmh; | executable/support line; behavior impact depends o
urrounding block context. |
| 8102 |         let path_v: Vec<_> = self | executable/support line; behavior impact depends on surrounding block co
xt. |
| 8103 |         .bench | executable/support line; behavior impact depends on surrounding block context. |
| 8104 |         .ad | executable/support line; behavior impact depends on surrounding block context. |

```

```
| 8105 |         .planned_path | executable/support line; behavior impact depends on surrounding block context. |  
| 8106 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |  
| 8107 |         .map(\w\| (w.x, w.y)) | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 8108 |         .collect(); | executable/support line; behavior impact depends on surrounding block context. |  
| 8109 | (blank) | line is blank; visual separation only. |  
| 8110 |     let mut do_engage = false; | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 8111 |     let mut do_lka = false; | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 8112 |     let mut do_acc_up = false; | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 8113 |     let mut do_acc_dn = false; | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 8114 |     let mut do_hdw_up = false; | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 8115 |     let mut do_hdw_dn = false; | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 8116 |     let readiness = self.ad_readiness_issues(); | executable/support line; behavior impact depends on su  
unding block context. |  
| 8117 |     let can_engage_ad = self.bench.engine_running() && !matches!(self.bench.tcm.direction, Direction::Pa  
; | executable/support line; behavior impact depends on surrounding block context. |  
| 8118 | (blank) | line is blank; visual separation only. |  
| 8119 |     ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block conte  
|  
| 8120 |         .auto_shrink([false, false]) | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 8121 |         .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte  
|  
| 8122 |             ui.columns(3, \|cols\| { | executable/support line; behavior impact depends on surrounding b  
k context. |  
| 8123 |                 // ■■ Left: Controls ████████████████████████████████████████████████████ | comment li  
on intent, no runtime effect. |  
| 8124 |                 let ui = &mut cols[0]; | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 8125 |                 egui::Frame::group(ui.style()) | executable/support line; behavior impact depends on sur  
nding block context. |  
| 8126 |                     .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on  
rounding block context. |  
| 8127 |                     .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 8128 |                         ui.label( | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 8129 |                             RichText::new("■■ AD CONTROLLER") | executable/support line; behavior impact  
ends on surrounding block context. |  
| 8130 |                                 .size(12.0) | executable/support line; behavior impact depends on surround  
ng block context. |  
| 8131 |                                     .color(Color32::from_rgb(80, 155, 255)), | executable/support line; beha  
r impact depends on surrounding block context. |  
| 8132 |                                         ); | executable/support line; behavior impact depends on surrounding block conte  
|  
| 8133 |                         let lc = if lvl_s.contains("L2") { | executable/support line; behavior impact de  
ds on surrounding block context. |  
| 8134 |                             Color32::GREEN | executable/support line; behavior impact depends on surround  
g block context. |  
| 8135 |                         } else if lvl_s.contains("L3") { | executable/support line; behavior impact depende  
on surrounding block context. |  
| 8136 |                             Color32::from_rgb(80, 220, 255) | executable/support line; behavior impact d  
nds on surrounding block context. |  
| 8137 |                         } else { | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 8138 |                             Color32::YELLOW | executable/support line; behavior impact depends on surround  
ng block context. |  
| 8139 |                         }; | scope delimiter; structure-only line, but misplaced edits can change owners  
and lifetime scopes. |
```

```

| 8140 |                 ui.label( | executable/support line; behavior impact depends on surrounding block
ontext. |
| 8141 |                 RichText::new(format!("SAE {}", lvl_s)) | executable/support line; behavior
act depends on surrounding block context. |
| 8142 |                 .size(14.0) | executable/support line; behavior impact depends on surround
ng block context. |
| 8143 |                 .color(lc) | executable/support line; behavior impact depends on surround
g block context. |
| 8144 |                 .strong(), | executable/support line; behavior impact depends on surround
g block context. |
| 8145 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8146 |             if !readiness.is_empty() { | conditional gate; changing condition shifts runtime
th and side effects. |
| 8147 |                 ui.add_space(4.0); | executable/support line; behavior impact depends on surr
nding block context. |
| 8148 |                 ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 8149 |                 RichText::new("READINESS / WHY IT MAY LOOK INACTIVE") | executable/suppo
line; behavior impact depends on surrounding block context. |
| 8150 |                 .size(10.5) | executable/support line; behavior impact depends on su
unding block context. |
| 8151 |                 .color(Color32::from_rgb(255, 190, 90)) | executable/support line; b
vior impact depends on surrounding block context. |
| 8152 |                 .strong(), | executable/support line; behavior impact depends on surr
nding block context. |
| 8153 |             ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8154 |             for item in &readiness { | iteration control; changing bounds/body affects t
ughput, timing, and possibly safety guarantees. |
| 8155 |                 ui.label( | executable/support line; behavior impact depends on surround
block context. |
| 8156 |                 RichText::new(format!("• {}", item)) | executable/support line; beha
r impact depends on surrounding block context. |
| 8157 |                 .size(10.0) | executable/support line; behavior impact depends o
urrounding block context. |
| 8158 |                 .color(Color32::from_gray(210)), | executable/support line; beha
r impact depends on surrounding block context. |
| 8159 |             ); | executable/support line; behavior impact depends on surrounding blo
context. |
| 8160 |             } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 8161 |             ui.separator(); | executable/support line; behavior impact depends on surrou
ng block context. |
| 8162 |             } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 8163 |             let fill = if eng { | executable/support line; behavior impact depends on surrou
ng block context. |
| 8164 |                 Color32::from_rgb(100, 20, 20) | executable/support line; behavior impact de
ds on surrounding block context. |
| 8165 |             } else { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 8166 |                 Color32::from_rgb(20, 80, 20) | executable/support line; behavior impact dep
s on surrounding block context. |
| 8167 |             }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 8168 |             let mut engage_resp = ui.add_enabled( | executable/support line; behavior impact
ends on surrounding block context. |
| 8169 |                 eng \\| can_engage_ad, | executable/support line; behavior impact depends o
urrounding block context. |
| 8170 |             Button::new( | executable/support line; behavior impact depends on surroundi
block context. |
| 8171 |                 RichText::new(if eng { | executable/support line; behavior impact depend
n surrounding block context. |
| 8172 |                     "■ DISENGAGE AD" | executable/support line; behavior impact depends

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surrounding block context. |
| 8173 |                                     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 8174 |                                     "■ ENGAGE AD" | executable/support line; behavior impact depends on surrounding block context. |
| 8175 |                                     }) | executable/support line; behavior impact depends on surrounding block context. |
| 8176 |                                     .size(13.0), | executable/support line; behavior impact depends on surrounding block context. |
| 8177 |                                     ) | executable/support line; behavior impact depends on surrounding block context. |
| 8178 |                                     .fill(fill) | executable/support line; behavior impact depends on surrounding block context. |
| 8179 |                                     .min_size(Vec2::new(180.0, 34.0)), | executable/support line; behavior impact depends on surrounding block context. |
| 8180 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 8181 |                                     if !eng && !can_engage_ad { | conditional gate; changing condition shifts runtime behavior and side effects. |
| 8182 |                                     engage_resp = engage_resp.on_disabled_hover_text( | executable/support line; behavior impact depends on surrounding block context. |
| 8183 |                                     "Para engatar o AD, deixe o motor em RUN e saia de Park.", | executable/support line; behavior impact depends on surrounding block context. |
| 8184 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 8185 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 8186 |                                     if engage_resp.clicked() { | conditional gate; changing condition shifts runtime behavior and side effects. |
| 8187 |                                     do_engage = true; | executable/support line; behavior impact depends on surrounding block context. |
| 8188 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 8189 |                                     ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 8190 |                                     RichText::new(if lead.is_finite() { | executable/support line; behavior impact depends on surrounding block context. |
| 8191 |                                     "ACC esta usando alvo frontal e TTC/THW para modular torque/freio." | executable/support line; behavior impact depends on surrounding block context. |
| 8192 |                                     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 8193 |                                     "Sem alvo frontal: ACC acelera ate a velocidade configurada e LKA depende de lane confidence." | executable/support line; behavior impact depends on surrounding block context. |
| 8194 |                                     }) | executable/support line; behavior impact depends on surrounding block context. |
| 8195 |                                     .size(10.0) | executable/support line; behavior impact depends on surrounding block context. |
| 8196 |                                     .color(Color32::from_gray(175)), | executable/support line; behavior impact depends on surrounding block context. |
| 8197 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 8198 |                                     ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 8199 |                                     // ACC | comment line; documentation intent, no runtime effect. |
| 8200 |                                     let ac = match acc_s { | executable/support line; behavior impact depends on surrounding block context. |
| 8201 |                                     FeatureState::Active => Color32::GREEN, | executable/support line; behavior impact depends on surrounding block context. |
| 8202 |                                     FeatureState::Ready => Color32::YELLOW, | executable/support line; behavior impact depends on surrounding block context. |
| 8203 |                                     _ => Color32::from_gray(80), | executable/support line; behavior impact depends on surrounding block context. |
| 8204 |                                     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 8205 |                                     ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding block context. |

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nding block context. |
| 8206 |             ui.label(RichText::new("ACC:").size(12.0).color(ac).strong()); | executable/
port line; behavior impact depends on surrounding block context. |
| 8207 |             ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 8208 |                 RichText::new(format!("{}", Set:{:.0}km/h", acc_s, acc_spd)) | executable
pport line; behavior impact depends on surrounding block context. |
| 8209 |                 .size(11.0) | executable/support line; behavior impact depends on su
unding block context. |
| 8210 |                 .color(ac), | executable/support line; behavior impact depends on su
unding block context. |
| 8211 |             ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8212 |             if ui.small_button("+5").clicked() { | conditional gate; changing condition
fts runtime path and side effects. |
| 8213 |                 do_acc_up = true; | executable/support line; behavior impact depends on
rounding block context. |
| 8214 |             } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 8215 |             if ui.small_button("-5").clicked() { | conditional gate; changing condition
fts runtime path and side effects. |
| 8216 |                 do_acc_dn = true; | executable/support line; behavior impact depends on
rounding block context. |
| 8217 |             } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 8218 |             }); | executable/support line; behavior impact depends on surrounding block cont
. |
| 8219 |             ui.horizontal(\ui\ { | executable/support line; behavior impact depends on sur
nding block context. |
| 8220 |                 ui.label(RichText::new("Headway").size(11.0).color(Color32::GREEN)); | execu
le/support line; behavior impact depends on surrounding block context. |
| 8221 |                 if ui.small_button("-0.2s").clicked() { | conditional gate; changing conditi
shifts runtime path and side effects. |
| 8222 |                     do_hdw_dn = true; | executable/support line; behavior impact depends on
rounding block context. |
| 8223 |                 } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 8224 |                 if ui.small_button("+0.2s").clicked() { | conditional gate; changing conditi
shifts runtime path and side effects. |
| 8225 |                     do_hdw_up = true; | executable/support line; behavior impact depends on
rounding block context. |
| 8226 |                 } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 8227 |             }); | executable/support line; behavior impact depends on surrounding block cont
. |
| 8228 |             bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 8229 |                 ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 8230 |                 "Set spd", | executable/support line; behavior impact depends on surrounding
ock context. |
| 8231 |                 acc_spd, | executable/support line; behavior impact depends on surrounding b
k context. |
| 8232 |                 150.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 8233 |                 "km/h", | executable/support line; behavior impact depends on surrounding bl
context. |
| 8234 |                 120.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 8235 |                 Color32::LIGHT_BLUE, | executable/support line; behavior impact depends on s
ounding block context. |
| 8236 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8237 |             bar_gauge(ui, "Headway", acc_hdw, 4.0, "s", 120.0, Color32::GREEN); | executable
pport line; behavior impact depends on surrounding block context. |

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| 8238 |                 ui.separator(); | executable/support line; behavior impact depends on surrounding
lock context. |
| 8239 |                 // AEB | comment line; documentation intent, no runtime effect. |
| 8240 |                 let ab = match aeb_s { | executable/support line; behavior impact depends on sur
nding block context. |
| 8241 |                     FeatureState::Active => Color32::RED, | executable/support line; behavior im
t depends on surrounding block context. |
| 8242 |                     FeatureState::Ready => Color32::GREEN, | executable/support line; behavior i
ct depends on surrounding block context. |
| 8243 |                     _ => Color32::from_gray(80), | executable/support line; behavior impact depe
on surrounding block context. |
| 8244 |                 }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 8245 |                 ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on sur
nding block context. |
| 8246 |                     ui.label(RichText::new("AEB:").size(12.0).color(ab).strong()); | executable/
port line; behavior impact depends on surrounding block context. |
| 8247 |                     ui.label(RichText::new(format!("{}", aeb_s)).size(11.0).color(ab)); | execut
e/support line; behavior impact depends on surrounding block context. |
| 8248 |                     if aeb_pc { | conditional gate; changing condition shifts runtime path and s
effects. |
| 8249 |                         ui.label( | executable/support line; behavior impact depends on surround
block context. |
| 8250 |                             RichText::new("PRE-CHG").size(10.5).color(Color32::YELLOW), | execut
e/support line; behavior impact depends on surrounding block context. |
| 8251 |                             ); | executable/support line; behavior impact depends on surrounding blo
context. |
| 8252 |                     } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 8253 |                     ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 8254 |                         RichText::new(format!("{:.0}%", aeb_br * 100.0)) | executable/support li
behavior impact depends on surrounding block context. |
| 8255 |                             .size(11.0) | executable/support line; behavior impact depends on su
unding block context. |
| 8256 |                             .color(Color32::RED), | executable/support line; behavior impact dep
s on surrounding block context. |
| 8257 |                     ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8258 |                 }); | executable/support line; behavior impact depends on surrounding block cont
. |
| 8259 |                 ui.separator(); | executable/support line; behavior impact depends on surrounding
lock context. |
| 8260 |                 // LKA | comment line; documentation intent, no runtime effect. |
| 8261 |                 let lc2 = match lka_s { | executable/support line; behavior impact depends on su
unding block context. |
| 8262 |                     FeatureState::Active => Color32::GREEN, | executable/support line; behavior
act depends on surrounding block context. |
| 8263 |                     _ => Color32::from_gray(80), | executable/support line; behavior impact depe
on surrounding block context. |
| 8264 |                 }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 8265 |                 ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on sur
nding block context. |
| 8266 |                     ui.label(RichText::new("LKA:").size(12.0).color(lc2).strong()); | executable
pport line; behavior impact depends on surrounding block context. |
| 8267 |                     ui.label(RichText::new(format!("{}", lka_s)).size(11.0).color(lc2)); | execu
le/support line; behavior impact depends on surrounding block context. |
| 8268 |                     if ui.small_button("Toggle").clicked() { | conditional gate; changing condit
shifts runtime path and side effects. |
| 8269 |                         do_lka = true; | executable/support line; behavior impact depends on sur
nding block context. |
| 8270 |                     } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 8271 |                 }); | executable/support line; behavior impact depends on surrounding block cont

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```
. | digital_readout( | executable/support line; behavior impact depends on surrounding  
block context. |  
| 8273 | ui, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 8274 | "Steer cmd", | executable/support line; behavior impact depends on surrounding  
block context. |  
| 8275 | &format!("{:.2}°", lka_st), | executable/support line; behavior impact depende  
on surrounding block context. |  
| 8276 | Color32::YELLOW, | executable/support line; behavior impact depends on surro  
ing block context. |  
| 8277 | ); | executable/support line; behavior impact depends on surrounding block conte  
|  
| 8278 | if !det_s { | conditional gate; changing condition shifts runtime path and side  
ects. |  
| 8279 | ui.label( | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 8280 | RichText::new("LKA em espera: nao ha faixa confiavel detectada.") | execu  
ble/support line; behavior impact depends on surrounding block context. |  
| 8281 | .size(10.0) | executable/support line; behavior impact depends on su  
unding block context. |  
| 8282 | .color(Color32::YELLOW), | executable/support line; behavior impact  
ends on surrounding block context. |  
| 8283 | ); | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 8284 | } | scope delimiter; structure-only line, but misplaced edits can change ownersh  
and lifetime scopes. |  
| 8285 | let tj = match tja_s { | executable/support line; behavior impact depends on sur  
nding block context. |  
| 8286 | FeatureState::Active => Color32::GREEN, | executable/support line; behavior  
act depends on surrounding block context. |  
| 8287 | _ => Color32::from_gray(80), | executable/support line; behavior impact depe  
on surrounding block context. |  
| 8288 | }; | scope delimiter; structure-only line, but misplaced edits can change owners  
and lifetime scopes. |  
| 8289 | ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on sur  
nding block context. |  
| 8290 | ui.label(RichText::new("TJA:").size(12.0).color(tj).strong()); | executable/  
port line; behavior impact depends on surrounding block context. |  
| 8291 | ui.label(RichText::new(format!("{}", tja_s)).size(11.0).color(tj)); | execut  
e/support line; behavior impact depends on surrounding block context. |  
| 8292 | }); | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 8293 | }); | executable/support line; behavior impact depends on surrounding block context.  
| 8294 | (blank) | line is blank; visual separation only. |  
| 8295 | // ■■ Centre: 2D path canvas ████████████████████████████████████████ | comment line;  
n intent, no runtime effect. |  
| 8296 | let ui = &mut cols[1]; | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 8297 | egui::Frame::group(ui.style()) | executable/support line; behavior impact depends on sur  
nding block context. |  
| 8298 | .fill(Color32::from_gray(12)) | executable/support line; behavior impact depends on  
rounding block context. |  
| 8299 | .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 8300 | ui.label( | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 8301 | RichText::new("PATH PLANNING – Click to add waypoints") | executable/support  
ne; behavior impact depends on surrounding block context. |  
| 8302 | .size(11.0) | executable/support line; behavior impact depends on surrou  
ng block context. |  
| 8303 | .color(Color32::from_gray(150)), | executable/support line; behavior imp  
depends on surrounding block context. |  
| 8304 | ); | executable/support line; behavior impact depends on surrounding block conte
```

```

| 8305 |           let (rect, resp) = ui.allocate_exact_size( | executable/support line; behavior i
ct depends on surrounding block context. |
| 8306 |           Vec2::new(ui.available_width(), ui.available_height() - 80.0), | executable/s
port line; behavior impact depends on surrounding block context. |
| 8307 |           Sense::click(), | executable/support line; behavior impact depends on surrou
ng block context. |
| 8308 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8309 |           let p = ui.painter_at(rect); | executable/support line; behavior impact depends
surrounding block context. |
| 8310 |           p.rect_filled(rect, 4.0, Color32::from_gray(10)); | executable/support line; beh
or impact depends on surrounding block context. |
| 8311 |           // Grid lines | comment line; documentation intent, no runtime effect. |
| 8312 |           let cx = rect.center().x; | executable/support line; behavior impact depends on
rounding block context. |
| 8313 |           let cy = rect.center().y; | executable/support line; behavior impact depends on
rounding block context. |
| 8314 |           let scale = 3.0_f32; | executable/support line; behavior impact depends on surro
ing block context. |
| 8315 |           for i in -5..=5 { | iteration control; changing bounds/body affects throughput,
ing, and possibly safety guarantees. |
| 8316 |               let x = cx + i as f32 * 10.0 * scale; | executable/support line; behavior im
t depends on surrounding block context. |
| 8317 |               let y = cy + i as f32 * 10.0 * scale; | executable/support line; behavior im
t depends on surrounding block context. |
| 8318 |               p.line_segment( | executable/support line; behavior impact depends on surrou
ng block context. |
| 8319 |                   [Pos2::new(x, rect.top()), Pos2::new(x, rect.bottom())], | executable/su
rt line; behavior impact depends on surrounding block context. |
| 8320 |                   Stroke::new(0.5, Color32::from_gray(30)), | executable/support line; beh
or impact depends on surrounding block context. |
| 8321 |               ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8322 |               p.line_segment( | executable/support line; behavior impact depends on surrou
ng block context. |
| 8323 |                   [Pos2::new(rect.left(), y), Pos2::new(rect.right(), y)], | executable/su
rt line; behavior impact depends on surrounding block context. |
| 8324 |                   Stroke::new(0.5, Color32::from_gray(30)), | executable/support line; beh
or impact depends on surrounding block context. |
| 8325 |               ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8326 |           } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 8327 |           // Scale label | comment line; documentation intent, no runtime effect. |
| 8328 |           p.text( | executable/support line; behavior impact depends on surrounding block
text. |
| 8329 |               Pos2::new(rect.left() + 4.0, rect.bottom() - 14.0), | executable/support lin
behavior impact depends on surrounding block context. |
| 8330 |               Align2::LEFT_BOTTOM, | executable/support line; behavior impact depends on s
ounding block context. |
| 8331 |               "Grid: 10m", | executable/support line; behavior impact depends on surroundi
block context. |
| 8332 |               FontId::proportional(9.0), | executable/support line; behavior impact depend
n surrounding block context. |
| 8333 |               Color32::from_gray(80), | executable/support line; behavior impact depends o
urrounding block context. |
| 8334 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8335 |           // Ego vehicle | comment line; documentation intent, no runtime effect. |
| 8336 |           p.rect_filled( | executable/support line; behavior impact depends on surrounding
ock context. |
| 8337 |               Rect::from_center_size(Pos2::new(cx, cy), Vec2::new(8.0, 14.0)), | executabl
upport line; behavior impact depends on surrounding block context. |
| 8338 |               2.0, | executable/support line; behavior impact depends on surrounding block
ntext. |

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| 8339 |                                     Color32::LIGHT_BLUE, | executable/support line; behavior impact depends on s
ounding block context. |
| 8340 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8341 |                                     p.text( | executable/support line; behavior impact depends on surrounding block
text. |
| 8342 |                                     Pos2::new(cx, cy - 20.0), | executable/support line; behavior impact depends
surrounding block context. |
| 8343 |                                     Align2::CENTER_CENTER, | executable/support line; behavior impact depends on
rrounding block context. |
| 8344 |                                     "■", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 8345 |                                     FontId::proportional(14.0), | executable/support line; behavior impact depen
on surrounding block context. |
| 8346 |                                     Color32::WHITE, | executable/support line; behavior impact depends on surrou
ng block context. |
| 8347 |                                     ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8348 |                                     // Planned path (from AD system) | comment line; documentation intent, no runtime
ffect. |
| 8349 |                                     if !path_v.is_empty() { | conditional gate; changing condition shifts runtime pa
and side effects. |
| 8350 |                                     let pts: Vec<Pos2> = path_v | executable/support line; behavior impact depen
on surrounding block context. |
| 8351 |                                     .iter() | executable/support line; behavior impact depends on surroundin
lock context. |
| 8352 |                                     .map(\|(x, y)\| { | executable/support line; behavior impact depends on
rounding block context. |
| 8353 |                                     Pos2::new(cx + *y as f32 * scale, cy - *x as f32 * scale) | executab
support line; behavior impact depends on surrounding block context. |
| 8354 |                                     }) | executable/support line; behavior impact depends on surrounding blo
context. |
| 8355 |                                     .collect(); | executable/support line; behavior impact depends on surrou
ng block context. |
| 8356 |                                     for w in pts.windows(2) { | iteration control; changing bounds/body affects
oughput, timing, and possibly safety guarantees. |
| 8357 |                                     p.line_segment( | executable/support line; behavior impact depends on su
unding block context. |
| 8358 |                                     [w[0], w[1]], | executable/support line; behavior impact depends on
rounding block context. |
| 8359 |                                     Stroke::new(1.5, Color32::from_rgb(0, 200, 100)), | executable/suppo
line; behavior impact depends on surrounding block context. |
| 8360 |                                     ); | executable/support line; behavior impact depends on surrounding blo
context. |
| 8361 |                                     } | scope delimiter; structure-only line, but misplaced edits can change own
hip and lifetime scopes. |
| 8362 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 8363 |                                     // User waypoints | comment line; documentation intent, no runtime effect. |
| 8364 |                                     let mut add_wp: Option<[f64; 2]> = None; | executable/support line; behavior imp
depends on surrounding block context. |
| 8365 |                                     for (i, wp) in self.ad_waypoints.iter().enumerate() { | iteration control; chang
bounds/body affects throughput, timing, and possibly safety guarantees. |
| 8366 |                                     let px = cx + wp[1] as f32 * scale; | executable/support line; behavior impa
depends on surrounding block context. |
| 8367 |                                     let py = cy - wp[0] as f32 * scale; | executable/support line; behavior impa
depends on surrounding block context. |
| 8368 |                                     p.circle_filled( | executable/support line; behavior impact depends on surro
ing block context. |
| 8369 |                                     Pos2::new(px, py), | executable/support line; behavior impact depends on
rrounding block context. |
| 8370 |                                     5.0, | executable/support line; behavior impact depends on surrounding b
k context. |
| 8371 |                                     Color32::from_rgb(255, 150, 30), | executable/support line; behavior imp
depends on surrounding block context. |

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| 8372 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 8373 |                                     p.text( | executable/support line; behavior impact depends on surrounding block context. |
| 8374 |                                     Pos2::new(px + 6.0, py), | executable/support line; behavior impact depends on surrounding block context. |
| 8375 |                                     Align2::LEFT_CENTER, | executable/support line; behavior impact depends on surrounding block context. |
| 8376 |                                     format!("{}", i), | executable/support line; behavior impact depends on surrounding block context. |
| 8377 |                                     FontId::proportional(9.0), | executable/support line; behavior impact depends on surrounding block context. |
| 8378 |                                     Color32::from_rgb(255, 150, 30), | executable/support line; behavior impact depends on surrounding block context. |
| 8379 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 8380 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 8381 |                                     // Connect waypoints | comment line; documentation intent, no runtime effect. |
| 8382 |                                     let all_pts: Vec<Pos2> = self | executable/support line; behavior impact depends on surrounding block context. |
| 8383 |                                     .ad_waypoints | executable/support line; behavior impact depends on surrounding block context. |
| 8384 |                                     .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 8385 |                                     .map(\|wp\| { | executable/support line; behavior impact depends on surrounding block context. |
| 8386 |                                     Pos2::new(cx + wp[1] as f32 * scale, cy - wp[0] as f32 * scale) | executable/support line; behavior impact depends on surrounding block context. |
| 8387 |                                     }) | executable/support line; behavior impact depends on surrounding block context. |
| 8388 |                                     .collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 8389 |                                     for w in all_pts.windows(2) { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 8390 |                                     p.line_segment( | executable/support line; behavior impact depends on surrounding block context. |
| 8391 |                                     [w[0], w[1]], | executable/support line; behavior impact depends on surrounding block context. |
| 8392 |                                     Stroke::new(1.5, Color32::from_rgb(255, 100, 0)), | executable/support line; behavior impact depends on surrounding block context. |
| 8393 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 8394 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 8395 |                                     // Click to add waypoint | comment line; documentation intent, no runtime effect. |
| 8396 |                                     if resp.clicked() { | conditional gate; changing condition shifts runtime path and side effects. |
| 8397 |                                     if let Some(pos) = resp.interact_pointer_pos() { | conditional gate; changing condition shifts runtime path and side effects. |
| 8398 |                                     let world_x = (cy - pos.y) as f64 / scale as f64; | executable/support line; behavior impact depends on surrounding block context. |
| 8399 |                                     let world_y = (pos.x - cx) as f64 / scale as f64; | executable/support line; behavior impact depends on surrounding block context. |
| 8400 |                                     add_wp = Some([world_x, world_y]); | executable/support line; behavior impact depends on surrounding block context. |
| 8401 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 8402 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 8403 |                                     if let Some(wp) = add_wp { | conditional gate; changing condition shifts runtime path and side effects. |
| 8404 |                                     self.cmds.push(Cmd::AddWaypoint(wp[0], wp[1])); | executable/support line; behavior impact depends on surrounding block context. |
| 8405 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership

```

[illegible]

```

block context. |
| 8440 |             ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 8441 |             "Steer", | executable/support line; behavior impact depends on surrounding b
k context. |
| 8442 |             &format!("{:.1}°", str_c), | executable/support line; behavior impact depen
on surrounding block context. |
| 8443 |             Color32::LIGHT_BLUE, | executable/support line; behavior impact depends on s
urrounding block context. |
| 8444 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8445 |             ui.separator(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 8446 |             ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 8447 |             RichText::new("SENSOR FUSION:") | executable/support line; behavior impact d
nds on surrounding block context. |
| 8448 |             .size(11.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 8449 |             .color(Color32::from_gray(140)), | executable/support line; behavior imp
depends on surrounding block context. |
| 8450 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8451 |             let tc = if ttc < 2.0 { | executable/support line; behavior impact depends on su
urrounding block context. |
| 8452 |             Color32::RED | executable/support line; behavior impact depends on surroundi
block context. |
| 8453 |             } else if ttc < 4.0 { | executable/support line; behavior impact depends on surr
ding block context. |
| 8454 |             Color32::YELLOW | executable/support line; behavior impact depends on surrou
ng block context. |
| 8455 |             } else { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 8456 |             Color32::GREEN | executable/support line; behavior impact depends on surroun
g block context. |
| 8457 |             }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 8458 |             digital_readout(ui, "TTC front", &format!("{:.1}s", ttc.min(99.9)), tc); | execu
le/support line; behavior impact depends on surrounding block context. |
| 8459 |             digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 8460 |             ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 8461 |             "THW front", | executable/support line; behavior impact depends on surroundi
block context. |
| 8462 |             &format!("{:.1}s", thw.min(99.9)), | executable/support line; behavior impac
depends on surrounding block context. |
| 8463 |             Color32::LIGHT_BLUE, | executable/support line; behavior impact depends on s
urrounding block context. |
| 8464 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8465 |             digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 8466 |             ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 8467 |             "Lead dist", | executable/support line; behavior impact depends on surroundi
block context. |
| 8468 |             &format!("{:.1}m", lead.min(999.0)), | executable/support line; behavior imp
depends on surrounding block context. |
| 8469 |             Color32::from_gray(200), | executable/support line; behavior impact depends
surrounding block context. |
| 8470 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8471 |             digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |

```

```

| 8472 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 8473 | "Lead spd", | executable/support line; behavior impact depends on surrounding
lock context. |
| 8474 | &format!("{:.1}m/s", lead_v), | executable/support line; behavior impact dep
s on surrounding block context. |
| 8475 | Color32::from_gray(200), | executable/support line; behavior impact depends
surrounding block context. |
| 8476 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8477 | digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 8478 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 8479 | "BSM L", | executable/support line; behavior impact depends on surrounding b
k context. |
| 8480 | if bsml { "■ OBJECT" } else { "clear " }, | conditional gate; changing conc
on shifts runtime path and side effects. |
| 8481 | if bsml { | conditional gate; changing condition shifts runtime path and sid
ffects. |
| 8482 | Color32::YELLOW | executable/support line; behavior impact depends on su
unding block context. |
| 8483 | } else { | executable/support line; behavior impact depends on surrounding b
k context. |
| 8484 | Color32::GREEN | executable/support line; behavior impact depends on sur
nding block context. |
| 8485 | }, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8486 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8487 | digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 8488 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 8489 | "BSM R", | executable/support line; behavior impact depends on surrounding b
k context. |
| 8490 | if bsmr { "■ OBJECT" } else { "clear " }, | conditional gate; changing conc
on shifts runtime path and side effects. |
| 8491 | if bsmr { | conditional gate; changing condition shifts runtime path and sid
ffects. |
| 8492 | Color32::YELLOW | executable/support line; behavior impact depends on su
unding block context. |
| 8493 | } else { | executable/support line; behavior impact depends on surrounding b
k context. |
| 8494 | Color32::GREEN | executable/support line; behavior impact depends on sur
nding block context. |
| 8495 | }, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8496 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8497 | ui.separator(); | executable/support line; behavior impact depends on surroundi
lock context. |
| 8498 | ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 8499 | RichText::new("LANE INFO:") | executable/support line; behavior impact depen
on surrounding block context. |
| 8500 | .size(11.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 8501 | .color(Color32::from_gray(140)), | executable/support line; behavior imp
depends on surrounding block context. |
| 8502 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8503 | digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 8504 | ui, | executable/support line; behavior impact depends on surrounding block

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text. |
| 8505 | "Detected", | executable/support line; behavior impact depends on surrounding
lock context. |
| 8506 | if det_s { "YES" } else { "NO" }, | conditional gate; changing condition shifts
runtime path and side effects. |
| 8507 | if det_s { Color32::GREEN } else { Color32::RED }, | conditional gate; changing
condition shifts runtime path and side effects. |
| 8508 | ); | executable/support line; behavior impact depends on surrounding block context. |
|
| 8509 | digital_readout( | executable/support line; behavior impact depends on surrounding
block context. |
| 8510 | ui, | executable/support line; behavior impact depends on surrounding block context. |
text. |
| 8511 | "Conf", | executable/support line; behavior impact depends on surrounding block context. |
| 8512 | &format!("{:.0}%", conf_s * 100.0), | executable/support line; behavior impact
depends on surrounding block context. |
| 8513 | if conf_s > 0.7 { | conditional gate; changing condition shifts runtime path and
side effects. |
| 8514 | Color32::GREEN | executable/support line; behavior impact depends on surrounding
block context. |
| 8515 | } else { | executable/support line; behavior impact depends on surrounding block context. |
| 8516 | Color32::YELLOW | executable/support line; behavior impact depends on surrounding
block context. |
| 8517 | }, | executable/support line; behavior impact depends on surrounding block context. |
ext. |
| 8518 | ); | executable/support line; behavior impact depends on surrounding block context. |
|
| 8519 | digital_readout( | executable/support line; behavior impact depends on surrounding
block context. |
| 8520 | ui, | executable/support line; behavior impact depends on surrounding block context. |
text. |
| 8521 | "Offset", | executable/support line; behavior impact depends on surrounding block context. |
| 8522 | &format!("{:+.3}m", off_s), | executable/support line; behavior impact depends
on surrounding block context. |
| 8523 | if off_s.abs() > 0.8 { | conditional gate; changing condition shifts runtime path and
side effects. |
| 8524 | Color32::RED | executable/support line; behavior impact depends on surrounding
block context. |
| 8525 | } else { | executable/support line; behavior impact depends on surrounding block context. |
| 8526 | Color32::GREEN | executable/support line; behavior impact depends on surrounding
block context. |
| 8527 | }, | executable/support line; behavior impact depends on surrounding block context. |
ext. |
| 8528 | ); | executable/support line; behavior impact depends on surrounding block context. |
|
| 8529 | digital_readout( | executable/support line; behavior impact depends on surrounding
block context. |
| 8530 | ui, | executable/support line; behavior impact depends on surrounding block context. |
text. |
| 8531 | "Width", | executable/support line; behavior impact depends on surrounding block context. |
| 8532 | &format!("{:.1}m", w_s), | executable/support line; behavior impact depends
on surrounding block context. |
| 8533 | Color32::from_gray(180), | executable/support line; behavior impact depends
on surrounding block context. |
| 8534 | ); | executable/support line; behavior impact depends on surrounding block context. |
|
| 8535 | digital_readout(ui, "L-mark", &lt_s, Color32::from_gray(180)); | executable/support
line; behavior impact depends on surrounding block context. |
| 8536 | digital_readout(ui, "R-mark", &rt_s, Color32::from_gray(180)); | executable/support
line; behavior impact depends on surrounding block context. |

```

```

| 8537 | ui.separator(); | executable/support line; behavior impact depends on surrounding
lock context. |
| 8538 | ui.label( | executable/support line; behavior impact depends on surrounding block
ontext. |
| 8539 | RichText::new("DRIVER MONITORING:") | executable/support line; behavior impa
depends on surrounding block context. |
| 8540 | .size(11.0) | executable/support line; behavior impact depends on surround
ng block context. |
| 8541 | .color(Color32::from_gray(140)), | executable/support line; behavior imp
depends on surrounding block context. |
| 8542 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8543 | bar_gauge( | executable/support line; behavior impact depends on surrounding block
context. |
| 8544 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 8545 | "Drowsy", | executable/support line; behavior impact depends on surrounding
ck context. |
| 8546 | drows, | executable/support line; behavior impact depends on surrounding block
context. |
| 8547 | 100.0, | executable/support line; behavior impact depends on surrounding block
context. |
| 8548 | "%", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 8549 | 120.0, | executable/support line; behavior impact depends on surrounding block
context. |
| 8550 | if drows > 70.0 { | conditional gate; changing condition shifts runtime path
d side effects. |
| 8551 | Color32::RED | executable/support line; behavior impact depends on surro
ing block context. |
| 8552 | } else if drows > 40.0 { | executable/support line; behavior impact depends
surrounding block context. |
| 8553 | Color32::YELLOW | executable/support line; behavior impact depends on su
unding block context. |
| 8554 | } else { | executable/support line; behavior impact depends on surrounding b
k context. |
| 8555 | Color32::GREEN | executable/support line; behavior impact depends on sur
nding block context. |
| 8556 | }, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8557 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8558 | digital_readout( | executable/support line; behavior impact depends on surroundi
block context. |
| 8559 | ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 8560 | "Hands off", | executable/support line; behavior impact depends on surroundi
block context. |
| 8561 | &format!("{:.0}s", hands), | executable/support line; behavior impact depend
n surrounding block context. |
| 8562 | Color32::from_gray(180), | executable/support line; behavior impact depends
surrounding block context. |
| 8563 | ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8564 | if alert { | conditional gate; changing condition shifts runtime path and side e
cts. |
| 8565 | ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 8566 | RichText::new("■ DRIVER ATTENTION!") | executable/support line; behavior
pact depends on surrounding block context. |
| 8567 | .size(12.0) | executable/support line; behavior impact depends on su
unding block context. |
| 8568 | .color(Color32::RED) | executable/support line; behavior impact depe
on surrounding block context. |
| 8569 | .strong(), | executable/support line; behavior impact depends on sur

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```

ending block context. |
| 8570 |             ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8571 |             } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 8572 |             ui.separator(); | executable/support line; behavior impact depends on surroundin
lock context. |
| 8573 |             let sc = if safe_s.contains("Normal") { | executable/support line; behavior impa
depends on surrounding block context. |
| 8574 |                 Color32::GREEN | executable/support line; behavior impact depends on surround
g block context. |
| 8575 |             } else if safe_s.contains("Degraded") { | executable/support line; behavior impa
depends on surrounding block context. |
| 8576 |                 Color32::YELLOW | executable/support line; behavior impact depends on surround
ng block context. |
| 8577 |             } else { | executable/support line; behavior impact depends on surrounding block
ntext. |
| 8578 |                 Color32::RED | executable/support line; behavior impact depends on surroundi
block context. |
| 8579 |             }; | scope delimiter; structure-only line, but misplaced edits can change owners
and lifetime scopes. |
| 8580 |             ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 8581 |                 RichText::new(format!("{}", safe_s)) | executable/support line; beha
r impact depends on surrounding block context. |
| 8582 |                 .size(12.0) | executable/support line; behavior impact depends on surround
ng block context. |
| 8583 |                 .color(sc) | executable/support line; behavior impact depends on surround
g block context. |
| 8584 |                 .strong(), | executable/support line; behavior impact depends on surround
g block context. |
| 8585 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8586 |             if let Some(r) = deg_r { | conditional gate; changing condition shifts runtime p
and side effects. |
| 8587 |                 ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 8588 |                 RichText::new(format!("{}", r)) | executable/support line; behavior in
t depends on surrounding block context. |
| 8589 |                 .size(10.5) | executable/support line; behavior impact depends on su
unding block context. |
| 8590 |                 .color(Color32::RED), | executable/support line; behavior impact dep
s on surrounding block context. |
| 8591 |             ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8592 |             } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 8593 |                 }); | executable/support line; behavior impact depends on surrounding block context.
| 8594 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 8595 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 8596 | (blank) | line is blank; visual separation only. |
| 8597 |             // Deferred mutations | comment line; documentation intent, no runtime effect. |
| 8598 |             if do_engage { | conditional gate; changing condition shifts runtime path and side effects. |
| 8599 |                 self.cmds | executable/support line; behavior impact depends on surrounding block context. |
| 8600 |                 .push(if eng { Cmd::DisengageAD } else { Cmd::EngageAD }); | executable/support line; behavi
impact depends on surrounding block context. |
| 8601 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 8602 |             if do_lka { | conditional gate; changing condition shifts runtime path and side effects. |
| 8603 |                 self.cmds.push(Cmd::ToggleLka); | executable/support line; behavior impact depends on surroundin
lock context. |
| 8604 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 8605 |             if do_acc_up { | conditional gate; changing condition shifts runtime path and side effects. |
| 8606 |                 self.cmds.push(Cmd::AccSpeedSet(acc_spd + 5.0)); | executable/support line; behavior impact depe

```

```
| on surrounding block context. |  
| 8607 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 8608 |         if do_acc_dn { | conditional gate; changing condition shifts runtime path and side effects. |  
| 8609 |             self.cmds.push(Cmd::AccSpeedSet(acc_spd - 5.0)); | executable/support line; behavior impact depende  
on surrounding block context. |  
| 8610 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 8611 |         if do_hdw_up { | conditional gate; changing condition shifts runtime path and side effects. |  
| 8612 |             self.cmds.push(Cmd::AccHeadwaySet(acc_hdw + 0.2)); | executable/support line; behavior impact depends  
ds on surrounding block context. |  
| 8613 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 8614 |         if do_hdw_dn { | conditional gate; changing condition shifts runtime path and side effects. |  
| 8615 |             self.cmds.push(Cmd::AccHeadwaySet(acc_hdw - 0.2)); | executable/support line; behavior impact depends  
ds on surrounding block context. |  
| 8616 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop  
|  
| 8617 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 8618 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 8619 | (blank) | line is blank; visual separation only. |  
| 8620 | // ████████████████████████████████████████████████████████████████████████████████████████████████████████ | comment  
intent, no runtime effect. |  
| 8621 | // TAB V2X | comment line; documentation intent, no runtime effect. |  
| 8622 | // ████████████████████████████████████████████████████████████████████████████████████████████████████████ | comment  
intent, no runtime effect. |  
| 8623 | impl App { | implementation block; method behavior and trait conformance are defined here. |  
| 8624 |     fn tab_v2x(&mut self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter API, control flow,  
and outputs. |  
| 8625 |         let v_dsrc = self.bench.v2x.dsrc_active; | executable/support line; behavior impact depends on surrounding  
block context. |  
| 8626 |         let v_cv2x = self.bench.v2x.cv2x_active; | executable/support line; behavior impact depends on surrounding  
block context. |  
| 8627 |         let v_rng = self.bench.v2x.range_m; | executable/support line; behavior impact depends on surrounding  
block context. |  
| 8628 |         let v_loss = self.bench.v2x.packet_loss_pct; | executable/support line; behavior impact depends on surrounding  
block context. |  
| 8629 |         let v_lat = self.bench.v2x.latency_ms; | executable/support line; behavior impact depends on surrounding  
block context. |  
| 8630 |         let v_rng_s = self.v2x_range_ema; | executable/support line; behavior impact depends on surrounding  
block context. |  
| 8631 |         let v_loss_s = self.v2x_loss_ema; | executable/support line; behavior impact depends on surrounding  
block context. |  
| 8632 |         let v_lat_s = self.v2x_lat_ema; | executable/support line; behavior impact depends on surrounding  
block context. |  
| 8633 |         let mut link_score = 100.0; | executable/support line; behavior impact depends on surrounding  
block context. |  
| 8634 |         link_score -= (v_loss_s * 2.0).clamp(0.0, 60.0); | executable/support line; behavior impact depends  
on surrounding block context. |  
| 8635 |         link_score -= ((v_lat_s - 10.0) * 1.6).clamp(0.0, 40.0); | executable/support line; behavior impact  
depends on surrounding block context. |  
| 8636 |         link_score -= ((180.0 - v_rng_s) * 0.15).clamp(0.0, 20.0); | executable/support line; behavior impact  
depends on surrounding block context. |  
| 8637 |         let link_score = link_score.clamp(0.0, 100.0); | executable/support line; behavior impact depends on  
surrounding block context. |  
| 8638 |         let v_tx = self.bench.v2x.bsm_tx_count; | executable/support line; behavior impact depends on surrounding  
block context. |  
| 8639 |         let v_rx = self.bench.v2x.rx_count; | executable/support line; behavior impact depends on surrounding  
block context. |  
| 8640 |         let fca = self.bench.v2x.forward_collision_alert; | executable/support line; behavior impact depends  
on surrounding block context. |  
| 8641 |         let intx = self.bench.v2x.intersection_alert; | executable/support line; behavior impact depends on  
surrounding block context. |  
| 8642 |         let emrg = self.bench.v2x.emergency_vehicle_alert; | executable/support line; behavior impact depends  
on surrounding block context. |
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| 8643 |         let haz = self.bench.v2x.road_hazard_alert; | executable/support line; behavior impact depends on su
unding block context. |
| 8644 |         let gps_lat = self.bench.gps.latitude_deg; | executable/support line; behavior impact depends on sur
unding block context. |
| 8645 |         let gps_lon = self.bench.gps.longitude_deg; | executable/support line; behavior impact depends on su
unding block context. |
| 8646 |         let nearby: Vec<_> = self | executable/support line; behavior impact depends on surrounding block co
xt. |
| 8647 |         .bench | executable/support line; behavior impact depends on surrounding block context. |
| 8648 |         .v2x | executable/support line; behavior impact depends on surrounding block context. |
| 8649 |         .nearby_vehicles | executable/support line; behavior impact depends on surrounding block context. |
| 8650 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 8651 |         .map(\|v\| { | executable/support line; behavior impact depends on surrounding block context. |
| 8652 |             ( | executable/support line; behavior impact depends on surrounding block context. |
| 8653 |                 v.vehicle_id, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 8654 |                 v.lat_deg, | executable/support line; behavior impact depends on surrounding block conte
|
| 8655 |                 v.lon_deg, | executable/support line; behavior impact depends on surrounding block conte
|
| 8656 |                 v.speed_ms, | executable/support line; behavior impact depends on surrounding block cont
. |
| 8657 |                 v.heading_deg, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8658 |             ) | executable/support line; behavior impact depends on surrounding block context. |
| 8659 |         }) | executable/support line; behavior impact depends on surrounding block context. |
| 8660 |         .collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 8661 |         let spat: Vec<_> = self | executable/support line; behavior impact depends on surrounding block cont
. |
| 8662 |         .bench | executable/support line; behavior impact depends on surrounding block context. |
| 8663 |         .v2x | executable/support line; behavior impact depends on surrounding block context. |
| 8664 |         .spat_messages | executable/support line; behavior impact depends on surrounding block context. |
| 8665 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 8666 |         .map(\|s\| { | executable/support line; behavior impact depends on surrounding block context. |
| 8667 |             ( | executable/support line; behavior impact depends on surrounding block context. |
| 8668 |                 s.intersection_id, | executable/support line; behavior impact depends on surrounding blo
context. |
| 8669 |                 s.phase_state, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8670 |                 s.time_to_change_s, | executable/support line; behavior impact depends on surrounding bl
context. |
| 8671 |                 s.distance_m, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 8672 |             ) | executable/support line; behavior impact depends on surrounding block context. |
| 8673 |         }) | executable/support line; behavior impact depends on surrounding block context. |
| 8674 |         .collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 8675 |         let wz: Vec<_> = self | executable/support line; behavior impact depends on surrounding block contex
|
| 8676 |         .bench | executable/support line; behavior impact depends on surrounding block context. |
| 8677 |         .v2x | executable/support line; behavior impact depends on surrounding block context. |
| 8678 |         .work_zones | executable/support line; behavior impact depends on surrounding block context. |
| 8679 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 8680 |         .map(\|w\| (w.description.clone(), w.distance_m, w.speed_limit_kmh)) | executable/support line;
avior impact depends on surrounding block context. |
| 8681 |         .collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 8682 |         let tel_conn = self.bench.telematics.connect_state; | executable/support line; behavior impact depen
on surrounding block context. |
| 8683 |         let telBars = self.bench.telematics.signal_bars; | executable/support line; behavior impact depends
surrounding block context. |
| 8684 |         let tel_rsrp = self.bench.telematics.rsrp_dbm; | executable/support line; behavior impact depends on
rrounding block context. |
| 8685 |         let tel_ping = self.bench.telematics.ping_ms; | executable/support line; behavior impact depends on
rounding block context. |
| 8686 |         let tel_dl = self.bench.telematics.download_kbps; | executable/support line; behavior impact depends
surrounding block context. |

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| 8687 |         let tel_ul = self.bench.telematics.upload_kbps; | executable/support line; behavior impact depends on
surrounding block context. |
| 8688 |         let tel_data = self.bench.telematics.data_used_mb; | executable/support line; behavior impact depend
n surrounding block context. |
| 8689 |         let tel_srv = self.bench.telematics.fleet_server.clone(); | executable/support line; behavior impact
depends on surrounding block context. |
| 8690 |         let tel_vid = self.bench.telematics.vehicle_id.clone(); | executable/support line; behavior impact d
nds on surrounding block context. |
| 8691 |         let tel_sync = self.bench.telematics.last_sync_s; | executable/support line; behavior impact depends
surrounding block context. |
| 8692 |         let tel_gf = self.bench.telematics.inside_geofence; | executable/support line; behavior impact depen
on surrounding block context. |
| 8693 |         let tel_ota_a = self.bench.telematics.ota_available; | executable/support line; behavior impact depe
on surrounding block context. |
| 8694 |         let tel_ota_d = self.bench.telematics.ota_downloading; | executable/support line; behavior impact de
ds on surrounding block context. |
| 8695 |         let tel_ota_v = self.bench.telematics.ota_version.clone(); | executable/support line; behavior impac
depends on surrounding block context. |
| 8696 |         let tel_ota_mb = self.bench.telematics.ota_size_mb; | executable/support line; behavior impact depen
on surrounding block context. |
| 8697 |         let tel_ota_p = self.bench.telematics.ota_progress_pct; | executable/support line; behavior impact d
nds on surrounding block context. |
| 8698 |         let tel_cmds: Vec<_> = self | executable/support line; behavior impact depends on surrounding block
text. |
| 8699 |             .bench | executable/support line; behavior impact depends on surrounding block context. |
| 8700 |             .telematics | executable/support line; behavior impact depends on surrounding block context. |
| 8701 |             .pending_commands | executable/support line; behavior impact depends on surrounding block contex
|
| 8702 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 8703 |             .map(\|c\| (format!("{}", c.cmd_type), c.id, c.params.clone())) | executable/support line; behav
impact depends on surrounding block context. |
| 8704 |             .collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 8705 |         let tel_evs: Vec<_> = self | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8706 |             .bench | executable/support line; behavior impact depends on surrounding block context. |
| 8707 |             .telematics | executable/support line; behavior impact depends on surrounding block context. |
| 8708 |             .events | executable/support line; behavior impact depends on surrounding block context. |
| 8709 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 8710 |             .map(\|e\| { | executable/support line; behavior impact depends on surrounding block context. |
| 8711 |                 ( | executable/support line; behavior impact depends on surrounding block context. |
| 8712 |                     e.timestamp, | executable/support line; behavior impact depends on surrounding block con
t. |
| 8713 |                     e.ack, | executable/support line; behavior impact depends on surrounding block context.
| 8714 |                     e.sent, | executable/support line; behavior impact depends on surrounding block context.
| 8715 |                     format!("{}", e.event_type), | error-propagating operation; changing it impacts failur
emantics and caller recovery paths. |
| 8716 |                     e.description[..e.description.len().min(60)].to_string(), | executable/support line; beh
or impact depends on surrounding block context. |
| 8717 |                 }) | executable/support line; behavior impact depends on surrounding block context. |
| 8718 |             }) | executable/support line; behavior impact depends on surrounding block context. |
| 8719 |             .collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 8720 |         let tel_total = self.bench.telematics.total_events_sent; | executable/support line; behavior impact
ends on surrounding block context. |
| 8721 |         let elapsed = self.bench.elapsed; | executable/support line; behavior impact depends on surrounding
ck context. |
| 8722 |         let mut do_ota = false; | executable/support line; behavior impact depends on surrounding block conte
. |
| 8723 | (blank) | line is blank; visual separation only. |
| 8724 |         ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block conte
|
| 8725 |             .id_salt("v2x::page_scroll") | executable/support line; behavior impact depends on surrounding b
k context. |
| 8726 |             .auto_shrink([false, false]) | executable/support line; behavior impact depends on surrounding b
k context. |
| 8727 |             .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte

```

```

|
| 8728 |         ui.columns(2, \|cols\| { | executable/support line; behavior impact depends on surrounding block con
t. |
| 8729 |         let ui = &mut cols[0]; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8730 |         egui::Frame::group(ui.style()) | executable/support line; behavior impact depends on surrounding
ock context. |
| 8731 |             .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on surround
block context. |
| 8732 |             .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8733 |                 ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 8734 |                     RichText::new("■ V2X – DSRC 5.9GHz + C-V2X") | executable/support line; behavior imp
depends on surrounding block context. |
| 8735 |                     .size(12.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 8736 |                     .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior imp
depends on surrounding block context. |
| 8737 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 8738 |                 ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 8739 |                     let dc = if v_dsrc { | executable/support line; behavior impact depends on surroundi
block context. |
| 8740 |                         Color32::GREEN | executable/support line; behavior impact depends on surrounding
ock context. |
| 8741 |                     } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 8742 |                         Color32::from_gray(60) | executable/support line; behavior impact depends on sur
nding block context. |
| 8743 |                     }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |
| 8744 |                     let cc = if v_cv2x { | executable/support line; behavior impact depends on surroundi
block context. |
| 8745 |                         Color32::GREEN | executable/support line; behavior impact depends on surrounding
ock context. |
| 8746 |                     } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 8747 |                         Color32::from_gray(60) | executable/support line; behavior impact depends on sur
nding block context. |
| 8748 |                     }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |
| 8749 |                     ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 8750 |                         RichText::new(if v_dsrc { "DSRC:ON" } else { "DSRC:off" }) | executable/support
e; behavior impact depends on surrounding block context. |
| 8751 |                         .size(11.0) | executable/support line; behavior impact depends on surrounding
lock context. |
| 8752 |                         .color(dc), | executable/support line; behavior impact depends on surrounding
lock context. |
| 8753 |                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 8754 |                     ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 8755 |                         RichText::new(if v_cv2x { "C-V2X:ON" } else { "C-V2X:off" }) | executable/support
ine; behavior impact depends on surrounding block context. |
| 8756 |                         .size(11.0) | executable/support line; behavior impact depends on surrounding
lock context. |
| 8757 |                         .color(cc), | executable/support line; behavior impact depends on surrounding
lock context. |
| 8758 |                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 8759 |                     ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 8760 |                         RichText::new(format!( | executable/support line; behavior impact depends on sur
nding block context. |
| 8761 |                             "{:.0}s Loss:{:.0}% Lat:{:.0}ms", | executable/support line; behavior impa

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depends on surrounding block context. |
| 8762 |             v_rng, v_loss, v_lat | executable/support line; behavior impact depends on s
ounding block context. |
| 8763 |             )) | executable/support line; behavior impact depends on surrounding block conte
|
| 8764 |             .size(11.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 8765 |             .color(Color32::from_gray(180)), | executable/support line; behavior impact depe
on surrounding block context. |
| 8766 |             ); | executable/support line; behavior impact depends on surrounding block context.
| 8767 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 8768 |             let link_col = if link_score >= 80.0 { | executable/support line; behavior impact depend
n surrounding block context. |
| 8769 |                 Color32::GREEN | executable/support line; behavior impact depends on surrounding blo
context. |
| 8770 |             } else if link_score >= 60.0 { | executable/support line; behavior impact depends on surr
nding block context. |
| 8771 |                 Color32::YELLOW | executable/support line; behavior impact depends on surrounding bl
context. |
| 8772 |             } else { | executable/support line; behavior impact depends on surrounding block context
| 8773 |                 Color32::RED | executable/support line; behavior impact depends on surrounding block
ntext. |
| 8774 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 8775 |             ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 8776 |                 RichText::new(format!( | executable/support line; behavior impact depends on surroun
g block context. |
| 8777 |                     "Smoothed link: {:.0}% \\\ range {:.0}m \\\ loss {:.1}% \\\ latency {:.1}ms", | exe
able/support line; behavior impact depends on surrounding block context. |
| 8778 |                     link_score, v_rng_s, v_loss_s, v_lat_s | executable/support line; behavior impac
depends on surrounding block context. |
| 8779 |                 )) | executable/support line; behavior impact depends on surrounding block context.
| 8780 |                 .size(10.6) | executable/support line; behavior impact depends on surrounding block
text. |
| 8781 |                 .color(link_col), | executable/support line; behavior impact depends on surrounding
ck context. |
| 8782 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 8783 |             digital_readout( | executable/support line; behavior impact depends on surrounding block
ntext. |
| 8784 |                 ui, | executable/support line; behavior impact depends on surrounding block context.
| 8785 |                 "TX/RX", | executable/support line; behavior impact depends on surrounding block con
t. |
| 8786 |                 &format!("{}", v_tx, v_rx), | executable/support line; behavior impact depends on
rrounding block context. |
| 8787 |                 Color32::from_gray(180), | executable/support line; behavior impact depends on surro
ing block context. |
| 8788 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 8789 |             ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 8790 |             for (a, l, c) in [ | iteration control; changing bounds/body affects throughput, timing,
d possibly safety guarantees. |
| 8791 |                 (fca, "■ FORWARD COLLISION", Color32::RED), | executable/support line; behavior impa
depends on surrounding block context. |
| 8792 |                 (intx, "■ INTERSECTION RED", Color32::RED), | executable/support line; behavior impa
depends on surrounding block context. |
| 8793 |                 (emrg, "■ EMERGENCY VEH", Color32::from_rgb(255, 150, 0)), | executable/support line
ehavior impact depends on surrounding block context. |
| 8794 |                 (haz, "■ ROAD HAZARD", Color32::YELLOW), | executable/support line; behavior impact
ends on surrounding block context. |
| 8795 |                 ] { | executable/support line; behavior impact depends on surrounding block context. |
| 8796 |                     if a { | conditional gate; changing condition shifts runtime path and side effects.
| 8797 |                         ui.label(RichText::new(l).size(12.0).color(c).strong()); | executable/support li
behavior impact depends on surrounding block context. |
| 8798 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership a

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lifetime scopes. |
| 8799 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 8800 |           ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 8801 |           ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 8802 |           RichText::new("V2V NEARBY:") | executable/support line; behavior impact depends on s
ounding block context. |
| 8803 |           .size(10.5) | executable/support line; behavior impact depends on surrounding bl
context. |
| 8804 |           .color(Color32::from_gray(130)), | executable/support line; behavior impact depe
on surrounding block context. |
| 8805 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 8806 |           ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding
ock context. |
| 8807 |           .id_salt("v2x::nearby_scroll") | executable/support line; behavior impact depends on
rrounding block context. |
| 8808 |           .max_height(130.0) | executable/support line; behavior impact depends on surrounding
ock context. |
| 8809 |           .auto_shrink([false, false]) | executable/support line; behavior impact depends on s
ounding block context. |
| 8810 |           .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 8811 |           for (vid, vlat, vlon, vspd, vhdg) in &nearby { | iteration control; changing bou
/body affects throughput, timing, and possibly safety guarantees. |
| 8812 |           let d = ((*vlat - gps_lat).powi(2) + (*vlon - gps_lon).powi(2)) | executable
pport line; behavior impact depends on surrounding block context. |
| 8813 |           .sqrt() | executable/support line; behavior impact depends on surroundin
lock context. |
| 8814 |           * 111320.0; | executable/support line; behavior impact depends on surrou
ng block context. |
| 8815 |           ui.label( | executable/support line; behavior impact depends on surrounding
ck context. |
| 8816 |           RichText::new(format!( | executable/support line; behavior impact depend
n surrounding block context. |
| 8817 |           "ID:{:04X} {:5.0}m {:.1}m/s Hdg:{:.0}°", | executable/support line;
avior impact depends on surrounding block context. |
| 8818 |           vid, d, vspd, vhdg | executable/support line; behavior impact depend
n surrounding block context. |
| 8819 |           )) | executable/support line; behavior impact depends on surrounding blo
context. |
| 8820 |           .size(10.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 8821 |           .monospace() | executable/support line; behavior impact depends on surro
ing block context. |
| 8822 |           .color(Color32::from_gray(190)), | executable/support line; behavior imp
depends on surrounding block context. |
| 8823 |           ); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8824 |           } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 8825 |           }); | executable/support line; behavior impact depends on surrounding block context.
| 8826 |           ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 8827 |           ui.label( | executable/support line; behavior impact depends on surrounding block context
|
| 8828 |           RichText::new("SPaT SIGNALS:") | executable/support line; behavior impact depends on
rrounding block context. |
| 8829 |           .size(10.5) | executable/support line; behavior impact depends on surrounding bl
context. |
| 8830 |           .color(Color32::from_gray(130)), | executable/support line; behavior impact depe
on surrounding block context. |
| 8831 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 8832 |           for (id, ph, ttc, dist) in &spat { | iteration control; changing bounds/body affects thr

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hput, timing, and possibly safety guarantees. |
| 8833 |         let c = match ph { | executable/support line; behavior impact depends on surrounding
ock context. |
| 8834 |             auto_breaking::v2x_telematics::TrafficPhase::Green => Color32::GREEN, | executable
support line; behavior impact depends on surrounding block context. |
| 8835 |             auto_breaking::v2x_telematics::TrafficPhase::Yellow => Color32::YELLOW, | execut
e/support line; behavior impact depends on surrounding block context. |
| 8836 |             auto_breaking::v2x_telematics::TrafficPhase::Red => Color32::RED, | executable/s
ort line; behavior impact depends on surrounding block context. |
| 8837 |             _ => Color32::GRAY, | executable/support line; behavior impact depends on surrou
ng block context. |
| 8838 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |
| 8839 |         ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surroun
g block context. |
| 8840 |             ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 8841 |                 RichText::new(format!("{:04X}", id)) | executable/support line; behavior imp
depends on surrounding block context. |
| 8842 |                 .size(11.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 8843 |                 .color(Color32::from_gray(180)), | executable/support line; behavior imp
depends on surrounding block context. |
| 8844 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8845 |             ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 8846 |                 RichText::new(format!("{}", ph)) | executable/support line; behavior impact
ends on surrounding block context. |
| 8847 |                 .size(13.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 8848 |                 .color(c) | executable/support line; behavior impact depends on surround
block context. |
| 8849 |                 .strong(), | executable/support line; behavior impact depends on surroun
g block context. |
| 8850 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8851 |             ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 8852 |                 RichText::new(format!("{:.1}s {:.0}m", ttc, dist)) | executable/support lin
behavior impact depends on surrounding block context. |
| 8853 |                 .size(11.0) | executable/support line; behavior impact depends on surrou
ng block context. |
| 8854 |                 .color(Color32::from_gray(180)), | executable/support line; behavior imp
depends on surrounding block context. |
| 8855 |             ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8856 |         }); | executable/support line; behavior impact depends on surrounding block context.
| 8857 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 8858 |     ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 8859 |     for (d, wd, spd) in &wz { | iteration control; changing bounds/body affects throughput,
ing, and possibly safety guarantees. |
| 8860 |         let c = if *wd < 100.0 { | executable/support line; behavior impact depends on surro
ing block context. |
| 8861 |             Color32::RED | executable/support line; behavior impact depends on surrounding b
k context. |
| 8862 |         } else if *wd < 300.0 { | executable/support line; behavior impact depends on surrou
ng block context. |
| 8863 |             Color32::YELLOW | executable/support line; behavior impact depends on surroundin
lock context. |
| 8864 |         } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 8865 |             Color32::from_gray(180) | executable/support line; behavior impact depends on su

```

```

unding block context. |
| 8866 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |
| 8867 |           ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 8868 |           RichText::new(format!("{}", - {:.0}m - {:.0}km/h", d, wd, spd)) | executable/suppo
line; behavior impact depends on surrounding block context. |
| 8869 |           .size(10.5) | executable/support line; behavior impact depends on surrounding
lock context. |
| 8870 |           .color(c), | executable/support line; behavior impact depends on surrounding
ock context. |
| 8871 |           ); | executable/support line; behavior impact depends on surrounding block context.
| 8872 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 8873 |       }); | executable/support line; behavior impact depends on surrounding block context. |
| 8874 | (blank) | line is blank; visual separation only. |
| 8875 |       let ui = &mut cols[1]; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8876 |       egui::Frame::group(ui.style()) | executable/support line; behavior impact depends on surrounding
ock context. |
| 8877 |           .fill(Color32::from_gray(15)) | executable/support line; behavior impact depends on surround
block context. |
| 8878 |           .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 8879 |           ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 8880 |           RichText::new("■ TELEMATICS") | executable/support line; behavior impact depends on
rounding block context. |
| 8881 |           .size(12.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 8882 |           .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior imp
depends on surrounding block context. |
| 8883 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 8884 |       let cc = match tel_conn { | executable/support line; behavior impact depends on surround
block context. |
| 8885 |           ConnectionState::Connected5G => Color32::GREEN, | executable/support line; behavior imp
depends on surrounding block context. |
| 8886 |           ConnectionState::Connected4G => Color32::from_rgb(100, 220, 100), | executable/support
e; behavior impact depends on surrounding block context. |
| 8887 |           ConnectionState::Connecting => Color32::YELLOW, | executable/support line; behavior imp
depends on surrounding block context. |
| 8888 |           ConnectionState::Offline => Color32::RED, | executable/support line; behavior impact de
ds on surrounding block context. |
| 8889 |       }; | scope delimiter; structure-only line, but misplaced edits can change ownership and
etime scopes. |
| 8890 |           ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 8891 |           ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 8892 |           RichText::new(format!("{}", tel_conn)) | executable/support line; behavior impac
depends on surrounding block context. |
| 8893 |           .size(13.0) | executable/support line; behavior impact depends on surrounding
lock context. |
| 8894 |           .color(cc) | executable/support line; behavior impact depends on surrounding
ock context. |
| 8895 |           .strong(), | executable/support line; behavior impact depends on surrounding
ock context. |
| 8896 |           ); | executable/support line; behavior impact depends on surrounding block context.
| 8897 |           for _ in 0..telBars { | iteration control; changing bounds/body affects throughput,
ming, and possibly safety guarantees. |
| 8898 |           ui.label(RichText::new("■").size(13.0).color(cc)); | executable/support line; be
ior impact depends on surrounding block context. |
| 8899 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 8900 |           for _ in telBars..5 { | iteration control; changing bounds/body affects throughput,

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ming, and possibly safety guarantees. |
| 8901 |         ui.label(RichText::new("■").size(13.0).color(Color32::from_gray(35))); | executable/s
/support line; behavior impact depends on surrounding block context. |
| 8902 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 8903 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 8904 |         digital_readout(ui, "Server", &tel_srv, Color32::from_gray(180)); | executable/support l
; behavior impact depends on surrounding block context. |
| 8905 |         digital_readout(ui, "Vehicle", &tel_vid, Color32::YELLOW); | executable/support line; be
ior impact depends on surrounding block context. |
| 8906 |         digital_readout( | executable/support line; behavior impact depends on surrounding block
ntext. |
| 8907 |             ui, | executable/support line; behavior impact depends on surrounding block context.
| 8908 |             "RSRP", | executable/support line; behavior impact depends on surrounding block cont
. |
| 8909 |             &format!("{:.0} dBm", tel_rsrp), | executable/support line; behavior impact depends
surrounding block context. |
| 8910 |             Color32::from_gray(180), | executable/support line; behavior impact depends on surro
ing block context. |
| 8911 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 8912 |         digital_readout( | executable/support line; behavior impact depends on surrounding block
ntext. |
| 8913 |             ui, | executable/support line; behavior impact depends on surrounding block context.
| 8914 |             "Ping", | executable/support line; behavior impact depends on surrounding block cont
. |
| 8915 |             &format!("{:.0} ms", tel_ping), | executable/support line; behavior impact depends o
urrounding block context. |
| 8916 |             Color32::from_gray(180), | executable/support line; behavior impact depends on surro
ing block context. |
| 8917 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 8918 |         digital_readout( | executable/support line; behavior impact depends on surrounding block
ntext. |
| 8919 |             ui, | executable/support line; behavior impact depends on surrounding block context.
| 8920 |             "DL/UL", | executable/support line; behavior impact depends on surrounding block con
t. |
| 8921 |             &format!("{:.0}/{:.0} kbps", tel_dl, tel_ul), | executable/support line; behavior im
t depends on surrounding block context. |
| 8922 |             Color32::from_gray(180), | executable/support line; behavior impact depends on surro
ing block context. |
| 8923 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 8924 |         digital_readout( | executable/support line; behavior impact depends on surrounding block
ntext. |
| 8925 |             ui, | executable/support line; behavior impact depends on surrounding block context.
| 8926 |             "Data", | executable/support line; behavior impact depends on surrounding block cont
. |
| 8927 |             &format!("{:.2} MB", tel_data), | executable/support line; behavior impact depends o
urrounding block context. |
| 8928 |             Color32::from_gray(180), | executable/support line; behavior impact depends on surro
ing block context. |
| 8929 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 8930 |         digital_readout( | executable/support line; behavior impact depends on surrounding block
ntext. |
| 8931 |             ui, | executable/support line; behavior impact depends on surrounding block context.
| 8932 |             "Last sync", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 8933 |             &format!("{:.0}s ago", elapsed - tel_sync), | executable/support line; behavior impa
depends on surrounding block context. |
| 8934 |             Color32::from_gray(155), | executable/support line; behavior impact depends on surro
ing block context. |
| 8935 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 8936 |         if let Some(gf) = tel_gf { | conditional gate; changing condition shifts runtime path an
ide effects. |
| 8937 |             ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 8938 |                 RichText::new(format!("■ Inside Geofence #{}", gf)) | executable/support line; b

```

```

vior impact depends on surrounding block context. |
| 8939 |             .size(11.0) | executable/support line; behavior impact depends on surrounding
lock context. |
| 8940 |             .color(Color32::GREEN), | executable/support line; behavior impact depends on
urrounding block context. |
| 8941 |             ); | executable/support line; behavior impact depends on surrounding block context.
| 8942 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 8943 |         if tel_ota_a \\|\\| tel_ota_d { | conditional gate; changing condition shifts runtime path
d side effects. |
| 8944 |             ui.separator(); | executable/support line; behavior impact depends on surrounding bl
context. |
| 8945 |             ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 8946 |                 RichText::new("OTA UPDATE:") | executable/support line; behavior impact depends
surrounding block context. |
| 8947 |                 .size(11.0) | executable/support line; behavior impact depends on surrounding
lock context. |
| 8948 |                 .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior
pact depends on surrounding block context. |
| 8949 |                 ); | executable/support line; behavior impact depends on surrounding block context.
| 8950 |             ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 8951 |                 RichText::new(format!("{}", (:.1}MB)", tel_ota_v, tel_ota_mb)) | executable/suppo
line; behavior impact depends on surrounding block context. |
| 8952 |                 .size(11.0) | executable/support line; behavior impact depends on surrounding
lock context. |
| 8953 |                 .color(Color32::YELLOW), | executable/support line; behavior impact depends
surrounding block context. |
| 8954 |                 ); | executable/support line; behavior impact depends on surrounding block context.
| 8955 |             if tel_ota_d { | conditional gate; changing condition shifts runtime path and side e
cts. |
| 8956 |                 bar_gauge( | executable/support line; behavior impact depends on surrounding blo
context. |
| 8957 |                     ui, | executable/support line; behavior impact depends on surrounding block
text. |
| 8958 |                     "Progress", | executable/support line; behavior impact depends on surroundin
lock context. |
| 8959 |                     tel_ota_p, | executable/support line; behavior impact depends on surrounding
ock context. |
| 8960 |                     100.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 8961 |                     "%", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 8962 |                     130.0, | executable/support line; behavior impact depends on surrounding blo
context. |
| 8963 |                     Color32::from_rgb(0, 210, 210), | executable/support line; behavior impact d
nds on surrounding block context. |
| 8964 |                 ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8965 |             } else if ui | executable/support line; behavior impact depends on surrounding block
ntext. |
| 8966 |                 .add( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 8967 |                     Button::new(RichText::new("■ Install").size(12.0)) | executable/support line
ehavior impact depends on surrounding block context. |
| 8968 |                     .fill(Color32::from_rgb(25, 70, 130)), | executable/support line; behavi
impact depends on surrounding block context. |
| 8969 |                 ) | executable/support line; behavior impact depends on surrounding block context
|
| 8970 |                 .clicked() | executable/support line; behavior impact depends on surrounding blo
context. |
| 8971 |             { | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 8972 |                 do_ota = true; | executable/support line; behavior impact depends on surrounding

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ock context. |
| 8973 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 8974 |           ui.separator(); | executable/support line; behavior impact depends on surrounding bl
context. |
| 8975 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 8976 |           if !tel_cmds.is_empty() { | conditional gate; changing condition shifts runtime path and
de effects. |
| 8977 |           ui.label( | executable/support line; behavior impact depends on surrounding block co
xt. |
| 8978 |           RichText::new("PENDING COMMANDS:") | executable/support line; behavior impact de
ds on surrounding block context. |
| 8979 |           .size(10.5) | executable/support line; behavior impact depends on surrounding
lock context. |
| 8980 |           .color(Color32::YELLOW), | executable/support line; behavior impact depends
surrounding block context. |
| 8981 |           ); | executable/support line; behavior impact depends on surrounding block context.
| 8982 |           for (ct, id, p) in &tel_cmds { | iteration control; changing bounds/body affects thr
hput, timing, and possibly safety guarantees. |
| 8983 |           ui.label( | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 8984 |           RichText::new(format!("{}", ct, id, p)) | executable/support line
havior impact depends on surrounding block context. |
| 8985 |           .size(10.5) | executable/support line; behavior impact depends on surrou
ng block context. |
| 8986 |           .color(Color32::YELLOW), | executable/support line; behavior impact depe
on surrounding block context. |
| 8987 |           ); | executable/support line; behavior impact depends on surrounding block conte
|
| 8988 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 8989 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 8990 |           ui.separator(); | executable/support line; behavior impact depends on surrounding block
text. |
| 8991 |           ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 8992 |           RichText::new(format!("EVENTS ({}):", tel_total)) | executable/support line; behavio
mpact depends on surrounding block context. |
| 8993 |           .size(11.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 8994 |           .color(Color32::from_gray(140)), | executable/support line; behavior impact depe
on surrounding block context. |
| 8995 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 8996 |           ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding
ock context. |
| 8997 |           .id_salt("v2x::events_scroll") | executable/support line; behavior impact depends on
rrounding block context. |
| 8998 |           .max_height(200.0) | executable/support line; behavior impact depends on surrounding
ock context. |
| 8999 |           .auto_shrink([false, false]) | executable/support line; behavior impact depends on s
ounding block context. |
| 9000 |           .stick_to_bottom(true) | executable/support line; behavior impact depends on surroun
g block context. |
| 9001 |           .show(ui, \ui\ { | executable/support line; behavior impact depends on surrounding
ock context. |
| 9002 |           for (ts, ack, sent, et, desc) in &tel_evs { | iteration control; changing bounds
dy affects throughput, timing, and possibly safety guarantees. |
| 9003 |           let c = if et.contains("Dtc") { | executable/support line; behavior impact d
nds on surrounding block context. |
| 9004 |           Color32::RED | executable/support line; behavior impact depends on surro
ing block context. |
| 9005 |           } else if et.contains("Speed") { | executable/support line; behavior impact
ends on surrounding block context. |

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| 9006 | Color32::YELLOW | executable/support line; behavior impact depends on surrounding block context. |  
| 9007 | } else if et.contains("Ota") { | executable/support line; behavior impact depends on surrounding block context. |  
| 9008 | Color32::from_rgb(0, 210, 210) | executable/support line; behavior impact depends on surrounding block context. |  
| 9009 | } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 9010 | Color32::from_gray(190) | executable/support line; behavior impact depends on surrounding block context. |  
| 9011 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 9012 | let a = if *ack { | executable/support line; behavior impact depends on surrounding block context. |  
| 9013 | "✓" | executable/support line; behavior impact depends on surrounding block context. |  
| 9014 | } else if *sent { | executable/support line; behavior impact depends on surrounding block context. |  
| 9015 | "→" | executable/support line; behavior impact depends on surrounding block context. |  
| 9016 | } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 9017 | "?" | error-propagating operation; changing it impacts failure semantics and caller recovery paths. |  
| 9018 | }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 9019 | ui.label(| executable/support line; behavior impact depends on surrounding block context. |  
| 9020 | RichText::new(format!("{:.0}s [{}] {} {}", ts, a, et, desc)) | executable/support line; behavior impact depends on surrounding block context. |  
| 9021 | .size(10.0) | executable/support line; behavior impact depends on surrounding block context. |  
| 9022 | .monospace() | executable/support line; behavior impact depends on surrounding block context. |  
| 9023 | .color(c), | executable/support line; behavior impact depends on surrounding block context. |  
| 9024 | ); | executable/support line; behavior impact depends on surrounding block context. |  
| 9025 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 9026 | }); | executable/support line; behavior impact depends on surrounding block context. |  
| 9027 | }); | executable/support line; behavior impact depends on surrounding block context. |  
| 9028 | }); | executable/support line; behavior impact depends on surrounding block context. |  
| 9029 | }); | executable/support line; behavior impact depends on surrounding block context. |  
| 9030 | if do_ota { | conditional gate; changing condition shifts runtime path and side effects. |  
| 9031 | self.cmds.push(Cmd::StartOta); | executable/support line; behavior impact depends on surrounding block context. |  
| 9032 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 9033 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 9034 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 9035 | (blank) | line is blank; visual separation only. |  
| 9036 | // ████████████████████████████████████████████████████████████████ | comment intent, no runtime effect. |  
| 9037 | // TAB UDS | comment line; documentation intent, no runtime effect. |  
| 9038 | // ████████████████████████████████████████████████████████████████ | comment intent, no runtime effect. |  
| 9039 | impl App { | implementation block; method behavior and trait conformance are defined here. |  
| 9040 | fn tab_uds(&mut self, ui: &mut Ui) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 9041 | let uds = &self.bench.uds_ecm; | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |  
| 9042 | let sess_s = format!("{}", uds.session); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |  
| 9043 | let sec_s = format!("#{?}", uds.security); | error-propagating operation; changing it impacts failure
```

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emantics and caller recovery paths. |
| 9044 |         let vin_s = uds.vin.clone(); | I/O or protocol-critical logic; edits can affect integration, timing,
and diagnostic outcomes. |
| 9045 |         let sw_s = uds.sw_version.clone(); | I/O or protocol-critical logic; edits can affect integration, t
ng, and diagnostic outcomes. |
| 9046 |         let hw_s = uds.hw_version.clone(); | I/O or protocol-critical logic; edits can affect integration, t
ng, and diagnostic outcomes. |
| 9047 |         let cal_s = uds.cal_id.clone(); | I/O or protocol-critical logic; edits can affect integration, timi
and diagnostic outcomes. |
| 9048 |         let ser_s = uds.ecu_serial.clone(); | I/O or protocol-critical logic; edits can affect integration,
ing, and diagnostic outcomes. |
| 9049 |         let idle_s = format!("{:.0}", uds.idle_rpm_cal); | I/O or protocol-critical logic; edits can affect
egration, timing, and diagnostic outcomes. |
| 9050 |         let rated_s = format!("{:.0}", uds.rated_rpm_cal); | I/O or protocol-critical logic; edits can affect
ntegration, timing, and diagnostic outcomes. |
| 9051 |         let torq_s = format!("{:.0} Nm", uds.max_torque_cal); | I/O or protocol-critical logic; edits can af
t integration, timing, and diagnostic outcomes. |
| 9052 |         let dpf_s = format!("{:.0}%", uds.dpf_regen_threshold); | I/O or protocol-critical logic; edits can
ect integration, timing, and diagnostic outcomes. |
| 9053 |         let fm_s = ["Standard", "Economy", "Power"][uds.fuel_map_select as usize]; | I/O or protocol-critica
logic; edits can affect integration, timing, and diagnostic outcomes. |
| 9054 |         let uds_events: Vec<_> = uds | I/O or protocol-critical logic; edits can affect integration, timing,
d diagnostic outcomes. |
| 9055 |             .event_log | executable/support line; behavior impact depends on surrounding block context. |
| 9056 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 9057 |             .map(\|e\| { | executable/support line; behavior impact depends on surrounding block context. |
| 9058 |                 ( | executable/support line; behavior impact depends on surrounding block context. |
| 9059 |                     e.timestamp, | executable/support line; behavior impact depends on surrounding block conte
t. |
| 9060 |                     e.service, | executable/support line; behavior impact depends on surrounding block conte
|
| 9061 |                     e.detail.clone(), | executable/support line; behavior impact depends on surrounding block
ontext. |
| 9062 |                     format!("{:?}", e.result), | error-propagating operation; changing it impacts failure se
tics and caller recovery paths. |
| 9063 |                 ) | executable/support line; behavior impact depends on surrounding block context. |
| 9064 |             }) | executable/support line; behavior impact depends on surrounding block context. |
| 9065 |             .collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 9066 |         let elapsed = self.bench.elapsed; | executable/support line; behavior impact depends on surrounding
ck context. |
| 9067 |         let mut send_bytes: Option<(Vec<u8>, u8)> = None; | I/O or protocol-critical logic; edits can affect
tegration, timing, and diagnostic outcomes. |
| 9068 |         let mut do_pick_flash = false; | I/O or protocol-critical logic; edits can affect integration, timin
and diagnostic outcomes. |
| 9069 |         let mut do_flash = false; | I/O or protocol-critical logic; edits can affect integration, timing, an
iagnostic outcomes. |
| 9070 |         let mut do_clean = false; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 9071 | (blank) | line is blank; visual separation only. |
| 9072 |         ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block conte
|
| 9073 |             .id_salt("uds::page_scroll") | I/O or protocol-critical logic; edits can affect integration, tim
, and diagnostic outcomes. |
| 9074 |             .auto_shrink([false, false]) | executable/support line; behavior impact depends on surrounding b
k context. |
| 9075 |             .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block conte
|
| 9076 |             ui.columns(2, \|cols\| { | executable/support line; behavior impact depends on surrounding block con
t. |
| 9077 |                 let ui = &mut cols[0]; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 9078 |                 ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 9079 |                     RichText::new("UDS ISO 14229 CONSOLE") | I/O or protocol-critical logic; edits can affect in
ration, timing, and diagnostic outcomes. |
| 9080 |                     .size(12.0) | executable/support line; behavior impact depends on surrounding block cont

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. |
| 9081 |             .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior impact depends
on surrounding block context. |
| 9082 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 9083 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 9084 |         ui.horizontal(\|ui\| { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 9085 |             ui.label(RichText::new("SA:").size(11.0).color(Color32::GRAY)); | executable/support line; b
vior impact depends on surrounding block context. |
| 9086 |             let mut sa_s = format!("{}", self.uds_sa); | I/O or protocol-critical logic; edits can
fect integration, timing, and diagnostic outcomes. |
| 9087 |             if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 9088 |                 .add_sized([65.0, 20.0], TextEdit::singleline(&mut sa_s)) | executable/support line; beh
or impact depends on surrounding block context. |
| 9089 |                 .changed() | executable/support line; behavior impact depends on surrounding block conte
|
| 9090 |             { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 9091 |                 if let Ok(v) = u8::from_str_radix(sa_s.trim().trim_start_matches("0x"), 16) { | conditio
gate; changing condition shifts runtime path and side effects. |
| 9092 |                     self.uds_sa = v; | I/O or protocol-critical logic; edits can affect integration, tim
, and diagnostic outcomes. |
| 9093 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 9094 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 9095 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 9096 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 9097 |             RichText::new("Request (hex):") | executable/support line; behavior impact depends on surrou
ng block context. |
| 9098 |             .size(11.0) | executable/support line; behavior impact depends on surrounding block cont
. |
| 9099 |             .color(Color32::GRAY), | executable/support line; behavior impact depends on surrounding
ock context. |
| 9100 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 9101 |         ui.add_sized( | executable/support line; behavior impact depends on surrounding block context. |
| 9102 |             [ui.available_width() - 10.0, 22.0], | executable/support line; behavior impact depends on s
ounding block context. |
| 9103 |             TextEdit::singleline(&mut self.uds_input).font(FontId::monospace(12.0)), | I/O or protocol-c
ical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 9104 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 9105 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 9106 |             RichText::new("Presets:") | executable/support line; behavior impact depends on surrounding
ck context. |
| 9107 |             .size(11.0) | executable/support line; behavior impact depends on surrounding block cont
. |
| 9108 |             .color(Color32::from_gray(140)), | executable/support line; behavior impact depends on s
ounding block context. |
| 9109 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 9110 |         ui.horizontal_wrapped(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 9111 |             for (l, h) in [ | iteration control; changing bounds/body affects throughput, timing, and po
bly safety guarantees. |
| 9112 |                 ("ExtSess", "10 02"), | executable/support line; behavior impact depends on surrounding
ck context. |
| 9113 |                 ("Prog", "10 03"), | executable/support line; behavior impact depends on surrounding blo
context. |
| 9114 |                 ("Default", "10 01"), | executable/support line; behavior impact depends on surrounding
ck context. |
| 9115 |                 ("Seed", "27 01"), | executable/support line; behavior impact depends on surrounding blo
context. |
| 9116 |                 ("Keep", "3E 00"), | executable/support line; behavior impact depends on surrounding blo
context. |
| 9117 |                 ("ClrDTC", "14 FF FF FF"), | executable/support line; behavior impact depends on surroun
g block context. |

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| 9118 |          ("RdDTC", "19 02 FF"), | executable/support line; behavior impact depends on surrounding
ock context. |
| 9119 |          ("RdRPM", "22 DD 01"), | executable/support line; behavior impact depends on surrounding
ock context. |
| 9120 |          ("RdCool", "22 DD 03"), | executable/support line; behavior impact depends on surrounding
lock context. |
| 9121 |          ("RdDPF", "22 DD 08"), | executable/support line; behavior impact depends on surrounding
ock context. |
| 9122 |          ("RdVIN", "22 F1 90"), | executable/support line; behavior impact depends on surrounding
ock context. |
| 9123 |          ("DPFRgn", "31 01 DF 01"), | executable/support line; behavior impact depends on surround
g block context. |
| 9124 |          ] { | executable/support line; behavior impact depends on surrounding block context. |
| 9125 |              if ui.small_button(1).clicked() { | conditional gate; changing condition shifts runtime
h and side effects. |
| 9126 |                  self.uds_input = h.into(); | I/O or protocol-critical logic; edits can affect integr
on, timing, and diagnostic outcomes. |
| 9127 |              } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 9128 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 9129 |      }); | executable/support line; behavior impact depends on surrounding block context. |
| 9130 |      ui.add_space(4.0); | executable/support line; behavior impact depends on surrounding block conte
|
| 9131 |          if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 9132 |              .add( | executable/support line; behavior impact depends on surrounding block context. |
| 9133 |                  Button::new(RichText::new("■ SEND").size(12.0)) | executable/support line; behavior impa
depends on surrounding block context. |
| 9134 |                      .fill(Color32::from_rgb(25, 70, 130)) | executable/support line; behavior impact dep
s on surrounding block context. |
| 9135 |                      .min_size(Vec2::new(120.0, 28.0)), | executable/support line; behavior impact depend
n surrounding block context. |
| 9136 |                  ) | executable/support line; behavior impact depends on surrounding block context. |
| 9137 |                  .clicked() | executable/support line; behavior impact depends on surrounding block context.
| 9138 |              { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 9139 |                  let bytes: Vec<u8> = self | executable/support line; behavior impact depends on surrounding
ck context. |
| 9140 |                      .uds_input | I/O or protocol-critical logic; edits can affect integration, timing, and d
nostic outcomes. |
| 9141 |                      .split_whitespace() | executable/support line; behavior impact depends on surrounding bl
context. |
| 9142 |                      .filter_map(|s| u8::from_str_radix(s, 16).ok()) | executable/support line; behavior im
t depends on surrounding block context. |
| 9143 |                      .collect(); | executable/support line; behavior impact depends on surrounding block cont
. |
| 9144 |                  if !bytes.is_empty() { | conditional gate; changing condition shifts runtime path and side e
cts. |
| 9145 |                      send_bytes = Some((bytes, self.uds_sa)); | I/O or protocol-critical logic; edits can aff
integration, timing, and diagnostic outcomes. |
| 9146 |                  } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 9147 |              } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 9148 |          if ui.small_button("■ Clear").clicked() { | conditional gate; changing condition shifts runtime
h and side effects. |
| 9149 |              self.uds_log.clear(); | I/O or protocol-critical logic; edits can affect integration, timing
nd diagnostic outcomes. |
| 9150 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 9151 |          ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 9152 |          ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 9153 |              RichText::new("ECM FLASH / UPLOAD WORKFLOW") | executable/support line; behavior impact depe
on surrounding block context. |
| 9154 |              .size(11.0) | executable/support line; behavior impact depends on surrounding block cont

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. |
| 9155 |         .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior impact depends
on surrounding block context. |
| 9156 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 9157 |         ui.horizontal_wrapped(\|ui\| { | executable/support line; behavior impact depends on surrounding
ock context. |
| 9158 |             if ui.button("Select FW (.bin)").clicked() { | conditional gate; changing condition shifts r
ime path and side effects. |
| 9159 |                 do_pick_flash = true; | I/O or protocol-critical logic; edits can affect integration, ti
g, and diagnostic outcomes. |
| 9160 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 9161 |                 if ui | conditional gate; changing condition shifts runtime path and side effects. |
| 9162 |                 .add_enabled(!self.uds_flash_path.is_empty(), Button::new("Flash Upload")) | I/O or prot
l-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 9163 |                 .clicked() | executable/support line; behavior impact depends on surrounding block conte
|
| 9164 |             { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 9165 |                 do_flash = true; | I/O or protocol-critical logic; edits can affect integration, timing,
d diagnostic outcomes. |
| 9166 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 9167 |             if ui.button("Clean (ClrDTC + ResetAdap)").clicked() { | conditional gate; changing conditio
hifts runtime path and side effects. |
| 9168 |                 do_clean = true; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 9169 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 9170 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 9171 |         if self.uds_flash_path.is_empty() { | conditional gate; changing condition shifts runtime path a
side effects. |
| 9172 |             ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 9173 |                 RichText::new("FW file: none selected") | executable/support line; behavior impact dependen
on surrounding block context. |
| 9174 |                 .size(10.0) | executable/support line; behavior impact depends on surrounding block
text. |
| 9175 |                 .color(Color32::from_gray(140)), | executable/support line; behavior impact depends
surrounding block context. |
| 9176 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 9177 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 9178 |             ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 9179 |                 RichText::new(format!("FW file: {}", self.uds_flash_path)) | I/O or protocol-critical lo
; edits can affect integration, timing, and diagnostic outcomes. |
| 9180 |                 .size(9.5) | executable/support line; behavior impact depends on surrounding block c
ext. |
| 9181 |                 .color(Color32::LIGHT_BLUE), | executable/support line; behavior impact depends on s
ounding block context. |
| 9182 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 9183 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 9184 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context.
| 9185 |         ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block c
ext. |
| 9186 |             .id_salt("uds::txrx_scroll") | I/O or protocol-critical logic; edits can affect integration,
ming, and diagnostic outcomes. |
| 9187 |             .max_height(200.0) | executable/support line; behavior impact depends on surrounding block c
ext. |
| 9188 |             .auto_shrink([false, false]) | executable/support line; behavior impact depends on surroundi
block context. |
| 9189 |             .stick_to_bottom(true) | executable/support line; behavior impact depends on surrounding blo
context. |
| 9190 |             .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block c
ext. |
| 9191 |                 for (is_req, line) in &self.uds_log { | iteration control; changing bounds/body affects

```

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oughput, timing, and possibly safety guarantees. |
| 9192 |           let c = if line.contains("■") { | executable/support line; behavior impact depends on
surrounding block context. |
| 9193 |           Color32::RED | executable/support line; behavior impact depends on surrounding b
k context. |
| 9194 |           } else if !is_req { | executable/support line; behavior impact depends on surrounding
lock context. |
| 9195 |           Color32::GREEN | executable/support line; behavior impact depends on surrounding
ock context. |
| 9196 |           } else { | executable/support line; behavior impact depends on surrounding block con
t. |
| 9197 |           Color32::from_gray(200) | executable/support line; behavior impact depends on su
unding block context. |
| 9198 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |
| 9199 |           ui.label(RichText::new(line).size(10.5).monospace().color(c)); | executable/support
e; behavior impact depends on surrounding block context. |
| 9200 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 9201 |           }); | executable/support line; behavior impact depends on surrounding block context. |
| 9202 | (blank) | line is blank; visual separation only. |
| 9203 |           let ui = &mut cols[1]; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 9204 |           ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 9205 |           RichText::new("ECM UDS SERVER STATE") | I/O or protocol-critical logic; edits can affect int
ation, timing, and diagnostic outcomes. |
| 9206 |           .size(12.0) | executable/support line; behavior impact depends on surrounding block cont
. |
| 9207 |           .color(Color32::from_rgb(80, 155, 255)), | executable/support line; behavior impact depe
on surrounding block context. |
| 9208 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 9209 |           let sc = match sess_s.trim() { | executable/support line; behavior impact depends on surrounding
ock context. |
| 9210 |           "DEFAULT" => Color32::from_gray(180), | executable/support line; behavior impact depends on
rounding block context. |
| 9211 |           "EXTENDED" => Color32::YELLOW, | executable/support line; behavior impact depends on surroun
g block context. |
| 9212 |           _ => Color32::RED, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 9213 |           }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
opes. |
| 9214 |           digital_readout(ui, "Session", &sess_s, sc); | executable/support line; behavior impact depends
surrounding block context. |
| 9215 |           digital_readout( | executable/support line; behavior impact depends on surrounding block context
| 9216 |           ui, | executable/support line; behavior impact depends on surrounding block context. |
| 9217 |           "Security", | executable/support line; behavior impact depends on surrounding block context.
| 9218 |           &sec_s, | executable/support line; behavior impact depends on surrounding block context. |
| 9219 |           if sec_s.contains("Locked") { | conditional gate; changing condition shifts runtime path and
de effects. |
| 9220 |           Color32::from_gray(120) | executable/support line; behavior impact depends on surrounding
lock context. |
| 9221 |           } else { | executable/support line; behavior impact depends on surrounding block context. |
| 9222 |           Color32::GREEN | executable/support line; behavior impact depends on surrounding block c
ext. |
| 9223 |           }, | executable/support line; behavior impact depends on surrounding block context. |
| 9224 |           ); | executable/support line; behavior impact depends on surrounding block context. |
| 9225 |           digital_readout(ui, "VIN", &vin_s, Color32::from_rgb(0, 210, 210)); | executable/support line; b
vior impact depends on surrounding block context. |
| 9226 |           digital_readout(ui, "SW", &sw_s, Color32::from_gray(200)); | executable/support line; behavior i
ct depends on surrounding block context. |
| 9227 |           digital_readout(ui, "HW", &hw_s, Color32::from_gray(200)); | executable/support line; behavior i
ct depends on surrounding block context. |
| 9228 |           digital_readout(ui, "Cal ID", &cal_s, Color32::from_gray(200)); | executable/support line; behav
impact depends on surrounding block context. |
| 9229 |           digital_readout(ui, "Serial", &ser_s, Color32::from_gray(200)); | executable/support line; behav

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impact depends on surrounding block context. |
| 9230 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 9231 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 9232 |             RichText::new("CALIBRATION:") | executable/support line; behavior impact depends on surrounding
block context. |
| 9233 |                 .size(11.0) | executable/support line; behavior impact depends on surrounding block context. |
. |
| 9234 |                 .color(Color32::from_gray(140)), | executable/support line; behavior impact depends on surrounding
block context. |
| 9235 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 9236 |             digital_readout(ui, "Idle RPM", &idle_s, Color32::YELLOW); | executable/support line; behavior impact
depends on surrounding block context. |
| 9237 |             digital_readout(ui, "Rated RPM", &rated_s, Color32::YELLOW); | executable/support line; behavior impact
depends on surrounding block context. |
| 9238 |             digital_readout(ui, "Max Torq", &torq_s, Color32::YELLOW); | executable/support line; behavior impact
depends on surrounding block context. |
| 9239 |             digital_readout(ui, "DPF Regen", &dpf_s, Color32::YELLOW); | executable/support line; behavior impact
depends on surrounding block context. |
| 9240 |             digital_readout(ui, "Fuel Map", fm_s, Color32::YELLOW); | executable/support line; behavior impact
depends on surrounding block context. |
| 9241 |         ui.separator(); | executable/support line; behavior impact depends on surrounding block context. |
| 9242 |         ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 9243 |             RichText::new("UDS LOG:") | I/O or protocol-critical logic; edits can affect integration, timing,
g, and diagnostic outcomes. |
| 9244 |                 .size(11.0) | executable/support line; behavior impact depends on surrounding block context. |
. |
| 9245 |                 .color(Color32::from_gray(140)), | executable/support line; behavior impact depends on surrounding
block context. |
| 9246 |             ); | executable/support line; behavior impact depends on surrounding block context. |
| 9247 |             ScrollArea::vertical() | executable/support line; behavior impact depends on surrounding block context. |
ext. |
| 9248 |                 .id_salt("uds::event_log_scroll") | I/O or protocol-critical logic; edits can affect integration,
n, timing, and diagnostic outcomes. |
| 9249 |                 .max_height(160.0) | executable/support line; behavior impact depends on surrounding block context. |
ext. |
| 9250 |                 .auto_shrink([false, false]) | executable/support line; behavior impact depends on surrounding
block context. |
| 9251 |                 .stick_to_bottom(true) | executable/support line; behavior impact depends on surrounding block
context. |
| 9252 |                 .show(ui, \|ui\| { | executable/support line; behavior impact depends on surrounding block context. |
ext. |
| 9253 |                     for (ts, svc, det, res) in &uds_events { | iteration control; changing bounds/body affects
throughput, timing, and possibly safety guarantees. |
| 9254 |                         let c = if res.contains("Positive") { | executable/support line; behavior impact depends
s on surrounding block context. |
| 9255 |                             Color32::GREEN | executable/support line; behavior impact depends on surrounding
ock context. |
| 9256 |                         } else if res.contains("Security") { | executable/support line; behavior impact depends
on surrounding block context. |
| 9257 |                             Color32::RED | executable/support line; behavior impact depends on surrounding block
k context. |
| 9258 |                         } else { | executable/support line; behavior impact depends on surrounding block context. |
t. |
| 9259 |                             Color32::YELLOW | executable/support line; behavior impact depends on surrounding
lock context. |
| 9260 |                     }; | scope delimiter; structure-only line, but misplaced edits can change ownership
lifetime scopes. |
| 9261 |                     ui.label( | executable/support line; behavior impact depends on surrounding block context. |
xt. |
| 9262 |                         RichText::new(format!("{:.2}s 0x{:02X} {}", ts, svc, det)) | executable/support
e; behavior impact depends on surrounding block context. |
| 9263 |                         .size(10.5) | executable/support line; behavior impact depends on surrounding block context. |
lock context. |
| 9264 |                         .monospace() | executable/support line; behavior impact depends on surrounding block context. |
block context. |

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| 9265 |                                     .color(c), | executable/support line; behavior impact depends on surrounding
ock context. |
| 9266 |                                     ); | executable/support line; behavior impact depends on surrounding block context. |
| 9267 |                                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 9268 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 9269 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 9270 |                                     }); | executable/support line; behavior impact depends on surrounding block context. |
| 9271 | (blank) | line is blank; visual separation only. |
| 9272 |         // Execute UDS send AFTER column closure released borrows | comment line; documentation intent, no r
ime effect. |
| 9273 |         if let Some((bytes, sa)) = send_bytes { | conditional gate; changing condition shifts runtime path a
side effects. |
| 9274 |             let ts = elapsed; | executable/support line; behavior impact depends on surrounding block context
|
| 9275 |             let _ = self.uds_send_and_log(sa, bytes, ts); | I/O or protocol-critical logic; edits can affect
tegration, timing, and diagnostic outcomes. |
| 9276 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 9277 |         if do_pick_flash { | conditional gate; changing condition shifts runtime path and side effects. |
| 9278 |             if let Some(path) = Self::pick_open_path("bin") { | conditional gate; changing condition shifts
time path and side effects. |
| 9279 |                 self.uds_flash_path = path; | I/O or protocol-critical logic; edits can affect integration,
ing, and diagnostic outcomes. |
| 9280 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 9281 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 9282 |         if do_flash { | conditional gate; changing condition shifts runtime path and side effects. |
| 9283 |             let ts = elapsed; | executable/support line; behavior impact depends on surrounding block context
|
| 9284 |             if matches!(self.hw_cfg.mode, io::hw::HwMode::Live) { | conditional gate; changing condition shi
runtime path and side effects. |
| 9285 |                 if self.uds_flash_path.trim().is_empty() { | conditional gate; changing condition shifts run
e path and side effects. |
| 9286 |                     self.uds_log | I/O or protocol-critical logic; edits can affect integration, timing, and
agnostic outcomes. |
| 9287 |                         .push((false, "                flash failed: seleccione .bin".into())); | I/O or protocol
itical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 9288 |                     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 9289 |                         match std::fs::read(&self.uds_flash_path) { | branch dispatch; changing arms alters beha
r for variants and input classes. |
| 9290 |                             Ok(payload) => { | executable/support line; behavior impact depends on surrounding b
k context. |
| 9291 |                                 match io::live_runner::live_flash_ecm_firmware( | branch dispatch; changing arms
ters behavior for variants and input classes. |
| 9292 |                                     &self.hw_cfg, | executable/support line; behavior impact depends on surround
block context. |
| 9293 |                                     Some(self.uds_sa), | I/O or protocol-critical logic; edits can affect integr
on, timing, and diagnostic outcomes. |
| 9294 |                                     &payload, | executable/support line; behavior impact depends on surrounding
ck context. |
| 9295 |                                     ) { | executable/support line; behavior impact depends on surrounding block cont
. |
| 9296 |                                     Ok(s) => self.uds_log.push(( | I/O or protocol-critical logic; edits can aff
integration, timing, and diagnostic outcomes. |
| 9297 |                                     false, | executable/support line; behavior impact depends on surrounding
ock context. |
| 9298 |                                     format!( | executable/support line; behavior impact depends on surroundi
block context. |
| 9299 |                                     "                LIVE flash ok: {} bytes / {} blocks / crc32=0x{:08X} /
2)V / VIN={} / SW={} / HW={} / CAL={} / SN={} / 0x36_ack={} / {} / seq_err={} / fc_timeout={}", | I/O or protocol-criti
logic; edits can affect integration, timing, and diagnostic outcomes. |
| 9300 |                                     s.bytes_sent, | executable/support line; behavior impact depends on
rounding block context. |

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| 9301 | s.blocks_sent, | executable/support line; behavior impact depends on
rounding block context. |
| 9302 | s.crc32, | executable/support line; behavior impact depends on surro
ing block context. |
| 9303 | s.supply_voltage_v, | executable/support line; behavior impact depen
on surrounding block context. |
| 9304 | s.vin, | executable/support line; behavior impact depends on surroun
g block context. |
| 9305 | s.sw_version, | executable/support line; behavior impact depends on
rounding block context. |
| 9306 | s.hw_version, | executable/support line; behavior impact depends on
rounding block context. |
| 9307 | s.calibration_id, | executable/support line; behavior impact depends
surrounding block context. |
| 9308 | s.ecu_serial, | executable/support line; behavior impact depends on
rounding block context. |
| 9309 | s.transport_diagnostics.transfer_data_blocks_acked, | executable/sup
t line; behavior impact depends on surrounding block context. |
| 9310 | s.transport_diagnostics.transfer_data_blocks_attempted, | executable
pport line; behavior impact depends on surrounding block context. |
| 9311 | s.transport_diagnostics.sequence_error_count, | executable/support l
; behavior impact depends on surrounding block context. |
| 9312 | s.transport_diagnostics.flowcontrol_timeout_count | executable/suppo
line; behavior impact depends on surrounding block context. |
| 9313 | ), | executable/support line; behavior impact depends on surrounding blo
context. |
| 9314 | )), | executable/support line; behavior impact depends on surrounding block
text. |
| 9315 | Err(e) => self | executable/support line; behavior impact depends on surroun
g block context. |
| 9316 | .uds_log | I/O or protocol-critical logic; edits can affect integration,
ming, and diagnostic outcomes. |
| 9317 | .push((false, format!("{}", LIVE flash failed: {}", e))), | I/O o
rotocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 9318 | } | scope delimiter; structure-only line, but misplaced edits can change ownersh
and lifetime scopes. |
| 9319 | } | scope delimiter; structure-only line, but misplaced edits can change ownership a
lifetime scopes. |
| 9320 | Err(e) => self.uds_log.push(( | I/O or protocol-critical logic; edits can affect int
ation, timing, and diagnostic outcomes. |
| 9321 | false, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 9322 | format!("{}", flash failed: leitura firmware: {}", e), | I/O or protocol-
tical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 9323 | )), | executable/support line; behavior impact depends on surrounding block context.
| 9324 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |
| 9325 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 9326 | } else { | executable/support line; behavior impact depends on surrounding block context. |
| 9327 | match self.run_uds_flash_pipeline(self.uds_sa, ts) { | branch dispatch; changing arms alters
havior for variants and input classes. |
| 9328 | Ok(msg) => self.uds_log.push((false, format!("{}", msg))), | I/O or protocol
itical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 9329 | Err(e) => self | executable/support line; behavior impact depends on surrounding block c
ext. |
| 9330 | .uds_log | I/O or protocol-critical logic; edits can affect integration, timing, and
agnostic outcomes. |
| 9331 | .push((false, format!("{}", flash failed: {}", e))), | I/O or protocol-critic
logic; edits can affect integration, timing, and diagnostic outcomes. |
| 9332 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet
scopes. |
| 9333 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 9334 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop

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| 9376 |           &self.pl_boost, | executable/support line; behavior impact depends on surrounding block context. | | |
| 9377 |           "BOOST kPa", | executable/support line; behavior impact depends on surrounding block context. |
| 9378 |           Color32::from_rgb(180, 80, 255), | executable/support line; behavior impact depends on surrounding block context. |
| 9379 |           300.0, | executable/support line; behavior impact depends on surrounding block context. |
| 9380 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 9381 |       ]; | executable/support line; behavior impact depends on surrounding block context. |
| 9382 |       let right = [ | executable/support line; behavior impact depends on surrounding block context. |
| 9383 |           (&self.pl_spd, "SPD km/h", Color32::LIGHT_BLUE, 55.0f64), | executable/support line; behavior impact depends on surrounding block context. |
| 9384 |           (&self.pl_fuel, "FUEL L/h", Color32::YELLOW, 30.0), | executable/support line; behavior impact depends on surrounding block context. |
| 9385 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 9386 |           &self.pl_dpfp, | executable/support line; behavior impact depends on surrounding block context. |
| 9387 |           "DPF S00T%", | executable/support line; behavior impact depends on surrounding block context. |
| 9388 |           Color32::from_rgb(200, 80, 200), | executable/support line; behavior impact depends on surrounding block context. |
| 9389 |           100.0, | executable/support line; behavior impact depends on surrounding block context. |
| 9390 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 9391 |           ( | executable/support line; behavior impact depends on surrounding block context. |
| 9392 |           &self.pl_hyd, | executable/support line; behavior impact depends on surrounding block context. |
| 9393 |           "HYD press", | executable/support line; behavior impact depends on surrounding block context. |
| 9394 |           Color32::from_rgb(0, 210, 210), | executable/support line; behavior impact depends on surrounding block context. |
| 9395 |           350.0, | executable/support line; behavior impact depends on surrounding block context. |
| 9396 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 9397 |       ]; | executable/support line; behavior impact depends on surrounding block context. |
| 9398 |       // Clone plot data so we can move into closure without borrow issues | comment line; documentation is correct, no runtime effect. |
| 9399 |       let ld: Vec<(Vec<[f64; 2]>, &str, Color32, f64)> = left | executable/support line; behavior impact depends on surrounding block context. |
| 9400 |           .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 9401 |           .map(|(v, l, c, m)| ((*v).clone(), *l, *c, *m)) | executable/support line; behavior impact depends on surrounding block context. |
| 9402 |           .collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 9403 |       let rd: Vec<(Vec<[f64; 2]>, &str, Color32, f64)> = right | executable/support line; behavior impact depends on surrounding block context. |
| 9404 |           .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 9405 |           .map(|(v, l, c, m)| ((*v).clone(), *l, *c, *m)) | executable/support line; behavior impact depends on surrounding block context. |
| 9406 |           .collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 9407 |       ui.columns(2, |cols| { | executable/support line; behavior impact depends on surrounding block context. |
| 9408 |           for (i, (v, lbl, c, y_max)) in ld.iter().enumerate() { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 9409 |               let last = v.last().map(|p| p[1]).unwrap_or(0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 9410 |               cols[0].horizontal(|ui| { | executable/support line; behavior impact depends on surrounding block context. |
| 9411 |                   ui.label(RichText::new(*lbl).size(11.0).color(*c)); | executable/support line; behavior impact depends on surrounding block context. |
| 9412 |                   ui.label( | executable/support line; behavior impact depends on surrounding block context. |
| 9413 |                       RichText::new(format!("{:.1}", last)) | executable/support line; behavior impact depends on surrounding block context. |
| 9414 |                       .size(13.0) | executable/support line; behavior impact depends on surrounding block context. |
| 9415 |                       .color(*c) | executable/support line; behavior impact depends on surrounding block context. |
| 9416 |                       .strong() | executable/support line; behavior impact depends on surrounding block context. |
| 9417 |                       .monospace(), | executable/support line; behavior impact depends on surrounding

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ck context. |
| 9418 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 9419 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 9420 |     let pts_c = v.clone(); | executable/support line; behavior impact depends on surrounding blo
context. |
| 9421 |         let cc = *c; | executable/support line; behavior impact depends on surrounding block context
| 9422 |         Plot::new(format!("plots::left::{:}:{:}", i, lbl)) | executable/support line; behavior impact
depends on surrounding block context. |
| 9423 |             .height(h) | executable/support line; behavior impact depends on surrounding block conte
|
| 9424 |             .include_y(0.0) | executable/support line; behavior impact depends on surrounding block
text. |
| 9425 |             .include_y(*y_max) | executable/support line; behavior impact depends on surrounding blo
context. |
| 9426 |             .allow_drag(false) | executable/support line; behavior impact depends on surrounding blo
context. |
| 9427 |             .allow_zoom(false) | executable/support line; behavior impact depends on surrounding blo
context. |
| 9428 |             .allow_scroll(false) | executable/support line; behavior impact depends on surrounding b
k context. |
| 9429 |             .show_axes([false, true]) | executable/support line; behavior impact depends on surround
block context. |
| 9430 |             .show(&mut cols[0], \|pu\| { | executable/support line; behavior impact depends on surro
ing block context. |
| 9431 |                 let pts: PlotPoints = pts_c.iter().map(\|p\| [p[0], p[1]]).collect(); | executable/s
ort line; behavior impact depends on surrounding block context. |
| 9432 |                 pu.line(Line::new(pts).color(cc).width(1.8)); | executable/support line; behavior im
t depends on surrounding block context. |
| 9433 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 9434 |             cols[0].add_space(4.0); | executable/support line; behavior impact depends on surrounding bl
context. |
| 9435 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 9436 |         for (i, (v, lbl, c, y_max)) in rd.iter().enumerate() { | iteration control; changing bounds/body
fects throughput, timing, and possibly safety guarantees. |
| 9437 |             let last = v.last().map(\|p\| p[1]).unwrap_or(0.0); | executable/support line; behavior impa
depends on surrounding block context. |
| 9438 |             cols[1].horizontal(\|ui\| { | executable/support line; behavior impact depends on surroundin
lock context. |
| 9439 |                 ui.label(RichText::new(*lbl).size(11.0).color(*c)); | executable/support line; behavior
act depends on surrounding block context. |
| 9440 |                 ui.label( | executable/support line; behavior impact depends on surrounding block contex
|
| 9441 |                     RichText::new(format!("{:.1}", last)) | executable/support line; behavior impact dep
s on surrounding block context. |
| 9442 |                     .size(13.0) | executable/support line; behavior impact depends on surrounding bl
context. |
| 9443 |                     .color(*c) | executable/support line; behavior impact depends on surrounding blo
context. |
| 9444 |                     .strong() | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 9445 |                     .monospace(), | executable/support line; behavior impact depends on surrounding
ck context. |
| 9446 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 9447 |             }); | executable/support line; behavior impact depends on surrounding block context. |
| 9448 |             let pts_c = v.clone(); | executable/support line; behavior impact depends on surrounding blo
context. |
| 9449 |                 let cc = *c; | executable/support line; behavior impact depends on surrounding block context
| 9450 |                 Plot::new(format!("plots::right::{:}:{:}", i, lbl)) | executable/support line; behavior impac
depends on surrounding block context. |
| 9451 |                 .height(h) | executable/support line; behavior impact depends on surrounding block conte
|
| 9452 |                 .include_y(0.0) | executable/support line; behavior impact depends on surrounding block
text. |
| 9453 |                 .include_y(*y_max) | executable/support line; behavior impact depends on surrounding blo

```

```
context. |
| 9454 |          .allow_drag(false) | executable/support line; behavior impact depends on surrounding blo
context. |
| 9455 |          .allow_zoom(false) | executable/support line; behavior impact depends on surrounding blo
context. |
| 9456 |          .allow_scroll(false) | executable/support line; behavior impact depends on surrounding b
k context. |
| 9457 |          .show_axes([false, true]) | executable/support line; behavior impact depends on surround
block context. |
| 9458 |          .show(&mut cols[1], \|pu\| { | executable/support line; behavior impact depends on surro
ing block context. |
| 9459 |          let pts: PlotPoints = pts_c.iter().map(\|p\| [p[0], p[1]]).collect(); | executable/s
ort line; behavior impact depends on surrounding block context. |
| 9460 |          pu.line(Line::new(pts).color(cc).width(1.8)); | executable/support line; behavior im
t depends on surrounding block context. |
| 9461 |          }); | executable/support line; behavior impact depends on surrounding block context. |
| 9462 |          cols[1].add_space(4.0); | executable/support line; behavior impact depends on surrounding bl
context. |
| 9463 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 9464 |          }); | executable/support line; behavior impact depends on surrounding block context. |
| 9465 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
| 9466 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
```

File: src/network_mgmt.rs

Role: Domain module in simulation/control/protocol stack.
Lines: 340 | Non-empty: 307 | Symbols: 23

Symbol Inventory

Line	Kind	Symbol	Change Impact
22	enum	NmState	data/structure coupling
35	impl	std::fmt::Display for NmState	behavior binding and method semantics
36	fn	fmt	signature/API coupling
50	enum	NmFrameType	data/structure coupling
64	struct	NmNode	data/structure coupling
77	impl	NmNode	behavior binding and method semantics
78	fn	new	signature/API coupling
90	fn	is_online	signature/API coupling
98	enum	BusNmState	data/structure coupling
111	impl	std::fmt::Display for BusNmState	behavior binding and method semantics
112	fn	fmt	signature/API coupling
125	struct	NetworkManager	data/structure coupling
146	struct	NmEvent	data/structure coupling
155	impl	Default for NetworkManager	behavior binding and method semantics
156	fn	default	signature/API coupling
161	impl	NetworkManager	behavior binding and method semantics
162	fn	new	signature/API coupling
189	fn	tick	signature/API coupling
306	fn	request_sleep	signature/API coupling
316	fn	wake_all	signature/API coupling
329	fn	node_state	signature/API coupling
333	fn	active_count	signature/API coupling
337	fn	total_nodes	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	/// Network Management – OSEK NM / AUTOSAR NM (simplified)	comment line; documentation intent, no runtime eff
2	///	comment line; documentation intent, no runtime effect.
3	/// In a real vehicle, Network Management coordinates ECU lifecycle:	comment line; documentation intent, no r

[illegible]

[illegible]

[illegible]

```
| 144 | (blank) | line is blank; visual separation only. |  
| 145 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |  
| 146 | pub struct NmEvent { | declares data layout; field changes may break serialization, tests, and callers. |  
| 147 |     pub timestamp: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 148 |     pub sa: u8, | executable/support line; behavior impact depends on surrounding block context. |  
| 149 |     pub node_name: &'static str, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 150 |     pub from: NmState, | executable/support line; behavior impact depends on surrounding block context. |  
| 151 |     pub to: NmState, | executable/support line; behavior impact depends on surrounding block context. |  
| 152 |     pub reason: &'static str, | executable/support line; behavior impact depends on surrounding block context  
| 153 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 154 | (blank) | line is blank; visual separation only. |  
| 155 | impl Default for NetworkManager { | implementation block; method behavior and trait conformance are defined h  
. |  
| 156 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, an  
outputs. |  
| 157 |         Self::new() | executable/support line; behavior impact depends on surrounding block context. |  
| 158 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 159 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 160 | (blank) | line is blank; visual separation only. |  
| 161 | impl NetworkManager { | implementation block; method behavior and trait conformance are defined here. |  
| 162 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, an  
outputs. |  
| 163 |         // Register all ECUs that participate in NM | comment line; documentation intent, no runtime effect.  
| 164 |         // In real OSEK NM, each ECU has a "node ID" separate from J1939 SA | comment line; documentation int  
, no runtime effect. |  
| 165 |         let nodes = vec![ | executable/support line; behavior impact depends on surrounding block context. |  
| 166 |             NmNode::new(0x00, "ECM #1", false), | executable/support line; behavior impact depends on surroun  
g block context. |  
| 167 |             NmNode::new(0x03, "TCM", false), | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 168 |             NmNode::new(0x27, "BCM/CAB", true), // BCM is typically NM coordinator | executable/support line;  
havior impact depends on surrounding block context. |  
| 169 |             NmNode::new(0x1C, "ICM/DASH", false), | executable/support line; behavior impact depends on surro  
ing block context. |  
| 170 |             NmNode::new(0x1E, "HCM", false), | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 171 |             NmNode::new(0x0B, "ABS/ESP", false), | executable/support line; behavior impact depends on surrou  
ng block context. |  
| 172 |             NmNode::new(0x26, "ISOBUS-VT", false), | executable/support line; behavior impact depends on surr  
ding block context. |  
| 173 |         ]; | executable/support line; behavior impact depends on surrounding block context. |  
| 174 | (blank) | line is blank; visual separation only. |  
| 175 |     NetworkManager { | executable/support line; behavior impact depends on surrounding block context. |  
| 176 |         nodes, | executable/supportline; behavior impact depends on surrounding block context. |  
| 177 |         bus_state: BusNmState::Off, | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 178 |         node_timeout_s: 0.5, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 179 |         sleep_countdown_s: 2.0, | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 180 |         sleep_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context  
| 181 |         wakeup_timer: 0.0, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 182 |         elapsed: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 183 |         event_log: Vec::new(), | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 184 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 185 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 186 | (blank) | line is blank; visual separation only. |  
| 187 | // ████████████████████████████████████████████████████████████████████████████████ | comment  
ntent, no runtime effect. |  
| 188 | /// Tick.powered_sas` = list of SA addresses that currently have power. | comment line; documentation i  
nt, no runtime effect. |
```

```
| 189 |         pub fn tick(&mut self, powered_sas: &[u8], dt: f64) { | declares executable behavior; signature/body chan  
alter API, control flow, and outputs. |  
| 190 |             self.elapsed += dt; | executable/support line; behavior impact depends on surrounding block context.  
| 191 | (blank) | line is blank; visual separation only. |  
| 192 |             // ■ Power state transitions ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ | comment line;  
intent, no runtime effect. |  
| 193 |             for node in &mut self.nodes { | iteration control; changing bounds/body affects throughput, timing, a  
possibly safety guarantees. |  
| 194 |                 let powered = powered_sas.contains(&node.sa); | executable/support line; behavior impact depends  
surrounding block context. |  
| 195 |                 match node.state { | branch dispatch; changing arms alters behavior for variants and input classe  
|  
| 196 |                     NmState::Unpowered if powered => { | executable/support line; behavior impact depends on surr  
ding block context. |  
| 197 |                         node.state = NmState::Initializing; | executable/support line; behavior impact depends on  
rrounding block context. |  
| 198 |                         node.last_nm_ts = self.elapsed; | executable/support line; behavior impact depends on sur  
nding block context. |  
| 199 |                         node.sleep_confirmed = false; | executable/support line; behavior impact depends on surro  
ing block context. |  
| 200 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti  
scopes. |  
| 201 |                     NmState::Initializing => { | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 202 |                         // ~200 ms init time before sending first NM frame | comment line; documentation intent,  
runtime effect. |  
| 203 |                         if self.elapsed - node.last_nm_ts >= 0.2 { | conditional gate; changing condition shifts  
time path and side effects. |  
| 204 |                             node.state = NmState::NormalOperation; | executable/support line; behavior impact dep  
s on surrounding block context. |  
| 205 |                             node.last_nm_ts = self.elapsed; | executable/support line; behavior impact depends on  
rrounding block context. |  
| 206 |                         } | scope delimiter; structure-only line, but misplaced edits can change ownership and li  
ime scopes. |  
| 207 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti  
scopes. |  
| 208 |                     NmState::NormalOperation if !powered => { | executable/support line; behavior impact depends  
surrounding block context. |  
| 209 |                         node.state = NmState::Unpowered; | executable/support line; behavior impact depends on su  
unding block context. |  
| 210 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti  
scopes. |  
| 211 |                     NmState::BusSleep if powered => { | executable/support line; behavior impact depends on surro  
ing block context. |  
| 212 |                         // Wakeup: any event (ignition, CAN frame) brings node back | comment line; documentation  
tent, no runtime effect. |  
| 213 |                         node.state = NmState::Initializing; | executable/support line; behavior impact depends on  
rrounding block context. |  
| 214 |                         node.last_nm_ts = self.elapsed; | executable/support line; behavior impact depends on sur  
nding block context. |  
| 215 |                         node.wakeup_reason = Some("IGN ON"); | executable/support line; behavior impact depends o  
urrounding block context. |  
| 216 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti  
scopes. |  
| 217 |                         _ => {} | executable/support line; behavior impact depends on surrounding block context. |  
| 218 |                     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 219 | (blank) | line is blank; visual separation only. |  
| 220 |             // NM heartbeat: each online node sends NM frame every 200 ms | comment line; documentation inten  
no runtime effect. |  
| 221 |             if node.state == NmState::NormalOperation { | conditional gate; changing condition shifts runtime  
th and side effects. |  
| 222 |                 node.last_nm_ts = self.elapsed; // simulated heartbeat | executable/support line; behavior im  
t depends on surrounding block context. |  
| 223 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
```



```
| es. |  
| 224 | (blank) | line is blank; visual separation only. |  
| 225 | // Timeout detection: node went silent | comment line; documentation intent, no runtime effect. |  
| 226 | let silent_for = self.elapsed - node.last_nm_ts; | executable/support line; behavior impact depends  
on surrounding block context. |  
| 227 | if node.state == NmState::NormalOperation && silent_for > self.node_timeout_s { | conditional gate;  
changing condition shifts runtime path and side effects. |  
| 228 | let prev = node.state; | executable/support line; behavior impact depends on surrounding block  
context. |  
| 229 | node.state = NmState::BusSleep; // node seems dead | executable/support line; behavior impact  
depends on surrounding block context. |  
| 230 | self.event_log.push(NmEvent { | executable/support line; behavior impact depends on surrounding  
block context. |  
| 231 | timestamp: self.elapsed, | executable/support line; behavior impact depends on surrounding  
lock context. |  
| 232 | sa: node.sa, | executable/support line; behavior impact depends on surrounding block context.  
. |  
| 233 | node_name: node.name, | executable/support line; behavior impact depends on surrounding block  
context. |  
| 234 | from: prev, | executable/support line; behavior impact depends on surrounding block context.  
|  
| 235 | to: NmState::BusSleep, | executable/support line; behavior impact depends on surrounding  
lock context. |  
| 236 | reason: "NM timeout", | executable/support line; behavior impact depends on surrounding block  
context. |  
| 237 | }); | executable/support line; behavior impact depends on surrounding block context. |  
| 238 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 239 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 240 | (blank) | line is blank; visual separation only. |  
| 241 | // ■ Bus state machine ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ | comment line  
intent, no runtime effect. |  
| 242 | let any_active = self | executable/support line; behavior impact depends on surrounding block context  
| 243 | .nodes | executable/support line; behavior impact depends on surrounding block context. |  
| 244 | .iter() | executable/support line; behavior impact depends on surrounding block context. |  
| 245 | .any(|n| n.state == NmState::NormalOperation); | executable/support line; behavior impact depend  
on surrounding block context. |  
| 246 | let _all_ready = !self.nodes.is_empty() | executable/support line; behavior impact depends on surround  
g block context. |  
| 247 | && self | executable/support line; behavior impact depends on surrounding block context. |  
| 248 | .nodes | executable/support line; behavior impact depends on surrounding block context. |  
| 249 | .iter() | executable/support line; behavior impact depends on surrounding block context. |  
| 250 | .filter(|n| n.is_online()) | executable/support line; behavior impact depends on surrounding  
lock context. |  
| 251 | .all(|n| n.sleep_confirmed || n.state == NmState::ReadyToSleep); | executable/support lin  
behavior impact depends on surrounding block context. |  
| 252 | (blank) | line is blank; visual separation only. |  
| 253 | self.bus_state = match self.bus_state { | executable/support line; behavior impact depends on surroun  
g block context. |  
| 254 | BusNmState::Off => { | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 255 | if any_active { | conditional gate; changing condition shifts runtime path and side effects.  
| 256 | self.wakeup_timer = 0.0; | executable/support line; behavior impact depends on surrounding  
lock context. |  
| 257 | BusNmState::WakingUp | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 258 | } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 259 | BusNmState::Off | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 260 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti  
scopes. |  
| 261 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 262 | BusNmState::WakingUp => { | executable/support line; behavior impact depends on surrounding block
```

```

ntext. |
| 263 |         self.wakeup_timer += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 264 |         if self.wakeup_timer >= 0.1 { | conditional gate; changing condition shifts runtime path and side effects. |
| 265 |             BusNmState::Active | executable/support line; behavior impact depends on surrounding block context. |
| 266 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 267 |             BusNmState::WakingUp | executable/support line; behavior impact depends on surrounding block context. |
| 268 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 269 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 270 |     BusNmState::Active => { | executable/support line; behavior impact depends on surrounding block context. |
| 271 |         if !any_active { | conditional gate; changing condition shifts runtime path and side effects. |
| 272 |             self.sleep_timer = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 273 |             BusNmState::PrepShutdown | executable/support line; behavior impact depends on surrounding block context. |
| 274 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 275 |             BusNmState::Active | executable/support line; behavior impact depends on surrounding block context. |
| 276 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 277 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 278 |     BusNmState::PrepShutdown => { | executable/support line; behavior impact depends on surrounding block context. |
| 279 |         self.sleep_timer += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 280 |         if any_active { | conditional gate; changing condition shifts runtime path and side effects. |
| 281 |             BusNmState::Active // someone woke up again | executable/support line; behavior impact depends on surrounding block context. |
| 282 |         } else if self.sleep_timer >= self.sleep_countdown_s { | executable/support line; behavior impact depends on surrounding block context. |
| 283 |             // All nodes confirmed sleep | comment line; documentation intent, no runtime effect. |
| 284 |             for node in &mut self.nodes { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 285 |                 if node.state == NmState::NormalOperation { | conditional gate; changing condition shifts runtime path and side effects. |
| 286 |                     node.state = NmState::BusSleep; | executable/support line; behavior impact depends on surrounding block context. |
| 287 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 288 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 289 |             BusNmState::Sleep | executable/support line; behavior impact depends on surrounding block context. |
| 290 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 291 |             BusNmState::PrepShutdown | executable/support line; behavior impact depends on surrounding block context. |
| 292 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 293 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 294 |     BusNmState::Sleep => { | executable/support line; behavior impact depends on surrounding block context. |
| 295 |         if any_active { | conditional gate; changing condition shifts runtime path and side effects. |
| 296 |             BusNmState::WakingUp | executable/support line; behavior impact depends on surrounding block context. |
| 297 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 298 |             BusNmState::Sleep | executable/support line; behavior impact depends on surrounding block context. |
| 299 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

```
| 299 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. | | | | |
| 300 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 301 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 302 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 303 | (blank) | line is blank; visual separation only. |
| 304 |     // ████████████████████████████████████████████████████████████████████████████████████████████████████ | comment line; documentation intent, no runtime effect. |
| 305 |     /// Called when an ECU is done and ready for bus sleep | comment line; documentation intent, no runtime effect. |
| 306 | pub fn request_sleep(&mut self, sa: u8) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 307 |     if let Some(node) = self.nodes.iter_mut().find(|n| n.sa == sa) { | conditional gate; changing condition shifts runtime path and side effects. |
| 308 |         if node.state == NmState::NormalOperation { | conditional gate; changing condition shifts runtime path and side effects. |
| 309 |             node.state = NmState::ReadyToSleep; | executable/support line; behavior impact depends on surrounding block context. |
| 310 |             node.sleep_confirmed = true; | executable/support line; behavior impact depends on surrounding block context. |
| 311 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 312 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 313 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 314 | (blank) | line is blank; visual separation only. |
| 315 |     /// Wake the bus (e.g., key-on event) | comment line; documentation intent, no runtime effect. |
| 316 | pub fn wake_all(&mut self, reason: &'static str) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 317 |     for node in &mut self.nodes { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 318 |         if node.state == NmState::BusSleep { | conditional gate; changing condition shifts runtime path and side effects. |
| 319 |             node.state = NmState::Initializing; | executable/support line; behavior impact depends on surrounding block context. |
| 320 |             node.last_nm_ts = self.elapsed(); | executable/support line; behavior impact depends on surrounding block context. |
| 321 |             node.wakeup_reason = Some(reason); | executable/support line; behavior impact depends on surrounding block context. |
| 322 |             node.sleep_confirmed = false; | executable/support line; behavior impact depends on surrounding block context. |
| 323 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 324 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 325 |     self.bus_state = BusNmState::WakingUp; | executable/support line; behavior impact depends on surrounding block context. |
| 326 |     self.wakeup_timer = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 327 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 328 | (blank) | line is blank; visual separation only. |
| 329 | pub fn node_state(&self, sa: u8) -> Option<NmState> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 330 |     self.nodes.iter().find(|n| n.sa == sa).map(|n| n.state) | executable/support line; behavior impact depends on surrounding block context. |
| 331 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 332 | (blank) | line is blank; visual separation only. |
| 333 | pub fn active_count(&self) -> usize { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 334 |     self.nodes.iter().filter(|n| n.is_online()).count() | executable/support line; behavior impact depends on surrounding block context. |
| 335 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 336 | (blank) | line is blank; visual separation only. |
```

```
| 337 |      pub fn total_nodes(&self) -> usize { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 338 |          self.nodes.len() | executable/support line; behavior impact depends on surrounding block context. |
| 339 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 340 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
```

File: src/nvm.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 385 | Non-empty: 354 | Symbols: 40

Symbol Inventory

Line	Kind	Symbol	Change Impact
21	enum	NvmRegion	data/structure coupling
36	impl	std::fmt::Display for NvmRegion	behavior binding and method semantics
37	fn	fmt	signature/API coupling
53	struct	NvmBlock	data/structure coupling
67	impl	NvmBlock	behavior binding and method semantics
68	fn	new_f64	signature/API coupling
80	fn	new_u64	signature/API coupling
92	fn	new_str	signature/API coupling
104	fn	write_f64	signature/API coupling
109	fn	write_u64	signature/API coupling
114	fn	write_str	signature/API coupling
119	fn	is_worn	signature/API coupling
122	fn	wear_pct	signature/API coupling
130	struct	Adaptations	data/structure coupling
149	impl	Default for Adaptations	behavior binding and method semantics
150	fn	default	signature/API coupling
155	impl	Adaptations	behavior binding and method semantics
156	fn	new	signature/API coupling
170	fn	update_fuel_trim	signature/API coupling
182	struct	StoredDtcRecord	data/structure coupling
195	impl	StoredDtcRecord	behavior binding and method semantics
196	fn	new	signature/API coupling
209	fn	increment_cycle	signature/API coupling
217	struct	NvmStore	data/structure coupling
233	impl	Default for NvmStore	behavior binding and method semantics
234	fn	default	signature/API coupling
239	impl	NvmStore	behavior binding and method semantics
240	fn	new	signature/API coupling
287	fn	tick	signature/API coupling
297	fn	flush	signature/API coupling
309	fn	on_key_on	signature/API coupling
326	fn	on_key_off	signature/API coupling
331	fn	write_f64	signature/API coupling
336	fn	write_u64	signature/API coupling
341	fn	read_f64	signature/API coupling
347	fn	read_u64	signature/API coupling
353	fn	read_str	signature/API coupling
360	fn	worn_blocks	signature/API coupling
364	fn	add_stored_dtc	signature/API coupling
382	fn	noise	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	///! NVM – Non-Volatile Memory simulation (EEPROM / Flash emulation)	comment line; documentation intent, no runtime effect.
2	///!	comment line; documentation intent, no runtime effect.
3	///! In a real ECU, NVM stores:	comment line; documentation intent, no runtime effect.
4	///! • Odometer and engine hours (survive power cuts)	comment line; documentation intent, no runtime effect.

[illegible]

```
| 47 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 48 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 49 | (blank) | line is blank; visual separation only. |  
| 50 | // ■■ NvmBlock ████████████████████████████████████████████████████████████████████████ | comment  
intent, no runtime effect. |  
| 51 | (blank) | line is blank; visual separation only. |  
| 52 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |  
| 53 | pub struct NvmBlock { | declares data layout; field changes may break serialization, tests, and callers. |  
| 54 |     pub region: NvmRegion, | executable/support line; behavior impact depends on surrounding block context. |  
| 55 |     pub name: &'static str, | executable/support line; behavior impact depends on surrounding block context. |  
| 56 |     /// Stored value (64-bit, multi-purpose) | comment line; documentation intent, no runtime effect. |  
| 57 |     pub value_u64: u64, | executable/support line; behavior impact depends on surrounding block context. |  
| 58 |     pub value_f64: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 59 |     pub value_str: String, | executable/support line; behavior impact depends on surrounding block context. |  
| 60 |     /// Write counter – for wear tracking | comment line; documentation intent, no runtime effect. |  
| 61 |     pub write_count: u32, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnosti  
outcomes. |  
| 62 |     /// Max writes before wear-out (EEPROM: ~100K, Flash: ~10K) | comment line; documentation intent, no runti  
effect. |  
| 63 |     pub max_writes: u32, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnosti  
utcomes. |  
| 64 |     pub dirty: bool, // modified since last "flush" | executable/support line; behavior impact depends on surr  
ding block context. |  
| 65 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 66 | (blank) | line is blank; visual separation only. |  
| 67 | impl NvmBlock { | implementation block; method behavior and trait conformance are defined here. |  
| 68 |     pub fn new_f64(region: NvmRegion, name: &'static str, initial: f64, max_wr: u32) -> Self { | declares exec  
ble behavior; signature/body changes alter API, control flow, and outputs. |  
| 69 |         NvmBlock { | executable/support line; behavior impact depends on surrounding block context. |  
| 70 |             region, | executable/support line; behavior impact depends on surrounding block context. |  
| 71 |             name, | executable/support line; behavior impact depends on surrounding block context. |  
| 72 |             value_u64: 0, | executable/support line; behavior impact depends on surrounding block context. |  
| 73 |             value_f64: initial, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 74 |             value_str: String::new(), | executable/support line; behavior impact depends on surrounding block  
text. |  
| 75 |             write_count: 0, | I/O or protocol-critical logic; edits can affect integration, timing, and diagno  
c outcomes. |  
| 76 |             dirty: false, | executable/support line; behavior impact depends on surrounding block context. |  
| 77 |             max_writes: max_wr, | I/O or protocol-critical logic; edits can affect integration, timing, and di  
ostic outcomes. |  
| 78 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes  
| 79 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 80 |     pub fn new_u64(region: NvmRegion, name: &'static str, initial: u64, max_wr: u32) -> Self { | declares exec  
ble behavior; signature/body changes alter API, control flow, and outputs. |  
| 81 |         NvmBlock { | executable/support line; behavior impact depends on surrounding block context. |  
| 82 |             region, | executable/support line; behavior impact depends on surrounding block context. |  
| 83 |             name, | executable/support line; behavior impact depends on surrounding block context. |  
| 84 |             value_u64: initial, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 85 |             value_f64: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 86 |             value_str: String::new(), | executable/support line; behavior impact depends on surrounding block  
text. |  
| 87 |             write_count: 0, | I/O or protocol-critical logic; edits can affect integration, timing, and diagno  
c outcomes. |  
| 88 |             dirty: false, | executable/support line; behavior impact depends on surrounding block context. |  
| 89 |             max_writes: max_wr, | I/O or protocol-critical logic; edits can affect integration, timing, and di  
ostic outcomes. |  
| 90 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes  
| 91 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 92 |     pub fn new_str(region: NvmRegion, name: &'static str, initial: &str, max_wr: u32) -> Self { | declares exe  
able behavior; signature/body changes alter API, control flow, and outputs. |  
| 93 |         NvmBlock { | executable/support line; behavior impact depends on surrounding block context. |  
| 94 |             region, | executable/support line; behavior impact depends on surrounding block context. |  
| 95 |             name, | executable/support line; behavior impact depends on surrounding block context. |
```

```
| 96 |         value_u64: 0, | executable/support line; behavior impact depends on surrounding block context. |  
| 97 |         value_f64: 0.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 98 |         value_str: initial.into(), | executable/support line; behavior impact depends on surrounding block  
ncontext. |  
| 99 |         write_count: 0, | I/O or protocol-critical logic; edits can affect integration, timing, and diagno  
c outcomes. |  
| 100 |         dirty: false, | executable/support line; behavior impact depends on surrounding block context. |  
| 101 |         max_writes: max_wr, | I/O or protocol-critical logic; edits can affect integration, timing, and d  
nostic outcomes. |  
| 102 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 103 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 104 | pub fn write_f64(&mut self, v: f64) { | declares executable behavior; signature/body changes alter API, c  
rol flow, and outputs. |  
| 105 |     self.value_f64 = v; | executable/support line; behavior impact depends on surrounding block context.  
| 106 |     self.write_count += 1; | I/O or protocol-critical logic; edits can affect integration, timing, and di  
ostic outcomes. |  
| 107 |     self.dirty = true; | executable/support line; behavior impact depends on surrounding block context. |  
| 108 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 109 | pub fn write_u64(&mut self, v: u64) { | declares executable behavior; signature/body changes alter API, c  
rol flow, and outputs. |  
| 110 |     self.value_u64 = v; | executable/support line; behavior impact depends on surrounding block context.  
| 111 |     self.write_count += 1; | I/O or protocol-critical logic; edits can affect integration, timing, and di  
ostic outcomes. |  
| 112 |     self.dirty = true; | executable/support line; behavior impact depends on surrounding block context. |  
| 113 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 114 | pub fn write_str(&mut self, v: String) { | declares executable behavior; signature/body changes alter API  
ontrol flow, and outputs. |  
| 115 |     self.value_str = v; | executable/support line; behavior impact depends on surrounding block context.  
| 116 |     self.write_count += 1; | I/O or protocol-critical logic; edits can affect integration, timing, and di  
ostic outcomes. |  
| 117 |     self.dirty = true; | executable/support line; behavior impact depends on surrounding block context. |  
| 118 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 119 | pub fn is_worn(&self) -> bool { | declares executable behavior; signature/body changes alter API, control  
ow, and outputs. |  
| 120 |     self.write_count >= self.max_writes | I/O or protocol-critical logic; edits can affect integration, t  
ng, and diagnostic outcomes. |  
| 121 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 122 | pub fn wear_pct(&self) -> f64 { | declares executable behavior; signature/body changes alter API, control  
ow, and outputs. |  
| 123 |     self.write_count as f64 / self.max_writes as f64 * 100.0 | I/O or protocol-critical logic; edits can  
ect integration, timing, and diagnostic outcomes. |  
| 124 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 125 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 126 | (blank) | line is blank; visual separation only. |  
| 127 | // ■■ Adaptation Values ████████████████████████████████████████████████████████████████ | comment l  
intent, no runtime effect. |  
| 128 | (blank) | line is blank; visual separation only. |  
| 129 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |  
| 130 | pub struct Adaptations { | declares data layout; field changes may break serialization, tests, and callers. |  
| 131 |     /// Long-term fuel trim (%) – learned correction to injector duration | comment line; documentation inten  
no runtime effect. |  
| 132 |     pub fuel_trim_lt_pct: f64, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 133 |     /// Short-term fuel trim | comment line; documentation intent, no runtime effect. |  
| 134 |     pub fuel_trim_st_pct: f64, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 135 |     /// Cylinder balance offset per cylinder (injection time correction in µs) | comment line; documentatio  
ent, no runtime effect. |  
| 136 |     pub injector_offset_us: [f64; 6], | executable/support line; behavior impact depends on surrounding block  
ncontext. |  
| 137 |     /// Clutch fill time adaptation (ms) – learned from TCM | comment line; documentation intent, no runtime  
ect. |  
| 138 |     pub clutch_fill_time_ms: f64, | executable/support line; behavior impact depends on surrounding block con  
t. |
```

[illegible]

[illegible]

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| 225 |      /// Timestamp of last successful NVM flush (seconds uptime) | comment line; documentation intent, no runtime
effect. |
| 226 |      last_flush_ts: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 227 |      pub flush_interval_s: f64, | executable/support line; behavior impact depends on surrounding block context
|
| 228 | (blank) | line is blank; visual separation only. |
| 229 |      /// Total writes across all blocks | comment line; documentation intent, no runtime effect. |
| 230 |      pub total_writes: u64, | I/O or protocol-critical logic; edits can affect integration, timing, and diagno
c outcomes. |
| 231 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 232 | (blank) | line is blank; visual separation only. |
| 233 | impl Default for NvmStore { | implementation block; method behavior and trait conformance are defined here. |
| 234 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
outputs. |
| 235 |         Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 236 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 237 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 238 | (blank) | line is blank; visual separation only. |
| 239 | impl NvmStore { | implementation block; method behavior and trait conformance are defined here. |
| 240 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
outputs. |
| 241 |         let blocks = vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 242 |             // Counters region (high write frequency) | comment line; documentation intent, no runtime effect
| 243 |             NvmBlock::new_f64(NvmRegion::Counters, "engine_hours", 2347.5, 500_000), | executable/support line
behavior impact depends on surrounding block context. |
| 244 |             NvmBlock::new_f64(NvmRegion::Counters, "odometer_km", 12450.0, 500_000), | executable/support line
behavior impact depends on surrounding block context. |
| 245 |             NvmBlock::new_f64(NvmRegion::Counters, "total_fuel_l", 28650.0, 500_000), | executable/support line
behavior impact depends on surrounding block context. |
| 246 |             NvmBlock::new_u64(NvmRegion::Counters, "key_on_count", 1247, 200_000), | executable/support line;
havior impact depends on surrounding block context. |
| 247 |             NvmBlock::new_u64(NvmRegion::Counters, "dpf_regen_count", 47, 100_000), | executable/support line
ehavior impact depends on surrounding block context. |
| 248 |             NvmBlock::new_u64(NvmRegion::Counters, "abs_event_count", 8, 100_000), | executable/support line;
havior impact depends on surrounding block context. |
| 249 |             NvmBlock::new_u64(NvmRegion::Counters, "tcs_event_count", 12, 100_000), | executable/support line
ehavior impact depends on surrounding block context. |
| 250 |             NvmBlock::new_u64(NvmRegion::Counters, "overheat_count", 0, 100_000), | executable/support line;
avior impact depends on surrounding block context. |
| 251 |             // Calibration region (low write frequency) | comment line; documentation intent, no runtime effe
|
| 252 |             NvmBlock::new_f64(NvmRegion::Calibration, "idle_rpm_cal", 800.0, 10_000), | executable/support line
behavior impact depends on surrounding block context. |
| 253 |             NvmBlock::new_f64(NvmRegion::Calibration, "rated_rpm_cal", 2200.0, 10_000), | executable/support
e; behavior impact depends on surrounding block context. |
| 254 |             NvmBlock::new_f64(NvmRegion::Calibration, "max_torque_cal", 1050.0, 10_000), | executable/support
ne; behavior impact depends on surrounding block context. |
| 255 |             NvmBlock::new_f64(NvmRegion::Calibration, "dpf_regen_thr", 75.0, 10_000), | executable/support line
behavior impact depends on surrounding block context. |
| 256 |             NvmBlock::new_f64(NvmRegion::Calibration, "def_warn_pct", 10.0, 10_000), | executable/support line
behavior impact depends on surrounding block context. |
| 257 |             NvmBlock::new_u64(NvmRegion::Calibration, "fuel_map_sel", 0, 10_000), | executable/support line;
avior impact depends on surrounding block context. |
| 258 |             NvmBlock::new_f64(NvmRegion::Calibration, "svc_interval_h", 500.0, 10_000), | executable/support
e; behavior impact depends on surrounding block context. |
| 259 |             // Identification (written only at factory / programming) | comment line; documentation intent, n
untime effect. |
| 260 |             NvmBlock::new_str(NvmRegion::Identification, "vin", "1HD1KEM16FB123456", 100), | executable/suppo
line; behavior impact depends on surrounding block context. |
| 261 |             NvmBlock::new_str(NvmRegion::Identification, "sw_version", "SW_01.23.004", 500), | executable/sup
t line; behavior impact depends on surrounding block context. |
| 262 |             NvmBlock::new_str(NvmRegion::Identification, "hw_version", "HW_02.00", 10), | executable/support
e; behavior impact depends on surrounding block context. |
| 263 |             NvmBlock::new_str(NvmRegion::Identification, "cal_id", "CAL_TIER4_V3.1", 100), | executable/suppo
line; behavior impact depends on surrounding block context. |

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| 264 |           NvmBlock::new_str( | executable/support line; behavior impact depends on surrounding block context. |
| 265 |           NvmRegion::Identification, | executable/support line; behavior impact depends on surrounding
ck context. |
| 266 |           "ecu_serial", | executable/support line; behavior impact depends on surrounding block context. |
| 267 |           "ECU20240811001", | executable/support line; behavior impact depends on surrounding block con
t. |
| 268 |           10, | executable/support line; behavior impact depends on surrounding block context. |
| 269 |           ), | executable/support line; behavior impact depends on surrounding block context. |
| 270 |           NvmBlock::new_u64(NvmRegion::Identification, "flash_count", 3, 50), | I/O or protocol-critical lo
; edits can affect integration, timing, and diagnostic outcomes. |
| 271 |       ]; | executable/support line; behavior impact depends on surrounding block context. |
| 272 | (blank) | line is blank; visual separation only. |
| 273 |       NvmStore { | executable/support line; behavior impact depends on surrounding block context. |
| 274 |           blocks, | executable/support line; behavior impact depends on surrounding block context. |
| 275 |           adaptations: Adaptations::new(), | executable/support line; behavior impact depends on surroundin
lock context. |
| 276 |           stored_dtcs: Vec::new(), | executable/support line; behavior impact depends on surrounding block
text. |
| 277 |           drive_cycle_count: 1247, | executable/support line; behavior impact depends on surrounding block
text. |
| 278 |           battery_disconnect_count: 0, | executable/support line; behavior impact depends on surrounding bl
context. |
| 279 |           last_flush_ts: 0.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 280 |           flush_interval_s: 60.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 281 |           total_writes: 0, | I/O or protocol-critical logic; edits can affect integration, timing, and diag
tic outcomes. |
| 282 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 283 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 284 | (blank) | line is blank; visual separation only. |
| 285 |   // ████████████████████████████████████████████████████████████████████████████████████████████████████ | comment
ntent, no runtime effect. |
| 286 |   /// Call every tick – writes dirty blocks to "storage" at flush_interval | comment line; documentation in
t, no runtime effect. |
| 287 |   pub fn tick(&mut self, elapsed: f64, dt: f64) { | declares executable behavior; signature/body changes al
API, control flow, and outputs. |
| 288 |       // Periodic flush (simulates real EEPROM write cycles) | comment line; documentation intent, no runti
effect. |
| 289 |       if elapsed - self.last_flush_ts >= self.flush_interval_s { | conditional gate; changing condition shi
runtime path and side effects. |
| 290 |           self.flush(elapsed); | executable/support line; behavior impact depends on surrounding block cont
. |
| 291 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 292 | (blank) | line is blank; visual separation only. |
| 293 |       // Update adaptations with small fuel trim drift | comment line; documentation intent, no runtime eff
. |
| 294 |       self.adaptations.update_fuel_trim(0.1 * noise(elapsed), dt); | executable/support line; behavior impa
depends on surrounding block context. |
| 295 |   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 296 | (blank) | line is blank; visual separation only. |
| 297 |   fn flush(&mut self, ts: f64) { | declares executable behavior; signature/body changes alter API, control
w, and outputs. |
| 298 |       for b in &mut self.blocks { | iteration control; changing bounds/body affects throughput, timing, and
ssibly safety guarantees. |
| 299 |           if b.dirty { | conditional gate; changing condition shifts runtime path and side effects. |
| 300 |               b.dirty = false; | executable/support line; behavior impact depends on surrounding block cont
. |
| 301 |               self.total_writes += 1; | I/O or protocol-critical logic; edits can affect integration, timin
and diagnostic outcomes. |
| 302 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |

```

```

| 303 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 304 |         self.last_flush_ts = ts; | executable/support line; behavior impact depends on surrounding block context. |
| 305 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 306 | (blank) | line is blank; visual separation only. |
| 307 |     // ████████████████████████████████████████████████████████████████████████████████████████████████████ | comment
intent, no runtime effect. |
| 308 |     /// Called on key-on – increments drive cycle counter, ages stored DTCs | comment line; documentation intent
, no runtime effect. |
| 309 |     pub fn on_key_on(&mut self) { | declares executable behavior; signature/body changes alter API, control flow
, and outputs. |
| 310 |         self.drive_cycle_count += 1; | executable/support line; behavior impact depends on surrounding block context. |
| 311 |         self.write_u64("key_on_count", self.drive_cycle_count as u64); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 312 | (blank) | line is blank; visual separation only. |
| 313 |         // Age stored DTCs – auto-clear after 40 drive cycles | comment line; documentation intent, no runtime effect. |
| 314 |         // retain_mut not stable – use manual drain approach | comment line; documentation intent, no runtime effect. |
| 315 |         let mut i = 0; | executable/support line; behavior impact depends on surrounding block context. |
| 316 |         while i < self.stored_dtc.len() { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 317 |             if self.stored_dtc[i].increment_cycle() { | conditional gate; changing condition shifts runtime path and side effects. |
| 318 |                 self.stored_dtc.remove(i); | executable/support line; behavior impact depends on surrounding block context. |
| 319 |             } else { | executable/support line; behavior impact depends on surrounding block context. |
| 320 |                 i += 1; | executable/support line; behavior impact depends on surrounding block context. |
| 321 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 322 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 323 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 324 | (blank) | line is blank; visual separation only. |
| 325 |     /// Called on key-off – flush everything | comment line; documentation intent, no runtime effect. |
| 326 |     pub fn on_key_off(&mut self, elapsed: f64) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 327 |         self.flush(elapsed); | executable/support line; behavior impact depends on surrounding block context. |
| 328 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 329 | (blank) | line is blank; visual separation only. |
| 330 |     // ████████████████████████████████████████████████████████████████████████████████████████████████████ | comment
intent, no runtime effect. |
| 331 |     pub fn write_f64(&mut self, name: &str, value: f64) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. | | |
| 332 |         if let Some(b) = self.blocks.iter_mut().find(|&b| b.name == name) { | conditional gate; changing condition shifts runtime path and side effects. |
| 333 |             b.write_f64(value); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 334 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 335 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 336 |     pub fn write_u64(&mut self, name: &str, value: u64) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 337 |         if let Some(b) = self.blocks.iter_mut().find(|&b| b.name == name) { | conditional gate; changing condition shifts runtime path and side effects. |
| 338 |             b.write_u64(value); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 339 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 340 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 341 |     pub fn read_f64(&self, name: &str) -> Option<f64> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 342 |         self.blocks | executable/support line; behavior impact depends on surrounding block context. |

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| 343 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 344 |         .find(\|b\| b.name == name) | executable/support line; behavior impact depends on surrounding block
context. |
| 345 |         .map(\|b\| b.value_f64) | executable/support line; behavior impact depends on surrounding block c
ext. |
| 346 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 347 |     pub fn read_u64(&self, name: &str) -> Option<u64> { | declares executable behavior; signature/body change
alter API, control flow, and outputs. |
| 348 |         self.blocks | executable/support line; behavior impact depends on surrounding block context. |
| 349 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 350 |         .find(\|b\| b.name == name) | executable/support line; behavior impact depends on surrounding block
context. |
| 351 |         .map(\|b\| b.value_u64) | executable/support line; behavior impact depends on surrounding block c
ext. |
| 352 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 353 |     pub fn read_str(&self, name: &str) -> Option<&str> { | declares executable behavior; signature/body chang
alter API, control flow, and outputs. |
| 354 |         self.blocks | executable/support line; behavior impact depends on surrounding block context. |
| 355 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 356 |         .find(\|b\| b.name == name) | executable/support line; behavior impact depends on surrounding block
context. |
| 357 |         .map(\|b\| b.value_str.as_str()) | executable/support line; behavior impact depends on surroundin
lock context. |
| 358 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 359 | (blank) | line is blank; visual separation only. |
| 360 |     pub fn worn_blocks(&self) -> impl Iterator<Item = &NvmBlock> { | declares executable behavior; signature/
y changes alter API, control flow, and outputs. |
| 361 |         self.blocks.iter().filter(\|b\| b.is_worn()) | executable/support line; behavior impact depends on su
unding block context. |
| 362 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 363 | (blank) | line is blank; visual separation only. |
| 364 |     pub fn add_stored_dtc(&mut self, spn: u32, fmi: u8) { | declares executable behavior; signature/body chan
alter API, control flow, and outputs. |
| 365 |         if !self | conditional gate; changing condition shifts runtime path and side effects. |
| 366 |             .stored_dtcs | executable/support line; behavior impact depends on surrounding block context. |
| 367 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 368 |             .any(\|d\| d.spn == spn && d.fmi == fmi) | executable/support line; behavior impact depends on su
unding block context. |
| 369 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 370 |             self.stored_dtcs.push(StoredDtcRecord::new(spn, fmi)); | executable/support line; behavior impact
depends on surrounding block context. |
| 371 |         } else if let Some(d) = self | executable/support line; behavior impact depends on surrounding block
text. |
| 372 |             .stored_dtcs | executable/support line; behavior impact depends on surrounding block context. |
| 373 |             .iter_mut() | executable/support line; behavior impact depends on surrounding block context. |
| 374 |             .find(\|d\| d.spn == spn && d.fmi == fmi) | executable/support line; behavior impact depends on s
ounding block context. |
| 375 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 376 |             d.occurrence_count = d.occurrence_count.saturating_add(1); | executable/support line; behavior im
t depends on surrounding block context. |
| 377 |             d.drive_cycles_ago = 0; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 378 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 379 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 380 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 381 | (blank) | line is blank; visual separation only. |
| 382 | fn noise(seed: f64) -> f64 { | declares executable behavior; signature/body changes alter API, control flow,
outputs. |
| 383 |     let x = (seed * 127.1 + 311.7).sin() * 43758.545; | executable/support line; behavior impact depends on s
ounding block context. |
| 384 |     x - x.floor() - 0.5 | executable/support line; behavior impact depends on surrounding block context. |
| 385 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/observability.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 51 | Non-empty: 45 | Symbols: 7

Symbol Inventory

Line	Kind	Symbol	Change Impact
4	struct	StructuredEvent	data/structure coupling
14	struct	SimMetrics	data/structure coupling
21	impl	SimMetrics	behavior binding and method semantics
22	fn	on_step	signature/API coupling
31	fn	on_error	signature/API coupling
35	fn	on_replay_failure	signature/API coupling
40	fn	log_structured	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	use serde::Serialize;	imports symbols; changing path/name affects compile-time resolution.
2	(blank)	line is blank; visual separation only.
3	#[derive(Debug, Clone, Serialize)]	serialization contract touchpoint; edits can break report compatibility and tooling.
4	pub struct StructuredEvent {	declares data layout; field changes may break serialization, tests, and callers.
5	pub timestamp: f64,	executable/support line; behavior impact depends on surrounding block context.
6	pub level: &'static str,	executable/support line; behavior impact depends on surrounding block context.
7	pub module: &'static str,	executable/support line; behavior impact depends on surrounding block context.
8	pub correlation_id: u64,	executable/support line; behavior impact depends on surrounding block context.
9	pub event: &'static str,	executable/support line; behavior impact depends on surrounding block context.
10	pub details: String,	executable/support line; behavior impact depends on surrounding block context.
11	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
12	(blank)	line is blank; visual separation only.
13	#[derive(Debug, Clone, Default, Serialize)]	serialization contract touchpoint; edits can break report compatibility and tooling.
14	pub struct SimMetrics {	declares data layout; field changes may break serialization, tests, and callers.
15	pub loop_duration_ms: f64,	executable/support line; behavior impact depends on surrounding block context.
16	pub steps_completed: u64,	executable/support line; behavior impact depends on surrounding block context.
17	pub error_count: u64,	executable/support line; behavior impact depends on surrounding block context.
18	pub replay_failures: u64,	executable/support line; behavior impact depends on surrounding block context.
19	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
20	(blank)	line is blank; visual separation only.
21	impl SimMetrics {	implementation block; method behavior and trait conformance are defined here.
22	pub fn on_step(&mut self, step_duration_ms: f64) {	declares executable behavior; signature/body changes alter API, control flow, and outputs.
23	self.steps_completed = self.steps_completed.saturating_add(1);	executable/support line; behavior impact depends on surrounding block context.
24	self.loop_duration_ms = if self.steps_completed == 1 {	executable/support line; behavior impact depends on surrounding block context.
25	step_duration_ms	executable/support line; behavior impact depends on surrounding block context.
26	} else {	executable/support line; behavior impact depends on surrounding block context.
27	self.loop_duration_ms * 0.95 + step_duration_ms * 0.05	executable/support line; behavior impact depends on surrounding block context.
28	};	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
29	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
30	(blank)	line is blank; visual separation only.
31	pub fn on_error(&mut self) {	declares executable behavior; signature/body changes alter API, control flow, and outputs.
32	self.error_count = self.error_count.saturating_add(1);	executable/support line; behavior impact depends on surrounding block context.
33	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
34	(blank)	line is blank; visual separation only.

```

| 35 |      pub fn on_replay_failure(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 36 |          self.replay_failures = self.replay_failures.saturating_add(1); | executable/support line; behavior impact depends on surrounding block context. |
| 37 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 38 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 39 | (blank) | line is blank; visual separation only. |
| 40 | pub fn log_structured(ev: &StructuredEvent) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 41 |     if let Ok(line) = serde_json::to_string(ev) { | conditional gate; changing condition shifts runtime path and side effects. |
| 42 |         #[cfg(feature = "advanced_observability")] | executable/support line; behavior impact depends on surrounding block context. |
| 43 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 44 |             tracing::info!(target: "auto_breaking", "{}", line); | executable/support line; behavior impact depends on surrounding block context. |
| 45 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 46 |         #[cfg(not(feature = "advanced_observability"))] | executable/support line; behavior impact depends on surrounding block context. |
| 47 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 48 |             println!("{}", line); | executable/support line; behavior impact depends on surrounding block context. |
| 49 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 50 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 51 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/radar.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 352 | Non-empty: 321 | Symbols: 25

Symbol Inventory

Line	Kind	Symbol	Change Impact
9	enum	RadarPosition	data/structure coupling
18	impl	std::fmt::Display for RadarPosition	behavior binding and method semantics
19	fn	fmt	signature/API coupling
33	enum	RadarObjectClass	data/structure coupling
43	impl	std::fmt::Display for RadarObjectClass	behavior binding and method semantics
44	fn	fmt	signature/API coupling
59	struct	RadarTarget	data/structure coupling
77	impl	RadarTarget	behavior binding and method semantics
78	fn	new	signature/API coupling
110	fn	kalman_predict	signature/API coupling
121	fn	kalman_update	signature/API coupling
142	struct	RadarSensor	data/structure coupling
154	impl	RadarSensor	behavior binding and method semantics
155	fn	new	signature/API coupling
169	fn	update	signature/API coupling
246	fn	compute_detection	signature/API coupling
254	fn	closest_inpath	signature/API coupling
263	struct	SimTrafficObj	data/structure coupling
273	struct	RadarSuite	data/structure coupling
290	impl	Default for RadarSuite	behavior binding and method semantics
291	fn	default	signature/API coupling
296	impl	RadarSuite	behavior binding and method semantics
297	fn	new	signature/API coupling
314	fn	update	signature/API coupling
344	fn	total_targets	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
---	---	---

[illegible]

[illegible]

```

context. |
| 87 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 88 |         let ttc = if range_rate < -0.1 { | executable/support line; behavior impact depends on surrounding blo
context. |
| 89 |             -range / range_rate | executable/support line; behavior impact depends on surrounding block context
|
| 90 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 91 |             f64::INFINITY | executable/support line; behavior impact depends on surrounding block context. |
| 92 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 93 |         RadarTarget { | executable/support line; behavior impact depends on surrounding block context. |
| 94 |             id, | executable/support line; behavior impact depends on surrounding block context. |
| 95 |             range_m: range, | executable/support line; behavior impact depends on surrounding block context. |
| 96 |             azimuth_deg: azimuth, | executable/support line; behavior impact depends on surrounding block context
. |
| 97 |             elevation_deg: 0.0, | executable/support line; behavior impact depends on surrounding block context
|
| 98 |             range_rate_ms: range_rate, | executable/support line; behavior impact depends on surrounding block co
ncontext. |
| 99 |             lat_vel_ms: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 100 |             rcs_dbm: rcs, | executable/support line; behavior impact depends on surrounding block context. |
| 101 |             confidence: 0.9, | executable/support line; behavior impact depends on surrounding block context.
| 102 |             object_class: cls, | executable/support line; behavior impact depends on surrounding block context
|
| 103 |             tracked_cycles: 1, | executable/support line; behavior impact depends on surrounding block context
|
| 104 |             ttc_s: ttc, | executable/support line; behavior impact depends on surrounding block context. |
| 105 |             kf_x: [range, range_rate, azimuth * range * PI / 180.0, 0.0], | executable/support line; behavior
impact depends on surrounding block context. |
| 106 |             kf_p: [5.0, 1.0, 2.0, 0.5], | executable/support line; behavior impact depends on surrounding block
context. |
| 107 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 108 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 109 | (blank) | line is blank; visual separation only. |
| 110 |     fn kalman_predict(&mut self, dt: f64) { | declares executable behavior; signature/body changes alter API,
control flow, and outputs. |
| 111 |         // State transition: range += range_rate*dt, lateral_pos += lat_vel*dt | comment line; documentation
intent, no runtime effect. |
| 112 |         self.kf_x[0] += self.kf_x[1] * dt; | executable/support line; behavior impact depends on surrounding
block context. |
| 113 |         self.kf_x[2] += self.kf_x[3] * dt; | executable/support line; behavior impact depends on surrounding
block context. |
| 114 |         // Process noise | comment line; documentation intent, no runtime effect. |
| 115 |         self.kf_p[0] += self.kf_p[1] * dt + 0.1 * dt * dt; | executable/support line; behavior impact depends
on surrounding block context. |
| 116 |         self.kf_p[1] += 0.5 * dt; | executable/support line; behavior impact depends on surrounding block context
. |
| 117 |         self.kf_p[2] += self.kf_p[3] * dt + 0.1 * dt * dt; | executable/support line; behavior impact depends
on surrounding block context. |
| 118 |         self.kf_p[3] += 0.3 * dt; | executable/support line; behavior impact depends on surrounding block context
. |
| 119 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 120 | (blank) | line is blank; visual separation only. |
| 121 |     fn kalman_update(&mut self, z_range: f64, z_rate: f64, z_azimuth: f64) { | declares executable behavior;
signature/body changes alter API, control flow, and outputs. |
| 122 |         let r_cov = 1.0; // measurement noise | executable/support line; behavior impact depends on surrounding
block context. |
| 123 |         let k0 = self.kf_p[0] / (self.kf_p[0] + r_cov); | executable/support line; behavior impact depends on
surrounding block context. |
| 124 |         let k1 = self.kf_p[1] / (self.kf_p[1] + r_cov); | executable/support line; behavior impact depends on
surrounding block context. |
| 125 |         self.kf_x[0] += k0 * (z_range - self.kf_x[0]); | executable/support line; behavior impact depends on
surrounding block context. |

```

[illegible]

[illegible]

[illegible]

[illegible]

```

| 294 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 295 | (blank) | line is blank; visual separation only. |
| 296 | impl RadarSuite { | implementation block; method behavior and trait conformance are defined here. |
| 297 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
outputs. |
| 298 |         RadarSuite { | executable/support line; behavior impact depends on surrounding block context. |
| 299 |             front_center: RadarSensor::new(RadarPosition::FrontCenter, 250.0, 18.0), | executable/support line;
behavior impact depends on surrounding block context. |
| 300 |             rear_center: RadarSensor::new(RadarPosition::RearCenter, 80.0, 40.0), | executable/support line;
behavior impact depends on surrounding block context. |
| 301 |             front_left: RadarSensor::new(RadarPosition::FrontLeft, 120.0, 150.0), | executable/support line;
behavior impact depends on surrounding block context. |
| 302 |             front_right: RadarSensor::new(RadarPosition::FrontRight, 120.0, 150.0), | executable/support line;
behavior impact depends on surrounding block context. |
| 303 |             rear_left: RadarSensor::new(RadarPosition::RearLeft, 120.0, 150.0), | executable/support line;
behavior impact depends on surrounding block context. |
| 304 |             rear_right: RadarSensor::new(RadarPosition::RearRight, 120.0, 150.0), | executable/support line;
behavior impact depends on surrounding block context. |
| 305 |             ttc_front: f64::INFINITY, | executable/support line; behavior impact depends on surrounding block
context. |
| 306 |             closest_front_m: f64::INFINITY, | executable/support line; behavior impact depends on surrounding
block context. |
| 307 |             bsm_left: false, | executable/support line; behavior impact depends on surrounding block context. |
| 308 |             bsm_right: false, | executable/support line; behavior impact depends on surrounding block context. |
| 309 |             cross_traffic_left: false, | executable/support line; behavior impact depends on surrounding block
context. |
| 310 |             cross_traffic_right: false, | executable/support line; behavior impact depends on surrounding block
context. |
| 311 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 312 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 313 | (blank) | line is blank; visual separation only. |
| 314 |     pub fn update(&mut self, ego_speed_ms: f64, traffic: &[SimTrafficObj], dt: f64) { | declares executable
behavior; signature/body changes alter API, control flow, and outputs. |
| 315 |         self.front_center.update(ego_speed_ms, traffic, dt); | executable/support line; behavior impact depends
on surrounding block context. |
| 316 |         self.rear_center.update(ego_speed_ms, traffic, dt); | executable/support line; behavior impact depends
on surrounding block context. |
| 317 |         self.front_left.update(ego_speed_ms, traffic, dt); | executable/support line; behavior impact depends
on surrounding block context. |
| 318 |         self.front_right.update(ego_speed_ms, traffic, dt); | executable/support line; behavior impact depends
on surrounding block context. |
| 319 |         self.rear_left.update(ego_speed_ms, traffic, dt); | executable/support line; behavior impact depends
on surrounding block context. |
| 320 |         self.rear_right.update(ego_speed_ms, traffic, dt); | executable/support line; behavior impact depends
on surrounding block context. |
| 321 | (blank) | line is blank; visual separation only. |
| 322 |         // Aggregate | comment line; documentation intent, no runtime effect. |
| 323 |         if let Some(t) = self.front_center.closest_inpath() { | conditional gate; changing condition shifts
runtime path and side effects. |
| 324 |             self.ttc_front = t.ttc_s; | executable/support line; behavior impact depends on surrounding block
context. |
| 325 |             self.closest_front_m = t.range_m; | executable/support line; behavior impact depends on surrounding
block context. |
| 326 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 327 |             self.ttc_front = f64::INFINITY; | executable/support line; behavior impact depends on surrounding
block context. |
| 328 |             self.closest_front_m = f64::INFINITY; | executable/support line; behavior impact depends on
surrounding block context. |
| 329 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 330 |         self.bsm_left = self | executable/support line; behavior impact depends on surrounding block context. |
| 331 |         .rear_left | executable/support line; behavior impact depends on surrounding block context. |
| 332 |         .targets | executable/support line; behavior impact depends on surrounding block context. |
| 333 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 334 |         .any(\|t\| t.range_m < 80.0 && t.azimuth_deg.abs() < 45.0); | executable/support line; behavior impact depends on surrounding block context. |
| 335 |         self.bsm_right = self | executable/support line; behavior impact depends on surrounding block context. |
| 336 |         .rear_right | executable/support line; behavior impact depends on surrounding block context. |
| 337 |         .targets | executable/support line; behavior impact depends on surrounding block context. |
| 338 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 339 |         .any(\|t\| t.range_m < 80.0 && t.azimuth_deg.abs() < 45.0); | executable/support line; behavior impact depends on surrounding block context. |
| 340 |         self.cross_traffic_left = self.rear_left.targets.iter().any(\|t\| t.range_m < 40.0); | executable/support line; behavior impact depends on surrounding block context. |
| 341 |         self.cross_traffic_right = self.rear_right.targets.iter().any(\|t\| t.range_m < 40.0); | executable/support line; behavior impact depends on surrounding block context. |
| 342 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 343 | (blank) | line is blank; visual separation only. |
| 344 |     pub fn total_targets(&self) -> usize { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 345 |         self.front_center.targets.len() | executable/support line; behavior impact depends on surrounding block context. |
| 346 |         + self.rear_center.targets.len() | executable/support line; behavior impact depends on surrounding block context. |
| 347 |         + self.front_left.targets.len() | executable/support line; behavior impact depends on surrounding block context. |
| 348 |         + self.front_right.targets.len() | executable/support line; behavior impact depends on surrounding block context. |
| 349 |         + self.rear_left.targets.len() | executable/support line; behavior impact depends on surrounding block context. |
| 350 |         + self.rear_right.targets.len() | executable/support line; behavior impact depends on surrounding block context. |
| 351 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 352 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/sensors.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 168 | Non-empty: 150 | Symbols: 6

Symbol Inventory

Line	Kind	Symbol	Change Impact
5	struct	RadarTarget	data/structure coupling
14	struct	SensorSuite	data/structure coupling
48	impl	SensorSuite	behavior binding and method semantics
49	fn	new	signature/API coupling
74	fn	update	signature/API coupling
165	fn	noise	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	use crate::vehicle::TrafficObject;	imports symbols; changing path/name affects compile-time resolution.
2	(blank)	line is blank; visual separation only.
3	/// A radar return from the forward/rear 77 GHz radar	comment line; documentation intent, no runtime effect.
4	#[derive(Debug, Clone)]	executable/support line; behavior impact depends on surrounding block context.
5	pub struct RadarTarget {	declares data layout; field changes may break serialization, tests, and callers.
6	pub distance: f64,	// m executable/support line; behavior impact depends on surrounding block context.
7	pub relative_speed: f64,	// km/h (positive = approaching) executable/support line; behavior impact depends on surrounding block context.
8	pub angle_deg: f64,	// degrees from forward axis executable/support line; behavior impact depends on surrounding block context.
9	pub is_in_path: bool,	executable/support line; behavior impact depends on surrounding block context.
10	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
11	(blank)	line is blank; visual separation only.


```

| 12 | /// Complete sensor suite (all simulated) | comment line; documentation intent, no runtime effect. |
| 13 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 14 | pub struct SensorSuite { | declares data layout; field changes may break serialization, tests, and callers. |
| 15 |     // 77 GHz RADAR | comment line; documentation intent, no runtime effect. |
| 16 |     pub forward_targets: Vec<RadarTarget>, | executable/support line; behavior impact depends on surrounding block context. |
| 17 |     pub rear_targets: Vec<RadarTarget>, | executable/support line; behavior impact depends on surrounding block context. |
| 18 | (blank) | line is blank; visual separation only. |
| 19 |     // Mono camera – lane detection | comment line; documentation intent, no runtime effect. |
| 20 |     pub lane_offset: f64, // m from lane center (+ = right) | executable/support line; behavior impact depends on surrounding block context. |
| 21 |     pub lane_width: f64, // m | executable/support line; behavior impact depends on surrounding block context. |
| 22 |     pub lane_curvature: f64, // 1/m | executable/support line; behavior impact depends on surrounding block context. |
| 23 |     pub lane_confidence: f64, // 0-1 | executable/support line; behavior impact depends on surrounding block context. |
| 24 |     pub lane_departure_left: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 25 |     pub lane_departure_right: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 26 | (blank) | line is blank; visual separation only. |
| 27 |     // Blind spot monitoring (short-range radar) | comment line; documentation intent, no runtime effect. |
| 28 |     pub bsm_left: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 29 |     pub bsm_right: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 30 | (blank) | line is blank; visual separation only. |
| 31 |     // Ultrasonic parking sensors: [FL,FC,FR, BL,BC,BR + 2 corners] | comment line; documentation intent, no runtime effect. |
| 32 |     pub ultrasonic: [f64; 8], | executable/support line; behavior impact depends on surrounding block context. |
| 33 | (blank) | line is blank; visual separation only. |
| 34 |     // Traffic Sign Recognition (camera-based) | comment line; documentation intent, no runtime effect. |
| 35 |     pub speed_limit_kmh: Option<u32>, | executable/support line; behavior impact depends on surrounding block context. |
| 36 |     pub stop_sign_ahead: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 37 | (blank) | line is blank; visual separation only. |
| 38 |     // IMU | comment line; documentation intent, no runtime effect. |
| 39 |     pub longitudinal_g: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 40 |     pub lateral_g: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 41 |     pub yaw_rate_imu: f64, // deg/s | executable/support line; behavior impact depends on surrounding block context. |
| 42 | (blank) | line is blank; visual separation only. |
| 43 |     // GPS (simplified) | comment line; documentation intent, no runtime effect. |
| 44 |     pub gps_speed: f64, // km/h | executable/support line; behavior impact depends on surrounding block context. |
| 45 |     pub gps_heading: f64, // degrees | executable/support line; behavior impact depends on surrounding block context. |
| 46 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 47 | (blank) | line is blank; visual separation only. |
| 48 | impl SensorSuite { | implementation block; method behavior and trait conformance are defined here. |
| 49 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and lifetimes. |
| 50 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 51 |             forward_targets: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 52 |             rear_targets: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 53 |             lane_offset: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 54 |             lane_width: 3.5, | executable/support line; behavior impact depends on surrounding block context. |
| 55 |             lane_curvature: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 56 |             lane_confidence: 0.92, | executable/support line; behavior impact depends on surrounding block context. |
| 57 |             lane_departure_left: false, | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 58 |         lane_departure_right: false, | executable/support line; behavior impact depends on surrounding block context. |
| 59 |         bsm_left: false, | executable/support line; behavior impact depends on surrounding block context. |
| 60 |         bsm_right: false, | executable/support line; behavior impact depends on surrounding block context. |
| 61 |         ultrasonic: [8.0; 8], | executable/support line; behavior impact depends on surrounding block context. |
| 62 |         speed_limit_kmh: Some(80), | executable/support line; behavior impact depends on surrounding block context. |
| 63 |         stop_sign_ahead: false, | executable/support line; behavior impact depends on surrounding block context. |
| 64 |         longitudinal_g: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 65 |         lateral_g: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 66 |         yaw_rate_imu: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 67 |         gps_speed: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 68 |         gps_heading: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 69 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 70 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 71 | (blank) | line is blank; visual separation only. |
| 72 |     /// Update all sensor readings from vehicle state and traffic | comment line; documentation intent, no runtime effect. |
| 73 |     #[allow(clippy::too_many_arguments)] | executable/support line; behavior impact depends on surrounding block context. |
| 74 |     pub fn update( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 75 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 76 |         vehicle_speed: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 77 |         steering_angle: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 78 |         acceleration: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 79 |         heading: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 80 |         traffic: &[TrafficObject], | executable/support line; behavior impact depends on surrounding block context. |
| 81 |         elapsed: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 82 |         dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 83 |     ) { | executable/support line; behavior impact depends on surrounding block context. |
| 84 |         // --- Forward RADAR ----- | comment line; documentation intent, no runtime effect. |
| 85 |         self.forward_targets.clear(); | executable/support line; behavior impact depends on surrounding block context. |
| 86 |         for obj in traffic { | iteration control; changing bounds/body affects throughput, timing, and possible safety guarantees. |
| 87 |             if obj.distance > 0.0 && obj.distance < 200.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 88 |                 self.forward_targets.push(RadarTarget { | executable/support line; behavior impact depends on surrounding block context. |
| 89 |                     distance: obj.distance, | executable/support line; behavior impact depends on surrounding block context. |
| 90 |                     relative_speed: obj.closing_speed, | executable/support line; behavior impact depends on surrounding block context. |
| 91 |                     angle_deg: (obj.lateral_offset / obj.distance.max(1.0)) | executable/support line; behavior impact depends on surrounding block context. |
| 92 |                     .atan() | executable/support line; behavior impact depends on surrounding block context. |
| 93 |                     .to_degrees(), | executable/support line; behavior impact depends on surrounding block context. |
| 94 |                     is_in_path: obj.in_path(), | executable/support line; behavior impact depends on surrounding block context. |
| 95 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 96 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 97 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 98 | (blank) | line is blank; visual separation only. |
| 99 |         // --- Rear RADAR ----- | comment line; documentation intent, no runtime effect. |
| 100 |         self.rear_targets.clear(); | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 101 |         for obj in traffic { | iteration control; changing bounds/body affects throughput, timing, and possibl
safety guarantees. |
| 102 |             if obj.distance < 0.0 && obj.distance > -50.0 { | conditional gate; changing condition shifts run
e path and side effects. |
| 103 |                 self.rear_targets.push(RadarTarget { | executable/support line; behavior impact depends on su
unding block context. |
| 104 |                     distance: -obj.distance, | executable/support line; behavior impact depends on surroundin
lock context. |
| 105 |                     relative_speed: -obj.closing_speed, | executable/support line; behavior impact depends on
rrounding block context. |
| 106 |                     angle_deg: 0.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 107 |                     is_in_path: obj.lateral_offset.abs() < 1.5, | executable/support line; behavior impact de
ds on surrounding block context. |
| 108 |                 }); | executable/support line; behavior impact depends on surrounding block context. |
| 109 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 110 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 111 | (blank) | line is blank; visual separation only. |
| 112 |         // --- Lane detection ----- | comment line; documentation in
t, no runtime effect. |
| 113 |         // Drift lane_offset based on steering (simplified) | comment line; documentation intent, no runtime
ect. |
| 114 |         self.lane_offset += steering_angle * 0.001 * dt * vehicle_speed / 3.6; | executable/support line; beh
or impact depends on surrounding block context. |
| 115 |         self.lane_offset = self.lane_offset.clamp(-2.0, 2.0); | executable/support line; behavior impact depe
on surrounding block context. |
| 116 |         self.lane_confidence = (0.92 + noise(elapsed * 7.3) * 0.04).clamp(0.6, 1.0); | executable/support lin
behavior impact depends on surrounding block context. |
| 117 |         self.lane_departure_left = self.lane_offset < -1.1; | executable/support line; behavior impact depend
n surrounding block context. |
| 118 |         self.lane_departure_right = self.lane_offset > 1.1; | executable/support line; behavior impact depend
n surrounding block context. |
| 119 | (blank) | line is blank; visual separation only. |
| 120 |         // --- BSM (side radar, 70m range) ----- | comment line; documentation in
t, no runtime effect. |
| 121 |         // Simulate periodic vehicle on right side | comment line; documentation intent, no runtime effect. |
| 122 |         self.bsm_right = (elapsed / 15.0).sin() > 0.7; | executable/support line; behavior impact depends on
rounding block context. |
| 123 |         self.bsm_left = (elapsed / 22.0).sin() > 0.85; | executable/support line; behavior impact depends on
rounding block context. |
| 124 | (blank) | line is blank; visual separation only. |
| 125 |         // --- Ultrasonic (parking) ----- | comment line; documentation in
t, no runtime effect. |
| 126 |         // Front center sensor reacts to nearest forward target | comment line; documentation intent, no runt
effect. |
| 127 |         if let Some(nearest) = self | conditional gate; changing condition shifts runtime path and side effec
|
| 128 |             .forward_targets | executable/support line; behavior impact depends on surrounding block context.
| 129 |             .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 130 |             .filter(\|t\| t.is_in_path) | executable/support line; behavior impact depends on surrounding blo
context. |
| 131 |             .min_by(\|a, b\| a.distance.total_cmp(&b.distance)) | executable/support line; behavior impact de
ds on surrounding block context. |
| 132 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 133 |             self.ultrasonic[1] = nearest.distance.min(8.0); // front center | executable/support line; behavi
impact depends on surrounding block context. |
| 134 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 135 |             self.ultrasonic[1] = 8.0; | executable/support line; behavior impact depends on surrounding block
ntext. |
| 136 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 137 |         self.ultrasonic[0] = (self.ultrasonic[1] + 0.3 + noise(elapsed * 5.0) * 0.1).min(8.0); | executable/s

```

```

ort line; behavior impact depends on surrounding block context. |
| 138 |         self.ultrasonic[2] = (self.ultrasonic[1] + 0.2 + noise(elapsed * 6.0) * 0.1).min(8.0); | executable/s
ort line; behavior impact depends on surrounding block context. |
| 139 |         // Rear static | comment line; documentation intent, no runtime effect. |
| 140 |         for i in 3..8 { | iteration control; changing bounds/body affects throughput, timing, and possibly sa
y guarantees. |
| 141 |             self.ultrasonic[i] = | executable/support line; behavior impact depends on surrounding block cont
. |
| 142 |                 (5.0 + noise(elapsed * f64::from(i as u8) * 3.1) * 0.5).clamp(0.1, 8.0); | executable/support
ne; behavior impact depends on surrounding block context. |
| 143 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 144 | (blank) | line is blank; visual separation only. |
| 145 |         // --- TSR ----- | comment line; documentation in
t, no runtime effect. |
| 146 |         // Change speed limit sign periodically | comment line; documentation intent, no runtime effect. |
| 147 |         if ((elapsed / 30.0) as u32).is_multiple_of(2) { | conditional gate; changing condition shifts runtim
ath and side effects. |
| 148 |             self.speed_limit_kmh = Some(80); | executable/support line; behavior impact depends on surroundin
lock context. |
| 149 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 150 |             self.speed_limit_kmh = Some(120); | executable/support line; behavior impact depends on surroundi
block context. |
| 151 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 152 | (blank) | line is blank; visual separation only. |
| 153 |         // --- IMU ----- | comment line; documentation in
t, no runtime effect. |
| 154 |         self.longitudinal_g = acceleration / 9.81; | executable/support line; behavior impact depends on surr
ding block context. |
| 155 |         self.lateral_g = steering_angle * vehicle_speed / 3.6 / 9.81 * 0.08; | executable/support line; behav
impact depends on surrounding block context. |
| 156 |         self.yaw_rate_imu = steering_angle * vehicle_speed / 3.6 / 2.7; | executable/support line; behavior i
ct depends on surrounding block context. |
| 157 | (blank) | line is blank; visual separation only. |
| 158 |         // --- GPS ----- | comment line; documentation in
t, no runtime effect. |
| 159 |         self.gps_speed = vehicle_speed + noise(elapsed * 2.1) * 0.3; | executable/support line; behavior impa
depends on surrounding block context. |
| 160 |         self.gps_heading = heading; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 161 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 162 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 163 | (blank) | line is blank; visual separation only. |
| 164 | /// Cheap pseudo-random noise in [-1, 1] | comment line; documentation intent, no runtime effect. |
| 165 | fn noise(seed: f64) -> f64 { | declares executable behavior; signature/body changes alter API, control flow,
outputs. |
| 166 |     let x = (seed * 127.1 + 311.7).sin() * 43758.545; | executable/support line; behavior impact depends on s
ounding block context. |
| 167 |     x - x.floor() - 0.5 | executable/support line; behavior impact depends on surrounding block context. |
| 168 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/sim_core.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 71 | Non-empty: 62 | Symbols: 22

Symbol Inventory

Line	Kind	Symbol	Change Impact
4	trait	SimState	signature/API coupling
5	fn	len	signature/API coupling
6	fn	is_empty	signature/API coupling
9	fn	get	signature/API coupling

```

| 10 | fn | set | signature/API coupling |
| 13 | trait | DynamicsModel | signature/API coupling |
| 14 | fn | derivative | signature/API coupling |
| 17 | trait | Integrator | signature/API coupling |
| 18 | fn | step | signature/API coupling |
| 22 | struct | EulerIntegrator | data/structure coupling |
| 24 | impl | Integrator for EulerIntegrator | behavior binding and method semantics |
| 25 | fn | step | signature/API coupling |
| 38 | module | tests | data/structure coupling |
| 41 | struct | VecState | data/structure coupling |
| 42 | impl | SimState for VecState | behavior binding and method semantics |
| 43 | fn | len | signature/API coupling |
| 46 | fn | get | signature/API coupling |
| 49 | fn | set | signature/API coupling |
| 54 | struct | UnitDrift | data/structure coupling |
| 55 | impl | DynamicsModel for UnitDrift | behavior binding and method semantics |
| 56 | fn | derivative | signature/API coupling |
| 62 | fn | euler_integrator_advances_state | signature/API coupling |

```

Line-by-Line Impact Map

```

| Line | Source | Operational Impact If Changed |
|---:|---|---|
| 1 | //! Incremental architecture layer for solver/integrator responsibilities. | comment line; documentation intent
o runtime effect. |
| 2 | //! This module is API-preserving: public simulation APIs can migrate internally. | comment line; documentation
tent, no runtime effect. |
| 3 | (blank) | line is blank; visual separation only. |
| 4 | pub trait SimState { | defines interface contract; signature changes impact all implementers and call sites. |
| 5 |     fn len(&self) -> usize; | declares executable behavior; signature/body changes alter API, control flow, and
tputs. |
| 6 |     fn is_empty(&self) -> bool { | declares executable behavior; signature/body changes alter API, control flow
nd outputs. |
| 7 |         self.len() == 0 | executable/support line; behavior impact depends on surrounding block context. |
| 8 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 9 |     fn get(&self, idx: usize) -> f64; | declares executable behavior; signature/body changes alter API, control
ow, and outputs. |
| 10 |     fn set(&mut self, idx: usize, value: f64); | declares executable behavior; signature/body changes alter AP
control flow, and outputs. |
| 11 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 12 | (blank) | line is blank; visual separation only. |
| 13 | pub trait DynamicsModel { | defines interface contract; signature changes impact all implementers and call sit
|
| 14 |     fn derivative(&self, x: &[f64], t: f64) -> Vec<f64>; | declares executable behavior; signature/body change
lter API, control flow, and outputs. |
| 15 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 16 | (blank) | line is blank; visual separation only. |
| 17 | pub trait Integrator { | defines interface contract; signature changes impact all implementers and call sites.
| 18 |     fn step(&self, model: &dyn DynamicsModel, state: &mut dyn SimState, t: f64, dt: f64); | declares executabl
ehavior; signature/body changes alter API, control flow, and outputs. |
| 19 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 20 | (blank) | line is blank; visual separation only. |
| 21 | #[derive(Debug, Clone, Copy, Default)] | executable/support line; behavior impact depends on surrounding block
ntext. |
| 22 | pub struct EulerIntegrator; | declares data layout; field changes may break serialization, tests, and callers.
| 23 | (blank) | line is blank; visual separation only. |
| 24 | impl Integrator for EulerIntegrator { | implementation block; method behavior and trait conformance are define
ere. |
| 25 |     fn step(&self, model: &dyn DynamicsModel, state: &mut dyn SimState, t: f64, dt: f64) { | declares executab
behavior; signature/body changes alter API, control flow, and outputs. |
| 26 |         let mut x = Vec::with_capacity(state.len()); | executable/support line; behavior impact depends on sur
nding block context. |
| 27 |         for i in 0..state.len() { | iteration control; changing bounds/body affects throughput, timing, and po
bly safety guarantees. |
| 28 |             x.push(state.get(i)); | executable/support line; behavior impact depends on surrounding block cont

```

```

. |
| 29 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 30 |         let dx = model.derivative(&x, t); | executable/support line; behavior impact depends on surrounding block context. |
| 31 |         for (i, dxi) in dx.iter().enumerate().take(state.len()) { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 32 |             state.set(i, state.get(i) + dxi * dt); | executable/support line; behavior impact depends on surrounding block context. |
| 33 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 34 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 35 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 36 | (blank) | line is blank; visual separation only. |
| 37 | #[cfg(test)] | executable/support line; behavior impact depends on surrounding block context. |
| 38 | mod tests { | declares a module boundary; changing module name/path breaks references. |
| 39 |     use super::*; | imports symbols; changing path/name affects compile-time resolution. |
| 40 | (blank) | line is blank; visual separation only. |
| 41 |     struct VecState(Vec<f64>); | declares data layout; field changes may break serialization, tests, and callers. |
| 42 |     impl SimState for VecState { | implementation block; method behavior and trait conformance are defined here. |
| 43 |         fn len(&self) -> usize { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 44 |             self.0.len() | executable/support line; behavior impact depends on surrounding block context. |
| 45 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 46 |         fn get(&self, idx: usize) -> f64 { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 47 |             self.0[idx] | executable/support line; behavior impact depends on surrounding block context. |
| 48 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 49 |         fn set(&mut self, idx: usize, value: f64) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 50 |             self.0[idx] = value; | executable/support line; behavior impact depends on surrounding block context. |
| 51 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 52 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 53 | (blank) | line is blank; visual separation only. |
| 54 |     struct UnitDrift; | declares data layout; field changes may break serialization, tests, and callers. |
| 55 |     impl DynamicsModel for UnitDrift { | implementation block; method behavior and trait conformance are defined here. |
| 56 |         fn derivative(&self, x: &[f64], _t: f64) -> Vec<f64> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 57 |             vec![1.0; x.len()] | executable/support line; behavior impact depends on surrounding block context. |
| 58 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 59 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 60 | (blank) | line is blank; visual separation only. |
| 61 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 62 |     fn euler_integrator_advances_state() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 63 |         let model = UnitDrift; | executable/support line; behavior impact depends on surrounding block context. |
| 64 |         let integ = EulerIntegrator; | executable/support line; behavior impact depends on surrounding block context. |
| 65 |         let mut state = VecState(vec![0.0, 2.0, -1.0]); | executable/support line; behavior impact depends on surrounding block context. |
| 66 |         integ.step(&model, &mut state, 0.0, 0.2); | executable/support line; behavior impact depends on surrounding block context. |
| 67 |         assert!((state.get(0) - 0.2).abs() < 1e-9); | executable/support line; behavior impact depends on surrounding block context. |
| 68 |         assert!((state.get(1) - 2.2).abs() < 1e-9); | executable/support line; behavior impact depends on surrounding block context. |
| 69 |         assert!((state.get(2) - -0.8).abs() < 1e-9); | executable/support line; behavior impact depends on surrounding block context. |
| 70 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 71 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/transmission.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 141 | Non-empty: 127 | Symbols: 15

Symbol Inventory

Line	Kind	Symbol	Change Impact
---: --- --- ---			
2	enum	DriveMode	data/structure coupling
11	impl	std::fmt::Display for DriveMode	behavior binding and method semantics
12	fn	fmt	signature/API coupling
26	enum	ShiftEvent	data/structure coupling
33	struct	TransmissionEcu	data/structure coupling
45	const	UP_DRIVE	data/structure coupling
46	const	UP_SPORT	data/structure coupling
47	const	UP_ECO	data/structure coupling
48	const	DOWN_THR	data/structure coupling
50	impl	TransmissionEcu	behavior binding and method semantics
51	fn	new	signature/API coupling
64	fn	set_mode	signature/API coupling
77	fn	update	signature/API coupling
122	fn	begin_shift	signature/API coupling
133	fn	gear_label	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
---: --- ---		
1	#[derive(Debug, Clone, Copy, PartialEq)]	executable/support line; behavior impact depends on surrounding block context.
2	pub enum DriveMode {	declares state variants; adding/removing variants impacts pattern matches and logic branches.
3	Park,	executable/support line; behavior impact depends on surrounding block context.
4	Reverse,	executable/support line; behavior impact depends on surrounding block context.
5	Neutral,	executable/support line; behavior impact depends on surrounding block context.
6	Drive,	executable/support line; behavior impact depends on surrounding block context.
7	Sport,	executable/support line; behavior impact depends on surrounding block context.
8	Eco,	executable/support line; behavior impact depends on surrounding block context.
9	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
10	(blank)	line is blank; visual separation only.
11	impl std::fmt::Display for DriveMode {	implementation block; method behavior and trait conformance are defined here.
12	fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result {	declares executable behavior; signature/body changes alter API, control flow, and outputs.
13	let s = match self {	executable/support line; behavior impact depends on surrounding block context.
14	DriveMode::Park => "P",	executable/support line; behavior impact depends on surrounding block context.
15	DriveMode::Reverse => "R",	executable/support line; behavior impact depends on surrounding block context.
16	DriveMode::Neutral => "N",	executable/support line; behavior impact depends on surrounding block context.
17	DriveMode::Drive => "D",	executable/support line; behavior impact depends on surrounding block context.
18	DriveMode::Sport => "S",	executable/support line; behavior impact depends on surrounding block context.
19	DriveMode::Eco => "E",	executable/support line; behavior impact depends on surrounding block context.
20	};	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
21	write!(f, "{}", s)	I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes.
22	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
23	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
24	(blank)	line is blank; visual separation only.
25	#[derive(Debug, Clone, Copy, PartialEq)]	executable/support line; behavior impact depends on surrounding block context.

```

| 26 | pub enum ShiftEvent { | declares state variants; adding/removing variants impacts pattern matches and logic br
hes. |
| 27 |     None, | executable/support line; behavior impact depends on surrounding block context. |
| 28 |     Up, | executable/support line; behavior impact depends on surrounding block context. |
| 29 |     Down, | executable/support line; behavior impact depends on surrounding block context. |
| 30 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 31 | (blank) | line is blank; visual separation only. |
| 32 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 33 | pub struct TransmissionEcu { | declares data layout; field changes may break serialization, tests, and callers
| 34 |     pub gear: u8, // 1-8 | executable/support line; behavior impact depends on surrounding block context. |
| 35 |     pub mode: DriveMode, | executable/support line; behavior impact depends on surrounding block context. |
| 36 |     pub is_shifting: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 37 |     pub shift_timer: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 38 |     pub last_shift: ShiftEvent, | executable/support line; behavior impact depends on surrounding block context
|
| 39 |     pub tcu_temp: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 40 |     pub torque_converter_slip: f64, // 0-1 | executable/support line; behavior impact depends on surrounding b
k context. |
| 41 |     pub oil_temp: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 42 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 43 | (blank) | line is blank; visual separation only. |
| 44 | // [gear index 0..7] → upshift threshold RPM by mode | comment line; documentation intent, no runtime effect.
| 45 | const UP_DRIVE: [f64; 8] = [0.0, 3100.0, 3200.0, 3200.0, 3100.0, 3000.0, 3000.0, 3000.0]; | executable/support
ne; behavior impact depends on surrounding block context. |
| 46 | const UP_SPORT: [f64; 8] = [0.0, 5800.0, 5800.0, 5800.0, 5800.0, 5800.0, 5800.0, 5800.0]; | executable/support
ne; behavior impact depends on surrounding block context. |
| 47 | const UP_ECO: [f64; 8] = [0.0, 2100.0, 2200.0, 2200.0, 2000.0, 1900.0, 1900.0, 1900.0]; | executable/support l
; behavior impact depends on surrounding block context. |
| 48 | const DOWN_THR: f64 = 1300.0; // downshift below this RPM | executable/support line; behavior impact depends o
urrounding block context. |
| 49 | (blank) | line is blank; visual separation only. |
| 50 | impl TransmissionEcu { | implementation block; method behavior and trait conformance are defined here. |
| 51 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
tputs. |
| 52 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 53 |             gear: 1, | executable/support line; behavior impact depends on surrounding block context. |
| 54 |             mode: DriveMode::Drive, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 55 |             is_shifting: false, | executable/support line; behavior impact depends on surrounding block context
|
| 56 |             shift_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 57 |             last_shift: ShiftEvent::None, | executable/support line; behavior impact depends on surrounding bl
context. |
| 58 |             tcu_temp: 25.0, | executable/support line; behavior impact depends on surrounding block context. |
| 59 |             torque_converter_slip: 0.0, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 60 |             oil_temp: 25.0, | executable/support line; behavior impact depends on surrounding block context. |
| 61 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 62 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 63 | (blank) | line is blank; visual separation only. |
| 64 |     pub fn set_mode(&mut self, mode: DriveMode) { | declares executable behavior; signature/body changes alter
I, control flow, and outputs. |
| 65 |         self.mode = mode; | executable/support line; behavior impact depends on surrounding block context. |
| 66 |         match mode { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 67 |             DriveMode::Drive \ | DriveMode::Eco \ | DriveMode::Sport => { | executable/support line; behavior im
t depends on surrounding block context. |
| 68 |                 if self.gear == 0 { | conditional gate; changing condition shifts runtime path and side effect
|
| 69 |                     self.gear = 1; | executable/support line; behavior impact depends on surrounding block con
t. |
| 70 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetim
copes. |
| 71 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime sc
s. |
| 72 |             DriveMode::Park \ | DriveMode::Neutral => {} | executable/support line; behavior impact depends on

```



```

rounding block context. |
| 73 |         DriveMode::Reverse => self.gear = 0, | executable/support line; behavior impact depends on surround
g block context. |
| 74 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 75 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 76 | (blank) | line is blank; visual separation only. |
| 77 |     pub fn update(&mut self, rpm: f64, throttle: f64, speed_kmh: f64, dt: f64) { | declares executable behavior
signature/body changes alter API, control flow, and outputs. |
| 78 |         // Shift cooldown | comment line; documentation intent, no runtime effect. |
| 79 |         if self.is_shifting { | conditional gate; changing condition shifts runtime path and side effects. |
| 80 |             self.shift_timer -= dt; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 81 |             if self.shift_timer <= 0.0 { | conditional gate; changing condition shifts runtime path and side e
cts. |
| 82 |                 self.is_shifting = false; | executable/support line; behavior impact depends on surrounding bl
context. |
| 83 |                 self.last_shift = ShiftEvent::None; | executable/support line; behavior impact depends on surr
ding block context. |
| 84 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime sc
s. |
| 85 |             return; | executable/support line; behavior impact depends on surrounding block context. |
| 86 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 87 | (blank) | line is blank; visual separation only. |
| 88 |         if !matches!( | conditional gate; changing condition shifts runtime path and side effects. |
| 89 |             self.mode, | executable/support line; behavior impact depends on surrounding block context. |
| 90 |             DriveMode::Drive \| DriveMode::Sport \| DriveMode::Eco | executable/support line; behavior impact
ends on surrounding block context. |
| 91 |         ) { | executable/support line; behavior impact depends on surrounding block context. |
| 92 |             return; | executable/support line; behavior impact depends on surrounding block context. |
| 93 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 94 |         if speed_kmh < 2.0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 95 |             return; | executable/support line; behavior impact depends on surrounding block context. |
| 96 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 97 | (blank) | line is blank; visual separation only. |
| 98 |         let gi = (self.gear as usize).saturating_sub(1).min(7); | executable/support line; behavior impact dep
s on surrounding block context. |
| 99 |         let upshift_rpm = match self.mode { | executable/support line; behavior impact depends on surrounding
ck context. |
| 100 |             DriveMode::Sport => UP_SPORT[gi], | executable/support line; behavior impact depends on surroundi
block context. |
| 101 |             DriveMode::Eco => UP_ECO[gi], | executable/support line; behavior impact depends on surrounding b
k context. |
| 102 |             _ => UP_DRIVE[gi], | executable/support line; behavior impact depends on surrounding block contex
|
| 103 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 104 | (blank) | line is blank; visual separation only. |
| 105 |         if rpm > upshift_rpm && self.gear < 8 { | conditional gate; changing condition shifts runtime path and
ide effects. |
| 106 |             self.gear += 1; | executable/support line; behavior impact depends on surrounding block context.
| 107 |             self.begin_shift(ShiftEvent::Up); | executable/support line; behavior impact depends on surroundi
block context. |
| 108 |         } else if rpm < DOWN_THR && throttle > 0.1 && self.gear > 1 { | executable/support line; behavior imp
depends on surrounding block context. |
| 109 |             self.gear -= 1; | executable/support line; behavior impact depends on surrounding block context.
| 110 |             self.begin_shift(ShiftEvent::Down); | executable/support line; behavior impact depends on surround
g block context. |
| 111 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 112 | (blank) | line is blank; visual separation only. |
| 113 |         // TC slip is highest at low speeds / high throttle | comment line; documentation intent, no runtime
ect. |
| 114 |         self.torque_converter_slip = | executable/support line; behavior impact depends on surrounding block
text. |
| 115 |             ((1.0 - (speed_kmh / 40.0).min(1.0)) * throttle * 0.35).max(0.0); | executable/support line; beha

```

```

r impact depends on surrounding block context. |
| 116 | (blank) | line is blank; visual separation only. |
| 117 |         let tcu_target = 75.0 + throttle * 20.0; | executable/support line; behavior impact depends on surrounding block context. |
| 118 |         self.tcu_temp += (tcu_target - self.tcu_temp) * dt * 0.008; | executable/support line; behavior impact depends on surrounding block context. |
| 119 |         self.oil_temp = self.tcu_temp - 2.0; | executable/support line; behavior impact depends on surrounding block context. |
| 120 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 121 | (blank) | line is blank; visual separation only. |
| 122 |     fn begin_shift(&mut self, event: ShiftEvent) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 123 |         self.is_shifting = true; | executable/support line; behavior impact depends on surrounding block context. |
| 124 |         self.last_shift = event; | executable/support line; behavior impact depends on surrounding block context. |
| 125 |         // Sport shifts faster (~150ms), Drive ~220ms, Eco ~280ms | comment line; documentation intent, no runtime effect. |
| 126 |         self.shift_timer = match self.mode { | executable/support line; behavior impact depends on surrounding block context. |
| 127 |             DriveMode::Sport => 0.15, | executable/support line; behavior impact depends on surrounding block context. |
| 128 |             DriveMode::Eco => 0.28, | executable/support line; behavior impact depends on surrounding block context. |
| 129 |             _ => 0.22, | executable/support line; behavior impact depends on surrounding block context. |
| 130 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 131 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 132 | (blank) | line is blank; visual separation only. |
| 133 |     pub fn gear_label(&self) -> String { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 134 |         match self.mode { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 135 |             DriveMode::Park => "P".into(), | executable/support line; behavior impact depends on surrounding block context. |
| 136 |             DriveMode::Reverse => "R".into(), | executable/support line; behavior impact depends on surrounding block context. |
| 137 |             DriveMode::Neutral => "N".into(), | executable/support line; behavior impact depends on surrounding block context. |
| 138 |             _ => format!("{}", self.mode, self.gear), | executable/support line; behavior impact depends on surrounding block context. |
| 139 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 140 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 141 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/uds.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 962 | Non-empty: 898 | Symbols: 72

Symbol Inventory

Line	Kind	Symbol	Change Impact
27	const	MAX_DOWNLOAD_LEN	data/structure coupling
32	enum	DiagSession	data/structure coupling
41	impl	std::fmt::Display for DiagSession	behavior binding and method semantics
42	fn	fmt	signature/API coupling
54	enum	SecurityLevel	data/structure coupling
69	enum	Nrc	data/structure coupling
93	module	did	data/structure coupling
95	const	VIN	data/structure coupling
96	const	ECU_SERIAL	data/structure coupling
97	const	SW_VERSION	data/structure coupling
98	const	HW_VERSION	data/structure coupling

99	const	MANUF_DATE	data/structure coupling
100	const	PROGRAM_DATE	data/structure coupling
101	const	CALIBRATION_ID	data/structure coupling
102	const	FINGERPRINT	data/structure coupling
105	const	ENGINE_RPM	data/structure coupling
106	const	ENGINE_LOAD	data/structure coupling
107	const	COOLANT_TEMP	data/structure coupling
108	const	OIL_PRESSURE	data/structure coupling
109	const	BOOST_PRESSURE	data/structure coupling
110	const	FUEL_RATE	data/structure coupling
111	const	ENGINE_TORQUE	data/structure coupling
112	const	DPF_SOOT	data/structure coupling
113	const	DEF_LEVEL	data/structure coupling
114	const	SCR_EFFICIENCY	data/structure coupling
115	const	EXHAUST_TEMP	data/structure coupling
116	const	RAIL_PRESSURE	data/structure coupling
117	const	ENGINE_HOURS	data/structure coupling
118	const	BATTERY_VOLTAGE	data/structure coupling
119	const	VEHICLE_SPEED	data/structure coupling
122	const	IDLE_RPM_CAL	data/structure coupling
123	const	RATED_RPM_CAL	data/structure coupling
124	const	MAX_TORQUE_CAL	data/structure coupling
125	const	GOVERNOR_MODE_CAL	data/structure coupling
126	const	DPF_REGEN_THRESHOLD	data/structure coupling
127	const	DEF_WARNING_LEVEL	data/structure coupling
128	const	FUEL_MAP_SELECT	data/structure coupling
129	const	PTO_MAX_RPM	data/structure coupling
130	const	SERVICE_INTERVAL_H	data/structure coupling
133	const	ROUTINE_DPF_REGEN	data/structure coupling
134	const	ROUTINE_INJECTOR_TEST	data/structure coupling
135	const	ROUTINE_ACTUATOR_TEST	data/structure coupling
136	const	ROUTINE_RESET_ADAP	data/structure coupling
142	struct	FreezeFrame	data/structure coupling
161	enum	DownloadState	data/structure coupling
170	struct	UdsServer	data/structure coupling
220	struct	UdsEvent	data/structure coupling
230	enum	UdsEventResult	data/structure coupling
236	impl	UdsServer	behavior binding and method semantics
237	fn	new	signature/API coupling
276	fn	tick	signature/API coupling
293	fn	process	signature/API coupling
320	fn	svc_session_control	signature/API coupling
353	fn	svc_ecu_reset	signature/API coupling
395	fn	svc_clear_dtc	signature/API coupling
428	fn	svc_read_dtc	signature/API coupling
504	fn	svc_read_data_by_id	signature/API coupling
554	fn	svc_security_access	signature/API coupling
640	fn	svc_write_data_by_id	signature/API coupling
684	fn	svc_routine_control	signature/API coupling
733	fn	svc_request_download	signature/API coupling
767	fn	svc_transfer_data	signature/API coupling
809	fn	svc_request_transfer_exit	signature/API coupling
830	fn	svc_tester_present	signature/API coupling
850	fn	record_dtc_set	signature/API coupling
879	fn	nrc	signature/API coupling
883	fn	log	signature/API coupling
905	fn	service_name	signature/API coupling
926	fn	dtc_to_3bytes	signature/API coupling
935	module	tests	data/structure coupling
939	fn	session_default_relocks_security	signature/API coupling
949	fn	transfer_exit_rejects_incomplete_download	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
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[illegible]

```
| here. |  
| 42 |     fn fmt(&self, f: &ampmut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signa  
e/body changes alter API, control flow, and outputs. |  
| 43 |         match self { | branch dispatch; changing arms alters behavior for variants and input classes. |  
| 44 |             DiagSession::Default => write!(f, "DEFAULT    "), | I/O or protocol-critical logic; edits can affe  
integration, timing, and diagnostic outcomes. |  
| 45 |             DiagSession::Extended => write!(f, "EXTENDED   "), | I/O or protocol-critical logic; edits can aff  
integration, timing, and diagnostic outcomes. |  
| 46 |             DiagSession::Programming => write!(f, "PROGRAMMING"), | I/O or protocol-critical logic; edits can  
ect integration, timing, and diagnostic outcomes. |  
| 47 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 48 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 49 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 50 | (blank) | line is blank; visual separation only. |  
| 51 | // ■■ Security Level ████████████████████████████████████████████████████████ | comment l  
tent, no runtime effect. |  
| 52 | (blank) | line is blank; visual separation only. |  
| 53 | #[derive(Debug, Clone, Copy, PartialEq, PartialOrd)] | executable/support line; behavior impact depends on surr  
nding block context. |  
| 54 | pub enum SecurityLevel { | declares state variants; adding/removing variants impacts pattern matches and logic b  
anches. |  
| 55 |     /// Locked – no programming access | comment line; documentation intent, no runtime effect. |  
| 56 |     Locked, | executable/support line; behavior impact depends on surrounding block context. |  
| 57 |     /// Level 1 (0x01/0x02) – extended diagnostic access | comment line; documentation intent, no runtime effe  
|  
| 58 |     Level1, | executable/support line; behavior impact depends on surrounding block context. |  
| 59 |     /// Level 3(0x05/0x06) – ECU programming / calibration write| comment line; documentation intent, no run  
e effect. |  
| 60 |     Level3, | executable/support line; behavior impact depends on surrounding block context. |  
| 61 |     /// Level 9 (0x11/0x12) – Factory / supplier level (erases all data) | comment line; documentation intent,  
runtime effect. |  
| 62 |     Level9, | executable/support line; behavior impact depends on surrounding block context. |  
| 63 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 64 | (blank) | line is blank; visual separation only. |  
| 65 | // ■■ UDS NRC (Negative Response Code) ████████████████████████████████████████████████████████ | comment line; do  
tent, no runtime effect. |  
| 66 | (blank) | line is blank; visual separation only. |  
| 67 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 68 | #[repr(u8)] | executable/support line; behavior impact depends on surrounding block context. |  
| 69 | pub enum Nrc { | declares state variants; adding/removing variants impacts pattern matches and logic branches.  
| 70 |     GeneralReject = 0x10, | executable/support line; behavior impact depends on surrounding block context. |  
| 71 |     ServiceNotSupported = 0x11, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 72 |     SubFunctionNotSupported = 0x12, | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 73 |     IncorrectMessageLength = 0x13, | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 74 |     ResponseTooLong = 0x14, | executable/support line; behavior impact depends on surrounding block context. |  
| 75 |     BusyRepeatRequest = 0x21, | executable/support line; behavior impact depends on surrounding block context.  
| 76 |     ConditionsNotCorrect = 0x22, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 77 |     RequestSequenceError = 0x24, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 78 |     RequestOutOfRange = 0x31, | executable/support line; behavior impact depends on surrounding block context.  
| 79 |     SecurityAccessDenied = 0x33, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 80 |     InvalidKey = 0x35, | executable/support line; behavior impact depends on surrounding block context. |  
| 81 |     ExceededNumberOfAttempts = 0x36, | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 82 |     RequiredTimeDelayNotExpired = 0x37, | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 83 |     UploadDownloadNotAccepted = 0x70, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 84 |     TransferDataSuspended = 0x71, | executable/support line; behavior impact depends on surrounding block cont
```

[illegible]

[illegible]

```
| context. |  
| 165 | Complete, // 0x37 received, waiting for ECU to validate/flash | I/O or protocol-critical logic; edits  
n affect integration, timing, and diagnostic outcomes. |  
| 166 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 167 | (blank) | line is blank; visual separation only. |  
| 168 | // ■ UDS Server ████████████████████████████████████████████████████████████████████ | comment  
intent, no runtime effect. |  
| 169 | (blank) | line is blank; visual separation only. |  
| 170 | pub struct UdsServer { | declares data layout; field changes may break serialization, tests, and callers. |  
| 171 |     pub node_name: &'static str, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 172 |     pub sa: u8, | executable/support line; behavior impact depends on surrounding block context. |  
| 173 | (blank) | line is blank; visual separation only. |  
| 174 |     // ■ Session & security ████████████████████████████████████████████████████████████████████ | comment line  
ntent, no runtime effect. |  
| 175 |     pub session: DiagSession, | executable/support line; behavior impact depends on surrounding block context  
| 176 |     pub security: SecurityLevel, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 177 |     session_timer: f64, // seconds since last tester activity | executable/support line; behavior impact d  
nds on surrounding block context. |  
| 178 |     security_seed: u32, // current seed (changes each request) | executable/support line; behavior impact  
ends on surrounding block context. |  
| 179 |     security_attempts: u8, // failed attempts counter (lockout after 3) | executable/support line; behavior i  
ct depends on surrounding block context. |  
| 180 |     lockout_timer: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 181 | (blank) | line is blank; visual separation only. |  
| 182 |     // ■ DTC storage (persists key cycles in real ECU via NVM) ████████████████████████████████████████████ | comment line; documen  
tent, no runtime effect. |  
| 183 |     pub active_dtcs: Vec<crate::j1939:Dtc>, | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 184 |     pub stored_dtcs: Vec<crate::j1939:Dtc>, | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 185 |     pub freeze_frames: Vec<FreezeFrame>, | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 186 | (blank) | line is blank; visual separation only. |  
| 187 |     // ■ Calibration parameters ████████████████████████████████████████████████████████████████████ | comment line;  
tent, no runtime effect. |  
| 188 |     pub idle_rpm_cal: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 189 |     pub rated_rpm_cal: f64, | executable/support line; behavior impact depends on surrounding block context.  
| 190 |     pub max_torque_cal: f64, | executable/support line; behavior impact depends on surrounding block context.  
| 191 |     pub dpf_regen_threshold: f64, | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 192 |     pub def_warning_level: f64, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 193 |     pub fuel_map_select: u8, | executable/support line; behavior impact depends on surrounding block context.  
| 194 |     pub service_interval_h: f64, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 195 |     pub pto_max_rpm: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 196 | (blank) | line is blank; visual separation only. |  
| 197 |     // ■ Identification strings ████████████████████████████████████████████████████████████████████ | comment line;  
tent, no runtime effect. |  
| 198 |     pub vin: String, | executable/support line; behavior impact depends on surrounding block context. |  
| 199 |     pub sw_version: String, | executable/support line; behavior impact depends on surrounding block context.  
| 200 |     pub hw_version: String, | executable/support line; behavior impact depends on surrounding block context.  
| 201 |     pub cal_id: String, | executable/support line; behavior impact depends on surrounding block context. |  
| 202 |     pub ecu_serial: String, | executable/support line; behavior impact depends on surrounding block context.  
| 203 | (blank) | line is blank; visual separation only. |  
| 204 |     // ■ Download session ████████████████████████████████████████████████████████████████████ | comment lin  
tent, no runtime effect. |  
| 205 |     pub download_state: DownloadState, | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 206 |     download_address: u32, | executable/support line; behavior impact depends on surrounding block context. |  
| 207 |     download_expected_len: u32, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 208 |     download_received_len: u32, | executable/support line; behavior impact depends on surrounding block cont
```



```
|
| 209 |         download_block_num: u8, | executable/support line; behavior impact depends on surrounding block context.
| 210 | (blank) | line is blank; visual separation only. |
| 211 |     // ■ Routine states ████████████████████████████████████████████████████████████████████████████ | comment lin
ntent, no runtime effect. |
| 212 |     pub dpf_regen_routine_active: bool, | executable/support line; behavior impact depends on surrounding blo
context. |
| 213 |     pub injector_test_active: bool, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 214 | (blank) | line is blank; visual separation only. |
| 215 |     // ■ Event log ████████████████████████████████████████████████████████████████████████████ | comment l
ntent, no runtime effect. |
| 216 |     pub event_log: VecDeque<UdsEvent>, | I/O or protocol-critical logic; edits can affect integration, timing
nd diagnostic outcomes. |
| 217 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 218 | (blank) | line is blank; visual separation only. |
| 219 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 220 | pub struct UdsEvent { | declares data layout; field changes may break serialization, tests, and callers. |
| 221 |     pub timestamp: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 222 |     pub service: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 223 |     pub sub_func: Option<u8>, | executable/support line; behavior impact depends on surrounding block context
| 224 |     pub did: Option<u16>, | executable/support line; behavior impact depends on surrounding block context. |
| 225 |     pub result: UdsEventResult, | I/O or protocol-critical logic; edits can affect integration, timing, and d
nostic outcomes. |
| 226 |     pub detail: String, | executable/support line; behavior impact depends on surrounding block context. |
| 227 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 228 | (blank) | line is blank; visual separation only. |
| 229 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding bl
context. |
| 230 | pub enum UdsEventResult { | declares state variants; adding/removing variants impacts pattern matches and log
branches. |
| 231 |     Positive, | executable/support line; behavior impact depends on surrounding block context. |
| 232 |     Negative(u8), | executable/support line; behavior impact depends on surrounding block context. |
| 233 |     SecurityFail, | executable/support line; behavior impact depends on surrounding block context. |
| 234 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 235 | (blank) | line is blank; visual separation only. |
| 236 | impl UdsServer { | implementation block; method behavior and trait conformance are defined here. |
| 237 |     pub fn new(node_name: &'static str, sa: u8) -> Self { | declares executable behavior; signature/body chan
alter API, control flow, and outputs. |
| 238 |         UdsServer { | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic ou
mes. |
| 239 |             node_name, | executable/support line; behavior impact depends on surrounding block context. |
| 240 |             sa, | executable/support line; behavior impact depends on surrounding block context. |
| 241 |             session: DiagSession::Default, | executable/support line; behavior impact depends on surrounding
ck context. |
| 242 |             security: SecurityLevel::Locked, | executable/support line; behavior impact depends on surroundin
lock context. |
| 243 |             session_timer: 0.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 244 |             security_seed: 0xA5A5A5A5, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 245 |             security_attempts: 0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 246 |             lockout_timer: 0.0, | executable/support line; behavior impact depends on surrounding block conte
|
| 247 |             active_dtcs: Vec::new(), | executable/support line; behavior impact depends on surrounding block
text. |
| 248 |             stored_dtcs: Vec::new(), | executable/support line; behavior impact depends on surrounding block
text. |
| 249 |             freeze_frames: Vec::new(), | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 250 |             idle_rpm_cal: 800.0, | executable/support line; behavior impact depends on surrounding block cont
. |
| 251 |             rated_rpm_cal: 2200.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
```

```
| 252 | max_torque_cal: 1050.0, | executable/support line; behavior impact depends on surrounding block context.  
ext. |  
| 253 | dpf_regen_threshold: 75.0, | executable/support line; behavior impact depends on surrounding block  
ontext. |  
| 254 | def_warning_level: 10.0, | executable/support line; behavior impact depends on surrounding block  
text. |  
| 255 | fuel_map_select: 0, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 256 | service_interval_h: 500.0, | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 257 | pto_max_rpm: 1000.0, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 258 | vin: "1HD1KEM16FB123456".into(), | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 259 | sw_version: "SW_01.23.004".into(), | executable/support line; behavior impact depends on surround  
block context. |  
| 260 | hw_version: "HW_02.00".into(), | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 261 | cal_id: "CAL_TIER4_V3.1".into(), | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 262 | ecu_serial: "ECU20240811001".into(), | executable/support line; behavior impact depends on surrou  
ng block context. |  
| 263 | download_state: DownloadState::Idle, | executable/support line; behavior impact depends on surrou  
ng block context. |  
| 264 | download_address: 0, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 265 | download_expected_len: 0, | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 266 | download_received_len: 0, | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 267 | download_block_num: 0, | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 268 | dpf_regenRoutineActive: false, | executable/support line; behavior impact depends on surrounding  
lock context. |  
| 269 | injector_test_active: false, | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 270 | event_log: VecDeque::new(), | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 271 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 272 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.  
| 273 | (blank) | line is blank; visual separation only. |  
| 274 | // ████████████████████████████████████████████████████████████████████████████ | comment  
intent, no runtime effect. |  
| 275 | /// Tick: advance session timer, auto-expire Extended/Programming sessions. | comment line; documentation  
intent, no runtime effect. |  
| 276 | pub fn tick(&mut self, dt: f64) { | declares executable behavior; signature/body changes alter API, contr  
flow, and outputs. |  
| 277 |     self.session_timer += dt; | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 278 | (blank) | line is blank; visual separation only. |  
| 279 |     // Extended session times out after 5 s without TesterPresent | comment line; documentation intent, n  
untime effect. |  
| 280 |         if self.session != DiagSession::Default && self.session_timer > 5.0 { | conditional gate; changing co  
tion shifts runtime path and side effects. |  
| 281 |             self.session = DiagSession::Default; | executable/support line; behavior impact depends on surrou  
ng block context. |  
| 282 |             self.security = SecurityLevel::Locked; | executable/support line; behavior impact depends on surr  
ding block context. |  
| 283 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 284 | (blank) | line is blank; visual separation only. |  
| 285 |         // Lockout timer for security (penalty 10 s after 3 failed attempts) | comment line; documentation in  
t, no runtime effect. |  
| 286 |         if self.lockout_timer > 0.0 { | conditional gate; changing condition shifts runtime path and side eff
```



```

325 |         let new_session = match sub { | executable/support line; behavior impact depends on surrounding block
context. |
| 326 |             0x01 => DiagSession::Default, | executable/support line; behavior impact depends on surrounding b
k context. |
| 327 |             0x02 => DiagSession::Extended, | executable/support line; behavior impact depends on surrounding
ck context. |
| 328 |             0x03 => DiagSession::Programming, | executable/support line; behavior impact depends on surroundi
block context. |
| 329 |             _ => return self.nrc(0x10, Nrc::SubFunctionNotSupported), | executable/support line; behavior imp
depends on surrounding block context. |
| 330 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 331 |         // Programming session only allowed if already in Extended + security unlocked | comment line; docume
tion intent, no runtime effect. |
| 332 |         if new_session == DiagSession::Programming && self.security < SecurityLevel::Level3 { | conditional g
;changing condition shifts runtime path and side effects. |
| 333 |             return self.nrc(0x10, Nrc::SecurityAccessDenied); | executable/support line; behavior impact depe
on surrounding block context. |
| 334 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 335 |         if new_session == DiagSession::Default { | conditional gate; changing condition shifts runtime path a
side effects. |
| 336 |             self.security = SecurityLevel::Locked; | executable/support line; behavior impact depends on surr
ding block context. |
| 337 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 338 |         self.session = new_session; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 339 |         self.session_timer = 0.0; | executable/support line; behavior impact depends on surrounding block con
t. |
| 340 |         self.log( | executable/support line; behavior impact depends on surrounding block context. |
| 341 |             ts, | executable/support line; behavior impact depends on surrounding block context. |
| 342 |             0x10, | executable/support line; behavior impact depends on surrounding block context. |
| 343 |             Some(sub), | executable/support line; behavior impact depends on surrounding block context. |
| 344 |             None, | executable/support line; behavior impact depends on surrounding block context. |
| 345 |             UdsEventResult::Positive, | I/O or protocol-critical logic; edits can affect integration, timing,
d diagnostic outcomes. |
| 346 |             format!("{}", self.session), | error-propagating operation; changing it impacts failure sema
s and caller recovery paths. |
| 347 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 348 |         vec![0x50, sub, 0x00, 0x19, 0x01, 0xF4] // P2=25ms, P2*=500ms | executable/support line; behavior imp
depends on surrounding block context. |
| 349 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 350 | (blank) | line is blank; visual separation only. |
| 351 |     // ████████████████████████████████████████████████████████████████████████████████████████████████████ | comment
ntent, no runtime effect. |
| 352 |     // Service 0x11 – ECU Reset | comment line; documentation intent, no runtime effect. |
| 353 |     fn svc_ecu_reset(&mut self, req: &[u8], ts: f64) -> Vec<u8> { | declares executable behavior; signature/b
changes alter API, control flow, and outputs. |
| 354 |         if req.len() < 2 { | conditional gate; changing condition shifts runtime path and side effects. |
| 355 |             return self.nrc(0x11, Nrc::IncorrectMessageLength); | executable/support line; behavior impact de
ds on surrounding block context. |
| 356 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 357 |         if self.session == DiagSession::Default { | conditional gate; changing condition shifts runtime path
side effects. |
| 358 |             return self.nrc(0x11, Nrc::ServiceNotSupportedInActiveSession); | executable/support line; behavi
impact depends on surrounding block context. |
| 359 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 360 |         let sub = req[1]; | executable/support line; behavior impact depends on surrounding block context. |
| 361 |         match sub { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 362 |             0x01 => self.log( | executable/support line; behavior impact depends on surrounding block context
| 363 |                 ts, | executable/support line; behavior impact depends on surrounding block context. |
| 364 |                 0x11, | executable/support line; behavior impact depends on surrounding block context. |

```

```
|
| 365 |         Some(sub), | executable/support line; behavior impact depends on surrounding block context. |
| 366 |         None, | executable/support line; behavior impact depends on surrounding block context. |
| 367 |         UdsEventResult::Positive, | I/O or protocol-critical logic; edits can affect integration, tim
, and diagnostic outcomes. |
| 368 |         "Hard Reset".into(), | executable/support line; behavior impact depends on surrounding block
text. |
| 369 |     ), | executable/support line; behavior impact depends on surrounding block context. |
| 370 |     0x02 => self.log( | executable/support line; behavior impact depends on surrounding block context
ts, | executable/support line; behavior impact depends on surrounding block context. |
| 371 |     0x11, | executable/support line; behavior impact depends on surrounding block context. |
| 372 |     Some(sub), | executable/support line; behavior impact depends on surrounding block context. |
| 373 |     None, | executable/support line; behavior impact depends on surrounding block context. |
| 374 |     UdsEventResult::Positive, | I/O or protocol-critical logic; edits can affect integration, tim
, and diagnostic outcomes. |
| 375 |     "Key-Off/On Reset".into(), | executable/support line; behavior impact depends on surrounding
ck context. |
| 376 |     ), | executable/support line; behavior impact depends on surrounding block context. |
| 377 |     0x03 => self.log( | executable/support line; behavior impact depends on surrounding block context
ts, | executable/support line; behavior impact depends on surrounding block context. |
| 378 |     0x11, | executable/support line; behavior impact depends on surrounding block context. |
| 379 |     Some(sub), | executable/support line; behavior impact depends on surrounding block context. |
| 380 |     None, | executable/support line; behavior impact depends on surrounding block context. |
| 381 |     UdsEventResult::Positive, | I/O or protocol-critical logic; edits can affect integration, tim
, and diagnostic outcomes. |
| 382 |     "Soft Reset".into(), | executable/support line; behavior impact depends on surrounding block
text. |
| 383 |     ), | executable/support line; behavior impact depends on surrounding block context. |
| 384 |     _ => return self.nrc(0x11, Nrc::SubFunctionNotSupported), | executable/support line; behavior imp
depends on surrounding block context. |
| 385 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 386 |     self.session = DiagSession::Default; | executable/support line; behavior impact depends on surroundin
lock context. |
| 387 |     self.security = SecurityLevel::Locked; | executable/support line; behavior impact depends on surround
block context. |
| 388 |     vec![0x51, sub] | executable/support line; behavior impact depends on surrounding block context. |
| 389 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 390 | (blank) | line is blank; visual separation only. |
| 391 | // ████████████████████████████████████████████████████████████████████████████████████████████████ | comment
ntent, no runtime effect. |
| 392 | // Service 0x14 – Clear Diagnostic Information | comment line; documentation intent, no runtime effect. |
| 393 | fn svc_clear_dtc(&mut self, req: &[u8], ts: f64) -> Vec<u8> { | declares executable behavior; signature/b
changes alter API, control flow, and outputs. |
| 394 |     if req.len() < 4 { | conditional gate; changing condition shifts runtime path and side effects. |
| 395 |         return self.nrc(0x14, Nrc::IncorrectMessageLength); | executable/support line; behavior impact de
ds on surrounding block context. |
| 396 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 397 |     if self.session == DiagSession::Default { | conditional gate; changing condition shifts runtime path
side effects. |
| 398 |         return self.nrc(0x14, Nrc::ServiceNotSupportedInActiveSession); | executable/support line; behavi
impact depends on surrounding block context. |
| 399 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 400 |     let group = ((req[1] as u32) << 16) \ | ((req[2] as u32) << 8) \ | (req[3] as u32); | executable/suppor
ine; behavior impact depends on surrounding block context. |
| 401 |     match group { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 402 |         0xFFFFF => { | executable/support line; behavior impact depends on surrounding block context. |
| 403 |             // clear all groups | comment line; documentation intent, no runtime effect. |
| 404 |             self.active_dtcs.clear(); | executable/support line; behavior impact depends on surrounding b
k context. |
| 405 |             self.stored_dtcs.clear(); | executable/support line; behavior impact depends on surrounding b
k context. |
| 406 |             self.freeze_frames.clear(); | executable/support line; behavior impact depends on surrounding
ock context. |
```

```
| 409 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 410 |         spn => { | executable/support line; behavior impact depends on surrounding block context. |  
| 411 |             self.active_dtcs.retain(\|d\| d.spn != spn); | executable/support line; behavior impact depen  
on surrounding block context. |  
| 412 |             self.stored_dtcs.retain(\|d\| d.spn != spn); | executable/support line; behavior impact depen  
on surrounding block context. |  
| 413 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 414 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 415 |     self.log( | executable/support line; behavior impact depends on surrounding block context. |  
| 416 |         ts, | executable/support line; behavior impact depends on surrounding block context. |  
| 417 |         0x14, | executable/support line; behavior impact depends on surrounding block context. |  
| 418 |         None, | executable/support line; behavior impact depends on surrounding block context. |  
| 419 |         None, | executable/support line; behavior impact depends on surrounding block context. |  
| 420 |         UdsEventResult::Positive, | I/O or protocol-critical logic; edits can affect integration, timing,  
d diagnostic outcomes. |  
| 421 |         format!("Cleared group 0x{:06X}", group), | executable/support line; behavior impact depends on s  
ounding block context. |  
| 422 |     ); | executable/support line; behavior impact depends on surrounding block context. |  
| 423 |     vec![0x54] | executable/support line; behavior impact depends on surrounding block context. |  
| 424 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 425 | (blank) | line is blank; visual separation only. |  
| 426 | // ████████████████████████████████████████████████████████████████████████████████████████████████ | comment  
ntent, no runtime effect. |  
| 427 | // Service 0x19 – Read DTC Information | comment line; documentation intent, no runtime effect. |  
| 428 | fn svc_read_dtc(&mut self, req: &[u8], ts: f64) -> Vec<u8> { | declares executable behavior; signature/bo  
changes alter API, control flow, and outputs. |  
| 429 |     if req.len() < 2 { | conditional gate; changing condition shifts runtime path and side effects. |  
| 430 |         return self.nrc(0x19, Nrc::IncorrectMessageLength); | executable/support line; behavior impact de  
ds on surrounding block context. |  
| 431 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
  
| 432 |     let sub = req[1]; | executable/support line; behavior impact depends on surrounding block context. |  
| 433 |     let mut resp = vec![0x59, sub]; | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 434 | (blank) | line is blank; visual separation only. |  
| 435 |     match sub { | branch dispatch; changing arms alters behavior for variants and input classes. |  
| 436 |         // 0x01 – Report number of DTCs by status mask | comment line; documentation intent, no runtime e  
ct. |  
| 437 |         0x01 => { | executable/support line; behavior impact depends on surrounding block context. |  
| 438 |             let mask = if req.len() >= 3 { req[2] } else { 0xFF }; | executable/support line; behavior im  
t depends on surrounding block context. |  
| 439 |             let count = self.active_dtcs.len() as u16; | executable/support line; behavior impact depends  
surrounding block context. |  
| 440 |             resp.push(mask); // status availability mask | executable/support line; behavior impact depen  
on surrounding block context. |  
| 441 |             resp.push(0x01); // DTC format (ISO 14229 format 1) | executable/support line; behavior impac  
depends on surrounding block context. |  
| 442 |             resp.push((count >> 8) as u8); | executable/support line; behavior impact depends on surround  
block context. |  
| 443 |             resp.push((count & 0xFF) as u8); | executable/support line; behavior impact depends on surrou  
ng block context. |  
| 444 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 445 |         // 0x02 – Report DTC by status mask (active) | comment line; documentation intent, no runtime eff  
. |  
| 446 |         0x02 => { | executable/support line; behavior impact depends on surrounding block context. |  
| 447 |             let mask = if req.len() >= 3 { req[2] } else { 0xFF }; | executable/support line; behavior im  
t depends on surrounding block context. |  
| 448 |             resp.push(mask); | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 449 |             for dtc in &self.active_dtcs { | iteration control; changing bounds/body affects throughput,  
ing, and possibly safety guarantees. |
```

```

| 450 |           // 3 bytes DTC + 1 byte status | comment line; documentation intent, no runtime effect. |
| 451 |           let dtc_bytes = dtc_to_3bytes(dtc.spn, dtc.fmi); | executable/support line; behavior impact
depends on surrounding block context. |
| 452 |           resp.extend_from_slice(&dtc_bytes); | executable/support line; behavior impact depends on s
rrounding block context. |
| 453 |           resp.push(0x08); // status: confirmed + test failed | executable/support line; behavior i
ct depends on surrounding block context. |
| 454 |           } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 455 |       } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 456 |           // 0x0A – Report all supported DTCs | comment line; documentation intent, no runtime effect. |
| 457 |           0x0A => { | executable/support line; behavior impact depends on surrounding block context. |
| 458 |               resp.push(0xFF); // all status bits supported | executable/support line; behavior impact depe
on surrounding block context. |
| 459 |               for dtc in self.active_dtc.iter().chain(self.stored_dtc.iter()) { | iteration control; chan
g bounds/body affects throughput, timing, and possibly safety guarantees. |
| 460 |                   let dtc_bytes = dtc_to_3bytes(dtc.spn, dtc.fmi); | executable/support line; behavior impa
depends on surrounding block context. |
| 461 |                   resp.extend_from_slice(&dtc_bytes); | executable/support line; behavior impact depends on
rrounding block context. |
| 462 |                   resp.push(if dtc.active { 0x08 } else { 0x00 }); | executable/support line; behavior impa
depends on surrounding block context. |
| 463 |                   } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 464 |               } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 465 |           // 0x04 – Report DTC snapshot (freeze frame) by DTC number | comment line; documentation intent,
runtime effect. |
| 466 |           0x04 => { | executable/support line; behavior impact depends on surrounding block context. |
| 467 |               if req.len() >= 5 { | conditional gate; changing condition shifts runtime path and side effect
|
| 468 |                   let spn = ((req[2] as u32) << 16) \ | ((req[3] as u32) << 8) \ | (req[4] as u32); | executab
/support line; behavior impact depends on surrounding block context. |
| 469 |                   if let Some(ff) = self | conditional gate; changing condition shifts runtime path and sid
ffects. |
| 470 |                       .freeze_frames | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 471 |                       .iter() | executable/support line; behavior impact depends on surrounding block conte
|
| 472 |                       .find(\f\ | dtc_to_3bytes(f.dtc_spn, f.dtc_fmi) == dtc_to_3bytes(spn, 0)) | executabl
upport line; behavior impact depends on surrounding block context. |
| 473 |                       { | scope delimiter; structure-only line, but misplaced edits can change ownership and li
ime scopes. |
| 474 |                           let dtc_b = dtc_to_3bytes(ff.dtc_spn, ff.dtc_fmi); | executable/support line; behavio
mpact depends on surrounding block context. |
| 475 |                           resp.extend_from_slice(&dtc_b); | executable/support line; behavior impact depends on
rrounding block context. |
| 476 |                           resp.push(0x01); // record number | executable/support line; behavior impact depends
surrounding block context. |
| 477 |                               // Append snapshot data as DID 0xDD01-0xDD0F | comment line; document
ion intent, no runtime effect. |
| 478 |                               let rpm_raw = (ff.engine_rpm / 0.125) as u16; | executable/support line; behavior imp
depends on surrounding block context. |
| 479 |                               resp.push(0xDD); | executable/support line; behavior impact depends on surrounding bl
context. |
| 480 |                               resp.push(0x01); | executable/support line; behavior impact depends on surrounding bl
context. |
| 481 |                               resp.push((rpm_raw >> 8) as u8); | executable/support line; behavior impact depends o
urrounding block context. |
| 482 |                               resp.push((rpm_raw & 0xFF) as u8); | executable/support line; behavior impact depends
surrounding block context. |
| 483 |                               resp.push(0xDD); | executable/support line; behavior impact depends on surrounding bl
context. |
| 484 |                               resp.push(0x03); | executable/support line; behavior impact depends on surrounding bl

```

```
| context. |  
| 485 |             resp.push((ff.coolant_temp_c + 40.0) as u8); | executable/support line; behavior impact depends on surrounding block context. |  
| 486 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 487 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 488 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 489 | _ => return self.nrc(0x19, Nrc::SubFunctionNotSupported), | executable/support line; behavior impact depends on surrounding block context. |  
| 490 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 491 |     self.log(| executable/support line; behavior impact depends on surrounding block context. |  
| 492 |         ts, | executable/support line; behavior impact depends on surrounding block context. |  
| 493 |         0x19, | executable/support line; behavior impact depends on surrounding block context. |  
| 494 |         Some(sub), | executable/support line; behavior impact depends on surrounding block context. |  
| 495 |         None, | executable/support line; behavior impact depends on surrounding block context. |  
| 496 |         UdsEventResult::Positive, | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |  
| 497 |         format!("{}", DTCs", self.active_dtcs.len()), | executable/support line; behavior impact depends on surrounding block context. |  
| 498 |     ); | executable/support line; behavior impact depends on surrounding block context. |  
| 499 |     resp | executable/support line; behavior impact depends on surrounding block context. |  
| 500 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 501 | (blank) | line is blank; visual separation only. |  
| 502 | // ████████████████████████████████████████████████████████████████████████████████ | comment intent, no runtime effect. |  
| 503 | // Service 0x22 – Read Data By Identifier | comment line; documentation intent, no runtime effect. |  
| 504 | fn svc_read_data_by_id(&mut self, req: &[u8], ts: f64) -> Vec<u8> { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |  
| 505 |     if req.len() < 3 { | conditional gate; changing condition shifts runtime path and side effects. |  
| 506 |         return self.nrc(0x22, Nrc::IncorrectMessageLength); | executable/support line; behavior impact depends on surrounding block context. |  
| 507 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 508 |     let did = ((req[1] as u16) << 8) \| req[2] as u16; | executable/support line; behavior impact depends on surrounding block context. |  
| 509 |     let mut resp = vec![0x62, req[1], req[2]]; | executable/support line; behavior impact depends on surrounding block context. |  
| 510 | (blank) | line is blank; visual separation only. |  
| 511 |     match did { | branch dispatch; changing arms alters behavior for variants and input classes. |  
| 512 |         d if d == did::VIN => resp.extend_from_slice(self.vin.as_bytes()), | executable/support line; behavior impact depends on surrounding block context. |  
| 513 |         d if d == did::SW_VERSION => resp.extend_from_slice(self.sw_version.as_bytes()), | executable/support line; behavior impact depends on surrounding block context. |  
| 514 |         d if d == did::HW_VERSION => resp.extend_from_slice(self.hw_version.as_bytes()), | executable/support line; behavior impact depends on surrounding block context. |  
| 515 |         d if d == did::CALIBRATION_ID => resp.extend_from_slice(self.cal_id.as_bytes()), | executable/support line; behavior impact depends on surrounding block context. |  
| 516 |         d if d == did::ECU_SERIAL => resp.extend_from_slice(self.ecu_serial.as_bytes()), | executable/support line; behavior impact depends on surrounding block context. |  
| 517 |         d if d == did::IDLE_RPM_CAL => { | executable/support line; behavior impact depends on surrounding block context. |  
| 518 |             let v = (self.idle_rpm_cal / 0.125) as u16; | executable/support line; behavior impact depends on surrounding block context. |  
| 519 |             resp.push((v >> 8) as u8); | executable/support line; behavior impact depends on surrounding block context. |  
| 520 |             resp.push((v & 0xFF) as u8); | executable/support line; behavior impact depends on surrounding block context. |  
| 521 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 522 |         d if d == did::RATED_RPM_CAL => { | executable/support line; behavior impact depends on surrounding block context. |  
| 523 |             let v = (self.rated_rpm_cal / 0.125) as u16; | executable/support line; behavior impact depends
```



```

| 561 | (blank) | line is blank; visual separation only. |
| 562 |         let sub = req[1]; | executable/support line; behavior impact depends on surrounding block context. |
| 563 |         match sub { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 564 |             // Odd sub-functions = seed request | comment line; documentation intent, no runtime effect. |
| 565 |             sub if sub % 2 == 1 => { | executable/support line; behavior impact depends on surrounding block
text. |
| 566 |                 // Generate seed using cryptographic-quality RNG. | comment line; documentation intent, no ru
me effect. |
| 567 |                 self.security_seed = rand::thread_rng().next_u32(); | executable/support line; behavior impac
depends on surrounding block context. |
| 568 |                 let seed_bytes = self.security_seed.to_be_bytes(); | executable/support line; behavior impact
depends on surrounding block context. |
| 569 |                 let mut resp = vec![0x67, sub]; | executable/support line; behavior impact depends on surroun
g block context. |
| 570 |                 resp.extend_from_slice(&seed_bytes); | executable/support line; behavior impact depends on su
unding block context. |
| 571 |                 self.log( | executable/support line; behavior impact depends on surrounding block context. |
| 572 |                     ts, | executable/support line; behavior impact depends on surrounding block context. |
| 573 |                     0x27, | executable/support line; behavior impact depends on surrounding block context. |
| 574 |                     Some(sub), | executable/support line; behavior impact depends on surrounding block contex
|
| 575 |                     None, | executable/support line; behavior impact depends on surrounding block context. |
| 576 |                     UdsEventResult::Positive, | I/O or protocol-critical logic; edits can affect integration,
ming, and diagnostic outcomes. |
| 577 |                     format!("Seed 0x{:08X}", self.security_seed), | executable/support line; behavior impact
ends on surrounding block context. |
| 578 |                 ); | executable/support line; behavior impact depends on surrounding block context. |
| 579 |                 resp | executable/support line; behavior impact depends on surrounding block context. |
| 580 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 581 |             // Even sub-functions = key send | comment line; documentation intent, no runtime effect. |
| 582 |             sub if sub % 2 == 0 => { | executable/support line; behavior impact depends on surrounding block
text. |
| 583 |                 if req.len() < 6 { | conditional gate; changing condition shifts runtime path and side effect
|
| 584 |                     return self.nrc(0x27, Nrc::IncorrectMessageLength); | executable/support line; behavior i
ct depends on surrounding block context. |
| 585 |                 } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti
scopes. |
| 586 |                 let key = ((req[2] as u32) << 24) | executable/support line; behavior impact depends on surro
ing block context. |
| 587 |                     \ | ((req[3] as u32) << 16) | executable/support line; behavior impact depends on surround
block context. |
| 588 |                     \ | ((req[4] as u32) << 8) | executable/support line; behavior impact depends on surroundi
block context. |
| 589 |                     \ | (req[5] as u32); | executable/support line; behavior impact depends on surrounding blo
context. |
| 590 |                 // Algorithm: XOR seed with constant (simplified – real uses AES/SHA) | comment line; documen
ion intent, no runtime effect. |
| 591 |                 let expected_key = self.security_seed ^ 0xDEADBEEF; | executable/support line; behavior impac
depends on surrounding block context. |
| 592 |                 if key == expected_key { | conditional gate; changing condition shifts runtime path and side
ects. |
| 593 |                     self.security = match sub { | executable/support line; behavior impact depends on surroun
g block context. |
| 594 |                         0x02 => SecurityLevel::Level1, | executable/support line; behavior impact depends on
rounding block context. |
| 595 |                         0x06 => SecurityLevel::Level3, | executable/support line; behavior impact depends on
rounding block context. |
| 596 |                         0x12 => SecurityLevel::Level9, | executable/support line; behavior impact depends on
rounding block context. |
| 597 |                         _ => SecurityLevel::Level1, | executable/support line; behavior impact depends on sur
nding block context. |
| 598 |                     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and l
time scopes. |

```

[illegible]

```

ntent, no runtime effect. |
| 639 |      // Service 0x2E – Write Data By Identifier | comment line; documentation intent, no runtime effect. |
| 640 |      fn svc_write_data_by_id(&mut self, req: &[u8], ts: f64) -> Vec<u8> { | declares executable behavior; sign
re/body changes alter API, control flow, and outputs. |
| 641 |          if req.len() < 3 { | conditional gate; changing condition shifts runtime path and side effects. |
| 642 |              return self.nrc(0x2E, Nrc::IncorrectMessageLength); | executable/support line; behavior impact de
ds on surrounding block context. |
| 643 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 644 |          if self.session != DiagSession::Extended \\| self.security < SecurityLevel::Level1 { | conditional g
; changing condition shifts runtime path and side effects. |
| 645 |              return self.nrc(0x2E, Nrc::SecurityAccessDenied); | executable/support line; behavior impact depende
on surrounding block context. |
| 646 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 647 |          let did = ((req[1] as u16) << 8) \\| req[2] as u16; | executable/support line; behavior impact depends
surrounding block context. |
| 648 |          let data = &req[3..]; | executable/support line; behavior impact depends on surrounding block context
| 649 | (blank) | line is blank; visual separation only. |
| 650 |          match did { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 651 |              d if d == did::IDLE_RPM_CAL && data.len() >= 2 => { | executable/support line; behavior impact de
ds on surrounding block context. |
| 652 |                  self.idle_rpm_cal = (((data[0] as u16) << 8) \\| data[1] as u16) as f64 * 0.125; | executable/s
port line; behavior impact depends on surrounding block context. |
| 653 |              } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 654 |              d if d == did::RATED_RPM_CAL && data.len() >= 2 => { | executable/support line; behavior impact d
nds on surrounding block context. |
| 655 |                  self.rated_rpm_cal = (((data[0] as u16) << 8) \\| data[1] as u16) as f64 * 0.125; | executable
pport line; behavior impact depends on surrounding block context. |
| 656 |              } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 657 |              d if d == did::MAX_TORQUE_CAL && data.len() >= 2 => { | executable/support line; behavior impact
ends on surrounding block context. |
| 658 |                  self.max_torque_cal = (((data[0] as u16) << 8) \\| data[1] as u16) as f64; | executable/suppor
ine; behavior impact depends on surrounding block context. |
| 659 |              } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 660 |              d if d == did::DPF_REGEN_THRESHOLD && !data.is_empty() => { | executable/support line; behavior i
ct depends on surrounding block context. |
| 661 |                  self.dpf_regen_threshold = data[0] as f64 * 0.4; | executable/support line; behavior impact d
nds on surrounding block context. |
| 662 |              } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 663 |              d if d == did::DEF_WARNING_LEVEL && !data.is_empty() => { | executable/support line; behavior imp
depends on surrounding block context. |
| 664 |                  self.def_warning_level = data[0] as f64 * 0.4; | executable/support line; behavior impact dep
s on surrounding block context. |
| 665 |              } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 666 |              d if d == did::FUEL_MAP_SELECT && !data.is_empty() => { | executable/support line; behavior impac
depends on surrounding block context. |
| 667 |                  self.fuel_map_select = data[0].min(2); | executable/support line; behavior impact depends on
rounding block context. |
| 668 |              } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 669 |              _ => return self.nrc(0x2E, Nrc::RequestOutOfRange), | executable/support line; behavior impact de
ds on surrounding block context. |
| 670 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 671 |          self.log( | executable/support line; behavior impact depends on surrounding block context. |
| 672 |              ts, | executable/support line; behavior impact depends on surrounding block context. |
| 673 |              0x2E, | executable/support line; behavior impact depends on surrounding block context. |
| 674 |              None, | executable/support line; behavior impact depends on surrounding block context. |
| 675 |              Some(did), | executable/support line; behavior impact depends on surrounding block context. |

```

```
| 676 | UdsEventResult::Positive, | I/O or protocol-critical logic; edits can affect integration, timing,  
d diagnostic outcomes. |  
| 677 |         format!("DID 0x{:04X} written", did), | executable/support line; behavior impact depends on surro  
ing block context. |  
| 678 |     ); | executable/support line; behavior impact depends on surrounding block context. |  
| 679 |     vec![0x6E, req[1], req[2]] | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 680 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 681 | (blank) | line is blank; visual separation only. |  
| 682 | // ████████████████████████████████████████████████████████████████████████████████████████████████ | comment  
ntent, no runtime effect. |  
| 683 | // Service 0x31 – Routine Control | comment line; documentation intent, no runtime effect. |  
| 684 | fn svc_routine_control(&mut self, req: &[u8], ts: f64) -> Vece/body changes alter API, control flow, and outputs. |  
| 685 |     if req.len() < 4 { | conditional gate; changing condition shifts runtime path and side effects. |  
| 686 |         return self.nrc(0x31, Nrc::IncorrectMessageLength); | executable/support line; behavior impact de  
ds on surrounding block context. |  
| 687 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 688 |     if self.session == DiagSession::Default { | conditional gate; changing condition shifts runtime path  
side effects. |  
| 689 |         return self.nrc(0x31, Nrc::ServiceNotSupportedInActiveSession); | executable/support line; behavi  
impact depends on surrounding block context. |  
| 690 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 691 |     if self.security < SecurityLevel::Level1 { | conditional gate; changing condition shifts runtime path  
d side effects. |  
| 692 |         return self.nrc(0x31, Nrc::SecurityAccessDenied); | executable/support line; behavior impact depende  
on surrounding block context. |  
| 693 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 694 |     let sub = req[1]; // 0x01=start, 0x02=stop, 0x03=request result | executable/support line; behavior i  
ct depends on surrounding block context. |  
| 695 |     let rid_hi = req[2]; | executable/support line; behavior impact depends on surrounding block context.  
| 696 |     let rid_lo = req[3]; | executable/support line; behavior impact depends on surrounding block context.  
| 697 |     let rid: u16 = ((rid_hi as u16) << 8) \| rid_lo as u16; | executable/support line; behavior impact de  
ds on surrounding block context. |  
| 698 | (blank) | line is blank; visual separation only. |  
| 699 |     match rid { | branch dispatch; changing arms alters behavior for variants and input classes. |  
| 700 |         r if r == did::ROUTINE_DPF_REGEN => match sub { | executable/support line; behavior impact depend  
n surrounding block context. |  
| 701 |             0x01 => { | executable/support line; behavior impact depends on surrounding block context. |  
| 702 |                 self.dpf_regen_routine_active = true; | executable/support line; behavior impact depends  
surrounding block context. |  
| 703 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti  
scopes. |  
| 704 |             0x02 => { | executable/support line; behavior impact depends on surrounding block context. |  
| 705 |                 self.dpf_regen_routine_active = false; | executable/support line; behavior impact depends  
surrounding block context. |  
| 706 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti  
scopes. |  
| 707 |             _ => {} | executable/support line; behavior impact depends on surrounding block context. |  
| 708 |         }, | executable/support line; behavior impact depends on surrounding block context. |  
| 709 |         r if r == did::ROUTINE_INJECTOR_TEST => match sub { | executable/support line; behavior impact de  
ds on surrounding block context. |  
| 710 |             0x01 => { | executable/support line; behavior impact depends on surrounding block context. |  
| 711 |                 self.injector_test_active = true; | executable/support line; behavior impact depends on s  
ounding block context. |  
| 712 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti  
scopes. |  
| 713 |             0x02 => { | executable/support line; behavior impact depends on surrounding block context. |  
| 714 |                 self.injector_test_active = false; | executable/support line; behavior impact depends on  
rounding block context. |  
| 715 |             } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifeti  
scopes. |
```

```
| 716 |         _ => {} | executable/support line; behavior impact depends on surrounding block context. |  
| 717 |     }, | executable/support line; behavior impact depends on surrounding block context. |  
| 718 |     _ => return self.nrc(0x31, Nrc::RequestOutOfRange), | executable/support line; behavior impact de  
ds on surrounding block context. |  
| 719 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 720 |     self.log( | executable/support line; behavior impact depends on surrounding block context. |  
| 721 |         ts, | executable/support line; behavior impact depends on surrounding block context. |  
| 722 |         0x31, | executable/support line; behavior impact depends on surrounding block context. |  
| 723 |         Some(sub), | executable/support line; behavior impact depends on surrounding block context. |  
| 724 |         Some(rid), | executable/support line; behavior impact depends on surrounding block context. |  
| 725 |         UdsEventResult::Positive, | I/O or protocol-critical logic; edits can affect integration, timing  
d diagnostic outcomes. |  
| 726 |         format!("Routine 0x{:04X} sub={}", rid, sub), | executable/support line; behavior impact depends  
surrounding block context. |  
| 727 |     ); | executable/support line; behavior impact depends on surrounding block context. |  
| 728 |     vec![0x71, sub, rid_hi, rid_lo, 0x00] | executable/support line; behavior impact depends on surroundi  
block context. |  
| 729 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 730 | (blank) | line is blank; visual separation only. |  
| 731 |     // ████████████████████████████████████████████████████████████████████████████████████████████ | comment  
tent, no runtime effect. |  
| 732 |     // Service 0x34 – Request Download | comment line; documentation intent, no runtime effect. |  
| 733 |     fn svc_request_download(&mut self, req: &[u8], ts: f64) -> Vec<u8> { | declares executable behavior; sign  
re/body changes alter API, control flow, and outputs. |  
| 734 |         if self.security < SecurityLevel::Level3 { | conditional gate; changing condition shifts runtime path  
d side effects. |  
| 735 |             return self.nrc(0x34, Nrc::SecurityAccessDenied); | executable/support line; behavior impact depende  
on surrounding block context. |  
| 736 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 737 |         // Format used here: [0x34, format, addr32(4 bytes), len32(4 bytes)] | comment line; documentation in  
t, no runtime effect. |  
| 738 |         if req.len() < 10 { | conditional gate; changing condition shifts runtime path and side effects. |  
| 739 |             return self.nrc(0x34, Nrc::IncorrectMessageLength); | executable/support line; behavior impact de  
ds on surrounding block context. |  
| 740 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 741 |         self.download_address = ((req[2] as u32) << 24) | executable/support line; behavior impact depends on  
rrounding block context. |  
| 742 |             \ | ((req[3] as u32) << 16) | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 743 |             \ | ((req[4] as u32) << 8) | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 744 |             \ | req[5] as u32; | executable/support line; behavior impact depends on surrounding block context  
| 745 |         self.download_expected_len = ((req[6] as u32) << 24) | executable/support line; behavior impact depen  
on surrounding block context. |  
| 746 |             \ | ((req[7] as u32) << 16) | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 747 |             \ | ((req[8] as u32) << 8) | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 748 |             \ | req[9] as u32; | executable/support line; behavior impact depends on surrounding block context  
| 749 |         if self.download_expected_len == 0 \| \| self.download_expected_len > MAX_DOWNLOAD_LEN { | conditional  
te; changing condition shifts runtime path and side effects. |  
| 750 |             return self.nrc(0x34, Nrc::RequestOutOfRange); | executable/support line; behavior impact depends  
surrounding block context. |  
| 751 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 752 |         self.download_received_len = 0; | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 753 |         self.download_block_num = 1; | executable/support line; behavior impact depends on surrounding block  
text. |  
| 754 |         self.download_state = DownloadState::Requested; | executable/support line; behavior impact depends on  
rrounding block context. |  
| 755 |         self.log( | executable/support line; behavior impact depends on surrounding block context. |
```

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| 756 |         ts, | executable/support line; behavior impact depends on surrounding block context. |
| 757 |         0x34, | executable/support line; behavior impact depends on surrounding block context. |
| 758 |         None, | executable/support line; behavior impact depends on surrounding block context. |
| 759 |         None, | executable/support line; behavior impact depends on surrounding block context. |
| 760 |         UdsEventResult::Positive, | I/O or protocol-critical logic; edits can affect integration, timing,
d diagnostic outcomes. |
| 761 |         format!("Download @0x{:08X}", self.download_address), | executable/support line; behavior impact
ends on surrounding block context. |
| 762 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 763 |         vec![0x74, 0x20, 0x00, 0x81] // maxBlockLen = 0x0081 = 129 bytes | executable/support line; behavior
act depends on surrounding block context. |
| 764 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 765 | (blank) | line is blank; visual separation only. |
| 766 |     // Service 0x36 – Transfer Data | comment line; documentation intent, no runtime effect. |
| 767 |     fn svc_transfer_data(&mut self, req: &[u8], ts: f64) -> Vec<u8> { | declares executable behavior; signatu
body changes alter API, control flow, and outputs. |
| 768 |         if self.download_state != DownloadState::Requested | conditional gate; changing condition shifts runt
path and side effects. |
| 769 |             && self.download_state != DownloadState::Transferring | executable/support line; behavior impact
ends on surrounding block context. |
| 770 |         { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 771 |             return self.nrc(0x36, Nrc::RequestSequenceError); | executable/support line; behavior impact depe
on surrounding block context. |
| 772 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 773 |         if req.len() < 2 { | conditional gate; changing condition shifts runtime path and side effects. |
| 774 |             return self.nrc(0x36, Nrc::IncorrectMessageLength); | executable/support line; behavior impact de
ds on surrounding block context. |
| 775 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 776 |         let block_num = req[1]; | executable/support line; behavior impact depends on surrounding block conte
|
| 777 |         if block_num == 0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 778 |             return self.nrc(0x36, Nrc::RequestSequenceError); | executable/support line; behavior impact depe
on surrounding block context. |
| 779 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 780 |         if block_num != self.download_block_num { | conditional gate; changing condition shifts runtime path
side effects. |
| 781 |             return self.nrc(0x36, Nrc::RequestSequenceError); | executable/support line; behavior impact depe
on surrounding block context. |
| 782 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 783 |         let payload_len = (req.len() - 2) as u32; | executable/support line; behavior impact depends on surro
ing block context. |
| 784 |         let Some(next_total) = self.download_received_len.checked_add(payload_len) else { | executable/support
ine; behavior impact depends on surrounding block context. |
| 785 |             return self.nrc(0x36, Nrc::GeneralProgrammingFailure); | executable/support line; behavior impact
ends on surrounding block context. |
| 786 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scop
|
| 787 |         if next_total > self.download_expected_len { | conditional gate; changing condition shifts runtime pa
and side effects. |
| 788 |             return self.nrc(0x36, Nrc::RequestOutOfRange); | executable/support line; behavior impact depends
surrounding block context. |
| 789 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 790 |         self.download_received_len = next_total; | executable/support line; behavior impact depends on surrou
ng block context. |
| 791 |         let next_block = self.download_block_num.wrapping_add(1); | executable/support line; behavior impact
ends on surrounding block context. |
| 792 |         if next_block == 0 { | conditional gate; changing condition shifts runtime path and side effects. |
| 793 |             return self.nrc(0x36, Nrc::RequestSequenceError); | executable/support line; behavior impact depe
on surrounding block context. |

```

```

| 794 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 795 |         self.download_block_num = next_block; | executable/support line; behavior impact depends on surrounding
block context. |
| 796 |         self.download_state = DownloadState::Transferring; | executable/support line; behavior impact depends
surrounding block context. |
| 797 |         self.log( | executable/support line; behavior impact depends on surrounding block context. |
| 798 |             ts, | executable/support line; behavior impact depends on surrounding block context. |
| 799 |             0x36, | executable/support line; behavior impact depends on surrounding block context. |
| 800 |             None, | executable/support line; behavior impact depends on surrounding block context. |
| 801 |             None, | executable/support line; behavior impact depends on surrounding block context. |
| 802 |             UdsEventResult::Positive, | I/O or protocol-critical logic; edits can affect integration, timing,
d diagnostic outcomes. |
| 803 |             format!("Block {} ({} bytes)", block_num, req.len() - 2), | executable/support line; behavior imp
depends on surrounding block context. |
| 804 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 805 |         vec![0x76, block_num] | executable/support line; behavior impact depends on surrounding block context
| 806 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 807 | (blank) | line is blank; visual separation only. |
| 808 |     // Service 0x37 – Request Transfer Exit | comment line; documentation intent, no runtime effect. |
| 809 |     fn svc_request_transfer_exit(&mut self, _req: &[u8], ts: f64) -> Vec<u8> { | declares executable behavior
signature/body changes alter API, control flow, and outputs. |
| 810 |         if self.download_state != DownloadState::Transferring { | conditional gate; changing condition shifts
runtime path and side effects. |
| 811 |             return self.nrc(0x37, Nrc::RequestSequenceError); | executable/support line; behavior impact depe
on surrounding block context. |
| 812 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 813 |         if self.download_received_len != self.download_expected_len { | conditional gate; changing condition
fts runtime path and side effects. |
| 814 |             self.download_state = DownloadState::Idle; | executable/support line; behavior impact depends on
rounding block context. |
| 815 |             return self.nrc(0x37, Nrc::GeneralProgrammingFailure); | executable/support line; behavior impact
ends on surrounding block context. |
| 816 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 817 |         self.download_state = DownloadState::Complete; | executable/support line; behavior impact depends on
rounding block context. |
| 818 |         self.log( | executable/support line; behavior impact depends on surrounding block context. |
| 819 |             ts, | executable/support line; behavior impact depends on surrounding block context. |
| 820 |             0x37, | executable/support line; behavior impact depends on surrounding block context. |
| 821 |             None, | executable/support line; behavior impact depends on surrounding block context. |
| 822 |             None, | executable/support line; behavior impact depends on surrounding block context. |
| 823 |             UdsEventResult::Positive, | I/O or protocol-critical logic; edits can affect integration, timing,
d diagnostic outcomes. |
| 824 |             format!("{}", bytes transferred", self.download_received_len), | executable/support line; behavior
act depends on surrounding block context. |
| 825 |         ); | executable/support line; behavior impact depends on surrounding block context. |
| 826 |         vec![0x77] | executable/support line; behavior impact depends on surrounding block context. |
| 827 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 828 | (blank) | line is blank; visual separation only. |
| 829 |     // Service 0x3E – Tester Present (keep-alive) | comment line; documentation intent, no runtime effect. |
| 830 |     fn svc_tester_present(&mut self, req: &[u8], ts: f64) -> Vec<u8> { | declares executable behavior; signat
/body changes alter API, control flow, and outputs. |
| 831 |         self.session_timer = 0.0; | executable/support line; behavior impact depends on surrounding block con
t. |
| 832 |         let sub = if req.len() >= 2 { req[1] } else { 0x00 }; | executable/support line; behavior impact depe
on surrounding block context. |
| 833 |         // sub=0x80 = "zero sub function" (suppress positive response) | comment line; documentation intent,
runtime effect. |
| 834 |         if sub == 0x80 { | conditional gate; changing condition shifts runtime path and side effects. |
| 835 |             return Vec::new(); | executable/support line; behavior impact depends on surrounding block context
|
| 836 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|

```



```
| 837 | self.log( | executable/support line; behavior impact depends on surrounding block context. |  
| 838 | ts, | executable/support line; behavior impact depends on surrounding block context. |  
| 839 | 0x3E, | executable/support line; behavior impact depends on surrounding block context. |  
| 840 | Some(sub), | executable/support line; behavior impact depends on surrounding block context. |  
| 841 | None, | executable/support line; behavior impact depends on surrounding block context. |  
| 842 | UdsEventResult::Positive, | I/O or protocol-critical logic; edits can affect integration, timing,  
d diagnostic outcomes. |  
| 843 | "Keep-alive".into(), | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 844 | ); | executable/support line; behavior impact depends on surrounding block context. |  
| 845 | vec![0x7E, sub] | executable/support line; behavior impact depends on surrounding block context. |  
| 846 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 847 | (blank) | line is blank; visual separation only. |  
| 848 | // ████████████████████████████████████████████████████████████████████████████████████████████ | comment  
tent, no runtime effect. |  
| 849 | /// Called by ECM when a new DTC becomes active – stores freeze frame. | comment line; documentation inte  
no runtime effect. |  
| 850 | pub fn record_dtc_set(&mut self, spn: u32, fmi: u8, ff: FreezeFrame) { | declares executable behavior; si  
ture/body changes alter API, control flow, and outputs. |  
| 851 | use crate::j1939::{Dtc, DtcSeverity}; | imports symbols; changing path/name affects compile-time reso  
ion. |  
| 852 | if !self | conditional gate; changing condition shifts runtime path and side effects. |  
| 853 | .active_dtcs | executable/support line; behavior impact depends on surrounding block context. |  
| 854 | .iter() | executable/support line; behavior impact depends on surrounding block context. |  
| 855 | .any(|d| d.spn == spn && d.fmi == fmi) | executable/support line; behavior impact depends on su  
unding block context. |  
| 856 | { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 857 | self.active_dtcs.push(Dtc { | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 858 |     spn, | executable/support line; behavior impact depends on surrounding block context. |  
| 859 |     fmi, | executable/support line; behavior impact depends on surrounding block context. |  
| 860 |     count: 1, | executable/support line; behavior impact depends on surrounding block context. |  
| 861 |     active: true, | executable/support line; behavior impact depends on surrounding block context  
| 862 |     desc: "via UDS record", | I/O or protocol-critical logic; edits can affect integration, timin  
and diagnostic outcomes. |  
| 863 |     severity: DtcSeverity::Amber, | executable/support line; behavior impact depends on surroundi  
block context. |  
| 864 | }); | executable/support line; behavior impact depends on surrounding block context. |  
| 865 | self.freeze_frames.push(ff); | executable/support line; behavior impact depends on surrounding bl  
context. |  
| 866 | // Move to stored too (persists after clear) | comment line; documentation intent, no runtime eff  
. |  
| 867 | self.stored_dtcs.push(Dtc { | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 868 |     spn, | executable/support line; behavior impact depends on surrounding block context. |  
| 869 |     fmi, | executable/support line; behavior impact depends on surrounding block context. |  
| 870 |     count: 1, | executable/support line; behavior impact depends on surrounding block context. |  
| 871 |     active: false, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 872 |     desc: "stored", | executable/support line; behavior impact depends on surrounding block conte  
|  
| 873 |     severity: DtcSeverity::Amber, | executable/support line; behavior impact depends on surroundi  
block context. |  
| 874 | }); | executable/support line; behavior impact depends on surrounding block context. |  
| 875 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 876 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 877 | (blank) | line is blank; visual separation only. |  
| 878 | // ████████████████████████████████████████████████████████████████████████████████████████████ | comment  
tent, no runtime effect. |  
| 879 | fn nrc(&self, service: u8, code: Nrc) -> Vec<u8> { | declares executable behavior; signature/body changes  
ter API, control flow, and outputs. |  
| 880 |     vec![0x7F, service, code as u8] | executable/support line; behavior impact depends on surrounding blo  
context. |
```

```

| 881 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 882 | (blank) | line is blank; visual separation only. |
| 883 |     fn log( | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 884 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 885 |         ts: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 886 |         svc: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 887 |         sf: Option<u8>, | executable/support line; behavior impact depends on surrounding block context. |
| 888 |         did: Option<u16>, | executable/support line; behavior impact depends on surrounding block context. |
| 889 |         res: UdsEventResult, | I/O or protocol-critical logic; edits can affect integration, timing, and diag
tic outcomes. |
| 890 |         detail: String, | executable/support line; behavior impact depends on surrounding block context. |
| 891 |     ) { | executable/support line; behavior impact depends on surrounding block context. |
| 892 |         self.event_log.push_front(UdsEvent { | I/O or protocol-critical logic; edits can affect integration,
ing, and diagnostic outcomes. |
| 893 |             timestamp: ts, | executable/support line; behavior impact depends on surrounding block context. |
| 894 |             service: svc, | executable/support line; behavior impact depends on surrounding block context. |
| 895 |             sub_func: sf, | executable/support line; behavior impact depends on surrounding block context. |
| 896 |             did, | executable/support line; behavior impact depends on surrounding block context. |
| 897 |             result: res, | executable/support line; behavior impact depends on surrounding block context. |
| 898 |             detail, | executable/support line; behavior impact depends on surrounding block context. |
| 899 |         }); | executable/support line; behavior impact depends on surrounding block context. |
| 900 |         if self.event_log.len() > 50 { | conditional gate; changing condition shifts runtime path and side ef
ts. |
| 901 |             self.event_log.pop_back(); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 902 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 903 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 904 | (blank) | line is blank; visual separation only. |
| 905 |     pub fn service_name(svc: u8) -> &'static str { | declares executable behavior; signature/body changes alt
API, control flow, and outputs. |
| 906 |         match svc { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 907 |             0x10 => "DiagSessCtrl", | executable/support line; behavior impact depends on surrounding block c
ext. |
| 908 |             0x11 => "ECUReset", | executable/support line; behavior impact depends on surrounding block conte
|
| 909 |             0x14 => "ClearDTC", | executable/support line; behavior impact depends on surrounding block conte
|
| 910 |             0x19 => "ReadDTC", | executable/support line; behavior impact depends on surrounding block context
|
| 911 |             0x22 => "RdDataByID", | executable/support line; behavior impact depends on surrounding block con
t. |
| 912 |             0x27 => "SecurityAccess", | executable/support line; behavior impact depends on surrounding block
ntext. |
| 913 |             0x2E => "WrDataByID", | executable/support line; behavior impact depends on surrounding block con
t. |
| 914 |             0x31 => "RoutineCtrl", | executable/support line; behavior impact depends on surrounding block co
xt. |
| 915 |             0x34 => "ReqDownload", | executable/support line; behavior impact depends on surrounding block co
xt. |
| 916 |             0x36 => "TransferData", | executable/support line; behavior impact depends on surrounding block c
ext. |
| 917 |             0x37 => "XferExit", | executable/support line; behavior impact depends on surrounding block conte
|
| 918 |             0x3E => "TesterPresent", | executable/support line; behavior impact depends on surrounding block
text. |
| 919 |             0x7F => "NRC", | executable/support line; behavior impact depends on surrounding block context. |
| 920 |             _ => "Unknown", | executable/support line; behavior impact depends on surrounding block context. |
| 921 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 922 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 923 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 924 | (blank) | line is blank; visual separation only. |
| 925 | // Helper: pack SPN+FMI into J1939 DTC 3-byte format | comment line; documentation intent, no runtime effect.
| 926 | fn dtc_to_3bytes(spn: u32, fmi: u8) -> [u8; 3] { | declares executable behavior; signature/body changes alter

```

```

I, control flow, and outputs. |
| 927 |      [ | executable/support line; behavior impact depends on surrounding block context. |
| 928 |          (spn & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding block context. |
| 929 |          ((spn >> 8) & 0xFF) as u8, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 930 |          (((spn >> 16) & 0x7) as u8) \ | ((fmi & 0x1F) << 3)), | executable/support line; behavior impact depe
on surrounding block context. |
| 931 |      ] | executable/support line; behavior impact depends on surrounding block context. |
| 932 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 933 | (blank) | line is blank; visual separation only. |
| 934 | #[cfg(test)] | executable/support line; behavior impact depends on surrounding block context. |
| 935 | mod tests { | declares a module boundary; changing module name/path breaks references. |
| 936 |     use super::*; | imports symbols; changing path/name affects compile-time resolution. |
| 937 | (blank) | line is blank; visual separation only. |
| 938 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 939 |     fn session_default_relocks_security() { | declares executable behavior; signature/body changes alter API,
ntrol flow, and outputs. |
| 940 |         let mut s = UdsServer::new("ECM", 0x00); | I/O or protocol-critical logic; edits can affect integrati
timing, and diagnostic outcomes. |
| 941 |         let _ = s.process(&[0x27, 0x01], 0.0); | executable/support line; behavior impact depends on surround
block context. |
| 942 |         // Force unlocked state for transition behavior test. | comment line; documentation intent, no runtime
ffect. |
| 943 |         s.security = SecurityLevel::Level3; | executable/support line; behavior impact depends on surrounding
ock context. |
| 944 |         let _ = s.process(&[0x10, 0x01], 0.1); | executable/support line; behavior impact depends on surround
block context. |
| 945 |         assert_eq!(s.security, SecurityLevel::Locked); | executable/support line; behavior impact depends on
rounding block context. |
| 946 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 947 | (blank) | line is blank; visual separation only. |
| 948 |     #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 949 |     fn transfer_exit_rejects_incomplete_download() { | declares executable behavior; signature/body changes a
r API, control flow, and outputs. |
| 950 |         let mut s = UdsServer::new("ECM", 0x00); | I/O or protocol-critical logic; edits can affect integrati
timing, and diagnostic outcomes. |
| 951 |         s.security = SecurityLevel::Level3; | executable/support line; behavior impact depends on surrounding
ock context. |
| 952 |         let dl = s.process(&[0x34, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x10], 0.0); | executable/
port line; behavior impact depends on surrounding block context. |
| 953 |         assert_eq!(dl.first().copied(), Some(0x74)); | executable/support line; behavior impact depends on su
unding block context. |
| 954 | (blank) | line is blank; visual separation only. |
| 955 |         let td = s.process(&[0x36, 0x01, 0xAA, 0xBB, 0xCC], 0.1); | executable/support line; behavior impact
ends on surrounding block context. |
| 956 |         assert_eq!(td.first().copied(), Some(0x76)); | executable/support line; behavior impact depends on su
unding block context. |
| 957 | (blank) | line is blank; visual separation only. |
| 958 |         let exit = s.process(&[0x37, 0x00], 0.2); | executable/support line; behavior impact depends on surro
ing block context. |
| 959 |         assert_eq!(exit.first().copied(), Some(0x7F)); | executable/support line; behavior impact depends on
rounding block context. |
| 960 |         assert_eq!(exit.get(2).copied(), Some(Nrc::GeneralProgrammingFailure as u8)); | executable/support li
behavior impact depends on surrounding block context. |
| 961 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 962 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/v2x_telematics.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 634 | Non-empty: 580 | Symbols: 34

Symbol Inventory

Line	Kind	Symbol	Change Impact
------	------	--------	---------------


```

| 21 | pub accel_long: f64, // m/s² | executable/support line; behavior impact depends on surrounding block context. |
| 22 | pub accel_lat: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 23 | pub accel_vert: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 24 | pub yaw_rate: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 25 | pub brake_status: BrakeStatus, | executable/support line; behavior impact depends on surrounding block context. |
| 26 | pub size_length_m: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 27 | pub size_width_m: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 28 | pub timestamp_ms: u64, | executable/support line; behavior impact depends on surrounding block context. |
| 29 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 30 | (blank) | line is blank; visual separation only. |
| 31 | #[derive(Debug, Clone, Copy)] | executable/support line; behavior impact depends on surrounding block context. |
| 32 | pub struct BrakeStatus { | declares data layout; field changes may break serialization, tests, and callers. |
| 33 |     pub brake_applied: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 34 |     pub traction_control: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 35 |     pub abs_active: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 36 |     pub stability: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 37 |     pub aux_brakes: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 38 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 39 | (blank) | line is blank; visual separation only. |
| 40 | /// SAE J2735 Signal Phase and Timing (SPaT) from infrastructure | comment line; documentation intent, no runtime effect. |
| 41 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 42 | pub struct SignalPhaseAndTiming { | declares data layout; field changes may break serialization, tests, and callers. |
| 43 |     pub intersection_id: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 44 |     pub movement_id: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 45 |     pub phase_state: TrafficPhase, | executable/support line; behavior impact depends on surrounding block context. |
| 46 |     pub time_to_change_s: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 47 |     pub distance_m: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 48 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 49 | (blank) | line is blank; visual separation only. |
| 50 | #[derive(Debug, Clone, Copy, PartialEq)] | executable/support line; behavior impact depends on surrounding block context. |
| 51 | pub enum TrafficPhase { | declares state variants; adding/removing variants impacts pattern matches and logic changes. |
| 52 |     Green, | executable/support line; behavior impact depends on surrounding block context. |
| 53 |     Yellow, | executable/support line; behavior impact depends on surrounding block context. |
| 54 |     Red, | executable/support line; behavior impact depends on surrounding block context. |
| 55 |     Unknown, | executable/support line; behavior impact depends on surrounding block context. |
| 56 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 57 | (blank) | line is blank; visual separation only. |
| 58 | impl std::fmt::Display for TrafficPhase { | implementation block; method behavior and trait conformance are defined here. |
| 59 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signal/body changes alter API, control flow, and outputs. |
| 60 |         match self { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 61 |             TrafficPhase::Green => "GREEN", | executable/support line; behavior impact depends on surrounding block context. |
| 62 |             TrafficPhase::Yellow => "YELLOW", | executable/support line; behavior impact depends on surrounding block context. |
| 63 |             TrafficPhase::Red => "RED ", | executable/support line; behavior impact depends on surrounding block context. |
| 64 |             TrafficPhase::Unknown => "UNKNOWN", | executable/support line; behavior impact depends on surrounding block context. |
| 65 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 66 |         .fmt(f) | executable/support line; behavior impact depends on surrounding block context. |
| 67 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 68 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 69 | (blank) | line is blank; visual separation only. |
| 70 | /// Work Zone Alert (roadwork ahead) | comment line; documentation intent, no runtime effect. |
| 71 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |

```



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. |
| 120 | (blank) | line is blank; visual separation only. |
| 121 | // Internal | comment line; documentation intent, no runtime effect. |
| 122 | bsm_timer: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 123 | sim_timer: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 124 | msg_count: u8, | executable/support line; behavior impact depends on surrounding block context. |
| 125 | noise_t: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 126 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 127 | (blank) | line is blank; visual separation only. |
| 128 | impl Default for V2xModule { | implementation block; method behavior and trait conformance are defined here.
| 129 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
outputs. |
| 130 |         Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 131 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 132 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 133 | (blank) | line is blank; visual separation only. |
| 134 | impl V2xModule { | implementation block; method behavior and trait conformance are defined here. |
| 135 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and
outputs. |
| 136 |         V2xModule { | executable/support line; behavior impact depends on surrounding block context. |
| 137 |             bsm_tx_count: 0, | executable/support line; behavior impact depends on surrounding block context.
| 138 |             bsm_tx_rate_hz: 10.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 139 |             dsrc_active: true, | executable/support line; behavior impact depends on surrounding block context
|
| 140 |             cv2x_active: true, | executable/support line; behavior impact depends on surrounding block context
|
| 141 |             nearby_vehicles: Vec::new(), | executable/support line; behavior impact depends on surrounding bl
context. |
| 142 |             spat_messages: Vec::new(), | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 143 |             work_zones: Vec::new(), | executable/support line; behavior impact depends on surrounding block c
ext. |
| 144 |             emergency_alerts: Vec::new(), | executable/support line; behavior impact depends on surrounding b
k context. |
| 145 |             range_m: 300.0, | executable/support line; behavior impact depends on surrounding block context.
| 146 |             packet_loss_pct: 2.0, | executable/support line; behavior impact depends on surrounding block con
t. |
| 147 |             latency_ms: 15.0, | executable/support line; behavior impact depends on surrounding block context
| 148 |             rx_count: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 149 |             forward_collision_alert: false, | executable/support line; behavior impact depends on surrounding
ock context. |
| 150 |             intersection_alert: false, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 151 |             emergency_vehicle_alert: false, | executable/support line; behavior impact depends on surrounding
ock context. |
| 152 |             road_hazard_alert: false, | executable/support line; behavior impact depends on surrounding block
ntext. |
| 153 |             bsm_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 154 |             sim_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context.
| 155 |             msg_count: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 156 |             noise_t: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 157 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 158 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 159 | (blank) | line is blank; visual separation only. |
| 160 |     #[allow(clippy::too_many_arguments)] | executable/support line; behavior impact depends on surrounding bl
context. |
| 161 |     pub fn update( | declares executable behavior; signature/body changes alter API, control flow, and output
|
| 162 |         &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 163 |         lat: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 164 |         lon: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 165 |         speed_ms: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 166 |         heading: f64, | executable/support line; behavior impact depends on surrounding block context. |

```

```
| 167 | _accel_lon: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 168 | _accel_lat: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 169 | _abs_active: bool, | executable/support line; behavior impact depends on surrounding block context. |  
| 170 | elapsed: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 171 | dt: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 172 | ) { | executable/support line; behavior impact depends on surrounding block context. |  
| 173 |     self.noise_t += dt; | executable/support line; behavior impact depends on surrounding block context.  
| 174 |     self.bsm_timer += dt; | executable/support line; behavior impact depends on surrounding block context  
| 175 |     self.sim_timer += dt; | executable/support line; behavior impact depends on surrounding block context  
| 176 |     let n = \s: f64\ ((s * 127.1 + 311.7).sin() * 43758.5 % 1.0 - 0.5) * 2.0; | executable/support line  
behavior impact depends on surrounding block context. |  
| 177 |     let nt = self.noise_t; | executable/support line; behavior impact depends on surrounding block context  
|  
| 178 | (blank) | line is blank; visual separation only. |  
| 179 | // ■ Transmit BSM at 10 Hz ████████████████████████████████████████████████████████ | comment line;  
intent, no runtime effect. |  
| 180 | if self.bsm_timer >= 1.0 / self.bsm_tx_rate_hz { | conditional gate; changing condition shifts runtim  
ath and side effects. |  
| 181 |     self.bsm_timer = 0.0; | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 182 |     self.bsm_tx_count += 1; | executable/support line; behavior impact depends on surrounding block c  
ext. |  
| 183 |     self.msg_count = self.msg_count.wrapping_add(1); | executable/support line; behavior impact dependen  
on surrounding block context. |  
| 184 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 185 | (blank) | line is blank; visual separation only. |  
| 186 | // ■ Simulate receiving BSMS from nearby vehicles ████████████████████████████████ | comment line; docum  
intent, no runtime effect. |  
| 187 | self.nearby_vehicles.clear(); | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 188 | // 3-5 vehicles within range | comment line; documentation intent, no runtime effect. |  
| 189 | let n_vehicles = 3 + (n(nt * 0.1).abs() * 2.0) as usize; | executable/support line; behavior impact d  
nds on surrounding block context. |  
| 190 | for i in 0..n_vehicles { | iteration control; changing bounds/body affects throughput, timing, and po  
bly safety guarantees. |  
| 191 |     let dist = 30.0 + i as f64 * 40.0 + n(nt + i as f64) * 15.0; | executable/support line; behavior  
act depends on surrounding block context. |  
| 192 |     let az = (i as f64 * 60.0 + n(nt * 0.3 + i as f64) * 10.0) % 360.0; | executable/support line; be  
ior impact depends on surrounding block context. |  
| 193 |     self.nearby_vehicles.push(BasicSafetyMessage { | executable/support line; behavior impact depends  
surrounding block context. |  
| 194 |         msg_count: self.msg_count.wrapping_add(i as u8), | executable/support line; behavior impact d  
nds on surrounding block context. |  
| 195 |         vehicle_id: 0xA000 + i as u32, | executable/support line; behavior impact depends on surround  
block context. |  
| 196 |         lat_deg: lat + (az.to_radians()).cos() * dist / 111320.0), | executable/support line; behavior  
impact depends on surrounding block context. |  
| 197 |         lon_deg: lon + (az.to_radians()).sin() * dist / 111320.0), | executable/support line; behavior  
impact depends on surrounding block context. |  
| 198 |         elevation_m: 0.0, | executable/support line; behavior impact depends on surrounding block con  
t. |  
| 199 |         speed_ms: 15.0 + n(nt + i as f64 * 2.0) * 5.0, | executable/support line; behavior impact dep  
s on surrounding block context. |  
| 200 |         heading_deg: (heading + n(nt + i as f64) * 15.0 + 360.0) % 360.0, | executable/support line;  
avior impact depends on surrounding block context. |  
| 201 |         accel_long: n(nt + i as f64 * 3.0) * 1.0, | executable/support line; behavior impact depends  
surrounding block context. |  
| 202 |         accel_lat: n(nt + i as f64 * 4.0) * 0.5, | executable/support line; behavior impact depends o  
urrounding block context. |  
| 203 |         accel_vert: 0.0, | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 204 |         yaw_rate: n(nt + i as f64 * 5.0) * 0.1, | executable/support line; behavior impact depends on  
rrounding block context. |  
| 205 |         brake_status: BrakeStatus { | executable/support line; behavior impact depends on surrounding
```


[illegible]

```
|
| 242 | distance_m: 800.0, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 243 | speed_limit_kmh: 40.0, | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 244 | description: "Road work – 800m ahead".into(), | executable/support line; behavior impact depe
on surrounding block context. |
| 245 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 246 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 247 | // Work zone gets closer as we drive | comment line; documentation intent, no runtime effect. |
| 248 | for wz in &ampmut self.work_zones { | iteration control; changing bounds/body affects throughput, timing
nd possibly safety guarantees. |
| 249 |     wz.distance_m = (wz.distance_m - speed_ms * dt).max(50.0); | executable/support line; behavior im
t depends on surrounding block context. |
| 250 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 251 | (blank) | line is blank; visual separation only. |
| 252 | // ■ Emergency vehicle simulation (periodic) ████████████████████████████████████████ | comment line; docum
ntent, no runtime effect. |
| 253 | self.emergency_alerts.clear(); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 254 | if (elapsed % 30.0) < 5.0 { | conditional gate; changing condition shifts runtime path and side effect
|
| 255 |     self.emergency_alerts.push(EmergencyAlert { | executable/support line; behavior impact depends on
rounding block context. |
| 256 |         vehicle_id: 0xE001, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 257 |         vehicle_type: EmergencyType::Ambulance, | executable/support line; behavior impact depends on
rounding block context. |
| 258 |         distance_m: 200.0 - speed_ms * (elapsed % 5.0), | executable/support line; behavior impact de
ds on surrounding block context. |
| 259 |         bearing_deg: 180.0, | executable/support line; behavior impact depends on surrounding block c
ext. |
| 260 |         approaching: true, | executable/support line; behavior impact depends on surrounding block co
xt. |
| 261 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 262 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 263 | (blank) | line is blank; visual separation only. |
| 264 | // ■ Generate alerts ████████████████████████████████████████████████████████████████ | comment lin
intent, no runtime effect. |
| 265 | self.forward_collision_alert = self.nearby_vehicles.iter().any(|v| { | executable/support line; beh
or impact depends on surrounding block context. |
| 266 |     v.speed_ms < speed_ms * 0.5 && { | executable/support line; behavior impact depends on surroundin
lock context. |
| 267 |         let dx = v.lat_deg - lat; | executable/support line; behavior impact depends on surrounding b
k context. |
| 268 |         let dy = v.lon_deg - lon; | executable/support line; behavior impact depends on surrounding b
k context. |
| 269 |         let dist = (dx * dx + dy * dy).sqrt() * 111320.0; | executable/support line; behavior impact
ends on surrounding block context. |
| 270 |         dist < 30.0 | executable/support line; behavior impact depends on surrounding block context. |
| 271 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 272 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 273 | self.intersection_alert = self | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 274 | .spat_messages | executable/support line; behavior impact depends on surrounding block context. |
| 275 | .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 276 | .any(|s| s.distance_m < 80.0 && s.phase_state == TrafficPhase::Red); | executable/support line;
havior impact depends on surrounding block context. |
| 277 | self.emergency_vehicle_alert = !self.emergency_alerts.is_empty(); | executable/support line; behavior
impact depends on surrounding block context. |
| 278 | self.road_hazard_alert = self.work_zones.iter().any(|w| w.distance_m < 100.0); | executable/support
```

[illegible]

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| 327 | ConnectState::Connected5G => write!(f, "5G NR"), | I/O or protocol-critical logic; edits can
fect integration, timing, and diagnostic outcomes. |
| 328 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 329 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 330 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 331 | (blank) | line is blank; visual separation only. |
| 332 | pub struct TelematicsModule { | declares data layout; field changes may break serialization, tests, and calle
|
| 333 | // ■ Connectivity ████████████████████████████████████████████████████████████████████████████████ | comment li
ntent, no runtime effect. |
| 334 | pub connect_state: ConnectionState, | executable/support line; behavior impact depends on surrounding block
text. |
| 335 | pub signal_bars: u8, // 0-5 | executable/support line; behavior impact depends on surrounding block conte
|
| 336 | pub rsrp_dbm: f64, // Reference Signal Received Power | executable/support line; behavior impact depend
n surrounding block context. |
| 337 | pub rsrq_db: f64, // Reference Signal Received Quality | executable/support line; behavior impact depende
on surrounding block context. |
| 338 | pub ping_ms: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 339 | pub download_kbps: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 340 | pub upload_kbps: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 341 | pub data_used_mb: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 342 | (blank) | line is blank; visual separation only. |
| 343 | // ■ Fleet server ████████████████████████████████████████████████████████████████████████████████ | comment li
ntent, no runtime effect. |
| 344 | pub fleet_server: String, | executable/support line; behavior impact depends on surrounding block context
| 345 | pub vehicle_id: String, | executable/support line; behavior impact depends on surrounding block context. |
| 346 | pub fleet_unit_id: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 347 | pub last_sync_s: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 348 | pub report_interval_s: f64, | executable/support line; behavior impact depends on surrounding block conte
|
| 349 | (blank) | line is blank; visual separation only. |
| 350 | // ■ Remote commands received ████████████████████████████████████████████████████████████████████████████████ | comment line
intent, no runtime effect. |
| 351 | pub pending_commands: Vec<RemoteCommand>, | executable/support line; behavior impact depends on surroundi
block context. |
| 352 | (blank) | line is blank; visual separation only. |
| 353 | // ■ OTA Update ████████████████████████████████████████████████████████████████████████████████ | comment li
tent, no runtime effect. |
| 354 | pub ota_available: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 355 | pub ota_version: String, | executable/support line; behavior impact depends on surrounding block context. |
| 356 | pub ota_size_mb: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 357 | pub ota_progress_pct: f64, | executable/support line; behavior impact depends on surrounding block contex
|
| 358 | pub ota_downloading: bool, | executable/support line; behavior impact depends on surrounding block context
|
| 359 | (blank) | line is blank; visual separation only. |
| 360 | // ■ Geofencing ████████████████████████████████████████████████████████████████████████████████ | comment li
tent, no runtime effect. |
| 361 | pub geofences: Vec<Geofence>, | executable/support line; behavior impact depends on surrounding block con
t. |
| 362 | pub inside_geofence: Option<u32>, | executable/support line; behavior impact depends on surrounding block co
ntext. |
| 363 | (blank) | line is blank; visual separation only. |
| 364 | // ■ Event log ████████████████████████████████████████████████████████████████████████████████ | comment l
tent, no runtime effect. |
| 365 | pub events: VecDeque<TelematicsEvent>, | executable/support line; behavior impact depends on surrounding
ck context. |
| 366 | pub total_events_sent: u64, | executable/support line; behavior impact depends on surrounding block conte
|
| 367 | (blank) | line is blank; visual separation only. |
| 368 | // ■ Timers ████████████████████████████████████████████████████████████████████████████████ | comment t
tent, no runtime effect. |
| 369 | pub report_timer: f64, | executable/support line; behavior impact depends on surrounding block context. |

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| 370 |     connect_timer: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 371 |     noise_t: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 372 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 373 | (blank) | line is blank; visual separation only. |
| 374 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 375 | pub struct RemoteCommand { | declares data layout; field changes may break serialization, tests, and callers. |
| 376 |     pub id: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 377 |     pub cmd_type: RemoteCmdType, | executable/support line; behavior impact depends on surrounding block context. |
| 378 |     pub params: String, | executable/support line; behavior impact depends on surrounding block context. |
| 379 |     pub received_ts: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 380 |     pub executed: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 381 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 382 | (blank) | line is blank; visual separation only. |
| 383 | #[derive(Debug, Clone, Copy)] | executable/support line; behavior impact depends on surrounding block context. |
| 384 | pub enum RemoteCmdType { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 385 |     RequestDtcs, | executable/support line; behavior impact depends on surrounding block context. |
| 386 |     ClearDtcs, | executable/support line; behavior impact depends on surrounding block context. |
| 387 |     UpdateFirmware, | executable/support line; behavior impact depends on surrounding block context. |
| 388 |     SetSpeedLimit, | executable/support line; behavior impact depends on surrounding block context. |
| 389 |     Immobilise, | executable/support line; behavior impact depends on surrounding block context. |
| 390 |     Unlock, | executable/support line; behavior impact depends on surrounding block context. |
| 391 |     RequestLocation, | executable/support line; behavior impact depends on surrounding block context. |
| 392 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 393 | (blank) | line is blank; visual separation only. |
| 394 | impl std::fmt::Display for RemoteCmdType { | implementation block; method behavior and trait conformance are defined here. |
| 395 |     fn fmt(&self, f: &mut std::fmt::Formatter<'_>) -> std::fmt::Result { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 396 |         match self { | branch dispatch; changing arms alters behavior for variants and input classes. |
| 397 |             RemoteCmdType::RequestDtcs => write!(f, "REQUEST_DTC "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 398 |             RemoteCmdType::ClearDtcs => write!(f, "CLEAR_DTC "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 399 |             RemoteCmdType::UpdateFirmware => write!(f, "OTA_UPDATE "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 400 |             RemoteCmdType::SetSpeedLimit => write!(f, "SET_SPD_LIM "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 401 |             RemoteCmdType::Immobilise => write!(f, "IMMOBILISE "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 402 |             RemoteCmdType::Unlock => write!(f, "UNLOCK "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 403 |             RemoteCmdType::RequestLocation => write!(f, "REQ_LOCATION "), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 404 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 405 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 406 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 407 | (blank) | line is blank; visual separation only. |
| 408 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 409 | pub struct Geofence { | declares data layout; field changes may break serialization, tests, and callers. |
| 410 |     pub id: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 411 |     pub name: String, | executable/support line; behavior impact depends on surrounding block context. |
| 412 |     pub center_lat: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 413 |     pub center_lon: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 414 |     pub radius_m: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 415 |     pub alert_type: GeofenceAlert, | executable/support line; behavior impact depends on surrounding block context. |
| 416 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 417 | (blank) | line is blank; visual separation only. |
| 418 | #[derive(Debug, Clone, Copy)] | executable/support line; behavior impact depends on surrounding block context. |
| 419 | pub enum GeofenceAlert { | declares state variants; adding/removing variants impacts pattern matches and logic branches. |
| 420 |     Entry, | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 421 |      Exit, | executable/support line; behavior impact depends on surrounding block context. |
| 422 |      Both, | executable/support line; behavior impact depends on surrounding block context. |
| 423 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 424 | (blank) | line is blank; visual separation only. |
| 425 | impl Default for TelematicsModule { | implementation block; method behavior and trait conformance are defined here. |
| 426 |     fn default() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 427 |         Self::new() | executable/support line; behavior impact depends on surrounding block context. |
| 428 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 429 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 430 | (blank) | line is blank; visual separation only. |
| 431 | impl TelematicsModule { | implementation block; method behavior and trait conformance are defined here. |
| 432 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 433 |         let geofences = vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 434 |             Geofence { | executable/support line; behavior impact depends on surrounding block context. |
| 435 |                 id: 1, | executable/support line; behavior impact depends on surrounding block context. |
| 436 |                 name: "HQ Depot".into(), | executable/support line; behavior impact depends on surrounding block context. |
| 437 |                 center_lat: -22.810, | executable/support line; behavior impact depends on surrounding block context. |
| 438 |                 center_lon: -47.062, | executable/support line; behavior impact depends on surrounding block context. |
| 439 |                 radius_m: 500.0, | executable/support line; behavior impact depends on surrounding block context. |
| 440 |                 alert_type: GeofenceAlert::Both, | executable/support line; behavior impact depends on surrounding block context. |
| 441 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 442 |             Geofence { | executable/support line; behavior impact depends on surrounding block context. |
| 443 |                 id: 2, | executable/support line; behavior impact depends on surrounding block context. |
| 444 |                 name: "Field Zone A".into(), | executable/support line; behavior impact depends on surrounding block context. |
| 445 |                 center_lat: -22.850, | executable/support line; behavior impact depends on surrounding block context. |
| 446 |                 center_lon: -47.090, | executable/support line; behavior impact depends on surrounding block context. |
| 447 |                 radius_m: 1000.0, | executable/support line; behavior impact depends on surrounding block context. |
| 448 |                 alert_type: GeofenceAlert::Entry, | executable/support line; behavior impact depends on surrounding block context. |
| 449 |             }, | executable/support line; behavior impact depends on surrounding block context. |
| 450 |         ]; | executable/support line; behavior impact depends on surrounding block context. |
| 451 |         TelematicsModule { | executable/support line; behavior impact depends on surrounding block context. |
| 452 |             connect_state: ConnectState::Connected4G, | executable/support line; behavior impact depends on surrounding block context. |
| 453 |             signal_bars: 4, | executable/support line; behavior impact depends on surrounding block context. |
| 454 |             rsrp_dbm: -85.0, | executable/support line; behavior impact depends on surrounding block context. |
| 455 |             rsrq_db: -9.0, | executable/support line; behavior impact depends on surrounding block context. |
| 456 |             ping_ms: 28.0, | executable/support line; behavior impact depends on surrounding block context. |
| 457 |             download_kbps: 15000.0, | executable/support line; behavior impact depends on surrounding block context. |
| 458 |             upload_kbps: 5000.0, | executable/support line; behavior impact depends on surrounding block context. |
| 459 |             data_used_mb: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 460 |             fleet_server: "fleet.example.com:8443".into(), | executable/support line; behavior impact depends on surrounding block context. |
| 461 |             vehicle_id: "CAT-320-001".into(), | executable/support line; behavior impact depends on surrounding block context. |
| 462 |             fleet_unit_id: 1001, | executable/support line; behavior impact depends on surrounding block context. |
| 463 |             last_sync_s: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 464 |             report_interval_s: 30.0, | executable/support line; behavior impact depends on surrounding block context. |

```

```
| 465 | pending_commands: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 466 | ota_available: false, | executable/support line; behavior impact depends on surrounding block context. |
| 467 | ota_version: String::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 468 | ota_size_mb: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 469 | ota_progress_pct: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 470 | ota_downloading: false, | executable/support line; behavior impact depends on surrounding block context. |
| 471 | geofences, | executable/support line; behavior impact depends on surrounding block context. |
| 472 | inside_geofence: None, | executable/support line; behavior impact depends on surrounding block context. |
| 473 | events: VecDeque::new(), | executable/support line; behavior impact depends on surrounding block context. |
| 474 | total_events_sent: 0, | executable/support line; behavior impact depends on surrounding block context. |
| 475 | report_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 476 | connect_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 477 | noise_t: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 478 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 479 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 480 | (blank) | line is blank; visual separation only. |
| 481 | #[allow(clippy::too_many_arguments)] | executable/support line; behavior impact depends on surrounding block context. |
| 482 | pub fn update( | declares executable behavior; signature/body changes alter API, control flow, and output |
| 483 | &mut self, | executable/support line; behavior impact depends on surrounding block context. |
| 484 | lat: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 485 | lon: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 486 | speed_kmh: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 487 | engine_hours: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 488 | active_dtcs: usize, | executable/support line; behavior impact depends on surrounding block context. |
| 489 | elapsed: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 490 | dt: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 491 | ) { | executable/support line; behavior impact depends on surrounding block context. |
| 492 |     self.noise_t += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 493 |     self.report_timer += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 494 |     self.connect_timer += dt; | executable/support line; behavior impact depends on surrounding block context. |
| 495 |     let n = \|s: f64\| ((s * 127.1 + 311.7).sin() * 43758.5 % 1.0 - 0.5) * 2.0; | executable/support line; behavior impact depends on surrounding block context. |
| 496 |     let nt = self.noise_t; | executable/support line; behavior impact depends on surrounding block context. |
| 497 | (blank) | line is blank; visual separation only. |
| 498 | // ■ Signal quality simulation ████████████████████████████████████████████████████████████ | comment line; intent, no runtime effect. |
| 499 |     self.rsrp_dbm = -82.0 + n(nt * 0.3) * 8.0; | executable/support line; behavior impact depends on surrounding block context. |
| 500 |     self.rsrq_db = -9.0 + n(nt * 0.5) * 3.0; | executable/support line; behavior impact depends on surrounding block context. |
| 501 |     self.signal_bars = match self.rsrp_dbm as i32 { | executable/support line; behavior impact depends on surrounding block context. |
| 502 |         p if p > -70 => 5, | executable/support line; behavior impact depends on surrounding block context. |
| 503 |         p if p > -80 => 4, | executable/support line; behavior impact depends on surrounding block context. |
| 504 |         p if p > -90 => 3, | executable/support line; behavior impact depends on surrounding block context. |
| 505 |         p if p > -100 => 2, | executable/support line; behavior impact depends on surrounding block context.
```



```
|  
| 583 |         }); | executable/support line; behavior impact depends on surrounding block context. |  
| 584 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 585 | (blank) | line is blank; visual separation only. |  
| 586 |     // ■ Geofence check ████████████████████████████████████████████████████████████████ | comment lin  
intent, no runtime effect. |  
| 587 |     self.inside_geofence = None; | executable/support line; behavior impact depends on surrounding block  
text. |  
| 588 |     for gf in &self.geofences { | iteration control; changing bounds/body affects throughput, timing, and  
ssibly safety guarantees. |  
| 589 |         let dx = (lat - gf.center_lat) * 111320.0; | executable/support line; behavior impact depends on  
rounding block context. |  
| 590 |         let dy = (lon - gf.center_lon) * 111320.0 * (lat * PI / 180.0).cos(); | executable/support line;  
avior impact depends on surrounding block context. |  
| 591 |         if (dx * dx + dy * dy).sqrt() < gf.radius_m { | conditional gate; changing condition shifts runti  
path and side effects. |  
| 592 |             self.inside_geofence = Some(gf.id); | executable/support line; behavior impact depends on surr  
nding block context. |  
| 593 |             break; | executable/support line; behavior impact depends on surrounding block context. |  
| 594 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s  
es. |  
| 595 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 596 | (blank) | line is blank; visual separation only. |  
| 597 |     // ■ OTA simulation ████████████████████████████████████████████████████████████████ | comment line  
ntent, no runtime effect. |  
| 598 |     if elapsed > 60.0 && !self.ota_available && !self.ota_downloading { | conditional gate; changing cond  
on shifts runtime path and side effects. |  
| 599 |         self.ota_available = true; | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 600 |         self.ota_version = "SW_01.24.001".into(); | executable/support line; behavior impact depends on s  
ounding block context. |  
| 601 |         self.ota_size_mb = 48.5; | executable/support line; behavior impact depends on surrounding block  
text. |  
| 602 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 603 |     if self.ota_downloading { | conditional gate; changing condition shifts runtime path and side effects  
| 604 |         self.ota_progress_pct = (self.ota_progress_pct + dt * 0.5).min(100.0); | executable/support line;  
havior impact depends on surrounding block context. |  
| 605 |         self.data_used_mb += 0.5 * dt; | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 606 |         if self.ota_progress_pct >= 100.0 { | conditional gate; changing condition shifts runtime path an  
ide effects. |  
| 607 |             self.ota_downloading = false; | executable/support line; behavior impact depends on surroundi  
block context. |  
| 608 |             self.ota_available = false; | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 609 |             let ev = TelematicsEvent { | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 610 |                 timestamp: elapsed, | executable/support line; behavior impact depends on surrounding blo  
context. |  
| 611 |                 sent: true, | executable/support line; behavior impact depends on surrounding block conte  
|  
| 612 |                 ack: true, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 613 |                 event_type: TelEventType::OtaUpdateComplete, | executable/support line; behavior impact d  
nds on surrounding block context. |  
| 614 |                 description: format!("OTA {} installed", self.ota_version), | executable/support line; be  
ior impact depends on surrounding block context. |  
| 615 |             }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifet  
scopes. |  
| 616 |             self.push_event(ev); | executable/support line; behavior impact depends on surrounding block  
text. |  
| 617 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
```

```

es. |
| 618 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 619 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 620 | (blank) | line is blank; visual separation only. |
| 621 |      fn push_event(&mut self, ev: TelematicsEvent) { | declares executable behavior; signature/body changes al
API, control flow, and outputs. |
| 622 |          self.events.push_front(ev); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 623 |          if self.events.len() > 100 { | conditional gate; changing condition shifts runtime path and side effe
. |
| 624 |          self.events.pop_back(); | executable/support line; behavior impact depends on surrounding block c
ext. |
| 625 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 626 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 627 | (blank) | line is blank; visual separation only. |
| 628 |      pub fn start_ota(&mut self) { | declares executable behavior; signature/body changes alter API, control f
, and outputs. |
| 629 |          if self.ota_available { | conditional gate; changing condition shifts runtime path and side effects.
| 630 |              self.ota_downloading = true; | executable/support line; behavior impact depends on surrounding bl
context. |
| 631 |              self.ota_progress_pct = 0.0; | executable/support line; behavior impact depends on surrounding bl
context. |
| 632 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 633 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 634 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: src/vehicle.rs

Role: Domain module in simulation/control/protocol stack.

Lines: 148 | Non-empty: 134 | Symbols: 15

Symbol Inventory

Line	Kind	Symbol	Change Impact
3	struct	VehicleState	data/structure coupling
17	impl	VehicleState	behavior binding and method semantics
18	fn	new	signature/API coupling
35	fn	velocity_ms	signature/API coupling
39	fn	reset	signature/API coupling
46	struct	TrafficObject	data/structure coupling
56	impl	TrafficObject	behavior binding and method semantics
57	fn	new	signature/API coupling
70	fn	ttc	signature/API coupling
78	fn	in_path	signature/API coupling
84	struct	TrafficSimulator	data/structure coupling
90	impl	TrafficSimulator	behavior binding and method semantics
91	fn	new	signature/API coupling
104	fn	update	signature/API coupling
145	fn	lead_vehicle	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	/// Physical state of the ego vehicle	comment line; documentation intent, no runtime effect.
2	#[derive(Debug, Clone)]	executable/support line; behavior impact depends on surrounding block context.
3	pub struct VehicleState {	declares data layout; field changes may break serialization, tests, and callers.
4	pub velocity: f64,	// km/h longitudinal executable/support line; behavior impact depends on surrounding block context.
5	pub acceleration: f64,	// m/s ² executable/support line; behavior impact depends on surrounding block context.

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| 6 | pub steering_angle: f64, // degrees (-45..45) | executable/support line; behavior impact depends on surrounding block context. |
| 7 | pub heading: f64, // degrees 0-360 | executable/support line; behavior impact depends on surrounding block context. |
| 8 | pub yaw_rate: f64, // deg/s | executable/support line; behavior impact depends on surrounding block context. |
| 9 | pub position_x: f64, // meters | executable/support line; behavior impact depends on surrounding block context. |
| 10 | pub position_y: f64, // meters | executable/support line; behavior impact depends on surrounding block context. |
| 11 | pub lateral_g: f64, // g | executable/support line; behavior impact depends on surrounding block context. |
| 12 | pub longitudinal_g: f64, // g | executable/support line; behavior impact depends on surrounding block context. |
| 13 | pub wheel_speeds: [f64; 4], // FL,FR,RL,RR km/h | executable/support line; behavior impact depends on surrounding block context. |
| 14 | pub wheel_slip: [f64; 4], // -1..1 | executable/support line; behavior impact depends on surrounding block context. |
| 15 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 16 | (blank) | line is blank; visual separation only. |
| 17 | impl VehicleState { | implementation block; method behavior and trait conformance are defined here. |
| 18 | pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 19 | Self { | executable/support line; behavior impact depends on surrounding block context. |
| 20 | velocity: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 21 | acceleration: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 22 | steering_angle: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 23 | heading: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 24 | yaw_rate: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 25 | position_x: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 26 | position_y: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 27 | lateral_g: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 28 | longitudinal_g: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
| 29 | wheel_speeds: [0.0; 4], | executable/support line; behavior impact depends on surrounding block context. |
| 30 | wheel_slip: [0.0; 4], | executable/support line; behavior impact depends on surrounding block context. |
| 31 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 32 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 33 | (blank) | line is blank; visual separation only. |
| 34 | /// Returns speed in m/s | comment line; documentation intent, no runtime effect. |
| 35 | pub fn velocity_ms(&self) -> f64 { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 36 | self.velocity / 3.6 | executable/support line; behavior impact depends on surrounding block context. |
| 37 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 38 | (blank) | line is blank; visual separation only. |
| 39 | pub fn reset(&mut self) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 40 | *self = Self::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 41 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 42 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 43 | (blank) | line is blank; visual separation only. |
| 44 | /// A simulated traffic participant | comment line; documentation intent, no runtime effect. |
| 45 | #[derive(Debug, Clone)] | executable/support line; behavior impact depends on surrounding block context. |
| 46 | pub struct TrafficObject { | declares data layout; field changes may break serialization, tests, and callers. |
| 47 | pub id: u32, | executable/support line; behavior impact depends on surrounding block context. |
| 48 | pub distance: f64, // m ahead (positive) / behind (negative) | executable/support line; behavior impact depends on surrounding block context. |
| 49 | pub lateral_offset: f64, // m from ego lane center | executable/support line; behavior impact depends on surrounding block context. |
| 50 | pub speed: f64, // km/h | executable/support line; behavior impact depends on surrounding block context. |
| 51 | pub closing_speed: f64, // km/h (positive = approaching ego) | executable/support line; behavior impact depends on surrounding block context. |

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```

nds on surrounding block context. |
| 52 |     pub width: f64,           // m | executable/support line; behavior impact depends on surrounding block context. |
. |
| 53 |     pub is_vehicle: bool, | executable/support line; behavior impact depends on surrounding block context. |
| 54 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 55 | (blank) | line is blank; visual separation only. |
| 56 | impl TrafficObject { | implementation block; method behavior and trait conformance are defined here. |
| 57 |     pub fn new(id: u32, distance: f64, lateral_offset: f64, speed: f64) -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
signature/body changes alter API, control flow, and outputs. |
| 58 |         Self { | executable/support line; behavior impact depends on surrounding block context. |
| 59 |             id, | executable/support line; behavior impact depends on surrounding block context. |
| 60 |             distance, | executable/support line; behavior impact depends on surrounding block context. |
| 61 |             lateral_offset, | executable/support line; behavior impact depends on surrounding block context. |
| 62 |             speed, | executable/support line; behavior impact depends on surrounding block context. |
| 63 |             closing_speed: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
|
| 64 |             width: 1.8, | executable/support line; behavior impact depends on surrounding block context. |
| 65 |             is_vehicle: true, | executable/support line; behavior impact depends on surrounding block context. |
| 66 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 67 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 68 | (blank) | line is blank; visual separation only. |
| 69 |     /// Time to collision in seconds (0 = already colliding) | comment line; documentation intent, no runtime effect. |
| 70 |     pub fn ttc(&self) -> f64 { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
d outputs. |
| 71 |         if self.closing_speed <= 0.0 \|\| self.distance <= 0.0 { | conditional gate; changing condition shifts runtime path and side effects. |
runtime path and side effects. |
| 72 |             return f64::INFINITY; | executable/support line; behavior impact depends on surrounding block context. |
. |
| 73 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 74 |         self.distance / (self.closing_speed / 3.6) | executable/support line; behavior impact depends on surrounding block context. |
ing block context. |
| 75 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 76 | (blank) | line is blank; visual separation only. |
| 77 |     /// Is this object in the ego vehicle path? | comment line; documentation intent, no runtime effect. |
| 78 |     pub fn in_path(&self) -> bool { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
w, and outputs. |
| 79 |         self.lateral_offset.abs() < 1.5 && self.distance > 0.0 | executable/support line; behavior impact depends on surrounding block context. |
on surrounding block context. |
| 80 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 81 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 82 | (blank) | line is blank; visual separation only. |
| 83 | /// Simple traffic simulation: one lead vehicle + side threats | comment line; documentation intent, no runtime effect. |
ffect. |
| 84 | pub struct TrafficSimulator { | declares data layout; field changes may break serialization, tests, and caller expectations. |
|
| 85 |     pub objects: Vec<TrafficObject>, | executable/support line; behavior impact depends on surrounding block context. |
ext. |
| 86 |     lead_behavior_timer: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 87 |     lead_target_speed: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 88 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 89 | (blank) | line is blank; visual separation only. |
| 90 | impl TrafficSimulator { | implementation block; method behavior and trait conformance are defined here. |
| 91 |     pub fn new() -> Self { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
tputs. |
| 92 |         let mut sim = Self { | executable/support line; behavior impact depends on surrounding block context. |
| 93 |             objects: Vec::new(), | executable/support line; behavior impact depends on surrounding block context. |
|
| 94 |             lead_behavior_timer: 0.0, | executable/support line; behavior impact depends on surrounding block context. |
text. |
| 95 |             lead_target_speed: 80.0, | executable/support line; behavior impact depends on surrounding block context. |
ext. |
| 96 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
|
| 97 |         // Lead vehicle | comment line; documentation intent, no runtime effect. |

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```

| 98 |         sim.objects.push(TrafficObject::new(1, 80.0, 0.0, 80.0)); | executable/support line; behavior impact d
nds on surrounding block context. |
| 99 |         // Static obstacle on side | comment line; documentation intent, no runtime effect. |
| 100 |        sim.objects.push(TrafficObject::new(2, 30.0, 3.2, 0.0)); | executable/support line; behavior impact d
nds on surrounding block context. |
| 101 |        sim | executable/support line; behavior impact depends on surrounding block context. |
| 102 |    } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 103 | (blank) | line is blank; visual separation only. |
| 104 |    pub fn update(&mut self, ego_speed: f64, dt: f64) { | declares executable behavior; signature/body change
lter API, control flow, and outputs. |
| 105 |        self.lead_behavior_timer += dt; | executable/support line; behavior impact depends on surrounding blo
context. |
| 106 | (blank) | line is blank; visual separation only. |
| 107 |        // Lead vehicle behavior changes every 8 seconds | comment line; documentation intent, no runtime eff
. |
| 108 |        if self.lead_behavior_timer > 8.0 { | conditional gate; changing condition shifts runtime path and si
effects. |
| 109 |            self.lead_behavior_timer = 0.0; | executable/support line; behavior impact depends on surrounding
ock context. |
| 110 |            // Alternate: slow down / speed up | comment line; documentation intent, no runtime effect. |
| 111 |            self.lead_target_speed = if self.lead_target_speed > 60.0 { | executable/support line; behavior i
ct depends on surrounding block context. |
| 112 |                30.0 | executable/support line; behavior impact depends on surrounding block context. |
| 113 |            } else { | executable/support line; behavior impact depends on surrounding block context. |
| 114 |                90.0 | executable/support line; behavior impact depends on surrounding block context. |
| 115 |            }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime
pes. |
| 116 |        } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 117 | (blank) | line is blank; visual separation only. |
| 118 |        for obj in &mut self.objects { | iteration control; changing bounds/body affects throughput, timing,
possibly safety guarantees. |
| 119 |            if !obj.is_vehicle { | conditional gate; changing condition shifts runtime path and side effects.
| 120 |                continue; | executable/support line; behavior impact depends on surrounding block context. |
| 121 |            } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 122 |            // Lead vehicle (id=1) tracks target speed | comment line; documentation intent, no runtime effec
|
| 123 |            if obj.id == 1 { | conditional gate; changing condition shifts runtime path and side effects. |
| 124 |                let speed_diff = self.lead_target_speed - obj.speed; | executable/support line; behavior impa
depends on surrounding block context. |
| 125 |                obj.speed += speed_diff * dt * 0.5; | executable/support line; behavior impact depends on sur
nding block context. |
| 126 |                obj.speed = obj.speed.max(0.0); | executable/support line; behavior impact depends on surroun
g block context. |
| 127 |            } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 128 |            // Relative distance changes by closing speed | comment line; documentation intent, no runtime ef
t. |
| 129 |            let relative_speed_ms = (ego_speed - obj.speed) / 3.6; | executable/support line; behavior impact
depends on surrounding block context. |
| 130 |            obj.distance -= relative_speed_ms * dt; | executable/support line; behavior impact depends on sur
nding block context. |
| 131 |            obj.closing_speed = ego_speed - obj.speed; | executable/support line; behavior impact depends on
rounding block context. |
| 132 | (blank) | line is blank; visual separation only. |
| 133 |            // Teleport lead vehicle if it goes too far | comment line; documentation intent, no runtime effe
|
| 134 |            if obj.id == 1 { | conditional gate; changing condition shifts runtime path and side effects. |
| 135 |                if obj.distance < -20.0 { | conditional gate; changing condition shifts runtime path and side
fects. |
| 136 |                    obj.distance = 180.0; | executable/support line; behavior impact depends on surrounding b
k context. |
| 137 |                    obj.speed = 80.0; | executable/support line; behavior impact depends on surrounding block
ntext. |

```



```

| 17 |     ui: &mut Ui, | executable/support line; behavior impact depends on surrounding block context. |
| 18 |     value: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 19 |     min: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 20 |     max: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 21 |     label: &str, | executable/support line; behavior impact depends on surrounding block context. |
| 22 |     unit: &str, | executable/support line; behavior impact depends on surrounding block context. |
| 23 |     size: f32, | executable/support line; behavior impact depends on surrounding block context. |
| 24 |     warn: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 25 |     crit: f64, | executable/support line; behavior impact depends on surrounding block context. |
| 26 |     low_warn: Option<f64>, | executable/support line; behavior impact depends on surrounding block context. |
| 27 | ) -> Response { | executable/support line; behavior impact depends on surrounding block context. |
| 28 |     let (rect, response) = ui.allocate_exact_size(Vec2::splat(size), Sense::hover()); | executable/support line;
behavior impact depends on surrounding block context. |
| 29 |     if !ui.is_rect_visible(rect) { | conditional gate; changing condition shifts runtime path and side effects
| 30 |         return response; | executable/support line; behavior impact depends on surrounding block context. |
| 31 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 32 | (blank) | line is blank; visual separation only. |
| 33 |     let painter = ui.painter_at(rect); | executable/support line; behavior impact depends on surrounding block
context. |
| 34 |     let center = rect.center(); | executable/support line; behavior impact depends on surrounding block context
|
| 35 |     let r = size * 0.35; | executable/support line; behavior impact depends on surrounding block context. |
| 36 |     let frac = ((value - min) / (max - min).max(0.001)).clamp(0.0, 1.0) as f32; | executable/support line; beh
or impact depends on surrounding block context. |
| 37 | (blank) | line is blank; visual separation only. |
| 38 |     // Color logic | comment line; documentation intent, no runtime effect. |
| 39 |     let value_color = if value >= crit { | executable/support line; behavior impact depends on surrounding blo
context. |
| 40 |         Color32::RED | executable/support line; behavior impact depends on surrounding block context. |
| 41 |     } else if value >= warn && warn > 0.0 { | executable/support line; behavior impact depends on surrounding
ck context. |
| 42 |         Color32::YELLOW | executable/support line; behavior impact depends on surrounding block context. |
| 43 |     } else if let Some(lo) = low_warn { | executable/support line; behavior impact depends on surrounding blo
context. |
| 44 |         if value <= lo { | conditional gate; changing condition shifts runtime path and side effects. |
| 45 |             Color32::RED | executable/support line; behavior impact depends on surrounding block context. |
| 46 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 47 |             Color32::from_rgb(30, 210, 90) | executable/support line; behavior impact depends on surrounding b
k context. |
| 48 |         } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes
| 49 |     } else { | executable/support line; behavior impact depends on surrounding block context. |
| 50 |         Color32::from_rgb(30, 210, 90) | executable/support line; behavior impact depends on surrounding block
context. |
| 51 |     }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 52 | (blank) | line is blank; visual separation only. |
| 53 |     // Arc: 135° → 405° (270° clockwise sweep) | comment line; documentation intent, no runtime effect. |
| 54 |     const START: f32 = PI * 0.75; | executable/support line; behavior impact depends on surrounding block cont
. |
| 55 |     const SWEEP: f32 = PI * 1.50; | executable/support line; behavior impact depends on surrounding block cont
. |
| 56 |     const N: usize = 80; | executable/support line; behavior impact depends on surrounding block context. |
| 57 | (blank) | line is blank; visual separation only. |
| 58 |     // Background arc | comment line; documentation intent, no runtime effect. |
| 59 |     for i in 0..N { | iteration control; changing bounds/body affects throughput, timing, and possibly safety
rantees. |
| 60 |         let a0 = START + (i as f32 / N as f32) * SWEEP; | executable/support line; behavior impact depends on
rounding block context. |
| 61 |         let a1 = START + ((i + 1) as f32 / N as f32) * SWEEP; | executable/support line; behavior impact depen
on surrounding block context. |
| 62 |         let p0 = center + Vec2::new(a0.cos() * r, a0.sin() * r); | executable/support line; behavior impact de
ds on surrounding block context. |
| 63 |         let p1 = center + Vec2::new(a1.cos() * r, a1.sin() * r); | executable/support line; behavior impact de
ds on surrounding block context. |
| 64 |         painter.line_segment([p0, p1], Stroke::new(4.0, Color32::from_gray(42))); | executable/support line; b
vior impact depends on surrounding block context. |

```



```

| 65 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 66 | (blank) | line is blank; visual separation only. |
| 67 |     // Value arc (gradient green → yellow → red) | comment line; documentation intent, no runtime effect. |
| 68 |     let filled = ((frac * N as f32).floor() as usize).min(N); | executable/support line; behavior impact depends
on surrounding block context. |
| 69 |     for i in 0..filled { | iteration control; changing bounds/body affects throughput, timing, and possibly sa
y guarantees. |
| 70 |         let t = i as f32 / N as f32; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 71 |         let seg = if t < 0.60 { | executable/support line; behavior impact depends on surrounding block context
|
| 72 |             Color32::from_rgb(30, 210, 90) | executable/support line; behavior impact depends on surrounding b
k context. |
| 73 |         } else if t < 0.82 { | executable/support line; behavior impact depends on surrounding block context.
| 74 |             Color32::YELLOW | executable/support line; behavior impact depends on surrounding block context. |
| 75 |         } else { | executable/support line; behavior impact depends on surrounding block context. |
| 76 |             Color32::RED | executable/support line; behavior impact depends on surrounding block context. |
| 77 |         }; | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 78 |         let a0 = START + t * SWEEP; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 79 |         let a1 = START + ((i + 1) as f32 / N as f32) * SWEEP; | executable/support line; behavior impact dependen
on surrounding block context. |
| 80 |         let p0 = center + Vec2::new(a0.cos() * r, a0.sin() * r); | executable/support line; behavior impact de
nds on surrounding block context. |
| 81 |         let p1 = center + Vec2::new(a1.cos() * r, a1.sin() * r); | executable/support line; behavior impact de
nds on surrounding block context. |
| 82 |         painter.line_segment([p0, p1], Stroke::new(5.5, seg)); | executable/support line; behavior impact depende
on surrounding block context. |
| 83 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 84 | (blank) | line is blank; visual separation only. |
| 85 |     // Warning tick mark | comment line; documentation intent, no runtime effect. |
| 86 |     if warn > min && warn < max { | conditional gate; changing condition shifts runtime path and side effects.
| 87 |         let wf = ((warn - min) / (max - min)) as f32; | executable/support line; behavior impact depends on su
rounding block context. |
| 88 |         let wa = START + wf * SWEEP; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 89 |         let ti: Pos2 = center + Vec2::new(wa.cos() * (r - 8.0), wa.sin() * (r - 8.0)); | executable/support li
behavior impact depends on surrounding block context. |
| 90 |         let to: Pos2 = center + Vec2::new(wa.cos() * (r + 4.0), wa.sin() * (r + 4.0)); | executable/support li
behavior impact depends on surrounding block context. |
| 91 |         painter.line_segment([ti, to], Stroke::new(2.0, Color32::YELLOW)); | executable/support line; behavior
impact depends on surrounding block context. |
| 92 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 93 |     if crit > min && crit < max { | conditional gate; changing condition shifts runtime path and side effects.
| 94 |         let cf = ((crit - min) / (max - min)) as f32; | executable/support line; behavior impact depends on su
rounding block context. |
| 95 |         let ca = START + cf * SWEEP; | executable/support line; behavior impact depends on surrounding block c
ext. |
| 96 |         let ti: Pos2 = center + Vec2::new(ca.cos() * (r - 8.0), ca.sin() * (r - 8.0)); | executable/support li
behavior impact depends on surrounding block context. |
| 97 |         let to: Pos2 = center + Vec2::new(ca.cos() * (r + 4.0), ca.sin() * (r + 4.0)); | executable/support li
behavior impact depends on surrounding block context. |
| 98 |         painter.line_segment([ti, to], Stroke::new(2.0, Color32::RED)); | executable/support line; behavior im
t depends on surrounding block context. |
| 99 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 100 | (blank) | line is blank; visual separation only. |
| 101 |     // Needle | comment line; documentation intent, no runtime effect. |
| 102 |     let na = START + frac * SWEEP; | executable/support line; behavior impact depends on surrounding block co
xt. |
| 103 |     let tip: Pos2 = center + Vec2::new(na.cos() * r * 0.80, na.sin() * r * 0.80); | executable/support line;
avior impact depends on surrounding block context. |
| 104 |     let perp_a = na + PI * 0.5; | executable/support line; behavior impact depends on surrounding block conte
|
| 105 |     let bl: Pos2 = center + Vec2::new(perp_a.cos() * 4.0, perp_a.sin() * 4.0); | executable/support line; beh

```

```

or impact depends on surrounding block context. |
| 106 |     let br: Pos2 = center - Vec2::new(perp_a.cos() * 4.0, perp_a.sin() * 4.0); | executable/support line; beh
or impact depends on surrounding block context. |
| 107 |     painter.add(Shape::convex_polygon( | executable/support line; behavior impact depends on surrounding block
ontext. |
| 108 |         vec![tip, bl, br], | executable/support line; behavior impact depends on surrounding block context. |
| 109 |         Color32::WHITE, | executable/support line; behavior impact depends on surrounding block context. |
| 110 |         Stroke::NONE, | executable/support line; behavior impact depends on surrounding block context. |
| 111 |     )); | executable/support line; behavior impact depends on surrounding block context. |
| 112 | (blank) | line is blank; visual separation only. |
| 113 |     // Center cap | comment line; documentation intent, no runtime effect. |
| 114 |     painter.circle_filled(center, 7.0, Color32::from_gray(68)); | executable/support line; behavior impact de
ds on surrounding block context. |
| 115 |     painter.circle_stroke(center, 7.0, Stroke::new(1.5, Color32::from_gray(120))); | executable/support line;
havior impact depends on surrounding block context. |
| 116 | (blank) | line is blank; visual separation only. |
| 117 |     // Value text | comment line; documentation intent, no runtime effect. |
| 118 |     painter.text( | executable/support line; behavior impact depends on surrounding block context. |
| 119 |         center + Vec2::new(0.0, r * 0.28), | executable/support line; behavior impact depends on surrounding
ck context. |
| 120 |         Align2::CENTER_CENTER, | executable/support line; behavior impact depends on surrounding block context
|
| 121 |         format!("{:.0}", value), | executable/support line; behavior impact depends on surrounding block context
. |
| 122 |         FontId::proportional(size * 0.165), | executable/support line; behavior impact depends on surrounding
ock context. |
| 123 |         value_color, | executable/support line; behavior impact depends on surrounding block context. |
| 124 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 125 |     // Unit | comment line; documentation intent, no runtime effect. |
| 126 |     painter.text( | executable/support line; behavior impact depends on surrounding block context. |
| 127 |         center + Vec2::new(0.0, r * 0.56), | executable/support line; behavior impact depends on surrounding
ck context. |
| 128 |         Align2::CENTER_CENTER, | executable/support line; behavior impact depends on surrounding block context
|
| 129 |         unit, | executable/support line; behavior impact depends on surrounding block context. |
| 130 |         FontId::proportional(size * 0.080), | executable/support line; behavior impact depends on surrounding
ock context. |
| 131 |         Color32::from_gray(155), | executable/support line; behavior impact depends on surrounding block context
. |
| 132 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 133 |     // Label | comment line; documentation intent, no runtime effect. |
| 134 |     painter.text( | executable/support line; behavior impact depends on surrounding block context. |
| 135 |         center + Vec2::new(0.0, r * 0.85), | executable/support line; behavior impact depends on surrounding
ck context. |
| 136 |         Align2::CENTER_CENTER, | executable/support line; behavior impact depends on surrounding block context
|
| 137 |         label, | executable/support line; behavior impact depends on surrounding block context. |
| 138 |         FontId::proportional(size * 0.095), | executable/support line; behavior impact depends on surrounding
ock context. |
| 139 |         Color32::from_gray(200), | executable/support line; behavior impact depends on surrounding block context
. |
| 140 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 141 | (blank) | line is blank; visual separation only. |
| 142 |     // Scale min/max | comment line; documentation intent, no runtime effect. |
| 143 |     let min_p: Pos2 = center + Vec2::new(START.cos() * (r + 14.0), START.sin() * (r + 14.0)); | executable/su
rt line; behavior impact depends on surrounding block context. |
| 144 |     let max_a = START + SWEEP; | executable/support line; behavior impact depends on surrounding block context
|
| 145 |     let max_p: Pos2 = center + Vec2::new(max_a.cos() * (r + 14.0), max_a.sin() * (r + 14.0)); | executable/su
rt line; behavior impact depends on surrounding block context. |
| 146 |     painter.text( | executable/support line; behavior impact depends on surrounding block context. |
| 147 |         min_p, | executable/support line; behavior impact depends on surrounding block context. |
| 148 |         Align2::CENTER_CENTER, | executable/support line; behavior impact depends on surrounding block context
|
| 149 |         format!("{:.0}", min), | executable/support line; behavior impact depends on surrounding block context

```

```
| 150 | FontId::proportional(size * 0.075), | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 151 | Color32::from_gray(90), | executable/support line; behavior impact depends on surrounding block conte  
|  
| 152 | ); | executable/support line; behavior impact depends on surrounding block context. |  
| 153 | painter.text( | executable/support line; behavior impact depends on surrounding block context. |  
| 154 | max_p, | executable/support line; behavior impact depends on surrounding block context. |  
| 155 | Align2::CENTER_CENTER, | executable/support line; behavior impact depends on surrounding block contex  
|  
| 156 | format!("{:.0}", max), | executable/support line; behavior impact depends on surrounding block contex  
|  
| 157 | FontId::proportional(size * 0.075), | executable/support line; behavior impact depends on surrounding  
ock context. |  
| 158 | Color32::from_gray(90), | executable/support line; behavior impact depends on surrounding block conte  
|  
| 159 | ); | executable/support line; behavior impact depends on surrounding block context. |  
| 160 | (blank) | line is blank; visual separation only. |  
| 161 | response | executable/support line; behavior impact depends on surrounding block context. |  
| 162 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 163 | (blank) | line is blank; visual separation only. |  
| 164 | // ■■ Bar Gauge ████████████████████████████████████████████████████████████████████████████ | comment  
ntent, no runtime effect. |  
| 165 | (blank) | line is blank; visual separation only. |  
| 166 | pub fn bar_gauge( | declares executable behavior; signature/body changes alter API, control flow, and outputs  
| 167 | ui: &mut Ui, | executable/support line; behavior impact depends on surrounding block context. |  
| 168 | label: &str, | executable/support line; behavior impact depends on surrounding block context. |  
| 169 | value: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 170 | max: f64, | executable/support line; behavior impact depends on surrounding block context. |  
| 171 | unit: &str, | executable/support line; behavior impact depends on surrounding block context. |  
| 172 | bar_width: f32, | executable/support line; behavior impact depends on surrounding block context. |  
| 173 | color: Color32, | executable/support line; behavior impact depends on surrounding block context. |  
| 174 | ) { | executable/support line; behavior impact depends on surrounding block context. |  
| 175 | let frac = (value / max.max(0.001)).clamp(0.0, 1.0) as f32; | executable/support line; behavior impact de  
ds on surrounding block context. |  
| 176 | ui.horizontal(\ui\ { | executable/support line; behavior impact depends on surrounding block context. |  
| 177 | ui.label( | executable/support line; behavior impact depends on surrounding block context. |  
| 178 | egui::RichText::new(format!("{:<14}", label)) | executable/support line; behavior impact depends  
surrounding block context. |  
| 179 | .size(11.0) | executable/support line; behavior impact depends on surrounding block context.  
| 180 | .monospace() | executable/support line; behavior impact depends on surrounding block context.  
| 181 | .color(Color32::from_gray(175)), | executable/support line; behavior impact depends on surround  
ng block context. |  
| 182 | ); | executable/support line; behavior impact depends on surrounding block context. |  
| 183 | let (rect, _) = ui.allocate_exact_size(Vec2::new(bar_width, 14.0), Sense::hover()); | executable/supp  
line; behavior impact depends on surrounding block context. |  
| 184 | let p = ui.painter_at(rect); | executable/support line; behavior impact depends on surrounding block  
text. |  
| 185 | p.rect_filled(rect, 3.0, Color32::from_gray(22)); | executable/support line; behavior impact depends  
surrounding block context. |  
| 186 | if frac > 0.0 { | conditional gate; changing condition shifts runtime path and side effects. |  
| 187 | let fill = Rect::from_min_size(rect.min, Vec2::new(rect.width() * frac, rect.height())); | execut  
e/support line; behavior impact depends on surrounding block context. |  
| 188 | p.rect_filled(fill, 3.0, color); | executable/support line; behavior impact depends on surroundin  
lock context. |  
| 189 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope  
|  
| 190 | p.rect_stroke(rect, 3.0, Stroke::new(0.7, Color32::from_gray(65))); | executable/support line; behavi  
impact depends on surrounding block context. |  
| 191 | ui.label( | executable/support line; behavior impact depends on surrounding block context. |  
| 192 | egui::RichText::new(format!("{:>8.1} {}", value, unit)) | executable/support line; behavior impac  
depends on surrounding block context. |  
| 193 | .size(11.0) | executable/support line; behavior impact depends on surrounding block context.  
| 194 | .monospace() | executable/support line; behavior impact depends on surrounding block context.  
| 195 | .color(color), | executable/support line; behavior impact depends on surrounding block conte
```

```
| 196 |         ); | executable/support line; behavior impact depends on surrounding block context. |  
| 197 |     }); | executable/support line; behavior impact depends on surrounding block context. |  
| 198 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 199 | (blank) | line is blank; visual separation only. |  
| 200 | // ■■ Warning Lamp (LED style) ████████████████████████████████████████████████████ | comment line  
intent, no runtime effect. |  
| 201 | (blank) | line is blank; visual separation only. |  
| 202 | /// Returns Response so caller can check `.clicked()` or `.hovered()`. | comment line; documentation intent,  
runtime effect. |  
| 203 | pub fn warning_lamp(ui: &ampmut Ui, active: bool, label: &ampstr, color: Color32) -> Response { | declares executab  
behavior; signature/body changes alter API, control flow, and outputs. |  
| 204 |     let (rect, response) = ui.allocate_exact_size(Vec2::new(68.0, 48.0), Sense::click()); | executable/support  
ine; behavior impact depends on surrounding block context. |  
| 205 |     if !ui.is_rect_visible(rect) { | conditional gate; changing condition shifts runtime path and side effects  
|  
| 206 |         return response; | executable/support line; behavior impact depends on surrounding block context. |  
| 207 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 208 |     let p = ui.painter_at(rect); | executable/support line; behavior impact depends on surrounding block cont  
. |  
| 209 |     let lc = Pos2::new(rect.center().x, rect.top() + 16.0); | executable/support line; behavior impact depend  
n surrounding block context. |  
| 210 |     let lr = 11.0_f32; | executable/support line; behavior impact depends on surrounding block context. |  
| 211 | (blank) | line is blank; visual separation only. |  
| 212 |     if active { | conditional gate; changing condition shifts runtime path and side effects. |  
| 213 |         let gc = Color32::from_rgba_unmultiplied(color.r(), color.g(), color.b(), 38); | executable/support l  
; behavior impact depends on surrounding block context. |  
| 214 |         p.circle_filled(lc, lr + 6.0, gc); | executable/support line; behavior impact depends on surrounding  
ck context. |  
| 215 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 216 |     p.circle_filled( | executable/support line; behavior impact depends on surrounding block context. |  
| 217 |         lc, | executable/support line; behavior impact depends on surrounding block context. |  
| 218 |         lr, | executable/support line; behavior impact depends on surrounding block context. |  
| 219 |         if active { | conditional gate; changing condition shifts runtime path and side effects. |  
| 220 |             color | executable/support line; behavior impact depends on surrounding block context. |  
| 221 |         } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 222 |             Color32::from_gray(28) | executable/support line; behavior impact depends on surrounding block co  
xt. |  
| 223 |     }, | executable/support line; behavior impact depends on surrounding block context. |  
| 224 | ); | executable/support line; behavior impact depends on surrounding block context. |  
| 225 | p.circle_stroke( | executable/support line; behavior impact depends on surrounding block context. |  
| 226 |     lc, | executable/support line; behavior impact depends on surrounding block context. |  
| 227 |     lr, | executable/support line; behavior impact depends on surrounding block context. |  
| 228 |     Stroke::new( | executable/support line; behavior impact depends on surrounding block context. |  
| 229 |         1.0, | executable/support line; behavior impact depends on surrounding block context. |  
| 230 |         if active { | conditional gate; changing condition shifts runtime path and side effects. |  
| 231 |             color | executable/support line; behavior impact depends on surrounding block context. |  
| 232 |         } else { | executable/support line; behavior impact depends on surrounding block context. |  
| 233 |             Color32::from_gray(55) | executable/support line; behavior impact depends on surrounding bloc  
ontext. |  
| 234 |         }, | executable/support line; behavior impact depends on surrounding block context. |  
| 235 |     ), | executable/support line; behavior impact depends on surrounding block context. |  
| 236 | ); | executable/support line; behavior impact depends on surrounding block context. |  
| 237 | if active { | conditional gate; changing condition shifts runtime path and side effects. |  
| 238 |     p.circle_filled( | executable/support line; behavior impact depends on surrounding block context. |  
| 239 |         lc - Vec2::new(3.0, 3.0), | executable/support line; behavior impact depends on surrounding block  
ntext. |  
| 240 |         3.5, | executable/support line; behavior impact depends on surrounding block context. |  
| 241 |         Color32::from_rgba_unmultiplied(255, 255, 255, 70), | executable/support line; behavior impact de  
ds on surrounding block context. |  
| 242 |     ); | executable/support line; behavior impact depends on surrounding block context. |  
| 243 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |  
| 244 | let lbl_col = if active { | executable/support line; behavior impact depends on surrounding block context  
| 245 |     color | executable/support line; behavior impact depends on surrounding block context. |  
| 246 | } else { | executable/support line; behavior impact depends on surrounding block context. |
```

[illegible]

```

| 296 |          Color32::from_rgb(220, 80, 80), | executable/support line; behavior impact depends on surrounding
g block context. |
| 297 |          Color32::from_gray(50), | executable/support line; behavior impact depends on surrounding blo
context. |
| 298 |      ), | executable/support line; behavior impact depends on surrounding block context. |
| 299 |      ("N", 'N', Color32::YELLOW, Color32::from_gray(50)), | executable/support line; behavior impact d
nds on surrounding block context. |
| 300 |      ( | executable/support line; behavior impact depends on surrounding block context. |
| 301 |          "F", | executable/support line; behavior impact depends on surrounding block context. |
| 302 |          'F', | executable/support line; behavior impact depends on surrounding block context. |
| 303 |          Color32::from_rgb(80, 210, 80), | executable/support line; behavior impact depends on surround
g block context. |
| 304 |          Color32::from_gray(50), | executable/support line; behavior impact depends on surrounding blo
context. |
| 305 |      ), | executable/support line; behavior impact depends on surrounding block context. |
| 306 |      ] { | executable/support line; behavior impact depends on surrounding block context. |
| 307 |          let active = current == label; | executable/support line; behavior impact depends on surrounding
ck context. |
| 308 |          let fill = if active { color_on } else { color_off }; | executable/support line; behavior impact
ends on surrounding block context. |
| 309 |          let btn = egui::Button::new( | executable/support line; behavior impact depends on surrounding bl
context. |
| 310 |              egui::RichText::new(label) | executable/support line; behavior impact depends on surrounding
ck context. |
| 311 |              .size(16.0) | executable/support line; behavior impact depends on surrounding block conte
|
| 312 |              .color(if active { | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 313 |                  Color32::BLACK | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 314 |              } else { | executable/support line; behavior impact depends on surrounding block context.
| 315 |                  Color32::GRAY | executable/support line; behavior impact depends on surrounding block
ntext. |
| 316 |              }) | executable/support line; behavior impact depends on surrounding block context. |
| 317 |              .strong(), | executable/support line; behavior impact depends on surrounding block contex
|
| 318 |          ) | executable/support line; behavior impact depends on surrounding block context. |
| 319 |          .min_size(Vec2::new(36.0, 36.0)) | executable/support line; behavior impact depends on surroundin
lock context. |
| 320 |          .fill(fill); | executable/support line; behavior impact depends on surrounding block context. |
| 321 |          if ui.add(btn).clicked() { | conditional gate; changing condition shifts runtime path and side ef
ts. |
| 322 |              clicked = Some(key); | executable/support line; behavior impact depends on surrounding block
text. |
| 323 |          } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime s
es. |
| 324 |      } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scope
|
| 325 |      }); | executable/support line; behavior impact depends on surrounding block context. |
| 326 |      clicked | executable/support line; behavior impact depends on surrounding block context. |
| 327 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: tests/io_mock.rs

Role: Automated regression and integration validation surface.

Lines: 208 | Non-empty: 183 | Symbols: 9

Symbol Inventory

Line	Kind	Symbol	Change Impact
4	module	io	data/structure coupling
10	fn	write_test_allowlist	signature/API coupling
25	fn	hw_config_parsing_cli_flags	signature/API coupling
75	fn	mock_adapter_blocks_write_by_default_and_reads_injected_frame	signature/API coupling

```
| 97 | fn | mock_adapter_allows_write_when_policy_satisfied | signature/API coupling |
| 125 | fn | mock_adapter_dry_run_logs_without_physical_tx | signature/API coupling |
| 157 | fn | mock_adapter_rate_limit_blocks_burst | signature/API coupling |
| 189 | fn | allowlist_parse_mask_accepts_hex | signature/API coupling |
| 195 | fn | replay_can_record_keeps_explicit_timestamp | signature/API coupling |
```

Line-by-Line Impact Map

```
| Line | Source | Operational Impact If Changed |
|---:|---|---|
| 1 | #[allow(dead_code)] | executable/support line; behavior impact depends on surrounding block context. |
| 2 | (blank) | line is blank; visual separation only. |
| 3 | #[path = "../src/io/mod.rs"] | executable/support line; behavior impact depends on surrounding block context. |
| 4 | mod io; | declares a module boundary; changing module name/path breaks references. |
| 5 | (blank) | line is blank; visual separation only. |
| 6 | use io::hw::{CanFrame, Frame, HardwareInterface, HwConfig, HwError, HwMode}; | imports symbols; changing path/n affects compile-time resolution. |
| 7 | use io::mock::MockAdapter; | imports symbols; changing path/name affects compile-time resolution. |
| 8 | use std::fs; | imports symbols; changing path/name affects compile-time resolution. |
| 9 | (blank) | line is blank; visual separation only. |
| 10 | fn write_test_allowlist(path: &std::path::Path) { | declares executable behavior; signature/body changes alter I, control flow, and outputs. |
| 11 |     let content = r#"[ | executable/support line; behavior impact depends on surrounding block context. |
| 12 |     { | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 13 |         "type": "can", | executable/support line; behavior impact depends on surrounding block context. |
| 14 |         "id": 419385573, | executable/support line; behavior impact depends on surrounding block context. |
| 15 |         "mask": "0x1FFFFFFF", | executable/support line; behavior impact depends on surrounding block context. |
| 16 |         "allowed_bytes": [[0, 8]], | executable/support line; behavior impact depends on surrounding block context. |
| 17 |         "max_rate_per_sec": 10, | executable/support line; behavior impact depends on surrounding block context. |
| 18 |         "description": "allow test can id" | executable/support line; behavior impact depends on surrounding block context. |
| 19 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 20 | ]"#; | executable/support line; behavior impact depends on surrounding block context. |
| 21 |     fs::write(path, content).expect("allowlist write"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 22 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 23 | (blank) | line is blank; visual separation only. |
| 24 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 25 | fn hw_config_pares_cli_flags() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 26 |     let args = vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 27 |         "simubench".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 28 |         "--hw-mode=live".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 29 |         "--dry-run".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 30 |         "--allowlist=./allowlist.json".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 31 |         "--vendor-name=cat_comm".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 32 |         "--vendor-template-dir=./vendor_template".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 33 |         "--vendor-bridge-exe=./vendor_template/cat_comm_bridge.exe".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 34 |         "--vendor-bridge-timeout-ms=2200".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 35 |         "--enable-write".to_string(), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 36 |         "--noninteractive-approved".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 37 |         "--uds-retries=5".to_string(), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
```

```

| 38 |         "--uds-timeout-p2-ms=1300".to_string(), | I/O or protocol-critical logic; edits can affect integration
iming, and diagnostic outcomes. |
| 39 |         "--uds-timeout-p2star-ms=4500".to_string(), | I/O or protocol-critical logic; edits can affect integra
n, timing, and diagnostic outcomes. |
| 40 |         "--uds-st-min-ms=10".to_string(), | I/O or protocol-critical logic; edits can affect integration, timi
and diagnostic outcomes. |
| 41 |         "--uds-block-size=8".to_string(), | I/O or protocol-critical logic; edits can affect integration, timi
and diagnostic outcomes. |
| 42 |         "--j1939-idle-guard-ms=3000".to_string(), | executable/support line; behavior impact depends on surrou
ng block context. |
| 43 |         "--keep-channels-alive=false".to_string(), | executable/support line; behavior impact depends on surro
ing block context. |
| 44 |     ]; | executable/support line; behavior impact depends on surrounding block context. |
| 45 | (blank) | line is blank; visual separation only. |
| 46 |     let cfg = HwConfig::from_cli_args(&args).expect("valid args"); | panic-prone operation; changing/removing
ects crash behavior and robustness. |
| 47 |     assert_eq!(cfg.mode, HwMode::Live); | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 48 |     assert!(cfg.dry_run); | executable/support line; behavior impact depends on surrounding block context. |
| 49 |     assert_eq!(cfg.vendor_name.as_deref(), Some("cat_comm")); | executable/support line; behavior impact depen
on surrounding block context. |
| 50 |     assert_eq!( | executable/support line; behavior impact depends on surrounding block context. |
| 51 |         cfg.vendor_template_dir.as_deref(), | executable/support line; behavior impact depends on surrounding
ck context. |
| 52 |         Some(std::path::Path::new("./vendor_template")) | executable/support line; behavior impact depends on
rounding block context. |
| 53 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 54 |     assert_eq!( | executable/support line; behavior impact depends on surrounding block context. |
| 55 |         cfg.vendor_bridge_exe.as_deref(), | executable/support line; behavior impact depends on surrounding bl
context. |
| 56 |         Some(std::path::Path::new("./vendor_template/cat_comm_bridge.exe")) | executable/support line; behavio
mpact depends on surrounding block context. |
| 57 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 58 |     assert_eq!(cfg.vendor_bridge_timeout_ms, 2200); | executable/support line; behavior impact depends on surr
ding block context. |
| 59 |     assert_eq!( | executable/support line; behavior impact depends on surrounding block context. |
| 60 |         cfg.allowlist_path.as_deref(), | executable/support line; behavior impact depends on surrounding block
ntext. |
| 61 |         Some(std::path::Path::new("./allowlist.json")) | executable/support line; behavior impact depends on s
ounding block context. |
| 62 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 63 |     // dry-run must keep writes disabled effectively | comment line; documentation intent, no runtime effect.
| 64 |     assert!(!cfg.write_effectively_enabled()); | I/O or protocol-critical logic; edits can affect integration,
ming, and diagnostic outcomes. |
| 65 |     assert_eq!(cfg.uds_retry_count, 5); | I/O or protocol-critical logic; edits can affect integration, timing
nd diagnostic outcomes. |
| 66 |     assert_eq!(cfg.uds_timeout_p2_ms, 1300); | I/O or protocol-critical logic; edits can affect integration, t
ng, and diagnostic outcomes. |
| 67 |     assert_eq!(cfg.uds_timeout_p2star_ms, 4500); | I/O or protocol-critical logic; edits can affect integratio
timing, and diagnostic outcomes. |
| 68 |     assert_eq!(cfg.uds_st_min_ms, 10); | I/O or protocol-critical logic; edits can affect integration, timing,
d diagnostic outcomes. |
| 69 |     assert_eq!(cfg.uds_block_size, 8); | I/O or protocol-critical logic; edits can affect integration, timing,
d diagnostic outcomes. |
| 70 |     assert_eq!(cfg.j1939_idle_guard_ms, 3000); | executable/support line; behavior impact depends on surroundi
block context. |
| 71 |     assert!(!cfg.keep_channels_alive); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 72 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 73 | (blank) | line is blank; visual separation only. |
| 74 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 75 | fn mock_adapter_blocks_write_by_default_and_reads_injected_frame() { | declares executable behavior; signature
dy changes alter API, control flow, and outputs. |
| 76 |     let cfg = HwConfig::default(); | executable/support line; behavior impact depends on surrounding block con
t. |

```



```

| 77 | let mut ad = MockAdapter::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 78 | ad.init(&cfg).expect("init ok"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 79 | (blank) | line is blank; visual separation only. |
| 80 | let f = Frame::Can(CanFrame { | executable/support line; behavior impact depends on surrounding block context. |
| 81 |     id: 0x18FF50E5, | executable/support line; behavior impact depends on surrounding block context. |
| 82 |     dlc: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 83 |     data: [1, 2, 3, 4, 5, 6, 7, 8], | executable/support line; behavior impact depends on surrounding block context. |
| 84 |     len: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 85 |     timestamp_ms: Some(1), | executable/support line; behavior impact depends on surrounding block context. |
| 86 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 87 | (blank) | line is blank; visual separation only. |
| 88 | let err = ad.send_frame(f.clone()).expect_err("write must be blocked"); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 89 | assert!(matches!(err, HwError::WriteBlockedAllowlist)); | executable/support line; behavior impact depends on surrounding block context. |
| 90 | (blank) | line is blank; visual separation only. |
| 91 | ad.inject_rx(f.clone()); | executable/support line; behavior impact depends on surrounding block context. |
| 92 | let got = ad.read_frame().expect("read injected frame"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 93 | assert_eq!(got, f); | executable/support line; behavior impact depends on surrounding block context. |
| 94 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 95 | (blank) | line is blank; visual separation only. |
| 96 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 97 | fn mock_adapter_allows_write_when_policy_satisfied() { | declares executable behavior; signature/body changes error API, control flow, and outputs. |
| 98 |     let temp = std::env::temp_dir().join("allowlist_io_mock.json"); | executable/support line; behavior impact depends on surrounding block context. |
| 99 |     write_test_allowlist(&temp); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 100 | (blank) | line is blank; visual separation only. |
| 101 | let args = vec!["simubench".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 102 |     "--enable-write".to_string(), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 103 |     "--allowlist={}".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 104 |     "--noninteractive-approved".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 105 |     "--dry-run=false".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 106 | ]; | executable/support line; behavior impact depends on surrounding block context. |
| 107 | let cfg = HwConfig::from_cli_args(&args).expect("valid args"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 108 | assert!(cfg.write_effectively_enabled()); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 109 | (blank) | line is blank; visual separation only. |
| 110 | let mut ad = MockAdapter::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 111 | ad.init(&cfg).expect("init ok"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 112 | let f = Frame::Can(CanFrame { | executable/support line; behavior impact depends on surrounding block context. |
| 113 |     id: 0x18FF50E5, | executable/support line; behavior impact depends on surrounding block context. |
| 114 |     dlc: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 115 |     data: [0, 8], | executable/support line; behavior impact depends on surrounding block context. |
| 116 |     len: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 117 |     timestamp_ms: None, | executable/support line; behavior impact depends on surrounding block context. |
| 118 | }); | executable/support line; behavior impact depends on surrounding block context. |
| 119 | ad.send_frame(f).expect("write allowed"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 120 |

```

```

| 121 |     assert_eq!(ad.tx_queue.len(), 1); | executable/support line; behavior impact depends on surrounding block context. |
| 122 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 123 | (blank) | line is blank; visual separation only. |
| 124 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 125 | fn mock_adapter_dry_run_logs_without_physical_tx() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 126 |     let temp_allow = std::env::temp_dir().join("allowlist_io_mock_dry_run.json"); | executable/support line; behavior impact depends on surrounding block context. |
| 127 |     let temp_log_dir = std::env::temp_dir().join("autobreaking_hw_logs_dry_run"); | executable/support line; behavior impact depends on surrounding block context. |
| 128 |     write_test_allowlist(&temp_allow); | I/O or protocol-critical logic; edits can affect integration, timing and diagnostic outcomes. |
| 129 | (blank) | line is blank; visual separation only. |
| 130 |     let args = vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 131 |         "simubench".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 132 |         "--enable-write".to_string(), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 133 |         format!("--allowlist={}", temp_allow.display()), | executable/support line; behavior impact depends on surrounding block context. |
| 134 |         "--noninteractive-approved".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 135 |         "--dry-run".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 136 |         format!("--log-dir={}", temp_log_dir.display()), | executable/support line; behavior impact depends on surrounding block context. |
| 137 |     ]; | executable/support line; behavior impact depends on surrounding block context. |
| 138 |     let cfg = HwConfig::from_cli_args(&args).expect("valid args"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 139 |     assert!(!cfg.write_effectively_enabled()); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 140 |     assert!(cfg.write_intent_enabled()); | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 141 | (blank) | line is blank; visual separation only. |
| 142 |     let mut ad = MockAdapter::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 143 |     ad.init(&cfg).expect("init ok"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 144 |     let f = Frame::Can(CanFrame { | executable/support line; behavior impact depends on surrounding block context. |
| 145 |         id: 0x18FF50E5, | executable/support line; behavior impact depends on surrounding block context. |
| 146 |         dlc: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 147 |         data: [0; 8], | executable/support line; behavior impact depends on surrounding block context. |
| 148 |         len: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 149 |         timestamp_ms: None, | executable/support line; behavior impact depends on surrounding block context. |
| 150 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 151 | (blank) | line is blank; visual separation only. |
| 152 |     ad.send_frame(f).expect("dry-run should allow intent"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 153 |     assert_eq!(ad.tx_queue.len(), 0); | executable/support line; behavior impact depends on surrounding block context. |
| 154 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 155 | (blank) | line is blank; visual separation only. |
| 156 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 157 | fn mock_adapter_rate_limit_blocks_burst() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 158 |     let temp_allow = std::env::temp_dir().join("allowlist_io_mock_rate_limit.json"); | executable/support line; behavior impact depends on surrounding block context. |
| 159 |     write_test_allowlist(&temp_allow); | I/O or protocol-critical logic; edits can affect integration, timing and diagnostic outcomes. |
| 160 | (blank) | line is blank; visual separation only. |
| 161 |     let args = vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 162 |         "simubench".to_string(), | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 163 |         "--enable-write".to_string(), | I/O or protocol-critical logic; edits can affect integration, timing,
d diagnostic outcomes. |
| 164 |         format!("--allowlist={}", temp_allow.display()), | executable/support line; behavior impact depends on
urrounding block context. |
| 165 |         "--noninteractive-approved".to_string(), | executable/support line; behavior impact depends on surroun
ng block context. |
| 166 |         "--dry-run=false".to_string(), | executable/support line; behavior impact depends on surrounding bloc
ontext. |
| 167 |         "--rate-limit-global=1".to_string(), | executable/support line; behavior impact depends on surroundin
lock context. |
| 168 |         "--rate-limit-per-id=1".to_string(), | executable/support line; behavior impact depends on surroundin
lock context. |
| 169 |     ]; | executable/support line; behavior impact depends on surrounding block context. |
| 170 |     let cfg = HwConfig::from_cli_args(&args).expect("valid args"); | panic-prone operation; changing/removing
ffects crash behavior and robustness. |
| 171 | (blank) | line is blank; visual separation only. |
| 172 |     let mut ad = MockAdapter::new(); | executable/support line; behavior impact depends on surrounding block
text. |
| 173 |     ad.init(&cfg).expect("init ok"); | panic-prone operation; changing/removing affects crash behavior and ro
tness. |
| 174 | (blank) | line is blank; visual separation only. |
| 175 |     let f = Frame::Can(CanFrame { | executable/support line; behavior impact depends on surrounding block con
t. |
| 176 |         id: 0x18FF50E5, | executable/support line; behavior impact depends on surrounding block context. |
| 177 |         dlc: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 178 |         data: [0; 8], | executable/support line; behavior impact depends on surrounding block context. |
| 179 |         len: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 180 |         timestamp_ms: None, | executable/support line; behavior impact depends on surrounding block context.
| 181 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 182 | (blank) | line is blank; visual separation only. |
| 183 |     ad.send_frame(f.clone()).expect("first send ok"); | panic-prone operation; changing/removing affects cras
ehavior and robustness. |
| 184 |     let err = ad.send_frame(f).expect_err("second send should rate-limit"); | I/O or protocol-critical logic;
its can affect integration, timing, and diagnostic outcomes. |
| 185 |     assert!(matches!(err, HwError::RateLimited)); | executable/support line; behavior impact depends on surro
ing block context. |
| 186 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 187 | (blank) | line is blank; visual separation only. |
| 188 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 189 | fn allowlist_parse_mask_accepts_hex() { | declares executable behavior; signature/body changes alter API, con
l flow, and outputs. |
| 190 |     let mask = io::allowlist::parse_mask("0x1FFFFFFF").expect("mask parse"); | panic-prone operation; changin
emoving affects crash behavior and robustness. |
| 191 |     assert_eq!(mask, 0x1FFF_FFFF); | executable/support line; behavior impact depends on surrounding block co
xt. |
| 192 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 193 | (blank) | line is blank; visual separation only. |
| 194 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 195 | fn replay_can_record_keeps_explicit_timestamp() { | declares executable behavior; signature/body changes alte
PI, control flow, and outputs. |
| 196 |     let f = Frame::Can(CanFrame { | executable/support line; behavior impact depends on surrounding block con
t. |
| 197 |         id: 0x18FF50E5, | executable/support line; behavior impact depends on surrounding block context. |
| 198 |         dlc: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 199 |         data: [1, 2, 3, 4, 5, 6, 7, 8], | executable/support line; behavior impact depends on surrounding blo
context. |
| 200 |         len: 8, | executable/support line; behavior impact depends on surrounding block context. |
| 201 |         timestamp_ms: Some(999), | executable/support line; behavior impact depends on surrounding block cont
. |
| 202 |     }); | executable/support line; behavior impact depends on surrounding block context. |
| 203 | (blank) | line is blank; visual separation only. |
| 204 |     let rec = io::replay::frame_to_record(&f, "rx", None, None); | executable/support line; behavior impact d
nds on surrounding block context. |
| 205 |     assert_eq!(rec.ts, 999); | executable/support line; behavior impact depends on surrounding block context.
| 206 |     assert!(rec.raw_hex.is_none()); | executable/support line; behavior impact depends on surrounding block c

```

```

ext. |
| 207 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 208 | (blank) | line is blank; visual separation only. |

## File: tests/leak_system_integration.rs

Role: Automated regression and integration validation surface.
Lines: 84 | Non-empty: 74 | Symbols: 2

### Symbol Inventory

| Line | Kind | Symbol | Change Impact |
|---:|---|---|---|
| 4 | fn | e2e_hydraulic_plus_ac_stress_generates_reports | signature/API coupling |
| 75 | fn | monte_carlo_distribution_has_valid_bounds | signature/API coupling |

### Line-by-Line Impact Map

| Line | Source | Operational Impact If Changed |
|---:|---|---|
| 1 | use auto_breaking::{HeavyMachinery, ManualCircuitParams, OilType}; | imports symbols; changing path/name affects compile-time resolution. |
| 2 | (blank) | line is blank; visual separation only. |
| 3 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 4 | fn e2e_hydraulic_plus_ac_stress_generates_reports() { | declares executable behavior; signature/body changes API, control flow, and outputs. |
| 5 |     let mut bench = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 6 | (blank) | line is blank; visual separation only. |
| 7 |     bench.apply_leak_manual_params( | executable/support line; behavior impact depends on surrounding block context. |
| 8 |         "HYD_MAIN", | executable/support line; behavior impact depends on surrounding block context. |
| 9 |         ManualCircuitParams { | executable/support line; behavior impact depends on surrounding block context. |
| 10 |             oil_type: Some(OilType::HydraulicIso68), | executable/support line; behavior impact depends on surrounding block context. |
| 11 |             piston_pressure_bar: Some(210.0), | executable/support line; behavior impact depends on surrounding block context. |
| 12 |             operation_pressure_bar: Some(180.0), | executable/support line; behavior impact depends on surrounding block context. |
| 13 |             pressure_min_bar: Some(45.0), | executable/support line; behavior impact depends on surrounding block context. |
| 14 |             pressure_mean_bar: Some(155.0), | executable/support line; behavior impact depends on surrounding block context. |
| 15 |             pressure_ideal_bar: Some(175.0), | executable/support line; behavior impact depends on surrounding block context. |
| 16 |             pressure_max_bar: Some(230.0), | executable/support line; behavior impact depends on surrounding block context. |
| 17 |             pressure_rupture_bar: Some(280.0), | executable/support line; behavior impact depends on surrounding block context. |
| 18 |             oring_squeeze_pct: Some(19.0), | executable/support line; behavior impact depends on surrounding block context. |
| 19 |             compression_set_pct: Some(14.0), | executable/support line; behavior impact depends on surrounding block context. |
| 20 |             base_leak_area_mm2: Some(0.0009), | executable/support line; behavior impact depends on surrounding block context. |
| 21 |             max_supported_temp_c: Some(115.0), | executable/support line; behavior impact depends on surrounding block context. |
| 22 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 23 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 24 | (blank) | line is blank; visual separation only. |
| 25 |     bench.bcm.hvac_on = true; | executable/support line; behavior impact depends on surrounding block context. |
| 26 |     bench.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 27 |     bench.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 28 |     bench.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 29 | (blank) | line is blank; visual separation only. |

```

```

| 30 |     for i in 0..900 { | iteration control; changing bounds/body affects throughput, timing, and possibly safety
uarantees. |
| 31 |         bench.throttle_pct = if i % 120 < 60 { 72.0 } else { 48.0 }; | executable/support line; behavior impact
depends on surrounding block context. |
| 32 |         bench.brake_pct = if i % 180 > 150 { 22.0 } else { 0.0 }; | executable/support line; behavior impact d
nds on surrounding block context. |
| 33 |         bench.loader_lift_cmd = if i % 90 < 45 { 0.8 } else { -0.6 }; | executable/support line; behavior impa
depends on surrounding block context. |
| 34 |         bench.loader_tilt_cmd = if i % 70 < 35 { 0.5 } else { -0.4 }; | executable/support line; behavior impa
depends on surrounding block context. |
| 35 |         bench.hitch_joystick = if i % 80 < 40 { 0.7 } else { -0.3 }; | executable/support line; behavior impac
depends on surrounding block context. |
| 36 |         bench.tick(1.0 / 60.0); | executable/support line; behavior impact depends on surrounding block context
|
| 37 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 38 | (blank) | line is blank; visual separation only. |
| 39 |     assert(!bench.leak_reports.is_empty()); | executable/support line; behavior impact depends on surrounding
ock context. |
| 40 | (blank) | line is blank; visual separation only. |
| 41 |     let pred = bench.predict_leak_scenarios(300.0, 0.05); | executable/support line; behavior impact depends o
urrounding block context. |
| 42 |     assert(!pred.is_empty()); | executable/support line; behavior impact depends on surrounding block context
| 43 | (blank) | line is blank; visual separation only. |
| 44 |     let root = std::env::temp_dir().join("autobreaking_e2e"); | executable/support line; behavior impact depen
on surrounding block context. |
| 45 |     bench | executable/support line; behavior impact depends on surrounding block context. |
| 46 |         .export_leak_report_json(root.join("runtime_report.json")) | executable/support line; behavior impact
ends on surrounding block context. |
| 47 |         .expect("runtime json export"); | panic-prone operation; changing/removing affects crash behavior and
ustness. |
| 48 |     bench | executable/support line; behavior impact depends on surrounding block context. |
| 49 |         .export_leak_report_csv(root.join("runtime_report.csv")) | executable/support line; behavior impact de
ds on surrounding block context. |
| 50 |         .expect("runtime csv export"); | panic-prone operation; changing/removing affects crash behavior and r
stness. |
| 51 |     bench | executable/support line; behavior impact depends on surrounding block context. |
| 52 |         .export_leak_predictions_json(root.join("pred_report.json"), &pred) | executable/support line; behavior
mpact depends on surrounding block context. |
| 53 |         .expect("pred json export"); | panic-prone operation; changing/removing affects crash behavior and rob
ness. |
| 54 |     bench | executable/support line; behavior impact depends on surrounding block context. |
| 55 |         .export_leak_predictions_csv(root.join("pred_report.csv"), &pred) | executable/support line; behavior
act depends on surrounding block context. |
| 56 |         .expect("pred csv export"); | panic-prone operation; changing/removing affects crash behavior and robu
ess. |
| 57 |     bench | executable/support line; behavior impact depends on surrounding block context. |
| 58 |         .export_leak_material_catalog_json(root.join("materials_catalog.json")) | executable/support line; beh
or impact depends on surrounding block context. |
| 59 |         .expect("materials catalog json export"); | panic-prone operation; changing/removing affects crash beh
or and robustness. |
| 60 |     bench | executable/support line; behavior impact depends on surrounding block context. |
| 61 |         .export_leak_oil_catalog_csv(root.join("oils_catalog.csv")) | executable/support line; behavior impact
pend on surrounding block context. |
| 62 |         .expect("oils catalog csv export"); | panic-prone operation; changing/removing affects crash behavior
robustness. |
| 63 | (blank) | line is blank; visual separation only. |
| 64 |     let hyd = bench | executable/support line; behavior impact depends on surrounding block context. |
| 65 |         .leak_reports | executable/support line; behavior impact depends on surrounding block context. |
| 66 |         .iter() | executable/support line; behavior impact depends on surrounding block context. |
| 67 |         .find(|r| r.name == "HYD_MAIN") | executable/support line; behavior impact depends on surrounding bl
context. |
| 68 |         .expect("HYD_MAIN report exists"); | panic-prone operation; changing/removing affects crash behavior a
robustness. |
| 69 |     assert!(hyd.recommended_hold_max_bar > hyd.recommended_hold_min_bar); | executable/support line; behavior
act depends on surrounding block context. |

```

```

| 70 |      assert!(!hyd.rca_hint.is_empty()); | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 71 |      assert!(!hyd.pca_recommendation.is_empty()); | executable/support line; behavior impact depends on surround
g block context. |
| 72 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 73 | (blank) | line is blank; visual separation only. |
| 74 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 75 | fn monte_carlo_distribution_has_valid_bounds() { | declares executable behavior; signature/body changes alter
, control flow, and outputs. |
| 76 |     let bench = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding block
ncontext. |
| 77 |     let sims = bench.monte_carlo_leak_predictions(40, 180.0, 0.05, 30.0); | executable/support line; behavior
act depends on surrounding block context. |
| 78 | (blank) | line is blank; visual separation only. |
| 79 |     assert!(!sims.is_empty()); | executable/support line; behavior impact depends on surrounding block context
| 80 |     for s in sims.iter().take(20) { | iteration control; changing bounds/body affects throughput, timing, and
sibly safety guarantees. |
| 81 |         assert!((0.0..=100.0).contains(&s.final_rupture_probability_pct)); | executable/support line; behavior
pact depends on surrounding block context. |
| 82 |         assert!(s.peak_pressure_bar >= 0.0); | executable/support line; behavior impact depends on surrounding
ock context. |
| 83 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 84 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: tests/property_invariants.rs

Role: Automated regression and integration validation surface.

Lines: 38 | Non-empty: 33 | Symbols: 2

Symbol Inventory

Line	Kind	Symbol	Change Impact
6	fn	physical_levels_stay_within_bounds	signature/API coupling
26	fn	can_health_score_is_always_bounded	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	use auto_breaking::HeavyMachinery;	imports symbols; changing path/name affects compile-time resolution.
2	use proptest::prelude::*;	imports symbols; changing path/name affects compile-time resolution.
3	(blank)	line is blank; visual separation only.
4	proptest! {	executable/support line; behavior impact depends on surrounding block context.
5	#[test]	executable/support line; behavior impact depends on surrounding block context.
6	fn physical_levels_stay_within_bounds(throttle in 0.0f64..100.0, brake in 0.0f64..100.0, steps in 1usize..1	size) { declares executable behavior; signature/body changes alter API, control flow, and outputs.
7	let mut sim = HeavyMachinery::new();	executable/support line; behavior impact depends on surrounding block context.
8	sim.key_advance();	executable/support line; behavior impact depends on surrounding block context.
9	sim.key_advance();	executable/support line; behavior impact depends on surrounding block context.
10	sim.key_advance();	executable/support line; behavior impact depends on surrounding block context.
11	(blank)	line is blank; visual separation only.
12	for _ in 0..steps {	iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees.
13	sim.throttle_pct = throttle;	executable/support line; behavior impact depends on surrounding block context.
14	sim.brake_pct = brake;	executable/support line; behavior impact depends on surrounding block context.
15	sim.tick(1.0 / 60.0);	executable/support line; behavior impact depends on surrounding block context.
16	(blank)	line is blank; visual separation only.
17	prop_assert!((0.0..=100.0).contains(&sim.ecm.fuel_level_pct));	executable/support line; behavior impact depends on surrounding block context.
18	prop_assert!((0.0..=100.0).contains(&sim.ecm.def_level_pct));	executable/support line; behavior impact depends on surrounding block context.

```

act depends on surrounding block context. |
| 19 |         prop_assert!((0.0..=100.0).contains(&sim.hcm.fluid_level_pct)); | executable/support line; behavior impact depends on surrounding block context. |
| 20 |         prop_assert!(sim.hcm.system_pressure_bar >= 0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 21 |         prop_assert!(sim.ecm.oil_pressure_kpa >= 0.0); | executable/support line; behavior impact depends on surrounding block context. |
| 22 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 23 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 24 | (blank) | line is blank; visual separation only. |
| 25 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 26 | fn can_health_score_is_always_bounded(steps in 1usize..240usize) { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 27 |     let mut sim = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 28 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 29 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 30 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 31 | (blank) | line is blank; visual separation only. |
| 32 |     for _ in 0..steps { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 33 |         sim.tick(1.0 / 60.0); | executable/support line; behavior impact depends on surrounding block context. |
| 34 |         let h = sim.can_net.network_health_score_01(); | executable/support line; behavior impact depends on surrounding block context. |
| 35 |         prop_assert!((0.0..=1.0).contains(&h)); | executable/support line; behavior impact depends on surrounding block context. |
| 36 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 37 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 38 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: tests/speed_regression.rs

Role: Automated regression and integration validation surface.

Lines: 66 | Non-empty: 53 | Symbols: 4

Symbol Inventory

Line	Kind	Symbol	Change Impact
4	fn	bootstrap_running_machine	signature/API coupling
13	fn	speed_increases_with_throttle_in_forward	signature/API coupling
28	fn	partial_throttle_launch_moves_machine_promptly	signature/API coupling
46	fn	brake_reduces_speed_after_acceleration	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	use auto_breaking::ecu_tcm::Direction;	imports symbols; changing path/name affects compile-time resolution.
2	use auto_breaking::HeavyMachinery;	imports symbols; changing path/name affects compile-time resolution.
3	(blank)	line is blank; visual separation only.
4	fn bootstrap_running_machine(sim: &mut HeavyMachinery) {	declares executable behavior; signature/body changes alter API, control flow, and outputs.
5	// OFF -> ACC -> ON -> START/RUN sequence	comment line; documentation intent, no runtime effect.
6	sim.key_advance();	executable/support line; behavior impact depends on surrounding block context.
7	sim.key_advance();	executable/support line; behavior impact depends on surrounding block context.
8	sim.key_advance();	executable/support line; behavior impact depends on surrounding block context.
9	sim.tcm.set_direction(Direction::Forward);	executable/support line; behavior impact depends on surrounding block context.
10	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
11	(blank)	line is blank; visual separation only.
12	#[test]	executable/support line; behavior impact depends on surrounding block context.
13	fn speed_increases_with_throttle_in_forward() {	declares executable behavior; signature/body changes alter API, control flow, and outputs.

```

| 14 |     let mut sim = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 15 |     bootstrap_running_machine(&mut sim); | executable/support line; behavior impact depends on surrounding block context. |
| 16 | (blank) | line is blank; visual separation only. |
| 17 |     sim.throttle_pct = 55.0; | executable/support line; behavior impact depends on surrounding block context. |
| 18 |     sim.brake_pct = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 19 | (blank) | line is blank; visual separation only. |
| 20 |     for _ in 0..600 { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 21 |         sim.tick(1.0 / 60.0); | executable/support line; behavior impact depends on surrounding block context. |
| 22 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 23 | (blank) | line is blank; visual separation only. |
| 24 |     assert!(sim.tcm.ground_speed_kmh > 3.0); | executable/support line; behavior impact depends on surrounding block context. |
| 25 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 26 | (blank) | line is blank; visual separation only. |
| 27 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 28 | fn partial_throttle_launch_moves_machine_promptly() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 29 |     let mut sim = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 30 |     bootstrap_running_machine(&mut sim); | executable/support line; behavior impact depends on surrounding block context. |
| 31 | (blank) | line is blank; visual separation only. |
| 32 |     sim.throttle_pct = 30.0; | executable/support line; behavior impact depends on surrounding block context. |
| 33 |     sim.brake_pct = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 34 | (blank) | line is blank; visual separation only. |
| 35 |     for _ in 0..240 { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 36 |         sim.tick(1.0 / 60.0); | executable/support line; behavior impact depends on surrounding block context. |
| 37 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 38 | (blank) | line is blank; visual separation only. |
| 39 |     assert!( | executable/support line; behavior impact depends on surrounding block context. |
| 40 |         sim.tcm.ground_speed_kmh > 2.0, | executable/support line; behavior impact depends on surrounding block context. |
| 41 |         "partial throttle should still move the machine visibly from launch" | executable/support line; behavior impact depends on surrounding block context. |
| 42 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 43 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 44 | (blank) | line is blank; visual separation only. |
| 45 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 46 | fn brake_reduces_speed_after_acceleration() { | declares executable behavior; signature/body changes alter API, control flow, and outputs. |
| 47 |     let mut sim = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding block context. |
| 48 |     bootstrap_running_machine(&mut sim); | executable/support line; behavior impact depends on surrounding block context. |
| 49 | (blank) | line is blank; visual separation only. |
| 50 |     sim.throttle_pct = 65.0; | executable/support line; behavior impact depends on surrounding block context. |
| 51 |     sim.brake_pct = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 52 |     for _ in 0..500 { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 53 |         sim.tick(1.0 / 60.0); | executable/support line; behavior impact depends on surrounding block context. |
| 54 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 55 |     let v_before = sim.tcm.ground_speed_kmh; | executable/support line; behavior impact depends on surrounding block context. |
| 56 | (blank) | line is blank; visual separation only. |
| 57 |     sim.throttle_pct = 0.0; | executable/support line; behavior impact depends on surrounding block context. |
| 58 |     sim.brake_pct = 70.0; | executable/support line; behavior impact depends on surrounding block context. |
| 59 |     for _ in 0..250 { | iteration control; changing bounds/body affects throughput, timing, and possibly safety guarantees. |
| 60 |         sim.tick(1.0 / 60.0); | executable/support line; behavior impact depends on surrounding block context. |
| 61 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 62 |     let v_after = sim.tcm.ground_speed_kmh; | executable/support line; behavior impact depends on surrounding

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```
ck context. |
| 63 | (blank) | line is blank; visual separation only. |
| 64 |     assert!(v_before > 4.0); | executable/support line; behavior impact depends on surrounding block context.
| 65 |     assert!(v_after < v_before); | executable/support line; behavior impact depends on surrounding block conte
|
| 66 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
```

File: tests/system_failure_scenarios.rs

Role: Automated regression and integration validation surface.

Lines: 81 | Non-empty: 65 | Symbols: 6

Symbol Inventory

```
| Line | Kind | Symbol | Change Impact |
|---:|---|---|---|
| 3 | fn | boot_running | signature/API coupling |
| 10 | fn | valve_stuck_scenario_keeps_hydraulic_bounds | signature/API coupling |
| 25 | fn | sensor_bias_scenario_keeps_controller_stable | signature/API coupling |
| 39 | fn | can_message_loss_fault_transitions_bus_state | signature/API coupling |
| 51 | fn | message_storm_scenario_degrades_but_bounds_health | signature/API coupling |
| 68 | fn | rapid_setpoint_change_preserves_runtime_safety_bounds | signature/API coupling |
```

Line-by-Line Impact Map

```
| Line | Source | Operational Impact If Changed |
|---:|---|---|
| 1 | use auto_breaking::HeavyMachinery; | imports symbols; changing path/name affects compile-time resolution. |
| 2 | (blank) | line is blank; visual separation only. |
| 3 | fn boot_running(sim: &mut HeavyMachinery) { | declares executable behavior; signature/body changes alter API, c
rol flow, and outputs. |
| 4 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 5 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 6 |     sim.key_advance(); | executable/support line; behavior impact depends on surrounding block context. |
| 7 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 8 | (blank) | line is blank; visual separation only. |
| 9 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 10 | fn valve_stuck_scenario_keeps_hydraulic_bounds() { | declares executable behavior; signature/body changes alter
PI, control flow, and outputs. |
| 11 |     let mut sim = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding blo
context. |
| 12 |     boot_running(&mut sim); | executable/support line; behavior impact depends on surrounding block context. |
| 13 | (blank) | line is blank; visual separation only. |
| 14 |     sim.loader_lift_cmd = 1.0; | executable/support line; behavior impact depends on surrounding block context
| 15 |     for _ in 0..300 { | iteration control; changing bounds/body affects throughput, timing, and possibly safet
uarantees. |
| 16 |         sim.loader_lift_cmd = 1.0; | executable/support line; behavior impact depends on surrounding block con
t. |
| 17 |         sim.tick(1.0 / 60.0); | executable/support line; behavior impact depends on surrounding block context.
| 18 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 19 | (blank) | line is blank; visual separation only. |
| 20 |     assert!(sim.hcm.system_pressure_bar >= 0.0); | executable/support line; behavior impact depends on surroun
g block context. |
| 21 |     assert!((0.0..=100.0).contains(&sim.hcm.fluid_level_pct)); | executable/support line; behavior impact depe
on surrounding block context. |
| 22 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 23 | (blank) | line is blank; visual separation only. |
| 24 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 25 | fn sensor_bias_scenario_keeps_controller_stable() { | declares executable behavior; signature/body changes alt
API, control flow, and outputs. |
| 26 |     let mut sim = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding blo
context. |
| 27 |     boot_running(&mut sim); | executable/support line; behavior impact depends on surrounding block context. |
| 28 | (blank) | line is blank; visual separation only. |
| 29 |     for _ in 0..200 { | iteration control; changing bounds/body affects throughput, timing, and possibly safet
```

```

uarantees. |
| 30 |      sim.ecm.coolant_temp_c += 0.03; // persistent positive bias | executable/support line; behavior impact
depends on surrounding block context. |
| 31 |      sim.tick(1.0 / 60.0); | executable/support line; behavior impact depends on surrounding block context.
| 32 |    } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 33 | (blank) | line is blank; visual separation only. |
| 34 |    assert!(sim.ecm.rpm >= 0.0); | executable/support line; behavior impact depends on surrounding block conte
|
| 35 |    assert!(sim.ecm.coolant_temp_c < 140.0); | executable/support line; behavior impact depends on surrounding
ock context. |
| 36 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 37 | (blank) | line is blank; visual separation only. |
| 38 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 39 | fn can_message_loss_fault_transitions_bus_state() { | declares executable behavior; signature/body changes alt
API, control flow, and outputs. |
| 40 |     let mut sim = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding blo
context. |
| 41 |     boot_running(&mut sim); | executable/support line; behavior impact depends on surrounding block context. |
| 42 | (blank) | line is blank; visual separation only. |
| 43 |     sim.can_net | executable/support line; behavior impact depends on surrounding block context. |
| 44 |     .inject_bus_off_once(auto_breaking::BusId::PowertrainHs); | executable/support line; behavior impact d
nds on surrounding block context. |
| 45 |     sim.tick(1.0 / 60.0); | executable/support line; behavior impact depends on surrounding block context. |
| 46 | (blank) | line is blank; visual separation only. |
| 47 |     assert!(sim.can_net.powertrain.state != auto_breaking::CanBusState::ErrorActive); | executable/support lin
behavior impact depends on surrounding block context. |
| 48 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 49 | (blank) | line is blank; visual separation only. |
| 50 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 51 | fn message_storm_scenario_degrades_but_bounds_health() { | declares executable behavior; signature/body change
lter API, control flow, and outputs. |
| 52 |     let mut sim = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding blo
context. |
| 53 |     boot_running(&mut sim); | executable/support line; behavior impact depends on surrounding block context. |
| 54 | (blank) | line is blank; visual separation only. |
| 55 |     sim.can_net.inject_error_once( | executable/support line; behavior impact depends on surrounding block con
t. |
| 56 |         auto_breaking::BusId::PowertrainHs, | executable/support line; behavior impact depends on surrounding
ck context. |
| 57 |         auto_breaking::CanErrorKind::BabblingIdiot, | executable/support line; behavior impact depends on surro
ing block context. |
| 58 |     ); | executable/support line; behavior impact depends on surrounding block context. |
| 59 |     for _ in 0..180 { | iteration control; changing bounds/body affects throughput, timing, and possibly safet
uarantees. |
| 60 |         sim.tick(1.0 / 60.0); | executable/support line; behavior impact depends on surrounding block context.
| 61 |     } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 62 | (blank) | line is blank; visual separation only. |
| 63 |     let h = sim.can_net.network_health_score_01(); | executable/support line; behavior impact depends on surro
ing block context. |
| 64 |     assert!((0.0..=1.0).contains(&h)); | executable/support line; behavior impact depends on surrounding block
ntext. |
| 65 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 66 | (blank) | line is blank; visual separation only. |
| 67 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 68 | fn rapid_setpoint_change_preserves_runtime_safety_bounds() { | declares executable behavior; signature/body ch
es alter API, control flow, and outputs. |
| 69 |     let mut sim = HeavyMachinery::new(); | executable/support line; behavior impact depends on surrounding blo
context. |
| 70 |     boot_running(&mut sim); | executable/support line; behavior impact depends on surrounding block context. |
| 71 | (blank) | line is blank; visual separation only. |
| 72 |     for i in 0..500 { | iteration control; changing bounds/body affects throughput, timing, and possibly safet
uarantees. |
| 73 |         sim.throttle_pct = if i % 20 < 10 { 90.0 } else { 5.0 }; | executable/support line; behavior impact de
ds on surrounding block context. |
| 74 |         sim.brake_pct = if i % 35 < 5 { 35.0 } else { 0.0 }; | executable/support line; behavior impact depend

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n surrounding block context. |
| 75 |      sim.tick(1.0 / 60.0); | executable/support line; behavior impact depends on surrounding block context.
| 76 |    } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 77 | (blank) | line is blank; visual separation only. |
| 78 |    assert!((0.0..=100.0).contains(&sim.ecm.fuel_level_pct)); | executable/support line; behavior impact depends
on surrounding block context. |
| 79 |    assert!(sim.hcm.system_pressure_bar >= 0.0); | executable/support line; behavior impact depends on surroundin
g block context. |
| 80 |    assert!(sim.ecm.oil_pressure_kpa >= 0.0); | executable/support line; behavior impact depends on surroundin
lock context. |
| 81 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

File: tests/vendor_bridge_e2e.rs

Role: Automated regression and integration validation surface.

Lines: 81 | Non-empty: 68 | Symbols: 3

Symbol Inventory

Line	Kind	Symbol	Change Impact
9	fn	bin_path	signature/API coupling
22	fn	unique_temp_dir	signature/API coupling
33	fn	cat_comm_bridge_probe_and_live_phases_flash_e2e	signature/API coupling

Line-by-Line Impact Map

Line	Source	Operational Impact If Changed
1	#![cfg(all(target_os = "windows", feature = "vendor-windows"))]	executable/support line; behavior impact depends on surrounding block context.
2	(blank)	line is blank; visual separation only.
3	use std::fs;	imports symbols; changing path/name affects compile-time resolution.
4	use std::path::{Path, PathBuf};	imports symbols; changing path/name affects compile-time resolution.
5	(blank)	line is blank; visual separation only.
6	use auto_breaking::io::hw::{probe_live_adapter, HwConfig};	imports symbols; changing path/name affects compile-time resolution.
7	use auto_breaking::io::production_program::{run_all_phases, ProgramOptions};	imports symbols; changing path/name affects compile-time resolution.
8	(blank)	line is blank; visual separation only.
9	fn bin_path(bin_name: &str) -> PathBuf {	declares executable behavior; signature/body changes alter API, control flow, and outputs.
10	let env_key = format!("CARGO_BIN_EXE_{bin_name}");	executable/support line; behavior impact depends on surrounding block context.
11	if let Ok(p) = std::env::var(&env_key) {	conditional gate; changing condition shifts runtime path and side effects.
12	return PathBuf::from(p);	executable/support line; behavior impact depends on surrounding block context.
13	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
14	(blank)	line is blank; visual separation only.
15	let mut fallback = PathBuf::from(env!("CARGO_MANIFEST_DIR"));	executable/support line; behavior impact depends on surrounding block context.
16	fallback.push("target");	executable/support line; behavior impact depends on surrounding block context.
17	fallback.push("debug");	executable/support line; behavior impact depends on surrounding block context.
18	fallback.push(format!("{bin_name}.exe"));	executable/support line; behavior impact depends on surrounding block context.
19	fallback	executable/support line; behavior impact depends on surrounding block context.
20	}	scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes.
21	(blank)	line is blank; visual separation only.
22	fn unique_temp_dir(prefix: &str) -> PathBuf {	declares executable behavior; signature/body changes alter API, control flow, and outputs.
23	let ts = std::time::SystemTime::now();	executable/support line; behavior impact depends on surrounding block context.
24	.duration_since(std::time::UNIX_EPOCH)	executable/support line; behavior impact depends on surrounding block context.

```

| 25 |         .map(\|d\| d.as_millis()) | executable/support line; behavior impact depends on surrounding block context. |
| 26 |         .unwrap_or(0); | executable/support line; behavior impact depends on surrounding block context. |
| 27 |     let p = std::env::temp_dir().join(format!("{prefix}_{ts}")); | executable/support line; behavior impact depends on surrounding block context. |
| 28 |     fs::create_dir_all(&p).expect("create temp dir"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 29 |     p | executable/support line; behavior impact depends on surrounding block context. |
| 30 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |
| 31 | (blank) | line is blank; visual separation only. |
| 32 | #[test] | executable/support line; behavior impact depends on surrounding block context. |
| 33 | fn cat_comm_bridge_probe_and_live_phases_flash_e2e() { | declares executable behavior; signature/body changes per API, control flow, and outputs. |
| 34 |     let template_dir = unique_temp_dir("autobreaking_cat_template"); | executable/support line; behavior impact depends on surrounding block context. |
| 35 |     let bridge_src = bin_path("cat_comm_bridge"); | executable/support line; behavior impact depends on surrounding block context. |
| 36 |     assert!(bridge_src.exists(), "cat_comm_bridge binary not found at {}", bridge_src.display()); | executable/support line; behavior impact depends on surrounding block context. |
| 37 | (blank) | line is blank; visual separation only. |
| 38 |     let bridge_dst = template_dir.join("cat_comm_bridge.exe"); | executable/support line; behavior impact depends on surrounding block context. |
| 39 |     fs::copy(&bridge_src, &bridge_dst).expect("copy bridge exe"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 40 | (blank) | line is blank; visual separation only. |
| 41 |     let allowlist_path = template_dir.join("allowlist.json"); | executable/support line; behavior impact depends on surrounding block context. |
| 42 |     fs::write(&allowlist_path, "").expect("write allowlist"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 43 | (blank) | line is blank; visual separation only. |
| 44 |     let firmware_path = template_dir.join("firmware.bin"); | executable/support line; behavior impact depends on surrounding block context. |
| 45 |     let firmware_payload: Vec = (0..512).map(\|i\| (i % 251) as u8).collect(); | executable/support line; behavior impact depends on surrounding block context. |
| 46 |     fs::write(&firmware_path, &firmware_payload).expect("write firmware"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 47 | (blank) | line is blank; visual separation only. |
| 48 |     let args = vec![ | executable/support line; behavior impact depends on surrounding block context. |
| 49 |         "simulator_cli".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 50 |         "--hw-mode=live".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 51 |         "--vendor-name=cat_comm".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 52 |         format!("--vendor-template-dir={}", template_dir.display()), | executable/support line; behavior impact depends on surrounding block context. |
| 53 |         "--enable-write".to_string(), | I/O or protocol-critical logic; edits can affect integration, timing, and diagnostic outcomes. |
| 54 |         "--noninteractive-approved".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 55 |         "--dry-run=false".to_string(), | executable/support line; behavior impact depends on surrounding block context. |
| 56 |         format!("--allowlist={}", allowlist_path.display()), | executable/support line; behavior impact depends on surrounding block context. |
| 57 |     ]; | executable/support line; behavior impact depends on surrounding block context. |
| 58 | (blank) | line is blank; visual separation only. |
| 59 |     let cfg = HwConfig::from_cli_args(&args).expect("parse cfg"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 60 |     let info = probe_live_adapter(&cfg).expect("probe live adapter"); | panic-prone operation; changing/removing affects crash behavior and robustness. |
| 61 |     assert!(info.contains("bridge_protocol"), "unexpected adapter info: {info}"); | executable/support line; behavior impact depends on surrounding block context. |
| 62 | (blank) | line is blank; visual separation only. |
| 63 |     let report_dir = template_dir.join("reports"); | executable/support line; behavior impact depends on surrounding block context. |

```

```

| 64 |     let report_path = run_all_phases( | executable/support line; behavior impact depends on surrounding block
text. |
| 65 |         &cfg, | executable/support line; behavior impact depends on surrounding block context. |
| 66 |         Some(0x00), | executable/support line; behavior impact depends on surrounding block context. |
| 67 |         Some(Path::new(&firmware_path)), | executable/support line; behavior impact depends on surrounding blo
context. |
| 68 |         &report_dir, | executable/support line; behavior impact depends on surrounding block context. |
| 69 |         ProgramOptions { | executable/support line; behavior impact depends on surrounding block context. |
| 70 |             strict_mode: true, | executable/support line; behavior impact depends on surrounding block context
| 71 |             execute_flash: true, | I/O or protocol-critical logic; edits can affect integration, timing, and d
nostic outcomes. |
| 72 |         }, | executable/support line; behavior impact depends on surrounding block context. |
| 73 |     ) | executable/support line; behavior impact depends on surrounding block context. |
| 74 |     .expect("run all phases"); | panic-prone operation; changing/removing affects crash behavior and robustnes
|
| 75 | (blank) | line is blank; visual separation only. |
| 76 |     let report = fs::read_to_string(&report_path).expect("read report"); | panic-prone operation; changing/rem
ng affects crash behavior and robustness. |
| 77 |     assert!(report.contains("\n\"overall_passed\": true"), "report did not pass gate: {report}"); | executable/s
ort line; behavior impact depends on surrounding block context. |
| 78 |     assert!(report.contains("\n\"conformance_summary\"", "missing conformance summary: {report}"); | executable
pport line; behavior impact depends on surrounding block context. |
| 79 |     assert!(report.contains("\n\"service_sid\": \"0x36\"", "missing SID 0x36 evidence: {report}"); | executable
pport line; behavior impact depends on surrounding block context. |
| 80 |     assert!(report.contains("request_transfer_exit_positive=true"), "missing transfer exit evidence: {report}")
| executable/support line; behavior impact depends on surrounding block context. |
| 81 | } | scope delimiter; structure-only line, but misplaced edits can change ownership and lifetime scopes. |

```

Change Safety Guide

- Prefer additive changes in protocol structs; preserve serde field names for backward compatibility.
- In flashing paths, preserve positive-response validation and do not weaken gate thresholds.
- For trait/interface changes in io/hw, run full workspace tests plus vendor-windows E2E.
- For any change touching line maps in live_runner/vendor_cat_comm/production_program, re-validate conformance eviden output.